Microlocal Analysis and Index Theory

A streamlined edition of three sets of lecture notes by Richard Melrose

After R. B. Melrose

Introduction to Microlocal Analysis (MIT, 2003),
Lectures on Pseudodifferential Operators (MIT 18.157, 2005–2006) and
From Microlocal to Global Analysis (MIT, 2008)

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Preface

This book combines three sets of lecture notes by Richard Melrose: *Introduction to Microlocal Analysis* (MIT, version 0.5B, 2003), a revised version of the notes from several courses he taught at MIT; *Lectures on Pseudodifferential Operators* (MIT course 18.157, Fall 2005, revised 2006); and *From Microlocal to Global Analysis* (MIT, version 0.7I, 2008), a later expansion of the first.

Microlocal analysis is a geometric theory of distributions, or equivalently a theory of geometric distributions. Rather than general distributions, which are like general continuous functions only worse, it studies the distributions that actually arise from differential and integral equations. Conormal distributions and, more generally, Lagrangian distributions are smooth away from a well-defined set and have a precisely described singularity there. Their singularities are the main subject.

Prerequisites and suggested reading

The reader should already know some distribution theory and some basic analysis, and some familiarity with analysis on manifolds will help. If one or two of these are missing they can be picked up along the way. A reader who has none of them should first do some preliminary reading. Friedlander's book [3] is a good introduction to distributions, and Volume I of Hörmander's treatise [7] gives a much fuller treatment that also serves as a general reference. For analysis on manifolds, most of what we need can be found in Munkres' book [9] or in Spivak's short book [12]. Chapter 1 reviews tempered distributions and the Fourier transform, so that the book is reasonably self-contained.

About this edition

This is a streamlined edition. The material from the three sources has been merged and reorganized into five parts.

- Part I builds the calculus: tempered distributions and compactifications, pseudodifferential operators on $\mathbb{R}^n$, microlocalization, conormal distributions and operators on manifolds, and the semiclassical and adiabatic calculus.
- Part II covers the global calculi on $\mathbb{R}^n$: the isotropic calculus, and traces and determinants.
- Part III covers boundary problems and propagation: boundary problems, the Lagrangian distributions, the Hamilton–Jacobi and transport equations, the wave equation and Weyl asymptotics.
- Part IV is index theory, taken mainly from the 18.157 notes: homotopy invariance of the index, Toeplitz operators, K-theory and Bott periodicity, product-type operators, the families index theorem (with a second proof by semiclassical and adiabatic limits, from the 2008 notes) and eta forms.
- Part V covers the algebraic structure of the calculus: Hochschild homology.

Notation has been simplified and made uniform throughout. Proofs that were only sketched, or left as exercises but needed later, have been completed, and errors have been corrected. The "addenda" to the 18.157 lectures have been moved into the main text or the exercises. Draft material has been removed, including chapters that existed in the originals only as headings. Each chapter ends with exercises and a short note on its main departures from the original notes.

How to read this book

The book is written for readers at the beginning of graduate school. Each chapter opens by saying what problem it solves and how it connects to what came before. Short *Discussion* paragraphs explain the idea behind a definition or a proof, and *Question* and *Answer* pairs take up points that often puzzle a first-time reader. These can be skipped by readers who already know the material.

Figure 1 shows how the chapters depend on each other. Part I is the foundation for everything else. The other four parts depend on Part I but are largely independent of each other. The main exception is Part IV: the proof of Bott periodicity in Chapter 13 uses the isotropic calculus of Chapter 7, and traces and determinants from Chapter 8 appear occasionally. Within each of Parts III and IV the chapters are best read in order.

For a first one-semester course we suggest the following path through Part I.

- Chapter 1, possibly quickly if distributions are familiar. The radial compactification (Section 1.7) is used throughout and should not be skipped; Section 1.8 can wait.
- Chapter 2 up to and including Sobolev spaces (Section 2.8). The later sections on topologies and holomorphic families are needed only for complex powers and determinants.
- Chapter 3: wavefront sets, the characteristic variety and microlocal elliptic regularity, followed by propagation of singularities (Section 3.20).
- Chapter 5 up to the Fredholm theory of elliptic operators (Section 5.9). The facts about conormal distributions that it uses are recalled in Section 5.3, so Chapter 4 can be read lightly on a first pass.

A second course can then follow either the analytic route (Part III) or the topological route (Part II and Part IV).

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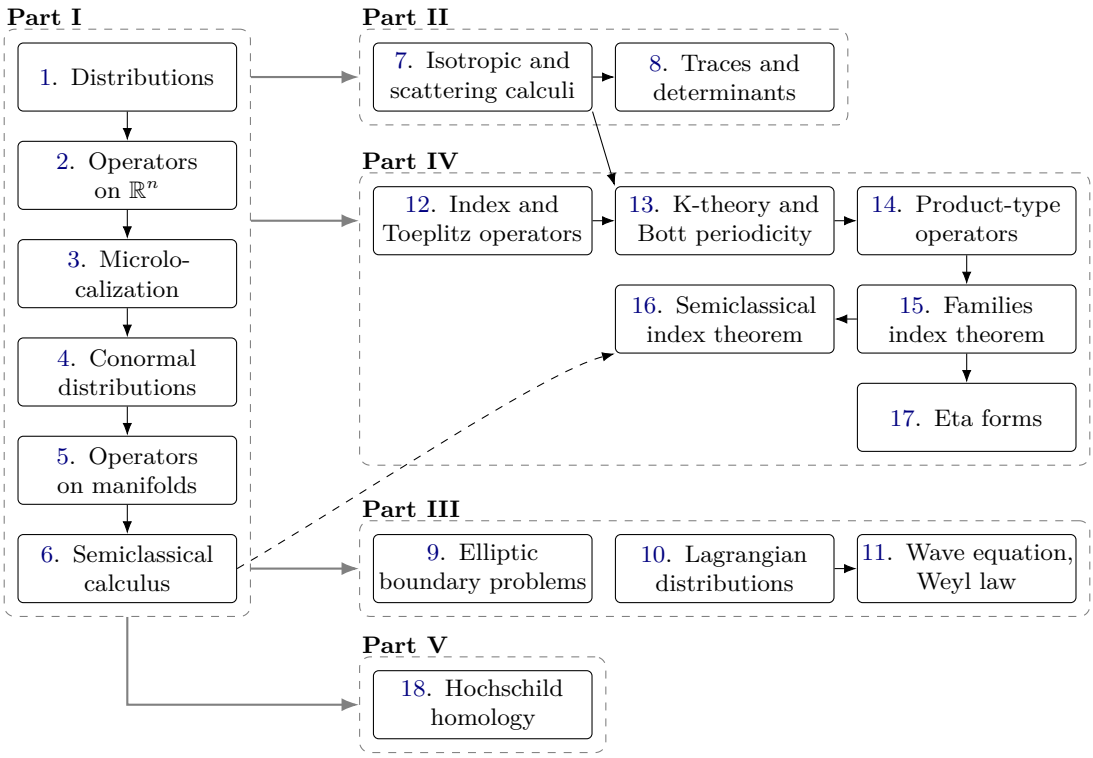


Figure 1: Dependencies between the chapters. Grey arrows mean that a part uses Part I; black arrows show the main dependencies between individual chapters. The dashed arrow indicates that the semiclassical proof of the index theorem in Chapter 16 uses the semiclassical calculus of Chapter 6.

Part I

The calculus

Chapter 1

Tempered distributions and the Fourier transform

Why do we need distributions at all? A linear differential equation $Pu = f$ usually has solutions that are not smooth, and the fundamental solutions of the simplest operators (the Laplacian, the wave operator) are not even functions. Distributions give a space large enough to contain all such solutions while still allowing us to differentiate, multiply by smooth functions and take Fourier transforms. The Fourier transform turns differentiation into multiplication by a polynomial. This simple fact is the seed of the whole theory of pseudodifferential operators.

This chapter sets out the analytic framework for the rest of the book. Microlocal analysis studies distributions with special singularities, and we work in the space $\mathcal{S}'(\mathbb{R}^n)$ of tempered distributions, the dual of the Schwartz space $\mathcal{S}(\mathbb{R}^n)$. The Fourier transform is an isomorphism of both spaces, and it is the main tool for everything that follows.

We review the Schwartz space and its topology (Section 1.1), define tempered distributions and the basic operations on them (Section 1.2), and prove two elementary lemmas (Section 1.3). These lemmas give a proof of the Fourier inversion formula that only uses the structure of distributions (Section 1.4). Section 1.5 applies the Fourier transform to differential operators. With the Fourier transform available, we prove the two main structural results of the theory in Section 1.6: the representation of tempered distributions as derivatives of bounded continuous functions, and the Schwartz kernel theorem. The last two sections cover compactifications of a vector space. The radial compactification $\overline{\mathbb{R}^n}$ is a compact manifold with boundary. Schwartz functions on $\mathbb{R}^n$ are exactly the smooth functions on $\overline{\mathbb{R}^n}$ that vanish to infinite order at the boundary, and smooth functions on $\overline{\mathbb{R}^n}$ are classical symbols of order zero. This is the geometric picture behind the symbol spaces of Chapter 2.

Notation. Multi-indices are $\alpha \in \mathbb{N}_0^n$, with $|\alpha| = \alpha_1 + \cdots + \alpha_n$, $\alpha! = \alpha_1! \cdots \alpha_n!$, $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$. We write $\partial_j = \partial/\partial x_j$, $D_j = -i\partial_j$, $D^\alpha = D_1^{\alpha_1} \cdots D_n^{\alpha_n}$ and $\langle x \rangle = (1 + |x|^2)^{1/2}$. Note that

$$\max(1, |x|) \le \langle x \rangle \le 1 + |x|, \qquad |x^\alpha| \le \langle x \rangle^{|\alpha|}, \qquad \langle x \rangle^{2N} = \sum_{|\gamma| \le N} \frac{N!}{(N - |\gamma|)!\, \gamma!}\, x^{2\gamma}. \tag{1.0.1}$$

1.1 Schwartz test functions

1.1.1 Smoothness

A function $\phi\colon \mathbb{R}^n \to \mathbb{C}$ is differentiable at $\bar{x}$ if there is an affine function $\ell_{\bar{x}}(x) = c + \sum_j d_j(x_j - \bar{x}_j)$ with $|\phi(x) - \ell_{\bar{x}}(x)| = o(|x - \bar{x}|)$. Then $c = \phi(\bar{x})$ and $d_j = \partial_j\phi(\bar{x})$. Iterating, ϕ is k times differentiable if its partial derivatives are $(k-1)$ times differentiable. In that case Taylor's theorem gives, for each $\bar{x}$,

$$\phi(x) = \sum_{|\alpha|\le k} \frac{i^{|\alpha|}(D^\alpha\phi)(\bar{x})}{\alpha!}(x-\bar{x})^\alpha + o(|x-\bar{x}|^k), \qquad x \to \bar{x},$$

because $\partial^\alpha = i^{|\alpha|}D^\alpha$. Having such polynomial approximations of all orders at every point does *not* by itself imply smoothness (Exercise 1.1). We write $\mathcal{C}^\infty(\mathbb{R}^n)$ for the infinitely differentiable functions. Its subspace

$$\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n) = \{u \in \mathcal{C}^\infty(\mathbb{R}^n);\ \sup_x |D^\alpha u(x)| < \infty\ \forall\,\alpha\}$$

consists of the smooth functions all of whose derivatives are bounded.

1.1.2 The Schwartz space

Definition 1.1.1. The *Schwartz space* $\mathcal{S}(\mathbb{R}^n)$ of smooth functions of rapid decrease consists of those $\phi \in \mathcal{C}^\infty(\mathbb{R}^n)$ such that

$$\sup_{x\in\mathbb{R}^n} |x^\alpha D^\beta\phi(x)| < \infty \quad \text{for all } \alpha,\beta \in \mathbb{N}_0^n.$$

For $k \in \mathbb{N}_0$ we set

$$\|\phi\|_k = \max_{|\alpha|+|\beta|\le k} \sup_{x\in\mathbb{R}^n} |x^\alpha D^\beta\phi(x)|, \tag{1.1.1}$$

and define the distance

$$d(\phi,\psi) = \sum_{k=0}^{\infty} 2^{-k}\frac{\|\phi-\psi\|_k}{1+\|\phi-\psi\|_k}. \tag{1.1.2}$$

Each $\|\cdot\|_k$ is a norm: it is a seminorm, and $\|\phi\|_k \ge \sup|\phi|$. The norms increase with k, that is $\|\phi\|_k \le \|\phi\|_{k+1}$. By (1.0.1) an equivalent family of norms is $\max_{|\beta|\le k}\sup\langle x\rangle^k|D^\beta\phi|$. More precisely, each $\sup\langle x\rangle^p|x^\alpha D^\beta\phi|$ is bounded by a constant times $\|\phi\|_{p+|\alpha|+|\beta|}$, and we use this freely. That d is a metric is Exercise 1.3. A sequence converges, or is Cauchy, for d if and only if it converges, or is Cauchy, for every norm $\|\cdot\|_k$ (Exercise 1.4). Two elementary inclusions relate the balls of the metric and the norms. If $d(\phi,0) < 2^{-k-1}$ then $\|\phi\|_k < 1$. Conversely, for every p,

$$d(\phi,0) \le \sum_{k\le p} 2^{-k}\|\phi\|_k + \sum_{k>p} 2^{-k} \le 2\|\phi\|_p + 2^{-p}. \tag{1.1.3}$$

Proposition 1.1.2. *With the metric d, $\mathcal{S}(\mathbb{R}^n)$ is a complete metric space; it is a Fréchet space.*

Proof. Let (ϕ_j) be Cauchy for d, so it is Cauchy in each norm $\|\cdot\|_k$. Then for all α,β the functions $x^\alpha D^\beta\phi_j$ converge uniformly on $\mathbb{R}^n$ to bounded continuous functions $g_{\alpha\beta}$. If a sequence of C^1 functions converges uniformly and its first derivatives also converge uniformly, then the limit is C^1 and its derivatives are the limits of the derivatives. Applying this repeatedly shows that $\phi = g_{00}$ is smooth and $D^\beta\phi = g_{0\beta}$. Hence $g_{\alpha\beta} = x^\alpha D^\beta\phi$, which is

bounded, so $\phi \in \mathcal{S}(\mathbb{R}^n)$. Finally $\|\phi_j - \phi\|_k \to 0$ for each k, so $d(\phi_j, \phi) \to 0$. The metric is translation invariant and given by a countable family of seminorms, so $\mathcal{S}(\mathbb{R}^n)$ is a Fréchet space. □

The space is not trivial. For example $\exp(-|x|^2) \in \mathcal{S}(\mathbb{R}^n)$, since every derivative is a polynomial times $\exp(-|x|^2)$. Every $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ is in $\mathcal{S}(\mathbb{R}^n)$.

Discussion. Why this particular space of test functions? The conditions defining $\mathcal{S}(\mathbb{R}^n)$ are symmetric in two operations: multiplication by x_j and differentiation D_j. The Fourier transform exchanges these two operations (Section 1.4), so it maps $\mathcal{S}(\mathbb{R}^n)$ to itself. The space $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ does not have this property: the Fourier transform of a non-zero compactly supported function is never compactly supported. Choosing $\mathcal{S}$ as test functions is what makes the Fourier transform available on the dual space.

Lemma 1.1.3. *Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\chi = 1$ on $|x| \le 1$, and set $\chi_R(x) = \chi(x/R)$. Then for every k there is C_k such that*

$$\|(1 - \chi_R)\phi\|_k \le C_k R^{-1} \|\phi\|_{k+1}, \qquad \phi \in \mathcal{S}(\mathbb{R}^n),\ R \ge 1.$$

In particular $\chi_R \phi \to \phi$ in $\mathcal{S}(\mathbb{R}^n)$ as $R \to \infty$, so $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ is dense in $\mathcal{S}(\mathbb{R}^n)$.

Proof. By Leibniz' formula, $x^\alpha D^\beta((1-\chi_R)\phi)$ is a sum of terms $x^\alpha D^\gamma(1-\chi_R)\, D^{\beta-\gamma}\phi$. Each $D^\gamma(1 - \chi_R)$ is bounded uniformly in $R \ge 1$, because $D^\gamma \chi_R = R^{-|\gamma|}(D^\gamma \chi)(x/R)$. All of these terms vanish in $|x| \le R$. There $1 \le |x|/R \le \langle x \rangle / R$, so each term is bounded by $CR^{-1} \sup \langle x \rangle |x^\alpha D^{\beta-\gamma}\phi| \le C' R^{-1} \|\phi\|_{k+1}$. □

1.1.3 Continuous linear maps

Since the topology is given by the increasing family of norms $\|\cdot\|_k$, continuity of linear maps reduces to estimates.

Lemma 1.1.4. *A linear map $T\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^m)$ is continuous if and only if for every k there are N and C with $\|T\phi\|_k \le C\|\phi\|_N$ for all $\phi \in \mathcal{S}(\mathbb{R}^n)$. A linear functional $u\colon \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$ is continuous if and only if there are N and C with $|u(\phi)| \le C\|\phi\|_N$ for all ϕ.*

Proof. If the estimates hold, then $\phi_j \to \phi$ implies $\|T\phi_j - T\phi\|_k \to 0$ for all k, hence $T\phi_j \to T\phi$. Conversely, suppose T is continuous at 0 and fix k. There is $r > 0$ such that $d(\phi, 0) < r$ implies $d(T\phi, 0) < 2^{-k-1}$, and hence $\|T\phi\|_k < 1$. Choose p with $2^{-p} < r/2$ and $\epsilon = r/4$. By (1.1.3), $\|\phi\|_p < \epsilon$ implies $d(\phi, 0) < r$. By homogeneity it follows that $\|T\phi\|_k \le \epsilon^{-1}\|\phi\|_p$ for all ϕ. The functional case is the same argument with $|\cdot|$ in place of $\|\cdot\|_k$. □

Using this criterion it is straightforward to check that the following maps are continuous.

- *Multiplication and differentiation.* The maps $x_j, D_j\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ satisfy $\|x_j\phi\|_k + \|D_j\phi\|_k \le C\|\phi\|_{k+1}$. Indeed, $x^\alpha D^\beta(x_j\phi) = x^\alpha x_j D^\beta \phi - i\beta_j x^\alpha D^{\beta - e_j}\phi$.
- *Multiplication by $\mathcal{C}^\infty_{\mathrm{b}}$ functions.* If $a \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ then, by Leibniz' formula

$$D^\alpha(a\phi) = \sum_{\beta \le \alpha} \binom{\alpha}{\beta} D^\beta a\, D^{\alpha-\beta}\phi, \tag{1.1.4}$$

  we have $\|a\phi\|_k \le C_k\|\phi\|_k$, with C_k depending on finitely many derivatives of a. So $\mathcal{S}(\mathbb{R}^n)$ is a module over $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$. In particular $\langle x \rangle^{-s} \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ for $s \ge 0$ acts on $\mathcal{S}(\mathbb{R}^n)$.

- *Products.* Pointwise multiplication $\mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ makes $\mathcal{S}(\mathbb{R}^n)$ into a commutative algebra without identity.
- *Exterior product.* The bilinear map

$$\mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^m) \ni (\phi, \psi) \longmapsto \phi \boxtimes \psi \in \mathcal{S}(\mathbb{R}^{n+m}), \qquad \phi \boxtimes \psi(x,y) = \phi(x)\psi(y),$$

  satisfies $\|\phi\boxtimes\psi\|_k \leq \|\phi\|_k\|\psi\|_k$, since $(x,y)^{(\alpha,\alpha')}D^{(\beta,\beta')}(\phi\boxtimes\psi) = (x^\alpha D^\beta\phi)\boxtimes(y^{\alpha'}D^{\beta'}\psi)$.
- *Restriction.* If $\mathbb{R}^k \subset \mathbb{R}^n$ is the subspace $x_j = 0$, $j > k$, then $\pi_k^*\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^k)$, $\pi_k^* f(y) = f(y_1, \ldots, y_k, 0, \ldots, 0)$, is continuous and surjective (Exercise 1.6).

1.2 Tempered distributions

1.2.1 Definition

A tempered distribution is a "generalized function" that we only know through its averages $\int u\phi\, dx$ against Schwartz functions. We first see how ordinary functions give such averages, then take the abstract definition: any continuous linear functional on $\mathcal{S}(\mathbb{R}^n)$.

Integration $\int\colon \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$ is continuous, since $|\int \phi\, dx| \leq \int\langle x\rangle^{-n-1}dx \cdot \sup\langle x\rangle^{n+1}|\phi|$. Combined with multiplication it gives the bilinear pairing $(\phi,\psi) \mapsto \int \phi\psi\, dx$. For fixed $\phi \in \mathcal{S}(\mathbb{R}^n)$ the functional

$$T_\phi\colon \mathcal{S}(\mathbb{R}^n) \ni \psi \longmapsto \int_{\mathbb{R}^n} \phi(x)\psi(x)\, dx \tag{1.2.1}$$

satisfies $|T_\phi(\psi)| \leq C\|\psi\|_{n+1}$.

Definition 1.2.1. The space $\mathcal{S}'(\mathbb{R}^n)$ of *tempered distributions* is the space of continuous linear functionals $u\colon \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$. By Lemma 1.1.4 a linear functional u is in $\mathcal{S}'(\mathbb{R}^n)$ if and only if there are k and C with

$$|u(\phi)| \leq C\|\phi\|_k \quad \forall\, \phi \in \mathcal{S}(\mathbb{R}^n). \tag{1.2.2}$$

We also write $\langle u, \phi\rangle = u(\phi)$.

Lemma 1.2.2. *The map $\mathcal{S}(\mathbb{R}^n) \ni \phi \mapsto T_\phi \in \mathcal{S}'(\mathbb{R}^n)$ is injective.*

Proof. $T_\phi(\bar\phi) = \int |\phi|^2 dx$, so $T_\phi = 0$ implies $\phi = 0$. □

More generally, if u is a measurable function with $\langle x\rangle^{-N}u \in L^1(\mathbb{R}^n)$ for some N, then $T_u(\psi) = \int u\psi\, dx$ satisfies $|T_u(\psi)| \leq \|\langle x\rangle^{-N}u\|_{L^1}\sup\langle x\rangle^N|\psi|$, so $T_u \in \mathcal{S}'(\mathbb{R}^n)$. This applies to bounded continuous functions, to $L^2(\mathbb{R}^n)$ and to polynomials. If u is continuous and $T_u = 0$ then $u = 0$: otherwise $\operatorname{Re} u$ or $\operatorname{Im} u$ has a strict sign near some point, and pairing with a non-negative bump function supported there gives a contradiction. For $u \in L^2(\mathbb{R}^n)$ see Exercise 1.8. We therefore identify such functions with the distributions they define, and simply write u for T_u.

1.2.2 The weak topology

We give $\mathcal{S}'(\mathbb{R}^n)$ the *weak topology*. This is the weakest topology in which all the maps $\mathcal{S}'(\mathbb{R}^n) \ni u \mapsto u(\phi) \in \mathbb{C}$, $\phi \in \mathcal{S}(\mathbb{R}^n)$, are continuous. A basis of neighbourhoods of u_0 is given by the sets

$$\{u \in \mathcal{S}'(\mathbb{R}^n);\ |u(\phi_j) - u_0(\phi_j)| < \epsilon_j,\ j = 1, \ldots, p\}, \tag{1.2.3}$$

where $\{\phi_1, \ldots, \phi_p\} \subset \mathcal{S}(\mathbb{R}^n)$ is finite and $\epsilon_j > 0$. A map into $\mathcal{S}'(\mathbb{R}^n)$ is continuous if and only if its composites with all the evaluations $u \mapsto u(\phi)$ are continuous.

Proposition 1.2.3. *$\mathcal{S}(\mathbb{R}^n)$, embedded by $\phi \mapsto T_\phi$, is dense in $\mathcal{S}'(\mathbb{R}^n)$.*

Proof. It suffices to show the following. For $u \in \mathcal{S}'(\mathbb{R}^n)$ and $\phi_1, \dots, \phi_p \in \mathcal{S}(\mathbb{R}^n)$ there is $\psi \in \mathcal{S}(\mathbb{R}^n)$ with $T_\psi(\phi_j) = u(\phi_j)$ for all j. Then T_ψ lies in every neighbourhood (1.2.3) of u. Let $\phi'_1, \dots, \phi'_q$ be a basis of $\operatorname{span}\{\phi_j\}$ that is orthonormal for the L^2 inner product, and set

$$\psi = \sum_{l=1}^{q} u(\phi'_l)\,\overline{\phi'_l} \in \mathcal{S}(\mathbb{R}^n).$$

Then $T_\psi(\phi'_m) = \sum_l u(\phi'_l) \int \overline{\phi'_l}\phi'_m\,dx = u(\phi'_m)$ for each m. By linearity, $T_\psi = u$ on the span of the ϕ_j. □

1.2.3 Operations by duality

If $T\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is continuous and linear and has a continuous *transpose* T^t, meaning $\int (T\phi)\psi = \int \phi\,(T^t\psi)$, then $(Tu)(\psi) = u(T^t\psi)$ defines a weakly continuous extension $T\colon \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$. It is consistent with the inclusion $\mathcal{S} \subset \mathcal{S}'$, and it is the unique continuous extension by Proposition 1.2.3. Integration by parts shows that $D_j^t = -D_j$, and directly from the definition $x_j^t = x_j$ and $a^t = a$ for multiplication by $a \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$. Thus for $u \in \mathcal{S}'(\mathbb{R}^n)$, $\psi \in \mathcal{S}(\mathbb{R}^n)$ and $a \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ we define

$$D_j u(\psi) = u(-D_j\psi), \qquad x_j u(\psi) = u(x_j\psi), \qquad au(\psi) = u(a\psi). \tag{1.2.4}$$

These are continuous maps $\mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ which extend the corresponding operations on $\mathcal{S}(\mathbb{R}^n)$. Consequently, if u is a bounded continuous function then $x^\beta D^\alpha u \in \mathcal{S}'(\mathbb{R}^n)$ for all α, β.

Discussion. This is the basic principle for extending operations to distributions. We never "evaluate" a distribution at a point. Instead we move the operation onto the test function, where it makes sense, by the rule we would get from integration by parts if u were a function. Every extension in this book (differentiation, change of variables, Fourier transform, and later pseudodifferential operators) follows this pattern: find the transpose on test functions, check that it is continuous, and dualize.

Similarly, the exterior product extends to distributions. For $u \in \mathcal{S}'(\mathbb{R}^n)$ and $v \in \mathcal{S}'(\mathbb{R}^m)$ there is a unique $u \boxtimes v \in \mathcal{S}'(\mathbb{R}^{n+m})$ with $(u \boxtimes v)(\phi \boxtimes \psi) = u(\phi)v(\psi)$. We only need the special case $1 \boxtimes v$, which is constructed directly in Section 1.3. The general case follows from the kernel theorem in Section 1.6 (Exercise 1.13).

1.2.4 Linear changes of variables

A linear map $L\colon \mathbb{R}^n \to \mathbb{R}^n$, $(Lx)_j = \sum_k L_{jk}x_k$, is invertible if and only if $\det L \neq 0$, if and only if there is $c > 0$ with $c|x| \le |Lx| \le c^{-1}|x|$ for all x. For $L \in \mathrm{GL}(n, \mathbb{R})$ the pull-back is $L^*\phi = \phi \circ L$. By the chain rule

$$D_j(L^*\phi) = \sum_{k=1}^{n} L_{kj}\, L^*(D_k\phi), \tag{1.2.5}$$

so $x^\alpha D^\beta(L^*\phi)$ is a linear combination of the functions $x^\alpha L^*(D^\gamma\phi)$, $|\gamma| = |\beta|$. Since $\langle x\rangle \le C\langle Lx\rangle$, $L^*\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is continuous, and it is an isomorphism with inverse $(L^{-1})^*$. Changing variables $y = Lx$ in $\int \phi(Lx)\psi(x)\,dx$ gives

$$T_{L^*\phi}(\psi) = T_\phi(|\det L|^{-1}(L^{-1})^*\psi), \tag{1.2.6}$$

and so we define, for $u \in \mathcal{S}'(\mathbb{R}^n)$,

$$L^*u(\psi) = u(|\det L|^{-1}(L^{-1})^*\psi), \qquad \psi \in \mathcal{S}(\mathbb{R}^n).$$

This is an isomorphism $L^*\colon \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ extending L^* on $\mathcal{S}(\mathbb{R}^n)$. By density, (1.2.5) and the identity $L^*(x_j u) = (L^*x_j)(L^*u)$ continue to hold for $u \in \mathcal{S}'(\mathbb{R}^n)$.

1.2.5 Support

Definition 1.2.4. A distribution $u \in \mathcal{S}'(\mathbb{R}^n)$ *vanishes* on an open set $O \subset \mathbb{R}^n$ if $u(\phi) = 0$ for all $\phi \in \mathcal{C}_c^\infty(O)$. The *support* $\operatorname{supp} u$ is the complement of the union of all open sets on which u vanishes.

If u vanishes on each O_i, then it vanishes on $\bigcup_i O_i$. Indeed, a $\phi \in \mathcal{C}_c^\infty(\bigcup_i O_i)$ has compact support, which is covered by finitely many O_i, and a finite partition of unity splits ϕ into pieces supported in these sets. So u vanishes on $\mathbb{R}^n \setminus \operatorname{supp} u$. For continuous functions this agrees with the usual support (Exercise 1.10). If $\operatorname{supp} u \subset F$, F closed, and $\phi \in \mathcal{S}(\mathbb{R}^n)$ vanishes in a neighbourhood of F, then $u(\phi) = 0$. To see this, note that $\chi_R\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n \setminus F)$ in the notation of Lemma 1.1.3, and $\chi_R\phi \to \phi$ in $\mathcal{S}(\mathbb{R}^n)$.

1.3 Two little lemmas

The two lemmas of this section solve the simplest distributional equations, $D_j u = 0$ and $x_j u = 0$. They are good exercises in the definitions. More importantly, they are exactly what we need for the proof of the Fourier inversion formula in Section 1.4. Each is proved by a decomposition of test functions that is dual to the equation.

Discussion. Heuristically, $D_n u = 0$ says that u is constant in x_n, and $x_j u = 0$ for all j says that u lives only at the origin. The proofs make these intuitions precise in the only way available for distributions: by testing. To show $D_n u = 0$ forces u to be independent of x_n, we split every test function into a part that u sees only through $\int \phi\, dx_n$ and a part of the form $D_n\psi$, which u annihilates.

Write $x = (x', x_n) \in \mathbb{R}^{n-1} \times \mathbb{R}$ (by relabelling, any variable x_j can be treated in the same way). Define the integration map

$$I\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^{n-1}), \qquad I\phi(x') = \int_{\mathbb{R}} \phi(x', x_n)\, dx_n. \tag{1.3.1}$$

Differentiating under the integral and using $\langle x'\rangle^p \le \langle x\rangle^p$ gives $\|I\phi\|_k \le C\|\phi\|_{k+2}$, so I is continuous. For $v \in \mathcal{S}'(\mathbb{R}^{n-1})$ we may therefore define

$$(1 \boxtimes v)(\phi) = v(I\phi), \qquad \phi \in \mathcal{S}(\mathbb{R}^n). \tag{1.3.2}$$

Then $1 \boxtimes v \in \mathcal{S}'(\mathbb{R}^n)$, and $(1 \boxtimes v)(\phi' \boxtimes \phi_n) = v(\phi')\int \phi_n$. This is the "distribution $v(x')$ regarded as a function of x independent of x_n". (In the original notes it is written π^*v, with $\pi(x) = x'$.) Since $I(\theta \boxtimes \rho) = \theta$ whenever $\int \rho = 1$, the map I is surjective and $v \mapsto 1 \boxtimes v$ is injective.

Lemma 1.3.1. *Let $u \in \mathcal{S}'(\mathbb{R}^n)$. Then $D_n u = 0$ if and only if $u = 1 \boxtimes v$ for some $v \in \mathcal{S}'(\mathbb{R}^{n-1})$, and then v is unique.*

Proof. If $\psi \in \mathcal{S}(\mathbb{R}^n)$ then $I(D_n\psi) = 0$, since $\psi \to 0$ as $|x_n| \to \infty$. Hence $D_n(1 \boxtimes v)(\psi) = v(I(-D_n\psi)) = 0$.

For the converse, fix $\rho \in \mathcal{S}(\mathbb{R})$ with $\int \rho = 1$. We claim that every $\phi \in \mathcal{S}(\mathbb{R}^n)$ can be written

$$\phi = (I\phi) \boxtimes \rho + D_n\psi, \qquad \psi \in \mathcal{S}(\mathbb{R}^n). \tag{1.3.3}$$

Set $\zeta = \phi - (I\phi) \boxtimes \rho \in \mathcal{S}(\mathbb{R}^n)$, so that $I\zeta = 0$. Define

$$\psi(x', x_n) = i \int_{-\infty}^{x_n} \zeta(x', t)\, dt = -i \int_{x_n}^{\infty} \zeta(x', t)\, dt,$$

where the two expressions agree because $I\zeta = 0$. Then $D_n\psi = \zeta$, and ψ is smooth. Every derivative of ψ is either a derivative of ζ (if it involves x_n) or of the form $i \int_{-\infty}^{x_n} D_{x'}^{\beta'} \zeta\, dt$, where $D_{x'}^{\beta'}\zeta$ again has integral zero in x_n. For $g \in \mathcal{S}(\mathbb{R}^n)$ with $\int g(x', t)\, dt = 0$ and every N,

$$\Big| \int_{-\infty}^{x_n} g(x', t)\, dt \Big| \le \sup \left(\langle x' \rangle^N \langle t \rangle^{N+2} |g| \right) \langle x' \rangle^{-N} \int_{|t| \ge |x_n|} \langle t \rangle^{-N-2} dt \le C \langle x' \rangle^{-N} \langle x_n \rangle^{-N-1}.$$

Here we use the first integral for $x_n \le 0$ and the second for $x_n \ge 0$. Since $\langle x \rangle \le \langle x' \rangle \langle x_n \rangle$, all $\langle x \rangle^N D^\beta \psi$ are bounded, so $\psi \in \mathcal{S}(\mathbb{R}^n)$. This proves (1.3.3).

Now suppose $D_n u = 0$. Then $u(D_n\psi) = -(D_n u)(\psi) = 0$, so by (1.3.3) $u(\phi) = u((I\phi)\boxtimes\rho)$. The map $\theta \mapsto \theta \boxtimes \rho$ is continuous $\mathcal{S}(\mathbb{R}^{n-1}) \to \mathcal{S}(\mathbb{R}^n)$, so $v(\theta) = u(\theta \boxtimes \rho)$ defines $v \in \mathcal{S}'(\mathbb{R}^{n-1})$, and $u(\phi) = v(I\phi)$, that is $u = 1 \boxtimes v$. Uniqueness follows from the injectivity of $v \mapsto 1 \boxtimes v$. □

Applying the lemma repeatedly gives the version we need. For $x \in \mathbb{R}^k$ and $z \in \mathbb{R}^m$, define $(1 \boxtimes v)(\Phi) = v\big(\int_{\mathbb{R}^k} \Phi(x, \cdot)\, dx \big)$ for $v \in \mathcal{S}'(\mathbb{R}^m)$ and $\Phi \in \mathcal{S}(\mathbb{R}^{k+m})$.

Corollary 1.3.2. *If $u \in \mathcal{S}'(\mathbb{R}^{k+m}_{x,z})$ satisfies $D_{x_j} u = 0$ for $j = 1, \dots, k$, then $u = 1 \boxtimes v$ for a unique $v \in \mathcal{S}'(\mathbb{R}^m)$. Moreover $z_j(1 \boxtimes v) = 1 \boxtimes (z_j v)$.*

Proof. Induct on k. By Lemma 1.3.1 (in the variable x_1), $u = 1 \boxtimes v_1$ with $v_1 \in \mathcal{S}'(\mathbb{R}^{k-1+m})$. Integration in x_1 commutes with D_{x_j} for $j \ge 2$, so $0 = D_{x_j}(1 \boxtimes v_1) = 1 \boxtimes (D_{x_j} v_1)$. By injectivity, $D_{x_j} v_1 = 0$ for $j \ge 2$, and the inductive hypothesis applies to v_1. The last identity holds because integration in x commutes with multiplication by z_j. □

The second lemma is a special case of Schwartz' description of distributions supported at a point (Theorem 1.5.1).

Lemma 1.3.3. *If $u \in \mathcal{S}'(\mathbb{R}^n)$ and $x_j u = 0$ for $j = 1, \dots, n$, then $u = c\delta$ for some $c \in \mathbb{C}$. Here $\delta(\phi) = \phi(0)$.*

Proof. Choose $\rho \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\rho = 1$ on $|x| \le \frac{1}{2}$ and $\rho = 0$ on $|x| \ge \frac{3}{4}$. We claim that every $\phi \in \mathcal{S}(\mathbb{R}^n)$ can be written

$$\phi = \phi(0)\rho + \sum_{j=1}^{n} x_j \psi_j, \qquad \psi_j \in \mathcal{S}(\mathbb{R}^n). \tag{1.3.4}$$

By Taylor's formula $\phi(x) = \phi(0) + \sum_j x_j \zeta_j(x)$ with $\zeta_j(x) = \int_0^1 (\partial_j \phi)(tx)\, dt \in \mathcal{C}^\infty(\mathbb{R}^n)$, so $\rho\phi = \phi(0)\rho + \sum_j x_j \rho \zeta_j$ with $\rho\zeta_j \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$. The function $(1-\rho)\phi$ vanishes for $|x| \le \frac{1}{2}$. The function $|x|^{-2}(1-\rho)$ is smooth with all derivatives bounded, so $\zeta = |x|^{-2}(1-\rho)\phi \in \mathcal{S}(\mathbb{R}^n)$ and $(1-\rho)\phi = \sum_j x_j (x_j \zeta)$. Thus (1.3.4) holds with $\psi_j = \rho\zeta_j + x_j\zeta$. If $x_j u = 0$ for all j, then $u(x_j\psi_j) = (x_j u)(\psi_j) = 0$, so $u(\phi) = \phi(0) u(\rho) = c\,\delta(\phi)$ with $c = u(\rho)$. □

1.4 The Fourier transform

1.4.1 On Schwartz functions

For $\phi \in \mathcal{S}(\mathbb{R}^n)$ we set

$$\mathcal{F}\phi(\xi) = \hat{\phi}(\xi) = \int_{\mathbb{R}^n} e^{-ix\cdot\xi}\phi(x)\,dx, \qquad \mathcal{G}\psi(x) = \int_{\mathbb{R}^n} e^{ix\cdot\xi}\psi(\xi)đ\xi, \tag{1.4.1}$$

where $đ\xi = (2\pi)^{-n}d\xi$. We will show that $\mathcal{G} = \mathcal{F}^{-1}$. The integrals converge absolutely, with $|\hat{\phi}(\xi)| \le \int \langle x\rangle^{-n-1}dx\, \sup\langle x\rangle^{n+1}|\phi|$. Differentiation under the integral sign is justified because $x^\beta\phi$ is again integrable, and gives $D_\xi^\beta\hat{\phi} = \mathcal{F}((-x)^\beta\phi)$. Integration by parts gives $\xi_j\hat{\phi} = \mathcal{F}(D_j\phi)$. Together:

$$\mathcal{F}(D_j\phi) = \xi_j\mathcal{F}\phi, \qquad \mathcal{F}(x_j\phi) = -D_{\xi_j}\mathcal{F}\phi, \qquad \xi^\alpha D_\xi^\beta\hat{\phi} = \mathcal{F}\big(D^\alpha((-x)^\beta\phi)\big). \tag{1.4.2}$$

Proposition 1.4.1. *$\mathcal{F}$ and $\mathcal{G}$ are continuous linear maps $\mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$, with $\|\mathcal{F}\phi\|_k \le C_k\|\phi\|_{k+n+1}$.*

Proof. For $|\alpha| + |\beta| \le k$, by (1.4.2),

$$|\xi^\alpha D_\xi^\beta\hat{\phi}(\xi)| \le \int \langle x\rangle^{-n-1}dx\ \sup_x \langle x\rangle^{n+1}|D^\alpha(x^\beta\phi)| \le C\|\phi\|_{k+n+1}.$$

The function $D^\alpha(x^\beta\phi)$ is a combination of terms $x^{\beta'}D^{\alpha'}\phi$ with $\beta' \le \beta$, $\alpha' \le \alpha$. Since $\mathcal{G}\psi(x) = (2\pi)^{-n}\mathcal{F}\psi(-x)$, the same holds for $\mathcal{G}$. □

The same argument applies to partial Fourier transforms. For example $\mathcal{F}_2\Phi(x,\xi) = \int e^{-iy\cdot\xi}\Phi(x,y)\,dy$ is continuous on $\mathcal{S}(\mathbb{R}^{2n})$: the weights in x simply pass through the integral.

Lemma 1.4.2. *$\mathcal{F}(e^{-|x|^2})(\xi) = \pi^{n/2}e^{-|\xi|^2/4}$ and $\mathcal{G}(e^{-|\xi|^2/4})(x) = \pi^{-n/2}e^{-|x|^2}$.*

Proof. Since $e^{-|x|^2} = \prod_j e^{-x_j^2}$, the integral factorizes and it suffices to take $n = 1$. Let $g(\xi) = \int e^{-ix\xi}e^{-x^2}dx$. Differentiating and integrating by parts,

$$g'(\xi) = \frac{i}{2}\int e^{-ix\xi}\,\frac{d}{dx}e^{-x^2}dx = -\frac{i}{2}\int \Big(\frac{d}{dx}e^{-ix\xi}\Big)e^{-x^2}dx = -\frac{\xi}{2}g(\xi).$$

So $g(\xi) = g(0)e^{-\xi^2/4}$, and $g(0) = \int e^{-x^2}dx = \pi^{1/2}$. For the second formula, change variables $\xi = 2\eta$: $\int e^{ix\xi}e^{-|\xi|^2/4}d\xi = 2^n\int e^{-i(-2x)\cdot\eta}e^{-|\eta|^2}d\eta = 2^n\pi^{n/2}e^{-|x|^2}$. Multiplying by $(2\pi)^{-n}$ gives the result. □

The usual proofs of the inversion formula compute $\mathcal{G}\mathcal{F}\phi$ with a Gaussian cut-off and let the cut-off go away. We take a more structural route, which previews the way kernels are used throughout the book. The composite $\mathcal{G}\mathcal{F}$ commutes with translations and with multiplication by x_j, so its kernel must be a multiple of $\delta(x-y)$; a single Gaussian computation fixes the multiple. Figure 1.1 shows the logic.

Theorem 1.4.3 (Fourier inversion)**.** *$\mathcal{F}\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is an isomorphism with inverse $\mathcal{G}$.*

Proof. We use the *idea* of the Schwartz kernel theorem, not the theorem itself, since the Fourier transform is used to prove it. Define $K\colon \mathcal{S}(\mathbb{R}^{2n}) \to \mathbb{C}$ by

$$K(\Phi) = \int_{\mathbb{R}^n}\int_{\mathbb{R}^n} e^{ix\cdot\xi}\,\mathcal{F}_2\Phi(x,\xi)đ\xi\,dx.$$

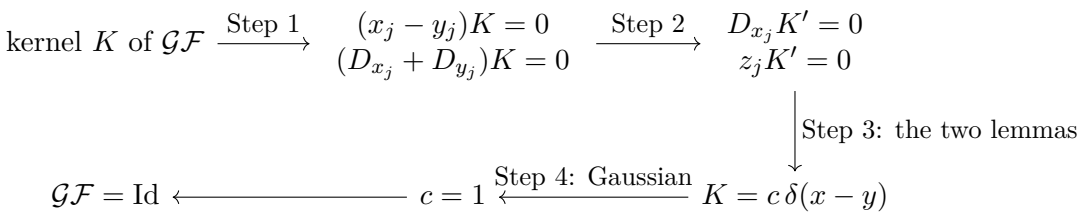


Figure 1.1: The logic of the proof of the Fourier inversion formula. The two little lemmas of Section 1.3 pin the kernel down up to a constant, and the Gaussian computation (Lemma 1.4.2) shows the constant is 1.

The integrand is in $\mathcal{S}(\mathbb{R}^{2n})$, and $\mathcal{F}_2$ is continuous, so $K \in \mathcal{S}'(\mathbb{R}^{2n})$. For $\zeta, \psi \in \mathcal{S}(\mathbb{R}^n)$ we have $\mathcal{F}_2(\zeta \boxtimes \psi) = \zeta \boxtimes \hat{\psi}$, so

$$K(\zeta \boxtimes \psi) = \int \zeta(x)\,(\mathcal{GF}\psi)(x)\,dx. \tag{1.4.3}$$

That is, K is the Schwartz kernel of $\mathcal{G} \circ \mathcal{F}$.

Step 1: the equations satisfied by K. We claim that

$$(x_j - y_j)K = 0, \qquad (D_{x_j} + D_{y_j})K = 0, \qquad j = 1, \dots, n. \tag{1.4.4}$$

By (1.4.2) applied in y, $\mathcal{F}_2((x_j - y_j)\Phi) = (x_j + D_{\xi_j})\mathcal{F}_2\Phi$. All integrands below are Schwartz functions of (x, ξ), so integrating by parts in ξ gives

$$K((x_j - y_j)\Phi) = \iint e^{ix\cdot\xi}(x_j + D_{\xi_j})\mathcal{F}_2\Phi đ\xi\,dx = \iint (x_j e^{ix\cdot\xi} - D_{\xi_j}e^{ix\cdot\xi})\mathcal{F}_2\Phi đ\xi\,dx = 0.$$

Similarly $\mathcal{F}_2((D_{x_j} + D_{y_j})\Phi) = (D_{x_j} + \xi_j)\mathcal{F}_2\Phi$, and integrating by parts in x shows that $K((D_{x_j} + D_{y_j})\Phi) = 0$. This gives (1.4.4).

Step 2: change of variables. Let $\Lambda(x, z) = (x, x - z)$. This is a linear involution of $\mathbb{R}^{2n}$ with $|\det \Lambda| = 1$. Set $K' = \Lambda^* K$, so that $K'(\Phi) = K(\Phi \circ \Lambda)$. Since $(D_{x_j} + D_{y_j})(\Phi \circ \Lambda) = (D_{x_j}\Phi) \circ \Lambda$ and $(x_j - y_j)(\Phi \circ \Lambda) = (z_j\Phi) \circ \Lambda$, the equations (1.4.4) become

$$D_{x_j}K' = 0, \qquad z_jK' = 0, \qquad j = 1, \dots, n.$$

Step 3: the two lemmas. By Corollary 1.3.2, $K' = 1 \boxtimes f$ with $f \in \mathcal{S}'(\mathbb{R}^n_z)$, and $1 \boxtimes (z_j f) = z_jK' = 0$, so $z_j f = 0$ for all j. By Lemma 1.3.3, $f = c\delta$. Hence $K'(\Phi) = c\int \Phi(x, 0)\,dx$, and since Λ is an involution, $K(\Phi) = K'(\Phi \circ \Lambda) = c\int \Phi(x, x)\,dx$. By (1.4.3), $\int \zeta\,\mathcal{GF}\psi = c\int \zeta\psi$ for all ζ, so $\mathcal{GF}\psi = c\psi$ for every $\psi \in \mathcal{S}(\mathbb{R}^n)$.

Step 4: the constant. By Lemma 1.4.2, $\mathcal{GF}(e^{-|x|^2}) = \pi^{n/2}\mathcal{G}(e^{-|\xi|^2/4}) = e^{-|x|^2}$, so $c = 1$.

Thus $\mathcal{GF} = \mathrm{Id}$. With $R\phi(x) = \phi(-x)$ we have $\mathcal{G} = (2\pi)^{-n}R\mathcal{F} = (2\pi)^{-n}\mathcal{F}R$, and therefore $\mathcal{FG} = (2\pi)^{-n}\mathcal{F}R\mathcal{F} = R\mathcal{GF}R = \mathrm{Id}$ as well. □

1.4.2 On tempered distributions

Fubini's theorem gives the symmetry

$$\int \hat{\phi}\,\psi\,dx = \int \phi\,\hat{\psi}\,dx, \qquad \phi, \psi \in \mathcal{S}(\mathbb{R}^n), \tag{1.4.5}$$

so $\mathcal{F}^t = \mathcal{F}$. As in Section 1.2.3 we define

$$\mathcal{F}u(\phi) = \hat{u}(\phi) = u(\hat{\phi}), \qquad \mathcal{G}u(\phi) = u(\mathcal{G}\phi), \qquad u \in \mathcal{S}'(\mathbb{R}^n),\ \phi \in \mathcal{S}(\mathbb{R}^n). \tag{1.4.6}$$

Corollary 1.4.4. *The map $\mathcal{F}$ of (1.4.6) is the unique continuous extension of the Fourier transform to $\mathcal{S}'(\mathbb{R}^n)$. It is an isomorphism with inverse $\mathcal{G}$, both maps are continuous for the weak topology, and for $u \in \mathcal{S}'(\mathbb{R}^n)$*

$$\mathcal{F}(D_j u) = \xi_j \mathcal{F}u, \qquad \mathcal{F}(x_j u) = -D_{\xi_j}\mathcal{F}u.$$

Proof. Weak continuity is immediate from (1.4.6). The definition is consistent with (1.4.5), and $\mathcal{G}\mathcal{F}u(\phi) = u(\mathcal{F}\mathcal{G}\phi) = u(\phi)$, and similarly $\mathcal{F}\mathcal{G} = \mathrm{Id}$. For the identities, $\mathcal{F}(D_j u)(\phi) = u(-D_j\hat{\phi}) = u(\mathcal{F}(x_j\phi)) = (\xi_j\hat{u})(\phi)$. Likewise $\mathcal{F}(x_j u)(\phi) = u(x_j\hat{\phi}) = u(\mathcal{F}(D_j\phi)) = \hat{u}(D_j\phi) = (-D_j\hat{u})(\phi)$. □

1.4.3 Plancherel's theorem

Complex conjugation of (1.4.1) gives $\overline{\hat{\psi}} = (2\pi)^n \mathcal{G}\bar{\psi}$. Combining this with (1.4.5) gives the adjoint identity

$$\int \phi\, \overline{\hat{\psi}}\, d\xi = (2\pi)^n \int (\mathcal{G}\phi)\, \bar{\psi}\, dx, \qquad \phi, \psi \in \mathcal{S}(\mathbb{R}^n). \tag{1.4.7}$$

Proposition 1.4.5 (Plancherel)**.** *The Fourier transform extends by continuity from $\mathcal{S}(\mathbb{R}^n)$ to an isomorphism $\mathcal{F}\colon L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$, consistent with its action on $\mathcal{S}'(\mathbb{R}^n)$, and*

$$\int \hat{u}\, \overline{\hat{v}}\, d\xi = (2\pi)^n \int u\bar{v}\, dx, \qquad \|\hat{u}\|_{L^2} = (2\pi)^{n/2}\|u\|_{L^2}.$$

Proof. Taking $\phi = \hat{u}$ in (1.4.7) with $u, v \in \mathcal{S}(\mathbb{R}^n)$ gives Parseval's formula, and $v = u$ gives the norm identity on $\mathcal{S}(\mathbb{R}^n)$. Now $\mathcal{C}_c^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ (a standard fact of measure theory), so $\mathcal{F}$ extends uniquely to a map $L^2 \to L^2$ that is $(2\pi)^{n/2}$ times an isometry. Its range is closed and contains $\mathcal{S}(\mathbb{R}^n)$, so it is all of $L^2(\mathbb{R}^n)$. The extension agrees with (1.4.6). Indeed, if $u_j \to u$ in L^2 with $u_j \in \mathcal{S}$, then $u_j \to u$ and $\hat{u}_j \to \hat{u}$ weakly in $\mathcal{S}'$, by the Cauchy–Schwarz inequality. □

1.5 Differential operators

The simplest Fourier transforms of distributions follow directly from the definitions. Since $\hat{\delta}(\phi) = \delta(\hat{\phi}) = \int \phi\, dx$, Corollary 1.4.4 gives

$$\mathcal{F}\delta = 1, \qquad \mathcal{F}(D^\alpha\delta) = \xi^\alpha, \quad \alpha \in \mathbb{N}_0^n. \tag{1.5.1}$$

Here $D^\alpha\delta(\phi) = \delta((-D)^\alpha\phi) = (-1)^{|\alpha|}D^\alpha\phi(0)$.

Theorem 1.5.1. *If $u \in \mathcal{S}'(\mathbb{R}^n)$ and $\operatorname{supp} u \subset \{0\}$, then $u = \sum_{|\alpha|\le k} c_\alpha D^\alpha\delta$ for some k and constants c_α. Consequently the Fourier transform is an isomorphism*

$$\mathcal{F}\colon \{u \in \mathcal{S}'(\mathbb{R}^n);\ \operatorname{supp} u \subset \{0\}\} \longrightarrow \mathbb{C}[\xi]$$

onto the polynomials in ξ.

Proof. Let $|u(\phi)| \le C\|\phi\|_k$. First suppose that $\phi \in \mathcal{S}(\mathbb{R}^n)$ vanishes to order k at 0, meaning $D^\mu\phi(0) = 0$ for $|\mu| \le k$. Applying Taylor's formula to $D^\mu\phi$ gives $|D^\mu\phi(x)| \le C|x|^{k+1-|\mu|}$ for $|x| \le 1$ and $|\mu| \le k$. Let $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\chi = 1$ near 0 and $\operatorname{supp}\chi \subset \{|x| \le 1\}$, and let $\chi_\epsilon(x) = \chi(x/\epsilon)$ for $0 < \epsilon \le 1$. Then $(1-\chi_\epsilon)\phi$ vanishes near 0, so $u((1-\chi_\epsilon)\phi) = 0$ and $u(\phi) = u(\chi_\epsilon\phi)$. For $|\alpha| + |\beta| \le k$, the function $x^\alpha D^\beta(\chi_\epsilon\phi)$ is supported in $|x| \le \epsilon$, and it is a sum of terms $\epsilon^{-|\gamma|}(D^\gamma\chi)(x/\epsilon)\, x^\alpha D^{\beta-\gamma}\phi$, each bounded by $C\epsilon^{-|\gamma|}\epsilon^{k+1-|\beta|+|\gamma|} \le C\epsilon$. Hence $|u(\phi)| \le C\|\chi_\epsilon\phi\|_k \le C'\epsilon \to 0$, and $u(\phi) = 0$.

For general ϕ, the function $\phi - \chi \sum_{|\alpha|\le k} \partial^\alpha\phi(0)x^\alpha/\alpha!$ vanishes to order k at 0. So $u(\phi) = \sum_{|\alpha|\le k} a_\alpha \partial^\alpha\phi(0)$ with $a_\alpha = u(\chi x^\alpha)/\alpha!$. This is of the stated form, since $\partial^\alpha\phi(0) = i^{|\alpha|}D^\alpha\phi(0) = (-i)^{|\alpha|}(D^\alpha\delta)(\phi)$. The final statement follows from (1.5.1), since $\mathcal{F}$ is injective and $D^\alpha\delta$ has support $\{0\}$. □

Lemma 1.3.3 is a special case. If $x_j u = 0$ for all j, then u vanishes on $\mathbb{R}^n \setminus \{0\}$, because there $\phi = \sum_j x_j(x_j|x|^{-2}\phi)$ for $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n \setminus 0)$. So $\operatorname{supp} u \subset \{0\}$, and then $x_j \sum c_\alpha D^\alpha\delta = 0$ forces $c_\alpha = 0$ for $\alpha \ne 0$. Indeed, applying $\mathcal{F}$ gives $D_{\xi_j} \sum c_\alpha \xi^\alpha = 0$ for all j, so the polynomial is constant.

1.5.1 Constant coefficients

A differential operator with constant coefficients $P(D) = \sum_{|\alpha|\le m} c_\alpha D^\alpha$ acts on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$. By Corollary 1.4.4,

$$\widehat{P(D)u}(\xi) = P(\xi)\hat{u}(\xi), \qquad P(\xi) = \sum_{|\alpha|\le m} c_\alpha \xi^\alpha, \qquad u \in \mathcal{S}'(\mathbb{R}^n). \tag{1.5.2}$$

The polynomial $P(\xi)$ is the *(full) characteristic polynomial*, or symbol, of $P(D)$, and it determines $P(D)$. By the inversion formula, for $u \in \mathcal{S}(\mathbb{R}^n)$,

$$P(D)u(x) = \int e^{ix\cdot\xi} P(\xi)\hat{u}(\xi)đ\xi = \int \Big(\int e^{i(x-y)\cdot\xi} P(\xi)u(y)\,dy\Big)đ\xi. \tag{1.5.3}$$

The last integral is *iterated*: the y-integral is done first. The case $P \equiv 1$ is the inversion formula.

1.5.2 Variable coefficients

By the module property (1.1.4), a differential operator

$$P(x,D) = \sum_{|\alpha|\le m} p_\alpha(x)D^\alpha, \qquad p_\alpha \in \mathcal{C}_b^\infty(\mathbb{R}^n),$$

is continuous on $\mathcal{S}(\mathbb{R}^n)$ and, by (1.2.4), on $\mathcal{S}'(\mathbb{R}^n)$. Inserting the inversion formula for $D^\alpha\phi$ gives

$$P(x,D)\phi(x) = \int e^{ix\cdot\xi} P(x,\xi)\hat{\phi}(\xi)đ\xi = \int \Big(\int e^{i(x-y)\cdot\xi} P(x,\xi)\phi(y)\,dy\Big)đ\xi, \tag{1.5.4}$$

again as an iterated integral, where $P(x,\xi) = \sum_{|\alpha|\le m} p_\alpha(x)\xi^\alpha$ is the *(full) symbol* of $P(x,D)$. In the notation of Chapter 2, $P(x,D) = \operatorname{Op}(P)$. Pseudodifferential operators are obtained by replacing the polynomial $P(x,\xi)$ in (1.5.4) by more general symbols.

1.6 Two big theorems

We now prove two structure theorems. The first says that tempered distributions are "not too bad": they are finite sums of derivatives of bounded continuous functions multiplied by polynomials. The second, the Schwartz kernel theorem, says that every continuous linear operator $\mathcal{S}(\mathbb{R}^m) \to \mathcal{S}'(\mathbb{R}^n)$ is given by a distributional kernel. The kernel theorem was long considered hard to prove, but the Fourier transform gives a fairly direct proof. Both proofs rest on the same estimate.

1.6.1 A basic estimate

For $N, M \in \mathbb{N}_0$ consider the differential operator with polynomial coefficients

$$A_{N,M} = (1 + |D|^2)^M \langle x\rangle^{2N} \colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n), \qquad |D|^2 = D_1^2 + \cdots + D_n^2.$$

It is an isomorphism, with continuous inverse $A_{N,M}^{-1}\psi = \langle x\rangle^{-2N}\mathcal{G}\big(\langle\xi\rangle^{-2M}\hat{\psi}\big)$. This uses (1.5.2) and the fact that $\langle\cdot\rangle^{-2N}$ and $\langle\cdot\rangle^{-2M}$ are in $\mathcal{C}^\infty_{\mathrm{b}}$. Its transpose is $A_{N,M}^t = \langle x\rangle^{2N}(1 + |D|^2)^M$. For $s \in \mathbb{R}$ we use the weighted L^2 norm of the Fourier transform,

$$\|\psi\|_{(-s)} = \big\|\langle\xi\rangle^{-s}\hat{\psi}\big\|_{L^2}.$$

Lemma 1.6.1. *Let $k \in \mathbb{N}_0$ and choose $2N \geq k$ and $2M \geq k + 2n$. Then there is C such that*

$$\|A_{N,M}^{-1}\psi\|_k \leq C\|\psi\|_{(-n)}, \qquad \psi \in \mathcal{S}(\mathbb{R}^n).$$

Proof. Let $f = \mathcal{G}(\langle\xi\rangle^{-2M}\hat{\psi})$. For $|\alpha|+|\beta| \leq k$, the function $x^\alpha D^\beta(\langle x\rangle^{-2N} f)$ is a sum of terms $x^\alpha D^{\beta-\gamma}(\langle x\rangle^{-2N})D^\gamma f$, and $|x^\alpha D^\mu\langle x\rangle^{-2N}| \leq C\langle x\rangle^{|\alpha|-2N} \leq C$. Also, by the Cauchy–Schwarz inequality,

$$|D^\gamma f(x)| = \Big|\int e^{ix\cdot\xi}\xi^\gamma\langle\xi\rangle^{-2M}\hat{\psi}\,d\!\!\bar{}\,\xi\Big| \leq (2\pi)^{-n}\big\|\langle\xi\rangle^{k+n-2M}\big\|_{L^2}\,\|\psi\|_{(-n)},$$

and $\langle\xi\rangle^{k+n-2M} \leq \langle\xi\rangle^{-n}$ is square integrable. □

Lemma 1.6.2. *Let $s > n/2$ and let $\mathcal{H}_s$ be the Hilbert space of measurable f on $\mathbb{R}^n$ with $\langle\xi\rangle^{-s} f \in L^2$, with inner product $(f, g)_s = \int\langle\xi\rangle^{-2s} f\bar{g}\,d\!\!\bar{}\,\xi$. Then:*

(i) $\mathcal{S}(\mathbb{R}^n)$ is dense in $\mathcal{H}_s$. For each $x \in \mathbb{R}^n$, $e_x(\xi) = e^{-ix\cdot\xi}$ is in $\mathcal{H}_s$, and $x \mapsto e_x \in \mathcal{H}_s$ is bounded and continuous.

(ii) If $\lambda\colon \mathcal{H}_s \to \mathbb{C}$ is a bounded linear functional, then $w(x) = \lambda(e_x)$ is a bounded continuous function and $\lambda(\hat{\psi}) = \int w(x)\psi(x)\,dx$ for all $\psi \in \mathcal{S}(\mathbb{R}^n)$.

Proof. Multiplication by $\langle\xi\rangle^{-s}$ is a unitary map $\mathcal{H}_s \to L^2$ which maps $\mathcal{S}$ onto $\mathcal{S}$, so density follows from density of $\mathcal{S}$ in L^2. Next, $\|e_x\|_s^2 = \int\langle\xi\rangle^{-2s}d\!\!\bar{}\,\xi < \infty$, since $2s > n$. Also $\|e_x - e_{x'}\|_s^2 = \int\langle\xi\rangle^{-2s}|e^{-ix\cdot\xi} - e^{-ix'\cdot\xi}|^2 d\!\!\bar{}\,\xi \to 0$ as $x' \to x$, by dominated convergence. For (ii), the Riesz representation theorem gives $h \in \mathcal{H}_s$ with $\lambda(f) = \int\langle\xi\rangle^{-2s} f\bar{h}\,d\!\!\bar{}\,\xi$. Then $g = \langle\xi\rangle^{-2s}\bar{h} = \langle\xi\rangle^{-s}\cdot\langle\xi\rangle^{-s}\bar{h}$ is a product of two L^2 functions, so $g \in L^1$. By Fubini,

$$\lambda(\hat{\psi}) = \int g(\xi)\int e^{-ix\cdot\xi}\psi(x)\,dx\,d\!\!\bar{}\,\xi = \int\Big(\int g(\xi)e^{-ix\cdot\xi}d\!\!\bar{}\,\xi\Big)\psi(x)\,dx,$$

and $\int g(\xi)e^{-ix\cdot\xi}d\!\!\bar{}\,\xi = \lambda(e_x) = w(x)$. This is bounded by $\|g\|_{L^1}$, and continuous by (i). □

1.6.2 The structure theorem

Theorem 1.6.3 (Schwartz representation theorem)**.** *Let $u \in \mathcal{S}'(\mathbb{R}^n)$ satisfy $|u(\phi)| \leq C\|\phi\|_k$. Then there are finitely many bounded continuous functions $u_{\alpha\beta}$ such that*

$$u = \sum_{|\alpha|+|\beta|\leq 2k+2n+2} x^\beta D^\alpha u_{\alpha\beta}.$$

More precisely, $u = \langle x\rangle^{2N}(1 + |D|^2)^M w$ with w bounded and continuous and N, M as in Lemma 1.6.1. Conversely, every such expression is in $\mathcal{S}'(\mathbb{R}^n)$.

Proof. Take $N = \lceil k/2 \rceil$ and $M = \lceil k/2 \rceil + n$, so that $2N + 2M \le 2k + 2n + 2$, and write $A = A_{N,M}$. By Lemma 1.6.1, $|u(A^{-1}\psi)| \le C'\|\psi\|_{(-n)}$. So $\lambda(\hat{\psi}) = u(A^{-1}\psi)$ is bounded on the dense subspace $\mathcal{S} \subset \mathcal{H}_n$ and extends to a bounded functional on $\mathcal{H}_n$. By Lemma 1.6.2 there is a bounded continuous w with $u(A^{-1}\psi) = \int w\psi\, dx$. Putting $\psi = A\phi$,

$$u(\phi) = \int w\, A\phi\, dx = (A^t w)(\phi) = \big(\langle x\rangle^{2N}(1 + |D|^2)^M w\big)(\phi).$$

Expanding $\langle x\rangle^{2N}(1 + |D|^2)^M$ by (1.0.1) gives the sum, with $u_{\alpha\beta}$ constant multiples of w. The converse was noted after (1.2.4). □

1.6.3 The Schwartz kernel theorem

If $K \in \mathcal{S}'(\mathbb{R}^{n+m})$, define $O_K \colon \mathcal{S}(\mathbb{R}^m) \to \mathcal{S}'(\mathbb{R}^n)$ by

$$O_K(\psi)(\phi) = K(\phi \boxtimes \psi), \qquad \phi \in \mathcal{S}(\mathbb{R}^n),\ \psi \in \mathcal{S}(\mathbb{R}^m). \tag{1.6.1}$$

Formally $O_K\psi(x) = \int K(x,y)\psi(y)\, dy$. If $|K(\Phi)| \le C\|\Phi\|_k$ then $|O_K(\psi)(\phi)| \le C\|\phi\|_k\|\psi\|_k$. Hence $O_K\psi \in \mathcal{S}'(\mathbb{R}^n)$, and O_K is continuous from $\mathcal{S}(\mathbb{R}^m)$ to $\mathcal{S}'(\mathbb{R}^n)$ with the weak topology. The kernel theorem is the converse. Two lemmas are needed first.

Lemma 1.6.4. *Let $B \colon \mathcal{S}(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^m) \to \mathbb{C}$ be bilinear and separately continuous. Then there are C and p with $|B(\phi,\psi)| \le C\|\phi\|_p\|\psi\|_p$.*

Proof. For $j \in \mathbb{N}$ let $E_j = \{\phi;\ |B(\phi,\psi)| \le j\|\psi\|_j\ \forall \psi \in \mathcal{S}(\mathbb{R}^m)\}$. Each E_j is closed, being an intersection of closed sets, since $B(\cdot,\psi)$ is continuous. The E_j cover $\mathcal{S}(\mathbb{R}^n)$, because each $B(\phi,\cdot)$ is a continuous functional and so satisfies an estimate (1.2.2). By the Baire category theorem and Proposition 1.1.2, some E_j contains a d-ball around some ϕ_0. By (1.1.3) it then contains $\{\phi_0 + \phi;\ \|\phi\|_q < \epsilon\}$ for some q and $\epsilon > 0$. For $\|\phi\|_q < \epsilon$, $|B(\phi,\psi)| \le |B(\phi_0 + \phi,\psi)| + |B(\phi_0,\psi)| \le 2j\|\psi\|_j$. By homogeneity $|B(\phi,\psi)| \le 2j\epsilon^{-1}\|\phi\|_q\|\psi\|_j$, and we can take $p = \max(q, j)$. □

Lemma 1.6.5. *Finite sums of exterior products $\phi \boxtimes \psi$, $\phi \in \mathcal{S}(\mathbb{R}^n)$, $\psi \in \mathcal{S}(\mathbb{R}^m)$, are dense in $\mathcal{S}(\mathbb{R}^{n+m})$.*

Proof. By Lemma 1.1.3 it suffices to approximate $\Phi \in \mathcal{C}_c^\infty(\mathbb{R}^{n+m})$, say with support in the open cube $(-L, L)^{n+m}$. Extend Φ to a smooth function Φ_{per} that is $2L$-periodic in each variable. Its Fourier coefficients c_κ, $\kappa \in \mathbb{Z}^{n+m}$, decay rapidly (integrate by parts), so the Fourier series $\sum_\kappa c_\kappa e^{i\pi\kappa\cdot(x,y)/L}$ converges to Φ_{per} uniformly with all derivatives. Choose $\chi \in \mathcal{C}_c^\infty((-L,L)^n)$ and $\chi' \in \mathcal{C}_c^\infty((-L,L)^m)$ with $\chi\boxtimes\chi' = 1$ on $\operatorname{supp}\Phi$. Then $\chi\boxtimes\chi'\cdot\Phi_{\text{per}} = \Phi$. Each term $\chi \boxtimes \chi' \cdot e^{i\pi\kappa\cdot(x,y)/L}$ is the exterior product of $\chi e^{i\pi\kappa'\cdot x/L}$ and $\chi' e^{i\pi\kappa''\cdot y/L}$, where $\kappa = (\kappa', \kappa'')$. The partial sums, multiplied by $\chi \boxtimes \chi'$, converge to Φ in every $\mathcal{C}^k$ norm with supports in a fixed compact set, and hence in $\mathcal{S}(\mathbb{R}^{n+m})$. □

Theorem 1.6.6 (Schwartz kernel theorem)**.** *For every linear map $A \colon \mathcal{S}(\mathbb{R}^m) \to \mathcal{S}'(\mathbb{R}^n)$ that is continuous for the weak topology on $\mathcal{S}'(\mathbb{R}^n)$, there is a unique $K \in \mathcal{S}'(\mathbb{R}^{n+m})$ with $A = O_K$. Thus $K \mapsto O_K$ is a bijection between $\mathcal{S}'(\mathbb{R}^{n+m})$ and the continuous linear maps $\mathcal{S}(\mathbb{R}^m) \to \mathcal{S}'(\mathbb{R}^n)$. Moreover $K = \langle x\rangle^{2N}\langle y\rangle^{2N}(1 + |D_x|^2)^M(1 + |D_y|^2)^M k$ for some bounded continuous function k on $\mathbb{R}^{n+m}$ and some N, M.*

We call $K = K_A$ the *Schwartz kernel* of A, and write formally $A\psi(x) = \int K_A(x,y)\psi(y)\, dy$.

The proof has three steps. First, separate continuity of the bilinear form $B(\phi,\psi) = (A\psi)(\phi)$ is upgraded to a joint estimate (Lemma 1.6.4). Second, the operators $A_{N,M}$ in each set of variables convert this estimate into a bound by weighted L^2 norms of Fourier transforms (Lemma 1.6.1). Third, the Riesz representation argument of Lemma 1.6.2 produces a bounded continuous function k, and K is a derivative of k.

Proof. Uniqueness follows from Lemma 1.6.5. For existence, let $B(\phi,\psi) = (A\psi)(\phi)$. This is separately continuous: in ϕ because $A\psi \in \mathcal{S}'$, and in ψ by the weak continuity of A. By Lemma 1.6.4, $|B(\phi,\psi)| \le C\|\phi\|_p\|\psi\|_p$. Choose $2N \ge p$ and $2M \ge p + 2\max(n,m)$, and let $A_x = A_{N,M}$ on $\mathbb{R}^n$ and $A_y = A_{N,M}$ on $\mathbb{R}^m$. By Lemma 1.6.1, applied in each set of variables, $B'(\phi,\psi) = B(A_x^{-1}\phi, A_y^{-1}\psi)$ satisfies $|B'(\phi,\psi)| \le C'\|\phi\|_{(-n)}\|\psi\|_{(-m)}$. So $b(\hat\phi,\hat\psi) = B'(\phi,\psi)$ extends to a bounded bilinear form b on $\mathcal{H}_n(\mathbb{R}^n) \times \mathcal{H}_m(\mathbb{R}^m)$. Set

$$k(x,y) = b(e_x, e_y).$$

By Lemma 1.6.2(i) and the bilinear bound, k is bounded and continuous. For fixed ϕ, Lemma 1.6.2(ii) applied to $b(\hat\phi,\cdot)$ gives $B'(\phi,\psi) = \int b(\hat\phi, e_y)\psi(y)\,dy$. For fixed y, applied to $b(\cdot, e_y)$, it gives $b(\hat\phi, e_y) = \int k(x,y)\phi(x)\,dx$. Hence $B'(\phi,\psi) = \iint k(x,y)\phi(x)\psi(y)\,dx\,dy$, and so

$$(A\psi)(\phi) = B'(A_x\phi, A_y\psi) = \iint k\; A_xA_y(\phi \boxtimes \psi)\,dx\,dy.$$

The operator A_xA_y is a differential operator with polynomial coefficients on $\mathbb{R}^{n+m}$, hence continuous on $\mathcal{S}(\mathbb{R}^{n+m})$, and k is bounded. So $K(\Phi) = \iint k\, A_xA_y\Phi$ defines $K \in \mathcal{S}'(\mathbb{R}^{n+m})$ with $O_K = A$, and $K = (A_xA_y)^t k$ is of the stated form. □

1.6.4 Examples of kernels

Amongst tempered distributions, $\mathcal{S}(\mathbb{R}^n)$ consists of the "trivial" examples, and studying the singularities of tempered distributions amounts to studying the quotient $\mathcal{S}'(\mathbb{R}^n)/\mathcal{S}(\mathbb{R}^n)$. This could reasonably be called the space of *tempered microfunctions*. A typical singular distribution is δ. Its definition says that δ is the Schwartz kernel of the map $\mathcal{S}(\mathbb{R}^n) \ni \phi \mapsto \phi(0) \in \mathbb{C} = \mathcal{S}(\mathbb{R}^0)$. More generally, $\phi \mapsto D^\alpha\phi(0)$ has kernel $(-D)^\alpha\delta$, since $(-D)^\alpha\delta(\phi) = \delta(D^\alpha\phi)$. For the identity operator on $\mathcal{S}(\mathbb{R}^n)$, $K_{\mathrm{Id}}(\phi\boxtimes\psi) = \int \phi\psi\,dx$, so $K_{\mathrm{Id}} = \delta(x-y)$. This notation means the pull-back of $1\boxtimes\delta$ under the change of variables of Step 2 of Theorem 1.4.3; the proof of that theorem identified $K_{\mathcal{GF}} = \delta(x-y)$. Figure 1.2 shows the supports of these kernels. If K_A is the kernel of A, then the kernel of $D^\alpha \circ A$ is $D_x^\alpha K_A$ and the kernel of $A \circ D^\alpha$ is $(-D_y)^\alpha K_A$. In particular the operator $P(x,D)$ of (1.5.4) has kernel

$$K_{P(x,D)}(x,y) = \sum_{|\alpha|\le m} p_\alpha(x) D_x^\alpha \delta(x-y) = \int e^{i(x-y)\cdot\xi} P(x,\xi)đ\xi.$$

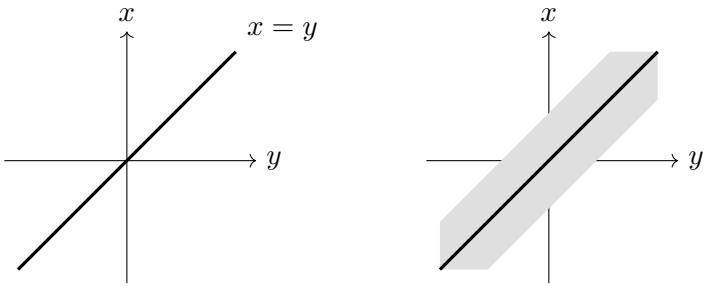
The second expression is an oscillatory integral, interpreted via (1.5.4). The kernels of pseudodifferential operators in Chapter 2 are of the same form with more general symbols $P(x,\xi)$.

Question. Where does the Fourier transform enter the proof of the kernel theorem, which is a statement about operators and not about frequencies?

Answer. Only in Lemma 1.6.1. The Schwartz norms $\|\cdot\|_k$ are not Hilbert norms, so Riesz representation cannot be applied to them directly. After applying $A_{N,M}^{-1}$, a Schwartz-norm estimate becomes a bound by $\|\langle\xi\rangle^{-n}\hat\psi\|_{L^2}$, a Hilbert norm. In that Hilbert space the point evaluations e_x are bounded, and this is what produces the continuous function k. □

1.7 Radial compactification

A real vector space W of dimension n is a smooth manifold, but $\mathcal{C}^\infty(W)$ puts no restriction on behaviour at infinity. A good way to describe functions with controlled behaviour at



differential operator pseudodifferential operator

Figure 1.2: Supports and singularities of Schwartz kernels in the (y, x)-plane. A differential operator, such as Id or $P(x, D)$, has kernel supported on the diagonal $\{x = y\}$ (left). A pseudodifferential operator of Chapter 2 has a kernel that is still singular only on the diagonal, but is no longer supported there: it is smooth off the diagonal and decays rapidly as $|x - y| \to \infty$ (right; the shading indicates where the kernel is concentrated).

infinity is to embed W as a dense open subset of a compact manifold X, possibly with boundary or corners, by a diffeomorphism onto its image. Restriction then identifies $\mathcal{C}^\infty(X)$ with a subspace of $\mathcal{C}^\infty(W)$, since a smooth function is determined by its values on a dense set. Several such compactifications exist. We construct them first for $\mathbb{R}^n$ and then check how they behave under linear (and affine) changes of coordinates. A compactification that is invariant under $\mathrm{GL}(n, \mathbb{R})$ is well defined for every vector space. For us the most important one is the *radial compactification*. Smooth functions on it are the classical symbols of order 0, and Schwartz functions are the smooth functions on it that vanish to infinite order at the boundary.

Discussion. Why compactify at all? Estimates such as $|D^\alpha a(z)| \le C\langle z\rangle^{-|\alpha|}$, or "rapid decay with all derivatives", are conditions on the behaviour of a function at infinity. On a compact manifold the analogous conditions are just smoothness, or vanishing to infinite order, at a boundary. Compactifying turns asymptotic analysis at infinity into local analysis near a boundary, where Taylor's formula and the other tools of calculus apply. The price is that we must choose the compactification carefully, so that the linear maps we care about extend smoothly.

1.7.1 The one-point compactification and its defect

Identify $\mathbb{R}^n$ with the hyperplane $\{1\} \times \mathbb{R}^n \subset \mathbb{R}^{n+1}$, and let $\mathbb{S}^n \subset \mathbb{R}^{n+1}$ be the unit sphere. The line through the south pole $(-1, 0)$ and $(1, z)$ is $t \mapsto (2t - 1, tz)$. It meets $\mathbb{S}^n$ again at $t = 4/(4 + |z|^2)$, which gives the stereographic projection

$$P(z) = \Big(\frac{4 - |z|^2}{4 + |z|^2}, \frac{4z}{4 + |z|^2}\Big), \qquad P^{-1}(Z_0, Z) = \frac{2Z}{1 + Z_0}, \quad Z_0 > -1.$$

P is a diffeomorphism of $\mathbb{R}^n$ onto $\mathbb{S}^n$ minus the south pole. The *one-point compactification* ${}^1\overline{\mathbb{R}^n} = \mathbb{R}^n \cup \{\infty\}$ is $\mathbb{S}^n$ with this identification. The reflection $Z_0 \mapsto -Z_0$ corresponds to the inversion $z \mapsto \iota(z) = 4z/|z|^2$. Hence a function $f \in \mathcal{C}^\infty(\mathbb{R}^n)$ extends smoothly to ${}^1\overline{\mathbb{R}^n}$ if and only if $f(\iota(w))$ extends smoothly across $w = 0$. Orthogonal transformations $O \in \mathrm{O}(n)$ lift to the diffeomorphisms $(Z_0, Z) \mapsto (Z_0, OZ)$. General linear transformations do not.

Proposition 1.7.1. *$G \in \mathrm{GL}(n, \mathbb{R})$ extends to a $\mathcal{C}^1$ map of ${}^1\overline{\mathbb{R}^n}$ fixing ∞ if and only if G is conformal, that is $G = cO$ with $c > 0$ and $O \in \mathrm{O}(n)$. In particular, for $n \ge 2$ the scaling $(z_1, z') \mapsto (sz_1, z')$, $s \ne 1$, does not extend.*

Proof. In the coordinate $w = \iota(z)$ near ∞, G becomes $h(w) = \iota(G\iota(w)) = |w|^2 Gw/|Gw|^2$ for $w \neq 0$, with $h(0) = 0$. If $G = cO$ then $h(w) = c^{-1}Ow$, which is smooth. Conversely, suppose h is differentiable at 0. Since $h(tw) = th(w)$ for $t > 0$, we get $h(w) = \lim_{t\downarrow 0} h(tw)/t = h'(0)w$, so h is linear, say $h(w) = Hw$. Then $G^{-1}Hw = q(w)w$ with $q(w) = |w|^2/|Gw|^2$. So every vector is an eigenvector of $G^{-1}H$, which is therefore a multiple cI of the identity. Hence $q \equiv c$, that is $|Gw|^2 = c^{-1}|w|^2$, and G is conformal. □

Thus the one-point compactification depends on a conformal structure on W, and not only on its linear structure (Exercise 1.14). It reappears in the proof of the index theorem, but it is not the right compactification for symbols.

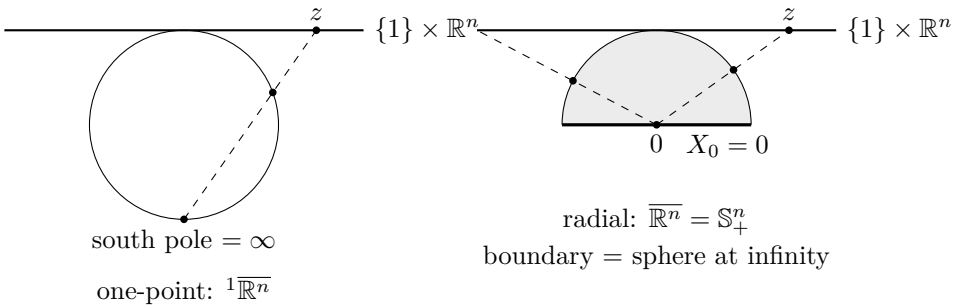


Figure 1.3: Two compactifications of $\mathbb{R}^n$, drawn for $n = 1$ with $\mathbb{R}^n$ placed at height $X_0 = 1$. Left: stereographic projection from the south pole adds a single point ∞. Right: projection from the origin maps $\mathbb{R}^n$ onto the open upper hemisphere, and each direction ω in $\mathbb{R}^n$ acquires its own point at infinity on the equator $\{X_0 = 0\}$.

Question. Why not use the one-point compactification for symbols? It is simpler, and it is a smooth manifold without boundary.

Answer. For two reasons. First, Proposition 1.7.1 shows that general linear changes of variables do not extend to it, while symbols must be invariant under them. Second, the typical symbol $\xi/|\xi|$ (for $|\xi| > 1$) has a different limit along each ray, so it does not even extend continuously to a single point at infinity. The radial compactification keeps one point at infinity for each direction, which is exactly what homogeneous functions of degree 0 need. □

1.7.2 The radial compactification

Instead of the south pole, project from the origin (Figure 1.3). The ray from 0 through $(1, z)$ meets the upper half of the unit sphere at the point

$$\mathrm{RC}\colon \mathbb{R}^n \ni z \longmapsto \Big(\frac{1}{\langle z\rangle}, \frac{z}{\langle z\rangle}\Big) \in \mathbb{S}^n_+ = \{X = (X_0, X') \in \mathbb{R}^{n+1};\ X_0 \geq 0,\ |X| = 1\}. \quad (1.7.1)$$

This is a diffeomorphism onto the open hemisphere $\{X_0 > 0\}$, with inverse $X \mapsto X'/X_0$. The closed hemisphere $\mathbb{S}^n_+$ is a compact manifold with boundary $\{X_0 = 0\} \cong \mathbb{S}^{n-1}$, diffeomorphic to a closed ball.

Definition 1.7.2. The *radial compactification* $\overline{\mathbb{R}^n}$ is $\mathbb{S}^n_+$ with $\mathbb{R}^n$ identified with its interior by RC. Its boundary is the *sphere at infinity*. $\mathcal{C}^\infty(\overline{\mathbb{R}^n})$ denotes the functions f on $\mathbb{R}^n$ such that $f \circ \mathrm{RC}^{-1}$ extends to a smooth function on $\mathbb{S}^n_+$, smooth up to the boundary. We write $\dot{\mathcal{C}}^\infty(\overline{\mathbb{R}^n})$ for the subspace of those that vanish to infinite order at the boundary, with all derivatives.

The function $\rho = X_0$, pulled back to $\langle z\rangle^{-1}$, is a *boundary defining function*: it is smooth, positive in the interior, and vanishes simply on the boundary. Near the boundary there is a convenient product structure.

Lemma 1.7.3. *The map $z \mapsto (s,\omega) = (|z|^{-1}, z/|z|)$, $|z|>1$, extends to a diffeomorphism of a neighbourhood $\{X_0 < |X'|\}$ of the boundary of $\overline{\mathbb{R}^n}$ onto $[0,1)_s \times \mathbb{S}^{n-1}$. Hence $f \in \mathcal{C}^\infty(\mathbb{R}^n)$ is in $\mathcal{C}^\infty(\overline{\mathbb{R}^n})$ if and only if $f(z) = F(|z|^{-1}, z/|z|)$ in $|z|>1$ for some $F \in \mathcal{C}^\infty([0,1)\times\mathbb{S}^{n-1})$. Moreover*

$$\langle z\rangle^{-1} = s(1+s^2)^{-1/2}, \qquad \frac{z}{\langle z\rangle} = (1+s^2)^{-1/2}\omega$$

are in $\mathcal{C}^\infty(\overline{\mathbb{R}^n})$.

Proof. In terms of X, $s = X_0/|X'|$ and $\omega = X'/|X'|$, which are smooth where $X' \neq 0$. The inverse is $X = (s,\omega)/(1+s^2)^{1/2}$, which is smooth on $[0,\infty)\times\mathbb{S}^{n-1}$. The region $|z|>1$ corresponds to $X_0 < |X'|$, that is $s<1$. □

Remark 1.7.4. As a set, $\overline{\mathbb{R}^n} = \mathbb{R}^n \sqcup \mathbb{S}^{n-1}$, where the boundary point $(0,\omega)$ is the limit of $\mathrm{RC}(t\omega)$ as $t\to\infty$, the "end" of the ray in direction ω. Lemma 1.7.3 shows that the smooth structure is generated by the functions on $\mathbb{R}^n$ which, outside a compact set, are homogeneous of degree 0 (such as $z/|z|$), or homogeneous of degree -1 (such as $|z|^{-1}$). A function homogeneous of degree 0 for $|z|>1$ extends to $\overline{\mathbb{R}^n}$ with boundary value its restriction to the unit sphere. A function homogeneous of negative integral degree $-k$ is s^k times such a function, and so vanishes at the boundary. For an abstract vector space W we set $\mathbb{S}W = (W\setminus 0)/\mathbb{R}^+$ and $\overline{W} = W \sqcup \mathbb{S}W$. The smooth structure is defined through any linear isomorphism $W\to\mathbb{R}^n$, and it does not depend on the choice by Proposition 1.7.5 below.

1.7.3 Invariance

Proposition 1.7.5. *Every affine map $T(z) = Gz + a$, $G\in\mathrm{GL}(n,\mathbb{R})$, $a\in\mathbb{R}^n$, extends to a diffeomorphism $\tilde{T}$ of $\overline{\mathbb{R}^n}$. Explicitly,*

$$\tilde{T}(X) = \frac{MX}{|MX|}, \qquad M(X_0,X') = (X_0, GX' + aX_0).$$

In particular the radial compactification $\overline{W}$ of any real vector space is well defined, and translations act on it smoothly.

Proof. M is an invertible linear map of $\mathbb{R}^{n+1}$ that preserves $\{X_0>0\}$ and $\{X_0=0\}$. If $X\in\mathbb{S}^n_+$ then $MX\neq 0$, so $\tilde{T}$ is a smooth map of a neighbourhood of $\mathbb{S}^n_+$ into $\mathbb{S}^n_+$. Its inverse is the same construction for M^{-1}, which corresponds to T^{-1}. Since $\mathrm{RC}(z)$ is a positive multiple of $(1,z)$ and $M(1,z) = (1, Gz+a)$, we get $\tilde{T}(\mathrm{RC}(z)) = \mathrm{RC}(Tz)$. □

The infinitesimal version is also useful. It shows which vector fields on $\mathbb{R}^n$ extend smoothly, and with what vanishing at infinity.

Lemma 1.7.6. *In the coordinates (s,ω) of Lemma 1.7.3,*

$$\partial_{z_j} = s(-\omega_j\, s\partial_s + W_j), \qquad z_i\partial_{z_j} = -\omega_i\omega_j\, s\partial_s + \omega_i W_j,$$

where W_j is the smooth vector field on $\mathbb{S}^{n-1}$ given by orthogonal projection of e_j onto the tangent space, $W_j = \sum_k(\delta_{jk} - \omega_j\omega_k)\partial_{\omega_k}$. Thus the linear vector fields $z_i\partial_{z_j}$ extend to smooth vector fields on $\overline{\mathbb{R}^n}$ tangent to the boundary. The constant vector fields ∂_{z_j} extend to ρ times smooth vector fields tangent to the boundary. Conversely, the radial field $s\partial_s = -\sum_j z_j\partial_{z_j}$ and the rotation fields $\Omega_{jk} = z_j\partial_{z_k} - z_k\partial_{z_j}$ span, over $\mathcal{C}^\infty([0,1)\times\mathbb{S}^{n-1})$, the smooth vector fields tangent to the boundary near the boundary.

Proof. From $s = |z|^{-1}$ and $\omega = z/|z|$: $\partial_{z_j} s = -z_j|z|^{-3} = -s^2\omega_j$ and $\partial_{z_j}\omega_k = s(\delta_{jk} - \omega_j\omega_k)$. This gives the first formula, and the second follows on multiplying by $z_i = \omega_i/s$. The fields Ω_{jk} are tangent to the spheres $|z| = \text{const}$, are homogeneous of degree 0, and span the tangent space of $\mathbb{S}^{n-1}$ at every point. □

1.7.4 Schwartz functions

Proposition 1.7.7. *Pull-back by* RC *identifies* $\dot{\mathcal{C}}^\infty(\overline{\mathbb{R}^n})$ *with* $\mathcal{S}(\mathbb{R}^n)$*, as sets and as Fréchet spaces.*

Proof. Both conditions are about behaviour in $|z| > 1$, so we work in the collar $[0,1)\times\mathbb{S}^{n-1}$ and write $f(z) = F(s,\omega)$.

Suppose $F \in \dot{\mathcal{C}}^\infty$. By Lemma 1.7.6, applying the first formula repeatedly and using $V(s^p g) = s^p(V - p\omega_j)g$ for $V = -\omega_j s\partial_s + W_j$, we get $\partial_z^\beta f = s^{|\beta|}Q_\beta F$, where Q_β is a differential operator with smooth coefficients on $[0,1)\times\mathbb{S}^{n-1}$. Also $z^\alpha = s^{-|\alpha|}\omega^\alpha$. Since F and all its derivatives vanish to infinite order at $s = 0$, $z^\alpha\partial^\beta f = s^{|\beta|-|\alpha|}\omega^\alpha Q_\beta F$ is bounded. So $f \in \mathcal{S}(\mathbb{R}^n)$.

Conversely, let $f \in \mathcal{S}(\mathbb{R}^n)$. Every differential operator with smooth coefficients on $[0,1)\times\mathbb{S}^{n-1}$ is a combination, with smooth coefficients, of products of ∂_s and the Ω_{jk}. On $s > 0$ we have $\partial_s = -s^{-1}\sum_j z_j\partial_{z_j} = -|z|\sum_j z_j\partial_{z_j}$, and Ω_{jk} has polynomial coefficients. Hence every derivative of F is a finite sum of smooth bounded coefficients times $|z|^p\, z^\alpha\partial_z^\beta f$, which is $O(|z|^{-N}) = O(s^N)$ for every N. A function on $(0,1)\times\mathbb{S}^{n-1}$ all of whose derivatives are $O(s^N)$ for all N extends by zero to an element of $\dot{\mathcal{C}}^\infty([0,1)\times\mathbb{S}^{n-1})$. Indeed, each derivative extends continuously by 0, and the difference quotient in s at $s = 0$ is $O(s^{N-1}) \to 0$, so by induction the extension is $\mathcal{C}^k$ for all k with vanishing derivatives at $s = 0$. The estimates in both directions are uniform, so the identification is a topological isomorphism. □

Hence the dual space $\mathcal{S}'(\mathbb{R}^n)$ is the dual of $\dot{\mathcal{C}}^\infty(\overline{\mathbb{R}^n})$. The dual of $\dot{\mathcal{C}}^\infty$ on a compact manifold with boundary is the space of *extendible distributions*, so tempered distributions on $\mathbb{R}^n$ are exactly the extendible distributions on $\overline{\mathbb{R}^n}$. This is one way of "bringing infinity in".

1.7.5 Symbols as functions on the radial compactification

Proposition 1.7.8. *Let* $a \in \mathcal{C}^\infty(\overline{\mathbb{R}^n})$*. Then:*

(i) $a \in \mathcal{C}^\infty(\mathbb{R}^n)$ *and* $|D^\alpha a(z)| \le C_\alpha\langle z\rangle^{-|\alpha|}$ *for all* α*, so* a *is a symbol of order* 0 *in the sense of Chapter 2;*

(ii) there are $a_j \in \mathcal{C}^\infty(\mathbb{S}^{n-1})$ *(the Taylor coefficients of* a *at* $s = 0$*) such that for every* N

$$r_N(z) = a(z) - \sum_{j<N} |z|^{-j}a_j(z/|z|) \quad \text{satisfies} \quad |D^\alpha r_N(z)| \le C_{N,\alpha}|z|^{-N-|\alpha|}, \quad |z| > 1;$$

(iii) for each polynomial p *of degree* m*,* $\langle z\rangle^{-m}p \in \mathcal{C}^\infty(\overline{\mathbb{R}^n})$*, with boundary value* $p_m(\omega)$*, where* p_m *is the homogeneous part of degree* m*.*

Proof. For (i), the proof of Proposition 1.7.7 gives $\partial^\alpha a = s^{|\alpha|}Q_\alpha F$ in $|z| > 1$, with $Q_\alpha F$ bounded. Since $s \le 2\langle z\rangle^{-1}$ there, this gives the estimate. In $|z| \le 1$ it is trivial. For (ii), Taylor's formula in s gives $F(s,\omega) = \sum_{j<N} s^j a_j(\omega) + s^N R_N(s,\omega)$ with R_N smooth, and $\partial^\alpha(s^N R_N) = s^{N+|\alpha|}Q'_\alpha R_N$ as before. For (iii), $\langle z\rangle^{-m}z^\alpha = (z/\langle z\rangle)^\alpha\langle z\rangle^{|\alpha|-m}$ is a product of components of X' and the power $X_0^{m-|\alpha|}$, so it is smooth. At $X_0 = 0$ only the terms with $|\alpha| = m$ survive, and there $X' = \omega$. □

More generally, if $\rho \in \mathcal{C}^\infty(\overline{W})$ is any boundary defining function, the space

$$S^z_{\mathrm{ph}}(W) = \rho^{-z}\mathcal{C}^\infty(\overline{W}), \qquad z \in \mathbb{C},$$

is the space of *classical* (or *polyhomogeneous*) symbols of order z on W. It does not depend on the choice of ρ, since two boundary defining functions differ by a positive smooth factor. By Proposition 1.7.5 it is invariant under affine changes of variables. Part (ii) of Proposition 1.7.8 is the asymptotic expansion in homogeneous terms of decreasing degree that defines polyhomogeneous symbols in Chapter 2. Using Borel's lemma one shows there that, conversely, every symbol of order 0 with such an expansion is in $\mathcal{C}^\infty(\overline{\mathbb{R}^n})$.

1.8 Further compactifications

1.8.1 The quadratic compactification

Projecting the hemisphere $\mathbb{S}^n_+$ onto its last n coordinates gives a homeomorphism onto the closed unit ball $\mathbb{B}^n$. Composing with RC gives the *quadratic compactification*

$$\mathrm{Q}\colon \mathbb{R}^n \ni z \longmapsto \frac{z}{\langle z\rangle} \in \mathbb{B}^n = \{y \in \mathbb{R}^n;\ |y| \leq 1\}, \tag{1.8.1}$$

a diffeomorphism onto the open ball. We write ${}^{\mathrm{q}}\overline{\mathbb{R}^n}$ for $\mathbb{B}^n$ with this identification. The projection $\mathbb{S}^n_+ \to \mathbb{B}^n$ is smooth, but its inverse $y \mapsto ((1-|y|^2)^{1/2}, y)$ is not smooth at $|y| = 1$. So the identity of $\mathbb{R}^n$ extends to a smooth map $\overline{\mathbb{R}^n} \to {}^{\mathrm{q}}\overline{\mathbb{R}^n}$ but not conversely, and $\mathcal{C}^\infty({}^{\mathrm{q}}\overline{\mathbb{R}^n}) \subsetneq \mathcal{C}^\infty(\overline{\mathbb{R}^n})$. Concretely, $\langle z\rangle^{-1} = (1-|y|^2)^{1/2}$ is not smooth on ${}^{\mathrm{q}}\overline{\mathbb{R}^n}$. The function $1-|y|^2 = \langle z\rangle^{-2}$ is a boundary defining function there, and near the boundary $(t,\omega) = (|z|^{-2}, z/|z|)$ are smooth coordinates, which explains the name.

Proposition 1.8.1. *(i) Pull-back by* Q *identifies* $\dot{\mathcal{C}}^\infty({}^{\mathrm{q}}\overline{\mathbb{R}^n})$*, the smooth functions on* $\mathbb{R}^n$ *supported in* $\mathbb{B}^n$*, with* $\mathcal{S}(\mathbb{R}^n)$*.*

(ii) Every $G \in \mathrm{GL}(n,\mathbb{R})$ *extends to a diffeomorphism of* ${}^{\mathrm{q}}\overline{\mathbb{R}^n}$*, given by* $y \mapsto Gy/(1-|y|^2+|Gy|^2)^{1/2}$*.*

(iii) For $n \geq 2$ *and* $a \neq 0$*, the translation* $z \mapsto z+a$ *does not extend to a* $\mathcal{C}^1$ *map of* ${}^{\mathrm{q}}\overline{\mathbb{R}^n}$*.*

Proof. (i) Let $\pi(X) = X'$. If $g \in \dot{\mathcal{C}}^\infty(\mathbb{B}^n)$ then $g\circ\pi \in \dot{\mathcal{C}}^\infty(\mathbb{S}^n_+)$, because π is smooth and maps the boundary to the boundary. Conversely, let $f \in \dot{\mathcal{C}}^\infty(\mathbb{S}^n_+)$ and $g = f\circ\pi^{-1}$. A derivative of order k of g is a sum of products of derivatives of f at $\pi^{-1}(y)$ with derivatives of $X_0 = (1-|y|^2)^{1/2}$ of order at most k, which are $O(X_0^{1-2k})$. The derivatives of f are $O(X_0^N)$ for all N. So all derivatives of g tend to 0 at $|y| = 1$, and g extended by 0 is smooth. Now apply Proposition 1.7.7.

(ii) With $y = z/\langle z\rangle$ we have $\langle z\rangle^2 = (1-|y|^2)^{-1}$, so $Gz/\langle Gz\rangle = Gy/(1-|y|^2+|Gy|^2)^{1/2}$. The denominator is positive on $\mathbb{B}^n$, since $Gy \neq 0$ when $|y| = 1$, and the inverse is given by G^{-1}.

(iii) Let $\rho = (1-|y|^2)^{1/2} = \langle z\rangle^{-1}$. Since $\langle z+a\rangle^2 = \langle z\rangle^2 + 2z\cdot a + |a|^2$, the translation becomes

$$y \longmapsto \frac{y+\rho a}{(1+2\rho\, y\cdot a + \rho^2|a|^2)^{1/2}} = y + \rho(a - (y\cdot a)y) + \rho^2 h(y,\rho),$$

with h smooth in (y,ρ). Choose a unit vector $y_0 \perp a$, which is possible as $n \geq 2$, and set $e = a/|a|$. Along $y = ty_0$, $t \uparrow 1$, the e-component of the image is $|a|\rho + \rho^2 h_e$ with $\rho = (1-t^2)^{1/2}$. Its derivative in t is unbounded as $t \to 1$, so the map is not $\mathcal{C}^1$ up to the boundary. □

Thus the quadratic compactification is well defined for a vector space, but not for an affine space. In terms of vector fields, near the boundary $z_i\partial_{z_j} = -2\omega_i\omega_j\, t\partial_t + \omega_i W_j$ is smooth, while $\partial_{z_j} = t^{1/2}(-2\omega_j\, t\partial_t + W_j)$ is not. This is one reason to prefer the radial compactification, on which translations act smoothly. The other is Proposition 1.7.8: classical symbols with terms of odd degree, such as $\langle z\rangle^{-1}$, are smooth on $\overline{\mathbb{R}^n}$ but not on ${}^{q}\overline{\mathbb{R}^n}$.

1.8.2 Relation to the one-point compactification

In the coordinate $w = \iota(z) = 4z/|z|^2$ near $\infty \in {}^{1}\overline{\mathbb{R}^n}$ we have $w = 4s\omega$ in terms of the coordinates of Lemma 1.7.3, and this is smooth on $[0,1)\times\mathbb{S}^{n-1}$. So the identity extends to a smooth map

$$\beta\colon \overline{\mathbb{R}^n} \longrightarrow {}^{1}\overline{\mathbb{R}^n},$$

which collapses the sphere at infinity to the point ∞ and is a diffeomorphism elsewhere. In particular $\mathcal{C}^\infty({}^{1}\overline{\mathbb{R}^n}) \subset \mathcal{C}^\infty(\overline{\mathbb{R}^n})$. Near the boundary, β is the introduction of polar coordinates $(s,\omega)\mapsto s\omega$ around ∞. In the language of later chapters, $\overline{\mathbb{R}^n}$ is the *blow-up* $[{}^{1}\overline{\mathbb{R}^n}, \{\infty\}]$. In general, a manifold with boundary X together with a smooth map $\beta\colon X\to M$ is the blow-up of M at p if β is a diffeomorphism of $X\setminus\partial X$ onto $M\setminus\{p\}$, maps ∂X to p, and polar coordinates around p (in some local coordinates) lift to a diffeomorphism of a neighbourhood of ∂X onto $[0,\epsilon)\times\mathbb{S}^{n-1}$. This does not depend on the choice of local coordinates (Exercise 1.19).

1.8.3 Subspaces and relative compactifications

Proposition 1.8.2. *If $V\subset W$ is a linear subspace, then the inclusion extends to a smooth embedding $\overline{V}\hookrightarrow\overline{W}$, and likewise ${}^{q}\overline{V}\hookrightarrow{}^{q}\overline{W}$. Every injective linear map $V\to W$ extends to a smooth map $\overline{V}\to\overline{W}$.*

Proof. By Proposition 1.7.5 we may choose coordinates with $V = \mathbb{R}^k\times\{0\}$. Then $\mathrm{RC}_n(v,0) = (\langle v\rangle^{-1}, v/\langle v\rangle, 0)$, which is the standard embedding $\mathbb{S}^k_+\hookrightarrow\mathbb{S}^n_+$ composed with RC_k. The quadratic case is the same. An injective linear map is an isomorphism onto its image followed by an inclusion. □

Linear maps that are not injective do not extend in general (Exercise 1.16): the points at infinity "do not know where to go". The remedy is to compactify relative to a subspace. Let $V\subset W$ and choose a complement U, so $W = V\times U$, together with Euclidean metrics on V and U. Consider

$$R_V\colon W\ni(v,u)\longmapsto \big(\mathrm{RC}_U(u),\ \mathrm{RC}_V(v/\langle u\rangle)\big)\in\overline{U}\times\overline{V}. \tag{1.8.2}$$

Explicitly, $R_V(v,u) = (t,u',s,v')$ with

$$t = \frac{1}{\langle u\rangle},\quad u' = \frac{u}{\langle u\rangle},\quad s = \frac{\langle u\rangle}{(1+|u|^2+|v|^2)^{1/2}},\quad v' = \frac{v}{(1+|u|^2+|v|^2)^{1/2}},$$

satisfying $t^2+|u'|^2 = 1 = s^2+|v'|^2$. Here we used $\langle v/\langle u\rangle\rangle = (1+|u|^2+|v|^2)^{1/2}/\langle u\rangle$. The map $(v,u)\mapsto(v/\langle u\rangle,u)$ is a diffeomorphism of W, so R_V is a diffeomorphism onto the interior of the compact manifold with corners $\overline{U}\times\overline{V}$, with inverse $(t,u',s,v')\mapsto(v'/(st),u'/t)$. We write ${}^{V}\overline{W}$ for $\overline{U}\times\overline{V}$ with this identification, and call it the *radial compactification of W relative to V*. Figure 1.4 shows the simplest case.

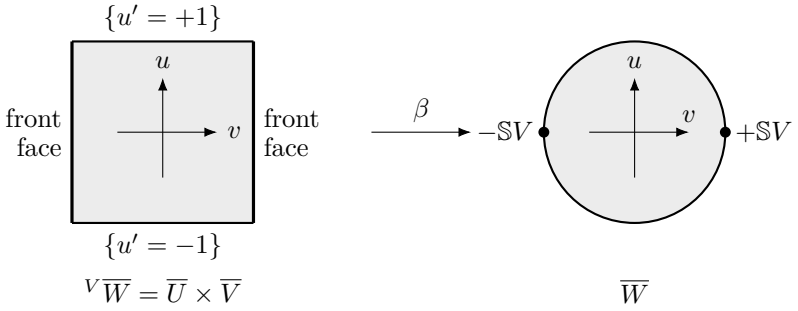


Figure 1.4: The relative compactification for $W = \mathbb{R}^2$, $V = \mathbb{R} \times \{0\}$ and $U = \{0\} \times \mathbb{R}$. The square $\overline{U} \times \overline{V}$ maps onto the disc $\overline{W}$ by a map β extending the identity of W. The faces $\{s = 0\}$ (heavy vertical sides), where $|v| \to \infty$ with u bounded, collapse to the two points $\pm\mathbb{S}V$ of the sphere at infinity of V; the other two sides map to the two arcs of the circle. This is the blow-up $[\overline{W}, \mathbb{S}V]$ of Remark 1.8.4.

Proposition 1.8.3. *Every affine map of W whose linear part maps V into itself extends to a diffeomorphism of ${}^V\overline{W}$. In particular ${}^V\overline{W}$ does not depend on the choice of complement U or of the metrics. The short exact sequence $0 \to V \to W \to W/V \to 0$ extends to smooth maps*

$$\overline{V} \longrightarrow {}^V\overline{W} \longrightarrow \overline{W/V}.$$

If $A\colon V \to W$ is any linear map with null space N, then A extends to a smooth map ${}^N\overline{V} \to \overline{W}$.

Proof. An affine map preserving the direction V has the form $T(v, u) = (Gv+Su+a,\ Hu+b)$, with $G \in \mathrm{GL}(V)$, $H \in \mathrm{GL}(U)$, $S\colon U \to V$ linear, $a \in V$ and $b \in U$. In the coordinates $(\nu, u) = (v/\langle u\rangle, u)$ it becomes

$$(\nu, u) \longmapsto \Big(c(u)(G\nu + S\tfrac{u}{\langle u\rangle} + a\langle u\rangle^{-1}),\ Hu + b\Big), \qquad c(u) = \frac{\langle u\rangle}{\langle Hu + b\rangle}.$$

By Proposition 1.7.5, $u \mapsto Hu + b$ extends to a diffeomorphism of $\overline{U}$, and $\langle Hu + b\rangle^{-1}$ is the pull-back of the boundary defining function $\langle u\rangle^{-1}$ under this map. Hence c is a positive function in $\mathcal{C}^\infty(\overline{U})$. The functions $u/\langle u\rangle$ and $\langle u\rangle^{-1}$ are also in $\mathcal{C}^\infty(\overline{U})$. Thus the first component is an affine map of V, with invertible linear part, depending smoothly on the parameter $p \in \overline{U}$. The formula of Proposition 1.7.5, with M depending smoothly on p, shows that it extends smoothly to $\overline{U} \times \overline{V}$. The same holds for the inverse. Changing the complement or the metrics is such a linear map, so ${}^V\overline{W}$ does not depend on these choices.

The inclusion $v \mapsto (v, 0)$ becomes $v \mapsto (\mathrm{RC}_U(0), \mathrm{RC}_V(v))$, which extends to $\overline{V} \cong \{\mathrm{RC}_U(0)\} \times \overline{V}$. The projection $(v, u) \mapsto u$, with $U \cong W/V$, extends as the projection $\overline{U} \times \overline{V} \to \overline{U}$. Finally, write A as the projection $V \to V/N$ followed by the injective map $V/N \to W$ induced by A, and use the first part of this proposition together with Proposition 1.8.2. □

Remark 1.8.4. In the trivial cases $V = \{0\}$ and $V = W$ the relative compactification reduces to $\overline{W}$ (Exercise 1.17). For a non-trivial subspace, ${}^V\overline{W}$ is canonically the blow-up $[\overline{W}, \mathbb{S}V]$ of $\overline{W}$ along the sphere at infinity of V, which is a submanifold of the boundary. Similarly, for a product $W = V \times U$ the identity map extends to neither $\overline{V} \times \overline{U} \to \overline{W}$ nor its inverse. However, the doubly-relative compactification (Exercise 1.18) is canonically both $[\overline{V \times U}; \mathbb{S}U, \mathbb{S}V]$ and $[\overline{V} \times \overline{U}; \mathbb{S}V \times \mathbb{S}U]$. We do not prove these identifications here. Blow-up of submanifolds is taken up when it is needed later in the book.

1.9 Exercises

Exercise 1.1. Suppose $\phi\colon \mathbb{R}^n \to \mathbb{C}$ is such that for each $\bar{x} \in \mathbb{R}^n$ and each $k \in \mathbb{N}_0$ there are $\epsilon_k > 0$ and a polynomial $p_k(x;\bar{x})$ with $|\phi(x) - p_k(x;\bar{x})| \le \epsilon_k^{-1}|x-\bar{x}|^{k+1}$ for $|x-\bar{x}| \le \epsilon_k$. Must ϕ be infinitely differentiable? (Consider $\phi(x) = e^{-1/x^2}\sin(e^{1/x^2})$ on $\mathbb{R}$.)

Exercise 1.2. Show that $u(x) = e^x\cos(e^x)$ is a tempered distribution on $\mathbb{R}$, in the sense that there is $T \in \mathcal{S}'(\mathbb{R})$ with $T(\phi) = \int u\phi\,dx$ for all $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$. Show that T is unique. (Write u as the derivative of a bounded function.) Note that $\int u\phi\,dx$ need not converge absolutely for $\phi \in \mathcal{S}(\mathbb{R})$.

Exercise 1.3. Show that d in (1.1.2) is a metric on $\mathcal{S}(\mathbb{R}^n)$. (If $\|\cdot\|$ is a norm, show that $\frac{\|u+v\|}{1+\|u+v\|} \le \frac{\|u\|}{1+\|u\|} + \frac{\|v\|}{1+\|v\|}$.)

Exercise 1.4. Show that a sequence $\phi_j \in \mathcal{S}(\mathbb{R}^n)$ is Cauchy, respectively converges to ϕ, for d if and only if it is Cauchy, respectively converges to ϕ, for each norm $\|\cdot\|_k$.

Exercise 1.5. Show that a bilinear map $B\colon \mathcal{S}(\mathbb{R}^n)\times\mathcal{S}(\mathbb{R}^p) \to \mathbb{C}$ is (jointly) continuous if and only if $|B(\phi,\psi)| \le C\|\phi\|_N\|\psi\|_N$ for some C and N. Combined with Lemma 1.6.4, this shows that separate continuity implies joint continuity.

Exercise 1.6. Show that the restriction map $\pi_k^*\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^k)$ is continuous and surjective.

Exercise 1.7. Give a second proof of Lemma 1.4.2. For $n = 1$ write $\int e^{-ix\xi}e^{-x^2}dx = e^{-\xi^2/4}\int e^{-(x+\frac{i}{2}\xi)^2}dx$, interpret the integral as a contour integral, and shift the contour to $\operatorname{Im} x = -\xi/2$.

Exercise 1.8. Show that (1.2.1) makes sense for $\phi \in L^2(\mathbb{R}^n)$, and that the resulting map $L^2(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ is injective.

Exercise 1.9. (A weak Sobolev embedding.)

(a) Show that if $f \in L^1(\mathbb{R}^n)$ then $v(x) = \int_{-\infty}^{x_1}\cdots\int_{-\infty}^{x_n} f(t)\,dt$ is bounded and continuous and $D_1\cdots D_n v = (-i)^n f$ in $\mathcal{S}'(\mathbb{R}^n)$. Show that $(1+|x|)^{-n-1}L^2(\mathbb{R}^n) \subset L^1(\mathbb{R}^n)$.

(b) Show that if $u \in \mathcal{S}'(\mathbb{R}^n)$ and $\langle\xi\rangle^s\hat{u} \in L^2(\mathbb{R}^n)$ for some $s > n/2 + k$, then $u \in \mathcal{C}^k(\mathbb{R}^n)$ and $D^\alpha u$ is bounded and continuous for $|\alpha| \le k$.

Exercise 1.10. Show that for $\phi \in \mathcal{S}(\mathbb{R}^n)$ (indeed for ϕ continuous and polynomially bounded), the support of T_ϕ in the sense of Definition 1.2.4 is the closure of $\{x;\ \phi(x) \ne 0\}$. Show that $\operatorname{supp}\delta = \{0\}$.

Exercise 1.11. Show that a differential operator P with coefficients in $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ is *local*: $\operatorname{supp}(Pu) \subset \operatorname{supp} u$ for all $u \in \mathcal{S}'(\mathbb{R}^n)$. The converse, for a linear operator $P\colon \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ with $\operatorname{supp}(Pu) \subset \operatorname{supp} u$ and (for simplicity) $\operatorname{supp}(Pu) \subset K$ for a fixed compact K, is Peetre's theorem. How would you try to prove it? (No full proof required.)

Exercise 1.12. Find a bounded continuous function w on $\mathbb{R}$ with $\delta = (1+D^2)w$, as predicted by Theorem 1.6.3. (Try $w = \frac{1}{2}e^{-|x|}$.) Can δ be written as Dw with w bounded and continuous?

Exercise 1.13. Use Theorem 1.6.6 to show that for $u \in \mathcal{S}'(\mathbb{R}^n)$ and $v \in \mathcal{S}'(\mathbb{R}^m)$ there is a unique $u \boxtimes v \in \mathcal{S}'(\mathbb{R}^{n+m})$ with $(u\boxtimes v)(\phi\boxtimes\psi) = u(\phi)v(\psi)$.

Exercise 1.14. Formulate precisely the notion of a conformal structure on a real vector space W (a Euclidean metric up to positive multiples). Show that the one-point compactification of W depends only on such a structure. If $V \subset W$ carries the induced conformal structure, show that the inclusion extends to a smooth map ${}^{1}\overline{V} \to {}^{1}\overline{W}$.

Exercise 1.15. Show that the quadratic compactification can be defined for any real vector space, and that as a set it is again $W \sqcup \mathbb{S}W$, with a different smooth structure. (Use Proposition 1.8.1(ii).)

Exercise 1.16. Show that the projection $\mathbb{R}^2 \ni (z_1, z_2) \mapsto z_1 \in \mathbb{R}$ does not extend to a continuous map $\overline{\mathbb{R}^2} \to \overline{\mathbb{R}}$. (Consider the curves $z = (c, t)$, $t \to \infty$.) Show more generally that a linear map between vector spaces of positive dimension extends continuously to the radial (or quadratic) compactifications if and only if it is injective.

Exercise 1.17. Show from (1.8.2) that ${}^{\{0\}}\overline{W}$ and ${}^{W}\overline{W}$ are both canonically $\overline{W}$.

Exercise 1.18. (Doubly-relative compactification.) For $W = V \times U$ with metrics on V and U, consider

$$W \ni (v,u) \longmapsto \Big(\frac{1}{\langle u\rangle}, \frac{1}{\langle v\rangle}, \frac{\langle u\rangle}{\langle (u,v)\rangle}, \frac{\langle v\rangle}{\langle (u,v)\rangle}, \frac{v}{\langle v\rangle}, \frac{(v,u)}{\langle (u,v)\rangle}, \frac{u}{\langle u\rangle}\Big) \in \mathbb{R}^4 \times V \times W \times U.$$

Show that this is a diffeomorphism onto its image, and that the closure of the image is a compact manifold with corners. It has three boundary hypersurfaces if $\dim U, \dim V \geq 2$. Denote it by ${}^{U,V}\overline{U \times V}$. Show that translations, and linear maps preserving both U and V, lift to diffeomorphisms. Show that the inclusions of $\overline{U}$ and $\overline{V}$ extend smoothly, and that the identity of W extends to smooth maps from ${}^{U,V}\overline{U \times V}$ to both $\overline{U \times V}$ and $\overline{U} \times \overline{V}$. Draw a picture for $\dim U = \dim V = 1$.

Exercise 1.19. Show that a change of local coordinates fixing $0 \in \mathbb{R}^n$ induces a diffeomorphism of $[0, \epsilon) \times \mathbb{S}^{n-1}$, for some $\epsilon > 0$, under polar coordinates $(r, \omega) \mapsto r\omega$. (Treat linear changes first, using Proposition 1.7.5. Then reduce to the case where the Jacobian at 0 is the identity.) Conclude that the blow-up of a point is well defined.

Notes

This chapter merges Chapter 1 of *Introduction to Microlocal Analysis* with Lecture 1 (and its addenda) of *Lectures on Pseudodifferential Operators*. The "two big theorems" were only stated in the original, with outlines of proofs in the problems. Here they are proved in full, using a single estimate (Lemma 1.6.1) and a Riesz representation argument in place of the Hilbert–Schmidt argument suggested originally. Separate continuity is upgraded to joint continuity by a Baire category argument, and uniqueness of the kernel uses the density of tensor products. The classification of distributions supported at a point is proved in all dimensions. Both compactification discussions have been unified around the explicit formulae (1.7.1) and Proposition 1.7.5, which replace the Lie algebra arguments of the original. The relative compactification is written as a product $\overline{U} \times \overline{V}$ by the change of variables $v \mapsto v/\langle u\rangle$. For the kernel theorem and the structure of distributions see [7, Chapter 5 and §7.1] and [3]. Compactifications, blow-ups and manifolds with corners are treated systematically in Melrose's work on the *b*-calculus, see [16].

Chapter 2

Pseudodifferential operators on Euclidean space

Differential operators form an algebra, but it is too small for most purposes: the inverse of an elliptic differential operator, such as $(1+\Delta)^{-1}$, is never a differential operator. On the Fourier side, $1+\Delta$ is multiplication by $\langle\xi\rangle^2$, and its inverse is multiplication by $\langle\xi\rangle^{-2}$, a smooth function that behaves like a polynomial of negative degree. Pseudodifferential operators are what we get by allowing all such "polynomial-like" multipliers, with coefficients depending on x. They form an algebra that contains approximate inverses of elliptic operators, and this is the main reason to introduce them.

This chapter constructs the basic algebra of pseudodifferential operators on $\mathbb{R}^n$, the "traditional" calculus with symbols of type $(1,0)$ whose coefficients are uniformly smooth in the base variables. The starting point is the Fourier representation of a differential operator with coefficients in $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$,

$$P(x,D)u(x) = \int e^{i(x-y)\cdot\xi} P(x,\xi)u(y)\,dy đ\xi = \int e^{ix\cdot\xi} P(x,\xi)\hat{u}(\xi) đ\xi, \qquad u \in \mathcal{S}(\mathbb{R}^n),$$

where $P(x,\xi) = \sum_{|\alpha|\le m} p_\alpha(x)\xi^\alpha$ is the characteristic polynomial. We generalize this formula by replacing the polynomial $P(x,\xi)$ by a *symbol*, a smooth function satisfying uniform estimates at infinity, and we even allow it to depend on the "incoming" variable y, so that operators are given by

$$Au(x) = \int e^{i(x-y)\cdot\xi} a(x,y,\xi)u(y)\,dy đ\xi, \qquad u \in \mathcal{S}(\mathbb{R}^n). \tag{2.0.1}$$

Such integrals are in general not absolutely convergent. Section 2.1 develops the symbol spaces, including asymptotic summation. In Section 2.2 the integral (2.0.1) is interpreted as an oscillatory integral and shown to define a continuous operator on $\mathcal{S}(\mathbb{R}^n)$. Section 2.3 shows that every such operator can be written with an amplitude independent of y (reduction to the left symbol), and Section 2.4 deduces from this that the operators form an order-filtered $*$-algebra. The remaining sections develop the consequences: principal symbols, ellipticity and parametrices, elliptic regularity, boundedness on L^2 and on (weighted) Sobolev spaces, and polyhomogeneous (classical) operators, including those of complex order. Section 2.10 topologizes the operator spaces and shows that reduction, adjoints and composition are continuous; Section 2.11 uses this to discuss smooth and holomorphic families of operators, including holomorphic families whose order varies with the parameter. Finally we show invariance under affine changes of variables and, for operators with compactly supported kernels, under diffeomorphisms of open subsets of $\mathbb{R}^n$ (Sections 2.12 and 2.13); a full treatment of the latter, together with the wavefront set, is given in Chapter 3, and operators

on manifolds are the subject of Chapter 5. The semiclassical version of the calculus is developed in Chapter 6. Figure 2.1 shows how the main results depend on each other; a first reading can stop after Section 2.8.

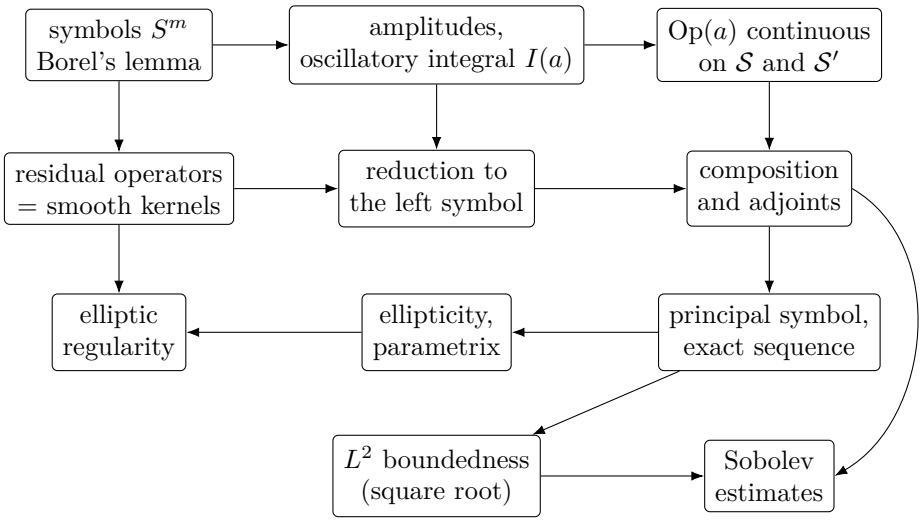


Figure 2.1: Logical structure of the chapter, up to Sobolev estimates. The polyhomogeneous calculus, the topologies, families and coordinate invariance (Sections 2.9 to 2.13) refine this core.

Throughout, $\langle\xi\rangle = (1+|\xi|^2)^{1/2}$, $D = -i\partial$, $đ\xi = (2\pi)^{-n}d\xi$, and the Schwartz space carries the norms $\|\phi\|_k = \max_{|\alpha|+|\beta|\le k}\sup_x|x^\alpha D^\beta\phi(x)|$ of Chapter 1. The bilinear pairing of a distribution u with a test function ϕ is written $u(\phi)$ or $\langle u,\phi\rangle$, and $(u,v) = \int u\bar v\,dx$ is the L^2 inner product.

2.1 Symbols

A polynomial $p(\xi)$ of degree at most m satisfies $|p(\xi)| \le C\langle\xi\rangle^m$, and since $D_\xi^\alpha p$ has degree at most $m-|\alpha|$, also $|D_\xi^\alpha p(\xi)| \le C_\alpha\langle\xi\rangle^{m-|\alpha|}$. If $P(x,\xi)$ is the characteristic polynomial of a differential operator of order m with coefficients in $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ (all derivatives bounded) then

$$|D_x^\alpha D_\xi^\beta P(x,\xi)| \le C_{\alpha\beta}\langle\xi\rangle^{m-|\beta|} \quad \text{on } \mathbb{R}^n\times\mathbb{R}^n,\ \alpha,\beta\in\mathbb{N}_0^n.$$

There is no reason to have the same number of base and fibre variables, so in general we define:

Definition 2.1.1. For $m\in\mathbb{R}$ the space $S^m(\mathbb{R}^p;\mathbb{R}^n)$ of *symbols of order* m consists of the functions $a\in\mathcal{C}^\infty(\mathbb{R}^p\times\mathbb{R}^n)$ satisfying

$$|D_z^\alpha D_\xi^\beta a(z,\xi)| \le C_{\alpha\beta}\langle\xi\rangle^{m-|\beta|} \quad \text{on } \mathbb{R}^p\times\mathbb{R}^n,\ \text{for all } \alpha\in\mathbb{N}_0^p,\ \beta\in\mathbb{N}_0^n. \tag{2.1.1}$$

It is given the topology of the norms

$$\|a\|_{N,m} = \max_{|\alpha|+|\beta|\le N}\ \sup_{(z,\xi)\in\mathbb{R}^p\times\mathbb{R}^n}\langle\xi\rangle^{-m+|\beta|}|D_z^\alpha D_\xi^\beta a(z,\xi)|, \qquad N\in\mathbb{N}_0.$$

We set $S^\infty(\mathbb{R}^p;\mathbb{R}^n) = \bigcup_m S^m(\mathbb{R}^p;\mathbb{R}^n)$ and $S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n) = \bigcap_m S^m(\mathbb{R}^p;\mathbb{R}^n)$, the latter with the topology given by all the norms $\|\cdot\|_{N,m}$, $N\in\mathbb{N}_0$, $m\in\mathbb{R}$; its elements are called *residual* symbols.

Remark 2.1.2. The estimates are uniform in the "base" variables z; this is the analogue of the condition that the coefficients be in $\mathcal{C}^\infty_{\mathrm{b}}$. For an open set $\Omega \subset \mathbb{R}^p$ one defines $S^m(\Omega;\mathbb{R}^n)$ in the same way, requiring (2.1.1) on $\Omega\times\mathbb{R}^n$. The class of symbols is not the only reasonable one (see Exercise 2.13), but it is large enough to contain everything we need at first and small enough for the calculus to work easily.

Discussion. The estimate (2.1.1) says two things. The size of a is at most $\langle\xi\rangle^m$, like a polynomial of degree m. More importantly, each derivative in ξ gains a factor $\langle\xi\rangle^{-1}$. This means that $a(z,\cdot)$ is roughly constant on balls of radius comparable to $\langle\xi\rangle$ around ξ (Figure 2.2): a symbol only varies on the scale of the frequency itself. Functions homogeneous of degree m in ξ, cut off near $\xi = 0$, are the model examples. No such gain is required in z: the coefficients need only be smooth with bounded derivatives.

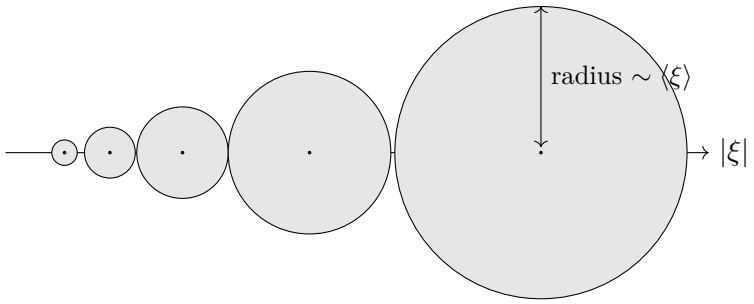


Figure 2.2: A symbol $a \in S^m$ is roughly constant on balls of radius proportional to $\langle\xi_0\rangle$ centred at ξ_0: by Taylor's formula and the symbol estimates, $|a(z,\xi) - a(z,\xi_0)| \le C\epsilon\langle\xi_0\rangle^m$ when $|\xi-\xi_0| \le \epsilon\langle\xi_0\rangle$ with $\epsilon \le \frac{1}{2}$. The balls grow linearly with the frequency.

Since $\langle\xi\rangle^m \le \langle\xi\rangle^{m'}$ for $m \le m'$ we have $\|a\|_{N,m'} \le \|a\|_{N,m}$, so there are continuous inclusions $S^m(\mathbb{R}^p;\mathbb{R}^n) \hookrightarrow S^{m'}(\mathbb{R}^p;\mathbb{R}^n)$ for $m \le m'$; the symbol spaces form a filtration of $S^\infty(\mathbb{R}^p;\mathbb{R}^n)$. A residual symbol is a smooth function all of whose derivatives are bounded by $C_{\alpha\beta N}\langle\xi\rangle^{-N}$ for every N.

We shall use repeatedly the following elementary inequality.

Lemma 2.1.3 (Peetre's inequality)**.** *For all $x, y \in \mathbb{R}^n$ and $s \in \mathbb{R}$,*

$$\langle x+y\rangle^s \le 2^{|s|/2}\langle x\rangle^s\langle y\rangle^{|s|}.$$

Proof. Since $|x+y|^2 \le 2|x|^2+2|y|^2$, $1+|x+y|^2 \le 2(1+|x|^2)(1+|y|^2)$, i.e. $\langle x+y\rangle \le \sqrt{2}\langle x\rangle\langle y\rangle$. This gives the case $s \ge 0$. Applied to $x = (x+y)+(-y)$ it gives $\langle x\rangle \le \sqrt{2}\langle x+y\rangle\langle y\rangle$, i.e. $\langle x+y\rangle^{-1} \le \sqrt{2}\langle x\rangle^{-1}\langle y\rangle$, and raising this to the power $|s|$ gives the case $s < 0$. □

Lemma 2.1.4. *Let $m, m' \in \mathbb{R}$.*

(i) $\langle\xi\rangle^m \in S^m(\mathbb{R}^p;\mathbb{R}^n)$.

(ii) *Multiplication defines a continuous bilinear map*

$$S^m(\mathbb{R}^p;\mathbb{R}^n) \times S^{m'}(\mathbb{R}^p;\mathbb{R}^n) \to S^{m+m'}(\mathbb{R}^p;\mathbb{R}^n).$$

(iii) *D_z^α maps $S^m(\mathbb{R}^p;\mathbb{R}^n)$ continuously into itself and D_ξ^β maps it continuously into $S^{m-|\beta|}(\mathbb{R}^p;\mathbb{R}^n)$.*

(iv) *$S^m(\mathbb{R}^p;\mathbb{R}^n)$ is a Fréchet space.*

(v) *If $a \in \mathcal{C}^\infty(\mathbb{R}^p \times \mathbb{R}^n)$ has all derivatives bounded and $a(z,\xi) = 0$ for $|\xi| \geq R$, then $a \in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$.*

Proof. (i) By induction on $|\beta|$, $D_\xi^\beta \langle\xi\rangle^m = \langle\xi\rangle^{m-2|\beta|} P_\beta(\xi)$ with P_β a polynomial of degree at most $|\beta|$: indeed $D_{\xi_j}(\langle\xi\rangle^s P) = \langle\xi\rangle^{s-2}(-is\xi_j P + \langle\xi\rangle^2 D_{\xi_j} P)$ and the polynomial in brackets has degree at most $\deg P + 1$. Hence $|D_\xi^\beta \langle\xi\rangle^m| \leq C_\beta \langle\xi\rangle^{m-|\beta|}$.

(ii) By Leibniz' formula,

$$\langle\xi\rangle^{-m-m'+|\beta|} |D_z^\alpha D_\xi^\beta (ab)| \leq \sum_{\mu\leq\alpha,\ \gamma\leq\beta} \binom{\alpha}{\mu}\binom{\beta}{\gamma} \langle\xi\rangle^{-m+|\gamma|} |D_z^\mu D_\xi^\gamma a|\, \langle\xi\rangle^{-m'+|\beta-\gamma|} |D_z^{\alpha-\mu} D_\xi^{\beta-\gamma} b|,$$

so $\|ab\|_{N,m+m'} \leq C_N \|a\|_{N,m} \|b\|_{N,m'}$.

(iii) is immediate from the definition: $\|D_z^\alpha a\|_{N,m} \leq \|a\|_{N+|\alpha|,m}$ and $\|D_\xi^\beta a\|_{N,m-|\beta|} \leq \|a\|_{N+|\beta|,m}$.

(iv) If (a_k) is Cauchy in S^m then, for each α, β, $\langle\xi\rangle^{-m+|\beta|} D_z^\alpha D_\xi^\beta a_k$ converges uniformly. Hence $a_k \to a$ in $\mathcal{C}^\infty$ with all derivatives converging locally uniformly, and the uniform limits are $\langle\xi\rangle^{-m+|\beta|} D_z^\alpha D_\xi^\beta a$; it follows that $a \in S^m$ and $a_k \to a$ in S^m.

(v) $\langle\xi\rangle^N |D_z^\alpha D_\xi^\beta a| \leq \langle R\rangle^N \sup |D_z^\alpha D_\xi^\beta a|$. □

In particular $S^m \cdot S^{m'} = S^{m+m'}$, since $a = \langle\xi\rangle^m \cdot (\langle\xi\rangle^{-m} a)$. The inclusion $S^m \hookrightarrow S^{m'}$, $m' > m$, is never dense. Instead we have the following technical but very useful result.

Lemma 2.1.5. *Let $\chi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $0 \leq \chi \leq 1$, $\chi(\xi) = 1$ for $|\xi| \leq 1$ and $\chi(\xi) = 0$ for $|\xi| \geq 2$. For $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ and $0 < \epsilon \leq 1$ set $a_\epsilon(z,\xi) = \chi(\epsilon\xi) a(z,\xi)$. Then $a_\epsilon \in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$, the family (a_ϵ) is bounded in $S^m(\mathbb{R}^p;\mathbb{R}^n)$, and $a_\epsilon \to a$ in $S^{m'}(\mathbb{R}^p;\mathbb{R}^n)$ as $\epsilon \to 0$ for every $m' > m$. In particular $S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$ is dense in $S^m(\mathbb{R}^p;\mathbb{R}^n)$ in the topology of $S^{m'}(\mathbb{R}^p;\mathbb{R}^n)$ for every $m' > m$.*

Proof. The function $\chi(\epsilon\xi)$ is bounded in $S^0(\mathbb{R}^n)$ uniformly in $\epsilon \in (0,1]$: for $\beta \neq 0$, $D_\xi^\beta(\chi(\epsilon\xi)) = \epsilon^{|\beta|}(D^\beta\chi)(\epsilon\xi)$ is supported in $1/\epsilon \leq |\xi| \leq 2/\epsilon$, where $\langle\xi\rangle \leq 1 + |\xi| \leq 3/\epsilon$, so it is bounded by $3^{|\beta|} \sup|D^\beta\chi|\, \langle\xi\rangle^{-|\beta|}$. By Lemma 2.1.4(ii) the family a_ϵ is bounded in S^m, and $a_\epsilon \in S^{-\infty}$ by Lemma 2.1.4(v). Now $D_z^\alpha D_\xi^\beta(a - a_\epsilon)$ is bounded by $C_{\alpha\beta}\langle\xi\rangle^{m-|\beta|}$ uniformly in ϵ and vanishes for $|\xi| < 1/\epsilon$. Hence, for $m' > m$,

$$\sup \langle\xi\rangle^{-m'+|\beta|} |D_z^\alpha D_\xi^\beta (a - a_\epsilon)| \leq C_{\alpha\beta} \sup_{|\xi|\geq 1/\epsilon} \langle\xi\rangle^{m-m'} \leq C_{\alpha\beta}\, \epsilon^{m'-m} \to 0. \qquad \square$$

One cannot take $m' = m$ here, for essentially the same reason that $\mathcal{S}(\mathbb{R}^n)$ is not dense in $\mathcal{C}_b^\infty(\mathbb{R}^n)$: a limit in S^m of residual symbols satisfies $\langle\xi\rangle^{-m} a \to 0$ as $|\xi| \to \infty$, so for instance $1 \in S^0(\mathbb{R}^p;\mathbb{R}^n)$ is not in the closure of $S^{-\infty}$ if $n > 0$.

2.1.1 Ellipticity of symbols

Lemma 2.1.6. *Let $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ and let $U \subset \mathbb{R}^p \times \mathbb{R}^n$ be an open set on which $|a(z,\xi)| \geq \epsilon\langle\xi\rangle^m$ for some $\epsilon > 0$. Then on U*

$$|D_z^\alpha D_\xi^\beta (a^{-1})| \leq C_{\alpha\beta} \langle\xi\rangle^{-m-|\beta|},$$

and if moreover $a > 0$ on U then, for every $s \in \mathbb{R}$, $|D_z^\alpha D_\xi^\beta (a^s)| \leq C_{\alpha\beta s}\langle\xi\rangle^{sm-|\beta|}$ on U. The constants depend only on ϵ, s and the norms of a. In particular, if $U = \mathbb{R}^p \times \mathbb{R}^n$, then $a^{-1} \in S^{-m}(\mathbb{R}^p;\mathbb{R}^n)$ and, when $a > 0$, $a^s \in S^{sm}(\mathbb{R}^p;\mathbb{R}^n)$.

Proof. Let $s \in \mathbb{R}$ (with $s = -1$ if a is not positive). We claim that on U

$$D_z^\alpha D_\xi^\beta(a^s) = \sum_{j=0}^{|\alpha|+|\beta|} a^{s-j} G_j, \qquad G_j = G_j^{\alpha\beta} \in S^{jm-|\beta|}(\mathbb{R}^p;\mathbb{R}^n),$$

where the G_j are universal polynomials in the derivatives of a (hence defined on all of $\mathbb{R}^p \times \mathbb{R}^n$). This holds for $\alpha = \beta = 0$ with $G_0 = 1$. If it holds for (α, β) then

$$D_{z_l}(a^{s-j}G_j) = a^{s-j-1}((s-j)(D_{z_l}a)G_j) + a^{s-j}D_{z_l}G_j,$$

with $(D_{z_l}a)G_j \in S^{(j+1)m-|\beta|}$ and $D_{z_l}G_j \in S^{jm-|\beta|}$; similarly for D_{ξ_l}, where $(D_{\xi_l}a)G_j \in S^{(j+1)m-|\beta|-1}$ and $D_{\xi_l}G_j \in S^{jm-|\beta|-1}$. This proves the claim. On U, $|a|^{s-j} \le C\langle\xi\rangle^{(s-j)m}$: for $s-j \ge 0$ by the upper bound $|a| \le C\langle\xi\rangle^m$, for $s-j < 0$ by the lower bound. Hence $|a^{s-j}G_j| \le C\langle\xi\rangle^{(s-j)m+jm-|\beta|} = C\langle\xi\rangle^{sm-|\beta|}$. □

Definition 2.1.7. A symbol $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ is *elliptic* if there are $\epsilon > 0$ and C such that

$$|a(z,\xi)| \ge \epsilon\langle\xi\rangle^m \quad \text{for all } z \in \mathbb{R}^p,\ |\xi| \ge C. \tag{2.1.2}$$

Note that ϵ and C are independent of z, so this is a notion of *uniform* ellipticity. Condition (2.1.2) is equivalent to the existence of $\epsilon > 0$ and C' with $|a(z,\xi)| \ge \epsilon\langle\xi\rangle^m - C'\langle\xi\rangle^{m-1}$ everywhere. Indeed, the latter gives $|a| \ge \frac{\epsilon}{2}\langle\xi\rangle^m$ once $\langle\xi\rangle \ge 2C'/\epsilon$; conversely, if (2.1.2) holds then for $|\xi| \le C$ we have $\epsilon\langle\xi\rangle^m \le \epsilon\langle C\rangle\langle\xi\rangle^{m-1}$, so $|a| \ge \epsilon\langle\xi\rangle^m - \epsilon\langle C\rangle\langle\xi\rangle^{m-1}$ everywhere.

Lemma 2.1.8. *If $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ is elliptic there exists $b \in S^{-m}(\mathbb{R}^p;\mathbb{R}^n)$ such that $ab - 1 \in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$. Conversely, if $ab - 1 \in S^{-\infty}$ for some $b \in S^{-m}$, then a is elliptic.*

Proof. Let C be as in (2.1.2) and choose $\psi \in \mathcal{C}^\infty(\mathbb{R}^n)$ with $\psi(\xi) = 0$ for $|\xi| \le C+1$ and $\psi(\xi) = 1$ for $|\xi| \ge C+2$; then $\psi \in S^0(\mathbb{R}^n)$. Set $b = \psi/a$ where $\psi \ne 0$ and $b = 0$ elsewhere; b is smooth since it vanishes on a neighbourhood of $\{|\xi| \le C+1\}$ and $a \ne 0$ on $|\xi| > C$. Applying Lemma 2.1.6 on $U = \{|\xi| > C\}$ and Leibniz' formula shows $b \in S^{-m}$, and $ab - 1 = \psi - 1$ is smooth, independent of z and supported in $|\xi| \le C+2$, hence residual.

Conversely, if $ab = 1 + e$ with $e \in S^{-\infty}$, choose C with $|e| \le \frac{1}{2}$ for $|\xi| \ge C$. Since $|b| \le C'\langle\xi\rangle^{-m}$, for $|\xi| \ge C$ we get $\frac{1}{2} \le |a||b| \le C'\langle\xi\rangle^{-m}|a|$, which is (2.1.2) with $\epsilon = 1/(2C')$. □

2.1.2 Asymptotic summation

Suppose $a_j \in S^{m-j}(\mathbb{R}^p;\mathbb{R}^n)$, $j \in \mathbb{N}_0$. The decreasing orders mean that these symbols become smaller and smaller as $|\xi| \to \infty$, but the series $\sum_j a_j$ need not converge. Nevertheless it can be "summed" asymptotically.

Definition 2.1.9. Let $m_j \in \mathbb{R}$ be a sequence decreasing to $-\infty$ and $a_j \in S^{m_j}(\mathbb{R}^p;\mathbb{R}^n)$. We write

$$a \sim \sum_{j=0}^{\infty} a_j$$

if $a \in S^{m_0}(\mathbb{R}^p;\mathbb{R}^n)$ and $a - \sum_{j<N} a_j \in S^{m_N}(\mathbb{R}^p;\mathbb{R}^n)$ for every N; a is then called an *asymptotic sum* of the series. Most often $m_j = m - j$.

Proposition 2.1.10 (Borel's lemma for symbols)**.** *Every series $\sum_j a_j$ with $a_j \in S^{m_j}(\mathbb{R}^p;\mathbb{R}^n)$, $m_j \downarrow -\infty$, has an asymptotic sum, and the asymptotic sum is unique up to an additive term in $S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$.*

Proof. Uniqueness: if a and a' are both asymptotic sums then $a - a' = (a - \sum_{j<N} a_j) - (a' - \sum_{j<N} a_j) \in S^{m_N}$ for every N, so $a - a' \in S^{-\infty}$.

Existence: we cut off each term near the origin in ξ, more and more drastically as j increases (Figure 2.3). Choose $\phi \in \mathcal{C}^\infty(\mathbb{R}^n)$ with $\phi(\xi) = 0$ for $|\xi| \leq 1$ and $\phi(\xi) = 1$ for $|\xi| \geq 2$ (e.g. $\phi = 1 - \chi$ with χ as in Lemma 2.1.5). As in the proof of Lemma 2.1.5, $\phi(\epsilon\xi)$ is bounded in $S^0(\mathbb{R}^n)$ for $\epsilon \in (0, 1]$, so $\|\phi(\epsilon\xi)a_j\|_{N,m_j} \leq C_{N,j}$ with $C_{N,j}$ independent of ϵ. Moreover $\phi(\epsilon\xi)a_j$ vanishes for $|\xi| < 1/\epsilon$, where $\langle\xi\rangle^{-1} \leq \epsilon$; replacing m_j by $m_j + 1$ in the norm therefore gains a factor ϵ:

$$\|\phi(\epsilon\xi)a_j\|_{N,m_j+1} \leq C_{N,j}\,\epsilon. \tag{2.1.3}$$

Choose $\epsilon_j \in (0, 1]$ with $C_{N,j}\epsilon_j \leq 2^{-j}$ for all $N \leq j$, and set

$$a(z, \xi) = \sum_{j=0}^{\infty} \phi(\epsilon_j\xi)a_j(z, \xi).$$

The ϵ_j may be taken decreasing to 0, so on each set $\{|\xi| \leq R\}$ only finitely many terms are non-zero and a is a well-defined smooth function. Fix $M \in \mathbb{N}_0$. Since $m_j \downarrow -\infty$ there is J_M with $m_j + 1 \leq m_M$ for $j \geq J_M$; for such j and $j \geq N$, $\|\phi(\epsilon_j\xi)a_j\|_{N,m_M} \leq \|\phi(\epsilon_j\xi)a_j\|_{N,m_j+1} \leq 2^{-j}$. Hence $\sum_{j\geq M} \phi(\epsilon_j\xi)a_j$ converges absolutely in each norm of the Fréchet space S^{m_M} and so converges there; its limit is the pointwise sum. Thus

$$a - \sum_{j<M} a_j = \sum_{j<M} (\phi(\epsilon_j\xi) - 1)a_j + \sum_{j\geq M} \phi(\epsilon_j\xi)a_j \in S^{-\infty} + S^{m_M},$$

where the first sum is residual by Lemma 2.1.4(v). □

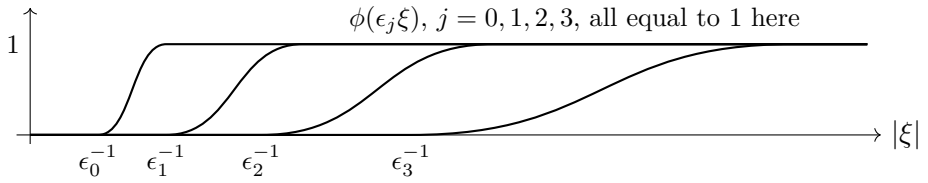


Figure 2.3: The cut-offs in the proof of Borel's lemma. The j-th term a_j is switched on only for $|\xi| \geq \epsilon_j^{-1}$. Since $\epsilon_j \to 0$, on any bounded set of frequencies only finitely many terms are non-zero, and far out each term is small relative to the preceding ones.

The following simple observations allow asymptotic sums to be manipulated formally.

Lemma 2.1.11. (i) *If T is a linear map with $T(S^k(\mathbb{R}^p; \mathbb{R}^n)) \subset S^{k+\mu}(\mathbb{R}^{p'}; \mathbb{R}^n)$ for all k (e.g. multiplication by an element of S^μ, differentiation, complex conjugation, restriction to a linear subspace of $\mathbb{R}^p$) and $a \sim \sum_j a_j$, then $Ta \sim \sum_j Ta_j$.*

(ii) *If $a \sim \sum_j a_j$ and $b \sim \sum_k b_k$, with $a_j \in S^{m-j}$, $b_k \in S^{m'-k}$, then $ab \sim \sum_l \sum_{j+k=l} a_j b_k$.*

(iii) (Rearrangement) *Let $a_{j,k} \in S^{m-j-k}(\mathbb{R}^p; \mathbb{R}^n)$ for $j, k \in \mathbb{N}_0$. If $c_j \sim \sum_k a_{j,k}$ for each j and $c \sim \sum_j c_j$, then $c \sim \sum_l (\sum_{j+k=l} a_{j,k})$.*

Proof. (i) $Ta - \sum_{j<N} Ta_j = T(a - \sum_{j<N} a_j) \in S^{m_N+\mu}$. (ii) $ab - \sum_{j+k<N} a_j b_k = (a - \sum_{j<N} a_j)b + \sum_{j<N} a_j(b - \sum_{k<N-j} b_k) \in S^{m+m'-N}$. (iii) $c - \sum_{j+k<N} a_{j,k} = (c - \sum_{j<N} c_j) + \sum_{j<N}(c_j - \sum_{k<N-j} a_{j,k}) \in S^{m-N}$. □

2.2 Amplitudes and pseudodifferential operators

We now interpret (2.0.1). It turns out to be convenient to allow the amplitude $a(x,y,\xi)$ to grow polynomially in the direction of $x-y$; this extra freedom makes no difference to the class of operators (Theorem 2.3.1) but is very useful, for instance for weighted estimates.

Definition 2.2.1. For $m, w \in \mathbb{R}$ let $\mathcal{A}^m_w(\mathbb{R}^n)$ be the space of $a \in \mathcal{C}^\infty(\mathbb{R}^n_x \times \mathbb{R}^n_y \times \mathbb{R}^n_\xi)$ such that

$$|D_x^\alpha D_y^\beta D_\xi^\gamma a(x,y,\xi)| \le C_{\alpha\beta\gamma}\langle x-y\rangle^w \langle\xi\rangle^{m-|\gamma|} \quad \text{for all } \alpha,\beta,\gamma \in \mathbb{N}_0^n, \tag{2.2.1}$$

with the norms $\|a\|_{N,m,w}$, the best constants in (2.2.1) for $|\alpha|+|\beta|+|\gamma| \le N$. Elements of $\mathcal{A}^m_w(\mathbb{R}^n)$ are called *amplitudes* of order m and weight w. We set $\mathcal{A}^{-\infty}_w(\mathbb{R}^n) = \bigcap_m \mathcal{A}^m_w(\mathbb{R}^n)$.

Lemma 2.2.2. *Multiplication by $\langle x-y\rangle^w$ is an isomorphism of Fréchet spaces from $S^m(\mathbb{R}^{2n}_{(x,y)};\mathbb{R}^n_\xi)$ onto $\mathcal{A}^m_w(\mathbb{R}^n)$. More generally, call $f \in \mathcal{C}^\infty(\mathbb{R}^{2n})$ of weight w' if $|D_x^\alpha D_y^\beta f| \le C_{\alpha\beta}\langle x-y\rangle^{w'}$ for all α,β; then multiplication by such f maps $\mathcal{A}^m_w(\mathbb{R}^n)$ continuously into $\mathcal{A}^m_{w+w'}(\mathbb{R}^n)$.*

Proof. The second statement is immediate from Leibniz' formula. The function $\langle z\rangle^w$ is a symbol of order w in $z \in \mathbb{R}^n$ (with no base variables) by Lemma 2.1.4(i), so $|D_z^\alpha \langle z\rangle^w| \le C\langle z\rangle^{w-|\alpha|} \le C\langle z\rangle^w$. Hence $\langle x-y\rangle^{\pm w}$ is of weight $\pm w$, and multiplication by $\langle x-y\rangle^w$ and by $\langle x-y\rangle^{-w}$ are mutually inverse continuous maps between $S^m(\mathbb{R}^{2n};\mathbb{R}^n) = \mathcal{A}^m_0(\mathbb{R}^n)$ and $\mathcal{A}^m_w(\mathbb{R}^n)$. □

Thus $\mathcal{A}^m_w(\mathbb{R}^n) = \langle x-y\rangle^w S^m(\mathbb{R}^{2n};\mathbb{R}^n)$, and the results of Section 2.1 transfer to amplitudes. In particular, by Lemma 2.1.5, if $a \in \mathcal{A}^m_w(\mathbb{R}^n)$ and χ is as there, then $\chi(\epsilon\xi)a \in \mathcal{A}^{-\infty}_w(\mathbb{R}^n)$ is bounded in $\mathcal{A}^m_w(\mathbb{R}^n)$ and converges to a in $\mathcal{A}^{m'}_w(\mathbb{R}^n)$ as $\epsilon \to 0$, for every $m' > m$. Note also the following elementary facts:

- $\mathcal{A}^m_w \subset \mathcal{A}^{m'}_{w'}$ if $m \le m'$ and $w \le w'$; $\mathcal{A}^m_w \cdot \mathcal{A}^{m'}_{w'} \subset \mathcal{A}^{m+m'}_{w+w'}$;
- D_{x_j}, D_{y_j} preserve $\mathcal{A}^m_w$, D_{ξ_j} maps it to $\mathcal{A}^{m-1}_w$ and multiplication by ξ_j maps it to $\mathcal{A}^{m+1}_w$;
- $(x-y)^\alpha$ is of weight $|\alpha|$, so $(x-y)^\alpha \mathcal{A}^m_w \subset \mathcal{A}^m_{w+|\alpha|}$;
- if $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ then both $a(x,\xi)$ and $a(y,\xi)$ lie in $\mathcal{A}^m_0(\mathbb{R}^n)$; and if $a \in \mathcal{A}^m_w$ then $a(x,x,\xi) \in S^m(\mathbb{R}^n;\mathbb{R}^n)$.

2.2.1 The kernel as an oscillatory integral

If $a \in \mathcal{A}^m_w(\mathbb{R}^n)$ with $m < -n$ then

$$K(x,y) = \int e^{i(x-y)\cdot\xi} a(x,y,\xi)\,đ\xi$$

converges absolutely and locally uniformly, so defines a continuous function with $|K(x,y)| \le C\langle x-y\rangle^w$. For general m we integrate by parts. The idea is that the factor $e^{i(x-y)\cdot\xi}$ oscillates faster and faster as $|\xi| \to \infty$, and these oscillations cancel against the growth of a, once the integral is paired with a smooth test function. Each integration by parts in x turns a derivative of the exponential into a factor ξ and so lets us divide by $\langle\xi\rangle^2$, at the cost of differentiating the amplitude and the test function. We repeat this until the ξ-integral converges absolutely. Let $\Delta_x = D_{x_1}^2 + \cdots + D_{x_n}^2$ be the (positive) Laplacian. Then

$$(1+\Delta_x)^q e^{i(x-y)\cdot\xi} = \langle\xi\rangle^{2q} e^{i(x-y)\cdot\xi}, \qquad (1+\Delta_y)^q e^{i(x-y)\cdot\xi} = \langle\xi\rangle^{2q} e^{i(x-y)\cdot\xi}. \tag{2.2.2}$$

Proposition 2.2.3. *Let $a \in \mathcal{A}_w^m(\mathbb{R}^n)$ and let $q \in \mathbb{N}_0$ satisfy $m - 2q < -n$. For $\phi \in \mathcal{S}(\mathbb{R}^{2n})$ the integral*

$$I(a)(\phi) = \int_{\mathbb{R}^{3n}} e^{i(x-y)\cdot\xi}\langle\xi\rangle^{-2q}(1+\Delta_x)^q\big[a(x,y,\xi)\phi(x,y)\big]\,dx\,dy\text{đ}\xi \tag{2.2.3}$$

converges absolutely, is independent of the choice of such q, and satisfies

$$|I(a)(\phi)| \le C\|a\|_{2q,m,w}\|\phi\|_k \tag{2.2.4}$$

for some $k = k(q,w,n)$ and $C = C(q,w,n,m)$. Thus $I(a) \in \mathcal{S}'(\mathbb{R}^{2n})$ and $a \mapsto I(a)$ is continuous in the sense of (2.2.4). Moreover:

(i) *if $m < -n$, $I(a)$ is the continuous function $K(x,y) = \int e^{i(x-y)\cdot\xi}a(x,y,\xi)\text{đ}\xi$;*

(ii) *if a_k is bounded in $\mathcal{A}_w^m(\mathbb{R}^n)$ and $a_k \to a$ in $\mathcal{A}_w^{m'}(\mathbb{R}^n)$ for some $m' > m$, then $I(a_k)(\phi) \to I(a)(\phi)$ for every $\phi \in \mathcal{S}(\mathbb{R}^{2n})$; in particular, for χ as in Lemma 2.1.5,*

$$I(a)(\phi) = \lim_{\epsilon\to 0}\int e^{i(x-y)\cdot\xi}\chi(\epsilon\xi)a(x,y,\xi)\phi(x,y)\,dx\,dy\text{đ}\xi;$$

(iii) *the same value is obtained if $(1+\Delta_x)^q$ is replaced by $(1+\Delta_y)^q$ in (2.2.3).*

Part (ii) justifies calling $I(a)$ an *oscillatory integral*: it is the limit of the absolutely convergent integrals obtained by cutting off the amplitude at large frequencies, and it is the unique extension of the absolutely convergent case that is continuous in the sense of (ii).

Proof. By Leibniz' formula $(1+\Delta_x)^q[a\phi]$ is a linear combination of terms $D_x^\mu a\, D_x^\nu\phi$ with $|\mu|+|\nu| \le 2q$, and $|D_x^\mu a| \le \|a\|_{2q,m,w}\langle x-y\rangle^w\langle\xi\rangle^m$. Since $\langle x-y\rangle \le \sqrt{2}\langle(x,y)\rangle$ we have $\langle x-y\rangle^w \le 2^{|w|/2}\langle(x,y)\rangle^{|w|}$. Hence the integrand in (2.2.3) is bounded by

$$C\|a\|_{2q,m,w}\langle\xi\rangle^{m-2q}\langle(x,y)\rangle^{|w|}\sum_{|\nu|\le 2q}|D^\nu\phi(x,y)| \le C'\|a\|_{2q,m,w}\|\phi\|_k\langle\xi\rangle^{m-2q}\langle(x,y)\rangle^{-2n-1}$$

for $k \ge 2q + |w| + 2n + 1$, which is integrable since $m - 2q < -n$. This proves absolute convergence and (2.2.4).

If q and $q+1$ are both admissible, put $F = \langle\xi\rangle^{-2q-2}(1+\Delta_x)^q[a\phi]$. For fixed (y,ξ), F and all its x-derivatives are rapidly decreasing in x, so integration by parts and (2.2.2) give

$$\int e^{i(x-y)\cdot\xi}(1+\Delta_x)F\,dx = \int((1+\Delta_x)e^{i(x-y)\cdot\xi})F\,dx = \langle\xi\rangle^2\int e^{i(x-y)\cdot\xi}F\,dx.$$

Integrating in (y,ξ) (both sides being absolutely integrable) shows that the integrals for $q+1$ and q agree. So $I(a)$ is independent of q.

(i) If $m < -n$ we may take $q = 0$, and then Fubini's theorem gives $I(a)(\phi) = \int K\phi$.

(ii) Choose q with $m' - 2q < -n$ and apply (2.2.4) with m' in place of m to $a - a_k$. The particular case follows from the remarks after Lemma 2.2.2 and (i).

(iii) Denote by $I'(a)(\phi)$ the integral with Δ_y in place of Δ_x. Exactly the same arguments show that it converges absolutely, is independent of q and satisfies (2.2.4), hence (ii). For $a \in \mathcal{A}_w^{-\infty}$ both $I(a)(\phi)$ and $I'(a)(\phi)$ equal $\int e^{i(x-y)\cdot\xi}a\phi\,dx\,dy\text{đ}\xi$ (take $q = 0$). By (ii), applied to $\chi(\epsilon\xi)a$, they agree for all a. □

Distributional operations on $I(a)$ correspond to simple operations on the amplitude.

Lemma 2.2.4. *For $a \in \mathcal{A}^m_w(\mathbb{R}^n)$ and $j = 1, \dots, n$, the following identities hold in $\mathcal{S}'(\mathbb{R}^{2n})$:*

$$D_{x_j} I(a) = I((\xi_j + D_{x_j})a), \qquad D_{y_j} I(a) = I((D_{y_j} - \xi_j)a),$$
$$(x_j - y_j) I(a) = I((x_j - y_j)a) = I(-D_{\xi_j} a).$$

Moreover $fI(a) = I(fa)$ for every f of weight w' in the sense of Lemma 2.2.2. In particular, for every $\alpha \in \mathbb{N}_0^n$, $(x-y)^\alpha I(a) = I((-D_\xi)^\alpha a)$.

Proof. Such an f and all its derivatives are polynomially bounded, so $f\phi \in \mathcal{S}(\mathbb{R}^{2n})$ for $\phi \in \mathcal{S}(\mathbb{R}^{2n})$, and the integrals defining $I(a)(f\phi)$ and $I(fa)(\phi)$ by (2.2.3) (with the same q) are identical. This gives $fI(a) = I(fa)$, and in particular the first equality in the second line.

For the remaining identities, suppose first that $a \in \mathcal{A}^{-\infty}_w$. Then all integrals below converge absolutely and

$$D_{x_j} I(a)(\phi) = -\int e^{i(x-y)\cdot\xi} a\, D_{x_j}\phi = \int D_{x_j}(e^{i(x-y)\cdot\xi} a)\phi = \int e^{i(x-y)\cdot\xi}(\xi_j a + D_{x_j} a)\phi,$$

integrating by parts in x. The identity for D_{y_j} is the same. Since $(x_j - y_j)e^{i(x-y)\cdot\xi} = D_{\xi_j} e^{i(x-y)\cdot\xi}$, integration by parts in ξ (the boundary terms vanish because a is rapidly decreasing in ξ) gives $I((x_j - y_j)a) = I(-D_{\xi_j} a)$. For general a, apply these identities to $\chi(\epsilon\xi)a$ and let $\epsilon \to 0$, using Proposition 2.2.3(ii) for both sides: the amplitudes involved are bounded in $\mathcal{A}^{m+1}_{w+1}$ and converge in $\mathcal{A}^{m'+1}_{w+1}$ for $m' > m$. The last statement follows by induction. □

2.2.2 Pseudodifferential operators

Proposition 2.2.5. *Let $a \in \mathcal{A}^m_w(\mathbb{R}^n)$. For $u \in \mathcal{S}(\mathbb{R}^n)$ define $\operatorname{Op}(a)u \in \mathcal{S}'(\mathbb{R}^n)$ by*

$$\langle \operatorname{Op}(a)u, v\rangle = I(a)(v \otimes u), \qquad v \in \mathcal{S}(\mathbb{R}^n), \tag{2.2.5}$$

where $(v \otimes u)(x,y) = v(x)u(y)$. Then $\operatorname{Op}(a)u \in \mathcal{S}(\mathbb{R}^n)$ and $\operatorname{Op}(a) : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ is continuous. For $q \in \mathbb{N}_0$ with $m - 2q < -n$,

$$\operatorname{Op}(a)u(x) = \int e^{i(x-y)\cdot\xi} \langle\xi\rangle^{-2q} (1 + \Delta_y)^q [a(x,y,\xi)u(y)]\, dy đ\xi, \tag{2.2.6}$$

an absolutely convergent integral; if $m < -n$ this is just $\int e^{i(x-y)\cdot\xi} a(x,y,\xi)u(y)\, dy đ\xi$. In general

$$\operatorname{Op}(a)u(x) = \lim_{\epsilon\to 0} \int e^{i(x-y)\cdot\xi} \chi(\epsilon\xi) a(x,y,\xi)u(y)\, dy đ\xi.$$

Proof. Let $F(x)$ denote the right side of (2.2.6). As in the proof of Proposition 2.2.3, but now using Peetre's inequality $\langle x - y\rangle^w \le 2^{|w|/2}\langle x\rangle^{|w|}\langle y\rangle^{|w|}$, its integrand is bounded by $C\|a\|_{2q,m,w}\langle x\rangle^{|w|}\langle\xi\rangle^{m-2q}\langle y\rangle^{|w|}\sum_{|\nu|\le 2q}|D^\nu u(y)|$. Hence the integral converges absolutely, F is continuous by dominated convergence, and

$$|F(x)| \le C\|a\|_{2q,m,w}\|u\|_k \langle x\rangle^{|w|} \tag{2.2.7}$$

for some k. By Proposition 2.2.3(iii) and Fubini's theorem, $I(a)(v \otimes u) = \int v(x)F(x)\, dx$, so $\operatorname{Op}(a)u = F$. The limit formula follows in the same way from Proposition 2.2.3(ii), the convergence being locally uniform in x by dominated convergence.

To see that $\operatorname{Op}(a)u \in \mathcal{S}$, note that (2.2.5) and Lemma 2.2.4 give

$$D_{x_j} \operatorname{Op}(a)u = \operatorname{Op}((\xi_j + D_{x_j})a)u, \qquad x_j \operatorname{Op}(a)u = \operatorname{Op}(a)(x_j u) + \operatorname{Op}(-D_{\xi_j} a)u.$$

For instance $\langle D_{x_j}\operatorname{Op}(a)u, v\rangle = -I(a)(D_{x_j}v\otimes u) = (D_{x_j}I(a))(v\otimes u)$, and $(x_j - y_j)(v\otimes u) = (x_j v)\otimes u - v\otimes(x_j u)$. By induction, for all α,β, $x^\alpha D^\beta \operatorname{Op}(a)u$ is a finite sum of terms $\operatorname{Op}(a')(x^{\alpha'}u)$ with $a'\in\mathcal{A}_w^{m+|\beta|}$, $|\alpha'|\le|\alpha|$, the a' depending continuously on a. By (2.2.7) each such term is a continuous function bounded by $C\|a\|_{N,m,w}\|u\|_{k'}\langle x\rangle^{|w|}$. Since all distributional derivatives of $\operatorname{Op}(a)u$ are continuous, $\operatorname{Op}(a)u\in\mathcal{C}^\infty(\mathbb{R}^n)$; and since $\langle x\rangle^{2L}$ is a polynomial in x, we get $\sup\langle x\rangle^{2L-|w|}|D^\beta\operatorname{Op}(a)u| \le C\|a\|_{N,m,w}\|u\|_{k''}$ for all L and β. Thus $\operatorname{Op}(a)u\in\mathcal{S}(\mathbb{R}^n)$ and $\operatorname{Op}(a)$ is continuous on $\mathcal{S}(\mathbb{R}^n)$. □

Definition 2.2.6. For $m\in\mathbb{R}$, $\Psi^m(\mathbb{R}^n)$ is the space of operators $\operatorname{Op}(a):\mathcal{S}(\mathbb{R}^n)\to\mathcal{S}(\mathbb{R}^n)$ with $a\in\mathcal{A}_w^m(\mathbb{R}^n)$ for some $w\in\mathbb{R}$. Its elements are the *pseudodifferential operators of order* m (of type $(1,0)$, with uniformly bounded coefficients). We set $\Psi^\infty(\mathbb{R}^n)=\bigcup_m\Psi^m(\mathbb{R}^n)$ and $\Psi^{-\infty}(\mathbb{R}^n)=\bigcap_m\Psi^m(\mathbb{R}^n)$. The distribution $K_A = I(a)$ is the *Schwartz kernel* of $A=\operatorname{Op}(a)$.

Since $\mathcal{A}_w^m\subset\mathcal{A}_{w'}^{m'}$ for $m\le m'$, $w\le w'$, each $\Psi^m(\mathbb{R}^n)$ is a linear space and $\Psi^m(\mathbb{R}^n)\subset\Psi^{m'}(\mathbb{R}^n)$ for $m\le m'$. By (2.2.5), K_A is determined by A on the products $v\otimes u$, and these span a dense subspace of $\mathcal{S}(\mathbb{R}^{2n})$ (see Chapter 1), so K_A is determined by A even though the amplitude is not.

The two extreme cases are amplitudes independent of y and amplitudes independent of x. For $a,b\in S^m(\mathbb{R}^n;\mathbb{R}^n)$ we write

$$\operatorname{Op}(a)=\operatorname{Op}(a(x,\xi)),\qquad \operatorname{Op}_R(b)=\operatorname{Op}(b(y,\xi)),$$

the *left* and *right quantizations*; they lie in $\Psi^m(\mathbb{R}^n)$.

Question. If every operator turns out to be the left quantization of a symbol (Theorem 2.3.1), why allow amplitudes $a(x,y,\xi)$ depending on both x and y at all?

Answer. Because many operations are obvious for amplitudes and hard for left symbols. The adjoint of $\operatorname{Op}(a)$ has amplitude $\overline{a(y,x,\xi)}$ (Proposition 2.2.9). The product $\operatorname{Op}(a)\operatorname{Op}_R(b)$ has amplitude $a(x,\xi)b(y,\xi)$ (Proposition 2.2.7). Conjugation by weights or changes of variables multiply the amplitude by functions of (x,y). In each case the reduction theorem then converts the amplitude back into a left symbol, and its asymptotic formula computes that symbol. □

Proposition 2.2.7. *Let $a,b\in S^m(\mathbb{R}^n;\mathbb{R}^n)$ and $u\in\mathcal{S}(\mathbb{R}^n)$. Then*

$$\operatorname{Op}(a)u(x)=\int e^{ix\cdot\xi}a(x,\xi)\hat u(\xi)đ\xi, \tag{2.2.8}$$

$$\widehat{\operatorname{Op}_R(b)u}(\xi)=\int e^{-iy\cdot\xi}b(y,\xi)u(y)\,dy, \tag{2.2.9}$$

Moreover, if $a\in S^m(\mathbb{R}^n;\mathbb{R}^n)$ and $b\in S^{m'}(\mathbb{R}^n;\mathbb{R}^n)$ then $a(x,\xi)b(y,\xi)\in\mathcal{A}_0^{m+m'}(\mathbb{R}^n)$ and $\operatorname{Op}(a)\operatorname{Op}_R(b)=\operatorname{Op}(a(x,\xi)b(y,\xi))\in\Psi^{m+m'}(\mathbb{R}^n)$.

Proof. If $a\in S^{-\infty}$, (2.2.8) follows from the absolutely convergent formula of Proposition 2.2.5 and Fubini. In general, let $a_\epsilon=\chi(\epsilon\xi)a$. Then $\operatorname{Op}(a_\epsilon)u(x)\to\operatorname{Op}(a)u(x)$ by Proposition 2.2.5, while $\int e^{ix\cdot\xi}a_\epsilon\hat u đ\xi\to\int e^{ix\cdot\xi}a\hat u đ\xi$ by dominated convergence, since $|a_\epsilon|\le C\langle\xi\rangle^m$ uniformly and $\hat u\in\mathcal{S}$.

For (2.2.9) let $g(\xi)$ be the right side and $B=\operatorname{Op}_R(b)$. If $b\in S^{-\infty}$, g is integrable and Fubini gives $Bu(x)=\int e^{ix\cdot\xi}g(\xi)đ\xi$, so $\widehat{Bu}=g$ by Fourier inversion. In general, with $b_\epsilon=\chi(\epsilon\xi)b$ and $B_\epsilon=\operatorname{Op}_R(b_\epsilon)$, we have $\widehat{B_\epsilon u}=g_\epsilon\to g$ pointwise by dominated convergence; on the other hand $B_\epsilon u\to Bu$ in $\mathcal{S}'(\mathbb{R}^n)$, by Proposition 2.2.3(ii), so $\widehat{B_\epsilon u}\to\widehat{Bu}$ in $\mathcal{S}'$. Since g_ϵ is bounded by $C\langle\xi\rangle^m\|u\|_{L^1}$ uniformly in ϵ, also $g_\epsilon\to g$ in $\mathcal{S}'$, and so $\widehat{Bu}=g$.

Finally let $c(x,y,\xi) = a(x,\xi)b(y,\xi) \in \mathcal{A}_0^{m+m'}$ and choose q with $m+m'-2q<-n$. By (2.2.6) and Fubini,

$$\mathrm{Op}(c)u(x) = \int e^{ix\cdot\xi} a(x,\xi)\langle\xi\rangle^{-2q}\Big[\int e^{-iy\cdot\xi}(1+\Delta_y)^q(b(y,\xi)u(y))dy\Big]đ\xi.$$

Integrating by parts in y (for fixed ξ, bu is rapidly decreasing with all derivatives) the inner integral is $\langle\xi\rangle^{2q}g(\xi)$, so by (2.2.8) and (2.2.9), $\mathrm{Op}(c)u(x) = \int e^{ix\cdot\xi}a(x,\xi)\widehat{Bu}(\xi)đ\xi = \mathrm{Op}(a)Bu(x)$. □

Example 2.2.8. Let $P = \sum_{|\alpha|\le m} p_\alpha(x)D^\alpha$ be a differential operator with coefficients $p_\alpha \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$; we denote the space of these by $\mathrm{Diff}^m_{\mathrm{b}}(\mathbb{R}^n)$. Its characteristic polynomial $p(x,\xi) = \sum_\alpha p_\alpha(x)\xi^\alpha$ lies in $S^m(\mathbb{R}^n;\mathbb{R}^n)$ and, since $\widehat{D^\alpha u} = \xi^\alpha\hat{u}$, (2.2.8) and Fourier inversion give $P = \mathrm{Op}(p)$. Thus $\mathrm{Diff}^m_{\mathrm{b}}(\mathbb{R}^n) \subset \Psi^m(\mathbb{R}^n)$. Similarly, multiplication by $f \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ is $\mathrm{Op}(f) \in \Psi^0(\mathbb{R}^n)$, and the Fourier multiplier $\langle D\rangle^s = \mathrm{Op}(\langle\xi\rangle^s)$, $\widehat{\langle D\rangle^s u} = \langle\xi\rangle^s\hat{u}$, lies in $\Psi^s(\mathbb{R}^n)$ for every $s\in\mathbb{R}$.

2.2.3 Adjoints and the action on tempered distributions

Proposition 2.2.9. *If $A = \mathrm{Op}(a) \in \Psi^m(\mathbb{R}^n)$ with $a \in \mathcal{A}^m_w(\mathbb{R}^n)$, then $A^* = \mathrm{Op}(a^*)$ with $a^*(x,y,\xi) = \overline{a(y,x,\xi)} \in \mathcal{A}^m_w(\mathbb{R}^n)$ satisfies*

$$(Au,v) = (u,A^*v) \quad \textit{for all } u,v\in\mathcal{S}(\mathbb{R}^n).$$

In particular $A^*\in\Psi^m(\mathbb{R}^n)$, and the formal adjoint is determined by this identity. Consequently, the transpose $A^t\phi = \overline{A^*\bar\phi}$, which satisfies $\int(Au)\phi = \int u(A^t\phi)$ for $u,\phi\in\mathcal{S}$, is also continuous on $\mathcal{S}(\mathbb{R}^n)$.

Proof. The estimates (2.2.1) are symmetric in x and y, so $a^*\in\mathcal{A}^m_w$, depending continuously on a in each $\mathcal{A}^{m'}_w$. We must show $I(a)(\bar v\otimes u) = \overline{I(a^*)(\bar u\otimes v)}$, since the left side is (Au,v) and the right side is $\overline{(A^*v,u)} = (u,A^*v)$. For $a\in\mathcal{A}^{-\infty}_w$ this follows from the absolutely convergent expressions:

$$\overline{I(a^*)(\bar u\otimes v)} = \int e^{-i(x-y)\cdot\xi}a(y,x,\xi)u(x)\overline{v(y)}\,dx\,dyđ\xi = \int e^{i(x-y)\cdot\xi}a(x,y,\xi)\overline{v(x)}u(y)\,dx\,dyđ\xi,$$

after exchanging the names of x and y. The general case follows by replacing a by $\chi(\epsilon\xi)a$ and using Proposition 2.2.3(ii). Uniqueness of A^* is clear since $\mathcal{S}$ is dense in L^2. □

This shows one advantage of allowing general amplitudes: closure under adjoints is immediate. It also allows every $A\in\Psi^m(\mathbb{R}^n)$ to be extended to tempered distributions by duality:

$$(Au)(\phi) = u(A^t\phi), \qquad u\in\mathcal{S}'(\mathbb{R}^n),\ \phi\in\mathcal{S}(\mathbb{R}^n). \tag{2.2.10}$$

Since A^t is continuous on $\mathcal{S}$, this defines $Au\in\mathcal{S}'(\mathbb{R}^n)$, the map $A:\mathcal{S}'(\mathbb{R}^n)\to\mathcal{S}'(\mathbb{R}^n)$ is continuous (for the weak topology), and it extends A on $\mathcal{S}(\mathbb{R}^n)$; as $\mathcal{S}(\mathbb{R}^n)$ is dense in $\mathcal{S}'(\mathbb{R}^n)$ this extension is unique. Since $(AB)^t = B^tA^t$, the extensions satisfy $(AB)u = A(Bu)$ for $u\in\mathcal{S}'$. For example $\widehat{\langle D\rangle^s u} = \langle\xi\rangle^s\hat{u}$ continues to hold for $u\in\mathcal{S}'(\mathbb{R}^n)$.

2.3 Reduction to the left symbol

The main step towards the composition theorem is to show that every pseudodifferential operator can be written, uniquely, as the left quantization of a symbol independent of y:

Theorem 2.3.1 (Reduction)**.** *For each $A \in \Psi^m(\mathbb{R}^n)$ there is a unique $\sigma_L(A) \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ with $A = \mathrm{Op}(\sigma_L(A))$. If $A = \mathrm{Op}(a)$ with $a \in \mathcal{A}^m_w(\mathbb{R}^n)$ then*

$$\sigma_L(A)(x,\xi) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha D_y^\alpha a(x,y,\xi)\Big|_{y=x}. \tag{2.3.1}$$

Here the terms with $|\alpha| = j$ are grouped together; they have order $m - j$. The symbol $\sigma_L(A)$ is the *left* (or *full*) *symbol* of A. The proof occupies the rest of this section and has three steps. First, Taylor expansion of a in y around $y = x$ produces the terms of (2.3.1) plus a remainder of arbitrarily low order (Lemma 2.3.2). Second, Borel's lemma sums the terms. Third, the error is an operator of order $-\infty$, and we show that such operators are exactly the left quantizations of residual symbols (Proposition 2.3.3).

2.3.1 Taylor expansion of the amplitude

The key observation is the last identity of Lemma 2.2.4: multiplying the amplitude by $(x-y)^\alpha$ appears to raise its weight, but actually lowers the order of the operator by $|\alpha|$.

Lemma 2.3.2. *Let $a \in \mathcal{A}^m_w(\mathbb{R}^n)$ and $N \in \mathbb{N}$. Then*

$$\mathrm{Op}(a) = \sum_{j=0}^{N-1} \mathrm{Op}(a_j) + R_N, \qquad a_j(x,\xi) = \sum_{|\alpha|=j} \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha D_y^\alpha a(x,y,\xi)\Big|_{y=x} \in S^{m-j}(\mathbb{R}^n;\mathbb{R}^n),$$

with $R_N \in \Psi^{m-N}(\mathbb{R}^n)$; in fact $R_N = \mathrm{Op}(r_N)$ with $r_N \in \mathcal{A}^{m-N}_{w_+}(\mathbb{R}^n)$, $w_+ = \max(w,0)$.

Proof. Taylor's formula with integral remainder, applied to a in the variable y about $y = x$, gives

$$a(x,y,\xi) = \sum_{|\alpha|<N} \frac{(y-x)^\alpha}{\alpha!}(\partial_y^\alpha a)(x,x,\xi) + \sum_{|\alpha|=N} (y-x)^\alpha \rho_\alpha(x,y,\xi),$$

$$\rho_\alpha(x,y,\xi) = \frac{N}{\alpha!}\int_0^1 (1-t)^{N-1}(\partial_y^\alpha a)(x, x+t(y-x),\xi)\,dt.$$

We claim $\rho_\alpha \in \mathcal{A}^m_{w_+}$. Differentiating under the integral, $D_x^\mu D_y^\nu D_\xi^\gamma$ of the integrand is a linear combination, with coefficients that are polynomials in t bounded on $[0,1]$, of derivatives of $\partial_y^\alpha a$ evaluated at $(x, x+t(y-x),\xi)$, and these are bounded by $C\langle t(x-y)\rangle^w \langle\xi\rangle^{m-|\gamma|} \le C\langle x-y\rangle^{w_+}\langle\xi\rangle^{m-|\gamma|}$, since $\langle t(x-y)\rangle \le \langle x-y\rangle$ and $\langle t(x-y)\rangle \ge 1$. Now apply Lemma 2.2.4: for any $b \in \mathcal{A}^m_{w'}$,

$$I\big((y-x)^\alpha b\big) = (-1)^{|\alpha|}(x-y)^\alpha I(b) = I(D_\xi^\alpha b).$$

With $b = (\partial_y^\alpha a)(x,x,\xi)/\alpha!$, and $\partial_y^\alpha = i^{|\alpha|}D_y^\alpha$, this shows that the term of order α in the Taylor sum contributes $\mathrm{Op}(\frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha D_y^\alpha a(x,y,\xi)|_{y=x})$. With $b = \rho_\alpha$ it shows that the remainder term contributes $\mathrm{Op}(D_\xi^\alpha \rho_\alpha)$, with $D_\xi^\alpha \rho_\alpha \in \mathcal{A}^{m-N}_{w_+}$. □

2.3.2 Residual operators

Applying Borel's lemma to the series in Lemma 2.3.2 we will find a symbol b with $\mathrm{Op}(a) - \mathrm{Op}(b) \in \Psi^{m-N}$ for all N. To finish the proof of Theorem 2.3.1 we need to identify the residual operators. Note that *a priori* an element of $\Psi^{-\infty}(\mathbb{R}^n)$ is only known to be of the form $\mathrm{Op}(a_M)$ with $a_M \in \mathcal{A}^{-M}_{w_M}$ for each M, where the weight w_M may depend on M.

Proposition 2.3.3. *For a continuous linear operator $A : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ with Schwartz kernel K the following are equivalent:*

(a) $A \in \Psi^{-\infty}(\mathbb{R}^n)$;

(b) $K \in \mathcal{C}^\infty(\mathbb{R}^{2n})$ *and for all* $\alpha, \beta \in \mathbb{N}_0^n$ *and* $N \in \mathbb{N}_0$

$$|D_x^\alpha D_y^\beta K(x,y)| \le C_{\alpha\beta N}\langle x-y\rangle^{-N}; \tag{2.3.2}$$

(c) $A = \mathrm{Op}(c)$ *for some* $c \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$.

Proof. (c)⇒(a) is immediate, since $S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n) \subset \mathcal{A}_0^M(\mathbb{R}^n)$ for every M.

(a)⇒(b). For each $M > n$ write $A = \mathrm{Op}(a_M)$ with $a_M \in \mathcal{A}_{w_M}^{-M}$. By Proposition 2.2.3(i), $I(a_M)$ is a continuous, polynomially bounded function; two such functions which give the same integral against all $v \otimes u$ coincide, so $K = I(a_M)$ is a continuous function independent of M. By Lemma 2.2.4, for all α, β, γ,

$$(x-y)^\gamma D_x^\alpha D_y^\beta K = I(b), \qquad b = (-D_\xi)^\gamma(\xi + D_x)^\alpha (D_y - \xi)^\beta a_M \in \mathcal{A}_{w_M}^{-M+|\alpha|+|\beta|-|\gamma|},$$

where the powers of $\xi + D_x$ and $D_y - \xi$ are understood componentwise. Taking $\gamma = 0$ and $M > n + |\alpha| + |\beta|$ shows that all distributional derivatives of K are continuous functions, so $K \in \mathcal{C}^\infty(\mathbb{R}^{2n})$; then $|(x-y)^\gamma D_x^\alpha D_y^\beta K| \le C\langle x-y\rangle^{w_M}$ for all γ. Since $\langle x-y\rangle^{2L}$ is a polynomial in $x-y$, this gives (2.3.2).

(b)⇒(c). Set $g(x,z) = K(x, x-z)$. Since $D_x g = (D_x K + D_y K)(x, x-z)$ and $D_z g = -(D_y K)(x, x-z)$, (2.3.2) gives $|D_x^\alpha D_z^\beta g(x,z)| \le C_{\alpha\beta N}\langle z\rangle^{-N}$, so $\sup_{x,z}|z^\gamma D_x^\alpha D_z^\beta g| < \infty$ for all α, β, γ. Let

$$c(x,\xi) = \int e^{-iz\cdot\xi} g(x,z)\, dz.$$

Then $\xi^\beta D_\xi^\gamma D_x^\alpha c$ is the Fourier transform in z of $D_z^\beta((-z)^\gamma D_x^\alpha g)$, which is bounded in L_z^1 uniformly in x; hence $\sup_{x,\xi}|\xi^\beta D_\xi^\gamma D_x^\alpha c| < \infty$ for all α, β, γ, i.e. $c \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$. By Fourier inversion $g(x,z) = \int e^{iz\cdot\xi} c(x,\xi)\,đ\xi$, so by Proposition 2.2.3(i) the kernel of $\mathrm{Op}(c)$ is $\int e^{i(x-y)\cdot\xi} c(x,\xi)\,đ\xi = g(x, x-y) = K(x,y)$. Hence $A = \mathrm{Op}(c)$. □

Proof of Theorem 2.3.1. Let $A = \mathrm{Op}(a)$, $a \in \mathcal{A}_w^m$, and let $a_j \in S^{m-j}(\mathbb{R}^n;\mathbb{R}^n)$ be as in Lemma 2.3.2. By Proposition 2.1.10 choose $b \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ with $b \sim \sum_j a_j$. Then for every N

$$A - \mathrm{Op}(b) = R_N - \mathrm{Op}\Big(b - \sum_{j<N} a_j\Big) \in \Psi^{m-N}(\mathbb{R}^n),$$

so $A - \mathrm{Op}(b) \in \Psi^{-\infty}(\mathbb{R}^n)$ and, by Proposition 2.3.3, $A - \mathrm{Op}(b) = \mathrm{Op}(c)$ with $c \in S^{-\infty}$. Thus $A = \mathrm{Op}(b+c)$ and $\sigma_L(A) = b + c$ satisfies (2.3.1).

For uniqueness, suppose $\mathrm{Op}(e) = 0$ with $e \in S^m(\mathbb{R}^n;\mathbb{R}^n)$. By (2.2.8), $\int e^{ix\cdot\xi} e(x,\xi)\hat{u}(\xi)\,đ\xi = 0$ for all $u \in \mathcal{S}$. For fixed x, $\hat{u}$ ranges over all of $\mathcal{S}(\mathbb{R}^n)$, so the continuous, polynomially bounded function $\xi \mapsto e^{ix\cdot\xi} e(x,\xi)$ vanishes identically. Hence $e = 0$. □

Corollary 2.3.4. (i) $\mathrm{Op} : S^m(\mathbb{R}^n;\mathbb{R}^n) \to \Psi^m(\mathbb{R}^n)$ *is a bijection with inverse* σ_L*; in particular every* $A \in \Psi^m(\mathbb{R}^n)$ *has an amplitude in* $\mathcal{A}_0^m(\mathbb{R}^n)$*, so the weights* w *play no role in the definition of* $\Psi^m(\mathbb{R}^n)$.

(ii) $\Psi^{-\infty}(\mathbb{R}^n) = \mathrm{Op}\big(S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)\big)$*, and for* $A \in \Psi^m(\mathbb{R}^n)$ *and any* $k \in \mathbb{R}$, $A \in \Psi^k(\mathbb{R}^n)$ *if and only if* $\sigma_L(A) \in S^k(\mathbb{R}^n;\mathbb{R}^n)$.

(iii) (Asymptotic summation of operators) *If* $A_j \in \Psi^{m-j}(\mathbb{R}^n)$*, there exists* $A \in \Psi^m(\mathbb{R}^n)$*, unique up to* $\Psi^{-\infty}(\mathbb{R}^n)$*, with* $A - \sum_{j<N} A_j \in \Psi^{m-N}(\mathbb{R}^n)$ *for all* N*; namely any* A *with* $\sigma_L(A) \sim \sum_j \sigma_L(A_j)$.

Proof. (i) and (ii) are restatements of Theorem 2.3.1 and Proposition 2.3.3, using that σ_L is linear. (iii) follows from (i), (ii) and Proposition 2.1.10. □

2.4 Composition and adjoints

We can now prove the fundamental result of this chapter.

Theorem 2.4.1 (Composition)**.** *The space $\Psi^\infty(\mathbb{R}^n)$ is an order-filtered $*$-algebra of continuous operators on $\mathcal{S}(\mathbb{R}^n)$ (and on $\mathcal{S}'(\mathbb{R}^n)$):*

$$A \in \Psi^m(\mathbb{R}^n) \Longrightarrow A^* \in \Psi^m(\mathbb{R}^n), \qquad A \in \Psi^m(\mathbb{R}^n),\ B \in \Psi^{m'}(\mathbb{R}^n) \Longrightarrow AB \in \Psi^{m+m'}(\mathbb{R}^n).$$

Moreover the left symbols satisfy

$$\sigma_L(A^*)(x,\xi) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha D_x^\alpha \overline{\sigma_L(A)(x,\xi)}, \tag{2.4.1}$$

$$\sigma_L(AB)(x,\xi) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha \sigma_L(A)(x,\xi)\, D_x^\alpha \sigma_L(B)(x,\xi). \tag{2.4.2}$$

Formally, (2.4.2) reads $\sigma_L(AB) = e^{iD_\eta \cdot D_y} \sigma_L(A)(x,\eta)\sigma_L(B)(y,\xi)\big|_{y=x,\eta=\xi}$.

Proof. The statement about adjoints is Proposition 2.2.9. By that proposition, the adjoint of $\mathrm{Op}(a)$, $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$, has amplitude $\overline{a(y,\xi)}$, so Theorem 2.3.1 (with $D_y^\alpha \overline{a(y,\xi)}|_{y=x} = D_x^\alpha \overline{a(x,\xi)}$) gives (2.4.1).

Right symbols. Every $B \in \Psi^{m'}(\mathbb{R}^n)$ can be written as $B = \mathrm{Op}_R(\sigma_R(B))$ with $\sigma_R(B) = \overline{\sigma_L(B^*)} \in S^{m'}$: indeed $B = (B^*)^* = \mathrm{Op}(\sigma_L(B^*))^*$, whose amplitude is $\overline{\sigma_L(B^*)(y,\xi)}$. Taking complex conjugates in (2.4.1) applied to B, and using $\overline{D^\alpha f} = (-1)^{|\alpha|} D^\alpha \bar f$ together with Lemma 2.1.11(i),

$$\sigma_R(B) \sim \sum_\beta \frac{(-i)^{|\beta|}}{\beta!} D_\xi^\beta D_x^\beta \sigma_L(B). \tag{2.4.3}$$

Composition. Let $a = \sigma_L(A)$ and $b = \sigma_R(B)$. By Proposition 2.2.7, $AB = \mathrm{Op}(a)\,\mathrm{Op}_R(b) = \mathrm{Op}(a(x,\xi)b(y,\xi))$ with $a(x,\xi)b(y,\xi) \in \mathcal{A}_0^{m+m'}$, so $AB \in \Psi^{m+m'}(\mathbb{R}^n)$, and Theorem 2.3.1 gives

$$\sigma_L(AB) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha [a\, D_x^\alpha b].$$

The map $f \mapsto D_\xi^\alpha[a\, D_x^\alpha f]$ lowers orders by $|\alpha|$ and shifts them by m, so by (2.4.3) and Lemma 2.1.11(i),(iii), with $c = \sigma_L(B)$,

$$\sigma_L(AB) \sim \sum_\gamma \sum_{\alpha+\beta=\gamma} \frac{i^{|\alpha|}(-i)^{|\beta|}}{\alpha!\,\beta!} D_\xi^\alpha [a\, D_\xi^\beta D_x^\gamma c],$$

the terms with $|\gamma| = l$ having order $m + m' - l$. By Leibniz' formula $D_\xi^\alpha[aD_\xi^\beta f] = \sum_{\mu\le\alpha} \binom{\alpha}{\mu} D_\xi^\mu a\, D_\xi^{\gamma-\mu} f$ when $\alpha+\beta=\gamma$, and using $\frac{\gamma!}{\alpha!\beta!}\binom{\alpha}{\mu} = \binom{\gamma}{\mu}\binom{\gamma-\mu}{\alpha-\mu}$, the coefficient of $\frac{1}{\gamma!} D_\xi^\mu a\, D_\xi^{\gamma-\mu} D_x^\gamma c$ in the inner sum is

$$\binom{\gamma}{\mu} i^{|\mu|} \sum_{\delta\le\gamma-\mu} \binom{\gamma-\mu}{\delta} i^{|\delta|}(-i)^{|\gamma-\mu-\delta|} = \binom{\gamma}{\mu} i^{|\mu|} \prod_{l=1}^n (i-i)^{\gamma_l-\mu_l},$$

which vanishes unless $\mu = \gamma$, when it is $i^{|\gamma|}$. Hence the inner sum equals $\frac{i^{|\gamma|}}{\gamma!} D_\xi^\gamma a\, D_x^\gamma c$, which is (2.4.2).

The operators extend continuously to $\mathcal{S}'(\mathbb{R}^n)$ by (2.2.10). □

Applying Theorem 2.3.1 to the amplitude $b(y,\xi)$ shows that left and right symbols are related by

$$\sigma_L(\mathrm{Op}_R(b)) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha D_x^\alpha b, \tag{2.4.4}$$

so that formally $\sigma_L = e^{iD_x\cdot D_\xi}\sigma_R$ and, by (2.4.3), $\sigma_R = e^{-iD_x\cdot D_\xi}\sigma_L$. The map $\mathrm{Op}_R : S^m(\mathbb{R}^n;\mathbb{R}^n) \to \Psi^m(\mathbb{R}^n)$ is a bijection with inverse σ_R; see Exercise 2.6 for a "centrist" alternative.

2.5 The principal symbol

By (2.4.4) and (2.4.3), $\sigma_L(A) - \sigma_R(A) \in S^{m-1}(\mathbb{R}^n;\mathbb{R}^n)$ for $A \in \Psi^m(\mathbb{R}^n)$: the full symbol depends on the choice of quantization, but its leading part does not. For this reason we consider the quotient spaces

$$S^m(\mathbb{R}^p;\mathbb{R}^n)/S^{m-1}(\mathbb{R}^p;\mathbb{R}^n),$$

and write $[a]$ for the class of $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$.

Definition 2.5.1. The *principal symbol map* $\sigma_m : \Psi^m(\mathbb{R}^n) \to S^m(\mathbb{R}^n;\mathbb{R}^n)/S^{m-1}(\mathbb{R}^n;\mathbb{R}^n)$ is $\sigma_m(A) = [\sigma_L(A)] = [\sigma_R(A)]$.

Discussion. Why pass to a quotient? The full symbol carries all the information about A, but it depends on choices (left, right or Weyl quantization) and it transforms in a complicated way under composition and changes of variables. The principal symbol forgets everything except the leading behaviour as $|\xi| \to \infty$. In exchange it is multiplicative (2.5.1), and on manifolds it will be an invariantly defined function on the cotangent bundle (Section 2.13). Many qualitative questions, above all ellipticity, are decided by the principal symbol alone.

Unlike σ_L or σ_R, σ_m depends on m: one needs to know that the order is at most m before it is defined. If $A = \mathrm{Op}(a)$ with $a \in \mathcal{A}^m_w$, then $\sigma_m(A) = [a(x,x,\xi)]$ by Theorem 2.3.1. For a differential operator $P = \sum_{|\alpha|\le m} p_\alpha D^\alpha \in \mathrm{Diff}^m_{\mathrm{b}}(\mathbb{R}^n)$, $\sigma_m(P)$ is represented by $\sum_{|\alpha|=m} p_\alpha(x)\xi^\alpha$.

Proposition 2.5.2. *For every $m \in \mathbb{R}$ the sequence*

$$0 \longrightarrow \Psi^{m-1}(\mathbb{R}^n) \hookrightarrow \Psi^m(\mathbb{R}^n) \xrightarrow{\sigma_m} S^m(\mathbb{R}^n;\mathbb{R}^n)/S^{m-1}(\mathbb{R}^n;\mathbb{R}^n) \longrightarrow 0$$

is exact. Moreover, for $A \in \Psi^m(\mathbb{R}^n)$ and $B \in \Psi^{m'}(\mathbb{R}^n)$,

$$\sigma_{m+m'}(AB) = \sigma_m(A)\,\sigma_{m'}(B), \qquad \sigma_m(A^*) = \overline{\sigma_m(A)}, \tag{2.5.1}$$

and the commutator $[A,B] = AB - BA$ lies in $\Psi^{m+m'-1}(\mathbb{R}^n)$ with

$$\sigma_{m+m'-1}([A,B]) = \frac{1}{i}\{\sigma_m(A),\sigma_{m'}(B)\}, \qquad \{a,b\} = \sum_{j=1}^n \Big(\frac{\partial a}{\partial\xi_j}\frac{\partial b}{\partial x_j} - \frac{\partial a}{\partial x_j}\frac{\partial b}{\partial\xi_j}\Big). \tag{2.5.2}$$

Proof. Surjectivity of σ_m holds because $\sigma_m(\mathrm{Op}(a)) = [a]$, and its null space is Ψ^{m-1} by Corollary 2.3.4(ii). All terms with $|\alpha| \ge 1$ in (2.4.2) and (2.4.1) have order at most $m+m'-1$, respectively $m-1$, which gives (2.5.1). In particular $\sigma_{m+m'}([A,B]) = 0$, so $[A,B] \in \Psi^{m+m'-1}$; the Poisson bracket is well defined on the classes (changing a by an element of S^{m-1} changes $\{a,b\}$ by an element of $S^{m+m'-2}$). With $a = \sigma_L(A)$, $b = \sigma_L(B)$, the terms with $|\alpha| = 1$ in (2.4.2) give

$$\sigma_L([A,B]) - i\sum_j (D_{\xi_j} a\, D_{x_j} b - D_{\xi_j} b\, D_{x_j} a) \in S^{m+m'-2},$$

and $iD_{\xi_j} a\, D_{x_j} b = -i\,\partial_{\xi_j} a\,\partial_{x_j} b$, which gives (2.5.2). □

The simplicity of (2.5.1) compared with (2.4.2) is achieved at the expense of a great loss of information. Still, many problems can be solved using it; it says that the principal symbol gives a homomorphism from the filtered algebra $\Psi^0(\mathbb{R}^n)$ onto the *commutative* algebra $S^0(\mathbb{R}^n;\mathbb{R}^n)/S^{-1}(\mathbb{R}^n;\mathbb{R}^n)$.

2.6 Ellipticity and parametrices

We now use the calculus to invert operators. Exact inverses are usually out of reach, and also unnecessary: it is enough to invert up to an error in $\Psi^{-\infty}$, since such errors are smoothing (Lemma 2.6.5). The principal symbol tells us exactly when this is possible.

Definition 2.6.1. An operator $A \in \Psi^m(\mathbb{R}^n)$ is *elliptic* if it is invertible modulo $\Psi^{-\infty}(\mathbb{R}^n)$ with an approximate inverse of order $-m$, i.e. if there exists $B \in \Psi^{-m}(\mathbb{R}^n)$ with

$$AB - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n). \tag{2.6.1}$$

Such a B is called a (right) *parametrix* of A.

Thus ellipticity is invertibility in the quotient algebra $\Psi^\infty(\mathbb{R}^n)/\Psi^{-\infty}(\mathbb{R}^n)$, with the inverse in $\Psi^{-m}(\mathbb{R}^n)/\Psi^{-\infty}(\mathbb{R}^n)$. The point is that it is a property of the principal symbol alone. Both at the level of symbols and of operators we use the following Neumann series argument.

Lemma 2.6.2. *If $e \in S^{-1}(\mathbb{R}^p;\mathbb{R}^n)$ there is $f \in S^{-1}(\mathbb{R}^p;\mathbb{R}^n)$ with $(1+e)(1+f)-1 \in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$. If $E \in \Psi^{-1}(\mathbb{R}^n)$ there is $F \in \Psi^{-1}(\mathbb{R}^n)$ with $(\mathrm{Id}+E)(\mathrm{Id}+F)-\mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$ and $(\mathrm{Id}+F)(\mathrm{Id}+E)-\mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$.*

Proof. We give the operator version; the symbol version is identical, with Proposition 2.1.10 in place of Corollary 2.3.4(iii). By Theorem 2.4.1, $(-E)^j \in \Psi^{-j}$, so by Corollary 2.3.4(iii) there is $F \in \Psi^{-1}$ with $F - \sum_{j=1}^N (-E)^j \in \Psi^{-N-1}$ for all N. Since $(\mathrm{Id}+E)\sum_{j=0}^N (-E)^j = \mathrm{Id} - (-E)^{N+1} = \sum_{j=0}^N (-E)^j(\mathrm{Id}+E)$,

$$(\mathrm{Id}+E)(\mathrm{Id}+F)-\mathrm{Id} = -(-E)^{N+1} + (\mathrm{Id}+E)\Big(F - \sum_{j=1}^N (-E)^j\Big) \in \Psi^{-N-1}(\mathbb{R}^n)$$

for every N, and similarly for $(\mathrm{Id}+F)(\mathrm{Id}+E)-\mathrm{Id}$. □

Theorem 2.6.3. *For $A \in \Psi^m(\mathbb{R}^n)$ the following conditions are equivalent:*

(a) *A is elliptic;*

(b) *there exists $b \in S^{-m}(\mathbb{R}^n;\mathbb{R}^n)$ such that $\sigma_m(A)\cdot[b] = 1$ in $S^0(\mathbb{R}^n;\mathbb{R}^n)/S^{-1}(\mathbb{R}^n;\mathbb{R}^n)$;*

(c) *there exists $b \in S^{-m}(\mathbb{R}^n;\mathbb{R}^n)$ such that $\sigma_L(A)\,b - 1 \in S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n)$;*

(d) *$\sigma_L(A)$ is an elliptic symbol: there is $\epsilon > 0$ such that $|\sigma_L(A)(x,\xi)| \geq \epsilon\langle\xi\rangle^m$ for $|\xi| \geq 1/\epsilon$.*

Condition (d) *holds for one representative of $\sigma_m(A)$ if and only if it holds for all of them (with different ϵ).*

Proof. The last statement follows from the equivalent form $|a| \geq \epsilon\langle\xi\rangle^m - C\langle\xi\rangle^{m-1}$ of ellipticity noted after Definition 2.1.7. We prove (a)⇒(b)⇒(c)⇔(d) and (c)⇒(a).

(a)⇒(b): since $\mathrm{Id} = \mathrm{Op}(1)$, $\sigma_0(\mathrm{Id}) = 1$; applying (2.5.1) to (2.6.1) gives $\sigma_m(A)\sigma_{-m}(B) = 1$.

(b)⇒(c): (b) means that $\sigma_L(A)b = 1 + e$ with $e \in S^{-1}$. With f as in Lemma 2.6.2, $\sigma_L(A)\, b(1+f) = (1+e)(1+f) \in 1 + S^{-\infty}$, and $b(1+f) \in S^{-m}$.

(c)⇔(d) is Lemma 2.1.8.

(c)⇒(a): let $B_1 = \mathrm{Op}(b)$. By (2.4.2), $\sigma_L(AB_1) - \sigma_L(A)b \in S^{-1}$, so $E = AB_1 - \mathrm{Id} \in \Psi^{-1}(\mathbb{R}^n)$. With F as in Lemma 2.6.2, $B = B_1(\mathrm{Id}+F) \in \Psi^{-m}(\mathbb{R}^n)$ satisfies $AB - \mathrm{Id} = (\mathrm{Id}+E)(\mathrm{Id}+F) - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$. □

In Definition 2.6.1 we took B to be a right parametrix. We could equally well have taken a left parametrix:

Lemma 2.6.4. *$A \in \Psi^m(\mathbb{R}^n)$ is elliptic if and only if there exists $B' \in \Psi^{-m}(\mathbb{R}^n)$ with $B'A - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$. If B is a right and B' a left parametrix then $B - B' \in \Psi^{-\infty}(\mathbb{R}^n)$.*

Proof. If $B'A - \mathrm{Id} \in \Psi^{-\infty}$ then $\sigma_{-m}(B')\sigma_m(A) = 1$, and since multiplication of symbols is commutative, (b) of Theorem 2.6.3 holds, so A is elliptic. Conversely, if A is elliptic, let b be as in Theorem 2.6.3(c) and $B_1 = \mathrm{Op}(b)$. As in the proof above, $B_1A = \mathrm{Id}+E'$ with $E' \in \Psi^{-1}$, and with F' as in Lemma 2.6.2 (for E'), $B' = (\mathrm{Id}+F')B_1$ satisfies $B'A - \mathrm{Id} \in \Psi^{-\infty}$. Finally, if $AB = \mathrm{Id}+E$ and $B'A = \mathrm{Id}+E'$ with $E, E' \in \Psi^{-\infty}$, then $B' + B'E = B'(AB) = (B'A)B = B + E'B$, so $B - B' = B'E - E'B \in \Psi^{-\infty}$. □

Thus a parametrix of an elliptic operator is automatically two-sided, and it is unique modulo $\Psi^{-\infty}(\mathbb{R}^n)$.

2.6.1 Elliptic regularity

One of the main reasons that residual operators are "residual" is that they are smoothing.

Lemma 2.6.5. *Let $E \in \Psi^{-\infty}(\mathbb{R}^n)$ have kernel K. For $u \in \mathcal{S}'(\mathbb{R}^n)$, Eu is the smooth function $x \mapsto u\big(K(x,\cdot)\big)$, and all its derivatives are polynomially bounded. In particular $E : \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n) \cap \mathcal{C}^\infty(\mathbb{R}^n)$.*

Proof. Write $k_x = K(x,\cdot)$. By (2.3.2) and Peetre's inequality $\langle y\rangle^N \le 2^{N/2}\langle x\rangle^N\langle x-y\rangle^N$,

$$\sup_y |y^\gamma D_y^\beta D_x^\alpha K(x,y)| \le C_{\alpha\beta\gamma}\langle x\rangle^{|\gamma|}.$$

Hence $x \mapsto k_x$ is a smooth map into $\mathcal{S}(\mathbb{R}^n)$ with $D_x^\alpha k_x = (D_x^\alpha K)(x,\cdot)$ and $\|D_x^\alpha k_x\|_j \le C\langle x\rangle^j$: for instance $t^{-1}(k_{x+te_l} - k_x) - (\partial_{x_l}K)(x,\cdot) = \int_0^1 \big((\partial_{x_l}K)(x+ste_l,\cdot) - (\partial_{x_l}K)(x,\cdot)\big)ds$, whose $\mathcal{S}$-norms are $O(|t|)$ by the bounds on the second derivatives. Since $|u(\phi)| \le C\|\phi\|_j$ for some j, the function $f(x) = u(k_x)$ is smooth with $D^\alpha f(x) = u(D_x^\alpha k_x)$ and $|D^\alpha f(x)| \le C_\alpha\langle x\rangle^j$. Finally, by (2.2.10) and Proposition 2.2.9, $(Eu)(\phi) = u(E^t\phi)$ with $E^t\phi = \int \phi(x)k_x\,dx$; this integral converges in $\mathcal{S}(\mathbb{R}^n)$ (the integrand is continuous with norms bounded by $C|\phi(x)|\langle x\rangle^j$), so $u(E^t\phi) = \int \phi(x)u(k_x)\,dx$, i.e. $Eu = f$. □

From the existence of parametrices we deduce the regularity of solutions of elliptic equations.

Proposition 2.6.6. *If $A \in \Psi^m(\mathbb{R}^n)$ is elliptic and $u \in \mathcal{S}'(\mathbb{R}^n)$ satisfies $Au \in \mathcal{S}(\mathbb{R}^n)$ (e.g. $Au = 0$), then $u \in \mathcal{C}^\infty(\mathbb{R}^n)$.*

Proof. Let $B \in \Psi^{-m}(\mathbb{R}^n)$ be a left parametrix, $BA = \mathrm{Id}+E$ with $E \in \Psi^{-\infty}$. Then $u = B(Au) - Eu$, where $B(Au) \in \mathcal{S}(\mathbb{R}^n)$ and $Eu \in \mathcal{C}^\infty(\mathbb{R}^n)$ by Lemma 2.6.5. □

A more precise, quantitative version is given in Proposition 2.8.4; a local version requires the microlocal methods of Chapter 3.

Question. Where exactly was ellipticity used in the proof of Proposition 2.6.6?

Answer. Only to produce a left parametrix B with $BA = \mathrm{Id} + E$, $E \in \Psi^{-\infty}$. The rest of the argument works for any operator: B preserves $\mathcal{S}$, and E maps every tempered distribution to a smooth function. Without ellipticity the conclusion fails: for $A = D_{x_1}$ on $\mathbb{R}^2$, the distribution $u = \delta(x_2)$, defined by $u(\phi) = \int \phi(x_1, 0)\, dx_1$, satisfies $Au = 0$ but is not smooth. The characteristic set $\{\xi_1 = 0\}$ of D_{x_1} is where the singularity is allowed to live. □

2.6.2 The Laplacian

Let $g_{ij} \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$, $i, j = 1, \dots, n$, be the components of a *uniformly positive* Riemannian metric on $\mathbb{R}^n$: $g_{ij} = g_{ji}$ and, for some $\epsilon > 0$,

$$\sum_{i,j=1}^{n} g_{ij}(x)\xi_i\xi_j \geq \epsilon|\xi|^2 \quad \text{for all } x, \xi \in \mathbb{R}^n. \tag{2.6.2}$$

Let $g(x) = \det\big(g_{ij}(x)\big)$ and let $(g^{ij}(x))$ be the inverse matrix. Since the g_{ij} are bounded, the eigenvalues of (g_{ij}) lie in $[\epsilon, C]$; hence $\epsilon^n \leq g \leq C^n$, and by Cramer's rule $g^{ij}, \sqrt{g} \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$, with $\sum g^{ij}\xi_i\xi_j \geq C^{-1}|\xi|^2$. The *Laplacian* of the metric is the differential operator

$$\Delta_g = \sum_{i,j=1}^{n} \frac{1}{\sqrt{g}} D_{x_i} g^{ij} \sqrt{g}\, D_{x_j} = \sum_{i,j} g^{ij} D_{x_i} D_{x_j} + \sum_{i,j} \frac{1}{\sqrt{g}} (D_{x_i}(g^{ij}\sqrt{g})) D_{x_j}.$$

It is characterized by the integration by parts formula

$$\int_{\mathbb{R}^n} \sum_{i,j} g^{ij}(x)\, D_{x_j}\phi\, \overline{D_{x_i}\psi}\, dg = \int_{\mathbb{R}^n} \Delta_g \phi\, \overline{\psi}\, dg \quad \text{for all } \phi, \psi \in \mathcal{S}(\mathbb{R}^n), \qquad dg = \sqrt{g}\, dx,$$

which follows from $\int D_{x_i} f\, \bar{h}\, dx = \int f\, \overline{D_{x_i} h}\, dx$. In particular Δ_g is formally self-adjoint and non-negative with respect to the measure dg.

Proposition 2.6.7. *$\Delta_g \in \mathrm{Diff}^2_{\mathrm{b}}(\mathbb{R}^n) \subset \Psi^2(\mathbb{R}^n)$ is elliptic, with $\sigma_2(\Delta_g) = \big[\sum_{i,j} g^{ij}(x)\xi_i\xi_j\big]$. Hence there is $B \in \Psi^{-2}(\mathbb{R}^n)$ with $\Delta_g B - \mathrm{Id}$, $B\Delta_g - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$, and if $u \in \mathcal{S}'(\mathbb{R}^n)$ satisfies $\Delta_g u = 0$ then $u \in \mathcal{C}^\infty(\mathbb{R}^n)$.*

Proof. The coefficients are in $\mathcal{C}^\infty_{\mathrm{b}}$ by the remarks above, and the principal symbol satisfies $\sum g^{ij}\xi_i\xi_j \geq C^{-1}|\xi|^2 \geq (2C)^{-1}\langle\xi\rangle^2$ for $|\xi| \geq 1$. The rest follows from Theorem 2.6.3, Lemma 2.6.4 and Proposition 2.6.6. □

Note that the upper bound on the g_{ij}, which comes from the assumption $g_{ij} \in \mathcal{C}^\infty_{\mathrm{b}}$, is needed to obtain the lower bound on the principal symbol $\sum g^{ij}\xi_i\xi_j$.

2.7 Boundedness on L^2

So far pseudodifferential operators have been considered as continuous operators on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$. Using the composition theorem we can now prove "finite order" regularity results, the basic one being:

Theorem 2.7.1. *Every $A \in \Psi^0(\mathbb{R}^n)$ extends by continuity from $\mathcal{S}(\mathbb{R}^n)$ to a bounded linear operator $A : L^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$, which coincides with the restriction to $L^2(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$ of the action (2.2.10) on $\mathcal{S}'(\mathbb{R}^n)$.*

The proof is in two stages: residual operators are handled by direct estimation, and the general case is reduced to that one by Hörmander's trick of using an approximate square root.

Discussion. The idea of the square-root trick is simple. If $A \in \Psi^0$ and $C > \sup|\sigma_L(A)|^2$, then the symbol of $C - A^*A$ is positive. A positive symbol has a square root in the symbol calculus, so $C - A^*A = B^*B + G$ with G residual. Then $\|A\phi\|^2 = C\|\phi\|^2 - \|B\phi\|^2 - (G\phi, \phi) \le C\|\phi\|^2 + |(G\phi, \phi)|$. Positivity of a symbol becomes positivity of an operator, up to a residual error which Schur's lemma controls.

Lemma 2.7.2 (Schur)**.** *If K is a measurable function on $\mathbb{R}^{2n}$ such that*

$$M_1 = \sup_{x \in \mathbb{R}^n} \int_{\mathbb{R}^n} |K(x,y)|\, dy < \infty, \qquad M_2 = \sup_{y \in \mathbb{R}^n} \int_{\mathbb{R}^n} |K(x,y)|\, dx < \infty,$$

then $T\phi(x) = \int K(x,y)\phi(y)\, dy$ defines a bounded operator on $L^2(\mathbb{R}^n)$ with $\|T\| \le \sqrt{M_1 M_2}$.

Proof. By the Cauchy–Schwarz inequality applied to $|K|^{1/2} \cdot |K|^{1/2}|\phi|$,

$$|T\phi(x)|^2 \le \int |K(x,y)|\, dy \int |K(x,y)||\phi(y)|^2\, dy \le M_1 \int |K(x,y)||\phi(y)|^2\, dy,$$

and integrating in x (Tonelli) gives $\|T\phi\|^2 \le M_1 M_2 \|\phi\|^2$. □

By Proposition 2.3.3, the kernel of $G \in \Psi^{-\infty}(\mathbb{R}^n)$ satisfies $|K(x,y)| \le C\langle x - y\rangle^{-n-1}$, and $\int \langle z\rangle^{-n-1} dz < \infty$. Hence every $G \in \Psi^{-\infty}(\mathbb{R}^n)$ is bounded on $L^2(\mathbb{R}^n)$.

We say that a real symbol $a \in S^m(\mathbb{R}^n; \mathbb{R}^n)$ is *positive elliptic* if $a \ge \epsilon\langle\xi\rangle^m$ for some $\epsilon > 0$. By Lemma 2.1.6, $a^s \in S^{sm}$ for all $s \in \mathbb{R}$.

Proposition 2.7.3. *Let $A \in \Psi^m(\mathbb{R}^n)$ be self-adjoint, $A = A^*$, and suppose that $\sigma_m(A)$ has a positive elliptic representative. Then there exists $B \in \Psi^{m/2}(\mathbb{R}^n)$ with $B = B^*$ and*

$$A = B^2 + G, \qquad G \in \Psi^{-\infty}(\mathbb{R}^n).$$

Proof. Let a be a positive elliptic representative of $\sigma_m(A)$ and $b_0 = a^{1/2} \in S^{m/2}(\mathbb{R}^n; \mathbb{R}^n)$. Let $B_0 = \frac{1}{2}(\mathrm{Op}(b_0) + \mathrm{Op}(b_0)^*)$; then $B_0 = B_0^*$ and $\sigma_{m/2}(B_0) = [b_0]$ since b_0 is real. By (2.5.1), $\sigma_m(B_0^2) = [b_0^2] = [a]$, so $E_1 = A - B_0^2 \in \Psi^{m-1}(\mathbb{R}^n)$.

We proceed inductively. Suppose $B_j \in \Psi^{m/2-j}(\mathbb{R}^n)$, $B_j = B_j^*$, have been chosen for $j \le N$ such that, with $S_N = \sum_{j=0}^N B_j$,

$$E_{N+1} = A - S_N^2 \in \Psi^{m-N-1}(\mathbb{R}^n);$$

this has been done for $N = 0$. Note $E_{N+1}^* = E_{N+1}$. For $B_{N+1} \in \Psi^{m/2-N-1}(\mathbb{R}^n)$,

$$A - (S_N + B_{N+1})^2 = E_{N+1} - S_N B_{N+1} - B_{N+1} S_N - B_{N+1}^2.$$

Since $S_N - B_0 \in \Psi^{m/2-1}$, all terms on the right are of order $m - N - 2$ except $E_{N+1} - B_0 B_{N+1} - B_{N+1} B_0 \in \Psi^{m-N-1}$, whose principal symbol is

$$\sigma_{m-N-1}(E_{N+1}) - 2b_0\, \sigma_{m/2-N-1}(B_{N+1}).$$

Let e represent $\sigma_{m-N-1}(E_{N+1})$; by Lemma 2.1.6, $e/(2b_0) \in S^{m/2-N-1}$. Let $B_{N+1} = \frac{1}{2}(\mathrm{Op}(e/2b_0) + \mathrm{Op}(e/2b_0)^*)$. Since E_{N+1} is self-adjoint, $[e] = [\bar{e}]$ and so $\sigma_{m/2-N-1}(B_{N+1}) = [e/(2b_0)]$. Then $A - S_{N+1}^2 \in \Psi^{m-N-2}$, completing the induction.

By Corollary 2.3.4(iii) choose $B' \in \Psi^{m/2}$ with $B' - S_N \in \Psi^{m/2-N-1}$ for all N, and set $B = \frac{1}{2}(B' + B'^*)$. Since $S_N = S_N^*$, also $B - S_N = \frac{1}{2}((B' - S_N) + (B' - S_N)^*) \in \Psi^{m/2-N-1}$. Writing $B = S_N + R_N$,

$$A - B^2 = E_{N+1} - S_N R_N - R_N S_N - R_N^2 \in \Psi^{m-N-1}(\mathbb{R}^n)$$

for every N. Thus $G = A - B^2 \in \Psi^{-\infty}(\mathbb{R}^n)$. □

Proof of Theorem 2.7.1. Let $a = \sigma_L(A) \in S^0$ and $C = 1 + \sup|a|^2$. Then $C - A^*A \in \Psi^0$ is self-adjoint and by (2.5.1) its principal symbol is represented by $C - |a|^2 \geq 1$, which is positive elliptic of order 0. By Proposition 2.7.3, $C - A^*A = B^2 + G$ with $B = B^* \in \Psi^0$ and $G \in \Psi^{-\infty}$; G is bounded on L^2 by Schur's lemma. For $\phi \in \mathcal{S}(\mathbb{R}^n)$,

$$\|A\phi\|^2 = (A^*A\phi, \phi) = C\|\phi\|^2 - \|B\phi\|^2 - (G\phi, \phi) \leq \big(C + \|G\|_{L^2 \to L^2}\big)\|\phi\|^2.$$

Since $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ (Exercise 2.18), A extends uniquely to a bounded operator on L^2. If $\phi_k \to u$ in L^2 then $\phi_k \to u$ in $\mathcal{S}'$, so $A\phi_k \to Au$ in $\mathcal{S}'$ by the continuity of (2.2.10); hence the two extensions agree. □

2.8 Sobolev spaces

Recall that for $k \in \mathbb{N}_0$ the Sobolev space $H^k(\mathbb{R}^n)$ consists of the $u \in L^2(\mathbb{R}^n)$ with $D^\alpha u \in L^2(\mathbb{R}^n)$ for all $|\alpha| \leq k$. By Plancherel's theorem this means $\xi^\alpha \hat{u} \in L^2$ for $|\alpha| \leq k$, and since

$$\langle \xi \rangle^{2k} = (1 + \xi_1^2 + \cdots + \xi_n^2)^k = \sum_{|\beta| \leq k} c_\beta \, \xi^{2\beta}, \qquad c_\beta > 0, \tag{2.8.1}$$

these finitely many conditions are equivalent to the single condition $\langle \xi \rangle^k \hat{u} \in L^2$, i.e. $\langle D \rangle^k u \in L^2(\mathbb{R}^n)$, where $\langle D \rangle^s = \mathrm{Op}(\langle \xi \rangle^s) \in \Psi^s(\mathbb{R}^n)$ as in Example 2.2.8. For negative integers $-k$, one defines

$$H^{-k}(\mathbb{R}^n) = \Big\{ u \in \mathcal{S}'(\mathbb{R}^n);\ u = \sum_{|\beta| \leq k} D^\beta u_\beta,\ u_\beta \in L^2(\mathbb{R}^n) \Big\},$$

and again $u \in H^{-k}(\mathbb{R}^n)$ if and only if $\langle D \rangle^{-k} u \in L^2$. Indeed, if $u = \sum D^\beta u_\beta$ then $\langle \xi \rangle^{-k} \hat{u} = \sum \xi^\beta \langle \xi \rangle^{-k} \hat{u}_\beta \in L^2$ since $|\xi^\beta| \leq \langle \xi \rangle^k$. Conversely, if $\langle \xi \rangle^{-k} \hat{u} \in L^2$, (2.8.1) gives $u = \sum_{|\beta| \leq k} D^\beta u_\beta$ with $\hat{u}_\beta = c_\beta \xi^\beta \langle \xi \rangle^{-2k} \hat{u}$, and $|\hat{u}_\beta| \leq c_\beta \langle \xi \rangle^{-k} |\hat{u}|$, so $u_\beta \in L^2$. In view of this we define:

Definition 2.8.1. For $s \in \mathbb{R}$, the Sobolev space of order s is

$$H^s(\mathbb{R}^n) = \{ u \in \mathcal{S}'(\mathbb{R}^n);\ \langle D \rangle^s u \in L^2(\mathbb{R}^n) \}, \qquad \|u\|_{H^s}^2 = \|\langle D \rangle^s u\|_{L^2}^2 = \int \langle \xi \rangle^{2s} |\hat{u}(\xi)|^2 đ\xi.$$

For $q \in \mathbb{R}$, the weighted Sobolev space is $\langle x \rangle^q H^s(\mathbb{R}^n) = \{ u \in \mathcal{S}'(\mathbb{R}^n);\ \langle x \rangle^{-q} u \in H^s(\mathbb{R}^n) \}$, with norm $\|u\|_{\langle x \rangle^q H^s} = \|\langle x \rangle^{-q} u\|_{H^s}$.

$H^s(\mathbb{R}^n)$ is a Hilbert space, isometric via $\langle D \rangle^s$ to $L^2(\mathbb{R}^n)$, and $H^{s'}(\mathbb{R}^n) \subset H^s(\mathbb{R}^n)$ for $s' \geq s$. Unfortunately these spaces get *smaller* as s increases, whereas $\Psi^m(\mathbb{R}^n)$ gets bigger with m. We set

$$H^\infty(\mathbb{R}^n) = \bigcap_s H^s(\mathbb{R}^n), \qquad H^{-\infty}(\mathbb{R}^n) = \bigcup_s H^s(\mathbb{R}^n).$$

The Sobolev scale measures smoothness in L^2 terms: $u \in H^s$ means roughly that "s derivatives of u are square integrable". Then $\mathcal{S} \subsetneq H^\infty \subsetneq H^{-\infty} \subsetneq \mathcal{S}'$: for instance

$\langle x\rangle^{-n} \in H^\infty \setminus \mathcal{S}$ (all its derivatives are $O(\langle x\rangle^{-n})$, hence square integrable), $\delta_0 \in H^{-\infty} \setminus H^\infty$ (as $\hat\delta_0 = 1$), and $1 \notin H^{-\infty}$ (as $\hat 1 = (2\pi)^n\delta_0$). In particular $H^{-\infty}$ does not capture all tempered distributions; the weighted spaces remedy this (Lemma 2.8.3). Note also the Sobolev embedding: if $s > n/2$ then $H^s(\mathbb{R}^n)$ is contained in the bounded continuous functions, since $\int|\hat u|\,d\!\!\bar{}\,\xi \le \big(\int\langle\xi\rangle^{-2s}d\!\!\bar{}\,\xi\big)^{1/2}\|u\|_{H^s}$.

Theorem 2.8.2. *For all $q, s, M \in \mathbb{R}$, each $A \in \Psi^M(\mathbb{R}^n)$ defines a continuous linear map*

$$A : \langle x\rangle^q H^s(\mathbb{R}^n) \longrightarrow \langle x\rangle^q H^{s-M}(\mathbb{R}^n).$$

Proof. First let $q = 0$. The operator $\langle D\rangle^{s'}$ is an isometric isomorphism $H^t \to H^{t-s'}$ for all t, with inverse $\langle D\rangle^{-s'}$. By the composition theorem $B = \langle D\rangle^{s-M}A\langle D\rangle^{-s} \in \Psi^0(\mathbb{R}^n)$, which is bounded on L^2 by Theorem 2.7.1. For $u \in H^s$, $v = \langle D\rangle^s u \in L^2$ and $Au = \langle D\rangle^{M-s}Bv$ (as operators on $\mathcal{S}'$), so $\|Au\|_{H^{s-M}} = \|Bv\|_{L^2} \le \|B\|\|u\|_{H^s}$.

$$\begin{array}{ccc} H^s(\mathbb{R}^n) & \xrightarrow{\ A\ } & H^{s-M}(\mathbb{R}^n) \\ {\scriptstyle\langle D\rangle^s}\big\downarrow & & \big\downarrow{\scriptstyle\langle D\rangle^{s-M}} \\ L^2(\mathbb{R}^n) & \xrightarrow{\ B\ } & L^2(\mathbb{R}^n) \end{array}$$

For general q, multiplication by $\langle x\rangle^q$ is by definition an isometry of H^s onto $\langle x\rangle^q H^s$, so it suffices to show that $Q = \langle x\rangle^{-q}A\langle x\rangle^q \in \Psi^M(\mathbb{R}^n)$. Write $A = \mathrm{Op}(a)$ with $a \in S^M(\mathbb{R}^n;\mathbb{R}^n)$. Then the kernel of Q is $fI(a)$ with $f(x,y) = \langle x\rangle^{-q}\langle y\rangle^q$. Now $D_x^\alpha D_y^\beta f = (D^\alpha\langle x\rangle^{-q})(D^\beta\langle y\rangle^q)$ is bounded by $C\langle x\rangle^{-q}\langle y\rangle^q \le C2^{|q|/2}\langle x-y\rangle^{|q|}$, by Lemma 2.1.4(i) and Peetre's inequality; so f has weight $|q|$, and by Lemma 2.2.4, $fI(a) = I(fa)$ with $fa \in \mathcal{A}^M_{|q|}(\mathbb{R}^n)$. Hence $Q = \mathrm{Op}(fa) \in \Psi^M(\mathbb{R}^n)$. □

This is a typical instance where amplitudes of non-zero weight are useful; the left symbol of Q is given by Theorem 2.3.1.

Question. On a compact manifold an operator of negative order is compact on L^2. Is every $A \in \Psi^{-1}(\mathbb{R}^n)$, or even every $A \in \Psi^{-\infty}(\mathbb{R}^n)$, compact on $L^2(\mathbb{R}^n)$?

Answer. No. Let $\chi \in \mathcal{S}(\mathbb{R}^n)$ be non-zero and $A = \mathrm{Op}(\chi(\xi)) \in \Psi^{-\infty}(\mathbb{R}^n)$, the Fourier multiplier by χ. It commutes with translations $\tau_c u(x) = u(x-c)$. Pick $u \in L^2$ with $Au \ne 0$. As $|c| \to \infty$, $\tau_c u \to 0$ weakly in L^2, but $\|A\tau_c u\| = \|Au\|$ does not tend to zero, so A is not compact. Order measures decay in ξ only; to get compactness one also needs decay in x, as in Exercise 2.16. □

Lemma 2.8.3. $\mathcal{S}'(\mathbb{R}^n) = \bigcup_{M\in\mathbb{N}_0}\langle x\rangle^M H^{-M}(\mathbb{R}^n)$.

Proof. Let $u \in \mathcal{S}'(\mathbb{R}^n)$. There are C and k with $|u(\phi)| \le C\max_{|\alpha|,|\beta|\le k}\sup|x^\alpha D^\beta\phi|$. For $|\alpha|, |\beta| \le k$ the operator $x^\alpha D^\beta\langle x\rangle^{-k}$ is a differential operator of order $\le k$ with coefficients in $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ (by Leibniz' formula, as $x^\alpha D^\gamma\langle x\rangle^{-k}$ is bounded with all derivatives for $|\alpha| \le k$), so it maps H^{n+k} to H^n boundedly by Theorem 2.8.2. With the Sobolev embedding this gives $\sup|x^\alpha D^\beta\phi| \le C\|\langle x\rangle^k\phi\|_{H^N}$, $N = n+k$. Hence $\psi \mapsto u(\langle x\rangle^{-k}\langle D\rangle^{-N}\psi)$, $\psi \in \mathcal{S}(\mathbb{R}^n)$, is bounded in the L^2 norm, and by the Riesz representation theorem there is $h \in L^2(\mathbb{R}^n)$ with $u(\langle x\rangle^{-k}\langle D\rangle^{-N}\psi) = \int h\psi\,dx$. Since $\langle x\rangle^{-k}\langle D\rangle^{-N}$ is an isomorphism of $\mathcal{S}(\mathbb{R}^n)$ with inverse $\langle D\rangle^N\langle x\rangle^k$, and $\langle D\rangle^N$ is its own transpose,

$$u(\phi) = \int h\,\langle D\rangle^N(\langle x\rangle^k\phi)\,dx = (\langle x\rangle^k\langle D\rangle^N h)(\phi),$$

i.e. $u = \langle x\rangle^k\langle D\rangle^N h \in \langle x\rangle^k H^{-N} \subset \langle x\rangle^M H^{-M}$ with $M = N$. Here we used that $\langle x\rangle^k H^{-N} \subset \langle x\rangle^M H^{-M}$ for $k \le M$, $N \le M$, which holds because multiplication by $\langle x\rangle^{k-M} \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ preserves H^{-N} by Theorem 2.8.2. □

Alternatively this follows from the structure theorem for tempered distributions: every $u \in \mathcal{S}'(\mathbb{R}^n)$ is a finite sum $\sum x^\alpha D^\beta u_{\alpha\beta}$ with $u_{\alpha\beta}$ bounded and continuous, and bounded continuous functions lie in $\langle x\rangle^n L^2(\mathbb{R}^n)$.

The elliptic regularity result can now be refined.

Proposition 2.8.4. *Let $A \in \Psi^m(\mathbb{R}^n)$ be elliptic. If $u \in \langle x\rangle^{p'} H^{s'}(\mathbb{R}^n)$ and $Au \in \langle x\rangle^p H^s(\mathbb{R}^n)$, then $u \in \langle x\rangle^{p''} H^{s+m}(\mathbb{R}^n)$ with $p'' = \max(p, p')$. In particular, if $Au \in \langle x\rangle^p H^\infty$ then $u \in \langle x\rangle^{p''} H^\infty$.*

Proof. Let $B \in \Psi^{-m}$ be a parametrix, $BA = \mathrm{Id} + G$ with $G \in \Psi^{-\infty}$. Then by Theorem 2.8.2

$$u = B(Au) - Gu \in \langle x\rangle^p H^{s+m}(\mathbb{R}^n) + \langle x\rangle^{p'} H^\infty(\mathbb{R}^n) \subset \langle x\rangle^{p''} H^{s+m}(\mathbb{R}^n),$$

using that $\langle x\rangle^p H^t \subset \langle x\rangle^{p''} H^t$ for $p \le p''$ (as in the proof of Lemma 2.8.3). □

2.9 Polyhomogeneous operators

So far we have considered operators whose amplitudes satisfy the symbol estimates. Some weakenings of these estimates are indicated in Exercise 2.13. Here instead we consider a restriction, to symbols with asymptotic expansions in homogeneous functions; this is the most commonly used class of pseudodifferential operators.

A function $a \in \mathcal{C}^\infty(\mathbb{R}^p \times \mathbb{R}^n)$ is *homogeneous of degree $\mu \in \mathbb{R}$ in $|\xi| \ge 1$* if

$$a(z, t\xi) = t^\mu a(z, \xi) \quad \text{for all } z,\ t \ge 1,\ |\xi| \ge 1.$$

If such a function also satisfies the uniform regularity estimates

$$\sup_{z\in\mathbb{R}^p,\ |\xi|\le 2} |D_z^\alpha D_\xi^\beta a(z,\xi)| < \infty \quad \text{for all } \alpha, \beta, \tag{2.9.1}$$

then $a \in S^\mu(\mathbb{R}^p; \mathbb{R}^n)$. Indeed, differentiating the homogeneity relation shows that $D_z^\alpha D_\xi^\beta a$ is homogeneous of degree $\mu - |\beta|$ in $|\xi| > 1$, hence by continuity in $|\xi| \ge 1$, so for $|\xi| \ge 1$

$$|D_z^\alpha D_\xi^\beta a(z,\xi)| = |\xi|^{\mu-|\beta|} |D_z^\alpha D_\xi^\beta a(z, \xi/|\xi|)| \le C\langle\xi\rangle^{\mu-|\beta|},$$

while for $|\xi| \le 1$ the estimates are given by (2.9.1). Conversely (2.9.1) holds for any symbol. A function homogeneous of degree μ in $|\xi| \ge 1$ which lies in $S^{\mu-1}$ vanishes in $|\xi| \ge 1$, since $|a(z,\xi)| = t^{-\mu}|a(z,t\xi)| \le Ct^{-1}\langle\xi\rangle^{\mu-1}$ for all $t \ge 1$.

Definition 2.9.1. For $m \in \mathbb{R}$, the space $S^m_{\mathrm{ph}}(\mathbb{R}^p; \mathbb{R}^n) \subset S^m(\mathbb{R}^p; \mathbb{R}^n)$ of (*one-step*) *polyhomogeneous* or *classical* symbols consists of those a for which there exist $a_{m-j} \in S^{m-j}(\mathbb{R}^p; \mathbb{R}^n)$, homogeneous of degree $m - j$ in $|\xi| \ge 1$, $j \in \mathbb{N}_0$, such that

$$a \sim \sum_{j=0}^\infty a_{m-j}.$$

By the remark preceding the definition, the terms a_{m-j} are uniquely determined by a in $|\xi| \ge 1$ (if $\sum_{j<N}(a_{m-j} - a'_{m-j}) \in S^{m-N}$ then inductively each difference is homogeneous and of lower order, so vanishes in $|\xi| \ge 1$). Also $S^{-\infty} \subset S^m_{\mathrm{ph}}$ for every m (take all $a_{m-j} = 0$), and $S^m_{\mathrm{ph}} \cap S^{m-1} = S^{m-1}_{\mathrm{ph}}$: if $a \in S^m_{\mathrm{ph}} \cap S^{m-1}$ then $a_m \in S^{m-1}$ vanishes in $|\xi| \ge 1$, so $a_m \in S^{-\infty}$ and $a \sim a'_{m-1} + \sum_{j\ge 2} a_{m-j}$ with $a'_{m-1} = a_{m-1} + a_m$, which is homogeneous of degree $m - 1$ in $|\xi| \ge 1$. The choice of 1 in "$|\xi| \ge 1$" is immaterial: if h is homogeneous of degree μ in $|\xi| \ge R$, then $h'(z,\xi) = \rho(\xi)|\xi|^\mu R^{-\mu} h(z, R\xi/|\xi|)$, with $\rho \in \mathcal{C}^\infty$ vanishing near 0 and equal to 1 in $|\xi| \ge 1$, is homogeneous of degree μ in $|\xi| \ge 1$ and $h - h'$ has compact support in ξ, uniformly in z, so is residual.

Lemma 2.9.2. (i) $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)\cdot S^{m'}_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)\subset S^{m+m'}_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$; *$D_{z_j}$ preserves S^m_{ph}, D_{ξ_j} maps S^m_{ph} into S^{m-1}_{ph}, and complex conjugation and restriction to linear subspaces of $\mathbb{R}^p$ preserve polyhomogeneity.*

(ii) *If $b_j\in S^{m-j}_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ and $b\sim\sum_j b_j$ then $b\in S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$.*

(iii) *Polynomials in ξ with $\mathcal{C}^\infty_{\mathrm{b}}$ coefficients are polyhomogeneous, and $\langle\xi\rangle^s\in S^s_{\mathrm{ph}}(\mathbb{R}^n)$ for all $s\in\mathbb{R}$; multiplication by $\langle\xi\rangle^s$ is an isomorphism $S^m_{\mathrm{ph}}\to S^{m+s}_{\mathrm{ph}}$.*

Proof. (i) Homogeneity is preserved by products, D_z, conjugation and restriction, and D_ξ lowers the degree of homogeneity by one (in $|\xi|>1$, hence by continuity in $|\xi|\geq 1$). The asymptotic expansions are then given by Lemma 2.1.11(i),(ii).

(ii) Write $b_j\sim\sum_k b_{j,k}$ with $b_{j,k}$ homogeneous of degree $m-j-k$. Then $c_l=\sum_{j+k=l}b_{j,k}$ is homogeneous of degree $m-l$ in $|\xi|\geq 1$, and $b\sim\sum_l c_l$ by Lemma 2.1.11(iii).

(iii) A polynomial $\sum_{|\alpha|\leq k}p_\alpha(z)\xi^\alpha$ is the finite sum of its homogeneous parts. For $\langle\xi\rangle^s$, let ρ be as above. For $|\xi|\geq 1$, $\langle\xi\rangle^s=|\xi|^s(1+|\xi|^{-2})^{s/2}$, and by the binomial series, for $K\in\mathbb{N}$,

$$\langle\xi\rangle^s-\sum_{k<K}\binom{s/2}{k}\rho(\xi)|\xi|^{s-2k}=|\xi|^{s-2K}h_K(|\xi|^{-2})\quad\text{in }|\xi|\geq 2,$$

where h_K is real-analytic on $[0,\frac14]$. Since $|\xi|^{-2}$ is homogeneous of degree -2 in $|\xi|\geq 1$, hence a symbol of order -2 there, the chain rule shows that $h_K(|\xi|^{-2})$ satisfies symbol estimates of order 0 in $|\xi|\geq 2$; so the left side is in S^{s-2K}. Thus $\langle\xi\rangle^s\sim\sum_k\binom{s/2}{k}\rho(\xi)|\xi|^{s-2k}$. The last statement follows from (i), with inverse multiplication by $\langle\xi\rangle^{-s}$. □

Definition 2.9.3. For $m\in\mathbb{R}$, $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ is the space of operators $\mathrm{Op}(a)$ with $a=\langle x-y\rangle^w\tilde a$, $\tilde a\in S^m_{\mathrm{ph}}(\mathbb{R}^{2n};\mathbb{R}^n)$, for some $w\in\mathbb{R}$.

Proposition 2.9.4. *An operator $A\in\Psi^m(\mathbb{R}^n)$ lies in $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ if and only if $\sigma_L(A)\in S^m_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$. The spaces $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ form an order-filtered $*$-subalgebra of $\Psi^\infty(\mathbb{R}^n)$ containing $\Psi^{-\infty}(\mathbb{R}^n)$ and $\mathrm{Diff}^m_{\mathrm{b}}(\mathbb{R}^n)$, and for every $A\in\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ also $\sigma_R(A)\in S^m_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$.*

Proof. If $\sigma_L(A)\in S^m_{\mathrm{ph}}$ then $A=\mathrm{Op}(\sigma_L(A))$ is of the required form with $w=0$. Conversely let $A=\mathrm{Op}(a)$, $a=\langle x-y\rangle^w\tilde a$. By Leibniz' formula

$$D_y^\alpha a(x,y,\xi)\big|_{y=x}=\sum_{\gamma\leq\alpha}\binom{\alpha}{\gamma}c_\gamma\,D_y^{\alpha-\gamma}\tilde a(x,y,\xi)\big|_{y=x},\qquad c_\gamma=D_y^\gamma\langle x-y\rangle^w\big|_{y=x}\in\mathbb{C},$$

so by Lemma 2.9.2(i) each term in (2.3.1) is polyhomogeneous, of order $m-|\alpha|$, and $\sigma_L(A)\in S^m_{\mathrm{ph}}$ by Lemma 2.9.2(ii). The formulas (2.4.1), (2.4.2) and (2.4.3), together with Lemma 2.9.2, show that A^*, AB and $\sigma_R(A)$ are polyhomogeneous when A and B are. The remaining statements follow from $S^{-\infty}\subset S^m_{\mathrm{ph}}$ and Lemma 2.9.2(iii). □

This subalgebra is often denoted simply $\Psi^m(\mathbb{R}^n)$, its elements being called *classical* pseudodifferential operators. For $A\in\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ the principal symbol has a preferred representative, the leading term a_m in the expansion of $\sigma_L(A)$, and it is natural to identify $\sigma_m(A)$ with the function a_m, homogeneous of degree m on $|\xi|\geq 1$ (or by extension on $\mathbb{R}^n\times(\mathbb{R}^n\setminus 0)$). Conversely, if $A\in\Psi^m(\mathbb{R}^n)$ and $\sigma_m(A)$ has a representative a_m homogeneous of degree m in $|\xi|\geq 1$, then $A-\mathrm{Op}(a_m)\in\Psi^{m-1}(\mathbb{R}^n)$; that is,

$$\sigma_m(A)\text{ has a homogeneous representative }\iff A\in\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)+\Psi^{m-1}(\mathbb{R}^n).$$

It is usually the case that a construction in $\Psi^\infty(\mathbb{R}^n)$ applied to elements of $\Psi^\infty_{\mathrm{ph}}(\mathbb{R}^n)$ yields elements of $\Psi^\infty_{\mathrm{ph}}(\mathbb{R}^n)$. As an illustration:

Proposition 2.9.5. *If $A \in \Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ is elliptic, then every parametrix of A lies in $\Psi^{-m}_{\mathrm{ph}}(\mathbb{R}^n)$.*

Proof. Since parametrices are unique modulo $\Psi^{-\infty}(\mathbb{R}^n) \subset \Psi^{-m}_{\mathrm{ph}}(\mathbb{R}^n)$, it suffices to construct one in Ψ^{-m}_{ph}. Let a_m be the leading term of $\sigma_L(A)$. By Theorem 2.6.3(d), $|\sigma_L(A)| \geq \epsilon\langle\xi\rangle^m$ for $|\xi| \geq R$, hence $|a_m| \geq \frac{\epsilon}{2}\langle\xi\rangle^m$ for $|\xi| \geq R' \geq 1$; by homogeneity $|a_m(x,\theta)| = R'^{-m}|a_m(x,R'\theta)| \geq \frac{\epsilon}{2}(\langle R'\rangle/R')^m \geq c > 0$ for $|\theta| = 1$, so $|a_m| \geq c|\xi|^m$ in $|\xi| \geq 1$. As $\nabla_\xi a_m$ is bounded on $|\xi| \leq 2$, uniformly in x, there is $\delta \in (0, \frac{1}{2})$ with $|a_m| \geq \frac{c}{2}$ on $|\xi| \geq 1-\delta$. Choose $\rho \in \mathcal{C}^\infty(\mathbb{R}^n)$ vanishing in $|\xi| \leq 1-\delta$ and equal to 1 in $|\xi| \geq 1$, and set $b = \rho/a_m$. By Lemma 2.1.6 (applied on $\{|\xi| > 1-\delta\}$) $b \in S^{-m}$, and b is homogeneous of degree $-m$ in $|\xi| \geq 1$, so $b \in S^{-m}_{\mathrm{ph}}$. Let $B_0 = \mathrm{Op}(b) \in \Psi^{-m}_{\mathrm{ph}}$. Then $E = AB_0 - \mathrm{Id} \in \Psi^0_{\mathrm{ph}}$ and $\sigma_L(E) \equiv a_m b - 1 = \rho - 1 \equiv 0$ modulo S^{-1}, so $\sigma_L(E) \in S^0_{\mathrm{ph}} \cap S^{-1} = S^{-1}_{\mathrm{ph}}$. The operator F of Lemma 2.6.2 has $\sigma_L(F) \sim \sum_{j\geq 1} \sigma_L((-E)^j)$, which is polyhomogeneous by Lemma 2.9.2(ii) and Proposition 2.9.4. Hence $B = B_0(\mathrm{Id}+F) \in \Psi^{-m}_{\mathrm{ph}}(\mathbb{R}^n)$ is a parametrix. □

2.9.1 Complex orders

For holomorphic families of operators (Section 2.11) and, later, for complex powers of elliptic operators, it is necessary to allow polyhomogeneous symbols of *complex* order. For $m \in \mathbb{C}$ and $t > 0$ we always take $t^m = e^{m \log t}$, so that $|t^m| = t^{\Re m}$. A function $a \in \mathcal{C}^\infty(\mathbb{R}^p \times \mathbb{R}^n)$ is homogeneous of degree $m \in \mathbb{C}$ in $|\xi| \geq 1$ if $a(z, t\xi) = t^m a(z,\xi)$ for all z, $t \geq 1$ and $|\xi| \geq 1$. The discussion at the beginning of this section goes through unchanged, with the real part of the degree in all estimates: such a function satisfying (2.9.1) lies in $S^{\Re m}(\mathbb{R}^p;\mathbb{R}^n)$, and if it lies in $S^{\Re m-1}(\mathbb{R}^p;\mathbb{R}^n)$ it vanishes in $|\xi| \geq 1$.

Definition 2.9.6. For $m \in \mathbb{C}$, $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ is the space of $a \in S^{\Re m}(\mathbb{R}^p;\mathbb{R}^n)$ for which there exist $a_{m-j} \in S^{\Re m-j}(\mathbb{R}^p;\mathbb{R}^n)$, homogeneous of degree $m-j$ in $|\xi| \geq 1$, $j \in \mathbb{N}_0$, with $a \sim \sum_j a_{m-j}$ (in the sense of Definition 2.1.9, with orders $\Re m - j$). The space $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$, $m \in \mathbb{C}$, is defined as in Definition 2.9.3.

For real m this is Definition 2.9.1, and the terms a_{m-j} are again unique in $|\xi| \geq 1$.

Lemma 2.9.7. *Let $m, m' \in \mathbb{C}$.*

(i) *$S^m_{\mathrm{ph}} \cdot S^{m'}_{\mathrm{ph}} \subset S^{m+m'}_{\mathrm{ph}}$; D_{z_j} preserves S^m_{ph}, D_{ξ_j} maps S^m_{ph} into S^{m-1}_{ph}, complex conjugation maps S^m_{ph} onto $S^{\bar m}_{\mathrm{ph}}$, and asymptotic sums of series $\sum_j b_j$ with $b_j \in S^{m-j}_{\mathrm{ph}}$ lie in S^m_{ph}.*

(ii) *$\langle\xi\rangle^s \in S^s_{\mathrm{ph}}(\mathbb{R}^n)$ for every $s \in \mathbb{C}$, and multiplication by $\langle\xi\rangle^s$ is a bijection $S^m_{\mathrm{ph}} \to S^{m+s}_{\mathrm{ph}}$.*

(iii) *$S^{m-k}_{\mathrm{ph}} \subset S^m_{\mathrm{ph}}$ for $k \in \mathbb{N}_0$, and $S^m_{\mathrm{ph}} \cap S^{\Re m-1} = S^{m-1}_{\mathrm{ph}}$.*

Proof. (i) and (ii) are proved exactly as Lemma 2.9.2; note that $\overline{t^m} = t^{\bar m}$, and that the binomial series $(1+u)^{s/2} = \sum_k \binom{s/2}{k} u^k$ converges, and is real-analytic, for $|u| < 1$ also when s is complex. (iii) If $a \sim \sum_j a_{m-k-j}$, set $a'_{m-j} = 0$ for $j < k$ and $a'_{m-j} = a_{m-j}$ for $j \geq k$; then $a \sim \sum_j a'_{m-j}$. The second statement is proved as in the real case: if $a \in S^m_{\mathrm{ph}} \cap S^{\Re m-1}$ then the leading term a_m lies in $S^{\Re m-1}$, hence vanishes in $|\xi| \geq 1$ and is residual, and a has an expansion starting with $a_{m-1} + a_m$. □

With these facts the proofs of Proposition 2.9.4 and Proposition 2.9.5 apply verbatim to complex orders. Thus $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n) \subset \Psi^{\Re m}(\mathbb{R}^n)$, an operator $A \in \Psi^{\Re m}(\mathbb{R}^n)$ lies in $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ if and only if $\sigma_L(A) \in S^m_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$, and for all $m, m' \in \mathbb{C}$

$$\Psi^m_{\mathrm{ph}}(\mathbb{R}^n) \circ \Psi^{m'}_{\mathrm{ph}}(\mathbb{R}^n) \subset \Psi^{m+m'}_{\mathrm{ph}}(\mathbb{R}^n), \qquad \left(\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)\right)^* = \Psi^{\bar m}_{\mathrm{ph}}(\mathbb{R}^n).$$

The principal symbol $\sigma_m(A)$ of $A \in \Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ is the leading term a_m of $\sigma_L(A)$, a function homogeneous of degree m in $|\xi| \geq 1$; then $\sigma_{m+m'}(AB) = \sigma_m(A)\sigma_{m'}(B)$, $\sigma_{\bar{m}}(A^*) = \overline{\sigma_m(A)}$, and by Lemma 2.9.7(iii) the null space of σ_m on $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ is $\Psi^{m-1}_{\mathrm{ph}}(\mathbb{R}^n)$. If $|\sigma_m(A)| \geq c|\xi|^{\Re m}$ in $|\xi| \geq 1$ for some $c > 0$, A is elliptic as an element of $\Psi^{\Re m}(\mathbb{R}^n)$ and all its parametrices lie in $\Psi^{-m}_{\mathrm{ph}}(\mathbb{R}^n)$. For example $\langle D\rangle^s = \mathrm{Op}(\langle\xi\rangle^s) \in \Psi^s_{\mathrm{ph}}(\mathbb{R}^n)$ is elliptic for every $s \in \mathbb{C}$, with parametrix (indeed inverse) $\langle D\rangle^{-s}$.

The natural topology on the space $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$, which is finer than the one induced from $S^{\Re m}(\mathbb{R}^p;\mathbb{R}^n)$, is discussed in Section 2.10.

2.10 Topologies and continuity of the calculus

The spaces of symbols are Fréchet spaces, and the left symbol identifies $\Psi^m(\mathbb{R}^n)$ with $S^m(\mathbb{R}^n;\mathbb{R}^n)$. We use this to topologize the operator spaces and show that all the operations of the calculus (reduction, adjoint, composition) are continuous. This is what is needed to speak of continuous, smooth or holomorphic dependence of operators on parameters (Section 2.11). Rather than tracking the constants through the proofs of Sections 2.3 and 2.4, we derive continuity from the closed graph theorem.

2.10.1 Two facts from functional analysis

Lemma 2.10.1. *Let E, E_1, E_2 and F be Fréchet spaces and let G be a Hausdorff topological vector space with $F \subset G$, the inclusion being continuous.*

(i) *A linear map $T : E \to F$ which is continuous as a map $E \to G$ is continuous as a map $E \to F$.*

(ii) *A bilinear map $B : E_1 \times E_2 \to F$ which is separately continuous is jointly continuous.*

(iii) *Consequently, a bilinear map $B : E_1 \times E_2 \to F$ which is separately continuous as a map into G is jointly continuous as a map into F.*

Proof. (i) By the closed graph theorem it suffices to show that the graph of T is closed in $E \times F$. If $e_k \to e$ in E and $Te_k \to f$ in F, then $Te_k \to f$ in G, while $Te_k \to Te$ in G by assumption; as G is Hausdorff, $f = Te$.

(ii) The spaces are metrizable, so it suffices to show that $B(x_k, y_k) \to B(x, y)$ when $x_k \to x$ and $y_k \to y$. Write $B(x_k, y_k) - B(x, y) = B(x_k, y_k - y) + B(x_k - x, y)$. The second term tends to 0 by continuity in the first variable. The linear maps $T_k = B(x_k, \cdot) : E_2 \to F$ are continuous, and for each $y' \in E_2$ the sequence $T_k y'$ converges (to $B(x, y')$), so is bounded. By the uniform boundedness principle for the Fréchet space E_2 the family $\{T_k\}$ is equicontinuous, and hence $T_k(y_k - y) \to 0$.

(iii) By (i), applied to $B(x, \cdot)$ and $B(\cdot, y)$, B is separately continuous into F; then apply (ii). □

The same holds for conjugate-linear maps (compose with complex conjugation on the target).

2.10.2 Topologies on symbols and operators

The spaces $S^m(\mathbb{R}^p;\mathbb{R}^n)$, $m \in \mathbb{R}$, and $S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$ are Fréchet spaces (Lemma 2.1.4), and so, by Lemma 2.2.2, are the spaces of amplitudes $\mathcal{A}^m_w(\mathbb{R}^n)$ with the norms $\|\cdot\|_{N,m,w}$ and $\mathcal{A}^{-\infty}_w(\mathbb{R}^n)$ with all these norms. By Corollary 2.3.4, $\sigma_L : \Psi^m(\mathbb{R}^n) \to S^m(\mathbb{R}^n;\mathbb{R}^n)$ is a linear bijection for every $m \in \mathbb{R} \cup \{-\infty\}$.

Definition 2.10.2. For $m \in \mathbb{R} \cup \{-\infty\}$, $\Psi^m(\mathbb{R}^n)$ is given the Fréchet topology for which $\sigma_L : \Psi^m(\mathbb{R}^n) \to S^m(\mathbb{R}^n;\mathbb{R}^n)$ is an isomorphism of topological vector spaces.

The inclusions $\Psi^m(\mathbb{R}^n) \hookrightarrow \Psi^{m'}(\mathbb{R}^n)$, $m \le m'$, are continuous, and a sequence converges in $\Psi^{-\infty}(\mathbb{R}^n)$ if and only if it converges in every $\Psi^m(\mathbb{R}^n)$. By the estimates in the proof of Proposition 2.2.5 (with $w = 0$ and $a = \sigma_L(A)$), the bilinear map $\Psi^m(\mathbb{R}^n) \times \mathcal{S}(\mathbb{R}^n) \ni (A,u) \mapsto Au \in \mathcal{S}(\mathbb{R}^n)$ is continuous.

Proposition 2.10.3. *For $m \in \mathbb{R} \cup \{-\infty\}$ and $w \in \mathbb{R}$, the map*

$$\mathcal{A}^m_w(\mathbb{R}^n) \ni a \longmapsto \sigma_L(\mathrm{Op}(a)) \in S^m(\mathbb{R}^n;\mathbb{R}^n)$$

is continuous; equivalently, $\mathrm{Op} : \mathcal{A}^m_w(\mathbb{R}^n) \to \Psi^m(\mathbb{R}^n)$ *is continuous.*

Proof. The map is linear and well defined by Theorem 2.3.1 and Corollary 2.3.4; we show that its graph is closed. Let $a_k \to a$ in $\mathcal{A}^m_w$ and $c_k = \sigma_L(\mathrm{Op}(a_k)) \to c$ in $S^m(\mathbb{R}^n;\mathbb{R}^n)$ (for $m = -\infty$, take $m = 0$ in what follows). By (2.2.4), $I(a_k) \to I(a)$ weakly in $\mathcal{S}'(\mathbb{R}^{2n})$. Since $S^m(\mathbb{R}^n;\mathbb{R}^n) \subset \mathcal{A}^m_0(\mathbb{R}^n)$ continuously, $c_k(x,\xi) \to c(x,\xi)$ in $\mathcal{A}^m_0(\mathbb{R}^n)$ and so $I(c_k) \to I(c)$ weakly. But $I(c_k) = I(a_k)$ is the kernel of $\mathrm{Op}(a_k)$, so $I(c) = I(a)$, i.e. $\mathrm{Op}(c) = \mathrm{Op}(a)$, and $c = \sigma_L(\mathrm{Op}(a))$ by uniqueness of the left symbol. The closed graph theorem gives continuity. □

2.10.3 Polyhomogeneous symbols

As a subspace, $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ is not closed in $S^{\Re m}(\mathbb{R}^p;\mathbb{R}^n)$: it contains $S^{-\infty}$, so by Lemma 2.1.5 its closure contains $S^{m'}(\mathbb{R}^p;\mathbb{R}^n)$ for every $m' < \Re m$ (see also Exercise 2.9). To capture its structure we strengthen the topology by adding norms on the terms of the expansion. Since the terms are only determined in $|\xi| \ge 1$ we normalize them. Fix $\rho \in \mathcal{C}^\infty(\mathbb{R}^n)$ with $\rho(\xi) = 0$ for $|\xi| \le \frac{1}{2}$ and $\rho(\xi) = 1$ for $|\xi| \ge 1$. For $a \in S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ with $a \sim \sum_j a_{m-j}$, the *normalized terms* are

$$a_{(m-j)}(z,\xi) = \rho(\xi)|\xi|^{m-j} a_{m-j}(z,\xi/|\xi|), \qquad j \in \mathbb{N}_0;$$

they coincide with a_{m-j} in $|\xi| \ge 1$, are uniquely determined by a, and $a \sim \sum_j a_{(m-j)}$ since $a_{m-j} - a_{(m-j)}$ is residual.

Definition 2.10.4. The topology of $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$, $m \in \mathbb{C}$, is defined by the norms

$$\|a\|_{N,\Re m}, \qquad \max_{|\alpha|+|\beta|\le N} \sup_{z\in\mathbb{R}^p,\ 1\le|\xi|\le 2} |D_z^\alpha D_\xi^\beta a_{(m-j)}(z,\xi)|, \qquad \Big\| a - \sum_{j=0}^{k} a_{(m-j)} \Big\|_{N,\Re m-k-1},$$

for $j,k,N \in \mathbb{N}_0$. The space $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ is given the topology for which $\sigma_L : \Psi^m_{\mathrm{ph}}(\mathbb{R}^n) \to S^m_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$ is an isomorphism.

Lemma 2.10.5. *Let $m, m' \in \mathbb{C}$.*

(i) *For each j, the map $a \mapsto a_{(m-j)}$ is continuous from $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ to $S^{\Re m-j}(\mathbb{R}^p;\mathbb{R}^n)$.*

(ii) *$S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ is a Fréchet space and the inclusion $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n) \hookrightarrow S^{\Re m}(\mathbb{R}^p;\mathbb{R}^n)$ is continuous.*

(iii) *Multiplication $S^m_{\mathrm{ph}} \times S^{m'}_{\mathrm{ph}} \to S^{m+m'}_{\mathrm{ph}}$ is jointly continuous; D_{z_j}, D_{ξ_j}, complex conjugation, multiplication by $\langle\xi\rangle^s$ and the inclusions $S^{m-k}_{\mathrm{ph}} \hookrightarrow S^m_{\mathrm{ph}}$, $k \in \mathbb{N}_0$, are continuous between the corresponding spaces.*

Proof. (i) Write $h = a_{(m-j)}$ and $G = \{1 \le |\xi| \le 2\}$. For $|\xi| \ge 1$, by homogeneity, $|D_z^\alpha D_\xi^\beta h(z,\xi)| = |\xi|^{\Re m - j - |\beta|} |D_z^\alpha D_\xi^\beta h(z, \xi/|\xi|)|$, and $\xi/|\xi| \in G$. For $\frac{1}{2} \le |\xi| \le 1$ we have $h = \rho \tilde{h}$ with $\tilde{h}(z,\xi) = 2^{-(m-j)} \tilde{h}(z, 2\xi)$, $\tilde{h}$ the homogeneous extension, and $2\xi \in G$; so the derivatives of h there are bounded by constants times those of h on G (Leibniz' formula for the factor ρ). Since $h = 0$ for $|\xi| \le \frac{1}{2}$, $\|h\|_{N,\Re m-j} \le C_N \max_{|\alpha|+|\beta|\le N} \sup_{\mathbb{R}^p \times G} |D_z^\alpha D_\xi^\beta h|$.

(ii) The norms are countable and include those of $S^{\Re m}$, so the topology is metrizable and the inclusion continuous. Let (a_k) be a Cauchy sequence. Then $a_k \to a$ in $S^{\Re m}$. For each j, by (i) and its proof, the normalized terms $a_{k,(m-j)}$ form a Cauchy sequence in $S^{\Re m - j}$, with limit h_j; pointwise limits of functions of the form $\rho(\xi) \times$(homogeneous of degree $m-j$) are of the same form, so h_j is homogeneous of degree $m-j$ in $|\xi| \ge 1$ and normalized. Finally, for each K, $a_k - \sum_{j \le K} a_{k,(m-j)}$ is Cauchy in $S^{\Re m - K - 1}$, so converges there, and its limit is the pointwise limit $a - \sum_{j \le K} h_j$. Hence $a \in S^m_{\mathrm{ph}}$ with normalized terms h_j, and $a_k \to a$ in every norm of Definition 2.10.4.

(iii) Each of these maps is well defined by Lemma 2.9.7, and is continuous as a map into the corresponding space S^μ by Lemma 2.1.4; for multiplication by $\langle \xi \rangle^s$, use Lemma 2.1.4(i),(ii). Now apply Lemma 2.10.1 with $G = S^\mu$. □

2.10.4 Continuity of adjoint and product

Proposition 2.10.6. *For $m, m' \in \mathbb{R} \cup \{-\infty\}$ the adjoint and the product,*

$$\Psi^m(\mathbb{R}^n) \ni A \longmapsto A^* \in \Psi^m(\mathbb{R}^n), \qquad \Psi^m(\mathbb{R}^n) \times \Psi^{m'}(\mathbb{R}^n) \ni (A,B) \longmapsto AB \in \Psi^{m+m'}(\mathbb{R}^n),$$

are continuous (the product jointly). The same is true for the polyhomogeneous spaces: $A \mapsto A^*$ is continuous from $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ to $\Psi^{\bar{m}}_{\mathrm{ph}}(\mathbb{R}^n)$ and the product is jointly continuous from $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n) \times \Psi^{m'}_{\mathrm{ph}}(\mathbb{R}^n)$ to $\Psi^{m+m'}_{\mathrm{ph}}(\mathbb{R}^n)$, for $m, m' \in \mathbb{C}$.

Proof. First let $m, m' \in \mathbb{R}$. By Proposition 2.2.9, $\sigma_L(A^*) = \sigma_L(\mathrm{Op}(\overline{a(y,\xi)}))$ with $a = \sigma_L(A)$. The map $a \mapsto \overline{a(y,\xi)}$ is continuous (and conjugate-linear) from $S^m(\mathbb{R}^n;\mathbb{R}^n)$ to $\mathcal{A}_0^m(\mathbb{R}^n)$, so continuity of the adjoint follows from Proposition 2.10.3. In particular $B \mapsto \sigma_R(B) = \overline{\sigma_L(B^*)}$ is continuous from $\Psi^{m'}$ to $S^{m'}$. By the proof of Theorem 2.4.1,

$$\sigma_L(AB) = \sigma_L\Big(\mathrm{Op}\big(\sigma_L(A)(x,\xi)\, \sigma_R(B)(y,\xi)\big)\Big).$$

The bilinear map $(a,b) \mapsto a(x,\xi)b(y,\xi)$ is continuous from $S^m(\mathbb{R}^n;\mathbb{R}^n) \times S^{m'}(\mathbb{R}^n;\mathbb{R}^n)$ to $\mathcal{A}_0^{m+m'}(\mathbb{R}^n)$, by Leibniz' formula as in Lemma 2.1.4(ii); composing it with the continuous linear maps $A \mapsto \sigma_L(A)$, $B \mapsto \sigma_R(B)$ and $c \mapsto \sigma_L(\mathrm{Op}(c))$ shows that the product is jointly continuous. If $m = -\infty$ (or $m' = -\infty$), the product $\Psi^{-\infty} \times \Psi^{m'} \to \Psi^{-M}$ factors through the continuous inclusion $\Psi^{-\infty} \hookrightarrow \Psi^{-M-m'}$ for every M, so it is continuous into $\Psi^{-\infty}$; similarly for the adjoint.

For the polyhomogeneous spaces, the adjoint and product map into the stated spaces by Section 2.9.1, and they are continuous into $\Psi^{\Re \bar{m}}(\mathbb{R}^n)$, respectively $\Psi^{\Re(m+m')}(\mathbb{R}^n)$, by the first part, since the inclusions $S^\mu_{\mathrm{ph}} \hookrightarrow S^{\Re\mu}$ are continuous. By Lemma 2.10.5(ii) the spaces involved are Fréchet, so Lemma 2.10.1 (with $G = S^{\Re\bar{m}}$, respectively $G = S^{\Re(m+m')}$) gives the continuity for the polyhomogeneous topologies. □

Corollary 2.10.7. *For $m, m' \in \mathbb{R}$ and $N \in \mathbb{N}_0$, the remainder in the composition formula,*

$$r_N(a,b) = \sigma_L(\mathrm{Op}(a)\,\mathrm{Op}(b)) - \sum_{|\alpha|<N} \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha a\, D_x^\alpha b,$$

is a continuous bilinear map $S^m(\mathbb{R}^n;\mathbb{R}^n) \times S^{m'}(\mathbb{R}^n;\mathbb{R}^n) \to S^{m+m'-N}(\mathbb{R}^n;\mathbb{R}^n)$.

Proof. By Theorem 2.4.1, r_N takes values in $S^{m+m'-N}$, and by Proposition 2.10.6 and Lemma 2.1.4 it is continuous into $S^{m+m'}$. Apply Lemma 2.10.1(iii) with $G = S^{m+m'}$. □

The following consequence of the off-diagonal smoothness of the kernels will be used repeatedly when operators are localized.

Lemma 2.10.8. *Let $\phi_1 \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ and $\phi_2 \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$, with $\phi_2 = 1$ on a neighbourhood of $\operatorname{supp}\phi_1$. Then for every $A \in \Psi^m(\mathbb{R}^n)$, $\phi_1 A(1-\phi_2) \in \Psi^{-\infty}(\mathbb{R}^n)$ and $(1-\phi_2)A\phi_1 \in \Psi^{-\infty}(\mathbb{R}^n)$, and the maps*

$$\Psi^m(\mathbb{R}^n) \ni A \longmapsto \phi_1 A(1-\phi_2),\ (1-\phi_2)A\phi_1 \in \Psi^{-\infty}(\mathbb{R}^n)$$

are continuous.

Figure 2.4 shows the geometry: the cut-off kernel lives in a region a fixed distance away from the diagonal, where the kernel of A is smooth and rapidly decreasing.

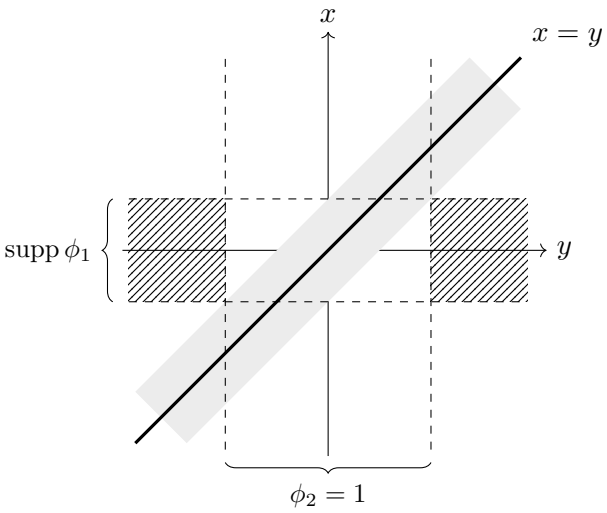


Figure 2.4: The kernel of $\phi_1 A(1-\phi_2)$ is supported in the hatched region $\operatorname{supp}\phi_1 \times \operatorname{supp}(1-\phi_2)$, which stays at distance at least $\delta > 0$ from the diagonal. The kernel of A is singular only on the diagonal and is smooth and rapidly decreasing in the rest of the plane; the shading indicates a neighbourhood of the diagonal.

Proof. Since $\operatorname{supp}\phi_1$ is compact and $\operatorname{supp}(1-\phi_2)$ is a closed set disjoint from it, $\delta = \operatorname{dist}(\operatorname{supp}\phi_1, \operatorname{supp}(1-\phi_2)) > 0$. Let $a = \sigma_L(A)$ and $f(x,y) = \phi_1(x)(1-\phi_2(y))$, which has weight 0 in the sense of Lemma 2.2.2. Thus $A' = \phi_1 A(1-\phi_2) = \operatorname{Op}(fa) \in \Psi^m(\mathbb{R}^n)$, with kernel $K' = fK$, $K = I(a)$, and f vanishes unless $|x-y| \geq \delta$.

By Lemma 2.2.4, as in the proof of Proposition 2.3.3, for all α, β, γ,

$$(x-y)^\gamma D_x^\alpha D_y^\beta K = I(b_{\alpha\beta\gamma}), \qquad b_{\alpha\beta\gamma} = (-D_\xi)^\gamma(\xi + D_x)^\alpha(D_y - \xi)^\beta a \in \mathcal{A}_0^{m+|\alpha|+|\beta|-|\gamma|}(\mathbb{R}^n),$$

which is a bounded continuous function if $m + |\alpha| + |\beta| - |\gamma| < -n$, by Proposition 2.2.3(i). Let $W = \{|x-y| > \delta/2\}$. Since $|x-y|^{2L} = \sum_{|\gamma|=L} \frac{L!}{\gamma!}(x-y)^{2\gamma}$, for $2L > m + n + |\alpha| + |\beta|$ we get, in W,

$$D_x^\alpha D_y^\beta K = |x-y|^{-2L} \sum_{|\gamma|=L} \frac{L!}{\gamma!} I(b_{\alpha\beta(2\gamma)}),$$

a continuous function bounded by $C|x-y|^{-2L}$. Thus all distributional derivatives of K in W are continuous, so K is smooth in W, which contains a neighbourhood of $\operatorname{supp} f$. Hence $K' = fK$ is smooth, and on $\operatorname{supp} f$, where $\langle x-y\rangle^{2L} \le C_\delta |x-y|^{2L}$,

$$|D_x^\alpha D_y^\beta K'(x,y)| \le C_{\alpha\beta L}\langle x-y\rangle^{-2L} \quad \text{for all } L,$$

by Leibniz' formula, the derivatives of f being bounded. By Proposition 2.3.3, $A' \in \Psi^{-\infty}(\mathbb{R}^n)$.

The map $A \mapsto A'$ is linear, and continuous into $\Psi^m(\mathbb{R}^n)$ by Proposition 2.10.3, since $a \mapsto fa$ is continuous from $S^m(\mathbb{R}^n;\mathbb{R}^n)$ to $\mathcal{A}_0^m(\mathbb{R}^n)$. By Lemma 2.10.1(i), with $G = S^m(\mathbb{R}^n;\mathbb{R}^n)$, it is continuous into $\Psi^{-\infty}(\mathbb{R}^n)$. The case of $(1-\phi_2)A\phi_1$ is the same with $f(x,y) = (1-\phi_2(x))\phi_1(y)$ (or follows by taking adjoints). □

The hypothesis is that the supports of ϕ_1 and $1-\phi_2$ are disjoint; it would not suffice to assume $\phi_1 = 1$ on $\operatorname{supp}\phi_2$: for $A = \mathrm{Id}$, $\phi_1 A(1-\phi_2)$ is multiplication by $\phi_1(1-\phi_2)$, which is residual only if $\phi_1(1-\phi_2) = 0$.

2.11 Smooth and holomorphic families

With the topologies of Section 2.10 the notions of continuous, smooth and holomorphic dependence of operators on parameters are defined. For operators of fixed order there is nothing new: these are just the corresponding properties of the left symbol, and products of such families have the same regularity. More interesting is that, for polyhomogeneous operators, the *order* itself may vary holomorphically with a complex parameter; families such as $\langle D\rangle^s$, $s \in \mathbb{C}$, and the complex powers of elliptic operators ([11]; see Chapter 8) are of this type.

2.11.1 Fréchet-valued functions

Let E be a Fréchet space. For $U \subset \mathbb{R}^k$ open (or an interval such as $[0,1]$, with one-sided difference quotients at the end points), a map $u : U \to E$ is $\mathcal{C}^1$ if the partial derivatives $\partial_{t_l} u(t) = \lim_{h\to 0} h^{-1}(u(t+he_l) - u(t))$ exist in E and are continuous maps $U \to E$; inductively u is $\mathcal{C}^{k+1}$ if it is $\mathcal{C}^1$ with all $\partial_{t_l} u \in \mathcal{C}^k$, and $\mathcal{C}^\infty(U;E)$ is the intersection of these spaces. For $U \subset \mathbb{C}$ open, $u : U \to E$ is *holomorphic* if $u'(s) = \lim_{h\to 0} h^{-1}(u(s+h) - u(s))$ exists in E for every $s \in U$, the limit being over complex $h \neq 0$. A differentiable map is continuous: $u(t+h) - u(t) = h\,(h^{-1}(u(t+h) - u(t))) \to 0$, as a convergent sequence is bounded.

Lemma 2.11.1. *Let $B : E_1 \times E_2 \to F$ be a continuous bilinear map of Fréchet spaces and let $u_i : U \to E_i$ be $\mathcal{C}^k$, $k \le \infty$ (respectively holomorphic). Then $B(u_1,u_2)$ is $\mathcal{C}^k$ (respectively holomorphic), with $\partial_{t_l} B(u_1,u_2) = B(\partial_{t_l}u_1, u_2) + B(u_1, \partial_{t_l}u_2)$. The same holds for continuous linear maps $T : E \to F$ and Tu.*

Proof. With $Q_h u = h^{-1}(u(t+he) - u(t))$, $Q_h B(u_1,u_2) = B(Q_h u_1, u_2(t+he)) + B(u_1(t), Q_h u_2)$. As $h \to 0$, $Q_h u_i$ converges and $u_2(t+he) \to u_2(t)$, so by joint continuity of B the right side converges to $B(\partial u_1, u_2) + B(u_1, \partial u_2)$. This expression is continuous in t, and the higher derivatives follow by induction. The holomorphic case is the same with complex h. □

Lemma 2.11.2. *Let $U \subset \mathbb{R}^k$ be open (or an interval), $m \in \mathbb{R}$, and let $a \in \mathcal{C}^\infty(U \times \mathbb{R}^p \times \mathbb{R}^n)$ satisfy, for every compact $K \subset U$ and all γ, α, β,*

$$\sup_{t\in K} |D_t^\gamma D_z^\alpha D_\xi^\beta a(t,z,\xi)| \le C_{K,\gamma\alpha\beta}\langle\xi\rangle^{m-|\beta|}. \tag{2.11.1}$$

Then $t \mapsto a(t,\cdot)$ lies in $\mathcal{C}^\infty(U; S^m(\mathbb{R}^p;\mathbb{R}^n))$, its derivatives being the pointwise derivatives $\partial_t^\gamma a$. Conversely, every element of $\mathcal{C}^\infty(U; S^m(\mathbb{R}^p;\mathbb{R}^n))$ is of this form.

Proof. Fix $t \in U$, l, and a compact neighbourhood K of t which is convex (or a convex neighbourhood relative to the interval). For small h, by Taylor's formula in t_l,

$$h^{-1}(a(t+he_l)-a(t)) - \partial_{t_l} a(t) = h\int_0^1 (1-\sigma)\,\partial_{t_l}^2 a(t+\sigma h e_l)\,d\sigma,$$

and applying $D_z^\alpha D_\xi^\beta$ under the integral and using (2.11.1) shows that the S^m-norms of the left side are $O(|h|)$. Similarly $\|\partial_{t_l}a(t+h)-\partial_{t_l}a(t)\|_{N,m} = O(|h|)$, so $t\mapsto a(t)$ is $\mathcal{C}^1$ with derivative $\partial_{t_l}a$. Since $\partial_{t_l}a$ again satisfies (2.11.1), induction gives smoothness.

Conversely, if $t \mapsto a(t) \in S^m$ is $\mathcal{C}^\infty$, then, since evaluation of $D_z^\alpha D_\xi^\beta$ at a point is a continuous linear functional on S^m, all functions $D_z^\alpha D_\xi^\beta \partial_t^\gamma a$ exist and are given by $D_z^\alpha D_\xi^\beta(\partial_t^\gamma a(t))$. They are jointly continuous: for instance $|F(t,z,\xi)-F(t',z,\xi)| \le \|\partial_t^\gamma a(t) - \partial_t^\gamma a(t')\|_{N,m}\langle\xi\rangle^m$ for $F = D_z^\alpha D_\xi^\beta \partial_t^\gamma a$, $N \ge |\alpha|+|\beta|$. Hence $a \in \mathcal{C}^\infty(U\times\mathbb{R}^p\times\mathbb{R}^n)$, and (2.11.1) holds because the continuous image $\{\partial_t^\gamma a(t); t\in K\}$ of a compact set is bounded in S^m. □

Lemma 2.11.3. *Let $U\subset\mathbb{C}$ be open, $m\in\mathbb{R}$, and let $a\in\mathcal{C}^\infty(U\times\mathbb{R}^p\times\mathbb{R}^n)$ be holomorphic in $s\in U$ for each fixed (z,ξ), and satisfy*

$$\sup_{s\in K}|D_z^\alpha D_\xi^\beta a(s,z,\xi)| \le C_{K,\alpha\beta}\langle\xi\rangle^{m-|\beta|}$$

for every compact $K\subset U$ and all α,β. Then $s\mapsto a(s,\cdot)$ is a holomorphic map $U\to S^m(\mathbb{R}^p;\mathbb{R}^n)$. Conversely, if $s\mapsto a(s)\in S^m$ is holomorphic then $a(s,z,\xi)$ is holomorphic in s and satisfies these estimates.

Proof. Since a is smooth, $\bar\partial_s$ commutes with $D_z^\alpha D_\xi^\beta$, so $D_z^\alpha D_\xi^\beta a$ is holomorphic in s. Fix s_0 and $r>0$ with $\{|s-s_0|\le 2r\}\subset U$, and let $|s-s_0|<r$, $0<|h|<r/2$. Cauchy's integral formula over $|t-s_0|=2r$ gives

$$\frac{a(s+h)-a(s)}{h} - \partial_s a(s) = \frac{h}{2\pi i}\oint_{|t-s_0|=2r}\frac{a(t)\,dt}{(t-s-h)(t-s)^2},$$

pointwise in (z,ξ), and the same with $D_z^\alpha D_\xi^\beta a$ in place of a. Since $|t-s-h|\ge r/2$ and $|t-s|\ge r$ on the contour, the $\|\cdot\|_{N,m}$-norm of the left side is at most $8r^{-2}|h|\sup_{|t-s_0|=2r}\|a(t)\|_{N,m}$, and similarly $\partial_s a(s) = \frac{1}{2\pi i}\oint a(t)(t-s)^{-2}dt$ is bounded in S^m. Hence the difference quotients converge in S^m. The converse follows as in Lemma 2.11.2: evaluation at a point is continuous, and continuous images of compact sets are bounded. □

2.11.2 Families of fixed order

Definition 2.11.4. Let $m\in\mathbb{R}$ and U be as above. A map $U\ni t\mapsto A(t)\in\Psi^m(\mathbb{R}^n)$ is a *smooth* (respectively, for $U\subset\mathbb{C}$, *holomorphic*) *family* if $t\mapsto\sigma_L(A(t))$ is a smooth (respectively holomorphic) map into $S^m(\mathbb{R}^n;\mathbb{R}^n)$; for $m\in\mathbb{C}$, a smooth or holomorphic family in $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$ is defined in the same way using $S^m_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$.

By Lemma 2.11.2, $A(t)$ is a smooth family in $\Psi^m(\mathbb{R}^n)$ if and only if its left symbol is a smooth function of (t,x,ξ) satisfying the symbol estimates of order m, together with all its t-derivatives, locally uniformly in t. Similarly Lemma 2.11.3 characterizes holomorphic families.

Proposition 2.11.5. *Let U be as above.*

(i) *If $a \in \mathcal{C}^\infty(U; \mathcal{A}^m_w(\mathbb{R}^n))$ (respectively holomorphic), then $\operatorname{Op}(a(t))$ is a smooth (respectively holomorphic) family in $\Psi^m(\mathbb{R}^n)$.*

(ii) *If $A(t)$ and $B(t)$ are smooth families in $\Psi^m(\mathbb{R}^n)$ and $\Psi^{m'}(\mathbb{R}^n)$, then $A(t)B(t)$ and $A(t)^*$ are smooth families, and $t \mapsto A(t)u$ is smooth from U to $\mathcal{S}(\mathbb{R}^n)$ for each $u \in \mathcal{S}(\mathbb{R}^n)$. If $A(s)$ and $B(s)$ are holomorphic families, so is $A(s)B(s)$. The same holds for polyhomogeneous families of fixed complex orders.*

Proof. (i) follows from Proposition 2.10.3 and Lemma 2.11.1 (for the continuous linear map $a \mapsto \sigma_L(\operatorname{Op}(a))$). (ii) follows from Proposition 2.10.6, the continuity of $\Psi^m \times \mathcal{S} \to \mathcal{S}$, and Lemma 2.11.1; the adjoint is conjugate-linear, which does not affect real derivatives. □

In particular, for $A \in \Psi^m(\mathbb{R}^n)$ the constant family $A(t) = A$ is smooth, so $t \mapsto B(t)A$ is smooth whenever $B(t)$ is. Note that $s \mapsto A(s)^*$ is *anti*-holomorphic if $A(s)$ is holomorphic; the family $A(\bar{s})^*$ is holomorphic.

2.11.3 Holomorphic families of variable order

Let $U \subset \mathbb{C}$ be open and $f : U \to \mathbb{C}$ holomorphic. We wish to consider families $A(s) \in \Psi^{f(s)}_{\mathrm{ph}}(\mathbb{R}^n)$. These do not lie in a fixed space, so we normalize the order by the elliptic family $\langle\xi\rangle^{f(s)}$.

Definition 2.11.6. For $f : U \to \mathbb{C}$ holomorphic, $\Psi^f_{\mathrm{hol}}(\mathbb{R}^n)$ denotes the space of families $(A(s))_{s\in U}$ of operators of the form

$$A(s) = \operatorname{Op}(\langle\xi\rangle^{f(s)} b(s)), \qquad b : U \to S^0_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n) \text{ holomorphic.}$$

Thus $A(s) \in \Psi^{f(s)}_{\mathrm{ph}}(\mathbb{R}^n)$ for each s, by Lemma 2.9.7(ii), and $b(s) = \langle\xi\rangle^{-f(s)}\sigma_L(A(s))$ is determined by A. Since $\langle\xi\rangle^{f(s)}$ does not depend on x, (2.2.8) gives the exact factorization $A(s) = \operatorname{Op}(b(s))\,\langle D\rangle^{f(s)}$. For example $\langle D\rangle^s \in \Psi^s_{\mathrm{hol}}(\mathbb{R}^n)$, $U = \mathbb{C}$, and if $f \equiv m$ is constant then, by Lemma 2.10.5(iii), $\Psi^m_{\mathrm{hol}}(\mathbb{R}^n)$ is just the space of holomorphic families in $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$. (The subscript "ph" is dropped from the notation, as the definition makes little sense without polyhomogeneity.)

Lemma 2.11.7. *Let $f : U \to \mathbb{C}$ be holomorphic.*

(i) *For $s_0 \in U$ and $\epsilon > 0$ there is a neighbourhood V of s_0 such that $s \mapsto \langle\xi\rangle^{f(s)}$ is holomorphic from V to $S^{\Re f(s_0)+\epsilon}(\mathbb{R}^n)$.*

(ii) *For every $\mu \in \mathbb{N}_0^n$ there are $q_{\mu,k} \in S^{-|\mu|}_{\mathrm{ph}}(\mathbb{R}^n)$, $0 \le k \le |\mu|$, such that $\langle\xi\rangle^{-w} D^\mu_\xi \langle\xi\rangle^w = \sum_{k\le|\mu|} w^k q_{\mu,k}(\xi)$ for all $w \in \mathbb{C}$.*

Proof. (ii) The induction in the proof of Lemma 2.1.4(i) is valid for complex exponents: $D^\mu_\xi\langle\xi\rangle^w = \langle\xi\rangle^{w-2|\mu|}P_\mu(w,\xi)$, where P_μ is a polynomial of degree at most $|\mu|$ in ξ and in w, since $D_{\xi_j}(\langle\xi\rangle^t P) = \langle\xi\rangle^{t-2}(-it\xi_j P + \langle\xi\rangle^2 D_{\xi_j}P)$ with $t = w - 2|\mu|$. Writing $P_\mu(w,\xi) = \sum_k w^k p_{\mu,k}(\xi)$, the function $q_{\mu,k} = \langle\xi\rangle^{-2|\mu|}p_{\mu,k}$ lies in $S^{-2|\mu|}_{\mathrm{ph}} \cdot S^{|\mu|}_{\mathrm{ph}} \subset S^{-|\mu|}_{\mathrm{ph}}$ by Lemma 2.9.7.

(i) The function $\langle\xi\rangle^{f(s)} = \exp(f(s)\log\langle\xi\rangle)$ is smooth in (s,ξ) and holomorphic in s, and by (ii), $|D^\beta_\xi\langle\xi\rangle^{f(s)}| \le C_\beta(1+|f(s)|)^{|\beta|}\langle\xi\rangle^{\Re f(s)-|\beta|}$. Let V be an open disc about s_0, with closure in U, on which $|\Re f(s) - \Re f(s_0)| < \epsilon$. Then the estimates of Lemma 2.11.3 hold on V with order $\Re f(s_0) + \epsilon$. □

Proposition 2.11.8. *Let $f, g : U \to \mathbb{C}$ be holomorphic. If $A \in \Psi^f_{\mathrm{hol}}(\mathbb{R}^n)$ and $B \in \Psi^g_{\mathrm{hol}}(\mathbb{R}^n)$, then the family $A(s)B(s)$ lies in $\Psi^{f+g}_{\mathrm{hol}}(\mathbb{R}^n)$.*

Proof. Write $\sigma_L(A(s)) = \langle\xi\rangle^{f(s)}a(s)$ and $\sigma_L(B(s)) = \langle\xi\rangle^{g(s)}b(s)$ with $a, b : U \to S^0_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$ holomorphic, and set

$$c(s) = \langle\xi\rangle^{-f(s)-g(s)}\sigma_L\big(A(s)B(s)\big).$$

We must show that $c : U \to S^0_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$ is holomorphic; this is a local property, so fix $s_0 \in U$. By Lemma 2.11.7(i), with $\epsilon = \frac{1}{8}$, there is a neighbourhood V of s_0 on which $\langle\xi\rangle^{f(s)}$, $\langle\xi\rangle^{g(s)}$ and $\langle\xi\rangle^{-f(s)-g(s)}$ are holomorphic into S^{μ}, $S^{\mu'}$ and $S^{-\mu-\mu'+3\epsilon}$, where $\mu = \Re f(s_0) + \epsilon$, $\mu' = \Re g(s_0) + \epsilon$. Since $S^0_{\mathrm{ph}} \hookrightarrow S^0$ is continuous and multiplication of symbols is continuous, Lemma 2.11.1 shows that $p(s) = \langle\xi\rangle^{f(s)}a(s)$ and $q(s) = \langle\xi\rangle^{g(s)}b(s)$ are holomorphic from V into $S^{\mu}(\mathbb{R}^n;\mathbb{R}^n)$ and $S^{\mu'}(\mathbb{R}^n;\mathbb{R}^n)$.

For $N \in \mathbb{N}_0$, with r_N as in Corollary 2.10.7,

$$\sigma_L\big(A(s)B(s)\big) = \sum_{|\alpha|<N} \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha p(s)\, D_x^\alpha q(s) + r_N\big(p(s), q(s)\big).$$

By Leibniz' formula, $\langle\xi\rangle^{-f-g} D_\xi^\alpha p\, D_x^\alpha q = \sum_{\nu\le\alpha} \binom{\alpha}{\nu} \big(\langle\xi\rangle^{-f} D_\xi^\nu \langle\xi\rangle^{f}\big) D_\xi^{\alpha-\nu} a\, D_x^\alpha b$, so

$$c(s) = \sum_{|\alpha|<N} c_\alpha(s) + \langle\xi\rangle^{-f(s)-g(s)} r_N\big(p(s), q(s)\big),$$

$$c_\alpha(s) = \frac{i^{|\alpha|}}{\alpha!} \sum_{\nu\le\alpha} \binom{\alpha}{\nu} \Big(\sum_{k\le|\nu|} f(s)^k q_{\nu,k} \Big) D_\xi^{\alpha-\nu} a(s)\, D_x^\alpha b(s),$$

with $q_{\nu,k}$ as in Lemma 2.11.7(ii). By Lemma 2.10.5(iii) and Lemma 2.11.1, each c_α is holomorphic from U into $S^{-|\alpha|}_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$. By Corollary 2.10.7 and Lemma 2.11.1, the last term is holomorphic from V into $S^{-N+3\epsilon} \subset S^{-N+\frac{1}{2}}$. In particular, for each $s \in V$, $c(s) - \sum_{|\alpha|<N} c_\alpha(s) \in S^{-N+\frac{1}{2}}$ for every N, so $c(s) \sim \sum_l \big(\sum_{|\alpha|=l} c_\alpha(s)\big)$, and by Lemma 2.9.7(i) and Lemma 2.1.11(iii), $c(s) \in S^0_{\mathrm{ph}}$ with normalized terms $c(s)_{(-j)} = \sum_{|\alpha|\le j} c_\alpha(s)_{(-j)}$, where $c_\alpha(s)_{(-j)}$ is the normalized term of degree $-j$ of $c_\alpha(s) \in S^{-|\alpha|}_{\mathrm{ph}}$.

It remains to check holomorphy in the topology of Definition 2.10.4. If each of the maps $s \mapsto c(s) \in S^0$, $s \mapsto c(s)_{(-j)} \in S^{-j}$ and $s \mapsto c(s) - \sum_{j\le k} c(s)_{(-j)} \in S^{-k-1}$ is holomorphic on V, then the difference quotients of c are Cauchy in every norm of S^0_{ph}, hence converge since S^0_{ph} is complete (Lemma 2.10.5(ii)), and c is holomorphic on V. Now:

- $c = c_0 + \langle\xi\rangle^{-f-g} r_1(p,q)$ is holomorphic into S^0, as c_0 is holomorphic into $S^0_{\mathrm{ph}} \hookrightarrow S^0$ and the second term into $S^{-1/2} \subset S^0$;
- $c_{(-j)} = \sum_{|\alpha|\le j} (c_\alpha)_{(-j)}$ is holomorphic into S^{-j} by Lemma 2.10.5(i);
- with $N = k + 2$,

$$c - \sum_{j\le k} c_{(-j)} = \langle\xi\rangle^{-f-g} r_N(p,q) + \sum_{|\alpha|\le k} \Big(c_\alpha - \sum_{j=|\alpha|}^{k} (c_\alpha)_{(-j)} \Big) + \sum_{|\alpha|=k+1} c_\alpha,$$

  where the first term is holomorphic into $S^{-k-3/2}$, the second into S^{-k-1} by the definition of the topology of $S^{-|\alpha|}_{\mathrm{ph}}$, and the third into $S^{-k-1}_{\mathrm{ph}} \hookrightarrow S^{-k-1}$.

Hence c is holomorphic into $S^0_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^n)$, and $A(s)B(s) = \mathrm{Op}(\langle\xi\rangle^{f(s)+g(s)}c(s)) \in \Psi^{f+g}_{\mathrm{hol}}(\mathbb{R}^n)$. □

These families do not have very interesting global examples on $\mathbb{R}^n$ beyond $\langle D\rangle^s$ and its products with fixed operators; on a compact manifold, however, the complex powers A^s of a positive self-adjoint elliptic operator A of order one form a holomorphic family of order s, by Seeley's theorem [11], and this is the starting point of the theory of ζ-functions and regularized determinants (Chapter 8).

2.12 Affine invariance

The algebra $\Psi^\infty(\mathbb{R}^n)$ is invariant under affine changes of coordinates. Let $T:\mathbb{R}^n\to\mathbb{R}^n$, $Tx = Lx+c$, with $L\in \mathrm{GL}(n,\mathbb{R})$ and $c\in\mathbb{R}^n$, and let $T^*f = f\circ T$, an isomorphism of $\mathcal{S}(\mathbb{R}^n)$ and of $\mathcal{S}'(\mathbb{R}^n)$.

Proposition 2.12.1. *If $A\in\Psi^m(\mathbb{R}^n)$ then $A_T = T^*A(T^*)^{-1}\in\Psi^m(\mathbb{R}^n)$, and*

$$\sigma_L(A_T)(x,\xi) = \sigma_L(A)(Tx,(L^t)^{-1}\xi), \tag{2.12.1}$$

where L^t is the transpose of L. The same holds for $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$.

Proof. Let $a=\sigma_L(A)$, $f\in\mathcal{S}(\mathbb{R}^n)$ and $g=(T^*)^{-1}f = f\circ T^{-1}$. Substituting $y = Ly'+c$, $\hat g(\xi) = |\det L| e^{-ic\cdot\xi}\hat f(L^t\xi)$. Hence, by (2.2.8),

$$\begin{aligned} A_Tf(x) = Ag(Tx) &= \int e^{i(Lx+c)\cdot\xi} a(Tx,\xi)|\det L| e^{-ic\cdot\xi}\hat f(L^t\xi)đ\xi \\ &= \int e^{ix\cdot\eta} a\big(Tx,(L^t)^{-1}\eta\big)\hat f(\eta)đ\eta, \end{aligned}$$

with $\eta = L^t\xi$. The function $a(Tx,(L^t)^{-1}\eta)$ is in $S^m(\mathbb{R}^n;\mathbb{R}^n)$, since $C^{-1}\langle\eta\rangle\le\langle(L^t)^{-1}\eta\rangle\le C\langle\eta\rangle$ and derivatives transform linearly; so $A_T = \mathrm{Op}\big(a(Tx,(L^t)^{-1}\eta)\big)$. If a is polyhomogeneous, so is $a(Tx,(L^t)^{-1}\eta)$: its terms are homogeneous in $|(L^t)^{-1}\eta|\ge 1$, which contains $|\eta|\ge R$ for some R, and by the remark after Definition 2.9.1 this suffices. □

In particular, for a translation $Tx = x+c$ we get $\sigma_L(A_T)(x,\xi) = \sigma_L(A)(x+c,\xi)$, and for a linear map there is no Jacobian factor in (2.12.1). This invariance means that the spaces $\Psi^m(V)$ and $\Psi^m_{\mathrm{ph}}(V)$ of operators on $\mathcal{S}(V)$ are well defined for any finite-dimensional real vector space (or affine space) V, with symbols defined on $V\times V^*$. Since $A\mapsto A_T$ is linear and maps symbols continuously, it is an isomorphism of the Fréchet spaces $\Psi^m(\mathbb{R}^n)$ and $\Psi^m_{\mathrm{ph}}(\mathbb{R}^n)$. Invariance under general changes of coordinates, which is needed to define pseudodifferential operators on manifolds, is more subtle; it is the subject of the next section.

2.13 Local coordinate invariance

To transfer pseudodifferential operators to manifolds one needs invariance not only under affine maps but under diffeomorphisms $F:\Omega\to\Omega'$ between open subsets of $\mathbb{R}^n$. For this to make sense we consider operators on $\mathbb{R}^n$ which only "see" Ω':

$$\Psi^m_{\mathrm{c}}(\Omega') = \big\{A\in\Psi^m(\mathbb{R}^n);\ \operatorname{supp}K_A \text{ is a compact subset of } \Omega'\times\Omega'\big\},\qquad m\in\mathbb{R}\cup\{-\infty\},$$

and similarly $\Psi^m_{\mathrm{ph,c}}(\Omega')$, $m\in\mathbb{C}$. There are plenty of such operators: if $\phi,\psi\in\mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ and $B\in\Psi^m(\mathbb{R}^n)$ then $\phi B\psi = \mathrm{Op}(\phi(x)\psi(y)\sigma_L(B))\in\Psi^m_{\mathrm{c}}(\Omega')$. Moreover every symbol $a\in S^m(\mathbb{R}^n;\mathbb{R}^n)$ with $\operatorname{supp}a\subset K\times\mathbb{R}^n$, $K\subset\Omega'$ compact, is the left symbol of an element of $\Psi^m_{\mathrm{c}}(\Omega')$ modulo $S^{-\infty}$: choose $\phi_1,\phi\in\mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ with $\phi_1=1$ near K and $\phi=1$ near $\operatorname{supp}\phi_1$. Then $\mathrm{Op}(a) = \phi_1\,\mathrm{Op}(a)$, so by Lemma 2.10.8

$$\mathrm{Op}(a) - \phi\,\mathrm{Op}(a)\phi = \mathrm{Op}(a)(1-\phi) = \phi_1\,\mathrm{Op}(a)(1-\phi)\in\Psi^{-\infty}(\mathbb{R}^n),$$

using $\phi\phi_1 = \phi_1$.

Let $F : \Omega \to \Omega'$ be a diffeomorphism with inverse G, and $A \in \Psi^m_{\mathrm{c}}(\Omega')$. Choose $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ equal to 1 near the projection of $\operatorname{supp} K_A$ onto the second factor, and define for $u \in \mathcal{S}(\mathbb{R}^n)$

$$A_F u = F^*\big(A(\phi\, G^*(u|_\Omega))\big),$$

extended by zero outside Ω; here $G^*v = v \circ G$ and the argument of A lies in $\mathcal{C}^\infty_{\mathrm{c}}(\Omega')$, while A maps it to a function with compact support in Ω', so that F^* of it lies in $\mathcal{C}^\infty_{\mathrm{c}}(\Omega)$. The result does not depend on ϕ, and by the change of variables $y' = F(y)$ the kernel of A_F is

$$K_{A_F}(x,y) = K_A\big(F(x),F(y)\big)\,|\det DF(y)|,$$

extended by zero, which has compact support in $\Omega\times\Omega$. For a linear map this is the operator A_T of Section 2.12.

Proposition 2.13.1. *If $F : \Omega \to \Omega'$ is a diffeomorphism of open subsets of $\mathbb{R}^n$, then $A \mapsto A_F$ is a linear isomorphism $\Psi^m_{\mathrm{c}}(\Omega') \to \Psi^m_{\mathrm{c}}(\Omega)$, with inverse $B \mapsto B_G$, $G = F^{-1}$, and the same holds for $\Psi^m_{\mathrm{ph,c}}$. The principal symbols are related by*

$$\sigma_L(A_F)(x,\xi) - \sigma_L(A)(F(x),(DF(x)^t)^{-1}\xi) \in S^{m-1}(\mathbb{R}^n;\mathbb{R}^n),$$

where the second term is extended by zero for $x \notin \Omega$ (it vanishes for x outside a compact subset of Ω).

Sketch of proof. A complete proof, which also gives the transformation law for the essential support, is given in the coordinate invariance theorem of Chapter 3 (with the roles of F and G interchanged); we indicate the main steps. Let $a = \sigma_L(A)$, $K \subset \Omega'$ compact with $\operatorname{supp} K_A \subset K \times K$, and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ equal to 1 near K, so that $A = \operatorname{Op}(b)$ with $b(x',y',\xi') = \psi(x')\psi(y')a(x',\xi')$. Choose $\mu \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to 1 near 0, with small support, and split $A = A_1 + A_2$ where $K_{A_1} = \mu(x'-y')K_A$. By Lemma 2.10.8 (applied with a partition of unity on $\operatorname{supp}\psi$), or directly from the proof of that lemma, K_{A_2} is smooth with compact support, so $K_{(A_2)_F}$ is smooth with compact support in $\Omega\times\Omega$ and $(A_2)_F \in \Psi^{-\infty}(\mathbb{R}^n)$ by Proposition 2.3.3.

For A_1, write $F(x)-F(y) = T(x,y)(x-y)$ with $T(x,y) = \int_0^1 DF(y+t(x-y))\,dt$; if $\operatorname{supp}\mu$ is small enough, the kernel of $(A_1)_F$ is supported where x,y lie in a fixed compact subset of Ω and $|x-y|$ is so small that the segment $[x,y]$ lies in Ω and $T(x,y)$ is invertible, with $|\det T|$ and $\|T^{-1}\|$ bounded. Since $(F(x)-F(y))\cdot\xi' = (x-y)\cdot T(x,y)^t\xi'$, the substitution $\xi = T(x,y)^t\xi'$ (first for residual b, then in general by approximation, Proposition 2.2.3(ii)) gives $K_{(A_1)_F} = I(c)$ with

$$c(x,y,\xi) = \mu\big(F(x)-F(y)\big)\,b\big(F(x),F(y),(T(x,y)^t)^{-1}\xi\big)\frac{|\det DF(y)|}{|\det T(x,y)|}.$$

By the chain rule, $D_x^\alpha D_y^\beta D_\xi^\gamma$ of $b(F(x),F(y),S(x,y)\xi)$, $S = (T^t)^{-1}$, is a finite sum of terms $M(x,y)\xi^\nu(D_{x'}^{\alpha'}D_{y'}^{\beta'}D_{\xi'}^{\kappa}b)(F(x),F(y),S\xi)$ with smooth bounded coefficients and $|\nu| \le |\kappa|-|\gamma|$: each ξ-derivative produces one ξ'-derivative of b, and an x- or y-derivative falling on $S\xi$ produces one ξ'-derivative and at most one factor of ξ. As $\langle S\xi\rangle$ is comparable to $\langle\xi\rangle$ on the support, each term is $O(\langle\xi\rangle^{|\nu|+m-|\kappa|}) = O(\langle\xi\rangle^{m-|\gamma|})$, so $c \in S^m(\mathbb{R}^{2n};\mathbb{R}^n) = \mathcal{A}^m_0(\mathbb{R}^n)$ and $(A_1)_F \in \Psi^m(\mathbb{R}^n)$. If a is polyhomogeneous, so is c, since $\xi \mapsto S(x,y)\xi$ is linear and $|S\xi| \ge 1$ for $|\xi| \ge R$, uniformly on the support (compare the proof of Proposition 2.12.1). Finally, by Theorem 2.3.1 $\sigma_L(A_F) \equiv c(x,x,\xi)$ modulo S^{m-1}, and since $T(x,x) = DF(x)$ and $\psi a = a$, $c(x,x,\xi) = a(F(x),(DF(x)^t)^{-1}\xi)$. The kernel formula shows that $(A_F)_G = A$. □

Thus, after cutting off to compact sets, the principal symbol transforms as a function on the cotangent bundle: $(x,\xi)\mapsto (F(x),(DF(x)^t)^{-1}\xi)$ is the inverse of the cotangent map of F. This is the basis of the definition of pseudodifferential operators on manifolds in Chapter 5.

2.14 Exercises

Exercise 2.1. Show in detail that $\langle\xi\rangle^m \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ for every $m\in\mathbb{R}$ and every p, and deduce that $S^m(\mathbb{R}^p;\mathbb{R}^n)\cdot S^{m'}(\mathbb{R}^p;\mathbb{R}^n) = S^{m+m'}(\mathbb{R}^p;\mathbb{R}^n)$.

Exercise 2.2. Let $n=2$ and identify $\mathbb{R}^2_\xi$ with $\mathbb{C}$ via $\zeta=\xi_1+i\xi_2$. Let $a\in S^1(\mathbb{R}^2)$ (a symbol in ξ alone) be elliptic. Show that for $r>0$ sufficiently large

$$\frac{1}{2\pi}\int_0^{2\pi}\frac{1}{a(re^{i\theta})}\frac{d}{d\theta}a(re^{i\theta})\,d\theta$$

exists, is an integer, and is independent of r. Conclude that there is an elliptic symbol $a\in S^1(\mathbb{R}^2)$ such that there is no $b\in S^1(\mathbb{R}^2)$ with $b\neq 0$ on all of $\mathbb{R}^2$ and $b(\xi)=a(\xi)$ for $|\xi|>r$, for any r. *Hint:* consider a equal to $\xi_1+i\xi_2$ for $|\xi|\geq 1$.

Exercise 2.3. Show that a symbol $a\in S^m(\mathbb{R}^p;\mathbb{R}^n)$ which satisfies $|a(z,\xi)|\leq C\langle\xi\rangle^{m'}$ for some $m'<m$ is necessarily in $S^{m'+\epsilon}(\mathbb{R}^p;\mathbb{R}^n)$ for every $\epsilon>0$. *Hint:* use the interpolation inequality $\sup_I|f'|^2\leq C\sup_I|f|(\sup_I|f''|+\delta^{-2}\sup_I|f|)$ for functions on an interval I of length δ, on balls of radius comparable to $\langle\xi\rangle$ in ξ and of radius 1 in z, and iterate.

Exercise 2.4. Let $\psi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\psi(\xi)=1$ in $|\xi|<1$. Show that if $\phi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^p\times\mathbb{R}^n)$ then $c_\phi(z,\xi)=\phi(z,\xi/|\xi|)(1-\psi(\xi))\in S^0(\mathbb{R}^p;\mathbb{R}^n)$. For $a\in S^m(\mathbb{R}^p;\mathbb{R}^n)$ define the *cone support* of a by its complement: $(\bar z,\bar\xi)\in\mathbb{R}^p\times(\mathbb{R}^n\setminus 0)$ is *not* in $\operatorname{cone\,supp}(a)$ if there exists $\phi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^p\times\mathbb{R}^n)$ with $\phi(\bar z,\bar\xi/|\bar\xi|)\neq 0$ such that $c_\phi a\in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$. Show that for $a\in S^m$ and $b\in S^{m'}$

$$\operatorname{cone\,supp}(ab)\subset\operatorname{cone\,supp}(a)\cap\operatorname{cone\,supp}(b).$$

If $\operatorname{cone\,supp}(a)=\emptyset$, does it follow that $a\in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$? (Consider $p=n=1$ and $a(z,\xi)=\chi(\xi/(1+z^2))$ with $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$ supported in $1\leq|t|\leq 2$.)

Exercise 2.5. Show that the Schwartz kernel of any $A\in\Psi^\infty(\mathbb{R}^n)$ is smooth away from the diagonal $\{x=y\}$, and that it is rapidly decreasing, with all derivatives, as $|x-y|\to\infty$. *Hint:* by Lemma 2.2.4, $(x-y)^\alpha K_A$ becomes increasingly smooth as $|\alpha|$ increases.

Exercise 2.6. For $a\in S^m(\mathbb{R}^n;\mathbb{R}^n)$ let $\operatorname{Op}_W(a)=\operatorname{Op}(a(\frac{x+y}{2},\xi))$ be its *Weyl quantization*. Show that $\operatorname{Op}_W(a)\in\Psi^m(\mathbb{R}^n)$, that $\operatorname{Op}_W(a)^*=\operatorname{Op}_W(\bar a)$, and that $\sigma_L(\operatorname{Op}_W(a))\sim\sum_\alpha\frac{(i/2)^{|\alpha|}}{\alpha!}D_\xi^\alpha D_x^\alpha a$. Deduce that $\operatorname{Op}_W: S^m(\mathbb{R}^n;\mathbb{R}^n)\to\Psi^m(\mathbb{R}^n)$ is a bijection.

Exercise 2.7. (Polyhomogeneous symbols as smooth functions.) Show that $a\in\mathcal{C}^\infty(\mathbb{R}^p\times\mathbb{R}^n)$ is in $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ if and only if a satisfies (2.9.1) and the function $(z,\theta,r)\mapsto r^m a(z,\theta/r)$ on $\mathbb{R}^p\times\mathbb{S}^{n-1}\times(0,1]$ extends to a smooth function on $\mathbb{R}^p\times\mathbb{S}^{n-1}\times[0,1]$ all of whose derivatives are bounded. Relate the Taylor series at $r=0$ to the terms a_{m-j}.

Exercise 2.8. Show that the topology of $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^n)$ in Definition 2.10.4 does not depend on the choice of the cut-off ρ. *Hint:* if ρ' is another choice, the corresponding normalized terms differ by $(\rho-\rho')$ times a homogeneous function, which is supported in $\frac12\leq|\xi|\leq 1$; estimate its residual norms by the norms of the term on $1\leq|\xi|\leq 2$.

Exercise 2.9. Let $m \in \mathbb{R}$ and $n \geq 1$. Show that $a(\xi) = \langle \xi \rangle^m \cos(\log\langle \xi \rangle) \in S^m(\mathbb{R}^n)$, but that a is not in the closure of $S^m_{\mathrm{ph}}(\mathbb{R}^n)$ in $S^m(\mathbb{R}^n)$, so that S^m_{ph} is not dense in S^m. *Hint:* if $b \in S^m_{\mathrm{ph}}$ then $\langle r\theta \rangle^{-m} b(r\theta)$ converges as $r \to \infty$ for each $\theta \in \mathbb{S}^{n-1}$, whereas $\langle r\theta \rangle^{-m} a(r\theta)$ takes the values ± 1 for arbitrarily large r. On the other hand, show that $S^m_{\mathrm{ph}}(\mathbb{R}^p; \mathbb{R}^n)$ is dense in $S^m(\mathbb{R}^p; \mathbb{R}^n)$ for the topology of $S^{m'}(\mathbb{R}^p; \mathbb{R}^n)$, for every $m' > m$.

Exercise 2.10. (Asymptotic summation of holomorphic families.) Let $U \subset \mathbb{C}$ be open and let $a_j : U \to S^{-j}_{\mathrm{ph}}(\mathbb{R}^p; \mathbb{R}^n)$ be holomorphic, $j \in \mathbb{N}_0$. Show that there is a holomorphic $a : U \to S^0_{\mathrm{ph}}(\mathbb{R}^p; \mathbb{R}^n)$ such that $a - \sum_{j<N} a_j$ is holomorphic from U into $S^{-N}_{\mathrm{ph}}(\mathbb{R}^p; \mathbb{R}^n)$ for every N. *Hint:* follow the proof of Proposition 2.1.10 with $\phi(\epsilon_j \xi) a_j(s)$, choosing ϵ_j so that the relevant norms are at most 2^{-j} uniformly for s in the j-th set of an exhaustion of U by compact sets; show that $\phi(\epsilon\xi) a_j(s)$ is again holomorphic into S^{-j}_{ph}, and that a locally uniform limit of holomorphic maps into a Fréchet space is holomorphic (use the Cauchy integral formula applied to continuous linear functionals, or the difference-quotient formula in the proof of Lemma 2.11.3).

Exercise 2.11. Let $f : U \to \mathbb{C}$ be holomorphic and $A \in \Psi^f_{\mathrm{hol}}(\mathbb{R}^n)$. Set $U^* = \{\bar{s};\ s \in U\}$ and $f^*(s) = \overline{f(\bar{s})}$. Show that the family $s \mapsto A(\bar{s})^*$, $s \in U^*$, lies in $\Psi^{f^*}_{\mathrm{hol}}(\mathbb{R}^n)$. *Hint:* adapt the proof of Proposition 2.11.8, using (2.4.1).

Exercise 2.12. Let $U \subset \mathbb{R}^k$ be open and let $A(t)$ be a smooth family in $\Psi^m(\mathbb{R}^n)$ which is *locally uniformly elliptic*: for each compact $K \subset U$ there are $\epsilon, C > 0$ with $|\sigma_L(A(t))(x,\xi)| \geq \epsilon \langle \xi \rangle^m$ for $t \in K$, $|\xi| \geq C$. Show that there is a smooth family $B(t)$ in $\Psi^{-m}(\mathbb{R}^n)$ such that $A(t)B(t) - \mathrm{Id}$ and $B(t)A(t) - \mathrm{Id}$ are smooth families in $\Psi^{-\infty}(\mathbb{R}^n)$. *Hint:* construct B locally in t as in Theorem 2.6.3, using Lemma 2.11.2 and a parameter-dependent version of Proposition 2.1.10; then use a partition of unity in t.

Exercise 2.13. For $0 \leq \delta < \rho \leq 1$ let $S^m_{\rho,\delta}(\mathbb{R}^p; \mathbb{R}^n)$ be the space of $a \in \mathcal{C}^\infty(\mathbb{R}^p \times \mathbb{R}^n)$ with $|D_z^\alpha D_\xi^\beta a| \leq C_{\alpha\beta} \langle \xi \rangle^{m - \rho|\beta| + \delta|\alpha|}$. Show that these spaces are closed under multiplication (orders adding), that $D_\xi^\beta D_z^\alpha$ maps $S^m_{\rho,\delta}$ into $S^{m-\rho|\beta|+\delta|\alpha|}_{\rho,\delta}$, and that series $\sum_j a_j$ with $a_j \in S^{m-j(\rho-\delta)}_{\rho,\delta}$ have asymptotic sums. The symbols of this chapter are those of type $(1,0)$. Which of the steps in the proof of Theorem 2.3.1 use $\rho = 1$, $\delta = 0$, and how must they be modified? (See [7, Section 7.8] and [8, Section 18.1].)

Exercise 2.14. Show that for any $s \geq 0$ the unit ball in $H^s(\mathbb{R}^n)$ is *not* precompact in $L^2(\mathbb{R}^n)$, i.e. there is a sequence $f_j \in H^s(\mathbb{R}^n)$ with $\|f_j\|_{H^s} \leq 1$ and no subsequence convergent in $L^2(\mathbb{R}^n)$.

Exercise 2.15. Show that for any $R > 0$ and $N > 0$ the subspace $\{u \in H^N(\mathbb{R}^n);\ u(x) = 0 \text{ in } |x| > R\}$ of $H^N(\mathbb{R}^n)$ is compactly included in $L^2(\mathbb{R}^n)$, i.e. the intersection of the unit ball of $H^N(\mathbb{R}^n)$ with this subspace is precompact in $L^2(\mathbb{R}^n)$. *Hint:* for $N > n/2$ one can use the Sobolev embedding and the Arzelà–Ascoli theorem; in general, with $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to 1 on $|x| \leq R$, write $\hat{u} = \widehat{\phi u} = (2\pi)^{-n} \hat{\phi} * \hat{u}$ to see that the Fourier transforms are equicontinuous.

Exercise 2.16. Using Exercise 2.15, or otherwise, show that for any $\epsilon > 0$ the inclusion $\langle x \rangle^{-\epsilon} H^\epsilon(\mathbb{R}^n) \hookrightarrow L^2(\mathbb{R}^n)$ is compact, i.e. any sequence f_j such that $\langle x \rangle^\epsilon f_j$ is bounded in $H^\epsilon(\mathbb{R}^n)$ has a subsequence convergent in $L^2(\mathbb{R}^n)$. *Hint:* choose $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\phi(x) = 1$ in $|x| < 1$ and, for each k, consider the sequence $\phi(x/k) f_j$. Show that its Fourier transforms are equicontinuous and bounded in $\langle \xi \rangle^{-\epsilon} L^2$, deduce that it has a subsequence convergent in $L^2(\mathbb{R}^n)$, and estimate $\|(1 - \phi(x/k)) f_j\|_{L^2} \leq C \langle k \rangle^{-\epsilon}$. Conclude by diagonalization.

Exercise 2.17. Prove (2.12.1) directly from the definition of $\mathrm{Op}(a)$ via amplitudes, and show that the right symbol transforms in the same way: $\sigma_R(A_T)(y,\xi) = \sigma_R(A)(Ty, (L^t)^{-1}\xi)$. *Hint:* show that $(A_T)^* = (A^*)_T$.

Exercise 2.18. Show that $\mathcal{S}(\mathbb{R}^n)$, and even $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, is dense in $L^2(\mathbb{R}^n)$.

Exercise 2.19. Show that any continuous linear operator $\mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n)$ has Schwartz kernel in $\mathcal{S}(\mathbb{R}^{2n})$.

Notes

The main departures from Melrose's original text are as follows. The oscillatory integral $I(a)$ is defined directly by the integration by parts formula (2.2.3) rather than by extension by continuity, and all identities between kernels are derived from Lemma 2.2.4. The composition theorem is proved right after the reduction theorem, with the passage from right to left symbols in (2.4.2) carried out explicitly. The residual characterization (Proposition 2.3.3) takes into account that the weight of the amplitudes may depend on the order. The weighted Sobolev estimates use the weight $\langle x\rangle^{-q}\langle y\rangle^{q}$ as a multiplier of amplitudes, avoiding a case distinction. We have also added the commutator formula (2.5.2).

The material on complex orders, topologies, families and local coordinate invariance follows the revised version of these notes (*From microlocal to global analysis*). There the continuity of the reduction map is argued by following the Taylor expansion and the asymptotic summation; we have instead derived all continuity statements from the closed graph theorem and the uniform boundedness principle for Fréchet spaces (see, e.g., W. Rudin, *Functional Analysis*, Chapter 2), which requires the completeness of the polyhomogeneous symbol spaces (Lemma 2.10.5); for this we normalized the homogeneous terms by a fixed cut-off. The localization lemma Lemma 2.10.8 is stated in the original with the roles of the two cut-offs interchanged, which is not correct. The original gives no proof of the composition law for holomorphic families of variable order (Proposition 2.11.8); the proof given here reduces it, via the remainder estimate of Corollary 2.10.7, to holomorphy of finitely many explicit terms. The treatment of L^2 boundedness via an approximate square root is due to Hörmander; for alternative treatments of the calculus see [8], [15] and [11].

Chapter 3

Microlocalization

A distribution can fail to be smooth at a point, and the singular support records where this happens. Microlocal analysis asks a finer question: in which *frequency directions* is it singular? The Heaviside function $H(x_1)$ on $\mathbb{R}^2$ is singular along the line $\{x_1 = 0\}$, but only in the directions $\pm dx_1$ normal to that line; after a cut-off its Fourier transform decays rapidly in every other direction. Recording points together with directions gives a subset of the cotangent bundle, and pseudodifferential operators turn out to be exactly the right tools to detect and manipulate such sets.

The pseudodifferential operators of Chapter 2 are not local: the support of Au is in general not contained in the support of u. They are, however, *pseudolocal*: they do not create new singularities. This chapter refines that statement in two directions. First, singularities of a distribution are not only located at points of $\mathbb{R}^n$ but also have directions: the *wavefront set* $\mathrm{WF}(u)$ is a closed conic subset of $T^*\mathbb{R}^n \setminus 0 = \mathbb{R}^n \times (\mathbb{R}^n \setminus 0)$ which projects onto the singular support. Secondly, an operator $A \in \Psi^m(\mathbb{R}^n)$ has an *essential support* $\mathrm{WF}'(A) \subset T^*\mathbb{R}^n \setminus 0$, the set where its symbol is not rapidly decreasing, and A is *microlocal*: $\mathrm{WF}(Au) \subset \mathrm{WF}'(A) \cap \mathrm{WF}(u)$. Conversely, where A is elliptic it has a microlocal parametrix, so $\mathrm{WF}(u) \subset \mathrm{WF}(Au) \cup \mathrm{Char}_m(A)$.

Sections 3.1 and 3.2 deal with supports and singular supports, Sections 3.3 to 3.5 with the essential support, ellipticity at a point and microlocal parametrices, and Section 3.6 with the invariance of the calculus under changes of coordinates; this is what makes it possible to transfer everything to manifolds in Chapter 5. The wavefront set is introduced in Section 3.7 and characterized in terms of the Fourier transform in Section 3.8. The remaining sections develop the calculus of wavefront sets: invariance under diffeomorphisms, the wavefront set of the kernel of a pseudodifferential operator, and the behaviour of wavefront sets under pairing, exterior products, restriction, projection, multiplication, pull-back and push-forward, culminating in the wavefront relation of an operator with a distributional kernel.

The last part of the chapter is devoted to Hörmander's theorem on the propagation of singularities. Section 3.17 refines the wavefront set to the wavefront sets $\mathrm{WF}_s(u)$ relative to the Sobolev spaces H^s, Section 3.18 shows that the estimates of the calculus are uniform on bounded families of operators, and Section 3.19 introduces Hamilton vector fields and the null bicharacteristics of an operator with real homogeneous principal symbol. In Section 3.20 we prove, by a positive commutator argument, that for such an operator P the set $\mathrm{WF}(u) \setminus \mathrm{WF}(Pu)$ is a union of maximal null bicharacteristics (Theorem 3.20.1). This result is coordinate invariant and so holds on manifolds; it is used for the wave equation in Chapter 11.

Conventions

We use freely the results of Chapters 1 and 2. We write $u(\phi)$ for the (bilinear) pairing of a distribution with a test function and $\hat{u} = \mathcal{F}u$ for the Fourier transform. A function f on a set $U \subset \mathbb{R}^p \times \mathbb{R}^n$ is *rapidly decreasing on* U if $\sup_{(x,\xi)\in U} \langle\xi\rangle^N |f(x,\xi)| < \infty$ for every N; similarly for functions of ξ alone. For $\bar{\xi} \in \mathbb{R}^n \setminus 0$, $\bar{x} \in \mathbb{R}^p$ and $\epsilon > 0$ we use the open cones

$$V_\epsilon(\bar{\xi}) = \Big\{\xi \in \mathbb{R}^n \setminus 0;\ \Big|\frac{\xi}{|\xi|} - \frac{\bar{\xi}}{|\bar{\xi}|}\Big| < \epsilon\Big\}, \qquad \Gamma_\epsilon(\bar{x},\bar{\xi}) = \{x; |x-\bar{x}| < \epsilon\} \times V_\epsilon(\bar{\xi}).$$

A set $W \subset \mathbb{R}^p \times (\mathbb{R}^n \setminus 0)$ is *conic* if $(x,\xi) \in W \Rightarrow (x, t\xi) \in W$ for all $t > 0$. Every conic neighbourhood of $(\bar{x},\bar{\xi})$ contains some $\Gamma_\epsilon(\bar{x},\bar{\xi})$ (see Figure 3.2). For a conic set W we put $W^- = \{(x,-\xi); (x,\xi) \in W\}$. We shall often use the elementary inequality

$$|se - tf| \geq \tfrac{1}{2}|e-f|\,(s+t), \qquad s,t \geq 0,\ |e| = |f| = 1, \tag{3.0.1}$$

which follows from $|se-tf|^2 = (s-t)^2 + st|e-f|^2 \geq \frac{1}{2}|e-f|^2(s^2+t^2)$ (note $|e-f| \leq 2$), and the inequality $\big|\frac{v}{|v|} - \frac{w}{|w|}\big| \leq \frac{2|v-w|}{|w|}$ for $v, w \neq 0$.

From Chapter 2 we recall in particular: every $A \in \Psi^m(\mathbb{R}^n)$ is $A = \mathrm{Op}(\sigma_L(A)) = \mathrm{Op}_R(\sigma_R(A))$ with unique left and right symbols in $S^m(\mathbb{R}^n;\mathbb{R}^n)$; if $A = \mathrm{Op}(a)$ for an amplitude $a \in S^m(\mathbb{R}^{2n};\mathbb{R}^n)$ (an amplitude of weight 0), then its kernel is the oscillatory integral $K_A = I(a) = \int e^{i(x-y)\cdot\xi} a(x,y,\xi)\,đ\xi$ and

$$\sigma_L(A)(x,\xi) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha D_y^\alpha a(x,y,\xi)\big|_{y=x}; \tag{3.0.2}$$

$\mathrm{Op}(a)\,\mathrm{Op}_R(b) = \mathrm{Op}(a(x,\xi)b(y,\xi))$ for symbols a, b; $\Psi^{-\infty}(\mathbb{R}^n)$ consists of the operators whose kernels are smooth and satisfy $|D_x^\alpha D_y^\beta K(x,y)| \leq C_{\alpha\beta N}\langle x-y\rangle^{-N}$ for all α,β,N; such operators map $\mathcal{S}'(\mathbb{R}^n)$ into $\mathcal{C}^\infty(\mathbb{R}^n)$, with $Eu(x) = u(K_E(x,\cdot))$; every $A \in \Psi^m(\mathbb{R}^n)$ acts continuously on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$, the latter by $(Au)(\phi) = u(A^t\phi)$; and an operator $A \in \Psi^m(\mathbb{R}^n)$ which is *(globally) elliptic*, in the sense that $|\sigma_L(A)(x,\xi)| \geq \epsilon\langle\xi\rangle^m$ for $|\xi| \geq R$ and some $\epsilon, R > 0$, has a parametrix $B \in \Psi^{-m}(\mathbb{R}^n)$ with $AB - \mathrm{Id},\ BA - \mathrm{Id} \in \Psi^{-\infty}(\mathbb{R}^n)$.

3.1 Supports and singular supports

Before we refine singularities by directions, we record how an operator moves supports, and introduce the singular support, which the wavefront set will later refine.

Recall from Chapter 1 that the support $\operatorname{supp} u$ of $u \in \mathcal{S}'(\mathbb{R}^n)$ is the complement of the largest open set on which u vanishes; equivalently, $\bar{x} \notin \operatorname{supp} u$ if and only if there exists $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\phi(\bar{x}) \neq 0$ and $\phi u = 0$. We write

$$\mathcal{C}_c^{-\infty}(\mathbb{R}^n) = \{u \in \mathcal{S}'(\mathbb{R}^n);\ \operatorname{supp} u \text{ compact}\}.$$

An element $u \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ acts on all of $\mathcal{C}^\infty(\mathbb{R}^n)$ by $u(\phi) = u(\psi\phi)$, where $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equals 1 near $\operatorname{supp} u$; this does not depend on ψ. Since u is tempered, $|u(\phi)| \leq C\|\psi\phi\|_k$, so u has *finite order*: for a compact neighbourhood L of $\operatorname{supp} u$,

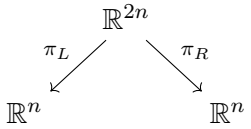
$$|u(\phi)| \leq C \sum_{|\beta|\leq k} \sup_L |D^\beta\phi|, \qquad \phi \in \mathcal{C}^\infty(\mathbb{R}^n). \tag{3.1.1}$$

3.1.1 The calculus of supports

If $A : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ is continuous and linear, its Schwartz kernel $K_A \in \mathcal{S}'(\mathbb{R}^{2n})$ has a support, and we set $\operatorname{supp}(A) = \operatorname{supp}(K_A)$. A subset $G \subset \mathbb{R}^n \times \mathbb{R}^n$ defines a *relation*, that is a map on subsets,

$$G \circ S = \{x \in \mathbb{R}^n;\ \exists\, y \in S \text{ with } (x,y) \in G\}, \qquad S \subset \mathbb{R}^n.$$

In terms of the projections $\pi_L(x,y) = x$, $\pi_R(x,y) = y$,



$$G \circ S = \pi_L\big(\pi_R^{-1}(S) \cap G\big).$$

If S is compact and G is closed then $G \circ S$ is closed: the restriction of π_L to the closed set $\pi_R^{-1}(S) \cap G$ is proper (the preimage of a compact set K is a closed subset of $K \times S$), and proper continuous maps into $\mathbb{R}^n$ are closed.

Proposition 3.1.1 (Calculus of supports)**.** *If $A : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ is continuous and linear then*

$$\operatorname{supp}(A\phi) \subset \operatorname{supp}(A) \circ \operatorname{supp}(\phi) \quad \textit{for all } \phi \in \mathcal{C}_c^\infty(\mathbb{R}^n).$$

Proof. Since $\operatorname{supp}(A) \circ \operatorname{supp}(\phi)$ is closed, a point $\bar{x}$ outside it has a neighbourhood disjoint from it; choose $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ supported in this neighbourhood with $\psi(\bar{x}) \neq 0$. If $(x,y) \in \operatorname{supp}(A)$ with $y \in \operatorname{supp}\phi$ then $x \in \operatorname{supp}(A) \circ \operatorname{supp}(\phi)$, so $x \notin \operatorname{supp}\psi$. Thus $\operatorname{supp}(A) \cap (\operatorname{supp}\psi \times \operatorname{supp}\phi) = \emptyset$. For every $\chi \in \mathcal{S}(\mathbb{R}^n)$ the function $(\chi\psi) \boxtimes \phi$ vanishes near $\operatorname{supp} K_A$, hence $(\psi A\phi)(\chi) = K_A\big((\chi\psi) \boxtimes \phi\big) = 0$. So $\psi A\phi = 0$ and $\bar{x} \notin \operatorname{supp}(A\phi)$. □

For instance, the kernel of a differential operator is supported in the diagonal $\mathrm{Diag} = \{(x,x)\}$, so differential operators are *local*: $\operatorname{supp}(P\phi) \subset \operatorname{supp}\phi$. A pseudodifferential operator, on the other hand, generally has $\operatorname{supp}(A) = \mathbb{R}^{2n}$; the kernel of $\langle D\rangle^{-2}$ on $\mathbb{R}$, for example, is $\frac{1}{2}e^{-|x-y|}$.

3.1.2 Singular support

Definition 3.1.2. The *singular support* of $u \in \mathcal{S}'(\mathbb{R}^n)$ is the set $\operatorname{sing\,supp} u$ of points $\bar{x}$ such that there is no $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\phi(\bar{x}) \neq 0$ and $\phi u \in \mathcal{C}^\infty(\mathbb{R}^n)$.

If $\phi u \in \mathcal{C}^\infty$ and $\phi(\bar{x}) \neq 0$ then $\phi \neq 0$ on a ball B around $\bar{x}$, and for every $\psi \in \mathcal{C}_c^\infty(B)$, $\psi u = (\psi/\phi)\phi u \in \mathcal{C}^\infty$. Hence the complement of $\operatorname{sing\,supp} u$ is open, $\operatorname{sing\,supp} u$ is closed, and $\bar{x} \notin \operatorname{sing\,supp} u$ if and only if $\phi u \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ for some $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\phi = 1$ near $\bar{x}$. Directly from the definitions,

$$\operatorname{sing\,supp} u \subset \operatorname{supp} u, \qquad \operatorname{sing\,supp} u = \emptyset \iff u \in \mathcal{C}^\infty(\mathbb{R}^n),$$

the second because u is smooth if it is smooth near each point.

3.2 Pseudolocality

A pseudodifferential operator may spread supports, but it cannot create singularities. The reason is visible in the Schwartz kernel: all of its singularity sits on the diagonal (Figure 3.1). The first observation is that the kernel of a pseudodifferential operator is smooth, and rapidly decreasing, away from the diagonal.

Lemma 3.2.1. *Let $a \in S^m(\mathbb{R}^{2n};\mathbb{R}^n)$ be an amplitude and let $\rho \in \mathcal{C}^\infty(\mathbb{R}^n)$ have all derivatives bounded and vanish near 0. Then $\rho(x-y)I(a)(x,y)$ is the kernel of an element of $\Psi^{-\infty}(\mathbb{R}^n)$: it is smooth and*

$$|D_x^\beta D_y^\gamma(\rho(x-y)I(a))| \le C_{\beta\gamma N}\langle x-y\rangle^{-N} \quad \text{for all } \beta,\gamma,N.$$

Proof. By the kernel identities of Chapter 2,

$$D_x^\beta D_y^\gamma[(x-y)^\alpha I(a)] = I((\xi+D_x)^\beta(D_y-\xi)^\gamma(-D_\xi)^\alpha a), \tag{3.2.1}$$

and the amplitude on the right has order $m+|\beta|+|\gamma|-|\alpha|$. If this is less than $-n$ the oscillatory integral converges absolutely and uniformly, so for $|\alpha| = N > m+n+k$ the function $k_\alpha = (x-y)^\alpha I(a)$ is of class $\mathcal{C}^k$ with all derivatives of order at most k bounded.

Suppose $\rho(z) = 0$ for $|z| < \delta$. Using $|z|^{2N} = \sum_{|\alpha|=N} \frac{N!}{\alpha!} z^{2\alpha}$ we can write

$$\rho(z) = \sum_{|\alpha|=N} \frac{N!}{\alpha!} h_\alpha(z) z^\alpha, \qquad h_\alpha(z) = \frac{\rho(z)z^\alpha}{|z|^{2N}},$$

where $h_\alpha \in \mathcal{C}^\infty(\mathbb{R}^n)$ and, since $z^\alpha|z|^{-2N}$ is homogeneous of degree $-N$ on $|z| \ge \delta$, $|D^\mu h_\alpha(z)| \le C_\mu\langle z\rangle^{-N}$. Then $\rho(x-y)I(a) = \sum_{|\alpha|=N} \frac{N!}{\alpha!} h_\alpha(x-y)k_\alpha$ as distributions, and by Leibniz' formula this is a $\mathcal{C}^k$ function whose derivatives of order at most k are bounded by $C\langle x-y\rangle^{-N}$. Since k and $N > m+n+k$ are arbitrary, the lemma follows. □

Proposition 3.2.2. *If $A \in \Psi^m(\mathbb{R}^n)$ then* $\operatorname{sing\,supp}(K_A) \subset \mathrm{Diag}$. *Moreover, if $\phi,\psi \in \mathcal{C}_{\mathrm{b}}^\infty(\mathbb{R}^n)$ have supports at positive distance from each other, then $\phi A\psi \in \Psi^{-\infty}(\mathbb{R}^n)$.*

Proof. Write $K_A = I(a)$ with $a = \sigma_L(A)(x,\xi)$. If $\bar{x} \ne \bar{y}$, choose ρ as in Lemma 3.2.1 with $\rho = 1$ near $\bar{x}-\bar{y}$; then $K_A = \rho(x-y)K_A$ near $(\bar{x},\bar{y})$, which is smooth. For the second statement, if $|x-y| \ge \delta > 0$ whenever $x \in \operatorname{supp}\phi$, $y \in \operatorname{supp}\psi$, choose ρ with $\rho(z) = 1$ for $|z| \ge \delta$ and $\rho(z) = 0$ for $|z| \le \delta/2$. The kernel of $\phi A\psi$ is $\phi(x)\psi(y)K_A = \phi(x)\psi(y)\rho(x-y)K_A$, and multiplication by $\phi(x)\psi(y)$, a function with bounded derivatives, preserves the estimates of Lemma 3.2.1. □

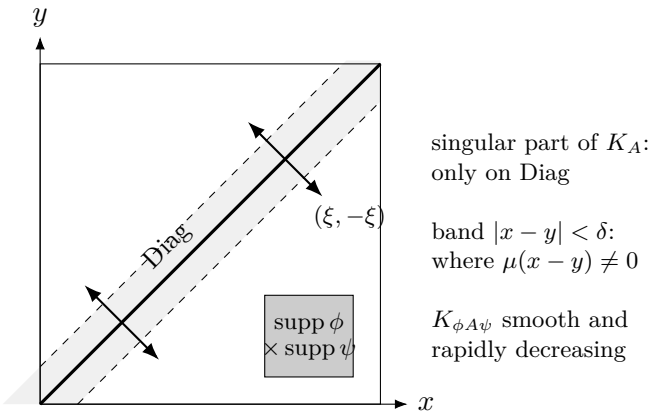


Figure 3.1: The Schwartz kernel K_A of $A \in \Psi^m(\mathbb{R}^n)$, drawn schematically for $n = 1$. It is singular only on the diagonal Diag. Cutting off to the band $|x-y| < \delta$ (shaded) changes A by a smoothing operator (Lemma 3.2.1), and cut-offs ϕ, ψ with separated supports give $\phi A\psi \in \Psi^{-\infty}$ (Proposition 3.2.2). The arrows indicate the wavefront set of K_A: covectors $(\xi,-\xi)$, normal to the diagonal (Proposition 3.10.1).

We also note how residual operators act on distributions of compact support.

Lemma 3.2.3. *If $E \in \Psi^{-\infty}(\mathbb{R}^n)$ and $u \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ then $Eu \in \mathcal{S}(\mathbb{R}^n)$.*

Proof. $Eu(x) = u(K_E(x,\cdot))$, and differentiating under u, $x^\alpha D^\gamma Eu(x) = u\big(x^\alpha D_x^\gamma K_E(x,\cdot)\big)$. By (3.1.1) this is bounded by $C \sup_{y\in L,|\beta|\le k} |x^\alpha D_x^\gamma D_y^\beta K_E(x,y)|$, which is finite since $\langle x\rangle \le C_L\langle x-y\rangle$ for y in the compact set L and $|D_x^\gamma D_y^\beta K_E| \le C_N\langle x-y\rangle^{-N}$. □

Definition 3.2.4. A continuous linear operator $A : \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ is *pseudolocal* if $\operatorname{sing\,supp}(Au) \subset \operatorname{sing\,supp}(u)$ for all $u \in \mathcal{S}'(\mathbb{R}^n)$.

Proposition 3.2.5. *Pseudodifferential operators are pseudolocal.*

Proof. Let $A \in \Psi^m(\mathbb{R}^n)$, $u \in \mathcal{S}'(\mathbb{R}^n)$ and $\bar{x} \notin \operatorname{sing\,supp} u$. Choose $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\phi = 1$ on a neighbourhood U of $\bar{x}$ and $\phi u \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, and $\psi \in \mathcal{C}_c^\infty(U)$ with $\psi = 1$ near $\bar{x}$. Then

$$\psi Au = \psi A(\phi u) + \psi A(1-\phi)u.$$

The first term is in $\mathcal{S}(\mathbb{R}^n)$, since A preserves $\mathcal{S}$. Since $\operatorname{supp}\psi$ is a compact subset of the open set U and $1-\phi = 0$ on U, the supports of ψ and $1-\phi$ are at positive distance, so $\psi A(1-\phi) \in \Psi^{-\infty}(\mathbb{R}^n)$ by Proposition 3.2.2, and the second term is smooth. Hence $\psi Au \in \mathcal{C}^\infty$ and $\bar{x} \notin \operatorname{sing\,supp}(Au)$. □

Question. Differential operators are local, and pseudodifferential operators preserve smoothness. Why are pseudodifferential operators not local as well?

Answer. Locality concerns supports, pseudolocality concerns singular supports. The kernel of $\langle D\rangle^{-2}$ on $\mathbb{R}$ is $\frac{1}{2}e^{-|x-y|}$, which vanishes nowhere, so $\langle D\rangle^{-2}$ moves supports everywhere. But this kernel is smooth off the diagonal, so what A adds far from $\operatorname{supp} u$ is smooth. The proof above splits u into a piece near $\bar{x}$ and a piece far away, and the far piece only meets the smooth part of the kernel. □

3.3 Cone support and essential support

The symbol of an operator tells us how strongly it acts at each position and frequency. If the symbol is rapidly decreasing in a cone around $(\bar{x},\bar{\xi})$, the operator is negligible there. The cone support collects the points where this fails.

Discussion. Why cones, and not ordinary neighbourhoods in $\mathbb{R}^p \times \mathbb{R}^n$? Smoothness is a statement about large frequencies: changing a symbol on a bounded set of ξ only changes the operator by a smoothing one. As $|\xi| \to \infty$ the only information that survives is the direction $\xi/|\xi|$. The cones $\Gamma_\epsilon(\bar{x},\bar{\xi})$ are the neighbourhoods of a point in the "sphere at infinity" of each fibre, and conic sets are those which see only this direction.

Definition 3.3.1. The *cone support* of $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ is the set $\operatorname{cone\,supp}(a) \subset \mathbb{R}^p \times (\mathbb{R}^n \setminus 0)$ of points $(\bar{x},\bar{\xi})$ such that a is *not* rapidly decreasing on any set $\Gamma_\epsilon(\bar{x},\bar{\xi})$, $\epsilon > 0$.

The complement of $\operatorname{cone\,supp}(a)$ is open and conic, so $\operatorname{cone\,supp}(a)$ is a closed conic subset of $\mathbb{R}^p \times (\mathbb{R}^n \setminus 0)$. Only the behaviour of a for large $|\xi|$ matters: changing a on a set where $|\xi|$ is bounded does not change the cone support. A priori the definition only involves a itself, but in fact all derivatives decay as well.

Lemma 3.3.2. *Let $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ be rapidly decreasing on $\Gamma_\epsilon(\bar{x},\bar{\xi})$. Then for every $\epsilon' < \epsilon$ and all α,β, $D_x^\alpha D_\xi^\beta a$ is rapidly decreasing on $\Gamma_{\epsilon'}(\bar{x},\bar{\xi})$. In particular* $\operatorname{cone\,supp}(D_x^\alpha D_\xi^\beta a) \subset \operatorname{cone\,supp}(a)$.

Proof. Since $D_{x_j}a \in S^m$ and $D_{\xi_j}a \in S^{m-1}$, by induction (with finitely many intermediate values of ϵ between ϵ' and ϵ) it suffices to show that $\partial_{x_j}a$ and $\partial_{\xi_j}a$ are rapidly decreasing on $\Gamma_{\epsilon''} = \Gamma_{\epsilon''}(\bar{x},\bar{\xi})$ for any $\epsilon'' < \epsilon$. Fix $M \geq |m|$; we show that both derivatives are $O(\langle\xi\rangle^{(m-M)/2})$ on $\Gamma_{\epsilon''}$, which suffices as M is arbitrary. Since all derivatives of a are $O(\langle\xi\rangle^m)$ we may restrict to $|\xi| \geq R$, for an R to be chosen.

Let $(x,\xi) \in \Gamma_{\epsilon''}$ and $0 < t \leq \epsilon - \epsilon''$. Then $(x + se_j, \xi) \in \Gamma_\epsilon$ for $|s| \leq t$, and Taylor's formula gives

$$|t\,\partial_{x_j}a(x,\xi)| \leq |a(x+te_j,\xi)| + |a(x,\xi)| + \tfrac{1}{2}t^2 \sup_{|s|\leq t}|\partial_{x_j}^2 a(x+se_j,\xi)| \leq 2C_M\langle\xi\rangle^{-M} + Ct^2\langle\xi\rangle^m.$$

Taking $t = (\epsilon - \epsilon'')\langle\xi\rangle^{-(M+m)/2}$ gives $|\partial_{x_j}a(x,\xi)| \leq C'\langle\xi\rangle^{(m-M)/2}$.

For the ξ-derivative let $0 < t \leq 1$ and $|s| \leq t$. Then $|\frac{\xi+se_j}{|\xi+se_j|} - \frac{\xi}{|\xi|}| \leq \frac{2}{|\xi|}$, so $(x,\xi+se_j) \in \Gamma_\epsilon$ if $|\xi| \geq R = 2/(\epsilon-\epsilon'') + 1$; also $\langle\xi + se_j\rangle \geq \frac{1}{2}\langle\xi\rangle$. As before

$$|t\,\partial_{\xi_j}a(x,\xi)| \leq C\langle\xi\rangle^{-M} + Ct^2\langle\xi\rangle^m,$$

and $t = \langle\xi\rangle^{-(M+m)/2}$ gives $|\partial_{\xi_j}a| \leq C'\langle\xi\rangle^{(m-M)/2}$. □

Corollary 3.3.3. *Let $a \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ and let $K \subset \mathbb{R}^p$ be compact with* $\operatorname{cone\,supp}(a) \cap (K \times (\mathbb{R}^n \setminus 0)) = \emptyset$. *Then $\phi a \in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$ for every $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^p)$ with* $\operatorname{supp}\phi \subset K$. *In particular, if a has compact support in x and* $\operatorname{cone\,supp}(a) = \emptyset$, *then $a \in S^{-\infty}(\mathbb{R}^p;\mathbb{R}^n)$.*

Proof. Each point of the compact set $K \times \mathbb{S}^{n-1}$ has a neighbourhood $\Gamma_\epsilon(\bar{x},\bar{\xi})$ on which a is rapidly decreasing, hence (by Lemma 3.3.2) with all derivatives on $\Gamma_{\epsilon/2}(\bar{x},\bar{\xi})$. Finitely many of the latter cover $K \times (\mathbb{R}^n \setminus 0)$, so every $D_x^\alpha D_\xi^\beta a$ is rapidly decreasing on $K \times \mathbb{R}^n$, and by Leibniz' formula $\phi a \in S^{-\infty}$. □

Remark 3.3.4. The compactness in x cannot be dropped: a symbol may have empty cone support without being in $S^{-\infty}$, because the definition only controls it locally uniformly in x; see Exercise 3.6.

Definition 3.3.5. The *essential support* (or *operator wavefront set*) of $A \in \Psi^m(\mathbb{R}^n)$ is the closed conic set

$$\mathrm{WF}'(A) = \operatorname{cone\,supp}(\sigma_L(A)) \subset T^*\mathbb{R}^n \setminus 0.$$

The essential support can be computed from any amplitude.

Lemma 3.3.6. *If $A = \mathrm{Op}(a)$ with $a \in S^m(\mathbb{R}^{2n};\mathbb{R}^n)$ then*

$$\mathrm{WF}'(A) \subset \{(x,\xi);\ (x,x,\xi) \in \operatorname{cone\,supp}(a)\}.$$

Proof. Suppose $(\bar{x},\bar{x},\bar{\xi}) \notin \operatorname{cone\,supp}(a)$, so a is rapidly decreasing on $\{|x-\bar{x}| < \epsilon, |y-\bar{x}| < \epsilon, \xi \in V_\epsilon(\bar{\xi})\}$ for some $\epsilon > 0$. By Lemma 3.3.2 (applied with (x,y) as base variables) all derivatives of a are rapidly decreasing on the corresponding set with $\epsilon/2$ in place of ϵ, so each term $D_\xi^\alpha D_y^\alpha a(x,y,\xi)|_{y=x}$ in (3.0.2) is rapidly decreasing on $\Gamma_{\epsilon/2}(\bar{x},\bar{\xi})$. By (3.0.2), $\sigma_L(A) - \sum_{|\alpha|<N} \frac{i^{|\alpha|}}{\alpha!} D_\xi^\alpha D_y^\alpha a|_{y=x} \in S^{m-N}$, so $|\sigma_L(A)| \leq C_N\langle\xi\rangle^{m-N}$ on $\Gamma_{\epsilon/2}(\bar{x},\bar{\xi})$ for every N, that is $(\bar{x},\bar{\xi}) \notin \mathrm{WF}'(A)$. □

Proposition 3.3.7. *For $A \in \Psi^m(\mathbb{R}^n)$ and $B \in \Psi^{m'}(\mathbb{R}^n)$:*

(i) $\mathrm{WF}'(A) = \operatorname{cone\,supp}(\sigma_R(A))$;

(ii) $\mathrm{WF}'(A+B) \subset \mathrm{WF}'(A) \cup \mathrm{WF}'(B)$ *(if* $m = m'$*) and* $\mathrm{WF}'(AB) \subset \mathrm{WF}'(A) \cap \mathrm{WF}'(B)$*;*

(iii) $\mathrm{WF}'(A^*) = \mathrm{WF}'(A)$ *and* $\mathrm{WF}'(A^t) = \mathrm{WF}'(A)^-$*;*

(iv) $\mathrm{WF}'(\mathrm{Id}) = T^*\mathbb{R}^n \setminus 0$, $\mathrm{WF}'(E) = \emptyset$ *for* $E \in \Psi^{-\infty}(\mathbb{R}^n)$*, and for multiplication by* $\phi \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$, $\mathrm{WF}'(\phi) = \operatorname{supp}\phi \times (\mathbb{R}^n \setminus 0)$.

Proof. (i) $A = \mathrm{Op}(b(y,\xi))$ with $b = \sigma_R(A)$, and $(x,x,\xi) \in \operatorname{cone\,supp}(b(y,\xi))$ if and only if $(x,\xi) \in \operatorname{cone\,supp}(b)$, so $\mathrm{WF}'(A) \subset \operatorname{cone\,supp}(\sigma_R(A))$ by Lemma 3.3.6. Conversely $\sigma_R(A) \sim \sum_\beta \frac{(-i)^{|\beta|}}{\beta!} D_\xi^\beta D_x^\beta \sigma_L(A)$, and the argument in the proof of Lemma 3.3.6 shows $\operatorname{cone\,supp}(\sigma_R(A)) \subset \operatorname{cone\,supp}(\sigma_L(A))$.

(ii) The first statement is clear. For the second, let $a = \sigma_L(A)$ and $b = \sigma_R(B)$, so that $AB = \mathrm{Op}(a(x,\xi)b(y,\xi))$. If $(\bar x,\bar\xi) \notin \mathrm{WF}'(A)$ then $a(x,\xi)b(y,\xi)$ is rapidly decreasing on $\{|x-\bar x| < \epsilon, \xi \in V_\epsilon(\bar\xi)\}$ for some ϵ, since b is polynomially bounded; if $(\bar x,\bar\xi) \notin \mathrm{WF}'(B) = \operatorname{cone\,supp}(b)$, the same holds with the roles of x and y exchanged. In either case $(\bar x,\bar x,\bar\xi) \notin \operatorname{cone\,supp}(a(x,\xi)b(y,\xi))$, and Lemma 3.3.6 applies.

(iii) The adjoint of $\mathrm{Op}(a)$, $a = \sigma_L(A)$, is $\mathrm{Op}(\overline{a(y,\xi)})$, and its transpose is $\mathrm{Op}(a(y,-\xi))$ (the kernel of A^t is $K_A(y,x) = \int e^{i(x-y)\cdot\xi} a(y,-\xi)\text{đ}\xi$). By Lemma 3.3.6, $\mathrm{WF}'(A^*) \subset \mathrm{WF}'(A)$ and $\mathrm{WF}'(A^t) \subset \mathrm{WF}'(A)^-$; applying this to A^* and A^t gives equality.

(iv) is immediate from the definition, since $\sigma_L(\mathrm{Id}) = 1$, $\sigma_L(E) \in S^{-\infty}$ and $\sigma_L(\phi) = \phi(x)$. □

Lemma 3.3.8. *Let* $A \in \Psi^m(\mathbb{R}^n)$ *and* $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ *with* $\mathrm{WF}'(A) \cap (\operatorname{supp}\phi \times (\mathbb{R}^n \setminus 0)) = \emptyset$*. Then* ϕA *and* $A\phi$ *lie in* $\Psi^{-\infty}(\mathbb{R}^n)$*. In particular, if* $\mathrm{WF}'(A) = \emptyset$ *then* $\phi A, A\phi \in \Psi^{-\infty}(\mathbb{R}^n)$ *for all* $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$*.*

Proof. $\sigma_L(\phi A) = \phi\sigma_L(A) \in S^{-\infty}$ by Corollary 3.3.3. Similarly $A\phi = \mathrm{Op}(\sigma_R(A)(y,\xi)\phi(y)) = \mathrm{Op}_R(\phi\sigma_R(A))$ and $\phi\sigma_R(A) \in S^{-\infty}$, by Proposition 3.3.7(i) and Corollary 3.3.3. □

Note that $\mathrm{WF}'(A) = \emptyset$ does *not* imply $A \in \Psi^{-\infty}(\mathbb{R}^n)$ (Remark 3.3.4); this is why compactly supported cut-offs appear in many arguments below.

3.3.1 Microlocal cut-offs

We shall constantly use operators which are "microlocally the identity" near a given closed cone and vanish microlocally away from it. Fix $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $0 \le \chi \le 1$, $\chi(t) = 1$ for $|t| \le \frac12$ and $\chi(t) = 0$ for $|t| \ge 1$. For $z = (\bar x,\bar\xi) \in T^*\mathbb{R}^n \setminus 0$ and $0 < \epsilon < 1$ set

$$\gamma_{z,\epsilon}(x,\xi) = \chi\Big(\frac{x-\bar x}{\epsilon}\Big)\chi\Big(\frac{1}{\epsilon}\Big(\frac{\xi}{|\xi|} - \frac{\bar\xi}{|\bar\xi|}\Big)\Big)\big(1-\chi(\epsilon\xi)\big). \tag{3.3.1}$$

The last factor vanishes for $|\xi| \le 1/(2\epsilon)$, and functions which are smooth and homogeneous of degree 0 in $|\xi| \ge c > 0$ are symbols of order 0 there, so $\gamma_{z,\epsilon} \in S^0(\mathbb{R}^n;\mathbb{R}^n)$. It takes values in $[0,1]$, is supported in $\overline{\Gamma_\epsilon(z)}$ and equals 1 on $\Gamma_{\epsilon/2}(z) \cap \{|\xi| \ge 1/\epsilon\}$. Hence $G_{z,\epsilon} = \mathrm{Op}(\gamma_{z,\epsilon}) \in \Psi^0(\mathbb{R}^n)$ satisfies

$$\mathrm{WF}'(G_{z,\epsilon}) \subset \overline{\Gamma_\epsilon(z)}, \qquad \mathrm{WF}'(\mathrm{Id} - G_{z,\epsilon}) \cap \Gamma_{\epsilon/2}(z) = \emptyset, \qquad G_{z,\epsilon} = \psi G_{z,\epsilon} \tag{3.3.2}$$

for any $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to 1 on $\{|x-\bar x| \le \epsilon\}$. Figure 3.2 shows the support of $\gamma_{z,\epsilon}$ in one fibre.

Lemma 3.3.9 (Microlocal cut-off)**.** *Let* $W_1, W_2 \subset T^*\mathbb{R}^n \setminus 0$ *be disjoint closed conic sets with* $\pi(W_1)$ *bounded, where* $\pi(x,\xi) = x$*. Then there is* $a \in S^0(\mathbb{R}^n;\mathbb{R}^n)$*, real-valued with compact support in* x*, such that* $C = \mathrm{Op}(a)$ *satisfies*

$$\mathrm{WF}'(C) \cap W_2 = \emptyset, \qquad \mathrm{WF}'(\mathrm{Id} - C) \cap W_1 = \emptyset.$$

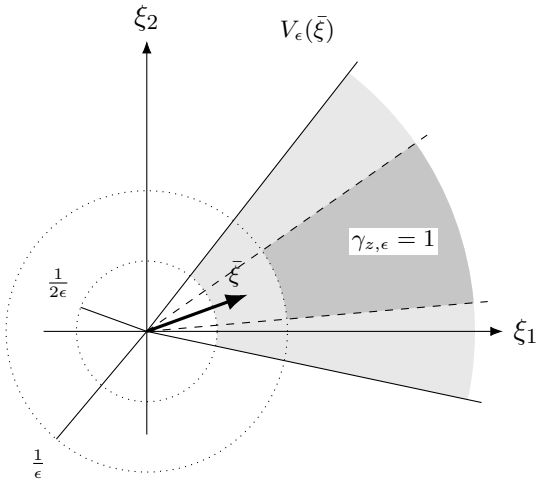


Figure 3.2: One fibre $\{x\} \times \mathbb{R}^2_\xi$, $|x - \bar x| < \epsilon/2$, of the cut-off $\gamma_{z,\epsilon}$ of (3.3.1). The open cone $V_\epsilon(\bar\xi)$ is bounded by the solid rays, $V_{\epsilon/2}(\bar\xi)$ by the dashed rays. The symbol $\gamma_{z,\epsilon}$ vanishes outside the light gray region and equals 1 on the dark region $V_{\epsilon/2}(\bar\xi) \cap \{|\xi| \geq 1/\epsilon\}$. Only the behaviour for large $|\xi|$ matters, so near infinity the support is a cone.

Proof. The set $W_1 \cap (\mathbb{R}^n \times \mathbb{S}^{n-1})$ is compact. For each of its points z there is $\epsilon_z > 0$ with $\overline{\Gamma_{2\epsilon_z}(z)} \cap W_2 = \emptyset$, since W_2 is closed and conic. Choose finitely many $z_1, \dots, z_N$ such that the cones $\Gamma_{\epsilon_j/2}(z_j)$, $\epsilon_j = 2\epsilon_{z_j}$, cover W_1, and put $g = \sum_j \gamma_{z_j,\epsilon_j} \in S^0$. Then $g \geq 0$, $g \geq 1$ on $\bigcup_j \Gamma_{\epsilon_j/2}(z_j) \cap \{|\xi| \geq R\}$ with $R = \max_j 1/\epsilon_j$, and $\operatorname{supp} g \subset \bigcup_j \overline{\Gamma_{\epsilon_j}(z_j)}$. Let $\Phi \in \mathcal{C}^\infty(\mathbb{R})$ with $\Phi(t) = 0$ for $t \leq 0$ and $\Phi(t) = 1$ for $t \geq 1$, and set $a = \Phi \circ g$. By the chain rule $a \in S^0$, and a has compact support in x. Since a vanishes outside $\bigcup_j \overline{\Gamma_{\epsilon_j}(z_j)}$, which is a closed cone disjoint from W_2, $\mathrm{WF}'(C) \cap W_2 = \emptyset$. Since $1 - a$ vanishes on the open cone $\bigcup_j \Gamma_{\epsilon_j/2}(z_j)$ for $|\xi| \geq R$, and this cone contains W_1, $\mathrm{WF}'(\mathrm{Id} - C) \cap W_1 = \emptyset$. □

3.4 The characteristic variety

Global ellipticity asks the symbol to be large in every direction at every point. Microlocally we only ask this near one point of $T^*\mathbb{R}^n \setminus 0$. The set where even this weak condition fails is the characteristic variety; it is the only place where an equation $Au = f$ can hide singularities of u (Theorem 3.7.8).

Definition 3.4.1. $A \in \Psi^m(\mathbb{R}^n)$ is *elliptic at* $(\bar x, \bar\xi) \in T^*\mathbb{R}^n \setminus 0$ if there is $\epsilon > 0$ such that

$$|\sigma_L(A)(x,\xi)| \geq \epsilon|\xi|^m \quad \text{on } \Gamma_\epsilon(\bar x, \bar\xi) \cap \{|\xi| \geq 1/\epsilon\}.$$

The set of such points is the *elliptic set* $\mathrm{Ell}_m(A)$; its complement $\mathrm{Char}_m(A) = T^*\mathbb{R}^n \setminus 0 \setminus \mathrm{Ell}_m(A)$ is the *characteristic variety* of A.

If $a' - \sigma_L(A) \in S^{m-1}$ then $|a' - \sigma_L(A)| \leq C\langle\xi\rangle^{m-1} \leq \frac{\epsilon}{2}|\xi|^m$ for $|\xi|$ large, so the condition holds for a' with $\epsilon/2$ in place of ϵ (after shrinking the cone). Thus ellipticity at a point depends only on the principal symbol $\sigma_m(A)$; in particular $\sigma_R(A)$ may be used instead of $\sigma_L(A)$. By definition $\mathrm{Ell}_m(A)$ is open and conic, so $\mathrm{Char}_m(A)$ is a closed conic subset of $T^*\mathbb{R}^n \setminus 0$. If A is elliptic at z then $\sigma_L(A)$ is not rapidly decreasing near z, so

$$\mathrm{Ell}_m(A) \subset \mathrm{WF}'(A).$$

For example, multiplication by $\phi \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ is elliptic (of order 0) at $(\bar{x}, \bar{\xi})$ if and only if $\phi(\bar{x}) \neq 0$, and $G_{z,\epsilon}$ in (3.3.2) is elliptic at z. The operator A is globally elliptic in the sense of Chapter 2 if and only if the estimate in Definition 3.4.1 holds uniformly, with one ϵ for all $(\bar{x}, \bar{\xi})$.

Discussion. Two examples show the range of possibilities. The operator $\langle D\rangle^2$, with symbol $\langle\xi\rangle^2 = 1 + |\xi|^2$, is elliptic everywhere: $\mathrm{Char}_2 = \emptyset$. The operator D_{x_1} on $\mathbb{R}^2$ has symbol ξ_1, and $|\xi_1| \geq \epsilon|\xi|$ near $(\bar{x}, \bar{\xi})$ exactly when $\bar{\xi}_1 \neq 0$. So $\mathrm{Char}_1(D_{x_1}) = \{\xi_1 = 0\}$: the operator controls every direction except the ones conormal to the lines $\{x_2 = \mathrm{const}\}$. For such operators the behaviour on the characteristic variety is the subject of Section 3.20.

Proposition 3.4.2. *For $A \in \Psi^m(\mathbb{R}^n)$ and $B \in \Psi^{m'}(\mathbb{R}^n)$,*

$$\mathrm{Char}_{m+m'}(AB) = \mathrm{Char}_m(A) \cup \mathrm{Char}_{m'}(B).$$

Proof. We may use the representatives $a = \sigma_L(A)$, $b = \sigma_L(B)$ and ab of the principal symbols, since $\sigma_L(AB) - ab \in S^{m+m'-1}$. If A and B are elliptic at z then $|ab| \geq \epsilon^2|\xi|^{m+m'}$ on a suitable $\Gamma_\epsilon(z) \cap \{|\xi| \geq 1/\epsilon\}$, so AB is elliptic at z. Conversely if $|ab| \geq \epsilon|\xi|^{m+m'}$ there, then since $|b| \leq C\langle\xi\rangle^{m'} \leq 2^{|m'|}C|\xi|^{m'}$ for $|\xi| \geq 1$, we get $|a| \geq \epsilon'|\xi|^m$, so A is elliptic at z, and similarly B is. □

3.5 Microlocal parametrices

The essential support allows us to refine the notion of a parametrix of an elliptic operator to that of a parametrix at a point.

Proposition 3.5.1. *If $A \in \Psi^m(\mathbb{R}^n)$ and $z \notin \mathrm{Char}_m(A)$, there exists a* microlocal parametrix *for A at z: an operator $B \in \Psi^{-m}(\mathbb{R}^n)$ with*

$$z \notin \mathrm{WF}'(AB - \mathrm{Id}) \quad \textit{and} \quad z \notin \mathrm{WF}'(BA - \mathrm{Id}).$$

Moreover there is then an open conic neighbourhood Γ of z with $\Gamma \cap (\mathrm{WF}'(AB - \mathrm{Id}) \cup \mathrm{WF}'(BA - \mathrm{Id})) = \emptyset$.

Proof. The last statement holds for any B with the first property, as essential supports are closed. The idea is to reduce to the global parametrix of Chapter 2. Away from z we add to A^*A a harmless elliptic term, which makes it globally elliptic; inverting it gives a left parametrix near z. A right parametrix is obtained in the same way, and a standard algebraic argument shows that the two agree near z.

Let $a = \sigma_L(A)$ and choose $\epsilon \in (0, \frac{1}{2})$ and $c_0 > 0$ such that $|a| \geq c_0|\xi|^m$ on $\Gamma_{2\epsilon}(z) \cap \{|\xi| \geq 1/(2\epsilon)\}$. Let $\gamma = \gamma_{z,\epsilon}$ and $L = \mathrm{Op}(\gamma)$. Then $z \notin \mathrm{WF}'(\mathrm{Id} - L)$ by (3.3.2), and $\operatorname{supp}\gamma \subset \overline{\Gamma_\epsilon(z)} \cap \{|\xi| \geq 1/(2\epsilon)\}$, so that $|a| \geq c_0|\xi|^m$ on $\operatorname{supp}\gamma$.

Let $\Lambda = \mathrm{Op}(\langle\xi\rangle^{2m}) \in \Psi^{2m}(\mathbb{R}^n)$ and consider

$$J = (\mathrm{Id} - L)\Lambda + A^*A \in \Psi^{2m}(\mathbb{R}^n).$$

Its principal symbol is represented by $(1 - \gamma)\langle\xi\rangle^{2m} + |a|^2$. Since $0 \leq \gamma \leq 1$, at a point with $|\xi| \geq R$ either $\gamma \leq \frac{1}{2}$, and then this is at least $\frac{1}{2}\langle\xi\rangle^{2m}$, or $\gamma > \frac{1}{2}$, and then the point is in $\operatorname{supp}\gamma$ and the expression is at least $c_0^2|\xi|^{2m} \geq c\langle\xi\rangle^{2m}$. Since $\sigma_L(J)$ differs from this representative by a symbol of order $2m - 1$, J is globally elliptic and has a parametrix $H \in \Psi^{-2m}(\mathbb{R}^n)$, $HJ - \mathrm{Id} = E_1 \in \Psi^{-\infty}$. Set $B_1 = HA^* \in \Psi^{-m}(\mathbb{R}^n)$. Then

$$B_1A - \mathrm{Id} = H\big(J - (\mathrm{Id} - L)\Lambda\big) - \mathrm{Id} = E_1 - H(\mathrm{Id} - L)\Lambda,$$

and $\mathrm{WF}'(H(\mathrm{Id}-L)\Lambda) \subset \mathrm{WF}'(\mathrm{Id}-L) \not\ni z$ by Proposition 3.3.7. So $z \notin \mathrm{WF}'(B_1A-\mathrm{Id})$: B_1 is a left microlocal parametrix.

In the same way $J' = (\mathrm{Id}-L)\Lambda + AA^*$ is globally elliptic; if H' is a parametrix, $J'H' - \mathrm{Id} = E_2 \in \Psi^{-\infty}$, then $B_2 = A^*H'$ satisfies $AB_2 - \mathrm{Id} = E_2 - (\mathrm{Id}-L)\Lambda H'$, so $z \notin \mathrm{WF}'(AB_2 - \mathrm{Id})$. Write $R_1 = B_1A - \mathrm{Id}$ and $R_2 = AB_2 - \mathrm{Id}$. Then

$$B_2 = (B_1A - R_1)B_2 = B_1(\mathrm{Id}+R_2) - R_1B_2 = B_1 + B_1R_2 - R_1B_2,$$

so $z \notin \mathrm{WF}'(B_2 - B_1)$, and $AB_1 - \mathrm{Id} = R_2 + A(B_1 - B_2)$ shows $z \notin \mathrm{WF}'(AB_1 - \mathrm{Id})$. Thus $B = B_1$ is a two-sided microlocal parametrix. □

The same argument shows that a microlocal parametrix at z is unique up to an operator whose essential support does not contain z (Exercise 3.11).

3.6 Coordinate invariance

To use pseudodifferential operators on manifolds we must know that a change of coordinates turns them into pseudodifferential operators, and how their symbols transform. The answer is that the principal symbol behaves like a function on the cotangent bundle. We first set up pull-backs of compactly supported distributions by diffeomorphisms.

For an open set $\Omega \subset \mathbb{R}^n$ let $\mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ be the smooth functions with compact support in Ω and

$$\mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega) = \{u \in \mathcal{S}'(\mathbb{R}^n);\ \operatorname{supp} u \text{ is a compact subset of } \Omega\}.$$

An element $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega)$ acts on $\mathcal{C}^\infty(\Omega)$ by $u(\phi) = u(\psi\phi)$, $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ equal to 1 near $\operatorname{supp} u$.

Let $F : \Omega \to \Omega'$ be a diffeomorphism between open subsets of $\mathbb{R}^n$, with inverse G, and let $F^*\phi = \phi \circ F$, so $F^* : \mathcal{C}^\infty_{\mathrm{c}}(\Omega') \to \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$. If $u \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ and $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$, the change of variables $y = F(x)$ gives

$$\int u(F(x))\phi(x)\,dx = \int u(y)\,\phi(G(y))\,|\det DG(y)|\,dy. \tag{3.6.1}$$

Lemma 3.6.1. *For $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega')$ the formula*

$$(F^*u)(\phi) = u\big(|\det DG|\,G^*\phi\big), \qquad \phi \in \mathcal{S}(\mathbb{R}^n),$$

(where $G^*\phi = \phi \circ G \in \mathcal{C}^\infty(\Omega')$) defines $F^*u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega)$ with $\operatorname{supp} F^*u = G(\operatorname{supp} u)$. It extends F^* on $\mathcal{C}^\infty_{\mathrm{c}}(\Omega')$, is continuous in the sense that $F^*u_j \to F^*u$ weakly if $u_j \to u$ weakly and the supports of the u_j lie in a fixed compact subset of Ω', and satisfies $(F_1 \circ F_2)^* = F_2^*F_1^*$; in particular $G^*F^* = \mathrm{Id}$.

Proof. u acts on $\mathcal{C}^\infty(\Omega')$, and $\phi \mapsto |\det DG|G^*\phi$ is continuous from $\mathcal{S}(\mathbb{R}^n)$ to $\mathcal{C}^\infty(\Omega')$, so $F^*u \in \mathcal{S}'(\mathbb{R}^n)$. If ϕ vanishes near $G(\operatorname{supp} u)$ then $|\det DG|G^*\phi$ vanishes near $\operatorname{supp} u$, so $\operatorname{supp} F^*u \subset G(\operatorname{supp} u)$, and equality follows from $G^*F^*u = u$. The consistency with smooth functions is (3.6.1), continuity is clear, and the composition law follows from the chain rule for Jacobians. □

If $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega')$ and $\rho_\epsilon(x) = \epsilon^{-n}\rho(x/\epsilon)$ with $\rho \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, $\int \rho = 1$, then $\rho_\epsilon * u \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ for small ϵ, with supports in a fixed compact set, and $\rho_\epsilon * u \to u$ weakly. So $\mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ is weakly dense in $\mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega')$ in this strong sense (see also Exercise 3.5), and F^* is the unique such continuous extension.

Now let $A : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ have kernel with compact support in $\Omega \times \Omega$. By Proposition 3.1.1, A maps $\mathcal{C}^\infty_{\mathrm{c}}(\Omega)$, indeed $\mathcal{S}(\mathbb{R}^n)$, into $\mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega)$, and we define

$$A_F = G^* \circ A \circ F^* : \mathcal{C}^\infty_{\mathrm{c}}(\Omega') \to \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega'). \tag{3.6.2}$$

By (3.6.1) its kernel is

$$K_{A_F}(x', y') = K_A(G(x'), G(y'))\, |\det DG(y')|,$$

that is, $K_{A_F} = |\det DG(y')|\,(G \times G)^* K_A$, where $(G \times G)^*$ is the pull-back of Lemma 3.6.1 for the diffeomorphism $G \times G : \Omega' \times \Omega' \to \Omega \times \Omega$; it has compact support in $\Omega' \times \Omega'$, and we extend it by zero to $\mathbb{R}^{2n}$, so $A_F : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$. The diffeomorphism F induces the map

$$\chi_F : \Omega \times \mathbb{R}^n \to \Omega' \times \mathbb{R}^n, \qquad \chi_F(x, \xi) = \big(F(x), (DF(x)^t)^{-1}\xi\big),$$

which is a diffeomorphism, linear and invertible on each fibre. (In the language of Chapter 5, χ_F is the inverse of the cotangent map $F^* : T^*_{F(x)}\Omega' \to T^*_x\Omega$, $\eta \mapsto DF(x)^t\eta$.)

Discussion. Why does ξ transform by $(DF^t)^{-1}$ rather than by DF? In the oscillatory factor $e^{i(x-y)\cdot\xi}$ the frequency variable is paired with a displacement $x - y$. Near the diagonal $F(x) - F(y) \approx DF(x)(x - y)$, so keeping the phase $(x - y)\cdot\xi$ unchanged forces $\xi = DF(x)^t\eta$. Thus ξ is a covector, a linear form on displacements, and (x, ξ) is a point of the cotangent bundle. This is the computation behind the matrix $T(x', y')$ in the proof below.

Theorem 3.6.2 (Coordinate invariance)**.** *Let $F : \Omega \to \Omega'$ be a diffeomorphism of open subsets of $\mathbb{R}^n$ and $A \in \Psi^m(\mathbb{R}^n)$ an operator whose kernel has compact support in $\Omega \times \Omega$. Then $A_F \in \Psi^m(\mathbb{R}^n)$, its kernel has compact support in $\Omega' \times \Omega'$, and*

$$\sigma_m(A_F) \circ \chi_F = \sigma_m(A), \qquad \mathrm{WF}'(A_F) = \chi_F\big(\mathrm{WF}'(A)\big),$$
$$\mathrm{Char}_m(A_F) \cap \pi^{-1}(\Omega') = \chi_F\big(\mathrm{Char}_m(A) \cap \pi^{-1}(\Omega)\big),$$

where $\pi(x, \xi) = x$ and the first identity is understood as $\sigma_L(A_F)(F(x), (DF(x)^t)^{-1}\xi) - \sigma_L(A)(x, \xi) \in S^{m-1}$ for x in a neighbourhood of the support, both symbols vanishing for x outside compact subsets of Ω, Ω' respectively.

Proof. The proof has three steps. First we split $A = A_1 + A_2$, where A_2 is smoothing and A_1 has kernel supported near the diagonal. Then we change variables in the oscillatory integral for A_1 and check that the new amplitude is a symbol. Finally we read off the principal symbol and the essential support from the new amplitude.

Let $K \subset \Omega$ be compact with $\operatorname{supp} K_A \subset K \times K$, and let $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ equal 1 near K. Then $\psi A = A$, so $a = \sigma_L(A)$ satisfies $a = \psi a$ by uniqueness of the left symbol, and $A = \psi A \psi = \mathrm{Op}(b_0)$ with $b_0(x, y, \xi) = \psi(x)\psi(y)a(x, \xi)$. Similarly $\sigma_L(A_F)$ is supported in $F(K) \times \mathbb{R}^n$.

Let $L = \operatorname{supp}\psi$. Choose $\delta' > 0$ so small that the closed $2\delta'$-neighbourhood of $F(L)$ lies in Ω' and that, whenever $\operatorname{dist}(x', F(L)) \le \delta'$ and $|x' - y'| \le \delta'$, the matrix

$$T(x', y') = \int_0^1 DG(y' + t(x' - y'))\, dt$$

is invertible with $|\det T| \ge c > 0$ and $\|T^{-1}\| \le C$; this is possible since $T(x', x') = DG(x')$ and $F(L)$ is compact. Then $G(x') - G(y') = T(x', y')(x' - y')$. By uniform continuity of F on a compact neighbourhood of L, choose $\delta > 0$ such that $x, y \in L$, $|x - y| \le 2\delta$ imply $|F(x) - F(y)| < \delta'$.

Let $\mu \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal 1 on $|z| \le \delta$ and vanish for $|z| \ge 2\delta$, and split $A = A_1 + A_2$ with $K_{A_1} = \mu(x-y)K_A$, $K_{A_2} = (1-\mu(x-y))K_A$. By Lemma 3.2.1, $A_2 \in \Psi^{-\infty}(\mathbb{R}^n)$, so K_{A_2} is smooth, and it has compact support in $\Omega \times \Omega$. Hence $K_{(A_2)_F}$ is smooth with compact support in $\Omega' \times \Omega'$, in particular it satisfies the estimates characterizing $\Psi^{-\infty}(\mathbb{R}^n)$, and $(A_2)_F \in \Psi^{-\infty}(\mathbb{R}^n)$.

The kernel of A_1 is $I(b)$ with $b(x,y,\xi) = \mu(x-y)\psi(x)\psi(y)a(x,\xi) \in S^m(\mathbb{R}^{2n};\mathbb{R}^n)$, which vanishes unless $x, y \in L$ and $|x-y| \le 2\delta$. If $b \in S^{-\infty}$ the integral converges absolutely and the substitution $\xi = S(x',y')\eta$, $S = (T^t)^{-1}$, gives

$$\begin{aligned} K_{(A_1)_F}(x',y') &= \int e^{i(x'-y')\cdot\eta} c(x',y',\eta)\, đ\eta, \\ c(x',y',\eta) &= b(G(x'),G(y'),S(x',y')\eta)\,\frac{|\det DG(y')|}{|\det T(x',y')|}, \end{aligned} \tag{3.6.3}$$

where c is extended by zero outside the compact set $\{(x',y');\ G(x'),G(y') \in L,\ |G(x') - G(y')| \le 2\delta\}$; on this set $|x'-y'| < \delta'$, so c is smooth. For general b we apply this to $\chi(\epsilon\xi)b$ and let $\epsilon \to 0$: both sides of (3.6.3) converge in $\mathcal{S}'(\mathbb{R}^{2n})$, by the continuity of I on amplitudes and of the pull-back on compactly supported distributions. Thus $(A_1)_F = \mathrm{Op}(c)$, provided we show $c \in S^m(\mathbb{R}^{2n};\mathbb{R}^n)$.

Write $f(x',y')$ for the smooth factor in c. By the chain rule, $D_{x'}^\alpha D_{y'}^\beta D_\eta^\gamma [b(G(x'),G(y'),S\eta)]$ is a finite sum of terms

$$M(x',y')\,\eta^\nu \,(D_x^{\alpha'} D_y^{\beta'} D_\xi^\mu b)(G(x'),G(y'),S\eta), \qquad |\nu| \le |\mu| - |\gamma|,$$

with smooth coefficients M: each η-derivative produces one ξ-derivative of b and no power of η, while an x'- or y'-derivative falling on $S(x',y')\eta$ produces one ξ-derivative and at most one factor of η. On the support, $\langle S\eta\rangle$ is comparable to $\langle\eta\rangle$, so each term is bounded by $C\langle\eta\rangle^{|\nu|}\langle\eta\rangle^{m-|\mu|} \le C\langle\eta\rangle^{m-|\gamma|}$. With Leibniz' formula for the factor f, this shows $c \in S^m(\mathbb{R}^{2n};\mathbb{R}^n)$.

Thus $A_F = (A_1)_F + (A_2)_F \in \Psi^m(\mathbb{R}^n)$. By Chapter 2, $\sigma_m(A_F)$ is represented by $c(x',x',\eta)$. Since $T(x',x') = DG(x')$,

$$c(x',x',\eta) = \mu(0)\psi(G(x'))^2 a\big(G(x'),(DG(x')^t)^{-1}\eta\big) = a(x, DF(x)^t\eta), \quad x = G(x'),$$

using $DG(x')^{-1} = DF(x)$ and $\psi a = a$. With $(x',\eta) = \chi_F(x,\xi)$ this is $a(x,\xi)$, proving the statement about principal symbols. The same chain rule estimates show that composition with χ_F^{-1} maps symbols of order $m-1$ supported over L to symbols of order $m-1$, so the class is well defined.

For the essential support we use Lemma 3.3.6. Let $(\bar x',\bar\eta) \in \Omega' \times (\mathbb{R}^n \setminus 0)$ with $\chi_F^{-1}(\bar x',\bar\eta) = (\bar x,\bar\xi) \notin \mathrm{WF}'(A)$, so a, and hence $b(x,y,\xi)$ for y arbitrary, is rapidly decreasing for $(x,\xi) \in \Gamma_\epsilon(\bar x,\bar\xi)$. Since $S(\bar x',\bar x')\bar\eta = \bar\xi$ and S, G are continuous, for (x',y') close to $(\bar x',\bar x')$ and $\eta \in V_{\epsilon'}(\bar\eta)$ with ϵ' small we have $(G(x'),S(x',y')\eta) \in \Gamma_\epsilon(\bar x,\bar\xi)$; as $\langle S\eta\rangle \ge c\langle\eta\rangle$, c is rapidly decreasing there. Hence $(\bar x',\bar x',\bar\eta) \notin \operatorname{cone}\operatorname{supp}(c)$ and $(\bar x',\bar\eta) \notin \mathrm{WF}'(A_F)$. Points over $\mathbb{R}^n \setminus F(K)$ are not in $\mathrm{WF}'(A_F)$, since $\sigma_L(A_F)$ vanishes there. Thus $\mathrm{WF}'(A_F) \subset \chi_F(\mathrm{WF}'(A))$. Since $A = (A_F)_G$, the reverse inclusion follows by symmetry. The statement about characteristic varieties follows from that about principal symbols, since χ_F is homogeneous of degree one in the fibres and satisfies $c^{-1}|\xi| \le |(DF(x)^t)^{-1}\xi| \le c|\xi|$ locally uniformly in x; so it maps the sets $\Gamma_\epsilon(z) \cap \{|\xi| \ge 1/\epsilon\}$ into and onto sets of the same type. □

This invariance shows that the principal symbol, essential support and characteristic variety of a pseudodifferential operator are invariantly defined on the cotangent bundle; this is the basis of the extension of the calculus to manifolds in Chapter 5. It also gives us the freedom to choose local coordinates adapted to a particular problem.

3.7 The wavefront set

We now use operators as probes. An operator elliptic at z "sees" the frequencies of u near z; if some such probe returns a smooth function, u is smooth microlocally at z. We first define the wavefront set for distributions of compact support and then localize.

Definition 3.7.1. For $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$,

$$\mathrm{WF}(u) = \bigcap \{ \mathrm{Char}_0(A);\ A \in \Psi^0(\mathbb{R}^n),\ Au \in \mathcal{C}^\infty(\mathbb{R}^n) \}.$$

Thus $z \notin \mathrm{WF}(u)$ if and only if there is $A \in \Psi^0(\mathbb{R}^n)$, elliptic at z, with $Au \in \mathcal{C}^\infty(\mathbb{R}^n)$. As an intersection of closed conic sets, $\mathrm{WF}(u) \subset T^*\mathbb{R}^n \setminus 0$ is closed and conic.

Discussion. A good test operator is the microlocal cut-off $G_{z,\epsilon}$ of (3.3.2): it keeps the part of u with x near $\bar{x}$ and frequency in a narrow cone around $\bar{\xi}$, and discards the rest. So $z \notin \mathrm{WF}(u)$ says, roughly, that this "microlocalized piece" of u is smooth. In Section 3.8 we make this precise: it means that the Fourier transform of a localization of u decays rapidly in a cone around $\bar{\xi}$. Figure 3.3 shows three typical wavefront sets.

The wavefront set refines the singular support. Let $\pi : T^*\mathbb{R}^n \setminus 0 \to \mathbb{R}^n$ be the projection.

Proposition 3.7.2. *For $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$, $\pi(\mathrm{WF}(u)) = \operatorname{sing\,supp}(u)$.*

Proof. If $\bar{x} \notin \operatorname{sing\,supp} u$ there is $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\phi(\bar{x}) \neq 0$ and $\phi u \in \mathcal{C}^\infty$. As a multiplication operator $\phi \in \Psi^0(\mathbb{R}^n)$ is elliptic at $(\bar{x}, \bar{\xi})$ for every $\bar{\xi} \neq 0$, so $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}(u)$. Thus $\pi(\mathrm{WF}(u)) \subset \operatorname{sing\,supp}(u)$.

Conversely suppose $(\bar{x}, \omega) \notin \mathrm{WF}(u)$ for every $\omega \in \mathbb{S}^{n-1}$. For each ω there is $A_\omega \in \Psi^0$ with $A_\omega u \in \mathcal{C}^\infty$ and $|\sigma_L(A_\omega)| \geq \epsilon_\omega$ on $\Gamma_{\epsilon_\omega}(\bar{x}, \omega) \cap \{|\xi| \geq 1/\epsilon_\omega\}$. By compactness of the sphere there are finitely many ω_j such that the $V_{\epsilon_j}(\omega_j)$, $\epsilon_j = \epsilon_{\omega_j}$, cover $\mathbb{R}^n \setminus 0$. Let $A_j = A_{\omega_j}$, $\epsilon_0 = \min_j \epsilon_j$, $R_0 = \max_j 1/\epsilon_j$ and

$$B = \sum_j A_j^* A_j \in \Psi^0(\mathbb{R}^n).$$

Since $A_j u \in \mathcal{C}^\infty \cap \mathcal{S}'$ and A_j^* is pseudolocal (Proposition 3.2.5), $Bu \in \mathcal{C}^\infty(\mathbb{R}^n)$. Moreover $\sigma_L(B) = \sum_j |\sigma_L(A_j)|^2 + r$ with $r \in S^{-1}$, and $\sum_j |\sigma_L(A_j)|^2 \geq \epsilon_0^2$ when $|x - \bar{x}| < \epsilon_0$ and $|\xi| \geq R_0$. Let $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, $0 \leq \phi \leq 1$, with support in $\{|x - \bar{x}| < \epsilon_0\}$ and $\phi = 1$ near $\bar{x}$. Then $\sigma_L(B + (1-\phi)) = \sum_j |\sigma_L(A_j)|^2 + (1 - \phi) + r$ and, for $|\xi| \geq R_0$, the sum of the first two terms is at least $\min(\epsilon_0^2, 1)$ (consider separately $|x - \bar{x}| < \epsilon_0$ and $|x - \bar{x}| \geq \epsilon_0$). As $|r| \leq C\langle \xi \rangle^{-1}$, the operator $B + (1 - \phi)$ is globally elliptic, so has a parametrix $Q \in \Psi^0(\mathbb{R}^n)$:

$$\mathrm{Id} = QB + Q(1 - \phi) - E, \qquad E \in \Psi^{-\infty}(\mathbb{R}^n).$$

Let $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\psi(\bar{x}) \neq 0$ and $\operatorname{supp} \psi$ contained in the interior of $\{\phi = 1\}$. Then $\psi u = \psi QBu + \psi Q(1 - \phi)u - \psi Eu$. The first term is smooth by pseudolocality, the second because $\psi Q(1 - \phi) \in \Psi^{-\infty}$ by Proposition 3.2.2, and the third because $E \in \Psi^{-\infty}$. Hence $\bar{x} \notin \operatorname{sing\,supp} u$. □

Lemma 3.7.3. *Let $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$ and $z \notin \mathrm{WF}(u)$. Then there is an open conic neighbourhood Γ of z such that $Cu \in \mathcal{C}^\infty(\mathbb{R}^n)$ for every $C \in \Psi^m(\mathbb{R}^n)$, $m \in \mathbb{R}$, with $\mathrm{WF}'(C) \subset \Gamma$.*

Proof. Let $A \in \Psi^0$ be elliptic at z with $Au \in \mathcal{C}^\infty$ and let Q be a microlocal parametrix for A at z (Proposition 3.5.1), so $QA = \mathrm{Id} + R$ with $z \notin \mathrm{WF}'(R)$. Let Γ be the complement of $\mathrm{WF}'(R)$, an open conic neighbourhood of z, and let $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal 1 near $\operatorname{supp} u$. If $\mathrm{WF}'(C) \subset \Gamma$ then

$$Cu = CQ(Au) - CR\psi u.$$

The first term is smooth by pseudolocality. Since $\mathrm{WF}'(CR) \subset \mathrm{WF}'(C) \cap \mathrm{WF}'(R) = \emptyset$, $CR\psi \in \Psi^{-\infty}$ by Lemma 3.3.8, so the second term is smooth too. □

In particular, $\mathrm{WF}(u+v) \subset \mathrm{WF}(u) \cup \mathrm{WF}(v)$ for $u, v \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^n)$: if z is in neither set, take $C = G_{z,\epsilon}$ with $\overline{\Gamma_\epsilon(z)}$ inside both of the cones given by the lemma.

Proposition 3.7.4. *If $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^n)$, $A \in \Psi^m(\mathbb{R}^n)$ and $\phi \in \mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^n)$ then*

$$\mathrm{WF}(\phi Au) \subset \mathrm{WF}'(A) \cap \mathrm{WF}(u).$$

Proof. Let $\psi \in \mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^n)$ equal 1 near $\operatorname{supp} u$.

Suppose first $z \notin \mathrm{WF}'(A)$. Since $\mathrm{WF}'(A)$ is closed and conic, there is $\epsilon > 0$ with $\overline{\Gamma_\epsilon(z)} \cap \mathrm{WF}'(A) = \emptyset$; let $C = G_{z,\epsilon}$, which is elliptic at z. Then $\mathrm{WF}'(C\phi A) \subset \mathrm{WF}'(C) \cap \mathrm{WF}'(A) = \emptyset$, and $C\phi A = \psi' C\phi A$ for a suitable $\psi' \in \mathcal{C}_{\mathrm{c}}^{\infty}$ by (3.3.2), so $C\phi A \in \Psi^{-\infty}$ by Lemma 3.3.8. Hence $C(\phi Au) \in \mathcal{C}^\infty$ and $z \notin \mathrm{WF}(\phi Au)$.

Suppose next $z \notin \mathrm{WF}(u)$ and let Γ be as in Lemma 3.7.3. Choose ϵ with $\overline{\Gamma_\epsilon(z)} \subset \Gamma$ and let $D = G_{z,\epsilon}$, so $\mathrm{WF}'(D) \subset \Gamma$ and $Du \in \mathcal{C}^\infty$; and let $C = G_{z,\epsilon/4}$, which is elliptic at z with $\mathrm{WF}'(C) \subset \overline{\Gamma_{\epsilon/4}(z)}$, a set disjoint from $\mathrm{WF}'(\mathrm{Id} - D)$ by (3.3.2). Then

$$C\phi Au = C\phi A(Du) + C\phi A(\mathrm{Id} - D)\psi u.$$

The first term is smooth by pseudolocality, as $Du \in \mathcal{C}^\infty \cap \mathcal{S}'$. In the second term, $\mathrm{WF}'(C\phi A(\mathrm{Id} - D)) \subset \mathrm{WF}'(C) \cap \mathrm{WF}'(\mathrm{Id} - D) = \emptyset$, so $C\phi A(\mathrm{Id} - D)\psi \in \Psi^{-\infty}$ by Lemma 3.3.8. Hence $C(\phi Au) \in \mathcal{C}^\infty$ and $z \notin \mathrm{WF}(\phi Au)$. □

Taking $A = \mathrm{Id}$ we see that $\mathrm{WF}(\phi u) \subset \mathrm{WF}(u)$ for $\phi \in \mathcal{C}_{\mathrm{c}}^{\infty}$ and $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}$, and $\mathrm{WF}(\phi u) = \mathrm{WF}(u)$ if $\phi = 1$ near $\operatorname{supp} u$. This makes the following extension consistent.

Definition 3.7.5. For $u \in \mathcal{S}'(\mathbb{R}^n)$,

$$\mathrm{WF}(u) = \bigcup_{\phi \in \mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^n)} \mathrm{WF}(\phi u) \subset T^*\mathbb{R}^n \setminus 0.$$

Proposition 3.7.6. *Let $u, v \in \mathcal{S}'(\mathbb{R}^n)$.*

(i) *If $\phi \in \mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^n)$ and $\phi(\bar{x}) \neq 0$ then $\mathrm{WF}(u) \cap \pi^{-1}(\bar{x}) = \mathrm{WF}(\phi u) \cap \pi^{-1}(\bar{x})$.*

(ii) *$\mathrm{WF}(u)$ is closed and conic, and $\pi(\mathrm{WF}(u)) = \operatorname{sing\,supp}(u)$.*

(iii) *$\mathrm{WF}(u+v) \subset \mathrm{WF}(u) \cup \mathrm{WF}(v)$.*

Proof. (i) One inclusion is the definition. Conversely let $z = (\bar{x}, \bar{\xi}) \notin \mathrm{WF}(\phi u)$ and $\psi \in \mathcal{C}_{\mathrm{c}}^{\infty}(\mathbb{R}^n)$. Choose $\phi' \in \mathcal{C}_{\mathrm{c}}^{\infty}$ supported in $\{\phi \neq 0\}$ with $\phi' = 1$ near $\bar{x}$, and write $\psi u = \theta(\phi u) + \psi(1 - \phi')u$ with $\theta = \psi\phi'/\phi \in \mathcal{C}_{\mathrm{c}}^{\infty}$. Then $z \notin \mathrm{WF}(\theta\phi u)$ by Proposition 3.7.4, while $\psi(1 - \phi')u$ vanishes near $\bar{x}$, so its wavefront set does not meet $\pi^{-1}(\bar{x})$ by Proposition 3.7.2. Hence $z \notin \mathrm{WF}(\psi u)$.

(ii) By (i), $\mathrm{WF}(u) \cap \pi^{-1}(U) = \mathrm{WF}(\phi u) \cap \pi^{-1}(U)$ on the open set $U = \{\phi \neq 0\}$, so $\mathrm{WF}(u)$ is closed, being locally closed; it is conic as a union of conic sets. Also $\bar{x} \notin \operatorname{sing\,supp} u$ if and only if $\bar{x} \notin \operatorname{sing\,supp}(\phi u)$ for some (any) ϕ as in (i), so the second statement follows from Proposition 3.7.2.

(iii) follows from (i) and the corresponding statement for $\mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^n)$. □

Theorem 3.7.7 (Microlocality)**.** *If $A \in \Psi^m(\mathbb{R}^n)$ and $u \in \mathcal{S}'(\mathbb{R}^n)$ then*

$$\mathrm{WF}(Au) \subset \mathrm{WF}'(A) \cap \mathrm{WF}(u).$$

Proof. Let $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ and choose $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to 1 near $\operatorname{supp}\phi$. Then $\phi Au = \phi A(\psi u) + \phi A(1-\psi)u$, and the second term is smooth by Proposition 3.2.2. By Proposition 3.7.4, $\mathrm{WF}(\phi Au) \subset \mathrm{WF}'(A) \cap \mathrm{WF}(\psi u) \subset \mathrm{WF}'(A) \cap \mathrm{WF}(u)$. Taking the union over ϕ gives the result. □

Combined with microlocal parametrices this gives the basic regularity result.

Theorem 3.7.8 (Microlocal elliptic regularity)**.** *If $A \in \Psi^m(\mathbb{R}^n)$ and $u \in \mathcal{S}'(\mathbb{R}^n)$ then*

$$\mathrm{WF}(u) \subset \mathrm{WF}(Au) \cup \mathrm{Char}_m(A).$$

Proof. Let $z \notin \mathrm{Char}_m(A)$ and let B be a microlocal parametrix at z, so $BA = \mathrm{Id} + R$ with $z \notin \mathrm{WF}'(R)$. Then $u = B(Au) - Ru$, and by Theorem 3.7.7 and Proposition 3.7.6(iii), $\mathrm{WF}(u) \subset \mathrm{WF}(Au) \cup \mathrm{WF}'(R)$, which does not contain z if $z \notin \mathrm{WF}(Au)$. □

In particular, if A is elliptic at every point (that is, $\mathrm{Char}_m(A) = \emptyset$), then $\mathrm{WF}(u) = \mathrm{WF}(Au)$. For instance $\mathrm{WF}(u)$ is unchanged by the Fourier multipliers $\langle D\rangle^s$.

Discussion. Theorems 3.7.7 and 3.7.8 together say

$$\mathrm{WF}(Au) \subset \mathrm{WF}(u) \subset \mathrm{WF}(Au) \cup \mathrm{Char}_m(A).$$

A pseudodifferential operator cannot create singularities, nor move them in position or direction; and where it is elliptic it cannot destroy them either. All the room for u to be singular while Au is smooth lies in $\mathrm{Char}_m(A)$. For $A = D_{x_1}$ on $\mathbb{R}^2$ this is the set $\{\xi_1 = 0\}$, and indeed $u = H(x_2)$ has $D_{x_1}u = 0$ while $\mathrm{WF}(u)$ is non-empty and lies in $\{\xi_1 = 0\}$. What happens inside the characteristic variety is the subject of Section 3.20.

3.8 The wavefront set and the Fourier transform

We now give the more classical characterization of the wavefront set, in terms of the Fourier transform. We use the following standard facts about $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$, which follow from (3.1.1) and the Fourier inversion formula by approximating the integrals below by Riemann sums in $\mathcal{S}(\mathbb{R}^n)$: $\hat{u}(\xi) = u(e^{-ix\cdot\xi})$ is a smooth function with $|\hat{u}(\xi)| \le C\langle\xi\rangle^k$; for $\phi \in \mathcal{S}(\mathbb{R}^n)$,

$$u(\phi) = \int \hat{u}(\xi)\,\hat{\phi}(-\xi)đ\xi, \qquad \widehat{\phi u}(\xi) = \int \hat{\phi}(\xi-\eta)\,\hat{u}(\eta)đ\eta. \tag{3.8.1}$$

Conversely, a continuous polynomially bounded function is the Fourier transform of a unique element of $\mathcal{S}'(\mathbb{R}^n)$.

For $v \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$ let the *frequency set* of v be

$$\Sigma_{\mathrm{fr}}(v) = \{\xi \in \mathbb{R}^n \setminus 0;\ (x,\xi) \in \mathrm{WF}(v) \text{ for some } x\}.$$

Since $\mathrm{WF}(v)$ is closed, conic and lies over the compact set $\operatorname{supp} v$, $\mathrm{WF}(v) \cap (\mathbb{R}^n \times \mathbb{S}^{n-1})$ is compact, so $\Sigma_{\mathrm{fr}}(v)$ is a closed cone in $\mathbb{R}^n \setminus 0$, and $\mathrm{WF}(v) \subset \operatorname{supp} v \times \Sigma_{\mathrm{fr}}(v)$.

Lemma 3.8.1. *Let $v \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$.*

(i) *If $\Gamma \subset \mathbb{R}^n \setminus 0$ is a closed cone with $\Gamma \cap \Sigma_{\mathrm{fr}}(v) = \emptyset$, then $\hat{v}$ is rapidly decreasing on Γ.*

(ii) *If $\hat{v}$ is rapidly decreasing on an open cone V then $\mathrm{WF}(v) \cap (\mathbb{R}^n \times V) = \emptyset$.*

(iii) *If $\bar{x} \in \mathbb{R}^n$ and V is an open cone containing $\{\xi;\ (\bar{x},\xi) \in \mathrm{WF}(v)\}$, there is $r > 0$ such that $\Sigma_{\mathrm{fr}}(\phi v) \subset V$ for all $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ supported in $\{|x - \bar{x}| < r\}$.*

Proof. (i) By compactness there is an open cone $\Gamma' \supset \Gamma$ with $\overline{\Gamma'} \cap \Sigma_{\mathrm{fr}}(v) = \emptyset$ (the distance between $\Gamma \cap \mathbb{S}^{n-1}$ and the compact set $\Sigma_{\mathrm{fr}}(v) \cap \mathbb{S}^{n-1}$ is positive). Choose $g \in S^0(\mathbb{R}^n)$, a function of ξ only, with $g = 1$ on $\Gamma \cap \{|\xi| \geq 1\}$ and $\operatorname{supp} g \subset \Gamma'$; for instance $g = h(\xi/|\xi|)(1 - \chi(\xi))$ with $h \in \mathcal{C}^\infty(\mathbb{S}^{n-1})$. Then $g(D) = \operatorname{Op}(g)$ has $\mathrm{WF}'(g(D)) \subset \mathbb{R}^n \times \overline{\Gamma'}$, which is disjoint from $\mathrm{WF}(v)$, so $g(D)v \in \mathcal{C}^\infty$ by Theorem 3.7.7. Let $\psi_1, \psi_2 \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\psi_1 = 1$ near $\operatorname{supp} v$ and $\psi_2 = 1$ near $\operatorname{supp} \psi_1$. Then

$$g(D)v = \psi_2 g(D)v + (1 - \psi_2)g(D)\psi_1 v,$$

where the first term is in $\mathcal{C}^\infty_{\mathrm{c}}$ and the second is in $\mathcal{S}$ by Proposition 3.2.2 and Lemma 3.2.3. Thus $g\hat{v} = \widehat{g(D)v} \in \mathcal{S}$, and $\hat{v}$ is rapidly decreasing on Γ.

(ii) Let $z = (\bar{x}, \bar{\xi})$ with $\bar{\xi} \in V$, and choose $\epsilon > 0$ with $\overline{V_\epsilon(\bar{\xi})} \subset V$ and $g \in S^0(\mathbb{R}^n)$ a function of ξ with $g = 1$ on $V_{\epsilon/2}(\bar{\xi}) \cap \{|\xi| \geq 1\}$, supported in $V_\epsilon(\bar{\xi})$. Then $g(D)$ is elliptic at z, and $\widehat{g(D)v} = g\hat{v}$ is rapidly decreasing, hence integrable together with $\xi^\alpha g\hat{v}$ for all α, so $g(D)v \in \mathcal{C}^\infty$ by Fourier inversion. Thus $z \notin \mathrm{WF}(v)$.

(iii) Otherwise there are $x_j \to \bar{x}$ and unit vectors $\xi_j \notin V$ with $(x_j, \xi_j) \in \mathrm{WF}(v)$; a subsequence has $\xi_j \to \xi$, and $(\bar{x}, \xi) \in \mathrm{WF}(v)$ by closedness, $\xi \notin V$ since the complement of V is closed, a contradiction. Since $\mathrm{WF}(\phi v) \subset \mathrm{WF}(v) \cap (\operatorname{supp} \phi \times \mathbb{R}^n)$, the claim follows. □

Proposition 3.8.2 (Fourier characterization)**.** *Let $u \in \mathcal{S}'(\mathbb{R}^n)$ and $(\bar{x}, \bar{\xi}) \in T^*\mathbb{R}^n \setminus 0$. Then $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}(u)$ if and only if there exist $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\phi(\bar{x}) \neq 0$ and $\epsilon > 0$ such that $\widehat{\phi u}$ is rapidly decreasing on $V_\epsilon(\bar{\xi})$:*

$$|\widehat{\phi u}(\xi)| \leq C_N \langle \xi \rangle^{-N} \quad \text{for } \xi \in V_\epsilon(\bar{\xi}),\ N \in \mathbb{N}.$$

More precisely, if $K \subset \mathbb{R}^n$ is compact, $\Gamma \subset \mathbb{R}^n \setminus 0$ is a closed cone and $\mathrm{WF}(u) \cap (K \times \Gamma) = \emptyset$, then $\widehat{\phi u}$ is rapidly decreasing on Γ for every $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\operatorname{supp} \phi \subset K$.

Proof. If $\widehat{\phi u}$ is rapidly decreasing on $V_\epsilon(\bar{\xi})$ then $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}(\phi u)$ by Lemma 3.8.1(ii), and so $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}(u)$ by Proposition 3.7.6(i).

For the last statement, $\mathrm{WF}(\phi u) \subset \mathrm{WF}(u) \cap (K \times \mathbb{R}^n)$, so $\Sigma_{\mathrm{fr}}(\phi u) \cap \Gamma = \emptyset$ and Lemma 3.8.1(i) applies. Finally, if $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}(u)$, closedness of $\mathrm{WF}(u)$ and a compactness argument as in Lemma 3.8.1(iii) give $r, \epsilon > 0$ with $\mathrm{WF}(u) \cap (K \times \overline{V_\epsilon(\bar{\xi})}) = \emptyset$ for $K = \{|x - \bar{x}| \leq r\}$, and any ϕ supported in K with $\phi(\bar{x}) \neq 0$ will do. □

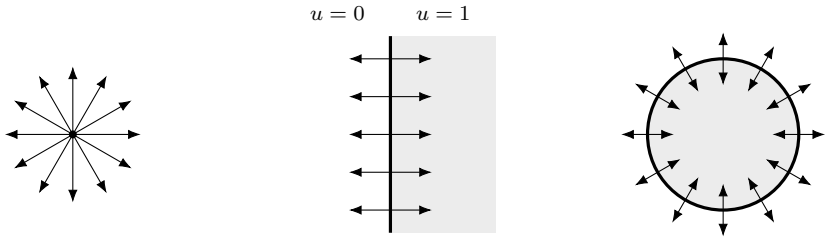
Example 3.8.3. Since $\hat{\delta} = 1$ and $\operatorname{sing\,supp} \delta = \{0\}$, $\mathrm{WF}(\delta) = \{0\} \times (\mathbb{R}^n \setminus 0)$: indeed $\widehat{\phi\delta} = \phi(0)$ is not rapidly decreasing in any direction if $\phi(0) \neq 0$. More generally, if $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$ is not smooth then $\Sigma_{\mathrm{fr}}(u) \neq \emptyset$, since otherwise $\hat{u} \in \mathcal{S}$ by Lemma 3.8.1(i).

Question. The Fourier characterization is concrete. Why did we define the wavefront set through pseudodifferential operators instead?

Answer. Invariance. A diffeomorphism does not act simply on Fourier transforms, so it is not clear from Proposition 3.8.2 that $\mathrm{WF}(u)$ transforms as a subset of the cotangent bundle. With the operator definition this follows almost at once from the coordinate invariance of the calculus (Theorem 3.6.2), as we see in the next section. The Fourier description, on the other hand, is the convenient tool for products, restrictions and push-forwards. □

Corollary 3.8.4. *For $u \in \mathcal{S}'(\mathbb{R}^n)$, $\mathrm{WF}(\bar{u}) = \mathrm{WF}(u)^-$.*

Proof. $\widehat{\phi\bar{u}}(\xi) = \overline{\hat{\phi} u(-\xi)}$, so this follows from Proposition 3.8.2 applied with $\bar{\phi}$. □



(a) δ_0 (b) $H(x_1)$ (c) $\chi_{\mathbb{B}}$

Figure 3.3: Wavefront sets of three distributions on $\mathbb{R}^2$, drawn as covectors attached to points of the base. (a) $\mathrm{WF}(\delta_0) = \{0\} \times (\mathbb{R}^2 \setminus 0)$: all directions at the origin (Example 3.8.3). (b) For the Heaviside function $H(x_1)$ the wavefront set is $\{(0, x_2; \xi_1, 0);\ \xi_1 \neq 0\}$: only the directions normal to the jump. (c) For the characteristic function of the unit disc, the covectors normal to the circle. In (b) and (c) the wavefront set is the conormal bundle of the curve of discontinuity minus its zero section; such distributions are studied in Chapter 4.

3.9 Invariance under diffeomorphisms

For a diffeomorphism $F : \Omega \to \Omega'$ between open subsets of $\mathbb{R}^n$ we use the map $\chi_F(x, \xi) = (F(x), (DF(x)^t)^{-1}\xi)$ of Section 3.6. Its inverse is

$$\chi_F^{-1}(y, \eta) = (G(y), DF(G(y))^t \eta), \qquad G = F^{-1},$$

the cotangent map $\eta \mapsto DF(x)^t\eta$, $x = G(y)$, which we denote by F^* when applied to subsets: for $W \subset T^*\Omega' \setminus 0$,

$$F^*W = \{(x, DF(x)^t\eta);\ (F(x), \eta) \in W\}.$$

Theorem 3.9.1. *If $F : \Omega \to \Omega'$ is a diffeomorphism between open subsets of $\mathbb{R}^n$ and $u \in \mathcal{C}_c^{-\infty}(\Omega')$, then*

$$\mathrm{WF}(F^*u) = F^* \,\mathrm{WF}(u) = \{(x, DF(x)^t\eta);\ (F(x), \eta) \in \mathrm{WF}(u)\}.$$

Proof. Let $(x, \xi) \in T^*\Omega \setminus 0$ with $w = \chi_F(x, \xi) \notin \mathrm{WF}(u)$; we show $(x, \xi) \notin \mathrm{WF}(F^*u)$. Let Γ be an open conic neighbourhood of w as in Lemma 3.7.3, choose $\epsilon > 0$ with $\overline{\Gamma_\epsilon(w)} \subset \Gamma$ and $\{|y - F(x)| \leq \epsilon\} \subset \Omega'$, and let $\theta \in \mathcal{C}_c^\infty(\Omega')$ equal 1 on $\{|y - F(x)| \leq \epsilon\}$. The operator $B = \theta G_{w,\epsilon}\theta \in \Psi^0(\mathbb{R}^n)$ has kernel with compact support in $\Omega' \times \Omega'$, $\mathrm{WF}'(B) \subset \overline{\Gamma_\epsilon(w)} \subset \Gamma$ and is elliptic at w. So $Bu \in \mathcal{C}_c^\infty(\Omega')$.

By Theorem 3.6.2, applied to the diffeomorphism $G : \Omega' \to \Omega$, the operator $\tilde{B} = B_G = F^*BG^*$ is in $\Psi^0(\mathbb{R}^n)$ and is elliptic at $\chi_G(w) = (x, \xi)$. Both $\tilde{B}$ (as an element of $\Psi^0(\mathbb{R}^n)$ acting on $\mathcal{S}'$) and $F^* \circ B \circ G^*$ are weakly continuous on $\mathcal{C}_c^{-\infty}(\Omega)$ and agree on $\mathcal{C}_c^\infty(\Omega)$, so by density they agree on $\mathcal{C}_c^{-\infty}(\Omega)$. Hence $\tilde{B}(F^*u) = F^*BG^*F^*u = F^*(Bu) \in \mathcal{C}^\infty$, and $(x, \xi) \notin \mathrm{WF}(F^*u)$.

We have shown $\mathrm{WF}(F^*u) \subset \chi_F^{-1}(\mathrm{WF}(u)) = F^* \,\mathrm{WF}(u)$. Applying this to G and F^*u, and using $G^*F^*u = u$, gives the reverse inclusion. □

If L is an invertible linear map of $\mathbb{R}^n$ and $u \in \mathcal{S}'(\mathbb{R}^n)$, then by localization (Proposition 3.7.6) the theorem also applies to L^*u: $\mathrm{WF}(L^*u) = \{(x, L^t\eta);\ (Lx, \eta) \in \mathrm{WF}(u)\}$.

3.10 The wavefront set of a kernel

The kernels of pseudodifferential operators form an interesting class of distributions. Their wavefront sets are determined by, and determine, the essential supports of the operators.

Proposition 3.10.1. *If $A \in \Psi^m(\mathbb{R}^n)$ then*

$$\mathrm{WF}(K_A) = \{(x, x, \xi, -\xi) \in T^*\mathbb{R}^{2n} \setminus 0;\ (x, \xi) \in \mathrm{WF}'(A)\}.$$

Proof. By Proposition 3.2.2 and Proposition 3.7.6(ii), $\mathrm{WF}(K_A)$ lies over the diagonal. Fix $\bar{x} \in \mathbb{R}^n$, let $\mu \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal 1 near 0 and let A_μ have kernel $\mu(x-y)K_A$; by Lemma 3.2.1, $A - A_\mu \in \Psi^{-\infty}$, so $a_\mu = \sigma_L(A_\mu)$ differs from $a = \sigma_L(A)$ by an element of $S^{-\infty}$. For $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\phi(\bar{x}) \neq 0$ consider

$$k(x, y) = \phi(x)\mu(x-y)K_A(x, y) \in \mathcal{C}_c^{-\infty}(\mathbb{R}^{2n}), \qquad w = L^*k, \quad L(x, z) = (x, x-z),$$

so $w(x, z) = \phi(x)\mu(z)K_A(x, x-z)$, and $k = K_A$ near $(\bar{x}, \bar{x})$. Since $K_{A_\mu} = \int e^{i(x-y)\cdot\xi} a_\mu(x, \xi) đ\xi$, the distribution $L^*K_{A_\mu}$ is the inverse Fourier transform of $a_\mu(x, \xi)$ in ξ, and so

$$\hat{w}(\zeta, \xi) = \int e^{-ix\cdot\zeta} \phi(x) a_\mu(x, \xi)\, dx,$$

the Fourier transform being taken in $(x, z) \mapsto (\zeta, \xi)$. Since ϕa_μ is a symbol with compact support in x, integration by parts in x gives

$$|\hat{w}(\zeta, \xi)| \leq C_N \langle\zeta\rangle^{-N} \langle\xi\rangle^m \quad \text{for all } N. \tag{3.10.1}$$

Points of the wavefront set and the covectors transform under L by Theorem 3.9.1: since $DL^t(\xi, \eta) = (\xi + \eta, -\eta)$,

$$(x, 0; \zeta, \eta') \in \mathrm{WF}(w) \iff (x, x; \zeta + \eta', -\eta') \in \mathrm{WF}(k). \tag{3.10.2}$$

Inclusion $\subset$. By (3.10.1), $\hat{w}$ is rapidly decreasing on every closed cone $\{|\xi| \leq c|\zeta|\}$, so $\Sigma_{\mathrm{fr}}(w) \subset \{\zeta = 0\}$ by Lemma 3.8.1(ii). Since w is smooth for $z \neq 0$, $\mathrm{WF}(w) \subset \{(x, 0; 0, \xi)\}$, and (3.10.2) gives $\mathrm{WF}(k) \subset \{(x, x; \xi, -\xi)\}$.

Now suppose $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}'(A)$. Then a_μ and all its derivatives are rapidly decreasing on $\Gamma_\epsilon(\bar{x}, \bar{\xi})$ for some ϵ (Lemma 3.3.2); take $\operatorname{supp}\phi \subset \{|x - \bar{x}| < \epsilon\}$. Then integration by parts gives $|\hat{w}(\zeta, \xi)| \leq C_N \langle\zeta\rangle^{-N} \langle\xi\rangle^{-N}$ for $\xi \in V_\epsilon(\bar{\xi})$, which with (3.10.1) shows that $\hat{w}$ is rapidly decreasing on the cone $\{(\zeta, \xi);\ \xi \in V_\epsilon(\bar{\xi}),\ |\zeta| < |\xi|\}$, a conic neighbourhood of $(0, \bar{\xi})$. Hence $(\bar{x}, 0; 0, \bar{\xi}) \notin \mathrm{WF}(w)$, and by (3.10.2), $(\bar{x}, \bar{x}; \bar{\xi}, -\bar{\xi}) \notin \mathrm{WF}(k)$, so not in $\mathrm{WF}(K_A)$.

Inclusion $\supset$. Suppose $(\bar{x}, \bar{x}, \bar{\xi}, -\bar{\xi}) \notin \mathrm{WF}(K_A)$. By closedness there are $r, \theta > 0$ such that $(x, x, \xi, -\xi) \notin \mathrm{WF}(K_A)$ for $|x - \bar{x}| < r$, $\xi \in V_\theta(\bar{\xi})$. Take $\operatorname{supp}\phi \subset \{|x - \bar{x}| < r\}$ and $\phi = 1$ near $\bar{x}$. Then, by (3.10.2) and the first part, $\Sigma_{\mathrm{fr}}(w) \subset \overline{\{(0, \xi);\ \xi \notin V_\theta(\bar{\xi})\}}$, which is disjoint from the closed cone $C = \{(\zeta, \xi) \neq 0;\ |\zeta| \leq |\xi|,\ \xi \in \overline{V_{\theta/2}(\bar{\xi})}\}$. By Lemma 3.8.1(i), $\hat{w}$ is rapidly decreasing on C. By Fourier inversion in ζ, for $\xi \in V_{\theta/2}(\bar{\xi})$,

$$|\phi(x)a_\mu(x, \xi)| \leq \int_{|\zeta| \leq |\xi|} |\hat{w}(\zeta, \xi)| đ\zeta + \int_{|\zeta| > |\xi|} |\hat{w}(\zeta, \xi)| đ\zeta \leq C_N \langle\xi\rangle^{n-N} + C_N \langle\xi\rangle^{m+n-N},$$

using (3.10.1) in the second integral. So a_μ, and hence a, is rapidly decreasing on $\Gamma_{r'}(\bar{x}, \bar{\xi})$ for small r', that is $(\bar{x}, \bar{\xi}) \notin \mathrm{WF}'(A)$. □

In particular $\mathrm{WF}'(A)$ and $\mathrm{WF}(K_A)$ determine each other, and $\mathrm{WF}'(A) = \emptyset$ if and only if $K_A \in \mathcal{C}^\infty(\mathbb{R}^{2n})$.

Discussion. The covectors $(\xi, -\xi)$ at (x, x) are exactly those which vanish on the tangent vectors (v, v) of the diagonal, that is, the conormal bundle of Diag (Figure 3.1). This is the sign which makes the kernel $\int e^{i(x-y)\cdot\xi} a đ\xi$ oscillate in x with frequency ξ and in y with frequency $-\xi$. To make the identity operator correspond to the identity relation one flips the sign of the second covector; this is the convention behind $\mathrm{WF}'(K)$ in Section 3.16. Kernels of pseudodifferential operators are thus distributions whose wavefront set lies in the conormal bundle of a submanifold; Chapter 4 studies such distributions in general.

3.11 The calculus of wavefront sets: pairing

We now wish to understand, in terms of wavefront sets, three fundamental operations on distributions: pull-back F^*u by a smooth map, push-forward F_*u, and multiplication $u_1 u_2$. None of them is defined on all distributions. To analyse them we first discuss some more elementary operations:

pairing:	$(u, v) \longmapsto \int u(x)v(x)\,dx,$
exterior product:	$(u, v) \longmapsto (u \boxtimes v)(x, y) = u(x)v(y),$
restriction:	$u(x, y) \longmapsto u(x, 0),$
projection:	$u(x, y) \longmapsto \int u(x, y)\,dy,$
invariance:	$u \longmapsto F^*u,\ F$ a diffeomorphism.

The last was treated in Section 3.9. Restriction and diffeomorphisms are special cases of pull-back and projection is a special case of push-forward; conversely, we shall express the three fundamental operations in terms of the elementary ones (Figure 3.4). Throughout, all distributions have compact support; by the local nature of the wavefront set this is not a serious restriction.

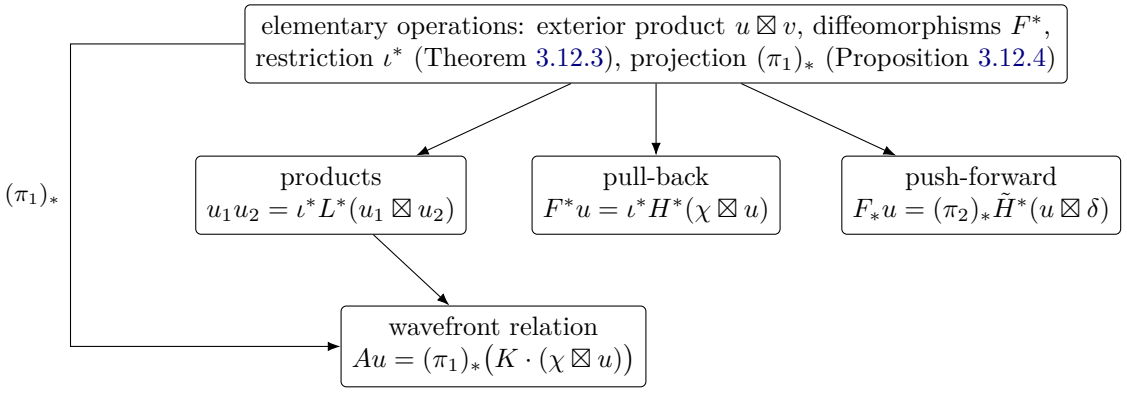


Figure 3.4: How the operations of Sections 3.11 to 3.16 are built from each other. Products and pull-backs are restrictions, after a linear change of variables, of exterior products; push-forward is a projection of the pull-back of $u \boxtimes \delta$; and the action of a general kernel is a projection of a product.

The pairing $\int u_1 u_2\,dx$ of two distributions is the first example of an operation which requires a condition on wavefront sets. We give Melrose's operator-theoretic definition; in Section 3.13 it is recovered as the integral of the product (Exercise 3.15). We need a transposition identity.

Lemma 3.11.1. *Let $D \in \Psi^m(\mathbb{R}^n)$ and $u, v \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ with $\mathrm{WF}'(D) \cap \mathrm{WF}(u) = \emptyset$ and $\mathrm{WF}'(D^t) \cap \mathrm{WF}(v) = \emptyset$. Then Du and $D^t v$ are smooth and, for any $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal to 1 near $\operatorname{supp} u \cup \operatorname{supp} v$,*

$$v(\psi Du) = u(\psi D^t v).$$

Proof. Smoothness follows from Theorem 3.7.7. Both sides are independent of the choice of ψ. Let $\psi_1 \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal 1 near $\operatorname{supp} u \cup \operatorname{supp} v$ and let $\psi = \psi_2$ equal 1 near $\operatorname{supp} \psi_1$. The closed conic sets $W_1 = \mathrm{WF}'(D) \cap (\operatorname{supp} \psi_1 \times \mathbb{R}^n)$ and $W_2 = \mathrm{WF}(u)$ are disjoint and $\pi(W_1)$ is bounded, so Lemma 3.3.9 gives $B \in \Psi^0(\mathbb{R}^n)$ with $\mathrm{WF}'(B) \cap \mathrm{WF}(u) = \emptyset$ and $\mathrm{WF}'(\mathrm{Id} - B) \cap W_1 = \emptyset$. Write $u = u_s + u_r$ with $u_s = \psi_1 B u$ and $u_r = \psi_1(\mathrm{Id} - B)u$. By microlocality Bu is smooth, so $u_s \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, and since $D^t v$ is smooth, the definition of D on $\mathcal{S}'$ gives

$$v(\psi D u_s) = v(D u_s) = (D^t v)(u_s) = \int (D^t v) u_s \, dx = u_s(\psi D^t v).$$

For u_r, let $E = \psi_2 D \psi_1 (\mathrm{Id} - B)\psi_1$. Since $\mathrm{WF}'(D\psi_1) \subset W_1$, $\mathrm{WF}'(E) \subset W_1 \cap \mathrm{WF}'(\mathrm{Id} - B) = \emptyset$, and by Lemma 3.3.8 $E \in \Psi^{-\infty}(\mathbb{R}^n)$ with kernel $K_E \in \mathcal{C}_c^\infty(\mathbb{R}^{2n})$. As $u = \psi_1 u$, $\psi_2 D u_r = E u$, while

$$u_r(\psi_2 D^t v) = u((\mathrm{Id} - B^t)\psi_1 \psi_2 D^t v) = u(\psi_1(\mathrm{Id} - B^t)\psi_1 D^t \psi_2 v) = u(E^t v),$$

using $u = \psi_1 u$, $v = \psi_2 v$ and $\psi_1 \psi_2 = \psi_1$. Finally $v(Eu) = u(E^t v)$ for E with kernel in $\mathcal{C}_c^\infty(\mathbb{R}^{2n})$: both sides equal $(v \boxtimes u)(K_E)$, since $Eu(x) = u(K_E(x, \cdot))$, $E^t v(y) = v(K_E(\cdot, y))$, and the identity $v_x(u_y(\Phi(x,y))) = u_y(v_x(\Phi(x,y)))$ holds for $\Phi = \phi \boxtimes \phi'$, hence by continuity and density for all $\Phi \in \mathcal{S}(\mathbb{R}^{2n})$. Adding the two identities proves the lemma. □

Proposition 3.11.2 (Pairing)**.** *Let $u_1, u_2 \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ satisfy*

$$(x, \xi) \in \mathrm{WF}(u_1) \Longrightarrow (x, -\xi) \notin \mathrm{WF}(u_2), \quad \textit{i.e.} \quad \mathrm{WF}(u_1) \cap \mathrm{WF}(u_2)^- = \emptyset. \tag{3.11.1}$$

Then there exists $A \in \Psi^0(\mathbb{R}^n)$ with

$$\mathrm{WF}'(A) \cap \mathrm{WF}(u_1) = \emptyset, \qquad \mathrm{WF}'(\mathrm{Id} - A) \cap \mathrm{WF}(u_2)^- = \emptyset, \tag{3.11.2}$$

and for any such A and any $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal to 1 near $\operatorname{supp} u_1 \cup \operatorname{supp} u_2$ the number

$$P(u_1, u_2) = u_2(\psi A u_1) + u_1(\psi(\mathrm{Id} - A)^t u_2)$$

is independent of A and ψ. The pairing P is bilinear, symmetric, and $P(u_1, u_2) = u_2(u_1) = \int u_1 u_2 \, dx$ if $u_1 \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ (and similarly if $u_2 \in \mathcal{C}_c^\infty$).

Proof. Existence of A follows from Lemma 3.3.9 with $W_1 = \mathrm{WF}(u_2)^-$, $W_2 = \mathrm{WF}(u_1)$. By microlocality and $\mathrm{WF}'((\mathrm{Id} - A)^t) = \mathrm{WF}'(\mathrm{Id} - A)^-$ (Proposition 3.3.7), Au_1 and $(\mathrm{Id} - A)^t u_2$ are smooth, so $P(u_1, u_2)$ is defined; it does not depend on ψ, since $\psi = 1$ near the supports. If A' is another choice, $D = A - A'$ satisfies $\mathrm{WF}'(D) \cap \mathrm{WF}(u_1) = \emptyset$, and since $D = (\mathrm{Id} - A') - (\mathrm{Id} - A)$ also $\mathrm{WF}'(D) \cap \mathrm{WF}(u_2)^- = \emptyset$, that is $\mathrm{WF}'(D^t) \cap \mathrm{WF}(u_2) = \emptyset$. The difference of the two expressions is $u_2(\psi D u_1) - u_1(\psi D^t u_2)$, which vanishes by Lemma 3.11.1.

For linearity in the first argument, suppose (u_1, u_2) and (u_1', u_2) both satisfy (3.11.1). Then $(\mathrm{WF}(u_1) \cup \mathrm{WF}(u_1')) \cap \mathrm{WF}(u_2)^- = \emptyset$, and an A obtained from Lemma 3.3.9 with $W_2 = \mathrm{WF}(u_1) \cup \mathrm{WF}(u_1')$ satisfies (3.11.2) for the three pairs (u_1, u_2), (u_1', u_2) and $(u_1 + u_1', u_2)$, since $\mathrm{WF}(u_1 + u_1') \subset \mathrm{WF}(u_1) \cup \mathrm{WF}(u_1')$; with this A the expressions are visibly additive. Linearity in the second argument is the same. For symmetry note that $A' = \mathrm{Id} - A^t$ satisfies (3.11.2) for the pair (u_2, u_1), and the corresponding expression is $u_1(\psi(\mathrm{Id} - A^t)u_2) + u_2(\psi A u_1)$. If $u_1 \in \mathcal{C}_c^\infty$ we may take $A = \mathrm{Id}$, and if $u_2 \in \mathcal{C}_c^\infty$, $A = 0$. □

The *sesquilinear* pairing $\langle u_1, u_2\rangle = \int u_1\overline{u_2}\,dx$ is then defined as $P(u_1, \overline{u_2})$. Since $\mathrm{WF}(\overline{u_2}) = \mathrm{WF}(u_2)^-$ (Corollary 3.8.4), the condition becomes simply $\mathrm{WF}(u_1)\cap\mathrm{WF}(u_2) = \emptyset$, and $\langle u_1, u_2\rangle = \langle Au_1, u_2\rangle + \langle u_1, (\mathrm{Id}-A^*)u_2\rangle$ for any A with $\mathrm{WF}'(A)\cap\mathrm{WF}(u_1) = \emptyset = \mathrm{WF}'(\mathrm{Id}-A)\cap\mathrm{WF}(u_2)$.

Discussion. The idea of the definition is to split the frequencies. The operator A removes the singular directions of u_1, and $\mathrm{Id}-A$ those of u_2; in each of the two terms one factor is smooth, so the pairing makes sense. The condition (3.11.1) is what allows such a split. The minus sign has a Fourier explanation: $\int u_1u_2\,dx$ is $\int \hat u_1(\xi)\hat u_2(-\xi)đ\xi$ by (3.8.1), so trouble arises only if $\hat u_1$ is large at ξ and $\hat u_2$ is large at $-\xi$. For example $\int \delta_0\cdot\delta_0$ is meaningless, and indeed $\mathrm{WF}(\delta_0)$ contains both ξ and $-\xi$.

3.12 Exterior products, restriction and projection

These three operations are the building blocks of Figure 3.4. Exterior products are always defined; restriction needs a condition on the wavefront set; projection is always defined but can only keep singularities that are conormal to the fibres.

3.12.1 Exterior product

The exterior product $u_1\boxtimes u_2$ of $u_1\in\mathcal{C}_c^{-\infty}(\mathbb{R}^p)$ and $u_2\in\mathcal{C}_c^{-\infty}(\mathbb{R}^k)$ is always defined (Chapter 1); it lies in $\mathcal{C}_c^{-\infty}(\mathbb{R}^{p+k})$, $(u_1\boxtimes u_2)(\Phi) = u_1(u_2(\Phi(x,\cdot)))$, and its Fourier transform is

$$\widehat{u_1\boxtimes u_2}(\xi,\eta) = \hat u_1(\xi)\,\hat u_2(\eta).$$

Proposition 3.12.1. *For $u_1\in\mathcal{C}_c^{-\infty}(\mathbb{R}^p)$ and $u_2\in\mathcal{C}_c^{-\infty}(\mathbb{R}^k)$,*

$$\begin{aligned}\mathrm{WF}(u_1\boxtimes u_2)\subset\big(\mathrm{WF}(u_1)\times\mathrm{WF}(u_2)\big)\cup\{(x,y,0,\eta);\ x\in\operatorname{supp}u_1,\ (y,\eta)\in\mathrm{WF}(u_2)\}\\ \cup\{(x,y,\xi,0);\ (x,\xi)\in\mathrm{WF}(u_1),\ y\in\operatorname{supp}u_2\},\end{aligned}$$

where $\mathrm{WF}(u_1)\times\mathrm{WF}(u_2)$ means $\{(x,y,\xi,\eta);(x,\xi)\in\mathrm{WF}(u_1),(y,\eta)\in\mathrm{WF}(u_2)\}$.

Proof. Let $(\bar x,\bar y,\bar\xi,\bar\eta)$ be outside the right side. We localize with $\phi_1(x)\phi_2(y)$, $\phi_1(\bar x)\neq 0\neq\phi_2(\bar y)$, using $\phi_1\phi_2(u_1\boxtimes u_2) = (\phi_1u_1)\boxtimes(\phi_2u_2)$, whose Fourier transform is $\widehat{\phi_1u_1}(\xi)\widehat{\phi_2u_2}(\eta)$. If $\bar x\notin\operatorname{supp}u_1$ or $\bar y\notin\operatorname{supp}u_2$ we can make this zero. Otherwise one of the following holds.

(a) $\bar\xi\neq 0$ and $(\bar x,\bar\xi)\notin\mathrm{WF}(u_1)$. Choose ϕ_1 by Proposition 3.8.2 with $\widehat{\phi_1u_1}$ rapidly decreasing on $V_\epsilon(\bar\xi)$. On the conic neighbourhood $\{(\xi,\eta);\ \xi\in V_\epsilon(\bar\xi),\ |\eta|<C|\xi|\}$ of $(\bar\xi,\bar\eta)$, where $C>|\bar\eta|/|\bar\xi|$, we have $\langle(\xi,\eta)\rangle\le(1+C)\langle\xi\rangle$ and $|\widehat{\phi_2u_2}(\eta)|\le C'\langle\xi\rangle^M$, so the product is rapidly decreasing.

(b) $\bar\eta\neq 0$ and $(\bar y,\bar\eta)\notin\mathrm{WF}(u_2)$: the same with the factors exchanged.

These cases exhaust the possibilities: if both $\bar\xi\neq 0$ and $\bar\eta\neq 0$, the point is not in $\mathrm{WF}(u_1)\times\mathrm{WF}(u_2)$, so (a) or (b) holds; if $\bar\xi = 0$ then $\bar\eta\neq 0$ and, $\bar x$ being in $\operatorname{supp}u_1$, $(\bar y,\bar\eta)\notin\mathrm{WF}(u_2)$, so (b) holds; similarly if $\bar\eta = 0$. □

3.12.2 Restriction

Write $\mathbb{R}^{p+k}\ni(x,y)$, let $Y=\{y=0\}$ and $\iota:\mathbb{R}^p\to\mathbb{R}^{p+k}$, $\iota(x)=(x,0)$. We want to define $\iota^*u = u\restriction Y$. No reasonable such map exists on all distributions (think of $\delta(y)$), but it does under a condition on the wavefront set: $\mathrm{WF}(u)$ should not meet the *conormal bundle*

$$N^*Y = \{(x,0;0,\eta);\ x\in\mathbb{R}^p,\ \eta\in\mathbb{R}^k\setminus 0\}.$$

For $\delta > 0$ consider the closed cone $C_\delta = \{(\xi,\eta) \in \mathbb{R}^{p+k} \setminus 0;\ |\xi| \le \delta|\eta|\}$ and let $\mathcal{R}_\delta$ be the set of $w \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^{p+k})$ with $\hat{w}$ rapidly decreasing on C_δ. If $w \in \mathcal{R}_\delta$ then, since $|\hat{w}| \le C\langle(\xi,\eta)\rangle^M$,

$$\int |\hat{w}(\xi,\eta)|\,d\eta \le \int_{|\eta|\le|\xi|/\delta} C\langle(\xi,\eta)\rangle^M\,d\eta + \int_{|\eta|>|\xi|/\delta} C_N\langle\eta\rangle^{-N}\,d\eta \le C'\langle\xi\rangle^{M+k}, \tag{3.12.1}$$

locally uniformly in ξ, so

$$g_w(\xi) = \int \hat{w}(\xi,\eta)đ\eta \qquad (đ\eta = (2\pi)^{-k}d\eta)$$

is continuous and polynomially bounded, and we define $\iota^*w = \mathcal{F}^{-1}g_w \in \mathcal{S}'(\mathbb{R}^p)$.

Lemma 3.12.2. *Let $w \in \mathcal{R}_\delta$, $\rho \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^k)$ with $\int\rho = 1$, and $\rho_\epsilon(y) = \epsilon^{-k}\rho(y/\epsilon)$.*

(i) *For $\phi \in \mathcal{S}(\mathbb{R}^p)$, $\iota^*w(\phi) = \lim_{\epsilon\to0} w(\phi\boxtimes\rho_\epsilon)$.*

(ii) *If $w \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^{p+k})$ then $\iota^*w = w(\cdot,0)$.*

(iii) $\operatorname{supp}\iota^*w \subset \{x;\ (x,0) \in \operatorname{supp}w\}$*; in particular $\iota^*w \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^p)$, and $\iota^*w = 0$ if* $\operatorname{supp}w \cap Y = \emptyset$.

(iv) *For $\phi \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^p)$, $\phi(x)w \in \mathcal{R}_{\delta/2}$ and $\iota^*(\phi w) = \phi\,\iota^*w$.*

Proof. (i) By (3.8.1) in $\mathbb{R}^{p+k}$, $w(\phi\boxtimes\rho_\epsilon) = \iint \hat{w}(\xi,\eta)\hat{\phi}(-\xi)\hat{\rho}(-\epsilon\eta)đ\xi đ\eta$. As $\epsilon\to0$, $\hat{\rho}(-\epsilon\eta)\to\hat{\rho}(0) = 1$ with $|\hat{\rho}| \le \|\rho\|_{L^1}$, and $|\hat{w}(\xi,\eta)\hat{\phi}(-\xi)|$ is integrable by (3.12.1); by dominated convergence the limit is $\int g_w(\xi)\hat{\phi}(-\xi)đ\xi = \iota^*w(\phi)$.

(ii) is clear from (i), and (iii) follows from (i) too: if $(\bar{x},0) \notin \operatorname{supp}w$ and ϕ is supported near $\bar{x}$, then $\phi\boxtimes\rho_\epsilon$ vanishes near $\operatorname{supp}w$ for small ϵ.

(iv) By Lemma 3.8.1(ii), $\Sigma_{\mathrm{fr}}(w)$ does not meet the interior of C_δ, which contains $C_{\delta/2}$; as $\Sigma_{\mathrm{fr}}(\phi w) \subset \Sigma_{\mathrm{fr}}(w)$, Lemma 3.8.1(i) shows $\phi w \in \mathcal{R}_{\delta/2}$. The identity follows from (i), since $(\phi w)(f\boxtimes\rho_\epsilon) = w(\phi f\boxtimes\rho_\epsilon)$. □

Theorem 3.12.3 (Restriction)**.** *Let $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^{p+k})$ satisfy $\mathrm{WF}(u)\cap N^*Y = \emptyset$. There are $r,\delta > 0$ such that $\chi u \in \mathcal{R}_\delta$ for every $\chi = \chi(y) \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^k)$ supported in $\{|y| < r\}$, and*

$$\iota^*u = \iota^*(\chi u) \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^p), \qquad \chi = 1 \text{ near } 0,$$

is independent of the choice of χ. The map $u \mapsto \iota^*u$ is linear, coincides with $u \mapsto u(\cdot,0)$ on $\mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^{p+k})$, satisfies $\iota^*u(\phi) = \lim_{\epsilon\to0} u(\phi\boxtimes\rho_\epsilon)$, $\iota^*(\phi u) = \phi\,\iota^*u$ for $\phi \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^p)$, and

$$\mathrm{WF}(\iota^*u) \subset \iota^*\,\mathrm{WF}(u) = \{(x,\xi) \in T^*\mathbb{R}^p \setminus 0;\ (x,0,\xi,\eta) \in \mathrm{WF}(u) \text{ for some } \eta \in \mathbb{R}^k\}.$$

Proof. The compact sets $Z_\delta = \mathrm{WF}(u)\cap\{|y| \le \delta\}\cap(\mathbb{R}^{p+k}\times C_\delta)\cap\{|(\xi,\eta)| = 1\}$ decrease as $\delta\downarrow0$, and their intersection is contained in $\mathrm{WF}(u)\cap N^*Y = \emptyset$. Hence $Z_\delta = \emptyset$ for some $\delta > 0$; take $r = \delta$. If $\operatorname{supp}\chi \subset \{|y| < r\}$ then $\mathrm{WF}(\chi u) \subset \mathrm{WF}(u)\cap\{|y| < r\}$, so $\Sigma_{\mathrm{fr}}(\chi u)\cap C_\delta = \emptyset$ and $\chi u \in \mathcal{R}_\delta$ by Lemma 3.8.1(i). If χ,χ' are two choices, $(\chi-\chi')u \in \mathcal{R}_\delta$ vanishes near Y, so $\iota^*(\chi u) = \iota^*(\chi' u)$ by Lemma 3.12.2(iii). Linearity is clear, and the other properties, except the last, follow from Lemma 3.12.2, since $\chi(y)\rho_\epsilon(y) = \rho_\epsilon(y)$ for small ϵ. If $u \in \mathcal{R}_{\delta'}$ for some $\delta' > 0$ then ι^*u as defined before Lemma 3.12.2 agrees with $\iota^*(\chi u)$, by the limit formula.

For the wavefront set, let $(\bar{x},\bar{\xi}) \in T^*\mathbb{R}^p \setminus 0$ with $(\bar{x},0,\bar{\xi},\eta) \notin \mathrm{WF}(u)$ for all η. For $\theta \in (0,1)$ let

$$C'_\theta = C_\theta \cup \{(\xi,\eta);\ \xi \in \overline{V_\theta(\bar{\xi})}\},$$

a closed cone in $\mathbb{R}^{p+k} \setminus 0$. As $\theta \downarrow 0$ the compact sets $C'_\theta \cap \mathbb{S}^{p+k-1}$ decrease to the set of unit vectors (ξ,η) with $\xi = 0$ or $\xi \in \mathbb{R}_+\bar\xi$, which is disjoint from the closed cone $\{(\xi,\eta);\ (\bar x,0,\xi,\eta) \in \mathrm{WF}(u)\}$ by hypothesis and since $\mathrm{WF}(u) \cap N^*Y = \emptyset$. Hence for some θ the latter cone lies in the open cone $\mathbb{R}^{p+k} \setminus (C'_\theta \cup \{0\})$, and by Lemma 3.8.1(iii) there are $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^p)$, $\phi(\bar x) \neq 0$, and $\chi(y)$ as above (with small supports) such that $\Sigma_{\mathrm{fr}}(\phi\chi u) \cap C'_\theta = \emptyset$. Then $\widehat{\phi\chi u}$ is rapidly decreasing on C'_θ; in particular $\phi\chi u \in \mathcal{R}_\theta$, and for $\xi \in V_\theta(\bar\xi)$, where $(\xi,\eta) \in C'_\theta$ for all η,

$$|\widehat{\phi\,\iota^*u}(\xi)| = \Big|\int \widehat{\phi\chi u}(\xi,\eta)đ\eta\Big| \le C_N \int \langle(\xi,\eta)\rangle^{-N-k-1}\,d\eta \le C'_N\langle\xi\rangle^{-N}.$$

Here we used $\phi\iota^*u = \iota^*(\phi\chi u)$, by the independence of χ and Lemma 3.12.2(iv). By Proposition 3.8.2, $(\bar x,\bar\xi) \notin \mathrm{WF}(\iota^*u)$. □

One can picture this as follows: of $\mathrm{WF}(u)$ only the part over Y survives, and the covector (ξ,η) is projected to its component ξ along Y; the hypothesis guarantees that no non-zero covector is projected to zero (Figure 3.5).

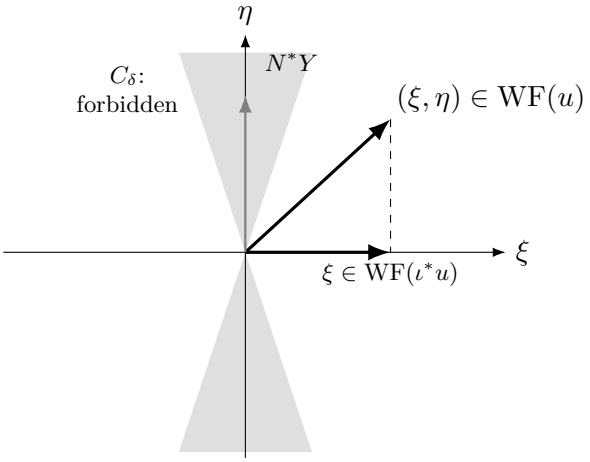


Figure 3.5: The fibre of $T^*\mathbb{R}^{p+k}$ over a point $(x,0) \in Y = \{y = 0\}$, schematically with $p = k = 1$. The vertical axis is the conormal direction N^*_xY. If $\mathrm{WF}(u)$ does not meet N^*Y, then near Y it avoids a whole cone $C_\delta = \{|\xi| \le \delta|\eta|\}$ (shaded), and by Theorem 3.12.3 the set $\mathrm{WF}(\iota^*u)$ is contained in the set of projections ξ of covectors $(\xi,\eta) \in \mathrm{WF}(u)$ over Y.

3.12.3 Projection

Let $\pi_1 : \mathbb{R}^{p+k} \to \mathbb{R}^p$, $\pi_1(x,y) = x$. For $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^{p+k})$ define the push-forward, or "integral over the fibres", $(\pi_1)_*u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^p)$ by

$$\Big(\int u(x,y)\,dy\Big)(\phi) = (\pi_1)_*u(\phi) = u(\pi_1^*\phi), \qquad \pi_1^*\phi(x,y) = \phi(x),$$

which makes sense because u acts on $\mathcal{C}^\infty(\mathbb{R}^{p+k})$. Thus projection is dual to pull-back of functions. By duality, $\mathrm{supp}(\pi_1)_*u \subset \pi_1(\mathrm{supp}\,u)$, since $\mathrm{supp}\,\pi_1^*\phi = \pi_1^{-1}(\mathrm{supp}\,\phi)$. Moreover $(\pi_1)_*$ maps $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{p+k})$ to $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^p)$, and

$$\widehat{(\pi_1)_*u}(\xi) = u(e^{-ix\cdot\xi}) = \hat u(\xi,0), \qquad \phi\,(\pi_1)_*u = (\pi_1)_*\big((\pi_1^*\phi)u\big). \tag{3.12.2}$$

Proposition 3.12.4 (Projection)**.** *For $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^{p+k})$,*

$$\mathrm{WF}\big((\pi_1)_*u\big) \subset \{(x,\xi) \in T^*\mathbb{R}^p \setminus 0;\ (x,y,\xi,0) \in \mathrm{WF}(u) \text{ for some } y \in \mathbb{R}^k\}.$$

Proof. Let $(\bar{x},\bar{\xi})$ be outside the right side. We claim there are $r,\theta>0$ such that $\mathrm{WF}(u)$ does not meet $\{|x-\bar{x}|<r\}\times\mathbb{R}^k\times C''_\theta$, where $C''_\theta=\{(\xi,\eta);\ \xi\in\overline{V_\theta(\bar{\xi})},\ |\eta|\le\theta|\xi|\}$. Otherwise there are $(x_j,y_j,\xi_j,\eta_j)\in\mathrm{WF}(u)$ with $|(\xi_j,\eta_j)|=1$, $x_j\to\bar{x}$ and $(\xi_j,\eta_j)\in C''_{1/j}$; since y_j lies in the compact set $\pi_2(\operatorname{supp}u)$, a subsequence converges to a point $(\bar{x},y,\bar{\xi}/|\bar{\xi}|,0)$ of the closed set $\mathrm{WF}(u)$, a contradiction. Now let $\phi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^p)$ be supported in $\{|x-\bar{x}|<r\}$ with $\phi(\bar{x})\neq0$. Then $\Sigma_{\mathrm{fr}}((\pi_1^*\phi)u)\cap C''_\theta=\emptyset$, so by Lemma 3.8.1(i) and (3.12.2), $\widehat{\phi(\pi_1)_*u}(\xi)=\widehat{(\pi_1^*\phi)u}(\xi,0)$ is rapidly decreasing on $V_\theta(\bar{\xi})$. By Proposition 3.8.2, $(\bar{x},\bar{\xi})\notin\mathrm{WF}((\pi_1)_*u)$. □

Corollary 3.12.5. *If $u\in\mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^{p+k})$ and $\mathrm{WF}(u)\cap\{(x,y,\xi,0);\ \xi\neq0\}=\emptyset$ then $(\pi_1)_*u\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^p)$.*

The covectors $(x,y,\xi,0)$ are exactly those which are conormal to the fibres $\{x\}\times\mathbb{R}^k$ over which we integrate. Thus the only singularities which survive integration are those conormal to the fibres; those which are even partially in the direction of integration are wiped out. This fits with the fact that integration in *all* variables leaves no singularities at all.

3.13 Products of distributions

The product of two distributions is not defined in general: δ_0^2 has no sensible meaning. The wavefront set gives a simple sufficient condition, the same as for the pairing. We express the product as "$u_1(x)u_2(x+y)$ restricted to $y=0$". Let $L:\mathbb{R}^{2n}\to\mathbb{R}^{2n}$ be the linear map $L(x,y)=(x,x+y)$, with $\det L=1$ and transpose $DL^t(\xi,\eta)=(\xi+\eta,\eta)$.

Theorem 3.13.1 (Products)**.** *Let $u_1,u_2\in\mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$ satisfy*

$$(x,\xi)\in\mathrm{WF}(u_1)\Longrightarrow(x,-\xi)\notin\mathrm{WF}(u_2).$$

Then $w=L^*(u_1\boxtimes u_2)\in\mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^{2n})$ satisfies $\mathrm{WF}(w)\cap N^*\{y=0\}=\emptyset$, and the product

$$u_1u_2=\iota^*L^*(u_1\boxtimes u_2)\in\mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$$

has the following properties.

(i) $\operatorname{supp}(u_1u_2)\subset\operatorname{supp}u_1\cap\operatorname{supp}u_2$, *and for $\phi\in\mathcal{S}(\mathbb{R}^n)$ and ρ, ρ_ϵ as in Lemma 3.12.2 (now on $\mathbb{R}^n$), with $\check{\rho}_\epsilon(t)=\rho_\epsilon(-t)$,*

$$(u_1u_2)(\phi)=\lim_{\epsilon\to0}u_1(\phi\cdot(u_2*\check{\rho}_\epsilon)).$$

(ii) *If u_1 or u_2 is smooth, u_1u_2 is the usual product.*

(iii) $(\phi u_1)u_2=\phi(u_1u_2)$ *for $\phi\in\mathcal{C}^\infty(\mathbb{R}^n)$.*

(iv) *The wavefront set satisfies*

$$\begin{aligned}\mathrm{WF}(u_1u_2)\subset{}&\{(x,\xi);\ x\in\operatorname{supp}u_1,\ (x,\xi)\in\mathrm{WF}(u_2)\}\\&\cup\{(x,\xi);\ x\in\operatorname{supp}u_2,\ (x,\xi)\in\mathrm{WF}(u_1)\}\\&\cup\{(x,\xi_1+\xi_2);\ (x,\xi_1)\in\mathrm{WF}(u_1),\ (x,\xi_2)\in\mathrm{WF}(u_2)\}.\end{aligned}$$

Note that $\xi_1+\xi_2\neq0$ in the last set, by the hypothesis.

Proof. By Theorem 3.9.1 (for the linear map L),

$$\mathrm{WF}(w) = \{(x, y; \xi + \eta, \eta);\ (x, x + y; \xi, \eta) \in \mathrm{WF}(u_1 \boxtimes u_2)\}.$$

A point of $N^*\{y = 0\}$ has $y = 0$ and $\xi + \eta = 0$, $\eta \neq 0$; by Proposition 3.12.1, $(x, x; -\eta, \eta)$ with both components non-zero can only lie in $\mathrm{WF}(u_1 \boxtimes u_2)$ if $(x, -\eta) \in \mathrm{WF}(u_1)$ and $(x, \eta) \in \mathrm{WF}(u_2)$, which is excluded. So Theorem 3.12.3 applies. The support statement follows from that theorem, since $(x, 0) \in \operatorname{supp} w$ means $(x, x) \in \operatorname{supp} u_1 \times \operatorname{supp} u_2$. For the formula in (i), by Theorem 3.12.3 and the definition of L^*,

$$(u_1 u_2)(\phi) = \lim_{\epsilon \to 0} (u_1 \boxtimes u_2)(\phi(x)\rho_\epsilon(z - x)) = \lim_{\epsilon \to 0} u_1\big(\phi(x)\, u_2(\rho_\epsilon(\cdot - x))\big),$$

and $u_2(\rho_\epsilon(\cdot - x)) = (u_2 * \check{\rho}_\epsilon)(x)$.

(ii) If u_2 is smooth, $u_2 * \check{\rho}_\epsilon \to u_2$ in $\mathcal{C}^\infty$, so (i) gives $u_1(\phi u_2)$. If u_1 is smooth, $u_1(\phi \cdot (u_2 * \check{\rho}_\epsilon)) = u_2((u_1\phi) * \rho_\epsilon) \to u_2(u_1\phi)$.

(iii) is immediate from (i).

(iv) By Theorem 3.12.3 and the formula for $\mathrm{WF}(w)$,

$$\mathrm{WF}(u_1 u_2) \subset \{(x, \xi + \eta);\ (x, x; \xi, \eta) \in \mathrm{WF}(u_1 \boxtimes u_2)\},$$

and Proposition 3.12.1 gives the three sets. □

Question. On $\mathbb{R}^2$, δ_0^2 is not defined. Why is the product $\delta(x_1)\delta(x_2)$ of two distributions which are both singular at the origin defined?

Answer. Up to a cut-off, $u_1 = \delta(x_1)$ has $\mathrm{WF}(u_1) = \{(0, x_2; \xi_1, 0);\ \xi_1 \neq 0\}$, and $u_2 = \delta(x_2)$ has $\mathrm{WF}(u_2) = \{(x_1, 0; 0, \xi_2);\ \xi_2 \neq 0\}$. At the origin the covectors $(\xi_1, 0)$ and $-(0, \xi_2)$ are never equal, so the hypothesis of Theorem 3.13.1 holds. The product is δ_0, and part (iv) allows all covectors $(\xi_1, \xi_2) \neq 0$ at the origin, as it must. For δ_0 times itself, every covector ξ at the origin is matched by $-\xi$, and the hypothesis fails. □

Remark 3.13.2. If the stronger condition $\Sigma_{\mathrm{fr}}(u_1) \cap (-\Sigma_{\mathrm{fr}}(u_2)) = \emptyset$ holds, for example if u_1 and u_2 have small supports near a point $\bar{x}$ at which the hypothesis of the theorem holds (Lemma 3.8.1(iii)), then the familiar formula

$$\widehat{u_1 u_2}(\xi) = \int \hat{u}_1(\xi - \eta)\hat{u}_2(\eta)đ\eta$$

holds, with an absolutely convergent integral. Indeed $\hat{w}(\xi, \eta) = \hat{u}_1(\xi - \eta)\hat{u}_2(\eta)$, and $w \in \mathcal{R}_\delta$ for small δ: choose open cones $V_i \supset \Sigma_{\mathrm{fr}}(u_i)$ with $\overline{V_1} \cap (-\overline{V_2}) = \emptyset$; if $|\xi| \leq \delta|\eta|$ then either $\eta \notin V_2$, where $\hat{u}_2(\eta)$ is rapidly decreasing and $|\hat{u}_1(\xi - \eta)| \leq C\langle\eta\rangle^M$, or $\eta \in V_2$, and then $\xi - \eta$, whose direction differs from that of $-\eta$ by at most 2δ, lies in a closed cone disjoint from $\Sigma_{\mathrm{fr}}(u_1)$ if δ is small, so $\hat{u}_1(\xi - \eta)$ is rapidly decreasing in $\langle\eta\rangle$. The formula is then the definition of $\iota^* w$.

3.14 Pull-back

Now let $F : \Omega_1 \to \Omega_2$ be a smooth map between open sets $\Omega_1 \subset \mathbb{R}^n$ and $\Omega_2 \subset \mathbb{R}^m$; the dimensions may differ. When can we define F^*u for $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\Omega_2)$, and what is its wavefront set? For a general F there is no consistent definition for all distributions (consider F constant and $u = \delta$). We assume that F is *proper*, i.e. $F^{-1}(K)$ is compact for every compact

$K \subset \Omega_2$, mostly for convenience: then $F^*\phi = \phi \circ F \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega_1)$ for $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega_2)$. The differential of F defines the cotangent maps

$$F_x^* : \mathbb{R}^m = T^*_{F(x)}\Omega_2 \to T^*_x\Omega_1 = \mathbb{R}^n, \qquad \eta \mapsto DF(x)^t\eta, \quad \text{i.e. } \xi_i = \sum_j \frac{\partial F_j}{\partial x_i}(x)\eta_j,$$

and we set

$$N_F = \{(F(x),\eta) \in \Omega_2 \times (\mathbb{R}^m \setminus 0);\ x \in \Omega_1,\ DF(x)^t\eta = 0\},$$
$$F^*W = \{(x, DF(x)^t\eta);\ (F(x),\eta) \in W\} \quad \text{for } W \subset T^*\Omega_2 \setminus 0.$$

Note that F^* is a relation rather than a map between cotangent spaces. N_F is the set of covectors that are mapped to zero; it is empty if F is a submersion, and for ι of the previous section it is N^*Y.

The construction decomposes F into elementary maps. Let $U = \{(x,y) \in \Omega_1 \times \mathbb{R}^m;\ y + F(x) \in \Omega_2\}$ and

$$H : U \to \Omega_1 \times \Omega_2, \qquad H(x,y) = (x, y + F(x)),$$

a diffeomorphism with inverse $(x,z) \mapsto (x, z - F(x))$ and $\det DH = 1$; it maps $U \cap \{y = 0\}$ onto the graph of F. For a function, $F^*\phi = \iota^* H^*(1 \boxtimes \phi)$, $\iota(x) = (x,0)$.

Theorem 3.14.1 (Pull-back)**.** *Let $F : \Omega_1 \to \Omega_2$ be a proper smooth map and let $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_2)$ satisfy $\mathrm{WF}(u) \cap N_F = \emptyset$. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega_1)$ equal 1 near the compact set $F^{-1}(\operatorname{supp} u)$. Then $H^*(\chi \boxtimes u) \in \mathcal{C}^{-\infty}_{\mathrm{c}}(U)$ satisfies the hypothesis of Theorem 3.12.3, and*

$$F^*u = \iota^* H^*(\chi \boxtimes u) \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega_1)$$

is independent of χ, with $\operatorname{supp} F^*u \subset F^{-1}(\operatorname{supp} u)$. This defines a linear map which extends $\phi \mapsto \phi \circ F$ on $\mathcal{C}^\infty_{\mathrm{c}}(\Omega_2)$, satisfies

$$F^*u(\phi) = \lim_{\epsilon \to 0} u\Big(\int \chi(x)\phi(x)\rho_\epsilon(\cdot - F(x))\,dx\Big), \qquad \phi \in \mathcal{S}(\mathbb{R}^n),$$

for $\rho \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^m)$, $\int \rho = 1$, agrees with the pull-back of Lemma 3.6.1 if F is a diffeomorphism, and

$$\mathrm{WF}(F^*u) \subset F^* \mathrm{WF}(u) = \bigcup_{x \in \Omega_1} \{x\} \times F_x^*(\mathrm{WF}(u) \cap T^*_{F(x)}\Omega_2).$$

Proof. By Proposition 3.12.1, since $\mathrm{WF}(\chi) = \emptyset$, $\mathrm{WF}(\chi \boxtimes u) \subset \{(x,z;0,\eta);\ x \in \operatorname{supp}\chi,\ (z,\eta) \in \mathrm{WF}(u)\}$. Since $DH(x,y)^t(\xi,\eta) = (\xi + DF(x)^t\eta, \eta)$, Theorem 3.9.1 gives

$$\mathrm{WF}\big(H^*(\chi \boxtimes u)\big) \subset \{(x,y; DF(x)^t\eta, \eta);\ x \in \operatorname{supp}\chi,\ (y + F(x),\eta) \in \mathrm{WF}(u)\}.$$

A point of this set over $y = 0$ lies in $N^*\{y = 0\}$ only if $DF(x)^t\eta = 0$ and $(F(x),\eta) \in \mathrm{WF}(u)$, i.e. if $(F(x),\eta) \in N_F \cap \mathrm{WF}(u) = \emptyset$. Thus Theorem 3.12.3 applies, and gives $\operatorname{supp} F^*u \subset \{x \in \operatorname{supp}\chi;\ F(x) \in \operatorname{supp} u\}$ and $\mathrm{WF}(F^*u) \subset \{(x, DF(x)^t\eta);\ (F(x),\eta) \in \mathrm{WF}(u)\}$.

If χ' is another choice, $H^*((\chi - \chi') \boxtimes u)$ is supported in $\{(x,y);\ x \in \operatorname{supp}(\chi - \chi'),\ y + F(x) \in \operatorname{supp} u\}$, which does not meet $\{y = 0\}$ because $\chi = \chi'$ near $F^{-1}(\operatorname{supp} u)$; hence its restriction vanishes (Lemma 3.12.2(iii)). By Theorem 3.12.3, the formula for H^* in Lemma 3.6.1 (with Jacobian 1) and the definition of the exterior product,

$$F^*u(\phi) = \lim_{\epsilon \to 0} H^*(\chi \boxtimes u)(\phi \boxtimes \rho_\epsilon) = \lim_{\epsilon \to 0} (\chi \boxtimes u)\big(\phi(x)\rho_\epsilon(z - F(x))\big),$$

which is the stated formula. If u is smooth, the expression under the limit is $\int \chi(x)\phi(x)(u * \check{\rho}_\epsilon)(F(x))\,dx$, with $\check{\rho}_\epsilon(t) = \rho_\epsilon(-t)$, which converges to $\int \chi\phi\,(u \circ F)\,dx = \int \phi\,(u \circ F)\,dx$. If

F is a diffeomorphism with inverse G, substituting $x = G(z')$ shows that the function $\int \chi\phi(x)\rho_\epsilon(z - F(x))\,dx$ is $\rho_\epsilon * f$ with $f = |\det DG|\,(\chi\phi)\circ G \in \mathcal{C}_c^\infty(\Omega_2)$, which converges to f in $\mathcal{C}^\infty$; so $F^*u(\phi) = u(f)$, which is the definition in Lemma 3.6.1 since $\chi \circ G = 1$ near $\operatorname{supp} u$. □

The hypothesis $\mathrm{WF}(u) \cap N_F = \emptyset$ is natural: without it the right side of the wavefront set bound would contain zero covectors. Restriction to a submanifold, products ($u_1u_2 = \Delta^*(u_1 \boxtimes u_2)$ with $\Delta(x) = (x, x)$, for which $N_\Delta = \{(x, x; \xi, -\xi)\}$) and diffeomorphisms are all special cases; see Exercise 3.16.

3.15 Push-forward

The dual operation is push-forward. Pull-back maps $\mathcal{C}_c^\infty$ to $\mathcal{C}_c^\infty$ (for proper F) but is not always defined on distributions. Dually:

Theorem 3.15.1 (Push-forward)**.** *Let $F : \Omega_1 \to \Omega_2$ be a smooth map from an open subset of $\mathbb{R}^n$ to an open subset of $\mathbb{R}^m$. For $u \in \mathcal{C}_c^{-\infty}(\Omega_1)$,*

$$F_*u(\phi) = u(F^*\phi), \qquad \phi \in \mathcal{C}^\infty(\Omega_2),$$

defines $F_*u \in \mathcal{C}_c^{-\infty}(\Omega_2)$, with $\operatorname{supp} F_*u \subset F(\operatorname{supp} u)$, and

$$\mathrm{WF}(F_*u) \subset \{(F(x), \eta);\ x \in \operatorname{supp} u,\ \eta \neq 0,\ DF(x)^t\eta = 0\} \cup \{(F(x), \eta);\ \eta \neq 0,\ (x, DF(x)^t\eta) \in \mathrm{WF}(u)\}.$$

Thus F_* is always defined, but even if $u \in \mathcal{C}_c^\infty(\Omega_1)$ in general $F_*u \notin \mathcal{C}_c^\infty(\Omega_2)$; if F is a submersion the first set is empty and F_* maps $\mathcal{C}_c^\infty(\Omega_1)$ into $\mathcal{C}_c^\infty(\Omega_2)$. The two sets in the bound are really the same, if one regards $(x, 0)$ as belonging to the "wavefront set" of u at every point of $\operatorname{supp} u$.

Proof. F_*u is a distribution since $\phi \mapsto \psi F^*\phi$, $\psi \in \mathcal{C}_c^\infty(\Omega_1)$ equal to 1 near $\operatorname{supp} u$, is continuous; the support statement is clear. Formally,

$$F_*u(\phi) = \int u(x)\phi(F(x))\,dx = \iint u(x)\delta(y - F(x))\phi(y)\,dy\,dx,$$

so we write $F_*u = (\pi_2)_*\tilde{H}^*(u \boxtimes \delta)$, where $\delta \in \mathcal{C}_c^{-\infty}(\mathbb{R}^m)$, $\tilde{H} : \Omega_1 \times \mathbb{R}^m \to \Omega_1 \times \mathbb{R}^m$, $\tilde{H}(x, y) = (x, y - F(x))$, is a diffeomorphism with $\det D\tilde{H} = 1$, and $\pi_2(x, y) = y$. To check this identity, note that $\tilde{H}^*(u \boxtimes \delta) \in \mathcal{C}_c^{-\infty}(\Omega_1 \times \mathbb{R}^m)$ is supported in the compact graph $\{(x, F(x));\ x \in \operatorname{supp} u\}$ and, by Lemma 3.6.1,

$$(\pi_2)_*\tilde{H}^*(u \boxtimes \delta)(\phi) = (u \boxtimes \delta)(\psi(x)\phi(z + F(x))) = u(\psi\,\phi \circ F) = F_*u(\phi).$$

Now we estimate the wavefront set using the elementary operations. Since $\mathrm{WF}(\delta) = \{0\} \times (\mathbb{R}^m \setminus 0)$ (Example 3.8.3), Proposition 3.12.1 gives

$$\mathrm{WF}(u \boxtimes \delta) \subset \{(x, 0; \xi, \eta);\ (\xi, \eta) \neq 0,\ (x, \xi) \in \mathrm{WF}(u) \text{ or } (\xi = 0,\ x \in \operatorname{supp} u)\}.$$

Since $D\tilde{H}(x, y)^t(\xi, \eta) = (\xi - DF(x)^t\eta, \eta)$, Theorem 3.9.1 gives

$$\mathrm{WF}(\tilde{H}^*(u \boxtimes \delta)) \subset \{(x, F(x); \xi - DF(x)^t\eta, \eta);\ (x, \xi) \in \mathrm{WF}(u) \text{ or } (\xi = 0,\ x \in \operatorname{supp} u)\}.$$

Finally, by Proposition 3.12.4 (with the roles of the factors exchanged), $(y, \eta) \in \mathrm{WF}(F_*u)$ only if $\eta \neq 0$ and $(x, y; 0, \eta)$ lies in the last set for some x, i.e. $y = F(x)$ and $\xi = DF(x)^t\eta$ with either $(x, DF(x)^t\eta) \in \mathrm{WF}(u)$, or $DF(x)^t\eta = 0$ and $x \in \operatorname{supp} u$. □

The operator F_* has Schwartz kernel $K(y,x)=\delta(y-F(x))$: $F_*u(y)=\int K(y,x)u(x)\,dx$. Locally K is the pull-back of $\delta\boxtimes\chi$, $\chi\in\mathcal{C}_c^\infty(\Omega_1)$, under the diffeomorphism $(y,x)\mapsto(y-F(x),x)$, so by Proposition 3.12.1 and Theorem 3.9.1,

$$\mathrm{WF}(K)\subset\{(F(x),x;\eta,-DF(x)^t\eta);\ \eta\neq0\}.$$

In the notation of the next section (with the roles of the two factors interchanged), $\mathrm{WF}'(K)$ is contained in the relation $\{((F(x),\eta),(x,DF(x)^t\eta))\}$, and the bound of Theorem 3.15.1 reads $\mathrm{WF}(F_*u)\subset\mathrm{WF}'(K)\circ\mathrm{WF}(u)\cup\{(y,\eta);\ (y,x;\eta,0)\in\mathrm{WF}(K),\ x\in\operatorname{supp}u\}$.

3.16 The wavefront relation

A pseudodifferential operator keeps every singularity where it is. A general operator, such as the solution operator of the wave equation, moves singularities around. The wavefront set of its kernel, read as a relation on T^*, says where each singularity can go. One serious application of these results is to general operators with distributional kernels. Let $\Omega_1\subset\mathbb{R}^n$, $\Omega_2\subset\mathbb{R}^m$ be open and $K\in\mathcal{C}_c^{-\infty}(\Omega_1\times\Omega_2)$. Then K defines an operator

$$A:\mathcal{C}_c^\infty(\Omega_2)\to\mathcal{C}_c^{-\infty}(\Omega_1),\qquad Au(\phi)=K(\phi\boxtimes u),$$

formally $Au(x)=\int K(x,y)u(y)\,dy$. We set

$$\begin{aligned}\mathrm{WF}'(K)&=\{((x,\xi),(y,\eta));\ (x,y;\xi,-\eta)\in\mathrm{WF}(K)\}\subset(\Omega_1\times\mathbb{R}^n)\times(\Omega_2\times\mathbb{R}^m),\\ \mathrm{WF}_{\Omega_1}(K)&=\{(x,\xi)\in T^*\Omega_1\setminus0;\ (x,y;\xi,0)\in\mathrm{WF}(K)\text{ for some }y\},\\ \mathrm{WF}'_{\Omega_2}(K)&=\{(y,\eta)\in T^*\Omega_2\setminus0;\ (x,y;0,-\eta)\in\mathrm{WF}(K)\text{ for some }x\}.\end{aligned}$$

We regard $\mathrm{WF}'(K)$ as a relation: for $W\subset T^*\Omega_2\setminus0$,

$$\mathrm{WF}'(K)\circ W=\{(x,\xi)\in T^*\Omega_1\setminus0;\ ((x,\xi),(y,\eta))\in\mathrm{WF}'(K)\text{ for some }(y,\eta)\in W\}.$$

Theorem 3.16.1 (Wavefront relation)**.** *Let $K\in\mathcal{C}_c^{-\infty}(\Omega_1\times\Omega_2)$ and let $\chi\in\mathcal{C}_c^\infty(\Omega_1)$ equal 1 near the projection of* $\operatorname{supp}K$*. For every $u\in\mathcal{C}_c^{-\infty}(\Omega_2)$ with*

$$\mathrm{WF}(u)\cap\mathrm{WF}'_{\Omega_2}(K)=\emptyset\tag{3.16.1}$$

the product $K\cdot(\chi\boxtimes u)$ is defined, and

$$Au=(\pi_1)_*(K\cdot(\chi\boxtimes u))\in\mathcal{C}_c^{-\infty}(\Omega_1)$$

extends the operator A above. It satisfies

$$\mathrm{WF}(Au)\subset\mathrm{WF}'(K)\circ\mathrm{WF}(u)\ \cup\ \{(x,\xi);\ (x,y;\xi,0)\in\mathrm{WF}(K)\text{ for some }y\in\operatorname{supp}u\}.$$

Proof. By Proposition 3.12.1, $\mathrm{WF}(\chi\boxtimes u)\subset\{(x,y;0,\eta);\ (y,\eta)\in\mathrm{WF}(u)\}$. The product condition of Theorem 3.13.1 fails only at points $(x,y;0,-\eta)\in\mathrm{WF}(K)$ with $(y,\eta)\in\mathrm{WF}(u)$, which are excluded by (3.16.1). If u is smooth the product is the usual product of K with the smooth function $\chi\boxtimes u$, and $(\pi_1)_*(K\cdot(\chi\boxtimes u))(\phi)=K(\chi\phi\boxtimes u)=K(\phi\boxtimes u)$, so Au agrees with the previous definition. By Theorem 3.13.1(iv), a point of $\mathrm{WF}(K\cdot(\chi\boxtimes u))$ has the form $(x,y;\zeta_1+\zeta_2)$ where either $\zeta_1=(\xi_1,\eta_1)$ with $(x,y;\zeta_1)\in\mathrm{WF}(K)$, or $\zeta_1=0$ and $(x,y)\in\operatorname{supp}K$; and either $\zeta_2=(0,\eta_2)$ with $(y,\eta_2)\in\mathrm{WF}(u)$, or $\zeta_2=0$ and $y\in\operatorname{supp}u$. By Proposition 3.12.4, $(x,\xi)\in\mathrm{WF}(Au)$ requires such a point with $\zeta_1+\zeta_2=(\xi,0)$, $\xi\neq0$. If $\zeta_1=0$ the first component of $\zeta_1+\zeta_2$ vanishes, which is excluded. If $\zeta_2=0$ then $(x,y;\xi,0)\in\mathrm{WF}(K)$ with $y\in\operatorname{supp}u$. Otherwise $\xi_1=\xi$ and $\eta_1=-\eta_2$, so $(x,y;\xi,-\eta_2)\in\mathrm{WF}(K)$ with $(y,\eta_2)\in\mathrm{WF}(u)$, that is $(x,\xi)\in\mathrm{WF}'(K)\circ\mathrm{WF}(u)$. □

Discussion. The simplest example is a translation, $Au(x) = u(x-a)$, with kernel $K = \delta(x-y-a)$ (cut off to have compact support). As for δ composed with a linear map, $\mathrm{WF}(K) = \{(x, x-a; \xi, -\xi)\}$, so $\mathrm{WF}'(K) = \{((x,\xi),(x-a,\xi))\}$, and the theorem says that a singularity of u at (y,η) can only produce a singularity of Au at $(y+a,\eta)$. The sign change in the definition of $\mathrm{WF}'(K)$ is what makes this the graph of a map, and not of its composition with $\xi \mapsto -\xi$. The second set in the theorem, coming from covectors $(\xi, 0)$ in $\mathrm{WF}(K)$, is empty here.

Remark 3.16.2. If $K = K_B$ for some $B \in \Psi^m(\mathbb{R}^n)$ with kernel of compact support in $\Omega \times \Omega$, then $Au = Bu$ for every $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\Omega)$. Indeed, by Theorem 3.13.1(i) with a product mollifier $\rho = \rho^{(1)} \boxtimes \rho^{(2)}$ on $\mathbb{R}^{2n}$, and $\phi \in \mathcal{C}_{\mathrm{c}}^{\infty}$, $\psi \in \mathcal{C}_{\mathrm{c}}^{\infty}$ equal to 1 near $\pi_2(\operatorname{supp} K)$,

$$Au(\phi) = \lim_{\epsilon\to 0} K\big((\phi\,(\chi * \check{\rho}_\epsilon^{(1)})) \boxtimes (\psi\,(u * \check{\rho}_\epsilon^{(2)}))\big) = \lim_{\epsilon\to 0}(Bu_\epsilon)(\phi) = \lim_{\epsilon\to 0} u_\epsilon(B^t\phi) = u(B^t\phi),$$

where $u_\epsilon = u * \check{\rho}_\epsilon^{(2)} \to u$ in $\mathcal{S}'$, since $\chi * \check{\rho}_\epsilon^{(1)} = 1$ near the projection of $\operatorname{supp} K$ for small ϵ.

Corollary 3.16.3. *Let $K \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\Omega_1 \times \Omega_2)$.*

(i) *If $\mathrm{WF}(K) \cap \{(x,y;\xi,0)\} = \emptyset$ then A maps $\mathcal{C}_{\mathrm{c}}^{\infty}(\Omega_2)$ into $\mathcal{C}_{\mathrm{c}}^{\infty}(\Omega_1)$.*

(ii) *If $\mathrm{WF}(K) \cap \{(x,y;0,\eta)\} = \emptyset$ then A extends to a linear map $A : \mathcal{C}_{\mathrm{c}}^{-\infty}(\Omega_2) \to \mathcal{C}_{\mathrm{c}}^{-\infty}(\Omega_1)$ with $\mathrm{WF}(Au) \subset \mathrm{WF}'(K) \circ \mathrm{WF}(u) \cup \mathrm{WF}_{\Omega_1}(K)$.*

Proof. Immediate from Theorem 3.16.1: in (i), $\mathrm{WF}(u) = \emptyset$ for smooth u, and in (ii), condition (3.16.1) is void. □

3.16.1 Applications

For a pseudodifferential operator $B \in \Psi^m(\mathbb{R}^n)$ with kernel of compact support, Proposition 3.10.1 gives

$$\mathrm{WF}'(K_B) = \{((x,\xi),(x,\xi));\ (x,\xi) \in \mathrm{WF}'(B)\},$$

a subset of the identity relation. In particular $\mathrm{WF}(K_B)$ contains no points of the form $(x,y;\xi,0)$ or $(x,y;0,\eta)$, so both parts of Corollary 3.16.3 apply, and by Remark 3.16.2 Theorem 3.16.1 reduces to $\mathrm{WF}(Bu) \subset \mathrm{WF}'(B) \cap \mathrm{WF}(u)$; this recovers microlocality (Theorem 3.7.7) for such operators. The next chapters will show that many other natural operators, such as the solution operators of hyperbolic equations, have kernels whose wavefront sets are well-behaved relations, and Theorem 3.16.1 then describes how they propagate singularities.

Remark 3.16.4. The restriction to compactly supported kernels is only for simplicity. If K is a distribution on $\Omega_1 \times \Omega_2$ with *proper* support, i.e. both projections $\operatorname{supp} K \to \Omega_i$ are proper, then A maps $\mathcal{C}_{\mathrm{c}}^{\infty}(\Omega_2)$ to distributions of compact support in Ω_1, and the results of this section hold for $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\Omega_2)$, since only K on the compact set $\operatorname{supp} K \cap (\Omega_1 \times \operatorname{supp} u)$ is involved.

3.17 Sobolev wavefront sets

The wavefront set records where, and in which directions, a distribution fails to be smooth. For operators of finite order, and in particular for the propagation theorem of Section 3.20, regularity has to be measured more finely, on the scale of Sobolev spaces. Recall from Chapter 2 that $H^s(\mathbb{R}^n) = \{u \in \mathcal{S}'(\mathbb{R}^n);\ \langle D\rangle^s u \in L^2\}$ with $\|u\|_{H^s} = \|\langle\xi\rangle^s \hat{u}\|_{L^2}$ (norms of Fourier transforms taken with respect to $d\!\!\bar{}\xi$), that every $A \in \Psi^m(\mathbb{R}^n)$ is bounded from H^s to H^{s-m}, and we put $H^{-\infty}(\mathbb{R}^n) = \bigcup_s H^s(\mathbb{R}^n)$, $H^{\infty}(\mathbb{R}^n) = \bigcap_s H^s(\mathbb{R}^n) \subset \mathcal{C}^{\infty}(\mathbb{R}^n)$. Every $u \in \mathcal{C}_{\mathrm{c}}^{-\infty}(\mathbb{R}^n)$ lies in $H^{-\infty}(\mathbb{R}^n)$: by Section 3.8, $|\hat{u}(\xi)| \le C\langle\xi\rangle^k$ for some k, so $u \in H^t$ for $t < -k - n/2$.

The Sobolev pairing

For $v \in H^t(\mathbb{R}^n)$ and $w \in H^{-t}(\mathbb{R}^n)$ we set

$$(v,w) = \int \hat v(\xi)\,\overline{\hat w(\xi)}đ\xi.$$

Writing the integrand as $\langle\xi\rangle^t\hat v \cdot \langle\xi\rangle^{-t}\overline{\hat w}$, the Cauchy–Schwarz inequality gives $|(v,w)| \le \|v\|_{H^t}\|w\|_{H^{-t}}$; for $t=0$ this is the L^2 inner product (Plancherel), and $(w,v) = \overline{(v,w)}$. If $T \in \Psi^k(\mathbb{R}^n)$ then

$$(Tv,w) = (v,T^*w), \qquad v \in H^t(\mathbb{R}^n),\ w \in H^{k-t}(\mathbb{R}^n). \tag{3.17.1}$$

Indeed both sides are continuous on $H^t \times H^{k-t}$, they agree on $\mathcal{S}\times\mathcal{S}$ by the definition of the adjoint, and $\mathcal{S}(\mathbb{R}^n)$ is dense in every $H^t(\mathbb{R}^n)$, since $\langle D\rangle^t$ maps $\mathcal{S}$ onto $\mathcal{S}$ and is unitary from H^t onto L^2. Finally, *if $v \in H^{-\infty}(\mathbb{R}^n)$ and $|(v,\phi)| \le C\|\phi\|_{L^2}$ for all $\phi \in \mathcal{S}(\mathbb{R}^n)$, then* $v \in L^2(\mathbb{R}^n)$: $\hat v$ is locally square integrable, and the estimate says that $\psi \mapsto \int \hat v\bar\psi đ\xi$ is bounded in the L^2 norm on $\mathcal{C}^\infty_{\mathrm c}(\mathbb{R}^n) \subset \mathcal{F}(\mathcal{S})$; by the Riesz representation theorem there is $f \in L^2$ with $\int(\hat v - f)\bar\psi đ\xi = 0$ for all $\psi \in \mathcal{C}^\infty_{\mathrm c}$, so $\hat v = f \in L^2$.

3.17.1 Distributions of compact support

Definition 3.17.1. For $u \in \mathcal{C}^{-\infty}_{\mathrm c}(\mathbb{R}^n)$ and $s \in \mathbb{R}$, the *wavefront set of order s* (or *H^s-wavefront set*) of u is

$$\mathrm{WF}_s(u) = \bigcap\{\mathrm{Char}_0(A);\ A \in \Psi^0(\mathbb{R}^n),\ Au \in H^s(\mathbb{R}^n)\}.$$

Thus $z \notin \mathrm{WF}_s(u)$ if and only if some $A \in \Psi^0(\mathbb{R}^n)$ which is elliptic at z satisfies $Au \in H^s$. The definition is Definition 3.7.1 with $\mathcal{C}^\infty$ replaced by H^s, and the proofs below follow those of Section 3.7, with the Sobolev boundedness of pseudodifferential operators added.

Lemma 3.17.2. *Let $u \in \mathcal{C}^{-\infty}_{\mathrm c}(\mathbb{R}^n)$ and $z \notin \mathrm{WF}_s(u)$. There is an open conic neighbourhood Γ of z such that $Cu \in H^{s-m}(\mathbb{R}^n)$ for every $C \in \Psi^m(\mathbb{R}^n)$, $m \in \mathbb{R}$, with $\mathrm{WF}'(C) \subset \Gamma$.*

Proof. Let $A \in \Psi^0$ be elliptic at z with $Au \in H^s$, and let $Q \in \Psi^0$ be a microlocal parametrix for A at z (Proposition 3.5.1), $QA = \mathrm{Id} + R$ with $z \notin \mathrm{WF}'(R)$. Let Γ be the complement of $\mathrm{WF}'(R)$ and $\psi \in \mathcal{C}^\infty_{\mathrm c}(\mathbb{R}^n)$ equal to 1 near $\operatorname{supp} u$. If $\mathrm{WF}'(C) \subset \Gamma$ then $Cu = CQ(Au) - CR\psi u$. Here $CQ \in \Psi^m$ maps H^s to H^{s-m}, and $\mathrm{WF}'(CR) = \emptyset$, so $CR\psi \in \Psi^{-\infty}$ by Lemma 3.3.8; it maps $H^{-\infty}$ into H^∞. □

Proposition 3.17.3. *Let $u, v \in \mathcal{C}^{-\infty}_{\mathrm c}(\mathbb{R}^n)$ and $s, t \in \mathbb{R}$.*

(i) *$\mathrm{WF}_s(u)$ is a closed cone; $\mathrm{WF}_t(u) \subset \mathrm{WF}_s(u)$ if $t \le s$; $\mathrm{WF}_s(u) \subset \mathrm{WF}(u)$; and $\mathrm{WF}_s(u+v) \subset \mathrm{WF}_s(u) \cup \mathrm{WF}_s(v)$.*

(ii) *If $A \in \Psi^m(\mathbb{R}^n)$ and $\phi \in \mathcal{C}^\infty_{\mathrm c}(\mathbb{R}^n)$ then $\mathrm{WF}_{s-m}(\phi Au) \subset \mathrm{WF}'(A) \cap \mathrm{WF}_s(u)$.*

(iii) *$\mathrm{WF}_s(u) = \emptyset$ if and only if $u \in H^s(\mathbb{R}^n)$.*

Proof. (i) $\mathrm{WF}_s(u)$ is an intersection of closed cones, and the monotonicity in s follows from $H^s \subset H^t$. If $z \notin \mathrm{WF}(u)$, let $A \in \Psi^0$ be elliptic at z with $Au \in \mathcal{C}^\infty$, and let $\psi, \psi' \in \mathcal{C}^\infty_{\mathrm c}(\mathbb{R}^n)$ with $\psi = 1$ near $\operatorname{supp} u$, $\psi' = 1$ near $\operatorname{supp}\psi$. Then $Au = \psi' Au + (1-\psi')A\psi u$, where the first term is in $\mathcal{C}^\infty_{\mathrm c}$ and the second in $\mathcal{S}$ by Proposition 3.2.2 and Lemma 3.2.3. So $Au \in \mathcal{S} \subset H^s$ and $z \notin \mathrm{WF}_s(u)$. For the sum, if z is in neither set, take $C = G_{z,\epsilon}$ (see (3.3.2)) with $\overline{\Gamma_\epsilon(z)}$ inside both of the cones given by Lemma 3.17.2; C is elliptic at z and $C(u+v) \in H^s$.

(ii) Let $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal 1 near $\operatorname{supp} u$. If $z \notin \mathrm{WF}'(A)$, choose ϵ with $\overline{\Gamma_\epsilon(z)} \cap \mathrm{WF}'(A) = \emptyset$ and let $C = G_{z,\epsilon}$; as in the proof of Proposition 3.7.4, $C\phi A \in \Psi^{-\infty}$, so $C\phi Au \in H^\infty$ and $z \notin \mathrm{WF}_{s-m}(\phi Au)$. If $z \notin \mathrm{WF}_s(u)$, let Γ be as in Lemma 3.17.2, choose ϵ with $\overline{\Gamma_\epsilon(z)} \subset \Gamma$, and let $D = G_{z,\epsilon}$, $C = G_{z,\epsilon/4}$. Then $Du \in H^s$ and

$$C\phi Au = C\phi A(Du) + C\phi A(\mathrm{Id} - D)\psi u,$$

where $C\phi A \in \Psi^m$ maps H^s to H^{s-m}, and $C\phi A(\mathrm{Id} - D)\psi \in \Psi^{-\infty}$ because $\mathrm{WF}'(C) \cap \mathrm{WF}'(\mathrm{Id} - D) = \emptyset$ (Lemma 3.3.8). Since C is elliptic at z, $z \notin \mathrm{WF}_{s-m}(\phi Au)$.

(iii) If $u \in H^s$ take $A = \mathrm{Id}$. Conversely, suppose $\mathrm{WF}_s(u) = \emptyset$ and let L be a compact neighbourhood of $\operatorname{supp} u$. For each point (x, ω) of $L \times \mathbb{S}^{n-1}$ there is $A \in \Psi^0$ with $Au \in H^s$ and $|\sigma_L(A)| \geq \epsilon$ on $\Gamma_\epsilon(x, \omega) \cap \{|\xi| \geq 1/\epsilon\}$, for some $\epsilon > 0$. Finitely many of these cones, with operators A_j and constants ϵ_j, cover $L \times (\mathbb{R}^n \setminus 0)$; let $\epsilon_0 = \min \epsilon_j$, $R_0 = \max 1/\epsilon_j$. Choose $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $0 \leq \phi \leq 1$, $\operatorname{supp}\phi$ in the interior of L and $\phi = 1$ near $\operatorname{supp} u$, and let

$$B = \sum_j A_j^* A_j + (1 - \phi) \in \Psi^0(\mathbb{R}^n).$$

Then $\sigma_L(B) = \sum_j |\sigma_L(A_j)|^2 + (1 - \phi) + r$ with $r \in S^{-1}$, and for $|\xi| \geq R_0$ the sum of the first two terms is at least $\min(\epsilon_0^2, 1)$: if $x \in \operatorname{supp}\phi \subset L$ the point lies in one of the cones, and otherwise $1 - \phi(x) = 1$. So B is globally elliptic; let Q be a parametrix, $QB = \mathrm{Id} + E$ with $E \in \Psi^{-\infty}$. Since $(1 - \phi)u = 0$, $Bu = \sum_j A_j^*(A_j u) \in H^s$, and $u = QBu - Eu \in H^s$. □

3.17.2 General tempered distributions

By Proposition 3.17.3(ii) with $A = \mathrm{Id}$, $\mathrm{WF}_s(\phi u) \subset \mathrm{WF}_s(u)$ for $\phi \in \mathcal{C}^\infty_{\mathrm{c}}$ and $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}$, with equality if $\phi = 1$ near $\operatorname{supp} u$. So the following definition is consistent with Definition 3.17.1.

Definition 3.17.4. For $u \in \mathcal{S}'(\mathbb{R}^n)$ and $s \in \mathbb{R}$, $\mathrm{WF}_s(u) = \bigcup_{\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)} \mathrm{WF}_s(\phi u) \subset T^*\mathbb{R}^n \setminus 0$.

Notice that one could not simply use the definition for compactly supported distributions here: $Au \in H^s(\mathbb{R}^n)$ also requires decay of u at infinity, whereas $\mathrm{WF}_s(u)$ is meant to measure local regularity only.

Proposition 3.17.5. *Let $u, v \in \mathcal{S}'(\mathbb{R}^n)$ and $s, t \in \mathbb{R}$.*

(i) *If $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ and $\phi(\bar{x}) \neq 0$ then $\mathrm{WF}_s(u) \cap \pi^{-1}(\bar{x}) = \mathrm{WF}_s(\phi u) \cap \pi^{-1}(\bar{x})$. The set $\mathrm{WF}_s(u)$ is a closed cone, $\mathrm{WF}_t(u) \subset \mathrm{WF}_s(u) \subset \mathrm{WF}(u)$ for $t \leq s$, and $\mathrm{WF}_s(u + v) \subset \mathrm{WF}_s(u) \cup \mathrm{WF}_s(v)$.*

(ii) *$\mathrm{WF}_s(u) = \emptyset$ if and only if $u \in H^s_{\mathrm{loc}}(\mathbb{R}^n)$, i.e. $\phi u \in H^s(\mathbb{R}^n)$ for every $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$.*

(iii) (Relative microlocality.) *If $A \in \Psi^m(\mathbb{R}^n)$ then $\mathrm{WF}_{s-m}(Au) \subset \mathrm{WF}'(A) \cap \mathrm{WF}_s(u)$. In particular $\mathrm{WF}_{s-t}(\langle D \rangle^t u) = \mathrm{WF}_s(u)$.*

(iv) (Elliptic regularity.) *If $A \in \Psi^m(\mathbb{R}^n)$ then $\mathrm{WF}_s(u) \subset \mathrm{WF}_{s-m}(Au) \cup \mathrm{Char}_m(A)$.*

(v) *If $u \in H^{-\infty}(\mathbb{R}^n)$ and $G \in \Psi^0(\mathbb{R}^n)$ is such that $\sigma_L(G)(x, \xi) = 0$ for $|x| \geq R$, for some R, and $\mathrm{WF}'(G) \cap \mathrm{WF}_s(u) = \emptyset$, then $Gu \in H^s(\mathbb{R}^n)$.*

(vi) *If $\Gamma \subset T^*\mathbb{R}^n \setminus 0$ is an open cone with $\Gamma \cap \mathrm{WF}_s(u) = \emptyset$ for every $s \in \mathbb{R}$, then $\Gamma \cap \mathrm{WF}(u) = \emptyset$. Consequently $\mathrm{WF}(u)$ is the closure of $\bigcup_s \mathrm{WF}_s(u)$.*

Proof. (i) The localization statement is proved exactly as Proposition 3.7.6(i), using Proposition 3.17.3(i),(ii) and the fact that a compactly supported distribution vanishing near $\bar{x}$ has no WF_s, indeed no WF, over $\bar{x}$. As in Proposition 3.7.6(ii), it follows that $\mathrm{WF}_s(u)$ is closed, and the remaining statements follow from the corresponding ones in Proposition 3.17.3(i).

(ii) If $\phi u \in H^s$ for all ϕ then all $\mathrm{WF}_s(\phi u)$ are empty. Conversely, if $\mathrm{WF}_s(u) = \emptyset$ and $\phi \in \mathcal{C}^\infty_{\mathrm{c}}$, choose $\psi \in \mathcal{C}^\infty_{\mathrm{c}}$ equal to 1 near $\operatorname{supp}\phi$; then $\mathrm{WF}_s(\phi u) = \mathrm{WF}_s(\phi\psi u) \subset \mathrm{WF}_s(\psi u) = \emptyset$, so $\phi u \in H^s$ by Proposition 3.17.3(iii).

(iii) Let $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to 1 near $\operatorname{supp}\phi$. Then $\phi A u = \phi A(\psi u) + \phi A(1-\psi)u$, where $\phi A(1-\psi) \in \Psi^{-\infty}$ (Proposition 3.2.2), so the second term is smooth with compact support. By Proposition 3.17.3(ii), $\mathrm{WF}_{s-m}(\phi A u) \subset \mathrm{WF}'(A) \cap \mathrm{WF}_s(\psi u) \subset \mathrm{WF}'(A) \cap \mathrm{WF}_s(u)$; take the union over ϕ. The last statement follows by applying this to $\langle D\rangle^t \in \Psi^t$ and to its inverse $\langle D\rangle^{-t}$, whose essential supports are all of $T^*\mathbb{R}^n \setminus 0$.

(iv) If $z \notin \mathrm{Char}_m(A)$, let B be a microlocal parametrix at z, $BA = \mathrm{Id} + R$ with $z \notin \mathrm{WF}'(R)$. Then $u = B(Au) - Ru$, and by (i) and (iii), $\mathrm{WF}_s(u) \subset \mathrm{WF}_s(BAu) \cup \mathrm{WF}_s(Ru) \subset \mathrm{WF}_{s-m}(Au) \cup \mathrm{WF}'(R)$.

(v) Let $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ be 1 on $\{|x| \leq R\}$, so that $G = \psi G$ (as $\psi\sigma_L(G) = \sigma_L(G)$), and let $\psi' \in \mathcal{C}^\infty_{\mathrm{c}}$ be 1 near $\operatorname{supp}\psi$. Then $Gu = \psi G(\psi' u) + \psi G(1-\psi')u$. The operator $\psi G(1-\psi')$ is in $\Psi^{-\infty}$, so the second term is in H^∞ since $u \in H^{-\infty}$. By Proposition 3.17.3(ii), $\mathrm{WF}_s(\psi G\psi' u) \subset \mathrm{WF}'(G) \cap \mathrm{WF}_s(u) = \emptyset$, so the first term is in H^s by Proposition 3.17.3(iii).

(vi) Let $z = (\bar{x}, \bar{\xi}) \in \Gamma$, choose ϵ with $\overline{\Gamma_\epsilon(z)} \subset \Gamma$, let $C = G_{z,\epsilon}$ and let $\psi' \in \mathcal{C}^\infty_{\mathrm{c}}$ equal 1 near $\{|x - \bar{x}| \leq \epsilon\}$. Then $\mathrm{WF}'(C) \subset \Gamma$, so by (iii) $\mathrm{WF}_t(C\psi' u) \subset \mathrm{WF}'(C) \cap \mathrm{WF}_t(u) = \emptyset$ for all t; as $C\psi' u$ has compact support ($C = \psi C$ for suitable $\psi \in \mathcal{C}^\infty_{\mathrm{c}}$), it lies in $H^\infty \subset \mathcal{C}^\infty$ by Proposition 3.17.3(iii). Since C is elliptic at z, $z \notin \mathrm{WF}(\psi' u)$, hence $z \notin \mathrm{WF}(u)$ by Proposition 3.7.6(i). For the last statement: the union is contained in the closed set $\mathrm{WF}(u)$, and if $z \in \mathrm{WF}(u)$ then every open conic neighbourhood of z meets some $\mathrm{WF}_s(u)$, by what we have just shown. □

The inclusion $\bigcup_s \mathrm{WF}_s(u) \subset \mathrm{WF}(u)$ may be strict (Exercise 3.17); this is why the passage from Sobolev regularity to smoothness in Section 3.20 needs an argument which is uniform in s.

3.18 Bounded families of operators

The proof of the propagation theorem uses a regularization: one works with sequences of operators of order $-\infty$ which are bounded in Ψ^0, and needs to know that the estimates of the calculus are uniform along such sequences. A sequence (c_k) in $S^m(\mathbb{R}^n;\mathbb{R}^n)$ is *bounded* if $\sup_k \|c_k\|_{N,m} < \infty$ for every N, and a sequence (C_k) in $\Psi^m(\mathbb{R}^n)$ is *bounded in* Ψ^m if $(\sigma_L(C_k))$ is bounded in S^m. Recall that $S^m(\mathbb{R}^n;\mathbb{R}^n)$ is a Fréchet space: if (c_k) is Cauchy for all the norms $\|\cdot\|_{N,m}$ then all derivatives converge locally uniformly and the weighted estimates pass to the limit.

Lemma 3.18.1 (Continuity of the calculus)**.** *Let $m, m', s \in \mathbb{R}$ and $N \in \mathbb{N}_0$.*

(i) *The map $S^m(\mathbb{R}^n;\mathbb{R}^n) \ni c \mapsto \mathrm{Op}(c) \in \mathcal{B}(H^s, H^{s-m})$ is continuous.*

(ii) *The maps*

$$S^m \ni c \longmapsto \sigma_L(\mathrm{Op}(c)^*) - \sum_{|\alpha|<N} \frac{1}{\alpha!}\partial_\xi^\alpha D_x^\alpha \bar{c} \in S^{m-N},$$

$$S^m \times S^{m'} \ni (c,d) \longmapsto \sigma_L(\mathrm{Op}(c)\,\mathrm{Op}(d)) - \sum_{|\alpha|<N} \frac{1}{\alpha!}\partial_\xi^\alpha c\, D_x^\alpha d \in S^{m+m'-N}$$

(symbol spaces on $\mathbb{R}^n \times \mathbb{R}^n$) are continuous.

Consequently, if (C_k) is bounded in Ψ^m and (D_k) in $\Psi^{m'}$, then (C_k) is bounded in $\mathcal{B}(H^s, H^{s-m})$ for every s, (C_k^*) is bounded in Ψ^m, $(C_k D_k)$ is bounded in $\Psi^{m+m'}$, and the remainders in the expansions of $\sigma_L(C_k^*)$ and $\sigma_L(C_k D_k)$ after N terms are bounded in S^{m-N} and $S^{m+m'-N}$.

The expansions are those of the composition theorem of Chapter 2, written with $i^{|\alpha|} D_\xi^\alpha = \partial_\xi^\alpha$; the maps in (ii) are well defined by that theorem.

Proof. We use the closed graph theorem for linear maps between Fréchet spaces, and the fact that a bilinear map between Fréchet spaces which is continuous in each variable separately is continuous (a consequence of the uniform boundedness principle).

Step 1. For $\phi, \psi \in \mathcal{S}(\mathbb{R}^n)$ the linear form $c \mapsto (\mathrm{Op}(c)\phi, \psi)$ is continuous on S^m. Indeed, by the limit formula in Chapter 2 and dominated convergence, $\mathrm{Op}(c)\phi(x) = \int e^{ix\cdot\xi} c(x,\xi)\hat{\phi}(\xi)đ\xi$, so $|(\mathrm{Op}(c)\phi, \psi)| \le \|c\|_{0,m}\|\psi\|_{L^1} \int \langle\xi\rangle^m |\hat{\phi}| đ\xi$.

Step 2: (i). Both spaces are Fréchet (the second Banach) and the map is well defined by the Sobolev boundedness theorem. Suppose $c_k \to c$ in S^m and $\mathrm{Op}(c_k) \to T$ in $\mathcal{B}(H^s, H^{s-m})$. For $\phi, \psi \in \mathcal{S}$, $(\mathrm{Op}(c_k)\phi, \psi) \to (\mathrm{Op}(c)\phi, \psi)$ by Step 1, and $(\mathrm{Op}(c_k)\phi, \psi) \to (T\phi, \psi)$. Hence $T\phi = \mathrm{Op}(c)\phi$ for $\phi \in \mathcal{S}$, and by density $T = \mathrm{Op}(c)$. The graph is closed, so the map is continuous.

Step 3: adjoints. Let $\tau(c) = \sigma_L(\mathrm{Op}(c)^*)$, a linear map $S^m \to S^m$. If $c_k \to c$ and $\tau(c_k) \to d$ in S^m, then for $\phi, \psi \in \mathcal{S}$, by Step 1,

$$(\mathrm{Op}(d)\phi, \psi) = \lim(\mathrm{Op}(\tau(c_k))\phi, \psi) = \lim(\phi, \mathrm{Op}(c_k)\psi) = (\phi, \mathrm{Op}(c)\psi) = (\mathrm{Op}(\tau(c))\phi, \psi),$$

so $d = \tau(c)$ by uniqueness of the left symbol, and τ is continuous. The remainder map $\rho_N : S^m \to S^{m-N}$ in (ii) is τ minus a finite sum of continuous maps $S^m \to S^m$, so it is continuous into S^m. If $c_k \to c$ in S^m and $\rho_N(c_k) \to r$ in S^{m-N}, then $\rho_N(c_k) \to \rho_N(c)$ and $\rho_N(c_k) \to r$ in S^m, so $r = \rho_N(c)$; by the closed graph theorem ρ_N is continuous into S^{m-N}.

Step 4: compositions. Let $\beta(c,d) = \sigma_L(\mathrm{Op}(c)\,\mathrm{Op}(d))$. For fixed d, if $c_k \to c$ and $\beta(c_k, d) \to e$ in $S^{m+m'}$, then for $\phi, \psi \in \mathcal{S}$, with $\phi' = \mathrm{Op}(d)\phi \in \mathcal{S}$, Step 1 gives $(\mathrm{Op}(e)\phi, \psi) = \lim(\mathrm{Op}(c_k)\phi', \psi) = (\mathrm{Op}(c)\,\mathrm{Op}(d)\phi, \psi)$, so $e = \beta(c,d)$. For fixed c, if $d_k \to d$ and $\beta(c, d_k) \to e$ then, with $\psi' = \mathrm{Op}(c)^*\psi \in \mathcal{S}$, $(\mathrm{Op}(e)\phi, \psi) = \lim(\mathrm{Op}(d_k)\phi, \psi') = (\mathrm{Op}(c)\,\mathrm{Op}(d)\phi, \psi)$. So β is separately continuous, hence continuous, and the argument at the end of Step 3 applies to the remainders, which are therefore separately continuous into $S^{m+m'-N}$, hence continuous.

The last statement holds because continuous linear maps between locally convex spaces map bounded sets to bounded sets, and so do continuous bilinear maps: for each norm q on the target there are norms p_1, p_2 and C with $q(\beta(c,d)) \le C p_1(c) p_2(d)$. For the Sobolev bound apply (i) to $\sigma_L(C_k)$. □

We also need a uniform version of the essential support. In the following, $K \subset \mathbb{R}^n \times \mathbb{R}^n$ is a closed set.

Definition 3.18.2. A sequence (C_k), bounded in $\Psi^m(\mathbb{R}^n)$, is *uniformly supported in K* if for every $N \in \mathbb{N}$ there are symbols $c_k^{(N)}$ with $\operatorname{supp} c_k^{(N)} \subset K$ such that $(\sigma_L(C_k) - c_k^{(N)})$ is bounded in $S^{-N}(\mathbb{R}^n; \mathbb{R}^n)$.

For instance, if (c_k) is bounded in S^m and $\operatorname{supp} c_k \subset K$ for all k, then $(\mathrm{Op}(c_k))$ is uniformly supported in K.

Lemma 3.18.3. *Let (C_k) be bounded in $\Psi^m(\mathbb{R}^n)$ and uniformly supported in K, and let (D_k) be bounded in $\Psi^{m'}(\mathbb{R}^n)$.*

(i) (C_k^*), (C_kD_k) *and* (D_kC_k) *are uniformly supported in* K.

(ii) *If* $g \in S^0(\mathbb{R}^n;\mathbb{R}^n)$ *equals* 1 *on an open set containing* K *and* $G = \mathrm{Op}(g)$, *then* $(C_k(\mathrm{Id}-G))$ *and* $((\mathrm{Id}-G^*)C_k)$ *are bounded in* $\Psi^{-N}(\mathbb{R}^n)$ *for every* N.

Proof. (i) By Lemma 3.18.1, $\sigma_L(C_kD_k) = \sum_{|\alpha|<N} \frac{1}{\alpha!}\partial_\xi^\alpha \sigma_L(C_k)\, D_x^\alpha \sigma_L(D_k) + r_k$ with (r_k) bounded in $S^{m+m'-N}$. Substituting $\sigma_L(C_k) = c_k^{(N)} + (\sigma_L(C_k) - c_k^{(N)})$, the terms containing $c_k^{(N)}$ are supported in K, and the others are bounded in $S^{m'-N}$. Since N is arbitrary, (C_kD_k) is uniformly supported in K. The same argument applies to D_kC_k and to C_k^*.

(ii) In the expansion of $\sigma_L(C_k(\mathrm{Id}-G))$ the terms containing $c_k^{(N)}$ are $\frac{1}{\alpha!}\partial_\xi^\alpha c_k^{(N)}\, D_x^\alpha(1-g)$, which vanish identically because $1-g$ vanishes on an open set containing $\operatorname{supp} c_k^{(N)}$. The other terms and the remainder are bounded in $S^{\max(m,0)-N}$. Since $(\mathrm{Id}-G^*)C_k = (C_k^*(\mathrm{Id}-G))^*$, the second statement follows from (i), the first, and Lemma 3.18.1. □

Lemma 3.18.4. *Let* $C \subset T^*\mathbb{R}^n \setminus 0$ *be a closed cone with compact base (i.e.* $C \cap \{|\xi| = 1\}$ *is compact) and* $\Gamma \supset C$ *an open cone. There is a real* $g \in S^0(\mathbb{R}^n;\mathbb{R}^n)$, *vanishing for* $|x|$ *large, with* $\operatorname{cone}\operatorname{supp}(g) \subset \Gamma$ *and* $g = 1$ *on an open set containing* $C \cap \{|\xi| \geq 1\}$.

Proof. The compact set $C_1 = C \cap \{|\xi| = 1\}$ has positive distance from the closed set $\{|\xi| = 1\} \setminus \Gamma$; let 3ϵ be smaller than this distance. Choose $\theta \in \mathcal{C}_c^\infty(\mathbb{R}^{2n})$ with $\theta = 1$ on the ϵ-neighbourhood of C_1 and $\operatorname{supp}\theta$ in its 2ϵ-neighbourhood, and set $g(x,\xi) = \theta(x,\xi/|\xi|)(1-\chi(2\xi))$, with χ as in (3.3.1). This is smooth, homogeneous of degree 0 for $|\xi| \geq \frac{1}{2}$ and vanishes for large $|x|$, hence lies in S^0; it equals 1 on the open set of (x,ξ) with $|\xi| > \frac{1}{2}$ and $(x,\xi/|\xi|)$ in the ϵ-neighbourhood of C_1, and its cone support lies in the cone over the closed 2ϵ-neighbourhood of C_1 in $\{|\xi| = 1\}$, which is contained in Γ. □

Lemma 3.18.5 (Uniform microlocal bounds)**.** *Let* $\Gamma \subset T^*\mathbb{R}^n \setminus 0$ *be an open cone,* $C \subset \Gamma$ *a closed cone with compact base, and* (C_k) *a sequence bounded in* $\Psi^m(\mathbb{R}^n)$ *and uniformly supported in* $K = C \cap \{|\xi| \geq 1\}$. *Let* $v, w \in H^{-\infty}(\mathbb{R}^n)$ *and* $s, t \in \mathbb{R}$.

(i) *If* $\Gamma \cap \mathrm{WF}_s(v) = \emptyset$ *then* (C_kv) *is bounded in* $H^{s-m}(\mathbb{R}^n)$.

(ii) *If moreover* $\Gamma \cap \mathrm{WF}_t(w) = \emptyset$ *and* $s + t \geq m$, *then the sequence* (C_kv, w) *is bounded.*

Proof. Let g be as in Lemma 3.18.4 and $G = \mathrm{Op}(g)$. By Proposition 3.17.5(v), $Gv \in H^s$ and $Gw \in H^t$. Let $v \in H^{-M}$, $w \in H^{-M}$.

(i) $C_kv = C_kGv + C_k(\mathrm{Id}-G)v$. The first term is bounded in H^{s-m} by Lemma 3.18.1, and so is the second, by Lemma 3.18.3(ii) with $N \geq M + s - m$.

(ii) By (3.17.1),

$$(C_kv, w) = (G^*C_kv, w) + ((\mathrm{Id}-G^*)C_kv, w) = (C_kv, Gw) + ((\mathrm{Id}-G^*)C_kv, w).$$

The first term is bounded by $\|C_kv\|_{H^{s-m}}\|Gw\|_{H^{m-s}}$, which is bounded by (i) since $t \geq m-s$. The second is bounded by $\|(\mathrm{Id}-G^*)C_kv\|_{H^M}\|w\|_{H^{-M}}$, which is bounded by Lemma 3.18.3(ii) and Lemma 3.18.1. □

3.19 Hamilton vector fields and bicharacteristics

In the Hamiltonian formulation of classical mechanics the motion of a particle is governed by an energy function, the *Hamiltonian*, $h(x,\xi)$ of position x and momentum ξ, through Hamilton's equations

$$\frac{dx_j}{dt} = \frac{\partial h}{\partial \xi_j}(x,\xi), \qquad \frac{d\xi_j}{dt} = -\frac{\partial h}{\partial x_j}(x,\xi), \qquad j = 1,\dots,n. \tag{3.19.1}$$

Thus the trajectory $(x(t), \xi(t))$ is an integral curve of the *Hamilton vector field*

$$H_h = \sum_{j=1}^{n} \Big(\frac{\partial h}{\partial \xi_j} \frac{\partial}{\partial x_j} - \frac{\partial h}{\partial x_j} \frac{\partial}{\partial \xi_j} \Big).$$

For two functions, $H_h g = \{h, g\} = -\{g, h\} = -H_g h$ is the Poisson bracket of Chapter 2, which appeared there in the commutator formula $\sigma_{m+m'-1}([A, B]) = \frac{1}{i}\{\sigma_m(A), \sigma_{m'}(B)\}$. In particular $H_h h = 0$: *h is constant along the integral curves of H_h* (conservation of energy). See Exercise 3.19 for the Jacobi identity.

More invariantly, H_h is determined by the symplectic form

$$\omega = \sum_j d\xi_j \wedge dx_j = d\alpha, \qquad \alpha = \sum_j \xi_j \, dx_j,$$

through the identity $\omega(\cdot, H_h) = dh$. Indeed for a tangent vector $v = (v_x, v_\xi)$, $\omega(v, H_h) = \sum_j (v_{\xi_j} \partial_{\xi_j} h + v_{x_j} \partial_{x_j} h) = dh(v)$, and ω is non-degenerate: $\omega(v, w) = v_\xi \cdot w_x - v_x \cdot w_\xi$ vanishes for all v only if $w = 0$. The *tautological one-form* α is invariant under changes of coordinates, in the following sense.

Lemma 3.19.1. *Let $F : \Omega \to \Omega'$ be a diffeomorphism of open subsets of $\mathbb{R}^n$ and $\chi_F(x, \xi) = (F(x), (DF(x)^t)^{-1}\xi)$ as in Section 3.6. Then $\chi_F^* \alpha = \alpha$, $\chi_F^* \omega = \omega$, and for every smooth function h on an open subset of $\Omega' \times \mathbb{R}^n$,*

$$d\chi_F(H_{h \circ \chi_F}(z)) = H_h(\chi_F(z)).$$

Hence χ_F maps integral curves of $H_{h \circ \chi_F}$ to integral curves of H_h.

Proof. Write $(y, \eta) = \chi_F(x, \xi)$, so $y = F(x)$ and $\xi = DF(x)^t \eta$. Then $\sum_k \eta_k \, dy_k = \sum_{j,k} \eta_k \frac{\partial F_k}{\partial x_j} dx_j = \sum_j \xi_j dx_j$, i.e. $\chi_F^* \alpha = \alpha$, and $\chi_F^* \omega = d\chi_F^* \alpha = \omega$. For a tangent vector v at z, with $w = d\chi_F(H_{h \circ \chi_F}(z))$,

$$\omega(d\chi_F v, w) = (\chi_F^* \omega)(v, H_{h \circ \chi_F}) = d(h \circ \chi_F)(v) = dh(d\chi_F v).$$

Since $d\chi_F$ is onto, $\omega(\cdot, w) = dh$ at $\chi_F(z)$, so $w = H_h(\chi_F(z))$ by non-degeneracy. □

Since χ_F is the cotangent map of F (Section 3.6), the lemma shows that α, ω and Hamilton vector fields are well defined on the cotangent bundle of a manifold; see Chapter 5.

3.19.1 Homogeneity

Let $M_\lambda(x, \xi) = (x, \lambda\xi)$, $\lambda > 0$, be the dilations of $T^*\mathbb{R}^n \setminus 0$ and $\varrho = \sum_j \xi_j \partial_{\xi_j}$ the radial vector field, its infinitesimal generator; note that $dM_\lambda(\varrho(z)) = \varrho(M_\lambda z)$ and that $\omega(\cdot, \varrho) = -\alpha$. If $p \in \mathcal{C}^\infty(T^*\mathbb{R}^n \setminus 0)$ is homogeneous of degree m, i.e. $p \circ M_\lambda = \lambda^m p$, then $\partial_\xi p$ and $\partial_x p$ are homogeneous of degrees $m - 1$ and m, so

$$dM_\lambda(H_p(z)) = \lambda^{1-m} \, H_p(M_\lambda z). \tag{3.19.2}$$

In particular, *if $m = 1$ then H_p is invariant under dilations and its flow commutes with them*: if $c(t)$ is an integral curve of H_p, so is $M_\lambda c(t)$. By $\omega(\cdot, H_p) = dp$ and $\omega(\cdot, \varrho) = -\alpha$, $H_p(z)$ is a multiple of $\varrho(z)$ if and only if $dp(z)$ is a multiple of $\alpha(z)$.

Lemma 3.19.2. *Let $p \in \mathcal{C}^\infty(T^*\mathbb{R}^n \setminus 0)$ be homogeneous of degree 1 and suppose $H_p(z_1) = c\varrho(z_1)$ for some $c \in \mathbb{R}$. Then the integral curve of H_p through z_1 is $t \mapsto M_{e^{ct}} z_1$, $t \in \mathbb{R}$; it is contained in the ray $\{M_\lambda z_1;\ \lambda > 0\}$.*

Proof. The curve $c_1(t) = M_{e^{ct}} z_1$ satisfies $c_1'(t) = c\varrho(c_1(t))$ and, by (3.19.2), $H_p(c_1(t)) = dM_{e^{ct}}(c\varrho(z_1)) = c\varrho(c_1(t))$. By uniqueness of integral curves it is the integral curve through z_1. □

3.19.2 Principal symbols and bicharacteristics

Definition 3.19.3. An operator $P \in \Psi^m(\mathbb{R}^n)$ has *real homogeneous principal symbol* if there is a real $p \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ with $p(x,\lambda\xi) = \lambda^m p(x,\xi)$ for $|\xi| \geq 1$, $\lambda \geq 1$, such that $\sigma_L(P) - p \in S^{m-1}(\mathbb{R}^n;\mathbb{R}^n)$. We then also write p for its unique extension to a smooth function on $T^*\mathbb{R}^n \setminus 0$ homogeneous of degree m, $p(x,\xi) = |\xi|^m p(x,\xi/|\xi|)$, and H_p for its Hamilton vector field on $T^*\mathbb{R}^n \setminus 0$. The *characteristic variety* is $\Sigma(P) = \{z \in T^*\mathbb{R}^n \setminus 0;\ p(z) = 0\}$, and a *null bicharacteristic* of P is an integral curve $\gamma : J \to \Sigma(P)$ of H_p, $J \subset \mathbb{R}$ an interval.

Then $\Sigma(P) = \mathrm{Char}_m(P)$: if $p(\bar{x},\bar{\xi}) \neq 0$ then, by continuity and homogeneity, $|p| \geq c|\xi|^m$ on a cone $\Gamma_\epsilon(\bar{x},\bar{\xi})$, and $|\sigma_L(P) - p| \leq C\langle\xi\rangle^{m-1}$, so P is elliptic there; if $p(\bar{x},\bar{\xi}) = 0$ then $\sigma_L(P)(\bar{x},t\bar{\xi}) = O(t^{m-1})$ as $t \to \infty$, so P is not elliptic at $(\bar{x},\bar{\xi})$. Since $H_p p = 0$, an integral curve of H_p which meets $\Sigma(P)$ stays in $\Sigma(P)$. If γ is a null bicharacteristic, so is $t \mapsto M_\lambda\gamma(\lambda^{m-1}t)$, by (3.19.2). The principal symbol of a differential operator $\sum_{|\alpha|\leq m} a_\alpha(x)D^\alpha$ with real top order coefficients in $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ is of this type, with $p = \sum_{|\alpha|=m} a_\alpha \xi^\alpha$.

Note also that if p is as above with $m = 1$ and $a \in S^M(\mathbb{R}^n;\mathbb{R}^n)$ vanishes for $|\xi| \leq 1$, then $H_p a = \{p,a\} \in S^M$, since $\partial_\xi p \in S^0$ and $\partial_x p \in S^1$ there; this is the symbol-level version of the fact that commutators with first order operators preserve the order. Moreover the flow of H_p is then complete on $T^*\mathbb{R}^n \setminus 0$: along an integral curve $|\dot{x}| \leq C$ and $|\dot{\xi}| \leq C|\xi|$, by the uniform symbol estimates, so on a bounded time interval x stays bounded and $|\xi|$ stays between two positive constants.

The operator P is said to be of *real principal type* at $z \in \Sigma(P)$ if $dp(z)$ and $\alpha(z)$ are linearly independent; when $m = 1$ this means that $H_p(z)$ is not a multiple of $\varrho(z)$. By Lemma 3.19.2, at the remaining points (including those where $dp = 0$) the null bicharacteristic stays in a single ray; since wavefront sets are conic, the propagation theorem below then holds trivially.

Discussion. Why should singularities follow the Hamilton flow of p? Think of a wave packet u concentrated near x_0 with frequency near ξ_0, where $p(x_0,\xi_0) = 0$. Then Pu is small only if u moves in the way prescribed by Hamilton's equations (3.19.1) with Hamiltonian p: the position moves with velocity $\partial_\xi p$, the frequency changes at rate $-\partial_x p$. This is the "geometric optics" picture: high-frequency waves behave like classical particles. The propagation theorem is the precise version of this heuristic for wavefront sets.

3.19.3 A homogeneous flow box

Near a point where H_p is not radial we can straighten out H_p in a way compatible with dilations. For $z = (x,\xi) \in T^*\mathbb{R}^n \setminus 0$ we write $\hat{z} = (x,\xi/|\xi|)$.

Lemma 3.19.4. *Let $p \in \mathcal{C}^\infty(T^*\mathbb{R}^n \setminus 0)$ be real and homogeneous of degree 1, and let $z_0 \in T^*\mathbb{R}^n \setminus 0$ with $H_p(z_0) \notin \mathbb{R}\varrho(z_0)$. There are $\epsilon_0 > 0$, a conic hypersurface $S \subset T^*\mathbb{R}^n \setminus 0$ containing z_0 and a smooth map*

$$F : (-2\epsilon_0, 2\epsilon_0) \times S \to T^*\mathbb{R}^n \setminus 0,$$

which is a diffeomorphism onto an open cone W, such that $F(0,w) = w$, $\partial_t F(t,w) = H_p(F(t,w))$ and $F(t,M_\lambda w) = M_\lambda F(t,w)$. Moreover, if $\gamma(t) = F(t,z_0)$, then for every compact interval $I \subset (-2\epsilon_0, 2\epsilon_0)$ and every open cone $U \supset \gamma(I)$ there is a cone $O \subset S$, open in S and containing z_0, whose closure $\bar{O}$ in $T^*\mathbb{R}^n \setminus 0$ is contained in S, has compact base, and satisfies $F(I \times \bar{O}) \subset U$.

Thus on W we have the smooth functions $\tau = \mathrm{pr}_1 \circ F^{-1}$, homogeneous of degree 0, the map $\pi_0 = \mathrm{pr}_2 \circ F^{-1} : W \to S$, commuting with dilations, and the function $\Theta = |\xi| \circ \pi_0$, homogeneous of degree 1 and positive, which satisfy

$$H_p\tau = 1, \qquad H_p(f \circ \pi_0) = 0 \text{ for } f \in \mathcal{C}^\infty(S), \qquad H_p\Theta = 0, \tag{3.19.3}$$

since along the integral curves $t \mapsto F(t,w)$ of H_p the function τ is t while π_0 is constant. The sets $F(I \times \bar{O})$ are closed cones with compact base, contained in W.

Proof. Since $H_p(z_0)$ and $\varrho(z_0)$ are linearly independent, there is a linear form λ on $\mathbb{R}^{2n}$ with $\lambda(\varrho(z_0)) = 0$ and $\lambda(H_p(z_0)) = 1$. Let $z_0 = (x_0, \xi_0)$ and $\ell(x,\xi) = \lambda((x, |\xi_0|\xi/|\xi|) - z_0)$, a smooth function on $T^*\mathbb{R}^n \setminus 0$, homogeneous of degree 0, with $\ell(z_0) = 0$. The differential at ξ_0 of $\xi \mapsto |\xi_0|\xi/|\xi|$ is $v \mapsto v - |\xi_0|^{-2}(\xi_0 \cdot v)\xi_0$, so $d\ell(z_0)(H_p) = \lambda(H_p(z_0)) - c\lambda(\varrho(z_0)) = 1$ for some c. As $H_p\ell$ is homogeneous of degree 0, it is $> \frac{1}{2}$ on an open cone $V_0 \ni z_0$, and $S_0 = \{z \in V_0;\ \ell(z) = 0\}$ is a conic hypersurface to which H_p is transversal.

Let Φ_t be the (local) flow of H_p. It commutes with dilations, and its domain is open, so by compactness and homogeneity there are $\epsilon_1 > 0$ and an open cone $V_1 \ni z_0$ such that $\Phi_t(z)$ is defined for $|t| \le 3\epsilon_1$, $z \in V_1$. Put $F(t,w) = \Phi_t(w)$ for $|t| < 3\epsilon_1$, $w \in S_0 \cap V_1$. Then $F(0,w) = w$, $\partial_t F = H_p \circ F$ and $F(t, M_\lambda w) = M_\lambda F(t,w)$. Since $d\Phi_t(H_p(w)) = H_p(\Phi_t w)$,

$$dF_{(t,w)}(c,v) = d\Phi_t\big(cH_p(w) + v\big), \qquad v \in T_w S_0;$$

if this vanishes then $cH_p(w) + v = 0$, and applying $d\ell$ (which kills v) gives $c\,H_p\ell(w) = 0$, so $c = 0$ and $v = 0$. Thus dF is everywhere injective, hence invertible, and F is a local diffeomorphism.

We claim F is injective on $(-2\epsilon_0, 2\epsilon_0) \times S_r$ for some $\epsilon_0 \in (0, \epsilon_1]$ and $r > 0$, where $S_r = \{w \in S_0 \cap V_1;\ |\hat{w} - \hat{z}_0| < r\}$. Otherwise there are $(t_k, w_k) \neq (t_k', w_k')$ with $F(t_k, w_k) = F(t_k', w_k')$, $t_k, t_k' \to 0$ and $\hat{w}_k, \hat{w}_k' \to \hat{z}_0$. Applying the same dilation to w_k and w_k' we may assume $|\xi(w_k)| = 1$, so $w_k \to \hat{z}_0$; then $w_k' = \Phi_{t_k - t_k'}(w_k) \to \hat{z}_0$ by continuity of the flow. Both (t_k, w_k) and (t_k', w_k') tend to $(0, \hat{z}_0)$, near which F is injective, a contradiction. Taking r smaller we may also assume that $\{w \in S_0;\ |\xi(w)| = 1,\ |w - \hat{z}_0| \le r\}$ is compact and contained in V_1, since $S_0 \cap \{|\xi| = 1\}$ is a submanifold, hence locally closed. Let $S = S_r$ and $W = F((-2\epsilon_0, 2\epsilon_0) \times S)$; F is an injective local diffeomorphism, hence a diffeomorphism onto the open cone W.

Finally let I and U be given. The set $F(I \times \{\hat{z}_0\}) = \{M_{1/|\xi_0|}\gamma(t);\ t \in I\}$ lies in U, I is compact and F is continuous, so there is $r' \in (0, r)$ such that $F(t,w) \in U$ for $t \in I$ and $w \in S$ with $|\xi(w)| = 1$, $|w - \hat{z}_0| \le r'$. Then $O = \{w \in S;\ |\hat{w} - \hat{z}_0| < r'\}$ has the required properties, by homogeneity. □

3.20 Propagation of singularities

Elliptic regularity says nothing on the characteristic variety. There, singularities of u can exist although Pu is smooth, as for travelling waves. The theorem below says that they cannot appear or disappear at random: they travel along curves determined by the principal symbol alone. We can now state and prove Hörmander's theorem: for an operator with real homogeneous principal symbol, singularities of solutions of $Pu = f$ which are not caused by singularities of f propagate along null bicharacteristics (Figure 3.6).

Theorem 3.20.1 (Propagation of singularities)**.** *Let $P \in \Psi^m(\mathbb{R}^n)$ have real homogeneous principal symbol p, and let $u \in \mathcal{S}'(\mathbb{R}^n)$, $s \in \mathbb{R}$.*

(i) $\mathrm{WF}(u) \setminus \mathrm{WF}(Pu) \subset \Sigma(P)$ *and* $\mathrm{WF}_s(u) \setminus \mathrm{WF}_{s-m}(Pu) \subset \Sigma(P)$.

(ii) *If $\gamma : J \to \Sigma(P)$ is a null bicharacteristic with $\gamma(J) \cap \mathrm{WF}_{s-m+1}(Pu) = \emptyset$, then either $\gamma(J) \subset \mathrm{WF}_s(u)$ or $\gamma(J) \cap \mathrm{WF}_s(u) = \emptyset$.*

(iii) *If $\gamma : J \to \Sigma(P)$ is a null bicharacteristic with $\gamma(J) \cap \mathrm{WF}(Pu) = \emptyset$, then either $\gamma(J) \subset \mathrm{WF}(u)$ or $\gamma(J) \cap \mathrm{WF}(u) = \emptyset$.*

Note the loss of one derivative in (ii) compared with elliptic regularity (i): $Pu \in H^{s-m+1}$ microlocally along γ only gives $u \in H^s$, not $u \in H^{s+1}$, and this is optimal (Exercise 3.18). Since p is constant along integral curves of H_p, the theorem can be rephrased as follows.

Corollary 3.20.2. *Under the hypotheses of Theorem 3.20.1, the set* $\mathrm{WF}(u) \setminus \mathrm{WF}(Pu)$ *is contained in* $\Sigma(P)$ *and is a union of maximally extended null bicharacteristics of* P *in* $\Sigma(P) \setminus \mathrm{WF}(Pu)$*: if* $z \in \mathrm{WF}(u) \setminus \mathrm{WF}(Pu)$ *and* J *is the connected component containing* 0 *of the open set of* t *for which the integral curve* γ *of* H_p *with* $\gamma(0) = z$ *is defined at* t *and* $\gamma(t) \notin \mathrm{WF}(Pu)$*, then* $\gamma(J) \subset \mathrm{WF}(u)$*. The same holds for* $\mathrm{WF}_s(u) \setminus \mathrm{WF}_{s-m+1}(Pu)$*.*

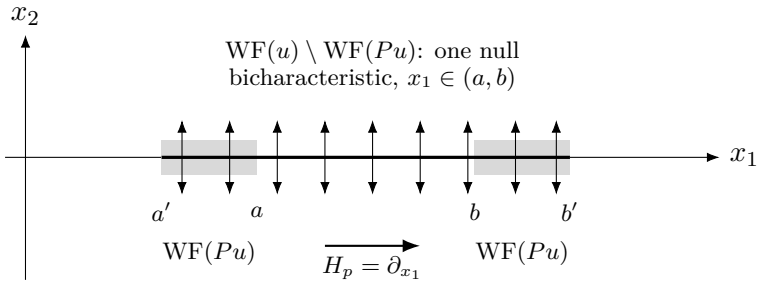


Figure 3.6: Propagation of singularities for $P = D_{x_1}$ on $\mathbb{R}^2$, with $p = \xi_1$, $\Sigma(P) = \{\xi_1 = 0\}$ and null bicharacteristics $t \mapsto (x_1 + t, x_2; 0, \xi_2)$. Let $u = \phi(x_1)H(x_2)$ with $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$, $\phi = 1$ on $[a, b]$, $\operatorname{supp}\phi = [a', b']$, and $\phi' \neq 0$ on $(a', a) \cup (b, b')$. The arrows show $\mathrm{WF}(u)$, the covectors $(0, \pm\xi_2)$ over the segment $[a', b'] \times \{0\}$. Since $Pu = -i\phi'(x_1)H(x_2)$, $\mathrm{WF}(Pu)$ lies over the two gray pieces. As Corollary 3.20.2 predicts, a singularity of u can begin or end only where Pu is singular; in between it runs along a whole null bicharacteristic.

The heart of the matter is the following local statement for first order operators, proved by a positive commutator argument. It says that regularity of order s can be pushed a short distance backwards along γ, onto the point z_0, provided we already know regularity of order $s - \frac{1}{2}$ at z_0. Iterating in steps of $\frac{1}{2}$ then gives the theorem. The idea of the proof is explained in Remark 3.20.4, which the reader may wish to read first.

Proposition 3.20.3. *Let* $P \in \Psi^1(\mathbb{R}^n)$ *have real homogeneous principal symbol* p*, let* $z_0 \in T^*\mathbb{R}^n \setminus 0$ *with* $H_p(z_0) \notin \mathbb{R}\varrho(z_0)$*, let* γ *be the integral curve of* H_p *with* $\gamma(0) = z_0$*, and let* $s \in \mathbb{R}$*. There is* $\delta_0 > 0$ *such that* γ *is defined on* $[-\delta_0, \delta_0]$ *and the following holds for* $0 < \delta \leq \delta_0$ *and* $u \in H^{-\infty}(\mathbb{R}^n)$*: if*

$$\gamma([0,\delta]) \cap \mathrm{WF}_s(Pu) = \emptyset, \qquad \gamma((0,\delta]) \cap \mathrm{WF}_s(u) = \emptyset, \qquad z_0 \notin \mathrm{WF}_{s-\frac{1}{2}}(u), \tag{3.20.1}$$

then $z_0 \notin \mathrm{WF}_s(u)$*. The same is true with* $[0, \delta]$ *and* $(0, \delta]$ *replaced by* $[-\delta, 0]$ *and* $[-\delta, 0)$*.*

Proof. The proof has six steps. We reduce to $s = 0$ and to the forward direction (Step 1), straighten H_p near γ (Step 2), build a symbol a with $H_p a + qa$ equal to a square plus an error supported where u is already known to be regular (Step 3, see Figure 3.7), quantize (Step 4), derive an L^2 estimate for regularized operators (Step 5) and pass to the limit (Step 6).

Step 1: reductions. The operator $-P$ has real principal symbol $-p$, $H_{-p} = -H_p$, and its integral curve through z_0 is $t \mapsto \gamma(-t)$; since $\mathrm{WF}_s(-Pu) = \mathrm{WF}_s(Pu)$, the last statement follows from the first applied to $-P$. Next, let $P' = \langle D\rangle^s P \langle D\rangle^{-s} \in \Psi^1$ and $u' = \langle D\rangle^s u \in H^{-\infty}$. By the multiplicativity of principal symbols, $\sigma_L(P') - p \in S^0$, so P' has the same principal symbol p; moreover $P'u' = \langle D\rangle^s Pu$, and by Proposition 3.17.5(iii)

$\mathrm{WF}_t(u') = \mathrm{WF}_{t+s}(u)$ and $\mathrm{WF}_t(P'u') = \mathrm{WF}_{t+s}(Pu)$ for all t. So the statement for (P,u,s) is the statement for $(P',u',0)$, and we may assume $s=0$ (the δ_0 found below for P' then depends on s).

Step 2: geometry. Since $\sigma_1(P^*) = \overline{\sigma_1(P)} = [p]$, $P_I = \frac{1}{2i}(P-P^*) \in \Psi^0$; it is self-adjoint, so $\sigma_0(P_I) = \overline{\sigma_0(P_I)}$ and $q = 2\operatorname{Re}\sigma_L(P_I) \in S^0$ is a real symbol with $2\sigma_L(P_I) - q \in S^{-1}$. Let $Q_0 = \sup|q|$. Apply Lemma 3.19.4 to p and z_0, obtaining ϵ_0, S, F, W, τ, π_0, Θ, with $\gamma(t) = F(t,z_0)$ for $|t| < 2\epsilon_0$, and set

$$\delta_0 = \min\bigl(\tfrac{1}{2}\epsilon_0,\ (18(1+Q_0))^{-1/2}\bigr).$$

Fix $\delta \in (0,\delta_0]$ and u satisfying (3.20.1) with $s=0$. The sets

$$\Gamma_1 = T^*\mathbb{R}^n \setminus 0 \setminus \bigl(\mathrm{WF}_{-\frac{1}{2}}(u) \cup \mathrm{WF}_0(Pu)\bigr), \qquad \Gamma_2 = T^*\mathbb{R}^n \setminus 0 \setminus \bigl(\mathrm{WF}_0(u) \cup \mathrm{WF}_0(Pu)\bigr)$$

are open cones, $\gamma([0,\delta]) \subset \Gamma_1$ (recall $\mathrm{WF}_{-1/2} \subset \mathrm{WF}_0$) and $\gamma((0,\delta]) \subset \Gamma_2$. Choose $\delta' \in (0,\delta/2]$ with $\gamma([-\delta',0]) \subset \Gamma_1$. Apply the last part of Lemma 3.19.4 to $I_1 = [-\delta',\delta]$, $U = \Gamma_1$, to $I_2 = [\delta/2,\delta]$, $U = \Gamma_2$, and to $I_3 = [-2\delta',2\delta]$, $U = W$; intersecting the resulting cones we get an open cone $O \ni z_0$ in S with closure $\bar O \subset S$ of compact base such that the closed cones with compact base

$$C_j = F(I_j \times \bar O), \qquad j = 1,2,3,$$

satisfy $C_1 \subset \Gamma_1$, $C_2 \subset \Gamma_2$, $C_3 \subset W$; note $C_2 \subset C_1 \subset C_3$. Choose smooth functions θ, θ_1 on S, homogeneous of degree 0 with values in $[0,1]$, with $\theta(z_0) = 1$, $\theta_1 = 1$ on a neighbourhood of $\operatorname{supp}\theta$ and $\operatorname{supp}\theta_1 \subset \bar O$ (for instance functions of $\hat w$), and set $g = \theta\circ\pi_0$, $g_1 = \theta_1\circ\pi_0$ on W. Then $H_p g = H_p g_1 = 0$. Finally let $c_\Theta, C_\Theta > 0$ be such that $c_\Theta|\xi| \le \Theta \le C_\Theta|\xi|$ on C_3 (a positive function homogeneous of degree 0 on a cone with compact base is bounded above and below), and $R_1 = 2C_\Theta$, so that $\Theta \ge R_1$ implies $|\xi| \ge 2$ on C_3.

Step 3: the symbols. We use the following functions of one variable: $h(t) = e^{-1/(t+\delta')}$ for $t > -\delta'$ and $h(t) = 0$ for $t \le -\delta'$, which satisfies $h' = (t+\delta')^{-2}h$; $k_\pm(t) = (t+\delta')^{\pm 1}e^{-1/(2(t+\delta'))}$ for $t > -\delta'$, $k_\pm(t) = 0$ for $t \le -\delta'$, which are smooth and satisfy $k_+k_- = h$; $\chi_0 \in \mathcal{C}^\infty(\mathbb{R})$ with $0 \le \chi_0 \le 1$, $\chi_0 = 1$ on $(-\infty,\delta/2]$, $\chi_0 = 0$ on $[\delta,\infty)$; $\vartheta \in \mathcal{C}^\infty_{\mathrm{c}}((-2\delta',2\delta))$ with $\vartheta = 1$ on $[-\delta',\delta]$; and $\mu \in \mathcal{C}^\infty_{\mathrm{c}}((-2,2))$ with $0 \le \mu \le 1$, $\mu = 1$ on $[-1,1]$. On W define

$$a_0 = g^2\chi_0(\tau)^2h(\tau), \quad f_0 = g\chi_0(\tau)k_+(\tau), \quad b_0 = g\chi_0(\tau)k_-(\tau), \quad e_0 = 2g^2h(\tau)\chi_0(\tau)\chi_0'(\tau),$$
$$r = g_1\vartheta(\tau)(\tau+\delta')^2, \qquad \nu_\ell = (1-\mu(\Theta/R_1))\mu(\Theta/\ell)\ (\ell\in\mathbb{N}), \qquad \nu_\infty = 1-\mu(\Theta/R_1).$$

The functions a_0, f_0, b_0 are supported in $\tau^{-1}(I_1) \cap \operatorname{supp} g \subset C_1$, e_0 in $\tau^{-1}(I_2)\cap\operatorname{supp} g \subset C_2$ (as χ_0' is supported in $[\delta/2,\delta]$) and r in C_3; extended by zero they are smooth on $T^*\mathbb{R}^n \setminus 0$ and homogeneous of degree 0. Moreover $0 \le r \le (2\delta+\delta')^2 \le 9\delta^2$, so with $\zeta(\xi) = 1-\chi(\xi)$ (χ as in (3.3.1)) the symbol $\tilde r = \zeta r \in S^0$ satisfies $|q\tilde r| \le 9\delta^2 Q_0 \le \frac{1}{2}$. By the chain rule $\rho_\pm = (1+q\tilde r)^{\pm 1/2} \in S^0(\mathbb{R}^n;\mathbb{R}^n)$. For $\ell \in \mathbb{N}\cup\{\infty\}$ let

$$a_\ell = \nu_\ell^2 a_0, \qquad b_\ell = \nu_\ell\rho_+ b_0, \qquad f_\ell = \nu_\ell\rho_- f_0, \qquad e_\ell = \nu_\ell^2 e_0,$$

extended by zero outside W. These are real and have the following properties.

(a) *Supports.* Since $\nu_\ell = 0$ where $\Theta < R_1$, a_ℓ, b_ℓ, f_ℓ are supported in $K_1 = C_1 \cap \{|\xi| \ge 2\}$ and e_ℓ in $K_2 = C_2 \cap \{|\xi| \ge 2\}$; for $\ell < \infty$ they also vanish where $\Theta > 2\ell$, hence have compact support in $\mathbb{R}^{2n}$ and lie in $S^{-\infty}$.

(b) *Uniform bounds.* The sequences (a_ℓ), (b_ℓ), (f_ℓ), (e_ℓ) are bounded in S^0, and $a_\ell \to a_\infty$, …, $e_\ell \to e_\infty$ in S^ϵ for every $\epsilon > 0$. Indeed, if ϕ is smooth, homogeneous of degree 0, and supported in C_3, then $\zeta\phi \in S^0$ (its derivatives $D_x^\alpha D_\xi^\beta$ are homogeneous

of degree $-|\beta|$ for $|\xi| \geq 1$ and ϕ has compact base). For $\mu(\Theta/\ell)$, the chain rule writes $D_x^\alpha D_\xi^\beta[\mu(\Theta/\ell)]$, $(\alpha,\beta) \neq 0$, as a sum of terms $\ell^{-j}\mu^{(j)}(\Theta/\ell)\prod_{i=1}^j D_x^{\alpha_i}D_\xi^{\beta_i}\Theta$ with $j \geq 1$, $\sum \alpha_i = \alpha$, $\sum \beta_i = \beta$. On C_3, $|D_x^{\alpha_i}D_\xi^{\beta_i}\Theta| \leq C|\xi|^{1-|\beta_i|}$ by homogeneity, and on the support of $\mu^{(j)}(\Theta/\ell)$ we have $c_\Theta|\xi| \leq \Theta \leq 2\ell$, so $\ell^{-j} \leq C|\xi|^{-j}$; each term is therefore $O(|\xi|^{-|\beta|})$ uniformly in ℓ. The same holds for $\mu(\Theta/R_1)$, and Leibniz' formula gives the boundedness. Finally $\nu_\ell - \nu_\infty = -\nu_\infty(1-\mu(\Theta/\ell))$, and $(1-\mu(\Theta/\ell))\zeta\phi$ is bounded in S^0 and supported in $\{\Theta \geq \ell\} \subset \{|\xi| \geq \ell/C_\Theta\}$, so its $\|\cdot\|_{N,\epsilon}$ norms are $O(\ell^{-\epsilon})$.

(c) *Factorization and the transport identity.* Where $b_0 \neq 0$ and $\nu_\ell \neq 0$ we have $g_1 = 1$, $\vartheta(\tau) = 1$, $\zeta = 1$, so $\tilde{r} = (\tau+\delta')^2$. Hence $\rho_+\rho_- = 1$ and $b_\ell f_\ell = \nu_\ell^2 g^2 \chi_0^2 k_+ k_- = a_\ell$. By (3.19.3), H_p annihilates g and ν_ℓ (a function of Θ) and $H_p\phi(\tau) = \phi'(\tau)$, so

$$H_p a_\ell + q a_\ell = \nu_\ell^2 g^2 \chi_0^2 h((\tau+\delta')^{-2} + q) + e_\ell = \nu_\ell^2 g^2 \chi_0^2 k_-^2 (1 + q(\tau+\delta')^2) + e_\ell,$$

that is,

$$H_p a_\ell + q a_\ell = b_\ell^2 + e_\ell, \qquad a_\ell = b_\ell f_\ell. \tag{3.20.2}$$

Here $H_p a_\ell = \{p, a_\ell\}$ with $p \in S^1$ the symbol of Definition 3.19.3, since it equals its homogeneous extension where $|\xi| \geq 1$.

(d) *Ellipticity.* b_∞ is elliptic at z_0: the function $g\chi_0(\tau)k_-(\tau)$ is homogeneous of degree 0 and equals $k_-(0) = e^{-1/(2\delta')}/\delta' > 0$ at z_0, so it is $\geq c_0 > 0$ on an open cone $\Gamma_0 \ni z_0$ contained in the open cone $F((-\delta',\delta)\times O) \subset C_3$; there $\rho_+ \geq 2^{-1/2}$ and $\nu_\infty = 1$ if $|\xi| \geq 2R_1/c_\Theta$, so $b_\infty \geq c_0/\sqrt{2}$ on $\Gamma_0 \cap \{|\xi| \geq 2R_1/c_\Theta\}$.

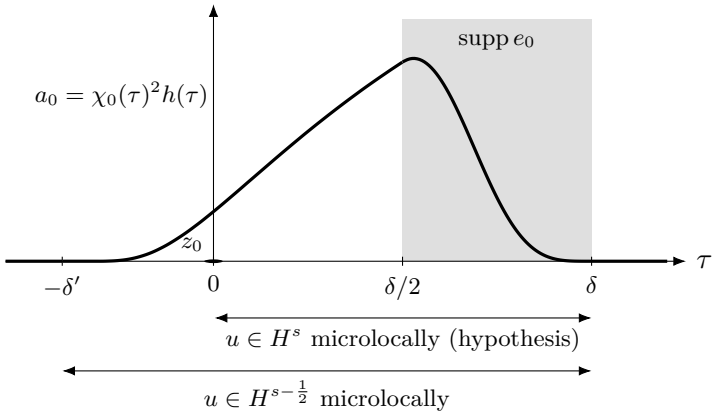


Figure 3.7: The symbol a_0 of Step 3 along the bicharacteristic through z_0 (where $g = 1$). The weight $h(\tau) = e^{-1/(\tau+\delta')}$ increases so fast that $H_p a_0 + q a_0 > 0$ wherever $\chi_0(\tau) = 1$; this gives the square b^2, which controls u at z_0. The cut-off χ_0 makes a_0 decrease on $[\delta/2, \delta]$, producing the error e_0 of the wrong sign, but there u is already known to be in H^s.

Step 4: the operators. For a real symbol $c \in S^0$ let $\mathrm{Op}_s(c) = \frac{1}{2}(\mathrm{Op}(c) + \mathrm{Op}(c)^*)$, which is self-adjoint and has $\sigma_L(\mathrm{Op}_s(c)) - c \in S^{-1}$. Let

$$A_\ell = \mathrm{Op}_s(a_\ell), \quad B_\ell = \mathrm{Op}_s(b_\ell), \quad F_\ell = \mathrm{Op}_s(f_\ell), \quad E_\ell = \mathrm{Op}_s(e_\ell), \qquad B = B_\infty.$$

For $\ell < \infty$ these are in $\Psi^{-\infty}(\mathbb{R}^n)$. By (a), (b), Lemma 3.18.1 and Lemma 3.18.3, the sequences $(A_\ell), (B_\ell), (F_\ell)$ are bounded in Ψ^0 and uniformly supported in K_1, and (E_ℓ) is uniformly supported in $K_2 \subset K_1$; moreover $\sigma_L(A_\ell) = a_\ell + a'_\ell$ with (a'_ℓ) bounded in S^{-1}, and similarly for the others. Define

$$R_\ell = \frac{1}{i}(A_\ell P - P^* A_\ell) - B_\ell^2 - E_\ell, \qquad R'_\ell = A_\ell - B_\ell F_\ell.$$

By Lemma 3.18.3(i) both sequences are uniformly supported in K_1; we claim they are bounded in Ψ^{-1}. For R'_ℓ this follows from Lemma 3.18.1, since $\sigma_L(B_\ell F_\ell) - b_\ell f_\ell$ is bounded in S^{-1} and $a_\ell = b_\ell f_\ell$. For R_ℓ, since $P^* = P - 2iP_I$,

$$\frac{1}{i}(A_\ell P - P^* A_\ell) = i[P, A_\ell] + 2P_I A_\ell.$$

Write $\sigma_L(P) = p + p'$ with $p' \in S^0$. By Lemma 3.18.1 with $N = 2$, the terms with $\alpha = 0$ cancelling,

$$\sigma_L([P, A_\ell]) = \sum_{j=1}^{n} (\partial_{\xi_j}\sigma_L(P)\, D_{x_j}\sigma_L(A_\ell) - \partial_{\xi_j}\sigma_L(A_\ell)\, D_{x_j}\sigma_L(P)) + (\text{bounded in } S^{-1}).$$

The contributions of p' and of a'_ℓ are bounded in S^{-1} (e.g. $\partial_\xi a'_\ell\, D_x p$ is bounded in S^{-2+1}), and the remaining terms are $-i\sum_j(\partial_{\xi_j} p\, \partial_{x_j} a_\ell - \partial_{\xi_j} a_\ell\, \partial_{x_j} p) = -iH_p a_\ell$. Hence $\sigma_L(i[P, A_\ell]) - H_p a_\ell$ is bounded in S^{-1}; similarly $\sigma_L(2P_I A_\ell) - q a_\ell$, $\sigma_L(B_\ell^2) - b_\ell^2$ and $\sigma_L(E_\ell) - e_\ell$ are bounded in S^{-1}. By (3.20.2), $(\sigma_L(R_\ell))$ is bounded in S^{-1}.

Step 5: the estimate. Let $\ell < \infty$ and $u \in H^{-M}$. All operators in the identity

$$\frac{1}{i}(A_\ell P - P^* A_\ell) = B_\ell^2 + E_\ell + R_\ell$$

are in $\Psi^{-\infty}$, so we may apply it to u and pair with u. By (3.17.1) and $A_\ell^* = A_\ell$, $(P^* A_\ell u, u) = (A_\ell u, Pu) = (u, A_\ell P u) = \overline{(A_\ell P u, u)}$, and $(B_\ell^2 u, u) = \|B_\ell u\|_{L^2}^2$. Hence

$$\|B_\ell u\|_{L^2}^2 = 2\,\mathrm{Im}(A_\ell P u, u) - (E_\ell u, u) - (R_\ell u, u), \tag{3.20.3}$$

and, using $A_\ell = B_\ell F_\ell + R'_\ell$ and $B_\ell^* = B_\ell$,

$$(A_\ell P u, u) = (F_\ell P u, B_\ell u) + (R'_\ell P u, u).$$

We now apply Lemma 3.18.5, with $v = Pu$ and $w = u$, both in $H^{-\infty}$. Since $K_1 \subset C_1 \cap \{|\xi| \geq 1\}$, $C_1 \subset \Gamma_1$, and Γ_1 does not meet $\mathrm{WF}_0(Pu)$ or $\mathrm{WF}_{-1/2}(u)$: $\|F_\ell P u\|_{L^2} \leq C$ (with $m = s = 0$); $|(R'_\ell P u, u)| \leq C$ (with $m = -1$, $s = 0$, $t = -\frac{1}{2}$); and $|(R_\ell u, u)| \leq C$ (with $m = -1$, $s = t = -\frac{1}{2}$). Since $K_2 \subset C_2 \cap \{|\xi| \geq 1\}$, $C_2 \subset \Gamma_2$ and $\Gamma_2 \cap \mathrm{WF}_0(u) = \emptyset$: $|(E_\ell u, u)| \leq C$ (with $m = s = t = 0$). With constants independent of ℓ, (3.20.3) gives

$$\|B_\ell u\|_{L^2}^2 \leq 2C\|B_\ell u\|_{L^2} + C',$$

so $\|B_\ell u\|_{L^2} \leq C + \sqrt{C^2 + C'}$ for all ℓ.

Step 6: the limit. Let $\phi \in \mathcal{S}(\mathbb{R}^n)$. By (b), Lemma 3.18.1 (applied in S^ϵ, to Op and to the adjoint map) gives $B_\ell \phi \to B\phi$ in H^M. Hence, by (3.17.1),

$$(Bu, \phi) = (u, B\phi) = \lim_{\ell\to\infty}(u, B_\ell\phi) = \lim_{\ell\to\infty}(B_\ell u, \phi),$$

and $|(Bu, \phi)| \leq \sup_\ell \|B_\ell u\|_{L^2}\|\phi\|_{L^2}$. So $Bu \in L^2(\mathbb{R}^n)$, and $\mathrm{WF}_0(Bu) = \emptyset$. Since $\sigma_L(B) - b_\infty \in S^{-1}$, B is elliptic at z_0 by (d), and Proposition 3.17.5(iv) gives $z_0 \notin \mathrm{WF}_0(u)$. □

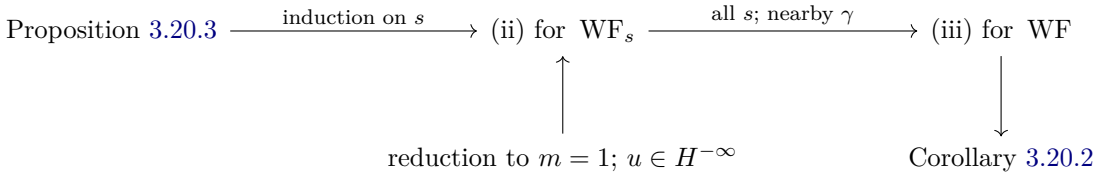
Remark 3.20.4. The idea of the proof is transparent if one ignores regularity issues. For a self-adjoint A of order 0, $2\,\mathrm{Im}(APu, u)$ is the expectation of $\frac{1}{i}(AP - P^*A)$, an operator of order 0 whose principal symbol is $H_p a + qa$. The symbol a is chosen to increase along H_p near z_0 and to be cut off further along the bicharacteristic, where u is already known to be regular; then $H_p a + qa$ is a square b^2 near z_0, up to an error e supported where u is regular, and the identity bounds $\|Bu\|$ in terms of Pu. The factor $h = e^{-1/(\tau+\delta')}$ grows so fast

$(h'/h = (\tau+\delta')^{-2})$ that it absorbs the term qa, which comes from the non-self-adjoint part of P, on a short enough interval; the factorization $a = bf$ is what allows the hypothesis on Pu to be of order s rather than $s+\frac{1}{2}$. Since $\|Bu\|$ is not known to be finite, the argument is applied to the regularized operators A_ℓ; it is essential that the regularizing factor $\mu(\Theta/\ell)$ is annihilated by H_p, so that the identity (3.20.2) holds exactly, uniformly in ℓ.

Question. Where was it used that the principal symbol p is real?

Answer. In Step 2. Because $\sigma_1(P^*) = \bar{p} = p$, the operator $P_I = \frac{1}{2i}(P - P^*)$ has order 0, so the term $2P_I A_\ell$ in $\frac{1}{i}(A_\ell P - P^* A_\ell)$ has the same order as the commutator term and can be absorbed by the weight h. If p had a non-zero imaginary part, $2P_I A_\ell$ would be of order 1, larger than $i[P, A_\ell]$, and its sign would govern the estimate. The conclusion then changes: for the elliptic operator $D_{x_1} + iD_{x_2}$, for example, there is no characteristic set and nothing propagates at all. □

The proof of the theorem combines the local step with two further arguments: an induction on the Sobolev order, and an approximation by nearby bicharacteristics to pass from WF_s to WF.



Proof of Theorem 3.20.1. (i) is elliptic regularity (Theorem 3.7.8 and Proposition 3.17.5(iv)), since $\mathrm{Char}_m(P) = \Sigma(P)$.

Reduction to $m = 1$. Let $\lambda \in \mathcal{C}^\infty(\mathbb{R}^n)$ be positive with $\lambda(\xi) = |\xi|^{1-m}$ for $|\xi| \geq 1$, and $\Lambda = \mathrm{Op}(\lambda)$; it is invertible with inverse $\mathrm{Op}(1/\lambda) \in \Psi^{m-1}$. The operator $P_1 = \Lambda P \in \Psi^1$ has real homogeneous principal symbol $p_1 = |\xi|^{1-m} p$ (by the composition formula, $\sigma_L(\Lambda P) - \lambda\sigma_L(P) \in S^0$), and $\Sigma(P_1) = \Sigma(P)$. On $\Sigma(P)$, $H_{p_1} = |\xi|^{1-m} H_p + p\, H_{|\xi|^{1-m}} = |\xi|^{1-m} H_p$. By Proposition 3.17.5(iii) applied to Λ and Λ^{-1}, $\mathrm{WF}_t(P_1 u) = \mathrm{WF}_{t+1-m}(Pu)$ for all t, and similarly $\mathrm{WF}(P_1 u) = \mathrm{WF}(Pu)$ by Theorem 3.7.7. If $\gamma : J \to \Sigma(P)$ is a null bicharacteristic of P, fix $t_* \in J$ and let $\kappa(t) = \int_{t_*}^t |\xi(\gamma(t'))|^{m-1} dt'$, a smooth increasing diffeomorphism of J onto an interval J_1; then $\gamma_1 = \gamma \circ \kappa^{-1}$ satisfies $\gamma_1' = |\xi(\gamma_1)|^{1-m} H_p(\gamma_1) = H_{p_1}(\gamma_1)$, so it is a null bicharacteristic of P_1 with the same image. Thus (ii) and (iii) for P follow from (ii) and (iii) for P_1, and we assume $m = 1$ from now on.

Localization. Let $I \subset J$ be a compact interval, U' an open set containing the projection of $\gamma(I)$ to $\mathbb{R}^n$, with compact closure, and $\psi \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^n)$ equal to 1 on a neighbourhood of $\overline{U'}$. Then $\tilde{u} = \psi u \in \mathcal{C}_{\mathrm{c}}^{-\infty} \subset H^{-\infty}$ and $P\tilde{u} = \psi P u + [P, \psi]u$. The terms of order $< N$ in the expansion of $\sigma_L([P, \psi])$ all contain derivatives of ψ, so $\sigma_L([P, \psi])$ is rapidly decreasing on $U' \times \mathbb{R}^n$ and $\mathrm{WF}'([P, \psi]) \cap \pi^{-1}(U') = \emptyset$. By Proposition 3.17.5(i),(iii), over U' the sets $\mathrm{WF}_t(\tilde{u})$, $\mathrm{WF}_t(P\tilde{u})$ coincide with $\mathrm{WF}_t(u)$, $\mathrm{WF}_t(Pu)$, for every t, and likewise for WF. So when dealing with $\gamma|_I$ we may replace u by $\tilde{u} \in H^{-\infty}$.

(ii) (with $m = 1$). It suffices to show: if $\gamma(t_1) \notin \mathrm{WF}_s(u)$ for some $t_1 \in J$, then $\gamma(t_2) \notin \mathrm{WF}_s(u)$ for every $t_2 \in J$. Fix t_2, say $t_2 > t_1$ (the other case is symmetric), let $I = [t_1, t_2]$, and localize as above, so $u \in H^{s'}$ for some s'. If some point of $\gamma(I)$ is a radial point, i.e. H_p is a multiple of ϱ there, then by Lemma 3.19.2 $\gamma(I)$ lies in one ray; as $\mathrm{WF}_s(u)$ is conic, $\gamma(t_2) \notin \mathrm{WF}_s(u)$. Otherwise choose $K \in \mathbb{N}$ with $s - K/2 \leq s'$ and put $s_k = s - (K-k)/2$. We show by induction on k that $\gamma(I) \cap \mathrm{WF}_{s_k}(u) = \emptyset$; for $k = 0$ this holds since $\mathrm{WF}_{s_0}(u) \subset \mathrm{WF}_{s'}(u) = \emptyset$. Suppose it holds for $k-1$, and let $C = \{t \in I;\ \gamma(t) \in \mathrm{WF}_{s_k}(u)\}$, a closed set not containing t_1. If $C \neq \emptyset$, let $t_0 = \min C > t_1$, so $\gamma([t_1, t_0)) \cap \mathrm{WF}_{s_k}(u) = \emptyset$.

Apply the backward form of Proposition 3.20.3 at $z_0 = \gamma(t_0)$, to the curve $t \mapsto \gamma(t_0 + t)$, with order s_k and $\delta = \min(\delta_0, t_0 - t_1)$: $\gamma([t_0 - \delta, t_0])$ does not meet $\mathrm{WF}_{s_k}(Pu) \subset \mathrm{WF}_s(Pu)$, $\gamma([t_0 - \delta, t_0))$ does not meet $\mathrm{WF}_{s_k}(u)$, and $z_0 \notin \mathrm{WF}_{s_k - 1/2}(u) = \mathrm{WF}_{s_{k-1}}(u)$ by the induction hypothesis. Hence $z_0 \notin \mathrm{WF}_{s_k}(u)$, contradicting $t_0 \in C$. So $C = \emptyset$, which completes the induction; for $k = K$ we get $\gamma(t_2) \notin \mathrm{WF}_s(u)$.

(iii) Suppose $\gamma(t_1) \notin \mathrm{WF}(u)$ and let $t_2 \in J$; let Γ_0 be an open cone containing $\gamma(t_1)$ and disjoint from $\mathrm{WF}(u)$, and let I be the closed interval between 0 and $t_1 - t_2$. The flow Φ_t of H_p is continuous on its open domain, $\Phi_t(\gamma(t_2)) = \gamma(t_2 + t)$ for $t \in I$, and the compact set $\gamma(t_2 + I)$ does not meet the closed set $\mathrm{WF}(Pu)$. By compactness of I there is an open neighbourhood V of $\gamma(t_2)$ such that for $z \in V$, $\Phi_t(z)$ is defined and not in $\mathrm{WF}(Pu)$ for $t \in I$, and $\Phi_{t_1 - t_2}(z) \in \Gamma_0$. Let $z \in V$. If $z \notin \Sigma(P)$ then $z \notin \mathrm{WF}(u)$ by (i). If $z \in \Sigma(P)$ then $t \mapsto \Phi_t(z)$, $t \in I$, is a null bicharacteristic which does not meet $\mathrm{WF}(Pu) \supset \mathrm{WF}_s(Pu)$ and whose point $\Phi_{t_1 - t_2}(z) \in \Gamma_0$ is not in $\mathrm{WF}_s(u)$, for every s; by (ii), $z \notin \mathrm{WF}_s(u)$ for every s. Thus the open cone generated by V does not meet any $\mathrm{WF}_s(u)$, and by Proposition 3.17.5(vi) it does not meet $\mathrm{WF}(u)$. In particular $\gamma(t_2) \notin \mathrm{WF}(u)$. □

Proof of Corollary 3.20.2. The first statement is (i). The set of t in the statement is open since the domain of the flow is open and $\mathrm{WF}(Pu)$ is closed, and $\gamma(J) \subset \Sigma(P)$ as p is constant along γ. Since $z = \gamma(0) \in \mathrm{WF}(u)$, (iii) gives $\gamma(J) \subset \mathrm{WF}(u)$. The Sobolev version follows in the same way from (ii). □

Remark 3.20.5 (Invariance; manifolds). The statement of Theorem 3.20.1(iii) is invariant under changes of coordinates. If $F : \Omega \to \Omega'$ is a diffeomorphism and P has kernel with compact support in $\Omega \times \Omega$, then P_F has principal symbol $p \circ \chi_F^{-1}$ (Theorem 3.6.2), so by Lemma 3.19.1 χ_F maps the null bicharacteristics of P to those of P_F, and by Theorem 3.9.1 it maps wavefront sets to wavefront sets. Consequently the theorem holds for a properly supported $P \in \Psi^m(X)$ with real homogeneous principal symbol on a manifold X (Chapter 5) and $u \in \mathcal{C}^{-\infty}(X)$: given a null bicharacteristic $\gamma : J \to \Sigma(P) \setminus \mathrm{WF}(Pu)$, the set of $t \in J$ with $\gamma(t) \in \mathrm{WF}(u)$ is closed, and it is open because near each t_1 the curve lies over a coordinate chart, in which, after multiplying u by a cut-off equal to 1 near the projection of a piece of γ and P by cut-offs on both sides, the theorem on $\mathbb{R}^n$ applies. See Chapter 11 for this argument in the case of the wave operator.

3.21 Exercises

Exercise 3.1. Is it true that every $E \in \Psi^{-\infty}(\mathbb{R}^n)$ defines a map from $\mathcal{S}'(\mathbb{R}^n)$ to $\mathcal{S}(\mathbb{R}^n)$? (Consider $E = \mathrm{Op}(e)$ with $e \in \mathcal{S}(\mathbb{R}^n_\xi)$ and $e(0) \neq 0$, acting on the constant function 1.) Compare with Lemma 3.2.3.

Exercise 3.2. Show that $\mathcal{S}(\mathbb{R}^n)$ is dense in $L^2(\mathbb{R}^n)$ by proving that if $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ is identically equal to 1 near the origin then

$$u_k(x) = \phi\Big(\frac{x}{k}\Big) \int e^{ix\cdot\xi} \phi\Big(\frac{\xi}{k}\Big) \hat{u}(\xi) d\!\!\bar{}\,\xi \in \mathcal{S}(\mathbb{R}^n) \quad \text{for } u \in L^2(\mathbb{R}^n),$$

and $u_k \to u$ in $L^2(\mathbb{R}^n)$. Can you see any relation to pseudodifferential operators here?

Exercise 3.3. For $k \in \mathbb{N}$ show that $u \in \mathcal{S}'(\mathbb{R}^n)$ can be written $u = \sum_{|\alpha| \leq k} D^\alpha u_\alpha$ with $u_\alpha \in L^2(\mathbb{R}^n)$ if and only if $\langle D \rangle^{-k} u \in L^2(\mathbb{R}^n)$, that is, if and only if $u \in H^{-k}(\mathbb{R}^n)$.

Exercise 3.4. Let $a \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ be real with $a \geq 0$, and $A = \sum_{j=1}^n D_{x_j}^2 + a(x)$. Show that $Au = 0$ has no non-trivial solution $u \in L^2(\mathbb{R}^n)$. (Show first that $u \in H^2(\mathbb{R}^n)$, then compute (Au, u).)

Exercise 3.5. Show that for any open set $\Omega \subset \mathbb{R}^n$, $\mathcal{C}_c^\infty(\Omega)$ is dense in $\mathcal{C}_c^{-\infty}(\Omega)$ in the weak topology, i.e. every $u \in \mathcal{C}_c^{-\infty}(\Omega)$ is the limit of a sequence in $\mathcal{C}_c^\infty(\Omega)$ in $\mathcal{S}'(\mathbb{R}^n)$.

Exercise 3.6. Let $h \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ be supported in $\{1 < |\eta| < 2\}$ and not identically zero, and let $a(x,\xi) = h(\xi/\langle x\rangle)$. Show that $a \in S^0(\mathbb{R}^n;\mathbb{R}^n)$, that $\operatorname{cone\,supp}(a) = \emptyset$, so $\mathrm{WF}'(\mathrm{Op}(a)) = \emptyset$, but that $a \notin S^{-1}(\mathbb{R}^n;\mathbb{R}^n)$, so $\mathrm{Op}(a) \notin \Psi^{-\infty}(\mathbb{R}^n)$.

Exercise 3.7. Show that for $u \in \mathcal{S}'(\mathbb{R}^n)$

$$\mathrm{WF}(u) = \bigcap\{\mathrm{Char}_0(A);\ A \in \Psi^0(\mathbb{R}^n),\ Au \in \mathcal{C}^\infty(\mathbb{R}^n)\},$$

so that Definition 3.7.1 extends verbatim.

Exercise 3.8. Compute the wavefront sets of the following distributions: $\delta \in \mathcal{S}'(\mathbb{R}^n)$; $|x| \in \mathcal{S}'(\mathbb{R}^n)$; the characteristic function $\chi_{\mathbb{B}}$ of the unit ball, $\chi_{\mathbb{B}}(x) = 1$ for $|x| \leq 1$ and 0 for $|x| > 1$. (For the last, use Theorem 3.9.1 to reduce to the Heaviside function of one variable, regarded as a function on $\mathbb{R}^n$.)

Exercise 3.9. Let $\Gamma \subset T^*\mathbb{R}^n \setminus 0$ be a closed cone and define

$$\mathcal{C}^{-\infty}_{c,\Gamma}(\mathbb{R}^n) = \{u \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n);\ Au \in \mathcal{C}^\infty(\mathbb{R}^n)\ \forall\, A \in \Psi^0(\mathbb{R}^n) \text{ with } \mathrm{WF}'(A) \cap \Gamma = \emptyset\}.$$

Show that this is the space of $u \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ with $\mathrm{WF}(u) \subset \Gamma$. Describe a complete topology on this space with respect to which $\mathcal{C}_c^\infty(\mathbb{R}^n)$ is a dense subspace.

Exercise 3.10. Give an alternative proof of Proposition 3.5.1, not using global parametrices, along the following lines. Let $z \notin \mathrm{Char}_m(A)$, $a = \sigma_L(A)$, and choose $\epsilon > 0$ so small that $\gamma_\epsilon = \gamma_{z,\epsilon}$ is supported where $|a| \geq c|\xi|^m$. Then $b_0 = \gamma_\epsilon/a \in S^{-m}(\mathbb{R}^n;\mathbb{R}^n)$. With $B_0 = \mathrm{Op}(b_0)$, show that $G = \mathrm{Id} - AB_0 \in \Psi^0$ has $\sigma_0(G) = [1 - \gamma_\epsilon]$, and that with $C = \mathrm{Op}(\gamma_{z,\epsilon/4})$, $CG \in \Psi^{-1}$ and $z \notin \mathrm{WF}'((\mathrm{Id} - C)G)$. Writing $\mathrm{Id} - AB_0 = CG + R_1$, let $B_1 \in \Psi^{-1}$ be an asymptotic sum of $\sum_{j\geq 1}(CG)^j$ and $B = B_0(\mathrm{Id} + B_1)$. Show that $AB = \mathrm{Id} + R_2$ with $z \notin \mathrm{WF}'(R_2)$. Construct a left parametrix in the same way, or by applying the construction to A^* and taking adjoints.

Exercise 3.11. Show that microlocal parametrices are essentially unique: if B and B' satisfy $z \notin \mathrm{WF}'(AB' - \mathrm{Id})$ and $z \notin \mathrm{WF}'(BA - \mathrm{Id})$ then $z \notin \mathrm{WF}'(B - B')$.

Exercise 3.12. Given $(\bar{x}, \bar{\xi}) \in T^*\mathbb{R}^n \setminus 0$, construct $u \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ with

$$\mathrm{WF}(u) = \{(\bar{x}, t\bar{\xi});\ t > 0\}.$$

(Hint: take $\phi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ with $\int \phi \neq 0$ and $u(x) = \sum_{k\geq 1} k^{-2}\phi(k(x - \bar{x}))e^{ik^3 x\cdot\bar{\xi}}$, and use the Fourier transform. See [7, Section 8.1] for the construction of a distribution with any prescribed closed conic wavefront set.)

Exercise 3.13. Suppose $A \in \Psi^m(\mathbb{R}^n)$ has Schwartz kernel of compact support. If $u \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ use the elementary operations (and Proposition 3.10.1) to investigate under what conditions $\kappa(x,y) = K_A(x,y)u(y)$, and then $\gamma = (\pi_1)_*\kappa$, make sense. What can you say about $\mathrm{WF}(\gamma)$?

Exercise 3.14. Consider the projection $\pi_1 : \mathbb{R}^p \times \mathbb{R}^k \to \mathbb{R}^p$. Show that $(\pi_1)_*$ can be extended to some distributions which do not have compact support, for example to $\{u \in \mathcal{S}'(\mathbb{R}^{p+k});\ \operatorname{supp} u \cap (K \times \mathbb{R}^k)$ is compact for each compact $K \subset \mathbb{R}^p\}$, with values in distributions on $\mathbb{R}^p$ (that is, continuous linear functionals on $\mathcal{C}_c^\infty(\mathbb{R}^p)$), and that Proposition 3.12.4 remains valid.

Exercise 3.15. Let $u_1, u_2 \in \mathcal{C}_c^{-\infty}(\mathbb{R}^n)$ satisfy (3.11.1). Show that $P(u_1, u_2) = (u_1 u_2)(\psi)$ for any $\psi \in \mathcal{C}_c^\infty(\mathbb{R}^n)$ equal to 1 near $\operatorname{supp} u_1 \cup \operatorname{supp} u_2$. (Use Theorem 3.13.1(i): replace u_2 by $u_2 * \check{\rho}_\epsilon$ in P, and use Lemma 3.11.1.) Deduce that $u_1 u_2 = u_2 u_1$.

Exercise 3.16. Check that Theorem 3.14.1 contains Theorem 3.12.3 (with $F = \iota$) and Theorem 3.13.1 (with $F = \Delta$, $\Delta(x) = (x, x)$, applied to $u_1 \boxtimes u_2$): show that the definitions agree, using the limit formulas, and that the wavefront set bounds coincide.

Exercise 3.17. Show that the inclusion $\bigcup_s \mathrm{WF}_s(u) \subset \mathrm{WF}(u)$ of Proposition 3.17.5(vi) can be strict. (Hint: on $\mathbb{R}$, let $\phi \in \mathcal{C}_c^\infty(\mathbb{R})$ equal 1 near $[0, 1]$ and $v_k(x) = \phi(x)|x - 1/k|^{k+1/2}$, which lies in H^s exactly for $s < k + 1$. Choose $\epsilon_k > 0$ so small that $u = \sum_k \epsilon_k v_k$ converges in $H^{k_0+1/2}(\mathbb{R})$ for every k_0 when the sum is restricted to $k \geq k_0$. Show that u is in H^s near 0 for every s, so that $(0, \pm 1) \notin \mathrm{WF}_s(u)$ for all s, but that $1/k \in \operatorname{sing\,supp} u$ for every k, so $0 \in \operatorname{sing\,supp} u$.)

Exercise 3.18. Let $P = D_{x_1}$ on $\mathbb{R}^2$. Determine $\Sigma(P)$ and the null bicharacteristics. Let $s \geq 0$, let $w \in H^s(\mathbb{R})$ have compact support and satisfy $w \notin H^{s+\epsilon}(\mathbb{R})$ for every $\epsilon > 0$, let $\phi \in \mathcal{C}_c^\infty((-1, 1))$ with $\int \phi = 1$, and let $u(x) = \Phi(x_1) w(x_2)$ with $\Phi(x_1) = i \int_{-\infty}^{x_1} \phi$. Show that $Pu = \phi \otimes w \in H^s(\mathbb{R}^2)$, that u vanishes for $x_1 < -1$, and that u is not in $H^{s+\epsilon}_{\mathrm{loc}}$ on the half-plane $\{x_1 > 1\}$, for any $\epsilon > 0$. Deduce that some null bicharacteristic meets $\mathrm{WF}_{s+\epsilon}(u)$ without being contained in it, although it does not meet $\mathrm{WF}_s(Pu) = \emptyset$. Thus in Theorem 3.20.1(ii) the order $s - m + 1$ cannot be replaced by $s - m + 1 - \epsilon$: compared with elliptic regularity, one derivative is lost.

Exercise 3.19. Check the Jacobi identity for the Poisson bracket,

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = 0, \qquad f, g, h \in \mathcal{C}^\infty(\mathbb{R}^{2n}),$$

and deduce that $[H_f, H_g] = H_{\{f,g\}}$.

Exercise 3.20. The fact that $\omega(\cdot, H_h) = dh$ determines H_h is equivalent to the non-degeneracy of ω: $\omega(v, w) = 0$ for all w implies $v = 0$. Let ω' be any non-degenerate 2-form on an open set of $\mathbb{R}^{2n}$, define H'_f by $\omega'(\cdot, H'_f) = df$ and $\{f, g\}' = \omega'(H'_f, H'_g)$. Show that $\{f, g\}' = H'_f g$, and that the Jacobi identity holds for $\{\cdot, \cdot\}'$ if and only if ω' is closed. (Use the formula for $d\omega'$ evaluated on three Hamilton vector fields.)

Exercise 3.21. For $\bar{\xi} \in \mathbb{R}^n \setminus 0$ choose j with $\bar{\xi}_j \neq 0$ and consider the conic region

$$\left\{\xi;\ \xi_j/\bar{\xi}_j > 0,\ |\xi_k/\xi_j - \bar{\xi}_k/\bar{\xi}_j| < \epsilon \text{ for } k \neq j\right\}$$

with coordinates $r = |\xi_j|$, $t_k = \xi_k/|\xi_j|$ $(k \neq j)$. Check that finitely many such regions cover $\mathbb{R}^n \setminus \{0\}$. If $a \in \mathcal{C}^\infty(\mathbb{R}^n \times \mathbb{R}^n)$ has compact support in x, show that $a \in S^M(\mathbb{R}^n; \mathbb{R}^n)$ if and only if, in each of finitely many such regions covering $\{|\xi| \geq 1\}$, the function $b(x, t, r) = a(x, \xi)$ satisfies $|D_x^\alpha D_t^\gamma D_r^k b| \leq C_{\alpha\gamma k} r^{M-k}$ for $r \geq 1$. (This is the form in which homogeneity, and the fact that H_p maps symbols to symbols, are most easily checked.)

Notes

The definition of the wavefront set via pseudodifferential operators follows Melrose's notes, while the Fourier transform characterization of Section 3.8 is Hörmander's original definition; see [7, Chapter 8] for the classical treatment of the calculus of wavefront sets (pull-back,

products, push-forward and the wavefront relation of kernels), including continuity statements in the spaces $\mathcal{D}'_\Gamma$ which we have not discussed, and [8, Chapter 18] for the essential support of pseudodifferential operators.

Compared with the original notes, the main changes are as follows. Essential supports are used together with compactly supported cut-offs throughout, because with uniformly bounded symbols $\mathrm{WF}'(A) = \emptyset$ does not imply $A \in \Psi^{-\infty}(\mathbb{R}^n)$. The operations on wavefront sets are all proved via the Fourier transform characterization and a single restriction theorem; products are defined as restrictions of $u_1(x)u_2(x+y)$ to $y = 0$ (rather than through the pairing), and the pull-back by a general map is defined through restriction to a graph, with a limit formula showing consistency with all the special cases. The wavefront relation theorem includes the term $\mathrm{WF}_{\Omega_1}(K)$, which is needed in general.

The propagation theorem of Section 3.20 is due to Hörmander (for operators of real principal type; it was developed further by Duistermaat and Hörmander, *Fourier integral operators II*, Acta Math. 128 (1972)); see also Hörmander, *The analysis of linear partial differential operators IV*, Chapter 26, where it is proved by reduction to D_{x_1} with Fourier integral operators. The positive commutator proof given here follows the revised version of Melrose's notes, with some changes: the non-self-adjoint part of P is absorbed by the rapidly growing weight $e^{-1/(\tau+\delta')}$ on a short bicharacteristic segment (instead of an integrating factor, which would require a homogeneous representative of the subprincipal term), the factorization $a = bf$ gives the sharp hypothesis $Pu \in H^{s-m+1}$ directly, and the uniformity of the calculus along the regularizing sequence is deduced from the closed graph theorem (Lemma 3.18.1; for the functional analysis see e.g. Rudin, *Functional analysis*, Theorems 2.15 and 2.17). The passage from Sobolev to $\mathcal{C}^\infty$ regularity uses nearby bicharacteristics instead of a straightening of the whole family.

Chapter 4

Conormal distributions

After smooth functions, the simplest distributions are those whose singularities are as regular as possible: they sit on a smooth submanifold Y, point only in the directions normal to Y, and look the same at all small scales up to a power. In Chapter 3 we met examples: the Heaviside function $H(x_1)$, the delta function and the kernel of a pseudodifferential operator, whose wavefront sets lie in the conormal bundle of a submanifold (Figures 3.1 and 3.3). This chapter builds the class of all such distributions.

The Dirac delta function, the Schwartz kernels of pseudodifferential operators and the travelling-wave solutions of hyperbolic equations are all examples of distributions that are smooth outside a submanifold Y and have a very particular, "symbolic", singularity at Y. In this chapter we introduce the corresponding spaces $I^m(X,Y)$ of *conormal distributions*. We start with the simplest case, Y a point in $\mathbb{R}^n$, where conormal distributions are simply inverse Fourier transforms of symbols; pass to a linear subspace $Y = \{0\} \times \mathbb{R}^d \subset \mathbb{R}^n$ (equivalently the zero section of a trivial vector bundle), where the Fourier transform is taken only in the normal variables; prove the invariance of these spaces under diffeomorphisms preserving Y; and then define $I^m(X,Y;\Omega^{\frac{1}{2}})$ for a closed embedded submanifold of a manifold, either through coordinates or through a normal fibration. The principal symbol is an invariantly defined half-density on the conormal bundle N^*Y, and gives a short exact sequence. Finally we show that the conormal distributions are characterized, up to an arbitrarily small loss in the order, by *iterated regularity*: stability of a fixed Sobolev regularity under repeated application of vector fields tangent to Y.

We use the symbol spaces of Chapter 2, the radial compactification of Chapter 1 and, for wavefront sets, Chapter 3. Pseudodifferential operators on manifolds are discussed only in Chapter 5; their action on conormal distributions is derived here in local coordinates and globalized in Chapter 10.

4.1 Symbols

Conormal distributions will be Fourier transforms of symbols in the normal variables, so we first collect what we need about symbols, and in particular about classical symbols, which are smooth functions on the radial compactification.

4.1.1 Symbols with bounds

Recall from Chapter 2 that for $m \in \mathbb{R}$, $S^m(\mathbb{R}^p;\mathbb{R}^k)$ is the space of $a \in \mathcal{C}^\infty(\mathbb{R}^p \times \mathbb{R}^k)$ such that

$$|D_z^\alpha D_\xi^\beta a(z,\xi)| \leq C_{\alpha\beta}\langle\xi\rangle^{m-|\beta|} \quad \forall\ \alpha,\beta, \tag{4.1.1}$$

with the Fréchet topology given by the best constants; $S^m(\mathbb{R}^k)$ denotes the case without parameters, $S^{-\infty} = \bigcap_m S^m$, and every sequence $a_j \in S^{m-j}$ has an asymptotic sum $a \sim \sum_j a_j$, unique modulo $S^{-\infty}$. We shall also write $S^m_{\mathrm{c}}(\mathbb{R}^p;\mathbb{R}^k)$ for the subspace of symbols vanishing for z outside a compact subset of $\mathbb{R}^p$. Directly from (4.1.1), $\partial_{\xi_j} : S^m \to S^{m-1}$, $\xi_j : S^m \to S^{m+1}$, and $S^m \cdot S^{m'} \subset S^{m+m'}$. Note that $S^{-\infty}(\mathbb{R}^k) = \mathcal{S}(\mathbb{R}^k)$.

The symbol spaces are invariant under linear changes of the variable: if $A(z) \in \mathrm{GL}(k,\mathbb{R})$ depends smoothly on z, with A, A^{-1} and all their derivatives bounded, then $a(z, A(z)\xi) \in S^m(\mathbb{R}^p;\mathbb{R}^k)$ whenever $a \in S^m(\mathbb{R}^p;\mathbb{R}^k)$, since $\langle A(z)\xi\rangle$ and $\langle\xi\rangle$ are uniformly comparable and each z-derivative falling on $A(z)\xi$ produces a factor of order one together with a ξ-derivative of a. In particular the symbol spaces $S^m(W)$ are well defined for any finite-dimensional real vector space W.

The following density statement allows many identities to be proved first for Schwartz functions.

Lemma 4.1.1. *If $a \in S^m(\mathbb{R}^p;\mathbb{R}^k)$, $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^k)$ equals 1 near 0 and $a_\epsilon(z,\xi) = \phi(\epsilon\xi)a(z,\xi)$, then a_ϵ is bounded in S^m for $0 < \epsilon \le 1$ and $a_\epsilon \to a$ in $S^{m'}$ for every $m' > m$ as $\epsilon \to 0$.*

Proof. The derivative $D^\gamma_\xi[\phi(\epsilon\xi)] = \epsilon^{|\gamma|}(D^\gamma\phi)(\epsilon\xi)$ is supported where $|\xi| \le C/\epsilon$, so it is bounded by $C_\gamma\langle\xi\rangle^{-|\gamma|}$ uniformly in ϵ; hence a_ϵ is bounded in S^m by Leibniz' formula. Moreover $a - a_\epsilon = (1-\phi(\epsilon\xi))a$ and its derivatives are supported in $|\xi| \ge c/\epsilon$, where $\langle\xi\rangle^{m-|\beta|} \le C\epsilon^{m'-m}\langle\xi\rangle^{m'-|\beta|}$; so $a - a_\epsilon \to 0$ in $S^{m'}$. □

4.1.2 Classical symbols

Let W be a real vector space of dimension k and $\overline{W}$ its radial compactification (Chapter 1), a compact manifold with boundary whose interior is W; the action of $\mathrm{GL}(W)$ and of translations extends smoothly to $\overline{W}$. We call the boundary $\mathbb{S}W = \partial\overline{W}$ the *sphere at infinity*; it is canonically $(W \setminus 0)/\mathbb{R}^+$. Let $\rho \in \mathcal{C}^\infty(\overline{W})$ be a boundary defining function, for instance $\rho = \langle\xi\rangle^{-1}$ in linear coordinates. Any two are related by $\rho' = f\rho$ with $0 < f \in \mathcal{C}^\infty(\overline{W})$, so

$$S^m_{\mathrm{ph}}(W) = \rho^{-m}\mathcal{C}^\infty(\overline{W}) = \{a \in \mathcal{C}^\infty(W);\ \rho^m a \in \mathcal{C}^\infty(\overline{W})\}$$

is independent of the choice; this is the space of *classical* (or polyhomogeneous) symbols of order m. (One may regard it as the space of sections of a trivial line bundle N_{-m} over $\overline{W}$ of which ρ^{-m} is a global trivializing section.)

Discussion. Why compactify? A classical symbol has an expansion in homogeneous terms of decreasing degree at infinity. After the substitution $r = |\xi|^{-1}$ this is simply a Taylor expansion in r at $r = 0$, the sphere at infinity (Figure 4.1). So statements about expansions, such as asymptotic summation and the principal part, become statements about smooth functions on a manifold with boundary and their Taylor series at the boundary. The one-point compactification would collapse the sphere at infinity to a point and forget the direction $\xi/|\xi|$, which is exactly where the principal part lives.

Lemma 4.1.2. *For all $m, m' \in \mathbb{R}$:*

(i) $S^m_{\mathrm{ph}}(W) \cdot S^{m'}_{\mathrm{ph}}(W) = S^{m+m'}_{\mathrm{ph}}(W)$ *and* $S^m_{\mathrm{ph}}(W)$ *is invariant under* $\mathrm{GL}(W)$ *and translations;*

(ii) $S^m_{\mathrm{ph}}(W) \subset S^m(W)$*; in linear coordinates* $\xi^\alpha\partial^\beta_\xi : S^m_{\mathrm{ph}} \to S^{m+|\alpha|-|\beta|}_{\mathrm{ph}}$*;*

(iii) $\bigcap_N S^{m-N}_{\mathrm{ph}}(W) = \mathcal{S}(W)$*;*

(iv) $a \in S^m_{\mathrm{ph}}(W)$ *if and only if there are functions* $a_{m-j} \in \mathcal{C}^\infty(W \setminus 0)$*, homogeneous of degree* $m-j$*, such that for one (hence any)* $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$ *equal to one near* 0*,*

$$a - \sum_{j<N} (1-\chi) a_{m-j} \in S^{m-N}(W) \quad \forall\ N. \tag{4.1.2}$$

Proof. (i) is immediate from the definition and the invariance of $\overline{W}$. For (ii) and (iv) use linear coordinates and, near the boundary, the coordinates $r = |\xi|^{-1} \in [0,1)$, $\omega = \xi/|\xi| \in \mathbb{S}^{k-1}$ on $\overline{W}$; then $\rho = r\,g$ with $g > 0$ smooth up to $r = 0$, so $a \in S^m_{\mathrm{ph}}$ means $a = r^{-m} f(r,\omega)$ with f smooth up to $r = 0$. In these coordinates $\xi_j \partial_{\xi_l}$ is a combination, with coefficients smooth up to $r = 0$, of $r\partial_r$ and vector fields on $\mathbb{S}^{k-1}$, while ∂_{ξ_l} is r times such a combination, as one sees from homogeneity ($\xi_j\partial_{\xi_l}$ is homogeneous of degree 0 and ∂_{ξ_l} of degree -1). Since $r\partial_r(r^{-m}f) = r^{-m}(r\partial_r - m)f$, repeated application of such vector fields to $r^{-m}f$ yields r^{-m} times bounded functions, so the estimates (4.1.1) hold in $|\xi| \geq 1$ (where $r \sim \langle\xi\rangle^{-1}$), and trivially in $|\xi| \leq 1$; the mapping property in (ii) follows as well. Taylor's formula in r at $r = 0$, $f = \sum_{j<N} r^j f_j(\omega) + r^N f'_N(r,\omega)$, gives (4.1.2) with $a_{m-j}(\xi) = |\xi|^{m-j} f_j(\xi/|\xi|)$, the remainder $r^{N-m} f'_N$ being in S^{m-N} by the argument just given. Conversely, if (4.1.2) holds, then near $r = 0$ the function $f = r^m a$ satisfies $f - \sum_{j<N} r^j f_j = O(r^N)$ together with all derivatives in $r\partial_r$ and ∂_ω, and $\partial_r = r^{-1}(r\partial_r)$; so f is $\mathcal{C}^{N-1}$ up to $r = 0$ for every N, i.e. smooth. Finally (iii): the intersection is contained in $S^{-\infty}(W) = \mathcal{S}(W)$ by (ii), and conversely a Schwartz function is r^∞ times a smooth function of (r,ω), by the same change of variables. □

The leading term in (4.1.2) is the *principal part* $\sigma_m(a) = a_m$; it is a smooth function on $W \setminus 0$ homogeneous of degree m, equivalently (multiplying by ρ^m and restricting) a smooth section of N_{-m} over $\mathbb{S}W$. Denote by $\mathcal{C}^\infty_{(m)}(W \setminus 0)$ the space of smooth functions homogeneous of degree m.

Proposition 4.1.3. *For every* m *the sequence*

$$0 \longrightarrow S^{m-1}_{\mathrm{ph}}(W) \longrightarrow S^m_{\mathrm{ph}}(W) \xrightarrow{\sigma_m} \mathcal{C}^\infty_{(m)}(W \setminus 0) \longrightarrow 0$$

is exact. Equivalently $\rho\mathcal{C}^\infty(\overline{W}) \to \mathcal{C}^\infty(\overline{W}) \to \mathcal{C}^\infty(\mathbb{S}W)$ *is exact: restriction to the boundary is surjective and a smooth function vanishing at the boundary is* ρ *times a smooth function.*

Proof. Multiplication by ρ^m reduces the statement to the case $m = 0$, where it is Taylor's theorem at the boundary (in the coordinates (r,ω) above) and the existence of extensions: $f(\omega) \mapsto (1-\chi) f(\xi/|\xi|)$. □

Proposition 4.1.4 (Asymptotic completeness)**.** *If* $a_j \in S^{m-j}_{\mathrm{ph}}(W)$*,* $j \in \mathbb{N}_0$*, there exists* $a \in S^m_{\mathrm{ph}}(W)$ *such that* $a - \sum_{j<N} a_j \in S^{m-N}_{\mathrm{ph}}(W)$ *for all* N*;* a *is unique modulo* $\mathcal{S}(W)$*. The same holds, with the same proof, for symbols depending smoothly on parameters in a compact set.*

Proof. Uniqueness follows from Lemma 4.1.2(iii). Multiplying by ρ^m, it suffices to show: if X is a compact manifold with boundary, ρ a boundary defining function and $b_j \in \rho^j\mathcal{C}^\infty(X)$, there is $b \in \mathcal{C}^\infty(X)$ with $b - \sum_{j<N} b_j \in \rho^N\mathcal{C}^\infty(X)$ for all N. Using a collar $[0,1)_r \times \partial X$ and a partition of unity this reduces to the following form of Borel's lemma: given $u_j \in \mathcal{C}^\infty(\partial X)$ there is $u \in \mathcal{C}^\infty_{\mathrm{c}}([0,1)\times\partial X)$ with $\partial_r^j u(0,\cdot) = u_j$ for all j. (Indeed, write $b_j = \sum_l r^{j+l} c_{jl}(\omega)/(j+l)! + O(r^\infty)$ near the boundary and take the u with $u_p = \sum_{j\leq p} c_{j,p-j}$;

then $b - \sum_{j<N} b_j$ vanishes to order N at $r = 0$.) Choose $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$ with $\chi = 1$ on $|r| < \frac{1}{2}$, $\chi = 0$ on $|r| > 1$, and set

$$u(r,\omega) = \sum_j \frac{r^j}{j!} u_j(\omega)\chi(r/\epsilon_j), \qquad 0 < \epsilon_j \le 1.$$

Each term vanishes for $r > \epsilon_j$, and for $l \le j$ and any differential operator Q of order q on ∂X, $|\partial_r^l Q(r^j u_j \chi(r/\epsilon_j)/j!)| \le C_{j,l,Q}\, \epsilon_j^{j-l}$, with $C_{j,l,Q}$ independent of ϵ_j, since $\partial_r^p[\chi(r/\epsilon_j)] = \epsilon_j^{-p}\chi^{(p)}(r/\epsilon_j)$ and $r \le \epsilon_j$ on the support. Fix an exhausting sequence $Q_1, Q_2, \dots$ of differential operators on ∂X. Choosing $\epsilon_j \le 2^{-j}/(1 + \max_{l+q\le j-1} C_{j,l,Q_q})$ imposes only finitely many conditions on each ϵ_j, and makes every differentiated series converge uniformly. Hence u is smooth and, since only the term j contributes to $\partial_r^j u(0,\cdot)$, $\partial_r^j u(0,\cdot) = u_j$. Parameters are handled by including the parameter derivatives among the Q_q. □

The symbol estimates (4.1.1) do not imply that a is classical: $S^m_{\mathrm{ph}}(W)$ is a closed subspace of $S^m(W)$ in the topology of $\rho^{-m}\mathcal{C}^\infty(\overline{W})$, and $\mathcal{S}(W)$ is not dense in it in that topology, though by Lemma 4.1.1 it is dense in the weaker topology of $S^{m'}$, $m' > m$. The relation between the two classes is explored in Exercises 4.1 and 4.2. For symbols depending on parameters $z \in \mathbb{R}^p$ we write $S^m_{\mathrm{ph}}(\mathbb{R}^p;\mathbb{R}^k)$ for those $a \in S^m(\mathbb{R}^p;\mathbb{R}^k)$ having an expansion (4.1.2) with $a_{m-j} \in \mathcal{C}^\infty(\mathbb{R}^p \times (\mathbb{R}^k \setminus 0))$ homogeneous in ξ, uniformly bounded with all derivatives on $\mathbb{R}^p \times \mathbb{S}^{k-1}$, the remainders being in $S^{m-N}(\mathbb{R}^p;\mathbb{R}^k)$. All statements in this chapter hold for the classical spaces as well as for the spaces of symbols with bounds; we indicate the classical versions by a subscript "ph".

4.2 Conormal distributions at a point

The model is the delta function: $\hat\delta_0 = 1$ is the simplest symbol. We allow any symbol on the Fourier side.

Definition 4.2.1. For $m \in \mathbb{R}$ set

$$I^m_{\mathcal{S}}(\mathbb{R}^n,\{0\}) = \{u \in \mathcal{S}'(\mathbb{R}^n);\ \hat u \in S^{m-\frac{n}{4}}(\mathbb{R}^n)\}, \qquad I^m_{\mathcal{S},\mathrm{ph}}(\mathbb{R}^n,\{0\}) = \mathcal{F}^{-1}(S^{m-\frac{n}{4}}_{\mathrm{ph}}(\mathbb{R}^n)).$$

The shift $n/4$ in the order is a normalization, which is explained by the behaviour of the order under the operations considered later (it makes the order of the Schwartz kernel of a pseudodifferential operator of order m, as a conormal distribution at the diagonal, equal to m). For example $\hat\delta_0 = 1$, so

$$\delta_0 \in I^{\frac{n}{4}}_{\mathcal{S},\mathrm{ph}}(\mathbb{R}^n,\{0\}).$$

Since $S^{m-1} \subset S^m$, $I^{m-1}_{\mathcal{S}} \subset I^m_{\mathcal{S}}$, and $\bigcap_m I^m_{\mathcal{S}}(\mathbb{R}^n,\{0\}) = \mathcal{F}^{-1}\mathcal{S}(\mathbb{R}^n) = \mathcal{S}(\mathbb{R}^n)$.

Question. Why ask $\hat u$ to be a symbol, and not just polynomially bounded?

Answer. Every $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n)$ has a polynomially bounded Fourier transform, so that alone says nothing about the singularity. The extra information in the symbol estimates is that each ζ-derivative gains a power of $\langle\zeta\rangle^{-1}$. Since D_ζ corresponds to multiplication by $-z$, this says that $z^\alpha u$ becomes smoother as $|\alpha|$ grows: the singularity is concentrated at $z = 0$ in a uniform, scale-invariant way. This is the content of Lemma 4.2.2(i)–(iii) below. □

Lemma 4.2.2. *Let $u \in I^m_{\mathcal{S}}(\mathbb{R}^n,\{0\})$ (resp. the classical space). Then*

(i) $z^\alpha D_z^\beta u \in I^{m+|\beta|-|\alpha|}_{\mathcal{S}}(\mathbb{R}^n,\{0\})$ (resp. classical);

(ii) *if* $m < -\frac{3n}{4} - j$, $j \in \mathbb{N}_0$, *then* $u \in \mathcal{C}^j(\mathbb{R}^n)$ *and* $D^\beta u \in \mathcal{C}^0_0(\mathbb{R}^n)$ *for* $|\beta| \le j$;

(iii) *if* $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ *equals one near* 0, *then* $(1-\chi)u \in \mathcal{S}(\mathbb{R}^n)$; *in particular* $\operatorname{sing\,supp} u \subset \{0\}$;

(iv) *the integral* $\int u\,dz$, *defined as* $\int \chi u + \int (1-\chi)u$ *(independent of* χ*), equals* $\hat u(0)$;

(v) *if* $v \in I^{m'}_{\mathcal{S}}(\mathbb{R}^n, \{0\})$ *then* $u * v \in I^{m+m'-\frac{n}{4}}_{\mathcal{S}}(\mathbb{R}^n, \{0\})$.

Proof. (i) follows from $\mathcal{F}(z^\alpha D^\beta u) = (-D_\zeta)^\alpha(\zeta^\beta \hat u)$ and Lemma 4.1.2(ii). (ii): if $m < -3n/4 - j$ then $\zeta^\beta \hat u \in S^{m-\frac{n}{4}+|\beta|} \subset L^1$ for $|\beta| \le j$, so $D^\beta u$ is the inverse Fourier transform of an integrable function, hence continuous and vanishing at infinity by the Riemann–Lebesgue lemma. (iii): by (i) and (ii), $|z|^{2N} D^\beta u = \sum_{|\gamma|=N} c_\gamma z^{2\gamma} D^\beta u$ is bounded if $2N > m + |\beta| + 3n/4$; since $1-\chi$ vanishes near 0, $|D^\beta((1-\chi)u)| \le C_{N,\beta}\langle z\rangle^{-2N}$ for all N and β. (iv): the definition is independent of χ since the two definitions agree on $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$; and $\int \chi u = \widehat{\chi u}(0)$, $\int (1-\chi)u = \widehat{(1-\chi)u}(0)$ add up to $\hat u(0)$. (v): $\widehat{u * v} = \hat u \hat v \in S^{m-\frac{n}{4}+m'-\frac{n}{4}} = S^{(m+m'-\frac{n}{4})-\frac{n}{4}}$. □

4.2.1 Densities and invariance

To see that these spaces are naturally associated to the origin of an abstract vector space we need densities. Let W be a real vector space of dimension n. For $t \in \mathbb{R}$, the space $\Omega^t W$ of *t-densities* consists of the maps $\nu : \Lambda^n W \setminus 0 \to \mathbb{R}$ with $\nu(cv) = |c|^t \nu(v)$ for $c \in \mathbb{R} \setminus 0$; it is a line, and $\Omega W = \Omega^1 W$. If $G \in \mathrm{GL}(W)$ then $G^*\nu = |\det G|^t \nu$; for $W = \mathbb{R}^n$ we write $|dz|^t$ for the t-density taking the value 1 on $e_1 \wedge \cdots \wedge e_n$. Densities are what one integrates invariantly: $\int G^*\mu = \int \mu$ for $\mu \in \mathcal{S}(W; \Omega W)$. The pairing $\Omega W \otimes \Omega W' \to \mathbb{R}$, $|dz| \otimes |d\zeta| \mapsto 1$ for dual bases, is independent of the basis, so $\Omega W' = (\Omega W)'$.

For $u \in \mathcal{S}(\mathbb{R}^n)$ and $G \in \mathrm{GL}(n, \mathbb{R})$ the change of variables $y = Gz$ gives

$$\widehat{G^*u}(\zeta) = \int e^{-iz\cdot\zeta} u(Gz)\,dz = |\det G|^{-1} \hat u((G^{-1})^t \zeta).$$

Thus if we define the Fourier transform to take values in densities, $\mathcal{F}u = \hat u(\zeta)|d\zeta|$, then $\mathcal{F}(G^*u) = ((G^{-1})^t)^* \mathcal{F}u$: the Fourier transform is a linear-invariant isomorphism

$$\mathcal{F} : \mathcal{S}(W) \longrightarrow \mathcal{S}(W'; \Omega W'), \qquad \mathcal{F}^{-1}\mu(w) = (2\pi)^{-n} \int_{W'} e^{i\langle w,\zeta\rangle} \mu(\zeta), \tag{4.2.1}$$

extending to $\mathcal{S}'(W) \to \mathcal{S}'(W'; \Omega W')$. Consequently

$$I^m(W, \{0\}) = \mathcal{F}^{-1}\big(S^{m-\frac{n}{4}}(W'; \Omega W')\big),$$

where $S^r(W'; \Omega W') = S^r(W') \cdot |d\zeta|$, is well defined independently of any choice of basis, and similarly for the classical space; the integral of Lemma 4.2.2(iv) becomes the invariant map $I^m(W, \{0\}; \Omega W) \ni u \mapsto (\mathcal{F}u)(0)$ if u is density-valued. The *principal symbol* of u is the class of $\mathcal{F}u$ in $S^{m-\frac{n}{4}}(W'; \Omega W')/S^{m-\frac{n}{4}-1}(W'; \Omega W')$ and, in the classical case, its principal part, a density on $W' \setminus 0$ homogeneous of degree $m - \frac{n}{4} + n = m + \frac{3n}{4}$ (the density factor $|d\zeta|$ being homogeneous of degree n). By Proposition 4.1.3 there are short exact sequences

$$I^{m-1}(W, \{0\}) \hookrightarrow I^m(W, \{0\}) \xrightarrow{\sigma_m} S^{m-\frac{n}{4}}(W'; \Omega W')/S^{m-\frac{n}{4}-1}(W'; \Omega W'),$$

and, in the classical case, onto the homogeneous densities of degree $m + \frac{3n}{4}$ on $W' \setminus 0$.

Invariance under non-linear changes of coordinates fixing 0 is a special case of Theorem 4.4.4 below.

4.2.2 Homogeneity and conormality

For classical conormal distributions the singularity at the origin can be described quite explicitly. We use two facts about homogeneous distributions [7, Theorems 3.2.3, 7.1.16, 7.1.18]: if $f \in \mathcal{C}^\infty(\mathbb{R}^n \setminus 0)$ is homogeneous of degree μ and $\mu \notin \{-n, -n-1, \dots\}$, then f has a unique extension to a distribution $\tilde f \in \mathcal{S}'(\mathbb{R}^n)$ homogeneous of degree μ; and the Fourier transform of a homogeneous distribution of degree μ which is smooth outside 0 is homogeneous of degree $-n-\mu$ and smooth outside 0.

Proposition 4.2.3. *Let $u \in I^m_{S,\mathrm{ph}}(\mathbb{R}^n, \{0\})$, so $\hat u \in S^\mu_{\mathrm{ph}}(\mathbb{R}^n)$ with $\mu = m - \frac{n}{4}$, and suppose $\mu \notin \mathbb{Z}$. Then there are functions $U_j \in \mathcal{C}^\infty(\mathbb{R}^n \setminus 0)$, homogeneous of degree $-n-\mu+j$ and determined by the homogeneous terms $\hat u_{\mu-j}$ in the expansion of $\hat u$, such that*

$$u - \sum_{j<N} U_j \in \mathcal{C}^r(\mathbb{R}^n) \quad \text{for } 0 \le r < N - \mu - n.$$

Conversely the U_j determine the terms $\hat u_{\mu-j}$, so u is determined modulo $\mathcal{S}(\mathbb{R}^n)$ by its expansion.

Proof. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal one near 0. Since $\mu - j$ is not an integer, $\hat u_{\mu-j}$ has a homogeneous extension $\tilde a_j$, and $U_j := \mathcal{F}^{-1}\tilde a_j$ is homogeneous of degree $-n-\mu+j$ and smooth outside 0. The difference $(1-\chi)\hat u_{\mu-j} - \tilde a_j = -\chi\tilde a_j$ has compact support, so its inverse Fourier transform is smooth. The remainder $\hat u - \sum_{j<N}(1-\chi)\hat u_{\mu-j} \in S^{\mu-N}$ has inverse Fourier transform in $\mathcal{C}^r$ if $\mu - N + r < -n$. The converse follows by applying the Fourier transform to the homogeneous terms, which are distinguished by their degrees. □

When μ is an integer the expansion of $\hat u$ is unchanged, but the expansion of u may contain terms $p(z)\log|z|$ with p a homogeneous polynomial (for $\mu \le -n$), coming from the homogeneous terms of degree $-n-l$, $l \in \mathbb{N}_0$, which have no homogeneous extension; and for $\mu \in \mathbb{N}_0$ the polynomial part of $\hat u$ corresponds to derivatives of the delta function at 0, invisible in $z \neq 0$.

4.2.3 Blow-up of the origin

A construction dual to radial compactification is the blow-up of the origin, which resolves the singularity of such homogeneous functions. Polar coordinates give a smooth surjective map

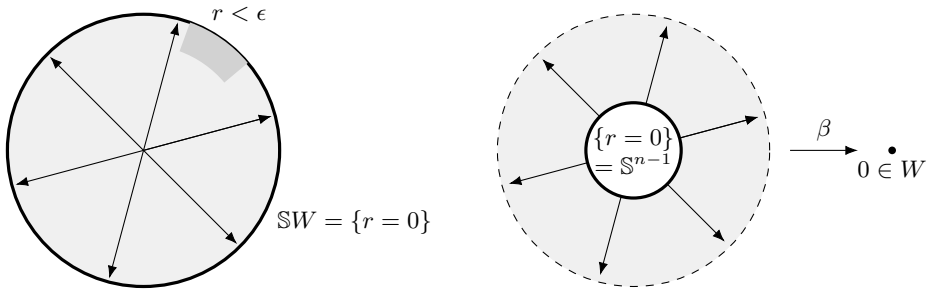
$$\beta : [0,\infty) \times \mathbb{S}^{n-1} \ni (r,\omega) \longmapsto r\omega \in \mathbb{R}^n,$$

a diffeomorphism of $(0,\infty)\times\mathbb{S}^{n-1}$ onto $\mathbb{R}^n \setminus 0$ which collapses $\{0\}\times\mathbb{S}^{n-1}$ to 0. Each U_j in Proposition 4.2.3 is of the form $r^{-n-\mu+j}$ times a smooth function on the sphere. Radial compactification adds a sphere at infinity; blowing up replaces the origin by a sphere (Figure 4.1). In both cases a homogeneous function becomes a power of the boundary defining function times a smooth function.

Lemma 4.2.4. *The action of $\mathrm{GL}(n,\mathbb{R})$ on $\mathbb{R}^n$ lifts under β to a smooth action on $[0,\infty)\times\mathbb{S}^{n-1}$.*

Proof. For $G \in \mathrm{GL}(n,\mathbb{R})$ the lift is $\tilde G(r,\omega) = (r|G\omega|,\, G\omega/|G\omega|)$, which is smooth up to $r=0$, satisfies $\beta\circ\tilde G = G\circ\beta$, and $\widetilde{GH} = \tilde G\tilde H$. (Infinitesimally: the linear vector fields $z_i\partial_{z_j}$ are homogeneous of degree 0, so on $(0,\infty)\times\mathbb{S}^{n-1}$ they are of the form $a(\omega)r\partial_r + V(\omega)$ with V a vector field on the sphere, which extends smoothly to $r=0$, tangent to the boundary.) □

Thus for a vector space W the blown-up space $[W,\{0\}]$, a manifold with boundary $\mathbb{S}W$ with a blow-down map onto W, is well defined; see Exercise 4.3.



(a) radial compactification $\overline{W}$ (b) blow-up $[W, \{0\}]$

Figure 4.1: (a) The radial compactification $\overline{W}$ adds the sphere at infinity $\mathbb{S}W$, with boundary defining function $r = |\xi|^{-1}$ near it. Rays become arcs ending at the boundary, and a neighbourhood of a boundary point (dark) is a conic region $\{|\xi| > 1/\epsilon\}$ around a direction. Classical symbols of order m are r^{-m} times smooth functions on $\overline{W}$. (b) The blow-up of the origin replaces 0 by the sphere $\{r = 0\}$ of directions, with $r = |z|$; the blow-down map β collapses it to 0. The terms U_j of Proposition 4.2.3 are powers of r times smooth functions on $[W, \{0\}]$.

4.3 The local model

For a submanifold Y of positive dimension we take the Fourier transform only in the variables normal to Y, and let the symbol depend smoothly on the point of Y. Thus a conormal distribution is, roughly, a smooth family, parametrized by Y, of conormal distributions at a point.

4.3.1 Notation

Throughout the rest of the chapter n is the dimension of the ambient space, k the codimension of the submanifold and $d = n - k$ its dimension. In the local model

$$\mathbb{R}^n = \mathbb{R}^k_x \times \mathbb{R}^d_y, \qquad Y = \{x = 0\}, \qquad z = (x, y),\ \zeta = (\xi, \eta),$$

and the order of a conormal distribution is related to the order of its symbol by

$$M = m + \frac{n}{4} - \frac{k}{2}. \tag{4.3.1}$$

For a point ($k = n$) this is $M = m - \frac{n}{4}$ as in Definition 4.2.1; for the diagonal in $\mathbb{R}^n \times \mathbb{R}^n$ ($n \to 2n$, $k = n$) it is $M = m$.

For an amplitude $c \in S^M(\mathbb{R}^n; \mathbb{R}^k)$, i.e. $c = c(x, y, \xi)$ satisfying the estimates (4.1.1) uniformly in (x, y), we define the distribution

$$\mathcal{I}[c](x, y) = \int_{\mathbb{R}^k} e^{ix\cdot\xi} c(x, y, \xi)\, đ\xi, \qquad \langle \mathcal{I}[c], \phi \rangle = \int \Big(\int e^{ix\cdot\xi} c(x, y, \xi)\phi(x, y)\, dx\, dy \Big) đ\xi, \tag{4.3.2}$$

for $\phi \in \mathcal{C}^\infty_c(\mathbb{R}^n)$; here $đ\xi = (2\pi)^{-k} d\xi$. Writing $e^{ix\cdot\xi} = \langle \xi \rangle^{-2N} (1 - \Delta_x)^N e^{ix\cdot\xi}$ and integrating by parts shows that the inner integral is $O(\langle \xi \rangle^{M-2N})$, so the outer integral converges for $2N > M + k$ and $\mathcal{I}[c] \in \mathcal{C}^{-\infty}(\mathbb{R}^n)$. When $M < -k$ the integral in (4.3.2) converges absolutely and $\mathcal{I}[c]$ is continuous. By Lemma 4.1.1, $\mathcal{I}[c_\epsilon] \to \mathcal{I}[c]$ in $\mathcal{C}^{-\infty}$ if $c_\epsilon \to c$ in $S^{M'}$, which justifies the formal manipulations below.

Lemma 4.3.1. *For $c \in S^M(\mathbb{R}^n; \mathbb{R}^k)$:*

(i) $D_{y_l}\mathcal{I}[c] = \mathcal{I}[D_{y_l}c]$, $D_{x_j}\mathcal{I}[c] = \mathcal{I}[\xi_j c + D_{x_j}c]$, $x_j\mathcal{I}[c] = \mathcal{I}[-D_{\xi_j}c]$ *and* $\psi\mathcal{I}[c] = \mathcal{I}[\psi c]$ *for* $\psi \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$;

(ii) if $M < -k - r$ *then* $\mathcal{I}[c] \in \mathcal{C}^r(\mathbb{R}^n)$ *with bounded derivatives of order* $\le r$;

(iii) $\mathcal{I}[c]$ *is smooth in* $x \neq 0$ *and, for every* $\epsilon > 0$, $\sup_{|x|\ge\epsilon}\langle x\rangle^N |D_x^\alpha D_y^\gamma \mathcal{I}[c]| < \infty$ *for all* N, α, γ.

Proof. (i) holds for c of order $< -k$ by differentiation under the integral and, for x_j, by $x_j e^{ix\cdot\xi} = D_{\xi_j}e^{ix\cdot\xi}$ and integration by parts; it extends by continuity. (ii) holds since then $D_x^\alpha D_y^\gamma \mathcal{I}[c]$, $|\alpha+\gamma| \le r$, is $\mathcal{I}[c']$ with c' of order $< -k$. (iii): by (i), $|x|^{2N} D_x^\alpha D_y^\gamma \mathcal{I}[c] = \mathcal{I}[c']$ with c' of order $M + |\alpha| - 2N$, which is bounded with all derivatives up to order r once $2N > M + |\alpha| + k + r$. □

Lemma 4.3.2 (Reduction)**.** *If* $c \in S^M(\mathbb{R}^n;\mathbb{R}^k)$ *and* $a \in S^M(\mathbb{R}^d;\mathbb{R}^k)$ *satisfies*

$$a(y,\xi) \sim \sum_\alpha \frac{1}{\alpha!}(-D_\xi)^\alpha \partial_x^\alpha c(0,y,\xi), \tag{4.3.3}$$

then $\mathcal{I}[c] - \mathcal{I}[a] \in \mathcal{C}^\infty(\mathbb{R}^n)$. *If* c *is classical, so is* a.

Proof. By Taylor's formula in x, $c(x,y,\xi) = \sum_{|\alpha|<N} \frac{x^\alpha}{\alpha!}\partial_x^\alpha c(0,y,\xi) + \sum_{|\alpha|=N} x^\alpha r_\alpha(x,y,\xi)$ with $r_\alpha = \frac{N}{\alpha!}\int_0^1 (1-t)^{N-1}\partial_x^\alpha c(tx,y,\xi)\,dt \in S^M(\mathbb{R}^n;\mathbb{R}^k)$. By Lemma 4.3.1(i), $\mathcal{I}[x^\alpha f] = \mathcal{I}[(-D_\xi)^\alpha f]$, so

$$\mathcal{I}[c] - \mathcal{I}[a] = \mathcal{I}[a_N - a] + \sum_{|\alpha|=N} \mathcal{I}[(-D_\xi)^\alpha r_\alpha], \quad a_N = \sum_{|\alpha|<N} \frac{1}{\alpha!}(-D_\xi)^\alpha \partial_x^\alpha c(0,y,\xi).$$

Both terms on the right have amplitudes of order $M - N$, so are in $\mathcal{C}^r$ if $N > M + k + r$. As N is arbitrary, $\mathcal{I}[c] - \mathcal{I}[a]$ is smooth. □

Definition 4.3.3. Let $\Omega \subset \mathbb{R}^n$ be open. The space $I^m_{\mathrm{c}}(\Omega, Y)$ consists of the $u \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\Omega)$ of the form $u = \mathcal{I}[c] + f$ with $c \in S^M(\mathbb{R}^n;\mathbb{R}^k)$, M given by (4.3.1), and $f \in \mathcal{C}^\infty(\mathbb{R}^n)$; $I^m_{\mathrm{c,ph}}(\Omega, Y)$ is defined with $c \in S^M_{\mathrm{ph}}(\mathbb{R}^n;\mathbb{R}^k)$. Further, $I^m(\Omega, Y)$ is the space of $u \in \mathcal{C}^{-\infty}(\Omega)$ such that $\phi u \in I^m_{\mathrm{c}}(\Omega, Y)$ for all $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$, and similarly in the classical case.

If $u = \mathcal{I}[c] + f \in I^m_{\mathrm{c}}(\Omega, Y)$ and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ equals one near $\operatorname{supp} u$, then $u = \mathcal{I}[\chi c] + \chi f$, so the amplitude can be taken with compact support in (x,y). By Lemma 4.3.2 it can also be taken independent of x: $u = \mathcal{I}[a] + f'$ with $a \in S^M_{\mathrm{c}}(\mathbb{R}^d;\mathbb{R}^k)$ and f' smooth (not compactly supported in general). We call such an a a *full symbol* of u.

Discussion. Two examples with $k = 1$ fix the ideas. The delta function $\delta(x_1)$ has full symbol 1, so $M = 0$ (see Example 4.8.1). The Heaviside function satisfies $H(x_1) - \frac12 = \frac12 \operatorname{sgn}(x_1)$, and the inverse Fourier transform in x_1 of $(1-\chi(\xi_1))/(i\xi_1)$, $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$ equal to one near 0, differs from $\frac12\operatorname{sgn}(x_1)$ by a smooth function. So, after a cut-off, $H(x_1)$ is conormal to $Y = \{x_1 = 0\}$ with full symbol $(1-\chi(\xi_1))/(i\xi_1)$, of order $M = -1$: integrating once lowers the order by one. In both cases the full symbol does not depend on y; in general it may, smoothly.

Proposition 4.3.4. *Let* $\Omega \subset \mathbb{R}^n$ *be open. Then (and likewise for the classical spaces):*

(i) $I^m(\Omega, Y)$ *is a* $\mathcal{C}^\infty(\Omega)$*-module,* $I^{m-1} \subset I^m$, $x_j : I^m \to I^{m-1}$, $D_{x_j} : I^m \to I^{m+1}$, *and every smooth vector field tangent to* Y *maps* I^m *to itself;*

(ii) each $u \in I^m(\Omega, Y)$ *is smooth in* $\Omega \setminus Y$, *and* $\bigcap_m I^m(\Omega, Y) = \mathcal{C}^\infty(\Omega)$;

(iii) *a full symbol of $u \in I^m_{\mathrm{c}}(\Omega, Y)$ is determined modulo $S^{-\infty}$; and $u \in I^{m-1}_{\mathrm{c}}(\Omega, Y)$ if and only if its full symbols lie in S^{M-1};*

(iv) *if $u_j \in I^{m-j}_{\mathrm{c}}(\Omega, Y)$ have supports in a fixed compact set, there is $u \in I^m_{\mathrm{c}}(\Omega, Y)$ with $u - \sum_{j<N} u_j \in I^{m-N}_{\mathrm{c}}(\Omega, Y)$ for all N;*

(v) *if $P = \sum_{|\alpha|\le q} p_\alpha(z) D_z^\alpha$ with $p_\alpha \in \mathcal{C}^\infty(\Omega)$, then $P : I^m(\Omega, Y) \to I^{m+q}(\Omega, Y)$ and, if $u \in I^m_{\mathrm{c}}$ has full symbol a, then Pu has full symbol congruent to $p_q(0, y; \xi, 0) a(y, \xi)$ modulo S^{M+q-1}, where $p_q(z, \zeta) = \sum_{|\alpha|=q} p_\alpha(z) \zeta^\alpha$.*

Proof. (i): for $\psi \in \mathcal{C}^\infty(\Omega)$ and $u \in I^m_{\mathrm{c}}$, $\psi u = \psi\chi u = \mathcal{I}[\psi\chi c] + \psi\chi f$ with χ as above. The rest follows from Lemma 4.3.1(i), since the vector fields tangent to Y are spanned over $\mathcal{C}^\infty$ by $x_i\partial_{x_j}$ and ∂_{y_l} (see Lemma 4.7.1 below) and $x_i\partial_{x_j}$ maps an amplitude of order M to $(-D_{\xi_i})(i\xi_j c + \partial_{x_j} c)$, of order M. (ii): smoothness off Y is Lemma 4.3.1(iii). If u lies in every I^m, then for each ϕ and r, ϕu is the sum of a smooth function and $\mathcal{I}[c]$ with c of order $< -k - r$, so $\phi u \in \mathcal{C}^r$. Conversely $\mathcal{C}^\infty_{\mathrm{c}}(\Omega) \subset I^m_{\mathrm{c}}$ trivially. (iii): suppose $\mathcal{I}[a] \in \mathcal{C}^\infty(\mathbb{R}^n)$ with $a \in S^M_{\mathrm{c}}(\mathbb{R}^d; \mathbb{R}^k)$. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^k)$ equal one near 0. For each y, $\mathcal{I}[a](\cdot, y)$ is the inverse Fourier transform of $a(y, \cdot)$, and it is the sum of $\chi(x)\mathcal{I}[a](x, y)$, smooth with compact support, and $(1 - \chi(x))\mathcal{I}[a](x, y)$, which is in $\mathcal{S}(\mathbb{R}^k)$ by Lemma 4.3.1(iii), both with seminorms bounded with all y-derivatives. Hence $a \in S^{-\infty}$. If now $u = \mathcal{I}[a] + f = \mathcal{I}[b] + g$ then $\mathcal{I}[a - b]$ is smooth, so $a - b \in S^{-\infty}$. The second statement follows: if $u \in I^{m-1}_{\mathrm{c}}$ it has a full symbol $b \in S^{M-1}$, and any other full symbol differs from b by an element of $S^{-\infty}$. (iv): with full symbols $a_j \in S^{M-j}_{\mathrm{c}}$ supported in a fixed compact set of y, take $u = \chi\mathcal{I}[a]$ with $a \sim \sum a_j$ (with support in the same set) and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ equal one near all the supports. (v): by (i) and Lemma 4.3.1(i), $D^\alpha \mathcal{I}[a] = \mathcal{I}[(\xi, 0)^\alpha a] + \mathcal{I}[c']$ with c' of order $M + |\alpha| - 1$ (the y-derivatives do not raise the order), and $p_\alpha \mathcal{I}[b] = \mathcal{I}[p_\alpha b]$, whose reduction is $p_\alpha(0, y) b$ modulo order $M' - 1$ by (4.3.3). □

Remark 4.3.5. For $k = n$ (so $Y = \{0\}$) we recover the spaces of Section 4.2: $I^m_{\mathcal{S}}(\mathbb{R}^n, \{0\}) = I^m_{\mathrm{c}}(\mathbb{R}^n, \{0\}) + \mathcal{S}(\mathbb{R}^n)$. Indeed if $\hat{u} \in S^{m-\frac{n}{4}}$ then $u = \chi u + (1 - \chi)u$ with $(1 - \chi)u \in \mathcal{S}$ by Lemma 4.2.2(iii) and $\chi u = \mathcal{I}[\chi\hat{u}]$. Conversely $\mathcal{I}[c]$ with c compactly supported in z has Fourier transform $\int \hat{c}(\zeta - \xi; \xi) d\!\!\bar{}\xi$, where $\hat{c}(\cdot; \xi)$ is the Fourier transform of $c(\cdot, \xi)$; this is in $S^{m-\frac{n}{4}}$ by Peetre's inequality $\langle\xi\rangle^M \le 2^{|M|}\langle\zeta\rangle^M \langle\zeta - \xi\rangle^{|M|}$ (moving derivatives onto c after the substitution $\xi \mapsto \zeta - \xi$). In particular $I^m_{\mathcal{S}}(\mathbb{R}^n, \{0\})$ is a $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$-module, and in particular an $\mathcal{S}(\mathbb{R}^n)$-module.

Example 4.3.6 (An elliptic problem at a point)**.** Let $P = \sum_{|\alpha|\le q} p_\alpha(z) D^\alpha$ with $p_\alpha \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$, and suppose P is *elliptic at* 0: $p_q(0, \zeta) = \sum_{|\alpha|=q} p_\alpha(0)\zeta^\alpha \ne 0$ for $\zeta \ne 0$. Then for every $f \in I^m_{\mathcal{S}}(\mathbb{R}^n, \{0\})$ there is $u \in I^{m-q}_{\mathcal{S}}(\mathbb{R}^n, \{0\})$ with $Pu - f \in \mathcal{S}(\mathbb{R}^n)$. Indeed, by Proposition 4.3.4(v) and Remark 4.3.5, $P : I^r_{\mathcal{S}} \to I^{r+q}_{\mathcal{S}}$ and $Pv - p_q(0, D)v \in I^{r+q-1}_{\mathcal{S}}$ for $v \in I^r_{\mathcal{S}}$. Choose $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to one near 0 and set $\hat{u}_0 = (1 - \chi)\hat{f}/p_q(0, \cdot) \in S^{m-q-\frac{n}{4}}$, using that $(1 - \chi)/p_q(0, \cdot) \in S^{-q}_{\mathrm{ph}}$. Then $p_q(0, D)u_0 - f \in \mathcal{S}$, so $Pu_0 = f - f_1$ with $f_1 \in I^{m-1}_{\mathcal{S}}$. Repeating with f replaced by f_1, we find $u_j \in I^{m-q-j}_{\mathcal{S}}$ with $P(u_0 + \cdots + u_j) - f \in I^{m-j-1}_{\mathcal{S}}$; an asymptotic sum $u \sim \sum u_j$ (Proposition 4.3.4(iv)) then satisfies $Pu - f \in \bigcap_j I^{m-j}_{\mathcal{S}} = \mathcal{S}(\mathbb{R}^n)$. The same works in the classical spaces.

4.3.2 Pseudodifferential operators

We now let pseudodifferential operators on $\mathbb{R}^n$, as in Chapter 2, act on conormal distributions. For $a \in S^M_{\mathrm{c}}(\mathbb{R}^d; \mathbb{R}^k)$ the distribution $\mathcal{I}[a]$ is tempered, with Fourier transform

$$\widehat{\mathcal{I}[a]}(\xi, \eta) = \tilde{a}(\xi, \eta) = \int e^{-iy\cdot\eta} a(y, \xi)\, dy, \qquad |D^\beta_\xi D^\gamma_\eta \tilde{a}(\xi, \eta)| \le C_{N\beta\gamma} \langle\eta\rangle^{-N} \langle\xi\rangle^{M-|\beta|}, \tag{4.3.4}$$

the estimate following by integration by parts in y.

Proposition 4.3.7. *Let $A = \mathrm{Op}(b)$, $b \in S^{m'}(\mathbb{R}^n;\mathbb{R}^n)$, and $u \in I^m_{\mathrm{c}}(\mathbb{R}^n, Y)$ with full symbol $a \in S^M_{\mathrm{c}}(\mathbb{R}^d;\mathbb{R}^k)$. Then $Au \in I^{m+m'}(\mathbb{R}^n, Y)$; more precisely $Au - \mathcal{I}[c] \in \mathcal{S}(\mathbb{R}^n)$ where $c \in S^{M+m'}(\mathbb{R}^n;\mathbb{R}^k)$ is given by*

$$c(x,y,\xi) = \int e^{iy\cdot\eta} b(x,y,\xi,\eta)\tilde{a}(\xi,\eta)đ\eta \sim \sum_\gamma \frac{1}{\gamma!}\partial_\eta^\gamma b(x,y,\xi,0)\, D_y^\gamma a(y,\xi). \tag{4.3.5}$$

In particular Au has full symbol (if b has compact support in z)

$$a'(y,\xi) \sim \sum_{\alpha,\gamma} \frac{1}{\alpha!\gamma!}(-D_\xi)^\alpha \partial_x^\alpha \Big(\partial_\eta^\gamma b(x,y,\xi,0) D_y^\gamma a(y,\xi)\Big)\Big|_{x=0}, \tag{4.3.6}$$

and $a' - b(0,y,\xi,0)a \in S^{M+m'-1}$. If b and a are classical, so is c.

Proof. Write $u = \mathcal{I}[a] - g$. Then g is smooth, and outside a compact set equals $\mathcal{I}[a]$, which vanishes for y outside the support of a and is rapidly decreasing in x by Lemma 4.3.1(iii); so $g \in \mathcal{S}(\mathbb{R}^n)$ and $Ag \in \mathcal{S}(\mathbb{R}^n)$. By (4.3.4), $A\mathcal{I}[a](x,y) = \int e^{ix\cdot\xi + iy\cdot\eta} b(x,y,\xi,\eta)\tilde{a}(\xi,\eta)đ\xi đ\eta$, the η-integral being absolutely convergent; this is $\mathcal{I}[c]$ with c as in (4.3.5). To estimate c, note that derivatives in (x,y) and ξ may be taken under the integral, and by Peetre's inequality

$$|\partial_z^\delta \partial_\xi^\beta b(z,\xi,\eta)| \le C\langle(\xi,\eta)\rangle^{m'-|\beta|} \le C'\langle\xi\rangle^{m'-|\beta|}\langle\eta\rangle^{|m'|+|\beta|};$$

combined with (4.3.4) (and y-derivatives falling on $e^{iy\cdot\eta}$ producing powers of η) this gives $c \in S^{M+m'}$. For the expansion, Taylor's formula in η gives $b(z,\xi,\eta) = \sum_{|\gamma|<L} \frac{\eta^\gamma}{\gamma!}\partial_\eta^\gamma b(z,\xi,0) + \sum_{|\gamma|=L} \eta^\gamma r_\gamma(z,\xi,\eta)$ with

$$|\partial_z^\delta \partial_\xi^\beta r_\gamma| \le C \sup_{0\le t\le 1} \langle(\xi,t\eta)\rangle^{m'-L-|\beta|} \le C\langle\xi\rangle^{m'-L-|\beta|}\langle\eta\rangle^{|m'-L|+|\beta|}.$$

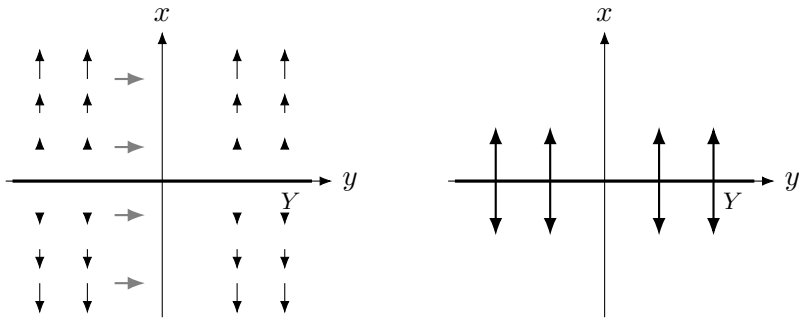
Since $\int e^{iy\cdot\eta}\eta^\gamma \tilde{a} đ\eta = D_y^\gamma a$, the terms with $|\gamma| < L$ give the expansion and the remainder is of order $M + m' - L$ by the same estimate as before. Finally (4.3.6) follows from Lemma 4.3.2. In the classical case each term of (4.3.5) is classical, so the asymptotic sum is. □

The microlocal content of this computation is that u is singular only in the conormal directions to Y. Recall from Chapter 3 that $(\bar{z},\bar{\zeta}) \notin \mathrm{WF}(u)$ if and only if there is $A \in \Psi^0(\mathbb{R}^n)$ elliptic at $(\bar{z},\bar{\zeta})$ with $Au \in \mathcal{C}^\infty$. The conormal bundle of Y is $N^*Y = \{(0,y;\xi,0)\} \subset T^*\mathbb{R}^n$.

Proposition 4.3.8. *If $u \in I^m(\Omega, Y)$ then $\mathrm{WF}(u) \subset N^*Y\backslash 0$. More precisely, if $u \in I^m_{\mathrm{c}}(\Omega, Y)$ has full symbol a, then $(0,\bar{y};\bar{\xi},0) \notin \mathrm{WF}(u)$ if and only if a satisfies the estimates of $S^{-\infty}$ in $U \times \Gamma$, for some neighbourhood U of $\bar{y}$ and open cone $\Gamma \ni \bar{\xi}$.*

Proof. We may assume $u \in I^m_{\mathrm{c}}$. Over $\Omega \setminus Y$, u is smooth. Let $\bar{z} = (0,\bar{y})$ and $\bar{\zeta} = (\bar{\xi},\bar{\eta})$ with $\bar{\eta} \ne 0$. Choose $\delta > 0$ with $|\bar{\eta}| > 2\delta|\bar{\xi}|$, and $b = \phi(z)\psi(\zeta)$ with $\phi \in \mathcal{C}^\infty_{\mathrm{c}}$, $\phi(\bar{z}) \ne 0$, $\psi \in S^0_{\mathrm{ph}}(\mathbb{R}^n)$ homogeneous of degree 0 in $|\zeta| \ge 1$, equal to one near $\bar{\zeta}/|\bar{\zeta}|$ there, and supported in $\{|\eta| \ge \delta|\xi|,\ |\zeta| \ge \frac{1}{2}\}$. On the support of ψ, $\langle\eta\rangle \ge c\langle\xi\rangle$, so by (4.3.4) the amplitude c of (4.3.5) satisfies $|c| \le C_N \int_{\langle\eta\rangle\ge c\langle\xi\rangle} \langle\eta\rangle^{-N}\langle\xi\rangle^M d\eta \le C'_N\langle\xi\rangle^{M-N+d}$ for all N, and similarly for derivatives; thus $c \in S^{-\infty}$ and $\mathrm{Op}(b)u \in \mathcal{C}^\infty$. Hence $\mathrm{WF}(u) \subset N^*Y$.

For the refined statement let $\bar{\zeta} = (\bar{\xi},0)$. If a is of order $-\infty$ in $U \times \Gamma$, take $b = \phi(z)\psi(\xi,\eta)$ as before but with ϕ supported in $\mathbb{R}^k \times U'$, $U' \Subset U$, and ψ supported in a small conic neighbourhood of $(\bar{\xi},0)$, so that $(y,\xi) \in U' \times \Gamma$ on the support of $\phi(x,y)\psi(\xi,0)$ for $|\xi| \ge 1$. Each term in (4.3.6) then vanishes to infinite order (the terms involve a only at



(a) $x\partial_x$ (black) and ∂_y (gray) (b) $N^*Y = \{(0, y; \xi, 0)\}$

Figure 4.2: The local model with $k = d = 1$. (a) Vector fields tangent to $Y = \{x = 0\}$ are generated by $x_i\partial_{x_j}$ and ∂_{y_l} (Lemma 4.7.1); they preserve each $I^m(\Omega, Y)$ (Proposition 4.3.4(i)), and repeated application of them characterizes conormality (Section 4.7). (b) The conormal bundle, which contains the wavefront set of every conormal distribution (Proposition 4.3.8).

points of $U' \times \Gamma$), so $Au \in \mathcal{C}^\infty$ and A is elliptic at $(\bar{z}, \bar{\zeta})$. Conversely, if $(\bar{z}, \bar{\zeta}) \notin \mathrm{WF}(u)$, choose $B = \mathrm{Op}(b)$ as just described with $b = 1$ on a conic neighbourhood of $(\bar{z}, \bar{\zeta})$ and $\mathrm{WF}'(B) \cap \mathrm{WF}(u) = \emptyset$; then $Bu \in \mathcal{C}^\infty$ by Chapter 3, so its full symbol (4.3.6) is in $S^{-\infty}$, and near $(\bar{y}, \bar{\xi})$ all terms with $(\alpha, \gamma) \neq 0$ vanish because b is constant there, leaving $a \in S^{-\infty}$ near $(\bar{y}, \bar{\xi})$. □

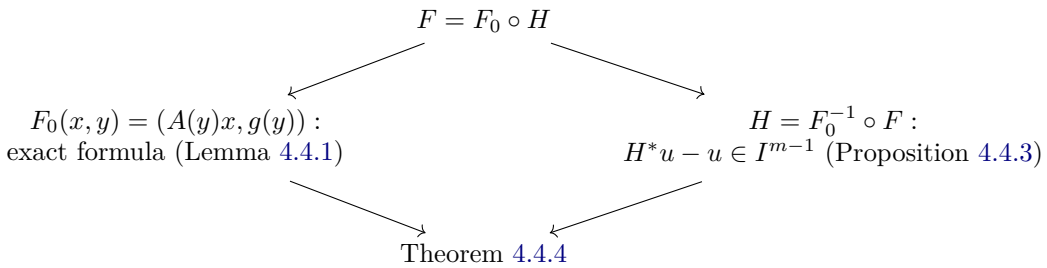
4.4 Invariance

The definition of the local model uses linear coordinates and a Fourier transform, so it is not clear that it makes sense on a manifold. We now show that the spaces $I^m_{\mathrm{c}}(\Omega, Y)$ are invariant under diffeomorphisms preserving Y, and compute the effect on the full symbol modulo lower order terms. Let $F : \Omega' \to \Omega$ be a diffeomorphism between open subsets of $\mathbb{R}^n$ with $F(\Omega' \cap Y) = \Omega \cap Y$. Write $F(x, y) = (F'(x, y), F''(x, y))$; then $F'(0, y) = 0$, so $\partial_y F'(0, y) = 0$, and

$$g(y) = F''(0, y), \qquad A(y) = \partial_x F'(0, y) \in \mathrm{GL}(k, \mathbb{R}) \tag{4.4.1}$$

define a diffeomorphism g of $\Omega' \cap Y$ onto $\Omega \cap Y$ and the linear maps $A(y)$ induced by F_* on the normal spaces.

The proof of invariance splits F into two pieces of very different nature:



The fibre-linear part acts on the symbol by a linear change of variables, and the remaining part, which is the identity to first order on Y, only changes lower order terms.

Lemma 4.4.1. *Let $V \subset \mathbb{R}^d$ be open, let g be a diffeomorphism of V onto $g(V)$ and let $A \in \mathcal{C}^\infty(V; \mathrm{GL}(k,\mathbb{R}))$. Set $F_0(x,y) = (A(y)x, g(y))$, a diffeomorphism of $\mathbb{R}^k \times V$ onto $\mathbb{R}^k \times g(V)$. If $a \in S^M_{\mathrm{c}}(\mathbb{R}^d;\mathbb{R}^k)$ has support in $g(V)\times\mathbb{R}^k$ then*

$$F_0^* \mathcal{I}[a] = \mathcal{I}[a_F], \qquad a_F(y,\eta) = |\det A(y)|^{-1} a(g(y), A(y)^{-t}\eta) \in S^M_{\mathrm{c}}(\mathbb{R}^d;\mathbb{R}^k),$$

where $A^{-t} = (A^t)^{-1}$; a_F is classical if a is.

Proof. For fixed y this is the linear change of variables $\xi = A(y)^{-t}\eta$ in $\int e^{iA(y)x\cdot\xi} a(g(y),\xi)đ\xi$. That a_F is a symbol follows from the invariance of symbol estimates under smooth families of linear maps, the support of a_F in y being compact. □

Lemma 4.4.2. *A vector field of the form*

$$V = \sum_{i,j,l} x_i x_j \alpha_{ijl}(z)\partial_{x_l} + \sum_{i,l} x_i \beta_{il}(z)\partial_{y_l}, \qquad \alpha_{ijl}, \beta_{il} \in \mathcal{C}^\infty(\Omega), \tag{4.4.2}$$

maps $I^m(\Omega, Y)$ to $I^{m-1}(\Omega, Y)$.

Proof. By Proposition 4.3.4(i), $x_i x_j \partial_{x_l}$ maps I^m into I^{m+1-2} and $x_i\partial_{y_l}$ maps I^m into I^{m-1}, and the coefficients are harmless. □

We also need conormal distributions depending smoothly on a parameter $s \in [0,1]$. A *smooth family* $u(s) \in I^m_{\mathrm{c}}(\Omega, Y)$ is one of the form $u(s) = \mathcal{I}[c(s)] + f(s)$ with $c \in S^M(\mathbb{R}_s \times \mathbb{R}^n; \mathbb{R}^k)$ and $f \in \mathcal{C}^\infty(\mathbb{R}\times\mathbb{R}^n)$, the supports of $u(s)$ lying in a fixed compact subset of Ω; that is, u is an element of I^m_{c} on $\mathbb{R}_s \times \mathbb{R}^n$ with respect to $\mathbb{R}_s \times Y$, restricted to $[0,1]$. By Proposition 4.3.4 applied in $n+1$ variables, $\partial_s u(s)$ and $\int_s^1 u(s')\,ds'$ are smooth families of order m, $V_s u(s)$ is a smooth family of order $m-1$ if V_s is of the form (4.4.2) with coefficients smooth in s, and families of orders $m-j$ have asymptotic sums.

Proposition 4.4.3. *Let H be a diffeomorphism from a neighbourhood O of $\bar z = (0,\bar y)$ onto its image such that $H(0,y) = (0,y)$ and $\partial_x H'(0,y) = \mathrm{Id}$ for $(0,y) \in O$. Then there is a neighbourhood O' of $\bar z$ such that $H^*u - u \in I^{m-1}_{\mathrm{c}}(\mathbb{R}^n, Y)$ for all $u \in I^m_{\mathrm{c}}(\mathbb{R}^n, Y)$ with support in O'; similarly in the classical case.*

Discussion. The method is Moser's homotopy trick. Join H to the identity by the family H_s. The derivative $\frac{d}{ds}H_s^*$ is H_s^* composed with a vector field V_s, and because H agrees with the identity to first order on Y, V_s has the special form (4.4.2), which lowers the conormal order by one (Lemma 4.4.2). Instead of integrating $\frac{d}{ds}H_s^*u$ directly, we solve the transport equation $\partial_s u(s) + V_s u(s) \approx 0$ with $u(1) = u$; then $H_s^* u(s)$ is almost constant in s, and $H^*u \approx u(0)$, which differs from u by terms of lower order.

Proof. Set $H_s(z) = z + s(H(z) - z)$, $s \in [0,1]$, so $H_0 = \mathrm{Id}$, $H_1 = H$ and $H_s(0,y) = (0,y)$. At points of Y, $dH_s = \left(\begin{smallmatrix} \mathrm{Id} & 0 \\ sB & \mathrm{Id}\end{smallmatrix}\right)$ with $B = \partial_x H''(0,y)$, which is invertible. By the inverse function theorem, applied uniformly for s in the compact set $[0,1]$, there is a ball $O_1 \ni \bar z$ such that each H_s is a diffeomorphism of O_1 onto its image, and a smaller ball O' contained in $H_s(O_1)$ for all s. For $w \in O'$ let

$$V_s(w) = (H - \mathrm{Id})(H_s^{-1}(w)),$$

a vector field depending smoothly on s. For smooth v the chain rule gives

$$\frac{d}{ds}H_s^* v = H_s^*(V_s v), \tag{4.4.3}$$

and the same holds for a smooth family of distributions $v(s)$ in the form $\frac{d}{ds}H_s^*v(s) = H_s^*(\partial_s v + V_s v)$: pairing with a test function ϕ, $\langle H_s^*v(s),\phi\rangle = \langle v(s),\phi_s\rangle$ with $\phi_s = |\det dH_s^{-1}|\,(H_s^{-1})^*\phi$ smooth in s, and the identity $\langle w, \partial_s\phi_s\rangle = \langle H_s^*(V_s w),\phi\rangle$, valid for smooth w by (4.4.3), extends to distributions by density.

Since $H'(x,y) - x = O(|x|^2)$, $H''(x,y) - y = O(|x|)$ and the x-component of $H_s^{-1}(w)$ vanishes on Y (so is a combination $\sum x_i(\dots)$), V_s is of the form (4.4.2) with coefficients smooth in s.

Now let $u \in I_{\mathrm{c}}^m$ have support in O'. We construct a smooth family $u(s)$, with supports in a fixed compact subset of O', such that

$$u(1) = u, \qquad \partial_s u(s) + V_s u(s) = r(s) \in \mathcal{C}_{\mathrm{c}}^\infty([0,1]\times O'), \qquad u(0) - u \in I_{\mathrm{c}}^{m-1}. \tag{4.4.4}$$

Set $u_0(s) = u$ and $f_1(s) = V_s u \in I^{m-1}$. Inductively, given smooth families $u_0,\dots,u_N$ with u_j of order $m-j$, $u_j(1) = 0$ for $j \geq 1$, and $\partial_s(u_0+\dots+u_N) + V_s(u_0+\dots+u_N) = f_{N+1}$ of order $m-N-1$, put $u_{N+1}(s) = \int_s^1 f_{N+1}(s')\,ds'$. Then u_{N+1} has order $m-N-1$, vanishes at $s=1$, and the new error is $f_{N+2} = V_s u_{N+1}$, of order $m-N-2$. Let $\tilde u(s) \sim \sum_j u_j(s)$ be an asymptotic sum, multiplied by a cut-off equal to one near the supports. Then $\partial_s\tilde u + V_s\tilde u$ is a smooth family of every order, hence a smooth function of (s,z), and $\tilde u(1) - u = \tilde u(1) - \sum_{j<N} u_j(1)$ has order $m-N$ for every N, so is smooth. Thus $u(s) = \tilde u(s) - (\tilde u(1) - u)$ satisfies (4.4.4); note that $u(0) - u \equiv \sum_{j\geq 1} u_j(0)$ modulo $\mathcal{C}^\infty$ has order $m-1$. Then

$$H^*u = H_1^*u(1) = H_0^*u(0) + \int_0^1 \frac{d}{ds}H_s^*u(s)\,ds = u(0) + \int_0^1 H_s^*r(s)\,ds,$$

and the last integral is smooth. Hence $H^*u - u = (u(0)-u) + \mathcal{C}^\infty \subset I_{\mathrm{c}}^{m-1}$. □

Theorem 4.4.4. *Let $F:\Omega'\to\Omega$ be a diffeomorphism of open subsets of $\mathbb{R}^n$ with $F(\Omega'\cap Y) = \Omega\cap Y$, and g, A as in (4.4.1). Then $F^*: I_{\mathrm{c}}^m(\Omega,Y) \to I_{\mathrm{c}}^m(\Omega',Y)$, and if $u \in I_{\mathrm{c}}^m(\Omega,Y)$ has full symbol a, then F^*u has full symbol congruent modulo S^{M-1} to*

$$|\det A(y)|^{-1} a(g(y), A(y)^{-t}\eta). \tag{4.4.5}$$

The same holds for the classical spaces.

Proof. Using a partition of unity and the module property we may assume that u has support in a small neighbourhood of $F(\bar z)$, $\bar z \in Y\cap\Omega'$ (away from Y, u and F^*u are smooth). Let $F_0(x,y) = (A(y)x, g(y))$, a diffeomorphism of $\mathbb{R}^k\times(Y\cap\Omega')$ onto $\mathbb{R}^k\times(Y\cap\Omega)$ (identifying Y with $\mathbb{R}^d$), and $H = F_0^{-1}\circ F$, defined near $\bar z$. Then $H(0,y) = (0,y)$ and $\partial_x H'(0,y) = A(y)^{-1}\partial_x F'(0,y) = \mathrm{Id}$, so Proposition 4.4.3 applies to H. With $u = \mathcal{I}[a] + f$ we have, by Lemma 4.4.1, $F_0^*u = \mathcal{I}[a_F] + F_0^*f$ (after cutting off), and $F^*u = H^*(F_0^*u) \in F_0^*u + I_{\mathrm{c}}^{m-1}$. □

4.5 Conormality at the zero section of a vector bundle

The zero section of a vector bundle is the intermediate case between the local model and a general submanifold: the fibres carry a linear structure, so the Fourier transform in the fibres makes sense, but there are no global coordinates. By the tubular neighbourhood theorem every submanifold looks like this near Y (Section 4.6).

Let $\pi: W\to Y$ be a real vector bundle of rank k over a manifold Y of dimension d, so W has dimension $n = d+k$, and let $0_W \subset W$ be the zero section and W' the dual bundle. The linear invariance of the radial compactification allows it to be performed fibrewise: $\overline{W}$

is a manifold with boundary, fibred over Y with fibres $\overline{W_y}$ (formally, the bundle associated to the frame bundle of W with fibre $\overline{\mathbb{R}^k}$), and $\rho^{-M}\mathcal{C}^\infty(\overline{W})$ is well defined. Symbols with bounds $S^M(W)$ on a bundle over a compact base, or with compact support in the base, are defined via local trivializations, by the linear invariance of the estimates.

Let $\Omega_{\mathrm{fib}} = \Omega_{\mathrm{fib}}(W')$ be the line bundle over W' whose fibre at $\zeta \in W'_y$ is $\Omega(W'_y)$, the densities on the fibre. The fibrewise version of (4.2.1) is an isomorphism

$$\mathcal{F}_{\mathrm{fib}} : \mathcal{S}(W) \longrightarrow \mathcal{S}(W';\Omega_{\mathrm{fib}}),$$

defined for (fibrewise) Schwartz functions; in a local trivialization $W|_U \cong U \times \mathbb{R}^k$ it is the partial Fourier transform in the fibre variables followed by multiplication by $|d\xi|$, and it does not depend on the trivialization. Its inverse is defined on symbols as an oscillatory integral.

Definition 4.5.1. For Y compact, $I^m(W, 0_W)$ is the space of distributions $u = \mathcal{F}_{\mathrm{fib}}^{-1}b + f$ with $b \in S^M(W';\Omega_{\mathrm{fib}})$ (that is, $b = \beta|d\xi|$ with $\beta \in S^M$ in local trivializations), $M = m + \frac{n}{4} - \frac{k}{2}$, and $f \in \mathcal{C}^\infty(W)$; similarly $I^m_{\mathrm{ph}}(W, 0_W)$ with b classical.

In a local trivialization, with fibre coordinates x and base coordinates y, $\mathcal{F}_{\mathrm{fib}}^{-1}(\beta|d\xi|) = \mathcal{I}[\beta]$, so $I^m(W, 0_W)$ consists of the distributions which are smooth outside 0_W and lie in the spaces of Definition 4.3.3 in every such local trivialization. Changes of trivialization are of the fibre-linear form of Lemma 4.4.1 (with g a change of coordinates on Y), under which $\beta|d\xi|$ transforms as a fibre density, so we obtain a well-defined symbol map with values in $S^M(W';\Omega_{\mathrm{fib}})/S^{M-1}(W';\Omega_{\mathrm{fib}})$, and, by Proposition 4.3.4(iii) and the surjectivity of the inverse Fourier transform, a short exact sequence

$$I^{m-1}(W, 0_W) \hookrightarrow I^m(W, 0_W) \xrightarrow{\sigma_m} S^M(W';\Omega_{\mathrm{fib}})/S^{M-1}(W';\Omega_{\mathrm{fib}}), \tag{4.5.1}$$

and in the classical case the corresponding sequence onto the smooth fibre densities on $W' \setminus 0$ homogeneous of degree $M + k$. For a product $W = Y \times \mathbb{R}^k$ the definition says that $I^m(Y \times \mathbb{R}^k, Y \times \{0\})$ consists of smooth functions of $y \in Y$ with values in $I^{m-\frac{d}{4}}(\mathbb{R}^k, \{0\})$ (up to smooth terms), the topology of $\mathcal{C}^\infty(Y; \rho^{-M}\mathcal{C}^\infty(\overline{\mathbb{R}^k})) = \rho^{-M}\mathcal{C}^\infty(Y \times \overline{\mathbb{R}^k})$ making precise what it means for a conormal distribution to depend smoothly on parameters.

Bundle isomorphisms, possibly covering a diffeomorphism of the base, act on $I^m(W, 0_W)$ by Lemma 4.4.1. For the extension to submanifolds, the essential point is invariance under diffeomorphisms which do not preserve the linear structure. If h is a diffeomorphism between neighbourhoods of 0_W which maps 0_W to itself, then at each point of 0_W the differential h_* preserves $T0_W$ and so induces a map on $T_yW/T_y0_W = W_y$.

Lemma 4.5.2. *Any diffeomorphism h of a neighbourhood of 0_W onto a neighbourhood of 0_W with $h(0_W) = 0_W$ factorizes, near 0_W, as $h = h_0 \circ h_1$ where h_0 is a bundle isomorphism (covering $h|_{0_W}$) and h_1 is the identity on 0_W with $(h_1)_* = \mathrm{Id}$ on $W_y = T_yW/T_y0_W$ for all y.*

Proof. Take h_0 to be the bundle map induced by h on the normal spaces, i.e. $h_0(w) = h_*(w)$ for $w \in W_y = N_y0_W$, covering $h|_{0_W}$; then $h_1 = h_0^{-1} \circ h$ has the stated properties. In local trivializations this is the factorization $F = F_0 \circ H$ used in the proof of Theorem 4.4.4. □

Proposition 4.5.3. *Let Y be compact and $h : N \to N'$ a diffeomorphism between open neighbourhoods of 0_W which is the identity on 0_W and induces the identity on W_y for all $y \in Y$. If $u \in I^m(W, 0_W)$ has support in N' then $h^*u - u \in I^{m-1}(W, 0_W)$; in particular $\sigma_m(h^*u) = \sigma_m(u)$. Combined with Lemma 4.5.2, every diffeomorphism h between neighbourhoods of 0_W mapping 0_W to itself acts on conormal distributions supported in its range, with $\sigma_m(h^*u) = h_0^*\sigma_m(u)$.*

Proof. Cover Y by finitely many open sets over which W is trivial, take a partition of unity ϕ_j of a neighbourhood of 0_W subordinate to their preimages and apply Proposition 4.4.3 (in the form of the proof of Theorem 4.4.4) to each $\phi_j u$, noting that $u - \sum \phi_j u$ is smooth. □

4.6 Conormal distributions on manifolds

4.6.1 Preliminaries

Manifolds are discussed systematically in Chapter 5; here we need only a few notions. Let X be a smooth manifold of dimension n. The density bundles $\Omega^t(X)$, $t \in \mathbb{R}$, are the line bundles with fibres $\Omega^t(T_pX)$; in local coordinates their sections are $v|dz|^t$, and under a change of coordinates $z = F(z')$, $v|dz|^t$ pulls back to $(v \circ F)|\det F'|^t|dz'|^t$. Smooth compactly supported densities can be integrated, and $\mathcal{C}^{-\infty}(X;\Omega^{\frac12})$ is the dual of $\mathcal{C}^\infty_{\mathrm{c}}(X;\Omega^{\frac12})$, the pairing of two half-densities being a density; locally an element of $\mathcal{C}^{-\infty}(X;\Omega^{\frac12})$ is $v|dz|^{1/2}$ with v a distribution in coordinates, transforming as above. A smooth vector field V acts on half-densities by the Lie derivative: in coordinates, with $V = \sum v_j\partial_{z_j}$,

$$\mathcal{L}_V(\phi|dz|^t) = (V\phi + t(\textstyle\sum_j \partial_{z_j} v_j)\phi)|dz|^t. \tag{4.6.1}$$

Let $Y \subset X$ be a closed embedded submanifold of codimension k. An *adapted chart* is a chart $\psi : U \to \psi(U) \subset \mathbb{R}^n = \mathbb{R}^k_x \times \mathbb{R}^d_y$ with $\psi(U \cap Y) = \psi(U) \cap \{x = 0\}$; such charts cover Y. The *normal bundle* NY has fibres $N_yY = T_yX/T_yY$, and the *conormal bundle* $N^*Y \subset T^*X$, with fibres $N^*_yY = \{\zeta \in T^*_yX;\ \zeta(T_yY) = 0\}$, is its dual. In an adapted chart, $N^*Y = \{(0, y; \xi, 0)\}$, and (y, ξ) are coordinates on N^*Y. If $F : \Omega' \to \Omega$ is a change of adapted charts as in Section 4.4, the pull-back $F^*(\xi \cdot dx) = (A(y)^t\xi) \cdot dx'$ at $(0, y)$, since $\partial_y F'(0, y) = 0$; so the induced identification of conormal bundles is

$$F^\sharp : N^*Y \ni (y, \eta) \longmapsto (g(y), A(y)^{-t}\eta) \in N^*Y, \tag{4.6.2}$$

(primed coordinates on the left, unprimed on the right) which is linear on the fibres.

Definition 4.6.1. The space $I^m(X, Y; \Omega^{\frac12})$ consists of those $u \in \mathcal{C}^{-\infty}(X;\Omega^{\frac12})$ which are smooth on $X \setminus Y$ and such that for every adapted chart $\psi : U \to \psi(U)$ and every $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$, $(\psi^{-1})^*(\phi u) = v|dz|^{1/2}$ with $v \in I^m_{\mathrm{c}}(\psi(U), \{x = 0\})$. The classical subspace $I^m_{\mathrm{ph}}(X, Y; \Omega^{\frac12})$ is defined in the same way. For a vector bundle E over X, $I^m(X, Y; E \otimes \Omega^{\frac12})$ is defined by requiring the coefficients in local trivializations to be in $I^m(X, Y; \Omega^{\frac12})$; similarly with $\Omega^{\frac12}$ omitted.

By Theorem 4.4.4 (the factor $|\det F'|^{1/2}$ being smooth) it suffices to check the condition for a family of adapted charts covering Y, and the definition agrees with Definition 4.3.3 for $X = \mathbb{R}^n$ and with Definition 4.5.1 for the zero section of a vector bundle; it is independent of the choice of transition functions for E by the module property.

4.6.2 Normal fibrations

Alternatively, $I^m(X, Y)$ can be defined by transferring Definition 4.5.1 from the normal bundle to X (Figure 4.3).

Theorem 4.6.2 (Tubular neighbourhood theorem)**.** *If $Y \subset X$ is a closed embedded submanifold, there are an open neighbourhood D of Y in X, an open neighbourhood D' of the zero section of NY, and a diffeomorphism $f : D \to D'$ (a* normal fibration*) such that*

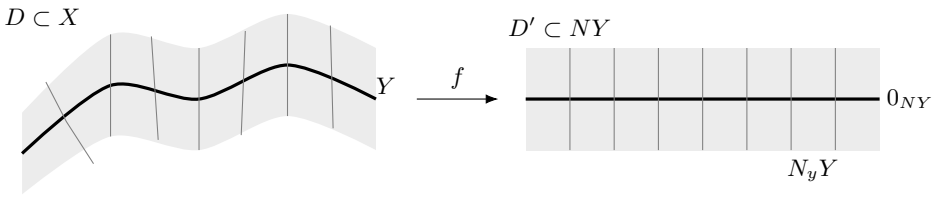


Figure 4.3: A normal fibration (Theorem 4.6.2) identifies a neighbourhood D of Y in X with a neighbourhood D' of the zero section in the normal bundle; the gray curves on the left (for instance normal geodesics) go to the fibres N_yY. Since its differential along Y induces the identity on N_yY, two choices differ by a map of the type in Proposition 4.5.3, which preserves conormal distributions and their principal symbols.

$f|_Y$ is the natural identification of Y with 0_{NY} and, for each $y \in Y$, the map induced by f_* from $N_yY = T_yX/T_yY$ to $T_y(NY)/T_y0_{NY} = N_yY$ is the identity. If f_1, f_2 are two normal fibrations then $f_2 \circ f_1^{-1}$ is a diffeomorphism between neighbourhoods of 0_{NY} satisfying the hypotheses of Proposition 4.5.3.

Sketch of proof. Choose a Riemannian metric on X; it identifies N_yY with the orthogonal complement of T_yY in T_yX. The exponential map $NY \ni (y, v) \mapsto \exp_y(v)$ is smooth, is the identity on the zero section and has differential the identity at points of the zero section (with respect to the splitting $T_y(NY) = T_yY \oplus N_yY$), so it is a local diffeomorphism near 0_{NY}; a standard argument (using that it is injective on 0_{NY} and Y is closed) shows it is a diffeomorphism of a neighbourhood D' of 0_{NY} onto a neighbourhood D of Y, and f is its inverse. The last statement is immediate from the definition. □

Proposition 4.6.3. *Let $f : D \to D'$ be a normal fibration and Y compact. Then $u \in \mathcal{C}^{-\infty}(X)$ lies in $I^m(X, Y)$ if and only if u is smooth outside Y and, for some (hence any) $\phi \in \mathcal{C}^\infty_c(D)$ equal to one near Y, $(f^{-1})^*(\phi u) \in I^m(NY, 0_{NY})$.*

Proof. Local trivializations of NY composed with f are adapted charts for Y, and local trivializations of NY are adapted charts for 0_{NY}, so both conditions amount to the condition of Definition 4.6.1 in a particular covering family of adapted charts. Independence of f also follows from Proposition 4.5.3 and Theorem 4.6.2. □

4.6.3 The principal symbol

Discussion. Why should the principal symbol be a half-density? Under a change of adapted coordinates the full symbol picks up the factor $|\det A(y)|^{-1}$ from the Fourier transform in the normal variables (Theorem 4.4.4). If u itself is a half-density, the Jacobian factor $|\det F'|^{1/2}$ of u combines with it to give exactly the transformation law of a half-density on N^*Y in the coordinates (y, ξ). With this convention the principal symbol is intrinsic, with no choices of densities on X or Y; this is also the form needed for Lagrangian distributions in Chapter 10.

For $r \in \mathbb{R}$ let $S^r(N^*Y; \Omega^{\frac12})$ be the space of smooth half-densities on N^*Y which in each adapted chart are of the form $\beta(y, \xi)|dy\, d\xi|^{1/2}$ with $\phi\beta \in S^{r-\frac{k}{2}}(\mathbb{R}^d; \mathbb{R}^k)$ for all $\phi \in \mathcal{C}^\infty_c$ of the chart. This is invariant under the fibre-linear maps (4.6.2), so is well defined; the shift by $k/2$ reflects the fact that $|d\xi|^{1/2}$ is homogeneous of degree $k/2$ under fibre dilations, so that the order r is the total homogeneity. Write $S^{r-[1]}(N^*Y; \Omega^{\frac12}) = S^r(N^*Y; \Omega^{\frac12})/S^{r-1}(N^*Y; \Omega^{\frac12})$. In the classical case the principal parts are smooth half-densities on $N^*Y \setminus 0$ homogeneous of degree r; we denote this space $\mathcal{C}^\infty_{(r)}(N^*Y \setminus 0; \Omega^{\frac12})$.

Theorem 4.6.4. *There is a unique map*

$$\sigma_m : I^m(X,Y;\Omega^{\frac12}) \longrightarrow S^{m+\frac{n}{4}-[1]}(N^*Y;\Omega^{\frac12})$$

such that if $u = (\mathcal{I}[a]+f)|dx\,dy|^{1/2}$ *in an adapted chart then* $\sigma_m(u) = a(y,\xi)|dy\,d\xi|^{1/2}$ *there. It gives a short exact sequence*

$$0 \longrightarrow I^{m-1}(X,Y;\Omega^{\frac12}) \longrightarrow I^m(X,Y;\Omega^{\frac12}) \xrightarrow{\sigma_m} S^{m+\frac{n}{4}-[1]}(N^*Y;\Omega^{\frac12}) \longrightarrow 0, \tag{4.6.3}$$

and, in the classical case, a short exact sequence

$$0 \to I^{m-1}_{\mathrm{ph}}(X,Y;\Omega^{\frac12}) \to I^m_{\mathrm{ph}}(X,Y;\Omega^{\frac12}) \xrightarrow{\sigma_m} \mathcal{C}^\infty_{(m+\frac{n}{4})}(N^*Y\setminus 0;\Omega^{\frac12}) \to 0.$$

Moreover $\sigma_m(\psi u) = (\psi|_Y)\sigma_m(u)$ *for* $\psi \in \mathcal{C}^\infty(X)$*, where* $\psi|_Y$ *is lifted to* N^*Y*.*

Proof. We must check that the local definition is consistent. Let F be a change of adapted coordinates. If $u = v|dz|^{1/2}$ then $F^*u = (F^*v)|\det F'|^{1/2}|dz'|^{1/2}$. At $x=0$, F' is block triangular with diagonal blocks $A(y)$ and $g'(y)$, so $|\det F'(0,y)|^{1/2} = |\det A|^{1/2}|\det g'|^{1/2}$; by Theorem 4.4.4 and the multiplication rule in Proposition 4.3.4(v), the full symbol of F^*u is congruent modulo S^{M-1} to

$$|\det A(y)|^{-\frac12}|\det g'(y)|^{\frac12} a(g(y), A(y)^{-t}\eta).$$

On the other hand the Jacobian determinant of $F^\sharp$ in (4.6.2) is $\det g'(y)\det A(y)^{-1}$, so $(F^\sharp)^*(a|dy\,d\xi|^{1/2})$ is exactly this function times $|dy\,d\eta|^{1/2}$. Hence σ_m is well defined, with $\sigma_m(F^*u) = (F^\sharp)^*\sigma_m(u)$; the classical case is the same. The multiplicative property holds because $\psi(x,y)-\psi(0,y) = \sum_j x_j\psi_j$ and x_j lowers the order. Exactness in the middle is local and is Proposition 4.3.4(iii); injectivity is trivial. For surjectivity, given $b \in S^{m+\frac{n}{4}}(N^*Y;\Omega^{\frac12})$, cover Y by adapted charts U_j (a locally finite cover) with a partition of unity ϕ_j on Y, write $\phi_j b = \beta_j|dy\,d\xi|^{1/2}$ and let $u = \sum_j \chi_j \mathcal{I}[\beta_j]|dz|^{1/2}$, with $\chi_j \in \mathcal{C}^\infty_{\mathrm{c}}(U_j)$ equal to one near $\operatorname{supp}\phi_j \times \{0\}$; then $\sigma_m(u) = \sum_j \phi_j b = b$. □

4.6.4 Properties

We collect the main properties of the spaces $I^m(X,Y;\Omega^{\frac12})$; all hold equally for the classical spaces.

Theorem 4.6.5. *Let* $Y \subset X$ *be a closed embedded submanifold of codimension* k*,* $\dim X = n$*.*

(i) Elements of $I^m(X,Y;\Omega^{\frac12})$ *are smooth outside* Y*,* $I^{m-1} \subset I^m$ *and* $\bigcap_m I^m(X,Y;\Omega^{\frac12}) = \mathcal{C}^\infty(X;\Omega^{\frac12})$*.*

(ii) (Asymptotic completeness) *If* $u_j \in I^{m-j}(X,Y;\Omega^{\frac12})$*, there exists* $u \in I^m(X,Y;\Omega^{\frac12})$ *with* $u - \sum_{j<N} u_j \in I^{m-N}$ *for all* N*.*

(iii) (Invariance) *If* $F : X' \to X$ *is a diffeomorphism and* $Y' = F^{-1}(Y)$*, then* $F^* : I^m(X,Y;\Omega^{\frac12}) \to I^m(X',Y';\Omega^{\frac12})$ *and* $\sigma_m(F^*u) = (F^\sharp)^*\sigma_m(u)$*, where* $F^\sharp : N^*Y' \to N^*Y$ *is the inverse of the restriction of* $F^* : T^*X \to T^*X'$*. The same holds for diffeomorphisms between open subsets, for distributions supported in the range.*

(iv) (Differential operators) *If* $P \in \mathrm{Diff}^q(X;\Omega^{\frac12})$ *then* $P : I^m(X,Y;\Omega^{\frac12}) \to I^{m+q}(X,Y;\Omega^{\frac12})$ *and* $\sigma_{m+q}(Pu) = \sigma_q(P)|_{N^*Y}\,\sigma_m(u)$*, where* $\sigma_q(P)$ *is the principal symbol, characterized by* $P(e^{itf}v) = e^{itf}(t^q\sigma_q(P)(df)v + O(t^{q-1}))$ *as* $t \to \infty$ *for real* $f \in \mathcal{C}^\infty(X)$ *and* $v \in \mathcal{C}^\infty_{\mathrm{c}}(X;\Omega^{\frac12})$*.*

(v) (Wavefront set) $\mathrm{WF}(u) \subset N^*Y \setminus 0$ *for* $u \in I^m(X,Y;\Omega^{\frac12})$.

Proof. All statements are local and invariant, so follow from the local results: (i) from Proposition 4.3.4(ii); (ii) from Proposition 4.3.4(iv) and a partition of unity; (iii) from Theorems 4.4.4 and 4.6.4; (iv) from Proposition 4.3.4(v), since in coordinates $\sigma_q(P)(z,\zeta) = p_q(z,\zeta)$ (the half-density factor and the lower order terms only contribute terms of order $q-1$); (v) from Proposition 4.3.8. □

Remark 4.6.6. If one works with functions rather than half-densities, the computation in the proof of Theorem 4.6.4 shows that the full symbol transforms by (4.4.5), i.e. $a|d\xi|$ transforms as a fibre density on N^*Y: the principal symbol of $u \in I^m(X,Y)$ is then a section of $\Omega_{\mathrm{fib}}(N^*Y)$ with coefficient of order M, as in (4.5.1). Half-densities make the symbol a half-density on the symplectic manifold N^*Y, which is the form used in Chapter 10.

4.7 Iterated regularity

We now show that conormal distributions are characterized by a regularity property which is manifestly invariant and does not involve the Fourier transform. The idea is that differentiating along Y, or applying $x_i\partial_{x_j}$, which is homogeneous of degree 0 under dilations normal to Y, does not make a conormal singularity worse (Figure 4.2(a)), whereas a general derivative raises the order by one. Let $\mathcal{V}(X,Y)$ denote the space of smooth vector fields on X tangent to Y; it is a Lie algebra and a $\mathcal{C}^\infty(X)$-module.

Lemma 4.7.1. *In the local model,* $\mathcal{V}(\Omega,Y)$ *is spanned over* $\mathcal{C}^\infty(\Omega)$ *by the vector fields* $x_i\partial_{x_j}$, $1 \le i,j \le k$, *and* ∂_{y_l}, $1 \le l \le d$.

Proof. A vector field $V = \sum_j a_j\partial_{x_j} + \sum_l b_l\partial_{y_l}$ is tangent to $\{x=0\}$ if and only if $a_j(0,y) = 0$ for all j, that is, $a_j = \sum_i x_i a_{ij}$ with $a_{ij} \in \mathcal{C}^\infty$ (Taylor's formula with integral remainder in x, valid in a neighbourhood of each point of Y; away from Y use $\partial_{x_j} = \sum_i (x_i/|x|^2)x_i\partial_{x_j}$ and a partition of unity). □

Definition 4.7.2. For $s \in \mathbb{R}$ let

$$IH^s(X,Y;\Omega^{\frac12}) = \{u \in H^s_{\mathrm{loc}}(X;\Omega^{\frac12});\ \mathcal{L}_{V_1}\cdots\mathcal{L}_{V_j}u \in H^s_{\mathrm{loc}}(X;\Omega^{\frac12})\ \forall\ j,\ V_i \in \mathcal{V}(X,Y)\},$$

and $I^*(X,Y;\Omega^{\frac12}) = \bigcup_s IH^s(X,Y;\Omega^{\frac12})$.

Here H^s_{loc} is defined via coordinates and is invariant under diffeomorphisms (Chapter 3); the zeroth order terms in (4.6.1) are irrelevant since H^s_{loc} is a $\mathcal{C}^\infty$-module, so in coordinates we may replace $\mathcal{L}_V$ by V acting on coefficients. Thus IH^s is manifestly invariant under diffeomorphisms preserving Y, and $\bigcap_s IH^s(X,Y;\Omega^{\frac12}) = \mathcal{C}^\infty(X;\Omega^{\frac12})$ by Sobolev embedding.

Lemma 4.7.3. *Let* $u \in H^s(\mathbb{R}^n)$ *have compact support. Then* $u \in IH^s(\mathbb{R}^n,Y)$ *if and only if*

$$x^\alpha D_x^\beta D_y^\gamma u \in H^s(\mathbb{R}^n) \quad \text{for all } \alpha,\beta,\gamma \text{ with } |\beta| \le |\alpha|, \tag{4.7.1}$$

or equivalently if and only if $D_x^\beta D_y^\gamma x^\alpha u \in H^s$ *for all* $|\beta| \le |\alpha|$.

Proof. If $V_i \in \mathcal{V}$, then by Lemma 4.7.1, moving coefficients to the left using $[W,\psi] = W\psi$, $V_1\cdots V_j u$ is a combination with smooth coefficients of products of at most j of the generators $x_i\partial_{x_j}$, ∂_{y_l} applied to u; and such products are linear combinations with constant coefficients of $x^\alpha D_x^\beta D_y^\gamma$ with $|\alpha| = |\beta|$. Conversely, by induction on $|\alpha|$, $x^\alpha D_x^\beta$ with $|\alpha| = |\beta|$ is a combination of products of the $x_iD_{x_j}$, since $x_iD_{x_j}x^{\alpha'}D_x^{\beta'} = x_ix^{\alpha'}D_{x_j}D_x^{\beta'} + x_i(D_{x_j}x^{\alpha'})D_x^{\beta'}$ and the last term is shorter. The cases $|\beta| < |\alpha|$ follow by multiplying by $\chi x^{\alpha''}$, with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}$ equal to one on $\operatorname{supp} u$; and $D^\beta x^\alpha$ is a combination of $x^{\alpha-\delta}D^{\beta-\delta}$, which preserves the condition $|\beta| \le |\alpha|$. □

The key analytic step converts weighted L^2 bounds on all derivatives into symbol estimates.

Lemma 4.7.4. *Let $t \in \mathbb{R}$, $j \in \mathbb{N}_0$, $L > j + \frac{k}{2}$, and let $b \in \mathcal{C}^\infty(\mathbb{R}^k)$ satisfy $\langle\xi\rangle^{t+|\alpha|} D^\alpha b \in L^2(\mathbb{R}^k)$ for $|\alpha| \le L$. Then*

$$|D^\beta b(\xi)| \le C\langle\xi\rangle^{-t-\frac{k}{2}-|\beta|}, \quad |\beta| \le j, \qquad C = C_{k,j,t,L} \sum_{|\alpha|\le L} \|\langle\xi\rangle^{t+|\alpha|} D^\alpha b\|_{L^2}.$$

Conversely, if $b \in S^\mu(\mathbb{R}^k)$ then $\langle\xi\rangle^{t+|\alpha|} D^\alpha b \in L^2$ for all α provided $t < -\mu - \frac{k}{2}$.

Proof. The converse is immediate since $\langle\xi\rangle^{t+\mu} \in L^2$ if $2(t+\mu) < -k$. For the direct statement, let $A = \{\frac{1}{2} < |\zeta| < 2\}$ and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(A)$ equal to one on $|\zeta| = 1$. For $R \ge 1$ set $b_R(\zeta) = b(R\zeta)$. On RA we have $R/2 \le \langle\xi\rangle \le 3R$, so

$$\|D^\alpha b_R\|^2_{L^2(A)} = R^{2|\alpha|-k} \int_{RA} |D^\alpha b|^2 d\xi \le C R^{2|\alpha|-k} R^{-2(t+|\alpha|)} \|\langle\xi\rangle^{t+|\alpha|} D^\alpha b\|^2_{L^2}.$$

Hence $\|\chi b_R\|_{H^L} \le C R^{-t-\frac{k}{2}}$, and by Sobolev's embedding $|D^\beta b_R(\zeta)| \le C R^{-t-\frac{k}{2}}$ for $|\zeta| = 1$, $|\beta| \le j$. Given $|\xi| \ge 1$ take $R = |\xi|$, $\zeta = \xi/|\xi|$: then $D^\beta b(\xi) = R^{-|\beta|}(D^\beta b_R)(\zeta)$, which gives the estimate. For $|\xi| \le 1$ apply Sobolev's embedding to $\chi_0 b$ with $\chi_0 \in \mathcal{C}^\infty_{\mathrm{c}}$ equal to one on the unit ball. □

Theorem 4.7.5. *In the local model, for compactly supported distributions,*

$$I^m_{\mathrm{c}}(\mathbb{R}^n, Y) \subset IH^s(\mathbb{R}^n, Y) \quad \text{if } s < -m - \frac{n}{4}, \qquad IH^s(\mathbb{R}^n, Y) \cap \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n) \subset I^{-s-\frac{n}{4}}_{\mathrm{c}}(\mathbb{R}^n, Y). \tag{4.7.2}$$

Consequently, for any closed embedded submanifold $Y \subset X$,

$$IH^s(X, Y; \Omega^{\frac12}) \subset I^{-s-\frac{n}{4}}(X, Y; \Omega^{\frac12}) \subset IH^{s'}(X, Y; \Omega^{\frac12}) \quad \forall\ s' < s,$$

so $I^*(X, Y; \Omega^{\frac12}) = \bigcup_m I^m(X, Y; \Omega^{\frac12})$.

Proof. First inclusion. Let $u = \mathcal{I}[c] + f$ with $c \in S^M$ compactly supported in (x, y) and $f \in \mathcal{C}^\infty_{\mathrm{c}}$. By Lemma 4.3.1(i), each $x^\alpha D_x^\beta D_y^\gamma$ with $|\alpha| = |\beta|$ maps $\mathcal{I}[c]$ to some $\mathcal{I}[c']$ with c' of order M and compact support, so by Lemma 4.7.3 it suffices to show $\mathcal{I}[c] \in H^s$. Let $\hat c(\zeta', \eta; \xi)$ be the Fourier transform of $c(\cdot, \cdot, \xi)$ in (x, y); since c has compact support in (x, y), $|\hat c(\zeta', \eta; \xi)| \le C_N \langle\zeta'\rangle^{-N} \langle\eta\rangle^{-N} \langle\xi\rangle^M$. Then (first for $M < -k$, by Fubini, then in general by Lemma 4.1.1)

$$\widehat{\mathcal{I}[c]}(\zeta, \eta) = \int \hat c(\zeta - \xi, \eta; \xi) đ\xi,$$
$$|\widehat{\mathcal{I}[c]}(\zeta, \eta)| \le C_N \langle\eta\rangle^{-N} \int \langle\zeta - \xi\rangle^{-N} \langle\xi\rangle^M \, d\xi \le C'_N \langle\eta\rangle^{-N} \langle\zeta\rangle^M$$

by Peetre's inequality, for $N > k + |M|$. Since $\langle(\zeta, \eta)\rangle^s \le \langle\zeta\rangle^s \langle\eta\rangle^{|s|}$, $\langle(\zeta, \eta)\rangle^s \widehat{\mathcal{I}[c]} \in L^2$ provided $2(s + M) < -k$, i.e. $s < -M - \frac{k}{2} = -m - \frac{n}{4}$.

Second inclusion. Let $u \in IH^s$ have compact support, so $\hat u$ is smooth. By Lemma 4.7.3 and $\mathcal{F}(x^\alpha u) = (-D_\zeta)^\alpha \hat u$, $\zeta^\beta \eta^\gamma D^\alpha_\zeta \hat u \in \langle(\zeta, \eta)\rangle^{-s} L^2$ for $|\beta| \le |\alpha|$ and all γ. Since $\langle\zeta\rangle^{|\alpha|} \le C \sum_{|\beta|\le|\alpha|} |\zeta^\beta|$, similarly for $\langle\eta\rangle^N$, and $\langle(\zeta, \eta)\rangle^s \ge c\langle\zeta\rangle^s \langle\eta\rangle^{-|s|}$, we get

$$\langle\zeta\rangle^{s+|\alpha|} \langle\eta\rangle^N D^\alpha_\zeta \hat u(\zeta, \eta) \in L^2(\mathbb{R}^n) \quad \forall\ \alpha, N. \tag{4.7.3}$$

Let $a(y,\zeta) = \int e^{iy\cdot\eta}\hat{u}(\zeta,\eta)đ\eta$ be the partial Fourier transform of u in x; it is smooth, has compact support in y, and $u = \mathcal{I}[a]$. By Minkowski's and the Cauchy–Schwarz inequalities, for $N > |\gamma| + \frac{d}{2}$,

$$\|\langle\zeta\rangle^{s+|\alpha|}D_\zeta^\alpha D_y^\gamma a(y,\cdot)\|_{L^2} \le \int |\eta|^{|\gamma|}\|\langle\zeta\rangle^{s+|\alpha|}D_\zeta^\alpha\hat{u}(\cdot,\eta)\|_{L^2}đ\eta \le C\|\langle\eta\rangle^N\langle\zeta\rangle^{s+|\alpha|}D_\zeta^\alpha\hat{u}\|_{L^2(\mathbb{R}^n)},$$

uniformly in y. Applying Lemma 4.7.4 with $t = s$ to $D_y^\gamma a(y,\cdot)$ gives $|D_\zeta^\beta D_y^\gamma a| \le C\langle\zeta\rangle^{-s-\frac{k}{2}-|\beta|}$, so $a \in S_{\mathrm{c}}^{-s-\frac{k}{2}}$, which by (4.3.1) means $u \in I_{\mathrm{c}}^{-s-\frac{n}{4}}$.

The global statement follows by localization in adapted charts, using the invariance of IH^s and of I^m. Note that the middle space in the global chain is $I^{-s-\frac{n}{4}}$, and $I^{-s-\frac{n}{4}} \subset IH^{s'}$ for $s' < s$ by the first inclusion. □

Question. Why do the two inclusions in Theorem 4.7.5 not match exactly?

Answer. Sobolev spaces measure L^2 norms, symbol classes measure supremum norms, and L^2 cannot see a borderline power exactly. Already for δ_0, of order $\frac{n}{4}$, we have $\delta_0 \in H^s$ if and only if $s < -\frac{n}{2}$, since $\langle\zeta\rangle^s \in L^2(\mathbb{R}^n)$ if and only if $s < -\frac{n}{2}$. So $\delta_0 \in IH^s$ for $s < -\frac{n}{2} = -\frac{n}{4} - \frac{n}{4}$ but not for $s = -\frac{n}{2}$. Conversely the passage from weighted L^2 bounds to pointwise bounds in Lemma 4.7.4 costs exactly $\frac{k}{2}$ in the order. The net effect is an arbitrarily small loss. □

Remark 4.7.6. Theorem 4.7.5 gives a second proof that $I^*(X,Y)$ is invariant under diffeomorphisms, and that the order is invariant up to an arbitrarily small loss; the sharp invariance of the order and of the principal symbol requires Theorem 4.4.4. For $Y = \{0\}$ the theorem is the statement that $D_\zeta^\alpha\hat{u} \in \langle\zeta\rangle^{-s-|\alpha|}L^2$ for all α implies $\hat{u} \in S^{-s-\frac{n}{2}}$, and conversely $S^\mu \subset \langle\zeta\rangle^{-s-|\alpha|}L^2$-type spaces for $s < -\mu - \frac{n}{2}$.

4.8 Examples

Example 4.8.1 (Delta sections)**.** In the local model, $\delta(x) = \delta(x_1)\cdots\delta(x_k) = \mathcal{I}[1]$, the full symbol being $1 \in S^0$; so $\delta(x) \in I_{\mathrm{ph}}^m(\mathbb{R}^n, Y)$ with $M = 0$, i.e. $m = \frac{k}{2} - \frac{n}{4} = \frac{1}{4}(\operatorname{codim} Y - \dim Y)$. Invariantly: the restriction to Y of the density bundle of X splits as $\Omega_Y X = \Omega Y \otimes \Omega(NY)$, where $\Omega(NY)$ is the bundle of densities on the fibres of the normal bundle. If $0 < \mu \in \mathcal{C}^\infty(Y;\Omega(NY))$ is a positive normal density, then

$$u_\mu : \mathcal{C}_{\mathrm{c}}^\infty(X;\Omega X) \ni \nu \longmapsto \int_Y \nu|_Y/\mu$$

is a well-defined distribution, and in adapted coordinates in which $\mu = \mu(y)|dx|$ it is $\mu(y)^{-1}\delta(x)$. So $u_\mu \in I_{\mathrm{ph}}^{\frac{1}{4}(k-d)}(X,Y)$ with full symbol $\mu(y)^{-1}$; with the normalization of Remark 4.6.6, its principal symbol is $\mu(y)^{-1}|d\xi|$, the fibre density on N^*Y dual to μ.

Example 4.8.2 (Poincaré dual forms)**.** Conormal sections of vector bundles are defined in Definition 4.6.1. Let X be a compact oriented manifold of dimension n and $Y \subset X$ a closed embedded oriented submanifold of dimension d. Near any point of Y choose oriented coordinates in which $Y = \{z_{d+1} = \cdots = z_n = 0\}$ and $(z_1,\dots,z_d)$ are oriented coordinates on Y, and set

$$p_Y = \delta(z_{d+1})\cdots\delta(z_n)\,dz_{d+1}\wedge\cdots\wedge dz_n \in I_{\mathrm{ph}}^{\frac{1}{4}(n-2d)}(X,Y;\Lambda^{n-d}).$$

This is independent of the choices, because it is characterized by

$$\int_X \phi\wedge p_Y = \int_Y \phi \qquad \forall\ \phi \in \mathcal{C}^\infty(X;\Lambda^d),$$

as one checks in the given coordinates ($\phi \wedge p_Y$ only involves the coefficient of $dz_1 \wedge \cdots \wedge dz_d$ in ϕ). It follows that $dp_Y = 0$ in the distributional sense: for $\psi \in \mathcal{C}^\infty(X; \Lambda^{d-1})$, $\int_X \psi \wedge dp_Y = \pm \int_X d\psi \wedge p_Y = \pm \int_Y d\psi = 0$ by Stokes' theorem. The distributional de Rham cohomology, defined by the complex $d : \mathcal{C}^{-\infty}(X; \Lambda^j) \to \mathcal{C}^{-\infty}(X; \Lambda^{j+1})$, is canonically isomorphic to the smooth de Rham cohomology (this follows, for instance, from the Hodge theorem), and the class of p_Y in $H^{n-d}_{\mathrm{dR}}(X)$ is, by the displayed identity, the Poincaré dual of Y.

4.9 Exercises

Exercise 4.1. Show by induction that $a \in S^0(\mathbb{R}^n)$ if and only if $a \in \mathcal{C}^\infty(\mathbb{R}^n)$ and $V_1 \cdots V_N a \in L^\infty(\mathbb{R}^n)$ for all N and all choices of $V_j \in \{\xi_k \partial_{\xi_l}\}$ (including $N = 0$). (Hint: in $|\xi| \geq 1$, $\partial_{\xi_l} = \sum_k \xi_k |\xi|^{-2} \xi_k \partial_{\xi_l}$.)

Exercise 4.2. For a compact manifold with boundary X let $\mathcal{V}_{\mathrm{b}}(X)$ be the Lie algebra of smooth vector fields tangent to ∂X and

$$\mathcal{A}(X) = \{a \in \mathcal{C}^\infty(X \setminus \partial X) \cap L^\infty;\ V_1 \cdots V_N a \in L^\infty \ \forall\ N,\ V_i \in \mathcal{V}_{\mathrm{b}}(X)\}.$$

Using that the linear vector fields $\xi_k \partial_{\xi_l}$ span $\mathcal{V}_{\mathrm{b}}(\overline{\mathbb{R}^n})$ near the boundary, show that $S^m(\mathbb{R}^n) = \rho^{-m} \mathcal{A}(\overline{\mathbb{R}^n})$.

Exercise 4.3. Let W be a real vector space of dimension n. Define the sphere of W as $\mathbb{S}W = (W \setminus 0)/\mathbb{R}^+$. Show that $[W, \{0\}] = (W \setminus 0) \sqcup \mathbb{S}W$ has a unique structure of manifold with boundary, diffeomorphic to $[0, \infty) \times \mathbb{S}^{n-1}$, such that the blow-down map $\beta : [W, \{0\}] \to W$ (the identity on $W \setminus 0$, mapping $\mathbb{S}W$ to 0) is smooth and the action of $\mathrm{GL}(W)$ lifts to a smooth action (use Lemma 4.2.4).

Exercise 4.4. Let U be a finite-dimensional complex vector space and set $I^m(W, \{0\}; U) = I^m(W, \{0\}) \otimes U$. Show that the Fourier transform gives an isomorphism $I^m(W, \{0\}; U) \to S^{m - \frac{n}{4}}(W'; U \otimes \Omega W')$ (with $n = \dim W$), that $\Omega U' \otimes \Omega U$ is canonically trivial for a real vector space U, and that the integral of Lemma 4.2.2(iv) defines a linearly invariant map $I^m(W, \{0\}; \Omega W) \to \mathbb{C}$.

Exercise 4.5. Let $u \in \mathcal{C}^{-\infty}(\mathbb{R}^n)$ be smooth outside 0 and homogeneous modulo $\mathcal{C}^\infty$ of degree h: $u(tz) = t^h u(z) + F(t, z)$ for $t > 0$, $z \neq 0$, with $F \in \mathcal{C}^\infty((0, \infty) \times \mathbb{R}^n)$. Show that χu is a classical conormal distribution at 0 for $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, and determine its order.

Exercise 4.6. Show that every $u \in I^m(X, Y)$ with $\operatorname{supp} u \subset Y$ is of the form $P u_\mu$ for some $P \in \mathrm{Diff}^q(X)$, with u_μ as in Example 4.8.1, at least locally. Which orders m can occur?

Exercise 4.7. Let $u \in I^m(X, Y; \Omega^{\frac{1}{2}})$ be classical, with principal symbol nonvanishing on $N^*Y \setminus 0$. Show that $\mathrm{WF}(u) = N^*Y \setminus 0$, and that if $A \in \Psi^0(\mathbb{R}^n)$ satisfies $Au \in \mathcal{C}^\infty$ then the principal symbol of A vanishes on N^*Y (locally).

Notes

This chapter merges the treatment of conormal distributions in the 1998 lecture notes (where they are introduced through iterated regularity) with that of the 2005 lectures (where they are defined by Fourier transform of classical symbols, first at a point, then at the zero section of a vector bundle, then at a submanifold via a normal fibration). The opening argument of the earlier notes, relating iterated regularity to symbol estimates, has been completed (Lemma 4.7.4 and Theorem 4.7.5), and the invariance under diffeomorphisms is

proved by Moser's homotopy method in the sharp form $H^*u - u \in I^{m-1}$, which yields the invariance of the principal symbol without the loss ϵ inherent in the Sobolev approach. We use Hörmander's normalization of the order, for which $\delta_0 \in I^{n/4}(\mathbb{R}^n, \{0\})$ and the kernels of pseudodifferential operators of order m have order m; some formulas for the order in the 2005 lectures contain sign slips which have been corrected. The standard reference is [8, Section 18.2]; see also [6] for the more general Lagrangian distributions, of which conormal distributions are the special case associated to the Lagrangian submanifold N^*Y.

Chapter 5

Pseudodifferential operators on manifolds

The index theorem is a statement about elliptic operators on compact manifolds: the Laplacian on forms, the signature operator, Dirac operators. To handle such operators, and their parametrices, we need a calculus of pseudodifferential operators that does not depend on coordinates. The good news is that almost no new analysis is required. The Euclidean calculus is local and coordinate invariant, so it can be glued together with partitions of unity; the only genuinely global ingredient is compactness, which turns smoothing errors into compact operators and hence gives the Fredholm property.

In this chapter pseudodifferential operators are transferred from $\mathbb{R}^n$ to a compact manifold X, acting between sections of vector bundles. Two definitions are given and shown to agree: the invariant one, in which an operator of order m is identified with its Schwartz kernel, a distribution on X^2 conormal of order m to the diagonal (using the conormal distributions of Chapter 4); and the local one, in which an operator is required to be, in every coordinate patch, an element of the calculus on $\mathbb{R}^n$ of Chapter 2, and to have a kernel smooth away from the diagonal. With the coordinate invariance established in Chapters 2 and 3, the whole symbol calculus carries over. We then construct parametrices for elliptic operators, introduce Sobolev spaces on X, and prove that elliptic operators are Fredholm, with generalized inverses which are again pseudodifferential. Smoothing operators, which carry all the "errors" of the calculus, are discussed in some detail: they form an ideal of "infinite matrices" with a trace and a group of invertible elements, $G^{-\infty}$, which is the starting point for the index theory of later chapters. The main applications here are the Hodge theorem for elliptic complexes, in particular for the de Rham complex (including its distributional form and the decomposition of the spectrum of the Laplacians into exact finite-dimensional complexes), and the construction of pseudodifferential projections: every smooth family of projections on the cosphere bundle is the principal symbol of a pseudodifferential projection, and a projection P of this kind carries a *Toeplitz algebra* $P\Psi P$ with its own symbol calculus and Fredholm theory, generalizing the Toeplitz operators on the circle used in Chapter 12. Along the way we prove the commutator formula for principal symbols, an L^2 bound for operators of order zero in terms of the principal symbol alone, and the analytic Fredholm alternative for holomorphic families of smoothing operators. The facts about bounded operators on Hilbert space which are used are collected, with proofs, in Appendix A.

Throughout, unless explicitly stated otherwise, X is a *compact* manifold of dimension n; the changes needed for non-compact manifolds are described in Section 5.7.

5.1 Smooth manifolds

We begin by fixing conventions. The reader who is comfortable with manifolds, vector bundles and densities can skim this section and the next one; the only non-standard item is the partition of unity of Lemma 5.1.3, which is used to localize operators.

5.1.1 $\mathcal{C}^\infty$ structures

Let X be a paracompact Hausdorff topological space. We describe the smooth structure of X through its algebra of smooth functions.

Definition 5.1.1. A $\mathcal{C}^\infty$ *structure* on X is a subspace $\mathcal{F} \subset \mathcal{C}^0(X;\mathbb{R})$ with the following property: for each $\bar{x} \in X$ there exist $f_1, \dots, f_n \in \mathcal{F}$ and an open neighbourhood $\Omega \ni \bar{x}$ such that

$$\kappa : \Omega \ni x \longmapsto (f_1(x), \dots, f_n(x)) \in \mathbb{R}^n \tag{5.1.1}$$

is a homeomorphism onto an open subset of $\mathbb{R}^n$ and every $f \in \mathcal{F}$ satisfies $f|_\Omega = g \circ \kappa$ for some $g \in \mathcal{C}^\infty(\kappa(\Omega))$. The map κ is then called a *coordinate system* (or chart) near $\bar{x}$. Two $\mathcal{C}^\infty$ structures $\mathcal{F}_1$, $\mathcal{F}_2$ are *compatible* if $\mathcal{F}_1 \cup \mathcal{F}_2$ is also a $\mathcal{C}^\infty$ structure.

Compatibility is an equivalence relation, and each equivalence class contains a unique maximal element, the union of the class (Exercise 5.1).

Definition 5.1.2. A $\mathcal{C}^\infty$ *manifold* is a paracompact Hausdorff space X with a maximal $\mathcal{C}^\infty$ structure, which is denoted $\mathcal{C}^\infty(X) \subset \mathcal{C}^0(X)$.

The maximal structure is an algebra (Exercise 5.2), and the number n of coordinate functions is locally constant; we always assume it is constant, $n = \dim X$. If $\kappa : \Omega \to \kappa(\Omega)$ is a coordinate system in X and $\mathcal{C}^\infty_{\mathrm{c}}(\Omega) \subset \mathcal{C}^\infty(X)$ denotes the space of smooth functions vanishing outside a compact subset of Ω, then

$$\kappa^* : \mathcal{C}^\infty_{\mathrm{c}}(\kappa(\Omega)) \longrightarrow \mathcal{C}^\infty_{\mathrm{c}}(\Omega) = \{u \in \mathcal{C}^\infty(X);\ u = 0 \text{ on } X \setminus K,\ K \Subset \Omega\}$$

is an isomorphism. Moreover $\mathcal{C}^\infty(X)$ is *local*: if $u : X \to \mathbb{R}$ and each $\bar{x} \in X$ has an open neighbourhood $\Omega_{\bar{x}}$ on which $u = f_{\bar{x}}$ for some $f_{\bar{x}} \in \mathcal{C}^\infty(X)$, then $u \in \mathcal{C}^\infty(X)$. The transition maps $\kappa' \circ \kappa^{-1}$ between two coordinate systems are diffeomorphisms between open subsets of $\mathbb{R}^n$, so this definition agrees with the usual one by atlases. A map $G : X \to Y$ between manifolds is $\mathcal{C}^\infty$ if $G^*\mathcal{C}^\infty(Y) \subset \mathcal{C}^\infty(X)$, that is, $u \circ G \in \mathcal{C}^\infty(X)$ for all $u \in \mathcal{C}^\infty(Y)$. Complex-valued smooth functions are those with real and imaginary parts in $\mathcal{C}^\infty(X)$.

5.1.2 Partitions of unity

Any $\mathcal{C}^\infty$ manifold admits partitions of unity subordinate to any open cover (Exercise 5.3). For compact manifolds this is elementary: for each point p choose a coordinate system near p inside an element of the cover and pull back a non-negative bump function on $\mathbb{R}^n$ which is positive at the image of p; by locality this gives an element of $\mathcal{C}^\infty(X)$. Finitely many of these are positive on a set covering X, and dividing each of them by their (positive) sum gives a finite partition of unity subordinate to the cover. For localizing operators a slightly stronger property is convenient.

Lemma 5.1.3. *If $\{U_i\}$ is an open cover of a compact manifold X there is a finite partition of unity $\{\phi_a\}$, $0 \le \phi_a \le 1$, $\sum_a \phi_a = 1$, each $\operatorname{supp}\phi_a$ contained in some U_i, such that*

$$\operatorname{supp}\phi_a \cap \operatorname{supp}\phi_b \neq \emptyset \Longrightarrow \operatorname{supp}\phi_a \cup \operatorname{supp}\phi_b \subset U_i \quad \textit{for some } i. \tag{5.1.2}$$

Proof. Choose a finite partition of unity $\{\chi_i\}$ with $\operatorname{supp}\chi_i \subset U_i$ (passing to a finite subcover). For $p \in X$ set

$$V_p = \bigcap_{i:\, p\in\operatorname{supp}\chi_i} U_i \;\setminus \bigcup_{i:\, p\notin\operatorname{supp}\chi_i} \operatorname{supp}\chi_i,$$

an open neighbourhood of p with the property that $V_p \cap \operatorname{supp}\chi_i \neq \emptyset$ implies $p \in \operatorname{supp}\chi_i$ and hence $V_p \subset U_i$. Choose finitely many p_j such that the $V_j = V_{p_j}$ cover X, a partition of unity $\{\psi_j\}$ with $\operatorname{supp}\psi_j \subset V_j$, and set $\phi_{ij} = \chi_i\psi_j$. If $\operatorname{supp}\phi_{ij} \cap \operatorname{supp}\phi_{i'j'} \neq \emptyset$ then $\operatorname{supp}\psi_j$ meets $\operatorname{supp}\chi_{i'}$, so $V_j \subset U_{i'}$, and therefore $\operatorname{supp}\phi_{ij} \subset U_{i'}$; since also $\operatorname{supp}\phi_{i'j'} \subset \operatorname{supp}\chi_{i'} \subset U_{i'}$, the union of the two supports lies in $U_{i'}$. □

5.2 Vector bundles, densities and distributions

5.2.1 Vector bundles

A (complex) *vector bundle* of rank N over X is a manifold E with a smooth surjection $\pi : E \to X$, each fibre $E_x = \pi^{-1}(x)$ carrying the structure of an N-dimensional complex vector space, such that each point of X has an open neighbourhood U and a diffeomorphism $\pi^{-1}(U) \to U \times \mathbb{C}^N$ which is fibrewise linear and covers the identity (a *local trivialization*). Two trivializations differ by a smooth map $U \cap U' \to \mathrm{GL}(N, \mathbb{C})$. Real vector bundles are defined in the same way. The space of smooth sections is

$$\mathcal{C}^\infty(X; E) = \{u : X \to E \text{ smooth};\ \pi \circ u = \mathrm{Id}\};$$

it is a module over $\mathcal{C}^\infty(X)$, and over a trivializing open set a section is just an N-tuple of smooth functions. We shall often use charts $\kappa : W \to \kappa(W)$ over which the bundles under consideration are trivial; we call these *trivializing charts*. Since X is compact, it is covered by finitely many of them.

The cotangent bundle

For $\bar{x} \in X$ let $\mathcal{I}_{\bar{x}} = \{u \in \mathcal{C}^\infty(X); u(\bar{x}) = 0\}$ and let $\mathcal{I}_{\bar{x}}^2$ be the space of finite sums of products of pairs of elements of $\mathcal{I}_{\bar{x}}$. The cotangent space is

$$T^*_{\bar{x}}X = \mathcal{I}_{\bar{x}}/\mathcal{I}_{\bar{x}}^2, \qquad T^*X = \bigcup_{\bar{x}\in X} T^*_{\bar{x}}X,$$

and the differential of $f \in \mathcal{C}^\infty(X)$ is the map $df : X \to T^*X$, $df(\bar{x}) = [f - f(\bar{x})] \in T^*_{\bar{x}}X$. In a coordinate system $x_1, \dots, x_n$ near $\bar{x}$, Taylor's formula with integral remainder shows that $f - f(\bar{x}) - \sum_j \partial_j f(\bar{x})(x_j - \bar{x}_j) \in \mathcal{I}_{\bar{x}}^2$ (after multiplication by a cutoff equal to one near $\bar{x}$), so $dx_1, \dots, dx_n$ is a basis of $T^*_{\bar{x}}X$ and $df = \sum_j \partial_j f\, dx_j$. The coordinates $(x, \xi) \mapsto \sum_j \xi_j\, dx_j$ make T^*X a real vector bundle; if $y = y(x)$ is another coordinate system, the corresponding fibre coordinates are related by

$$\xi = \Big(\frac{\partial y}{\partial x}\Big)^T \eta, \quad \text{i.e. } \sum_j \xi_j\, dx_j = \sum_k \eta_k\, dy_k, \tag{5.2.1}$$

which is linear in the fibre variables with smooth invertible coefficients. Each df is a smooth section of T^*X.

Operations on bundles

Any functorial operation on finite-dimensional vector spaces produces new bundles from old, the transition functions being transformed accordingly. In particular we have the tangent bundle $T_xX = (T_x^*X)^*$, the dual bundle E', the bundles $E \otimes F$, $\mathrm{Hom}(E,F) = F \otimes E'$, $\mathrm{End}(E) = \mathrm{Hom}(E,E)$, and the exterior powers

$$\Lambda^k_x X = \{u : (T_xX)^k \to \mathbb{C} \text{ multilinear and antisymmetric}\}, \qquad \Lambda^1 X = T^*X \otimes \mathbb{C},$$

whose sections are the k-forms. If $E \to Y$ is a vector bundle and $G : X \to Y$ is smooth, the *pull-back bundle* $G^*E = \bigcup_{x\in X} E_{G(x)}$ is a smooth vector bundle over X; if $\phi \in \mathcal{C}^\infty(Y;E)$ then $G^*\phi \in \mathcal{C}^\infty(X;G^*E)$ and $\mathcal{C}^\infty(X;G^*E)$ is spanned over $\mathcal{C}^\infty(X)$ by $G^*\mathcal{C}^\infty(Y;E)$. If $E \to X$ and $F \to Y$ are bundles, the *exterior tensor product* $F \boxtimes E \to X \times Y$ has fibre $(F \boxtimes E)_{(x,y)} = F_x \otimes E_y$. On X^2 we use the projections $\pi_L(x,y) = x$ and $\pi_R(x,y) = y$, and the "big" homomorphism bundle

$$\mathrm{Hom}(E,F) = \pi_L^*F \otimes \pi_R^*E' = F \boxtimes E', \qquad \mathrm{Hom}(E,F)_{(x,y)} = \hom(E_y, F_x), \tag{5.2.2}$$

whose restriction to the diagonal is $\hom(E,F)$. A *hermitian metric* on E is a smoothly varying inner product on the fibres; one exists by a partition of unity argument.

5.2.2 Densities and integration

For $\alpha \in \mathbb{R}$ the bundle of α-densities is

$$\Omega^\alpha_x X = \{u : \Lambda^n T_xX \setminus 0 \to \mathbb{C};\ u(t\,v) = |t|^\alpha u(v)\ \forall t \in \mathbb{R}\setminus 0\},$$

a line bundle, and $\Omega = \Omega^1$. In local coordinates an α-density is $u = g(x)|dx|^\alpha$, and under a change of coordinates $y = y(x)$ the coefficient transforms by $g \mapsto g\,|\det(\partial y/\partial x)|^\alpha$. Thus, by the change of variables formula, if $u \in \mathcal{C}^\infty_{\mathrm{c}}(W;\Omega)$ is supported in a coordinate patch then $\int u = \int_{\mathbb{R}^n} g(x)\,dx$ is independent of the coordinates, and using a partition of unity the integral

$$\int_X : \mathcal{C}^\infty(X;\Omega) \longrightarrow \mathbb{C}$$

is well defined. Densities, unlike forms, require no orientation. A *smooth positive density* ν (one which is a positive multiple of $|dx|$ in every coordinate system) exists, again by a partition of unity, and any other density is $f\nu$ with $f \in \mathcal{C}^\infty(X)$. Given ν and a hermitian metric on E, the space $L^2(X;E)$ is the completion of $\mathcal{C}^\infty(X;E)$ with respect to

$$(u,v) = \int_X \langle u(x), v(x)\rangle_{E_x}\,\nu(x). \tag{5.2.3}$$

Different choices give the same space with equivalent norms. On X^2 we use the *right density bundle* $\Omega_R = \pi_R^*\Omega$; a section of $\mathrm{Hom}(E,F)\otimes\Omega_R$ is in local coordinates and trivializations a matrix $k(x,y)|dy|$.

Discussion. Why carry densities around? An operator is to be written as $Au(x) = \int_X A(x,y)u(y)$, and on a manifold there is no preferred measure to integrate against. Letting the kernel itself carry a density factor in y makes the integral meaningful without choices, and makes the correspondence between operators and kernels coordinate free. Once a smooth positive density ν is fixed, $A(x,y) = k(x,y)\nu(y)$ and we are back to ordinary functions; we do this whenever L^2 norms are needed.

5.2.3 Distributional sections

The topology on $\mathcal{C}^\infty(X;E)$ is given by the seminorms $\sup|D^\alpha(\kappa_i^{-1})^*(\phi_i u)|$, $\alpha\in\mathbb{N}_0^n$, for a finite cover by trivializing charts κ_i and a subordinate partition of unity ϕ_i (the N-tuples of functions being estimated componentwise); this makes $\mathcal{C}^\infty(X;E)$ a Fréchet space and is independent of choices. The space of distributional sections is the dual space

$$\mathcal{C}^{-\infty}(X;E)=\bigl(\mathcal{C}^\infty(X;E'\otimes\Omega)\bigr)', \tag{5.2.4}$$

with the weak topology. Integration of the pairing gives the inclusion $\mathcal{C}^\infty(X;E)\hookrightarrow\mathcal{C}^{-\infty}(X;E)$, $u\mapsto(\phi\mapsto\int_X\phi(u))$, and $\mathcal{C}^\infty(X;E)$ is dense. Multiplication by $\mathcal{C}^\infty(X)$ and the restriction to open sets are defined by duality; if $u\in\mathcal{C}^{-\infty}(X;E)$ and $\kappa:W\to\kappa(W)$ is a trivializing chart, then for $\psi\in\mathcal{C}^\infty_{\mathrm{c}}(W)$ the localization ψu is represented by an N-tuple of compactly supported distributions on $\kappa(W)$, which we write $\kappa_*(\psi u)=(\kappa^{-1})^*(\psi u)$.

Distributional sections can equally be defined "algebraically".

Proposition 5.2.1. *For any vector bundle U over the compact manifold Y there is a natural isomorphism*

$$\mathcal{C}^{-\infty}(Y)\otimes_{\mathcal{C}^\infty(Y)}\mathcal{C}^\infty(Y;U)\cong\mathcal{C}^{-\infty}(Y;U).$$

Proof. The map $\sum_i u_i\otimes g_i\mapsto\sum_i u_ig_i$, where $(ug)(\psi)=u(g\cdot\psi)$ for $\psi\in\mathcal{C}^\infty(Y;U'\otimes\Omega)$ (so that $g\cdot\psi\in\mathcal{C}^\infty(Y;\Omega)$), is well defined on the tensor product since it is $\mathcal{C}^\infty(Y)$-balanced. To invert it, choose finitely many sections $g_i\in\mathcal{C}^\infty(Y;U)$ and $q_i\in\mathcal{C}^\infty(Y;U')$ with

$$\sum_i g_i\otimes q_i=\mathrm{Id}_U\in\mathcal{C}^\infty(Y;\mathrm{End}(U)). \tag{5.2.5}$$

These exist: take a finite partition of unity ρ_k subordinate to a cover by trivializing open sets U_k, local frames $e_{k,l}$ over U_k with dual frames $e^{k,l}$, functions $\psi_k\in\mathcal{C}^\infty_{\mathrm{c}}(U_k)$ equal to one on $\operatorname{supp}\rho_k$, and use the sections $\rho_ke_{k,l}$ and $\psi_ke^{k,l}$, extended by zero. Then $\sum_k\sum_l\rho_ke_{k,l}\otimes\psi_ke^{k,l}=\sum_k\rho_k\,\mathrm{Id}=\mathrm{Id}$. Define $u\mapsto\sum_i(u\cdot q_i)\otimes g_i$ for $u\in\mathcal{C}^{-\infty}(Y;U)$, where $u\cdot q_i\in\mathcal{C}^{-\infty}(Y)$ is the pairing. By (5.2.5), $\sum_i(u\cdot q_i)g_i=u$, while for $u=\sum_j u_j\otimes h_j$, $\sum_i\sum_j u_j(h_j\cdot q_i)\otimes g_i=\sum_j u_j\otimes\sum_i(h_j\cdot q_i)g_i=\sum_j u_j\otimes h_j$, using the $\mathcal{C}^\infty(Y)$-balancing. So the two maps are mutually inverse. □

The same argument shows that any $\mathcal{C}^\infty(Y)$-module of "distributions" which is local and invariant under multiplication by smooth functions, such as the conormal spaces below, can be extended to sections of bundles by tensoring.

5.3 Euclidean and conormal preliminaries

Everything in this chapter is built from two earlier ingredients: the calculus on $\mathbb{R}^n$, which supplies the local analysis, and conormal distributions, which supply the invariant language for kernels. We list exactly what is used, so that the rest of the chapter can refer to it by number.

5.3.1 The calculus on $\mathbb{R}^n$

We collect the results from Chapters 2 and 3 which are used below. Here $\Psi^m(\mathbb{R}^n)$ is the space of operators $\mathrm{Op}(a)$, $a\in S^m(\mathbb{R}^n;\mathbb{R}^n)$, with kernels $K(x,y)=\int e^{i(x-y)\cdot\xi}a(x,\xi)\text{đ}\xi$.

(E1) $\Psi^m(\mathbb{R}^n) \subset \Psi^{m'}(\mathbb{R}^n)$ for $m \leq m'$; $\Psi^{-\infty}(\mathbb{R}^n) = \bigcap_m \Psi^m(\mathbb{R}^n)$ consists of operators with smooth kernels, and every operator with kernel in $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$ lies in $\Psi^{-\infty}(\mathbb{R}^n)$. Kernels of elements of $\Psi^m(\mathbb{R}^n)$ are smooth off the diagonal.

(E2) (Amplitudes) If $a(x,y,\xi)$ satisfies $|D_x^\alpha D_y^\beta D_\xi^\gamma a| \leq C\langle\xi\rangle^{m-|\gamma|}$ then the operator with kernel $\int e^{i(x-y)\cdot\xi} a(x,y,\xi)\text{đ}\xi$ lies in $\Psi^m(\mathbb{R}^n)$, with principal symbol $a(x,x,\xi)$. Equivalently, every $A \in \Psi^m(\mathbb{R}^n)$ has a *left* symbol $a_L(x,\xi)$ and a *right* symbol $a_R(y,\xi)$, both in S^m, with $\sigma_m(A) = [a_L] = [a_R]$. In particular $\phi A\psi \in \Psi^m(\mathbb{R}^n)$ for $\phi,\psi \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$, and the kernel of A may be multiplied by $\chi(x-y)$, $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to one near 0, modulo $\Psi^{-\infty}(\mathbb{R}^n)$.

(E3) (Composition and adjoints) $\Psi^m(\mathbb{R}^n) \circ \Psi^{m'}(\mathbb{R}^n) \subset \Psi^{m+m'}(\mathbb{R}^n)$ with $\sigma_{m+m'}(AB) = \sigma_m(A)\sigma_{m'}(B)$; the L^2 adjoint satisfies $A^* \in \Psi^m(\mathbb{R}^n)$, $\sigma_m(A^*) = \overline{\sigma_m(A)}$.

(E4) (Principal symbol) The sequence $0 \to \Psi^{m-1}(\mathbb{R}^n) \to \Psi^m(\mathbb{R}^n) \xrightarrow{\sigma_m} S^m/S^{m-1} \to 0$ is exact, with $\sigma_m(\mathrm{Op}(a)) = [a]$.

(E5) (Asymptotic completeness) If $A_j \in \Psi^{m-j}(\mathbb{R}^n)$ there exists $A \in \Psi^m(\mathbb{R}^n)$ with $A - \sum_{j<N} A_j \in \Psi^{m-N}(\mathbb{R}^n)$ for all N. If the kernels of the A_j have support in a fixed compact set K, A may be chosen with kernel supported in any given neighbourhood of K (replace A by $\chi A\chi$ with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to one near both projections of K).

(E6) (Sobolev boundedness) Each $A \in \Psi^m(\mathbb{R}^n)$ is bounded $H^s(\mathbb{R}^n) \to H^{s-m}(\mathbb{R}^n)$ for every $s \in \mathbb{R}$; in particular $\Psi^0(\mathbb{R}^n)$ acts boundedly on $L^2(\mathbb{R}^n)$, and multiplication by $\phi \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ is bounded on every $H^s(\mathbb{R}^n)$.

(E7) (Coordinate invariance) Let $\Phi : \Omega \to \Omega'$ be a diffeomorphism between open subsets of $\mathbb{R}^n$ and $A \in \Psi^m(\mathbb{R}^n)$ have kernel with compact support in $\Omega \times \Omega$. Then $\Phi_* A = (\Phi^{-1})^* \circ A \circ \Phi^*$, extended by zero, lies in $\Psi^m(\mathbb{R}^n)$, and

$$\sigma_m(\Phi_* A)(\Phi(x),\eta) = \sigma_m(A)(x, (d\Phi_x)^T\eta). \tag{5.3.1}$$

The symbol formula (5.3.1) follows from the proof of coordinate invariance: writing $G = \Phi^{-1}$ and $G(x) - G(y) = T(x,y)(x-y)$, the kernel of $\Phi_* A$ near the diagonal is given by the amplitude $c(x,y,\eta) = a(G(x), T(x,y)^{-T}\eta)\,|\det dG(y)|\,|\det T(x,y)|^{-1}$, and since $T(x,x) = dG(x)$ its value on the diagonal is $a(G(x),(dG_x)^{-T}\eta)$, which is (5.3.1) because $(dG_{\Phi(x')})^{-T} = (d\Phi_{x'})^T$. Formula (5.3.1) says precisely that the principal symbol is invariantly a function on the cotangent bundle: it is the pull-back under the induced map $T^*\Omega' \to T^*\Omega$.

Discussion. Why is it the *cotangent* bundle? The frequency variable ξ is paired with $x-y$ in the phase $(x-y)\cdot\xi$. A change of coordinates acts on $x-y$, to first order near the diagonal, by the Jacobian $d\Phi$; to keep the phase unchanged, ξ must transform by the inverse transpose. That is exactly the transformation law (5.2.1) of covectors. The higher terms in the Taylor expansion of Φ only produce terms of lower order in the symbol, which is why only the principal part is invariant.

5.3.2 Conormal distributions

We recall the definition from Chapter 4. Let M be a manifold of dimension N and $Y \subset M$ a closed embedded submanifold of codimension k. The space $I^m(M,Y)$ of distributions (or distributional densities) conormal to Y of order m consists of those u which are smooth

on $M \setminus Y$ and such that, near each point of Y, in local coordinates $(w, z) \in \mathbb{R}^{N-k} \times \mathbb{R}^k$ in which $Y = \{z = 0\}$,

$$u = \int_{\mathbb{R}^k} e^{iz\cdot\zeta} a(w,\zeta) đ\zeta + v, \quad a \in S^{m+\frac{N}{4}-\frac{k}{2}}(\mathbb{R}^{N-k};\mathbb{R}^k),\ v \text{ smooth}. \tag{5.3.2}$$

(For distributional densities the formula is multiplied by the coordinate density $|dw\,dz|$.) The subspace in which the amplitudes a are classical symbols is denoted $I^m_{\mathrm{ph}}(M,Y)$. We use the following properties.

(C1) $I^m(M,Y)$ is a $\mathcal{C}^\infty(M)$-module, is local, and is invariant under diffeomorphisms: if $\Phi : M \to M'$ is a diffeomorphism with $\Phi(Y) = Y'$ then $\Phi^* I^m(M',Y') = I^m(M,Y)$. Moreover $I^m \subset I^{m'}$ for $m \le m'$ and $\bigcap_m I^m(M,Y) = \mathcal{C}^\infty(M)$.

(C2) The amplitude a in (5.3.2) may be allowed to depend on z as well, with symbol estimates uniform on compact sets in (w,z), without changing the space.

(C3) Differential operators of order p map $I^m(M,Y)$ into $I^{m+p}(M,Y)$, and multiplication by a smooth function vanishing on Y maps $I^m(M,Y)$ into $I^{m-1}(M,Y)$.

The order normalization is such that the Dirac delta at a point of $\mathbb{R}^N$ has order $N/4$. For a vector bundle E over M we set

$$I^m(M,Y;E) = I^m(M,Y) \otimes_{\mathcal{C}^\infty(M)} \mathcal{C}^\infty(M;E),$$

the space of finite sums $\sum_i u_i e_i$, $u_i \in I^m(M,Y)$, $e_i \in \mathcal{C}^\infty(M;E)$, modulo the relations generated by $\mathcal{C}^\infty(M)$-linearity. As in Proposition 5.2.1, if u has support in an open set over which E is trivial, with local frame e_i, then $u = \sum_i u_i e_i$ with $u_i \in I^m(M,Y)$, and there are natural inclusions

$$\mathcal{C}^\infty(M;E) \hookrightarrow I^m(M,Y;E) \hookrightarrow \mathcal{C}^{-\infty}(M;E).$$

Integration, restriction and push-forward

If M is compact and V is a fixed vector space, the integral of a distributional density, i.e. its pairing with the constant function 1, gives a linear map $\int_M : I^m(M,Y;V\otimes\Omega_M) \to V$.

Proposition 5.3.1. *Let $Z \subset M$ be an embedded submanifold transversal to Y, i.e. $T_pY + T_pZ = T_pM$ for all $p \in Y \cap Z$. Then restriction $\mathcal{C}^\infty(M;E) \to \mathcal{C}^\infty(Z;E|_Z)$ extends to a map*

$$I^m(M,Y;E) \longrightarrow I^{m+\frac{1}{4}\operatorname{codim} Z}(Z, Y\cap Z; E|_Z).$$

Proof. Elements supported away from Y are smooth, so by a partition of unity it suffices to consider u supported near a point $p \in Y \cap Z$, where E may be taken trivial. Let $z_1,\dots,z_k$ be local defining functions for Y and $t_1,\dots,t_q$ for Z, $q = \operatorname{codim} Z$. Transversality means precisely that the differentials of the z_j and t_l are linearly independent at p, so they can be completed by functions w' to a coordinate system (w',t,z) near p, in which $Y = \{z=0\}$ and $Z = \{t = 0\}$. In these coordinates $u = \int e^{iz\cdot\zeta} a(w',t,\zeta) đ\zeta + v$ with $a \in S^{m+\frac{N}{4}-\frac{k}{2}}$, and u depends smoothly on t as a distribution in (w',z). We set $u|_Z = \int e^{iz\cdot\zeta} a(w',0,\zeta) đ\zeta + v|_{t=0}$; this agrees with the restriction if $a \in S^{-\infty}$, and in general is the limit of the restrictions of the smooth functions obtained by replacing a by $a(w',t,\zeta)\chi(\zeta/R)$, $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^k)$ equal to one near 0, as $R \to \infty$; hence it is independent of choices. Since $\dim Z = N - q$, the order m'' of $u|_Z$ is fixed by $m + \frac{N}{4} - \frac{k}{2} = m'' + \frac{N-q}{4} - \frac{k}{2}$, i.e. $m'' = m + \frac{q}{4}$. □

Now let $\phi : M \to B$ be a fibration, locally the projection of a product, transversal to Y in the sense that each fibre is transversal to Y: $T_pY + T_p\phi^{-1}(\phi(p)) = T_pM$ for $p \in Y$. Let Ω_ϕ be the bundle of densities on the fibres, so that $\Omega_M = \Omega_\phi \otimes \phi^*\Omega_B$.

Proposition 5.3.2. *If $\phi : M \to B$ is a fibration with compact fibres transversal to Y, and E is a vector bundle over B, then fibre integration extends from smooth sections to a linear map*

$$\phi_* : I^m(M, Y; \phi^*E \otimes \Omega_\phi) \longrightarrow \mathcal{C}^\infty(B; E).$$

Proof. For smooth sections, locally in B where $M = U \times Z$, $\phi_* u(b) = \int_Z u(b, z)$, which is smooth in b. By a partition of unity it suffices to consider u supported near a point $p \in Y$ and E trivial. By transversality, the defining functions of Y restrict to each fibre near p to functions with independent differentials, so we may choose coordinates (b, z', z'') near p in which ϕ is $(b, z', z'') \mapsto b$ and $Y = \{z' = 0\}$. Then, modulo a smooth term, $u = \int e^{iz'\cdot\zeta} a(b, z'', \zeta) đ\zeta \, |dz'\, dz''|$ with compact support in (b, z''). Choosing $\chi \in \mathcal{C}^\infty_{\mathrm{c}}$ equal to one on the support of u in z',

$$\phi_* u(b) = \iint \chi(z') \int e^{iz'\cdot\zeta} a(b, z'', \zeta) đ\zeta \, dz'\, dz'' = \iint a(b, z'', \zeta) \hat{\chi}(-\zeta) đ\zeta \, dz'',$$

which is smooth in b since $\hat{\chi} \in \mathcal{S}$ and a is a symbol. (Formally, $\int e^{iz'\cdot\zeta} dz' = (2\pi)^k \delta(\zeta)$ and $\phi_* u(b) = \int a(b, z'', 0)\, dz''$.) □

5.4 Pseudodifferential operators on a compact manifold

On $\mathbb{R}^n$ a pseudodifferential operator was defined through its symbol, but the full symbol depends on coordinates. Its Schwartz kernel, on the other hand, is an honest distribution on $X \times X$, and the two characteristic properties of the Euclidean kernels make sense on any manifold: they are smooth away from the diagonal (pseudolocality) and, near the diagonal, they are inverse Fourier transforms of symbols in the directions normal to it. These are exactly the properties of a distribution conormal to the diagonal, and we take them as the definition (Figure 5.1).

5.4.1 Definition

Let E and F be complex vector bundles over the compact manifold X. The diagonal $\mathrm{Diag} = \{(x, x)\} \subset X^2$ is an embedded submanifold of codimension n, and the fibres $\{x\} \times X$ of π_L are transversal to it.

Definition 5.4.1. For $m \in \mathbb{R}$, the space of pseudodifferential operators of order m from sections of E to sections of F is

$$\Psi^m(X; E, F) = I^m(X^2, \mathrm{Diag}; \mathrm{Hom}(E, F) \otimes \Omega_R), \tag{5.4.1}$$

with $\mathrm{Hom}(E, F)$ as in (5.2.2). The operator defined by $A \in \Psi^m(X; E, F)$ is

$$Au(x) = \int_X A(x, y) u(y), \quad \text{i.e.} \quad Au = (\pi_L)_*(A \cdot \pi_R^* u), \qquad u \in \mathcal{C}^\infty(X; E). \tag{5.4.2}$$

We write $\Psi^m(X; E) = \Psi^m(X; E, E)$, $\Psi^m(X)$ for trivial line bundles, and Ψ^m_{ph} for the classical operators, defined using I^m_{ph}.

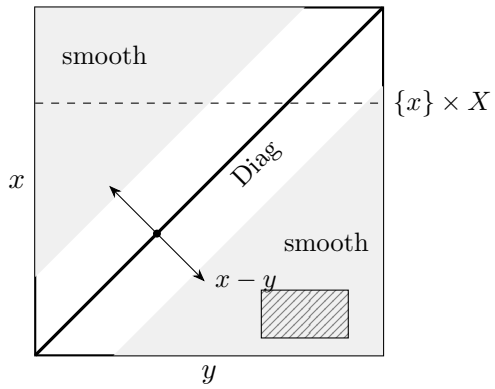


Figure 5.1: The kernel of $A \in \Psi^m(X; E, F)$ lives on $X \times X$. It is smooth off the diagonal Diag (shaded); near Diag it oscillates in the normal variable $x - y$ with frequency ξ, i.e. it is conormal to Diag. Integrating along the fibres $\{x\} \times X$ of π_L (dashed), which are transversal to Diag, defines the action $u \mapsto Au$. The hatched box is the support of $\phi(x)\psi(y)$ for ϕ, ψ with disjoint supports: there the kernel of $\phi A \psi$ is smooth.

In (5.4.2), $\pi_R^* u \in \mathcal{C}^\infty(X^2; \pi_R^* E)$, the product with the kernel contracts E' with E and so gives an element of $I^m(X^2, \mathrm{Diag}; \pi_L^* F \otimes \Omega_R)$; since Ω_R is the density bundle of the fibres of π_L, which are transversal to Diag, Proposition 5.3.2 shows that $Au \in \mathcal{C}^\infty(X; F)$. Thus $A : \mathcal{C}^\infty(X; E) \to \mathcal{C}^\infty(X; F)$, and the operator determines its kernel, by the Schwartz kernel theorem; we identify the two, and also write K_A for the kernel. For $\phi, \psi \in \mathcal{C}^\infty(X)$, the kernel of $\phi A \psi$ is $\phi(x) A(x, y) \psi(y)$.

The space of *smoothing operators* is

$$\Psi^{-\infty}(X; E, F) = \bigcap_m \Psi^m(X; E, F) = \mathcal{C}^\infty(X^2; \mathrm{Hom}(E, F) \otimes \Omega_R), \tag{5.4.3}$$

by (C1). Such an operator, $Au(x) = \int_X A(x, y) u(y)$, extends to a continuous map $\mathcal{C}^{-\infty}(X; E) \to \mathcal{C}^\infty(X; F)$, by letting the distribution u act in the variable y.

5.4.2 Local description

We now connect Definition 5.4.1 with the calculus on $\mathbb{R}^n$. If $\kappa : W \to \kappa(W)$ is a chart over which E and F are trivialized, and A has kernel with compact support in $W \times W$, its *local representative* is the matrix of operators

$$\kappa_* A = (\kappa^{-1})^* \circ A \circ \kappa^* : \mathcal{C}_{\mathrm{c}}^\infty(\kappa(W))^{N_E} \longrightarrow \mathcal{C}_{\mathrm{c}}^\infty(\kappa(W))^{N_F},$$

where the right density factor is trivialized by the coordinate density.

Theorem 5.4.2. *A linear operator $A : \mathcal{C}^\infty(X; E) \to \mathcal{C}^\infty(X; F)$ is given by an element of $\Psi^m(X; E, F)$ if and only if*

(i) *if $\phi, \psi \in \mathcal{C}^\infty(X)$ have disjoint supports then $\phi A \psi$ has kernel in $\mathcal{C}^\infty(X^2; \mathrm{Hom}(E, F) \otimes \Omega_R)$, and*

(ii) *for every trivializing chart $\kappa : W \to \kappa(W)$ and every $\psi \in \mathcal{C}_{\mathrm{c}}^\infty(W)$, the entries of $\kappa_*(\psi A \psi)$ are elements of $\Psi^m(\mathbb{R}^n)$ (with kernels supported in $\kappa(\mathrm{supp}\, \psi)^2$).*

Moreover it suffices to require (ii) *for the charts of one finite cover of X by trivializing charts.*

Proof. Suppose $A \in \Psi^m(X; E, F)$. Its kernel is smooth outside the diagonal, which gives (i). For (ii), $\psi(x)\psi(y)A(x,y)$ is conormal of order m with compact support in $W \times W$, so by (C1) the kernel of $\kappa_*(\psi A\psi)$ is a matrix of compactly supported elements of $I^m(\mathbb{R}^{2n}, \mathrm{Diag})$. Use the linear coordinates $(x,z) = (x, x-y)$ on $\mathbb{R}^{2n}$, in which $\mathrm{Diag} = \{z = 0\}$; here $N = 2n$, $k = n$ and the symbol order in (5.3.2) is $m + \frac{2n}{4} - \frac{n}{2} = m$. By (C1) the local representation (5.3.2) holds in these coordinates, so with a partition of unity in x and a cutoff $\chi(z)$ equal to one near $z = 0$, each entry of the kernel is a finite sum of terms $\chi_j(x)\chi(x-y)\int e^{i(x-y)\cdot\xi} a_j(x,\xi)\text{đ}\xi$, $a_j \in S^m$, plus a function in $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$. By (E2) and (E1) these are kernels of elements of $\Psi^m(\mathbb{R}^n)$.

Conversely suppose (i) holds and (ii) holds for the charts $\kappa_i : W_i \to \kappa_i(W_i)$ of a finite cover. By (i), near any point $(\bar{x}, \bar{y})$ with $\bar{x} \neq \bar{y}$ the kernel of A coincides with that of $\phi A\psi$, where $\phi = 1$ near $\bar{x}$, $\psi = 1$ near $\bar{y}$ and $\operatorname{supp}\phi \cap \operatorname{supp}\psi = \emptyset$; so K_A is smooth off the diagonal. Near $(\bar{x}, \bar{x})$, with $\bar{x} \in W_i$ and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W_i)$ equal to one near $\bar{x}$, K_A coincides with the kernel of $\psi A\psi$, whose local representative is $\int e^{i(x-y)\cdot\xi} a(x,\xi)\text{đ}\xi$ with $a \in S^m$: this is of the form (5.3.2) in the coordinates $(x, x-y)$. Since conormality is a local property, $K_A \in I^m(X^2, \mathrm{Diag}; \mathrm{Hom}(E,F) \otimes \Omega_R)$. In particular (ii) then holds for all trivializing charts. □

Question. Why is condition (i) needed? Does (ii) not already say everything?

Answer. Condition (ii) only sees the kernel on sets $W \times W$, for coordinate patches W. If x and y are far apart, there may be no chart containing both, and then (ii) says nothing about the kernel near (x,y). Condition (i) fills this gap: it is pseudolocality, imposed globally. For instance the operator $u \mapsto u(p)\, f$, with $p \in X$ fixed and $f \in \mathcal{C}^\infty(X)$ supported far from p, satisfies (ii) for every chart that does not meet both p and $\operatorname{supp} f$ (for those $\psi A\psi = 0$), hence for the charts of a suitable finite cover; but its kernel $f(x)\delta_p(y)$ is singular at points (x,p) far from the diagonal. □

The second half of the proof, combined with the first, shows again that the compactly supported elements of $\Psi^m(\mathbb{R}^n)$ are invariant under diffeomorphisms, as in (E7). It also gives the following, which we use repeatedly to build operators on X out of operators on $\mathbb{R}^n$.

Corollary 5.4.3. *If $\kappa : W \to \kappa(W)$ is a trivializing chart and B is an $N_F \times N_E$ matrix of elements of $\Psi^m(\mathbb{R}^n)$ with kernels of compact support in $\kappa(W) \times \kappa(W)$, then $\kappa^* \circ B \circ (\kappa^{-1})^*$, extended by zero, is an element of $\Psi^m(X; E, F)$.*

5.4.3 Normal fibrations and reduced forms

To write the kernel of an operator supported in a coordinate patch as an inverse Fourier transform, one identifies a neighbourhood of the diagonal with its normal bundle. The normal bundle of $\mathrm{Diag} \subset \mathbb{R}^{2n}$ is identified with $T\mathbb{R}^n = \mathbb{R}^n \times \mathbb{R}^n$ via the left tangent vectors $(w, 0)$; there remain several natural choices of a *normal fibration*, i.e. a linear map $\mathbb{R}^{2n} \to T\mathbb{R}^n$ inducing this identification. The *left*, *right* and *Weyl* fibrations are

$$f_L(x,y) = (x, x-y), \quad f_R(x,y) = (y, x-y), \quad f_W(x,y) = (\tfrac{x+y}{2}, x-y).$$

In each case the fibre variable is $x - y$ and one of x, y or $\frac{x+y}{2}$ is held fixed. Correspondingly an operator $A \in \Psi^m(\mathbb{R}^n)$ with compactly supported kernel has the three representations

$$K_A = \int e^{i(x-y)\cdot\xi} a_L(x,\xi)\text{đ}\xi = \int e^{i(x-y)\cdot\xi} a_R(y,\xi)\text{đ}\xi = \int e^{i(x-y)\cdot\xi} a_W(\tfrac{x+y}{2},\xi)\text{đ}\xi, \qquad (5.4.4)$$

with $a_L, a_R, a_W \in S^m$ (classical if A is): the first two exist by (E2), and the Weyl symbol is obtained in the same way by Taylor expansion of an amplitude about $x = y$ (see [8,

Section 18.5]). All three have the same principal symbol, and they may be taken to have compact support in the base variable.

Lemma 5.4.4. *If $A \in \Psi^m(\mathbb{R}^n)$ has compactly supported kernel, then for $f \in \mathcal{S}(\mathbb{R}^n)$*

$$Af(x) = \int e^{ix\cdot\xi} a_L(x,\xi)\hat{f}(\xi)đ\xi, \qquad \widehat{Af}(\xi) = \int e^{-iy\cdot\xi} a_R(y,\xi) f(y)\, dy. \tag{5.4.5}$$

Proof. The first formula is the definition of $\mathrm{Op}(a_L)$ after inserting $\hat{f}(\xi) = \int e^{-iy\cdot\xi} f(y)dy$. For the second, let $g(\xi)$ denote the right side, with a_R compactly supported in y; integration by parts in y shows that $|\xi^\alpha g(\xi)| \le C_\alpha \langle\xi\rangle^m$, so g is smooth and polynomially bounded, and by Fubini (first for a_R replaced by $a_R(y,\xi)\chi(\xi/R)$, then letting $R\to\infty$) $Af(x) = \int e^{ix\cdot\xi} g(\xi) đ\xi$, i.e. $\widehat{Af} = g$. □

5.4.4 Differential operators

A differential operator $P \in \mathrm{Diff}^k(X;E,F)$ is a linear map $\mathcal{C}^\infty(X;E) \to \mathcal{C}^\infty(X;F)$ which in any local coordinates and trivializations takes the form

$$P = \sum_{|\alpha|\le k} p_\alpha(x) D_x^\alpha, \quad p_\alpha \text{ smooth } N_F \times N_E \text{ matrices.} \tag{5.4.6}$$

By Theorem 5.4.2, $\mathrm{Diff}^k(X;E,F) \subset \Psi^k_{\mathrm{ph}}(X;E,F)$: condition (i) holds trivially since $\phi P\psi = 0$ if $\operatorname{supp}\phi \cap \operatorname{supp}\psi = \emptyset$, and $\psi P\psi = \sum \psi p_\alpha D^\alpha \psi$ in local coordinates is a composite of elements of $\Psi(\mathbb{R}^n)$. The kernel of P is $\sum_\alpha p_\alpha(x) D_x^\alpha \delta(x-y)|dy|$, conormal to the diagonal.

The leading part of (5.4.6) has an invariant meaning. Let $\bar{x} \in X$, $\Xi \in T^*_{\bar{x}}X$, choose a real-valued $f \in \mathcal{C}^\infty(X)$ with $df(\bar{x}) = \Xi$ and $u \in \mathcal{C}^\infty(X;E)$. Then by Leibniz' formula

$$e^{-itf} P(e^{itf}u) = \sum_{j=0}^{k} t^j U_j, \qquad U_k(\bar{x}) = \sum_{|\alpha|=k} p_\alpha(\bar{x})\, \Xi^\alpha\, u(\bar{x}), \tag{5.4.7}$$

where $\Xi^\alpha = (\partial f(\bar{x}))^\alpha$ in local coordinates; the left side is defined invariantly, so the coefficient of t^k is too. Hence the *principal symbol*

$$\sigma_k(P)(x,\xi) = \sum_{|\alpha|=k} p_\alpha(x)\xi^\alpha \in \mathcal{C}^\infty(T^*X; \hom(E,F))$$

is a well-defined section, homogeneous polynomial of degree k on the fibres; P is called *elliptic* if $\sigma_k(P)(x,\xi)$ is invertible for all $\xi \neq 0$. For example, for the exterior derivative $d(e^{itf}u) = e^{itf}(it\, df \wedge u + du)$, so $\sigma_1(d)(x,\xi) = i\,\xi\wedge$.

5.5 The calculus

We now show that the operators just defined form an algebra with a principal symbol, exactly as on $\mathbb{R}^n$. All proofs follow one pattern: cut the operator into pieces supported in coordinate patches, plus a smoothing error; apply the Euclidean result to each piece; and check that the smoothing errors are harmless because they form an ideal.

5.5.1 Localization

Lemma 5.5.1. *Let $\{U_i\}$ be a finite open cover of X and $A \in \Psi^m(X;E,F)$. Then*

$$A = \sum_i A_i + A', \qquad A_i \in \Psi^m(X;E,F),\ \operatorname{supp} K_{A_i} \Subset U_i \times U_i,\ A' \in \Psi^{-\infty}(X;E,F). \tag{5.5.1}$$

Proof. Let $\{\phi_a\}$ be a partition of unity as in Lemma 5.1.3. Then $A = \sum_{a,b} \phi_a A \phi_b$, the kernel of $\phi_a A \phi_b$ being $\phi_a(x)A(x,y)\phi_b(y)$. If $\operatorname{supp}\phi_a \cap \operatorname{supp}\phi_b = \emptyset$ this kernel vanishes near the diagonal, so is smooth. Otherwise $\operatorname{supp}\phi_a \cup \operatorname{supp}\phi_b \subset U_i$ for some i, and the term has kernel supported in $U_i \times U_i$. Collecting terms gives (5.5.1). □

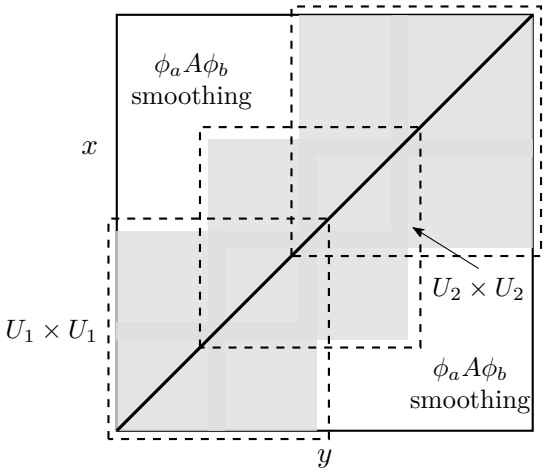


Figure 5.2: Localization (Lemma 5.5.1). The kernel of $A = \sum_{a,b} \phi_a A \phi_b$ is cut into the pieces $\phi_a(x)A(x,y)\phi_b(y)$. Pieces whose supports meet the diagonal (shaded) lie, by Lemma 5.1.3, inside a single product $U_i \times U_i$ of a chart with itself (dashed), where the Euclidean calculus applies. All other pieces vanish near the diagonal and are smoothing.

Figure 5.2 shows why the stronger property (5.1.2) of the partition of unity is useful: an ordinary partition of unity would produce pieces $\phi_a A \phi_b$ which meet the diagonal but are not contained in any single chart.

Taking the U_i to be trivializing charts, each A_i in (5.5.1) is given by a matrix of compactly supported elements of $\Psi^m(\mathbb{R}^n)$, and conversely by Corollary 5.4.3.

Proposition 5.5.2. *Each $A \in \Psi^m(X;E,F)$ is continuous $\mathcal{C}^\infty(X;E) \to \mathcal{C}^\infty(X;F)$. If $R \in \Psi^{-\infty}(X;G,E)$ then $AR \in \Psi^{-\infty}(X;G,F)$.*

Proof. In (5.5.1), A' has a smooth kernel, so is continuous, and each A_i is, in local coordinates, a matrix of elements of $\Psi^m(\mathbb{R}^n)$ with compactly supported kernels; these are continuous on $\mathcal{S}(\mathbb{R}^n)$ and hence on smooth functions supported in a fixed compact set. The kernel of AR is $(x,z) \mapsto A(K_R(\cdot,z))(x)$; since $z \mapsto K_R(\cdot,z)$ is a smooth map into $\mathcal{C}^\infty(X;E)$ (after trivializing the bundles in z locally) and A is continuous and linear, this is smooth in (x,z). □

5.5.2 Adjoints, transposes and action on distributions

Fix a smooth positive density ν on X and hermitian metrics on E and F, so that $L^2(X;E)$ and $L^2(X;F)$ are defined by (5.2.3).

Proposition 5.5.3. *If $A \in \Psi^m(X;E,F)$ there is a unique $A^* \in \Psi^m(X;F,E)$ such that $(Au,v) = (u,A^*v)$ for all $u \in \mathcal{C}^\infty(X;E)$, $v \in \mathcal{C}^\infty(X;F)$, and $\sigma_m(A^*) = \sigma_m(A)^*$ (pointwise adjoint with respect to the metrics).*

Proof. Write $A(x,y) = k(x,y)\nu(y)$ and set $A^*(x,y) = k(y,x)^*\nu(y)$, where $k(y,x)^*$ is the adjoint of $k(y,x) \in \hom(E_x,F_y)$. The reflection $(x,y) \mapsto (y,x)$ preserves the diagonal,

and I^m is invariant under complex conjugation (replace $a(w,\zeta)$ by $\overline{a(w,-\zeta)}$ in (5.3.2)), so by (C1) $A^* \in \Psi^m(X;F,E)$. The identity $(Au,v) = (u,A^*v)$ is the statement that both sides equal the pairing of the kernel with $\overline{v}(x)\otimes u(y)$. Uniqueness is clear. In local coordinates where $\nu = \nu_0(x)|dx|$ and the metrics are given by positive matrices h_E, h_F, the local representative of A^* is $\nu_0^{-1}h_E^{-1}B^*h_F\nu_0$ where B^* is the Euclidean adjoint of that of A, so (E3) gives $\sigma_m(A^*) = h_E^{-1}\sigma_m(A)^\dagger h_F$, which is the adjoint with respect to the metrics. □

Corollary 5.5.4. *If $A \in \Psi^m(X;F,G)$ and $R \in \Psi^{-\infty}(X;E,F)$ then AR and $R'A$ are smoothing, for any $R' \in \Psi^{-\infty}(X;G,E)$.*

Proof. AR is smoothing by Proposition 5.5.2, and $R'A = (A^*R'^*)^*$, the adjoint of a smoothing operator being smoothing. □

Similarly the *transpose* of A, with kernel $A^t(x,y) = A(y,x)^t$, is an element of $\Psi^m(X;F'\otimes\Omega, E'\otimes\Omega)$ (the density factors match, since $A(y,x)\in F_y\otimes E'_x\otimes\Omega_x$), and $\int_X (Au)\cdot\phi = \int_X u\cdot A^t\phi$ for smooth u and ϕ. We may therefore extend A to distributions by

$$A : \mathcal{C}^{-\infty}(X;E) \longrightarrow \mathcal{C}^{-\infty}(X;F), \qquad (Au)(\phi) = u(A^t\phi), \tag{5.5.2}$$

which is continuous and, by density, the unique continuous extension of A on $\mathcal{C}^\infty(X;E)$. All the identities below between operators on smooth sections therefore hold on distributions.

5.5.3 Composition

Theorem 5.5.5. *If E, F, G are vector bundles over the compact manifold X then*

$$\Psi^m(X;F,G)\circ\Psi^{m'}(X;E,F) \subset \Psi^{m+m'}(X;E,G).$$

Proof. Let $A\in\Psi^m(X;F,G)$, $B\in\Psi^{m'}(X;E,F)$. Choose a finite cover by charts U_c over which E, F and G are trivial, a subordinate partition of unity ϕ_c, and $\chi_c, \chi'_c, \tilde\chi_c \in \mathcal{C}^\infty_{\mathrm{c}}(U_c)$ with $\chi_c = \chi'_c = 1$ near $\operatorname{supp}\phi_c$ and $\tilde\chi_c = 1$ near $\operatorname{supp}\chi_c\cup\operatorname{supp}\chi'_c$. Then

$$AB = \sum_c A\phi_c B = \sum_c \big(\chi_c A\phi_c B\chi'_c + (1-\chi_c)A\phi_c B + \chi_c A\phi_c B(1-\chi'_c)\big).$$

Since $(1-\chi_c)$ and ϕ_c have disjoint supports, $(1-\chi_c)A\phi_c$ is smoothing, and so is its product with B by Corollary 5.5.4; similarly for the last term. Finally $\chi_c A\phi_c B\chi'_c = \chi_c(\tilde\chi_c A\tilde\chi_c)\phi_c(\tilde\chi_c B\tilde\chi_c)\chi'_c$ is, in the chart U_c, a product of matrices of elements of $\Psi^m(\mathbb{R}^n)$ and $\Psi^{m'}(\mathbb{R}^n)$ with compactly supported kernels, so lies in $\Psi^{m+m'}(\mathbb{R}^n)$ by (E3), and the result follows from Corollary 5.4.3. □

The local step can be seen directly from Lemma 5.4.4: writing the first factor in left-reduced and the second in right-reduced form, for $f\in\mathcal{S}(\mathbb{R}^n)$,

$$\begin{aligned} ABf(x) &= \int e^{ix\cdot\xi} a_L(x,\xi)\Big(\int e^{-iy\cdot\xi} b_R(y,\xi)f(y)\,dy\Big)đ\xi, \\ K_{AB} &= \int e^{i(x-y)\cdot\xi} a_L(x,\xi)b_R(y,\xi)đ\xi. \end{aligned} \tag{5.5.3}$$

The amplitude $a_L(x,\xi)b_R(y,\xi)$ satisfies the estimates of (E2) with order $m+m'$, so $AB\in\Psi^{m+m'}(\mathbb{R}^n)$ with principal symbol $a_L(x,\xi)b_R(x,\xi)$. Equivalently, K_{AB} is the restriction to $z=y$ of $\int e^{i(x-y)\cdot\xi}a_L(x,\xi)b_R(z,\xi)đ\xi$, which is conormal in $\mathbb{R}^{3n}$ to $\{x=y\}$; the submanifold $\{z=y\}$ is transversal to it, and Proposition 5.3.1 shows again that the result is conormal.

Remark 5.5.6. When one factor is smoothing, the composite can also be treated globally. With $\pi_S(x,y,z)=(x,y)$, $\pi_F(x,y,z)=(y,z)$ and $\pi_C(x,y,z)=(x,z)$ on X^3, the kernel of AR, R smoothing, is $(\pi_C)_*(\pi_S^*A\cdot\pi_F^*R)$, where the density factor of A in y is used for the integration. The pull-back π_S^*A is conormal to $\pi_S^{-1}(\mathrm{Diag})=\{x=y\}$, of order $m-\frac{n}{4}$ by the behaviour of conormal distributions under pull-back by submersions (Chapter 4), the product with the smooth section π_F^*R preserves this, and the fibres of π_C (on which y varies) are transversal to $\{x=y\}$, so the push-forward is smooth by Proposition 5.3.2. For general A and B, however, the corresponding product $\pi_S^*A\cdot\pi_F^*B$ is not defined in this simple way, since both factors are singular at the triple diagonal; this is why the proof above localizes.

5.5.4 Symbols and the principal symbol

Let $V\to X$ be a real vector bundle (we only need $V=T^*X$) and W a complex vector bundle over X, pulled back to V. The space $S^m(V;W)$ consists of the smooth sections a of W over V such that, in any local trivializations of V and W over a coordinate patch, and on any compact subset K of the patch,

$$|D_x^\alpha D_\xi^\beta a(x,\xi)|\le C_{K,\alpha,\beta}\langle\xi\rangle^{m-|\beta|}.$$

This is independent of the trivializations: under $(x,\xi)\mapsto(x,L(x)\xi)$, with L smooth and invertible, and a change $a\mapsto g(x)a$ of trivialization of W, the chain rule shows that each ξ-derivative of the new expression is a combination of ξ-derivatives of the old one with smooth coefficients, and each x-derivative produces terms $(\partial_x L)\xi\cdot\partial_\xi a$ which are of the same order; since $\langle L\xi\rangle$ and $\langle\xi\rangle$ are comparable on compact sets the estimates follow by induction. Since X is compact, $S^m(V;W)$ is a Fréchet space. Set $S^{m-[1]}(V;W)=S^m(V;W)/S^{m-1}(V;W)$. The classical symbols S^m_{ph} are defined in the same way; for these the leading part is a smooth section over $V\setminus 0$, homogeneous of degree m, i.e. a section over the sphere bundle $SV=(V\setminus 0)/\mathbb{R}^+$ of $N_m\otimes W$, where N_m is the line bundle whose sections are the functions homogeneous of degree m on $V\setminus 0$.

Proposition 5.5.7. *There is a unique linear map*

$$\sigma_m:\Psi^m(X;E,F)\longrightarrow S^{m-[1]}(T^*X;\hom(E,F))$$

such that for every trivializing chart $\kappa:W\to\kappa(W)$ and every $\psi\in\mathcal{C}^\infty_{\mathrm{c}}(W)$, $\sigma_m(\kappa_*(\psi A\psi))$ is the local expression of $\psi^2\sigma_m(A)$, using the coordinates (5.2.1) on T^*W. It has the following properties.

(a) *The sequence $0\to\Psi^{m-1}(X;E,F)\hookrightarrow\Psi^m(X;E,F)\xrightarrow{\sigma_m}S^{m-[1]}(T^*X;\hom(E,F))\to 0$ is exact.*

(b) *$\sigma_{m+m'}(AB)=\sigma_m(A)\sigma_{m'}(B)$ and $\sigma_m(A^*)=\sigma_m(A)^*$.*

(c) *For $P\in\mathrm{Diff}^k(X;E,F)$, $\sigma_k(P)$ is the class of the polynomial symbol defined by (5.4.7).*

(d) *For classical operators $\sigma_m(A)$ is a homogeneous section and there is a short exact sequence*

$$\Psi^{m-1}_{\mathrm{ph}}(X;E,F)\hookrightarrow\Psi^m_{\mathrm{ph}}(X;E,F)\xrightarrow{\sigma_m}\mathcal{C}^\infty(S^*X;N_m\otimes\hom(E,F))\to 0.$$

Proof. Choose a finite cover by trivializing charts $\kappa_i:W_i\to\kappa_i(W_i)$ and $\psi_i\in\mathcal{C}^\infty_{\mathrm{c}}(W_i)$ with $\sum_i\psi_i^2=1$ (e.g. $\psi_i=\rho_i(\sum_k\rho_k^2)^{-1/2}$ for a partition of unity ρ_i). The principal symbol of $\kappa_{i*}(\psi_iA\psi_i)\in\Psi^m(\mathbb{R}^n)$ has a representative supported, in x, in a compact subset of $\kappa_i(W_i)$,

well defined modulo S^{m-1} symbols with the same property; transferring it to T^*W_i by (5.2.1) and extending by zero gives $a_i \in S^{m-[1]}(T^*X; \hom(E,F))$. Set $\sigma_m(A) = \sum_i a_i$.

To check the required property, let $\kappa : W \to \kappa(W)$ and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$ be arbitrary. Then $\psi A\psi = \sum_i \psi_i^2 \psi A \psi$ and $\psi_i^2(\psi A\psi) - \psi_i(\psi A\psi)\psi_i = \psi_i[\psi_i, \psi A\psi]$, which in the chart κ has principal symbol zero by (E3). The operator $\psi_i \psi A \psi \psi_i = \psi(\psi_i A \psi_i)\psi$ has kernel supported in $(W \cap W_i)^2$; in the chart κ_i its principal symbol is ψ^2 times that of $\kappa_{i*}(\psi_i A \psi_i)$, and by (E7) its principal symbol in the chart κ is the same invariant function on T^*X, namely $\psi^2 a_i$. Summing over i gives $\sigma_m(\kappa_*(\psi A\psi)) = \psi^2 \sigma_m(A)$. Uniqueness follows by applying the property to the ψ_i: two such maps differ by b with $\psi_i^2 b \in S^{m-1}$ for all i, so $b = \sum_i \psi_i^2 b \in S^{m-1}$.

(a) Clearly σ_m vanishes on Ψ^{m-1}. If $\sigma_m(A) = 0$ then for every trivializing chart and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$, $\kappa_*(\psi A\psi)$ has vanishing principal symbol, so lies in $\Psi^{m-1}(\mathbb{R}^n)$ by (E4); by Theorem 5.4.2, $A \in \Psi^{m-1}(X; E, F)$. For surjectivity define, with the notation above, the *quantization map*

$$q(a) = \sum_i \kappa_i^* \circ (\tilde\chi_i \operatorname{Op}(a_i) \tilde\chi_i) \circ (\kappa_i^{-1})^*, \tag{5.5.4}$$

where a_i is the local expression of $\psi_i^2 a$ in the chart κ_i (extended by zero outside $\kappa_i(W_i)$) and $\tilde\chi_i \in \mathcal{C}^\infty_{\mathrm{c}}(\kappa_i(W_i))$ is equal to one near the support of $\psi_i \circ \kappa_i^{-1}$. By Corollary 5.4.3, $q(a) \in \Psi^m(X; E, F)$, and applying the defining property in an arbitrary chart, using (E7) as above, gives $\sigma_m(q(a)) = \sum_i \psi_i^2 a = a$. Note that q is linear and maps $S^{-\infty}$ into $\Psi^{-\infty}$.

(b) Let $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$ and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$ equal to one near $\operatorname{supp}\psi$. Then $\psi A B \psi - (\psi A \chi)(\chi B \psi) = \psi A(1-\chi^2) B \psi$ is smoothing, by the argument in the proof of Theorem 5.5.5, and the local representatives of $\psi A \chi$ and $\chi B \psi$ have principal symbols $\psi\chi\sigma_m(A)$ and $\chi\psi\sigma_{m'}(B)$ (compare $\psi A\chi$ with $\tilde\psi A \tilde\psi$ for $\tilde\psi = 1$ on $\operatorname{supp}\chi$, using (E2)). So (E3) gives the multiplicativity. The statement for adjoints was proved in Proposition 5.5.3.

(c) is immediate from (5.4.6), and (d) follows from (a), since the quantization of a classical symbol is classical. □

$$0 \longrightarrow \Psi^{m-1} \hookrightarrow \Psi^m \xrightarrow{\sigma_m} S^{m-[1]}(T^*X; \hom(E,F)) \longrightarrow 0, \qquad q : S^{m-[1]} \dashrightarrow \Psi^m$$

Figure 5.3: The symbol exact sequence of Proposition 5.5.7(a), with the bundles E, F suppressed from the notation for the operator spaces. The quantization map q of (5.5.4) (dashed) is a linear right inverse of σ_m, built from charts and a partition of unity. It is not canonical and not multiplicative, but σ_m is: this is what makes symbolic inversion work.

The exact sequence (Figure 5.3) is the basic tool of the chapter. It says that the principal symbol captures an operator up to an error of one order lower, and that every symbol arises. Combined with multiplicativity, it lets us solve operator equations one order at a time: solve the equation for symbols, quantize, and pass the error, now one order better, to the next step.

Question. The full symbol of an operator on $\mathbb{R}^n$ determines the operator. Why does Proposition 5.5.7 only define a principal symbol on X?

Answer. Under a change of coordinates the full symbol transforms by a formula involving all derivatives of the coordinate change (see the proof of coordinate invariance in Chapter 2). Only the leading term transforms simply, as a function on T^*X; the lower-order terms mix with derivatives of the leading one. So there is no invariant full symbol, unless one fixes extra structure such as a connection or a choice of quantization map q. □

Remark 5.5.8. Since $\Psi^m(X) = I^m(X^2, \mathrm{Diag}; \Omega_R)$ is a $\mathcal{C}^\infty(X^2)$-module, (5.4.1) may be written $\Psi^m(X; E, F) = \Psi^m(X) \otimes_{\mathcal{C}^\infty(X^2)} \mathcal{C}^\infty(X^2; \mathrm{Hom}(E, F))$. Thus an operator between sections of bundles is a finite sum $\sum_i A_i G_i$ with $A_i \in \Psi^m(X)$ and $G_i \in \mathcal{C}^\infty(X^2; \mathrm{Hom}(E, F))$, and in local trivializations it is simply a matrix of scalar operators.

Proposition 5.5.9. *If $A_j \in \Psi^{m-j}(X; E, F)$ for $j \in \mathbb{N}_0$, there exists $A \in \Psi^m(X; E, F)$, unique modulo $\Psi^{-\infty}(X; E, F)$, such that $A - \sum_{j<N} A_j \in \Psi^{m-N}(X; E, F)$ for all N; we write $A \sim \sum_j A_j$.*

Proof. Choose a finite cover by trivializing charts U_c, decompose each A_j as in Lemma 5.5.1 with respect to it, $A_j = \sum_c A_{j,c} + A'_j$, and note that the supports of the kernels of the $A_{j,c}$ may be taken to lie in a fixed compact subset K_c of $U_c \times U_c$ independent of j (use the same partition of unity for all j). By (E5) there are $B_c \in \Psi^m(X; E, F)$ with kernels supported in $U_c \times U_c$ and $B_c - \sum_{j<N} A_{j,c} \in \Psi^{m-N}$ for all N. Then $A = \sum_c B_c$ satisfies $A - \sum_{j<N} A_j = \sum_c (B_c - \sum_{j<N} A_{j,c}) - \sum_{j<N} A'_j \in \Psi^{m-N}$. If A and $\tilde{A}$ both have the property, $A - \tilde{A} \in \bigcap_N \Psi^{m-N} = \Psi^{-\infty}$. □

Thus $\Psi(X; E) = \bigcup_m \Psi^m(X; E)$ is a filtered $*$-algebra, the principal symbol map is a homomorphism onto the symbol algebra, and $\Psi^{-\infty}(X; E)$ is an ideal. Moreover, from (C3) or directly from Theorem 5.5.5,

$$\mathrm{Diff}^k(X; F, G) \circ \Psi^m(X; E, F) \subset \Psi^{k+m}(X; E, G), \qquad \sigma_{k+m}(PA) = \sigma_k(P)\sigma_m(A).$$

5.5.5 The symbol calculus

We collect some consequences of the multiplicativity of the principal symbol which are used constantly. Recall that T^*X carries the canonical one-form, given in local coordinates by $\sum_j \xi_j \, dx_j$; by (5.2.1) this expression is independent of the coordinates. Its differential is the symplectic form $\omega = \sum_j d\xi_j \wedge dx_j$, and the *Poisson bracket*

$$\{a, b\} = \sum_{j=1}^{n} \Big(\frac{\partial a}{\partial \xi_j} \frac{\partial b}{\partial x_j} - \frac{\partial a}{\partial x_j} \frac{\partial b}{\partial \xi_j} \Big) = H_a b$$

of two functions on T^*X is the derivative of b along the Hamilton vector field H_a of a, defined invariantly by $\omega(\cdot, H_a) = da$ (with the sign convention fixed by $H_{\xi_j} = \partial_{x_j}$). In particular the Poisson bracket does not depend on the coordinates used to compute it.

Proposition 5.5.10. *Let $A \in \Psi^m(X; E, F)$.*

(a) *If $\phi \in \mathcal{C}^\infty(X)$ then $[\phi, A] = \phi A - A\phi \in \Psi^{m-1}(X; E, F)$ and $\sigma_m(\phi A) = \sigma_m(A\phi) = \phi \, \sigma_m(A)$.*

(b) *If $\rho_i \in \mathcal{C}^\infty(X)$ are finitely many real functions with $\sum_i \rho_i^2 = 1$ then $A - \sum_i \rho_i A \rho_i \in \Psi^{m-1}(X; E, F)$.*

(c) *Let $A \in \Psi^m(X; E)$ and $B \in \Psi^{m'}(X; E)$ have scalar principal symbols, $\sigma_m(A) = [a] \, \mathrm{Id}_E$ and $\sigma_{m'}(B) = [b] \, \mathrm{Id}_E$, where $a \in S^m(T^*X)$ and $b \in S^{m'}(T^*X)$. Then $[A, B] \in \Psi^{m+m'-1}(X; E)$ and*

$$\sigma_{m+m'-1}([A, B]) = \frac{1}{i} \{a, b\} \, \mathrm{Id}_E \, .$$

Proof. (a) Multiplication by ϕ is a differential operator of order zero with principal symbol ϕ, so by the composition theorem $\sigma_m(\phi A) = \phi \sigma_m(A) = \sigma_m(A\phi)$; hence $\sigma_m([\phi, A]) = 0$ and $[\phi, A] \in \Psi^{m-1}$ by Proposition 5.5.7(a).

(b) $A - \sum_i \rho_i A \rho_i = \sum_i \rho_i [\rho_i, A]$, which lies in Ψ^{m-1} by (a).

(c) Since the principal symbols are scalar, they commute, so $\sigma_{m+m'}([A,B]) = 0$ and $[A,B] \in \Psi^{m+m'-1}$. The class $\frac{1}{i}\{a,b\} \in S^{m+m'-1}/S^{m+m'-2}$ is independent of the representatives a, b, and by the remark above is invariantly defined on T^*X. Since $\sigma_{m+m'-1}$ is determined by its local expressions (Proposition 5.5.7), it suffices to show that for a trivializing chart $\kappa : W \to \kappa(W)$ and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$ the local representative of $\psi[A,B]\psi$ has principal symbol $\psi^2 \frac{1}{i}\{a,b\}$. Choose $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$ equal to one near $\operatorname{supp}\psi$. As in the proof of Proposition 5.5.7(b), $\psi A B \psi - \psi(\chi A \chi)(\chi B \chi)\psi = \psi A (1-\chi^2) B \psi$ is smoothing, and similarly with A and B interchanged, so $\psi[A,B]\psi \equiv \psi[A_\chi, B_\chi]\psi$ modulo $\Psi^{-\infty}$, with $A_\chi = \chi A \chi$, $B_\chi = \chi B \chi$. In the chart, the local representatives of A_χ and B_χ are matrices of elements of $\Psi^m(\mathbb{R}^n)$ and $\Psi^{m'}(\mathbb{R}^n)$ of the form $A_\chi = \operatorname{Op}(\chi^2 a)\operatorname{Id} + A_1$, $B_\chi = \operatorname{Op}(\chi^2 b)\operatorname{Id} + B_1$ with $A_1 \in \Psi^{m-1}$, $B_1 \in \Psi^{m'-1}$ (by the local exact sequence (E4)). Now $[\operatorname{Op}(\chi^2 a)\operatorname{Id}, B_1]$ has vanishing principal symbol of order $m+m'-1$ because $\chi^2 a \operatorname{Id}$ commutes with every matrix, so lies in $\Psi^{m+m'-2}$; similarly for $[A_1, \operatorname{Op}(\chi^2 b)\operatorname{Id}]$, and $[A_1, B_1] \in \Psi^{m+m'-2}$. The remaining term has principal symbol $\frac{1}{i}\{\chi^2 a, \chi^2 b\}\operatorname{Id}$ by the commutator formula of Chapter 2, and $\psi^2\{\chi^2 a, \chi^2 b\} = \psi^2\{a,b\}$ since $\chi = 1$ near $\operatorname{supp}\psi$. Finally, the coordinate changes act on T^*W by the maps (5.2.1), which preserve $\sum_j \xi_j dx_j$ and hence the Poisson bracket, so the local expressions fit together. □

Part (c) has no analogue when $\sigma_{m'}(B) = b$ is a genuine matrix, even if $\sigma_m(A)$ is scalar: then the term a_{m-1} of order $m-1$ in the local symbol of A, which is not invariantly defined, contributes $[a_{m-1}, b]$ to the symbol of order $m+m'-1$ of $[A,B]$.

5.6 Ellipticity and parametrices

An operator cannot be inverted by inverting its symbol, but it can be inverted *up to smoothing errors*, provided the principal symbol is invertible at infinity. Such an approximate inverse, a parametrix, is the main output of the calculus. It immediately gives elliptic regularity, and, after Sobolev spaces are introduced, the Fredholm property.

Definition 5.6.1. An operator $A \in \Psi^m(X; E, F)$ is *elliptic* if there is a symbol $b \in S^{-m}(T^*X; \hom(F,E))$ such that, for a representative a of $\sigma_m(A)$,

$$ba - \operatorname{Id}_E \in S^{-1}(T^*X; \hom(E)), \qquad ab - \operatorname{Id}_F \in S^{-1}(T^*X; \hom(F)).$$

Lemma 5.6.2. *Fix a metric on T^*X. Then $A \in \Psi^m(X; E, F)$ is elliptic if and only if a (any) representative a of $\sigma_m(A)$ is invertible for $|\xi| \geq R$, for some R, with $|a(x,\xi)^{-1}| \leq C\langle\xi\rangle^{-m}$ there. For classical operators this holds if and only if the homogeneous principal symbol is invertible on $T^*X \setminus 0$, and for differential operators it is the condition given in Section 5.4.*

Proof. If $ba = \operatorname{Id} + r$, $ab = \operatorname{Id} + r'$ with $r, r' \in S^{-1}$ then for $|\xi|$ large $|r|, |r'| < \frac{1}{2}$, so a has a left and a right inverse, hence is invertible (in particular the ranks of E and F coincide), and $a^{-1} = (\operatorname{Id} + r)^{-1} b$ satisfies the bound. Conversely, if the bound holds, let $\chi \in \mathcal{C}^\infty(T^*X)$ be equal to one for $|\xi| \leq R+1$ and vanish for $|\xi| \geq R+2$, and set $b = (1-\chi)a^{-1}$. Differentiating $a\, a^{-1} = \operatorname{Id}$ repeatedly, each derivative of a^{-1} is a sum of products of a^{-1} with derivatives of a, and the symbol estimates for $b \in S^{-m}$ follow by induction; $ab - \operatorname{Id} = -\chi \operatorname{Id}$ and $ba - \operatorname{Id}$ have compact support in ξ, so lie in $S^{-\infty}$. For a homogeneous symbol invertible on the sphere bundle the bound follows from compactness of S^*X and homogeneity. □

Theorem 5.6.3. *If $A \in \Psi^m(X;E,F)$ is elliptic, there exists $B \in \Psi^{-m}(X;F,E)$, unique modulo $\Psi^{-\infty}$, such that*

$$BA = \mathrm{Id}_E - R_E, \quad AB = \mathrm{Id}_F - R_F, \qquad R_E \in \Psi^{-\infty}(X;E),\ R_F \in \Psi^{-\infty}(X;F). \tag{5.6.1}$$

If A is classical, so is B.

The proof has three steps: quantize an inverse of the symbol to get B_0, with error $S = \mathrm{Id} - AB_0$ of order -1; sum the Neumann series $B_0 \sum_j S^j$ asymptotically to push the error to order $-\infty$; and compare left and right parametrices. Figure 5.4 shows the logic.

$$\begin{array}{ccccc} A & \xrightarrow{\quad\sigma_m\quad} & a & \xrightarrow{\quad\text{invert at }\infty\quad} & b \approx a^{-1} \\ & & & & \Big\downarrow q \\ \text{Fredholm on } H^s & \xleftarrow{\quad\text{Rellich}\quad} & B \sim B_0 \textstyle\sum_j S^j & \xleftarrow[\text{asymptotic sum}]{\text{Neumann series}} & B_0 = q(b) \end{array}$$

Figure 5.4: The logic of the parametrix construction (Theorem 5.6.3). Ellipticity is used only once, to invert the symbol; all other steps are the symbol calculus. Elliptic regularity (Corollary 5.6.4) follows directly from B. The Fredholm property (Theorem 5.9.1) uses in addition that smoothing operators are compact (Rellich's theorem), which needs compactness of X.

Proof. With b as in Definition 5.6.1 let $B_0 = q(b) \in \Psi^{-m}(X;F,E)$. By Proposition 5.5.7, $AB_0 = \mathrm{Id}_F - S$ with $S \in \Psi^{-1}(X;F)$. By Proposition 5.5.9 choose $B_r \in \Psi^{-m}(X;F,E)$ with $B_r \sim \sum_{j\geq 0} B_0 S^j$. Then for each N,

$$AB_r \equiv AB_0 \sum_{j<N} S^j = (\mathrm{Id} - S) \sum_{j<N} S^j = \mathrm{Id} - S^N \mod \Psi^{-N},$$

so $AB_r - \mathrm{Id} \in \Psi^{-N}$ for all N, i.e. $AB_r = \mathrm{Id} - R_F$ with R_F smoothing. In the same way, starting from $B_0 A = \mathrm{Id} - S'$, there is B_l with $B_l A = \mathrm{Id} - R_E$. Then

$$B_l = B_l(AB_r + R_F) = (\mathrm{Id} - R_E)B_r + B_l R_F = B_r - R_E B_r + B_l R_F,$$

so $B_l - B_r \in \Psi^{-\infty}$ by Corollary 5.5.4, and $B = B_r$ satisfies both identities in (5.6.1). The same computation shows that any left parametrix and any right parametrix differ by a smoothing operator, which gives uniqueness. In the classical case each step stays in the classical calculus. □

Corollary 5.6.4 (Elliptic regularity)**.** *If $A \in \Psi^m(X;E,F)$ is elliptic, $u \in \mathcal{C}^{-\infty}(X;E)$ and $Au \in \mathcal{C}^\infty(X;F)$, then $u \in \mathcal{C}^\infty(X;E)$. In particular the null space of A on distributions,* $\mathrm{Nul}(A) = \{u \in \mathcal{C}^{-\infty}(X;E);\, Au = 0\}$*, consists of smooth sections.*

Proof. By (5.6.1), $u = BAu + R_E u$; here $BAu \in \mathcal{C}^\infty$ by Proposition 5.5.2 and $R_E u \in \mathcal{C}^\infty$ since R_E is smoothing. □

Discussion. Elliptic regularity is the prototype of a microlocal argument. Since $u = BAu + R_E u$, any singularity of u must already be visible in Au: the parametrix B cannot create singularities, and R_E destroys them all. Without ellipticity this fails. For the wave operator $\partial_t^2 - \partial_x^2$ on the torus $\mathbb{T}^2$, the function $u(t,x) = f(t-x)$ with f singular is a non-smooth solution of $Pu = 0$; its singularities sit exactly where the principal symbol $\xi^2 - \tau^2$ vanishes. The characteristic set, studied in Chapter 3, is where regularity can be lost.

The parametrix reduces the study of A to that of operators $\mathrm{Id}+R$ with R smoothing: $\mathrm{Nul}(A) \subset \mathrm{Nul}(\mathrm{Id}-R_E)$ and $\mathrm{Ran}(A) \supset \mathrm{Ran}(\mathrm{Id}-R_F)$. In Section 5.9 we use this to show that elliptic operators are Fredholm, on smooth sections as well as on Sobolev spaces. The *index* of an elliptic operator is then

$$\mathrm{ind}(A) = \dim \mathrm{Nul}(A) - \dim\big(\mathcal{C}^\infty(X;F)/A\,\mathcal{C}^\infty(X;E)\big). \tag{5.6.2}$$

For a linear map between finite-dimensional spaces the corresponding integer is just the difference of the dimensions of the spaces; the index is a genuinely infinite-dimensional invariant. Its computation in terms of the principal symbol is the content of the Atiyah–Singer index theorem, the subject of Chapter 12 and the following chapters.

5.7 Non-compact manifolds and properly supported operators

On a non-compact manifold the local theory is unchanged, but there is a new problem: an operator with a kernel spread out to infinity need not map compactly supported functions to compactly supported ones, so two operators cannot always be composed. The cure is to cut the kernel off near the diagonal, which changes the operator only by a smoothing error. In this section X is an arbitrary (paracompact, not necessarily compact) manifold, and E, F, G are vector bundles over it. The definition

$$\Psi^m(X;E,F) = I^m(X^2, \mathrm{Diag}; \mathrm{Hom}(E,F)\otimes\Omega_R)$$

makes sense without change, and so does the action (5.4.2) on *compactly supported* sections: if $u \in \mathcal{C}^\infty_{\mathrm{c}}(X;E)$ then $A\cdot\pi_R^*u$ has support in $X\times\operatorname{supp}u$, which meets each fibre of π_L in a compact set, so fibre integration (Proposition 5.3.2, applied after a locally finite partition of unity in the base) gives

$$A : \mathcal{C}^\infty_{\mathrm{c}}(X;E) \longrightarrow \mathcal{C}^\infty(X;F).$$

Theorem 5.4.2 and its proof hold verbatim, with "finite cover" replaced by "locally finite cover"; for X open in $\mathbb{R}^n$ this recovers the operators of Chapter 2 with kernels conormal to the diagonal. Since A need not map $\mathcal{C}^\infty_{\mathrm{c}}$ into $\mathcal{C}^\infty_{\mathrm{c}}$, and need not act on $\mathcal{C}^\infty(X;E)$, two such operators cannot in general be composed. This is remedied by restricting the supports.

Definition 5.7.1. An operator $A \in \Psi^m(X;E,F)$ is *properly supported* if both projections $\pi_L, \pi_R : \operatorname{supp}K_A \to X$ are proper maps, i.e. if $\operatorname{supp}K_A \cap (L\times X)$ and $\operatorname{supp}K_A\cap(X\times L)$ are compact for every compact $L\subset X$. We write $\Psi^m_{\mathrm{p}}(X;E,F)$ for the space of these operators.

If A is properly supported then $\operatorname{supp}(Au) \subset \pi_L\big(\operatorname{supp}K_A \cap \pi_R^{-1}(\operatorname{supp}u)\big)$ is compact for $u\in\mathcal{C}^\infty_{\mathrm{c}}(X;E)$, so $A : \mathcal{C}^\infty_{\mathrm{c}}(X;E)\to\mathcal{C}^\infty_{\mathrm{c}}(X;F)$. Moreover A acts on $\mathcal{C}^\infty(X;E)$: for a relatively compact open set $U\subset X$ and $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(X)$ equal to one on the compact set $\pi_R(\operatorname{supp}K_A\cap\pi_L^{-1}(\overline{U}))$, set $Au|_U = A(\chi u)|_U$, which is independent of χ and consistent on overlaps. The transpose and adjoint of a properly supported operator are properly supported, so by duality, as in (5.5.2), A acts on $\mathcal{C}^{-\infty}(X;E)$ and on $\mathcal{C}^{-\infty}_{\mathrm{c}}(X;E)$. When X is compact every operator is properly supported.

Lemma 5.7.2. *Every $A\in\Psi^m(X;E,F)$ can be written $A = A_{\mathrm{p}} + R$ with A_{p} properly supported and $R\in\mathcal{C}^\infty(X^2;\mathrm{Hom}(E,F)\otimes\Omega_R)$ smoothing. Moreover A_{p} may be chosen with kernel supported in any given neighbourhood of the diagonal.*

Proof. Let N be the given open neighbourhood of Diag. Each point of X has an open neighbourhood U with $U \times U \subset N$. Since X is paracompact (and metrizable), this cover has an open star refinement $\{V\}$: for each V the union of all members of the refinement meeting V lies in one of the sets U. Let $\{\phi_a\}$ be a partition of unity with each $\operatorname{supp}\phi_a$ compact and contained in some V_a, such that every compact set meets only finitely many of the supports. Then if $\operatorname{supp}\phi_a \cap \operatorname{supp}\phi_b \neq \emptyset$, both supports lie in the star of V_a, hence in some U, and $\operatorname{supp}\phi_a \times \operatorname{supp}\phi_b \subset U \times U \subset N$. Put

$$\chi(x,y) = \sum_{(a,b):\ \operatorname{supp}\phi_a \cap \operatorname{supp}\phi_b \neq \emptyset} \phi_a(x)\phi_b(y).$$

The sum is locally finite, so $\chi \in \mathcal{C}^\infty(X^2)$. If L is compact, only finitely many ϕ_a are non-zero on L, and each of them is paired with finitely many ϕ_b; so $\operatorname{supp}\chi \cap (L \times X)$ is contained in a finite union of the compact sets $\operatorname{supp}\phi_a \times \operatorname{supp}\phi_b$, and similarly for $X \times L$: χ has proper support. The set $C = \bigcup\{\operatorname{supp}\phi_a \times \operatorname{supp}\phi_b;\ \operatorname{supp}\phi_a \cap \operatorname{supp}\phi_b = \emptyset\}$ is a locally finite union of closed sets, hence closed, and disjoint from the diagonal; on the open neighbourhood $X^2 \setminus C$ of the diagonal, $\chi(x,y) = \sum_{a,b}\phi_a(x)\phi_b(y) = 1$. Set $A_{\mathrm{p}} = \chi K_A$ and $R = (1-\chi)K_A$; the latter vanishes near the diagonal, so is smooth, and $\operatorname{supp}K_{A_{\mathrm{p}}} \subset \operatorname{supp}\chi \subset N$ by the choice of the partition. □

Proposition 5.7.3. *The properly supported operators form a filtered algebra:*

$$\Psi^m_{\mathrm{p}}(X;F,G) \circ \Psi^{m'}_{\mathrm{p}}(X;E,F) \subset \Psi^{m+m'}_{\mathrm{p}}(X;E,G),$$

the principal symbol is defined, with values in $S^{m-[1]}(T^*X;\hom(E,F))$ *(symbol estimates uniform on compact subsets of X), is multiplicative and gives the exact sequence of Proposition 5.5.7(a), and asymptotic sums exist. If $A \in \Psi^m_{\mathrm{p}}(X;E,F)$ is elliptic, it has a properly supported parametrix $B \in \Psi^{-m}_{\mathrm{p}}(X;F,E)$, with $AB - \mathrm{Id}$ and $BA - \mathrm{Id}$ smoothing, and $u \in \mathcal{C}^{-\infty}(X;E)$, $Au \in \mathcal{C}^\infty(X;F)$ imply $u \in \mathcal{C}^\infty(X;E)$.*

Proof. The support of the kernel of AB is contained in the composite relation

$$\{(x,z);\ (x,y) \in \operatorname{supp}K_A \text{ and } (y,z) \in \operatorname{supp}K_B \text{ for some } y\},$$

which has proper projections. To see that $AB \in \Psi^{m+m'}$, a local property, let $\phi, \psi \in \mathcal{C}^\infty_{\mathrm{c}}(X)$ and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(X)$ be equal to one on $\pi_R(\operatorname{supp}K_A \cap \pi_L^{-1}(\operatorname{supp}\phi))$. Then $\phi A(1-\chi) = 0$ and $\phi AB\psi = (\phi A\chi)(\chi B\psi)$ is a product of operators with compactly supported kernels. The proofs of Theorem 5.5.5 and Proposition 5.5.7 only use finitely many charts covering the supports of the kernels, so apply to such operators, and give the composition and symbol properties. The quantization map (5.5.4), now a locally finite sum over a locally finite cover by charts, takes values in properly supported operators. Asymptotic sums are constructed as in Proposition 5.5.9, applying (E5) in each chart of a locally finite cover and multiplying the result by the cutoff χ of Lemma 5.7.2. With these ingredients the proof of Theorem 5.6.3 goes through unchanged (all operators being properly supported, so that all composites are defined), and elliptic regularity follows as in Corollary 5.6.4. □

Local Sobolev spaces $H^s_{\mathrm{loc}}(X;E) = \{u \in \mathcal{C}^{-\infty};\ \phi u \in H^s\ \forall \phi \in \mathcal{C}^\infty_{\mathrm{c}}(X)\}$ and $H^s_{\mathrm{c}}(X;E)$ are defined using Lemma 5.8.1, and the proof of Theorem 5.8.4 shows that properly supported operators of order m map $H^s_{\mathrm{loc}} \to H^{s-m}_{\mathrm{loc}}$ and $H^s_{\mathrm{c}} \to H^{s-m}_{\mathrm{c}}$. The global Fredholm theory below, however, depends on the compactness of X.

5.8 Sobolev spaces

To turn the parametrix into a statement about kernels and ranges we need Hilbert spaces on which pseudodifferential operators act boundedly and on which smoothing operators are compact. These are the Sobolev spaces, defined by patching together the Euclidean ones. The key facts are boundedness (Theorem 5.8.4) and Rellich's compactness theorem (Proposition 5.8.3(d)); the latter is where compactness of X enters.

5.8.1 Local facts

Lemma 5.8.1. *Let $K \subset \mathbb{R}^n$ be compact and $H^s_K = \{u \in H^s(\mathbb{R}^n); \operatorname{supp} u \subset K\}$.*

(a) *If $\Phi : \Omega \to \Omega'$ is a diffeomorphism of open sets and $K \subset \Omega$, then $\Phi_* = (\Phi^{-1})^* : H^s_K \to H^s_{\Phi(K)}$ is bounded, for every $s \in \mathbb{R}$.*

(b) *An operator with kernel in $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$ is bounded $H^t(\mathbb{R}^n) \to H^{t'}(\mathbb{R}^n)$ for all t, t'.*

(c) (Rellich) *If $t < s$ the inclusion $H^s_K \hookrightarrow H^t(\mathbb{R}^n)$ is compact.*

(d) *$\mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^n) = \bigcup_s H^s_{\mathrm{c}}(\mathbb{R}^n)$, and $H^s(\mathbb{R}^n) \subset \mathcal{C}^k(\mathbb{R}^n)$, with bounded inclusion into bounded $\mathcal{C}^k$ functions, if $s > k + \frac{n}{2}$.*

Proof. (a) Let $\Lambda_t = \operatorname{Op}(\langle\xi\rangle^t)$ and choose $\chi, \chi' \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ with $\chi = 1$ near K and $\chi' = 1$ near $\operatorname{supp}\chi$. For $u \in H^s_K$ put $f = \Lambda_s u \in L^2$; then

$$u = \chi\Lambda_{-s} f = \chi\Lambda_{-s}\chi' f + \chi\Lambda_{-s}(1-\chi')f.$$

The kernel of Λ_{-s} is $k(x-y)$ where k is smooth outside 0 and, together with all its derivatives, rapidly decreasing at infinity (since $|z|^{2N} D^\alpha k = \mathcal{F}^{-1}(\Delta^N_\xi(\xi^\alpha\langle\xi\rangle^{-s}))$ is bounded when $2N > n - s + |\alpha|$). So the second term is given by a smooth kernel $\chi(x)k(x-y)(1-\chi'(y))$, supported in $\operatorname{supp}\chi \times \mathbb{R}^n$ and with x-derivatives bounded by $C_N\langle y\rangle^{-N}$; it maps L^2 boundedly into smooth functions supported in $\operatorname{supp}\chi$, with each $\mathcal{C}^k$ norm bounded, and these are preserved by Φ_*. For the first term, $\Phi_*(\chi\Lambda_{-s}\chi' f) = \Phi_*(\chi\Lambda_{-s}\chi')\,\Phi_*(\chi' f)$, where $\Phi_*(\chi\Lambda_{-s}\chi') \in \Psi^{-s}(\mathbb{R}^n)$ by (E7), which is bounded $L^2 \to H^s$ by (E6), and $\|\Phi_*(\chi' f)\|_{L^2} \le C\|f\|_{L^2}$ since the Jacobian of Φ^{-1} is bounded on $\Phi(\operatorname{supp}\chi')$.

(b) Let T have kernel $k \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$. Then for each multi-index α the derivative $D^\alpha Tu(x) = u(D^\alpha_x k(x,\cdot))$ is bounded by the product $\|u\|_{H^t} \sup_x \|D^\alpha_x k(x,\cdot)\|_{H^{-t}}$ and Tu has support in a fixed compact set.

(c) Let $u_j \in H^s_K$, $\|u_j\|_{H^s} \le 1$, and $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to one near K. Then $\hat{u}_j(\xi) = u_j(\chi e^{-ix\cdot\xi})$, so $\hat{u}_j$ and all its derivatives are bounded on compact sets uniformly in j. By the Arzelà–Ascoli theorem and a diagonal argument a subsequence (still denoted u_j) has $\hat{u}_j$ converging uniformly on compact sets. Then for any R,

$$\|u_j - u_k\|^2_{H^t} \le \int_{|\xi|\le R} \langle\xi\rangle^{2t} |\hat{u}_j - \hat{u}_k|^2 đ\xi + \langle R\rangle^{2(t-s)} \int \langle\xi\rangle^{2s} |\hat{u}_j - \hat{u}_k|^2 đ\xi,$$

where the last term is at most $4\langle R\rangle^{2(t-s)}$; choosing R large and then j, k large shows that (u_j) is Cauchy in H^t.

(d) A compactly supported distribution has finite order, so $|\hat{u}(\xi)| \le C\langle\xi\rangle^N$ for some N and $u \in H^{-N-n}$. If $s > k + \frac{n}{2}$ and $|\alpha| \le k$ then $\int |\xi^\alpha \hat{u}| đ\xi \le \|\langle\xi\rangle^{|\alpha|-s}\|_{L^2}\|u\|_{H^s}$, so $D^\alpha u$ is continuous and bounded. □

5.8.2 Sobolev spaces on a compact manifold

Fix a finite cover of X by charts $\kappa_i : W_i \to \kappa_i(W_i)$ over which E is trivial, and a subordinate partition of unity ρ_i.

Definition 5.8.2. For $s \in \mathbb{R}$,

$$H^s(X;E) = \{u \in \mathcal{C}^{-\infty}(X;E);\ \kappa_{i*}(\rho_i u) \in H^s(\mathbb{R}^n)^{N_E}\ \forall i\}, \qquad \|u\|_s^2 = \sum_i \|\kappa_{i*}(\rho_i u)\|_{H^s}^2.$$

Proposition 5.8.3. (a) *$H^s(X;E)$ is independent of the choices made, with equivalent norms. A distribution u lies in $H^s(X;E)$ if and only if $\kappa_*(\psi u) \in H^s(\mathbb{R}^n)^{N_E}$ for every trivializing chart κ and $\psi \in \mathcal{C}_c^\infty(W)$; and if w has support in a compact subset of W and $\kappa_* w \in H^s$, then $w \in H^s(X;E)$, with $\|w\|_s \le C\|\kappa_* w\|_{H^s}$, C depending only on the support.*

(b) *$H^s(X;E)$ is a Hilbert space in which $\mathcal{C}^\infty(X;E)$ is dense, and $H^0(X;E) = L^2(X;E)$ with equivalent norms.*

(c) *$\bigcap_s H^s(X;E) = \mathcal{C}^\infty(X;E)$ and $\bigcup_s H^s(X;E) = \mathcal{C}^{-\infty}(X;E)$.*

(d) *For $t < s$ the inclusion $H^s(X;E) \hookrightarrow H^t(X;E)$ is compact.*

(e) *The L^2 pairing (5.2.3) extends by continuity to a sesquilinear pairing $H^s(X;E) \times H^{-s}(X;E) \to \mathbb{C}$, and every continuous linear functional on $H^s(X;E)$ is of the form $u \mapsto (u,w)$ for a unique $w \in H^{-s}(X;E)$.*

Proof. (a) Let κ be a trivializing chart and $\psi \in \mathcal{C}_c^\infty(W)$. Then $\psi u = \sum_i \psi\rho_i u$, and $\kappa_{i*}(\psi\rho_i u) = (\kappa_{i*}\psi)\kappa_{i*}(\rho_i u) \in H^s$ (by (E6), after changing trivializations, which is multiplication by a smooth matrix) is supported in a compact subset of $\kappa_i(W \cap W_i)$, so by Lemma 5.8.1(a) applied to $\kappa \circ \kappa_i^{-1}$, $\kappa_*(\psi\rho_i u) \in H^s$, with norm estimates. This proves that membership for one choice of cover and partition implies the condition for all charts, which in turn implies membership for any other choice; the norm equivalences come with the estimates. The last statement follows in the same way.

(b) The map $u \mapsto (\kappa_{i*}(\rho_i u))_i$ is an isometry onto a subspace of $\bigoplus_i H^s(\mathbb{R}^n)^{N_E}$. Its range is closed: if $v_i^{(k)} = \kappa_{i*}(\rho_i u^{(k)}) \to v_i$ in H^s, then $u^{(k)} = \sum_i \kappa_i^* v_i^{(k)}$ converges in $\mathcal{C}^{-\infty}(X;E)$ to $u = \sum_i \kappa_i^* v_i$ (the v_i being supported in $\kappa_i(\operatorname{supp}\rho_i)$), and so $v_i = \kappa_{i*}(\rho_i u)$. For density, approximate each $\kappa_{i*}(\rho_i u)$ in H^s by smooth functions supported in a fixed compact subset of $\kappa_i(W_i)$ (by convolution with an approximate identity) and use (a). The identification of H^0 with L^2 follows since the density ν and the metric on E are given by smooth, positive, locally bounded factors.

(c) If $u \in H^s$ for all s, each $\kappa_{i*}(\rho_i u)$ is smooth by Lemma 5.8.1(d), so $u = \sum_i \rho_i u$ is smooth. Conversely, each $\kappa_{i*}(\rho_i u)$ is a compactly supported distribution, so lies in some H^{s_i}.

(d) Under the isometry of (b), $H^s(X;E)$ maps into $\bigoplus_i (H^s_{K_i})^{N_E}$ with $K_i = \kappa_i(\operatorname{supp}\rho_i)$, and the inclusion of this space into $\bigoplus_i H^t(\mathbb{R}^n)^{N_E}$ is compact by Lemma 5.8.1(c).

(e) Choose $\psi_i \in \mathcal{C}_c^\infty(W_i)$ equal to one on $\operatorname{supp}\rho_i$. In local coordinates $\nu = \nu_i|dx|$ and the metric is h_i, so $(u,w) = \sum_i \int \langle h_i\nu_i\kappa_{i*}(\rho_i u), \kappa_{i*}(\psi_i w)\rangle dx$, which is bounded by $C\|u\|_s\|w\|_{-s}$ using $|\int f\bar g| \le \|f\|_{H^s}\|g\|_{H^{-s}}$ on $\mathbb{R}^n$ and (a). Conversely, let ℓ be a continuous linear functional on $H^s(X;E)$. By the Riesz representation theorem, $\ell(u)$ is the H^s inner product of u with some $w' \in H^s$, i.e.

$$\ell(u) = \sum_i \int \langle \kappa_{i*}(\rho_i u), \Lambda_{2s} v_i\rangle dx = \sum_i \int \langle \kappa_{i*}(\rho_i u), \chi_i \Lambda_{2s} v_i\rangle dx,$$

where $v_i = \kappa_{i*}(\rho_i w')$, $\Lambda_{2s} = \mathrm{Op}(\langle\xi\rangle^{2s})$ and $\chi_i \in \mathcal{C}^\infty_{\mathrm{c}}(\kappa_i(W_i))$ equals one on $\kappa_i(\operatorname{supp}\rho_i)$. Here $\chi_i\Lambda_{2s}v_i \in H^{-s}$ has compact support in $\kappa_i(W_i)$, so $w = \sum_i \rho_i\kappa_i^*(\nu_i^{-1}h_i^{-1}\chi_i\Lambda_{2s}v_i) \in H^{-s}(X;E)$ by (a), and $\ell(u) = (u,w)$. Uniqueness follows from density of $\mathcal{C}^\infty(X;E)$ in $H^s(X;E)$. □

Theorem 5.8.4. *Each $A \in \Psi^m(X;E,F)$ extends to a bounded operator $A : H^s(X;E) \to H^{s-m}(X;F)$ for every $s \in \mathbb{R}$. In particular smoothing operators map H^s into H^t for all s,t, and are compact on each H^s; operators of negative order are compact on each H^s.*

Proof. Use the decomposition (5.5.1) with respect to a cover by charts trivializing E and F. The smoothing part A' maps $H^s(X;E)$ to $H^t(X;F)$ for all t: decomposing $A' = \sum_{j,k}\rho_j A'\rho_k$, each term is, in the charts κ_j and κ_k, an operator with kernel in $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$, so Lemma 5.8.1(b) and Proposition 5.8.3(a) apply. For A_i, with kernel supported in $K' \times K'$, $K' \Subset W_i$, choose $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W_i)$ equal to one near K'. Then $A_iu = A_i(\psi u)$, $\kappa_{i*}(\psi u) \in H^s$, and the local representative of A_i is a matrix of elements of $\Psi^m(\mathbb{R}^n)$, so maps it into H^{s-m}, with support in $\kappa_i(K')$; by Proposition 5.8.3(a), $A_iu \in H^{s-m}(X;F)$ with the appropriate bound. The compactness statements follow from Proposition 5.8.3(d), since an operator of order $m<0$ maps H^s boundedly into H^{s-m}, which includes compactly into H^s. □

Proposition 5.8.5. *Let $A \in \Psi^m(X;E,F)$ be elliptic. If $u \in \mathcal{C}^{-\infty}(X;E)$ and $Au \in H^{s-m}(X;F)$ then $u \in H^s(X;E)$, and for any $t<s$*

$$\|u\|_s \le C(\|Au\|_{s-m} + \|u\|_t).$$

Consequently, for any elliptic $\Lambda \in \Psi^s(X;E)$,

$$H^s(X;E) = \{u \in \mathcal{C}^{-\infty}(X;E);\ \Lambda u \in L^2\} = \{u \in \mathcal{C}^{-\infty}(X;E);\ Pu \in L^2\ \forall P \in \Psi^s(X;E)\}.$$

Proof. By (5.6.1), $u = BAu + R_Eu$, with $BAu \in H^s$ by Theorem 5.8.4 and $R_Eu \in \mathcal{C}^\infty$ (note $u \in H^t$ for some t by Proposition 5.8.3(c)); the estimate follows from the bounds on B and R_E. For the last statement, an elliptic $\Lambda \in \Psi^s(X;E)$ exists (e.g. $q(\langle\xi\rangle^s \mathrm{Id}_E)$ for a metric on T^*X), and the inclusions follow from Theorem 5.8.4 and the first part. □

5.8.3 An L^2 bound from the principal symbol

The boundedness of $\Psi^0(X;E,F)$ on L^2 was deduced above from the corresponding Euclidean result. There is a more intrinsic argument, due to Hörmander, which uses only the symbol calculus and the boundedness of smoothing operators, and which gives more: the norm of A modulo compact operators is controlled by the size of its principal symbol at infinity. Fix a positive density, hermitian metrics on E and F, and a metric on T^*X; for a representative a of $\sigma_0(A)$ put

$$M(A) = \lim_{R\to\infty} \sup\{\|a(x,\xi)\|_{\hom(E_x,F_x)};\ (x,\xi) \in T^*X,\ |\xi| \ge R\}.$$

This does not depend on the representative, since elements of S^{-1} tend to zero at infinity, nor on the metric on T^*X. For a classical operator, $M(A)$ is the maximum over S^*X of the norm of the homogeneous principal symbol.

Proposition 5.8.6. *Let $A \in \Psi^0(X;E,F)$ and $C > M(A)^2$. Then there exist a formally self-adjoint $B \in \Psi^0(X;E)$ and a smoothing $G \in \Psi^{-\infty}(X;E)$ such that*

$$A^*A + B^2 = C\ \mathrm{Id}_E - G. \tag{5.8.1}$$

Consequently $\|Au\|^2_{L^2} \le (C+\|G\|)\|u\|^2_{L^2}$, and for every $\epsilon>0$ there is a finite-rank smoothing operator K with $\|A-K\|_{\mathcal{B}(L^2)} \le M(A)+\epsilon$. In particular the essential norm satisfies $\inf\{\|A+K\|;\ K \in \mathcal{K}(L^2(X;E),L^2(X;F))\} \le M(A)$.

The idea is that (5.8.1) immediately gives $\|Au\|^2 = C\|u\|^2 - \|Bu\|^2 - (Gu,u) \le C\|u\|^2 + |(Gu,u)|$, so it suffices to bound the smoothing operator G. To find B we take a square root of the positive symbol $C - a^*a$ and then correct it order by order, solving at each step a linear equation for the next term of the symbol.

Proof. Let a represent $\sigma_0(A)$, and choose $\epsilon > 0$ and R with $\|a(x,\xi)\|^2 \le C - 2\epsilon$ for $|\xi| \ge R$, so that $a^*a \le (C-2\epsilon)\,\mathrm{Id}$ there (here a^* is the pointwise adjoint). Let $\chi \in \mathcal{C}^\infty(T^*X)$, $0 \le \chi \le 1$, be equal to one for $|\xi| \le R$ and to zero for $|\xi| \ge R+1$, and

$$h = (1-\chi)(C - a^*a) + \chi\,\epsilon\,\mathrm{Id}_E \in S^0(T^*X; \hom(E)).$$

At each point h is a convex combination of self-adjoint endomorphisms which are $\ge \epsilon\,\mathrm{Id}$, so $\epsilon\,\mathrm{Id} \le h \le C'\,\mathrm{Id}$ for some C'. Let $b_0 = h^{1/2}$ be the positive square root; it is given by the Cauchy integral $b_0 = \frac{1}{2\pi i}\oint_\Gamma \lambda^{1/2}(\lambda - h)^{-1}d\lambda$ over a fixed contour in $\{\Re\lambda > 0\}$ enclosing $[\epsilon, C']$, and since $(\lambda - h)^{-1}$ satisfies symbol estimates of order 0 uniformly for $\lambda \in \Gamma$ (its derivatives are sums of products of $(\lambda-h)^{-1}$ with derivatives of h), $b_0 \in S^0$. Moreover $C - a^*a - b_0^2 = \chi(C - a^*a - \epsilon)$ has compact support, so lies in $S^{-\infty}$.

Let $B_0 = \frac12(q(b_0) + q(b_0)^*)$, with q the quantization map (5.5.4); it is self-adjoint with $\sigma_0(B_0) = b_0$, and $R_1 = C - A^*A - B_0^2 \in \Psi^{-1}(X;E)$ is self-adjoint. Suppose inductively that $B_k = B_0 + D_1 + \cdots + D_k$ with $D_j \in \Psi^{-j}(X;E)$ self-adjoint and $R_{k+1} = C - A^*A - B_k^2 \in \Psi^{-k-1}(X;E)$. For a self-adjoint endomorphism $b \ge c\,\mathrm{Id}$, $c > 0$, the linear map $L_b : d \mapsto bd + db$ on $\hom(E_x)$ satisfies $\operatorname{tr}(d^*L_b(d)) = \operatorname{tr}(d^*bd) + \operatorname{tr}(dbd^*) \ge 2c\operatorname{tr}(d^*d)$, so it is invertible with $\|L_b^{-1}\| \le (2c)^{-1}$ in the Hilbert–Schmidt norm, and it commutes with $d \mapsto d^*$. Hence, if r is a self-adjoint representative of $\sigma_{-k-1}(R_{k+1})$, then $d = L_{b_0}^{-1}(r)$ is self-adjoint, and $d \in S^{-k-1}$ by the same argument as in Lemma 5.6.2 (the inverse of the order-zero symbol L_{b_0} is uniformly bounded). Let $D_{k+1} = \frac12(q(d) + q(d)^*)$. Then

$$R_{k+2} = C - A^*A - (B_k + D_{k+1})^2 = R_{k+1} - B_kD_{k+1} - D_{k+1}B_k - D_{k+1}^2$$

has symbol of order $-k-1$ equal to $r - b_0d - db_0 = 0$, so lies in Ψ^{-k-2}. Taking $B' \sim B_0 + \sum_j D_j$ by Proposition 5.5.9 and $B = \frac12(B' + B'^*)$, we obtain (5.8.1) with $G = C - A^*A - B^2 \in \bigcap_k \Psi^{-k} = \Psi^{-\infty}$, self-adjoint.

For $u \in \mathcal{C}^\infty(X;E)$, (5.8.1) gives $\|Au\|^2 = C\|u\|^2 - \|Bu\|^2 - (Gu,u) \le (C + \|G\|)\|u\|^2$, where $\|G\| < \infty$ by (5.10.1); by density this bounds A on L^2. Now G is compact and self-adjoint; given $\delta > 0$, let π be the orthogonal projection onto the (finite-dimensional) sum of the eigenspaces of G with eigenvalues of absolute value $\ge \delta$ (Theorem A.2.3). These eigenvectors lie in the range of G, so are smooth, and π is a finite-rank smoothing operator (Lemma 5.10.5). For $u \perp \operatorname{Ran}\pi$, $(Gu,u) \ge -\delta\|u\|^2$, so $\|A(\mathrm{Id} - \pi)u\|^2 \le (C+\delta)\|u\|^2$ for all u. Thus $K = A\pi$, a finite-rank smoothing operator, satisfies $\|A - K\| \le (C+\delta)^{1/2}$, and C and δ may be chosen with $(C+\delta)^{1/2} \le M(A) + \epsilon$. □

For classical operators the bound on the essential norm is in fact an equality (Exercise 5.11). Using Proposition 5.8.5, the Sobolev spaces may also be *defined* as $\{u \in \mathcal{C}^{-\infty}(X;E);\ Pu \in L^2\ \forall P \in \Psi^s(X;E,E')\ \forall E'\}$; with this definition the boundedness $\Psi^m : H^s \to H^{s-m}$ reduces, by composition with an elliptic element of Ψ^s and its parametrix, to L^2-boundedness of Ψ^0, for which Proposition 5.8.6 provides a proof independent of the Euclidean theory.

5.9 Fredholm theory of elliptic operators

We now combine the parametrix with Sobolev spaces. Since the parametrix inverts an elliptic operator up to smoothing operators, and smoothing operators are compact on a

compact manifold, Atkinson's theorem applies: an elliptic operator is Fredholm. The duality between H^s and H^{-s} then identifies the cokernel with the null space of the adjoint.

Theorem 5.9.1. *Let $A \in \Psi^m(X;E,F)$ be elliptic, and let $A^* \in \Psi^m(X;F,E)$ be its adjoint.*

(a) *$N = \mathrm{Nul}(A)$ and $N^* = \mathrm{Nul}(A^*)$ (on distributions) are finite-dimensional subspaces of $\mathcal{C}^\infty(X;E)$ and $\mathcal{C}^\infty(X;F)$ respectively.*

(b) *For every $s \in \mathbb{R}$, $A : H^s(X;E) \to H^{s-m}(X;F)$ is Fredholm, with null space N and range $\{f \in H^{s-m}(X;F);\ (f,v) = 0\ \forall v \in N^*\}$.*

(c) *$A : \mathcal{C}^\infty(X;E) \to \mathcal{C}^\infty(X;F)$ has null space N and closed range, with complement N^*: $\mathcal{C}^\infty(X;F) = A\,\mathcal{C}^\infty(X;E) \oplus N^*$, an L^2-orthogonal sum.*

(d) *$\mathrm{ind}(A) = \dim N - \dim N^*$ is the index on $\mathcal{C}^\infty$ and on every H^s. It depends only on $\sigma_m(A)$; moreover $\mathrm{ind}(A^*) = -\mathrm{ind}(A)$, $\mathrm{ind}(AB) = \mathrm{ind}(A) + \mathrm{ind}(B)$ for elliptic $B \in \Psi^{m'}(X;G,E)$, and the index is constant on any family of elliptic operators which is continuous in the norm of $\mathcal{B}(H^s(X;E), H^{s-m}(X;F))$ for some s.*

Proof. Let B be a parametrix as in (5.6.1). By Theorem 5.8.4, $A : H^s \to H^{s-m}$ and $B : H^{s-m} \to H^s$ are bounded and R_E, R_F are compact on H^s and H^{s-m}, so A is Fredholm on these spaces by the parametrix characterization of Fredholm operators (Theorem A.3.3). Its null space on H^s is contained in N, and $N \subset \mathcal{C}^\infty \subset H^s$ by Corollary 5.6.4; so it equals N, which is therefore finite-dimensional. The adjoint is elliptic, since $\sigma_m(A^*) = \sigma_m(A)^*$ is inverted by b^*, so the same applies to N^*. This proves (a).

For (b), let $R_s = AH^s(X;E)$, which is closed, and $Y_s = \{f \in H^{s-m}; (f,v) = 0\ \forall v \in N^*\}$. If $u \in H^s$ and $v \in N^*$, then $(Au, v) = (u, A^*v) = 0$, by continuity from smooth u; so $R_s \subset Y_s$. If $R_s \neq Y_s$ there is a continuous linear functional on H^{s-m} which vanishes on R_s but not on Y_s. By Proposition 5.8.3(e) it is $f \mapsto (f,w)$ for some $w \in H^{m-s}(X;F)$; then $(u, A^*w) = (Au, w) = 0$ for all $u \in \mathcal{C}^\infty(X;E)$, so $A^*w = 0$, $w \in N^*$, and the functional vanishes on Y_s, a contradiction.

(c) If $f \in \mathcal{C}^\infty(X;F)$ is orthogonal to N^* then by (b) $f = Au$ with $u \in H^s$ for any s, and u is smooth by Corollary 5.6.4. The range on $\mathcal{C}^\infty$ is thus the orthocomplement of N^* in $\mathcal{C}^\infty(X;F)$, which is closed, and since $N^* \subset \mathcal{C}^\infty$, $f = (f - \pi_{N^*} f) + \pi_{N^*} f$ gives the decomposition, where π_{N^*} is orthogonal projection onto N^*.

(d) By (b), the codimension of the range on H^{s-m} is $\dim N^*$, the elements of N^* being linearly independent functionals on H^{s-m}; similarly on $\mathcal{C}^\infty$ by (c). If $B' \in \Psi^{m-1}(X;E,F)$, then $B' : H^s \to H^{s-m}$ is compact (it maps into H^{s-m+1}), so $\mathrm{ind}(A + B') = \mathrm{ind}(A)$ by the stability of the index under compact perturbations (Theorem A.3.6). The formula for A^* is clear from (a), the product formula follows from Proposition A.3.5 applied to $H^s \to H^{s-m'} \to H^{s-m'-m}$, and the constancy along continuous families is Theorem A.3.6 again. □

In particular, for $R \in \Psi^{-\infty}(X;E)$ the operator $\mathrm{Id} + R$ is Fredholm of index zero on $L^2(X;E)$ and on $\mathcal{C}^\infty(X;E)$: it is elliptic of order 0, and $\mathrm{Id} + tR$, $t \in [0,1]$, is a continuous family connecting it to the identity. Note also that the range in (c) is closed in $\mathcal{C}^\infty(X;F)$ and has a finite-dimensional complement; closedness of the range is part of the Fredholm property, and is not implied by the finite codimension of its closure.

Question. Where exactly was compactness of X used, and what goes wrong without it?

Answer. Only in the fact that the parametrix errors R_E, R_F are compact on Sobolev spaces, through Rellich's theorem. On $\mathbb{R}^n$ this fails: convolution with a Schwartz function $\phi \neq 0$ is

smoothing but commutes with translations, so it is not compact on $L^2(\mathbb{R}^n)$. Accordingly, elliptic operators on $\mathbb{R}^n$ need not be Fredholm. The Laplacian $\Delta : H^2(\mathbb{R}^n) \to L^2(\mathbb{R}^n)$ is injective with dense range, but 0 lies in its spectrum, so the range is not closed. Recovering a Fredholm theory on non-compact spaces requires control at infinity, as in the isotropic calculus of Chapter 7. □

5.9.1 Generalized inverses

We need a lemma on composites with smoothing operators.

Lemma 5.9.2. *Let $R_1 \in \Psi^{-\infty}(X; G', G)$, $R_2 \in \Psi^{-\infty}(X; E, E')$, and let $T : H^t(X; E') \to H^{t'}(X; G')$ be bounded, for some $t, t' \in \mathbb{R}$. Then $R_1 T R_2 \in \Psi^{-\infty}(X; E, G)$, and its kernel depends continuously, in $\mathcal{C}^\infty(X^2; \operatorname{Hom}(E, G) \otimes \Omega_R)$, on T in the operator norm.*

Proof. The kernel of $R_1 T R_2$ is $K(x, z) = (R_1\, T\, k_z)(x)$, where $k_z = K_{R_2}(\cdot, z)$ (with the fibre of $E'_z \otimes \Omega_z$ trivialized locally in z). The map $z \mapsto k_z$ is smooth into $H^t(X; E')$, so $z \mapsto T k_z$ is smooth into $H^{t'}(X; G')$ with derivatives bounded by $\|T\|$ times those of k_z, and $R_1 : H^{t'}(X; G') \to \mathcal{C}^\infty(X; G)$ is continuous, with $D_x^\alpha (R_1 g)(x)$ the pairing of g with the smooth family $D_x^\alpha K_{R_1}(x, \cdot) \in H^{-t'}$. Hence K is smooth, and each $\mathcal{C}^k$ norm of K is bounded by $C_k \|T\|$. □

Proposition 5.9.3. *Let $A \in \Psi^m(X; E, F)$ be elliptic and let π_N, π_{N^*} be the orthogonal projections onto $N = \operatorname{Nul}(A)$ and $N^* = \operatorname{Nul}(A^*)$. These are smoothing operators of finite rank, and there is a unique $G \in \Psi^{-m}(X; F, E)$ such that*

$$GA = \operatorname{Id}_E - \pi_N, \qquad AG = \operatorname{Id}_F - \pi_{N^*}, \qquad G\pi_{N^*} = 0. \tag{5.9.1}$$

Moreover $G - B \in \Psi^{-\infty}$ for any parametrix B.

Proof. If $e_1, \dots, e_k$ is an orthonormal basis of $N \subset \mathcal{C}^\infty(X; E)$, then $\pi_N u = \sum_j (u, e_j) e_j$ has kernel $\sum_j e_j(x) \otimes \langle \cdot, e_j(y) \rangle \nu(y)$, which is smooth; similarly for π_{N^*}. For $f \in H^{s-m}(X; F)$ let Gf be the unique $u \in H^s(X; E)$ with $u \perp N$ and $Au = (\operatorname{Id} - \pi_{N^*}) f$; it exists by Theorem 5.9.1(b), and does not depend on s. The map A restricted to $N^\perp \cap H^s$ is a continuous bijection onto the closed range, so G is bounded $H^{s-m} \to H^s$ by the open mapping theorem. The identities (5.9.1) follow directly from the definition, and they determine G. Now

$$\begin{aligned} G &= G(AB + R_F) = (\operatorname{Id} - \pi_N) B + G R_F, \\ G R_F &= (BA + R_E) G R_F = B(\operatorname{Id} - \pi_{N^*}) R_F + R_E G R_F. \end{aligned}$$

Here $R_E G R_F$ is smoothing by Lemma 5.9.2, and the remaining terms other than B are smoothing by Corollary 5.5.4. Thus $G - B \in \Psi^{-\infty}$. □

The generalized inverse is the unique linear map from $\mathcal{C}^\infty(X; F)$ to $\mathcal{C}^\infty(X; E)$ which vanishes on N^*, has range $N^\perp$ (the orthocomplement in $\mathcal{C}^\infty(X; E)$) and inverts A as a map from $N^\perp$ to the range of A.

Corollary 5.9.4. *For each $s \in \mathbb{R}$ and each bundle E there exists an invertible, formally self-adjoint, elliptic $\Lambda_s \in \Psi^s(X; E)$ with $\Lambda_s^{-1} \in \Psi^{-s}(X; E)$ and $(\Lambda_s u, u) \geq 0$. In particular $\|\Lambda_s u\|_{L^2}$ is a norm on $H^s(X; E)$ equivalent to $\|u\|_s$, and $\Lambda_s : H^t \to H^{t-s}$ is an isomorphism for every t.*

Proof. Let $Q \in \Psi^{s/2}(X;E)$ be elliptic, and $\Lambda_s = Q^*Q + \pi$, where π is the orthogonal projection onto the finite-dimensional space $\mathrm{Nul}(Q) \subset \mathcal{C}^\infty$. Then Λ_s is self-adjoint and elliptic, with principal symbol $\sigma(Q)^*\sigma(Q)$. If $\Lambda_s u = 0$ then u is smooth and $0 = (\Lambda_s u, u) = \|Qu\|^2 + \|\pi u\|^2$, so $u \in \mathrm{Nul}(Q)$ and $u = \pi u = 0$. Thus $\mathrm{Nul}(\Lambda_s) = \mathrm{Nul}(\Lambda_s^*) = 0$, and Λ_s is invertible on $\mathcal{C}^\infty$ and on every H^t with inverse its generalized inverse, an element of Ψ^{-s}. □

Proposition 5.9.5. *Let $A \in \Psi^m(X;E,F)$ be elliptic. There exists a finite-rank $R \in \Psi^{-\infty}(X;E,F)$ such that $A + \epsilon R$ is invertible for all $\epsilon \neq 0$ if and only if* $\mathrm{ind}(A) = 0$. *If $A + R$ is invertible, its inverse lies in $\Psi^{-m}(X;F,E)$.*

Proof. If $A+\epsilon R$ is invertible for some ϵ then $\mathrm{ind}(A) = \mathrm{ind}(A+\epsilon R) = 0$, by Theorem 5.9.1(d). Conversely, if $\mathrm{ind}(A) = 0$, choose orthonormal bases e_j of N and f_j of N^* (of the same dimension) and set $Ru = \sum_j (u, e_j) f_j$. If $(A+\epsilon R)u = 0$ then $Au = -\epsilon Ru \in \mathrm{Ran}(A) \cap N^* = \{0\}$, so $u \in N$ and $Ru = 0$, hence $u = 0$. Thus $A + \epsilon R$ is injective, of index zero, hence surjective, and its inverse is its generalized inverse, an element of Ψ^{-m}. □

Theorem 5.9.6. *Let $A \in \Psi^m(X;E)$, $m > 0$, be elliptic and formally self-adjoint. Then $L^2(X;E)$ has an orthonormal basis of eigenfunctions $Ae_j = \lambda_j e_j$ with $e_j \in \mathcal{C}^\infty(X;E)$, $\lambda_j \in \mathbb{R}$, each eigenvalue of finite multiplicity, and $|\lambda_j| \to \infty$.*

Proof. The generalized inverse $G \in \Psi^{-m}(X;E)$ of A is self-adjoint (the characterizing identities (5.9.1) are preserved by taking adjoints, since $N = N^*$) and compact on L^2 by Theorem 5.8.4. By the spectral theorem for compact self-adjoint operators (Theorem A.2.3), L^2 is the orthogonal sum of $\mathrm{Nul}(G) = N$ and the eigenspaces of G for non-zero eigenvalues $\mu_j \to 0$, each finite-dimensional. If $Gu = \mu u$ with $\mu \neq 0$ then $u \perp N$ and $u = AGu/\mu = Au/\mu$, so $Au = \mu^{-1}u$ and u is smooth by elliptic regularity. Taking orthonormal bases of N (eigenvalue 0) and of these eigenspaces gives the result, with $\lambda_j = \mu_j^{-1}$. □

5.10 Smoothing operators

Most results about elliptic operators are ultimately reduced, via the parametrix, to properties of smoothing operators. We fix a smooth positive density ν and hermitian metrics, and write the kernel of $A \in \Psi^{-\infty}(X;E,F)$ as $A(x,y) = k_A(x,y)\nu(y)$ with $k_A \in \mathcal{C}^\infty(X^2; \mathrm{Hom}(E,F))$. By the Cauchy–Schwarz inequality,

$$\|Au\|_{L^2} \leq \|k_A\|_{L^2(X^2)} \|u\|_{L^2}, \tag{5.10.1}$$

and indeed A is Hilbert–Schmidt with $\|A\|_2 = \|k_A\|_{L^2(X^2)}$ (Proposition A.4.2); in particular it is compact on L^2.

Lemma 5.10.1. *If $A \in \Psi^m(X;E,F)$ with $m < -n/2$ then A is Hilbert–Schmidt on L^2.*

Proof. By Lemma 5.5.1 it suffices to consider an operator given in local coordinates by $\mathrm{Op}(a)$, $a \in S^m$, with kernel of compact support. Its kernel is $K(x, x-z) = \mathcal{F}^{-1}_{\xi \to z} a(x,\cdot)(z)$, so by Plancherel $\iint |K(x,y)|^2 dy\, dx = \iint |a(x,\xi)|^2 d\!\!\!^-\xi\, dx$ over the compact x-support, which is finite since $\langle \xi \rangle^{2m}$ is integrable. □

5.10.1 The group $G^{-\infty}$

Lemma 5.10.2. *If $A \in \Psi^{-\infty}(X;E)$ and $\mathrm{Id} + A$ is invertible on $L^2(X;E)$, then $(\mathrm{Id}+A)^{-1} = \mathrm{Id} + B$ with $B \in \Psi^{-\infty}(X;E)$. In particular this holds if $\|A\|_{\mathcal{B}(L^2)} < 1$, when $B = \sum_{l \geq 1} (-A)^l$.*

Proof. Let $T = (\mathrm{Id}+A)^{-1}$. From $T = \mathrm{Id}-AT = \mathrm{Id}-TA$ we get $T = \mathrm{Id}-A(\mathrm{Id}-TA) = \mathrm{Id}-A+ATA$, so $B = -A+ATA$, which is smoothing by Lemma 5.9.2. □

Definition 5.10.3. For a compact manifold X and a vector bundle E over it,

$$\begin{aligned} G^{-\infty}(X;E) &= \{A \in \Psi^{-\infty}(X;E);\ \mathrm{Id}+A \text{ is invertible on } L^2(X;E)\} \\ &= \{A;\ (\mathrm{Id}+A)^{-1} = \mathrm{Id}+B,\ B \in \Psi^{-\infty}(X;E)\}, \end{aligned}$$

topologized as a subset of the Fréchet space $\Psi^{-\infty}(X;E) = \mathcal{C}^\infty(X^2;\mathrm{Hom}(E)\otimes\Omega_R)$, with the group law $(A,B)\mapsto A+B+AB$, i.e. composition of $\mathrm{Id}+A$ and $\mathrm{Id}+B$.

By Theorem 5.9.1, $\mathrm{Id}+A$ is invertible on L^2 if and only if it is invertible on $\mathcal{C}^\infty(X;E)$, i.e. is injective there, since the index is zero.

Proposition 5.10.4. *$G^{-\infty}(X;E)$ is an open subset of $\Psi^{-\infty}(X;E)$ and a topological group.*

Proof. Let $A \in G^{-\infty}$ and $\delta = \|(\mathrm{Id}+A)^{-1}\|^{-1}$. If $\|B\|_{\mathcal{B}(L^2)} < \delta$ then $\mathrm{Id}+A+B = (\mathrm{Id}+A)(\mathrm{Id}+(\mathrm{Id}+A)^{-1}B)$ is invertible; by (5.10.1) this holds for B in a neighbourhood of zero in $\Psi^{-\infty}$, so $G^{-\infty}$ is open. Composition, with kernel $k_A+k_B+\int k_A(x,y)k_B(y,z)\nu(y)$, is continuous; the inverse $B = -A+A(\mathrm{Id}+A)^{-1}A$ depends continuously on A by Lemma 5.9.2, since $A\mapsto(\mathrm{Id}+A)^{-1}$ is continuous into $\mathcal{B}(L^2)$. □

Discussion. One should think of $G^{-\infty}(X;E)$ as the group of invertible operators on $L^2(X;E)$ which differ from the identity by an operator with smooth kernel; it plays the role of an infinite-dimensional general linear group, as the following results show. A smoothing operator is close, in every sense that matters, to a finite matrix (Lemma 5.10.6), so $G^{-\infty}$ behaves like the union $\bigcup_N \mathrm{GL}(N,\mathbb{C})$. This is why it is connected, why it carries a trace and a determinant, and why its homotopy classes of maps compute odd K-theory.

5.10.2 Finite-rank approximation

Lemma 5.10.5. *A smoothing operator $A \in \Psi^{-\infty}(X;E,F)$ has finite rank if and only if*

$$A(x,y) = \sum_{i=1}^{N} f_i(x)e_i(y)\nu(y), \qquad f_i \in \mathcal{C}^\infty(X;F),\ e_i \in \mathcal{C}^\infty(X;E'). \tag{5.10.2}$$

Proof. Such an operator has range in the span of the f_i. Conversely, if the range is finite-dimensional, it lies in $\mathcal{C}^\infty(X;F)$; let $f_1,\dots,f_N$ be an orthonormal basis of it. Then $Au = \sum_i (Au,f_i)f_i$ and $(Au,f_i) = (u,A^*f_i) = \int_X (A^*f_i)^\dagger(y)u(y)\nu(y)$, where $(A^*f_i)^\dagger \in \mathcal{C}^\infty(X;E')$ is the dual of A^*f_i under the metric; this is (5.10.2) with $e_i = (A^*f_i)^\dagger$. □

If the e_i and f_j in (5.10.2) are taken to be orthonormal, and $E = F$, one may orthonormalize all of them together, obtaining a finite orthonormal set $v_1,\dots,v_M \subset \mathcal{C}^\infty(X;E)$ and $Au = \sum_{j,k} a_{jk}(u,v_k)v_j$; the finite-rank smoothing operators with range and coefficients in a fixed such span form a subalgebra isomorphic, as an algebra, to $M(M,\mathbb{C})$.

Lemma 5.10.6. *The finite-rank operators are dense in $\Psi^{-\infty}(X;E,F)$. More precisely, for every A there are finite-rank A_N with $k_{A_N}\to k_A$ in $\mathcal{C}^\infty(X^2;\mathrm{Hom}(E,F))$, and for a smooth family $A(p)$ depending on parameters p in a compact manifold P the $A_N(p)$ may be chosen as in (5.10.2) with f_i, e_i independent of p and coefficients depending smoothly on p, converging uniformly with all derivatives.*

Proof. Consider first the torus $\mathbb{T}^n = \mathbb{R}^n/2\pi\mathbb{Z}^n$ with density $|d\theta|$ and trivial bundles. A kernel $a \in \mathcal{C}^\infty(\mathbb{T}^{2n})$ has a rapidly convergent Fourier expansion

$$a(\theta,\theta') = \sum_{I,J\in\mathbb{Z}^n} a_{IJ} e^{iI\cdot\theta} e^{-iJ\cdot\theta'}, \quad a_{IJ} = (2\pi)^{-2n} \int_{\mathbb{T}^{2n}} e^{-iI\cdot\theta+iJ\cdot\theta'} a(\theta,\theta')\, d\theta\, d\theta',$$

the coefficients being rapidly decreasing in (I,J) since a is smooth. Hence the truncations $a_N = \sum_{|I|,|J|\le N} a_{IJ} e^{iI\cdot\theta} e^{-iJ\cdot\theta'}$, which are finite-rank kernels, converge to a in $\mathcal{C}^\infty$, and continuously in parameters.

In general choose a finite cover of X by charts $\kappa_p : U_p \to \kappa_p(U_p)$ with bounded images, over which E and F are trivial, and a subordinate partition of unity ρ_p. Then $A = \sum_{p,q} \rho_p A \rho_q$. In the coordinates $\kappa_p(x)$, $\kappa_q(y)$ (rescaled and translated so that $\kappa_p(U_p)$ and $\kappa_q(U_q)$ lie in $(0,2\pi)^n$) the kernel of $\rho_p A \rho_q$, with respect to the coordinate density, is a matrix of functions $a \in \mathcal{C}^\infty_{\mathrm{c}}((0,2\pi)^{2n})$, which we regard as functions on $\mathbb{T}^{2n}$. Choose $\chi_p \in \mathcal{C}^\infty_{\mathrm{c}}(\kappa_p(U_p))$ equal to one near $\kappa_p(\operatorname{supp}\rho_p)$, similarly χ_q. Then $\chi_p(\theta) a_N(\theta,\theta')\chi_q(\theta')$ converges to a in $\mathcal{C}^\infty$, and is a finite sum of products of functions of θ and of θ', compactly supported in the coordinate patches. Transferring back to X and summing over p,q gives finite-rank smoothing operators converging to A; the functions involved depend only on N, not on parameters. □

Proposition 5.10.7. *$G^{-\infty}(X;E)$ is connected (by smooth paths).*

Proof. Let $A \in G^{-\infty}$ and $\delta = \|(\mathrm{Id}+A)^{-1}\|^{-1}_{\mathcal{B}(L^2)}$. As in the proof of Proposition 5.10.4, the convex set $\{A+B;\ \|B\|_{\mathcal{B}(L^2)} < \delta\} \cap \Psi^{-\infty}$ lies in $G^{-\infty}$. By Lemma 5.10.6 and (5.10.1) there is a finite-rank A_N in this set, and the segment from A to A_N lies in $G^{-\infty}$. Let $V \subset \mathcal{C}^\infty(X;E)$ be a finite-dimensional space containing the range of A_N and the range of A_N^*, with orthonormal basis $v_1,\dots,v_M$. Then A_N vanishes on $V^\perp$ and maps V to itself, with matrix a, so $\mathrm{Id}+A_N = (\mathrm{Id}_V + a) \oplus \mathrm{Id}_{V^\perp}$ and $\mathrm{Id}+a \in \mathrm{GL}(M,\mathbb{C})$. Since $\mathrm{GL}(M,\mathbb{C})$ is connected, there is a smooth path $g(t)$ in it from $\mathrm{Id}+a$ to Id, and $\sum_{j,k}(g(t)-\mathrm{Id})_{jk}(\cdot,v_k)v_j$ is a smooth path in $G^{-\infty}$ from A_N to 0. □

5.10.3 Odd K-theory

The homotopy classes of maps into $G^{-\infty}$ give a direct definition of odd K-theory: for a compact manifold X set

$$K^{-1}(X) = [X; G^{-\infty}(Y;E)], \tag{5.10.3}$$

the set of smooth homotopy classes of smooth maps $X \to G^{-\infty}(Y;E)$, where Y is any compact manifold of positive dimension and E any vector bundle over Y. Here a smooth map is an element

$$K \in \mathcal{C}^\infty(X \times Y^2; \mathrm{Hom}(E) \otimes \Omega_R) \ \text{ with } \ \mathrm{Id}+K(x,\cdot) \text{ invertible for every } x \in X,$$

and two such are homotopic if they are the ends of a smooth map $X \times [0,1] \to G^{-\infty}(Y;E)$. Implicit in (5.10.3) is the following fact; the group structure is induced by that of $G^{-\infty}$, and the relation with the usual definition by stable homotopy classes of maps into $\mathrm{GL}(N,\mathbb{C})$ is visible in the proof.

Proposition 5.10.8. *For any two choices (Y,E), (Z,F) with $\dim Y, \dim Z > 0$ there is a natural bijection $[X; G^{-\infty}(Y;E)] \cong [X; G^{-\infty}(Z;F)]$.*

Sketch of proof. It suffices to compare with the model $Y = \mathbb{S}$, $E = \mathbb{C}$. Embed $\mathrm{GL}(N,\mathbb{C})$ into $G^{-\infty}(\mathbb{S})$ by

$$(a_{kl}) \longmapsto \sum_{k,l=1}^{N} (a_{kl} - \delta_{kl})\, e^{ik\theta} e^{-il\theta'} \frac{|d\theta'|}{2\pi};$$

the identity is subtracted so that this is a group homomorphism. Given a smooth map $A : X \to G^{-\infty}(Y;E)$, Lemma 5.10.6 gives finite-rank approximations $A_{(N)}(x)$, with fixed functions f_i, e_i, converging uniformly with all derivatives; for N large the straight line from A to $A_{(N)}$ stays in $G^{-\infty}$, by the openness estimate, which is uniform since X is compact. Choosing an orthonormal basis of the span of the f_i and e_i, as in Proposition 5.10.7, identifies $\mathrm{Id} + A_{(N)}(x)$ with a smooth map into some $\mathrm{GL}(M,\mathbb{C})$, and hence, via the embedding, a map $\tilde{A} : X \to G^{-\infty}(\mathbb{S})$. A different basis conjugates $\tilde{A}$ by a fixed element of $G^{-\infty}(\mathbb{S})$, which can be connected to the identity by Proposition 5.10.7; adding redundant elements to the basis does not change $\tilde{A}$; and increasing N changes $A_{(N)}$ only within the convex neighbourhoods used above. So the class of $\tilde{A}$ is well defined, and applying the construction to $X \times [0,1]$ shows it depends only on the homotopy class of A. Reversing the roles, using an orthonormal set of smooth sections of E over Y (which exists in any finite number since $\dim Y > 0$) gives the inverse map. □

5.10.4 The trace

For $A \in \Psi^{-\infty}(X;E)$ the restriction of the kernel to the diagonal is a section of $\hom(E) \otimes \Omega$, and we set

$$\mathrm{Tr}(A) = \int_X \mathrm{tr}\,(A|_{\mathrm{Diag}}) = \int_X \mathrm{tr}\, k_A(x,x)\, \nu(x). \tag{5.10.4}$$

For a finite-rank operator $Au = \sum_{j,k} a_{jk}(u, v_k)v_j$ with orthonormal v_j as above, $k_A(x,x) = \sum_{j,k} a_{jk} v_j(x) \otimes v_k(x)^\dagger$, so $\mathrm{Tr}(A) = \sum_{j,k} a_{jk}(v_j, v_k) = \sum_j a_{jj}$, the trace of its matrix; since the matrix trace is invariant under change of basis, $\mathrm{tr}(b^{-1}ab) = \sum_{i,j,k} b^{-1}_{ij} a_{jk} b_{ki} = \mathrm{tr}\, a$, this is consistent.

Proposition 5.10.9. *The trace (5.10.4) is a continuous linear functional on $\Psi^{-\infty}(X;E)$, and*

(a) $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ *for $A \in \Psi^{-\infty}(X;E,F)$, $B \in \Psi^{-\infty}(X;F,E)$;*

(b) *each $A \in \Psi^{-\infty}(X;E)$ is of trace class on $L^2(X;E)$, with $\|A\|_1 \le C\|k_A\|_{\mathcal{C}^k}$ for some k, and (5.10.4) is its Hilbert space trace; in particular $\mathrm{Tr}(A) = \lim_N \mathrm{Tr}(A_N)$ for any finite-rank approximations as in Lemma 5.10.6;*

(c) $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ *for $A \in \Psi^{-\infty}(X;E)$ and $B \in \Psi^m(X;E)$, any m.*

Proof. Continuity is clear. (a) By Fubini,

$$\mathrm{Tr}(AB) = \int_X \int_X \mathrm{tr}\,(k_A(x,y)k_B(y,x))\nu(y)\nu(x)$$
$$= \int_X \int_X \mathrm{tr}\,(k_B(y,x)k_A(x,y))\nu(x)\nu(y) = \mathrm{Tr}(BA),$$

using $\mathrm{tr}(ab) = \mathrm{tr}(ba)$ for $a \in \hom(E_y, F_x)$, $b \in \hom(F_x, E_y)$.

(b) Let s be an even integer with $s > n/2$ and Λ_s as in Corollary 5.9.4. Then $A = \Lambda_s^{-1}(\Lambda_s A)$, where Λ_s^{-1} is Hilbert–Schmidt by Lemma 5.10.1, and $\Lambda_s A$ is smoothing with $\|\Lambda_s A\|_2^2 = \int \|\Lambda_s k_A(\cdot,y)\|_{L^2}^2 \nu(y) \le C \int \|k_A(\cdot,y)\|_s^2 \nu(y) \le C'\|k_A\|_{\mathcal{C}^s}^2$. So A is of trace class

with $\|A\|_1 \le \|\Lambda_s^{-1}\|_2 \|\Lambda_s A\|_2 \le C\|k_A\|_{\mathcal{C}^s}$ (Definition A.4.3). For finite-rank A, (5.10.2) gives Hilbert space trace $\sum_i (f_i, e_i^\dagger) = \int \sum_i e_i(x)(f_i(x))\nu(x)$, which is (5.10.4). Both traces are continuous on $\Psi^{-\infty}$ (the first by the estimate just proved), so they agree by Lemma 5.10.6.

(c) Write $B = \Lambda_m(\Lambda_m^{-1}B)$ with $\Lambda_m^{-1}B \in \Psi^0$ bounded. The operators $A\Lambda_m$ and BA are smoothing, hence of trace class, and using that $\mathrm{Tr}(ST) = \mathrm{Tr}(TS)$ for S of trace class and T bounded (Proposition A.4.4), $\mathrm{Tr}(AB) = \mathrm{Tr}\left((A\Lambda_m)(\Lambda_m^{-1}B)\right) = \mathrm{Tr}(\Lambda_m^{-1}BA\Lambda_m) = \mathrm{Tr}\left((BA\Lambda_m)\Lambda_m^{-1}\right) = \mathrm{Tr}(BA)$. □

Remark 5.10.10. Using the trace one can define the Fredholm determinant: there is a unique entire function $\det(\mathrm{Id}+A)$ on $\Psi^{-\infty}(X;E)$ such that

$$\det\left((\mathrm{Id}+A)(\mathrm{Id}+B)\right) = \det(\mathrm{Id}+A)\det(\mathrm{Id}+B), \qquad \tfrac{d}{ds}\det(\mathrm{Id}+sA)|_{s=0} = \mathrm{Tr}(A),$$

and $A \in G^{-\infty}(X;E)$ if and only if $\det(\mathrm{Id}+A) \neq 0$. It follows that $G^{-\infty}(X;E)$ is dense in $\Psi^{-\infty}(X;E)$. Since $G^{-\infty}$ is connected, the determinant can be defined on $G^{-\infty}$ by

$$\det(\mathrm{Id}+A) = \exp\Big(\int_0^1 \mathrm{Tr}\left((\mathrm{Id}+\gamma(t))^{-1}\tfrac{d\gamma}{dt}\right)dt\Big)$$

for a smooth curve γ in $G^{-\infty}$ from 0 to A; the main points are independence of the curve and the extension to all of $\Psi^{-\infty}$. This is carried out in Chapter 12; compare also Chapter 8.

Proposition 5.10.11. *Let $A \in \Psi^{-\infty}(X;E)$. The set*

$$\mathrm{Spec}(A)\setminus\{0\} = \{z \in \mathbb{C}\setminus\{0\};\ z\,\mathrm{Id}-A \text{ is not invertible on } L^2(X;E)\}$$

is contained in $\{|z| \le \|A\|\}$ and is discrete in $\mathbb{C}\setminus\{0\}$; each of its points is an eigenvalue, and the generalized eigenspace

$$E(z) = \{u \in L^2(X;E);\ (z\,\mathrm{Id}-A)^N u = 0 \text{ for some } N \in \mathbb{N}\}$$

is finite-dimensional and contained in $\mathcal{C}^\infty(X;E)$.

Proof. For $|z| > \|A\|$, $z - A = z(\mathrm{Id} - A/z)$ is invertible by the Neumann series. Since A is compact, the other statements, except smoothness, are the Riesz theory of compact operators (Theorem A.3.7). If $(z-A)^N u = 0$ with $z \neq 0$, expanding by the binomial theorem gives $z^N u = Ap(A)u$ for a polynomial p, so $u \in \mathrm{Ran}(A) \subset \mathcal{C}^\infty(X;E)$. □

Alternatively, $\mathrm{Spec}(A)\setminus\{0\}$ is the zero set of the entire function $z \mapsto \det(\mathrm{Id} - A/z)$ on $\mathbb{C}\setminus\{0\}$, which tends to one as $|z| \to \infty$.

5.10.5 The analytic Fredholm alternative

The same reasoning applies to holomorphic families. Let $\Omega \subset \mathbb{C}$ be open. A map $A : \Omega \to \Psi^{-\infty}(X;E)$ is *holomorphic* if it is holomorphic as a map into the Fréchet space $\mathcal{C}^\infty(X^2; \mathrm{Hom}(E)\otimes\Omega_R)$, i.e. if the kernel $A(z)(x,y)$ is smooth in all variables and holomorphic in z. Note that $z \mapsto A(z)$ is then holomorphic, in particular continuous, as a map into $\mathcal{B}(L^2(X;E))$, by (5.10.1).

Proposition 5.10.12. *Let $\Omega \subset \mathbb{C}$ be open and connected and $A : \Omega \to \Psi^{-\infty}(X;E)$ holomorphic. Then either $\mathrm{Id}+A(z)$ is invertible for no $z \in \Omega$, or there is a discrete subset $D \subset \Omega$ such that $\mathrm{Id}+A(z)$ is invertible for $z \in \Omega\setminus D$; in the second case $(\mathrm{Id}+A(z))^{-1} = \mathrm{Id}+B(z)$, where $B : \Omega\setminus D \to \Psi^{-\infty}(X;E)$ is holomorphic and near each $p \in D$*

$$B(z) = \sum_{j=1}^{N} (z-p)^{-j} B_j + B'(z),$$

with $B_j \in \Psi^{-\infty}(X;E)$ of finite rank and B' holomorphic near p.

The idea of the proof is to reduce to linear algebra. Near any point, $A(z)$ is a small perturbation of a finite-rank operator F. The small part can be inverted by a Neumann series, and what remains is an $N \times N$ matrix $M(z)$ depending holomorphically on z. Invertibility of $\mathrm{Id}+A(z)$ is then the non-vanishing of the holomorphic function $\det M(z)$, whose zeros are isolated unless it vanishes identically.

Proof. Fix $p \in \Omega$. By Lemma 5.10.6 and (5.10.1) there is a finite-rank smoothing operator F, $Fu = \sum_{i=1}^{N} (u, e_i) f_i$ with $e_i, f_i \in \mathcal{C}^\infty(X; E)$ and the f_i linearly independent, such that $\|A(p) - F\|_{\mathcal{B}(L^2)} < \frac{1}{2}$, and then a disc $U \ni p$ in which $\|A(z) - F\| < \frac{1}{2}$. For $z \in U$ set $C(z) = A(z) - F$. Then $\mathrm{Id}+C(z)$ is invertible, with inverse $\mathrm{Id}+C'(z)$ given by a convergent Neumann series, so holomorphic into $\mathcal{B}(L^2)$; and $C' = -C + C(\mathrm{Id}+C)^{-1}C$ is holomorphic into $\Psi^{-\infty}$, since its kernel $C(z)(\mathrm{Id}+C(z))^{-1}C(z)$ is, as in Lemma 5.9.2, a continuous trilinear expression in holomorphic functions. Now

$$\begin{gathered} \mathrm{Id}+A(z) = \big(\mathrm{Id}+K(z)\big)\big(\mathrm{Id}+C(z)\big), \\ K(z) = F(\mathrm{Id}+C'(z)), \quad K(z)u = \sum_{i=1}^{N} ((\mathrm{Id}+C'(z))u, e_i) f_i. \end{gathered} \tag{5.10.5}$$

So $\mathrm{Id}+A(z)$ is invertible if and only if $\mathrm{Id}+K(z)$ is. The range of $K(z)$ lies in $V = \mathrm{span}\{f_i\}$. If $(\mathrm{Id}+K)u = 0$ then $u = -Ku \in V$; since $\mathrm{Id}+K$ is Fredholm of index zero (Theorem 5.9.1), it is invertible if and only if its restriction to V is injective, i.e. if and only if the matrix

$$M(z)_{ij} = \delta_{ij} + ((\mathrm{Id}+C'(z))f_j, e_i), \qquad (\mathrm{Id}+K(z))f_j = \sum_i M(z)_{ij} f_i,$$

is invertible. Its determinant $d(z)$ is holomorphic in U, so the set of $z \in U$ where $\mathrm{Id}+A(z)$ is not invertible is either all of U or discrete in U.

Let Ω_1 be the set of $p \in \Omega$ having a neighbourhood on which $\mathrm{Id}+A(z)$ is nowhere invertible, and Ω_2 the set of p having a neighbourhood in which the non-invertible points are discrete. Both are open, they are disjoint (a punctured neighbourhood of a point of Ω_2 contains invertible points arbitrarily close to it), and by the above they cover Ω. Since Ω is connected, either $\Omega = \Omega_1$ or $\Omega = \Omega_2$.

In the second case, invertibility on $\Omega \setminus D$ with inverse $\mathrm{Id}+B(z)$, $B(z) \in \Psi^{-\infty}$, follows from Lemma 5.10.2, and $B = -A + A(\mathrm{Id}+A)^{-1}A$ is holomorphic as before. Near $p \in D$, with U as above, $(\mathrm{Id}+K)^{-1} = \mathrm{Id}-L(z)$, where $L(z) = (\mathrm{Id}+K|_V)^{-1}K(z)$: indeed $L(z)u \in V$ and $\mathrm{Id}+K$ acts on V by the matrix M, so $(\mathrm{Id}+K)(\mathrm{Id}-L) = \mathrm{Id}+K-K = \mathrm{Id}$. Explicitly

$$L(z)u = \sum_{i,j} \big(M(z)^{-1}\big)_{ji} ((\mathrm{Id}+C'(z))u, e_i) f_j,$$

and $M(z)^{-1} = \mathrm{adj}\, M(z)/d(z)$ is meromorphic at p. The kernel of $u \mapsto ((\mathrm{Id}+C'(z))u, e_i)$ is a holomorphic family of smooth sections in y, so the Laurent coefficients of L at p are smoothing with range in V. Finally $\mathrm{Id}+B(z) = (\mathrm{Id}+C'(z))(\mathrm{Id}-L(z))$ by (5.10.5), and multiplying the Laurent series of L by the Taylor series of the holomorphic $\Psi^{-\infty}$-valued function C' shows that the singular part of B at p has finite-rank smoothing coefficients. □

This is the manifold version of the analytic Fredholm theorem proved, for the isotropic residual algebra, in Chapter 8; there the set of non-invertible points is described as the zero set of a Fredholm determinant, which could be done here too using Remark 5.10.10.

5.11 Elliptic complexes and the Hodge theorem

The first application of the Fredholm theory is to cohomology. The de Rham complex is not made of elliptic operators (d has a large null space), but it is elliptic as a *complex*: its symbol sequence is exact. The Laplacians of such a complex are elliptic, and their null spaces, the harmonic sections, give canonical finite-dimensional representatives of the cohomology.

5.11.1 Elliptic complexes

Let $E_0, E_1, \dots, E_N$ be vector bundles over X and $P_i \in \mathrm{Diff}^k(X; E_i, E_{i+1})$ differential operators with $P_{i+1}P_i = 0$, so that

$$0 \longrightarrow \mathcal{C}^\infty(X; E_0) \xrightarrow{P_0} \mathcal{C}^\infty(X; E_1) \xrightarrow{P_1} \cdots \xrightarrow{P_{N-1}} \mathcal{C}^\infty(X; E_N) \longrightarrow 0 \tag{5.11.1}$$

is a complex (we set $P_{-1} = 0$, $P_N = 0$). Then $\sigma_k(P_{i+1})\sigma_k(P_i) = \sigma_{2k}(P_{i+1}P_i) = 0$, and the complex is called *elliptic* if the symbol sequence

$$0 \to E_{0,x} \xrightarrow{\sigma_k(P_0)(x,\xi)} E_{1,x} \to \cdots \to E_{N,x} \to 0$$

is exact for every $(x, \xi) \in T^*X \setminus 0$. Its cohomology is $H^i = \mathrm{Nul}(P_i)/P_{i-1}\mathcal{C}^\infty(X; E_{i-1})$, where P_i acts on smooth sections.

Choose a positive density and hermitian metrics, and let P_i^* be the formal adjoints; they form the adjoint complex, $P_{i-1}^*P_i^* = (P_iP_{i-1})^* = 0$. Set

$$\Delta_i = P_i^*P_i + P_{i-1}P_{i-1}^* \in \mathrm{Diff}^{2k}(X; E_i).$$

Theorem 5.11.1 (Hodge)**.** *For an elliptic complex* (5.11.1) *on a compact manifold:*

(a) *each Δ_i is elliptic and formally self-adjoint, and its null space*

$$\mathcal{H}^i = \{u \in \mathcal{C}^\infty(X; E_i);\ P_iu = 0,\ P_{i-1}^*u = 0\}$$

is finite-dimensional;

(b) *there is an L^2-orthogonal decomposition*

$$\mathcal{C}^\infty(X; E_i) = \mathcal{H}^i \oplus P_{i-1}\mathcal{C}^\infty(X; E_{i-1}) \oplus P_i^*\mathcal{C}^\infty(X; E_{i+1}); \tag{5.11.2}$$

(c) *the ranges of the P_i on smooth sections are closed, and we have $\mathrm{Nul}(P_i) = \mathcal{H}^i \oplus P_{i-1}\mathcal{C}^\infty(X; E_{i-1})$, so that the natural map $\mathcal{H}^i \to H^i$ is an isomorphism; in particular the cohomology of the complex is finite-dimensional.*

Proof. (a) The principal symbol is $\sigma_{2k}(\Delta_i) = \sigma_k(P_i)^*\sigma_k(P_i) + \sigma_k(P_{i-1})\sigma_k(P_{i-1})^*$, a self-adjoint endomorphism. If it annihilates $v \in E_{i,x}$, $\xi \neq 0$, then $0 = \langle \sigma_{2k}(\Delta_i)v, v\rangle = |\sigma_k(P_i)v|^2 + |\sigma_k(P_{i-1})^*v|^2$, so $v \in \mathrm{Nul}\,\sigma_k(P_i) = \mathrm{Ran}\,\sigma_k(P_{i-1})$ by exactness, and also $v \perp \mathrm{Ran}\,\sigma_k(P_{i-1})$; hence $v = 0$. Being homogeneous of degree $2k$ and invertible on $T^*X \setminus 0$, the symbol is elliptic (Lemma 5.6.2). For $u \in \mathcal{C}^\infty$, $(\Delta_i u, u) = \|P_iu\|^2 + \|P_{i-1}^*u\|^2$, which identifies the null space, and it is finite-dimensional by Theorem 5.9.1.

(b) Since Δ_i is self-adjoint, $\mathrm{Nul}(\Delta_i^*) = \mathcal{H}^i$, and the generalized inverse G_i of Proposition 5.9.3 satisfies $\Delta_iG_i = \mathrm{Id} - \pi_i$, where π_i is orthogonal projection onto $\mathcal{H}^i$. Thus for $u \in \mathcal{C}^\infty(X; E_i)$

$$u = \pi_iu + P_{i-1}(P_{i-1}^*G_iu) + P_i^*(P_iG_iu).$$

The three spaces are mutually orthogonal: $(h, P_{i-1}v) = (P_{i-1}^* h, v) = 0$ and $(h, P_i^* w) = (P_i h, w) = 0$ for $h \in \mathcal{H}^i$, and $(P_{i-1}v, P_i^* w) = (P_i P_{i-1} v, w) = 0$.

(c) If $P_i u = 0$, then $w = P_i^* P_i G_i u$ satisfies $P_i w = P_i(u - \pi_i u - P_{i-1}P_{i-1}^* G_i u) = 0$, so $\|w\|^2 = (P_i G_i u, P_i w) = 0$. Hence $\operatorname{Nul}(P_i) = \mathcal{H}^i \oplus P_{i-1}\mathcal{C}^\infty(X; E_{i-1})$, and $P_{i-1}\mathcal{C}^\infty(X; E_{i-1}) = \operatorname{Nul}(P_i) \cap (\mathcal{H}^i)^\perp$ is closed in $\mathcal{C}^\infty$ (and its closure in L^2 meets $\mathcal{C}^\infty$ in itself). The map $\mathcal{H}^i \to H^i$ is therefore surjective, and it is injective since $\mathcal{H}^i \perp P_{i-1}\mathcal{C}^\infty(X; E_{i-1})$. □

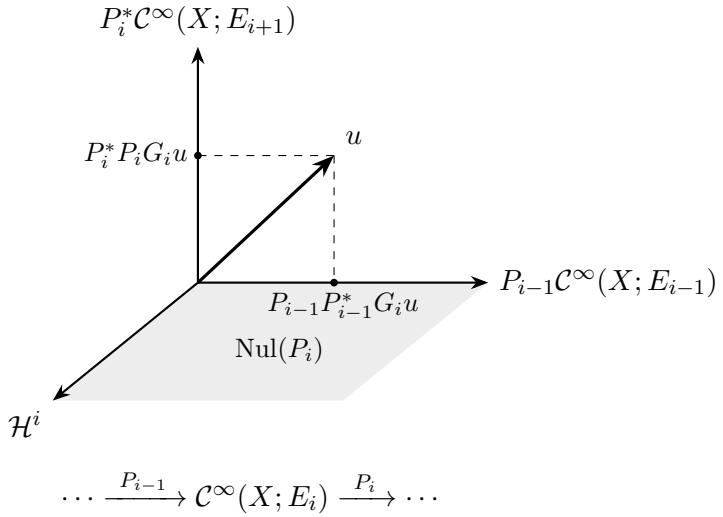


Figure 5.5: The Hodge decomposition (5.11.2) of $\mathcal{C}^\infty(X; E_i)$ into three mutually orthogonal pieces: harmonic sections, the range of P_{i-1}, and the range of P_i^*. The closed sections, $\operatorname{Nul}(P_i)$, form the shaded "plane" spanned by the first two, so each cohomology class $[u] \in H^i$ has exactly one harmonic representative $\pi_i u$. In the picture, u has no harmonic component.

Discussion. The harmonic representative has a variational meaning. If $P_i u = 0$, the elements of the class $u + P_{i-1}\mathcal{C}^\infty(X; E_{i-1})$ differ from $\pi_i u$ by elements orthogonal to $\mathcal{H}^i$, so $\pi_i u$ is the unique element of the class with smallest L^2 norm (Figure 5.5). Ellipticity of the complex is what makes Δ_i elliptic, and hence what makes the minimizer smooth and the space of minimizers finite-dimensional.

5.11.2 The rolled-up complex

The Laplacians can be combined into a single operator. On $E_* = \bigoplus_i E_i$ let $P = \bigoplus_i P_i$ and $P^* = \bigoplus_i P_i^*$, and put

$$D = P + P^* \in \operatorname{Diff}^k(X; E_*), \qquad E_{\text{ev}} = \bigoplus_i E_{2i}, \quad E_{\text{odd}} = \bigoplus_i E_{2i+1}.$$

Since $P^2 = 0$ and $(P^*)^2 = 0$, $D^2 = PP^* + P^*P = \bigoplus_i \Delta_i$. The operator D is formally self-adjoint and interchanges E_{ev} and E_{odd}; let $D_{\text{ev}} : \mathcal{C}^\infty(X; E_{\text{ev}}) \to \mathcal{C}^\infty(X; E_{\text{odd}})$ be its restriction, so that D_{ev}^* is the restriction to odd sections.

Corollary 5.11.2. *For an elliptic complex (5.11.1), D and D_{ev} are elliptic,* $\operatorname{Nul}(D) = \bigoplus_i \mathcal{H}^i$, *and*

$$\operatorname{ind}(D_{\text{ev}}) = \sum_i (-1)^i \dim H^i,$$

the Euler characteristic of the complex.

Proof. $\sigma_k(D)^2 = \sigma_{2k}(D^2) = \bigoplus_i \sigma_{2k}(\Delta_i)$ is invertible on $T^*X \setminus 0$ by Theorem 5.11.1(a), so $\sigma_k(D)$ is invertible there. For smooth u, $(D^2u, u) = \|Du\|^2$, so $\mathrm{Nul}(D) = \mathrm{Nul}(D^2) = \bigoplus_i \mathcal{H}^i$ (both consist of smooth sections by elliptic regularity). Hence $\mathrm{Nul}(D_{\mathrm{ev}}) = \bigoplus_i \mathcal{H}^{2i}$ and $\mathrm{Nul}(D^*_{\mathrm{ev}}) = \bigoplus_i \mathcal{H}^{2i+1}$, and the result follows from Theorem 5.9.1(d) and Theorem 5.11.1(c). □

5.11.3 Distributional cohomology

The operators P_i act on distributional sections, and the complex (5.11.1) with $\mathcal{C}^\infty$ replaced by $\mathcal{C}^{-\infty}$ has cohomology $H^i_{-\infty} = \{u \in \mathcal{C}^{-\infty}(X; E_i);\ P_i u = 0\}/P_{i-1}\mathcal{C}^{-\infty}(X; E_{i-1})$. The pairing (u, v) of $u \in \mathcal{C}^{-\infty}$ with $v \in \mathcal{C}^\infty$ (extending the L^2 inner product, as in Proposition 5.8.3(e)) satisfies $(P_i u, v) = (u, P_i^* v)$.

Proposition 5.11.3. *For an elliptic complex on a compact manifold,*

$$\mathcal{C}^{-\infty}(X; E_i) = \mathcal{H}^i \oplus P_{i-1}\mathcal{C}^{-\infty}(X; E_{i-1}) \oplus P_i^*\mathcal{C}^{-\infty}(X; E_{i+1}), \tag{5.11.3}$$

a direct sum, and $\{u \in \mathcal{C}^{-\infty}(X; E_i);\ P_i u = 0\} = \mathcal{H}^i \oplus P_{i-1}\mathcal{C}^{-\infty}(X; E_{i-1})$. *Consequently the inclusion* $\mathcal{C}^\infty \hookrightarrow \mathcal{C}^{-\infty}$ *induces isomorphisms* $H^i \cong H^i_{-\infty}$. *In particular, if* $u \in \mathcal{C}^\infty(X; E_i)$ *and* $u = P_{i-1}v$ *with* $v \in \mathcal{C}^{-\infty}(X; E_{i-1})$, *then* $u = P_{i-1}v'$ *for some* $v' \in \mathcal{C}^\infty(X; E_{i-1})$.

Proof. Let π_i and G_i be the orthogonal projection onto $\mathcal{H}^i$ and the generalized inverse of Δ_i; both are pseudodifferential, so act on distributions, and the identity $\Delta_i G_i = \mathrm{Id} - \pi_i$ extends by continuity. Hence every $u \in \mathcal{C}^{-\infty}(X; E_i)$ is

$$u = \pi_i u + P_{i-1}(P^*_{i-1} G_i u) + P_i^*(P_i G_i u).$$

To see that the sum is direct, suppose $h + P_{i-1}v + P_i^* w = 0$ with $h \in \mathcal{H}^i$ and v, w distributions. Applying P_i gives $P_i P_i^* w = 0$. Then $z = P_i^* w$ satisfies $P_i z = 0$ and $P^*_{i-1} z = 0$, so $\Delta_i z = 0$ and $z \in \mathcal{H}^i$ is smooth, and $\|z\|^2 = (P_i^* w, z) = (w, P_i z) = 0$. Similarly, applying P^*_{i-1} shows that $P_{i-1}v = 0$, and then $h = 0$.

If $P_i u = 0$, write $u = h + P_{i-1}v + P_i^* w$ as above; then $P_i P_i^* w = 0$, so $P_i^* w = 0$ by the argument just given. This proves the description of the null space, and hence that $\mathcal{H}^i \to H^i_{-\infty}$ is an isomorphism. Since $\mathcal{H}^i \to H^i$ is an isomorphism by Theorem 5.11.1, so is $H^i \to H^i_{-\infty}$. The last statement is injectivity of this map. □

In particular, for the de Rham complex below, the cohomology computed with distributional ("currents") or smooth forms is the same. The isomorphism $H^i \to H^i_{-\infty}$ is natural, although the harmonic spaces through which it was proved depend on the choice of density and metrics.

5.11.4 Spectral decomposition

The complex also splits along the spectrum of the Laplacians. By Theorem 5.9.6, each Δ_i (elliptic, formally self-adjoint, of positive order) has discrete spectrum $0 \le \lambda_0 < \lambda_1 < \cdots \to \infty$ with finite-dimensional eigenspaces $E^i_\lambda = \mathrm{Nul}(\Delta_i - \lambda) \subset \mathcal{C}^\infty(X; E_i)$; non-negativity follows from $(\Delta_i u, u) = \|P_i u\|^2 + \|P^*_{i-1} u\|^2$.

Proposition 5.11.4. *For an elliptic complex,* $P_i \Delta_i = \Delta_{i+1} P_i$ *and* $P_i^* \Delta_{i+1} = \Delta_i P_i^*$. *Hence* P_i *maps the orthocomplement* $(\mathcal{H}^i)^\perp \cap \mathcal{C}^\infty(X; E_i)$ *into* $(\mathcal{H}^{i+1})^\perp$, *the complex*

$$0 \to (\mathcal{H}^0)^\perp \xrightarrow{P_0} (\mathcal{H}^1)^\perp \xrightarrow{P_1} \cdots \xrightarrow{P_{N-1}} (\mathcal{H}^N)^\perp \to 0$$

is exact, and for each $\lambda > 0$ the finite-dimensional complex

$$0 \to E_\lambda^0 \xrightarrow{P_0} E_\lambda^1 \xrightarrow{P_1} \cdots \xrightarrow{P_{N-1}} E_\lambda^N \to 0$$

is exact, with $E_\lambda^i = P_{i-1}E_\lambda^{i-1} \oplus P_i^* E_\lambda^{i+1}$ orthogonally. In particular $\sum_i (-1)^i \dim E_\lambda^i = 0$ for every $\lambda > 0$.

Proof. Using $P_iP_{i-1} = 0$ and $P_{i+1}P_i = 0$, both $P_i\Delta_i$ and $\Delta_{i+1}P_i$ equal $P_iP_i^*P_i$; the second identity is the adjoint of the first. If $h \in \mathcal{H}^{i+1}$ and $u \in \mathcal{C}^\infty(X; E_i)$ then $(P_iu, h) = (u, P_i^*h) = 0$, so the range of P_i is orthogonal to $\mathcal{H}^{i+1}$. If $u \perp \mathcal{H}^i$ and $P_iu = 0$, then by Theorem 5.11.1(c) $u = P_{i-1}v$ with v smooth, and replacing v by $v - \pi_{i-1}v$ (note $P_{i-1}\pi_{i-1} = 0$) we may take $v \perp \mathcal{H}^{i-1}$; this is the exactness of the first complex. By the intertwining identities P_i maps E_λ^i into E_λ^{i+1} and P_{i-1}^* maps E_λ^i into E_λ^{i-1}. If $\lambda > 0$, $u \in E_\lambda^i$ and $P_iu = 0$, then $\lambda u = \Delta_i u = P_{i-1}(P_{i-1}^* u)$, so $u = P_{i-1}(\lambda^{-1}P_{i-1}^* u)$ with $\lambda^{-1}P_{i-1}^* u \in E_\lambda^{i-1}$. In general, for $u \in E_\lambda^i$, $u = \lambda^{-1}P_{i-1}(P_{i-1}^* u) + \lambda^{-1}P_i^*(P_iu)$ gives the decomposition, which is orthogonal as in Theorem 5.11.1(b). The alternating sum of the dimensions of an exact sequence of finite-dimensional spaces vanishes. □

Thus, although $D = P + P^*$ mixes the degrees, each of its non-zero eigenspaces decomposes into pieces of pure degree, and the contributions of the non-zero eigenvalues to $\sum_i (-1)^i \dim E_\lambda^i$ cancel. Combined with the heat kernel (not treated here) this cancellation gives the McKean–Singer formula $\sum_i (-1)^i \operatorname{Tr} e^{-t\Delta_i} = \sum_i (-1)^i \dim H^i$ for all $t > 0$.

5.11.5 The de Rham complex and the Laplacian on forms

The exterior derivative $d : \mathcal{C}^\infty(X; \Lambda^p) \to \mathcal{C}^\infty(X; \Lambda^{p+1})$ extends the differential of functions, satisfies $d^2 = 0$, and in local coordinates is $d = \sum_j dx_j \wedge \partial_{x_j} = i \sum_j dx_j \wedge D_{x_j}$. As computed in Section 5.4, $\sigma_1(d)(x, \xi) = i\,\xi\wedge$. The de Rham complex

$$0 \to \mathcal{C}^\infty(X; \Lambda^0) \xrightarrow{d} \mathcal{C}^\infty(X; \Lambda^1) \xrightarrow{d} \cdots \xrightarrow{d} \mathcal{C}^\infty(X; \Lambda^n) \to 0 \tag{5.11.4}$$

is elliptic: for $\xi \neq 0$, the forms $\alpha \in \Lambda_x^p$ with $\xi \wedge \alpha = 0$ are exactly those of the form $\xi \wedge \beta$, $\beta \in \Lambda_x^{p-1}$, as one sees by choosing a basis $e_1 = \xi, e_2, \dots, e_n$ of T_x^*X and expanding α in the induced basis of Λ_x^p. The cohomology of (5.11.4) is the de Rham cohomology $H^p_{\mathrm{dR}}(X)$.

Choose a Riemannian metric g on X, which induces hermitian metrics on the Λ^p and the Riemannian density. Let d^* be the formal adjoint of d and

$$\Delta = dd^* + d^*d \in \operatorname{Diff}^2(X; \Lambda^p), \qquad \sigma_2(\Delta)(x, \xi) = |\xi|_g^2 \operatorname{Id},$$

since $\sigma_1(d^*) = -i\,\iota_\xi$ (contraction with the dual vector of ξ) and $\xi \wedge \iota_\xi + \iota_\xi\,\xi\wedge = |\xi|^2$. On functions, Δ is the Laplace–Beltrami operator, non-negative with null space the locally constant functions.

Corollary 5.11.5 (Hodge theorem)**.** *On a compact Riemannian manifold, the space $\mathcal{H}^p$ of harmonic p-forms, $\Delta u = 0$ (equivalently $du = 0$ and $d^*u = 0$), is finite-dimensional, every de Rham cohomology class contains a unique harmonic representative, and*

$$\mathcal{C}^\infty(X; \Lambda^p) = \mathcal{H}^p \oplus d\,\mathcal{C}^\infty(X; \Lambda^{p-1}) \oplus d^*\mathcal{C}^\infty(X; \Lambda^{p+1}).$$

Thus the Betti numbers $b_p = \dim H^p_{\mathrm{dR}}(X)$ are finite. On the full form bundle $\Lambda^* = \bigoplus_p \Lambda^p$, the operator $d + d^*$ satisfies $(d + d^*)^2 = \Delta$, so is elliptic, and it maps even forms to odd forms.

Corollary 5.11.6. *The index of* $d+d^*:\mathcal{C}^\infty(X;\Lambda^{\mathrm{ev}})\to\mathcal{C}^\infty(X;\Lambda^{\mathrm{odd}})$ *is the Euler characteristic* $\chi(X)=\sum_p(-1)^p b_p$.

Proof. This is Corollary 5.11.2 for the de Rham complex: $D=d+d^*$, and the null space of $d+d^*$ on even (odd) forms is $\bigoplus_p\mathcal{H}^{2p}$ ($\bigoplus_p\mathcal{H}^{2p+1}$), whose dimension is $\sum_p b_{2p}$ ($\sum_p b_{2p+1}$) by Corollary 5.11.5. □

In particular, the index of this operator does not depend on the Riemannian metric, as also follows from the homotopy invariance of the index.

5.12 Pseudodifferential projections

In this section E is a hermitian bundle over the compact manifold X, with L^2 defined by a positive density. We show that every smooth family of projections on the cosphere bundle is the principal symbol of a pseudodifferential projection, in three steps: a symbolic construction, which makes P^2-P smoothing; a contour integral, which removes the smoothing error; and, in Proposition 5.12.4, a characterization of the closed subspaces which arise as ranges of such projections.

Lemma 5.12.1. *Let* $Q_0\in\Psi^0(X;E)$ *satisfy* $Q_0^2-Q_0\in\Psi^{-1}(X;E)$. *Then there exists* $Q\in\Psi^0(X;E)$ *with* $Q-Q_0\in\Psi^{-1}(X;E)$ *and* $Q^2-Q\in\Psi^{-\infty}(X;E)$. *If* Q_0 *is classical, or formally self-adjoint, then* Q *may be chosen with the same property.*

Proof. Define inductively

$$E_k=Q_k^2-Q_k,\qquad Q_{k+1}=Q_k+(\mathrm{Id}-2Q_k)E_k.$$

Each Q_k and E_k is a polynomial in Q_0, so they all commute. Writing $Q=Q_k$, $E=E_k$ and using $(\mathrm{Id}-2Q)^2=\mathrm{Id}+4E$,

$$E_{k+1}=E+(2Q-\mathrm{Id})(\mathrm{Id}-2Q)E+(\mathrm{Id}-2Q)^2E^2=E-(\mathrm{Id}+4E)E+(\mathrm{Id}+4E)E^2=4E^3-3E^2.$$

Hence, by induction, $E_k\in\Psi^{-2^k}$ and $Q_{k+1}-Q_k=(\mathrm{Id}-2Q_k)E_k\in\Psi^{-2^k}$ (this is Newton's iteration for the equation $Q^2=Q$). By Proposition 5.5.9 there exists $Q\in\Psi^0$ with $Q-Q_k\in\Psi^{-k-1}$ for all k, since $2^j\geq j+1$. Writing $Q=Q_k+T$,

$$Q^2-Q=E_k+Q_kT+TQ_k+T^2-T\in\Psi^{-k-1}$$

for every k, so Q^2-Q is smoothing. If Q_0 is classical, so are all Q_k and Q may be taken classical. If Q_0 is self-adjoint so are all Q_k, and $\frac{1}{2}(Q+Q^*)$ has the same properties as Q. □

Question. Why use Newton's iteration? Would it not be enough to correct one order at a time, as for the parametrix?

Answer. Correcting one order at a time also works, since only asymptotic summation is needed in the end. But the equation $Q^2=Q$ is non-linear, and a one-order correction requires solving a linear symbol equation at each step. Newton's iteration avoids this: each Q_k is an explicit polynomial in Q_0, so all the Q_k commute and inherit classicality and self-adjointness for free, and the order of the error doubles at each step. □

Proposition 5.12.2. *If* $P\in\Psi^0(X;E)$ *satisfies* $P^2-P\in\Psi^{-\infty}(X;E)$ *then there exists* $\Pi\in\Psi^0(X;E)$ *with* $\Pi^2=\Pi$ *and* $\Pi-P\in\Psi^{-\infty}(X;E)$.

Proof. Let $R = P^2 - P$, which commutes with P. For $\lambda \notin \{0,1\}$ a direct computation gives

$$(\lambda - P)(P + \lambda - 1) = \lambda(\lambda-1) - R = \lambda(\lambda-1)(\operatorname{Id} - S_\lambda), \qquad S_\lambda = \frac{R}{\lambda(\lambda-1)}, \tag{5.12.1}$$

and all these operators commute. Hence if $\lambda(\lambda-1) \notin \operatorname{Spec}(R)$, then $\lambda - P$ is invertible, with

$$(\lambda - P)^{-1} = \frac{P+\lambda-1}{\lambda(\lambda-1)}(\operatorname{Id} - S_\lambda)^{-1} = \frac{P+\lambda-1}{\lambda(\lambda-1)} + \frac{P+\lambda-1}{\lambda(\lambda-1)}\, S_\lambda(\operatorname{Id} - S_\lambda)^{-1}.$$

Since R is compact, $\operatorname{Spec}(R) \setminus \{0\}$ is countable (Theorem A.3.7), so the set of $r \in (\frac14, \frac34)$ such that the circle $\gamma_r = \{|\lambda - 1| = r\}$ meets $\{\lambda; \lambda(\lambda-1) \in \operatorname{Spec}(R)\}$ is countable, and we may fix r with $\gamma = \gamma_r$ disjoint from $\operatorname{Spec}(P)$. Let

$$\Pi = \frac{1}{2\pi i} \oint_\gamma (\lambda - P)^{-1} d\lambda.$$

The first term in the expansion of the resolvent contributes, by the residue theorem (the only pole inside γ is at $\lambda = 1$), exactly P. In the second term, $S_\lambda(\operatorname{Id} - S_\lambda)^{-1} = S_\lambda + S_\lambda(\operatorname{Id} - S_\lambda)^{-1} S_\lambda$ is smoothing, with kernel depending continuously on $\lambda \in \gamma$ by Lemma 5.9.2, and remains so after composition with $P + \lambda - 1$; so its integral is smoothing. Hence $\Pi - P \in \Psi^{-\infty}$.

That $\Pi^2 = \Pi$ is the standard property of Riesz projections: with $\gamma' = \gamma_{r'}$, $r' < r$ close to r and also disjoint from $\operatorname{Spec}(P)$, the resolvent identity $(\lambda - P)^{-1}(\mu - P)^{-1} = ((\lambda - P)^{-1} - (\mu - P)^{-1})/(\mu - \lambda)$ gives

$$\Pi^2 = \frac{1}{(2\pi i)^2} \oint_{\gamma'} \oint_\gamma \frac{(\lambda - P)^{-1} - (\mu - P)^{-1}}{\mu - \lambda}\, d\mu\, d\lambda = \frac{1}{2\pi i} \oint_{\gamma'} (\lambda - P)^{-1} d\lambda = \Pi,$$

since $\oint_\gamma (\mu - \lambda)^{-1} d\mu = 2\pi i$ for λ on γ' (inside γ) and $\oint_{\gamma'} (\mu - \lambda)^{-1} d\lambda = 0$ for μ on γ; the integral over γ' equals that over γ by Cauchy's theorem, the resolvent being holomorphic between the circles. □

If in Proposition 5.12.2 P is formally self-adjoint, then so is Π, which is therefore the orthogonal projection onto its range. Indeed $\big((\lambda - P)^{-1}\big)^* = (\bar\lambda - P)^{-1}$, so

$$\Pi^* = -\frac{1}{2\pi i} \oint_\gamma (\bar\lambda - P)^{-1}\, d\bar\lambda = \frac{1}{2\pi i} \oint_\gamma (\mu - P)^{-1}\, d\mu = \Pi,$$

since $\mu = \bar\lambda$ runs over the same circle γ (which is symmetric about the real axis) with the opposite orientation. Also, Π is classical if P is, since $\Pi - P$ is smoothing.

Theorem 5.12.3. *Let $p \in \mathcal{C}^\infty(S^*X; \hom(E))$ satisfy $p^2 = p$ (we use the same letter for the pull-back of $\hom(E)$ to S^*X). Then there exists $P \in \Psi^0_{\mathrm{ph}}(X; E)$ with*

$$P^2 = P, \qquad \sigma_0(P) = p.$$

If moreover $p^* = p$ pointwise, then P can be chosen with $P^* = P$, i.e. as an orthogonal projection on $L^2(X; E)$.

Proof. By Proposition 5.5.7(d) there is $Q_0 \in \Psi^0_{\mathrm{ph}}(X; E)$ with $\sigma_0(Q_0) = p$, and if $p^* = p$ we replace Q_0 by $\frac12(Q_0 + Q_0^*)$. Then $\sigma_0(Q_0^2 - Q_0) = p^2 - p = 0$, so $Q_0^2 - Q_0 \in \Psi^{-1}_{\mathrm{ph}}$. By Lemma 5.12.1 there is a classical Q, self-adjoint if p is, with $\sigma_0(Q) = p$ and $Q^2 - Q$ smoothing, and Proposition 5.12.2 and the remarks above give P. □

The range of a pseudodifferential projection is a closed subspace of $L^2(X;E)$, which "microlocally" consists of the sections taking values in the range of p. Three examples play a role later: the Szegő projector onto the Hardy space of the circle, for which p is 1 on the positive and 0 on the negative half of $S^*\mathbb{S}^1$ (Chapter 12); the Calderón projector onto the boundary values of solutions of an elliptic equation (Chapter 9); and the projection onto the sum of the eigenspaces with positive eigenvalue of a self-adjoint elliptic operator $A \in \Psi^1(X;E)$, whose symbol is the projection onto the positive eigenspaces of $\sigma_1(A)$ (that it is pseudodifferential follows from the theory of complex powers, $\Pi_+ = \frac{1}{2}(\mathrm{Id} + A|A|^{-1})$ modulo finite rank, see [11]; it gives the boundary condition of Atiyah, Patodi and Singer, see [16]). The following result characterizes the possible ranges.

Proposition 5.12.4. *Let $P \in \Psi^0(X;E)$ satisfy $P^2 - P \in \Psi^{-\infty}(X;E)$, and let $F \subset H^s(X;E)$ be a closed subspace for which there are $A, B \in \Psi^{-\infty}(X;E)$ with*

$$(\mathrm{Id} - P)u = Au \quad \forall u \in F \quad \text{and} \quad (P+B)H^s(X;E) \subset F.$$

Then there exists $B' \in \Psi^{-\infty}(X;E)$ such that $(P+B')^2 = P+B'$ and $F = (P+B')H^s(X;E)$.

Proof. Assume first that $s = 0$. All modifications below change P by smoothing operators, and the hypotheses are preserved, with new A and B, under such changes: if $P' = P + C$ with C smoothing, then $\mathrm{Id} - P' = A - C$ on F and $P' + (B - C) = P + B$.

Step 1. By Proposition 5.12.2 we may assume that P is a projection, $P^2 = P$. Then $\mathrm{Ran}(\mathrm{Id} - P) = \mathrm{Nul}(P)$.

Step 2. Let $E_0 = F \cap \mathrm{Nul}(P)$. On E_0, $u = (\mathrm{Id} - P)u = Au$, so $E_0 \subset \mathrm{Nul}(\mathrm{Id} - A)$, which is finite-dimensional and consists of smooth sections (Theorem 5.9.1). Let $e_1, \dots, e_k$ be an orthonormal basis of E_0 and $f_j = (\mathrm{Id} - P^*)e_j \in \mathcal{C}^\infty(X;E)$; then $(e_l, f_j) = ((\mathrm{Id} - P)e_l, e_j) = \delta_{lj}$, so $Qu = \sum_j (u, f_j)e_j$ is a smoothing projection onto E_0. Moreover $QP = 0$, since $P^* f_j = 0$, and $PQ = 0$, since $e_j \in \mathrm{Nul}(P)$; hence $P_1 = P + Q$ is a projection. If $u \in F$ and $P_1 u = 0$, then $Pu = -Qu \in E_0 \subset \mathrm{Nul}(P)$, so $Pu = P^2 u = 0$, then $Qu = 0$, $u \in E_0$ and $u = Qu = 0$. Thus, replacing P by P_1, we may assume $F \cap \mathrm{Nul}(P) = \{0\}$.

Step 3. Let $G = F + \mathrm{Nul}(P)$. Since $\mathrm{Id} + B = (P + B) + (\mathrm{Id} - P)$, the range of $\mathrm{Id} + B$ lies in G. By Theorem 5.9.1, $\mathrm{Ran}(\mathrm{Id} + B)$ is closed, with L^2-orthogonal complement $N' = \mathrm{Nul}(\mathrm{Id} + B^*) \subset \mathcal{C}^\infty(X;E)$ of finite dimension. Hence $G = \mathrm{Ran}(\mathrm{Id} + B) \oplus (G \cap N')$ is closed, and if C' is the orthocomplement of $G \cap N'$ in N' then $L^2(X;E) = G \oplus C'$ with $C' \subset \mathcal{C}^\infty$ finite-dimensional. Let $c_1, \dots, c_l$ be a basis of C'. Since $C' \cap \mathrm{Nul}(P) = \{0\}$, the $c'_j = Pc_j$ are linearly independent; let h_j be the dual basis of their span with respect to the L^2 inner product and $g_j = P^* h_j$. Then $Q'u = \sum_j (u, g_j)c'_j$ is a smoothing operator with $Q'P = Q'$, $PQ' = Q'$, $Q'c'_j = c'_j$, so $Q'^2 = Q'$ and $P_2 = P - Q'$ is a projection. One checks that $\mathrm{Nul}(P_2) = \mathrm{Nul}(P) \oplus C'$: indeed $P_2 c_j = c'_j - Q'Pc_j = 0$, $\mathrm{Nul}(P) \subset \mathrm{Nul}(P_2)$, and if $P_2 u = 0$ then $Pu = Q'u = \sum a_j Pc_j$, so $u - \sum a_j c_j \in \mathrm{Nul}(P)$. Consequently $L^2 = F \oplus \mathrm{Nul}(P_2)$: the sum is $G \oplus C'$, and if $f = v + c$ with $f \in F$, $v \in \mathrm{Nul}(P)$, $c \in C'$, then $c \in G \cap C' = \{0\}$ and $f \in F \cap \mathrm{Nul}(P) = \{0\}$. Replacing P by P_2, we may therefore assume

$$L^2(X;E) = F \oplus \mathrm{Nul}(P), \tag{5.12.2}$$

a direct, not necessarily orthogonal, sum of closed subspaces.

Step 4. Let P' be the projection with range F and null space $\mathrm{Nul}(P)$, which is bounded by (5.12.2) and the closed graph theorem. Since $\mathrm{Nul}(P') = \mathrm{Nul}(P) = \mathrm{Ran}(\mathrm{Id} - P)$, we have $P'(\mathrm{Id} - P) = 0$ and $P(\mathrm{Id} - P') = 0$, so $P' = P'P$ and $P = PP'$. Hence

$$P' - P = (\mathrm{Id} - P)P' = AP',$$

since $P'u \in F$ and $\mathrm{Id} - P = A$ on F. Also $P'(P+B) = P+B$ since $(P+B)L^2 \subset F$, so $P' - P = P'P - P = B - P'B = (\mathrm{Id} - P')B$. Combining,

$$P' - P = AP' = A(P + (\mathrm{Id} - P')B) = AP + A(\mathrm{Id} - P')B,$$

which is smoothing by Corollary 5.5.4 and Lemma 5.9.2. Thus $B' = P' - P$ (relative to the original P, plus the smoothing changes made in Steps 1–3) has the required properties.

For general s, let Λ_s be as in Corollary 5.9.4. Then $\tilde{P} = \Lambda_s P \Lambda_s^{-1} \in \Psi^0$ satisfies $\tilde{P}^2 - \tilde{P} \in \Psi^{-\infty}$, $\tilde{F} = \Lambda_s F \subset L^2$ is closed, and the hypotheses hold with $\tilde{A} = \Lambda_s A \Lambda_s^{-1}$, $\tilde{B} = \Lambda_s B \Lambda_s^{-1}$. The case $s = 0$ gives $\tilde{B}'$, and $B' = \Lambda_s^{-1} \tilde{B}' \Lambda_s$ has the required properties. □

Remark 5.12.5. The functional calculus of elliptic operators, through the heat kernel e^{-tA}, the resolvent $(A - \lambda)^{-1}$ with parameter, and the complex powers A^z of a positive elliptic operator, is not developed here; for these see [11] and [15].

5.13 The Toeplitz algebra of a pseudodifferential projection

Some natural operators act not on all sections of a bundle but on a closed subspace cut out by a projection: the Hardy space of the circle is the model example. If the projection P is pseudodifferential, the compressed operators PAP inherit a symbol calculus, with symbols now restricted to the range of the principal symbol of P. This gives a Fredholm theory for operators which are not elliptic in the usual sense.

Let E be a hermitian bundle over the compact manifold X, with L^2 defined by a positive density, and let $P \in \Psi^0_{\mathrm{ph}}(X; E)$ be an orthogonal projection, $P^2 = P = P^*$; such projections exist with any prescribed self-adjoint projection-valued principal symbol by Theorem 5.12.3. Its principal symbol $p = \sigma_0(P) \in \mathcal{C}^\infty(S^*X; \hom(E))$ satisfies $p^2 = p = p^*$, so its range

$$V = \operatorname{Ran} p \subset \pi^* E \longrightarrow S^*X$$

is a smooth subbundle of the pull-back of E to S^*X (its rank is locally constant, and p provides local frames), with orthocomplement $V^\perp = \operatorname{Ran}(\mathrm{Id} - p)$. Since P is bounded on every $H^s(X; E)$ and $P^2 = P$,

$$H^s_P = P H^s(X; E) = \operatorname{Nul}(\mathrm{Id} - P) \cap H^s(X; E), \qquad \mathcal{C}^\infty_P = P\,\mathcal{C}^\infty(X; E)$$

are closed subspaces, and $\mathcal{C}^\infty_P = \bigcap_s H^s_P$.

Definition 5.13.1. The *Toeplitz algebra* of P is $\mathcal{T}_P = \bigcup_m \mathcal{T}^m_P$,

$$\mathcal{T}^m_P = \{A \in \Psi^m_{\mathrm{ph}}(X; E);\ A = PAP\} = P\,\Psi^m_{\mathrm{ph}}(X; E)\,P,$$

whose elements we regard as operators on $\mathcal{C}^\infty_P$, or as bounded operators $H^s_P \to H^{s-m}_P$. The *Toeplitz symbol* of $A \in \mathcal{T}^m_P$ is the restriction $\sigma^P_m(A) = \sigma_m(A)|_V$, a section of $N_m \otimes \hom(V)$ over S^*X. An element $A \in \mathcal{T}^m_P$ is *elliptic* if $\sigma^P_m(A)$ is invertible at each point of S^*X.

The two descriptions of $\mathcal{T}^m_P$ agree since $P(PAP)P = PAP$. Note that an element of $\mathcal{T}^m_P$ is never elliptic as an element of $\Psi^m(X; E)$ unless $p = \mathrm{Id}$ (and then $\mathrm{Id} - P$, a compact projection, has finite rank), since its symbol vanishes on $V^\perp$.

Proposition 5.13.2. (a) *$\mathcal{T}_P$ is a filtered $*$-algebra with identity P: $\mathcal{T}^m_P \circ \mathcal{T}^{m'}_P \subset \mathcal{T}^{m+m'}_P$ and $(\mathcal{T}^m_P)^* = \mathcal{T}^m_P$. The residual elements $\mathcal{T}^{-\infty}_P = P\Psi^{-\infty}(X; E)P$ form an ideal, and asymptotic sums of elements $A_j \in \mathcal{T}^{m-j}_P$ exist in $\mathcal{T}^m_P$.*

(b) *The Toeplitz symbol gives a short exact sequence*

$$0 \longrightarrow \mathcal{T}_P^{m-1} \longrightarrow \mathcal{T}_P^m \xrightarrow{\sigma_m^P} \mathcal{C}^\infty(S^*X; N_m \otimes \hom(V)) \longrightarrow 0,$$

and $\sigma_{m+m'}^P(AB) = \sigma_m^P(A)\sigma_{m'}^P(B)$, $\sigma_m^P(A^*) = \sigma_m^P(A)^*$.

Proof. (a) If $A = PAP$ and $B = PBP$ then $AB = P(AB)P$, and $A^* = PA^*P$ since $P^* = P$. If $A' \sim \sum_j A_j$ in Ψ_{ph}^m then $A = PA'P \in \mathcal{T}_P^m$ and $A - \sum_{j<N} A_j = P(A' - \sum_{j<N} A_j)P \in \Psi^{m-N}$.

(b) For $A \in \mathcal{T}_P^m$, $a = \sigma_m(A) = p\,a\,p$, so a vanishes on $V^\perp$ and maps V into V; thus a is determined by $a|_V$, and σ_m^P is multiplicative and compatible with adjoints by Proposition 5.5.7. If $\sigma_m^P(A) = 0$ then $\sigma_m(A) = 0$, so $A \in \Psi_{\mathrm{ph}}^{m-1}$, and $A = PAP \in \mathcal{T}_P^{m-1}$. For surjectivity, given a section b of $N_m \otimes \hom(V)$, let $\tilde{b} = b \circ p$, a section of $N_m \otimes \hom(E)$ with $p\tilde{b} = \tilde{b} = \tilde{b}p$; if $A' \in \Psi_{\mathrm{ph}}^m$ has $\sigma_m(A') = \tilde{b}$, then $A = PA'P \in \mathcal{T}_P^m$ has $\sigma_m(A) = p\tilde{b}p = \tilde{b}$, so $\sigma_m^P(A) = b$. □

Theorem 5.13.3. *Let $A \in \mathcal{T}_P^m$ be elliptic. Then:*

(a) *there is $B \in \mathcal{T}_P^{-m}$ with $AB - P \in \mathcal{T}_P^{-\infty}$ and $BA - P \in \mathcal{T}_P^{-\infty}$;*

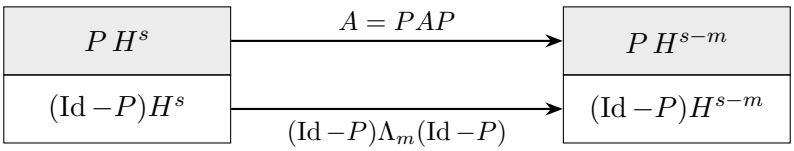
(b) *if $u \in \mathcal{C}^{-\infty}(X; E)$, $u = Pu$ and $Au \in \mathcal{C}^\infty$, then $u \in \mathcal{C}_P^\infty$;*

(c) *for every s, $A : H_P^s \to H_P^{s-m}$ is Fredholm, with null space $\mathrm{Nul}_P(A) = \{u \in \mathcal{C}_P^\infty;\ Au = 0\}$ and range $\{f \in H_P^{s-m};\ (f, v) = 0\ \forall v \in \mathrm{Nul}_P(A^*)\}$, where $A^* \in \mathcal{T}_P^m$ is elliptic; the same holds for $A : \mathcal{C}_P^\infty \to \mathcal{C}_P^\infty$;*

(d) *the Toeplitz index $\mathrm{ind}_P(A) = \dim \mathrm{Nul}_P(A) - \dim \mathrm{Nul}_P(A^*)$ is independent of s, depends only on $\sigma_m^P(A)$, and is constant along families of elliptic elements of $\mathcal{T}_P^m$ which are continuous in $\mathcal{B}(H_P^s, H_P^{s-m})$ for some s.*

The proof reduces everything to the ordinary elliptic theory by a trick: extend A from the range of P to all sections by putting an invertible elliptic operator on the complementary range of $\mathrm{Id} - P$ (Figure 5.6). The extended operator $\tilde{A}$ is elliptic in the usual sense and does not mix the two ranges.

$$\tilde{A} = A + (\mathrm{Id} - P)\Lambda_m(\mathrm{Id} - P)$$



$$\sigma_m(\tilde{A}) = \sigma_m^P(A) \oplus |\xi|^m \,\mathrm{Id}_{V^\perp} \text{ on } \pi^*E = V \oplus V^\perp$$

Figure 5.6: A Toeplitz operator $A = PAP$ acts only on the range of P (shaded). Adding an elliptic operator on the range of $\mathrm{Id} - P$ gives a block-diagonal operator $\tilde{A}$ whose principal symbol is invertible exactly when the Toeplitz symbol $\sigma_m^P(A)$ is. The ordinary theory applied to $\tilde{A}$ yields the Toeplitz parametrix and Fredholm property.

Proof. Fix a metric on T^*X and let $\Lambda_m \in \Psi_{\mathrm{ph}}^m(X; E)$ be self-adjoint and invertible, with principal symbol $|\xi|^m \,\mathrm{Id}_E$; it is constructed as in Corollary 5.9.4, starting from a classical Q with symbol $|\xi|^{m/2} \,\mathrm{Id}_E$. Consider

$$\tilde{A} = A + (\mathrm{Id} - P)\Lambda_m(\mathrm{Id} - P) \in \Psi_{\mathrm{ph}}^m(X; E).$$

Its principal symbol is $\sigma_m(A) + |\xi|^m(\mathrm{Id} - p)$, which in the decomposition $\pi^*E = V \oplus V^\perp$ is the block diagonal endomorphism $\sigma_m^P(A) \oplus |\xi|^m \mathrm{Id}_{V^\perp}$; so $\tilde{A}$ is elliptic. Moreover $\tilde{A}$ is block diagonal with respect to $\mathcal{C}^{-\infty}(X;E) = P\mathcal{C}^{-\infty} \oplus (\mathrm{Id} - P)\mathcal{C}^{-\infty}$: since $A = PAP$,

$$\tilde{A}P = AP = A = PA = P\tilde{A}, \qquad \tilde{A}(\mathrm{Id} - P) = (\mathrm{Id} - P)\Lambda_m(\mathrm{Id} - P) = (\mathrm{Id} - P)\tilde{A}.$$

In particular $\tilde{A}$ acts on H_P^s as A.

(a) If $\tilde{B}$ is a parametrix of $\tilde{A}$, $\tilde{B}\tilde{A} = \mathrm{Id} - R$, $\tilde{A}\tilde{B} = \mathrm{Id} - R'$, then $B = P\tilde{B}P \in \mathcal{T}_P^{-m}$ satisfies $BA = P\tilde{B}\tilde{A}P = P - PRP$ and $AB = P\tilde{A}\tilde{B}P = P - PR'P$.

(b) $\tilde{A}u = Au$ is smooth, so u is smooth by Corollary 5.6.4.

(c) By (a), $B : H_P^{s-m} \to H_P^s$ inverts A up to the operators PRP and $PR'P$, which are compact on H_P^s and H_P^{s-m} by Theorem 5.8.4, and P is the identity on these spaces; so A is Fredholm by Theorem A.3.3. Its null space consists of smooth sections by (b). For the range, if $f \in H_P^{s-m}$ then $f \in \mathrm{Ran}(\tilde{A})$ if and only if $f \in AH_P^s$: if $f = \tilde{A}(u_1 + u_2)$ with $u_1 = Pu$, $u_2 = (\mathrm{Id} - P)u$, then $\tilde{A}u_2 = (\mathrm{Id} - P)\tilde{A}u_2 = (\mathrm{Id} - P)f = 0$, so $f = Au_1$. By Theorem 5.9.1(b) the range of $\tilde{A}$ is the orthocomplement of $\mathrm{Nul}(\tilde{A}^*)$, and $\tilde{A}^* = A^* + (\mathrm{Id} - P)\Lambda_m(\mathrm{Id} - P)$ is block diagonal, with $\mathrm{Nul}(\tilde{A}^*) = \mathrm{Nul}_P(A^*) \oplus \big(\mathrm{Nul}(\tilde{A}^*) \cap (\mathrm{Id} - P)\mathcal{C}^\infty\big)$. Elements of H_P^{s-m} are orthogonal to $(\mathrm{Id} - P)\mathcal{C}^\infty$, since P is self-adjoint, which gives the description of the range. That $A^* \in \mathcal{T}_P^m$ is elliptic follows from Proposition 5.13.2(b). The statement on $\mathcal{C}_P^\infty$ follows as in Theorem 5.9.1(c), using (b).

(d) By (c), the codimension of the range is $\dim \mathrm{Nul}_P(A^*)$ for every s. If $A' \in \mathcal{T}_P^m$ has the same Toeplitz symbol then $A' - A \in \mathcal{T}_P^{m-1}$ is compact $H_P^s \to H_P^{s-m}$, and the last two statements follow from the stability of the index (Theorem A.3.6). □

The operator $\tilde{A}$ in the proof shows a little more: for $m = 0$ one may take $\Lambda_0 = \mathrm{Id}$, and then $\tilde{A} = A + \mathrm{Id} - P$ is an elliptic element of $\Psi^0(X;E)$ with $\mathrm{ind}(\tilde{A}) = \mathrm{ind}_P(A)$, since $\tilde{A}$ is the identity on $(\mathrm{Id} - P)L^2$. Thus the Toeplitz index is an ordinary index. In general, Toeplitz ellipticity concerns the symbol only on V, and the most important elliptic Toeplitz operators are those of the form

$$T_a = PaP, \qquad a \in \mathcal{C}^\infty(X; \mathrm{End}(E)),$$

with $\sigma_0^P(T_a) = p\,a\,p|_V$; T_a is elliptic if a is invertible, but more generally if only the compression of a to V is invertible.

Example 5.13.4. For $X = \mathbb{S}^1$, $E = \mathbb{C}^N$ and $P = S$ the Szegő projector onto the Hardy space of boundary values of holomorphic functions on the disc, $S^*\mathbb{S}^1$ consists of two circles, V is the trivial bundle $\mathbb{C}^N$ over the positive one and zero over the negative one, and $\mathcal{T}_S$ is the Toeplitz algebra of Chapter 12. There it is shown that $S \in \Psi^0(\mathbb{S}^1)$ and that, for $a \in \mathcal{C}^\infty(\mathbb{S}^1; \mathrm{GL}(N,\mathbb{C}))$, the Toeplitz operator $T_a = SaS$ has index $-\mathrm{wind}(\det a)$, minus the winding number of $\det a$; for instance $T_{e^{i\theta}}$ is the shift $\sum_{k\geq 0} c_k e^{ik\theta} \mapsto \sum_{k\geq 0} c_k e^{i(k+1)\theta}$, which is injective with range of codimension one. The general Toeplitz algebra above extends this to arbitrary pseudodifferential projections; on odd-dimensional manifolds, the spectral projections of Dirac operators lead in this way to the Toeplitz index theorem for Toeplitz operators of Baum and Douglas.

5.14 Exercises

Exercise 5.1. Show that compatibility, as defined in Definition 5.1.1, is an equivalence relation on $\mathcal{C}^\infty$ structures. Conclude that there is a unique maximal $\mathcal{C}^\infty$ structure containing any given $\mathcal{C}^\infty$ structure.

Exercise 5.2. Let $\mathcal{F}$ be a $\mathcal{C}^\infty$ structure on X and let O_a, $a \in \mathcal{A}$, be a covering of X by coordinate neighbourhoods as in (5.1.1). Show that the maximal $\mathcal{C}^\infty$ structure containing $\mathcal{F}$ consists of all functions on X which, on each O_a, are of the form $g \circ \kappa_a$ with g smooth. Conclude that the maximal $\mathcal{C}^\infty$ structure is an algebra.

Exercise 5.3 (Partitions of unity)**.** Show that any $\mathcal{C}^\infty$ manifold admits partitions of unity. That is, if O_a, $a \in \mathcal{A}$, is an open cover of X, then there exist $\rho_{a,i} \in \mathcal{C}^\infty(X)$, $a \in \mathcal{A}$, $i \in \mathbb{N}$, with $0 \le \rho_{a,i} \le 1$, each $\rho_{a,i}$ vanishing outside a compact subset $K_{a,i} \subset O_a$, such that each point has a neighbourhood meeting only finitely many of the $K_{a,i}$, and $\sum_{a,i} \rho_{a,i} = 1$.

Exercise 5.4. Fill in the details in the proof of Proposition 5.3.1: with u_R obtained by replacing a by $a(w', t, \zeta)\chi(\zeta/R)$, show that $u_R \to u$ in $\mathcal{C}^{-\infty}(M)$ and that $u_R|_Z$ converges in $\mathcal{C}^{-\infty}(Z)$ as $R \to \infty$. Deduce that $u|_Z$ does not depend on the coordinates used, and that $(\phi u)|_Z = \phi|_Z\, u|_Z$ for $\phi \in \mathcal{C}^\infty(M)$.

Exercise 5.5. Show that a continuous linear map $A : \mathcal{C}^\infty(X; E) \to \mathcal{C}^\infty(X; F)$ is smoothing if and only if it extends to a continuous linear map $\mathcal{C}^{-\infty}(X; E) \to \mathcal{C}^\infty(X; F)$. Show that smoothing operators are *not* characterized as the bounded operators $L^2(X) \to \mathcal{C}^\infty(X)$, by considering $Au = (u, g) \cdot 1$ with $g \in L^2(X) \setminus \mathcal{C}^\infty(X)$.

Exercise 5.6. Let $A \in \Psi^m_{\mathrm{ph}}(X; E, F)$ and let $f \in \mathcal{C}^\infty(X)$ be real-valued with $df(\bar{x}) \neq 0$. Show that for $u \in \mathcal{C}^\infty(X; E)$, as $t \to +\infty$, $e^{-itf} A(e^{itf} u)(\bar{x}) = t^m \sigma_m(A)(\bar{x}, df(\bar{x}))u(\bar{x}) + O(t^{m-1})$, where $\sigma_m(A)$ is the homogeneous principal symbol. (Use the left-reduced form in local coordinates and Taylor's formula for f.) Conclude that this gives an invariant definition of the principal symbol of classical operators extending (5.4.7).

Exercise 5.7. Show that an elliptic $A \in \Psi^m(X; E)$ which is formally self-adjoint has index zero, and that the same is true of any elliptic operator in $\Psi^m(X; E)$ whose principal symbol can be connected to that of a self-adjoint elliptic operator through elliptic symbols.

Exercise 5.8. Let X be compact, oriented and Riemannian, of dimension n, and let $* : \Lambda^p \to \Lambda^{n-p}$ be the Hodge star. Show that $d^* = \pm * d *$ on p-forms and that $*$ maps harmonic p-forms isomorphically onto harmonic $(n-p)$-forms. Deduce Poincaré duality $b_p = b_{n-p}$, and that $\chi(X) = 0$ if n is odd.

Exercise 5.9. Assuming the properties of the Fredholm determinant in Remark 5.10.10, show that $G^{-\infty}(X; E)$ is dense in $\Psi^{-\infty}(X; E)$. Prove this directly using Lemma 5.10.6 and the density of $\mathrm{GL}(M, \mathbb{C})$ in $M(M, \mathbb{C})$.

Exercise 5.10. For $s \in \mathbb{C}$ and $\omega \in \mathbb{S}^{n-1}$ let $N_{s,\omega}$ be the one-dimensional space of functions f on the ray $\mathbb{R}^+\omega$ with $f(r\omega) = r^s f(\omega)$. Show that $N_s = \bigcup_\omega N_{s,\omega}$ is a smooth line bundle over $\mathbb{S}^{n-1}$ whose smooth sections are the smooth functions on $\mathbb{R}^n \setminus 0$ positively homogeneous of degree s, and that it is trivial (any norm $|\cdot|$ gives the non-vanishing section $|z|^s$). Show that there is a non-vanishing section invariant under all of $\mathrm{GL}(n, \mathbb{R})$ only if $s = 0$, so that N_s is not canonically trivial for $s \neq 0$. Deduce the corresponding facts for the bundles N_m over S^*X used in Proposition 5.5.7.

Exercise 5.11. Let $A \in \Psi^0_{\mathrm{ph}}(X; E, F)$ and let $(\bar{x}, \bar{\xi}) \in T^*X \setminus 0$, $v \in E_{\bar{x}}$. In local coordinates near $\bar{x}$, trivializing E, set $u_t(x) = t^{n/4}\phi(t^{1/2}(x - \bar{x}))e^{it x \cdot \bar{\xi}} v$ with $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$, $\|\phi\|_{L^2} = 1$. Show that, as $t \to \infty$, $u_t \to 0$ weakly in L^2, $\|u_t\|_{L^2}$ converges to a positive limit, and $\|Au_t\|_{L^2}/\|u_t\|_{L^2} \to \|\sigma_0(A)(\bar{x}, \bar{\xi})v\|/\|v\|$ (the norms on the fibres being those of the coordinate trivialization, adjusted by the density and the metrics). Deduce that $\|A + K\| \ge M(A)$ for every compact K, so that the inequality in Proposition 5.8.6 is an equality for classical operators.

Exercise 5.12. Let $\dim X > 0$ and let $\Lambda \in \Psi^1(X;E)$ be elliptic, self-adjoint and positive, with an orthonormal basis of eigenfunctions e_j, $\Lambda e_j = \lambda_j e_j$, $0 < \lambda_1 \leq \lambda_2 \leq \cdots$ (Theorem 5.9.6 and Corollary 5.9.4).

(a) Show that a bounded operator A on $L^2(X;E)$ is smoothing if and only if $\Lambda^N A \Lambda^N$ is bounded for all N, if and only if the matrix $a_{jk} = (Ae_k, e_j)$ satisfies $\sup_{j,k}(\lambda_j\lambda_k)^N|a_{jk}| < \infty$ for all N. (Use Exercise 5.5.)

(b) Using that Λ^{-s} is of trace class for $s > n$ (Lemma 5.10.1), show that $\lambda_j \geq cj^{1/s}$. Assuming in addition the polynomial upper bound $\lambda_j \leq Cj^N$ (which follows from the Weyl asymptotics of Chapter 11), show that $A \mapsto (a_{jk})$ is an isomorphism of $\Psi^{-\infty}(X;E)$ onto the algebra of rapidly decreasing matrices $\{(a_{jk})_{j,k\in\mathbb{N}};\ \sup_{j,k}(jk)^N|a_{jk}| < \infty\ \forall N\}$.

(c) Using the eigenfunctions of the harmonic oscillator, show similarly that $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ (Chapter 7) is isomorphic to the same algebra. Conclude that the smoothing algebras of all compact manifolds of positive dimension are isomorphic to each other and to $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, although there is no natural isomorphism.

Exercise 5.13. Let M be a compact manifold with boundary, such as the radial compactification $\overline{\mathbb{R}^n}$ of Chapter 1, and let ν be a smooth positive density on M. Show that the operators $Au(x) = \int_M A(x,y)u(y)\nu(y)$ with $A \in \mathcal{C}^\infty(M^2)$ form an algebra under composition, and that those with kernels vanishing to infinite order at $(\partial M \times M) \cup (M \times \partial M)$ form a subalgebra, which is not an ideal. For $M = \overline{\mathbb{R}^n}$, using that $\dot{\mathcal{C}}^\infty(\overline{\mathbb{R}^n})$ is identified with $\mathcal{S}(\mathbb{R}^n)$, show that the operators $u \mapsto \int K(x,y)u(y)\,dy$ with $K \in \dot{\mathcal{C}}^\infty(\overline{\mathbb{R}^n} \times \overline{\mathbb{R}^n})$ are exactly the elements of $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, i.e. that $\dot{\mathcal{C}}^\infty(\overline{\mathbb{R}^n} \times \overline{\mathbb{R}^n}) = \mathcal{S}(\mathbb{R}^{2n})$ under the natural identification. (Note that the Lebesgue density is not smooth up to the boundary of $\overline{\mathbb{R}^n}$, but this does not matter for kernels vanishing to infinite order there.)

Exercise 5.14. Let P be as in Section 5.13 and let $A \in \mathcal{T}^m_P$ be elliptic. Show that there is $G \in \mathcal{T}^{-m}_P$ with $GA = P - \pi_N$, $AG = P - \pi_{N^*}$, where π_N, π_{N^*} are the orthogonal projections onto $\mathrm{Nul}_P(A)$ and $\mathrm{Nul}_P(A^*)$. In particular, if A is invertible on $\mathcal{C}^\infty_P$ its inverse lies in $\mathcal{T}^{-m}_P$. (Apply Proposition 5.9.3 to the operator $\tilde{A}$ in the proof of Theorem 5.13.3.)

Exercise 5.15. Let P be as in Section 5.13 and $a, b \in \mathcal{C}^\infty(X;\mathrm{End}(E))$. Show that $T_aT_b - T_{ab} = P[P,a](\mathrm{Id} - P)[P,b]P$, and deduce that $T_aT_b - T_{ab} \in \mathcal{T}^{-1}_P$ if a or b commutes with p on S^*X, in particular if one of them is scalar. Let $Y \subset X$ be the (compact) projection to X of the set of points of S^*X where $V \neq 0$. Show that for scalar $a \in \mathcal{C}^\infty(X)$, T_a is elliptic if and only if a vanishes nowhere on Y, and that then T_b is a parametrix of T_a modulo $\mathcal{T}^{-1}_P$ for any $b \in \mathcal{C}^\infty(X)$ equal to $1/a$ near Y.

Notes

This chapter combines Chapter 5 of the 2003 notes with Lectures 5–8 of the 2005 lectures. The invariant definition of pseudodifferential operators through conormal kernels comes from the latter, and the local definition from the former; Theorem 5.4.2 shows that they agree. The source text gave only a sketch of the Fredholm property, carried out on $\mathcal{C}^\infty$ with L^2 estimates; here it is proved on Sobolev spaces, which are defined in Section 5.8, and the range is characterized by duality. The Hodge theorem, only announced in the 2003 notes, is proved for general elliptic complexes as in the 2005 lectures. The existence of the projection Π in Proposition 5.12.2 was stated without proof; we use a contour integral. The functional calculus (heat kernel, resolvent, complex powers) announced in the original is

not included; see [11, 15]. For a full treatment of pseudodifferential operators on manifolds see [8, Chapter 18] and [15].

Several additions come from the later revision of the notes, *From microlocal to global analysis* (2008): the treatment of non-compact manifolds through properly supported operators (Section 5.7), the L^2 bound by the "approximate square root" construction (Proposition 5.8.6; the induction step there, which in the source is only indicated, requires solving the matrix equation $b_0 d + d b_0 = r$), the distributional Hodge decomposition and the exactness of the eigenspace complexes, the symbolic construction of projections (Lemma 5.12.1, where we use Newton's iteration $Q \mapsto Q + (\mathrm{Id} - 2Q)(Q^2 - Q)$, which squares the order of the error at each step), and the remark that the algebra of smoothing operators is, non-naturally, isomorphic to the residual isotropic algebra. The source contains only a heading for the Toeplitz algebra; Section 5.13 was written for this edition, following the model of the Szegő projector in Chapter 12; the general Toeplitz calculus goes back to L. Boutet de Monvel and V. Guillemin, *The spectral theory of Toeplitz operators* (Princeton, 1981), and the index theorem alluded to in Example 5.13.4 is due to P. Baum and R. G. Douglas. The commutator formula Proposition 5.5.10(c) and the analytic Fredholm alternative Proposition 5.10.12 are standard; the latter is proved in the source only for the isotropic algebra.

Chapter 6

Semiclassical and adiabatic calculus

On a compact manifold an elliptic operator is invertible only up to smoothing operators, and the obstruction to inverting it exactly is its index. In index theory one often wants the opposite situation: a way to turn symbolic data into operators which are *exactly* invertible, or exactly projections. The semiclassical calculus provides this. One introduces a small parameter ϵ and quantizes symbols at frequencies of size $1/\epsilon$; as $\epsilon \downarrow 0$ the operators behave more and more like their symbols, and the errors become small in norm, not merely compact.

In this chapter we study one-parameter families of pseudodifferential operators A_ϵ, $0 < \epsilon \leq 1$, which degenerate in a controlled way as $\epsilon \downarrow 0$: the *semiclassical* families. The model is the Schrödinger operator

$$-\tfrac{1}{2}\hbar \frac{d^2}{dx^2} + V(x) = -\tfrac{1}{2}\Big(\epsilon \frac{d}{dx}\Big)^2 + V(x), \qquad \hbar = \epsilon^2,$$

the quantization of the classical energy $E(x,\xi) = \frac{1}{2}\xi^2 + V(x)$. For $\epsilon > 0$ this is an ordinary elliptic operator, but as $\epsilon \downarrow 0$ it "becomes" the classical observable E; the derivatives only enter in the combination ϵD. The general semiclassical family is obtained by quantizing a symbol $a(\epsilon, x, \eta)$, depending smoothly on $\epsilon \in [0,1]$, at the rescaled frequency $\eta = \epsilon\xi$:

$$A_\epsilon u(x) = \int e^{i(x-y)\cdot\xi} a(\epsilon, x, \epsilon\xi) u(y)\, dy d\!\!\!^{-}\xi = (2\pi\epsilon)^{-n} \int e^{i(x-y)\cdot\eta/\epsilon} a(\epsilon, x, \eta) u(y)\, dy\, d\eta.$$

Such families form an algebra with *two* symbol maps: the *semiclassical symbol* $\sigma_{\mathrm{sl}}(A) = a(0,\cdot,\cdot)$, a function on the whole cotangent bundle which is multiplicative and records the leading behaviour as $\epsilon \downarrow 0$ (the operators "become commutative" in the limit), and a rescaled version of the usual principal symbol, uniform in ϵ. The two are compatible at $\epsilon = 0$ and together control the family modulo $\epsilon\Psi_{\mathrm{sl}}^{m-1}$. The residual ideal is that of smoothing families vanishing to infinite order at $\epsilon = 0$, so that elliptic semiclassical families are actually *invertible* for small ϵ. This is what makes the calculus useful in index theory: semiclassical quantization of symbols with values in invertible matrices, or in projections, produces invertible families, respectively families of projections, of operators on a compact manifold. These facts are the basis of the semiclassical proof of the Atiyah–Singer index theorem in Chapter 16.

Section 6.1 introduces the semiclassical symbol spaces and the relevant asymptotic summation. In Section 6.2 the operators on $\mathbb{R}^n$ are defined and, using an exact oscillatory integral formula for the symbol of a composite, it is shown that they form an order-filtered $*$-algebra with an expansion in powers of ϵ. Section 6.3 describes the residual families and the kernels, and Section 6.4 the symbol maps, ellipticity, uniform boundedness on L^2, oscillatory testing and coordinate invariance. In Section 6.5 everything is transferred to compact

manifolds, following Chapter 5, and in Section 6.6 projection-valued symbols are quantized to families of projections. Finally Section 6.7 treats the *adiabatic* families, in which only some of the variables are rescaled, and the two-parameter semiclassical-adiabatic families. We use freely the calculus on $\mathbb{R}^n$ of Chapter 2 and on compact manifolds of Chapter 5; notation is as there, in particular $D = -i\partial$, $đ\xi = (2\pi)^{-n} d\xi$, and $\Delta = D_1^2 + \cdots + D_n^2$ is the positive Laplacian, so that $(1 + \Delta_y)e^{-iy\cdot\zeta} = \langle\zeta\rangle^2 e^{-iy\cdot\zeta}$. The parameter ϵ always ranges over $[0, 1]$ (for symbols) or $(0, 1]$ (for operators).

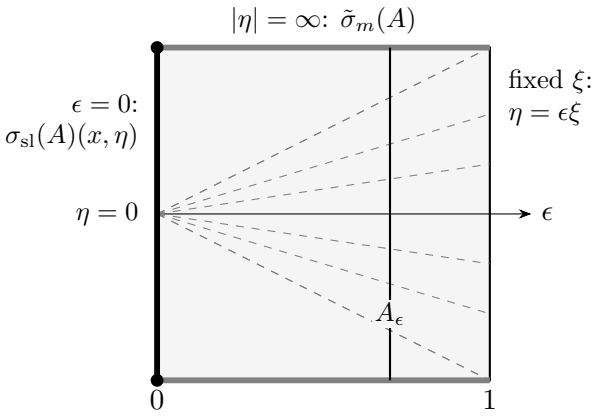


Figure 6.1: Semiclassical symbols as functions of ϵ and the rescaled frequency $\eta = \epsilon\xi$ (the η-axis compactified, x suppressed). At fixed $\epsilon > 0$ (vertical line) we see an ordinary operator A_ϵ, whose principal symbol is read off at $|\eta| = \infty$. A fixed classical frequency ξ moves along a ray (dashed) into $\eta = 0$ as $\epsilon \downarrow 0$, so the semiclassical regime is $|\xi| \sim 1/\epsilon$. The semiclassical symbol lives on the left edge $\epsilon = 0$, the uniform principal symbol on the top and bottom edges; they must agree at the corners (dots), see Proposition 6.4.2.

Discussion. There are two independent ways for a semiclassical symbol to be "small": vanishing at $\epsilon = 0$, and decay as $|\eta| \to \infty$ (Figure 6.1). The usual calculus only sees the second; its errors are smoothing, hence compact, but not small. The semiclassical calculus controls both, and the residual errors are smoothing *and* $O(\epsilon^\infty)$, so for small ϵ they can be removed by a Neumann series. This is the source of all the exact statements in the chapter.

6.1 Semiclassical symbols

We first set up symbols depending smoothly on ϵ up to $\epsilon = 0$, and the asymptotic summation that controls the two kinds of error simultaneously.

Definition 6.1.1. For $m \in \mathbb{R}$ the space $S^m_{\mathrm{sl}}(\mathbb{R}^p; \mathbb{R}^n)$ of *semiclassical symbols of order* m consists of the functions $a \in \mathcal{C}^\infty([0, 1]_\epsilon \times \mathbb{R}^p_z \times \mathbb{R}^n_\eta)$ such that for all $k \in \mathbb{N}_0$, $\alpha \in \mathbb{N}_0^p$, $\beta \in \mathbb{N}_0^n$,

$$|\partial_\epsilon^k D_z^\alpha D_\eta^\beta a(\epsilon, z, \eta)| \le C_{k\alpha\beta}\langle\eta\rangle^{m-|\beta|} \quad \text{on } [0, 1] \times \mathbb{R}^p \times \mathbb{R}^n, \tag{6.1.1}$$

topologized by the best constants in (6.1.1) as seminorms. We set $S^{-\infty}_{\mathrm{sl}}(\mathbb{R}^p; \mathbb{R}^n) = \bigcap_m S^m_{\mathrm{sl}}(\mathbb{R}^p; \mathbb{R}^n)$, and for $N \in \mathbb{N}_0$, $\epsilon^N S^m_{\mathrm{sl}}(\mathbb{R}^p; \mathbb{R}^n) = \{\epsilon^N a; a \in S^m_{\mathrm{sl}}(\mathbb{R}^p; \mathbb{R}^n)\}$. When $p = n$ we simply write $S^m_{\mathrm{sl}}(\mathbb{R}^n)$.

Thus $a \in S^m_{\mathrm{sl}}$ if and only if each $\partial^k_\epsilon a$ is bounded, as ϵ varies, in the Fréchet space $S^m(\mathbb{R}^p; \mathbb{R}^n)$ of Chapter 2. By Taylor's formula (applied pointwise, with the remainder estimated using the bounds on $\partial^{k+2}_\epsilon a$) this is equivalent to a being a $\mathcal{C}^\infty$ map $[0, 1] \to$

$S^m(\mathbb{R}^p;\mathbb{R}^n)$, so $S^m_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n) = \mathcal{C}^\infty([0,1];S^m(\mathbb{R}^p;\mathbb{R}^n))$, and we shall use the two descriptions interchangeably. Clearly $S^m_{\mathrm{sl}}\cdot S^{m'}_{\mathrm{sl}} \subset S^{m+m'}_{\mathrm{sl}}$, D_z preserves S^m_{sl} and D_η maps it to S^{m-1}_{sl}.

Lemma 6.1.2. *Let $a \in S^m_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$ and $N \in \mathbb{N}$. Then*

$$a(\epsilon,z,\eta) = \sum_{k<N} \epsilon^k a_k(z,\eta) + \epsilon^N r_N(\epsilon,z,\eta),$$

with $a_k = \frac{1}{k!}\partial_\epsilon^k a(0,\cdot,\cdot) \in S^m(\mathbb{R}^p;\mathbb{R}^n)$ and $r_N \in S^m_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$. In particular $a \in \epsilon^N S^m_{\mathrm{sl}}$ if and only if $\partial_\epsilon^k a(0,\cdot,\cdot) = 0$ for $k < N$.

Proof. Taylor's formula gives the expansion with

$$r_N(\epsilon,z,\eta) = \frac{1}{(N-1)!}\int_0^1 (1-s)^{N-1}(\partial_\epsilon^N a)(s\epsilon,z,\eta)\,ds,$$

and differentiating under the integral shows that r_N satisfies (6.1.1), since ∂_ϵ^k of the integrand is $s^k(\partial_\epsilon^{N+k}a)(s\epsilon,z,\eta)$. Conversely, if $a = \epsilon^N b$ then the first N derivatives of a at $\epsilon = 0$ vanish. □

The residual symbols which will appear as errors are those which are of order $-\infty$ *and* vanish to infinite order at $\epsilon = 0$.

Lemma 6.1.3. *For $a \in \mathcal{C}^\infty([0,1]\times\mathbb{R}^p\times\mathbb{R}^n)$ the following are equivalent:*

(i) $a \in \bigcap_N \epsilon^N S^{-N}_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$;

(ii) $a \in \bigcap_N \epsilon^N S^{m-N}_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$ *for some (and then every)* $m \in \mathbb{R}$;

(iii) *for all k,α,β and N, $|\partial_\epsilon^k D_z^\alpha D_\eta^\beta a| \le C_{k\alpha\beta N}\,\epsilon^N\langle\eta\rangle^{-N}$;*

(iv) $a \in S^{-\infty}_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$ *and $\partial_\epsilon^k a(0,\cdot,\cdot) = 0$ for all k.*

The space of such symbols is denoted $\epsilon^\infty S^{-\infty}_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$.

Proof. (i)⇒(ii) and (iii)⇒(iv) are clear, and (ii) for one m implies (i), since $\epsilon^{N'}S^{m-N'}_{\mathrm{sl}} \subset \epsilon^N S^{-N}_{\mathrm{sl}}$ if $N' \ge N + \max(m,0)$. If $a = \epsilon^{N'}b$ with $b \in S^{-N'}_{\mathrm{sl}}$ then by Leibniz' formula $|\partial_\epsilon^k D_z^\alpha D_\eta^\beta a| \le C\epsilon^{N'-k}\langle\eta\rangle^{-N'}$; taking $N' = N + k$ gives (iii). Finally (iv)⇒(i): by Lemma 6.1.2, applied with $m = -N$, $a = \epsilon^N r_N$ with $r_N \in S^{-N}_{\mathrm{sl}}$. □

Since we are going to quantize at the frequency $\eta = \epsilon\xi$, we need to know how the rescaled function $a(\epsilon,z,\epsilon\xi)$ behaves as an ordinary symbol.

Lemma 6.1.4. *Let $a \in S^m_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$, put $a^\flat(\epsilon,z,\xi) = a(\epsilon,z,\epsilon\xi)$ and $m_+ = \max(m,0)$.*

(i) *For each $\epsilon \in (0,1]$, $a^\flat(\epsilon,\cdot,\cdot) \in S^m(\mathbb{R}^p;\mathbb{R}^n)$.*

(ii) *For all k,α,β there are constants such that, for $0 < \epsilon \le 1$,*

$$|\partial_\epsilon^k D_z^\alpha D_\xi^\beta a^\flat(\epsilon,z,\xi)| \le C_{k\alpha\beta}\langle\xi\rangle^{m_+ + k - |\beta|};$$

that is, $\partial_\epsilon^k a^\flat$ is bounded in $S^{m_+ + k}(\mathbb{R}^p;\mathbb{R}^n)$ for $\epsilon \in (0,1]$.

Proof. The basic inequalities are, for $0 < \epsilon \le 1$,

$$\langle \epsilon\xi\rangle \le \langle\xi\rangle, \qquad \langle\epsilon\xi\rangle \ge \epsilon\langle\xi\rangle, \qquad \frac{\epsilon}{\langle\epsilon\xi\rangle} = (\epsilon^{-2}+|\xi|^2)^{-1/2} \le \langle\xi\rangle^{-1}.$$

Since $D_\xi^\beta a^\flat = \epsilon^{|\beta|}(D_\eta^\beta a)^\flat$,

$$|D_z^\alpha D_\xi^\beta a^\flat| \le C\epsilon^{|\beta|}\langle\epsilon\xi\rangle^{m-|\beta|} = C\Big(\frac{\epsilon}{\langle\epsilon\xi\rangle}\Big)^{|\beta|}\langle\epsilon\xi\rangle^m \le C\langle\xi\rangle^{-|\beta|}\langle\epsilon\xi\rangle^m.$$

If $m \ge 0$, $\langle\epsilon\xi\rangle^m \le \langle\xi\rangle^m$; if $m < 0$, $\langle\epsilon\xi\rangle^m \le 1$ and also $\langle\epsilon\xi\rangle^m \le \epsilon^m\langle\xi\rangle^m$. This proves (i) and the case $k = 0$ of (ii). For $k \ge 1$, by the chain rule $\partial_\epsilon a^\flat = (\partial_\epsilon a)^\flat + \sum_j \xi_j(\partial_{\eta_j}a)^\flat$, and by induction $\partial_\epsilon^k a^\flat$ is a finite sum of terms $\xi^\gamma\,(\partial_\epsilon^l\partial_\eta^\gamma a)^\flat$ with $l + |\gamma| = k$. Here $\partial_\epsilon^l\partial_\eta^\gamma a \in S^{m-|\gamma|}_{\mathrm{sl}}$, so by the case $k = 0$ the second factor is bounded in $S^{(m-|\gamma|)_+}$, while $\xi^\gamma \in S^{|\gamma|}$; since $|\gamma| + (m-|\gamma|)_+ \le m_+ + k$, (ii) follows. □

6.1.1 Asymptotic summation

For semiclassical symbols there are two independent ways of being small: vanishing at $\epsilon = 0$ and decay in η. We need to sum series which are asymptotic in both senses simultaneously.

Lemma 6.1.5. (i) *If $b_j \in S^{m-j}_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$ for $j \in \mathbb{N}_0$, there exists $b \in S^m_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$ such that*

$$b - \sum_{j<N}\epsilon^j b_j \in \epsilon^N S^{m-N}_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n) \quad \text{for every } N; \tag{6.1.2}$$

we write $b \sim \sum_j \epsilon^j b_j$. It is unique modulo $\epsilon^\infty S^{-\infty}_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$.

(ii) *If $b_j \in S^m_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$ for all j, there exists $b \in S^m_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$ with $b - \sum_{j<N}\epsilon^j b_j \in \epsilon^N S^m_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$ for all N, and if all $b_j \in S^{-\infty}_{\mathrm{sl}}$ then b may be chosen in $S^{-\infty}_{\mathrm{sl}}$ with $b - \sum_{j<N}\epsilon^j b_j \in \epsilon^N S^{-\infty}_{\mathrm{sl}}$.*

Proof. (i) Fix $\psi \in \mathcal{C}^\infty(\mathbb{R})$ with $\psi(s) = 1$ for $s \le \frac12$ and $\psi(s) = 0$ for $s \ge 1$, and for $0 < \delta \le 1$ set $\psi_\delta(\epsilon,\eta) = \psi(\epsilon/(\delta\langle\eta\rangle))$. Then $\epsilon \le \delta\langle\eta\rangle$ on $\operatorname{supp}\psi_\delta$ and $\epsilon \ge \frac12\delta\langle\eta\rangle$ on $\operatorname{supp}(1-\psi_\delta)$. Each ∂_ϵ applied to ψ_δ produces a factor $(\delta\langle\eta\rangle)^{-1}$, and η-derivatives of $\epsilon/(\delta\langle\eta\rangle)$ are bounded on $\operatorname{supp}\psi$ by $C\langle\eta\rangle^{-1}$ times powers of $\epsilon/(\delta\langle\eta\rangle) \le 1$, so

$$|\partial_\epsilon^k D_\eta^\beta \psi_\delta| \le C_{k\beta}\,\delta^{-k}\langle\eta\rangle^{-k-|\beta|}, \tag{6.1.3}$$

with $C_{k\beta}$ independent of δ. For $j \ge N$ consider $g_{j,N} = \epsilon^{j-N}\psi_\delta$. Since $|\partial_\epsilon^i\epsilon^{j-N}| \le C\epsilon^{j-N-i} \le C(\delta\langle\eta\rangle)^{j-N-i}$ on $\operatorname{supp}\psi_\delta$ (for $i \le j-N$, and the derivative vanishes for $i > j-N$), Leibniz' formula and (6.1.3) give

$$|\partial_\epsilon^k D_\eta^\beta g_{j,N}| \le C_{k\beta jN}\,\delta^{j-N-k}\langle\eta\rangle^{j-N-k-|\beta|}.$$

Hence each seminorm of $g_{j,N}b_j$ in S^{m-N}_{sl} involving at most k derivatives in ϵ is bounded by $C\delta^{j-N-k}$ times a seminorm of b_j in S^{m-j}_{sl}. Now choose $\delta_j \in (0,1]$ so small that for all $N < j$ the seminorms of $g_{j,N}b_j$ in S^{m-N}_{sl} with at most $j-N-1$ ϵ-derivatives and at most j derivatives in all are bounded by 2^{-j}, where $g_{j,N}$ is formed with $\delta = \delta_j$; this is possible since all these seminorms carry the factor δ_j^{j-N-k} with $j-N-k \ge 1$. Then for each N and each seminorm of S^{m-N}_{sl}, all but finitely many terms of $\sum_{j\ge N} g_{j,N}b_j$ are bounded by 2^{-j}, so the series converges in S^{m-N}_{sl}. Set

$$b = \sum_j \epsilon^j\psi_{\delta_j}b_j = \epsilon^N\sum_{j\ge N} g_{j,N}b_j + \sum_{j<N}\epsilon^j\psi_{\delta_j}b_j.$$

Taking $N = 0$ shows $b \in S^m_{\mathrm{sl}}$. For $j < N$, $\epsilon^j(\psi_{\delta_j} - 1)b_j = \epsilon^N h_{j,N} b_j$ with $h_{j,N} = \epsilon^{j-N}(\psi_{\delta_j} - 1)$, which vanishes for $\epsilon < \frac{1}{2}\delta_j$; on its support $\epsilon^{j-N} \le (\frac{1}{2}\delta_j\langle\eta\rangle)^{j-N}$ and, as above, $|\partial_\epsilon^k D_\eta^\beta h_{j,N}| \le C\langle\eta\rangle^{j-N-|\beta|}$, so $h_{j,N} \in S^{j-N}_{\mathrm{sl}}$ and $h_{j,N}b_j \in S^{m-N}_{\mathrm{sl}}$. Therefore $b - \sum_{j<N}\epsilon^j b_j = \epsilon^N\big(\sum_{j\ge N} g_{j,N}b_j + \sum_{j<N} h_{j,N}b_j\big) \in \epsilon^N S^{m-N}_{\mathrm{sl}}$. If b' is another such sum then $b - b' \in \bigcap_N \epsilon^N S^{m-N}_{\mathrm{sl}} = \epsilon^\infty S^{-\infty}_{\mathrm{sl}}$ by Lemma 6.1.3.

(ii) The same construction with $\psi(\epsilon/\delta_j)$ in place of ψ_{δ_j} works: now $g_{j,N} = \epsilon^{j-N}\psi(\epsilon/\delta_j)$ has $C^k([0,1])$-norm bounded by $C\delta_j^{j-N-k}$ and does not depend on η, so the series $\sum_{j\ge N} g_{j,N}b_j$ converges in S^m_{sl}, and in every S^{-K}_{sl} if the δ_j are also chosen so that the corresponding seminorms of b_j in S^{-K}_{sl}, $K \le j$, are dominated; and $h_{j,N} = \epsilon^{j-N}(\psi(\epsilon/\delta_j) - 1) \in \mathcal{C}^\infty([0,1])$. □

6.1.2 Classical symbols

A function f on $\mathbb{R}^p\times\mathbb{R}^n$ is said to be homogeneous of degree μ for $|\eta| \ge 1$ if $f(z,t\eta) = t^\mu f(z,\eta)$ for $t \ge 1$, $|\eta| \ge 1$.

Definition 6.1.6. A symbol $a \in S^m_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$ is *classical* if there are $a_j \in S^{m-j}_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$, each homogeneous of degree $m-j$ for $|\eta| \ge 1$ (for every ϵ), such that $a - \sum_{j<N} a_j \in S^{m-N}_{\mathrm{sl}}(\mathbb{R}^p;\mathbb{R}^n)$ for all N. The space of these is $S^m_{\mathrm{sl,ph}}(\mathbb{R}^p;\mathbb{R}^n)$.

For each ϵ, $a(\epsilon,\cdot,\cdot)$ is then a classical symbol in the sense of Chapter 2, and the leading term a_0 restricts to a smooth function on $[0,1]\times\mathbb{R}^p\times\mathbb{S}^{n-1}$. Products of classical symbols are classical, and so are asymptotic sums $\sum_j \epsilon^j b_j$ of classical $b_j \in S^{m-j}_{\mathrm{sl,ph}}$: the remainder in (6.1.2) lies in S^{m-N}_{sl}, so the homogeneous terms of b of degree $m-k$ are $\sum_{j\le k}\epsilon^j(b_j)_{k-j}$.

6.2 Semiclassical quantization on $\mathbb{R}^n$

Quantizing at the rescaled frequency is a simple substitution, but the composition formula has to be redone, because the Euclidean estimates are not uniform in ϵ. The key is an exact oscillatory integral formula for the symbol of a composite, from which the expansion in powers of ϵ follows.

Definition 6.2.1. For $a \in S^m_{\mathrm{sl}}(\mathbb{R}^n)$ and $0 < \epsilon \le 1$ the *semiclassical quantization* of a is

$$\mathrm{Op}_\epsilon(a) = \mathrm{Op}(a^\flat(\epsilon,\cdot,\cdot)) \in \Psi^m(\mathbb{R}^n), \qquad a^\flat(\epsilon,x,\xi) = a(\epsilon,x,\epsilon\xi),$$

so that for $u \in \mathcal{S}(\mathbb{R}^n)$

$$\mathrm{Op}_\epsilon(a)u(x) = \int e^{ix\cdot\xi}a(\epsilon,x,\epsilon\xi)\hat u(\xi)\,d\!\!\bar{}\,\xi = (2\pi\epsilon)^{-n}\int e^{i(x-y)\cdot\eta/\epsilon}a(\epsilon,x,\eta)u(y)\,dy\,d\eta, \quad (6.2.1)$$

the last integral being oscillatory. The space $\Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ of *semiclassical families of order* m consists of the families $A = (A_\epsilon)_{0<\epsilon\le1}$, $A_\epsilon = \mathrm{Op}_\epsilon(a)$, with $a \in S^m_{\mathrm{sl}}(\mathbb{R}^n)$. We write $\epsilon^N\Psi^m_{\mathrm{sl}}(\mathbb{R}^n) = \mathrm{Op}_\epsilon(\epsilon^N S^m_{\mathrm{sl}}(\mathbb{R}^n))$, $\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n) = \bigcap_m \Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$, $\Psi^m_{\mathrm{sl,ph}}(\mathbb{R}^n) = \mathrm{Op}_\epsilon(S^m_{\mathrm{sl,ph}}(\mathbb{R}^n))$ and

$$\epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n) = \mathrm{Op}_\epsilon(\epsilon^\infty S^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)) = \bigcap_N \epsilon^N\Psi^{-N}_{\mathrm{sl}}(\mathbb{R}^n).$$

By Lemma 6.1.4(i), $A_\epsilon \in \Psi^m(\mathbb{R}^n)$ for each $\epsilon > 0$; but note that the estimates on its left symbol $a^\flat(\epsilon)$ in S^m are uniform in ϵ only if $m \le 0$ and the derivatives in ξ are not counted. Since $\mathrm{Op} : S^m(\mathbb{R}^n;\mathbb{R}^n) \to \Psi^m(\mathbb{R}^n)$ is injective (Corollary 2.3.4), A determines $a(\epsilon,\cdot,\cdot)$ for every $\epsilon > 0$, hence by continuity also for $\epsilon = 0$: *the map* $\mathrm{Op}_\epsilon : S^m_{\mathrm{sl}}(\mathbb{R}^n) \to \Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ *is a bijection.* This justifies the notation in the definition (e.g. $\Psi^{-\infty}_{\mathrm{sl}} = \mathrm{Op}_\epsilon(S^{-\infty}_{\mathrm{sl}})$ by

Corollary 2.3.4(ii)). The Schwartz kernel of A_ϵ is, with $\mathcal{F}^{-1}$ the inverse Fourier transform in η,

$$K_{A_\epsilon}(x,y) = \epsilon^{-n} K\Big(\epsilon, x, \frac{x-y}{\epsilon}\Big), \qquad K(\epsilon,x,Z) = \int e^{iZ\cdot\eta} a(\epsilon,x,\eta)đ\eta = \mathcal{F}^{-1}_{\eta\to Z} a(\epsilon,x,\cdot), \tag{6.2.2}$$

as follows from (6.2.1) by the change of variables $\eta = \epsilon\xi$. So as $\epsilon \downarrow 0$ the kernel concentrates at the diagonal, on the scale $|x-y| \sim \epsilon$; this is studied in Section 6.3.

Discussion. Formula (6.2.2) is the heart of the matter. An ordinary pseudodifferential operator has a kernel singular on the diagonal and smooth away from it. A semiclassical family has, in addition, a length scale: its kernel is spread over a neighbourhood of width ϵ of the diagonal, like an approximate identity $\epsilon^{-n}\phi((x-y)/\epsilon)$. As $\epsilon \downarrow 0$ it therefore acts more and more locally, and only the values of the symbol near each point matter; this is why the symbols multiply in the limit.

Example 6.2.2. $\mathrm{Op}_\epsilon(1) = \mathrm{Id}$, $\mathrm{Op}_\epsilon(f(x))$ is multiplication by $f \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$, and $\mathrm{Op}_\epsilon(\eta^\alpha) = \epsilon^{|\alpha|} D^\alpha$. More generally, if $p_\alpha \in \mathcal{C}^\infty([0,1]; \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n))$ (all derivatives in (ϵ, x) bounded) then

$$P_\epsilon = \sum_{|\alpha|\le m} p_\alpha(\epsilon,x)(\epsilon D)^\alpha = \mathrm{Op}_\epsilon(p), \qquad p(\epsilon,x,\eta) = \sum_{|\alpha|\le m} p_\alpha(\epsilon,x)\eta^\alpha \in S^m_{\mathrm{sl,ph}}(\mathbb{R}^n).$$

For instance the Schrödinger operator $-\epsilon^2\Delta_{\mathrm{eucl}} + V$, with $\Delta_{\mathrm{eucl}} = \sum_j \partial^2_{x_j}$ and $V \in \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ real, is $\mathrm{Op}_\epsilon(|\eta|^2 + V(x))$.

As in Chapter 2 it is convenient to allow amplitudes depending on both x and y: for $p \in S^m_{\mathrm{sl}}(\mathbb{R}^{2n}; \mathbb{R}^n)$, i.e. $p = p(\epsilon, x, y, \eta)$, we set

$$\mathrm{Op}_\epsilon(p) = \mathrm{Op}(p(\epsilon, x, y, \epsilon\xi)),$$

which for each $\epsilon > 0$ is an element of $\Psi^m(\mathbb{R}^n)$ given by an amplitude in $\mathcal{A}^m_0(\mathbb{R}^n)$, by Lemma 6.1.4(i) with $p = 2n$. That such families are again in $\Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ is part of Theorem 6.2.6 below.

6.2.1 An oscillatory integral

The proof that the composition theorem of Chapter 2 survives the semiclassical rescaling with uniform control in ϵ rests on an exact formula for the symbol of a composite. It involves the following class of functions on $\mathbb{R}^n_x \times \mathbb{R}^n_\xi \times \mathbb{R}^n_y \times \mathbb{R}^n_\zeta$.

Definition 6.2.3. For $\mu, w \in \mathbb{R}$ and $k \ge 0$ let $\mathcal{Q}^\mu_{k,w}$ be the space of $q \in \mathcal{C}^\infty(\mathbb{R}^{4n})$ such that for all multi-indices

$$|\partial^\alpha_x \partial^\beta_\xi \partial^\gamma_y \partial^\delta_\zeta q(x,\xi,y,\zeta)| \le C_{\alpha\beta\gamma\delta} \langle\xi\rangle^{\mu-|\beta|} \langle\zeta\rangle^{k+|\beta|+|\delta|} \langle y\rangle^w, \tag{6.2.3}$$

with the best constants as seminorms.

The typical elements are as follows. If $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ and $0 \le t \le 1$ then, by Peetre's inequality (Lemma 2.1.3),

$$\langle\xi + t\zeta\rangle^s \le 2^{|s|/2} \langle\xi\rangle^s \langle\zeta\rangle^{|s|}, \tag{6.2.4}$$

so $|\partial^\beta_\xi \partial^\delta_\zeta [a(x,\xi+t\zeta)]| \le C t^{|\delta|} \langle\xi+t\zeta\rangle^{m-|\beta|-|\delta|} \le C\langle\xi\rangle^{m-|\beta|} \langle\zeta\rangle^{|m|+|\beta|+|\delta|}$, using $\langle\xi\rangle^{-|\delta|} \le 1$. Hence, uniformly in $t \in [0,1]$,

$$\begin{aligned} a(x,\xi+t\zeta)\, b(x+y,\xi) &\in \mathcal{Q}^{m+m'}_{|m|,0} \quad \text{if } a \in S^m(\mathbb{R}^n;\mathbb{R}^n),\ b \in S^{m'}(\mathbb{R}^n;\mathbb{R}^n), \\ p(x, x+y, \xi+t\zeta) &\in \mathcal{Q}^m_{|m|,0} \quad \text{if } p \in S^m(\mathbb{R}^{2n};\mathbb{R}^n), \end{aligned} \tag{6.2.5}$$

with seminorms bounded by products of seminorms of a and b, respectively by seminorms of p. Fix $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\chi = 1$ near 0.

Lemma 6.2.4. *Let $q \in \mathcal{Q}^\mu_{k,w}$. The limit*

$$J(q)(x,\xi) = \lim_{s\downarrow 0} (2\pi)^{-n} \iint e^{-iy\cdot\zeta}\chi(sy)\chi(s\zeta)q(x,\xi,y,\zeta)\,dy\,d\zeta$$

exists, and for integers M, L with $2M > n + w$ and $2L > n + k + 2M$,

$$J(q)(x,\xi) = (2\pi)^{-n} \iint e^{-iy\cdot\zeta}\langle y\rangle^{-2M}(1+\Delta_\zeta)^M\Big[\langle\zeta\rangle^{-2L}(1+\Delta_y)^L q(x,\xi,y,\zeta)\Big]\,dy\,d\zeta, \quad (6.2.6)$$

an absolutely convergent integral. Moreover:

(i) *J is a continuous linear map $\mathcal{Q}^\mu_{k,w} \to S^\mu(\mathbb{R}^n;\mathbb{R}^n)$, and $\partial_x J(q) = J(\partial_x q)$, $\partial_\xi J(q) = J(\partial_\xi q)$;*

(ii) *$J(\zeta^\alpha q) = J(D_y^\alpha q)$ for all α;*

(iii) *if q is independent of ζ then $J(q)(x,\xi) = q(x,\xi,0,\cdot)$; if q is independent of y then $J(q)(x,\xi) = q(x,\xi,\cdot,0)$;*

(iv) *if q depends on a parameter t in an interval, with $\partial_t^j q \in \mathcal{Q}^\mu_{k,w}$ bounded for each j, then $\partial_t^j J(q) = J(\partial_t^j q)$, and $J(q)$ satisfies symbol estimates of order μ uniformly in t, together with all its t-derivatives.*

(v) *if q_l is a bounded sequence in $\mathcal{Q}^\mu_{k,w}$ converging, with all derivatives, locally uniformly to q, then $J(q_l) \to J(q)$ pointwise, with $|J(q_l)| \le C\langle\xi\rangle^\mu$.*

Proof. For $s \in (0,1]$ let $q_s = \chi(sy)\chi(s\zeta)q$. Since $\partial_y^\gamma[\chi(sy)] = s^{|\gamma|}(\partial^\gamma\chi)(sy)$ is bounded uniformly in s, the functions q_s are bounded in $\mathcal{Q}^\mu_{k,w}$. They have compact support in (y,ζ), so using $(1+\Delta_y)^L e^{-iy\cdot\zeta} = \langle\zeta\rangle^{2L}e^{-iy\cdot\zeta}$ and $(1+\Delta_\zeta)^M e^{-iy\cdot\zeta} = \langle y\rangle^{2M}e^{-iy\cdot\zeta}$ we may integrate by parts (the operators are formally self-adjoint) to obtain (6.2.6) with q replaced by q_s, for the integral defining J before the limit. The function in square brackets is a sum of terms $\partial_\zeta^\delta(\langle\zeta\rangle^{-2L})\partial_\zeta^{\delta'}\partial_y^\gamma q_s$ with $|\gamma| \le 2L$, $|\delta| + |\delta'| \le 2M$, and $|\partial_\zeta^\delta\langle\zeta\rangle^{-2L}| \le C\langle\zeta\rangle^{-2L}$; so by (6.2.3) the integrand is bounded by

$$C\,\langle y\rangle^{w-2M}\langle\zeta\rangle^{k+2M-2L}\langle\xi\rangle^\mu, \quad (6.2.7)$$

uniformly in s, with C a seminorm of q. This is integrable by the choice of M and L. As $s \downarrow 0$ every derivative of $\chi(sy)\chi(s\zeta)$ of positive order tends to zero pointwise and $\chi(sy)\chi(s\zeta) \to 1$, so by dominated convergence the integrals converge to the right side of (6.2.6). In particular the limit exists and the formula does not depend on M and L.

(i) The bound (6.2.7) gives $|J(q)| \le C\langle\xi\rangle^\mu$. Differentiation under the integral in (6.2.6) is justified by the same bounds, and gives $\partial_x^\alpha\partial_\xi^\beta J(q) = J(\partial_x^\alpha\partial_\xi^\beta q)$. Since $\partial_x^\alpha\partial_\xi^\beta q \in \mathcal{Q}^{\mu-|\beta|}_{k+|\beta|,w}$, we get $|\partial_x^\alpha\partial_\xi^\beta J(q)| \le C\langle\xi\rangle^{\mu-|\beta|}$, with C a seminorm of q.

(ii) Note $\zeta^\alpha q \in \mathcal{Q}^\mu_{k+|\alpha|,w}$. Since $\zeta_j e^{-iy\cdot\zeta} = -D_{y_j}e^{-iy\cdot\zeta}$, integration by parts gives

$$\iint e^{-iy\cdot\zeta}\chi(sy)\chi(s\zeta)\zeta^\alpha q\,dy\,d\zeta = \iint e^{-iy\cdot\zeta}\chi(s\zeta)D_y^\alpha[\chi(sy)q]\,dy\,d\zeta.$$

Expanding by Leibniz' formula, the term in which no derivative falls on $\chi(sy)$ tends to $(2\pi)^n J(D_y^\alpha q)$, and the others are, up to constants, $s^{|\gamma|}$ times the regularized integrals of $(D^\gamma\chi)(sy)\chi(s\zeta)D_y^{\alpha-\gamma}q$, which are bounded uniformly in s by the argument above (with χ replaced by $D^\gamma\chi$ in the y variable). Hence they tend to zero.

(iii) If $q = f(x,\xi,y)$, then with $\check\chi(y) = (2\pi)^{-n}\int e^{-iy\cdot\zeta}\chi(\zeta)\,d\zeta \in \mathcal{S}(\mathbb{R}^n)$, which has integral $\chi(0) = 1$,

$$(2\pi)^{-n}\iint e^{-iy\cdot\zeta}\chi(sy)\chi(s\zeta)f\,dy\,d\zeta = \int \chi(sy)f(x,\xi,y)s^{-n}\check\chi(y/s)\,dy$$
$$= \int \chi(s^2Y)f(x,\xi,sY)\check\chi(Y)\,dY,$$

which tends to $f(x,\xi,0)$ by dominated convergence, since $|f(x,\xi,sY)| \le C\langle Y\rangle^{|w|}$. The case of q independent of y is the same with the roles of y and ζ exchanged.

(iv) and (v) follow from (6.2.6) by differentiation under the integral and dominated convergence, respectively, using (6.2.7). □

Proposition 6.2.5. (i) *If $p \in \mathcal{A}^m_w(\mathbb{R}^n)$ then the left symbol of $\mathrm{Op}(p)$ is*

$$\sigma_L(\mathrm{Op}(p))(x,\xi) = J\big(p(x,x+y,\xi+\zeta)\big)(x,\xi).$$

(ii) *If $a \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ and $b \in S^{m'}(\mathbb{R}^n;\mathbb{R}^n)$ then*

$$\sigma_L(\mathrm{Op}(a)\,\mathrm{Op}(b))(x,\xi) = J\big(a(x,\xi+\zeta)b(x+y,\xi)\big)(x,\xi).$$

Formally, these are obtained by inserting $u(y') = \int e^{iy'\cdot\theta}\hat u(\theta)đ\theta$ into the defining integrals and changing variables; the proof justifies this.

Proof. The function q in (i) lies in $\mathcal{Q}^m_{|m|,|w|}$ (as in (6.2.5), the weight $\langle x-y'\rangle^w$ of p becoming $\langle y\rangle^{|w|}$), and in (ii) it lies in $\mathcal{Q}^{m+m'}_{|m|,0}$. So in both cases $J(q) \in S^\mu(\mathbb{R}^n;\mathbb{R}^n)$ with $\mu = m$, resp. $m+m'$, and it suffices to show that the operators agree on each $u \in \mathcal{S}(\mathbb{R}^n)$.

(i) Suppose first that $p(x,y',\xi) = 0$ for $|\xi| \ge R$. Then for $u \in \mathcal{S}(\mathbb{R}^n)$ the integral $\mathrm{Op}(p)u(x) = \iint e^{i(x-y')\cdot\xi}p(x,y',\xi)u(y')\,dy'đ\xi$ converges absolutely, so equals the limit as $s \downarrow 0$ of the same integral with $u(y')$ replaced by $\chi(sy')u(y')$. In the latter we insert $u(y') = \int e^{iy'\cdot\theta}\hat u(\theta)đ\theta$; the resulting triple integral converges absolutely, and after the substitution $y' = x+y$, $\xi = \theta+\zeta$ (so that $(x-y')\cdot\xi + y'\cdot\theta = x\cdot\theta - y\cdot\zeta$) it becomes

$$\int e^{ix\cdot\theta}\hat u(\theta)\,F_s(x,\theta)đ\theta, \qquad F_s(x,\theta) = (2\pi)^{-n}\iint e^{-iy\cdot\zeta}\chi(s(x+y))q(x,\theta,y,\zeta)\,dy\,d\zeta.$$

For fixed θ, q has compact support in ζ, so we may integrate by parts with $(1+\Delta_\zeta)^M$ to gain the factor $\langle y\rangle^{-2M}$; dominated convergence (the ζ-integration being over the ball $|\theta+\zeta| \le R$) shows that $F_s(x,\theta)$ is bounded uniformly in s and θ and converges to the same limit as the regularization in Lemma 6.2.4, i.e. to $J(q)(x,\theta)$. Hence $\mathrm{Op}(p)u(x) = \int e^{ix\cdot\theta}J(q)(x,\theta)\hat u(\theta)đ\theta = \mathrm{Op}(J(q))u(x)$, by (2.2.8).

For general p let $p_l = \chi(\xi/l)p$ (with χ as in Lemma 2.1.5). By Proposition 2.2.5, $\mathrm{Op}(p_l)u(x) \to \mathrm{Op}(p)u(x)$. The corresponding $q_l = \chi((\xi+\zeta)/l)q$ are bounded in $\mathcal{Q}^m_{|m|,|w|}$ and converge with all derivatives locally uniformly to q, so by Lemma 6.2.4(v) and dominated convergence in $\int e^{ix\cdot\theta}J(q_l)(x,\theta)\hat u(\theta)đ\theta$, also $\mathrm{Op}(J(q_l))u(x) \to \mathrm{Op}(J(q))u(x)$. This proves (i).

(ii) Let $v = \mathrm{Op}(b)u \in \mathcal{S}(\mathbb{R}^n)$, so $v(x') = \int e^{ix'\cdot\theta}b(x',\theta)\hat u(\theta)đ\theta$. If a vanishes for $|\xi| \ge R$, then $\mathrm{Op}(a)v(x) = \iint e^{i(x-x')\cdot\xi}a(x,\xi)v(x')\,dx'đ\xi$ converges absolutely, and we may again insert $\chi(sx')$ and let $s \downarrow 0$. With the cutoff, the triple integral in (x',ξ,θ) converges absolutely, and the substitution $x' = x+y$, $\xi = \theta+\zeta$ turns it into $\int e^{ix\cdot\theta}\hat u(\theta)F_s(x,\theta)đ\theta$ with F_s as before but now with $q = a(x,\theta+\zeta)b(x+y,\theta)$. The same argument gives $|F_s(x,\theta)| \le C\langle\theta\rangle^{m'}$ uniformly and $F_s \to J(q)$, hence $\mathrm{Op}(a)\,\mathrm{Op}(b)u = \mathrm{Op}(J(q))u$. For general a, replace a by $a_l = \chi(\xi/l)a$; then $\mathrm{Op}(a_l)v(x) = \int e^{ix\cdot\xi}a_l(x,\xi)\hat v(\xi)đ\xi \to \mathrm{Op}(a)v(x)$, and $\mathrm{Op}(J(q_l))u(x) \to \mathrm{Op}(J(q))u(x)$ as in (i). □

6.2.2 The semiclassical calculus

We can now prove that semiclassical families form an algebra, with an expansion of the composition in powers of ϵ.

Theorem 6.2.6 (Reduction). *If $p \in S^m_{\mathrm{sl}}(\mathbb{R}^{2n};\mathbb{R}^n)$ then $\mathrm{Op}_\epsilon(p) = \mathrm{Op}_\epsilon(c)$, where*

$$c(\epsilon,x,\eta) = J\big(p(\epsilon,x,x+y,\eta+\epsilon\zeta)\big)(x,\eta) \in S^m_{\mathrm{sl}}(\mathbb{R}^n),$$

and for every N

$$c(\epsilon,x,\eta) - \sum_{|\alpha|<N} \frac{\epsilon^{|\alpha|}}{\alpha!}\partial_\eta^\alpha D_y^\alpha p(\epsilon,x,y,\eta)\Big|_{y=x} \in \epsilon^N S^{m-N}_{\mathrm{sl}}(\mathbb{R}^n). \tag{6.2.8}$$

In particular $c(0,x,\eta) = p(0,x,x,\eta)$ and $c - p(\epsilon,x,x,\eta) \in \epsilon S^{m-1}_{\mathrm{sl}}(\mathbb{R}^n)$.

Proof. For $0 \le t \le 1$ let $q_t(\epsilon;x,\eta,y,\zeta) = p(\epsilon,x,x+y,\eta+t\zeta)$. By (6.2.5), $\partial_\epsilon^j\partial_t^l q_t$ is bounded in $\mathcal{Q}^m_{|m|+l,0}$ (each ∂_t produces a factor $\zeta\cdot\partial_\eta$), so by Lemma 6.2.4(iv) the function $c(\epsilon,x,\eta) = J(q_\epsilon)(x,\eta)$ lies in $S^m_{\mathrm{sl}}(\mathbb{R}^n)$. For fixed $\epsilon>0$, $\mathrm{Op}_\epsilon(p) = \mathrm{Op}(p_\epsilon)$ with $p_\epsilon(x,y',\xi) = p(\epsilon,x,y',\epsilon\xi) \in \mathcal{A}^m_0$, and by Proposition 6.2.5(i) its left symbol at (x,ξ) is $J\big(p(\epsilon,x,x+y,\epsilon\xi+\epsilon\zeta)\big)(x,\xi) = c(\epsilon,x,\epsilon\xi)$, since J acts with ξ as a parameter. Hence $\mathrm{Op}_\epsilon(p) = \mathrm{Op}(c^\flat(\epsilon)) = \mathrm{Op}_\epsilon(c)$.

For the expansion, Taylor's formula in the third argument gives

$$\begin{aligned} p(\epsilon,x,x+y,\eta+\epsilon\zeta) = \sum_{|\alpha|<N}\frac{(\epsilon\zeta)^\alpha}{\alpha!}\partial_\eta^\alpha p(\epsilon,x,x+y,\eta) \\ + \epsilon^N \sum_{|\alpha|=N}\frac{N\zeta^\alpha}{\alpha!}\int_0^1 (1-s)^{N-1}\partial_\eta^\alpha p(\epsilon,x,x+y,\eta+s\epsilon\zeta)\,ds. \end{aligned}$$

Apply J. By Lemma 6.2.4(ii),(iii), $J(\zeta^\alpha\partial_\eta^\alpha p(\epsilon,x,x+y,\eta)) = D_y^\alpha\partial_\eta^\alpha p(\epsilon,x,y,\eta)|_{y=x}$, which gives the sum in (6.2.8). In the remainder, $J(\zeta^\alpha\partial_\eta^\alpha p(\epsilon,x,x+y,\eta+s\epsilon\zeta)) = J\big((D_y^\alpha\partial_\eta^\alpha p)(\epsilon,x,x+y,\eta+s\epsilon\zeta)\big)$, and $D_y^\alpha\partial_\eta^\alpha p \in S^{m-N}_{\mathrm{sl}}(\mathbb{R}^{2n};\mathbb{R}^n)$; by Lemma 6.2.4(iv), with parameters (ϵ,s), this is a symbol of order $m-N$ with all ϵ-derivatives bounded uniformly in $s\in[0,1]$, and integrating in s gives an element of S^{m-N}_{sl}. (We may interchange J and $\int_0^1 ds$ by Fubini in (6.2.6).) □

Theorem 6.2.7 (Composition and adjoints). *The families $\Psi^\infty_{\mathrm{sl}}(\mathbb{R}^n) = \bigcup_m \Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ form an order-filtered $*$-algebra. Precisely, for $a \in S^m_{\mathrm{sl}}(\mathbb{R}^n)$ and $b \in S^{m'}_{\mathrm{sl}}(\mathbb{R}^n)$,*

$$\mathrm{Op}_\epsilon(a)\,\mathrm{Op}_\epsilon(b) = \mathrm{Op}_\epsilon(a\#b), \qquad \mathrm{Op}_\epsilon(a)^* = \mathrm{Op}_\epsilon(a^*),$$

where

$$(a\#b)(\epsilon,x,\eta) = J\big(a(\epsilon,x,\eta+\epsilon\zeta)\,b(\epsilon,x+y,\eta)\big), \qquad a^*(\epsilon,x,\eta) = J\big(\overline{a(\epsilon,x+y,\eta+\epsilon\zeta)}\big)$$

lie in $S^{m+m'}_{\mathrm{sl}}(\mathbb{R}^n)$ and $S^m_{\mathrm{sl}}(\mathbb{R}^n)$ respectively, and for every N

$$a\#b - \sum_{|\alpha|<N}\frac{\epsilon^{|\alpha|}}{\alpha!}\,\partial_\eta^\alpha a\,D_x^\alpha b \in \epsilon^N S^{m+m'-N}_{\mathrm{sl}}(\mathbb{R}^n), \qquad a^* - \sum_{|\alpha|<N}\frac{\epsilon^{|\alpha|}}{\alpha!}\,\partial_\eta^\alpha D_x^\alpha \bar a \in \epsilon^N S^{m-N}_{\mathrm{sl}}(\mathbb{R}^n). \tag{6.2.9}$$

In particular $\epsilon^N\Psi^m_{\mathrm{sl}} \circ \epsilon^{N'}\Psi^{m'}_{\mathrm{sl}} \subset \epsilon^{N+N'}\Psi^{m+m'}_{\mathrm{sl}}$, and $\epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$ is a two-sided ideal in $\Psi^\infty_{\mathrm{sl}}(\mathbb{R}^n)$.

Proof. The statement about adjoints is Theorem 6.2.6 applied to $p(\epsilon,x,y,\eta)=\overline{a(\epsilon,y,\eta)}$: by Proposition 2.2.9, for each $\epsilon>0$ the adjoint of $\mathrm{Op}(a^\flat(\epsilon))$ is the operator with amplitude $\overline{a^\flat(\epsilon,y,\xi)}$, i.e. $\mathrm{Op}_\epsilon(a)^*=\mathrm{Op}_\epsilon(p)$. For the composite, fix $\epsilon>0$; by Proposition 6.2.5(ii) the left symbol of $\mathrm{Op}(a^\flat(\epsilon))\,\mathrm{Op}(b^\flat(\epsilon))$ at (x,ξ) is $J\big(a(\epsilon,x,\epsilon\xi+\epsilon\zeta)b(\epsilon,x+y,\epsilon\xi)\big)=(a\#b)(\epsilon,x,\epsilon\xi)$. That $a\#b\in S^{m+m'}_{\mathrm{sl}}$ and the expansion follow exactly as in the proof of Theorem 6.2.6, using (6.2.5) and Taylor's formula for $a(\epsilon,x,\eta+\epsilon\zeta)$, now with $J(\zeta^\alpha\partial_\eta^\alpha a(\epsilon,x,\eta)b(\epsilon,x+y,\eta))=\partial_\eta^\alpha a\,D_x^\alpha b$ and remainder terms $J\big(\partial_\eta^\alpha a(\epsilon,x,\eta+s\epsilon\zeta)\,D_x^\alpha b(\epsilon,x+y,\eta)\big)$, $|\alpha|=N$, which are of order $m+m'-N$ uniformly in s.

If $a\in\epsilon^N S^m_{\mathrm{sl}}$ and $b\in\epsilon^{N'}S^{m'}_{\mathrm{sl}}$ then $a\#b\in\epsilon^{N+N'}S^{m+m'}_{\mathrm{sl}}$ since $\#$ is bilinear and J commutes with multiplication by functions of ϵ. If $b\in\epsilon^\infty S^{-\infty}_{\mathrm{sl}}$ then $a\#b\in\bigcap_N\epsilon^N S^{m-N}_{\mathrm{sl}}=\epsilon^\infty S^{-\infty}_{\mathrm{sl}}$, and similarly in the other order, by Lemma 6.1.3. □

Since $\partial_\eta^\alpha=i^{|\alpha|}D_\eta^\alpha$, (6.2.9) is the rescaled form of the composition formula of Chapter 2: formally $a\#b=e^{i\epsilon D_\eta\cdot D_y}a(\epsilon,x,\eta)b(\epsilon,y,\eta)|_{y=x}$. The difference is that now the expansion is asymptotic in ϵ as well as in the order, with uniform remainders. The two leading terms give:

Corollary 6.2.8. *For $a\in S^m_{\mathrm{sl}}(\mathbb{R}^n)$, $b\in S^{m'}_{\mathrm{sl}}(\mathbb{R}^n)$,*

$$a\#b-ab\in\epsilon S^{m+m'-1}_{\mathrm{sl}},\qquad a\#b-b\#a-\frac{\epsilon}{i}\{a,b\}\in\epsilon^2S^{m+m'-2}_{\mathrm{sl}},$$

where $\{a,b\}$ is the Poisson bracket, $\{a,b\}=\sum_j(\partial_{\eta_j}a\,\partial_{x_j}b-\partial_{x_j}a\,\partial_{\eta_j}b)$. In particular $[\mathrm{Op}_\epsilon(a),\mathrm{Op}_\epsilon(b)]\in\epsilon\Psi^{m+m'-1}_{\mathrm{sl}}(\mathbb{R}^n)$.

Proof. The terms with $|\alpha|=1$ in (6.2.9) for $a\#b-b\#a$ are $\epsilon\sum_j(\partial_{\eta_j}a\,D_{x_j}b-\partial_{\eta_j}b\,D_{x_j}a)=\frac{\epsilon}{i}\{a,b\}$. □

6.3 Residual families and kernels

6.3.1 Off-diagonal behaviour and the residual ideal

By (6.2.2) the kernel of a semiclassical family is determined by the inverse Fourier transform $K(\epsilon,x,Z)$ of its symbol, which is singular only at $Z=0$ and rapidly decreasing as $|Z|\to\infty$. Away from the diagonal the variable $Z=(x-y)/\epsilon$ tends to infinity as $\epsilon\downarrow0$, and this gives:

Lemma 6.3.1. *Let $a\in S^m_{\mathrm{sl}}(\mathbb{R}^n)$ and $d>0$. The kernel $k(\epsilon,x,y)$ of $\mathrm{Op}_\epsilon(a)$, restricted to $\{|x-y|>d\}$, is smooth and extends smoothly to $[0,1]\times\{(x,y);|x-y|>d\}$, with*

$$|\partial_\epsilon^j\partial_x^\alpha\partial_y^\beta k(\epsilon,x,y)|\le C_{j\alpha\beta N}\,\epsilon^N\langle x-y\rangle^{-N}\quad\text{for all } N.$$

Proof. Let $K(\epsilon,x,Z)$ be as in (6.2.2). Since $\Delta_\eta e^{iZ\cdot\eta}=|Z|^2e^{iZ\cdot\eta}$, for $Z\neq0$ and any L,

$$\partial_\epsilon^j\partial_x^\alpha\partial_Z^\gamma K(\epsilon,x,Z)=|Z|^{-2L}\int e^{iZ\cdot\eta}\Delta_\eta^L\big[(i\eta)^\gamma\partial_\epsilon^j\partial_x^\alpha a(\epsilon,x,\eta)\big]d\eta,$$

in the sense of distributions; once $m+|\gamma|-2L<-n$ the integral converges absolutely and uniformly. Hence K is smooth on $[0,1]\times\mathbb{R}^n\times(\mathbb{R}^n\setminus0)$ with $|\partial_\epsilon^j\partial_x^\alpha\partial_Z^\gamma K|\le C_L|Z|^{-2L}$ for $|Z|\ge1$ and every L. Now $k(\epsilon,x,y)=\epsilon^{-n}K(\epsilon,x,(x-y)/\epsilon)$, and by the chain rule each derivative ∂_ϵ, ∂_x, ∂_y of such an expression is a sum of terms $\epsilon^{-p}Z^\beta(\partial_\epsilon^{j'}\partial_x^{\alpha'}\partial_Z^{\gamma'}K)(\epsilon,x,Z)$ at $Z=(x-y)/\epsilon$ (e.g. ∂_ϵ produces $-\epsilon^{-1}Z\cdot\partial_ZK$). On $|x-y|>d$ we have $|Z|>d/\epsilon\ge d$, and each term is bounded by

$$C\epsilon^{-n-p}|Z|^{|\beta|-2L}=C\epsilon^{2L-|\beta|-n-p}|x-y|^{|\beta|-2L},$$

which for L large is bounded by $C\epsilon^N\langle x-y\rangle^{-N}$. So all derivatives tend to zero as $\epsilon\downarrow0$, uniformly, and k extends smoothly by zero. □

We denote by $\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty}(\mathbb{R}^n))$ the space of families of operators whose kernels $k(\epsilon,x,y)$ extend to smooth functions on $[0,1]\times\mathbb{R}^{2n}$ satisfying

$$|\partial_\epsilon^j\partial_x^\alpha\partial_y^\beta k(\epsilon,x,y)|\le C_{j\alpha\beta N}\,\epsilon^N\langle x-y\rangle^{-N}\quad\text{for all } j,\alpha,\beta,N. \tag{6.3.1}$$

By Proposition 2.3.3 these are exactly the smooth families of elements of $\Psi^{-\infty}(\mathbb{R}^n)$ vanishing, with all ϵ-derivatives and in all the seminorms of that proposition, to infinite order at $\epsilon=0$.

Lemma 6.3.2. $\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}(\mathbb{R}^n)=\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty}(\mathbb{R}^n))$.

Proof. Let $a\in\epsilon^\infty S_{\mathrm{sl}}^{-\infty}$. By part (iii) of Lemma 6.1.3, the function K of (6.2.2) satisfies $|Z^\gamma\partial_\epsilon^j\partial_x^\alpha\partial_Z^\beta K|\le C\epsilon^N$ for all indices and N, since $Z^\gamma\partial_Z^\beta K$ is the inverse Fourier transform of $(-D_\eta)^\gamma[(i\eta)^\beta a]$. As in the previous proof, derivatives of $k=\epsilon^{-n}K(\epsilon,x,(x-y)/\epsilon)$ are sums of terms $\epsilon^{-p}Z^\beta\partial K$, and $\langle x-y\rangle=\langle\epsilon Z\rangle\le\langle Z\rangle$, so $\langle x-y\rangle^N|\partial_\epsilon^j\partial_x^\alpha\partial_y^\beta k|$ is bounded by $C\epsilon^{-n-p}\sup\langle Z\rangle^{N+|\beta|}|\partial K|\le C\epsilon^{N'}$ for any N'. Thus (6.3.1) holds.

Conversely, let k satisfy (6.3.1). As in the proof of Proposition 2.3.3, $b(\epsilon,x,\xi)=\int e^{-iz\cdot\xi}k(\epsilon,x,x-z)\,dz$ satisfies $|\partial_\epsilon^j\partial_x^\alpha\partial_\xi^\beta b|\le C\epsilon^N\langle\xi\rangle^{-N}$ for all indices and N, and the kernel of $\mathrm{Op}(b(\epsilon))$ is $k(\epsilon)$. Set $a(\epsilon,x,\eta)=b(\epsilon,x,\eta/\epsilon)$ for $\epsilon>0$ and $a(0,x,\eta)=0$, so that $\mathrm{Op}_\epsilon(a)=\mathrm{Op}(b(\epsilon))$. Since $\langle\eta/\epsilon\rangle\ge\langle\eta\rangle$, $|\partial_x^\alpha\partial_\eta^\beta a|=\epsilon^{-|\beta|}|(\partial_x^\alpha\partial_\xi^\beta b)(\epsilon,x,\eta/\epsilon)|\le C\epsilon^{N-|\beta|}\langle\eta\rangle^{-N}$. By the chain rule, $\partial_\epsilon a=\big(\partial_\epsilon b-\epsilon^{-1}\xi\cdot\partial_\xi b\big)(\epsilon,x,\eta/\epsilon)$, and $\xi\cdot\partial_\xi b$ satisfies the same estimates as b; inductively, $\partial_\epsilon^j\partial_x^\alpha\partial_\eta^\beta a$ is bounded by $C\epsilon^N\langle\eta\rangle^{-N}$ for every N, for $\epsilon>0$. Hence a extends smoothly by zero to $\epsilon=0$ and $a\in\epsilon^\infty S_{\mathrm{sl}}^{-\infty}$ by Lemma 6.1.3(iii). □

Question. Why is the residual ideal $\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}$ and not simply the smoothing families $\Psi_{\mathrm{sl}}^{-\infty}$?

Answer. Because $\Psi_{\mathrm{sl}}^{-\infty}$ is not small. For example $\mathrm{Op}_\epsilon(e^{-|\eta|^2})$ is smoothing for every ϵ, but its semiclassical symbol is not zero and its L^2 norm equals one for every ϵ: it is the Fourier multiplier $e^{-\epsilon^2|\xi|^2}$, which tends to the identity as $\epsilon\downarrow 0$. Only families which are both smoothing and $O(\epsilon^\infty)$ are negligible in every sense, and these are exactly the families with smooth kernels vanishing to infinite order at $\epsilon=0$. □

Corollary 6.3.3. *Let $A\in\Psi_{\mathrm{sl}}^m(\mathbb{R}^n)$ and $\phi,\psi\in\mathcal{C}_{\mathrm{b}}^\infty(\mathbb{R}^n)$. Then $\phi A\psi\in\Psi_{\mathrm{sl}}^m(\mathbb{R}^n)$; if $A=\mathrm{Op}_\epsilon(a)$ then $\phi A\psi=\mathrm{Op}_\epsilon(c)$ with $c-\phi\psi a\in\epsilon S_{\mathrm{sl}}^{m-1}(\mathbb{R}^n)$. If the supports of ϕ and ψ are at positive distance, then $\phi A\psi\in\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}(\mathbb{R}^n)$.*

Proof. $\phi A\psi=\mathrm{Op}_\epsilon(p)$ with $p(\epsilon,x,y,\eta)=\phi(x)a(\epsilon,x,\eta)\psi(y)\in S_{\mathrm{sl}}^m(\mathbb{R}^{2n};\mathbb{R}^n)$, so the first part is Theorem 6.2.6. The second follows from Lemmas 6.3.1 and 6.3.2. □

We also record how semiclassical families act on families of functions.

Lemma 6.3.4. *Let $A=\mathrm{Op}_\epsilon(a)\in\Psi_{\mathrm{sl}}^m(\mathbb{R}^n)$ and let $f\in\mathcal{C}^\infty([0,1];\mathcal{S}(\mathbb{R}^n))$, i.e. $f\in\mathcal{C}^\infty([0,1]\times\mathbb{R}^n)$ with $\sup|x^\gamma\partial_\epsilon^j\partial_x^\beta f|<\infty$ for all indices. Then $g(\epsilon)=A_\epsilon f(\epsilon)$, $\epsilon>0$, extends to an element of $\mathcal{C}^\infty([0,1];\mathcal{S}(\mathbb{R}^n))$, with $g(0)(x)=a(0,x,0)f(0,x)$, and $f\mapsto g$ is continuous. If f vanishes to infinite order at $\epsilon=0$, so does g.*

Proof. $g(\epsilon,x)=\int e^{ix\cdot\xi}a(\epsilon,x,\epsilon\xi)\hat f(\epsilon,\xi)d\!\!\!^-\xi$, and $\hat f\in\mathcal{C}^\infty([0,1];\mathcal{S}(\mathbb{R}^n))$. By Lemma 6.1.4(ii), all derivatives in (ϵ,x,ξ) of $a(\epsilon,x,\epsilon\xi)$ are bounded by powers of $\langle\xi\rangle$, uniformly in $\epsilon\in(0,1]$, so we may differentiate under the integral; moreover $x^\gamma e^{ix\cdot\xi}=D_\xi^\gamma e^{ix\cdot\xi}$ allows us to integrate by parts to show that $x^\gamma\partial_\epsilon^j\partial_x^\beta g$ is bounded, uniformly in ϵ, by a Schwartz seminorm of f and finitely many ϵ-derivatives of it. The integrand is continuous up to $\epsilon=0$, where it equals $e^{ix\cdot\xi}a(0,x,0)\hat f(0,\xi)$. If f vanishes to infinite order then so do all these bounds. □

6.3.2 Semiclassical smoothing families

For the families of order $-\infty$ the description (6.2.2) of the kernel becomes very explicit, since the inverse Fourier transform in η is an isomorphism of $S^{-\infty}(\mathbb{R}^n;\mathbb{R}^n) = \mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n_x;\mathcal{S}(\mathbb{R}^n_\eta))$ onto $\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n_x;\mathcal{S}(\mathbb{R}^n_Z))$ (the functions with all derivatives bounded by $C_N\langle Z\rangle^{-N}$). Let $\mathcal{K}(\mathbb{R}^n)$ denote the space of $K \in \mathcal{C}^\infty([0,1]\times\mathbb{R}^n_x\times\mathbb{R}^n_Z)$ such that

$$|\partial_\epsilon^j\partial_x^\alpha\partial_Z^\beta K(\epsilon,x,Z)| \le C_{j\alpha\beta N}\langle Z\rangle^{-N} \quad \text{for all } N.$$

Proposition 6.3.5. *A family of operators A_ϵ is in $\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$ if and only if its kernels are of the form*

$$K_{A_\epsilon}(x,y) = \epsilon^{-n}K\Big(\epsilon,x,\frac{x-y}{\epsilon}\Big), \qquad K\in\mathcal{K}(\mathbb{R}^n); \tag{6.3.2}$$

then K is uniquely determined, and $A = \mathrm{Op}_\epsilon(a)$ with $a(\epsilon,x,\eta) = \int e^{-iZ\cdot\eta}K(\epsilon,x,Z)\,dZ$. If B has kernel $\epsilon^{-n}L(\epsilon,x,(x-y)/\epsilon)$, $L\in\mathcal{K}(\mathbb{R}^n)$, then AB has kernel $\epsilon^{-n}M(\epsilon,x,(x-y)/\epsilon)$ with

$$M(\epsilon,x,Z) = \int_{\mathbb{R}^n} K(\epsilon,x,t)\,L(\epsilon,x-\epsilon t,Z-t)\,dt. \tag{6.3.3}$$

Proof. The first part is (6.2.2) together with the remark preceding the proposition, applied with ϵ (and its derivatives) as a parameter. For the composite, for $\epsilon>0$,

$$\begin{aligned} K_{A_\epsilon B_\epsilon}(x,y) &= \epsilon^{-2n}\int K\Big(\epsilon,x,\frac{x-z}{\epsilon}\Big)L\Big(\epsilon,z,\frac{z-y}{\epsilon}\Big)dz \\ &= \epsilon^{-n}\int K(\epsilon,x,t)L\Big(\epsilon,x-\epsilon t,\frac{x-y}{\epsilon}-t\Big)dt, \end{aligned}$$

by the substitution $z = x-\epsilon t$. Peetre's inequality $\langle Z\rangle^N \le 2^{N/2}\langle t\rangle^N\langle Z-t\rangle^N$ and differentiation under the integral show directly that $M\in\mathcal{K}(\mathbb{R}^n)$ (the ϵ-derivatives falling on the first argument of L produce factors of t, which are absorbed by the decay of K). □

At $\epsilon=0$, (6.3.3) is the convolution $M(0,x,\cdot) = K(0,x,\cdot)*L(0,x,\cdot)$ in Z, with x a parameter. Taking the Fourier transform in Z shows once more that the symbol of AB at $\epsilon=0$ is the product of the symbols at $\epsilon=0$: the semiclassical families "become commutative" in the limit. See Exercise 6.3 for the full expansion from this point of view.

Remark 6.3.6 (The semiclassical double space). The description (6.3.2) has a geometric meaning. On $[0,1]\times\mathbb{R}^{2n}$ the kernel $\epsilon^n K_{A_\epsilon}(x,y) = K(\epsilon,x,(x-y)/\epsilon)$ is singular only at $\{0\}\times\mathrm{Diag}$; it becomes smooth after this submanifold is *blown up*, i.e. after introducing as new coordinates (ϵ,x,Z), $Z=(x-y)/\epsilon$, near it. Globally, the blown-up space $[[0,1]\times\mathbb{R}^{2n};\{0\}\times\mathrm{Diag}]$ is obtained by replacing $\{0\}\times\mathrm{Diag}$ by its inward-pointing spherical normal bundle; the new boundary hypersurface, the *front face*, is a bundle of closed half-spheres over Diag, whose interior is identified with $T\mathbb{R}^n$ through $(\epsilon,x,y)\mapsto(x,(x-y)/\epsilon)$. The other boundary face is the lift of $\{\epsilon=0\}$, i.e. $\{\epsilon=0\}$ away from the diagonal, which is reached as $|Z|\to\infty$ in the front face. Thus Proposition 6.3.5 says: the kernels of the elements of $\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$ are ϵ^{-n} times smooth functions on the blown-up space which vanish to infinite order at the lifted face $\{\epsilon=0\}$ (with the appropriate uniformity as $x\to\infty$), and the semiclassical symbol at $\epsilon=0$ (Definition 6.4.1) is the fibrewise Fourier transform of the restriction of the kernel to the front face. On a compact manifold X the same description holds with $[[0,1]\times X^2;\{0\}\times\mathrm{Diag}]$, as follows from Proposition 6.4.10; this is the space used in Chapter 16. The identification of the interior of the front face with TX, and hence of the domain of the semiclassical symbol with T^*X, depends on the choice of the parameter ϵ up to a positive multiple; see Remark 6.5.4. The two spaces are sketched in Figure 6.2.

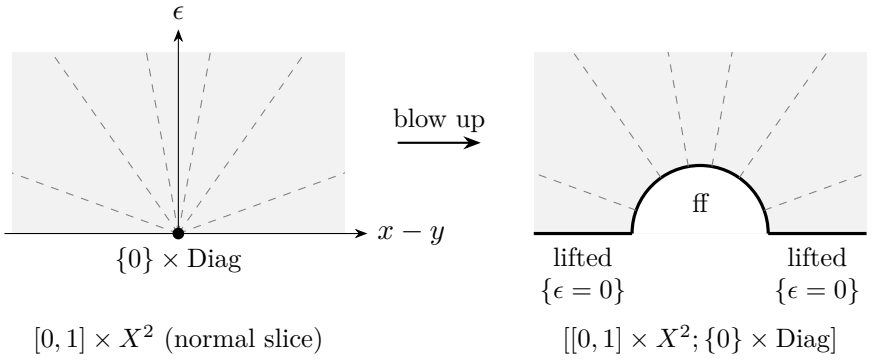


Figure 6.2: The semiclassical double space, in a two-dimensional slice transverse to the diagonal. Left: the kernel $\epsilon^{-n}K(\epsilon, x, (x-y)/\epsilon)$ is singular only at $\{0\} \times \mathrm{Diag}$ (dot); it is constant along the dashed rays $x - y = \epsilon Z$. Right: after blowing up, the rays are separated and end at different points of the front face ff, whose interior is parametrized by $Z = (x-y)/\epsilon \in \mathbb{R}^n$. Semiclassical smoothing kernels are smooth here, vanish to infinite order at the lifted face $\{\epsilon = 0\}$, and their restriction to ff is the inverse Fourier transform of the semiclassical symbol.

6.4 Symbols, ellipticity and boundedness

6.4.1 The symbol maps

Definition 6.4.1. For $A = \mathrm{Op}_\epsilon(a) \in \Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ we define

(i) the *semiclassical symbol* $\sigma_{\mathrm{sl}}(A) = a(0,\cdot,\cdot) \in S^m(\mathbb{R}^n;\mathbb{R}^n)$;

(ii) the *uniform principal symbol* $\tilde\sigma_m(A) = [a] \in S^m_{\mathrm{sl}}(\mathbb{R}^n)/S^{m-1}_{\mathrm{sl}}(\mathbb{R}^n)$;

(iii) the *joint symbol* $\sigma_{\epsilon,m}(A) = [a] \in S^m_{\mathrm{sl}}(\mathbb{R}^n)/\epsilon S^{m-1}_{\mathrm{sl}}(\mathbb{R}^n)$.

For each $\epsilon > 0$ the ordinary principal symbol of $A_\epsilon \in \Psi^m(\mathbb{R}^n)$ is the class of $a(\epsilon, x, \epsilon\xi)$, so $\tilde\sigma_m(A)$ is the principal symbol of A_ϵ *rescaled*: $\tilde\sigma_m(A)(\epsilon, x, \eta) = \sigma_m(A_\epsilon)(x, \eta/\epsilon)$, and it carries uniform information as $\epsilon \downarrow 0$. The semiclassical symbol is not a "leading part at infinity" but a full function on the cotangent bundle; e.g. for the Schrödinger operator of Example 6.2.2, $\sigma_{\mathrm{sl}}(-\epsilon^2\Delta_{\mathrm{eucl}} + V) = |\eta|^2 + V(x)$ and $\tilde\sigma_2 = [|\eta|^2]$.

Proposition 6.4.2. *For each m the sequences*

$$0 \longrightarrow \epsilon\Psi^m_{\mathrm{sl}}(\mathbb{R}^n) \hookrightarrow \Psi^m_{\mathrm{sl}}(\mathbb{R}^n) \xrightarrow{\sigma_{\mathrm{sl}}} S^m(\mathbb{R}^n;\mathbb{R}^n) \longrightarrow 0,$$
$$0 \longrightarrow \Psi^{m-1}_{\mathrm{sl}}(\mathbb{R}^n) \hookrightarrow \Psi^m_{\mathrm{sl}}(\mathbb{R}^n) \xrightarrow{\tilde\sigma_m} S^m_{\mathrm{sl}}(\mathbb{R}^n)/S^{m-1}_{\mathrm{sl}}(\mathbb{R}^n) \longrightarrow 0,$$
$$0 \longrightarrow \epsilon\Psi^{m-1}_{\mathrm{sl}}(\mathbb{R}^n) \hookrightarrow \Psi^m_{\mathrm{sl}}(\mathbb{R}^n) \xrightarrow{\sigma_{\epsilon,m}} S^m_{\mathrm{sl}}(\mathbb{R}^n)/\epsilon S^{m-1}_{\mathrm{sl}}(\mathbb{R}^n) \longrightarrow 0$$

are exact, and all three maps are multiplicative and $*$-preserving: $\sigma(AB) = \sigma(A)\sigma(B)$ and $\sigma(A^*) = \overline{\sigma(A)}$ for $\sigma = \sigma_{\mathrm{sl}}$, $\tilde\sigma$, σ_ϵ (with the appropriate orders). Moreover, the map $[a] \mapsto ([a], a(0,\cdot,\cdot))$ identifies $S^m_{\mathrm{sl}}(\mathbb{R}^n)/\epsilon S^{m-1}_{\mathrm{sl}}(\mathbb{R}^n)$ with the space of pairs

$$(\tilde a, b) \in S^m_{\mathrm{sl}}(\mathbb{R}^n)/S^{m-1}_{\mathrm{sl}}(\mathbb{R}^n) \times S^m(\mathbb{R}^n;\mathbb{R}^n) \quad \text{with} \quad \tilde a|_{\epsilon=0} = [b] \text{ in } S^m(\mathbb{R}^n;\mathbb{R}^n)/S^{m-1}(\mathbb{R}^n;\mathbb{R}^n);$$

that is, $\sigma_{\epsilon,m} = (\tilde\sigma_m, \sigma_{\mathrm{sl}})$, the two symbols are jointly surjective subject only to this compatibility condition, and the joint null space is $\epsilon\Psi^{m-1}_{\mathrm{sl}}(\mathbb{R}^n)$.

Proof. Exactness follows from the bijectivity of Op_ϵ together with Lemma 6.1.2: $a(0) = 0$ if and only if $a \in \epsilon S^m_{\mathrm{sl}}$. Multiplicativity follows from Corollary 6.2.8 ($a\#b - ab \in \epsilon S^{m+m'-1}_{\mathrm{sl}}$, which lies both in $\epsilon S^{m+m'}_{\mathrm{sl}}$ and in $S^{m+m'-1}_{\mathrm{sl}}$) and similarly for adjoints from (6.2.9). For the last statement, the map is well defined since $\epsilon S^{m-1}_{\mathrm{sl}} \subset S^{m-1}_{\mathrm{sl}}$ and elements of $\epsilon S^{m-1}_{\mathrm{sl}}$ vanish at $\epsilon = 0$; its range satisfies the compatibility condition. It is injective: if $a \in S^{m-1}_{\mathrm{sl}}$ and $a(0) = 0$ then by Lemma 6.1.2 (applied in S^{m-1}_{sl}) $a \in \epsilon S^{m-1}_{\mathrm{sl}}$. It is surjective onto compatible pairs: given $(\tilde a, b)$ choose a representative $a_1 \in S^m_{\mathrm{sl}}$ of $\tilde a$; then $b - a_1(0) \in S^{m-1}$, and $a = a_1 + (b - a_1(0))$, with the second term constant in ϵ, represents $\tilde a$ and satisfies $a(0) = b$. □

In terms of Figure 6.1, the joint symbol records a representative on the left edge $\epsilon = 0$ together with the leading behaviour along the top and bottom edges $|\eta| = \infty$, and the compatibility condition is that the two agree at the corners where these edges meet.

Remark 6.4.3. If $A \in \Psi^m_{\mathrm{sl,ph}}(\mathbb{R}^n)$, the uniform principal symbol is represented by the leading homogeneous term a_0 of Definition 6.1.6, a smooth function on $[0,1] \times \mathbb{R}^n \times (\mathbb{R}^n \setminus 0)$ homogeneous of degree m (for $|\eta| \geq 1$, and then extended by homogeneity), and the compatibility condition says that $a_0|_{\epsilon=0}$ is the leading homogeneous part of the classical symbol $\sigma_{\mathrm{sl}}(A)$. The classical families form a subalgebra: in (6.2.9) the terms of the expansion are classical, and the remainder after N terms is of order $m + m' - N$, so $a\#b$ is classical by the argument following Definition 6.1.6; similarly for a^* and for the reduction formula (6.2.8).

6.4.2 Ellipticity and the parametrix

Definition 6.4.4. $A \in \Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ (or an $N \times N$ matrix of such families) is *elliptic* if its joint symbol is invertible, i.e. there exists $b \in S^{-m}_{\mathrm{sl}}(\mathbb{R}^n)$ (matrix valued) with

$$a\,b - \mathrm{Id} \in \epsilon S^{-1}_{\mathrm{sl}}(\mathbb{R}^n), \qquad b\,a - \mathrm{Id} \in \epsilon S^{-1}_{\mathrm{sl}}(\mathbb{R}^n),$$

for a representative a of $\sigma_{\epsilon,m}(A)$.

Lemma 6.4.5. *$A = \mathrm{Op}_\epsilon(a) \in \Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ is elliptic if and only if the following two conditions hold:*

(i) *$\sigma_{\mathrm{sl}}(A)(x,\eta) = a(0,x,\eta)$ is invertible for* all *(x,η), with $|a(0,x,\eta)^{-1}| \leq C\langle\eta\rangle^{-m}$;*

(ii) *A is uniformly elliptic: there are R and C such that $a(\epsilon,x,\eta)$ is invertible and $|a(\epsilon,x,\eta)^{-1}| \leq C\langle\eta\rangle^{-m}$ for $|\eta| \geq R$, $\epsilon \in [0,1]$.*

Proof. If A is elliptic, then $a(0)b(0) = \mathrm{Id} = b(0)a(0)$ gives (i). Also $ab = \mathrm{Id} + r$ with $r \in S^{-1}_{\mathrm{sl}}$, so $|r| \leq \frac12$ for $|\eta| \geq R$, and there a is invertible with $a^{-1} = b(\mathrm{Id}+r)^{-1}$, which gives (ii). Conversely, assume (i) and (ii). Since $a(\epsilon) - a(0) = \epsilon r$ with $r \in S^m_{\mathrm{sl}}$, we have $a(\epsilon) = a(0)(\mathrm{Id} + \epsilon a(0)^{-1} r)$ with $a(0)^{-1}r$ bounded; so there is $\epsilon_0 > 0$ such that $a(\epsilon)$ is invertible with $|a(\epsilon)^{-1}| \leq C\langle\eta\rangle^{-m}$ for all (x,η) and $\epsilon \leq \epsilon_0$. Differentiating $a\,a^{-1} = \mathrm{Id}$ repeatedly (as in Lemma 2.1.6, whose proof applies verbatim to matrices and to ϵ-derivatives), a^{-1} satisfies the estimates of S^{-m}_{sl} on $[0,\epsilon_0] \times \mathbb{R}^{2n}$. By (ii) and the same argument, $\tilde b = (1 - \chi(\eta/R))a^{-1}$, with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to 1 on $\{|\eta| \leq 2\}$, lies in S^{-m}_{sl}, and $a\tilde b - \mathrm{Id} = \tilde b a - \mathrm{Id} = -\chi(\eta/R)\,\mathrm{Id} \in S^{-\infty}_{\mathrm{sl}}$. Let $\psi \in \mathcal{C}^\infty([0,1])$ equal 1 on $[0, \frac12\epsilon_0]$ and 0 on $[\epsilon_0, 1]$, and set $b = \psi(\epsilon)a^{-1} + (1-\psi(\epsilon))\tilde b \in S^{-m}_{\mathrm{sl}}$. Then $ab - \mathrm{Id} = (1 - \psi(\epsilon))(a\tilde b - \mathrm{Id})$, which lies in S^{-1}_{sl} and vanishes for $\epsilon \leq \frac12\epsilon_0$, hence lies in $\epsilon^N S^{-1}_{\mathrm{sl}}$ for every N; similarly for $ba - \mathrm{Id}$. □

Discussion. Semiclassical ellipticity is stronger than ellipticity of each A_ϵ, because (i) asks for invertibility of $\sigma_{\mathrm{sl}}(A)$ at *all* points of $T^*\mathbb{R}^n$, not only near infinity. For the positive Laplacian Δ, the family $\epsilon^2\Delta + 1 = \mathrm{Op}_\epsilon(|\eta|^2 + 1)$ is elliptic, and indeed invertible for every ϵ. The family $\epsilon^2\Delta - 1 = \mathrm{Op}_\epsilon(|\eta|^2 - 1)$ is not: each A_ϵ is elliptic in the usual sense, but $\sigma_{\mathrm{sl}}(A)$ vanishes on the sphere $|\eta| = 1$. This is the Helmholtz operator at frequency $1/\epsilon$, whose solutions oscillate at exactly the scale the calculus is designed to see.

Note that (i) alone implies (ii) for ϵ small; condition (ii) is only needed to control the family for ϵ away from 0. As usual, one may simply restrict attention to $\epsilon \in (0, \epsilon_0]$, rescaling the parameter if desired.

Theorem 6.4.6. *If $A \in \Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ is elliptic then there exists $B \in \Psi^{-m}_{\mathrm{sl}}(\mathbb{R}^n)$, unique modulo $\epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$, with*

$$AB - \mathrm{Id} \in \epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n), \qquad BA - \mathrm{Id} \in \epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n).$$

If $A - \mathrm{Id} \in \Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$ then $B - \mathrm{Id} \in \Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$.

Proof. Let b be as in Definition 6.4.4 and $B_0 = \mathrm{Op}_\epsilon(b)$. By Proposition 6.4.2, $AB_0 = \mathrm{Id} - \epsilon R$ with $R \in \Psi^{-1}_{\mathrm{sl}}$. By Lemma 6.1.5(i) (applied to the symbols) there is $S \in \Psi^0_{\mathrm{sl}}$ with $S - \sum_{j<N} \epsilon^j R^j \in \epsilon^N \Psi^{-N}_{\mathrm{sl}}$ for all N, and then $B_R = B_0 S$ satisfies

$$AB_R = (\mathrm{Id} - \epsilon R)S \equiv (\mathrm{Id} - \epsilon R) \sum_{j<N} \epsilon^j R^j = \mathrm{Id} - \epsilon^N R^N \equiv \mathrm{Id}$$

modulo $\epsilon^N \Psi^{-N}_{\mathrm{sl}}$, for every N; hence $AB_R - \mathrm{Id} \in \epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}$. In the same way there is a left parametrix B_L, and as usual $B_L \equiv B_L A B_R \equiv B_R$ modulo $\epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}$, which is an ideal. This also gives uniqueness. If $A = \mathrm{Id} + A'$ with $A' \in \Psi^{-\infty}_{\mathrm{sl}}$, then b may be taken with $b - \mathrm{Id} \in S^{-\infty}_{\mathrm{sl}}$ (in the proof of Lemma 6.4.5, $\tilde{b} = \mathrm{Id}$), $R \in \Psi^{-\infty}_{\mathrm{sl}}$, and all the terms stay in $\mathrm{Id} + \Psi^{-\infty}_{\mathrm{sl}}$, using Lemma 6.1.5(ii). □

6.4.3 Uniform boundedness on L^2

Theorem 6.4.7. *If $A \in \Psi^0_{\mathrm{sl}}(\mathbb{R}^n)$ then $\sup_{0<\epsilon\le 1} \|A_\epsilon\|_{\mathcal{B}(L^2(\mathbb{R}^n))} < \infty$. If $A \in \epsilon^N\Psi^0_{\mathrm{sl}}(\mathbb{R}^n)$ then $\|A_\epsilon\|_{\mathcal{B}(L^2)} = O(\epsilon^N)$.*

Proof. First let $A \in \Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$, with kernel as in (6.3.2). Then $|K(\epsilon, x, Z)| \le C\langle Z\rangle^{-n-1}$, so

$$\sup_x \int |K_{A_\epsilon}(x, y)|\, dy = \sup_x \int |K(\epsilon, x, Z)|\, dZ \le C',$$
$$\sup_y \int |K_{A_\epsilon}(x, y)|\, dx = \sup_y \int |K(\epsilon, y + \epsilon Z, Z)|\, dZ \le C',$$

and Schur's lemma (Lemma 2.7.2) gives $\|A_\epsilon\| \le C'$.

In general, let $A = \mathrm{Op}_\epsilon(a)$, $a \in S^0_{\mathrm{sl}}$, and $C = 2 + \sup|a|^2$, the supremum over $[0, 1] \times \mathbb{R}^{2n}$. We follow the proof of Theorem 2.7.1, now keeping track of ϵ. The family $Q = C - A^*A \in \Psi^0_{\mathrm{sl}}$ is self-adjoint and $\sigma_{\epsilon,0}(Q) = [g]$ with $g = C - |a|^2 \ge 2$. Let $b_0 = g^{1/2} \in S^0_{\mathrm{sl}}$ (a smooth function of a symbol of order 0 with values in $[2, \infty)$; the estimates follow from the chain rule) and $B_0 = \frac{1}{2}(\mathrm{Op}_\epsilon(b_0) + \mathrm{Op}_\epsilon(b_0)^*)$, so $B_0 = B_0^*$ and $\sigma_{\epsilon,0}(B_0) = [b_0]$. Then $E_1 = Q - B_0^2 \in \epsilon\Psi^{-1}_{\mathrm{sl}}$. Suppose inductively that $S_N = B_0 + \cdots + B_N$ with $B_j = B_j^* \in \epsilon^j\Psi^{-j}_{\mathrm{sl}}$ has been found with $E_{N+1} = Q - S_N^2 \in \epsilon^{N+1}\Psi^{-N-1}_{\mathrm{sl}}$. Then $E_{N+1} = E_{N+1}^*$, so $E_{N+1} = \epsilon^{N+1}\,\mathrm{Op}_\epsilon(e)$ with $e \in S^{-N-1}_{\mathrm{sl}}$ real modulo $\epsilon S^{-N-2}_{\mathrm{sl}}$. Let $h = e/(2b_0) \in S^{-N-1}_{\mathrm{sl}}$ and $B_{N+1} = \frac{1}{2}\epsilon^{N+1}(\mathrm{Op}_\epsilon(h) + \mathrm{Op}_\epsilon(h)^*)$. Then

$$Q - S_{N+1}^2 = E_{N+1} - S_N B_{N+1} - B_{N+1} S_N - B_{N+1}^2,$$

and since $S_N - B_0 \in \epsilon\Psi_{\mathrm{sl}}^{-1}$ and $\sigma_\epsilon(E_{N+1} - B_0B_{N+1} - B_{N+1}B_0) = \epsilon^{N+1}(e - 2b_0h) = 0$ modulo $\epsilon^{N+2}S_{\mathrm{sl}}^{-N-2}$, we get $Q - S_{N+1}^2 \in \epsilon^{N+2}\Psi_{\mathrm{sl}}^{-N-2}$. By Lemma 6.1.5(i) there is $B' \in \Psi_{\mathrm{sl}}^0$ with $B' - S_N \in \epsilon^{N+1}\Psi_{\mathrm{sl}}^{-N-1}$ for all N; replacing B' by $B = \frac{1}{2}(B' + B'^*)$ we may assume $B = B^*$. As in Proposition 2.7.3, $G = Q - B^2 \in \bigcap_N \epsilon^N\Psi_{\mathrm{sl}}^{-N} = \epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}$. For $u \in \mathcal{S}(\mathbb{R}^n)$,

$$\|A_\epsilon u\|^2 = C\|u\|^2 - \|B_\epsilon u\|^2 - (G_\epsilon u, u) \le (C + \|G_\epsilon\|)\|u\|^2,$$

and $\|G_\epsilon\|$ is bounded by the first part. The last statement follows by applying this to $\epsilon^{-N}A \in \Psi_{\mathrm{sl}}^0$. □

Corollary 6.4.8. *If $A \in \Psi_{\mathrm{sl}}^0(\mathbb{R}^n)$ then $\limsup_{\epsilon\downarrow 0}\|A_\epsilon\|_{\mathcal{B}(L^2)} \le \sup_{x,\eta}|\sigma_{\mathrm{sl}}(A)(x,\eta)|$.*

Proof. Let $C > \sup|a(0,\cdot,\cdot)|^2$. Since $a(\epsilon) - a(0) = O(\epsilon)$ uniformly, $g = C - |a|^2 \ge c_0 > 0$ for $\epsilon \le \epsilon_0$. Let ψ be as in the proof of Lemma 6.4.5 and $b_0 = \psi(\epsilon)g^{1/2} + (1 - \psi(\epsilon))$, a positive element of S_{sl}^0 bounded below. Then $g - b_0^2$ vanishes for $\epsilon \le \frac{1}{2}\epsilon_0$, so lies in $\epsilon^\infty S_{\mathrm{sl}}^0 = \bigcap_N \epsilon^N S_{\mathrm{sl}}^0$. Running the construction of the previous proof with this b_0, all errors are now in $\epsilon^N\Psi_{\mathrm{sl}}^{-N} + \epsilon^\infty\Psi_{\mathrm{sl}}^0$, and we obtain $C - A^*A = B^2 + G$ with $B = B^*$ and $G \in \epsilon^\infty\Psi_{\mathrm{sl}}^0$. By Theorem 6.4.7, $\|G_\epsilon\| = O(\epsilon^\infty)$, so $\|A_\epsilon\|^2 \le C + O(\epsilon^\infty)$. □

In fact equality holds (Exercise 6.5). Together with Theorem 6.4.6 this gives invertibility for small ϵ: if $A \in \Psi_{\mathrm{sl}}^0(\mathbb{R}^n)$ is elliptic and B is a parametrix, then $\|(AB - \mathrm{Id})_\epsilon\| + \|(BA - \mathrm{Id})_\epsilon\| = O(\epsilon^\infty)$, so for ϵ small A_ϵ is invertible on $L^2(\mathbb{R}^n)$ and $\|A_\epsilon^{-1} - B_\epsilon\| = O(\epsilon^\infty)$. On a compact manifold we shall see that the inverse is itself a semiclassical family (Theorem 6.5.8).

6.4.4 Oscillatory testing

The semiclassical symbol can be recovered from the action of the family on rapidly oscillating functions; this gives an invariant description of it.

Proposition 6.4.9. *Let $A \in \Psi_{\mathrm{sl}}^m(\mathbb{R}^n)$, let $f \in \mathcal{C}^\infty(\mathbb{R}^n)$ be real with $\partial_j f \in \mathcal{C}_{\mathrm{b}}^\infty(\mathbb{R}^n)$ for all j, and $u \in \mathcal{S}(\mathbb{R}^n)$. Then*

$$e^{-if/\epsilon}A_\epsilon(e^{if/\epsilon}u) = \mathrm{Op}_\epsilon(c_f)u, \qquad c_f \in S_{\mathrm{sl}}^m(\mathbb{R}^n), \quad c_f(0,x,\eta) = \sigma_{\mathrm{sl}}(A)(x,\eta + df(x)),$$

and consequently

$$\lim_{\epsilon\downarrow 0} e^{-if/\epsilon}A_\epsilon(e^{if/\epsilon}u) = \sigma_{\mathrm{sl}}(A)(x,df(x))\,u \quad \text{in } \mathcal{S}(\mathbb{R}^n).$$

Proof. Write $f(y) - f(x) = -(x-y)\cdot\Phi(x,y)$ with $\Phi(x,y) = \int_0^1 \nabla f(y + s(x-y))\,ds$, so that $\Phi \in \mathcal{C}_{\mathrm{b}}^\infty(\mathbb{R}^{2n};\mathbb{R}^n)$ and $\Phi(x,x) = \nabla f(x)$, the coordinate expression of df. If $A = \mathrm{Op}_\epsilon(a)$, the kernel of $u \mapsto e^{-if/\epsilon}A_\epsilon(e^{if/\epsilon}u)$ is

$$\int e^{i(x-y)\cdot(\xi - \Phi(x,y)/\epsilon)}a(\epsilon,x,\epsilon\xi)\,d\!\!\!^{-}\xi = \int e^{i(x-y)\cdot\xi}a(\epsilon,x,\epsilon\xi + \Phi(x,y))\,d\!\!\!^{-}\xi,$$

by the change of variables $\xi \mapsto \xi + \Phi(x,y)/\epsilon$. This is justified when a has compact support in η, the integrals then being absolutely convergent, and in general by replacing a by $\chi(\eta/l)a$ and letting $l \to \infty$, using Proposition 2.2.3(ii) on both sides. Thus the operator is $\mathrm{Op}_\epsilon(p)$ with $p(\epsilon,x,y,\eta) = a(\epsilon,x,\eta + \Phi(x,y))$, and $p \in S_{\mathrm{sl}}^m(\mathbb{R}^{2n};\mathbb{R}^n)$ by (6.2.4) and the chain rule. By Theorem 6.2.6 it equals $\mathrm{Op}_\epsilon(c_f)$ with $c_f(0,x,\eta) = p(0,x,x,\eta) = a(0,x,\eta + \nabla f(x))$. The limit follows from Lemma 6.3.4 applied to the ϵ-independent function u. □

6.4.5 Coordinate invariance

Let $F : \Omega \to \Omega'$ be a diffeomorphism between open subsets of $\mathbb{R}^n$ and $H = F^{-1}$. The canonical identification of cotangent spaces is

$$\tilde{F} : T^*\Omega' \to T^*\Omega, \qquad (x', \eta') \mapsto \big(H(x'), (DH(x')^T)^{-1}\eta'\big),$$

so that $\eta' = F^*\xi$ at $x' = F(x)$; for a function b on $T^*\Omega$ we write $\tilde{F}^*b = b \circ \tilde{F}$.

Proposition 6.4.10. *Let $A \in \Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ have kernels supported in $K \times K$ for all ϵ, where $K \subset \Omega$ is compact. Then the family $A_F = (F^{-1})^* \circ A \circ F^*$, i.e. $A_{F,\epsilon}v = \big(A_\epsilon(v \circ F)\big) \circ H$ on Ω' and 0 outside, is an element of $\Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ with kernels supported in $F(K) \times F(K)$, and*

$$\sigma_{\epsilon,m}(A_F) = \tilde{F}^*\sigma_{\epsilon,m}(A), \qquad \text{in particular} \quad \sigma_{\mathrm{sl}}(A_F) = \tilde{F}^*\sigma_{\mathrm{sl}}(A).$$

If A is classical so is A_F.

Proof. Let $A = \mathrm{Op}_\epsilon(a)$; since the kernel vanishes for $x \notin K$, so does $a(\epsilon, x, \eta)$. The kernel of A_F is $k'(\epsilon, x', y') = k(\epsilon, H(x'), H(y'))|\det DH(y')|$, where k is the kernel of A. Choose $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ equal to 1 near $F(K)$ and let $L = \operatorname{supp}\chi$. Write $H(x') - H(y') = G(x', y')(x' - y')$ with $G(x', y') = \int_0^1 DH(y' + s(x' - y'))\, ds$. By compactness there is $r > 0$ such that for $x', y' \in L$ with $|x' - y'| < r$ the segment $[y', x']$ lies in Ω' and $|\det G(x', y')| \geq c > 0$. Let $\rho \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ be supported in $\{|z| < r\}$ and equal to 1 for $|z| \leq r/2$. Then

$$k' = \rho(x' - y')\chi(x')\chi(y')k' + (1 - \rho(x' - y'))\chi(x')\chi(y')k'.$$

On the support of the second term $|x' - y'| \geq r/2$, hence $|H(x') - H(y')| \geq d > 0$; by Lemma 6.3.1 this term is smooth, compactly supported and $O(\epsilon^\infty)$ with all derivatives, so it is the kernel of an element of $\epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}$ by Lemma 6.3.2. In the first term we write $k(\epsilon, x, y) = \int e^{i(x-y)\cdot\xi}a(\epsilon, x, \epsilon\xi)d\!\!\!^-\xi$ and change variables $\xi = (G(x', y')^T)^{-1}\xi'$, so that $(H(x') - H(y')) \cdot \xi = (x' - y') \cdot \xi'$. (As in the proof of Proposition 6.4.9 this is first done for a with compact support in η and then justified by a limiting argument.) Since the map $\xi \mapsto \xi'$ is linear it commutes with the rescaling, and the first term becomes the kernel of $\mathrm{Op}_\epsilon(p)$ with

$$p(\epsilon, x', y', \eta') = \rho(x' - y')\chi(x')\chi(y')\frac{|\det DH(y')|}{|\det G(x', y')|}\, a\big(\epsilon, H(x'), (G(x', y')^T)^{-1}\eta'\big).$$

Here $p \in S^m_{\mathrm{sl}}(\mathbb{R}^{2n}; \mathbb{R}^n)$: it is compactly supported in (x', y'), and the symbol estimates follow from the chain rule since $(G^T)^{-1}$ is smooth and uniformly invertible on the support, so $\langle (G^T)^{-1}\eta'\rangle$ and $\langle \eta'\rangle$ are comparable. Thus $A_F = \mathrm{Op}_\epsilon(p) + R$ with $R \in \epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}$, and by Theorem 6.2.6, $\mathrm{Op}_\epsilon(p) = \mathrm{Op}_\epsilon(c)$ with $c - p(\epsilon, x', x', \eta') \in \epsilon S^{m-1}_{\mathrm{sl}}$. Since $G(x', x') = DH(x')$ and $\chi = 1$ on $F(K)$, which contains the x'-support of $a(\epsilon, H(x'), \cdot)$, $p(\epsilon, x', x', \eta') = a(\epsilon, H(x'), (DH(x')^T)^{-1}\eta') = \tilde{F}^*a$. The statement about supports is clear, and if a is classical then so is p and, by Remark 6.4.3, so is c. □

The proof shows more precisely that the full symbol of A_F has the expansion

$$\sum_\alpha \frac{\epsilon^{|\alpha|}}{\alpha!}\partial^\alpha_{\eta'}D^\alpha_{y'}p(\epsilon, x', y', \eta')\big|_{y'=x'},$$

but only the leading term is invariantly meaningful.

6.5 Semiclassical families on manifolds

The calculus on $\mathbb{R}^n$ is local and coordinate invariant, so it transfers to manifolds exactly as in Chapter 5. The only new feature is the behaviour of kernels away from the diagonal: they must not only be smooth, but also vanish to infinite order as $\epsilon \downarrow 0$.

Throughout this section X is a compact manifold of dimension n and E, F, G are complex vector bundles over X; notation is as in Chapter 5. In particular $\mathrm{Hom}(E,F)\otimes\Omega_R$ is the bundle over X^2 of which kernels of operators from sections of E to sections of F are sections. The definition follows the local characterization of pseudodifferential operators in Theorem 5.4.2; the only new point is what is required of the kernel away from the diagonal, which is dictated by Lemma 6.3.1.

We write $\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty}(X;E,F))$ for the space of families whose kernels are given by sections of $\mathrm{Hom}(E,F)\otimes\Omega_R$ over $[0,1]\times X^2$ which are smooth and vanish to infinite order at $\epsilon=0$; we also denote this space by $\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}(X;E,F)$, a notation justified by Theorem 6.5.3(e) below. By Lemma 6.3.2, if such a family has kernel supported in $W\times W$ for a coordinate chart W, then its local representative is in $\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}(\mathbb{R}^n)$.

Definition 6.5.1. For $m\in\mathbb{R}$, $\Psi_{\mathrm{sl}}^m(X;E,F)$ is the space of families $A=(A_\epsilon)_{0<\epsilon\le1}$ of linear operators $A_\epsilon:\mathcal{C}^\infty(X;E)\to\mathcal{C}^\infty(X;F)$ such that

(i) if $\phi,\psi\in\mathcal{C}^\infty(X)$ have disjoint supports, then $\phi A\psi\in\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty}(X;E,F))$;

(ii) for every coordinate chart $\kappa:W\to\kappa(W)\subset\mathbb{R}^n$ over which E and F are trivial, and every $\psi\in\mathcal{C}_{\mathrm{c}}^\infty(W)$, the local representative $\kappa_*(\psi A\psi)=(\kappa^{-1})^*\circ\psi A\psi\circ\kappa^*$ is a matrix of elements of $\Psi_{\mathrm{sl}}^m(\mathbb{R}^n)$.

We set $\Psi_{\mathrm{sl}}^{-\infty}(X;E,F)=\bigcap_m\Psi_{\mathrm{sl}}^m(X;E,F)$, $\epsilon^N\Psi_{\mathrm{sl}}^m(X;E,F)=\{\epsilon^N A;A\in\Psi_{\mathrm{sl}}^m(X;E,F)\}$, $\Psi_{\mathrm{sl}}^m(X;E)=\Psi_{\mathrm{sl}}^m(X;E,E)$, and define the classical families $\Psi_{\mathrm{sl,ph}}^m(X;E,F)$ by requiring the local representatives in (ii) to be classical.

Let $S^m(T^*X;W)$ be the symbol space of Chapter 5, for a bundle W over X pulled back to T^*X, and let $S_{\mathrm{sl}}^m(T^*X;W)$ be the space of smooth sections a of W over $[0,1]\times T^*X$ such that in local coordinates and trivializations, on compact subsets of the coordinate patches, $|\partial_\epsilon^k D_x^\alpha D_\xi^\beta a|\le C\langle\xi\rangle^{m-|\beta|}$; equivalently $S_{\mathrm{sl}}^m(T^*X;W)=\mathcal{C}^\infty([0,1];S^m(T^*X;W))$. As X is compact, $S^{-\infty}(T^*X;W)=\mathcal{S}(T^*X;W)$ is the space of smooth sections which are rapidly decreasing, with all derivatives, along the fibres. The results of Section 6.1 (Taylor expansion in ϵ, residual symbols, asymptotic summation) hold for these spaces, with the same proofs or by a partition of unity.

6.5.1 Localization

Lemma 6.5.2. (i) $\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty}(X;E,F))\subset\Psi_{\mathrm{sl}}^m(X;E,F)$ *for every* m.

(ii) *If* $\kappa:W\to\kappa(W)$ *is a trivializing chart and* B *is a matrix of elements of* $\Psi_{\mathrm{sl}}^m(\mathbb{R}^n)$ *with kernels supported in* $K\times K$ *for a fixed compact* $K\subset\kappa(W)$, *then* $\kappa^*\circ B\circ(\kappa^{-1})^*$, *extended by zero, is an element of* $\Psi_{\mathrm{sl}}^m(X;E,F)$.

(iii) (Localization) *If* $\{U_i\}$ *is a finite open cover of* X *by trivializing charts and* $A\in\Psi_{\mathrm{sl}}^m(X;E,F)$, *then* $A=\sum_i A_i+A'$ *with* $A'\in\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty}(X;E,F))$ *and* A_i *of the form in* (ii) *with kernels supported in compact subsets of* $U_i\times U_i$.

(iv) *In Definition 6.5.1, it suffices to require (ii) for the charts of one finite cover of* X.

(v) *Each* $A \in \Psi^m_{\mathrm{sl}}(X;E,F)$ *is a family of elements* $A_\epsilon \in \Psi^m(X;E,F)$*, and maps* $\mathcal{C}^\infty([0,1]\times X;E)$ *into* $\mathcal{C}^\infty([0,1]\times X;F)$ *and sections vanishing to infinite order at* $\epsilon = 0$ *to sections with the same property.*

Proof. (i) Condition (i) of the definition is clear, and condition (ii) holds by Lemma 6.3.2, since the local representatives have kernels in $\mathcal{C}^\infty_{\mathrm{c}}([0,1]\times\mathbb{R}^{2n})$ vanishing to infinite order at $\epsilon = 0$.

(ii) For condition (i), if ϕ,ψ have disjoint supports, let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\kappa(W))$ equal 1 near K; then the local representative of $\phi(\kappa^*B)\psi$ is $\tilde\phi B\tilde\psi$ with $\tilde\phi = \chi\cdot(\phi\circ\kappa^{-1})$, $\tilde\psi = \chi\cdot(\psi\circ\kappa^{-1})$ in $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with disjoint supports, which is in $\epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$ by Corollary 6.3.3, with compactly supported kernel. For condition (ii), let $\kappa' : W' \to \kappa'(W')$ be another trivializing chart and $\psi' \in \mathcal{C}^\infty_{\mathrm{c}}(W')$. The kernel of $\psi'(\kappa^*B)\psi'$ is supported in $L\times L$, $L = \operatorname{supp}\psi'\cap\kappa^{-1}(K)$, a compact subset of $W\cap W'$. In the chart κ it is $\tilde\psi B\tilde\psi$, with $\tilde\psi = \chi\cdot(\psi'\circ\kappa^{-1}) \in \mathcal{C}^\infty_{\mathrm{c}}(\kappa(W\cap W'))$, so in $\Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ by Corollary 6.3.3. Applying Proposition 6.4.10 to the diffeomorphism $\kappa'\circ\kappa^{-1} : \kappa(W\cap W') \to \kappa'(W\cap W')$ (the change of trivializations of E, F is multiplication by smooth matrices, which are in $\Psi^0_{\mathrm{sl}}(\mathbb{R}^n)$ after cutting off) shows that $\kappa'_*(\psi'(\kappa^*B)\psi') \in \Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$.

(iii) Let $\{\phi_a\}$ be a partition of unity such that any two supports which meet lie in a common U_i (Lemma 5.1.3). Then $A = \sum_{a,b}\phi_a A\phi_b$; the terms with disjoint supports are in $\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty})$ by (i) of the definition, and each remaining term $\phi_a A\phi_b$ has kernel supported in $U_i\times U_i$ for some i; it equals $\phi_a(\psi A\psi)\phi_b$ with $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(U_i)$ equal to one on $\operatorname{supp}\phi_a\cup\operatorname{supp}\phi_b$, so its local representative is in $\Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ by (ii) of the definition and Corollary 6.3.3.

(iv) If (ii) of the definition holds for the charts U_i of a cover, the decomposition in the proof of (iii) still applies (using only these charts), and then (ii) of the present lemma shows that A satisfies (ii) of the definition for all charts.

(v) By (iii) it suffices to consider A', for which this is clear, and the A_i, for which it follows from Lemma 6.1.4(i) and Lemma 6.3.4 in local coordinates. □

6.5.2 The calculus

Fix a smooth positive density on X and hermitian metrics on the bundles, so that adjoints are defined as in Proposition 5.5.3.

Theorem 6.5.3. *Let* X *be compact.*

(a) (Ideal) *Let* $A \in \Psi^m_{\mathrm{sl}}(X;F,G)$*. If* $R \in \epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}(X;E,F)$ *then* $AR \in \epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}(X;E,G)$*; similarly* $R'A \in \epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}(X;E,G)$ *if* $R' \in \epsilon^\infty\Psi^{-\infty}_{\mathrm{sl}}(X;G,E)$*.*

(b) (Composition and adjoints) $\Psi^m_{\mathrm{sl}}(X;F,G)\circ\Psi^{m'}_{\mathrm{sl}}(X;E,F) \subset \Psi^{m+m'}_{\mathrm{sl}}(X;E,G)$*, and* $A \in \Psi^m_{\mathrm{sl}}(X;E,F)$ *implies* $A^* \in \Psi^m_{\mathrm{sl}}(X;F,E)$*. The same holds for the classical families.*

(c) (Symbols) *There are linear maps*

$$\sigma_{\mathrm{sl}} : \Psi^m_{\mathrm{sl}}(X;E,F) \to S^m(T^*X;\hom(E,F)),$$
$$\sigma_{\epsilon,m} : \Psi^m_{\mathrm{sl}}(X;E,F) \to S^m_{\mathrm{sl}}(T^*X;\hom(E,F))/\epsilon S^{m-1}_{\mathrm{sl}}(T^*X;\hom(E,F)),$$

and $\tilde\sigma_m$ *with values in* $S^m_{\mathrm{sl}}/S^{m-1}_{\mathrm{sl}}$*, uniquely determined by the requirement that for every trivializing chart and* $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$ *the corresponding symbol of* $\kappa_*(\psi A\psi)$ *is the local expression of* ψ^2 *times the symbol of* A*. They are multiplicative,* $\sigma(A^*) = \sigma(A)^*$*, the three sequences of Proposition 6.4.2 are exact with* $\mathbb{R}^n$ *replaced by* X *and the symbol spaces by those over* T^*X*, and* $\sigma_{\epsilon,m} = (\tilde\sigma_m,\sigma_{\mathrm{sl}})$ *identifies the joint symbol with the compatible pairs.*

(d) (Quantization) *There is a linear map* $q_\epsilon : S^m_{\mathrm{sl}}(T^*X;\hom(E,F)) \to \Psi^m_{\mathrm{sl}}(X;E,F)$ *with* $\sigma_{\epsilon,m}(q_\epsilon(a)) = [a]$, *mapping* $\epsilon^N S^{m'}_{\mathrm{sl}}$ *into* $\epsilon^N\Psi^{m'}_{\mathrm{sl}}$ *and classical symbols to classical families.*

(e) (Asymptotic completeness) *If* $A_j \in \epsilon^j\Psi^{m-j}_{\mathrm{sl}}(X;E,F)$ *there is* $A \in \Psi^m_{\mathrm{sl}}(X;E,F)$ *with* $A - \sum_{j<N} A_j \in \epsilon^N\Psi^{m-N}_{\mathrm{sl}}(X;E,F)$ *for all* N, *unique modulo* $\bigcap_N \epsilon^N\Psi^{m-N}_{\mathrm{sl}}(X;E,F) = \dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty}(X;E,F))$.

(f) *For each* $\epsilon > 0$, $A_\epsilon \in \Psi^m(X;E,F)$ *and its principal symbol is represented by* $(x,\xi) \mapsto a(\epsilon,x,\epsilon\xi)$, *where* a *represents* $\tilde\sigma_m(A)$.

Proof. (a) By Lemma 6.5.2(iii) it suffices to consider $A'R$, which clearly has kernel in $\dot{\mathcal{C}}^\infty$, and A_iR. The kernel of A_iR is $(x,y) \mapsto A_i(R(\cdot,y))(x)$; in local coordinates, and after multiplying $R(\cdot,y)$ by a cutoff equal to one on the projection of the support of A_i, this is the action of a matrix of elements of $\Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ on a family of compactly supported smooth functions, depending smoothly on y and vanishing to infinite order at $\epsilon = 0$. By Lemma 6.3.4, applied with the derivatives in y, the result is smooth in (ϵ,x,y) and vanishes to infinite order at $\epsilon = 0$. For $R'A$ use $R'A = (A^*R'^*)^*$, once adjoints are known; the adjoint of an element of $\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty})$ clearly is of the same type.

(b) Adjoints: condition (i) is preserved since the kernel of $\phi A^*\psi$ is the adjoint of that of $\bar\psi A\bar\phi$. For (ii), in a chart where the density is $\nu_0|dx|$ and the metrics are given by positive matrices h_E, h_F, the local representative of $\psi A^*\psi$ is $\nu_0^{-1}h_E^{-1}C^*h_F\nu_0$ with C the representative of $\bar\psi A\bar\psi$ and C^* its Euclidean adjoint; this is in $\Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ by Theorem 6.2.7, multiplication by smooth functions being in $\Psi^0_{\mathrm{sl}}(\mathbb{R}^n)$ (after cutting off).

Composition: we follow the proof of Theorem 5.5.5. With ϕ_c, χ_c, χ'_c, $\tilde\chi_c$ as there,

$$AB = \sum_c (\chi_cA\phi_cB\chi'_c + (1-\chi_c)A\phi_cB + \chi_cA\phi_cB(1-\chi'_c)).$$

Now $(1-\chi_c)A\phi_c \in \dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty})$ by (i) of the definition, and its product with B is again of this type by (a); similarly for the last term. Finally $\chi_cA\phi_cB\chi'_c = \chi_c(\tilde\chi_cA\tilde\chi_c)\phi_c(\tilde\chi_cB\tilde\chi_c)\chi'_c$ is, in the chart, a product of matrices of elements of $\Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ and $\Psi^{m'}_{\mathrm{sl}}(\mathbb{R}^n)$, hence in $\Psi^{m+m'}_{\mathrm{sl}}(\mathbb{R}^n)$ by Theorem 6.2.7, with compact support, and Lemma 6.5.2(i),(ii) gives the result. Classical families are preserved by Remark 6.4.3.

(c) This is the proof of Proposition 5.5.7, with Proposition 6.4.10 in place of the coordinate invariance of the principal symbol, and Proposition 6.4.2 and Corollary 6.3.3 in place of the local symbol calculus. We indicate the main points for $\sigma = \sigma_{\mathrm{sl}}$ or $\sigma_{\epsilon,m}$. Choose a finite cover by trivializing charts $\kappa_i : W_i \to \kappa_i(W_i)$ and $\psi_i \in \mathcal{C}^\infty_{\mathrm{c}}(W_i)$ with $\sum_i \psi_i^2 = 1$, and set $\sigma(A) = \sum_i \tilde\kappa_i^*\sigma(\kappa_{i*}(\psi_iA\psi_i))$, where $\tilde\kappa_i^*$ transfers a symbol on $T^*\kappa_i(W_i)$, supported over a compact set, to T^*X as in Proposition 6.4.10, extended by zero. If $\kappa : W \to \kappa(W)$ is any chart and $\psi \in \mathcal{C}^\infty_{\mathrm{c}}(W)$, then $\psi A\psi - \sum_i \psi_i(\psi A\psi)\psi_i = \sum_i \psi_i[\psi_i,\psi A\psi]$, and by Proposition 6.4.2 (commutativity of the symbols) the local representative of each commutator has vanishing symbol σ. The operator $\psi_i\psi A\psi\psi_i = \psi(\psi_iA\psi_i)\psi$ has kernel supported in $(W\cap W_i)^2$; its symbol in the chart κ_i is ψ^2 times that of $\kappa_{i*}(\psi_iA\psi_i)$, by Corollary 6.3.3, and by Proposition 6.4.10 its symbol in the chart κ is the same function on T^*X. Summing gives the required property; uniqueness follows by applying the property to the ψ_i. Multiplicativity, the behaviour under adjoints and exactness are local, and follow from Proposition 6.4.2 as in the proof of Proposition 5.5.7: for instance if $\sigma_{\mathrm{sl}}(A) = 0$ then in the decomposition of Lemma 6.5.2(iii), each $A_i = \phi_aA\phi_b$ has vanishing semiclassical symbol, so $A_i \in \epsilon\Psi^m_{\mathrm{sl}}$, and $A' = \epsilon(\epsilon^{-1}A')$ with $\epsilon^{-1}A' \in \dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty})$.

(d) Define q_ϵ by the formula (5.5.4), with Op replaced by Op_ϵ; the proof that it has the stated properties is the same as there.

(e) Choose the A_j in the form of Lemma 6.5.2(iii) with respect to a fixed cover and partition of unity, so that the local pieces have kernels supported in fixed compact sets independent of j, and sum the local symbols with Lemma 6.1.5(i), as in the proof of Proposition 5.5.9. The identification of the intersection follows from Lemmas 6.1.3 and 6.3.2 in local coordinates and Lemma 6.5.2(iii).

(f) This is Lemma 6.5.2(v) together with the remark following Definition 6.4.1. □

Differential families $P_\epsilon = \sum_{|\alpha|\le m} p_\alpha(\epsilon, x)(\epsilon D)^\alpha$, with p_α smooth on $[0,1]\times X$ (in local coordinates and trivializations, the form being preserved under changes of coordinates), are elements of $\Psi^m_{\mathrm{sl,ph}}(X;E,F)$ with $\sigma_{\mathrm{sl}}(P)(x,\xi) = \sum_\alpha p_\alpha(0,x)\xi^\alpha$, an invariantly defined function on T^*X (compare Proposition 6.5.5 below). For example if g is a Riemannian metric and $V \in \mathcal{C}^\infty(X)$ is real, then $\epsilon^2\Delta_g + V \in \Psi^2_{\mathrm{sl}}(X)$ has semiclassical symbol $|\xi|_g^2 + V(x)$.

Remark 6.5.4. The identification of the semiclassical symbol with a function on T^*X depends on the choice of the parameter. If $\epsilon = \lambda\epsilon'$ with $\lambda > 0$ constant, then the kernels (6.3.2) written in terms of ϵ' have $K'(\epsilon', x, Z) = \lambda^{-n}K(\lambda\epsilon', x, Z/\lambda)$, so that $\sigma'_{\mathrm{sl}}(A)(x,\xi) = \sigma_{\mathrm{sl}}(A)(x,\lambda\xi)$. More generally the class Ψ^m_{sl} is unchanged if ϵ is replaced by $T(\epsilon)$, T a diffeomorphism of $[0,1]$ with $T(0) = 0$, and then the semiclassical symbol is dilated by $T'(0)$ (Exercise 6.2). Thus only the derivative of the parameter at $\epsilon = 0$ matters, and invariantly $\sigma_{\mathrm{sl}}(A)$ is a function on the *semiclassical cotangent bundle*

$$ {}^{\mathrm{sl}}T^*X = T^*X \otimes T_0[0,1], $$

the tensor product of T^*X with the (oriented) tangent line of the parameter space at 0: a choice of ϵ trivializes $T_0[0,1]$ by ∂_ϵ and so identifies ${}^{\mathrm{sl}}T^*X$ with T^*X, and replacing ϵ by ϵ' with $\epsilon = \lambda\epsilon' + O(\epsilon'^2)$ replaces $\xi \otimes \partial_\epsilon$ by $(\xi/\lambda)\otimes\partial_{\epsilon'}$, in accordance with the dilation above. In the same way the phases f/ϵ of Proposition 6.5.5 have differentials $df \otimes \partial_\epsilon$ in ${}^{\mathrm{sl}}T^*X$. The bundle ${}^{\mathrm{sl}}T^*X$ is isomorphic, but not canonically, to T^*X; we shall always fix ϵ and identify the two.

Proposition 6.5.5. *If $A \in \Psi^m_{\mathrm{sl}}(X;E,F)$, $u \in \mathcal{C}^\infty(X;E)$ and $f \in \mathcal{C}^\infty(X)$ is real, then*

$$ \lim_{\epsilon\downarrow 0} e^{-if/\epsilon} A_\epsilon(e^{if/\epsilon}u) = \sigma_{\mathrm{sl}}(A)(df)\,u \quad \text{in } \mathcal{C}^\infty(X;F), $$

where $\sigma_{\mathrm{sl}}(A)(df)(x) = \sigma_{\mathrm{sl}}(A)(x, df(x)) \in \hom(E_x, F_x)$. This property determines $\sigma_{\mathrm{sl}}(A)$.

Proof. Decompose A as in Lemma 6.5.2(iii). The term A' contributes $e^{-if/\epsilon}A'_\epsilon(e^{if/\epsilon}u)$, whose kernel is smooth and $O(\epsilon^\infty)$ with all derivatives, uniformly, even after multiplication by $e^{i(f(y)-f(x))/\epsilon}$ (each derivative of the phase factor costs at most ϵ^{-1}). Each A_i is treated in local coordinates by Proposition 6.4.9, after replacing f by a function with the same values near the support of the kernel of A_i and with bounded derivatives. Since every $(x,\xi) \in T^*X$ is of the form $(x, df(x))$, the limits determine $\sigma_{\mathrm{sl}}(A)$. □

6.5.3 Smoothing families

For the semiclassical smoothing families on X the results above specialize as follows; this is the algebra used in the definition of the semiclassical index in Chapter 16.

Proposition 6.5.6. *Let X be compact.*

(i) *$\Psi^{-\infty}_{\mathrm{sl}}(X;E)$ is a $*$-algebra, and there is a multiplicative short exact sequence*

$$ 0 \longrightarrow \epsilon\Psi^{-\infty}_{\mathrm{sl}}(X;E) \hookrightarrow \Psi^{-\infty}_{\mathrm{sl}}(X;E) \xrightarrow{\sigma_{\mathrm{sl}}} \mathcal{S}(T^*X;\hom(E)) \longrightarrow 0. $$

(ii) *Every element of* $\Psi_{\mathrm{sl}}^{-\infty}(X;E)$ *is the sum of an element of* $\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}(X;E)$, *i.e. of* $\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty}(X;E))$, *and finitely many families supported in coordinate charts, of the form* $\kappa^* B(\kappa^{-1})^*$ *with* B *a matrix of elements of* $\Psi_{\mathrm{sl}}^{-\infty}(\mathbb{R}^n)$ *with kernels supported in a fixed compact subset of* $\kappa(W)^2$.

(iii) *Equivalently, a family* A_ϵ *is in* $\Psi_{\mathrm{sl}}^{-\infty}(X;E)$ *if and only if its kernel is smooth on* $(0,1]\times X^2$, *extends smoothly to* $[0,1]\times(X^2\setminus\mathrm{Diag})$ *vanishing to infinite order at* $\epsilon=0$ *there, and near any point of the diagonal, in local coordinates, is of the form* $\epsilon^{-n}K(\epsilon,x,(x-y)/\epsilon)$ *with* K *smooth and, with all derivatives, rapidly decreasing in* Z *(as in Proposition 6.3.5). In the language of Remark 6.3.6, the kernels are* ϵ^{-n} *times smooth sections over* $[[0,1]\times X^2;\{0\}\times\mathrm{Diag}]$ *vanishing to infinite order at all boundary faces other than the front face.*

(iv) *The identity* $\mathrm{Id}\in\Psi_{\mathrm{sl}}^0(X;E)$ *has* $\sigma_{\mathrm{sl}}(\mathrm{Id})=\mathrm{Id}$, *so that* $\sigma_{\mathrm{sl}}(A)=\sigma_{\mathrm{sl}}(\mathrm{Id}\circ A)=\mathrm{Id}\cdot\sigma_{\mathrm{sl}}(A)$ *is consistent with multiplicativity.*

Proof. Parts (i) and (iv) are special cases of Theorem 6.5.3, using $S^{-\infty}(T^*X;\hom(E))=\mathcal{S}(T^*X;\hom(E))$; (ii) is Lemma 6.5.2(iii). For (iii), the conditions are necessary by (ii), Proposition 6.3.5 and Lemma 6.3.1. Conversely, if the kernel has these properties, then (i) of Definition 6.5.1 holds, and in a chart, $\psi A\psi$ has kernel $\epsilon^{-n}K'(\epsilon,x,(x-y)/\epsilon)$ near the diagonal plus a term in $\dot{\mathcal{C}}^\infty$ away from it, with K' of the stated type; by Proposition 6.3.5 and Lemma 6.3.2 this is an element of $\Psi_{\mathrm{sl}}^{-\infty}(\mathbb{R}^n)$. □

6.5.4 Ellipticity and invertibility

Definition 6.5.7. $A\in\Psi_{\mathrm{sl}}^m(X;E,F)$ is *elliptic* if there is $b\in S_{\mathrm{sl}}^{-m}(T^*X;\hom(F,E))$ with $ab-\mathrm{Id}$ and $ba-\mathrm{Id}$ in $\epsilon S_{\mathrm{sl}}^{-1}$, for a representative a of $\sigma_{\epsilon,m}(A)$.

As in Lemma 6.4.5, A is elliptic if and only if $\sigma_{\mathrm{sl}}(A)(x,\xi)$ is invertible for all $(x,\xi)\in T^*X$, with inverse bounded by $C\langle\xi\rangle^{-m}$, and $\tilde\sigma_m(A)$ is invertible for $|\xi|$ large, uniformly in $\epsilon\in[0,1]$. Since X is compact, the first condition just says that $\sigma_{\mathrm{sl}}(A)$ is invertible at every point and elliptic in the usual sense.

Theorem 6.5.8. *Let* $A\in\Psi_{\mathrm{sl}}^m(X;E,F)$ *be elliptic. Then:*

(i) *there is* $B\in\Psi_{\mathrm{sl}}^{-m}(X;F,E)$ *with* $AB-\mathrm{Id}$ *and* $BA-\mathrm{Id}$ *in* $\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}$;

(ii) *there are* $\epsilon_0>0$ *and* $G\in\Psi_{\mathrm{sl}}^{-m}(X;F,E)$ *such that for* $0<\epsilon<\epsilon_0$, $A_\epsilon:\mathcal{C}^\infty(X;E)\to\mathcal{C}^\infty(X;F)$ *is invertible with inverse* G_ϵ, *and* $G-B\in\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}(X;F,E)$;

(iii) *if* $E=F$ *and* $A-\mathrm{Id}\in\Psi_{\mathrm{sl}}^{-\infty}(X;E)$ *then* $G-\mathrm{Id}\in\Psi_{\mathrm{sl}}^{-\infty}(X;E)$.

Proof. (i) is proved exactly as Theorem 6.4.6, using Theorem 6.5.3. For (ii), let $R=AB-\mathrm{Id}\in\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty}(X;F))$. Its kernel, and hence its Hilbert–Schmidt norm, which bounds the operator norm by (5.10.1), is a smooth function of $\epsilon\in[0,1]$ vanishing to infinite order at 0; choose $\epsilon_1>0$ with $\|R_\epsilon\|_{\mathcal{B}(L^2)}\le\frac12$ for $\epsilon\le\epsilon_1$. Then $\mathrm{Id}+R_\epsilon$ is invertible on L^2, and $T_\epsilon=(\mathrm{Id}+R_\epsilon)^{-1}$ depends smoothly on $\epsilon\in[0,\epsilon_1]$ in the norm topology. As in Lemma 5.10.2, $T_\epsilon=\mathrm{Id}+S_\epsilon$ with $S_\epsilon=-R_\epsilon+R_\epsilon T_\epsilon R_\epsilon$. By (the proof of) Lemma 5.9.2, the kernel of $R_\epsilon T_\epsilon R_\epsilon$ is smooth in (ϵ,x,y), and since all derivatives in ϵ of R_ϵ vanish at $\epsilon=0$, so do those of the kernel of S_ϵ. Let $\psi\in\mathcal{C}^\infty([0,1])$ equal 1 on $[0,\frac12\epsilon_1]$ and vanish near $[\epsilon_1,1]$; then $\tilde S=\psi(\epsilon)S\in\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty}(X;F))$ and $(\mathrm{Id}+R)(\mathrm{Id}+\tilde S)=\mathrm{Id}$ for $\epsilon\le\frac12\epsilon_1$. Thus $G_R=B(\mathrm{Id}+\tilde S)$ is a right inverse of A_ϵ for small ϵ, and $G_R-B=B\tilde S\in\dot{\mathcal{C}}^\infty([0,1];\Psi^{-\infty})$ by Theorem 6.5.3(a). In the same way there is a left inverse G_L for small ϵ, and then $G_L=G_LAG_R=G_R$, so A_ϵ is invertible on $\mathcal{C}^\infty$ (and on L^2 if $m=0$) with inverse $G=G_R$. (iii) follows from the last part of Theorem 6.4.6, whose proof applies on X. □

Compare with the classical case (Theorem 5.6.3). There the parametrix errors are smoothing, hence compact, and one obtains only the Fredholm property. Here they are also $O(\epsilon^\infty)$ in norm, and a Neumann series removes them for small ϵ. The logic is:

$$\begin{array}{ccc} \sigma_{\epsilon,m}(A)^{-1} & \xrightarrow{\text{quantize, sum}} & B \\ & & \Big\downarrow {\scriptstyle AB=\mathrm{Id}+R,\ R\in\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}} \\ G=B(\mathrm{Id}+R)^{-1}=A_\epsilon^{-1} & \xleftarrow{\text{Neumann series}} & \|R_\epsilon\|=O(\epsilon^\infty) \end{array}$$

In particular the index of an elliptic semiclassical family vanishes for small ϵ; the topology now sits entirely in the symbol, and this is what the starting point of Chapter 16.

Corollary 6.5.9. *Let $a\in\mathcal{S}(T^*X;\hom(E))$ be such that $\mathrm{Id}+a(x,\xi)$ is invertible for all $(x,\xi)\in T^*X$, and let $A\in\Psi_{\mathrm{sl}}^{-\infty}(X;E)$ with $\sigma_{\mathrm{sl}}(A)=a$. Then there is $\epsilon_0>0$ such that $A_\epsilon\in G^{-\infty}(X;E)$ for $0<\epsilon<\epsilon_0$, i.e. $\mathrm{Id}+A_\epsilon$ is invertible, and $(\mathrm{Id}+A_\epsilon)^{-1}=\mathrm{Id}+B_\epsilon$ with $B\in\Psi_{\mathrm{sl}}^{-\infty}(X;E)$ and $\mathrm{Id}+\sigma_{\mathrm{sl}}(B)=(\mathrm{Id}+a)^{-1}$.*

Proof. $\mathrm{Id}+A\in\Psi_{\mathrm{sl}}^0(X;E)$ is elliptic: $\sigma_{\mathrm{sl}}(\mathrm{Id}+A)=\mathrm{Id}+a$ is invertible everywhere, with $(\mathrm{Id}+a)^{-1}-\mathrm{Id}=-a(\mathrm{Id}+a)^{-1}\in\mathcal{S}(T^*X;\hom(E))$ (the inverse is bounded since $a\to 0$ at infinity in the fibres and X is compact), and $\tilde\sigma_0(\mathrm{Id}+A)=[\mathrm{Id}]$. Now apply Theorem 6.5.8(ii),(iii) and the multiplicativity of σ_{sl}. □

Theorem 6.5.10. *If $A\in\Psi_{\mathrm{sl}}^0(X;E,F)$ then $\sup_{0<\epsilon\le1}\|A_\epsilon\|_{\mathcal{B}(L^2(X;E),L^2(X;F))}<\infty$, and $\|A_\epsilon\|=O(\epsilon^N)$ if $A\in\epsilon^N\Psi_{\mathrm{sl}}^0(X;E,F)$.*

Proof. In the decomposition of Lemma 6.5.2(iii), A' is uniformly bounded since its Hilbert–Schmidt norm is, and each A_i is, in local coordinates, a matrix of elements of $\Psi_{\mathrm{sl}}^0(\mathbb{R}^n)$ with kernels supported in a fixed compact set, on which the L^2 norms of X and $\mathbb{R}^n$ are equivalent; so Theorem 6.4.7 applies. □

Remark 6.5.11 (Parameters and fibrations). All the constructions of this chapter are explicit: they involve the symbol calculus, Borel summation, and, in Theorem 6.5.8 and in Section 6.6 below, the inversion of $\mathrm{Id}+R_\epsilon$ where $\|R_\epsilon\|\to0$. They therefore apply without change to families depending smoothly on additional parameters t in a compact manifold T (symbols in $\mathcal{C}^\infty(T;S_{\mathrm{sl}}^m)$, and operators satisfying the conditions of Definition 6.5.1 locally uniformly in t, with smooth dependence), and the constants, in particular ϵ_0, may be chosen uniformly in t, since $\sup_t\|R_{\epsilon,t}\|\to0$ as $\epsilon\downarrow0$ by compactness. More generally, if $\phi:M\to B$ is a fibration of compact manifolds and E a bundle over M, the space $\Psi_{\mathrm{sl}}^m(M/B;E)$ of semiclassical families on the fibres is defined by requiring that in each local trivialization $\phi^{-1}(U)\cong Z\times U$ the family is a smooth family, parametrized by U, of elements of $\Psi_{\mathrm{sl}}^m(Z;E)$; this is independent of the trivialization by Proposition 6.4.10 (applied with parameters), and the semiclassical symbol is then a section over the fibre cotangent bundle $T^*(M/B)$. All the results above, including Proposition 6.5.6, Theorem 6.5.8 and Corollary 6.5.9, hold for such families, with σ_{sl} taking values in $S^m(T^*(M/B);\hom(E))$.

6.6 Quantization of projections

A smooth family of projections in the symbol algebra can be quantized to a semiclassical family of *exact* projections, for small ϵ. This is the semiclassical replacement for the spectral projections of the harmonic oscillator used in the isotropic calculus, and it is what is needed

to define the even semiclassical index map in Chapter 16. Throughout, X is compact and E is a hermitian vector bundle over X; $L^2(X;E)$ is defined using a smooth positive density.

The construction follows the classical one of Section 5.12 (Figure 6.3): quantize the symbol, improve it by Newton's iteration until P^2-P is residual, and then remove the residual error by a contour integral, which is now possible because the error is small in norm for small ϵ.

$$\begin{array}{c} p,\ p^2=p \xrightarrow{\ q_\epsilon\ } P_1,\ P_1^2-P_1\in\epsilon\Psi_{\mathrm{sl}}^{-1} \xrightarrow{\ \text{Newton}\ } P_j,\ P_j^2-P_j\in\epsilon^{2^{j-1}}\Psi_{\mathrm{sl}}^{-2^{j-1}} \\ \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad \Big\downarrow \text{asymptotic sum} \\ \Pi_\epsilon^2=\Pi_\epsilon\ (\epsilon<\epsilon_0) \xleftarrow{\ \text{Riesz projection}\ } P_\infty,\ P_\infty^2-P_\infty\in\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty} \end{array}$$

Figure 6.3: Quantization of a projection-valued symbol (Theorem 6.6.4). Each step keeps the joint symbol equal to p. The last step (Lemma 6.6.1) uses that the error is $O(\epsilon^\infty)$ in norm, so the spectrum of P_∞ stays away from the circle $|\lambda-1|=\frac12$.

Lemma 6.6.1. *Let $P\in\Psi_{\mathrm{sl}}^0(X;E)$ be such that $P^2-P\in\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}(X;E)$. Then there are $\epsilon_0>0$ and $\Pi\in\Psi_{\mathrm{sl}}^0(X;E)$ with $\Pi-P\in\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}(X;E)$ and $\Pi_\epsilon^2=\Pi_\epsilon$ for $0<\epsilon<\epsilon_0$.*

Proof. Let $R=P^2-P$; then R commutes with P, its kernel is smooth on $[0,1]\times X^2$ and vanishes to infinite order at $\epsilon=0$, and there is $\epsilon_1>0$ with $\|R_\epsilon\|_{\mathcal{B}(L^2)}\le\frac{1}{32}$ for $\epsilon\le\epsilon_1$. As in (5.12.1), for $\lambda\notin\{0,1\}$,

$$(\lambda-P)(P+\lambda-1)=\lambda(\lambda-1)(\mathrm{Id}-S_\lambda),\qquad S_\lambda=\frac{R}{\lambda(\lambda-1)}.$$

On the annulus $\frac38\le|\lambda-1|\le\frac58$ we have $|\lambda(\lambda-1)|\ge\frac{9}{64}$, so $\|S_\lambda\|\le\frac29$ for $\epsilon\le\epsilon_1$; hence $\lambda-P_\epsilon$ is invertible there, with

$$(\lambda-P)^{-1}=\frac{P+\lambda-1}{\lambda(\lambda-1)}(\mathrm{Id}+T_\lambda),\qquad T_\lambda=S_\lambda+S_\lambda(\mathrm{Id}-S_\lambda)^{-1}S_\lambda.$$

As in the proof of Theorem 6.5.8, the kernel of T_λ is smooth in (ϵ,λ,x,y), $\epsilon\le\epsilon_1$, holomorphic in λ, and vanishes to infinite order at $\epsilon=0$, uniformly in λ. Let γ be the circle $|\lambda-1|=\frac12$ and, for $\epsilon\le\epsilon_1$,

$$\Pi_\epsilon=\frac{1}{2\pi i}\oint_\gamma(\lambda-P_\epsilon)^{-1}d\lambda=P_\epsilon+\frac{1}{2\pi i}\oint_\gamma\frac{P_\epsilon+\lambda-1}{\lambda(\lambda-1)}T_\lambda\,d\lambda,$$

where the first term has been evaluated by the residue theorem: the only pole of $(P+\lambda-1)/(\lambda(\lambda-1))$ inside γ is at $\lambda=1$, with residue P. By Theorem 6.5.3(a), applied with the parameter λ, the integrand of the second term has kernel smooth in all variables and vanishing to infinite order at $\epsilon=0$, so $\Pi-P$ is of this form on $[0,\epsilon_1]$. That $\Pi_\epsilon^2=\Pi_\epsilon$ is the standard property of the Riesz projection, proved as in Proposition 5.12.2, the resolvent being holomorphic on the annulus. Finally, replace $\Pi-P$ by $\psi(\epsilon)(\Pi-P)$, with $\psi\in\mathcal{C}^\infty([0,1])$ equal to 1 on $[0,\frac12\epsilon_1]$ and supported in $[0,\epsilon_1)$, and take $\epsilon_0=\frac12\epsilon_1$. □

Lemma 6.6.2. *Let P be an element of an associative algebra with unit and $E=P^2-P$. Then $PE=EP$, and $P'=P+(\mathrm{Id}-2P)E$ satisfies*

$$P'^2-P'=4E^3-3E^2.$$

Proof. $PE = P^3 - P^2 = EP$. Using this, and $(\mathrm{Id}-2P)^2 = \mathrm{Id}+4E$,

$$\begin{aligned} P'^2 - P' &= P^2 + 2P(\mathrm{Id}-2P)E + (\mathrm{Id}-2P)^2E^2 - P - (\mathrm{Id}-2P)E \\ &= E + (2P - 4P^2 - \mathrm{Id}+2P)E + (\mathrm{Id}+4E)E^2 \\ &= E - (4E+\mathrm{Id})E + E^2 + 4E^3 = 4E^3 - 3E^2. \end{aligned}$$ □

Thus the correction $P \mapsto P'$ squares the error, as in Newton's method. In the semiclassical calculus, if $E \in \epsilon^k \Psi_{\mathrm{sl}}^{-k}$, then $4E^3 - 3E^2 \in \epsilon^{2k}\Psi_{\mathrm{sl}}^{-2k}$.

Theorem 6.6.3. *Let $\pi_0 \in \mathcal{C}^\infty(X; \hom(E))$ be a family of projections, $\pi_0^2 = \pi_0$ (for instance a constant projection in $M(N,\mathbb{C})$ when $E = X \times \mathbb{C}^N$), and let $a \in \mathcal{S}(T^*X; \hom(E))$ be such that $(\pi_0 + a)^2 = \pi_0 + a$ on T^*X. Then there are $\epsilon_0 > 0$ and $A \in \Psi_{\mathrm{sl}}^{-\infty}(X;E)$ with $\sigma_{\mathrm{sl}}(A) = a$ such that*

$$(\pi_0 + A_\epsilon)^2 = \pi_0 + A_\epsilon \quad \text{for } 0 < \epsilon < \epsilon_0.$$

Proof. Let $A_1 = q_\epsilon(a) \in \Psi_{\mathrm{sl}}^{-\infty}(X;E)$ be a quantization of a (Theorem 6.5.3(d)) and $P_1 = \pi_0 + A_1 \in \Psi_{\mathrm{sl}}^0(X;E)$. Then $E_1 = P_1^2 - P_1 = \pi_0 A_1 + A_1\pi_0 + A_1^2 - A_1 \in \Psi_{\mathrm{sl}}^{-\infty}$, and $\sigma_{\mathrm{sl}}(E_1) = (\pi_0+a)^2 - (\pi_0+a) = 0$, so $E_1 \in \epsilon\Psi_{\mathrm{sl}}^{-\infty}$. Define inductively $P_{j+1} = P_j + (\mathrm{Id}-2P_j)E_j$ and $E_{j+1} = P_{j+1}^2 - P_{j+1}$. By Lemma 6.6.2, $E_{j+1} = 4E_j^3 - 3E_j^2$, so by induction $E_j \in \epsilon^{2^{j-1}}\Psi_{\mathrm{sl}}^{-\infty}$, $P_j - \pi_0 \in \Psi_{\mathrm{sl}}^{-\infty}$ and $P_{j+1} - P_j \in \epsilon^{2^{j-1}}\Psi_{\mathrm{sl}}^{-\infty}$. By asymptotic summation (Lemma 6.1.5(ii) in local coordinates, as in Theorem 6.5.3(e)) there is $A_\infty \in \Psi_{\mathrm{sl}}^{-\infty}$ with $\pi_0 + A_\infty - P_j \in \epsilon^{2^{j-1}}\Psi_{\mathrm{sl}}^{-\infty}$ for every j. Writing $P_\infty = \pi_0 + A_\infty = P_j + D_j$,

$$P_\infty^2 - P_\infty = E_j + P_jD_j + D_jP_j + D_j^2 - D_j \in \epsilon^{2^{j-1}}\Psi_{\mathrm{sl}}^{-\infty}$$

for all j, so $P_\infty^2 - P_\infty \in \epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}$. Now Lemma 6.6.1 gives a family of projections Π, for $\epsilon < \epsilon_0$, with $\Pi - P_\infty \in \epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}$, and $A = \Pi - \pi_0$ has the required properties, since $\sigma_{\mathrm{sl}}(A) = \sigma_{\mathrm{sl}}(A_\infty) = \sigma_{\mathrm{sl}}(A_1) = a$. □

Theorem 6.6.4. *Let $p \in S^0(T^*X; \hom(E))$ satisfy $p^2 = p$. Then there are $\epsilon_0 > 0$ and $\Pi \in \Psi_{\mathrm{sl}}^0(X;E)$ with $\sigma_{\epsilon,0}(\Pi) = [p]$ (so $\sigma_{\mathrm{sl}}(\Pi) = p$ and $\tilde\sigma_0(\Pi) = [p]$, independent of ϵ) such that $\Pi_\epsilon^2 = \Pi_\epsilon$ for $0 < \epsilon < \epsilon_0$. If p is classical then Π may be taken classical, and then for each $\epsilon \in (0,\epsilon_0)$, $\Pi_\epsilon \in \Psi_{\mathrm{ph}}^0(X;E)$ is a projection with principal symbol $\sigma_0(\Pi_\epsilon) = p_0$, the leading homogeneous part of p (a projection-valued function on S^*X).*

Proof. Let $P_1 = q_\epsilon(p)$, with p regarded as an element of S_{sl}^0 independent of ϵ. Then $\sigma_{\epsilon,0}(P_1^2 - P_1) = [p^2 - p] = 0$, so $E_1 = P_1^2 - P_1 \in \epsilon\Psi_{\mathrm{sl}}^{-1}(X;E)$ by Theorem 6.5.3(c). As in the previous proof, with $P_{j+1} = P_j + (\mathrm{Id}-2P_j)E_j$ we get $E_j \in \epsilon^{2^{j-1}}\Psi_{\mathrm{sl}}^{-2^{j-1}}$ and $P_{j+1} - P_j \in \epsilon^{2^{j-1}}\Psi_{\mathrm{sl}}^{-2^{j-1}}$. By Theorem 6.5.3(e) (with $A_0 = P_1$, $A_k = P_{j+1} - P_j$ if $k = 2^{j-1}$ and $A_k = 0$ otherwise) there is $P_\infty \in \Psi_{\mathrm{sl}}^0$ with $P_\infty - P_j \in \epsilon^{2^{j-1}}\Psi_{\mathrm{sl}}^{-2^{j-1}}$ for all j, and as before $P_\infty^2 - P_\infty \in \bigcap_k \epsilon^k\Psi_{\mathrm{sl}}^{-k} = \epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}$. Apply Lemma 6.6.1. The symbols are unchanged at each step, since $P_{j+1} - P_j \in \epsilon\Psi_{\mathrm{sl}}^{-1}$. In the classical case every step preserves classical families (Remark 6.4.3), and the leading homogeneous term of the symbol of Π is p_0, independent of ϵ; by Theorem 6.5.3(f) the principal symbol of Π_ϵ is $p_0(x,\epsilon\xi) = p_0(x,\xi)$, by homogeneity. □

The quantization is essentially unique.

Proposition 6.6.5. *Let $\Pi, \Pi' \in \Psi_{\mathrm{sl}}^0(X;E)$ be such that Π_ϵ and Π'_ϵ are projections for $\epsilon < \epsilon_0$, and $\sigma_{\mathrm{sl}}(\Pi) = \sigma_{\mathrm{sl}}(\Pi')$. Then there are $\epsilon_1 > 0$ and a smooth family $U_t \in \Psi_{\mathrm{sl}}^0(X;E)$, $t \in [0,1]$, with $U_0 = \mathrm{Id}$, $U_t - \mathrm{Id} \in \epsilon\Psi_{\mathrm{sl}}^0$, $U_{t,\epsilon}$ invertible with $U_t^{-1} \in \Psi_{\mathrm{sl}}^0(X;E)$ for $\epsilon < \epsilon_1$, and $U_1\Pi'U_1^{-1} = \Pi$ for $\epsilon < \epsilon_1$. Thus $\Pi_t = U_t\Pi'U_t^{-1}$ is a smooth homotopy of semiclassical*

families of projections from Π' *to* Π *with constant semiclassical symbol. If* $\Pi - \pi_0$ *and* $\Pi' - \pi_0$ *lie in* $\Psi_{\mathrm{sl}}^{-\infty}(X;E)$*, as in Theorem 6.6.3, then* $U_t - \mathrm{Id}$*,* $U_t^{-1} - \mathrm{Id}$ *and* $\Pi_t - \pi_0$ *lie in* $\Psi_{\mathrm{sl}}^{-\infty}(X;E)$*.*

Proof. For $\epsilon < \epsilon_0$ let $U = \Pi\Pi' + (\mathrm{Id} - \Pi)(\mathrm{Id} - \Pi')$. Then $U\Pi' = \Pi\Pi' = \Pi U$, and

$$U - \mathrm{Id} = 2\Pi\Pi' - \Pi - \Pi' = (2\Pi - \mathrm{Id})(\Pi' - \Pi) = \epsilon W, \qquad W \in \Psi_{\mathrm{sl}}^{0}(X;E),$$

since $\Pi' - \Pi \in \epsilon\Psi_{\mathrm{sl}}^{0}$ by Theorem 6.5.3(c). Let $w \in S_{\mathrm{sl}}^{0}$ represent $\sigma_{\epsilon,0}(W)$ and choose $\epsilon_2 > 0$ with $\epsilon_2 \sup|w| \leq \frac{1}{2}$, and $\psi \in \mathcal{C}^\infty([0,1])$ equal to 1 on $[0, \frac{1}{2}\epsilon_2]$ and supported in $[0, \epsilon_2)$. Set $U_t = \mathrm{Id} + t\psi(\epsilon)\epsilon W$. Its joint symbol is represented by $\mathrm{Id} + t\psi(\epsilon)\epsilon w$, which is invertible with inverse in S_{sl}^{0}, so U_t is elliptic, uniformly in t; by Theorem 6.5.8 and Remark 6.5.11 there is $\epsilon_1 > 0$ such that $U_{t,\epsilon}$ is invertible for $\epsilon < \epsilon_1$, with inverse given by a smooth family in Ψ_{sl}^{0}. For $\epsilon < \min(\epsilon_1, \frac{1}{2}\epsilon_2, \epsilon_0)$, $U_1 = U$ and so $U_1\Pi'U_1^{-1} = \Pi$. In the smoothing case $W \in \Psi_{\mathrm{sl}}^{-\infty}$, and Theorem 6.5.8(iii) applies. □

6.7 Adiabatic and semiclassical-adiabatic families

In the semiclassical limit *all* variables are rescaled and the operators become commutative at $\epsilon = 0$. In an *adiabatic* limit only some of the variables are rescaled: in the model $\mathbb{R}^{n+\tilde{n}} = \mathbb{R}^n_z \times \mathbb{R}^{\tilde{n}}_{\tilde{z}}$, the kernels are compressed in the $\tilde{z}$ variables only, and in the limit one obtains a symbol on $T^*\mathbb{R}^{\tilde{n}}$ with values in operators on $\mathbb{R}^n$. The semiclassical calculus is the case $n = 0$. We only consider smoothing families, which is what is needed in Chapter 16; the adiabatic parameter is denoted $\delta \in (0,1]$.

6.7.1 Adiabatic families

Let $\mathcal{K}_{\mathrm{ad}}$ be the space of functions $B \in \mathcal{C}^\infty([0,1]_\delta \times \mathbb{R}^n_z \times \mathbb{R}^n_{z'} \times \mathbb{R}^{\tilde{n}}_{\tilde{z}} \times \mathbb{R}^{\tilde{n}}_{\tilde{Z}})$ such that, for all multi-indices and all N,

$$|\partial_\delta^k \partial_z^\alpha \partial_{z'}^{\alpha'} \partial_{\tilde{z}}^\beta \partial_{\tilde{Z}}^\gamma B| \leq C\langle z - z'\rangle^{-N}\langle \tilde{Z}\rangle^{-N}. \tag{6.7.1}$$

Definition 6.7.1. $\Psi_{\mathrm{ad}}^{-\infty}(\mathbb{R}^n;\mathbb{R}^{\tilde{n}})$ is the space of families B_δ, $0 < \delta \leq 1$, of operators on $\mathcal{S}(\mathbb{R}^{n+\tilde{n}})$ with kernels

$$K_{B_\delta}(z,\tilde{z};z',\tilde{z}') = \delta^{-\tilde{n}} B\Big(\delta, z, z', \tilde{z}, \frac{\tilde{z} - \tilde{z}'}{\delta}\Big), \qquad B \in \mathcal{K}_{\mathrm{ad}}.$$

The *adiabatic symbol* of such a family is

$$\sigma_{\mathrm{ad}}(B)(z,z',\tilde{z},\tilde{\zeta}) = \int e^{-i\tilde{Z}\cdot\tilde{\zeta}} B(0,z,z',\tilde{z},\tilde{Z})\, d\tilde{Z},$$

regarded, for each $(\tilde{z},\tilde{\zeta}) \in T^*\mathbb{R}^{\tilde{n}}$, as the kernel of an element of $\Psi^{-\infty}(\mathbb{R}^n)$.

For $\delta > 0$ fixed, $B_\delta \in \Psi^{-\infty}(\mathbb{R}^{n+\tilde{n}})$, since $\langle z - z'\rangle^{-N}\langle(\tilde{z} - \tilde{z}')/\delta\rangle^{-N} \leq C\langle(z - z', \tilde{z} - \tilde{z}')\rangle^{-N}$; and B_δ determines $B(\delta,\cdot)$, hence, by continuity, B. Let $S^{-\infty}(T^*\mathbb{R}^{\tilde{n}};\Psi^{-\infty}(\mathbb{R}^n))$ denote the space of smooth $b(z,z',\tilde{z},\tilde{\zeta})$ satisfying the estimates (6.7.1) with $\tilde{Z}$ replaced by $\tilde{\zeta}$ (and no δ).

Proposition 6.7.2. $\Psi_{\mathrm{ad}}^{-\infty}(\mathbb{R}^n;\mathbb{R}^{\tilde{n}})$ *is a* $*$*-algebra, and there is a short exact sequence*

$$0 \longrightarrow \delta\Psi_{\mathrm{ad}}^{-\infty}(\mathbb{R}^n;\mathbb{R}^{\tilde{n}}) \hookrightarrow \Psi_{\mathrm{ad}}^{-\infty}(\mathbb{R}^n;\mathbb{R}^{\tilde{n}}) \xrightarrow{\sigma_{\mathrm{ad}}} S^{-\infty}(T^*\mathbb{R}^{\tilde{n}};\Psi^{-\infty}(\mathbb{R}^n)) \longrightarrow 0$$

which is multiplicative, where the product on the right is pointwise in $(\tilde{z},\tilde{\zeta})$ *and composition of operators on* $\mathbb{R}^n$*:* $\sigma_{\mathrm{ad}}(AB)(\tilde{z},\tilde{\zeta}) = \sigma_{\mathrm{ad}}(A)(\tilde{z},\tilde{\zeta}) \circ \sigma_{\mathrm{ad}}(B)(\tilde{z},\tilde{\zeta})$*; moreover* $\sigma_{\mathrm{ad}}(B^*) = \sigma_{\mathrm{ad}}(B)^*$ *pointwise.*

Proof. If A, B have kernels given by $A, B \in \mathcal{K}_{\mathrm{ad}}$, then for $\delta > 0$, substituting $\tilde{z}' = \tilde{z} - \delta\tilde{Z}'$ in the composition of kernels, the kernel of $A_\delta B_\delta$ is $\delta^{-\tilde{n}} C(\delta, z, z'', \tilde{z}, (\tilde{z} - \tilde{z}'')/\delta)$ with

$$C(\delta, z, z'', \tilde{z}, \tilde{Z}) = \iint A(\delta, z, z', \tilde{z}, \tilde{Z}')\, B(\delta, z', z'', \tilde{z} - \delta\tilde{Z}', \tilde{Z} - \tilde{Z}')\, dz'\, d\tilde{Z}'. \tag{6.7.2}$$

By Peetre's inequality, $\langle z - z''\rangle^N \leq C\langle z - z'\rangle^N \langle z' - z''\rangle^N$ and $\langle \tilde{Z}\rangle^N \leq C\langle \tilde{Z}'\rangle^N \langle \tilde{Z} - \tilde{Z}'\rangle^N$; differentiating under the integral (derivatives in δ produce factors $\tilde{Z}'$, absorbed by the decay of A) shows that $C \in \mathcal{K}_{\mathrm{ad}}$. The adjoint has kernel $\overline{K_{B_\delta}(z', \tilde{z}'; z, \tilde{z})}$, i.e. it is given by $B^*(\delta, z, z', \tilde{z}, \tilde{Z}) = \overline{B(\delta, z', z, \tilde{z} - \delta\tilde{Z}, -\tilde{Z})} \in \mathcal{K}_{\mathrm{ad}}$. At $\delta = 0$, (6.7.2) is composition in z and convolution in $\tilde{Z}$, with $\tilde{z}$ as a parameter; taking the Fourier transform in $\tilde{Z}$ gives the multiplicativity, and similarly for adjoints. The Fourier transform in $\tilde{Z}$ is an isomorphism from the restrictions $B(0, \cdot)$ onto $S^{-\infty}(T^*\mathbb{R}^{\tilde{n}}; \Psi^{-\infty}(\mathbb{R}^n))$, so σ_{ad} is surjective (extend $B(0, \cdot)$ to be independent of δ), and $\sigma_{\mathrm{ad}}(B) = 0$ if and only if $B(0, \cdot) = 0$, i.e. $B = \delta B'$ with $B' \in \mathcal{K}_{\mathrm{ad}}$ by Taylor's formula. □

Discussion. The adiabatic symbol is no longer a function but an operator: for each covector $(\tilde{z}, \tilde{\zeta})$ in the rescaled directions, it is a smoothing operator in the remaining variables z. In the limit $\delta \downarrow 0$ the family becomes "commutative" in $\tilde{z}$ only, and stays genuinely non-commutative in z. This is the right structure for a fibration: one rescales the base and keeps the operators on the fibres, and the adiabatic symbol is a family of fibre operators parametrized by T^* of the base.

The adiabatic calculus is invariant under *fibred* changes of coordinates, i.e. diffeomorphisms which preserve the projection onto the $\tilde{z}$ variables.

Lemma 6.7.3. *Let $\Omega, \Omega' \subset \mathbb{R}^{n+\tilde{n}}$ be open and $\Phi : \Omega' \to \Omega$ a diffeomorphism of the form $\Phi(w, \tilde{w}) = (\Gamma(w, \tilde{w}), \Theta(\tilde{w}))$. If $B \in \Psi^{-\infty}_{\mathrm{ad}}(\mathbb{R}^n; \mathbb{R}^{\tilde{n}})$ has kernels supported in $K \times K$, $K \subset \Omega$ compact, then $B_\Phi = \Phi^* \circ B \circ (\Phi^{-1})^* \in \Psi^{-\infty}_{\mathrm{ad}}(\mathbb{R}^n; \mathbb{R}^{\tilde{n}})$, and $\sigma_{\mathrm{ad}}(B_\Phi)(\tilde{w}, \tilde{\omega})$ is the operator $\sigma_{\mathrm{ad}}(B)(\Theta(\tilde{w}), (D\Theta(\tilde{w})^T)^{-1}\tilde{\omega})$ transported to the fibre $\mathbb{R}^n_w$ by the diffeomorphism $w \mapsto \Gamma(w, \tilde{w})$.*

Proof. The kernel of B_Φ is $K_{B_\delta}(\Phi(w, \tilde{w}); \Phi(w', \tilde{w}'))|\det D\Phi(w', \tilde{w}')|$. Write $\Theta(\tilde{w}) - \Theta(\tilde{w}') = M(\tilde{w}, \tilde{w}')(\tilde{w} - \tilde{w}')$ with $M = \int_0^1 D\Theta(\tilde{w}' + s(\tilde{w} - \tilde{w}'))ds$, and put $\tilde{W} = (\tilde{w} - \tilde{w}')/\delta$. As in the proof of Proposition 6.4.10, we split the kernel with a cutoff $\rho(\tilde{w} - \tilde{w}')$. Where $|\tilde{w} - \tilde{w}'| \geq r/2$ we have $|\Theta(\tilde{w}) - \Theta(\tilde{w}')| \geq d > 0$, so the last argument of B is at least d/δ in size, and this part of the kernel is smooth and $O(\delta^\infty)$ with all derivatives; it is of the form in Definition 6.7.1 with a function in $\delta^\infty \mathcal{K}_{\mathrm{ad}}$. Near the fibre diagonal the kernel is $\delta^{-\tilde{n}} B'(\delta, w, w', \tilde{w}, \tilde{W})$ with

$$B' = \rho(\delta\tilde{W})\, B(\delta, \Gamma(w, \tilde{w}), \Gamma(w', \tilde{w} - \delta\tilde{W}), \Theta(\tilde{w}), M(\tilde{w}, \tilde{w} - \delta\tilde{W})\tilde{W})|\det D\Phi(w', \tilde{w} - \delta\tilde{W})|,$$

which lies in $\mathcal{K}_{\mathrm{ad}}$: it is smooth up to $\delta = 0$, the matrix M is uniformly invertible on the support so that $|M\tilde{W}| \geq c|\tilde{W}|$, and the factors of $\tilde{W}$ produced by differentiation in δ are absorbed by the rapid decay of B (the decay in $w - w'$ is trivial since the support is compact). At $\delta = 0$, $B'(0) = B(0, \Gamma(w, \tilde{w}), \Gamma(w', \tilde{w}), \Theta(\tilde{w}), D\Theta(\tilde{w})\tilde{W})|\det D_w\Gamma(w', \tilde{w})||\det D\Theta(\tilde{w})|$, since $D\Phi$ is block triangular, and the Fourier transform in $\tilde{W}$ gives the stated formula for the adiabatic symbol. □

Remark 6.7.4 (Fibrations). Let $\phi : Y \to Z$ be a fibration of compact manifolds and E a bundle over Y. Using Lemma 6.7.3 one defines the adiabatic families $\Psi^{-\infty}_{\mathrm{ad}(\phi)}(Y; E)$, families $B_\delta \in \Psi^{-\infty}(Y; E)$, by the analogue of Definition 6.5.1: if $\chi, \chi' \in \mathcal{C}^\infty(Y)$ are such that $\phi(\operatorname{supp}\chi) \cap \phi(\operatorname{supp}\chi') = \emptyset$, then $\chi B \chi'$ has kernel smooth on $[0, 1] \times Y^2$ and vanishing to

infinite order at $\delta = 0$; and for fibred coordinate charts W, W' over the same coordinate chart of Z (with local coordinates $(z, \tilde z)$, $\tilde z$ the base coordinates), and $\chi \in \mathcal{C}^\infty_{\mathrm c}(W)$, $\chi' \in \mathcal{C}^\infty_{\mathrm c}(W')$, the kernel of $\chi B \chi'$ in these coordinates is of the form in Definition 6.7.1. The adiabatic symbol then becomes a Schwartz function on T^*Z with values in the smoothing operators on the fibres of the pull-back of ϕ to T^*Z, and Proposition 6.7.2 holds in the form

$$0 \longrightarrow \delta\Psi^{-\infty}_{\mathrm{ad}(\phi)}(Y;E) \hookrightarrow \Psi^{-\infty}_{\mathrm{ad}(\phi)}(Y;E) \xrightarrow{\sigma_{\mathrm{ad}}} \mathcal{S}\big(T^*Z; \Psi^{-\infty}(\pi^*Y/T^*Z;E)\big) \longrightarrow 0,$$

where $\pi : T^*Z \to Z$. As in Remark 6.5.4, T^*Z should properly be replaced by ${}^{\mathrm{sl}}T^*Z$, identified with it by the choice of δ.

6.7.2 Semiclassical-adiabatic families

Finally we combine the two limits: an overall semiclassical limit $\epsilon \downarrow 0$ and an adiabatic limit $\delta \downarrow 0$ in the variables $\tilde z$. Let $\mathcal{K}_{\mathrm{sl,ad}}$ be the space of $B \in \mathcal{C}^\infty([0,1]_\epsilon \times [0,1]_\delta \times \mathbb{R}^n_z \times \mathbb{R}^n_Z \times \mathbb{R}^{\tilde n}_{\tilde z} \times \mathbb{R}^{\tilde n}_{\tilde Z})$ with

$$|\partial_\epsilon^j \partial_\delta^k \partial_z^\alpha \partial_Z^{\alpha'} \partial_{\tilde z}^\beta \partial_{\tilde Z}^\gamma B| \le C \langle Z\rangle^{-N} \langle \tilde Z\rangle^{-N} \quad \text{for all indices and } N.$$

Definition 6.7.5. $\Psi^{-\infty}_{\mathrm{sl,ad}}(\mathbb{R}^n;\mathbb{R}^{\tilde n})$ is the space of two-parameter families $B_{\epsilon,\delta}$, $(\epsilon,\delta) \in (0,1]^2$, with kernels

$$\epsilon^{-n-\tilde n}\delta^{-\tilde n} B\Big(\epsilon, \delta, z, \frac{z-z'}{\epsilon}, \tilde z, \frac{\tilde z - \tilde z'}{\epsilon\delta}\Big), \qquad B \in \mathcal{K}_{\mathrm{sl,ad}}.$$

Its *semiclassical* and *adiabatic symbols* are

$$\sigma_{\mathrm{sl}}(B)(\delta, z, \tilde z, \zeta, \tilde\zeta) = \iint e^{-iZ\cdot\zeta - i\tilde Z\cdot\tilde\zeta} B(0,\delta,z,Z,\tilde z,\tilde Z)\, dZ\, d\tilde Z,$$

and, for $(\tilde z, \tilde\zeta) \in T^*\mathbb{R}^{\tilde n}$, $\sigma_{\mathrm{ad}}(B)(\tilde z,\tilde\zeta)$ is the family in $\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$ with kernel

$$\epsilon^{-n} b\Big(\epsilon, z, \frac{z-z'}{\epsilon}, \tilde z, \tilde\zeta\Big), \qquad b = \int e^{-i\tilde Z\cdot\tilde\zeta} B(\epsilon, 0, z, Z, \tilde z, \tilde Z)\, d\tilde Z.$$

Thus $\sigma_{\mathrm{sl}}(B) \in \mathcal{C}^\infty([0,1]_\delta; S^{-\infty}(\mathbb{R}^{n+\tilde n};\mathbb{R}^{n+\tilde n}))$ and $\sigma_{\mathrm{ad}}(B) \in S^{-\infty}(T^*\mathbb{R}^{\tilde n}; \Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n))$, the latter space consisting of the functions $b(\epsilon, z, Z, \tilde z, \tilde\zeta)$ with all derivatives bounded by $C_N\langle Z\rangle^{-N}\langle\tilde\zeta\rangle^{-N}$, each $(\tilde z,\tilde\zeta)$ defining a semiclassical smoothing family on $\mathbb{R}^n$ by Proposition 6.3.5. For fixed $\delta > 0$, B is a semiclassical smoothing family on $\mathbb{R}^{n+\tilde n}$: its kernel is $\epsilon^{-n-\tilde n} K_\delta(\epsilon, (z,\tilde z), ((z-z')/\epsilon, (\tilde z - \tilde z')/\epsilon))$ with $K_\delta(\epsilon, z, \tilde z, Z, \tilde Z'') = \delta^{-\tilde n} B(\epsilon, \delta, z, Z, \tilde z, \tilde Z''/\delta)$, whose semiclassical symbol in the sense of Definition 6.4.1 is $\sigma_{\mathrm{sl}}(B)(\delta, z, \tilde z, \zeta, \delta\tilde\zeta)$. For fixed $\epsilon > 0$, B is an adiabatic family on $\mathbb{R}^{n+\tilde n}$ with parameter $\epsilon\delta$.

Proposition 6.7.6. *$\Psi^{-\infty}_{\mathrm{sl,ad}}(\mathbb{R}^n;\mathbb{R}^{\tilde n})$ is a $*$-algebra. The two symbol maps are multiplicative (pointwise products, resp. pointwise composition in $\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$) and give short exact sequences*

$$0 \longrightarrow \epsilon\Psi^{-\infty}_{\mathrm{sl,ad}} \hookrightarrow \Psi^{-\infty}_{\mathrm{sl,ad}} \xrightarrow{\sigma_{\mathrm{sl}}} \mathcal{C}^\infty([0,1]_\delta; S^{-\infty}(\mathbb{R}^{n+\tilde n};\mathbb{R}^{n+\tilde n})) \longrightarrow 0,$$
$$0 \longrightarrow \delta\Psi^{-\infty}_{\mathrm{sl,ad}} \hookrightarrow \Psi^{-\infty}_{\mathrm{sl,ad}} \xrightarrow{\sigma_{\mathrm{ad}}} S^{-\infty}\big(T^*\mathbb{R}^{\tilde n}; \Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)\big) \longrightarrow 0.$$

They satisfy the compatibility condition

$$\sigma_{\mathrm{sl}}(B)\big|_{\delta=0} = \sigma_{\mathrm{sl}}(\sigma_{\mathrm{ad}}(B)), \tag{6.7.3}$$

where on the right σ_{sl} is applied to $\sigma_{\mathrm{ad}}(B)(\tilde z,\tilde\zeta) \in \Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$ for each $(\tilde z,\tilde\zeta)$. The joint symbol map $(\sigma_{\mathrm{sl}}, \sigma_{\mathrm{ad}})$ has null space $\epsilon\delta\Psi^{-\infty}_{\mathrm{sl,ad}}(\mathbb{R}^n;\mathbb{R}^{\tilde n})$, and its range consists precisely of the pairs satisfying (6.7.3).

Proof. For $\epsilon,\delta>0$, the substitutions $z'=z-\epsilon Z'$ and $\tilde z'=\tilde z-\epsilon\delta\tilde Z'$ in the composition of kernels show that $A_{\epsilon,\delta}B_{\epsilon,\delta}$ has kernel of the same form with

$$C(\epsilon,\delta,z,Z,\tilde z,\tilde Z)=\iint A(\epsilon,\delta,z,Z',\tilde z,\tilde Z')\,B(\epsilon,\delta,z-\epsilon Z',Z-Z',\tilde z-\epsilon\delta\tilde Z',\tilde Z-\tilde Z')\,dZ'\,d\tilde Z',$$

and as before $C\in\mathcal{K}_{\mathrm{sl,ad}}$; similarly for adjoints. At $\epsilon=0$ this is a convolution in $(Z,\tilde Z)$, with $(\delta,z,\tilde z)$ as parameters, so σ_{sl} is multiplicative. At $\delta=0$ it is a convolution in $\tilde Z$ combined, for each $\tilde z$, with the composition formula (6.3.3) in the variables (z,Z); after Fourier transform in $\tilde Z$ this is the composition of the semiclassical families $\sigma_{\mathrm{ad}}(A)(\tilde z,\tilde\zeta)$ and $\sigma_{\mathrm{ad}}(B)(\tilde z,\tilde\zeta)$. Both sides of (6.7.3) are the Fourier transform in $(Z,\tilde Z)$ of $B(0,0,\cdot)$. Exactness of the two sequences is proved as in Proposition 6.7.2, by Taylor's formula in ϵ, resp. δ.

If $\sigma_{\mathrm{sl}}(B)=0$ and $\sigma_{\mathrm{ad}}(B)=0$, then $B=\epsilon B_1$ with $B_1\in\mathcal{K}_{\mathrm{sl,ad}}$, and $\epsilon B_1(\epsilon,0,\cdot)=0$, so $B_1(\epsilon,0,\cdot)=0$ for $\epsilon>0$ and hence for all ϵ; thus $B_1=\delta B_2$ and $B=\epsilon\delta B_2$. Conversely, given a compatible pair (s,b), let $B_1\in\mathcal{K}_{\mathrm{sl,ad}}$ be the inverse Fourier transform of s in $(\zeta,\tilde\zeta)$, independent of ϵ, so $\sigma_{\mathrm{sl}}(B_1)=s$. By compatibility, $b-\sigma_{\mathrm{ad}}(B_1)$ has vanishing semiclassical symbol, so its kernel function is $\epsilon b'$ with b' in the same class; let B_2 be the inverse Fourier transform of $\epsilon b'$ in $\tilde\zeta$, independent of δ. Then $B=B_1+B_2$ has $\sigma_{\mathrm{sl}}(B)=s$, since B_2 vanishes at $\epsilon=0$, and $\sigma_{\mathrm{ad}}(B)=\sigma_{\mathrm{ad}}(B_1)+\epsilon b'=b$. □

The meaning of (6.7.3) is that the limits $\epsilon\downarrow 0$ and $\delta\downarrow 0$ may be taken in either order with the same result. For a fibration $\phi:Y\to Z$, the semiclassical-adiabatic families $\Psi^{-\infty}_{\mathrm{sl,ad}(\phi)}(Y;E)$ are defined by localization as in Remark 6.7.4, and Proposition 6.7.6 holds for them, the semiclassical symbol being a smooth family in $\delta\in[0,1]$ of Schwartz functions on (rescaled) cotangent spaces of Y and the adiabatic symbol taking values in $\mathcal{S}(T^*Z;\Psi^{-\infty}_{\mathrm{sl}}(\pi^*Y/T^*Z;E))$. These are the algebras used in Chapter 16 to prove the multiplicativity of the semiclassical index under iterated fibrations.

6.8 Exercises

Exercise 6.1. Let $P_\epsilon=-\epsilon^2\sum_j\partial_{x_j}^2+V(x)$ on $\mathbb{R}^n$, with $V\in\mathcal{C}^\infty_{\mathrm{b}}(\mathbb{R}^n)$ real, and let $f\in\mathcal{C}^\infty(\mathbb{R}^n)$ be real with $\nabla f\in\mathcal{C}^\infty_{\mathrm{b}}$. Show by direct computation that

$$e^{-if/\epsilon}P_\epsilon(e^{if/\epsilon}u)=(|\nabla f|^2+V)u-i\epsilon(2\nabla f\cdot\nabla u+(\textstyle\sum_j\partial_{x_j}^2 f)u)-\epsilon^2\sum_j\partial_{x_j}^2u,$$

and compare with Proposition 6.4.9. Conclude that a WKB ansatz $e^{if/\epsilon}(u_0+\epsilon u_1+\dots)$ solves $(P_\epsilon-\lambda)w=O(\epsilon^\infty)$ only if f satisfies the eikonal equation $|\nabla f|^2+V=\lambda$ on $\operatorname{supp}u_0$. Show that P is elliptic in the sense of Definition 6.4.4 if and only if $\inf V>0$.

Exercise 6.2. Let $T:[0,1]\to[0,1]$ be a diffeomorphism with $T(0)=0$. Show that if $A\in\Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$ then the family $A'_{\epsilon'}=A_{T^{-1}(\epsilon')}$ is again in $\Psi^m_{\mathrm{sl}}(\mathbb{R}^n)$, and that $\sigma_{\mathrm{sl}}(A')(x,\eta)=\sigma_{\mathrm{sl}}(A)(x,\eta/T'(0))$. (Write $T^{-1}(\epsilon')=\epsilon' g(\epsilon')$ with g smooth and positive.)

Exercise 6.3. Let $A,B\in\Psi^{-\infty}_{\mathrm{sl}}(\mathbb{R}^n)$ have kernels as in (6.3.2) with functions K, L and symbols a, b. Expanding $L(\epsilon,x-\epsilon t,Z-t)$ in (6.3.3) in Taylor series in its second argument, and using $\int e^{-it\cdot\eta}t^\alpha K\,dt=(i\partial_\eta)^\alpha a$, show directly that the symbol c of AB satisfies, in the sense of Taylor series at $\epsilon=0$,

$$c(\epsilon,x,\eta)\simeq\sum_\alpha\frac{\epsilon^{|\alpha|}}{\alpha!}\partial_\eta^\alpha a(\epsilon,x,\eta)\,D_x^\alpha b(\epsilon,x,\eta),$$

in agreement with Theorem 6.2.7. (Note the factor $(-i)^{|\alpha|}$ hidden in D_x^α.)

Exercise 6.4. Give an alternative proof of Lemma 6.3.1 by integrating by parts in the second form of (6.2.1), using the identity $\epsilon^2\Delta_\eta e^{i(x-y)\cdot\eta/\epsilon} = |x-y|^2 e^{i(x-y)\cdot\eta/\epsilon}$ (with $\Delta_\eta = \sum D_{\eta_j}^2$). Use this, Taylor expansion of the amplitude about $y = x$ and Lemma 6.1.5 to give a second proof of Theorem 6.2.6 along the lines of Theorem 2.3.1.

Exercise 6.5. Let $A \in \Psi_{\mathrm{sl}}^0(\mathbb{R}^n)$ and $(x_0,\eta_0) \in T^*\mathbb{R}^n$. Apply Proposition 6.4.9 with $f(x) = x\cdot\eta_0$ and $u = \phi \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^n)$ supported near x_0 to show that $\liminf_{\epsilon\downarrow 0}\|A_\epsilon\| \geq |\sigma_{\mathrm{sl}}(A)(x_0,\eta_0)|$. Conclude, using Corollary 6.4.8, that $\lim_{\epsilon\downarrow 0}\|A_\epsilon\|_{\mathcal{B}(L^2)} = \sup|\sigma_{\mathrm{sl}}(A)|$.

Exercise 6.6. Show that the family $A_\epsilon = e^{-1/\epsilon}\,\mathrm{Id}$ lies in $\bigcap_N \epsilon^N\Psi_{\mathrm{sl}}^0(\mathbb{R}^n)$ but not in $\epsilon^\infty\Psi_{\mathrm{sl}}^{-\infty}(\mathbb{R}^n)$, and that $\epsilon\,\mathrm{Id} \in \Psi_{\mathrm{sl}}^0(\mathbb{R}^n)$ is invertible for each $\epsilon > 0$ but is not elliptic, its inverse not being a semiclassical family. Show that $(1-\epsilon^2\sum_j \partial_{x_j}^2)^{-1} = \mathrm{Op}_\epsilon(\langle\eta\rangle^{-2})$ exactly.

Exercise 6.7. For $a \in S_{\mathrm{sl}}^m(\mathbb{R}^n)$ define the semiclassical Weyl quantization $\mathrm{Op}_\epsilon^W(a) = \mathrm{Op}_\epsilon(a(\epsilon,\frac{x+y}{2},\eta))$ (an amplitude in the sense of Theorem 6.2.6). Show that $\mathrm{Op}_\epsilon^W(a) \in \Psi_{\mathrm{sl}}^m(\mathbb{R}^n)$ with $\sigma_{\epsilon,m}(\mathrm{Op}_\epsilon^W(a)) = [a]$, that $\mathrm{Op}_\epsilon^W(a)^* = \mathrm{Op}_\epsilon^W(\bar a)$, and compute the term of order ϵ in the left symbol of $\mathrm{Op}_\epsilon^W(a)$.

Exercise 6.8. In the situation of Lemma 6.6.2 show that $(\mathrm{Id}-2P)^2 = \mathrm{Id}+4E$ and that if P is a projection then $\mathrm{Id}-2P$ is an involution. Show that the iteration of Lemma 6.6.2 applied to a complex number p_0 with $|p_0^2 - p_0|$ small converges to 0 or 1, quadratically.

Exercise 6.9. Show that for $n = 0$, $\Psi_{\mathrm{ad}}^{-\infty}(\mathbb{R}^0;\mathbb{R}^{\tilde n}) = \Psi_{\mathrm{sl}}^{-\infty}(\mathbb{R}^{\tilde n})$ with δ in place of ϵ, and that σ_{ad} is then the semiclassical symbol. Show that for $\tilde n = 0$, $\Psi_{\mathrm{sl,ad}}^{-\infty}(\mathbb{R}^n;\mathbb{R}^0) = \mathcal{C}^\infty([0,1]_\delta;\Psi_{\mathrm{sl}}^{-\infty}(\mathbb{R}^n))$.

Exercise 6.10. Let $A \in \Psi_{\mathrm{sl}}^m(X;E)$ and $B \in \Psi_{\mathrm{sl}}^{m'}(X;E)$ have scalar semiclassical symbols modulo ϵ, i.e. $\sigma_{\epsilon,m}(A) = [a\,\mathrm{Id}]$, $\sigma_{\epsilon,m'}(B) = [b\,\mathrm{Id}]$. Show that $[A,B] \in \epsilon\Psi_{\mathrm{sl}}^{m+m'-1}(X;E)$ and $\sigma_{\mathrm{sl}}(\epsilon^{-1}[A,B]) = \frac{1}{i}\{a(0),b(0)\}\,\mathrm{Id}$, the Poisson bracket being taken on T^*X.

Notes

This chapter combines §§2.19–2.20, 3.12–3.15 and 6.10 of Melrose's revised notes *From microlocal to global analysis*. The original treats the composition of finite order families by a continuity argument starting from the smoothing case; here instead we use the exact oscillatory integral formula for the symbol of a composite (Proposition 6.2.5), which makes the uniformity in ϵ and the expansion in powers of ϵ transparent; this approach is standard in semiclassical analysis (see e.g. the books of Dimassi–Sjöstrand, *Spectral asymptotics in the semi-classical limit*, and Zworski, *Semiclassical analysis*, which use the parameter $h = \epsilon$). The joint symbol $\sigma_{\epsilon,m}$, with values in $S_{\mathrm{sl}}^m/\epsilon S_{\mathrm{sl}}^{m-1}$, is equivalent to Melrose's pair consisting of the semiclassical symbol and the rescaled principal symbol with their compatibility condition (Proposition 6.4.2). We have added a proof of coordinate invariance, the double asymptotic summation needed to obtain residual errors which are both smoothing and $O(\epsilon^\infty)$, the invertibility of elliptic families on compact manifolds, and the uniqueness up to homotopy of quantized projections, all of which are used in Chapter 16. The quantization of projections uses a Newton iteration (Lemma 6.6.2) instead of the original linear iteration and the Riesz projection of Proposition 5.12.2. The sections on blow-up and on the resolution of semiclassical kernels were not written in the original; their content is summarized in Remark 6.3.6 and Proposition 6.5.6(iii). For the general theory of blow-up and of calculi defined by kernels on blown-up spaces see [16]. Holomorphic families of symbols, discussed briefly in the original after the semiclassical limit, are not needed here and are omitted.

Part II

Global calculi, traces and determinants

Chapter 7

Isotropic and scattering calculi

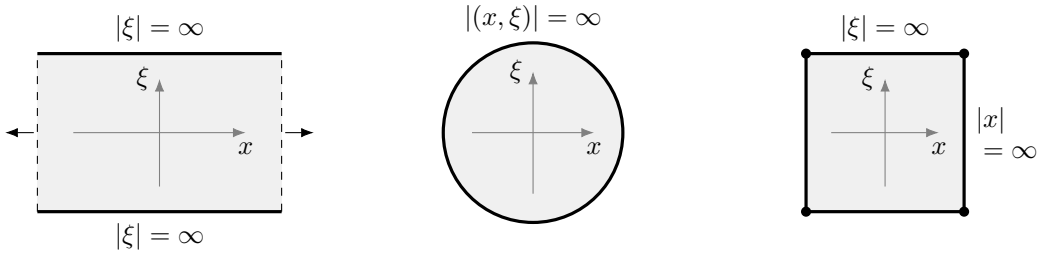
On a compact manifold the only way to "go to infinity" in phase space is along the fibres, $|\xi| \to \infty$, and the traditional calculus of Chapter 2 is built to control exactly this. On $\mathbb{R}^n$ the base variable x can also go to infinity. An operator such as the harmonic oscillator $D^2 + x^2$ grows in both directions at once, and no estimate in ξ alone can show that it has discrete spectrum. To obtain compactness and Fredholm properties on $\mathbb{R}^n$ we therefore need calculi in which large x is treated as seriously as large ξ.

In this chapter two *global* calculi of pseudodifferential operators on $\mathbb{R}^n$ are constructed and analysed. In the *isotropic* calculus the symbols are required to satisfy estimates in which the space variables x and the dual variables ξ play completely symmetric roles: each derivative, in x or in ξ, improves the decay by a factor of $(1 + |x| + |\xi|)^{-1}$. In the *scattering* calculus derivatives in x gain powers of $\langle x\rangle^{-1}$ and derivatives in ξ gain powers of $\langle \xi\rangle^{-1}$, separately. Both calculi are subalgebras of (weighted versions of) the traditional calculus $\Psi^m(\mathbb{R}^n)$ of Chapter 2, and we use the results proved there freely. The isotropic calculus is especially simple because its residual algebra $\Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ consists of the operators with kernels in $\mathcal{S}(\mathbb{R}^{2n})$; these operators are compact (indeed Hilbert–Schmidt) on $L^2(\mathbb{R}^n)$. This residual algebra, the associated group $G_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ of invertible perturbations of the identity, and the approximation of residual operators by finite rank operators built from the eigenfunctions of the harmonic oscillator, are the basic objects for the discussion of traces and determinants in Chapter 8. Thus this chapter is preparatory: the constructions (adjoints, composition, parametrices, generalized inverses, Sobolev spaces, L^2 estimates, complex powers) reappear later for operators on manifolds, and here they can be carried out without the complications of coordinate changes. Finally, the isotropic calculus has a large symmetry group: written in the Weyl quantization, it is invariant under all linear symplectic transformations of $\mathbb{R}^{2n}$, and these are implemented by conjugation with unitary operators on $L^2(\mathbb{R}^n)$ forming the *metaplectic group*, a double cover of the symplectic group generated by the exponentials of Weyl quantizations of quadratic forms (Sections 7.11 and 7.12).

We work with general (not necessarily polyhomogeneous) symbols by default and indicate the polyhomogeneous subclasses separately.

Discussion. The three calculi differ only in which variables a derivative of the symbol is allowed to gain decay in. Figure 7.1 shows the resulting pictures of "infinity" in phase space. In the traditional calculus only $|\xi| = \infty$ is a boundary; x ranges over a non-compact set, so nothing forces decay in x. The isotropic calculus compactifies $\mathbb{R}^{2n}$ to a ball with one boundary sphere. The scattering calculus compactifies x and ξ separately, giving a product with two boundary faces that meet in a corner. Compactness results (a residual operator is compact on L^2) hold exactly when phase space has been compactified in every direction.

Notation. Points of $\mathbb{R}^{2n} = \mathbb{R}^n_x \times \mathbb{R}^n_\xi$ are written $w = (x, \xi)$ and $\langle w\rangle = \langle (x, \xi)\rangle =$



(a) traditional S^m
∂_ξ gains $\langle\xi\rangle^{-1}$,
∂_x gains nothing

(b) isotropic Γ^m
every derivative
gains $\langle(x,\xi)\rangle^{-1}$

(c) scattering $S^{m,l}$
∂_x gains $\langle x\rangle^{-1}$,
∂_ξ gains $\langle\xi\rangle^{-1}$

Figure 7.1: Phase space $\mathbb{R}_x \times \mathbb{R}_\xi$ ($n = 1$) and its "boundary at infinity" (thick) for the three symbol classes. In (a) the x-direction is not compactified (dashed, arrows). In (b) the radial compactification gives one boundary sphere. In (c) the two variables are compactified separately; the dots are the corners $|x| = |\xi| = \infty$.

$(1 + |x|^2 + |\xi|^2)^{1/2}$. We repeatedly use the following elementary inequalities. For $v, w \in \mathbb{R}^N$ and $s \in \mathbb{R}$ (Peetre's inequality)

$$\langle w + v\rangle^s \le 2^{|s|/2}\langle w\rangle^s\langle v\rangle^{|s|}, \tag{7.0.1}$$

which follows from $1 + |w + v|^2 \le 2(1 + |w|^2)(1 + |v|^2)$, applied to (w, v) if $s \ge 0$ and to $(w + v, -v)$ if $s < 0$. For $z \in \mathbb{R}^p$, $\xi \in \mathbb{R}^n$ and $\theta \in [0, 1]$,

$$\max(\langle z\rangle, \langle\xi\rangle) \le \langle(z,\xi)\rangle \le \langle z\rangle\langle\xi\rangle, \qquad \langle(z,\xi)\rangle \ge \langle z\rangle^\theta\langle\xi\rangle^{1-\theta}, \tag{7.0.2}$$

the last by the arithmetic–geometric mean inequality, $\langle z\rangle^{2\theta}\langle\xi\rangle^{2(1-\theta)} \le \theta\langle z\rangle^2 + (1-\theta)\langle\xi\rangle^2 \le \langle(z,\xi)\rangle^2$. We write $m_+ = \max(m, 0)$.

7.1 Symbol classes

We begin with the symbol spaces, since all properties of the operators are read off from them. The model symbols are the polynomials $x^\beta\xi^\alpha$ of degree $|\alpha| + |\beta|$; the isotropic class is designed so that these have the "right" order.

7.1.1 Isotropic symbols

Definition 7.1.1. For $m \in \mathbb{R}$ and $N \in \mathbb{N}$ the space of *isotropic symbols* of order m is

$$\Gamma^m(\mathbb{R}^N) = \{a \in \mathcal{C}^\infty(\mathbb{R}^N);\ |\partial^\alpha a(w)| \le C_\alpha\langle w\rangle^{m-|\alpha|}\ \forall\,\alpha \in \mathbb{N}_0^N\}.$$

It is a Fréchet space with the norms $\|a\|_{m,k} = \max_{|\alpha|\le k}\sup_w\langle w\rangle^{-m+|\alpha|}|\partial^\alpha a(w)|$. We set $\Gamma^{-\infty}(\mathbb{R}^N) = \bigcap_m \Gamma^m(\mathbb{R}^N) = \mathcal{S}(\mathbb{R}^N)$ and $\Gamma^\infty(\mathbb{R}^N) = \bigcup_m \Gamma^m(\mathbb{R}^N)$. Vector- and matrix-valued symbols are defined in the same way.

Discussion. For $N = 2n$ and $w = (x, \xi)$, the condition says that a behaves like a polynomial of degree m in (x, ξ) jointly: a derivative in *any* direction of phase space lowers the order by one. The polynomial $x_1\xi_1$ lies in Γ^2, while in the traditional class it is only of order 1 in ξ with an unbounded coefficient. Because x and ξ enter symmetrically, any linear change of variables of $\mathbb{R}^{2n}$ preserves Γ^m (Lemma 7.1.2(iii) below); this is the source of the Fourier and symplectic invariance proved later in the chapter.

Lemma 7.1.2. (i) $\Gamma^m \cdot \Gamma^{m'} \subset \Gamma^{m+m'}$, $\partial^\alpha : \Gamma^m \to \Gamma^{m-|\alpha|}$, $\Gamma^m \subset \Gamma^{m'}$ *for* $m \le m'$, *and* $\langle w\rangle^s \in \Gamma^s$ *for all* $s \in \mathbb{R}$.

(ii) *If* $a \in \Gamma^0(\mathbb{R}^N;\mathbb{C}^M)$ *and* F *is smooth on a neighbourhood of the (compact) closure of* $a(\mathbb{R}^N)$, *then* $F \circ a \in \Gamma^0(\mathbb{R}^N)$.

(iii) *If* L *is a linear isomorphism of* $\mathbb{R}^N$ *then* $a \mapsto a \circ L$ *is an isomorphism of* $\Gamma^m(\mathbb{R}^N)$.

(iv) $\mathcal{S}(\mathbb{R}^N)$ *is dense in* $\Gamma^m(\mathbb{R}^N)$ *in the topology of* $\Gamma^{m'}(\mathbb{R}^N)$ *for any* $m' > m$.

Proof. (i) follows from Leibniz' formula and, for $\langle w\rangle^s$, from the fact that $\partial^\alpha\langle w\rangle^s$ is a sum of terms $p(w)\langle w\rangle^{s-2k}$ with p a polynomial of degree at most $2k-|\alpha|$. (ii) By the chain rule $\partial^\alpha(F\circ a)$ is a sum of terms $(\partial^k F)(a)\,\partial^{\alpha_1}a\cdots\partial^{\alpha_k}a$ with $\alpha_1+\cdots+\alpha_k=\alpha$, $|\alpha_i|\ge 1$, each bounded by $C\langle w\rangle^{-|\alpha|}$. (iii) holds since $c\langle w\rangle \le \langle Lw\rangle \le C\langle w\rangle$. (iv) Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^N)$ equal 1 near 0 and $a_k = \chi(w/k)a$. The functions $\chi(w/k)$ are bounded in Γ^0 uniformly in $k\ge 1$ and $1-\chi(w/k)$ vanishes for $|w|\le ck$, where $\langle w\rangle^{m-m'} \le Ck^{m-m'}$; hence $\|a-a_k\|_{m',j} \le C_j k^{m-m'} \to 0$. □

Lemma 7.1.3 (Asymptotic completeness)**.** *If* $a_j \in \Gamma^{m-j}(\mathbb{R}^N)$ *for* $j \in \mathbb{N}_0$, *there exists* $a \in \Gamma^m(\mathbb{R}^N)$ *such that*

$$a - \sum_{j<k} a_j \in \Gamma^{m-k}(\mathbb{R}^N) \quad \forall\, k \in \mathbb{N}_0;$$

a *is unique modulo* $\mathcal{S}(\mathbb{R}^N)$ *and we write* $a \sim \sum_j a_j$.

Proof. Choose $\psi \in \mathcal{C}^\infty(\mathbb{R}^N)$ with $\psi(w)=0$ for $|w|\le 1$, $\psi(w)=1$ for $|w|\ge 2$, and set $\psi_\epsilon(w)=\psi(\epsilon w)$ for $0<\epsilon\le 1$. Then $\partial^\alpha\psi_\epsilon = \epsilon^{|\alpha|}(\partial^\alpha\psi)(\epsilon w)$ is, for $\alpha\neq 0$, supported in $1/\epsilon \le |w| \le 2/\epsilon$, so $|\partial^\alpha\psi_\epsilon| \le C_\alpha\langle w\rangle^{-|\alpha|}$ uniformly in ϵ; moreover $\langle w\rangle \ge 1/\epsilon$ on $\operatorname{supp}\psi_\epsilon$. Hence for $k<j$ and $\ell\in\mathbb{N}_0$

$$\|\psi_\epsilon a_j\|_{m-k,\ell} \le C_{j,\ell} \sup_{\operatorname{supp}\psi_\epsilon} \langle w\rangle^{k-j} \le C_{j,\ell}\,\epsilon.$$

Choose $\epsilon_j \downarrow 0$ with $C_{j,j}\epsilon_j \le 2^{-j}$ (we may take $C_{j,\ell}$ increasing in ℓ and valid for all $k<j$). Then for each k and ℓ the series $\sum_{j>k}\psi_{\epsilon_j}a_j$ converges in the norm $\|\cdot\|_{m-k,\ell}$, since all but finitely many terms are bounded by 2^{-j}. Thus $a=\sum_j \psi_{\epsilon_j}a_j$ converges in Γ^m and

$$a - \sum_{j<k} a_j = \sum_{j<k}(\psi_{\epsilon_j}-1)a_j + \psi_{\epsilon_k}a_k + \sum_{j>k}\psi_{\epsilon_j}a_j \in \mathcal{C}^\infty_{\mathrm{c}} + \Gamma^{m-k} + \Gamma^{m-k}.$$

Uniqueness holds because $\bigcap_k \Gamma^{m-k} = \mathcal{S}(\mathbb{R}^N)$. □

Sums with steps of two, $a\sim\sum_j a_j$ with $a_j\in\Gamma^{m-2j}$, are covered by inserting zero terms.

7.1.2 Scattering symbols

Definition 7.1.4. For $m,l\in\mathbb{R}$ the space $S^{m,l}(\mathbb{R}^p;\mathbb{R}^n)$ of *scattering symbols* consists of the $a\in\mathcal{C}^\infty(\mathbb{R}^p_z\times\mathbb{R}^n_\xi)$ such that

$$|D_z^\alpha D_\xi^\beta a(z,\xi)| \le C_{\alpha\beta}\langle z\rangle^{l-|\alpha|}\langle\xi\rangle^{m-|\beta|} \quad \forall\,\alpha,\beta.$$

Here m is the order in the fibre variables and l the order in the base variables.

The following properties follow directly from the definition and Leibniz' formula: $S^{m,l}\cdot S^{m',l'} \subset S^{m+m',l+l'}$, $D_z : S^{m,l}\to S^{m,l-1}$, $D_\xi : S^{m,l}\to S^{m-1,l}$, $S^{m,l}(\mathbb{R}^p;\mathbb{R}^n) \subset \langle z\rangle^l S^m(\mathbb{R}^p;\mathbb{R}^n)$, and $\bigcap_{m,l} S^{m,l}(\mathbb{R}^n;\mathbb{R}^n) = \mathcal{S}(\mathbb{R}^{2n})$.

Lemma 7.1.5. *If $a_j \in S^{m-j,l-j}(\mathbb{R}^p;\mathbb{R}^n)$, $j \in \mathbb{N}_0$, there exists $a \in S^{m,l}(\mathbb{R}^p;\mathbb{R}^n)$ with $a - \sum_{j<k} a_j \in S^{m-k,l-k}(\mathbb{R}^p;\mathbb{R}^n)$ for all k; a is unique modulo $\mathcal{S}(\mathbb{R}^{p+n})$.*

Proof. Repeat the proof of Lemma 7.1.3 with $\psi_\epsilon(z,\xi) = 1-(1-\psi(\epsilon z))(1-\psi(\epsilon\xi))$, which is bounded in $S^{0,0}$ uniformly in ϵ, vanishes where $|z| \le 1/\epsilon$ and $|\xi| \le 1/\epsilon$ (so $1-\psi_\epsilon$ has compact support), and satisfies $\langle z\rangle^{-1}\langle\xi\rangle^{-1} \le \epsilon$ on its support. Thus for $k<j$ the $S^{m-k,l-k}$ norms of $\psi_\epsilon a_j$ are bounded by $C\sup_{\operatorname{supp}\psi_\epsilon}(\langle z\rangle\langle\xi\rangle)^{k-j} \le C\epsilon$. □

The two classes are related as follows.

Lemma 7.1.6. *For $z \in \mathbb{R}^p$, $\xi \in \mathbb{R}^n$,*

$$\Gamma^m(\mathbb{R}^{p+n}) \subset \begin{cases} S^{m,m}(\mathbb{R}^p;\mathbb{R}^n), & m \ge 0, \\ \bigcap_{m\le t\le 0} S^{m-t,t}(\mathbb{R}^p;\mathbb{R}^n), & m \le 0. \end{cases}$$

In particular $\Gamma^m(\mathbb{R}^{p+n}) \subset \langle z\rangle^{m_+} S^m(\mathbb{R}^p;\mathbb{R}^n)$.

Proof. By (7.0.2), $|\partial_z^\alpha\partial_\xi^\beta a| \le C\langle(z,\xi)\rangle^m\langle z\rangle^{-|\alpha|}\langle\xi\rangle^{-|\beta|}$. If $m \ge 0$, $\langle(z,\xi)\rangle^m \le \langle z\rangle^m\langle\xi\rangle^m$. If $m<0$ and $m \le t \le 0$, take $\theta = t/m \in [0,1]$ in (7.0.2) to get $\langle(z,\xi)\rangle^m \le \langle z\rangle^t\langle\xi\rangle^{m-t}$. □

7.1.3 Polyhomogeneous symbols and radial compactification

A function $a \in \mathcal{C}^\infty(\mathbb{R}^N)$ is *homogeneous of degree μ outside the unit ball* if $a(tw) = t^\mu a(w)$ for $|w| \ge 1$, $t \ge 1$. Such a function lies in $\Gamma^\mu(\mathbb{R}^N)$, since on $|w| \ge 1$ its derivatives of order k are homogeneous of degree $\mu - k$.

Definition 7.1.7. The space $\Gamma^m_{\mathrm{ph}}(\mathbb{R}^N)$ of *polyhomogeneous* (classical) isotropic symbols consists of those $a \in \Gamma^m(\mathbb{R}^N)$ for which there are a_{m-j}, homogeneous of degree $m-j$ outside the unit ball, with $a \sim \sum_j a_{m-j}$. The *principal symbol* of a is the restriction of a_m to the sphere $\mathbb{S}^{N-1}$.

The polyhomogeneous classes are closed under multiplication and differentiation ($\partial_{w_j} : \Gamma^m_{\mathrm{ph}} \to \Gamma^{m-1}_{\mathrm{ph}}$), contain $\langle w\rangle^s \in \Gamma^s_{\mathrm{ph}}$ (expand $\langle w\rangle^s = |w|^s(1+|w|^{-2})^{s/2}$ binomially for $|w| \ge 2$ and cut the terms off near 0), and if $a_j \in \Gamma^{m-j}_{\mathrm{ph}}$ and $a \sim \sum_j a_j$ then $a \in \Gamma^m_{\mathrm{ph}}$, as one sees by collecting the finitely many terms of each degree of homogeneity in the expansions of the a_j.

The *radial compactification* is the map

$$\mathrm{RC} : \mathbb{R}^N \ni w \longmapsto (\langle w\rangle^{-1}, w\langle w\rangle^{-1}) \in \mathbb{S}^N_+ = \{X = (X_0, X') \in \mathbb{R}^{1+N};\ |X| = 1,\ X_0 \ge 0\},$$

a diffeomorphism of $\mathbb{R}^N$ onto the interior of the closed half-sphere $\mathbb{S}^N_+$, a compact manifold with boundary $\partial\mathbb{S}^N_+ = \{X_0 = 0\} \cong \mathbb{S}^{N-1}$. The function $\rho = X_0$ is a boundary defining function and $\rho \circ \mathrm{RC} = \langle w\rangle^{-1}$. We denote by $\dot{\mathcal{C}}^\infty(\mathbb{S}^N_+)$ the smooth functions vanishing to infinite order at the boundary. The space $\mathcal{L}(\mathbb{S}^N_+) = \bigcup_{k\in\mathbb{N}_0}\rho^{-k}\mathcal{C}^\infty(\mathbb{S}^N_+)$ is the space of *Laurent functions* on the half-sphere; more generally $\rho^\mu\mathcal{C}^\infty(\mathbb{S}^N_+)$, $\mu \in \mathbb{C}$, is the space of products $\rho^\mu v$ with v smooth, regarded as functions on the interior.

Proposition 7.1.8. (i) *For every $m \in \mathbb{R}$, $\mathrm{RC}^*(\rho^{-m}\mathcal{C}^\infty(\mathbb{S}^N_+)) = \Gamma^m_{\mathrm{ph}}(\mathbb{R}^N)$, and the principal symbol of $\mathrm{RC}^*(\rho^{-m}F)$ is $F|_{\partial\mathbb{S}^N_+}$.*

(ii) $\mathrm{RC}^*\,\dot{\mathcal{C}}^\infty(\mathbb{S}^N_+) = \mathcal{S}(\mathbb{R}^N)$.

(iii) *If $f \in \Gamma^{-k}(\mathbb{R}^N)$, $k \in \mathbb{N}$, then $f \circ \mathrm{RC}^{-1}$ extends to a $\mathcal{C}^{k-1}$ function on $\mathbb{S}^N_+$ vanishing at the boundary, and $\Gamma^{-k}(\mathbb{R}^N) \ni f \mapsto f \circ \mathrm{RC}^{-1} \in \mathcal{C}^{k-1}(\mathbb{S}^N_+)$ is continuous.*

Sketch of proof. Near the boundary use the coordinates $\rho \in [0,1)$ and $\theta = X'/|X'| \in \mathbb{S}^{N-1}$; in terms of w, $\rho = \langle w\rangle^{-1}$ and $\theta = w/|w|$, so $|w| = (1-\rho^2)^{1/2}/\rho$. In polar coordinates $\partial_{w_j} = \theta_j \partial_r + r^{-1}\Omega_j$, where Ω_j is the vector field on $\mathbb{S}^{N-1}$ obtained by projecting e_j tangentially, and $\partial_r = -r\rho^3\partial_\rho$. Hence

$$V_j := \langle w\rangle \partial_{w_j} = -\theta_j(1-\rho^2)^{1/2}\rho\partial_\rho + (1-\rho^2)^{-1/2}\Omega_j. \tag{7.1.1}$$

These are smooth vector fields on $\mathbb{S}^N_+$ tangent to the boundary, and since $\sum_j \theta_j\Omega_j = 0$ they span, over $\mathcal{C}^\infty$, the fields $\rho\partial_\rho$ and Ω_j, that is all smooth vector fields tangent to the boundary. An induction shows that $\langle w\rangle^{|\alpha|}\partial^\alpha$ is a combination, with coefficients in Γ^0, of products of at most $|\alpha|$ of the V_j. Consequently $\mathrm{RC}^*\,\mathcal{C}^\infty(\mathbb{S}^N_+) \subset \Gamma^0$. Conversely, a derivative of order j of $f\circ \mathrm{RC}^{-1}$ in (ρ,θ) is ρ^{-j} times a combination of at most j-fold products of the V_j applied to f; this gives (iii), since such products applied to $f \in \Gamma^{-k}$ are $O(\langle w\rangle^{-k}) = O(\rho^k)$, and also (ii), since for $f \in \mathcal{S}$ these are $O(\rho^K)$ for every K (conversely, $F \in \dot{\mathcal{C}}^\infty$ can be written $F = \rho^K F_K$ with F_K smooth for every K, so $\mathrm{RC}^* F \in \bigcap_K \Gamma^{-K} = \mathcal{S}$ by the first argument).

For (i), if $F \in \mathcal{C}^\infty(\mathbb{S}^N_+)$, Taylor's formula in ρ gives $F = \sum_{j<K}\rho^j F_j(\theta) + \rho^K R_K$ near the boundary. Since $\rho = |w|^{-1}(1+|w|^{-2})^{-1/2}$ for $|w| \geq 1$, each $\rho^{j-m}F_j(\theta)$ pulls back to an asymptotic sum of functions homogeneous of degrees $m-j-2i$, $i \in \mathbb{N}_0$, while $\rho^{K-m}R_K$ pulls back into Γ^{m-K}. Thus $\mathrm{RC}^*(\rho^{-m}F) \in \Gamma^m_{\mathrm{ph}}$, with leading term $|w|^m F(0,\theta)$. Conversely, a function homogeneous of degree $m-j$ outside the unit ball is $|w|^{m-j}f(\theta) = \rho^{j-m}(1-\rho^2)^{(m-j)/2}f(\theta)$, which lies in $\rho^{-m}\rho^j\mathcal{C}^\infty$ near the boundary. Given $a \sim \sum_j a_{m-j}$, Borel's lemma gives $F \in \mathcal{C}^\infty(\mathbb{S}^N_+)$ with $\rho^{-m}F$ having the same expansion, and then $a - \mathrm{RC}^*(\rho^{-m}F) \in \mathcal{S}(\mathbb{R}^N) = \mathrm{RC}^*\,\dot{\mathcal{C}}^\infty$ by (ii). □

Thus $\Gamma^{\mathbb{Z}}_{\mathrm{ph}}(\mathbb{R}^N) = \bigcup_{k\in\mathbb{Z}}\Gamma^k_{\mathrm{ph}}(\mathbb{R}^N)$ is identified with the Laurent functions $\mathcal{L}(\mathbb{S}^N_+)$ and $\Gamma^0_{\mathrm{ph}}(\mathbb{R}^N)$ with $\mathcal{C}^\infty(\mathbb{S}^N_+)$.

7.2 Isotropic operators

Recall the left quantization $\mathrm{Op}(a)u(x) = \int e^{i(x-y)\cdot\xi}a(x,\xi)u(y)\,dy đ\xi$ and, for an amplitude $a(x,y,\xi)$, the same formula with $a(x,y,\xi)$.

Definition 7.2.1. For $m \in \mathbb{R}$ the space of *isotropic pseudodifferential operators* of order m on $\mathbb{R}^n$ is

$$\Psi^m_{\mathrm{iso}}(\mathbb{R}^n) = \{\mathrm{Op}(a);\ a \in \Gamma^m(\mathbb{R}^{2n})\}, \qquad \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n) = \{\mathrm{Op}(a);\ a \in \Gamma^m_{\mathrm{ph}}(\mathbb{R}^{2n})\}.$$

We set $\Psi^{-\infty}_{\mathrm{iso}} = \bigcap_m \Psi^m_{\mathrm{iso}}$ and $\Psi^{\infty}_{\mathrm{iso}} = \bigcup_m \Psi^m_{\mathrm{iso}}$.

By Lemma 7.1.6, if $a \in \Gamma^m(\mathbb{R}^{2n})$ then $a = \langle x\rangle^{m_+}b$ with $b \in S^m(\mathbb{R}^n;\mathbb{R}^n)$, so $\mathrm{Op}(a) = \langle x\rangle^{m_+}\mathrm{Op}(b) \in \langle x\rangle^{m_+}\Psi^m(\mathbb{R}^n)$. In particular isotropic operators act continuously on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$, by the results of Chapter 2. The kernel of $\mathrm{Op}(a)$ is $K(x,y) = \int e^{i(x-y)\cdot\xi}a(x,\xi)đ\xi$, i.e. the partial inverse Fourier transform of a in ξ evaluated at $(x, x-y)$; hence a is determined by $\mathrm{Op}(a)$ and we call $\sigma_L(\mathrm{Op}(a)) = a$ the *(full) left symbol*. Since partial Fourier transformation and linear changes of variables preserve $\mathcal{S}$:

Lemma 7.2.2. *An operator A is in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ if and only if it has Schwartz kernel in $\mathcal{S}(\mathbb{R}^{2n})$, and $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n) \ni A \mapsto K_A \in \mathcal{S}(\mathbb{R}^{2n})$ is a bijection.*

We topologize $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ by this identification.

Example 7.2.3. Since $\mathrm{Op}(x^\beta \xi^\alpha) = x^\beta D^\alpha$, every differential operator with polynomial coefficients, $P = \sum_{|\alpha|+|\beta|\le m} c_{\alpha\beta} x^\beta D^\alpha$, lies in $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$. Conversely, a differential operator lies in $\Psi^\infty_{\mathrm{iso}}(\mathbb{R}^n)$ only if its coefficients are polynomials (a smooth function with all derivatives of order k bounded by $C\langle x\rangle^{m-k}$ for all k is a polynomial). The basic example is the *harmonic oscillator* $H = \sum_j (D_j^2 + x_j^2)$, with symbol $|x|^2 + |\xi|^2$, studied in Section 7.8.

Question. Is every isotropic operator a pseudodifferential operator in the sense of Chapter 2?

Answer. Only up to a weight. For $m \le 0$, Lemma 7.1.6 gives $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n) \subset \Psi^m(\mathbb{R}^n)$. For $m > 0$ a symbol such as x_1 (the operator of multiplication by x_1, of isotropic order 1) is not bounded in x, so it lies in no traditional class $S^{m'}$; it lies in $\langle x\rangle S^0$. This is why the inclusion above reads $\Psi^m_{\mathrm{iso}} \subset \langle x\rangle^{m_+}\Psi^m$. In return the isotropic calculus gives information that the traditional one cannot see, such as decay of solutions as $|x| \to \infty$. □

7.2.1 Amplitudes and reduction

As in Chapter 2, adjoints and products are obtained by reducing operators defined by amplitudes to left-reduced form. Note that a left symbol $a(x,\xi) \in \Gamma^m(\mathbb{R}^{2n})$, regarded as a function of (x,y,ξ), is *not* in general in $\Gamma^m(\mathbb{R}^{3n})$, so we use a hybrid class of amplitudes.

Definition 7.2.4. For $m \in \mathbb{R}$, $\mathcal{A}^m(\mathbb{R}^{3n})$ is the space of $a \in \mathcal{C}^\infty(\mathbb{R}^{3n}_{x,y,\xi})$ for which there is $w \ge 0$ such that for all multi-indices $\alpha' \le \alpha$, $\beta' \le \beta$, $\gamma' \le \gamma$

$$|D_x^\alpha D_y^\beta D_\xi^\gamma a| \le C\langle (x,\xi)\rangle^{m-k'}\langle x-y\rangle^{w+k'}\langle \xi\rangle^{-|\gamma-\gamma'|}, \qquad k' = |\alpha'| + |\beta'| + |\gamma'|. \tag{7.2.1}$$

Thus any subset of the derivatives may be used to improve the decay in (x,ξ), at the cost of a corresponding growth in $|x-y|$, while the remaining ξ-derivatives still give decay in ξ.

Lemma 7.2.5. (i) *If $a \in \Gamma^m(\mathbb{R}^{2n})$ and $b \in \Gamma^{m'}(\mathbb{R}^{2n})$ then $a(x,\xi)b(y,\xi) \in \mathcal{A}^{m+m'}(\mathbb{R}^{3n})$.*

(ii) *If $a \in \Gamma^m(\mathbb{R}^{2n})$ and $\lambda \in [0,1]$ then $a(x+\lambda(y-x),\xi) \in \mathcal{A}^m(\mathbb{R}^{3n})$.*

(iii) *If $a \in \mathcal{A}^m(\mathbb{R}^{3n})$ then $|D_x^\alpha D_y^\beta D_\xi^\gamma a| \le C\langle x\rangle^{m_+}\langle x-y\rangle^w\langle\xi\rangle^{m-|\gamma|}$.*

Proof. (i) By Leibniz' formula $D_x^\alpha D_y^\beta D_\xi^\gamma(ab)$ is a sum of terms $(D_x^\alpha D_\xi^{\gamma_1} a)(x,\xi)(D_y^\beta D_\xi^{\gamma_2} b)(y,\xi)$ with $\gamma_1+\gamma_2 = \gamma$. Given $\gamma' \le \gamma$, choose $\gamma_i' \le \gamma_i$ with $|\gamma_1'|+|\gamma_2'| = |\gamma'|$. Using $\langle(x,\xi)\rangle^{-1} \le \langle\xi\rangle^{-1}$ we get $|D_x^\alpha D_\xi^{\gamma_1} a| \le C\langle(x,\xi)\rangle^{m-|\alpha'|-|\gamma_1'|}\langle\xi\rangle^{-|\gamma_1-\gamma_1'|}$, and similarly, followed by (7.0.1) with $v = (y-x,0)$,

$$\begin{aligned}|D_y^\beta D_\xi^{\gamma_2} b(y,\xi)| &\le C\langle(y,\xi)\rangle^{m'-|\beta'|-|\gamma_2'|}\langle\xi\rangle^{-|\gamma_2-\gamma_2'|}\\ &\le C'\langle(x,\xi)\rangle^{m'-|\beta'|-|\gamma_2'|}\langle x-y\rangle^{|m'|+|\beta'|+|\gamma_2'|}\langle\xi\rangle^{-|\gamma_2-\gamma_2'|}.\end{aligned}$$

Multiplying gives (7.2.1) with $w = |m'|$. (ii) With $\bar x = x+\lambda(y-x)$ the derivative is a constant times $(\partial_x^{\alpha+\beta} D_\xi^\gamma a)(\bar x,\xi)$, bounded by $C\langle(\bar x,\xi)\rangle^{m-k'}\langle\xi\rangle^{-|\gamma-\gamma'|}$, and (7.0.1) gives $\langle(\bar x,\xi)\rangle^{m-k'} \le C\langle(x,\xi)\rangle^{m-k'}\langle x-y\rangle^{|m|+k'}$. (iii) Take $\alpha' = \beta' = \gamma' = 0$ and use $\langle(x,\xi)\rangle^m \le \langle x\rangle^{m_+}\langle\xi\rangle^m$. □

By (iii), $\langle x\rangle^{-m_+} a$ is an amplitude of the traditional calculus of Chapter 2 (of order m with weight $\langle x-y\rangle^w$), so $\mathrm{Op}(a) = \langle x\rangle^{m_+}\mathrm{Op}(\langle x\rangle^{-m_+}a) \in \langle x\rangle^{m_+}\Psi^m(\mathbb{R}^n)$ is a well-defined continuous operator on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$.

Proposition 7.2.6 (Reduction). *If $a \in \mathcal{A}^m(\mathbb{R}^{3n})$ then $\mathrm{Op}(a) \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ and*

$$\sigma_L(\mathrm{Op}(a)) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!}(D_\xi^\alpha D_y^\alpha a)(x,x,\xi), \tag{7.2.2}$$

where the term of index α lies in $\Gamma^{m-2|\alpha|}(\mathbb{R}^{2n})$.

Proof. The proof has three steps: we check that the terms c_α of the expansion have the stated orders; we Taylor expand a in y about x and integrate by parts, which writes $\mathrm{Op}(a)$ as a finite sum plus a remainder; and we show that the remainder is small in both x and ξ by comparing with weighted traditional symbol classes.

Let $c_\alpha(x,\xi) = \frac{i^{|\alpha|}}{\alpha!}(D_\xi^\alpha D_y^\alpha a)(x,x,\xi)$. A derivative $D_x^\mu D_\xi^\gamma c_\alpha$ is a sum of values at $y=x$ of derivatives of a containing $|\alpha|$ derivatives in y, $|\alpha|+|\gamma|$ in ξ and $|\mu|$ in x or y; applying (7.2.1) with all derivatives used for decay, and $\langle x-y\rangle = 1$ at $y=x$, shows $c_\alpha \in \Gamma^{m-2|\alpha|}(\mathbb{R}^{2n})$.

Taylor's formula in y around x gives

$$a(x,y,\xi) = \sum_{|\alpha|<N} \frac{(y-x)^\alpha}{\alpha!}(\partial_y^\alpha a)(x,x,\xi) + \sum_{|\alpha|=N} \frac{N(y-x)^\alpha}{\alpha!}\int_0^1 (1-t)^{N-1}(\partial_y^\alpha a)(x,x+t(y-x),\xi)\,dt.$$

Since $(y-x)^\alpha e^{i(x-y)\cdot\xi} = (-D_\xi)^\alpha e^{i(x-y)\cdot\xi}$, integration by parts in ξ (legitimate for these oscillatory integrals, as in Chapter 2) gives

$$\begin{aligned} \mathrm{Op}(a) &= \sum_{|\alpha|<N} \mathrm{Op}(c_\alpha) + \mathrm{Op}(r_N), \\ r_N(x,y,\xi) &= \sum_{|\alpha|=N} \frac{Ni^N}{\alpha!}\int_0^1 (1-t)^{N-1}(D_\xi^\alpha D_y^\alpha a)(x,x+t(y-x),\xi)\,dt. \end{aligned} \tag{7.2.3}$$

We estimate r_N. A derivative $D_x^\mu D_y^\nu D_\xi^\gamma$ of the integrand is, by the chain rule, a combination with bounded coefficients of derivatives of a evaluated at $(x, x+t(y-x), \xi)$ which contain the N derivatives D_y^α, the $N+|\gamma|$ derivatives $D_\xi^{\alpha+\gamma}$ and further x- and y-derivatives. Applying (7.2.1) using exactly the $2N$ derivatives $D_y^\alpha D_\xi^\alpha$ for decay, and noting that the first argument is still x and $|x-(x+t(y-x))| \le |x-y|$, we find

$$|D_x^\mu D_y^\nu D_\xi^\gamma r_N| \le C\langle(x,\xi)\rangle^{m-2N}\langle x-y\rangle^{w+2N}\langle\xi\rangle^{-|\gamma|}.$$

Let $2N > m$ and $s = N - m/2$. By (7.0.2), $\langle(x,\xi)\rangle^{-2s} \le \langle x\rangle^{-s}\langle\xi\rangle^{-s}$, so $\langle x\rangle^s r_N$ is an amplitude of the traditional calculus of order $-s$ with weight $\langle x-y\rangle^{w+2N}$ (derivatives of $\langle x\rangle^s$ are bounded by $C\langle x\rangle^s$). By Chapter 2, $\mathrm{Op}(r_N) = \langle x\rangle^{-s}Q_N$ with $Q_N \in \Psi^{-s}(\mathbb{R}^n)$, so the left symbol of $\mathrm{Op}(r_N)$ lies in $\langle x\rangle^{-s}S^{-s}(\mathbb{R}^n;\mathbb{R}^n)$: all its derivatives are bounded by $C\langle x\rangle^{-s}\langle\xi\rangle^{-s}$.

Now $\mathrm{Op}(a)$ has a left symbol $\sigma \in \langle x\rangle^{m_+}S^m(\mathbb{R}^n;\mathbb{R}^n)$ and, by (7.2.3) and uniqueness of left symbols, $\sigma - \sum_{|\alpha|<N} c_\alpha \in \langle x\rangle^{-s}S^{-s}$. Choose $b \in \Gamma^m(\mathbb{R}^{2n})$ with $b \sim \sum_\alpha c_\alpha$ (Lemma 7.1.3). Then for every N

$$\sigma - b = \Big(\sigma - \sum_{|\alpha|<N} c_\alpha\Big) - \Big(b - \sum_{|\alpha|<N} c_\alpha\Big) \in \langle x\rangle^{-s}S^{-s} + \Gamma^{m-2N},$$

and all derivatives of elements of both spaces are bounded by $C\langle x\rangle^{-s}\langle\xi\rangle^{-s}$, with $s = N - m/2$ arbitrarily large. Hence $\sigma - b \in \mathcal{S}(\mathbb{R}^{2n})$, so $\sigma \in \Gamma^m(\mathbb{R}^{2n})$ and $\sigma \sim \sum_\alpha c_\alpha$. □

7.2.2 The algebra of isotropic operators

Theorem 7.2.7. *The spaces $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ form an order-filtered $*$-algebra with residual algebra $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, the operators with kernels in $\mathcal{S}(\mathbb{R}^{2n})$. Precisely, if $A = \mathrm{Op}(a) \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ and $B = \mathrm{Op}(b) \in \Psi^{m'}_{\mathrm{iso}}(\mathbb{R}^n)$ then*

(i) *the formal adjoint A^*, defined by $(Au, v) = (u, A^*v)$ for $u, v \in \mathcal{S}(\mathbb{R}^n)$, lies in $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ and $\sigma_L(A^*) \sim \sum_\alpha \frac{i^{|\alpha|}}{\alpha!} D^\alpha_\xi D^\alpha_x \bar{a}$;*

(ii) *$AB \in \Psi^{m+m'}_{\mathrm{iso}}(\mathbb{R}^n)$ and*

$$\sigma_L(AB) = a\#b \sim \sum_\alpha \frac{1}{\alpha!} \partial^\alpha_\xi a \, D^\alpha_x b, \qquad a\#b - ab \in \Gamma^{m+m'-2}(\mathbb{R}^{2n}); \tag{7.2.4}$$

(iii) *the subspaces $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ form a $*$-subalgebra;*

(iv) *the bilinear maps $\Gamma^m \times \Gamma^{m'} \ni (a, b) \mapsto a\#b \in \Gamma^{m+m'}$ and*

$$\Gamma^m \times \Gamma^{m'} \ni (a, b) \mapsto a\#b - \sum_{|\alpha|<N} \frac{1}{\alpha!} \partial^\alpha_\xi a \, D^\alpha_x b \in \Gamma^{m+m'-2N}$$

are continuous.

Proof. (i) The kernel of A^* is $\overline{K_A(y,x)} = \int e^{i(x-y)\cdot\xi} \bar{a}(y,\xi) d\!\!\bar{}\xi$, so $A^* = \mathrm{Op}(\bar{a}(y,\xi))$, and $\bar{a}(y,\xi) \in \mathcal{A}^m$ by Lemma 7.2.5(ii) with $\lambda = 1$; apply Proposition 7.2.6.

(ii) Applying (i) to $B^* = \mathrm{Op}(c)$ we find $B = \mathrm{Op}(c)^* = \mathrm{Op}(b_R(y,\xi))$ with $b_R = \bar{c} \in \Gamma^{m'}$; taking conjugates in the expansion of (i), $b_R \sim \sum_\gamma \frac{(-i)^{|\gamma|}}{\gamma!} D^\gamma_\xi D^\gamma_x b$. For $u \in \mathcal{S}$, $\widehat{Bu}(\xi) = \int e^{-iy\cdot\xi} b_R(y,\xi) u(y)\, dy$, hence $ABu(x) = \int e^{ix\cdot\xi} a(x,\xi) \widehat{Bu}(\xi) d\!\!\bar{}\xi$, i.e. $AB = \mathrm{Op}(a(x,\xi) b_R(y,\xi))$. By Lemma 7.2.5(i) and Proposition 7.2.6, $AB \in \Psi^{m+m'}_{\mathrm{iso}}$ with $\sigma_L(AB) \sim \sum_\beta \frac{i^{|\beta|}}{\beta!} D^\beta_\xi (a \, D^\beta_x b_R)$. Inserting the expansion of b_R (with errors in $\Gamma^{m'-2N}$, which contribute errors in $\Gamma^{m+m'-2N}$) we obtain a double series whose terms of index (β,γ) lie in $\Gamma^{m+m'-2|\beta|-2|\gamma|}$. Grouping the terms with $|\beta| + |\gamma| = j$ we get a finite sum which, by the identity of formal power series in commuting variables

$$e^{iD_y\cdot(D_\eta + D_\xi)} e^{-iD_y\cdot D_\xi} = e^{iD_y\cdot D_\eta}$$

applied to $a(x,\eta)b(y,\xi)$ and evaluated at $y = x$, $\eta = \xi$, equals $\sum_{|\alpha|=j} \frac{i^{|\alpha|}}{\alpha!} D^\alpha_\xi a \, D^\alpha_x b$. Since $i^{|\alpha|} D^\alpha_\xi = \partial^\alpha_\xi$ this is (7.2.4); the terms with $\alpha \neq 0$ lie in $\Gamma^{m+m'-2}$.

(iii) All terms in the expansions are polyhomogeneous when a, b are, and asymptotic sums of polyhomogeneous symbols are polyhomogeneous.

(iv) By the closed graph theorem for Fréchet spaces it suffices to check that $a \mapsto a\#b$ (for fixed b) and $b \mapsto a\#b$ have closed graphs (separately continuous bilinear maps of Fréchet spaces are continuous). If $a_k \to a$ in Γ^m and $a_k\#b \to c$ in $\Gamma^{m+m'}$, then for $u, v \in \mathcal{S}(\mathbb{R}^n)$, $\langle \mathrm{Op}(a_k\#b)u, v\rangle = \langle \mathrm{Op}(a_k)\,\mathrm{Op}(b)u, v\rangle$; the left side tends to $\langle \mathrm{Op}(c)u, v\rangle$ and the right side to $\langle \mathrm{Op}(a\#b)u, v\rangle$, since $\langle \mathrm{Op}(a)f, v\rangle = \iint e^{ix\cdot\xi} a(x,\xi) \hat{f}(\xi) v(x)\, dx d\!\!\bar{}\xi$ depends continuously on $a \in \Gamma^m$ for $f, v \in \mathcal{S}$. Thus $c = a\#b$. The second map is handled in the same way. □

Remark 7.2.8. The *principal symbol* $\sigma_m(A)$ of $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ is the class of $\sigma_L(A)$ in Γ^m/Γ^{m-1}; there is a short exact sequence $0 \to \Psi^{m-1}_{\mathrm{iso}} \to \Psi^m_{\mathrm{iso}} \to \Gamma^m/\Gamma^{m-1} \to 0$, and $\sigma_{m+m'}(AB) = \sigma_m(A)\sigma_{m'}(B)$, $\sigma_m(A^*) = \overline{\sigma_m(A)}$. For $A \in \Psi^m_{\mathrm{iso,ph}}$ we identify $\sigma_m(A)$ with the principal symbol of $\sigma_L(A)$, a function in $\mathcal{C}^\infty(\mathbb{S}^{2n-1})$. A feature of the isotropic calculus is that the expansions above proceed in steps of two: $AB - \mathrm{Op}(ab) \in \Psi^{m+m'-2}_{\mathrm{iso}}$ and, by

(7.2.4), $\sigma_L([A,B]) - (-i)\{a,b\} \in \Gamma^{m+m'-4}$, where $\{a,b\} = \sum_j (\partial_{\xi_j} a\, \partial_{x_j} b - \partial_{x_j} a\, \partial_{\xi_j} b)$. Under the identification of Proposition 7.1.8, the theorem shows that the Laurent functions $\mathcal{L}(\mathbb{S}^{2n}_+)$ carry a non-commutative product, for which $\mathcal{C}^\infty(\mathbb{S}^{2n}_+)$ is a subalgebra.

Discussion. Why do the expansions proceed in steps of two? The term of index α in (7.2.4) contains $|\alpha|$ derivatives in ξ of a and $|\alpha|$ derivatives in x of b. In the traditional calculus only the ξ-derivatives lower the order, so each term gains one order. In the isotropic calculus *every* derivative lowers the order, so each term gains two. In particular the commutator of two isotropic operators is determined to leading order by the Poisson bracket, with an error four (not two) orders lower than $m + m'$.

Remark 7.2.9. If E and F are finite-dimensional vector spaces, $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n; E, F)$ denotes the operators from $\mathcal{S}(\mathbb{R}^n; E)$ to $\mathcal{S}(\mathbb{R}^n; F)$ given, after a choice of bases, by matrices of elements of $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$; equivalently their left symbols are in $\Gamma^m(\mathbb{R}^{2n}; \mathrm{Hom}(E,F))$. All results of this chapter extend to this setting with only notational changes, and we use them freely; in Section 7.7 we work with $\mathbb{C}^N$-valued operators explicitly.

7.2.3 Invariance under the Fourier transform

A property which distinguishes the isotropic calculus sharply from the traditional one is its invariance under the Fourier transform.

Proposition 7.2.10. *If $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ then $\hat{A} = \mathcal{F} A \mathcal{F}^{-1} \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$, i.e. $\widehat{Au} = \hat{A}\hat{u}$; if $A = \mathrm{Op}(b(y,\xi))$ is written in right-reduced form, $b \in \Gamma^m(\mathbb{R}^{2n})$, then $\sigma_L(\hat{A})(\xi, z) = b(-z, \xi)$. The same holds for $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$.*

Proof. The right-reduced form exists by the proof of Theorem 7.2.7(ii). Suppose first $b \in \mathcal{S}(\mathbb{R}^{2n})$. Then $\widehat{Au}(\xi) = \int e^{-iy\cdot\xi} b(y,\xi) u(y)\, dy$, and substituting $u(y) = \int e^{iy\cdot\eta} \hat{u}(\eta) đ\eta$ and then $y = -z$,

$$\widehat{Au}(\xi) = \iint e^{i(\xi-\eta)\cdot z} b(-z,\xi) \hat{u}(\eta)\, d\eta đz,$$

which is $\mathrm{Op}(c)\hat{u}$ with $c(\xi, z) = b(-z, \xi)$, ξ now playing the role of the base variable and z that of the dual variable. Moreover $c \in \Gamma^m$ if $b \in \Gamma^m$, by Lemma 7.1.2(iii). For general $b \in \Gamma^m$ choose $b_k \in \mathcal{S}$ with $b_k \to b$ in Γ^{m+1} (Lemma 7.1.2(iv)); for $u, v \in \mathcal{S}$ both $\langle \mathcal{F}\,\mathrm{Op}(b_k(y,\xi))\mathcal{F}^{-1} u, v\rangle$ and $\langle \mathrm{Op}(c_k)u, v\rangle$ depend continuously on $b_k \in \Gamma^{m+1}$, so the identity persists in the limit. □

Discussion. In phase space, conjugation by $\mathcal{F}$ acts on symbols through the linear map $(x,\xi) \mapsto (\xi, -x)$, a rotation by a right angle. The traditional calculus cannot be invariant under this map, since it treats x and ξ differently, but the isotropic calculus is. This is the first instance of a general fact: the isotropic calculus is invariant under *all* linear symplectic maps of $\mathbb{R}^{2n}$ (Sections 7.11 and 7.12).

7.3 Scattering operators

The isotropic calculus ties the behaviour in x to that in ξ. Many natural operators do not fit this pattern. The Laplacian Δ has symbol $|\xi|^2$, which does not grow in x, so it is not elliptic in the isotropic sense. The scattering calculus keeps two independent orders, m in ξ and l in x, and is the right setting for such operators.

Definition 7.3.1. For $m, l \in \mathbb{R}$ set $\Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n) = \{\mathrm{Op}(a);\ a \in S^{m,l}(\mathbb{R}^n; \mathbb{R}^n)\}$.

Thus $\Psi_{\mathrm{sc}}^{m,l}(\mathbb{R}^n) \subset \langle x\rangle^l \Psi^m(\mathbb{R}^n)$, and by Lemma 7.1.6, $\Psi_{\mathrm{iso}}^m \subset \Psi_{\mathrm{sc}}^{m,m}$ if $m \geq 0$ and $\Psi_{\mathrm{iso}}^m \subset \Psi_{\mathrm{sc}}^{m-t,t}$ for $m \leq t \leq 0$. Since $\bigcap_{m,l} S^{m,l}(\mathbb{R}^n;\mathbb{R}^n) = \mathcal{S}(\mathbb{R}^{2n})$,

$$\bigcap_{m,l} \Psi_{\mathrm{sc}}^{m,l}(\mathbb{R}^n) = \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n). \tag{7.3.1}$$

Typical elements are the differential operators $\sum_{|\alpha|\leq m} c_\alpha(x) D^\alpha$ with $|\partial^\beta c_\alpha| \leq C_\beta \langle x\rangle^{l-|\beta|}$; for instance the Laplacian $\Delta = \sum_j D_j^2$ is in $\Psi_{\mathrm{sc}}^{2,0}(\mathbb{R}^n)$, and $\Delta + 1$ is elliptic in the scattering sense below, although it is not elliptic as an isotropic operator.

For scattering operators one uses the amplitudes $\mathcal{A}_{\mathrm{sc}}^{m,l}$: smooth functions $a(x,y,\xi)$ such that for some $w \geq 0$

$$|D_x^\alpha D_y^\beta D_\xi^\gamma a| \leq C\langle x\rangle^{l-|\alpha'|-|\beta'|}\langle x-y\rangle^{w+|\alpha'|+|\beta'|}\langle \xi\rangle^{m-|\gamma|} \qquad \forall\, \alpha' \leq \alpha,\ \beta' \leq \beta.$$

As in Lemma 7.2.5, $a(x,\xi)b(y,\xi) \in \mathcal{A}_{\mathrm{sc}}^{m+m',l+l'}$ if $a \in S^{m,l}$, $b \in S^{m',l'}$ (use (7.0.1) for $\langle y\rangle^{l'-|\beta'|}$), and $a(x+\lambda(y-x),\xi) \in \mathcal{A}_{\mathrm{sc}}^{m,l}$.

Theorem 7.3.2. *The spaces $\Psi_{\mathrm{sc}}^{m,l}(\mathbb{R}^n)$ form a bi-filtered $*$-algebra, $\Psi_{\mathrm{sc}}^{m,l} \circ \Psi_{\mathrm{sc}}^{m',l'} \subset \Psi_{\mathrm{sc}}^{m+m',l+l'}$, with residual space (7.3.1). The left symbols of adjoints and products are given by the formulæ of Theorem 7.2.7, the expansions now being asymptotic in the sense of Lemma 7.1.5; in particular $AB - \mathrm{Op}(ab) \in \Psi_{\mathrm{sc}}^{m+m'-1,l+l'-1}(\mathbb{R}^n)$.*

Proof. The proof of Proposition 7.2.6 applies with the following changes. For $a \in \mathcal{A}_{\mathrm{sc}}^{m,l}$ the terms $c_\alpha = \frac{i^{|\alpha|}}{\alpha!}(D_\xi^\alpha D_y^\alpha a)(x,x,\xi)$ lie in $S^{m-|\alpha|,l-|\alpha|}$, and the remainder in (7.2.3) satisfies $|D_x^\mu D_y^\nu D_\xi^\gamma r_N| \leq C\langle x\rangle^{l-N}\langle x-y\rangle^{w+N}\langle\xi\rangle^{m-N-|\gamma|}$. Hence $\mathrm{Op}(r_N) \in \langle x\rangle^{l-N}\Psi^{m-N}(\mathbb{R}^n)$ and the left symbol σ of $\mathrm{Op}(a)$ satisfies $\sigma - \sum_{|\alpha|<N} c_\alpha \in \langle x\rangle^{l-N} S^{m-N}(\mathbb{R}^n;\mathbb{R}^n)$. If $b \in S^{m,l}$ with $b \sim \sum c_\alpha$, then $\sigma - b \in \bigcap_N \langle x\rangle^{l-N} S^{m-N}(\mathbb{R}^n;\mathbb{R}^n) = \mathcal{S}(\mathbb{R}^{2n})$. Adjoints and products are then treated exactly as in Theorem 7.2.7. □

Remark 7.3.3. Polyhomogeneous scattering symbols are defined using the radial compactification in each factor: $S_{\mathrm{ph}}^{m,l}(\mathbb{R}^p;\mathbb{R}^n) = \langle z\rangle^l\langle\xi\rangle^m(\mathrm{RC}\times\mathrm{RC})^*\mathcal{C}^\infty(\mathbb{S}_+^p \times \mathbb{S}_+^n)$. By Proposition 7.1.8 applied in each factor these are scattering symbols, and they are preserved by D_z, D_ξ (which lower the respective orders) and by products. If $\Psi_{\mathrm{sc,ph}}^{m,l}(\mathbb{R}^n) = \mathrm{Op}(S_{\mathrm{ph}}^{m,l})$ then $\Psi_{\mathrm{sc,ph}}^{m,l} \circ \Psi_{\mathrm{sc,ph}}^{m',l'} \subset \Psi_{\mathrm{sc,ph}}^{m+m',l+l'}$: the terms of the composition formula are polyhomogeneous, an asymptotic sum $\sum_j c_j$ with $c_j \in S_{\mathrm{ph}}^{m-j,l-j}$ can be realized by a polyhomogeneous symbol using Borel's lemma for the series $\sum_j (\rho_1\rho_2)^j G_j$ on the manifold with corners $\mathbb{S}_+^p \times \mathbb{S}_+^n$ (ρ_1, ρ_2 the two boundary defining functions), and $\mathcal{S}(\mathbb{R}^{p+n})$ is contained in every $S_{\mathrm{ph}}^{m,l}$. Repeating the computation in the proof of Proposition 7.2.10 shows that $\mathcal{F}\,\Psi_{\mathrm{sc}}^{m,l}(\mathbb{R}^n)\,\mathcal{F}^{-1} = \Psi_{\mathrm{sc}}^{l,m}(\mathbb{R}^n)$: the Fourier transform interchanges the two orders.

7.4 The residual isotropic algebra

The residual algebra $\Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$, a subalgebra of the residual algebra $\Psi^{-\infty}(\mathbb{R}^n)$ of the traditional calculus, has two important properties not shared by the latter: its elements are compact on $L^2(\mathbb{R}^n)$, and it is a 'corner' in the bounded operators. We use the strong (dual) topology on $\mathcal{S}'(\mathbb{R}^n)$, in which a set is bounded if and only if it is weakly bounded (Banach–Steinhaus, $\mathcal{S}$ being Fréchet), and convergence means uniform convergence on bounded subsets of $\mathcal{S}$.

Lemma 7.4.1. *If $u \in \mathcal{S}'(\mathbb{R}^q)$ and $F \in \mathcal{S}(\mathbb{R}^p \times \mathbb{R}^q)$ then $f(z) = \langle u, F(z,\cdot)\rangle$ lies in $\mathcal{S}(\mathbb{R}^p)$, with $\partial_z^\alpha f = \langle u, \partial_z^\alpha F(z,\cdot)\rangle$, and $(u,F) \mapsto f$ is continuous (hypocontinuous) and bounded on products of bounded sets.*

Proof. The map $z \mapsto F(z,\cdot) \in \mathcal{S}(\mathbb{R}^q)$ is smooth, by Taylor's formula with remainder and the uniform Schwartz bounds on the derivatives of F. The distribution u has finite order: $|\langle u,\phi\rangle| \le C \sum_{|\beta|+|\gamma|\le k} \sup |y^\beta D^\gamma \phi|$. Hence

$$|z^\delta \partial_z^\alpha f(z)| \le C \sum_{|\beta|+|\gamma|\le k} \sup_y |z^\delta y^\beta D_y^\gamma \partial_z^\alpha F(z,y)|,$$

which is bounded. □

Proposition 7.4.2. *A continuous linear operator $A : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$ lies in $\Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ if and only if it extends to a continuous linear map*

$$A : \mathcal{S}'(\mathbb{R}^n) \to \mathcal{S}(\mathbb{R}^n).$$

In particular each $A \in \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ maps bounded subsets of $L^2(\mathbb{R}^n)$ into bounded subsets of $\mathcal{S}(\mathbb{R}^n)$.

Proof. If $K_A \in \mathcal{S}(\mathbb{R}^{2n})$ then $Au(x) = \langle u, K_A(x,\cdot)\rangle$ for $u \in \mathcal{S}'$ by Lemma 7.4.1, and $\sup_x |x^\gamma D^\alpha Au(x)| \le \sup_{\phi\in B} |\langle u,\phi\rangle|$ where $B = \{x^\gamma D_x^\alpha K_A(x,\cdot);\ x \in \mathbb{R}^n\}$ is bounded in $\mathcal{S}$, which is continuity for the strong topology.

Conversely suppose $A : \mathcal{S}' \to \mathcal{S}$ is continuous. The map $\mathbb{R}^n \ni y \mapsto \delta_y \in \mathcal{S}'$ is smooth, with $\partial_{y_j}\delta_y = -\partial_j \delta_y$, since difference quotients of $\phi \in \mathcal{S}$ converge uniformly for ϕ in bounded sets. For each N the set $\{\langle y\rangle^N \partial^\beta \delta_y;\ y \in \mathbb{R}^n,\ |\beta| \le N\}$ is weakly, hence strongly, bounded in $\mathcal{S}'$, because $|\langle \langle y\rangle^N \partial^\beta \delta_y, \phi\rangle| = \langle y\rangle^N |\partial^\beta \phi(y)|$ is bounded for each $\phi \in \mathcal{S}$. Its image under A is bounded in $\mathcal{S}$. Thus $K(x,y) = (A\delta_y)(x)$ is smooth and $\langle y\rangle^N |x^\gamma D_x^\alpha D_y^\beta K(x,y)|$ is bounded for all N, γ, α and $|\beta| \le N$, i.e. $K \in \mathcal{S}(\mathbb{R}^{2n})$. Finally, for $\phi \in \mathcal{S}$ the Riemann sums of $\int \phi(y)\delta_y\, dy$ converge to ϕ in $\mathcal{S}'$, so $A\phi = \int \phi(y) A\delta_y\, dy = \int K(\cdot,y)\phi(y)\, dy$ and K is the kernel of A. □

Proposition 7.4.3. *Each $A \in \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ is a Hilbert–Schmidt, hence compact, operator on $L^2(\mathbb{R}^n)$, and $\|A\|_{\mathcal{B}(L^2)} \le \|K_A\|_{L^2(\mathbb{R}^{2n})}$; in particular the inclusion $\Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n) \to \mathcal{B}(L^2(\mathbb{R}^n))$ is continuous.*

Proof. $K_A \in \mathcal{S}(\mathbb{R}^{2n}) \subset L^2(\mathbb{R}^{2n})$. (Alternatively: by Proposition 7.4.2, A maps the unit ball of L^2 into a bounded set in $\mathcal{S}$, and bounded subsets of $\mathcal{S}$ are precompact in L^2 by the Arzelà–Ascoli theorem applied to $\langle x\rangle^{n+1}u$.) □

Lemma 7.4.4. *If $A_1, A_2 \in \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ and B is a bounded operator on $L^2(\mathbb{R}^n)$, or more generally any continuous linear operator $B : \mathcal{S}(\mathbb{R}^n) \to \mathcal{S}'(\mathbb{R}^n)$, then $A_1 B A_2 \in \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$. Moreover for each k there are k' and C such that, for bounded B,*

$$p_k(K_{A_1BA_2}) \le C\, p_{k'}(K_{A_1})\, p_{k'}(K_{A_2})\, \|B\|_{\mathcal{B}(L^2)}, \tag{7.4.1}$$

where $p_k(K) = \sum_{|\alpha|+|\beta|+|\gamma|+|\delta|\le k} \sup |x^\alpha y^\beta D_x^\gamma D_y^\delta K(x,y)|$.

Proof. For bounded B the kernel of $C = A_1BA_2$ is

$$K_C(x,y) = \big(B[K_{A_2}(\cdot,y)], \overline{K_{A_1}(x,\cdot)}\big)_{L^2},$$

and differentiating under the inner product (the maps $y \mapsto K_{A_2}(\cdot, y)$, $x \mapsto K_{A_1}(x, \cdot)$ are smooth into L^2)

$$|x^\alpha y^\beta D_x^\gamma D_y^\delta K_C(x,y)| \le \|B\| \sup_y \|y^\beta D_y^\delta K_{A_2}(\cdot, y)\|_{L^2} \sup_x \|x^\alpha D_x^\gamma K_{A_1}(x, \cdot)\|_{L^2},$$

and $\|f(x,\cdot)\|_{L^2} \le C \sup_{x'} \langle x' \rangle^n |f(x,x')|$. For general B with kernel $K_B \in \mathcal{S}'(\mathbb{R}^{2n})$ (Schwartz kernel theorem), $K_C(x,y) = \langle K_B, K_{A_1}(x,\cdot) \otimes K_{A_2}(\cdot, y) \rangle$, the pairing being in the variables (x', y') of $K_{A_1}(x,x')K_{A_2}(y',y) \in \mathcal{S}(\mathbb{R}^{4n})$; this is in $\mathcal{S}(\mathbb{R}^{2n})$ by Lemma 7.4.1. □

Note that $\Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ is an ideal in $\Psi_{\mathrm{iso}}^{\infty}(\mathbb{R}^n)$, but not in $\mathcal{B}(L^2(\mathbb{R}^n))$.

Discussion. Proposition 7.4.2 and Lemma 7.4.4 say that a residual operator smooths *and* localizes: it turns any tempered distribution into a Schwartz function. So in a product $A_1 B A_2$ the outer factors absorb whatever roughness or growth the middle factor B produces. This "sandwich" property is what lets us treat $\mathrm{Id} + \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ almost like a group of matrices: inverses, exponentials and logarithms computed in the Banach algebra $\mathcal{B}(L^2)$ automatically land back in the residual class (Propositions 7.4.6 and 7.5.1).

7.4.1 Finite rank operators and perturbations of the identity

If B is a bounded operator on $L^2(\mathbb{R}^n)$ of finite rank, let $f_1, \dots, f_N$ be an orthonormal basis of its range; then $u \mapsto (Bu, f_j)$ are continuous functionals, given by $g_j \in L^2$, and $K_B = \sum_j f_j(x)\overline{g_j(y)}$. If $B \in \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ the range of B and of B^* lie in $\mathcal{S}$, so $f_j, g_j \in \mathcal{S}(\mathbb{R}^n)$; conversely such kernels are in $\mathcal{S}(\mathbb{R}^{2n})$. The finite rank elements of $\Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ form an ideal in $\Psi_{\mathrm{iso}}^{\infty}(\mathbb{R}^n)$.

Lemma 7.4.5. *If K is a compact operator on a Hilbert space $\mathcal{H}$ then for each $t \in \mathbb{C}$, $\mathrm{Id} + tK$ has finite-dimensional null space and closed range of finite codimension, and* $\mathrm{ind}(\mathrm{Id} + tK) = \dim \mathrm{Nul}(\mathrm{Id} + tK) - \dim \mathrm{Nul}(\mathrm{Id} + \bar{t}K^*) = 0$.

Proof. Let $T = \mathrm{Id} + K$ (replace K by tK). On $N_0 = \mathrm{Nul}(T)$ we have $u = -Ku$, so the unit ball of N_0 is contained in K(unit ball), which is precompact; hence $\dim N_0 < \infty$. Let $R_0 = N_0^\perp$ and $R_1 = \mathrm{Ran}(T)$. If $f_k = Tu_k \to f$ with $u_k \in R_0$, then (u_k) is bounded: otherwise along a subsequence $\|u_k\| \to \infty$, and $v_k = u_k/\|u_k\|$ satisfies $v_k = -Kv_k + f_k/\|u_k\|$; by compactness a subsequence converges to some $v \in R_0$ with $\|v\| = 1$ and $Tv = 0$, a contradiction. Then $u_k = -Ku_k + f_k$ has a convergent subsequence, with limit u satisfying $Tu = f$; so R_1 is closed, and $N_1 = R_1^\perp = \mathrm{Nul}(T^*)$ is finite-dimensional by the first argument applied to K^*.

To see that the index is locally constant in t, fix t_0, let $T = \mathrm{Id} + t_0 K$ and write $\mathrm{Id} + tK = T + (t - t_0)K$ as a 2×2 matrix with respect to the decompositions $\mathcal{H} = N_0 \oplus R_0$ (domain) and $\mathcal{H} = N_1 \oplus R_1$ (range):

$$\mathrm{Id} + tK = \begin{pmatrix} (t-t_0)K_{00} & (t-t_0)K_{01} \\ (t-t_0)K_{10} & D(t) \end{pmatrix}, \qquad D(t) = \tilde{T} + (t-t_0)K_{11},$$

where $\tilde{T} : R_0 \to R_1$ is the restriction of T, invertible by the closed graph theorem. For $|t - t_0|$ small $D(t)$ is invertible and

$$\begin{pmatrix} P & Q \\ R & D \end{pmatrix} = \begin{pmatrix} \mathrm{Id} & QD^{-1} \\ 0 & \mathrm{Id} \end{pmatrix} \begin{pmatrix} P - QD^{-1}R & 0 \\ 0 & D \end{pmatrix} \begin{pmatrix} \mathrm{Id} & 0 \\ D^{-1}R & \mathrm{Id} \end{pmatrix}.$$

The outer factors are invertible, so null space and cokernel of $\mathrm{Id} + tK$ are isomorphic to those of the linear map $G(t) = P - QD^{-1}R : N_0 \to N_1$ between finite-dimensional spaces, whose index is $\dim N_0 - \dim N_1 = \mathrm{ind}(\mathrm{Id} + t_0 K)$. Thus $t \mapsto \mathrm{ind}(\mathrm{Id} + tK)$ is locally constant on $\mathbb{C}$, hence constant, and it vanishes at $t = 0$. □

Proposition 7.4.6. *Let $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$. Then $\mathrm{Id}+A$ is Fredholm of index zero on $L^2(\mathbb{R}^n)$: its null space N_0 is finite-dimensional and contained in $\mathcal{S}(\mathbb{R}^n)$, and its range is closed and is the orthocomplement of the null space $N_1 \subset \mathcal{S}(\mathbb{R}^n)$ of $\mathrm{Id}+A^*$. If Π_0, Π_1 denote the orthogonal projections onto N_0, N_1, then $\Pi_0, \Pi_1 \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and there is $B \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ such that*

$$(\mathrm{Id}+A)(\mathrm{Id}+B) = \mathrm{Id}-\Pi_1, \qquad (\mathrm{Id}+B)(\mathrm{Id}+A) = \mathrm{Id}-\Pi_0. \tag{7.4.2}$$

Moreover there is a finite rank $A' \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, of rank $\dim N_0$, such that $\mathrm{Id}+A+sA'$ is invertible on $L^2(\mathbb{R}^n)$ for all $s \neq 0$.

Proof. By Proposition 7.4.3 and Lemma 7.4.5, $\mathrm{Id}+A$ is Fredholm of index zero. If $u \in N_0$ then $u = -Au \in \mathcal{S}$, and similarly $N_1 \subset \mathcal{S}$, so $\Pi_0 = \sum_j f_j \otimes \bar{f}_j$ and $\Pi_1 = \sum_j g_j \otimes \bar{g}_j$, for orthonormal bases $f_j \in N_0$, $g_j \in N_1$, are in $\Psi^{-\infty}_{\mathrm{iso}}$. The restriction of $\mathrm{Id}+A$ is a bijection $N_0^\perp \to N_1^\perp = \mathrm{Ran}(\mathrm{Id}+A)$, with bounded inverse; extending this inverse by 0 on N_1 gives a bounded operator $\mathrm{Id}+B$ satisfying (7.4.2). From the first identity $B = -A - AB - \Pi_1$, and from the second $AB = -A^2 - ABA - A\Pi_0$; hence

$$B = -A + A^2 + ABA + A\Pi_0 - \Pi_1 \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$$

by Lemma 7.4.4. Finally, since $\dim N_0 = \dim N_1 = N$, the operator $A' = \sum_{j=1}^N g_j \otimes \bar{f}_j$ maps N_0 isomorphically onto N_1 and vanishes on $N_0^\perp$; with respect to $L^2 = N_0 \oplus N_0^\perp \to N_1 \oplus N_1^\perp$, $\mathrm{Id}+A+sA'$ is the direct sum of $sA'|_{N_0}$ and $(\mathrm{Id}+A)|_{N_0^\perp}$, invertible for $s \neq 0$. □

Remark 7.4.7. Once the trace on $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ has been introduced (Chapter 8), the vanishing of the index also follows from (7.4.2): $\mathrm{ind}(\mathrm{Id}+A) = \mathrm{Tr}\,\Pi_0 - \mathrm{Tr}\,\Pi_1 = \mathrm{Tr}([A,B]) = 0$. The argument above is independent of the trace.

Corollary 7.4.8. *If $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and $\mathrm{Id}+A$ is injective, or surjective, on $L^2(\mathbb{R}^n)$, then $\mathrm{Id}+A$ is invertible and its inverse lies in $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$.*

Proof. By index zero both N_0 and N_1 vanish, and $(\mathrm{Id}+A)^{-1} = \mathrm{Id}+B$. □

Corollary 7.4.9. *For $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, $\mathrm{Id}+A$ is Fredholm of index zero as an operator on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$: in both cases its null space is N_0 and its range is $\{f;\ \Pi_1 f = 0\}$, a closed subspace of codimension $\dim N_1$.*

Proof. The identities (7.4.2) hold on $\mathcal{S}$ and, by continuity and density, on $\mathcal{S}'$. If $u \in \mathcal{S}'$ and $(\mathrm{Id}+A)u = 0$ then $u = -Au \in \mathcal{S}$, so $u \in N_0$. The range contains $\mathrm{Ran}(\mathrm{Id}-\Pi_1) = \mathrm{Nul}(\Pi_1)$ and is contained in it, since $((\mathrm{Id}+A)u, v) = (u, (\mathrm{Id}+A^*)v) = 0$ for $v \in N_1$. □

7.5 Exponential and logarithm

The determinant of Chapter 8 is built from the logarithm of $\mathrm{Id}+A$, so we need the exponential and logarithm on residual perturbations of the identity. The corner estimate (7.4.1) reduces their convergence to convergence in operator norm.

Proposition 7.5.1. *For $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ the series*

$$\exp(A) = \sum_{j \geq 0} \frac{1}{j!} A^j$$

converges in $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, *and* $\exp : \Psi^{-\infty}_{\mathrm{iso}} \to \mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}$ *is continuous and entire (holomorphic along every complex line, locally uniformly). Each* $\exp(A)$ *is invertible with inverse* $\exp(-A)$. *If* $\|A\|_{\mathcal{B}(L^2)} < 1$ *the series*

$$\log(\mathrm{Id}+A) = \sum_{j\geq 1} \frac{(-1)^{j+1}}{j} A^j$$

converges in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, $\exp(\log(\mathrm{Id}+A)) = \mathrm{Id}+A$, *and* $\log(\exp(A)) = A$ *if* $\|A\|_{\mathcal{B}(L^2)} < \log 2$. *In particular the range of* exp *contains the neighbourhood* $\{\mathrm{Id}+A;\ \|A\|_{\mathcal{B}(L^2)} < 1\}$ *of* Id *in* $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$.

Proof. For $j \geq 2$ we write $A^j = A\,A^{j-2}A$ (with $A^0 = \mathrm{Id}$); then by (7.4.1) $p_k(K_{A^j}) \leq C\,p_{k'}(K_A)^2\,\|A\|^{j-2}_{\mathcal{B}(L^2)}$. Thus both series converge absolutely in every seminorm, uniformly for A in sets on which $p_{k'}(K_A)$ and $\|A\|$ are bounded (respectively $\|A\| \leq r < 1$), which by Proposition 7.4.3 include neighbourhoods of each point. The partial sums are continuous polynomial maps, so the limits are continuous and holomorphic along complex lines. The identities $\exp(A)\exp(-A) = \mathrm{Id}$, $\exp(\log(\mathrm{Id}+A)) = \mathrm{Id}+A$ and $\log(\exp A) = A$ (where $\|\exp A - \mathrm{Id}\| \leq e^{\|A\|} - 1 < 1$) hold for the norm-convergent series in the Banach algebra $\mathcal{B}(L^2)$, and convergence in $\Psi^{-\infty}_{\mathrm{iso}}$ implies convergence in $\mathcal{B}(L^2)$. □

7.6 Elliptic operators

Ellipticity in the isotropic sense asks that the symbol be invertible near infinity in *all* directions of phase space, not only for large ξ. For the harmonic oscillator, with symbol $|x|^2+|\xi|^2$, this holds; for the Laplacian it fails along $\xi = 0$. The parametrix construction is the same as in Chapter 2, but its consequences are global.

Definition 7.6.1. An operator $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ with left symbol a is *elliptic* if there are $c, R > 0$ such that $a(w)$ is invertible and $|a(w)^{-1}| \leq c^{-1}\langle w\rangle^{-m}$ for $|w| \geq R$ (for $N = 1$: $|a(w)| \geq c\langle w\rangle^m$). For $A \in \Psi^m_{\mathrm{iso,ph}}$ this holds if and only if the principal symbol $\sigma_m(A)$ is invertible at every point of $\mathbb{S}^{2n-1}$. Similarly $A \in \Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)$ is elliptic if $|a(x,\xi)| \geq c\langle x\rangle^l\langle\xi\rangle^m$ for $|x|+|\xi| \geq R$.

Theorem 7.6.2. *If* $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ *is elliptic there is* $B \in \Psi^{-m}_{\mathrm{iso}}(\mathbb{R}^n)$, *unique modulo* $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, *with*

$$AB - \mathrm{Id},\ BA - \mathrm{Id} \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n); \tag{7.6.1}$$

if $A \in \Psi^m_{\mathrm{iso,ph}}$ *then* $B \in \Psi^{-m}_{\mathrm{iso,ph}}$. *Consequently, if* $u \in \mathcal{S}'(\mathbb{R}^n)$ *and* $Au \in \mathcal{S}(\mathbb{R}^n)$ *then* $u \in \mathcal{S}(\mathbb{R}^n)$. *The same holds for elliptic* $A \in \Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)$, *with* $B \in \Psi^{-m,-l}_{\mathrm{sc}}(\mathbb{R}^n)$.

Proof. Let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$ equal 1 on $|w| \leq R$. Choose $F \in \mathcal{C}^\infty(\mathbb{C})$ with $F(z) = 1/z$ for $|z| \geq c$; then $F(\langle w\rangle^{-m}a) \in \Gamma^0$ by Lemma 7.1.2(ii), and $b_0 = (1-\chi)\langle w\rangle^{-m}F(\langle w\rangle^{-m}a) \in \Gamma^{-m}$ satisfies $ab_0 = 1-\chi$. (For matrix symbols use a smooth function equal to the inverse on the closed set of matrices M with $|M^{-1}| \leq c^{-1}$.) If a is polyhomogeneous so is b_0, since $1/a = a_m^{-1}\sum_k(-r)^k$ asymptotically, with $r = (a-a_m)/a_m \in \Gamma^{-1}_{\mathrm{ph}}$ for large $|w|$. By (7.2.4), $B_0 = \mathrm{Op}(b_0)$ satisfies $AB_0 = \mathrm{Id}-E$ with $\sigma_L(E) = \chi + \Gamma^{-2}$, so $E \in \Psi^{-2}_{\mathrm{iso}}$. Choose $B_R \in \Psi^{-m}_{\mathrm{iso}}$ with $\sigma_L(B_R) \sim \sum_j \sigma_L(B_0E^j)$, the terms having orders $-m-2j$. Then for each N, $B_R - \sum_{j<N} B_0E^j \in \Psi^{-m-2N}_{\mathrm{iso}}$ and

$$AB_R = \sum_{j<N}(\mathrm{Id}-E)E^j + \Psi^{-2N}_{\mathrm{iso}} = \mathrm{Id}-E^N + \Psi^{-2N}_{\mathrm{iso}} = \mathrm{Id}+\Psi^{-2N}_{\mathrm{iso}}.$$

Similarly $B_0 A = \mathrm{Id} - E'$ with $E' \in \Psi_{\mathrm{iso}}^{-2}$, and $B_L \sim \sum_j E'^j B_0$ satisfies $B_L A - \mathrm{Id} \in \Psi_{\mathrm{iso}}^{-\infty}$. Then $B_L - B_R = B_L(\mathrm{Id} - AB_R) + (B_L A - \mathrm{Id})B_R \in \Psi_{\mathrm{iso}}^{-\infty}$, so $B = B_R$ satisfies (7.6.1); uniqueness follows in the same way. If $Au \in \mathcal{S}$ then $u = B(Au) - (BA - \mathrm{Id})u \in \mathcal{S}$, since residual operators map $\mathcal{S}'$ into $\mathcal{S}$ (Proposition 7.4.2). In the scattering case the same construction works with $E \in \Psi_{\mathrm{sc}}^{-1,-1}$ and asymptotic sums in the sense of Lemma 7.1.5, the residual space being (7.3.1). □

Question. On a compact manifold elliptic regularity says that $Au \in \mathcal{C}^\infty$ implies $u \in \mathcal{C}^\infty$. Why does the isotropic version give the much stronger conclusion $u \in \mathcal{S}(\mathbb{R}^n)$, which includes decay at infinity?

Answer. The error terms of the parametrix are residual isotropic operators, and these map $\mathcal{S}'(\mathbb{R}^n)$ into $\mathcal{S}(\mathbb{R}^n)$ (Proposition 7.4.2), not merely into smooth functions. Ellipticity at $|x| \to \infty$ controls growth in x exactly as ellipticity at $|\xi| \to \infty$ controls regularity. For instance an eigenfunction of the harmonic oscillator is automatically a Schwartz function. □

Combining the symbolic construction with the analysis of Section 7.4 we see that elliptic operators have generalized inverses with finite rank errors.

Proposition 7.6.3. *Let $A \in \Psi_{\mathrm{iso}}^m(\mathbb{R}^n)$ be elliptic. Then $N_0 = \mathrm{Nul}(A)$ and $N_1 = \mathrm{Nul}(A^*)$ (on $\mathcal{S}'(\mathbb{R}^n)$) are finite-dimensional subspaces of $\mathcal{S}(\mathbb{R}^n)$, and if Π_0, Π_1 are the L^2-orthogonal projections onto them, there is $G \in \Psi_{\mathrm{iso}}^{-m}(\mathbb{R}^n)$ such that*

$$AG = \mathrm{Id} - \Pi_1, \qquad GA = \mathrm{Id} - \Pi_0. \tag{7.6.2}$$

Hence A is Fredholm on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$, with range $\{f;\ \Pi_1 f = 0\}$ and index $\dim N_0 - \dim N_1$. If A is polyhomogeneous then so is G. The same holds for elliptic $A \in \Psi_{\mathrm{sc}}^{m,l}(\mathbb{R}^n)$, with $G \in \Psi_{\mathrm{sc}}^{-m,-l}(\mathbb{R}^n)$.

Proof. Let B be a parametrix, $AB = \mathrm{Id} - E_R$, $BA = \mathrm{Id} - E_L$ with $E_R, E_L \in \Psi_{\mathrm{iso}}^{-\infty}$. If $Au = 0$, $u \in \mathcal{S}'$, then $u = E_L u$, so $N_0 \subset \mathrm{Nul}(\mathrm{Id} - E_L)$, which is finite-dimensional and contained in $\mathcal{S}$ by Corollary 7.4.9; the same applies to A^*, which is elliptic.

We show that $\mathrm{Ran}_{\mathcal{S}}(A) = \{f \in \mathcal{S};\ f \perp N_1\}$. Applying Proposition 7.4.6 to $-E_R$ gives $B' \in \Psi_{\mathrm{iso}}^{-\infty}$ and a finite rank orthogonal projection Π, with range $W \subset \mathcal{S}$, such that $AB(\mathrm{Id} + B') = (\mathrm{Id} - E_R)(\mathrm{Id} + B') = \mathrm{Id} - \Pi$. Hence $V = A(\mathcal{S}) \supset \{f \in \mathcal{S};\ f \perp W\}$. Since $W \subset \mathcal{S}$ is finite-dimensional, every $v \in V$ decomposes as $v = (v - P_W v) + P_W v$ with $v - P_W v \in V$, so $V = (W^\perp \cap \mathcal{S}) \oplus (V \cap W)$, and with $W_1 = W \ominus (V \cap W)$ we get $V = \{f \in \mathcal{S};\ f \perp W_1\}$. On the other hand $W_1 = \{g \in \mathcal{S};\ g \perp V\}$ (using the density of $W^\perp \cap \mathcal{S}$ in $W^\perp$), and $g \in \mathcal{S}$ is orthogonal to $A(\mathcal{S})$ if and only if $A^* g = 0$. Thus $W_1 = N_1$, as claimed.

Next, if $f \in \mathcal{S}'$ and $\Pi_1 f = 0$, then $f = AB(\mathrm{Id} + B')f + \Pi f$ with $\Pi f \in \mathcal{S}$ and $\Pi f \perp N_1$ (as $(Au, v) = (u, A^* v) = 0$ for $v \in N_1$), so $\Pi f \in A(\mathcal{S})$ and $f \in A(\mathcal{S}')$. Conversely $A(\mathcal{S}') \perp N_1$. Hence we may define $G : \mathcal{S}' \to \mathcal{S}'$ by $G = 0$ on N_1 and, for $\Pi_1 f = 0$, $Gf = u$ where u is the unique solution of $Au = f$ with $\Pi_0 u = 0$. Then (7.6.2) holds on $\mathcal{S}'$, and G maps $\mathcal{S}$ to $\mathcal{S}$ (elliptic regularity). The restriction $G : \mathcal{S} \to \mathcal{S}$ has closed graph, hence is continuous: if $f_k \to f$ and $Gf_k \to u$ in $\mathcal{S}$ then $Au = (\mathrm{Id} - \Pi_1)f$ and $\Pi_0 u = 0$, so $u = Gf$. Multiplying (7.6.2) by B gives $(\mathrm{Id} - E_L)G = B - B\Pi_1$ and $G(\mathrm{Id} - E_R) = B - \Pi_0 B$, so

$$G = B - B\Pi_1 + E_L G = B - B\Pi_1 + E_L B - E_L \Pi_0 B + E_L G E_R.$$

Here $E_L G E_R$ is continuous from $\mathcal{S}'$ to $\mathcal{S}$, since $E_R : \mathcal{S}' \to \mathcal{S}$, $G : \mathcal{S} \to \mathcal{S}$ and $E_L : \mathcal{S} \to \mathcal{S}$ are; so it is in $\Psi_{\mathrm{iso}}^{-\infty}$ by Proposition 7.4.2. All other terms except B are residual, so $G \in B + \Psi_{\mathrm{iso}}^{-\infty} \subset \Psi_{\mathrm{iso}}^{-m}$. The statements about $\mathcal{S}$ follow in the same way, and the scattering case is identical. □

7.7 L^2 boundedness and compactness

Proposition 7.7.1. *Every $A \in \Psi^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ is bounded on $L^2(\mathbb{R}^n;\mathbb{C}^N)$, and every $A \in \Psi^{-\epsilon}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$, $\epsilon>0$, is compact. Elements of Ψ^m_{iso} with $m<-3n/2$ are Hilbert–Schmidt.*

Proof. By Lemma 7.1.6, $\Psi^0_{\mathrm{iso}}(\mathbb{R}^n)\subset\Psi^0(\mathbb{R}^n)$, which is bounded on L^2 by Chapter 2. If $a\in\Gamma^{-M}$, $M>n$, the kernel of $\mathrm{Op}(a)$ satisfies $(x-y)^\beta K(x,y)=\int e^{i(x-y)\cdot\xi}(-D_\xi)^\beta a(x,\xi)đ\xi$, and $\int\langle(x,\xi)\rangle^{-M}d\xi=C\langle x\rangle^{n-M}$ (substitute $\xi=\langle x\rangle\eta$), so $|K(x,y)|\le C_N\langle x\rangle^{n-M}\langle x-y\rangle^{-N}$ for all N; this is in $L^2(\mathbb{R}^{2n})$ if $M>3n/2$. Now let $A\in\Psi^{-\epsilon}_{\mathrm{iso}}$ and $T=A^*A\in\Psi^{-2\epsilon}_{\mathrm{iso}}$, a bounded self-adjoint operator. For j large, $T^{2^j}\in\Psi^{-2^{j+1}\epsilon}_{\mathrm{iso}}$ is Hilbert–Schmidt, hence compact. If S is bounded and self-adjoint with S^2 compact, then S is compact: for $u_k\rightharpoonup u$ weakly, $\|S(u_k-u)\|^2=(S^2(u_k-u),u_k-u)\le\|S^2(u_k-u)\|\|u_k-u\|\to0$. By induction T is compact, and then $\|A(u_k-u)\|^2=(T(u_k-u),u_k-u)\to0$, so A is compact. □

We next identify the norm of $A\in\Psi^0_{\mathrm{iso}}$ modulo compact operators. The lower bound uses *coherent states*. For $w=(x_0,\xi_0)\in\mathbb{R}^{2n}$ let $(T_wu)(x)=e^{ix\cdot\xi_0}u(x-x_0)$, a unitary operator on $L^2(\mathbb{R}^n)$ preserving $\mathcal{S}(\mathbb{R}^n)$; a direct change of variables shows

$$T_w^{-1}\,\mathrm{Op}(a)T_w=\mathrm{Op}(a(\cdot+w)). \tag{7.7.1}$$

Let $u_0=\pi^{-n/4}e^{-|x|^2/2}$, so $\|u_0\|=1$.

Discussion. A coherent state T_wu_0 is a Gaussian translated to the point x_0 and modulated to the frequency ξ_0; it is concentrated, up to an uncertainty of size one, near the point $w=(x_0,\xi_0)$ of phase space. On such a wave packet an operator of order 0 acts, approximately, as multiplication by the number $a(w)$ (Lemma 7.7.2). Moving w to infinity gives a sequence of unit vectors tending weakly to zero, on which compact operators become negligible. So the size of a near infinity can be read off from A modulo compact operators.

Lemma 7.7.2. *Let $w_k\in\mathbb{R}^{2n}$ with $|w_k|\to\infty$ and $\phi_k=T_{w_k}u_0$. Then $\|\phi_k\|=1$, $\phi_k\rightharpoonup0$ weakly in $L^2(\mathbb{R}^n)$, and for every $a\in\Gamma^0(\mathbb{R}^{2n};M(N,\mathbb{C}))$*

$$\sup_{v\in\mathbb{C}^N,|v|\le1}\|\mathrm{Op}(a)(\phi_kv)-\phi_k\,a(w_k)v\|_{L^2}\to0.$$

Proof. By (7.7.1) the norm in question is $\|\mathrm{Op}(b_k)(u_0v)\|$ with $b_k(w)=a(w+w_k)-a(w_k)$. All derivatives of b_k are bounded uniformly in k (as $\Gamma^0\subset\mathcal{C}^\infty_{\mathrm{b}}$), and

$$|b_k(w)|\le|w|\sup_{t\in[0,1]}|\nabla a(w_k+tw)|\le C|w|\sup_{t\in[0,1]}\langle w_k+tw\rangle^{-1}\to0$$

for each w. Now $\mathrm{Op}(b_k)u_0(x)=\int e^{ix\cdot\xi}b_k(x,\xi)\hat u_0(\xi)đ\xi$; integrating by parts, $x^\beta\,\mathrm{Op}(b_k)u_0=\int e^{ix\cdot\xi}(-D_\xi)^\beta(b_k\hat u_0)đ\xi$, so $|\,\mathrm{Op}(b_k)u_0(x)|\le C\langle x\rangle^{-n-1}$ uniformly in k, while $\mathrm{Op}(b_k)u_0(x)\to0$ pointwise by dominated convergence. Dominated convergence in L^2 gives $\|\mathrm{Op}(b_k)u_0\|\to0$.

For the weak convergence it suffices, by density, to show $(\phi_k,f)\to0$ for $f\in\mathcal{S}$, and for this it suffices to show that every subsequence has a further subsequence along which this holds. Write $w_k=(x_k,\xi_k)$. If $|x_k|\to\infty$ along a subsequence then $|(\phi_k,f)|\le\int u_0(x-x_k)|f(x)|\,dx\to0$. Otherwise $x_k\to x_*$ along a subsequence and $|\xi_k|\to\infty$; then $(\phi_k,f)=\hat g_k(-\xi_k)$ with $g_k=u_0(\cdot-x_k)\bar f\to g_*$ in L^1, and $|\hat g_k(-\xi_k)|\le\|g_k-g_*\|_{L^1}+|\hat g_*(-\xi_k)|\to0$ by the Riemann–Lebesgue lemma. □

Theorem 7.7.3. *If $A\in\Psi^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ has left symbol a then*

$$\inf_{K\ \text{compact}}\|A+K\|_{\mathcal{B}(L^2)}=\inf_{E\in\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)}\|A+E\|_{\mathcal{B}(L^2)}=\limsup_{|w|\to\infty}\|a(w)\|. \tag{7.7.2}$$

If $A\in\Psi^0_{\mathrm{iso,ph}}$ the right side is $\sup_{p\in\mathbb{S}^{2n-1}}\|\sigma_0(A)(p)\|$.

Proof. Let μ be the right side of (7.7.2). Since residual operators are compact, the first infimum is at most the second.

Lower bound. Let K be compact. Choose $w_k \to \infty$ with $\|a(w_k)\| \to \mu$ and unit vectors v_k with $|a(w_k)v_k| = \|a(w_k)\|$; passing to a subsequence $v_k \to v$. With ϕ_k as in Lemma 7.7.2, $K(\phi_k v) \to 0$ since K is compact and $\phi_k \rightharpoonup 0$, so

$$\|A+K\| \geq \liminf_k \|(A+K)(\phi_k v)\| \geq \liminf_k |a(w_k)v| = \mu.$$

Upper bound. Let $C > \mu^2$ and choose R and $\delta' > 0$ such that $p = C - a^*a \geq 2\delta'$ for $|w| \geq R$. With $\chi \in \mathcal{C}_c^\infty$ equal to 1 on $|w| \leq R$, $0 \leq \chi \leq 1$ and supported in $|w| \leq R+1$, the function $p' = (1-\chi)p + \chi C$ lies in Γ^0, satisfies $p' \geq 2\delta'$ everywhere and equals p for $|w| \geq R+1$. Since the square root is smooth on positive definite matrices, $q_0 = (p')^{1/2} \in \Gamma^0$ is hermitian, invertible, with $q_0^{-1} \in \Gamma^0$ (Lemma 7.1.2(ii)). Let $Q_0 = \operatorname{Op}(q_0)$. By Theorem 7.2.7, $G_1 = Q_0^*Q_0 - C + A^*A$ is self-adjoint and lies in Ψ^{-2}_{iso}, since its symbol is $p' - p$ (compactly supported) modulo Γ^{-2}. Suppose $Q = Q_0 + R_1 + \cdots + R_{k-1}$ has been found with $R_j \in \Psi^{-2j}_{\mathrm{iso}}$ and $G_k = Q^*Q - C + A^*A \in \Psi^{-2k}_{\mathrm{iso}}$, and let $g_k = \sigma_L(G_k)$, so that $g_k - g_k^* \in \Gamma^{-2k-2}$. Set $R_k = \operatorname{Op}(r_k)$, $r_k = -\frac{1}{2}q_0^{-1}g_k$. Then $(Q+R_k)^*(Q+R_k) - C + A^*A = G_k + Q^*R_k + R_k^*Q + R_k^*R_k$, whose symbol is, modulo Γ^{-2k-2},

$$g_k + q_0 r_k + r_k^* q_0 = g_k - \tfrac{1}{2}g_k - \tfrac{1}{2}g_k^* \equiv 0.$$

Taking $Q \in \Psi^0_{\mathrm{iso}}$ with $\sigma_L(Q) \sim q_0 + \sum_k r_k$ we obtain $Q^*Q = C - A^*A + G$ with $G \in \Psi^{-\infty}_{\mathrm{iso}}$ self-adjoint. Hence for $u \in \mathcal{S}$

$$\|Au\|^2 = C\|u\|^2 - \|Qu\|^2 + (Gu,u) \leq C\|u\|^2 + (Gu,u).$$

Given $\delta > 0$, approximate the compact operator G within $\delta/2$ by a finite rank operator $\sum_{j\leq J} f_j \otimes \bar{g}_j$ and then replace the g_j by nearby $g_j' \in \mathcal{S}$, to get a finite-dimensional $W = \operatorname{span}\{g_j'\} \subset \mathcal{S}$ with $\|Gu\| \leq \delta\|u\|$ for $u \perp W$. Let $\Pi_W \in \Psi^{-\infty}_{\mathrm{iso}}$ be the orthogonal projection onto W. Then for $u \in \mathcal{S}$, $u' = (\mathrm{Id} - \Pi_W)u \perp W$ and $\|A(\mathrm{Id} - \Pi_W)u\|^2 \leq (C+\delta)\|u\|^2$. Since $A\Pi_W \in \Psi^{-\infty}_{\mathrm{iso}}$ and $C > \mu^2$, $\delta > 0$ are arbitrary, the second infimum is at most μ.

For polyhomogeneous A, $a - a_0 \to 0$ at infinity, a_0 being homogeneous of degree 0 outside the unit ball, so $\mu = \sup_{\mathbb{S}^{2n-1}} \|a_0\|$. □

Corollary 7.7.4. *An operator $A \in \Psi^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ is compact on $L^2(\mathbb{R}^n;\mathbb{C}^N)$ if and only if $\sigma_L(A)(w) \to 0$ as $|w| \to \infty$. In particular $A \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ is compact if and only if $\sigma_0(A) = 0$, that is $A \in \Psi^{-1}_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$.*

Proof. If A is compact the left side of (7.7.2) vanishes. Conversely if the right side vanishes, A is a norm limit of residual, hence compact, operators. □

Corollary 7.7.5. *An operator $A \in \Psi^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ is Fredholm on $L^2(\mathbb{R}^n;\mathbb{C}^N)$ if and only if it is elliptic.*

Proof. If A is elliptic, (7.6.2) with $G \in \Psi^0_{\mathrm{iso}}$ bounded shows that A is Fredholm. Conversely, if A is Fredholm there is a bounded B with $BA = \mathrm{Id} - P$, P of finite rank. If A is not elliptic there are $w_k \to \infty$ and unit vectors v_k with $|a(w_k)v_k| \to 0$; passing to a subsequence $v_k \to v$. By Lemma 7.7.2, $\|A(\phi_k v)\| \to 0$, hence $\phi_k v = BA(\phi_k v) + P(\phi_k v) \to 0$, since P is compact and $\phi_k \rightharpoonup 0$. This contradicts $\|\phi_k v\| = 1$. □

In particular ellipticity is necessary in Proposition 7.6.3.

7.8 The harmonic oscillator

The harmonic oscillator is the basic elliptic operator of the isotropic calculus, and the one operator whose spectral theory can be computed completely by hand. We use it for two purposes: its eigenfunctions, the Hermite functions, give a concrete orthonormal basis in which residual operators become rapidly decreasing infinite matrices; and its ground state u_0 will later fix the phases of the metaplectic operators.

The *harmonic oscillator* is the differential operator

$$H = \sum_{j=1}^{n} (D_j^2 + x_j^2) = \Delta + |x|^2.$$

It is an elliptic element of $\Psi^2_{\mathrm{iso,ph}}(\mathbb{R}^n)$ with symbol $|w|^2$. Its spectral decomposition is computed explicitly using the *creation* and *annihilation* operators

$$C_j = D_j + ix_j, \qquad A_j = D_j - ix_j = C_j^*, \qquad j = 1, \dots, n.$$

Lemma 7.8.1. *The following identities hold for $j, k = 1, \dots, n$:*

$$[A_j, A_k] = [C_j, C_k] = 0, \qquad [A_j, C_k] = 2\delta_{jk},$$
$$H = \sum_j C_j A_j + n = \sum_j A_j C_j - n, \qquad [C_j, H] = -2C_j, \qquad [A_j, H] = 2A_j.$$

Hence $HC_j = C_j(H+2)$ and $HA_j = A_j(H-2)$.

Proof. Since $[D_j, x_k] = -i\delta_{jk}$, $C_jA_j = D_j^2 + x_j^2 - i[D_j, x_j] = D_j^2 + x_j^2 - 1$ and $A_jC_j = D_j^2 + x_j^2 + 1$, which gives the formulæ for H and $[A_j, C_j] = 2$; the other commutators of the A's and C's vanish by the same computation, using $[D_j, D_k] = [x_j, x_k] = 0$. Then $[C_j, H] = \sum_k C_k[C_j, A_k] = -2C_j$ and $[A_j, H] = \sum_k [A_j, C_k]A_k = 2A_j$. □

Proposition 7.8.2. *For $u \in \mathcal{S}(\mathbb{R}^n)$,*

$$(Hu, u) = \sum_j \|A_j u\|^2 + n\|u\|^2. \tag{7.8.1}$$

If $u \in \mathcal{S}'(\mathbb{R}^n)$ and $(H - \lambda)u = 0$ for some $\lambda \in \mathbb{C}$, then $u \in \mathcal{S}(\mathbb{R}^n)$, and $u \neq 0$ implies $\lambda \in \mathbb{R}$, $\lambda \geq n$. The operator H is invertible on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$, with inverse $H^{-1} \in \Psi^{-2}_{\mathrm{iso,ph}}(\mathbb{R}^n)$, which is a compact, self-adjoint, injective operator on $L^2(\mathbb{R}^n)$. More generally $H^p \in \Psi^{2p}_{\mathrm{iso,ph}}(\mathbb{R}^n)$ for all $p \in \mathbb{Z}$.

Proof. (7.8.1) follows from $H = \sum C_jA_j + n$ and $C_j^* = A_j$. Since $H - \lambda$ is elliptic, eigenfunctions in $\mathcal{S}'$ lie in $\mathcal{S}$ (Theorem 7.6.2); then $\lambda\|u\|^2 = (Hu, u)$ is real and at least $n\|u\|^2$. Thus $\mathrm{Nul}(H) = \mathrm{Nul}(H^*) = \{0\}$ on $\mathcal{S}'$, and Proposition 7.6.3 gives $G \in \Psi^{-2}_{\mathrm{iso,ph}}$ with $HG = GH = \mathrm{Id}$ on $\mathcal{S}'$. By Proposition 7.7.1, G is compact on L^2; it is injective since $HG = \mathrm{Id}$, and self-adjoint since $(Gf, g) = (Gf, HGg) = (HGf, Gg) = (f, Gg)$ for $f, g \in \mathcal{S}$. □

Proposition 7.8.3. *The eigenvalues of H (on $\mathcal{S}'(\mathbb{R}^n)$) are*

$$\mathrm{spec}(H) = \{n, n+2, n+4, \dots\} = n + 2\mathbb{N}_0.$$

The eigenvalue n has a one-dimensional eigenspace spanned by the ground state $u_0 = \pi^{-n/4}\exp(-|x|^2/2)$.

Proof. By (7.8.1), $Hu = nu$ if and only if $A_j u = 0$ for all j. Now $A_j u = -i(\partial_j + x_j)u = -ie^{-|x|^2/2}\partial_j(e^{|x|^2/2}u)$, so for a distribution u, $A_j u = 0$ for all j means that $e^{|x|^2/2}u$ is constant, i.e. $u \in \mathbb{C}u_0$. Similarly $C_j v = -ie^{|x|^2/2}\partial_j(e^{-|x|^2/2}v)$, so the only tempered solution of $C_j v = 0$ is $v = 0$ (as $e^{x_j^2/2}$ is not tempered); thus each C_j is injective on $\mathcal{S}'$.

If $Hu = \lambda u$ with $0 \neq u \in \mathcal{S}$, then by Lemma 7.8.1 $H(A_j u) = (\lambda - 2)A_j u$ and $H(C_j u) = (\lambda + 2)C_j u$, with $C_j u \neq 0$. So $\lambda + 2\mathbb{N}_0$ consists of eigenvalues. Since all eigenvalues are $\geq n$, $A^\alpha u = 0$ when $\lambda - 2|\alpha| < n$. Let k be the largest integer such that $A^\alpha u \neq 0$ for some $|\alpha| = k$, and $v = A^\alpha u$ for such α. Then $A_j v = 0$ for all j, so $v \in \mathbb{C}u_0$ and $\lambda - 2k = n$. Combined with $Hu_0 = nu_0$ and the ladder property, the eigenvalues are exactly $n + 2\mathbb{N}_0$. □

Proposition 7.8.4. *For $k \in \mathbb{N}_0$ let E_k be the eigenspace of H with eigenvalue $n+2k$. Then the map*

$$\{\textit{polynomials on } \mathbb{R}^n \textit{ homogeneous of degree } k\} \ni p \longmapsto p(C)u_0 \in E_k$$

is an isomorphism; thus $\{C^\alpha u_0;\ |\alpha| = k\}$ is a basis of E_k and $\dim E_k = \binom{n+k-1}{k}$.

Proof. The C_j commute, so $p(C)$ is well defined, and by Lemma 7.8.1, $HC^\alpha u_0 = C^\alpha(H + 2|\alpha|)u_0 = (n+2|\alpha|)C^\alpha u_0$. From $[A_j, C_k] = 2\delta_{jk}$ and $A_j u_0 = 0$ we get $A_j C^\alpha u_0 = 2\alpha_j C^{\alpha - e_j} u_0$ (zero if $\alpha_j = 0$), hence for $|\beta| = |\alpha|$

$$A^\beta C^\alpha u_0 = \begin{cases} 0, & \beta \neq \alpha, \\ 2^{|\alpha|}\alpha!\, u_0, & \beta = \alpha. \end{cases} \tag{7.8.2}$$

Applying A^β to $\sum_{|\alpha|=k} c_\alpha C^\alpha u_0 = 0$ gives $c_\beta = 0$, so the map is injective.

For surjectivity we show by induction on k: if $v \in E_k$ and $A^\beta v = 0$ for all $|\beta| = k$, then $v = 0$. For $k = 0$ this is trivial, as $A^0 = \mathrm{Id}$. For $k > 0$ and j fixed, $A_j v \in E_{k-1}$ (or is 0) and $A^\beta(A_j v) = A^{\beta + e_j} v = 0$ for $|\beta| = k - 1$, so $A_j v = 0$ by induction; then $v \in E_0 = \mathbb{C}u_0$ and $Hv = (n + 2k)v$ force $v = 0$. Now let $v \in E_k$ be orthogonal to all $C^\alpha u_0$, $|\alpha| = k$. Then $(A^\alpha v, u_0) = (v, C^\alpha u_0) = 0$, and $A^\alpha v \in E_0 = \mathbb{C}u_0$, so $A^\alpha v = 0$ for all $|\alpha| = k$, whence $v = 0$. Since $E_k \subset \mathcal{S}$ is finite-dimensional (Proposition 7.6.3 applied to $H - n - 2k$), E_k is spanned by the $C^\alpha u_0$. □

Eigenfunctions with different eigenvalues are orthogonal, H being symmetric on $\mathcal{S}$, and by (7.8.2), $(C^\alpha u_0, C^\beta u_0) = (A^\beta C^\alpha u_0, u_0) = \delta_{\alpha\beta} 2^{|\alpha|}\alpha!$ for $|\alpha| = |\beta|$, since $\|u_0\| = 1$. Hence the *Hermite functions*

$$u_\alpha = (2^{|\alpha|}\alpha!)^{-1/2} C^\alpha u_0, \qquad \alpha \in \mathbb{N}_0^n, \tag{7.8.3}$$

form an orthonormal set in $L^2(\mathbb{R}^n)$, with $Hu_\alpha = (n + 2|\alpha|)u_\alpha$. In one variable, $(\partial - x)f = e^{x^2/2}\partial(e^{-x^2/2}f)$ gives $C^k e^{-x^2/2} = i^k H_k(x) e^{-x^2/2}$, where $H_k(x) = (-1)^k e^{x^2} \frac{d^k}{dx^k} e^{-x^2}$ is the k-th Hermite polynomial, and in general $u_\alpha(x) = \prod_j u_{\alpha_j}(x_j)$. One also checks from (7.8.2) that

$$C_j u_\alpha = (2\alpha_j + 2)^{1/2} u_{\alpha + e_j}, \qquad A_j u_\alpha = (2\alpha_j)^{1/2} u_{\alpha - e_j}. \tag{7.8.4}$$

Discussion. The whole computation rests on the identities $HC_j = C_j(H + 2)$ and $HA_j = A_j(H - 2)$ of Lemma 7.8.1: a creation operator raises an eigenvalue by 2 and an annihilation operator lowers it by 2. Since $H \geq n$, lowering must stop, and it can only stop at a function killed by all A_j, namely the ground state u_0. Hence every eigenfunction is reached from u_0 by creation operators, and the spectrum is the ladder of Figure 7.2.

Lemma 7.8.5. *If K is a compact self-adjoint operator on a Hilbert space $V \neq \{0\}$ and $K \neq 0$, then K has an eigenvector with non-zero eigenvalue.*

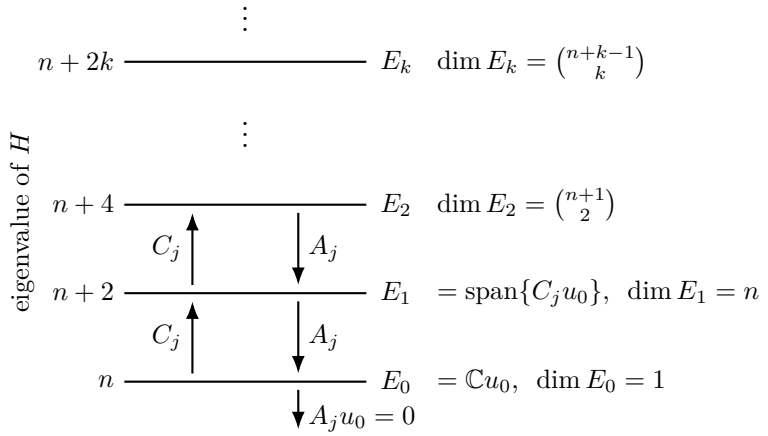


Figure 7.2: The spectrum of the harmonic oscillator $H = \sum_j (D_j^2 + x_j^2)$ on $\mathbb{R}^n$. The creation operators $C_j = D_j + ix_j$ move each eigenspace E_k into E_{k+1}; the annihilation operators $A_j = D_j - ix_j$ move it into E_{k-1}, and they all annihilate the ground state $u_0 = \pi^{-n/4} e^{-|x|^2/2}$.

Proof. Let $\tau = \|K\| > 0$ and choose unit vectors ϕ_j with $\|K\phi_j\| \to \tau$. Passing to a subsequence, $\phi_j \rightharpoonup \phi$ weakly with $\|\phi\| \le 1$, and $K\phi_j \to K\phi$ by compactness. Thus $\|K\phi\| = \tau \le \tau\|\phi\|$, so $\|\phi\| = 1$ and ϕ maximizes $\|K\psi\|^2 = (K^2\psi, \psi)$ on the unit sphere. For $\phi' \perp \phi$ the function $t \mapsto \|K(\phi + t\phi')\|^2 / \|\phi + t\phi'\|^2$, $t \in \mathbb{R}$, has a maximum at $t = 0$, so $\mathrm{Re}(K^2\phi, \phi') = 0$; replacing ϕ' by $i\phi'$, $(K^2\phi, \phi') = 0$. Hence $K^2\phi = \tau^2\phi$, i.e. $(K - \tau)(K + \tau)\phi = 0$. Either $(K + \tau)\phi \neq 0$ is an eigenvector with eigenvalue τ, or ϕ is one with eigenvalue $-\tau$. □

Theorem 7.8.6. *The Hermite functions u_α, $\alpha \in \mathbb{N}_0^n$, form an orthonormal basis of $L^2(\mathbb{R}^n)$; that is, for $u \in L^2(\mathbb{R}^n)$*

$$u = \sum_\alpha c_\alpha u_\alpha, \qquad c_\alpha = (u, u_\alpha), \tag{7.8.5}$$

with convergence in L^2.

Proof. Let W be the closed span of the u_α and $V = W^\perp$. The operator $G = H^{-1}$ of Proposition 7.8.2 satisfies $Gu_\alpha = (n + 2|\alpha|)^{-1} u_\alpha$, so $G(W) \subset W$ and, G being self-adjoint, $G(V) \subset V$. If $V \neq \{0\}$, the restriction of G to V is compact, self-adjoint and injective, so by Lemma 7.8.5 there is $0 \neq f \in V$ with $Gf = \mu f$, $\mu \neq 0$. Then $Hf = \mu^{-1} f$, so f is an eigenfunction of H, hence lies in some $E_k \subset W$ by Proposition 7.8.4, a contradiction. □

Proposition 7.8.7. *If $u \in \mathcal{S}(\mathbb{R}^n)$ then the coefficients in (7.8.5) are rapidly decreasing, $|c_\alpha| \le C_N (1 + |\alpha|)^{-N}$ for all N, and the series converges absolutely in $\mathcal{S}(\mathbb{R}^n)$. Conversely, every such series with rapidly decreasing coefficients converges in $\mathcal{S}(\mathbb{R}^n)$.*

Proof. Since $H^N u \in \mathcal{S}$, $(n + 2|\alpha|)^N |c_\alpha| = |(u, H^N u_\alpha)| = |(H^N u, u_\alpha)| \le \|H^N u\|$. By (7.8.4) and $2ix_j = C_j - A_j$, $2D_j = C_j + A_j$, each $x^\beta D^\gamma u_\alpha$ is a combination of at most $2^{|\beta|+|\gamma|}$ functions $u_{\alpha'}$ with coefficients bounded by $(2|\alpha| + 2|\beta| + 2|\gamma|)^{(|\beta|+|\gamma|)/2}$. By the Sobolev embedding $\sup |v| \le C \sum_{|\mu| \le n+1} \|D^\mu v\|_{L^2}$, the Schwartz seminorms of u_α are therefore bounded by $C_k (1 + |\alpha|)^{M_k}$, and the series converges absolutely in $\mathcal{S}$ when the coefficients are rapidly decreasing; its sum is u since it converges to u in L^2. □

For $N \in \mathbb{N}_0$ let Π_N be the orthogonal projection onto $E_0 \oplus \cdots \oplus E_N$, i.e. $K_{\Pi_N} = \sum_{|\alpha| \le N} u_\alpha(x)\overline{u_\alpha(y)}$; it is a finite rank element of $\Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$.

Corollary 7.8.8. *For every $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, $\Pi_N A \Pi_N \to A$ in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ as $N \to \infty$. In particular the finite rank elements are dense in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, and also in $\mathcal{B}(L^2)$ norm.*

Proof. The functions $u_\alpha(x)\overline{u_\beta(y)}$, $(\alpha,\beta) \in \mathbb{N}_0^{2n}$, are, up to factors ± 1 (note $\overline{u_\beta} = (-1)^{|\beta|}u_\beta$), the Hermite functions of $\mathbb{R}^{2n}$. Hence by Proposition 7.8.7 in $\mathbb{R}^{2n}$ the expansion $K_A = \sum_{\alpha,\beta} c_{\alpha\beta} u_\alpha(x)\overline{u_\beta(y)}$ converges absolutely in $\mathcal{S}(\mathbb{R}^{2n})$; its partial sum over $|\alpha|, |\beta| \le N$ is the kernel of $\Pi_N A \Pi_N$. The last statement follows from Proposition 7.4.3. □

Remark 7.8.9. The same arguments show that for $p \in \mathbb{Z}$ the eigenvalues of H^p are $(n+2k)^p$, with the same eigenspaces E_k. Thus H^p is compact if $p < 0$ and $H^p = \sum_\alpha (n+2|\alpha|)^p u_\alpha \otimes \bar{u}_\alpha$ is of trace class if $p < -n$.

7.9 Isotropic Sobolev spaces

The space $L^2(\mathbb{R}^n)$ is associated with the inclusion $\mathcal{S}(\mathbb{R}^n) \subset \mathcal{S}'(\mathbb{R}^n)$, and operators of order 0 are bounded on it. The isotropic Sobolev space of order s consists of the distributions mapped into L^2 by all isotropic operators of order s. For each $s \in \mathbb{R}$ we fix the elliptic operator $\Lambda_s = \mathrm{Op}(\langle w \rangle^s) \in \Psi^s_{\mathrm{iso,ph}}(\mathbb{R}^n)$, and let P_s be the orthogonal projection onto its (finite-dimensional) null space and $G_s \in \Psi^{-s}_{\mathrm{iso,ph}}$ its generalized inverse (Proposition 7.6.3), $G_s\Lambda_s = \mathrm{Id} - P_s$.

Definition 7.9.1. For $s \in \mathbb{R}$,

$$H^s_{\mathrm{iso}}(\mathbb{R}^n) = \{u \in \mathcal{S}'(\mathbb{R}^n);\ Au \in L^2(\mathbb{R}^n)\ \forall\, A \in \Psi^s_{\mathrm{iso}}(\mathbb{R}^n)\},$$

with the norm $\|u\|_s^2 = \|\Lambda_s u\|_{L^2}^2 + \|P_s u\|_{L^2}^2$.

Proposition 7.9.2. (i) *If $A \in \Psi^s_{\mathrm{iso}}(\mathbb{R}^n)$ is elliptic then $H^s_{\mathrm{iso}}(\mathbb{R}^n) = \{u \in \mathcal{S}';\ Au \in L^2\}$, and $\|Au\| + \|\Pi_0 u\|$, with Π_0 the projection onto $\mathrm{Nul}(A)$, is an equivalent norm.*

(ii) *$H^s_{\mathrm{iso}}(\mathbb{R}^n)$ is a Hilbert space in which $\mathcal{S}(\mathbb{R}^n)$ is dense; $H^0_{\mathrm{iso}} = L^2$.*

(iii) *For $k \in \mathbb{N}_0$, $H^k_{\mathrm{iso}}(\mathbb{R}^n) = \{u \in L^2;\ x^\alpha D^\beta u \in L^2\ \forall\, |\alpha| + |\beta| \le k\}$ with equivalent norm $\big(\sum_{|\alpha|+|\beta|\le k} \|x^\alpha D^\beta u\|^2\big)^{1/2}$.*

(iv) *The L^2 pairing extends to a non-degenerate pairing $H^s_{\mathrm{iso}} \times H^{-s}_{\mathrm{iso}} \to \mathbb{C}$ identifying H^{-s}_{iso} with the dual of H^s_{iso} (via $v \mapsto (\cdot, v)$, an antilinear isomorphism).*

(v) *Each $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ is bounded $H^s_{\mathrm{iso}} \to H^{s-m}_{\mathrm{iso}}$ for all s.*

(vi) *$\bigcap_s H^s_{\mathrm{iso}} = \mathcal{S}(\mathbb{R}^n)$, $\bigcup_s H^s_{\mathrm{iso}} = \mathcal{S}'(\mathbb{R}^n)$, and for $s > t$ the inclusion $H^s_{\mathrm{iso}} \to H^t_{\mathrm{iso}}$ is compact.*

Proof. (i) If A is elliptic of order s with parametrix B, any $A' \in \Psi^s_{\mathrm{iso}}$ can be written $A' = (A'B)A + A'(\mathrm{Id} - BA)$, with $A'B \in \Psi^0_{\mathrm{iso}}$ bounded on L^2 and $A'(\mathrm{Id} - BA)$ residual, mapping $\mathcal{S}'$ to $\mathcal{S}$. So $Au \in L^2$ implies $A'u \in L^2$. If G is the generalized inverse of A, $u = GAu + \Pi_0 u$, so $\|A'u\| \le \|A'G\|\|Au\| + \|A'\Pi_0\|\|\Pi_0 u\|$; by symmetry the norms are equivalent.

(ii) Let u_k be Cauchy. Then $\Lambda_s u_k \to f$ in L^2 and $P_s u_k \to g$. Since $\Pi_1 \Lambda_s u_k = 0$ (Π_1 the projection onto $\mathrm{Nul}\,\Lambda_s^*$), also $\Pi_1 f = 0$. As G_s is continuous on $\mathcal{S}'$, $u_k = G_s\Lambda_s u_k + P_s u_k \to u = G_s f + g$ in $\mathcal{S}'$, and $\Lambda_s u = (\mathrm{Id} - \Pi_1) f = f$, $P_s u = g$ (note $P_s G_s = 0$ by construction). Thus $u_k \to u$ in H^s_{iso}. For density, given $u \in H^s_{\mathrm{iso}}$ choose $f_k \in \mathcal{S}$ with $f_k \to \Lambda_s u$ in L^2; then $u_k = G_s f_k + P_s u \in \mathcal{S}$ and $u_k \to u$ in H^s_{iso}. That $H^0_{\mathrm{iso}} = L^2$ follows from (i) with $A = \mathrm{Id}$.

(iii) Since $x^\alpha D^\beta \in \Psi_{\mathrm{iso}}^{|\alpha|+|\beta|} \subset \Psi_{\mathrm{iso}}^k$ for $|\alpha|+|\beta| \le k$, elements of H_{iso}^k satisfy the conditions. Conversely the operator $Q_k = \sum_{|\alpha|+|\beta|\le k} (x^\alpha D^\beta)^* (x^\alpha D^\beta) \in \Psi_{\mathrm{iso,ph}}^{2k}$ has principal symbol $q_k = \sum_{|\alpha|+|\beta|=k} x^{2\alpha}\xi^{2\beta}$, which is positive on $\mathbb{S}^{2n-1}$ (if $w_i \neq 0$ the term w_i^{2k} occurs), so Q_k is elliptic. If B is a parametrix and $A \in \Psi_{\mathrm{iso}}^k$, then $Au = \sum (ABD^\beta x^\alpha)(x^\alpha D^\beta u) + A(\mathrm{Id} - BQ_k)u$ with $ABD^\beta x^\alpha \in \Psi_{\mathrm{iso}}^0$. The norm equivalence follows as in (i).

(iv) For $u \in \mathcal{S}$, $v \in H_{\mathrm{iso}}^{-s}$: $(u,v) = (G_s\Lambda_s u + P_s u, v) = (\Lambda_s u, G_s^* v) + (u, P_s v)$, and $G_s^* \in \Psi_{\mathrm{iso}}^{-s}$, so $|(u,v)| \le C\|u\|_s \|v\|_{-s}$; the pairing extends by density. Conversely a continuous linear functional ℓ on H_{iso}^s is, by the Riesz theorem, $\ell(u) = (u,w)_s$ for some $w \in H_{\mathrm{iso}}^s$, i.e. $\ell(u) = (u,v)$ with $v = \Lambda_s^*\Lambda_s w + P_s w$, and $v \in H_{\mathrm{iso}}^{-s}$ since $A\Lambda_s^* \in \Psi_{\mathrm{iso}}^0$ for $A \in \Psi_{\mathrm{iso}}^{-s}$. Non-degeneracy follows from the density of $\mathcal{S}$.

(v) If $A' \in \Psi_{\mathrm{iso}}^{s-m}$ then $A'A \in \Psi_{\mathrm{iso}}^s$, so A maps H_{iso}^s into H_{iso}^{s-m}, and boundedness follows from $\|\Lambda_{s-m}Au\| \le \|\Lambda_{s-m}AG_s\|\|\Lambda_s u\| + \|\Lambda_{s-m}AP_s\|\|P_s u\|$.

(vi) If $u \in H_{\mathrm{iso}}^k$ for all k then $x^\alpha D^\beta u \in L^2$ for all α, β, so $u \in \mathcal{S}$ by Sobolev embedding. A tempered distribution satisfies $|\langle u, \phi\rangle| \le C \sum_{|\alpha|+|\beta|\le k} \sup |x^\alpha D^\beta \phi| \le C'\|\phi\|_{k+n+1}$ (Sobolev embedding again), so by (iii), (iv) it lies in $H_{\mathrm{iso}}^{-k-n-1}$. For compactness, $\Lambda_t u = \Lambda_t G_s \Lambda_s u + \Lambda_t P_s u$ where $\Lambda_t G_s \in \Psi_{\mathrm{iso}}^{t-s}$ is compact on L^2 (Proposition 7.7.1) and $\Lambda_t P_s$ has finite rank. □

Proposition 7.9.3. *An operator $A \in \Psi_{\mathrm{iso}}^m(\mathbb{R}^n)$ is Fredholm as a map $H_{\mathrm{iso}}^s(\mathbb{R}^n) \to H_{\mathrm{iso}}^{s-m}(\mathbb{R}^n)$ if and only if it is elliptic; then its null space and index are those of Proposition 7.6.3, independent of s.*

Proof. If A is elliptic, (7.6.2) with G bounded $H_{\mathrm{iso}}^{s-m} \to H_{\mathrm{iso}}^s$ shows that A is Fredholm, with null space N_0 and range $\{f;\ \Pi_1 f = 0\}$. In particular $\Lambda_t : H_{\mathrm{iso}}^t \to L^2$ and $G_t : L^2 \to H_{\mathrm{iso}}^t$ are Fredholm. If $A : H_{\mathrm{iso}}^s \to H_{\mathrm{iso}}^{s-m}$ is Fredholm, then $T = \Lambda_{s-m} A G_s \in \Psi_{\mathrm{iso}}^0$ is Fredholm on L^2, so elliptic by Corollary 7.7.5. Since $\sigma_L(T) - \langle w\rangle^{s-m} a\, \sigma_L(G_s) \in \Gamma^{-2}$ and $\langle w\rangle^{s-m}$ and $\sigma_L(G_s)$ are elliptic of orders $s-m$ and $-s$, a is elliptic of order m. □

Remark 7.9.4. For the scattering calculus the natural spaces carry two orders, $H_{\mathrm{sc}}^{m,l}(\mathbb{R}^n) = \langle x\rangle^{-l} H^m(\mathbb{R}^n)$, and analogous results hold for $\Psi_{\mathrm{sc}}^{m,l}(\mathbb{R}^n)$ acting between them; we do not pursue this here.

7.10 The residual group

In finite dimensions the determinant is a function on the group $\mathrm{GL}(N,\mathbb{C})$. Its analogue in Chapter 8 lives on the group of invertible operators of the form $\mathrm{Id} + A$ with A residual. Here we record that this group is an open subset of a Fréchet space on which multiplication and inversion are smooth, so that differential forms such as $g^{-1}dg$ make sense on it.

Definition 7.10.1. The *residual isotropic group* is

$$G_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n) = \{\mathrm{Id} + A;\ A \in \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n),\ \mathrm{Id} + A \text{ invertible on } L^2(\mathbb{R}^n)\}.$$

By Corollary 7.4.8, $G_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ is a group, the inverse of $\mathrm{Id} + A$ being of the form $\mathrm{Id} + B$ with $B \in \Psi_{\mathrm{iso}}^{-\infty}$, and $\mathrm{Id} + A$ is in $G_{\mathrm{iso}}^{-\infty}$ as soon as it is injective (or surjective) on L^2. We identify $\mathrm{Id} + \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ with $\Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n) \cong \mathcal{S}(\mathbb{R}^{2n})$ and regard $G_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ as an open subset of this Fréchet space.

Proposition 7.10.2. *$G_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ is open in $\mathrm{Id} + \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$, and multiplication and inversion are smooth; the derivative of inversion at g is $v \mapsto -g^{-1}vg^{-1}$. The exponential map of Proposition 7.5.1 takes values in $G_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ and restricts to a homeomorphism of a neighbourhood of 0 onto a neighbourhood of Id.*

Proof. Openness follows from the openness of the invertible operators in $\mathcal{B}(L^2)$ and the continuity of $\Psi^{-\infty}_{\mathrm{iso}} \to \mathcal{B}(L^2)$ (Proposition 7.4.3). Multiplication $(\mathrm{Id}+A)(\mathrm{Id}+B) = \mathrm{Id}+A+B+AB$ is continuous and bilinear in (A,B), hence smooth. If $(\mathrm{Id}+A)^{-1} = \mathrm{Id}+B$, then as in the proof of Proposition 7.4.6, $B = -A + A^2 + ABA$. Since $A \mapsto B$ is continuous into $\mathcal{B}(L^2)$, (7.4.1) shows it is continuous into $\Psi^{-\infty}_{\mathrm{iso}}$. From $(g+tv)^{-1} - g^{-1} = -t(g+tv)^{-1}vg^{-1}$ the derivative of inversion is $-g^{-1}vg^{-1}$, continuous in (g,v), and by induction inversion is $\mathcal{C}^\infty$. The last statement is Proposition 7.5.1. □

7.10.1 Differential forms

Here a map $f : U \to F$ from an open subset U of a Fréchet space E into a Fréchet space F is $\mathcal{C}^1$ if the directional derivatives $Df(u;v) = \frac{d}{dt} f(u+tv)|_{t=0}$ exist and $(u,v) \mapsto Df(u;v)$ is continuous on $U \times E$; $\mathcal{C}^k$ and $\mathcal{C}^\infty$ are defined inductively. For $E = \mathcal{S}(\mathbb{R}^p)$, the continuous k-linear forms on E are identified, by the Schwartz kernel theorem, with $\mathcal{S}'(\mathbb{R}^{kp})$, and the alternating ones with

$$\Lambda^k\mathcal{S}(\mathbb{R}^p) = \{u \in \mathcal{S}'(\mathbb{R}^{kp});\ u(x_{e(1)}, \dots, x_{e(k)}) = \operatorname{sgn}(e)\, u(x_1, \dots, x_k)\ \forall\, e\},$$

e ranging over the permutations of $\{1, \dots, k\}$ and $x_i \in \mathbb{R}^p$. A *k-form* on an open set $U \subset \mathcal{S}(\mathbb{R}^p)$ is a map $\alpha : U \times E^k \to \mathbb{C}$, alternating and k-linear in the last k arguments (so $\alpha(u) \in \Lambda^k\mathcal{S}(\mathbb{R}^p)$ for each u), which is smooth on $U \times E^k$. The exterior derivative is defined by the usual formula for constant vector fields,

$$d\alpha(u; v_0, \dots, v_k) = \sum_{i=0}^{k} (-1)^i D_u\alpha(u; v_0, \dots, \widehat{v_i}, \dots, v_k)[v_i],$$

so that $df(u;v) = Df(u;v)$ for functions; the symmetry of second derivatives gives $d^2 = 0$, and $d(\alpha\wedge\beta) = d\alpha\wedge\beta + (-1)^{\deg\alpha}\alpha\wedge d\beta$. Applying this with $U = G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, $E = \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n) \cong \mathcal{S}(\mathbb{R}^{2n})$, we obtain the de Rham complex of $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$; for example the Maurer–Cartan form $g^{-1}dg$, $(g;v) \mapsto g^{-1}v$, is a $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$-valued 1-form. These forms are used in Chapter 8.

7.11 Symplectic invariance and the Weyl quantization

The composition law of the isotropic calculus has a more invariant formulation, which exhibits its invariance under linear symplectic maps. The left quantization Op is not suited to this: it puts every x to the left of every D, a choice that is not preserved by maps mixing x and ξ. The Weyl quantization makes the symmetric choice.

7.11.1 Symplectic vector spaces

A *symplectic vector space* is a real vector space W with an antisymmetric non-degenerate bilinear form ω:

$$\omega(w_1, w_2) + \omega(w_2, w_1) = 0, \qquad \omega(v, w) = 0\ \forall\, w \in W \implies v = 0.$$

A subspace V is *isotropic* if ω vanishes on $V \times V$, and *Lagrangian* if moreover $2\dim V = \dim W$.

Lemma 7.11.1. *Every symplectic vector space W has a Lagrangian subspace, and for every Lagrangian subspace U_1 there is a Lagrangian subspace U_2 with $W = U_1 \oplus U_2$.*

Proof. Every one-dimensional subspace is isotropic, by antisymmetry. Let V be an isotropic subspace of maximal dimension and U a complement of V. The pairing $\omega : V \times U \to \mathbb{R}$ is non-degenerate: if $u \in U$ and $\omega(v,u) = 0$ for all $v \in V$ then $V + \mathbb{R}u$ is isotropic, so $u = 0$ by maximality; if $v \in V$ and $\omega(v,u) = 0$ for all $u \in U$ then, since ω vanishes on V, $\omega(v,\cdot) = 0$ and $v = 0$. Hence $\dim U = \dim V$ and V is Lagrangian.

Now let U_1 be Lagrangian, $\dim W = 2n$, and let V be an isotropic subspace with $V \cap U_1 = \{0\}$, of maximal dimension among such subspaces. Suppose $\dim V < n$. The map $V \to U_1'$, $v \mapsto \omega(v,\cdot)|_{U_1}$, is injective, for if $\omega(v, U_1) = 0$ then $U_1 + \mathbb{R}v$ is isotropic, so $v \in U_1$ and $v = 0$. Hence $U_1 \to V'$, $u \mapsto \omega(u,\cdot)|_V$, is surjective. Choose $w \notin V \oplus U_1$, possible since $\dim V + \dim U_1 < 2n$, and $u_1 \in U_1$ with $\omega(u_1, v) = \omega(w, v)$ for all $v \in V$. Then $V \oplus \mathbb{R}(w - u_1)$ is isotropic and meets U_1 only in 0 (as $w \notin V + U_1$), contradicting maximality. Thus $\dim V = n$ and $U_2 = V$ is a Lagrangian complement. □

Given a Lagrangian decomposition $W = U_1 \oplus U_2$, the pairing $\omega : U_2 \times U_1 \to \mathbb{R}$ is non-degenerate. Choosing a basis $e_1, \dots, e_n$ of U_1 and the basis $f_1, \dots, f_n$ of U_2 with $\omega(f_i, e_j) = \delta_{ij}$, and writing $w = \sum x_i e_i + \sum \xi_i f_i$,

$$\omega(w, w') = \sum_{i=1}^{n} (\xi_i x_i' - x_i \xi_i'), \qquad \text{i.e.} \quad \omega = \sum_{i=1}^{n} d\xi_i \wedge dx_i. \tag{7.11.1}$$

These are *Darboux coordinates*. The *Liouville measure* $\nu = |\omega^n/n!|$ is, in Darboux coordinates, Lebesgue measure $dx\,d\xi$, and is invariant under the symplectic group $\mathrm{Sp}(W)$ of linear maps preserving ω.

7.11.2 Weyl quantization

For $a \in \Gamma^m(\mathbb{R}^{2n})$ the *Weyl quantization* of a is

$$\mathrm{Op^W}(a)u(x) = \iint e^{i(x-y)\cdot\xi} a\big(\tfrac{x+y}{2}, \xi\big) u(y)\, dy đ\xi,$$

i.e. $\mathrm{Op^W}(a) = \mathrm{Op}(a(\frac{x+y}{2}, \xi))$. Its kernel K and a are related by

$$K\big(s + \tfrac{t}{2}, s - \tfrac{t}{2}\big) = \int e^{it\cdot\xi} a(s,\xi) đ\xi, \qquad a(s,\xi) = \int e^{-it\cdot\xi} K\big(s + \tfrac{t}{2}, s - \tfrac{t}{2}\big)\, dt. \tag{7.11.2}$$

Discussion. In the left quantization the symbol is evaluated at the point x where the output is computed; in the Weyl quantization it is evaluated at the midpoint $\frac{x+y}{2}$ between output and input. For example $\mathrm{Op}(x\xi) = xD$, whereas $\mathrm{Op^W}(x\xi) = \frac{1}{2}(xD + Dx)$. The midpoint rule has two pleasant consequences, proved below: a real symbol gives a symmetric operator, and the composition law (7.11.3) is expressed through the symplectic form alone, so it is invariant under linear symplectic maps.

Proposition 7.11.2. *For $a \in \Gamma^m(\mathbb{R}^{2n})$, $\mathrm{Op^W}(a) \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ with*

$$\sigma_L(\mathrm{Op^W}(a)) \sim \sum_{\alpha} \frac{(i/2)^{|\alpha|}}{\alpha!} D_\xi^\alpha D_x^\alpha a,$$

and $\mathrm{Op^W}(a)^* = \mathrm{Op^W}(\bar a)$. Conversely every $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ is $\mathrm{Op^W}(a)$ for a unique $a \in \Gamma^m(\mathbb{R}^{2n})$, the Weyl symbol *of A; $A \in \Psi^m_{\mathrm{iso,ph}}$ if and only if $a \in \Gamma^m_{\mathrm{ph}}$, and $A \in \Psi^{-\infty}_{\mathrm{iso}}$ if and only if $a \in \mathcal{S}(\mathbb{R}^{2n})$.*

Proof. The first statement is Proposition 7.2.6 applied to the amplitude $a(\frac{x+y}{2},\xi)\in\mathcal{A}^m$ (Lemma 7.2.5(ii) with $\lambda=\frac12$), since $D_y^\alpha[a(\frac{x+y}{2},\xi)]|_{y=x}=2^{-|\alpha|}D_x^\alpha a(x,\xi)$. The adjoint formula is immediate from the kernel. By (7.11.2), a is determined by K, and $a\in\mathcal{S}(\mathbb{R}^{2n})$ if and only if $K\in\mathcal{S}(\mathbb{R}^{2n})$. Given A with left symbol b, choose $a'\in\Gamma^m$ with $a'\sim\sum_\alpha\frac{(-i/2)^{|\alpha|}}{\alpha!}D_\xi^\alpha D_x^\alpha b$; by the formal identity $e^{\frac{i}{2}D_x\cdot D_\xi}e^{-\frac{i}{2}D_x\cdot D_\xi}=1$, $A-\mathrm{Op^W}(a')\in\Psi^{-\infty}_{\mathrm{iso}}=\mathrm{Op^W}(\mathcal{S})$, so $A=\mathrm{Op^W}(a)$ with $a-a'\in\mathcal{S}$. The expansions preserve polyhomogeneity. □

Proposition 7.11.3. *If $a,b\in\mathcal{S}(\mathbb{R}^{2n})$ then $\mathrm{Op^W}(a)\,\mathrm{Op^W}(b)=\mathrm{Op^W}(a\#_{\mathrm W}b)$ where*

$$a\#_{\mathrm W}b(w)=\pi^{-2n}\iint_{\mathbb{R}^{2n}\times\mathbb{R}^{2n}}e^{-2i\omega(w_1,w_2)}a(w+w_1)b(w+w_2)\,dw_1\,dw_2, \tag{7.11.3}$$

ω being the standard symplectic form (7.11.1) on $\mathbb{R}^{2n}$.

Proof. Let $A=\mathrm{Op^W}(a)$, $B=\mathrm{Op^W}(b)$ with kernels K_A,K_B, and let c be the Weyl symbol of AB. By (7.11.2),

$$\begin{aligned}c(s,\zeta)&=\int e^{-it\cdot\zeta}\int K_A(s+\tfrac t2,z)K_B(z,s-\tfrac t2)\,dz\,dt\\&=(2\pi)^{-2n}\int e^{i\Phi}a(\tfrac s2+\tfrac t4+\tfrac z2,\xi)b(\tfrac z2+\tfrac s2-\tfrac t4,\eta)\,dt\,dz\,d\xi\,d\eta,\\\Phi&=-t\cdot\zeta+(s+\tfrac t2-z)\cdot\xi+(z-s+\tfrac t2)\cdot\eta,\end{aligned}$$

all integrals converging absolutely. Change variables to $X=\frac z2+\frac t4-\frac s2$, $Y=\frac z2-\frac t4-\frac s2$, $\Xi=\xi-\zeta$, $\mathrm H=\eta-\zeta$. Then $z-s=X+Y$, $t=2(X-Y)$, $dt\,dz=4^n\,dX\,dY$ and $\Phi=2(X\cdot\mathrm H-Y\cdot\Xi)$, so

$$c(s,\zeta)=\pi^{-2n}\int e^{2i(X\cdot\mathrm H-Y\cdot\Xi)}a(s+X,\zeta+\Xi)b(s+Y,\zeta+\mathrm H)\,dX\,d\Xi\,dY\,d\mathrm H.$$

With $w=(s,\zeta)$, $w_1=(X,\Xi)$, $w_2=(Y,\mathrm H)$ we have $\omega(w_1,w_2)=\Xi\cdot Y-X\cdot\mathrm H$, which gives (7.11.3). □

Theorem 7.11.4. *Let (W,ω) be a symplectic vector space of dimension $2n$ with Liouville measure ν. Then*

$$a\#b(w)=\pi^{-2n}\int_{W\times W}e^{-2i\omega(w_1,w_2)}a(w+w_1)b(w+w_2)\,d\nu(w_1)\,d\nu(w_2) \tag{7.11.4}$$

defines an associative product on $\mathcal{S}(W)$. For any Lagrangian decomposition $W=U_1\oplus U_2$ and Darboux coordinates as above, $(\mathcal{S}(W),\#)$ is isomorphic, via the Weyl quantization, to $(\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n),\circ)$. The product extends uniquely to continuous bilinear maps $\Gamma^m(W)\times\Gamma^{m'}(W)\to\Gamma^{m+m'}(W)$, which in any Darboux coordinates correspond to composition in $\Psi^\infty_{\mathrm{iso}}(\mathbb{R}^n)$, and for every $F\in\mathrm{Sp}(W)$

$$F^*(a\#b)=(F^*a)\#(F^*b). \tag{7.11.5}$$

Proof. In Darboux coordinates (7.11.4) becomes (7.11.3), so the isomorphism (and hence associativity) follows from Propositions 7.11.2 and 7.11.3. For $F\in\mathrm{Sp}(W)$, substituting $v_i=Fw_i$ in (7.11.4), which preserves ω and ν, gives (7.11.5) for $a,b\in\mathcal{S}(W)$. In Darboux coordinates the Weyl product of symbols in Γ^m, $\Gamma^{m'}$ is continuous into $\Gamma^{m+m'}$: it is the composite of the continuous maps $a\mapsto\sigma_L(\mathrm{Op^W}(a))$, its inverse (both continuous by the closed graph theorem, as in Theorem 7.2.7(iv)) and the product of Theorem 7.2.7(iv). Since $\mathcal{S}$ is dense in Γ^m for the topology of $\Gamma^{m+\epsilon}$ (Lemma 7.1.2(iv)) and $\epsilon>0$ is arbitrary, the extension is unique, and (7.11.5) extends by continuity because F^* is continuous on each $\Gamma^m(W)$. In particular, since two Darboux coordinate systems differ by an element of $\mathrm{Sp}(W)$, the extended product does not depend on the choice. □

Remark 7.11.5. The invariance (7.11.5) is *implemented* by unitary operators on $L^2(\mathbb{R}^n)$: for each $F \in \mathrm{Sp}(\mathbb{R}^{2n})$ there is a unitary operator U, unique up to a factor of modulus one, with $U^{-1}\operatorname{Op^W}(a)U = \operatorname{Op^W}(F^*a)$ for all $a \in \Gamma^m(\mathbb{R}^{2n})$. For example $\mathcal{F}^{-1}\operatorname{Op^W}(a)\mathcal{F} = \operatorname{Op^W}(a\circ J)$ with $J(x,\xi) = (\xi,-x)$, which is the Weyl form of Proposition 7.2.10. These operators, suitably normalized, form the metaplectic group, a double cover of the symplectic group; this is the subject of Section 7.12, where (7.11.5) is reproved in this way (Corollary 7.12.19).

7.12 The metaplectic group

By Theorem 7.11.4, every linear symplectic map S of $\mathbb{R}^{2n}$ induces an automorphism $a \mapsto a\circ S$ of the isotropic symbol algebra with the Weyl product. In this section we show that this automorphism is implemented by conjugation with a unitary operator U on $L^2(\mathbb{R}^n)$,

$$U^{-1}\operatorname{Op^W}(a)U = \operatorname{Op^W}(a\circ S),$$

and that U is unique up to a factor of modulus one. The operators so obtained form a central extension of the symplectic group by the circle. Fixing the factor by means of the action on the Gaussian u_0 singles out a subgroup, the *metaplectic group* $\mathrm{Mp}(2n,\mathbb{R})$, which is a double cover of $\mathrm{Sp}(2n,\mathbb{R})$ and is generated by the exponentials $\exp(-i\operatorname{Op^W}(q))$ of Weyl quantizations of real quadratic forms. Its elements are 'isotropic Fourier integral operators' with quadratic phase functions (Proposition 7.12.14); they are associated with linear symplectic maps only, and not with general (homogeneous) canonical transformations.

The construction has four steps, summarized in Figure 7.3. First we describe $\mathrm{Sp}(2n,\mathbb{R})$ and its generators (shears). Second, we show that a unitary operator which transforms the linear operators $\ell(x,D)$ correctly transforms every isotropic operator correctly (Theorem 7.12.7). Third, explicit operators realize each shear, and an operator commuting with all x_j and D_j is a constant, so the implementing operators form an extension of $\mathrm{Sp}(2n,\mathbb{R})$ by $\mathrm{U}(1)$. Fourth, the action on the Gaussian u_0 defines a character φ of this extension, and its kernel is the double cover $\mathrm{Mp}(2n,\mathbb{R})$.

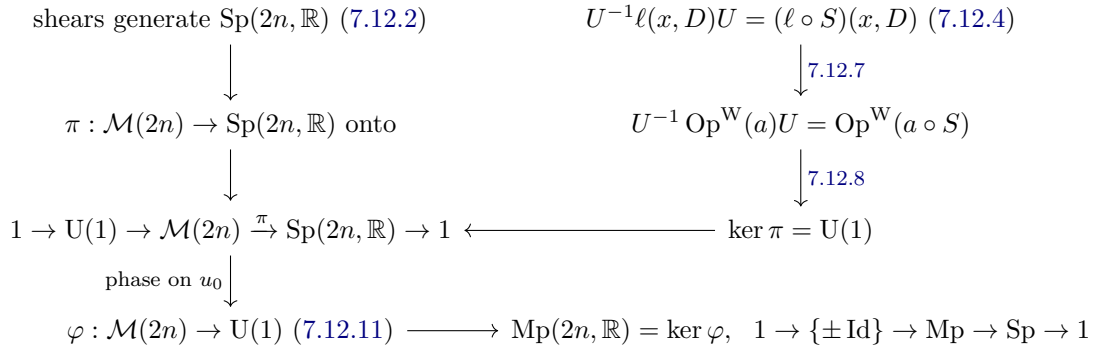


Figure 7.3: Logic of the construction of the metaplectic group. Numbers refer to the results used at each step. The second column is the analytic input; the first column is group theory.

7.12.1 The symplectic group

We use the standard symplectic form (7.11.1) on $\mathbb{R}^{2n} = \mathbb{R}^n_x \times \mathbb{R}^n_\xi$, written with the matrix

$$\omega(w,w') = \langle Jw, w'\rangle, \qquad J = \begin{pmatrix} 0 & \mathrm{Id} \\ -\mathrm{Id} & 0 \end{pmatrix}, \quad J(x,\xi) = (\xi,-x),$$

where $\langle\cdot,\cdot\rangle$ is the Euclidean inner product. Note that $J^T = -J = J^{-1}$. The *symplectic group* $\mathrm{Sp}(2n,\mathbb{R})$ consists of the real $2n\times 2n$ matrices S with $\omega(Sw,Sw') = \omega(w,w')$, i.e. $S^TJS = J$. We write symplectic matrices in block form, acting on (x,ξ),

$$S = \begin{pmatrix} A & B \\ C & E \end{pmatrix}, \qquad S(x,\xi) = (Ax + B\xi,\, Cx + E\xi). \tag{7.12.1}$$

For real quadratic forms on $\mathbb{R}^{2n}$ we use the notation

$$q(x,\xi) = \tfrac{1}{2}\langle \alpha x, x\rangle + \langle \beta x,\xi\rangle + \tfrac{1}{2}\langle \gamma\xi,\xi\rangle = \tfrac{1}{2}\langle Qw, w\rangle, \qquad Q = \begin{pmatrix} \alpha & \beta^T \\ \beta & \gamma \end{pmatrix}, \tag{7.12.2}$$

with α,γ symmetric and β arbitrary real $n\times n$ matrices. The Poisson bracket of Remark 7.2.8 is $\{a,b\} = \langle J\nabla a, \nabla b\rangle$, so for a quadratic form q

$$\{q,a\}(w) = \langle X_q w, \nabla a(w)\rangle = \frac{d}{dt} a(e^{tX_q}w)\Big|_{t=0}, \qquad X_q = JQ = \begin{pmatrix} \beta & \gamma \\ -\alpha & -\beta^T \end{pmatrix}; \tag{7.12.3}$$

that is, the Hamilton vector field of q is the linear vector field $X_q w$.

Lemma 7.12.1. (i) $\mathrm{Sp}(2n,\mathbb{R})$ *is a group containing* J*, closed under transposition, and* $S^{-1} = J^{-1}S^TJ$*. In terms of the blocks* (7.12.1)*,* $S\in\mathrm{Sp}(2n,\mathbb{R})$ *if and only if*

$$A^TC,\ B^TE \text{ are symmetric and } A^TE - C^TB = \mathrm{Id}, \tag{7.12.4}$$

and then also AB^T*,* CE^T *are symmetric,* $AE^T - BC^T = \mathrm{Id}$*, and* $S^{-1} = \left(\begin{smallmatrix} E^T & -B^T \\ -C^T & A^T \end{smallmatrix}\right)$.

(ii) *For every real quadratic form* q *and* $t\in\mathbb{R}$*,* $\Phi_t = e^{tX_q}$ *lies in* $\mathrm{Sp}(2n,\mathbb{R})$*, and* $q\circ\Phi_t = q$.

(iii) *If* $S\in\mathrm{Sp}(2n,\mathbb{R})$ *and* q *is a real quadratic form then* $SX_qS^{-1} = X_{q\circ S^{-1}}$.

Proof. (i) If $S^TJS = J$ then S is invertible and $S^{-1} = J^{-1}S^TJ$, so $SJ^{-1}S^TJ = \mathrm{Id}$, i.e. $SJ^{-1}S^T = J^{-1}$. Inverting gives $S^{-T}JS^{-1} = J$, so S^{-1} is symplectic, and multiplying by -1 (recall $J^{-1} = -J$) gives $SJS^T = J$, so S^T is symplectic. Products of symplectic matrices are symplectic, and $J^TJJ = J$. Writing out $S^TJS = J$ in blocks,

$$S^TJS = \begin{pmatrix} A^TC - C^TA & A^TE - C^TB \\ B^TC - E^TA & B^TE - E^TB \end{pmatrix} = \begin{pmatrix} 0 & \mathrm{Id} \\ -\mathrm{Id} & 0 \end{pmatrix},$$

which is (7.12.4) (the lower left entry is minus the transpose of the upper right one). Applying this to S^T gives the second set of conditions, and the formula for $S^{-1} = J^{-1}S^TJ$ follows by multiplying out.

(ii) $\frac{d}{dt}(\Phi_t^TJ\Phi_t) = \Phi_t^T(X_q^TJ + JX_q)\Phi_t = \Phi_t^T(QJ^TJ + J^2Q)\Phi_t = 0$, since $J^TJ = \mathrm{Id}$ and $J^2 = -\mathrm{Id}$. Moreover $\frac{d}{dt}q(\Phi_t w) = \{q,q\}(\Phi_t w) = 0$ by (7.12.3).

(iii) The form $q\circ S^{-1}$ has matrix $S^{-T}QS^{-1}$, and $JS^{-T} = J(JSJ^{-1}) = -SJ^{-1} = SJ$ by (i), so $X_{q\circ S^{-1}} = JS^{-T}QS^{-1} = SJQS^{-1}$. □

For symmetric α,γ we use the *shears*

$$L_\alpha = \begin{pmatrix} \mathrm{Id} & 0 \\ \alpha & \mathrm{Id} \end{pmatrix}, \qquad R_\gamma = \begin{pmatrix} \mathrm{Id} & \gamma \\ 0 & \mathrm{Id} \end{pmatrix},$$

which are symplectic by (7.12.4); note $L_\alpha L_{\alpha'} = L_{\alpha+\alpha'}$ and $R_\gamma R_{\gamma'} = R_{\gamma+\gamma'}$.

Lemma 7.12.2. (i) *If $S \in \mathrm{Sp}(2n,\mathbb{R})$ has invertible block A then CA^{-1} and $A^{-1}B$ are symmetric and*

$$S = L_{CA^{-1}} \begin{pmatrix} A & 0 \\ 0 & A^{-T} \end{pmatrix} R_{A^{-1}B}. \tag{7.12.5}$$

(ii) *For every $S \in \mathrm{Sp}(2n,\mathbb{R})$ the block $A+tB$ is invertible for all but at most n values of $t \in \mathbb{R}$; thus $SL_{t\,\mathrm{Id}}$ has invertible upper left block for such t.*

(iii) $\mathrm{Sp}(2n,\mathbb{R})$ *is generated by the shears L_α, R_γ, α,γ real symmetric.*

Proof. (i) By Lemma 7.12.1(i), $CA^{-1} = A^{-T}(A^TC)A^{-1}$ and $A^{-1}B = A^{-1}(BA^T)A^{-T}$ are symmetric. Multiplying out, the right side of (7.12.5) is $\left(\begin{smallmatrix} A & B \\ CA^{-1}A & CA^{-1}B+A^{-T} \end{smallmatrix}\right)$, and $E = A^{-T}(\mathrm{Id}+C^TB) = A^{-T} + (CA^{-1})^TB = A^{-T} + CA^{-1}B$ by (7.12.4).

(ii) The space $K = \{w;\ Sw \in \{0\}\times\mathbb{R}^n\} = S^{-1}(\{0\}\times\mathbb{R}^n)$ is Lagrangian, as $\{0\}\times\mathbb{R}^n$ is and S^{-1} is symplectic, and $(A+tB)y = 0$ if and only if $(y,ty) \in K$. If (y_1,t_1y_1), $(y_2,t_2y_2) \in K$ then $0 = \omega((y_1,t_1y_1),(y_2,t_2y_2)) = (t_1-t_2)\langle y_1,y_2\rangle$. Thus non-zero null vectors of $A+tB$ for distinct values of t are mutually orthogonal, so there are at most n such values. The upper left block of $SL_{t\,\mathrm{Id}}$ is $A+tB$.

(iii) Let $\mathcal{G}$ be the group generated by the shears. For E symmetric and invertible a direct computation gives

$$R_EL_{-E^{-1}}R_E = \begin{pmatrix} 0 & E \\ -E^{-1} & 0 \end{pmatrix}, \qquad (R_EL_{-E^{-1}}R_E)(R_{\mathrm{Id}}L_{-\,\mathrm{Id}}R_{\mathrm{Id}})^{-1} = \begin{pmatrix} E & 0 \\ 0 & E^{-1} \end{pmatrix},$$

so $\mathrm{diag}(E,E^{-1}) \in \mathcal{G}$. Every $A \in \mathrm{GL}(n,\mathbb{R})$ is a product $A = E_1\cdots E_k$ of symmetric invertible matrices: write $A = OP$ with $P = (A^TA)^{1/2}$ and $O = AP^{-1}$ orthogonal, and write O as a product of reflections $\mathrm{Id}-2vv^T$, $|v|=1$ (if $Oe_1 \neq e_1$, the reflection in $v = (Oe_1-e_1)/|Oe_1-e_1|$ maps Oe_1 to e_1, and one proceeds by induction on $e_1^\perp$). Since $(E_1\cdots E_k)^{-T} = E_1^{-1}\cdots E_k^{-1}$, we get $\mathrm{diag}(A,A^{-T}) = \prod_i \mathrm{diag}(E_i,E_i^{-1}) \in \mathcal{G}$. By (i), every S with invertible block A lies in $\mathcal{G}$, and by (ii) every S is of the form $S'L_{-t\,\mathrm{Id}}$ with such an S'. □

7.12.2 Quadratic forms and exact commutators

For a linear function $\ell(x,\xi) = p\cdot x + r\cdot\xi$, $p,r \in \mathbb{C}^n$, we write $\ell(x,D) = p\cdot x + r\cdot D$. This is $\mathrm{Op^W}(\ell)$ (and also $\mathrm{Op}(\ell)$), since the amplitude $\ell(\frac{x+y}{2},\xi)$ differs from $\ell(x,\xi)$ by $\frac12 p\cdot(y-x)$, which contributes $\frac12\,\mathrm{Op}(p\cdot D_\xi 1) = 0$ after integration by parts.

Lemma 7.12.3. *Let $a \in \Gamma^m(\mathbb{R}^{2n})$ and let ℓ,ℓ' be linear functions on $\mathbb{R}^{2n}$. Then*

$$\ell(x,D)\,\mathrm{Op^W}(a) = \mathrm{Op^W}(\ell a - \tfrac{i}{2}\{\ell,a\}), \qquad \mathrm{Op^W}(a)\ell(x,D) = \mathrm{Op^W}(\ell a + \tfrac{i}{2}\{\ell,a\}). \tag{7.12.6}$$

Consequently $\mathrm{Op^W}(\ell\ell') = \frac12(\ell(x,D)\ell'(x,D) + \ell'(x,D)\ell(x,D))$, and for every polynomial q of degree at most two on $\mathbb{R}^{2n}$

$$[\mathrm{Op^W}(q),\mathrm{Op^W}(a)] = -i\,\mathrm{Op^W}(\{q,a\}). \tag{7.12.7}$$

If q is real then $\mathrm{Op^W}(q)$ is a differential operator with polynomial coefficients which is symmetric on $\mathcal{S}(\mathbb{R}^n)$; for q as in (7.12.2),

$$\mathrm{Op^W}(q) = \tfrac12\langle\alpha x,x\rangle + \sum_{j,k}\beta_{jk}x_kD_j - \tfrac{i}{2}\,\mathrm{tr}\,\beta + \tfrac12\sum_{j,k}\gamma_{jk}D_jD_k. \tag{7.12.8}$$

Proof. Both sides of (7.12.6) are linear in ℓ, so it suffices to take $\ell = x_j$ and $\ell = \xi_j$. Let $\tilde a(x,y,\xi) = a(\frac{x+y}{2},\xi)$, so $\operatorname{Op}^{\mathrm W}(a) = \operatorname{Op}(\tilde a)$. Integrating by parts in the oscillatory integrals, as in Chapter 2,

$$x_j \operatorname{Op}(\tilde a) = \operatorname{Op}(x_j\tilde a), \quad \operatorname{Op}(\tilde a)x_j = \operatorname{Op}(y_j\tilde a), \quad \operatorname{Op}((x_j-y_j)\tilde a) = \operatorname{Op}(-D_{\xi_j}\tilde a),$$
$$D_j \operatorname{Op}(\tilde a) = \operatorname{Op}((\xi_j + D_{x_j})\tilde a), \qquad \operatorname{Op}(\tilde a)D_j = \operatorname{Op}((\xi_j - D_{y_j})\tilde a).$$

Writing $x_j = \frac12(x_j+y_j) + \frac12(x_j-y_j)$ and $y_j = \frac12(x_j+y_j) - \frac12(x_j-y_j)$, and using $D_{x_j}\tilde a = D_{y_j}\tilde a = \frac12(D_{x_j}a)(\frac{x+y}{2},\xi)$, we find

$$x_j \operatorname{Op}^{\mathrm W}(a) = \operatorname{Op}^{\mathrm W}(x_j a + \tfrac{i}{2}\partial_{\xi_j}a), \qquad \operatorname{Op}^{\mathrm W}(a)x_j = \operatorname{Op}^{\mathrm W}(x_j a - \tfrac{i}{2}\partial_{\xi_j}a),$$
$$D_j \operatorname{Op}^{\mathrm W}(a) = \operatorname{Op}^{\mathrm W}(\xi_j a - \tfrac{i}{2}\partial_{x_j}a), \qquad \operatorname{Op}^{\mathrm W}(a)D_j = \operatorname{Op}^{\mathrm W}(\xi_j a + \tfrac{i}{2}\partial_{x_j}a).$$

Since $\{x_j,a\} = -\partial_{\xi_j}a$ and $\{\xi_j,a\} = \partial_{x_j}a$, this is (7.12.6). Taking $a = \ell'$ and adding the two formulas (7.12.6) with ℓ and ℓ' interchanged gives the formula for $\operatorname{Op}^{\mathrm W}(\ell\ell')$, because $\{\ell,\ell'\} = -\{\ell',\ell\}$ is constant. Subtracting the two formulas (7.12.6) gives (7.12.7) for $q = \ell$, and it is trivial for constant q. Every quadratic form is a sum of products $\ell\ell'$; writing $L = \ell(x,D)$, $L' = \ell'(x,D)$ and $A = \operatorname{Op}^{\mathrm W}(a)$, and applying the linear case three times,

$$\begin{aligned}[\operatorname{Op}^{\mathrm W}(\ell\ell'), A] &= \tfrac12(L[L',A] + [L,A]L' + L'[L,A] + [L',A]L)\\ &= -\tfrac{i}{2}(L\operatorname{Op}^{\mathrm W}(\{\ell',a\}) + \operatorname{Op}^{\mathrm W}(\{\ell',a\})L + L'\operatorname{Op}^{\mathrm W}(\{\ell,a\}) + \operatorname{Op}^{\mathrm W}(\{\ell,a\})L')\\ &= -i\operatorname{Op}^{\mathrm W}(\ell\{\ell',a\} + \ell'\{\ell,a\}) = -i\operatorname{Op}^{\mathrm W}(\{\ell\ell',a\}),\end{aligned}$$

where in the third step we used the first part of the lemma with a replaced by $\{\ell',a\}$ and $\{\ell,a\}$. Finally $\operatorname{Op}^{\mathrm W}(x_jx_k) = x_jx_k$, $\operatorname{Op}^{\mathrm W}(\xi_j\xi_k) = D_jD_k$ and $\operatorname{Op}^{\mathrm W}(x_k\xi_j) = \frac12(x_kD_j + D_jx_k) = x_kD_j - \frac{i}{2}\delta_{jk}$, which gives (7.12.8); for real q, $\operatorname{Op}^{\mathrm W}(q)^* = \operatorname{Op}^{\mathrm W}(q)$ by Proposition 7.11.2. □

7.12.3 Unitary implementations of symplectic maps

Definition 7.12.4. Let $\mathcal M(2n)$ be the set of unitary operators U on $L^2(\mathbb R^n)$ such that $U(\mathcal S(\mathbb R^n)) = \mathcal S(\mathbb R^n)$, $U:\mathcal S(\mathbb R^n)\to\mathcal S(\mathbb R^n)$ is continuous, and there is a real linear map S of $\mathbb R^{2n}$ with

$$U^{-1}\ell(x,D)U = (\ell\circ S)(x,D) \quad \text{on } \mathcal S(\mathbb R^n) \tag{7.12.9}$$

for every linear function ℓ on $\mathbb R^{2n}$.

By the open mapping theorem U^{-1} is also continuous on $\mathcal S(\mathbb R^n)$. Each $U\in\mathcal M(2n)$ extends to a continuous linear isomorphism of $\mathcal S'(\mathbb R^n)$ by $\langle Uu,\phi\rangle = \langle u,\overline{U^{-1}\bar\phi}\rangle$, which is consistent with the action on L^2 since $U^* = U^{-1}$; we use this extension without comment.

Lemma 7.12.5. *$\mathcal M(2n)$ is a group. For $U\in\mathcal M(2n)$ the map S in (7.12.9) is unique and symplectic, and $\pi:\mathcal M(2n)\ni U\mapsto S\in\operatorname{Sp}(2n,\mathbb R)$ is a group homomorphism. Moreover $U\ell(x,D)U^{-1} = (\ell\circ S^{-1})(x,D)$.*

Proof. For real ℓ, the operator $\ell(x,D)$ determines ℓ, as $\ell(x,D)e^{-|x|^2/2} = (p+ir)\cdot x\,e^{-|x|^2/2}$; hence S is unique. If $U_1,U_2\in\mathcal M(2n)$ then $(U_1U_2)^{-1}\ell(x,D)U_1U_2 = U_2^{-1}(\ell\circ S_1)(x,D)U_2 = (\ell\circ S_1\circ S_2)(x,D)$, so $U_1U_2\in\mathcal M(2n)$ and $\pi(U_1U_2) = \pi(U_1)\pi(U_2)$. By Lemma 7.12.3, $[\ell(x,D),\ell'(x,D)] = -i\{\ell,\ell'\}$ for linear ℓ,ℓ'. For $\ell_\zeta(w) = \langle\zeta,w\rangle$ we have $\{\ell_\zeta,\ell_\eta\} = \langle J\zeta,\eta\rangle = \omega(\zeta,\eta)$ and $\ell_\zeta\circ S = \ell_{S^T\zeta}$. Conjugating the commutator relation by U gives $\omega(S^T\zeta,S^T\eta) = \omega(\zeta,\eta)$, so S^T, and hence S, is symplectic. In particular S is invertible, and applying (7.12.9) to $\ell\circ S^{-1}$ gives the last formula; it shows that $U^{-1}\in\mathcal M(2n)$ with $\pi(U^{-1}) = S^{-1}$. □

Lemma 7.12.6. *The following operators lie in $\mathcal{M}(2n)$:*

(i) *multiplication by $e^{-iq(x)}$, $q(x) = \frac{1}{2}\langle \alpha x, x\rangle$, α real symmetric, with $\pi(e^{-iq(x)}) = L_{-\alpha}$;*

(ii) *the normalized Fourier transform $\mathcal{F}_0 = (2\pi)^{-n/2}\mathcal{F}$, with $\pi(\mathcal{F}_0) = J$; more generally the normalized Fourier transform $\mathcal{F}_1$ in the variable x_1 alone, with $\pi(\mathcal{F}_1) = J_1$, $J_1(x,\xi) = (\xi_1, x_2, \dots, x_n, -x_1, \xi_2, \dots, \xi_n)$;*

(iii) *the Fourier multiplier $e^{-iq(D)} = \mathcal{F}^{-1}e^{-iq(\xi)}\mathcal{F}$, $q(\xi) = \frac{1}{2}\langle \gamma\xi, \xi\rangle$, γ real symmetric, with $\pi(e^{-iq(D)}) = R_\gamma$;*

(iv) *$T_L u(x) = |\det L|^{1/2} u(Lx)$, $L \in \mathrm{GL}(n,\mathbb{R})$, with $\pi(T_L) = \mathrm{diag}(L^{-1}, L^T)$;*

(v) *the constants $\lambda \,\mathrm{Id}$, $|\lambda| = 1$, with $\pi(\lambda\,\mathrm{Id}) = \mathrm{Id}$.*

Moreover, for $u \in \mathcal{S}(\mathbb{R}^n)$, the maps $t \mapsto e^{-itq(x)}u$ and $t \mapsto e^{-itq(D)}u$ are smooth from $\mathbb{R}$ to $\mathcal{S}(\mathbb{R}^n)$, with derivatives $-iq(x)e^{-itq(x)}u$ and $-iq(D)e^{-itq(D)}u$.

Proof. All these operators are unitary and are isomorphisms of $\mathcal{S}(\mathbb{R}^n)$. (i) Multiplication by $e^{-iq(x)}$ commutes with x_j, and $e^{iq(x)}D_j(e^{-iq(x)}u) = D_j u - (\alpha x)_j u$; thus $\ell \circ S = \ell$ for $\ell = x_j$ and $\xi_j \circ S = \xi_j - (\alpha x)_j$, i.e. $S(x,\xi) = (x, \xi - \alpha x)$. (ii) From $\mathcal{F}(D_j u) = \xi_j \hat{u}$ and $\mathcal{F}(x_j u) = -D_{\xi_j}\hat{u}$ we get $\mathcal{F}^{-1}x_j\mathcal{F} = D_j$ and $\mathcal{F}^{-1}D_j\mathcal{F} = -x_j$, i.e. $x_j \circ S = \xi_j$ and $\xi_j \circ S = -x_j$, which is $S = J$; the partial Fourier transform is treated in the same way. (iii) $e^{-iq(D)} = \mathcal{F}_0^{-1}e^{-iq(x)}\mathcal{F}_0$ with q regarded as a function of x, so $\pi(e^{-iq(D)}) = J^{-1}L_{-\gamma}J = R_\gamma$. (iv) $T_L^{-1}x_j T_L = (L^{-1}x)_j$ and $T_L^{-1}D_j T_L = (L^T D)_j$. (v) is clear. The last statement holds because $q(x)^k e^{-itq(x)}u$ is bounded in $\mathcal{S}$ uniformly for t in compact sets, for each k, and similarly on the Fourier transform side. □

The commutation relations (7.12.9) for linear symbols imply the same relation for all isotropic operators.

Theorem 7.12.7. *If $U \in \mathcal{M}(2n)$ and $S = \pi(U)$ then for every $m \in \mathbb{R}$ and $a \in \Gamma^m(\mathbb{R}^{2n})$*

$$U^{-1}\,\mathrm{Op^W}(a)U = \mathrm{Op^W}(a \circ S) \tag{7.12.10}$$

as operators on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$.

Proof. For $\zeta = (p,r) \in \mathbb{R}^{2n}$ let $\ell_\zeta(x,\xi) = p\cdot x + r\cdot\xi$ and define the unitary operator

$$\tau_\zeta u(x) = e^{ip\cdot x + \frac{i}{2}p\cdot r}u(x+r),$$

which preserves $\mathcal{S}(\mathbb{R}^n)$. For $u \in \mathcal{S}$, $s \mapsto \tau_{s\zeta}u$ is a smooth curve in $\mathcal{S}$, and differentiating $e^{isp\cdot x + \frac{i}{2}s^2 p\cdot r}u(x+sr)$ in s gives

$$\frac{d}{ds}\tau_{s\zeta}u = i\ell_\zeta(x,D)\tau_{s\zeta}u, \qquad \tau_0 u = u. \tag{7.12.11}$$

Step 1: $U^{-1}\tau_\zeta U = \tau_{S^T\zeta}$. Let $u \in \mathcal{S}$, $v(s) = U^{-1}\tau_{s\zeta}Uu$ and $\tilde{v}(s) = \tau_{sS^T\zeta}u$. Both are $\mathcal{C}^1$ curves in L^2 with values in $\mathcal{S}$, and by (7.12.11), (7.12.9) and $\ell_\zeta \circ S = \ell_{S^T\zeta}$ both satisfy $\frac{d}{ds}f = iLf$, $f(0) = u$, with $L = \ell_{S^T\zeta}(x,D)$. Since L is symmetric on $\mathcal{S}$, the difference $d = v - \tilde{v}$ satisfies $\frac{d}{ds}\|d\|^2 = 2\,\mathrm{Re}(iLd, d) = 0$, so $d \equiv 0$; take $s = 1$.

Step 2: $a \in \mathcal{S}(\mathbb{R}^{2n})$. Let $\hat{a}(\zeta) = \int e^{-i\langle\zeta,w\rangle}a(w)\,dw$, so that $a(w) = \int \hat{a}(\zeta)e^{i\ell_\zeta(w)}đ\zeta$ with $đ\zeta = (2\pi)^{-2n}d\zeta$. We claim that for $u, v \in \mathcal{S}(\mathbb{R}^n)$

$$(\mathrm{Op^W}(a)u, v) = \int \hat{a}(\zeta)(\tau_\zeta u, v)đ\zeta. \tag{7.12.12}$$

Indeed the right side is the absolutely convergent integral

$$\int \hat{a}(p,r)e^{ip\cdot(x+r/2)}u(x+r)\overline{v(x)}\,dx đp đr$$

(with $đp = (2\pi)^{-n}dp$). Carrying out the p-integration first gives $\check{a}(x+\frac{r}{2},r)$, where $\check{a}(s,r) = \int e^{-ir\cdot\xi}a(s,\xi)\,d\xi$, and substituting $y = x+r$ we obtain $\iint K(x,y)u(y)\overline{v(x)}\,dx\,dy$ with $K(x,y) = (2\pi)^{-n}\check{a}(\frac{x+y}{2},y-x) = \int e^{i(x-y)\cdot\xi}a(\frac{x+y}{2},\xi)đ\xi$, the kernel of $\mathrm{Op}^{\mathrm{W}}(a)$. By Step 1 and (7.12.12),

$$(U^{-1}\,\mathrm{Op}^{\mathrm{W}}(a)Uu,v) = (\mathrm{Op}^{\mathrm{W}}(a)Uu,Uv) = \int \hat{a}(\zeta)(\tau_{S^T\zeta}u,v)đ\zeta = \int \hat{a}(S^{-T}\eta)(\tau_\eta u,v)đ\eta,$$

using $|\det S| = 1$. Since $\widehat{a\circ S}(\eta) = \hat{a}(S^{-T}\eta)$, this is $(\mathrm{Op}^{\mathrm{W}}(a\circ S)u,v)$ by (7.12.12).

Step 3: general a. Let $a\in\Gamma^m$, $m'>m$, and choose $a_k\in\mathcal{S}$ with $a_k\to a$ in $\Gamma^{m'}$ (Lemma 7.1.2(iv)). For fixed $u,v\in\mathcal{S}$, the map $b\mapsto(\mathrm{Op}^{\mathrm{W}}(b)u,v)$ is continuous on $\Gamma^{m'}$, since $b\mapsto\sigma_L(\mathrm{Op}^{\mathrm{W}}(b))$ is continuous (proof of Theorem 7.11.4) and $c\mapsto(\mathrm{Op}(c)u,v)$ is continuous (proof of Theorem 7.2.7(iv)); and $b\mapsto b\circ S$ is continuous on $\Gamma^{m'}$. Applying this to $(U^{-1}\,\mathrm{Op}^{\mathrm{W}}(a_k)Uu,v) = (\mathrm{Op}^{\mathrm{W}}(a_k)Uu,Uv)$ and to $(\mathrm{Op}^{\mathrm{W}}(a_k\circ S)u,v)$ gives (7.12.10) on $\mathcal{S}$. Both sides are continuous on $\mathcal{S}'$ and $\mathcal{S}$ is dense in $\mathcal{S}'$, so the identity extends. □

Proposition 7.12.8. *If $U\in\mathcal{M}(2n)$ and $\pi(U)=\mathrm{Id}$ then $U=\lambda\,\mathrm{Id}$ with $|\lambda|=1$.*

Proof. By assumption U commutes on $\mathcal{S}$ with all x_j and D_j, hence with the annihilation and creation operators A_j, C_j of Section 7.8. Since $A_j(Uu_0) = UA_ju_0 = 0$ for all j and $Uu_0\in\mathcal{S}$, Proposition 7.8.3 (and its proof) gives $Uu_0=\lambda u_0$. Then $UC^\mu u_0 = C^\mu Uu_0 = \lambda C^\mu u_0$ for every multi-index μ, so $U=\lambda$ on all Hermite functions, which span a dense subspace of L^2 by Theorem 7.8.6. As U is bounded, $U=\lambda\,\mathrm{Id}$, and $|\lambda|=1$ since U is unitary. □

Corollary 7.12.9. *The homomorphism π is surjective, so there is an exact sequence of groups*

$$1\longrightarrow \mathrm{U}(1)\longrightarrow\mathcal{M}(2n)\xrightarrow{\pi}\mathrm{Sp}(2n,\mathbb{R})\longrightarrow 1.$$

Every linear symplectic map S is implemented, in the sense of (7.12.10), by a unitary operator, unique up to a factor of modulus one.

Proof. By Lemma 7.12.6 the image of π contains all shears, which generate $\mathrm{Sp}(2n,\mathbb{R})$ by Lemma 7.12.2. The kernel is described in Proposition 7.12.8. □

7.12.4 Gaussians and the double cover

To fix the factor of modulus one we look at the action on the ground state $u_0 = \pi^{-n/4}e^{-|x|^2/2}$ of the harmonic oscillator.

Discussion. Each S is implemented by a whole circle $\{\lambda U;\ |\lambda|=1\}$ of unitary operators, and there is no way to pick one point on each circle so that products are respected. The next best thing is to cut each circle down to two points. The idea is to attach to U a number $\varphi(U)\in\mathrm{U}(1)$, computed from the Gaussian Uu_0, such that $\varphi(\lambda U)=\lambda^2\varphi(U)$ and φ is multiplicative. Then $\varphi(U)=1$ fixes U up to the sign $\lambda=\pm1$, and these normalized operators form a group.

For $S\in\mathrm{Sp}(2n,\mathbb{R})$ with blocks (7.12.1) set

$$(M_S\ \ N_S) = (\mathrm{Id}\ \ i\,\mathrm{Id})S^T,\quad \text{i.e.}\quad M_S = A^T+iB^T,\ N_S = C^T+iE^T. \tag{7.12.13}$$

Lemma 7.12.10. *Let $S\in\mathrm{Sp}(2n,\mathbb{R})$.*

(i) *M_S is invertible, $Z_S = M_S^{-1}N_S$ is a complex symmetric matrix and* $\operatorname{Im} Z_S = M_S^{-1}(M_S^*)^{-1}$ *is positive definite.*

(ii) *If $U \in \mathcal{M}(2n)$ and $\pi(U) = S$, then*

$$Uu_0 = c_U\, e^{\frac{i}{2}\langle Z_S x, x\rangle}, \qquad c_U = (Uu_0)(0), \qquad |c_U|^2 = \pi^{-n/2}|\det M_S|^{-1}. \tag{7.12.14}$$

In particular $c_U \neq 0$, $c_{\lambda U} = \lambda c_U$, and U is determined by S and c_U.

Proof. (i) Write $M = M_S$, $N = N_S$. Using (7.12.4),

$$\begin{aligned} NM^* - MN^* &= (C^T + iE^T)(A - iB) - (A^T + iB^T)(C - iE) \\ &= (C^TA - A^TC) + (E^TB - B^TE) \\ &\quad + i((E^TA - B^TC) + (A^TE - C^TB)) = 2i\,\mathrm{Id}, \\ NM^T - MN^T &= (C^TA - A^TC) - (E^TB - B^TE) \\ &\quad + i((E^TA - B^TC) - (A^TE - C^TB)) = 0, \end{aligned}$$

since $A^TE - C^TB = \mathrm{Id} = E^TA - B^TC$. If $\mu^TM = 0$ for some $\mu \in \mathbb{C}^n$ then $2i\mu^T\bar\mu = \mu^T(NM^* - MN^*)\bar\mu = 0$, because $M^*\bar\mu = \overline{M^T\mu} = 0$; so $\mu = 0$ and M is invertible. The second identity says $Z = M^{-1}N$ is symmetric, and then $\bar Z = Z^*$ and the first identity gives $Z - \bar Z = M^{-1}(NM^* - MN^*)M^{-*} = 2iM^{-1}M^{-*}$. Thus $\operatorname{Im} Z = M^{-1}M^{-*}$, a positive definite hermitian matrix which is real since Z is symmetric.

(ii) Let $v = Uu_0 \in \mathcal{S}$. Since $A_ju_0 = 0$, where $A_j = a_j(x, D)$ with $a_j(x,\xi) = \xi_j - ix_j$, Lemma 7.12.5 gives $(a_j \circ S^{-1})(x, D)v = 0$. By Lemma 7.12.1(i), the vector of the linear functions $a_j \circ S^{-1}$ is $(-C^Tx + A^T\xi) - i(E^Tx - B^T\xi) = M\xi - Nx$. Hence $MDv = Nxv$, i.e. $Dv = Zxv$. The function $w = e^{-\frac{i}{2}\langle Zx,x\rangle}v$ is smooth and, Z being symmetric, $\partial_j w = e^{-\frac{i}{2}\langle Zx,x\rangle}(\partial_j v - i(Zx)_j v) = e^{-\frac{i}{2}\langle Zx,x\rangle}(iD_jv - i(Zx)_jv) = 0$. So w is constant, which is (7.12.14). Since $\|v\| = 1$ and $|v|^2 = |c_U|^2 e^{-\langle(\operatorname{Im} Z)x,x\rangle}$, $1 = |c_U|^2\pi^{n/2}\det(\operatorname{Im} Z)^{-1/2} = |c_U|^2\pi^{n/2}|\det M|$. If $\pi(U') = \pi(U)$ then $U' = \lambda U$ by Proposition 7.12.8, and $\lambda = c_{U'}/c_U$. □

By the lemma

$$\varphi(U) = \pi^{n/2}c_U^2 \det M_{\pi(U)} \in \mathrm{U}(1), \qquad U \in \mathcal{M}(2n), \tag{7.12.15}$$

and $\varphi(\lambda U) = \lambda^2\varphi(U)$. For example $\varphi(\mathrm{Id}) = 1$; for T_L in Lemma 7.12.6(iv), $c_{T_L} = |\det L|^{1/2}\pi^{-n/4}$ and $M = L^{-T}$, so $\varphi(T_L) = \operatorname{sgn}\det L$; and $\mathcal{F}_1u_0 = u_0$ while $M_{J_1} = \operatorname{diag}(i, 1, \dots, 1)$, so $\varphi(\mathcal{F}_1) = i$.

Lemma 7.12.11. *Let $U \in \mathcal{M}(2n)$ and let $W(t)$ be one of the families $e^{-itq(x)}$ or $e^{-itq(D)}$ of Lemma 7.12.6. Then $\varphi(UW(t)) = \varphi(U)$ for all $t \in \mathbb{R}$.*

Proof. Regard q as a quadratic form on $\mathbb{R}^{2n}$ (depending only on x, or only on ξ), so that $W(t)$ is multiplication by, or the Fourier multiplier, e^{-itP} with $P = \mathrm{Op^W}(q)$, and $\pi(W(t)) = e^{tX_q}$ by Lemma 7.12.6 and (7.12.3). Let $S_0 = \pi(U)$, $S(t) = S_0e^{tX_q} = \pi(UW(t))$ and $v(t) = UW(t)u_0$. By Lemma 7.12.6, v is a $\mathcal{C}^1$ curve in $\mathcal{S}$ with

$$v'(t) = -iUPW(t)u_0 = -i\,\mathrm{Op^W}(\tilde q)v(t), \qquad \tilde q = q \circ S_0^{-1},$$

by Theorem 7.12.7. Write $\tilde q = \frac12\langle\alpha x, x\rangle + \langle\beta x,\xi\rangle + \frac12\langle\gamma\xi,\xi\rangle$. By Lemma 7.12.10, $v(t) = c(t)e^{\frac{i}{2}\langle Z(t)x,x\rangle}$ with $c(t) = v(t)(0)$ and $Z(t) = Z_{S(t)}$, and c is $\mathcal{C}^1$ since evaluation at 0 is continuous on $\mathcal{S}$. Using (7.12.8) and $(D_jD_kv)(0) = -(\partial_j\partial_kv)(0) = -iZ_{jk}c$ we find

$$c'(t) = -i(\mathrm{Op^W}(\tilde q)v(t))(0) = -\tfrac12(\operatorname{tr}\beta + \operatorname{tr}(\gamma Z(t)))c(t).$$

On the other hand, by Lemma 7.12.1(iii), $S'(t) = S_0 X_q e^{tX_q} = X_{\tilde{q}} S(t)$, so by (7.12.13) and (7.12.3)

$$(M' \;\; N') = (M \;\; N) X_{\tilde{q}}^T, \qquad M' = M\beta^T + N\gamma, \quad N' = -M\alpha - N\beta, \tag{7.12.16}$$

where $M = M_{S(t)}$, $N = N_{S(t)}$. Hence $(\det M)'/\det M = \operatorname{tr}(M^{-1}M') = \operatorname{tr}\beta + \operatorname{tr}(Z\gamma)$, and

$$\frac{d}{dt}(c^2 \det M) = c^2 \det M\Big(\frac{2c'}{c} + \operatorname{tr}(M^{-1}M')\Big) = 0.$$

Thus $\varphi(UW(t)) = \pi^{n/2} c(t)^2 \det M_{S(t)}$ is constant. □

Theorem 7.12.12. *The map $\varphi : \mathcal{M}(2n) \to \mathrm{U}(1)$ is a group homomorphism with $\varphi(\lambda \operatorname{Id}) = \lambda^2$. Its kernel*

$$\mathrm{Mp}(2n, \mathbb{R}) = \{U \in \mathcal{M}(2n);\ \varphi(U) = 1\},$$

the metaplectic group*, is the subgroup of $\mathcal{M}(2n)$ generated by the operators $e^{-iq(x)}$ and $e^{-iq(D)}$, q a real quadratic form on $\mathbb{R}^n$, and there is an exact sequence*

$$1 \longrightarrow \{\pm \operatorname{Id}\} \longrightarrow \mathrm{Mp}(2n, \mathbb{R}) \xrightarrow{\pi} \mathrm{Sp}(2n, \mathbb{R}) \longrightarrow 1. \tag{7.12.17}$$

Moreover $\mathcal{M}(2n) = \mathrm{U}(1) \cdot \mathrm{Mp}(2n, \mathbb{R}) \cong (\mathrm{U}(1) \times \mathrm{Mp}(2n, \mathbb{R}))/\{\pm(1, \operatorname{Id})\}$.

Proof. Let $\mathcal{G}$ be the group generated by the operators $e^{-iq(x)}$ and $e^{-iq(D)}$; it is closed under inverses since q may be replaced by $-q$. By Lemma 7.12.11 and induction on the length of words, $\varphi(UW) = \varphi(U)$ for all $U \in \mathcal{M}(2n)$ and $W \in \mathcal{G}$; in particular $\mathcal{G} \subset \{\varphi = 1\}$, as $\varphi(\operatorname{Id}) = 1$. By Lemma 7.12.6 and Lemma 7.12.2, $\pi(\mathcal{G}) = \mathrm{Sp}(2n, \mathbb{R})$.

Next, $-\operatorname{Id} \in \mathcal{G}$. Indeed let $\mathcal{J}_1 = e^{-ix_1^2/2} e^{-iD_1^2/2} e^{-ix_1^2/2} \in \mathcal{G}$. In the (x_1, ξ_1)-plane, $\left(\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 0 \\ -1 & 1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, so $\pi(\mathcal{J}_1) = J_1 = \pi(\mathcal{F}_1)$. Hence $\mathcal{J}_1 = \lambda \mathcal{F}_1$ by Proposition 7.12.8, and $1 = \varphi(\mathcal{J}_1) = \lambda^2 \varphi(\mathcal{F}_1) = i\lambda^2$. Since $\mathcal{F}_1^4 = \operatorname{Id}$ (indeed $\mathcal{F}_1^2 u(x) = u(-x_1, x_2, \dots, x_n)$), $\mathcal{J}_1^4 = \lambda^4 \operatorname{Id} = (-i)^2 \operatorname{Id} = -\operatorname{Id}$.

Now let $U \in \mathcal{M}(2n)$. Choose $W \in \mathcal{G}$ with $\pi(W) = \pi(U)$; then $U = \lambda W$ with $|\lambda| = 1$ and $\varphi(U) = \lambda^2 \varphi(W) = \lambda^2$. If $\varphi(U) = 1$ then $\lambda = \pm 1$ and $U = \pm W \in \mathcal{G}$. Thus $\mathcal{G} = \{\varphi = 1\}$. If $U' = \lambda' W'$ is another element, with $W' \in \mathcal{G}$, then $\varphi(UU') = \lambda'^2 \varphi(UW') = \lambda'^2 \varphi(U) = \varphi(U)\varphi(U')$, so φ is a homomorphism. The kernel of π on $\mathrm{Mp}(2n, \mathbb{R})$ consists of the $\lambda \operatorname{Id}$ with $\lambda^2 = 1$, and $-\operatorname{Id} \in \mathcal{G}$; with the surjectivity of π on $\mathcal{G}$ this gives (7.12.17). Finally every $U \in \mathcal{M}(2n)$ is λW with $W \in \mathrm{Mp}(2n, \mathbb{R})$, and (λ, W) is determined up to the sign $\pm(1, \operatorname{Id})$. □

Thus over each $S \in \mathrm{Sp}(2n, \mathbb{R})$ there are exactly two elements of $\mathrm{Mp}(2n, \mathbb{R})$, differing in sign; by Lemma 7.12.10 they are the elements U over S with $c_U^2 = \pi^{-n/2} (\det M_S)^{-1}$. For instance $T_L \in \mathrm{Mp}(2n, \mathbb{R})$ if $\det L > 0$, while $\pm i T_L$ are the elements over $\operatorname{diag}(L^{-1}, L^T)$ if $\det L < 0$; and $e^{-in\pi/4} \mathcal{F}_0 \in \mathrm{Mp}(2n, \mathbb{R})$, since $\mathcal{F}_0 u_0 = u_0$ and $\det M_J = i^n$.

7.12.5 Explicit form: Fourier integral operators with quadratic phase

We need the Fourier transform of complex Gaussians.

Lemma 7.12.13. *Let K be a complex symmetric $n \times n$ matrix with $\operatorname{Re} K$ positive definite. Then $\det K \neq 0$ and*

$$\Big(\int_{\mathbb{R}^n} e^{-\frac{1}{2}\langle K\eta, \eta\rangle}\, d\eta\Big)^2 = \frac{(2\pi)^n}{\det K}.$$

Proof. Let $K_0 = \operatorname{Re} K$, $K_1 = \operatorname{Im} K$ and $K(z) = K_0 + zK_1$. There is $\delta > 0$ such that $K_0 + sK_1$ is positive definite for $|s| < \delta$, so on the strip $\Omega = \{|\operatorname{Re} z| < \delta\}$ the function $f(z) = \left(\int e^{-\frac{1}{2}\langle K(z)\eta,\eta\rangle} d\eta\right)^2 \det K(z)$ is holomorphic (differentiate under the integral, which converges locally uniformly). For real $z \in (-\delta, \delta)$, diagonalizing the real positive definite matrix $K(z)$ by an orthogonal change of variables shows $f(z) = (2\pi)^n$. Since Ω is connected, $f \equiv (2\pi)^n$; take $z = i$. In particular $\det K \neq 0$. □

Proposition 7.12.14. *Let $S \in \operatorname{Sp}(2n, \mathbb{R})$ have invertible block A, and let*

$$\begin{aligned} \Phi_S(x,\eta) &= \tfrac{1}{2}\langle CA^{-1}x, x\rangle + \langle A^{-1}x, \eta\rangle - \tfrac{1}{2}\langle A^{-1}B\eta, \eta\rangle, \\ U_S u(x) &= |\det A|^{-1/2} \int e^{i\Phi_S(x,\eta)} \hat{u}(\eta) đ\eta. \end{aligned} \tag{7.12.18}$$

Then $U_S \in \mathcal{M}(2n)$, $\pi(U_S) = S$ and $\varphi(U_S) = \operatorname{sgn} \det A$. Hence the two elements of $\operatorname{Mp}(2n, \mathbb{R})$ over S are $\pm U_S$ if $\det A > 0$ and $\pm iU_S$ if $\det A < 0$. The quadratic form Φ_S is a generating function for S: $S(\partial_\eta \Phi_S, \eta) = (x, \partial_x \Phi_S)$ for all (x, η).

Proof. With $q_1(x) = -\frac{1}{2}\langle CA^{-1}x, x\rangle$, $L = A^{-1}$ and $q_2(\xi) = \frac{1}{2}\langle A^{-1}B\xi, \xi\rangle$ (symmetric matrices by Lemma 7.12.2(i)), the definitions give $U_S = e^{-iq_1(x)} T_L e^{-iq_2(D)}$. By Lemma 7.12.6, $\pi(U_S) = L_{CA^{-1}} \operatorname{diag}(A, A^{-T}) R_{A^{-1}B} = S$ by (7.12.5). Since $\hat{u}_0 = (2\pi)^{n/2}\pi^{-n/4} e^{-|\eta|^2/2}$,

$$c_{U_S} = U_S u_0(0) = |\det A|^{-1/2} \pi^{-n/4} (2\pi)^{-n/2} \int e^{-\frac{1}{2}\langle (\mathrm{Id} + iA^{-1}B)\eta, \eta\rangle}\, d\eta,$$

so by Lemma 7.12.13, $c_{U_S}^2 = \pi^{-n/2} |\det A|^{-1} \det(\mathrm{Id} + iA^{-1}B)^{-1}$. As $\det M_S = \det(A + iB) = \det A \det(\mathrm{Id} + iA^{-1}B)$, $\varphi(U_S) = \operatorname{sgn} \det A$, and the description of the elements over S follows from $\varphi(\lambda U_S) = \lambda^2 \varphi(U_S)$. Finally, with $y = \partial_\eta \Phi_S = A^{-1}x - A^{-1}B\eta$ we have $Ay + B\eta = x$ and $Cy + E\eta = CA^{-1}x + (E - CA^{-1}B)\eta = CA^{-1}x + A^{-T}\eta = \partial_x \Phi_S$, using (7.12.5). □

For S near the identity A is near Id, and U_S is the element of $\operatorname{Mp}(2n, \mathbb{R})$ over S which is close to the identity (for instance in the sense that c_{U_S} is close to c_{Id}). By Lemma 7.12.2(ii) every element of $\operatorname{Mp}(2n, \mathbb{R})$ is the product of an operator $\pm U_S$ or $\pm iU_S$ and a multiplication operator $e^{-it|x|^2/2}$ (which lies over $L_{-t\,\mathrm{Id}}$); so the metaplectic group consists of 'isotropic Fourier integral operators' associated with the graphs of linear symplectic maps.

7.12.6 Exponentials of quadratic forms

Proposition 7.12.15. *Let q be a real quadratic form on $\mathbb{R}^{2n}$ and $P = \operatorname{Op}^{\mathrm{W}}(q)$. There is a unique strongly continuous one-parameter group $W_q(t)$, $t \in \mathbb{R}$, in $\operatorname{Mp}(2n, \mathbb{R})$ with $\pi(W_q(t)) = e^{tX_q}$. It commutes with P, and for every $u \in \mathcal{S}(\mathbb{R}^n)$ the curve $t \mapsto W_q(t)u$ is $\mathcal{C}^1$ in L^2 with*

$$\frac{d}{dt} W_q(t)u = -iPW_q(t)u, \qquad W_q(0)u = u. \tag{7.12.19}$$

Conversely, if $V(t)$, $t \in \mathbb{R}$, are bounded operators on L^2 preserving $\mathcal{S}(\mathbb{R}^n)$ such that $t \mapsto V(t)u$ is $\mathcal{C}^1$ in L^2 and satisfies (7.12.19) for every $u \in \mathcal{S}$, then $V(t) = W_q(t)$.

We write $W_q(t) = \exp(-it \operatorname{Op}^{\mathrm{W}}(q))$. By the uniqueness statement, the multiplication operator $e^{-iq(x)}$ is $\exp(-i \operatorname{Op}^{\mathrm{W}}(q))$ for a quadratic form $q = q(x)$ independent of ξ, and the Fourier multiplier $e^{-iq(D)}$ is $\exp(-i \operatorname{Op}^{\mathrm{W}}(q))$ for $q = q(\xi)$. So Theorem 7.12.12 shows that $\operatorname{Mp}(2n, \mathbb{R})$ is generated by the exponentials $\exp(-i \operatorname{Op}^{\mathrm{W}}(q))$ of Weyl quantizations of real quadratic forms. Since $\operatorname{Mp}(2n, \mathbb{R})$ contains all $\exp(-it \operatorname{Op}^{\mathrm{W}}(q))$, it is in fact generated by any family of such exponentials whose images generate $\operatorname{Sp}(2n, \mathbb{R})$.

Proof. Let $\Phi_t = e^{tX_q}$ and $m(t) = \det M_{\Phi_t}$, a smooth non-vanishing function with $m(0) = 1$. Write $m = e^h$ with h smooth and $h(0) = 0$, and set $c(t) = \pi^{-n/4} e^{-h(t)/2}$, so $\pi^{n/2} c(t)^2 m(t) = 1$. The two elements of $\mathrm{Mp}(2n, \mathbb{R})$ over Φ_t have $c_U = \pm c(t)$ by Lemma 7.12.10; let $W(t)$ be the one with $c_{W(t)} = c(t)$. Then $W(0) = \mathrm{Id}$.

Continuity. By Lemma 7.12.10, $W(t)u_0 = c(t) e^{\frac{i}{2}\langle Z(t)x, x\rangle}$, $Z(t) = Z_{\Phi_t}$, which is a smooth curve in $\mathcal{S}$. By Theorem 7.12.7, $W(t) C_j W(t)^{-1} = \mathrm{Op^W}(c_j \circ \Phi_{-t})$ where $c_j = \xi_j + i x_j$, so for a multi-index μ

$$W(t) C^\mu u_0 = \prod_j (\mathrm{Op^W}(c_j \circ \Phi_{-t}))^{\mu_j} W(t) u_0, \tag{7.12.20}$$

which is again a smooth curve in $\mathcal{S}$, the operators $\mathrm{Op^W}(c_j \circ \Phi_{-t})$ being first order with coefficients depending smoothly on t. Since the Hermite functions span a dense subspace and $\|W(t)\| = 1$, $W(t)$ is strongly continuous.

Group property. $W(t)W(s)W(t+s)^{-1} \in \ker \pi \cap \mathrm{Mp}(2n, \mathbb{R}) = \{\pm \mathrm{Id}\}$, and its sign is $\epsilon(t, s) = (W(t)W(s)u_0, W(t+s)u_0)$, a continuous function of (t, s) (products of uniformly bounded strongly continuous families are strongly continuous) with $\epsilon(0, 0) = 1$. So $\epsilon \equiv 1$. Uniqueness among strongly continuous one-parameter groups in $\mathrm{Mp}(2n, \mathbb{R})$ lifting Φ_t follows in the same way: another such group is $\epsilon(t) W(t)$ with $\epsilon(t) = \pm 1$ continuous and $\epsilon(0) = 1$. Since $q \circ \Phi_t = q$ (Lemma 7.12.1(ii)), Theorem 7.12.7 gives $W(t)^{-1} P W(t) = P$.

The derivative. Write q as in (7.12.2). Since $\Phi_t' = X_q \Phi_t$, (7.12.16) holds with $\tilde{q} = q$, $M = M_{\Phi_t}$, $N = N_{\Phi_t}$, so $c'/c = -\frac{1}{2} m'/m = -\frac{1}{2}(\mathrm{tr}\, \beta + \mathrm{tr}(Z\gamma))$ and

$$Z' = -M^{-1} M' Z + M^{-1} N' = -\beta^T Z - Z\gamma Z - \alpha - Z\beta.$$

At $t = 0$, $Z = i\,\mathrm{Id}$, so $c'(0) = -\frac{1}{2}(\mathrm{tr}\, \beta + i\, \mathrm{tr}\, \gamma) c(0)$ and $Z'(0) = -i(\beta + \beta^T) + \gamma - \alpha$. Hence

$$\frac{d}{dt} W(t) u_0 \Big|_{t=0} = \Big(-\tfrac{1}{2}\, \mathrm{tr}\, \beta - \tfrac{i}{2}\, \mathrm{tr}\, \gamma + \langle \beta x, x\rangle + \tfrac{i}{2} \langle (\gamma - \alpha) x, x\rangle \Big) u_0.$$

On the other hand $D_j u_0 = i x_j u_0$ and $D_j D_k u_0 = (\delta_{jk} - x_j x_k) u_0$, so by (7.12.8)

$$P u_0 = \Big(\tfrac{1}{2} \langle \alpha x, x\rangle + i \langle \beta x, x\rangle - \tfrac{i}{2}\, \mathrm{tr}\, \beta + \tfrac{1}{2}\, \mathrm{tr}\, \gamma - \tfrac{1}{2} \langle \gamma x, x\rangle \Big) u_0,$$

and the two displays show $\frac{d}{dt} W(t) u_0|_{t=0} = -i P u_0$. Next, $\frac{d}{dt}(c_j \circ \Phi_{-t})|_{t=0} = -\{q, c_j\}$ by (7.12.3), and $-\mathrm{Op^W}(\{q, c_j\}) = -i[P, C_j]$ by (7.12.7). Differentiating (7.12.20) at $t = 0$ by the product rule,

$$\frac{d}{dt} W(t) C^\mu u_0 \Big|_{t=0} = -i[P, C^\mu] u_0 - i C^\mu P u_0 = -i P C^\mu u_0.$$

Thus $\frac{d}{dt} W(t) u|_{t=0} = -i P u$ for every finite linear combination u of Hermite functions, and by the group property $\frac{d}{dt} W(t) u = -i W(t) P u$, so that

$$W(t) u - u = -i \int_0^t W(s) P u \, ds. \tag{7.12.21}$$

Both sides of (7.12.21) are continuous in u for the norm of $H^2_{\mathrm{iso}}(\mathbb{R}^n)$, as $P : H^2_{\mathrm{iso}} \to L^2$ is bounded (Proposition 7.9.2(v)); finite sums of Hermite functions are dense in $\mathcal{S}$ (Proposition 7.8.7), hence in H^2_{iso}. So (7.12.21) holds for $u \in \mathcal{S}$, and since $s \mapsto W(s) P u$ is continuous, $W(t) u$ is $\mathcal{C}^1$ with derivative $-i W(t) P u = -i P W(t) u$.

Uniqueness. Let $V(t)$ be as in the statement and $u, v \in \mathcal{S}$. Then $f(t) = (V(t)u, W(t)v)$ is $\mathcal{C}^1$ and, P being symmetric on $\mathcal{S}$, $f'(t) = -i(PV(t)u, W(t)v) + i(V(t)u, PW(t)v) = 0$. Hence $(W(t)^* V(t) u, v) = (u, v)$ for all $v \in \mathcal{S}$, so $W(t)^* V(t) u = u$, i.e. $V(t) u = W(t) u$. $\square$

Example 7.12.16. Let $q = \frac{1}{2}(x_1^2 + \xi_1^2)$, so $P = \frac{1}{2}H_1$ with $H_1 = D_1^2 + x_1^2$, and $e^{tX_q} = R_t$ is the rotation by the angle t in the (x_1, ξ_1)-plane, $R_t(x, \xi) = (x_1 \cos t + \xi_1 \sin t, x'', \xi_1 \cos t - x_1 \sin t, \xi'')$. Since $H_1 u_\mu = (2\mu_1 + 1)u_\mu$ for the Hermite functions (7.8.3), the operators

$$V(t)u = \sum_\mu e^{-it(2\mu_1+1)/2}(u, u_\mu)u_\mu$$

are unitary, preserve $\mathcal{S}$ (Proposition 7.8.7) and satisfy (7.12.19), the differentiated series converging in $\mathcal{S}$. By Proposition 7.12.15, $V(t) = W_q(t) = \exp(-\frac{it}{2}H_1)$. In particular $W_q(2\pi) = -\operatorname{Id}$ although $R_{2\pi} = \operatorname{Id}$; similarly $\exp(-\frac{it}{2}H)$ lies over the rotation e^{tJ} of $\mathbb{R}^{2n}$ and equals $(-1)^n \operatorname{Id}$ at $t = 2\pi$.

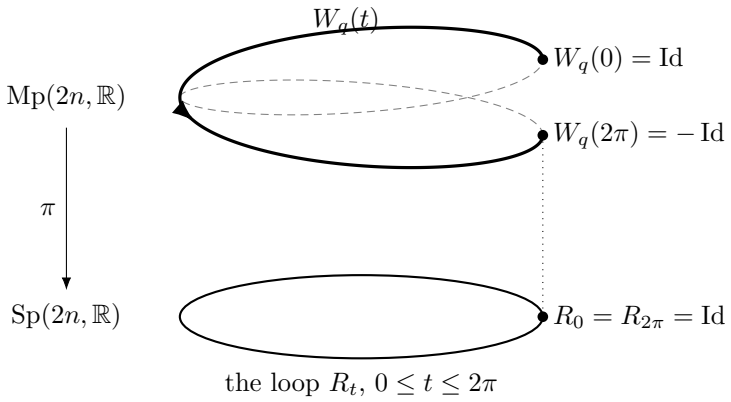


Figure 7.4: The double cover $\pi : \operatorname{Mp}(2n, \mathbb{R}) \to \operatorname{Sp}(2n, \mathbb{R})$ over the loop of rotations R_t of Example 7.12.16. Lifting R_t continuously, starting at Id, gives $W_q(t) = \exp(-\frac{it}{2}H_1)$ (thick), which ends at $-\operatorname{Id}$ after one turn; a second turn (dashed) returns to Id. Both $\pm\operatorname{Id}$ lie over $\operatorname{Id} \in \operatorname{Sp}(2n, \mathbb{R})$.

Corollary 7.12.17. $-\operatorname{Id}$ *lies on a strongly continuous path* $W_q(t)$, $t \in [0, 2\pi]$, *in* $\operatorname{Mp}(2n, \mathbb{R})$ *starting at* Id. *There is no homomorphism* $\sigma : \operatorname{Sp}(2n, \mathbb{R}) \to \operatorname{Mp}(2n, \mathbb{R})$ *with* $\pi \circ \sigma = \operatorname{Id}$ *such that* $t \mapsto \sigma(R_t)$ *is strongly continuous; the double cover* (7.12.17) *does not split.*

Proof. If σ were such a homomorphism, $t \mapsto \sigma(R_t)$ would be a strongly continuous one-parameter group in $\operatorname{Mp}(2n, \mathbb{R})$ lifting R_t, hence equal to $W_q(t)$ by Proposition 7.12.15, and then $\operatorname{Id} = \sigma(R_{2\pi}) = W_q(2\pi) = -\operatorname{Id}$. □

Question. Where does the sign -1 in $W_q(2\pi) = -\operatorname{Id}$ come from physically?

Answer. From the ground state energy. The operator $\frac{1}{2}H_1$ has eigenvalues $\mu_1 + \frac{1}{2}$, not integers, so $e^{-it(\mu_1+1/2)}$ returns to -1, not 1, at $t = 2\pi$. The classical flow R_t only sees the symbol $\frac{1}{2}(x_1^2 + \xi_1^2)$ and is periodic with period 2π; the quantum evolution has period 4π (Figure 7.4). The "zero-point energy" $\frac{1}{2}$ is exactly what forces the double cover. □

Remark 7.12.18. One can show that $\operatorname{Mp}(2n, \mathbb{R})$, with the topology of strong convergence, is a connected Lie group and that (7.12.17) is its unique connected double covering of $\operatorname{Sp}(2n, \mathbb{R})$ (the fundamental group of $\operatorname{Sp}(2n, \mathbb{R})$ being $\mathbb{Z}$, generated by the loop R_t, $t \in [0, 2\pi]$); see [8, Section 18.5]. We do not use these facts. Note also that the operator $P = \operatorname{Op}^{\mathrm{W}}(q)$ of Proposition 7.12.15 is in general not elliptic (e.g. for $q = x \cdot \xi$), so $\exp(-itP)$ is not obtained from the spectral theory of elliptic operators; the construction above uses the group structure instead, and avoids any discussion of self-adjoint extensions.

7.12.7 Conjugation and symplectic invariance

Theorem 7.12.7 gives a second proof of the symplectic invariance of the Weyl product, and shows that all the structures of this chapter are invariant under the metaplectic group.

Corollary 7.12.19. *Let $U \in \mathcal{M}(2n)$ and $S = \pi(U)$.*

(i) *For $a \in \Gamma^m(\mathbb{R}^{2n})$, $b \in \Gamma^{m'}(\mathbb{R}^{2n})$, if $a\#_{\mathrm{W}}b$ denotes the Weyl symbol of $\mathrm{Op}^{\mathrm{W}}(a)\,\mathrm{Op}^{\mathrm{W}}(b)$, then $(a\#_{\mathrm{W}}b)\circ S = (a\circ S)\#_{\mathrm{W}}(b\circ S)$; this is (7.11.5) for $W = \mathbb{R}^{2n}$ with the Weyl product.*

(ii) *$A \mapsto U^{-1}AU$ is an automorphism of the filtered $*$-algebra $\Psi^{\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ preserving each $\Psi^{m}_{\mathrm{iso}}(\mathbb{R}^n)$, $\Psi^{m}_{\mathrm{iso,ph}}(\mathbb{R}^n)$ and $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, and the group $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$. If A has Weyl symbol a then $U^{-1}AU$ has Weyl symbol $a\circ S$; in particular $\sigma_m(U^{-1}AU) = \sigma_m(A)\circ S$, and $U^{-1}AU$ is elliptic if and only if A is.*

(iii) *U is an isomorphism of $H^s_{\mathrm{iso}}(\mathbb{R}^n)$ for every $s \in \mathbb{R}$.*

Proof. (i) By (7.12.10), $\mathrm{Op}^{\mathrm{W}}((a\#_{\mathrm{W}}b)\circ S) = U^{-1}\,\mathrm{Op}^{\mathrm{W}}(a)\,\mathrm{Op}^{\mathrm{W}}(b)U = \mathrm{Op}^{\mathrm{W}}(a\circ S)\,\mathrm{Op}^{\mathrm{W}}(b\circ S)$, and the Weyl symbol is unique (Proposition 7.11.2). (ii) By Proposition 7.11.2, every element of Ψ^m_{iso} is $\mathrm{Op}^{\mathrm{W}}(a)$ with $a \in \Gamma^m$, polyhomogeneous or Schwartz exactly when the operator is, and these classes are invariant under composition with S by Lemma 7.1.2(iii) and the homogeneity of the terms. The left and Weyl symbols agree modulo Γ^{m-2}, which gives the statement about principal symbols and ellipticity. The group $G^{-\infty}_{\mathrm{iso}}$ is preserved since $U^{-1}(\mathrm{Id}+A)U = \mathrm{Id}+U^{-1}AU$ and invertibility is preserved. (iii) Let $a = \langle w\rangle^s$. Then $\mathrm{Op}^{\mathrm{W}}(a)$ and $\mathrm{Op}^{\mathrm{W}}(a\circ S)$ are elliptic elements of Ψ^s_{iso}, so by Proposition 7.9.2(i), $Uu \in H^s_{\mathrm{iso}}$ if and only if $\mathrm{Op}^{\mathrm{W}}(a)Uu = U\,\mathrm{Op}^{\mathrm{W}}(a\circ S)u \in L^2$, if and only if $u \in H^s_{\mathrm{iso}}$. Thus U is a bijection of H^s_{iso}, continuous by the closed graph theorem since it is continuous on $\mathcal{S}'$. □

The conjugation formula (7.12.10) is special to the Weyl quantization: for the left quantization, $U^{-1}\,\mathrm{Op}(a)U$ has left symbol $a\circ S$ in general only modulo $\Gamma^{m-2}(\mathbb{R}^{2n})$ (by Proposition 7.11.2), although the identity is exact for instance for $U = T_L$. Note also that the operators in $\mathcal{M}(2n)$ are in general not themselves isotropic pseudodifferential operators; for instance, by uniqueness of left symbols, multiplication by $e^{-iq(x)}$ with $q \neq 0$ would have to have left symbol $e^{-iq(x)}$, which is not in any $\Gamma^m(\mathbb{R}^{2n})$. So Corollary 7.12.19(ii) gives an action of $\mathrm{Sp}(2n,\mathbb{R})$ on $\Psi^{\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ by automorphisms which are implemented by unitary operators outside the algebra.

7.13 Complex order

In Chapter 8 traces of operators of high order are regularized by inserting complex powers $\mathrm{Op}(\langle w\rangle^z)$ and continuing analytically in z. For this we need isotropic operators of complex order and holomorphic families of them, which we now define.

The identification $\Gamma^0_{\mathrm{ph}}(\mathbb{R}^{2n}) = \mathrm{RC}^*\,\mathcal{C}^{\infty}(\mathbb{S}^{2n}_+)$ of Proposition 7.1.8 allows us to define isotropic operators of complex order. For $z \in \mathbb{C}$ set

$$\Gamma^z_{\mathrm{ph}}(\mathbb{R}^N) = \langle w\rangle^z\,\Gamma^0_{\mathrm{ph}}(\mathbb{R}^N) = \mathrm{RC}^*\big(\rho^{-z}\mathcal{C}^{\infty}(\mathbb{S}^N_+)\big), \qquad \Psi^z_{\mathrm{iso}}(\mathbb{R}^n) = \mathrm{Op}\big(\Gamma^z_{\mathrm{ph}}(\mathbb{R}^{2n})\big).$$

For real z this agrees with Γ^z_{ph} as defined before, since $\langle w\rangle^z \in \Gamma^z_{\mathrm{ph}}$. Since $|\partial^\alpha\langle w\rangle^z| \leq C_\alpha\langle w\rangle^{\mathrm{Re}\,z-|\alpha|}$, $\Gamma^z_{\mathrm{ph}} \subset \Gamma^{\mathrm{Re}\,z}$ and $\Psi^z_{\mathrm{iso}} \subset \Psi^{\mathrm{Re}\,z}_{\mathrm{iso}}$. The definition does not depend on the choice of boundary defining function: if $\rho' = f\rho$ with $0 < f \in \mathcal{C}^{\infty}(\mathbb{S}^N_+)$ then $f^z = e^{z\log f}$ is smooth, so $\rho'^{-z}\mathcal{C}^{\infty} = \rho^{-z}\mathcal{C}^{\infty}$.

Proposition 7.13.1. *For $z, z' \in \mathbb{C}$, $\Psi^z_{\mathrm{iso}}(\mathbb{R}^n) \circ \Psi^{z'}_{\mathrm{iso}}(\mathbb{R}^n) \subset \Psi^{z+z'}_{\mathrm{iso}}(\mathbb{R}^n)$ and $\Psi^z_{\mathrm{iso}}(\mathbb{R}^n)^* = \Psi^{\bar z}_{\mathrm{iso}}(\mathbb{R}^n)$.*

Proof. Differentiation maps $\langle w\rangle^z \Gamma^0_{\mathrm{ph}}$ into $\langle w\rangle^z \Gamma^{-1}_{\mathrm{ph}}$, since $\partial_{w_j}(\langle w\rangle^z f) = \langle w\rangle^z (z w_j \langle w\rangle^{-2} f + \partial_{w_j} f)$. Hence the terms of (7.2.4) satisfy $\frac{1}{\alpha!}\partial_\xi^\alpha a\, D_x^\alpha b \in \langle w\rangle^{z+z'}\Gamma^{-2|\alpha|}_{\mathrm{ph}}$. If $c = a\# b$ then $c \in \Gamma^\mu$, $\mu = \operatorname{Re}(z+z')$, and the remainders of the expansion lie in $\Gamma^{\mu-2N}$; multiplying by $\langle w\rangle^{-(z+z')} \in \Gamma^{-\mu}$, $\langle w\rangle^{-(z+z')}c$ has an asymptotic expansion with terms in $\Gamma^{-2j}_{\mathrm{ph}}$ and remainders in Γ^{-2N}, so it lies in Γ^0_{ph}. Adjoints are treated in the same way. □

Even more useful are holomorphic families. A map $\Omega \to \mathcal{C}^\infty(\mathbb{S}^N_+)$, with $\Omega \subset \mathbb{C}$ open, is *holomorphic* if it is given by an element of $\mathcal{C}^\infty(\Omega \times \mathbb{S}^N_+)$ satisfying the Cauchy–Riemann equation in the first variable. If $h : \Omega \to \mathbb{C}$ is holomorphic, a family $A(z) \in \Psi^{h(z)}_{\mathrm{iso}}(\mathbb{R}^n)$, $z \in \Omega$, is a *holomorphic family of order h* if $\sigma_L(A(z)) = \langle w\rangle^{h(z)} \mathrm{RC}^* F(z)$ with $F : \Omega \to \mathcal{C}^\infty(\mathbb{S}^{2n}_+)$ holomorphic.

Proposition 7.13.2. *If h, g are holomorphic on an open set $\Omega \subset \mathbb{C}$ and $A(z)$, $B(z)$ are holomorphic families of isotropic operators of orders $h(z)$ and $g(z)$, then $A(z) \circ B(z)$ is a holomorphic family of order $h(z) + g(z)$.*

Sketch of proof. It suffices to work on an open Ω' with compact closure in Ω, on which $\operatorname{Re} h, \operatorname{Re} g < M$ and $\operatorname{Re}(h+g) > -M$. The map $z \mapsto \langle w\rangle^{h(z)}$ is holomorphic into Γ^M (its derivative $h'(z)\log\langle w\rangle\,\langle w\rangle^{h(z)}$ lies in Γ^M, as $\operatorname{Re} h < M$ strictly), and $\mathrm{RC}^* : \mathcal{C}^\infty(\mathbb{S}^{2n}_+) \to \Gamma^0$ is continuous linear, so $a(z) = \sigma_L(A(z))$ and $b(z)$ are holomorphic Γ^M-valued. By Theorem 7.2.7(iv), $c(z) = a(z)\# b(z)$ is holomorphic into Γ^{2M} and

$$c(z) = \sum_{|\alpha|<N} \frac{1}{\alpha!}\partial_\xi^\alpha a(z)\, D_x^\alpha b(z) + R_N(z),$$

with R_N holomorphic into Γ^{2M-2N}. Using (7.1.1), $\partial_{w_j}(\langle w\rangle^h \mathrm{RC}^* F) = \langle w\rangle^{h-1}\mathrm{RC}^*(hX_jF + V_jF)$ with $X_j = w_j\langle w\rangle^{-1}$ and V_j smooth on $\mathbb{S}^{2n}_+$, so $\langle w\rangle^{-h-g}$ times each term of the sum is RC^* of a holomorphic family in $\mathcal{C}^\infty(\mathbb{S}^{2n}_+)$, while $\langle w\rangle^{-h-g}R_N(z)$ is holomorphic into Γ^{3M-2N}, hence, by Proposition 7.1.8(iii), into $\mathcal{C}^k(\mathbb{S}^{2n}_+)$ for $2N > 3M + k + 1$. Thus $\rho^{h+g}c(z)$ (pushed forward by RC) is holomorphic into $\mathcal{C}^k(\mathbb{S}^{2n}_+)$ for every k, which means that it is given by an element of $\mathcal{C}^\infty(\Omega' \times \mathbb{S}^{2n}_+)$ holomorphic in z. □

7.14 Exercises

Exercise 7.1. For $k \in \mathbb{N}$ define $H^{-k}_{\mathrm{iso}}(\mathbb{R}^n)$ as the space of $u \in \mathcal{S}'(\mathbb{R}^n)$ which can be written $u = \sum_{|\alpha|+|\beta|\le k} x^\alpha D^\beta u_{\alpha\beta}$ with $u_{\alpha\beta} \in L^2(\mathbb{R}^n)$. Show directly that each $A \in \Psi^p_{\mathrm{iso}}(\mathbb{R}^n)$, $p \in \mathbb{Z}$, maps H^k_{iso} to H^{k-p}_{iso} for all $k \in \mathbb{Z}$ (with H^k_{iso}, $k \ge 0$, as in Proposition 7.9.2(iii)), and that these definitions agree with Definition 7.9.1. Deduce that if $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ is elliptic, $u \in \mathcal{S}'(\mathbb{R}^n)$ and $Au \in H^{m'}_{\mathrm{iso}}(\mathbb{R}^n)$, then $u \in H^{m'+m}_{\mathrm{iso}}(\mathbb{R}^n)$.

Exercise 7.2. Show that for $\epsilon > 0$, $H^\epsilon_{\mathrm{iso}}(\mathbb{R}^n) \subset \langle x\rangle^{-\epsilon}L^2(\mathbb{R}^n) \cap H^\epsilon(\mathbb{R}^n)$, and that the inclusion is strict. [Hint: compose with $\langle D\rangle^\epsilon$ and $\langle x\rangle^\epsilon$, using the scattering calculus.] Deduce, without using Proposition 7.7.1, that $H^\epsilon_{\mathrm{iso}}(\mathbb{R}^n) \hookrightarrow L^2(\mathbb{R}^n)$ is compact, and hence that each element of $\Psi^{-\epsilon}_{\mathrm{iso}}(\mathbb{R}^n)$ is compact on $L^2(\mathbb{R}^n)$.

Exercise 7.3. Let $E \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$. Show that there is an operator $F \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ such that $(\mathrm{Id}+E)(\mathrm{Id}+F) = \mathrm{Id}+G$ with $G \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ of finite rank. Deduce that an elliptic $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ has a parametrix $B \in \Psi^{-m}_{\mathrm{iso}}(\mathbb{R}^n)$ up to finite rank errors, i.e. $AB - \mathrm{Id}$ and

$BA-\mathrm{Id}$ are finite rank elements of $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, and hence that $A : H^M_{\mathrm{iso}}(\mathbb{R}^n) \to H^{M-m}_{\mathrm{iso}}(\mathbb{R}^n)$ is Fredholm for every M. (Recall that a bounded operator between Hilbert spaces is Fredholm if it has finite-dimensional null space and closed range of finite codimension.)

Exercise 7.4. Suppose $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ is elliptic and has index zero as an operator on $\mathcal{S}'(\mathbb{R}^n)$. Show that there is a finite rank $E \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ such that $A+E$ is an isomorphism of $\mathcal{S}'(\mathbb{R}^n)$, of $\mathcal{S}(\mathbb{R}^n)$ and of $H^s_{\mathrm{iso}}(\mathbb{R}^n) \to H^{s-m}_{\mathrm{iso}}(\mathbb{R}^n)$ for every s, with inverse in $\Psi^{-m}_{\mathrm{iso}}(\mathbb{R}^n)$.

Exercise 7.5. [Koszul complex] Let $\Lambda^k = \Lambda^k\mathbb{R}^n$ with basis dx^I, I strictly increasing k-tuples, declared orthonormal, and let $\mathcal{S}'(\mathbb{R}^n;\Lambda^k)$ be the space of $u = \sum_I u_I dx^I$, $u_I \in \mathcal{S}'(\mathbb{R}^n)$. Using the annihilation operators define

$$D : \mathcal{S}'(\mathbb{R}^n;\Lambda^k) \to \mathcal{S}'(\mathbb{R}^n;\Lambda^{k+1}), \qquad Du = \sum_{j=1}^n \sum_I A_j u_I \, dx_j \wedge dx^I.$$

(a) Show that $D^2 = 0$ and that the formal adjoint is $D^*u = \sum_j \sum_I C_j u_I \, \iota_j dx^I$, where ι_j, contraction with e_j, is the adjoint of $dx_j\wedge$.

(b) Show that $D + D^* \in \Psi^1_{\mathrm{iso}}(\mathbb{R}^n;\Lambda^*)$ is elliptic, and that $(D+D^*)^2 = DD^* + D^*D$ acts on $\mathcal{S}'(\mathbb{R}^n;\Lambda^k)$ as $H - n + 2k$.

(c) Deduce that the null space of $D + D^*$ on $\mathcal{S}'(\mathbb{R}^n;\Lambda^*)$ is $\mathbb{C}\exp(-|x|^2/2) \subset \mathcal{S}'(\mathbb{R}^n;\Lambda^0)$, that $\{u \in \mathcal{S}'(\mathbb{R}^n;\Lambda^0);\ Du = 0\} = \mathbb{C}\exp(-|x|^2/2)$, and that for $k \geq 1$ $\{u \in \mathcal{S}'(\mathbb{R}^n;\Lambda^k);\ Du = 0\} = D\,\mathcal{S}'(\mathbb{R}^n;\Lambda^{k-1})$.

(d) Conclude that $D + D^*$, as an element of $\Psi^1_{\mathrm{iso}}(\mathbb{R}^n;\Lambda^{\mathrm{even}},\Lambda^{\mathrm{odd}})$, is elliptic of index 1.

Exercise 7.6. [Isotropic operator wavefront set] For $a \in \Gamma^m(\mathbb{R}^{2n})$ define the *cone support* $\operatorname{cone\,supp}(a) \subset \mathbb{R}^{2n}\setminus\{0\}$ as the complement of the set of $\bar w \neq 0$ having a conic neighbourhood Γ such that $\sup_{w\in\Gamma,|w|\geq 1}\langle w\rangle^N|\partial^\alpha a(w)| < \infty$ for all N and α. For $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ set $\mathrm{WF}'_{\mathrm{iso}}(A) = \operatorname{cone\,supp}(\sigma_L(A))$. Show that

$$\mathrm{WF}'_{\mathrm{iso}}(A+B) \subset \mathrm{WF}'_{\mathrm{iso}}(A) \cup \mathrm{WF}'_{\mathrm{iso}}(B), \quad \mathrm{WF}'_{\mathrm{iso}}(AB) \subset \mathrm{WF}'_{\mathrm{iso}}(A) \cap \mathrm{WF}'_{\mathrm{iso}}(B),$$

$\mathrm{WF}'_{\mathrm{iso}}(A^*) = \mathrm{WF}'_{\mathrm{iso}}(A)$, that $\mathrm{WF}'_{\mathrm{iso}}(A)$ is the same when defined using the Weyl symbol, and that $\mathrm{WF}'_{\mathrm{iso}}(A) = \emptyset$ if and only if $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$.

Exercise 7.7. [Isotropic partition of unity] Let U_i, $i = 1,\dots,N$, be an open cover of $\mathbb{S}^{2n-1}$ and $\tilde U_i = \{w \neq 0;\ w/|w| \in U_i\}$. Show that there are $A_i \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n)$ with $\mathrm{WF}'_{\mathrm{iso}}(A_i) \subset \tilde U_i$ and $\mathrm{Id} - \sum_i A_i \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$.

Exercise 7.8. [Isotropic wavefront set] For $u \in \mathcal{S}'(\mathbb{R}^n)$ let

$$\mathrm{WF}_{\mathrm{iso}}(u) = \bigcap\{\mathrm{Char}_{\mathrm{iso}}(A);\ A \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n),\ Au \in \mathcal{S}(\mathbb{R}^n)\},$$

where $\mathrm{Char}_{\mathrm{iso}}(A)$ is the cone over the zero set of $\sigma_0(A)$ in $\mathbb{S}^{2n-1}$. Show that $\mathrm{WF}_{\mathrm{iso}}(u)$ is a closed cone, that $\mathrm{WF}_{\mathrm{iso}}(u) = \emptyset$ if and only if $u \in \mathcal{S}(\mathbb{R}^n)$, that $\mathrm{WF}_{\mathrm{iso}}(Au) \subset \mathrm{WF}'_{\mathrm{iso}}(A) \cap \mathrm{WF}_{\mathrm{iso}}(u)$ for $A \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$, and that $\mathrm{WF}_{\mathrm{iso}}(\hat u) = \{(\xi,-x);\ (x,\xi) \in \mathrm{WF}_{\mathrm{iso}}(u)\}$.

Exercise 7.9. Verify in detail that $\mathcal{F}\,\Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)\,\mathcal{F}^{-1} = \Psi^{l,m}_{\mathrm{sc}}(\mathbb{R}^n)$, and compute the left symbol of $\mathcal{F}A\mathcal{F}^{-1}$ in terms of that of $A \in \Psi^{m,l}_{\mathrm{sc}}(\mathbb{R}^n)$ modulo $S^{l-1,m-1}$.

Exercise 7.10. Let $q = \frac12(|x|^2 + |\xi|^2)$, so $\mathrm{Op^W}(q) = \frac12 H$. Using Example 7.12.16 and Theorem 7.12.12, show that $\exp(-\frac{i\pi}{4}H) = W_q(\frac{\pi}{2}) = e^{-in\pi/4}(2\pi)^{-n/2}\mathcal{F}$, and deduce that the Hermite functions satisfy $\mathcal{F}u_\alpha = (2\pi)^{n/2}(-i)^{|\alpha|}u_\alpha$.

Exercise 7.11. Show that for $q(x,\xi) = x\cdot\xi$, $\exp(-it\operatorname{Op}^{\mathrm{W}}(q))u(x) = e^{-nt/2}u(e^{-t}x)$. More generally, for $q(x,\xi) = \langle \beta x, \xi\rangle$, show that $\exp(-it\operatorname{Op}^{\mathrm{W}}(q)) = T_L$ with $L = e^{-t\beta}$, in the notation of Lemma 7.12.6.

Exercise 7.12. Let $S \in \mathrm{Sp}(2n,\mathbb{R})$ have invertible block B in (7.12.1). Applying Proposition 7.12.14 to SJ^{-1}, show that the elements of $\mathrm{Mp}(2n,\mathbb{R})$ over S have Schwartz kernels

$$\lambda(2\pi)^{-n/2}|\det B|^{-1/2}\exp(i(\tfrac{1}{2}\langle EB^{-1}x,x\rangle - \langle B^{-1}x,y\rangle + \tfrac{1}{2}\langle B^{-1}Ay,y\rangle)),$$
$$\lambda^2 = (-i)^n \operatorname{sgn}\det B.$$

Deduce Mehler's formula: for $0 < t < \pi$ the kernel of $\exp(-\frac{it}{2}H)$ is

$$e^{-in\pi/4}(2\pi\sin t)^{-n/2}\exp\Big(\frac{i((|x|^2+|y|^2)\cos t - 2x\cdot y)}{2\sin t}\Big).$$

[Fix the sign of λ by continuity in t, or by applying the operator to u_0.]

Exercise 7.13. Show that $U \in \mathcal{M}(2n)$ commutes with the harmonic oscillator H if and only if $\pi(U)$ is orthogonal, and that the orthogonal symplectic matrices are those of the form $\left(\begin{smallmatrix} X & Y \\ -Y & X\end{smallmatrix}\right)$ with $g = X + iY \in \mathrm{U}(n)$. Show that such a U satisfies $Uu_0 = \mu u_0$ with $|\mu| = 1$, and that $U \in \mathrm{Mp}(2n,\mathbb{R})$ if and only if $\mu^2 = \overline{\det g}$. Conclude that the elements of $\mathrm{Mp}(2n,\mathbb{R})$ commuting with H form a group isomorphic to $\{(g,\mu) \in \mathrm{U}(n)\times\mathrm{U}(1);\ \mu^2 = \overline{\det g}\}$, a connected double cover of $\mathrm{U}(n)$.

Exercise 7.14. With the notation of Exercise 7.8, show that $\mathrm{WF}_{\mathrm{iso}}(Uu) = \pi(U)(\mathrm{WF}_{\mathrm{iso}}(u))$ for all $u \in \mathcal{S}'(\mathbb{R}^n)$ and $U \in \mathcal{M}(2n)$.

Notes

This chapter follows Sections 3.1–3.13 of the original notes, reorganised, with the metaplectic group added from the revised version of the notes. The general (non-classical) isotropic class Γ^m is the symbol class introduced by Shubin [15], where much of the material here (in particular the harmonic oscillator and the isotropic Sobolev spaces) can be found; the Weyl calculus and its symplectic invariance are treated in [8, Section 18.5]. The scattering calculus on $\mathbb{R}^n$ is the model for Melrose's scattering calculus on asymptotically Euclidean manifolds (R. B. Melrose, *Spectral and scattering theory for the Laplacian on asymptotically Euclidean spaces*, 1994); see also the work of H. O. Cordes and C. Parenti on SG-calculi.

The main departures from the original are as follows. The reduction theorem is proved with a new class of amplitudes (Definition 7.2.4) which contains the products $a(x,\xi)b(y,\xi)$ and the Weyl amplitudes, and the isotropic decay of the remainder is obtained by intersecting weighted traditional symbol spaces; this replaces the original argument with a two-region decomposition of $\mathbb{R}^{3n}$. The generalized inverse of an elliptic operator is constructed on $\mathcal{S}'$ by a finite-dimensional argument, the essential norm estimate is proved with coherent states (which also give the necessity of ellipticity for the Fredholm property), the harmonic oscillator section uses the generalized inverse H^{-1} to prove completeness of the Hermite functions, and the Sobolev spaces of all real orders are treated uniformly. Problems on traces and determinants from the original list are in Chapter 8.

The section on the metaplectic group is based on Section 4.8 of Melrose's revised notes (*From microlocal to global analysis*, 2008), but the construction is different. There, a group of (not necessarily unitary) operators generated by multiplications by $e^{iq(x)}$, linear changes of variables, partial Fourier transforms and constants is shown to be a central extension of

$\mathrm{Sp}(2n,\mathbb{R})$ by $\mathbb{C}^*$, a local lift near the identity is defined with generating functions, and its multiplicativity is checked by explicit one-dimensional Gaussian computations; several of these steps are only sketched there. Here the normalization is instead fixed by the action on the ground state u_0: the phase φ of (7.12.15) is shown to be a character by a differentiation argument (Lemma 7.12.11), which yields the double cover, the generation by exponentials of quadratic forms and the explicit generating-function form (Proposition 7.12.14) at once. For other treatments see [8, Section 18.5] and G. B. Folland, *Harmonic analysis in phase space* (1989), Chapter 4.

Chapter 8

Traces, determinants and the index

For a linear map between finite-dimensional spaces three numbers are always available: the trace, the determinant, and the index $\dim(\text{domain}) - \dim(\text{target})$. The trace vanishes on commutators, the determinant is multiplicative and detects invertibility, and the index of a square matrix is zero. This chapter asks how much of this survives for operators on $L^2(\mathbb{R}^n)$, and finds that the failures are as interesting as the successes: the trace does not extend to all operators, and the precise way in which it fails computes the index.

In this chapter we use the isotropic calculus of Chapter 7 as a laboratory for three closely related constructions: traces, determinants and the index. On the residual algebra $\Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$, whose elements have Schwartz kernels, there is a unique trace, and it gives rise to the Fredholm determinant on $\mathrm{Id} + \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$; the determinant leads directly to analytic Fredholm theory and so to the meromorphy of the resolvent of an elliptic operator. On the full algebra of polyhomogeneous operators of integral order the trace no longer extends, but complex powers allow us to define the *residue trace*, a genuine trace which only sees one term of the symbol expansion, and the *regularized trace*, which extends the trace but fails to be a trace; the failure is measured by the residue trace and an exterior derivation. These ingredients give a formula for the index of an elliptic operator, an obstruction to inverting families of elliptic operators by smoothing perturbations, and finally the determinant line bundle of such a family, together with a natural connection whose curvature is that obstruction. Figure 8.1 shows how these constructions depend on each other.

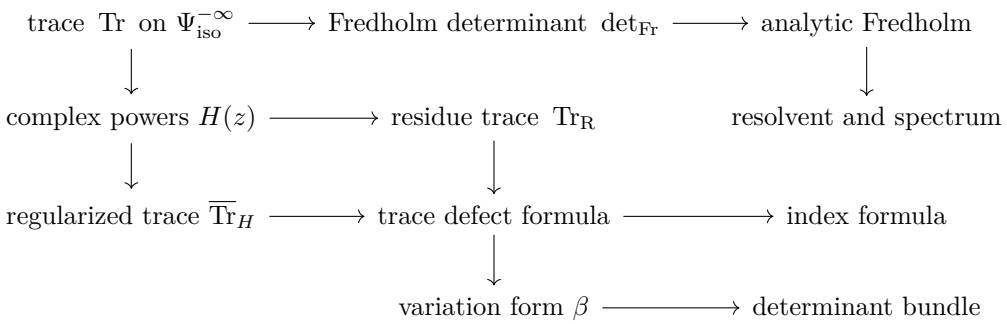


Figure 8.1: Roadmap of the chapter. The top row uses only the residual algebra; the lower rows use complex powers to go beyond it.

8.1 Setup

We recall from Chapter 7 what we need, and fix notation. Write $Z = (x, \xi) \in \mathbb{R}^{2n}$. For $m \in \mathbb{R}$ the space of *isotropic symbols* of order m is

$$\Gamma^m(\mathbb{R}^{2n}) = \{a \in \mathcal{C}^\infty(\mathbb{R}^{2n});\ |D_Z^\alpha a(Z)| \le C_\alpha \langle Z\rangle^{m-|\alpha|}\ \forall\, \alpha\},$$

a Fréchet space with the best constants C_α as seminorms, and $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n) = \mathrm{Op}(\Gamma^m(\mathbb{R}^{2n}))$, with the left quantization of the Introduction. The polyhomogeneous subspace $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ consists of the operators whose left symbol has an expansion $a \sim \sum_{j\ge 0} a_{m-j}$ with a_{m-j} homogeneous of degree $m-j$ in $|Z| \ge 1$; equivalently, $\langle Z\rangle^{-m} a$ is smooth on the radial compactification $\overline{\mathbb{R}^{2n}}$, i.e. in polar coordinates $Z = \theta/t$, $t = |Z|^{-1}$, it is smooth up to $t = 0$. The same definition with m replaced by $z \in \mathbb{C}$ gives the operators $\Psi^z_{\mathrm{iso,ph}}(\mathbb{R}^n) \subset \Psi^{\Re z}_{\mathrm{iso}}(\mathbb{R}^n)$ of complex order. We write $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n) = \bigcup_{m\in\mathbb{Z}} \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$. The *residual* operators are

$$\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n) = \bigcap_m \Psi^m_{\mathrm{iso}}(\mathbb{R}^n) = \bigcap_m \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n),$$

which are exactly the operators with Schwartz kernels in $\mathcal{S}(\mathbb{R}^{2n})$; we identify $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ with $\mathcal{S}(\mathbb{R}^{2n})$, including its topology.

All of this applies to operators on $\mathbb{C}^N$-valued functions: $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$ consists of $N \times N$ matrices of elements of $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ (see Exercise 8.9). Everything below holds in this generality, with tr denoting the trace of matrices; to keep the notation light we often suppress $\mathbb{C}^N$.

We use the following results from Chapter 7.

(I1) *Algebra.* Composition maps $\Psi^m_{\mathrm{iso}} \times \Psi^{m'}_{\mathrm{iso}}$ to $\Psi^{m+m'}_{\mathrm{iso}}$ continuously for the symbol topologies, and $\Psi^z_{\mathrm{iso,ph}} \times \Psi^{z'}_{\mathrm{iso,ph}}$ to $\Psi^{z+z'}_{\mathrm{iso,ph}}$; the left symbol of a composite is given by $\sigma_L(AB) \sim \sum_\alpha \frac{1}{\alpha!} \partial_\xi^\alpha a\, D_x^\alpha b$. Adjoints and transposes of elements of Ψ^m_{iso} (resp. $\Psi^z_{\mathrm{iso,ph}}$) lie in the same space. In particular $\Psi^{-\infty}_{\mathrm{iso}}$ is a two-sided ideal. The principal symbol is multiplicative.

(I2) *Boundedness.* Every element of Ψ^m_{iso} is continuous on $\mathcal{S}(\mathbb{R}^n)$ and on $\mathcal{S}'(\mathbb{R}^n)$; elements of Ψ^0_{iso} are bounded on $L^2(\mathbb{R}^n)$, with norm bounded by a symbol seminorm; elements of negative order are compact on L^2, and residual operators map $\mathcal{S}'(\mathbb{R}^n)$ continuously into $\mathcal{S}(\mathbb{R}^n)$.

(I3) *Ellipticity.* An element $A \in \Psi^z_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$ is elliptic if its principal symbol is invertible on the sphere. It then has a parametrix $B \in \Psi^{-z}_{\mathrm{iso,ph}}$, with $AB - \mathrm{Id}$ and $BA - \mathrm{Id}$ residual; any two parametrices differ by a residual operator. If $Au \in \mathcal{S}$ for some $u \in \mathcal{S}'$, then $u \in \mathcal{S}$. There is a *generalized inverse* $B \in \Psi^{-z}_{\mathrm{iso,ph}}$ with

$$AB = \mathrm{Id} - \Pi_1, \quad BA = \mathrm{Id} - \Pi_0, \tag{8.1.1}$$

where Π_0, Π_1 are the L^2-orthogonal projections onto the finite dimensional spaces $\mathrm{Nul}\, A \subset \mathcal{S}$ and $\mathrm{Nul}\, A^* \subset \mathcal{S}$ (null spaces on $\mathcal{S}'$). Replacing B by $(\mathrm{Id} - \Pi_0)B(\mathrm{Id} - \Pi_1)$, which changes neither identity since $A\Pi_0 = 0$ and $\Pi_1 A = 0$, we may and do assume $\Pi_0 B = 0 = B\Pi_1$. In particular an elliptic A which is a bijection on $\mathcal{S}'$ (we then say A is *invertible*) has inverse in $\Psi^{-z}_{\mathrm{iso,ph}}$. For order 0, A is invertible in this sense if and only if it is invertible on L^2.

(I4) *The residual algebra.* If $E \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ then $\mathrm{Id}+E$ is Fredholm on L^2 of index zero; if it is invertible on L^2 its inverse is $\mathrm{Id}+F$ with $F \in \Psi^{-\infty}_{\mathrm{iso}}$. If $\mathrm{Nul}(\mathrm{Id}+E) \neq \{0\}$ there is a finite rank $F' \in \Psi^{-\infty}_{\mathrm{iso}}$, mapping $\mathrm{Nul}(\mathrm{Id}+E)$ isomorphically onto $\mathrm{Nul}(\mathrm{Id}+E^*)$ and vanishing on its orthocomplement, such that $\mathrm{Id}+E+sF'$ is invertible for all $s \neq 0$. (This is also the special case $A = \mathrm{Id}+E$ of Proposition 8.4.2 below.) The series $\exp E = \sum_k E^k/k!$ converges in $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}$.

(I5) *The harmonic oscillator.* The eigenfunctions $u_\alpha = c_\alpha C^\alpha e^{-|x|^2/2}$, $\alpha \in \mathbb{N}_0^n$, of $H = |D|^2 + |x|^2$ (with $C_j = D_j + ix_j$ and normalizing constants $c_\alpha > 0$) are real-valued and form an orthonormal basis of $L^2(\mathbb{R}^n)$; $Hu_\alpha = (n+2|\alpha|)u_\alpha$; and each Schwartz seminorm of u_α is bounded by a polynomial in $|\alpha|$. See also Exercise 8.6.

Definition 8.1.1. Let Y be a manifold and $\Omega \subset \mathbb{C}$ open.

(a) A family $E_y \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, $y \in Y$, is *smooth* if its kernel $E(y,x,x')$ is in $\mathcal{C}^\infty(Y \times \mathbb{R}^{2n})$ and, in local coordinates, $\langle (x,x')\rangle^k \partial_y^\gamma \partial^\alpha_{x,x'} E$ is bounded on $K \times \mathbb{R}^{2n}$ for every compact K and all k, γ, α. A family $E(z)$, $z \in \Omega$, is *holomorphic* if it is smooth and E is holomorphic in z.

(b) Let h be holomorphic on Ω. A family $A(z) = \mathrm{Op}(a(z))$, $z \in \Omega$, is a *holomorphic family of order* $h(z)$ if $a \in \mathcal{C}^\infty(\Omega \times \mathbb{R}^{2n})$ is holomorphic in z and

$$a(z; r\theta) = r^{h(z)}\, b(z; 1/r, \theta), \quad r \geq 1,\ \theta \in \mathbb{S}^{2n-1}, \tag{8.1.2}$$

with $b \in \mathcal{C}^\infty(\Omega \times [0,1] \times \mathbb{S}^{2n-1})$ holomorphic in z. Then $A(z) \in \Psi^{h(z)}_{\mathrm{iso,ph}}$, and the term of homogeneity $h(z) - j$ in the expansion of $a(z)$ is $|Z|^{h(z)-j} b_j(z; Z/|Z|)$, with $b_j(z;\theta) = \frac{1}{j!}\partial_t^j b(z; 0, \theta)$. Families of fixed order m depending smoothly on $y \in Y$ are defined in the same way, with b smooth in (y,t,θ).

(I6) *Families.* Composites of holomorphic (or smooth) families are holomorphic (or smooth), with orders adding; this includes composites with residual families. If $A(z)$ is a holomorphic family of elliptic operators of order $h(z)$, then near each point of Ω there is a holomorphic family $B(z)$ of order $-h(z)$ such that $A(z)B(z) - \mathrm{Id}$ and $B(z)A(z) - \mathrm{Id}$ are holomorphic families in $\Psi^{-\infty}_{\mathrm{iso}}$ (the parametrix construction of Chapter 7, with Borel's lemma applied in the variable t uniformly in z); the same holds for smooth families. Finally

$$H(z) = \mathrm{Op}(\langle Z\rangle^z)\ \mathrm{Id}_N \tag{8.1.3}$$

is an entire holomorphic family of order z with $H(0) = \mathrm{Id}$.

We complete these facts with three elementary lemmas about residual operators. Fix seminorms on $\mathcal{S}(\mathbb{R}^{2n})$ by

$$\|K\|_{(k)} = \sum_{|\alpha|+|\beta|+|\gamma|+|\delta| \leq k} \sup_{x,x'} |x^\alpha D_x^\beta x'^\gamma D_{x'}^\delta K(x,x')|.$$

For $E \in \Psi^{-\infty}_{\mathrm{iso}}$ we have $\|E\|_{\mathcal{B}(L^2)} \leq \|E\|_{L^2(\mathbb{R}^{2n})} \leq C\|E\|_{(2n+1)}$.

Lemma 8.1.2. *If $E_1, E_2 \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and $T \in \mathcal{B}(L^2(\mathbb{R}^n))$, then $E_1 T E_2 \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and for each k there are C and k' such that $\|E_1 T E_2\|_{(k)} \leq C\|T\|_{\mathcal{B}(L^2)}\|E_1\|_{(k')}\|E_2\|_{(k')}$.*

Proof. The map $x' \mapsto E_2(\cdot, x')$ is smooth into $L^2(\mathbb{R}^n)$, so the kernel of $E_1 T E_2$ is $K(x,x') = \int E_1(x,w)\,(TE_2(\cdot,x'))(w)\,dw$, and we may differentiate under T: $x^\alpha D_x^\beta x'^\gamma D_{x'}^\delta K(x,x') = \int x^\alpha D_x^\beta E_1(x,w)\,(T[x'^\gamma D_{x'}^\delta E_2(\cdot,x')])(w)\,dw$. By Cauchy–Schwarz this is bounded by $\|T\|$ times $\sup_x \|x^\alpha D_x^\beta E_1(x,\cdot)\|_{L^2} \sup_{x'} \|x'^\gamma D_{x'}^\delta E_2(\cdot,x')\|_{L^2}$, and each factor is bounded by a seminorm $\|\cdot\|_{(k')}$. □

Lemma 8.1.3 (Finite rank approximation)**.** *Let π_N be the orthogonal projection onto $L_N = \operatorname{span}\{u_\alpha; |\alpha| \le N\}$. For $E \in \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$, $\pi_N E \pi_N \to E$ in $\Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ as $N \to \infty$. If E_y is a smooth family, the convergence holds locally uniformly in y together with all y-derivatives.*

Proof. The functions $u_\alpha \otimes u_\beta$ form an orthonormal basis of $L^2(\mathbb{R}^{2n})$ of eigenfunctions of the harmonic oscillator $H_{2n} = H_x + H_{x'}$ with eigenvalues $2n + 2|\alpha| + 2|\beta|$. Thus the coefficients $e_{\alpha\beta} = (E, u_\alpha \otimes u_\beta)$ satisfy $(2n + 2|\alpha| + 2|\beta|)^k |e_{\alpha\beta}| = |(H_{2n}^k E, u_\alpha \otimes u_\beta)| \le \|H_{2n}^k E\|_{L^2}$, a continuous seminorm of E. Since the seminorms of $u_\alpha \otimes u_\beta$ grow only polynomially, $\sum e_{\alpha\beta} u_\alpha \otimes u_\beta$ converges absolutely to E in $\mathcal{S}(\mathbb{R}^{2n})$, and its partial sum over $|\alpha|, |\beta| \le N$ is the kernel of $\pi_N E \pi_N$. For a smooth family the same estimates hold for $\partial_y^\gamma E_y$, locally uniformly. □

Lemma 8.1.4. *The set $G_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$ of $\mathrm{Id} + E$, $E \in \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$, which are invertible on L^2 is a group, open in $\mathrm{Id} + \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$. The inverse is $\mathrm{Id} + F$ with*

$$F = -E + E(\mathrm{Id}+E)^{-1}E \in \Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n), \tag{8.1.4}$$

and F depends continuously on E. If E_y is a smooth (holomorphic) family with $\mathrm{Id} + E_y$ invertible for all y, then F_y is a smooth (holomorphic) family.

Proof. From $(\mathrm{Id}+E)(\mathrm{Id}+F) = \mathrm{Id} = (\mathrm{Id}+F)(\mathrm{Id}+E)$ we get $F = -E - EF$ and $F = -E - FE$, so $F = -E + E^2 + EFE = -E + E(\mathrm{Id}+E)^{-1}E$, which is residual by Lemma 8.1.2. Openness holds because invertibility in $\mathcal{B}(L^2)$ is open and $\|E\|_{\mathcal{B}(L^2)} \le C\|E\|_{(2n+1)}$. Inversion is smooth (indeed analytic) on the invertible elements of the Banach algebra $\mathcal{B}(L^2)$, so $y \mapsto (\mathrm{Id}+E_y)^{-1}$ is smooth (holomorphic) in norm; the trilinear estimate of Lemma 8.1.2 and the product rule then show that F_y is smooth (holomorphic), and continuity follows in the same way. □

Lemma 8.1.5 (Local inversion)**.** *Let $A(z)$ be a holomorphic family of elliptic operators of order $h(z)$ on Ω, and suppose $A(p)$ is invertible. Then $A(z)$ is invertible for z near p and $A(z)^{-1}$ is a holomorphic family of order $-h(z)$ near p. The same holds for smooth families of fixed order.*

Proof. Let $B(z)$ be a holomorphic parametrix near p, $B(z)A(z) = \mathrm{Id} + E(z)$, $A(z)B(z) = \mathrm{Id} + E'(z)$. Set $B'(z) = B(z) - E(p)A(p)^{-1}$; then $B'(z)A(z) = \mathrm{Id} + E_1(z)$ with $E_1(z) = E(z) - E(p)A(p)^{-1}A(z)$ a holomorphic residual family vanishing at p. By Lemma 8.1.4, $\mathrm{Id} + E_1(z)$ is invertible near p with holomorphic inverse $\mathrm{Id} + S(z)$, so $L(z) = (\mathrm{Id} + S(z))B'(z)$ is a holomorphic family of order $-h(z)$ with $L(z)A(z) = \mathrm{Id}$. In the same way, with $B''(z) = B(z) - A(p)^{-1}E'(p)$, we find a right inverse $R(z)$. Then $L = LAR = R$ is a two-sided inverse. □

8.2 Traces on the residual algebra

We start with the simplest invariant. For matrices the trace is characterized by linearity, the commutator identity and normalization; we show that the same characterization singles out a unique continuous trace on $\Psi_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n)$, given by integrating the kernel over the diagonal.

Matrices

On the algebra $M(N;\mathbb{C})$ of $N\times N$ matrices, $\operatorname{tr} A=\sum_i A_{ii}$ satisfies $\operatorname{tr}(AB)=\sum_{i,k}A_{ik}B_{ki}=\operatorname{tr}(BA)$, that is,

$$\operatorname{tr}([A,B])=0\quad\forall\, A,B\in M(N;\mathbb{C}),$$

and the normalization $\operatorname{tr}\mathrm{Id}=N$. These properties characterize it.

Lemma 8.2.1. *If $F: M(N;\mathbb{C})\to\mathbb{C}$ is linear and $F([A,B])=0$ for all A,B, then $F=F(E_{11})\operatorname{tr}$.*

Proof. Let E_{jk} be the matrix units, $E_{jk}E_{pq}=\delta_{kp}E_{jq}$. For $j\neq k$, $E_{jk}=[E_{jj},E_{jk}]$, so $F(E_{jk})=0$; and $E_{jj}-E_{11}=[E_{j1},E_{1j}]$, so $F(E_{jj})=F(E_{11})$. Hence $F(A)=\sum_{j,k}A_{jk}F(E_{jk})=F(E_{11})\operatorname{tr}A$. □

The trace on residual operators

For $A\in\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ with kernel $A(x,x')$ we set

$$\operatorname{Tr}(A)=\int_{\mathbb{R}^n}\operatorname{tr}A(x,x)\,dx. \tag{8.2.1}$$

This is a continuous linear functional on $\Psi^{-\infty}_{\mathrm{iso}}$. It extends to operators of sufficiently negative order. If $a\in\Gamma^m(\mathbb{R}^{2n})$ with $m<-2n$, then $\int|a(x,\xi)|\,d\xi\le C\langle x\rangle^{m+n}$, so the kernel $K(x,x')=\int e^{i(x-x')\cdot\xi}a(x,\xi)d\!\!\!^{-}\xi$ of $\operatorname{Op}(a)$ is continuous, $|K(x,x)|\le C\langle x\rangle^{m+n}$ is integrable, and we define

$$\operatorname{Tr}(\operatorname{Op}(a))=\int K(x,x)\,dx=(2\pi)^{-n}\int_{\mathbb{R}^{2n}}\operatorname{tr}a(x,\xi)\,dx\,d\!\!\!^{-}\xi,\qquad a\in\Gamma^m(\mathbb{R}^{2n}),\ m<-2n. \tag{8.2.2}$$

This is continuous on Ψ^m_{iso} for the symbol topology and agrees with (8.2.1) on residual operators. In particular Tr is defined on $\Psi^z_{\mathrm{iso,ph}}$ for $\Re z<-2n$.

Lemma 8.2.2. *If $A\in\Psi^p_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$, $B\in\Psi^q_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ and $p+q<-2n$, then $\operatorname{Tr}(AB)=\operatorname{Tr}(BA)$. In particular this holds if one of A, B is residual and the other is arbitrary.*

Proof. For residual A and B this is Fubini's theorem,

$$\operatorname{Tr}(AB)=\iint\operatorname{tr}(A(x,w)B(w,x))\,dw\,dx=\operatorname{Tr}(BA).$$

In general, let $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{2n})$ equal 1 near 0, and let $a_k(Z)=\chi(Z/k)a(Z)$, $b_k(Z)=\chi(Z/k)b(Z)$. Then $a-a_k$ is supported in $|Z|\ge k$, where $\langle Z\rangle^{-\epsilon}\le k^{-\epsilon}$, and the derivatives of $\chi(Z/k)$ are bounded by $C\langle Z\rangle^{-|\beta|}$ uniformly in k; hence $a_k\to a$ in $\Gamma^{p+\epsilon}$ for every $\epsilon>0$, and similarly $b_k\to b$ in $\Gamma^{q+\epsilon}$. Choose ϵ with $p+q+2\epsilon<-2n$. The operators $A_k=\operatorname{Op}(a_k)$, $B_k=\operatorname{Op}(b_k)$ are residual, so $\operatorname{Tr}(A_kB_k)=\operatorname{Tr}(B_kA_k)$, and by the continuity of composition (see (I1)) and of the trace on $\Psi^{p+q+2\epsilon}_{\mathrm{iso}}$ we may let $k\to\infty$. □

Lemma 8.2.3. *Let $L\subset\mathcal{S}(\mathbb{R}^n;\mathbb{C}^N)$ be a finite dimensional subspace and Π the L^2-orthogonal projection onto L. Then $\Pi\in\Psi^{-\infty}_{\mathrm{iso}}$. If $T\in\Psi^{-\infty}_{\mathrm{iso}}$ satisfies $T=\Pi T\Pi$, then $\operatorname{Tr}(T)=\operatorname{tr}(T|_L)$, the trace of T as an endomorphism of L. More generally $\operatorname{Tr}(\Pi T)=\operatorname{Tr}(\Pi T\Pi)=\operatorname{tr}(\Pi T|_L)$ for all $T\in\Psi^m_{\mathrm{iso}}$.*

Proof. If $\varphi_1,\dots,\varphi_l$ is an orthonormal basis of L, the kernel of Π is $\sum_i\varphi_i(x)\overline{\varphi_i(x')}$ (a matrix-valued function when $N>1$) and that of $T=\Pi T\Pi$ is $\sum_{i,j}t_{ij}\varphi_i(x)\overline{\varphi_j(x')}$, $t_{ij}=(T\varphi_j,\varphi_i)$. Thus $\operatorname{Tr}(T)=\sum_{i,j}t_{ij}(\varphi_i,\varphi_j)=\sum_i t_{ii}$. Finally $\operatorname{Tr}(\Pi T(\mathrm{Id}-\Pi))=\operatorname{Tr}(T(\mathrm{Id}-\Pi)\Pi)=0$ by Lemma 8.2.2. □

Taking $L = L_N$ in Lemma 8.2.3 and using Lemma 8.1.3, we see that $\operatorname{Tr}(A) = \lim_N \sum_{|\alpha|\le N}(Au_\alpha, u_\alpha)$, so Tr coincides with the trace of A as a trace-class operator on L^2. The trace is essentially the only continuous trace on the residual algebra.

Proposition 8.2.4. *If $F : \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n) \to \mathbb{C}$ is continuous, linear and satisfies $F([A,B]) = 0$ for all $A, B \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$, then $F = c\operatorname{Tr}$ for some $c \in \mathbb{C}$. Consequently $F([A,B]) = 0$ also for all $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and $B \in \Psi^{m}_{\mathrm{iso}}(\mathbb{R}^n)$, $m \in \mathbb{R}$.*

Proof. For a finite dimensional $L \subset \mathcal{S}(\mathbb{R}^n)$ with orthonormal basis $\varphi_1, \dots, \varphi_l$, the operators P_{ij} with kernels $\varphi_i(x)\overline{\varphi_j(x')}$ satisfy $P_{ij}P_{kl} = \delta_{jk}P_{il}$, so they span a subalgebra of $\Psi^{-\infty}_{\mathrm{iso}}$ isomorphic to $M(l;\mathbb{C})$, on which Tr is the matrix trace by Lemma 8.2.3. By Lemma 8.2.1, $F = F(P_{11})\operatorname{Tr}$ on this subalgebra. Applying this to $L = \operatorname{span}\{\varphi, \psi\}$ for two unit vectors $\varphi, \psi \in \mathcal{S}$ shows that $c = F(\varphi \otimes \overline{\varphi})$ is independent of φ. Every finite rank $A \in \Psi^{-\infty}_{\mathrm{iso}}$ has range, and range of A^*, in $\mathcal{S}$, and so lies in the subalgebra associated with $L = \operatorname{Ran} A + \operatorname{Ran} A^*$; thus $F = c\operatorname{Tr}$ on finite rank operators, and hence everywhere by Lemma 8.1.3 and continuity. The last statement then follows from Lemma 8.2.2. □

Discussion. A useful way to think of $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ is through the Hermite basis u_α. By Lemma 8.1.3, a residual operator E is the same as an infinite matrix $(e_{\alpha\beta})$ whose entries decay faster than any power of $|\alpha| + |\beta|$. For such matrices the trace $\sum_\alpha e_{\alpha\alpha}$ converges absolutely and is the limit of the traces of the finite truncations $\pi_N E \pi_N$. Most statements in this and the next section are proved by exactly this reduction to finite matrices, followed by a limit.

8.3 The Fredholm determinant

The determinant of $\mathrm{Id} + E$ cannot be defined as a product over a basis, since there are infinitely many factors. Instead we use the derivative of $\log\det$, which is a trace, and integrate it along a path.

For matrices, $\det : M(N;\mathbb{C}) \to \mathbb{C}$ is multiplicative, $\det \mathrm{Id} = 1$, and $\mathrm{GL}(N;\mathbb{C}) = \{A; \det A \neq 0\}$. Multiplicativity and normalization do not determine det (consider $(\det)^k$), but they do together with the derivative condition $\frac{d}{ds}\det(\mathrm{Id} + sA)|_{s=0} = \operatorname{tr} A$, and for any smooth curve A_s in $\mathrm{GL}(N;\mathbb{C})$ one has Jacobi's formula

$$\frac{d}{ds}\det A_s = \det A_s \operatorname{tr}\Big(A_s^{-1}\frac{dA_s}{ds}\Big) \tag{8.3.1}$$

(Exercise 8.7). Locally along such a curve this reads $d\log\det A = \operatorname{tr}(A^{-1}dA)$; the right side is a closed 1-form on $\mathrm{GL}(N;\mathbb{C})$, independent of any choice of logarithm. We use the analogous 1-form

$$\alpha = \operatorname{Tr}(A^{-1}dA) \quad \text{on} \quad G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n),$$

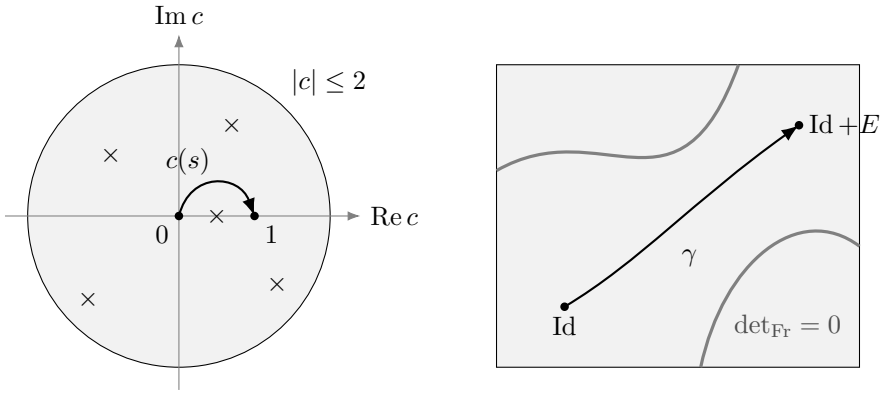
where, at $A = \mathrm{Id} + E$, dA is the identification of tangent vectors with elements of $\Psi^{-\infty}_{\mathrm{iso}}$ (so $A^{-1}dA$ takes values in $\Psi^{-\infty}_{\mathrm{iso}}$ and Tr applies). To avoid the differential geometry of Fréchet manifolds we only integrate α along paths. A *path* in $G^{-\infty}_{\mathrm{iso}}$ is a continuous map $\gamma : [s_0, s_1] \to G^{-\infty}_{\mathrm{iso}}$, $\gamma(s) = \mathrm{Id} + E(s)$, which is piecewise a smooth family in the sense of Definition 8.1.1, and we set

$$\int_\gamma \alpha = \int_{s_0}^{s_1} \operatorname{Tr}\big(\gamma(s)^{-1}\gamma'(s)\big)\, ds.$$

That α is closed is the content of Exercise 8.8; we shall not need it, since the following lemma, which compares α with its finite dimensional analogue, gives path independence directly.

Lemma 8.3.1. *$G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ is path connected: if $\mathrm{Id}+E \in G^{-\infty}_{\mathrm{iso}}$ there is a path in $G^{-\infty}_{\mathrm{iso}}$ from Id to $\mathrm{Id}+E$.*

Proof. E is compact on L^2, so its non-zero spectrum consists of isolated eigenvalues accumulating only at 0. For $c \neq 0$, $\mathrm{Id}+cE$ is invertible unless $-1/c$ is an eigenvalue, so the set Λ of c with $|c| \leq 2$ and $\mathrm{Id}+cE$ not invertible is finite, and $0, 1 \notin \Lambda$. If $c : [0,1] \to \{|c| \leq 2\} \setminus \Lambda$ is a smooth path from 0 to 1, then $\mathrm{Id}+c(s)E$ is the required path. □



(a) Λ (×) is finite (b) a path in $G^{-\infty}_{\mathrm{iso}}$

Figure 8.2: (a) The proof of Lemma 8.3.1: the values of c for which $\mathrm{Id}+cE$ is not invertible form a finite set in $\{|c| \leq 2\}$, which a path from 0 to 1 can avoid. (b) Schematic picture of $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$: the non-invertible operators form a (singular) hypersurface, the zero set of $\det_{\mathrm{Fr}}$, and $\det_{\mathrm{Fr}}(\mathrm{Id}+E)$ is defined by integrating $\mathrm{Tr}(\gamma^{-1}\gamma')$ along any path γ from Id that avoids it.

Lemma 8.3.2. *Let $\gamma(s) = \mathrm{Id}+E(s)$, $s \in [0,1]$, be a path in $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ with $E(0) = 0$. Then*

$$\exp\Big(\int_\gamma \alpha\Big) = \lim_{N\to\infty} \det\big(\mathrm{Id}+\pi_N E(1)|_{L_N}\big),$$

where $\pi_N E(1)|_{L_N}$ is regarded as an endomorphism of the finite dimensional space L_N. In particular the left side depends only on $\gamma(1)$.

Proof. Set $E_N(s) = \pi_N E(s)\pi_N$ and $\gamma_N = \mathrm{Id}+E_N$. By Lemma 8.1.3, $E_N \to E$ and $E_N' \to E'$ in $\Psi^{-\infty}_{\mathrm{iso}}$, uniformly on each interval of smoothness. Let $M = \sup_s \|\gamma(s)^{-1}\|_{\mathcal{B}(L^2)} < \infty$. For N large, $\|E_N(s) - E(s)\|_{\mathcal{B}(L^2)} < 1/(2M)$ for all s, so $\gamma_N(s)$ is invertible with $\gamma_N(s)^{-1} \to \gamma(s)^{-1}$ uniformly in norm. By (8.1.4) and Lemma 8.1.2, $\gamma_N^{-1} - \mathrm{Id} \to \gamma^{-1} - \mathrm{Id}$ uniformly in $\Psi^{-\infty}_{\mathrm{iso}}$, hence $\gamma_N^{-1}\gamma_N' \to \gamma^{-1}\gamma'$ uniformly in $\Psi^{-\infty}_{\mathrm{iso}}$, and, Tr being continuous, $\int_{\gamma_N} \alpha \to \int_\gamma \alpha$.

Now $\gamma_N(s)$ preserves the decomposition $L^2 = L_N \oplus L_N^\perp$, acting as $g_N(s) = \mathrm{Id}+\pi_N E(s)|_{L_N}$ on L_N and as the identity on $L_N^\perp$. Hence $\gamma_N^{-1}\gamma_N' = \pi_N \gamma_N^{-1}\gamma_N' \pi_N$ acts as $g_N^{-1}g_N'$ on L_N, and by Lemma 8.2.3 and Jacobi's formula (8.3.1),

$$\int_{\gamma_N} \alpha = \int_0^1 \mathrm{tr}(g_N^{-1}g_N')\, ds, \qquad \exp\int_{\gamma_N} \alpha = \frac{\det g_N(1)}{\det g_N(0)} = \det g_N(1).$$

Letting $N \to \infty$ proves the lemma. □

Definition 8.3.3. The *Fredholm determinant* $\det_{\mathrm{Fr}} : \mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N) \to \mathbb{C}$ is

$$\det_{\mathrm{Fr}}(\mathrm{Id}+E) = \begin{cases} \exp\bigl(\int_\gamma \alpha\bigr) & \text{if } \mathrm{Id}+E \in G^{-\infty}_{\mathrm{iso}}, \\ 0 & \text{otherwise,} \end{cases}$$

where γ is any path in $G^{-\infty}_{\mathrm{iso}}$ from Id to $\mathrm{Id}+E$.

By Lemmas 8.3.1 and 8.3.2 this is well defined (see Figure 8.2).

Question. Why not simply define $\det_{\mathrm{Fr}}(\mathrm{Id}+E) = \prod_j (1+\lambda_j)$, the product over the eigenvalues λ_j of E?

Answer. This is in fact correct for trace class E (a theorem of Lidskii), but proving it requires a good deal of spectral theory of non-self-adjoint compact operators, and the product formula is awkward for proving multiplicativity or holomorphy. The path definition uses only the trace property and finite-dimensional determinants, and the limit formula of Lemma 8.3.2 shows directly that it agrees with the determinants of the finite truncations $\pi_N E \pi_N$. □

Concatenating a path from Id to G_1 with a path δ in $G^{-\infty}_{\mathrm{iso}}$ from G_1 to G_2 gives

$$\det_{\mathrm{Fr}}(G_2) = \det_{\mathrm{Fr}}(G_1)\exp\Bigl(\int_\delta \alpha\Bigr). \tag{8.3.2}$$

In particular, if G_y is a smooth family in $G^{-\infty}_{\mathrm{iso}}$, then $\det_{\mathrm{Fr}}(G_y)$ is smooth and $d\log\det_{\mathrm{Fr}}(G_y) = \mathrm{Tr}(G_y^{-1}dG_y)$: applying (8.3.2) along curves shows that the directional derivatives are $\det_{\mathrm{Fr}}(G)\,\mathrm{Tr}(G^{-1}\partial G)$, which are continuous, and iterating gives smoothness. For a holomorphic family the Cauchy–Riemann equations follow in the same way, so $\det_{\mathrm{Fr}}(G(z))$ is holomorphic. Also $\det_{\mathrm{Fr}}$ is continuous on $G^{-\infty}_{\mathrm{iso}}$, since for G' close to G the segment from G to G' lies in $G^{-\infty}_{\mathrm{iso}}$.

Proposition 8.3.4. *The Fredholm determinant has the following properties.*

(i) $\det_{\mathrm{Fr}}(AB) = \det_{\mathrm{Fr}}(A)\det_{\mathrm{Fr}}(B)$ *for* $A, B \in \mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}$, *and* $\det_{\mathrm{Fr}}(\mathrm{Id}) = 1$.

(ii) $\det_{\mathrm{Fr}}(A) \neq 0$ *if and only if* A *is invertible.*

(iii) $\det_{\mathrm{Fr}}(\exp E) = \exp\mathrm{Tr}(E)$ *for all* $E \in \Psi^{-\infty}_{\mathrm{iso}}$; *if* $\|E\|_{\mathcal{B}(L^2)} < 1$, *then* $\log(\mathrm{Id}+E) = \sum_{k\geq 1}(-1)^{k+1}E^k/k$ *converges in* $\Psi^{-\infty}_{\mathrm{iso}}$ *and* $\det_{\mathrm{Fr}}(\mathrm{Id}+E) = \exp\mathrm{Tr}\log(\mathrm{Id}+E)$.

(iv) If $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ *is invertible (with inverse in* $\Psi^{-m}_{\mathrm{iso,ph}}$*) and* $B \in \mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}$, *then* $ABA^{-1} \in \mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}$ *and* $\det_{\mathrm{Fr}}(ABA^{-1}) = \det_{\mathrm{Fr}}(B)$.

Proof. (i) If A and B are invertible, let γ_1, γ_2 be paths from Id to A and B; then $\gamma = \gamma_1\gamma_2$ is a path from Id to AB and $\gamma^{-1}\gamma' = \gamma_2^{-1}(\gamma_1^{-1}\gamma_1')\gamma_2 + \gamma_2^{-1}\gamma_2'$. By Lemma 8.2.2, $\mathrm{Tr}(\gamma_2^{-1}X\gamma_2) = \mathrm{Tr}(X\gamma_2\gamma_2^{-1}) = \mathrm{Tr}(X)$ for residual X, so $\int_\gamma \alpha = \int_{\gamma_1}\alpha + \int_{\gamma_2}\alpha$. If AB is invertible then A is surjective and B is injective, so both are invertible since they have index zero (I4). Hence if one of A, B is not invertible, neither is AB, and both sides vanish.

(ii) is immediate from the definition.

(iii) The path $s \mapsto \exp(sE)$ has $\gamma^{-1}\gamma' = E$. If $\|E\| < 1$, the path $\mathrm{Id}+sE$, $s \in [0,1]$, lies in $G^{-\infty}_{\mathrm{iso}}$, and $(\mathrm{Id}+sE)^{-1}E = \sum_{k\geq 0}(-s)^k E^{k+1}$, the series converging in $\Psi^{-\infty}_{\mathrm{iso}}$ uniformly for $s \in [0,1]$ because, by Lemma 8.1.2, $\|E^{k+2}\|_{(j)} = \|E\,E^k E\|_{(j)} \leq C\|E\|^k_{\mathcal{B}(L^2)}\|E\|^2_{(j')}$. Integrating term by term gives $\int_\gamma \alpha = \sum_{k\geq 0}(-1)^k\,\mathrm{Tr}(E^{k+1})/(k+1) = \mathrm{Tr}\log(\mathrm{Id}+E)$.

(iv) $ABA^{-1} - \mathrm{Id} = A(B-\mathrm{Id})A^{-1}$ is residual and ABA^{-1} is invertible if and only if B is. If γ is a path from Id to B, then $A\gamma A^{-1}$ is a path from Id to ABA^{-1} and $\mathrm{Tr}(A\gamma^{-1}\gamma' A^{-1}) = \mathrm{Tr}(\gamma^{-1}\gamma')$ by Lemma 8.2.2. □

Lemma 8.3.5 (Finite dimensional reduction)**.** *Let* Π *be the orthogonal projection onto a finite dimensional subspace* $L \subset \mathcal{S}(\mathbb{R}^n;\mathbb{C}^N)$*, and let* $R \in \Psi^{-\infty}_{\mathrm{iso}}$ *satisfy* $R = R\Pi$*. Then* $\mathrm{Id}+R$ *is invertible if and only if* $\mathrm{Id}_L + \Pi R|_L$ *is invertible, and*

$$\det_{\mathrm{Fr}}(\mathrm{Id}+R) = \det\big(\mathrm{Id}_L + \Pi R|_L\big).$$

The same holds if $R = \Pi R$*.*

Proof. With respect to $L^2 = L \oplus L^\perp$ (columns corresponding to the domain), $\mathrm{Id}+cR = \begin{pmatrix} \mathrm{Id}+cr & 0 \\ c\rho & \mathrm{Id} \end{pmatrix}$ with $r = \Pi R|_L$ and $\rho = (\mathrm{Id}-\Pi)R|_L$. This is invertible if and only if $q(c) = \det(\mathrm{Id}_L + cr) \neq 0$, which proves the first statement for $c = 1$. If $q(1) = 0$ both sides vanish. Otherwise, since the polynomial q has $q(0) = 1$, it has finitely many zeros, and we may choose a smooth path $c(s)$ from 0 to 1 avoiding them; $\gamma(s) = \mathrm{Id}+c(s)R$ is then a path in $G^{-\infty}_{\mathrm{iso}}$. From the triangular form, $T = \gamma^{-1}\gamma' = \begin{pmatrix} (\mathrm{Id}+cr)^{-1}c'r & 0 \\ * & 0 \end{pmatrix}$, so $T = T\Pi$ and, by Lemma 8.2.3 and Lemma 8.2.2, $\mathrm{Tr}(T) = \mathrm{Tr}(\Pi T\Pi) + \mathrm{Tr}((\mathrm{Id}-\Pi)T\Pi) = \mathrm{tr}\big((\mathrm{Id}+cr)^{-1}c'r\big) + \mathrm{Tr}(\Pi(\mathrm{Id}-\Pi)T)$. The last term vanishes and by Jacobi's formula the first is $\frac{d}{ds}\log q(c(s))$, so $\exp\int_\gamma \alpha = q(1)/q(0) = \det(\mathrm{Id}_L + r)$. The case $R = \Pi R$ follows by taking adjoints, or by the same argument with upper triangular matrices. □

In particular, for finite rank residual operators the Fredholm determinant is an ordinary determinant, and $\det_{\mathrm{Fr}}(\mathrm{Id}+\pi_N E\pi_N) = \det(\mathrm{Id}+\pi_N E|_{L_N})$.

Theorem 8.3.6. *The Fredholm determinant is continuous on* $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$*. If* E_y *is a smooth (resp. holomorphic) family in* $\Psi^{-\infty}_{\mathrm{iso}}$*, then* $\det_{\mathrm{Fr}}(\mathrm{Id}+E_y)$ *is a smooth (resp. holomorphic) function. Moreover, for every* $E \in \Psi^{-\infty}_{\mathrm{iso}}$*,*

$$\det_{\mathrm{Fr}}(\mathrm{Id}+E) = \lim_{N\to\infty} \det\big(\mathrm{Id}+\pi_N E|_{L_N}\big). \tag{8.3.3}$$

Proof. Near invertible points this was shown above. Let $\mathrm{Id}+E_0$ be non-invertible, let Π_0 be the projection onto $\mathrm{Nul}(\mathrm{Id}+E_0) \subset \mathcal{S}$, and let F be as in (I4), so that $G_0 = \mathrm{Id}+E_0+F$ is invertible and $F = F\Pi_0$. For E near E_0 the operator $G_0 + E - E_0$ is invertible; let $K(E) = (G_0 + E - E_0)^{-1}$, which lies in $G^{-\infty}_{\mathrm{iso}}$ and depends continuously on E (Lemma 8.1.4). Then

$$\mathrm{Id}+E = K(E)^{-1}\big(\mathrm{Id}-K(E)F\big), \qquad \det_{\mathrm{Fr}}(\mathrm{Id}+E) = \det_{\mathrm{Fr}}(K(E))^{-1}\det\big(\mathrm{Id}-\Pi_0 K(E)F|_L\big),$$

using Proposition 8.3.4(i) and Lemma 8.3.5 with $L = \mathrm{Ran}\,\Pi_0$ and $R = -K(E)F = R\Pi_0$. The first factor is continuous and non-vanishing, and the second is a polynomial in the matrix entries $(K(E)Fe_j, e_i)$ for an orthonormal basis e_i of L, which are continuous in E. If $E = E_y$ is a smooth or holomorphic family, both factors are smooth or holomorphic. Finally (8.3.3) follows from $\det_{\mathrm{Fr}}(\mathrm{Id}+\pi_N E\pi_N) = \det(\mathrm{Id}+\pi_N E|_{L_N})$, the convergence $\pi_N E\pi_N \to E$ and continuity. □

Remark 8.3.7. Thus $\det_{\mathrm{Fr}}$ is an entire function on $\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}$: holomorphic along every holomorphic family, and in particular $c \mapsto \det_{\mathrm{Fr}}(\mathrm{Id}+cE)$ is entire. In view of Proposition 8.3.4, $G^{-\infty}_{\mathrm{iso}}$ is the complement of the (singular) hypersurface $\{\mathrm{Id}+E;\ \det_{\mathrm{Fr}}(\mathrm{Id}+E) = 0\}$. The trace and the determinant extend further, to operators of order less than $-2n$; see Exercise 8.16.

8.4 Index and invertible perturbations

For an elliptic $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ the null spaces of A and A^* on $\mathcal{S}'$ are finite dimensional subspaces of $\mathcal{S}$, and the *index* is

$$\operatorname{ind}(A) = \dim \operatorname{Nul}(A) - \dim \operatorname{Nul}(A^*).$$

For $m = 0$ this is the index of A as a Fredholm operator on L^2, since $\operatorname{Ran}(A)$ is closed with orthocomplement $\operatorname{Nul}(A^*)$ there.

Proposition 8.4.1. *If $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ is elliptic and B is any parametrix, then $[A,B] = (AB - \mathrm{Id}) - (BA - \mathrm{Id}) \in \Psi^{-\infty}_{\mathrm{iso}}$ and*

$$\operatorname{ind}(A) = \operatorname{Tr}([A,B]). \tag{8.4.1}$$

Consequently $\operatorname{ind}(A+E) = \operatorname{ind}(A)$ for $E \in \Psi^{-\infty}_{\mathrm{iso}}$, and if $A(z)$ is a holomorphic (or smooth) family of elliptic operators on a connected set, then $\operatorname{ind} A(z)$ is constant.

Proof. For the generalized inverse, (8.1.1) and Lemma 8.2.3 give $\operatorname{ind}(A) = \operatorname{Tr}\Pi_0 - \operatorname{Tr}\Pi_1 = \operatorname{Tr}(\mathrm{Id} - BA) - \operatorname{Tr}(\mathrm{Id} - AB) = \operatorname{Tr}([A,B])$. Any other parametrix is $B' = B + E'$ with E' residual, and $\operatorname{Tr}([A,B']) - \operatorname{Tr}([A,B]) = \operatorname{Tr}([A,E']) = 0$ by Lemma 8.2.2. A parametrix of A is also one of $A+E$, and $\operatorname{Tr}([E,B]) = 0$. For a family, choose locally a holomorphic parametrix $B(z)$ (I6); then $[A(z),B(z)]$ is a holomorphic residual family, so $\operatorname{ind} A(z)$ is continuous, hence locally constant, hence constant. □

Discussion. For matrices $\operatorname{Tr}([A,B]) = 0$, so formula (8.4.1) recovers the fact that a square matrix has index zero. In infinite dimensions neither A nor B need be of trace class, and $[A,B]$ can have non-zero trace even though it is a commutator. The index measures exactly this failure of the trace property. The rest of the chapter makes this precise: once the trace is extended (as the regularized trace) to all of $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}$, its failure to vanish on commutators is a local expression in the symbols, and so is the index (Theorem 8.10.1).

Proposition 8.4.2. *Let $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ be elliptic with $\operatorname{ind}(A) = 0$. Then there is a finite rank $F \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ such that $A + sF$ is invertible for every $s \neq 0$; its inverse lies in $\Psi^{-m}_{\mathrm{iso,ph}}$.*

Proof. Let $e_1,\dots,e_p$ and $f_1,\dots,f_p$ be orthonormal bases of $\operatorname{Nul}(A)$ and $\operatorname{Nul}(A^*)$, both in $\mathcal{S}$. Let F and G be the operators with kernels $\sum_j f_j(x)\overline{e_j(x')}$ and $\sum_j e_j(x)\overline{f_j(x')}$, so $F = \Pi_1 F = F\Pi_0$, $G = \Pi_0 G = G\Pi_1$, $FG = \Pi_1$, $GF = \Pi_0$. With B the generalized inverse, normalized as in (I3), and using $A\Pi_0 = 0$, $\Pi_1 A = 0$, $\Pi_0 B = 0$, $B\Pi_1 = 0$,

$$(A+sF)(B+s^{-1}G) = AB + s^{-1}A\Pi_0 G + sF\Pi_0 B + FG = \mathrm{Id} - \Pi_1 + \Pi_1 = \mathrm{Id},$$
$$(B+s^{-1}G)(A+sF) = BA + sB\Pi_1 F + s^{-1}G\Pi_1 A + GF = \mathrm{Id}\,.$$

Thus $A + sF$ is invertible with inverse $B + s^{-1}G \in \Psi^{-m}_{\mathrm{iso,ph}}$. □

Write $\mathrm{Ell}^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N) \subset \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$ for the elliptic elements, $\mathrm{Ell}^0_{\mathrm{iso},k}$ for those of index k, and $G^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ for the group of invertible elements (on L^2, or equivalently in $\Psi^0_{\mathrm{iso,ph}}$). Then $G^0_{\mathrm{iso}} \subset \mathrm{Ell}^0_{\mathrm{iso},0} \subset \mathrm{Ell}^0_{\mathrm{iso}} \subset \Psi^0_{\mathrm{iso,ph}}$.

Corollary 8.4.3. *$G^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ is dense in $\mathrm{Ell}^0_{\mathrm{iso},0}(\mathbb{R}^n;\mathbb{C}^N)$ for the topology of $\Psi^0_{\mathrm{iso,ph}}$: indeed $A + sF \to A$ as $s \to 0$.*

By Proposition 8.4.1 the index is defined on the quotient. Let

$$\mathcal{A}^m = \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)/\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N);$$

then $\mathcal{A}^{\mathbb{Z}} = \bigcup_m \mathcal{A}^m$ is an algebra, A is elliptic if and only if its class $a = [A]$ is invertible in $\mathcal{A}^{\mathbb{Z}}$ (the inverse being the class of a parametrix), and

$$\mathrm{ind} : \{a \in \mathcal{A}^0 \text{ invertible}\} \to \mathbb{Z}, \qquad \mathrm{ind}(a) = \mathrm{ind}(A) \ \text{ if } a = [A],$$

is well defined. We return to it in Section 8.10.

8.5 Analytic Fredholm theory

The determinant turns questions about invertibility of operators into questions about zeros of a scalar function. For a holomorphic family $\mathrm{Id} + A(z)$ that function is holomorphic, so its zeros are isolated, and the inverse can only blow up at isolated points.

A family $M(z)$ with values in $\Psi^{-\infty}_{\mathrm{iso}}$ is *meromorphic* near p if $M(z) = \sum_{k=1}^{K}(z-p)^{-k}M_{-k} + M_0(z)$ with $M_{-k} \in \Psi^{-\infty}_{\mathrm{iso}}$ and M_0 a holomorphic family; we say it has *finite rank singular part* if the M_{-k} have finite rank.

Lemma 8.5.1 (Fredholm alternative)**.** *Let $\Omega \subset \mathbb{C}$ be open and connected and let $A(z)$ be a holomorphic family in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$. Then either $\mathrm{Id} + A(z)$ is invertible for no $z \in \Omega$, or it is invertible outside a discrete set $D \subset \Omega$; in the latter case $(\mathrm{Id} + A(z))^{-1} = \mathrm{Id} + B(z)$ where B is meromorphic in $\Psi^{-\infty}_{\mathrm{iso}}$ with finite rank singular parts.*

Proof. By Theorem 8.3.6, $f(z) = \det_{\mathrm{Fr}}(\mathrm{Id} + A(z))$ is holomorphic on Ω, so either $f \equiv 0$ or its zero set D is discrete; and $\mathrm{Id} + A(z)$ is invertible exactly when $f(z) \neq 0$. Away from D, $B(z)$ is holomorphic by Lemma 8.1.4. Near $p \in D$ we proceed as in the proof of Theorem 8.3.6: with $L = \mathrm{Nul}(\mathrm{Id} + A(p))$, Π_0 the projection onto L and $G_0 = \mathrm{Id} + A(p) + F$ invertible, $F = F\Pi_0$ of finite rank, the family $K(z) = (G_0 + A(z) - A(p))^{-1}$ is holomorphic in $G^{-\infty}_{\mathrm{iso}}$ near p and $\mathrm{Id} + A(z) = K(z)^{-1}(\mathrm{Id} + R(z))$ with $R(z) = -K(z)F = R(z)\Pi_0$. The matrix $\mathrm{Id}_L + \Pi_0 R(z)|_L$ has determinant $\det_{\mathrm{Fr}}(\mathrm{Id} + R(z)) = f(z)\det_{\mathrm{Fr}}(K(z)) \not\equiv 0$, so its inverse $M(z)$ is a meromorphic matrix (Cramer's rule). A direct computation, using $R = R\Pi_0$ and $(\mathrm{Id}_L + \Pi_0 R \Pi_0)M = \mathrm{Id}_L$, shows $(\mathrm{Id} + R)(\mathrm{Id} - RM\Pi_0) = \mathrm{Id} + R - R\Pi_0 = \mathrm{Id}$, so

$$(\mathrm{Id} + A(z))^{-1} = K(z) - \sum_{i,j} m_{ij}(z)\, R(z) P_{ij} K(z),$$

where P_{ij} are the matrix units with respect to an orthonormal basis of L and m_{ij} the entries of M. Each $R(z)P_{ij}K(z)$ is a holomorphic family of residual operators of rank at most one, whose Taylor coefficients therefore have finite rank; multiplying by the meromorphic scalars m_{ij} gives a singular part of finite rank. □

This result for smoothing operators, which really follows from the corresponding result for matrices, implies the analogous statement for holomorphic families of elliptic operators. The idea of the proof is to perturb by a fixed finite rank operator F so as to make $A(z) + F$ invertible near a point; then $A(z) = (A(z) + F)(\mathrm{Id} - C(z)F)$, and the second factor is a residual perturbation of the identity, to which Lemma 8.5.1 applies.

Proposition 8.5.2. *Let $\Omega \subset \mathbb{C}$ be open and connected, h holomorphic on Ω, and $A(z)$ a holomorphic family of elliptic operators of order $h(z)$. If $A(z_0)$ is invertible for some $z_0 \in \Omega$, then $A(z)$ is invertible outside a discrete set $D \subset \Omega$, and near each $p \in \Omega$*

$$A(z)^{-1} = B(z) + H(z),$$

with $B(z)$ a holomorphic family of order $-h(z)$ and $H(z)$ meromorphic in $\Psi_{\mathrm{iso}}^{-\infty}$ with finite rank singular part ($H = 0$ if $p \notin D$).

Proof. By Proposition 8.4.1, $\operatorname{ind} A(z) = \operatorname{ind} A(z_0) = 0$ for all z. Fix $p \in \Omega$. By Proposition 8.4.2 there is a finite rank residual F with $A(p) + F$ invertible, so by Lemma 8.1.5 $A(z) + F$ is invertible on a disc U_p around p, with inverse $C(z)$ a holomorphic family of order $-h(z)$. Then $A(z) = (A(z) + F)(\mathrm{Id} - C(z)F)$, so on U_p, $A(z)$ is invertible exactly where $g_p(z) = \det_{\mathrm{Fr}}(\mathrm{Id} - C(z)F) \neq 0$.

Let W be the set of $p \in \Omega$ such that $A(z)$ is invertible at points arbitrarily close to p. If $p \in W$ then $g_p \not\equiv 0$ on the disc U_p, so its zeros are isolated, and every point of U_p is in W; thus W is open. Its complement is the interior of the set of non-invertible points, also open. Since $z_0 \in W$ and Ω is connected, $W = \Omega$, and the non-invertible points form a discrete set.

Near p, Lemma 8.5.1 gives $(\mathrm{Id} - C(z)F)^{-1} = \mathrm{Id} + M(z)$ with $M = \sum_{k=1}^{K}(z-p)^{-k}M_{-k} + M_0(z)$ as above, so $A(z)^{-1} = (\mathrm{Id} + M(z))C(z)$. Set $B = C + M_0C$, a holomorphic family of order $-h(z)$, and $H = \sum_k (z-p)^{-k}M_{-k}C(z)$. Each $M_{-k}C(z)$ is a holomorphic residual family with range in the finite dimensional space $\operatorname{Ran} M_{-k}$, so the singular part of H has finite rank. □

8.6 Resolvent and spectrum

One direct application of analytic Fredholm theory is to the resolvent of an elliptic operator of positive order. For simplicity we take the order to be a positive integer m, so that $A - \lambda \in \Psi_{\mathrm{iso,ph}}^{m}$ for $\lambda \in \mathbb{C}$; the case of non-integral order needs only notational changes, replacing $\Psi_{\mathrm{iso,ph}}^{m}$ by the class of operators with symbols in Γ^m having an expansion in terms of homogeneous functions of degrees $m - j$ and $-j$.

Proposition 8.6.1. *Let $A \in \Psi_{\mathrm{iso,ph}}^{m}(\mathbb{R}^n; \mathbb{C}^N)$, $m \in \mathbb{N}$, be elliptic, and suppose that for some $\lambda' \in \mathbb{C}$ both $A - \lambda'$ and $A^* - \overline{\lambda'}$ have trivial null space. Then the set $\operatorname{Spec}(A)$ of $\lambda \in \mathbb{C}$ for which $A - \lambda$ is not invertible is discrete, and $(A - \lambda)^{-1}$ is a meromorphic family in $\Psi_{\mathrm{iso,ph}}^{-m}$ with finite rank residual singular parts. At $\tilde{\lambda} \in \operatorname{Spec}(A)$ the generalized eigenspace*

$$E_{\tilde{\lambda}} = \{u \in \mathcal{S}'(\mathbb{R}^n; \mathbb{C}^N);\ (A - \tilde{\lambda})^p u = 0 \text{ for some } p \in \mathbb{N}\}$$

is a finite dimensional subspace of $\mathcal{S}$, and

$$(A - \lambda)^{-1} = -\sum_{j=1}^{k} (\lambda - \tilde{\lambda})^{-j} (A - \tilde{\lambda})^{j-1} P + (\text{holomorphic near } \tilde{\lambda}),$$

where P is a projection (not necessarily orthogonal) onto $E_{\tilde{\lambda}}$ commuting with A, and k is the least integer with $(A - \tilde{\lambda})^k P = 0$.

Proof. Since $m > 0$, $A - \lambda$ is, for every λ, elliptic with the same principal symbol as A, and $\lambda \mapsto A - \lambda$ is an entire family of order m. The hypothesis says $\operatorname{ind}(A - \lambda') = 0$ and $A - \lambda'$ is invertible, so Proposition 8.5.2 applies with $\Omega = \mathbb{C}$. Replacing A by $A - \tilde{\lambda}$ we may take $\tilde{\lambda} = 0$ and write $(A - \lambda)^{-1} = \sum_{j \geq -k} \lambda^j S_j$, with $S_{-1}, \ldots, S_{-k}$ finite rank residual. Comparing coefficients in $(A - \lambda)(A - \lambda)^{-1} = \mathrm{Id} = (A - \lambda)^{-1}(A - \lambda)$ gives $AS_j - S_{j-1} = \delta_{j0}\,\mathrm{Id} = S_jA - S_{j-1}$, so with $R_j = S_{-j}$,

$$R_{j+1} = AR_j = R_jA\ (1 \leq j \leq k), \quad R_{k+1} = 0, \qquad \text{hence } R_j = A^{j-1}R_1,\ A^kR_1 = 0.$$

Thus $\operatorname{Ran} R_1 \subset E_0$ and R_1 commutes with A. Conversely, if $A^p u = 0$, then $(A - \lambda)(-\sum_{j=0}^{p-1} \lambda^{-j-1} A^j u) = u$, so $(A - \lambda)^{-1}u = -\sum_{j<p} \lambda^{-j-1} A^j u$ and comparing the coefficients of λ^{-1}, $R_1 u = -u$. Hence $P = -R_1$ is a projection with range exactly E_0, which is therefore finite dimensional and contained in $\mathcal{S}$. □

Note also that $\operatorname{Ran} R_k \subset \operatorname{Ran} R_{k-1} \subset \cdots \subset \operatorname{Ran} R_1$ and $\operatorname{Nul} R_k \supset \cdots \supset \operatorname{Nul} R_1$; precisely, $u \in \operatorname{Ran} R_p$, $p \geq 1$, if and only if $u = A^{p-1}u_1$ for some $u_1 \in \operatorname{Ran} R_1$.

Discussion. In the notation of the proof, $P = -R_1$ is minus the residue of the resolvent at $\tilde{\lambda}$, so by the residue theorem

$$P = -\frac{1}{2\pi i} \oint_{|\lambda - \tilde{\lambda}| = \epsilon} (A - \lambda)^{-1} \, d\lambda$$

for small $\epsilon > 0$ (Figure 8.3). This is the usual Riesz projection of spectral theory. Here no self-adjointness is assumed, so eigenvalues may be complex and the eigenspaces need not be orthogonal; the analytic Fredholm theorem still guarantees that the spectrum is discrete and that each generalized eigenspace is finite dimensional.

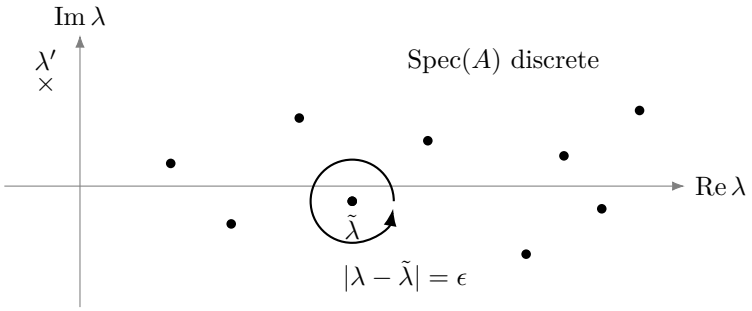


Figure 8.3: The spectrum of an elliptic A of positive order satisfying the hypothesis of Proposition 8.6.1 is a discrete set (dots) in $\mathbb{C}$; at λ' ($\times$) both $A - \lambda'$ and its adjoint are injective. The resolvent $(A - \lambda)^{-1}$ is meromorphic, and integrating it around a small circle about $\tilde{\lambda} \in \operatorname{Spec}(A)$ gives $-2\pi i$ times the projection P onto the generalized eigenspace $E_{\tilde{\lambda}}$.

8.7 The residue trace

We have shown that there is a unique trace on the residual algebra. Following ideas of Seeley [11], Guillemin [4, 5] and Wodzicki [13, 14], we now look for traces on the algebra $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n)$. For $A \in \Psi^{-\infty}_{\mathrm{iso}}$, by (8.2.2), $\operatorname{Tr}(A) = (2\pi)^{-n} \int a_L(x,\xi)\, dx\, d\xi$ is the integral of the left symbol; this is also true for any amplitude: if $A = \operatorname{Op}(a)$ with $a(x,x',\xi)$ an amplitude of order $-\infty$, then $\operatorname{Tr}(A) = (2\pi)^{-n} \int a(x,x,\xi)\, dx\, d\xi$. The integral extends to Ψ^m_{iso} for $m < -2n$, as we saw, but not beyond. To go further we regularize it by embedding operators in holomorphic families, which moves us from the symbol classes Γ^m to the polyhomogeneous operators.

Lemma 8.7.1. *Let $\Omega \subset \mathbb{C}$ be open and connected, $m \in \mathbb{C}$, and $A(z)$ a holomorphic family of order $m + z$ on Ω, with b, b_j as in (8.1.2). Suppose Ω contains points with $\Re z < -2n - \Re m$. Then $\operatorname{Tr}(A(z))$, defined where $\Re z < -2n - \Re m$, extends to a meromorphic function on Ω with at most simple poles, at the points $z_j = j - 2n - m \in \Omega$, $j \in \mathbb{N}_0$, and*

$$\operatorname{Res}_{z=z_j} \operatorname{Tr}(A(z)) = -(2\pi)^{-n} \int_{\mathbb{S}^{2n-1}} \operatorname{tr} b_j(z_j;\theta)\, d\theta. \tag{8.7.1}$$

If $A(z) = A_y(z)$ depends smoothly on a parameter y, so do the residues and the coefficients of the Laurent expansions.

Proof. For $\Re z < -2n - \Re m$ we have $\operatorname{Tr}(A(z)) = f_1(z) + f_2(z)$ with

$$f_1(z) = (2\pi)^{-n} \int_{|Z|\le 1} \operatorname{tr} a(z;Z)\, dZ,$$
$$f_2(z) = (2\pi)^{-n} \int_{\mathbb{S}^{2n-1}} \int_1^\infty r^{m+z+2n-1} \operatorname{tr} b(z;1/r,\theta)\, dr\, d\theta.$$

The first is holomorphic on Ω. Write $b(z;t,\theta) = \sum_{j<k} t^j b_j(z;\theta) + t^k b_{(k)}(z;t,\theta)$, with $b_{(k)}$ smooth and holomorphic in z (Taylor's formula with integral remainder). Since $\int_1^\infty r^{m+z+2n-1-j}\, dr = (j-2n-m-z)^{-1}$ for $\Re z < j - 2n - \Re m$,

$$f_2(z) = (2\pi)^{-n} \sum_{j<k} \frac{1}{j-2n-m-z} \int_{\mathbb{S}^{2n-1}} \operatorname{tr} b_j(z;\theta)\, d\theta + f_2^{(k)}(z),$$

where $f_2^{(k)}(z) = (2\pi)^{-n} \int_{\mathbb{S}^{2n-1}} \int_1^\infty r^{m+z+2n-1-k} \operatorname{tr} b_{(k)}(z;1/r,\theta)\, dr\, d\theta$ is holomorphic in $\Re z < k - 2n - \Re m$, the integral converging locally uniformly there. This identity holds pointwise for $\Re z < -2n - \Re m$, and its right side is meromorphic on $\Omega_k = \{z \in \Omega; \Re z < k - 2n - \Re m\}$; the right sides for k and $k+1$ agree on Ω_k (the difference is the $j = k$ term, computed by the same integral). Since Ω is connected, these define the meromorphic continuation, with the stated poles and residues. All the expressions depend smoothly on parameters if b does. □

We continue to write $\operatorname{Tr}(A(z))$ for the meromorphic continuation. By uniqueness of continuation, any identity between such functions which holds for $\Re z \ll 0$ holds on Ω. For a holomorphic family $A(z)$ of order $m + z$ we have $b_j(0;\theta)$ equal to the coefficient of the term of homogeneity $m - j$ in the symbol of $A(0)$. Taking $j = 2n + m$, the residue at $z = 0$ only depends on $A(0)$, and it vanishes unless $m \in \mathbb{Z}$, $m \ge -2n$.

Definition 8.7.2. For $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$, $m \in \mathbb{C}$, the *residue trace* is

$$\operatorname{Tr}_{\mathrm{R}}(A) = -\operatorname{Res}_{z=0} \operatorname{Tr}(A(z)),$$

where $A(z)$ is any holomorphic family of order $m + z$, on a connected neighbourhood of 0 containing points with $\Re z < -2n - \Re m$, with $A(0) = A$; for instance $A(z) = AH(z)$, with $H(z)$ from (8.1.3).

Proposition 8.7.3. *The residue trace is well defined, and*

$$\operatorname{Tr}_{\mathrm{R}}(A) = (2\pi)^{-n} \int_{\mathbb{S}^{2n-1}} \operatorname{tr} a_{-2n}(\theta)\, d\theta, \tag{8.7.2}$$

where a_{-2n} is the term of homogeneity $-2n$ in the expansion of the left symbol of A (so $\operatorname{Tr}_{\mathrm{R}}$ vanishes unless $m \in \mathbb{Z}$, $m \ge -2n$; in particular on $\Psi^{-2n-1}_{\mathrm{iso,ph}}$). It is a trace on $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}$:

$$\operatorname{Tr}_{\mathrm{R}}([A,B]) = 0 \quad \forall\, A, B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N). \tag{8.7.3}$$

Proof. The formula is (8.7.1) with $j = 2n + m$, and shows that $\operatorname{Tr}_{\mathrm{R}}(A)$ is independent of the family. If $A \in \Psi^{m_1}_{\mathrm{iso,ph}}$, $B \in \Psi^{m_2}_{\mathrm{iso,ph}}$, then $C(z) = [AH(z), B]$ is an entire family of order $m_1 + m_2 + z$ with $C(0) = [A,B]$. For $\Re z < -2n - m_1 - m_2$, $\operatorname{Tr}(C(z)) = \operatorname{Tr}(AH(z)B) - \operatorname{Tr}(BAH(z)) = 0$ by Lemma 8.2.2, since $AH(z) \in \Psi^{m_1+\Re z}_{\mathrm{iso}}$. Hence the continuation vanishes identically and $\operatorname{Tr}_{\mathrm{R}}([A,B]) = 0$. □

Remark 8.7.4. The left symbol in (8.7.2) may be replaced by the right symbol (or the Weyl symbol). Indeed the right symbol of A has the expansion $a_R \sim \sum_\alpha \frac{1}{\alpha!}(-D_\xi)^\alpha \partial_x^\alpha a_L$, and each term with $\alpha \neq 0$ is a derivative $\partial_{Z_k} g$ of a polyhomogeneous function. If g is homogeneous of degree $1-2n$ then, integrating $\partial_{Z_k} g$ over $1 \leq |Z| \leq R$ and using the divergence theorem,

$$\log R \int_{\mathbb{S}^{2n-1}} \partial_{Z_k} g \, d\theta = \int_{\mathbb{S}^{2n-1}} (R^{1-2n} R^{2n-1} - 1) g(\theta) \theta_k \, d\theta = 0,$$

so the homogeneous term of degree $-2n$ of each such derivative integrates to zero over the sphere. The residue trace is, up to a constant multiple, the unique trace on $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n)$; this is the isotropic analogue of Wodzicki's theorem [13], and we shall neither prove nor use it.

Discussion. Why does the residue trace see only one term of the symbol? Integrating a symbol term of homogeneity μ over $|Z| \geq 1$ in $\mathbb{R}^{2n}$ gives $\int_1^\infty r^{\mu+2n-1}\, dr$. This converges for $\mu < -2n$, diverges like a power for $\mu > -2n$, and diverges *logarithmically* exactly when $\mu = -2n$. Inserting the regularizer $\langle Z \rangle^z$ turns a power divergence into a pole at a non-zero value of z, harmless at $z = 0$, while the logarithmic divergence becomes a pole at $z = 0$. So the residue at $z = 0$ records the coefficient of the logarithmic divergence, which is the integral of the homogeneous term of degree $-2n$ over the sphere.

Question. Every residual operator, indeed every operator of order less than $-2n$, has residue trace zero. Is the residue trace then useless on trace class operators?

Answer. Yes, and this is the point. On operators of very negative order the ordinary trace is available, and the residue trace vanishes because $\mathrm{Tr}(A(z))$ has no pole at $z = 0$. The residue trace is a trace on the *quotient* $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}/\Psi^{-\infty}_{\mathrm{iso}}$, where no non-zero multiple of the ordinary trace makes sense. It complements the ordinary trace rather than extending it. □

8.8 Exterior derivation

The regularized trace of the next section depends on a choice of regularizing family $A(z)$, and its failure to be a trace is governed by the derivative $\dot{A}$ of the family at $z = 0$. Commutation with $\dot{A}$, which is essentially the logarithm of an elliptic operator of order one, is the "exterior derivation" introduced here.

Definition 8.8.1. A *regularizing family* is an entire holomorphic family $A(z)$ of order z, with values in $\Psi^z_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$, with scalar principal symbol and $A(0) = \mathrm{Id}$. Its derivative at 0 is denoted $\dot{A} = \frac{d}{dz} A(z)|_{z=0}$.

If $\sigma_L(A(z)) = \langle Z \rangle^z \tilde{a}(z; Z)$ with $\tilde{a}$ holomorphic in z with values in $\mathcal{C}^\infty(\overline{\mathbb{R}^{2n}})$, then $A(0) = \mathrm{Id}$ means $\tilde{a}(0) = 1$, so

$$\dot{A} = \mathrm{Op}(\log\langle Z \rangle + c), \qquad c = \partial_z \tilde{a}(0) \in S^0_{\mathrm{ph}}(\mathbb{R}^{2n}) \text{ with scalar leading term.}$$

Thus $\dot{A} \in \Psi^\epsilon_{\mathrm{iso}}$ for every $\epsilon > 0$, but $\dot{A} \notin \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}$ because of the logarithm. For the standard family $H(z)$ of (8.1.3), $\dot{H} = \mathrm{Op}(\log\langle Z \rangle) = \mathrm{Op}(\frac{1}{2}\log(1 + |x|^2 + |\xi|^2))$.

Proposition 8.8.2. *For a regularizing family $A(z)$, the map*

$$D_A B = [\dot{A}, B]$$

is a derivation of $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$ with $D_A : \Psi^m_{\mathrm{iso,ph}} \to \Psi^{m-1}_{\mathrm{iso,ph}}$ and $D_A(\Psi^{-\infty}_{\mathrm{iso}}) \subset \Psi^{-\infty}_{\mathrm{iso}}$. Equivalently, $D_A B = \frac{d}{dz}(A(z) B A(z)^{-1})\big|_{z=0}$. For two regularizing families, $D_{A_2} - D_{A_1} = [\dot{A}_2 - \dot{A}_1, \cdot\,]$ is an inner derivation, given by commutation with an element of $\Psi^0_{\mathrm{iso,ph}}$. Moreover

$$\mathrm{Tr_R}(D_A B) = 0 \quad \forall\, B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N). \tag{8.8.1}$$

Proof. A commutator is a derivation. For $B \in \Psi^m_{\mathrm{iso,ph}}$, $K(z) = [A(z), B]/z$ is an entire holomorphic family of order $m + z$ (the numerator is a holomorphic family vanishing at $z = 0$, and dividing its normalized symbol by z preserves holomorphy), with $K(0) = [\dot{A}, B]$; hence $D_A B \in \Psi^m_{\mathrm{iso,ph}}$. Since the principal symbol of $A(z)$ is scalar, $[A(z), B]$ has order $m + z - 1$, so in fact $D_A B \in \Psi^{m-1}_{\mathrm{iso,ph}}$. Residual operators form an ideal in $\bigcup_m \Psi^m_{\mathrm{iso}}$, so D_A preserves them. Near $z = 0$, $A(z)$ is invertible with holomorphic inverse by Lemma 8.1.5, and $\frac{d}{dz}(A(z)BA(z)^{-1})|_{z=0} = \dot{A}B - B\dot{A}$ by the product rule. The statement about two families follows since $\dot{A}_2 - \dot{A}_1 = \mathrm{Op}(c_2 - c_1) \in \Psi^0_{\mathrm{iso,ph}}$. Finally, $K(z)$ has value $D_A B$ at 0 and $\mathrm{Tr}(K(z)) = z^{-1}(\mathrm{Tr}(A(z)B) - \mathrm{Tr}(BA(z))) = 0$ for $\Re z \ll 0$ by Lemma 8.2.2, so the residue at 0 vanishes. □

Remark 8.8.3. D_H is not an inner derivation of $\Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}$, which is why we call it an *exterior* derivation. Suppose $[\dot{H} - P, B] = 0$ for all $B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}$, for some $P \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}$, and let c be the left symbol of $C = \dot{H} - P$. The composition formula is exact when one factor has polynomial symbol, and gives $[C, x_j] = \mathrm{Op}(-i\partial_{\xi_j} c)$ and $[C, D_j] = \mathrm{Op}(-D_{x_j} c)$; hence c is constant, which is impossible since $c - \log\langle Z\rangle$ is polyhomogeneous while $\log\langle Z\rangle$ is not.

8.9 The regularized trace

Let $A(z)$ be a regularizing family. For $B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$ the function $\mathrm{Tr}(BA(z))$ is meromorphic on $\mathbb{C}$ with at most a simple pole at 0, with residue $-\mathrm{Tr}_{\mathrm{R}}(B)$. We define the *regularized trace* as the finite part at $z = 0$, the constant term of the Laurent expansion:

$$\overline{\mathrm{Tr}}_A(B) = \operatorname*{FP}_{z=0} \mathrm{Tr}(BA(z)) = \lim_{z\to 0}\Big(\mathrm{Tr}(BA(z)) + \frac{\mathrm{Tr}_{\mathrm{R}}(B)}{z}\Big). \tag{8.9.1}$$

If B has order less than $-2n$ (in particular if B is residual), then $\mathrm{Tr}(BA(z))$ is holomorphic near 0 and $\overline{\mathrm{Tr}}_A(B) = \mathrm{Tr}(B)$; thus $\overline{\mathrm{Tr}}_A$ extends the trace. By Lemma 8.7.1, if B_y is a smooth family then $\overline{\mathrm{Tr}}_A(B_y)$ is smooth in y and $d\,\overline{\mathrm{Tr}}_A(B_y) = \overline{\mathrm{Tr}}_A(dB_y)$. In contrast with the residue trace, $\overline{\mathrm{Tr}}_A$ depends on the choice of regularizing family, though only through $\dot{A}$.

Lemma 8.9.1. *If $A_1(z)$, $A_2(z)$ are regularizing families, then for $B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$,*

$$\overline{\mathrm{Tr}}_{A_2}(B) - \overline{\mathrm{Tr}}_{A_1}(B) = -\mathrm{Tr}_{\mathrm{R}}(B(\dot{A}_2 - \dot{A}_1)).$$

Proof. $L(z) = (A_2(z) - A_1(z))/z$ is an entire family of order z with $L(0) = \dot{A}_2 - \dot{A}_1 \in \Psi^0_{\mathrm{iso,ph}}$, and $\mathrm{Tr}(BA_2(z)) - \mathrm{Tr}(BA_1(z)) = z\,\mathrm{Tr}(BL(z))$. The function $\mathrm{Tr}(BL(z))$ has at most a simple pole at 0 with residue $-\mathrm{Tr}_{\mathrm{R}}(BL(0))$, which is therefore the value at 0 of $z\,\mathrm{Tr}(BL(z))$. □

In particular, if B has non-integral order, $\overline{\mathrm{Tr}}_A(B)$ is independent of A. The regularized trace is *not* a trace; its failure to be one is measured by the residue trace and the exterior derivation.

Discussion. The trace defect formula below is easy to remember. For $\Re z \ll 0$ the trace property gives $\mathrm{Tr}([B_1, B_2]A(z)) = \mathrm{Tr}(B_1[B_2, A(z)])$. The commutator $[B_2, A(z)]$ vanishes at $z = 0$, so it is z times something, and that something is $-D_A B_2$ at $z = 0$. A factor z times a function with a simple pole leaves a finite value, the residue. So the defect is a residue trace: it is computed from finitely many terms of the symbols and is local.

Proposition 8.9.2 (Trace defect formula)**.** *For a regularizing family $A(z)$ and $B_1, B_2 \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$,*

$$\overline{\mathrm{Tr}}_A([B_1, B_2]) = \mathrm{Tr}_{\mathrm{R}}(B_1 D_A B_2) = -\mathrm{Tr}_{\mathrm{R}}(B_2 D_A B_1). \tag{8.9.2}$$

Proof. For $\Re z \ll 0$, Lemma 8.2.2 gives $\operatorname{Tr}(B_2B_1A(z)) = \operatorname{Tr}(B_1A(z)B_2)$, so

$$\operatorname{Tr}([B_1,B_2]A(z)) = \operatorname{Tr}(B_1[B_2,A(z)]) = -z\operatorname{Tr}(B_1K(z)),$$

where $K(z) = [A(z),B_2]/z$ is the entire family of order m_2+z with $K(0) = D_AB_2$ used in the proof of Proposition 8.8.2. By continuation this holds for all z; since $\operatorname{Tr}(B_1K(z))$ has at most a simple pole at 0 with residue $-\operatorname{Tr_R}(B_1D_AB_2)$, the right side is regular at 0 with value $\operatorname{Tr_R}(B_1D_AB_2)$. The second equality follows from (8.8.1) and the trace property: $0 = \operatorname{Tr_R}(D_A(B_1B_2)) = \operatorname{Tr_R}(B_2D_AB_1) + \operatorname{Tr_R}(B_1D_AB_2)$. □

Corollary 8.9.3. *If $G \in G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ and $B \in \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$, then $\overline{\operatorname{Tr}}_A(G^{-1}BG) = \overline{\operatorname{Tr}}_A(B)$.*

Proof. $\overline{\operatorname{Tr}}_A(G^{-1}BG) - \overline{\operatorname{Tr}}_A(BGG^{-1}) = \overline{\operatorname{Tr}}_A([G^{-1},BG]) = -\operatorname{Tr_R}(BG\,D_A(G^{-1}))$, which vanishes since $D_A(G^{-1}) = D_A(G^{-1}-\mathrm{Id})$ is residual and $\operatorname{Tr_R}$ vanishes on residual operators. □

8.10 The index formula

Let D_H be the exterior derivation of the standard family (8.1.3). It preserves $\Psi^{-\infty}_{\mathrm{iso}}$, and $\operatorname{Tr_R}$ vanishes there, so both descend to the quotient algebra $\mathcal{A}^{\mathbb{Z}} = \Psi^{\mathbb{Z}}_{\mathrm{iso,ph}}/\Psi^{-\infty}_{\mathrm{iso}}$: $D_H : \mathcal{A}^m \to \mathcal{A}^{m-1}$ and $\operatorname{Tr_R} : \mathcal{A}^{\mathbb{Z}} \to \mathbb{C}$.

Theorem 8.10.1. *If $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n;\mathbb{C}^N)$, $m \in \mathbb{Z}$, is elliptic with parametrix B, then*

$$\operatorname{ind}(A) = \operatorname{Tr_R}(AD_HB) = -\operatorname{Tr_R}(BD_HA). \tag{8.10.1}$$

Thus, for invertible $a \in \mathcal{A}^{\mathbb{Z}}$, $\operatorname{ind}(a) = \operatorname{Tr_R}(a\,D_H(a^{-1}))$. The index is locally constant along smooth families of elliptic operators, and $\operatorname{ind}(a_1a_2) = \operatorname{ind}(a_1) + \operatorname{ind}(a_2)$.

Proof. Since $[A,B]$ is residual, $\operatorname{ind}(A) = \operatorname{Tr}([A,B]) = \overline{\operatorname{Tr}}_H([A,B]) = \operatorname{Tr_R}(AD_HB)$ by (8.4.1) and (8.9.2). If A_t is a smooth family with smooth family of parametrices B_t (I6), the right side of (8.10.1) is an integral of finitely many terms of the symbol expansions, so it is continuous in t; being integer-valued it is locally constant. (Alternatively, $[A_t,B_t]$ is a smooth residual family.) For additivity,

$$a_1a_2\,D_H(a_2^{-1}a_1^{-1}) = a_1(a_2D_H(a_2^{-1}))a_1^{-1} + a_1D_H(a_1^{-1}),$$

and the residue trace of the first term is $\operatorname{Tr_R}(a_2D_H(a_2^{-1}))$ by the trace property. □

Both properties hold for general Fredholm operators, but the point here is that they follow from a simple algebraic formula.

8.11 The variation 1-form

The index is the obstruction to perturbing a single elliptic operator, by a smoothing operator, to be invertible. We now derive a closely related obstruction for families. Let Y be a manifold and

$$Y \ni y \longmapsto A_y \in \mathrm{Ell}^0_{\mathrm{iso},0}(\mathbb{R}^n;\mathbb{C}^N) \tag{8.11.1}$$

a smooth family of elliptic operators of order 0 and index 0. The *families perturbative invertibility question* asks whether there is a smooth family

$$Y \ni y \longmapsto E_y \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N) \quad \text{with} \quad A_y + E_y \in G^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)\ \forall\, y. \tag{8.11.2}$$

Since the index vanishes, such E_y exists for each fixed y (Proposition 8.4.2), and also locally: if $A_{y_0}+F$ is invertible then so is A_y+F for y near y_0, with inverse a smooth family in $\Psi^0_{\mathrm{iso,ph}}$ by Lemma 8.1.5. The issue is the global existence of a smooth (or even continuous) choice.

Question. If every A_y can be made invertible by a smoothing perturbation, why can we not just patch the local choices together with a partition of unity?

Answer. Invertibility is not preserved by taking convex combinations: if A_y+F_1 and A_y+F_2 are invertible, $A_y+\tfrac12(F_1+F_2)$ need not be. The local choices can be compared only through the group $G^{-\infty}_{\mathrm{iso}}$, by $A_y+F_2=(A_y+F_1)(\mathrm{Id}+X_y)$, and the obstruction to patching them lives in the topology of the transition functions $\mathrm{Id}+X_y$. Taking determinants of these turns them into $\mathbb{C}^*$-valued functions, that is, into the transition functions of a line bundle, and the obstruction becomes its first Chern class (Section 8.12). □

Operator-valued differential forms are handled coefficientwise: if $\omega=\sum\omega_j\,dy_j$ and $\eta=\sum\eta_k\,dy_k$ in local coordinates, $\omega\wedge\eta=\sum_{j,k}\omega_j\eta_k\,dy_j\wedge dy_k$, and Tr_{R} and $\overline{\mathrm{Tr}}_H$ are applied coefficientwise. Let $a_y=[A_y]\in\mathcal{A}^0$ and let B_y be a smooth family of parametrices, so $a_y^{-1}=[B_y]$. We set

$$\theta=a^{-1}da,\qquad \varphi=a^{-1}D_Ha,$$

represented by $B_y\,dA_y$ and $B_yD_HA_y$, and define

$$\beta=\tfrac12\,\mathrm{Tr}_{\mathrm{R}}(a^{-1}da\wedge a^{-1}da\;a^{-1}D_Ha)=\tfrac12\,\mathrm{Tr}_{\mathrm{R}}(\theta\wedge\theta\,\varphi),\qquad \gamma=\tfrac12\,\mathrm{Tr}_{\mathrm{R}}(\theta\varphi).\tag{8.11.3}$$

These are smooth forms on Y of degrees 2 and 1; they depend only on a_y, since Tr_{R} vanishes on residual operators.

Proposition 8.11.1. *The 2-form β is closed. If there is a smooth family E_y as in (8.11.2), then, with $\tilde A_y=A_y+E_y$ and*

$$\bar\alpha=\overline{\mathrm{Tr}}_H(\tilde A_y^{-1}d\tilde A_y),\tag{8.11.4}$$

we have $d\bar\alpha=-\beta+d\gamma$. In particular β is exact, so its de Rham class $[\beta]\in H^2(Y;\mathbb{C})$ is an obstruction to the existence of a smooth invertible perturbation.

Proof. Suppose first that $\tilde A=A+E$ is invertible on Y, and compute with $\theta=\tilde A^{-1}d\tilde A$ and $\varphi=\tilde A^{-1}D_H\tilde A$, which represent the classes above. Since $d(\tilde A^{-1})=-\tilde A^{-1}d\tilde A\,\tilde A^{-1}$,

$$d\theta=-\theta\wedge\theta=-\tfrac12\sum_{j,k}[\theta_j,\theta_k]\,dy_j\wedge dy_k.$$

Since d commutes with $\overline{\mathrm{Tr}}_H$, the trace defect formula (8.9.2) gives

$$d\bar\alpha=-\tfrac12\sum_{j,k}\mathrm{Tr}_{\mathrm{R}}(\theta_jD_H\theta_k)\,dy_j\wedge dy_k=-\tfrac12\,\mathrm{Tr}_{\mathrm{R}}(\theta\wedge D_H\theta).$$

As D_H is a derivation commuting with d (it is commutation with the fixed operator $\dot H$), $D_H(\tilde A^{-1})=-\tilde A^{-1}(D_H\tilde A)\tilde A^{-1}$ and

$$D_H\theta=-\varphi\theta+\tilde A^{-1}d(D_H\tilde A)=d\varphi+\theta\varphi-\varphi\theta.$$

By the trace property, $\mathrm{Tr}_{\mathrm{R}}(\theta\wedge\varphi\theta)=\sum\mathrm{Tr}_{\mathrm{R}}(\theta_k\theta_j\varphi)\,dy_j\wedge dy_k=-\,\mathrm{Tr}_{\mathrm{R}}(\theta\wedge\theta\varphi)$, so $\mathrm{Tr}_{\mathrm{R}}(\theta\wedge D_H\theta)=\mathrm{Tr}_{\mathrm{R}}(\theta\wedge d\varphi)+2\,\mathrm{Tr}_{\mathrm{R}}(\theta\wedge\theta\varphi)$. On the other hand $d(\theta\varphi)=d\theta\,\varphi-\theta\wedge d\varphi$, so $2\,d\gamma=-\,\mathrm{Tr}_{\mathrm{R}}(\theta\wedge\theta\varphi)-\mathrm{Tr}_{\mathrm{R}}(\theta\wedge d\varphi)$. Combining,

$$\mathrm{Tr}_{\mathrm{R}}(\theta\wedge D_H\theta)=2\beta-2\,d\gamma,\qquad d\bar\alpha=-\beta+d\gamma.$$

Thus $\beta=d(\gamma-\bar\alpha)$ is exact. In general, invertible perturbations exist locally on Y, as noted above, so β is locally exact and hence closed. □

The form γ is defined on all of Y for any family (8.11.1), so $[\beta] = [\beta - d\gamma]$. On G^0_{iso}, $\bar\alpha$ extends the 1-form $d\log\det_{\mathrm{Fr}} = \mathrm{Tr}(A^{-1}dA)$ from $G^{-\infty}_{\mathrm{iso}}$, but it is not closed in general, because the regularized trace is not a trace. To explain the topological meaning of $[\beta]$ we now construct the determinant line bundle.

8.12 The determinant bundle

The determinant bundle was introduced by Quillen [10] for families of Cauchy–Riemann operators. Recall that $\det_{\mathrm{Fr}}$ is a character of $G^{-\infty}_{\mathrm{iso}}$: $\det_{\mathrm{Fr}}(AB) = \det_{\mathrm{Fr}}(A)\det_{\mathrm{Fr}}(B)$. Instead of trying to extend it multiplicatively to G^0_{iso} (see Remark 8.12.4), we consider all its possible values at once.

Discussion. An invertible $A \in G^0$ has no preferred determinant. But if we change A to $A(\mathrm{Id}+E)$ with $\mathrm{Id}+E \in G^{-\infty}$, the "determinant" ought to be multiplied by $\det_{\mathrm{Fr}}(\mathrm{Id}+E)$. The relation (8.12.1) below records exactly this rule. Over each class $a = [A]$ the possible values then form a complex line, not a number. As a varies these lines fit together into a line bundle, and the question whether a multiplicative determinant exists becomes the question whether this bundle is trivial.

Write $G^0 = G^0_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ and $G^{-\infty} = G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$; the latter is a normal subgroup of G^0, and $\mathcal{G} = G^0/G^{-\infty}$ is the set of classes $a = [A] \in \mathcal{A}^0$ of invertible operators, which by Proposition 8.4.2 are exactly the invertible elements of $\mathcal{A}^0$ of index zero. On $G^0 \times \mathbb{C}$ define

$$(A,z) \sim (A',z') \iff A' = A(\mathrm{Id}+E),\ z' = \det_{\mathrm{Fr}}(\mathrm{Id}+E)\,z \ \text{ for some } \mathrm{Id}+E \in G^{-\infty}. \tag{8.12.1}$$

Lemma 8.12.1. (8.12.1) *is an equivalence relation. Let* $\mathcal{D} = (G^0\times\mathbb{C})/\!\sim$ *and* $\pi : \mathcal{D} \to \mathcal{G}$, $\pi[(A,z)] = [A]$. *Each fibre* $\mathcal{D}_a = \pi^{-1}(a)$ *is a one-dimensional complex vector space with the operations* $[(A,z)] + [(A,w)] = [(A,z+w)]$, $\lambda[(A,z)] = [(A,\lambda z)]$.

Proof. Reflexivity, symmetry and transitivity follow from $\det_{\mathrm{Fr}}(\mathrm{Id}) = 1$, $\det_{\mathrm{Fr}}((\mathrm{Id}+E)^{-1}) = \det_{\mathrm{Fr}}(\mathrm{Id}+E)^{-1}$ and multiplicativity. If $[A] = [A_0] = a$ then $A_0^{-1}A \in G^0 \cap (\mathrm{Id}+\Psi^{-\infty}_{\mathrm{iso}}) = G^{-\infty}$, so every element of $\mathcal{D}_a$ has a representative (A_0,w), and w is unique since $(A_0,w)\sim(A_0,w')$ forces $\mathrm{Id}+E = \mathrm{Id}$. Thus $[(A_0,w)] \mapsto w$ is a bijection $\mathcal{D}_a \to \mathbb{C}$. The operations above are well defined because the relation is linear in z, and they correspond to those of $\mathbb{C}$ under this bijection. □

Now let $y \mapsto A_y$ be a smooth family (8.11.1) over a manifold Y and $a_y = [A_y]$. Set $\mathcal{D}_Y = \bigsqcup_{y\in Y}\mathcal{D}_{a_y}$. A *local invertible lift* on $U \subset Y$ open is a smooth family $\tilde A_y = A_y + E_y \in G^0$, $y \in U$, with E_y residual; it defines the section $s_{\tilde A}(y) = [(\tilde A_y, 1)]$ of $\mathcal{D}_Y$ over U. Local invertible lifts exist near every point of Y.

Proposition 8.12.2. $\mathcal{D}_Y$ *has a unique structure of smooth complex line bundle over* Y *for which the sections* $s_{\tilde A}$ *are smooth. If* $\tilde A' = \tilde A(\mathrm{Id}+X)$ *is another local invertible lift, then* X_y *is a smooth residual family and*

$$s_{\tilde A'} = \det_{\mathrm{Fr}}(\mathrm{Id}+X)^{-1} s_{\tilde A}. \tag{8.12.2}$$

If (8.11.2) *has a solution,* $\mathcal{D}_Y$ *is trivial.*

Proof. $\mathrm{Id}+X = \tilde A^{-1}\tilde A' = \mathrm{Id} + \tilde A^{-1}(E'-E)$, and $\tilde A^{-1}$ is a smooth family in $\Psi^0_{\mathrm{iso,ph}}$ by Lemma 8.1.5, so X is a smooth residual family. By (8.12.1), $(\tilde A, \det_{\mathrm{Fr}}(\mathrm{Id}+X)^{-1}) \sim (\tilde A(\mathrm{Id}+X), 1)$, which is (8.12.2). The transition functions $\det_{\mathrm{Fr}}(\mathrm{Id}+X_y)^{-1}$ are smooth and

non-vanishing by Theorem 8.3.6, and the cocycle condition holds automatically since all sections live in the same space $\mathcal{D}_Y$. Hence the local trivializations $s_{\tilde{A}}$ define a smooth line bundle structure, which is unique since a line bundle structure is determined by a family of local non-vanishing smooth sections. A global invertible lift gives a global non-vanishing section. □

The projection to $\mathcal{G}$ and the quotient map fit into the commutative diagram

$$\begin{array}{ccc} G^0 \times \mathbb{C} & \xrightarrow{/\sim} & \mathcal{D} \\ \downarrow & & \downarrow \pi \\ G^0 & \xrightarrow{/G^{-\infty}} & \mathcal{G}, \end{array}$$

and the pull-back of $\mathcal{D}$ to G^0 is canonically trivial, with the section $A \mapsto [(A, 1)]$. Over Y, the form $\bar{\alpha}$ of (8.11.4) defines a connection on $\mathcal{D}_Y$.

Theorem 8.12.3. *There is a unique connection ∇ on $\mathcal{D}_Y$ such that, for every local invertible lift $\tilde{A}$,*

$$\nabla s_{\tilde{A}} = -\overline{\mathrm{Tr}}_H(\tilde{A}^{-1} d\tilde{A}) \otimes s_{\tilde{A}}.$$

Its curvature is the globally defined 2-form $\beta - d\gamma$ of (8.11.3).

Proof. Write $\bar{\alpha}_{\tilde{A}} = \overline{\mathrm{Tr}}_H(\tilde{A}^{-1} d\tilde{A})$. We must check that the local definitions agree. With $\tilde{A}' = \tilde{A}(\mathrm{Id} + X)$,

$$\tilde{A}'^{-1} d\tilde{A}' = (\mathrm{Id} + X)^{-1}(\tilde{A}^{-1} d\tilde{A})(\mathrm{Id} + X) + (\mathrm{Id} + X)^{-1} dX,$$

so by Corollary 8.9.3, and since $\overline{\mathrm{Tr}}_H = \mathrm{Tr}$ on residual operators,

$$\bar{\alpha}_{\tilde{A}'} = \bar{\alpha}_{\tilde{A}} + \mathrm{Tr}((\mathrm{Id} + X)^{-1} dX) = \bar{\alpha}_{\tilde{A}} + d \log \det{}_{\mathrm{Fr}}(\mathrm{Id} + X).$$

With $f = \det_{\mathrm{Fr}}(\mathrm{Id} + X)^{-1}$, the Leibniz rule and (8.12.2) give

$$\nabla s_{\tilde{A}'} = \nabla(f s_{\tilde{A}}) = (d \log f - \bar{\alpha}_{\tilde{A}})\, s_{\tilde{A}'} = -\bar{\alpha}_{\tilde{A}'}\, s_{\tilde{A}'},$$

as required. The curvature is $-d\bar{\alpha}_{\tilde{A}} = \beta - d\gamma$ by Proposition 8.11.1. □

Thus $\beta - d\gamma$ is a curvature form, and by Chern–Weil theory the class $\frac{i}{2\pi}[\beta]$ represents the first Chern class of $\mathcal{D}_Y$ in $H^2(Y; \mathbb{C})$; in particular it is the image of an integral class. This is the topological origin of the obstruction in Proposition 8.11.1: the family A_y admits a smooth invertible perturbation only if $\mathcal{D}_Y$ is trivial.

Remark 8.12.4. If $f : G^0 \to \mathbb{C}^*$ were a multiplicative extension of $\det_{\mathrm{Fr}}$ which is smooth along smooth families, then the map $[(A, z)] \mapsto z/f(A)$ would be well defined on $\mathcal{D}$, since

$$\det{}_{\mathrm{Fr}}(\mathrm{Id} + E)\, z/f(A(\mathrm{Id} + E)) = z/f(A),$$

and would give a smooth non-vanishing linear functional on each $\mathcal{D}_Y$; hence every $\mathcal{D}_Y$ would be trivial and every $[\beta]$ would vanish. So any family with $[\beta] \neq 0$ shows that no such extension exists.

8.13 Exercises

These exercises cover both this chapter and Chapter 7.

Exercise 8.1 (Isotropic Sobolev spaces)**.** For $k \in \mathbb{N}_0$ define

$$H^k_{\mathrm{iso}}(\mathbb{R}^n) = \{u \in L^2(\mathbb{R}^n);\ x^\alpha D_x^\beta u \in L^2(\mathbb{R}^n)\ \forall\, |\alpha| + |\beta| \leq k\},$$
$$H^{-k}_{\mathrm{iso}}(\mathbb{R}^n) = \Big\{u \in \mathcal{S}'(\mathbb{R}^n);\ u = \sum_{|\alpha|+|\beta|\leq k} x^\alpha D_x^\beta u_{\alpha\beta},\ u_{\alpha\beta} \in L^2(\mathbb{R}^n)\Big\}.$$

Show that if $A \in \Psi^p_{\mathrm{iso,ph}}(\mathbb{R}^n)$, $p \in \mathbb{Z}$, then $A : H^k_{\mathrm{iso}} \to H^{k-p}_{\mathrm{iso}}$ for all $k \in \mathbb{Z}$. Deduce, using elliptic isotropic operators, that the definition

$$H^s_{\mathrm{iso}}(\mathbb{R}^n) = \{u \in \mathcal{S}'(\mathbb{R}^n);\ Au \in L^2(\mathbb{R}^n)\ \forall\, A \in \Psi^{-s}_{\mathrm{iso,ph}}(\mathbb{R}^n)\}, \quad s \in \mathbb{R},$$

is consistent with the previous one, and that

$$A \in \Psi^M_{\mathrm{iso,ph}}(\mathbb{R}^n) \implies A : H^s_{\mathrm{iso}} \to H^{s-M}_{\mathrm{iso}}, \qquad \bigcap_s H^s_{\mathrm{iso}} = \mathcal{S}(\mathbb{R}^n), \quad \bigcup_s H^s_{\mathrm{iso}} = \mathcal{S}'(\mathbb{R}^n),$$
$$A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n) \text{ elliptic},\ u \in \mathcal{S}'(\mathbb{R}^n),\ Au \in H^s_{\mathrm{iso}} \implies u \in H^{s+m}_{\mathrm{iso}}.$$

Exercise 8.2. Show that if $\epsilon > 0$ then $H^\epsilon_{\mathrm{iso}}(\mathbb{R}^n) \subset \langle x\rangle^{-\epsilon} L^2(\mathbb{R}^n) \cap H^\epsilon(\mathbb{R}^n)$, and deduce that the inclusion $H^\epsilon_{\mathrm{iso}}(\mathbb{R}^n) \hookrightarrow L^2(\mathbb{R}^n)$ is compact.

Exercise 8.3. Using Exercise 8.2, or otherwise, show that each element of $\Psi^{-\epsilon}_{\mathrm{iso,ph}}(\mathbb{R}^n)$, $\epsilon > 0$, is compact on $L^2(\mathbb{R}^n)$.

Exercise 8.4. Without using the Fredholm theory of (I4), show that if $E \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ there is $F \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ such that $(\mathrm{Id}+E)(\mathrm{Id}+F) = \mathrm{Id}+G$ with $G \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ of finite rank, i.e. $G \cdot \mathcal{S}(\mathbb{R}^n)$ is finite dimensional. (Hint: write $E = E_1 + E_2$ with E_1 of finite rank and $\|E_2\|_{\mathcal{B}(L^2)} < 1$, and use Lemma 8.1.4.)

Exercise 8.5. Using Exercise 8.4, show that an elliptic $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ has a parametrix $B \in \Psi^{-m}_{\mathrm{iso,ph}}(\mathbb{R}^n)$ up to finite rank error, i.e. $AB - \mathrm{Id}$ and $BA - \mathrm{Id}$ are finite rank elements of $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$. Deduce that A is Fredholm as an operator $H^M_{\mathrm{iso}}(\mathbb{R}^n) \to H^{M-m}_{\mathrm{iso}}(\mathbb{R}^n)$ for every M. (An operator between Hilbert spaces is Fredholm if it is bounded, with finite dimensional null space and closed range of finite codimension.)

Exercise 8.6 (The harmonic oscillator)**.** Show that $H = |D|^2 + |x|^2 = \sum_j (D_j^2 + x_j^2)$ is an elliptic element of $\Psi^2_{\mathrm{iso,ph}}(\mathbb{R}^n)$. With the creation and annihilation operators $C_j = D_j + ix_j$ and $A_j = D_j - ix_j = C_j^*$, show that

$$H = \sum_{j=1}^n C_j A_j + n = \sum_{j=1}^n A_j C_j - n,$$
$$[A_j, H] = 2A_j, \quad [C_j, H] = -2C_j, \quad [C_l, C_j] = 0, \quad [A_l, A_j] = 0, \quad [A_l, C_j] = 2\delta_{lj}\,\mathrm{Id}\,.$$

Knowing that $(H - \lambda)u = 0$ with $u \in \mathcal{S}'(\mathbb{R}^n)$ implies $u \in \mathcal{S}(\mathbb{R}^n)$ (why?), show that

$$E_\lambda = \{u \in \mathcal{S}'(\mathbb{R}^n);\ (H-\lambda)u = 0\} \neq \{0\} \iff \lambda \in n + 2\mathbb{N}_0,$$
$$E_{n+2k} = \Big\{\sum_{|\alpha|=k} c_\alpha C^\alpha e^{-|x|^2/2};\ c_\alpha \in \mathbb{C}\Big\}, \quad k \in \mathbb{N}_0.$$

Exercise 8.7 (Determinants of matrices)**.** Show that there is a unique function det : $M(N;\mathbb{C}) \to \mathbb{C}$ which is linear in each column, vanishes when two columns are equal, and satisfies $\det \mathrm{Id} = 1$, and that $\det A = \sum_\sigma \mathrm{sgn}(\sigma) \prod_i A_{i\sigma(i)}$. Show that $\det(AB) = \det A \det B$ (consider $A \mapsto \det(AB)$ for fixed B), that A is invertible if and only if $\det A \neq 0$, that $\frac{d}{ds} \det(\mathrm{Id} + sA)|_{s=0} = \mathrm{tr}\, A$, and deduce Jacobi's formula (8.3.1). Show also that det is the unique multiplicative polynomial function on $M(N;\mathbb{C})$ with $\frac{d}{ds} \det(\mathrm{Id} + sA)|_{s=0} = \mathrm{tr}\, A$.

Exercise 8.8 (The 1-form α is closed)**.** Let $A(s,t)$ be a smooth two-parameter family in $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ and $\alpha = \mathrm{Tr}(A^{-1}\partial_s A)\, ds + \mathrm{Tr}(A^{-1}\partial_t A)\, dt$ the pull-back of $\mathrm{Tr}(A^{-1}dA)$. Show that

$$d\alpha = \mathrm{Tr}([A^{-1}\partial_t A, A^{-1}\partial_s A])\, ds \wedge dt,$$

and hence that $d\alpha = 0$. Explain why this implies that $\mathrm{Tr}(A^{-1}dA)$ is closed on $G^{-\infty}_{\mathrm{iso}}$, and show directly, using Lemma 8.3.2, that its integral over any closed path lies in $2\pi i\mathbb{Z}$.

Exercise 8.9 (Bundle coefficients)**.** If V and W are finite dimensional complex vector spaces, show that the space $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n; V, W)$ of operators from $\mathcal{S}(\mathbb{R}^n; V)$ to $\mathcal{S}'(\mathbb{R}^n; W)$ given by matrices with entries in $\Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$, with respect to bases of V and W, is independent of the choice of bases. Extend the results of Chapter 7 to these spaces.

Exercise 8.10 (Ellipticity is necessary for the Fredholm property)**.** Let $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n; \mathbb{C}^N)$ be Fredholm as an operator $H^s_{\mathrm{iso}} \to H^{s-m}_{\mathrm{iso}}$ for some s. Show that A is elliptic, as follows.

1. Reduce to the case $m = s = 0$, using invertible elliptic operators of appropriate orders.
2. If the principal symbol of A is not invertible at $\theta \in \mathbb{S}^{2n-1}$, construct a sequence $u_k \in \mathcal{S}$ with $\|u_k\|_{L^2} = 1$, $u_k \to 0$ weakly and $Au_k \to 0$ in L^2 (consider $u_k = e^{ik^2 x\cdot\xi_0}\phi(x/k - kx_0)v$, suitably normalized, with $(x_0, \xi_0) = \theta$ and v in the null space of the principal symbol at θ).
3. Show that a Fredholm operator satisfies $\|u\| \leq C\|Au\| + C\|\Pi u\|$ with Π the projection onto its finite dimensional null space, and derive a contradiction.

Exercise 8.11 (The Koszul complex)**.** Let $\Lambda^k = \Lambda^k\mathbb{C}^n$ with basis $dx^\alpha = dx^{\alpha_1} \wedge \cdots \wedge dx^{\alpha_k}$, $\alpha_1 < \cdots < \alpha_k$, declared orthonormal, and let $\mathcal{S}'(\mathbb{R}^n; \Lambda^k)$ be the space of $u = \sum_\alpha u_\alpha\, dx^\alpha$, $u_\alpha \in \mathcal{S}'(\mathbb{R}^n)$. Using the annihilation operators of Exercise 8.6 define

$$D : \mathcal{S}'(\mathbb{R}^n; \Lambda^k) \to \mathcal{S}'(\mathbb{R}^n; \Lambda^{k+1}), \qquad Du = \sum_{j=1}^n \sum_\alpha A_j u_\alpha\, dx^j \wedge dx^\alpha.$$

Show that $D^2 = 0$, and that the formal adjoint of D with respect to the L^2 pairing is $D^* u = \sum_{j,\alpha} C_j u_\alpha\, \iota_j dx^\alpha$, where ι_j is contraction with e_j, the adjoint of $dx^j\wedge$. Show that $D + D^* \in \Psi^1_{\mathrm{iso,ph}}(\mathbb{R}^n; \Lambda^*)$ is elliptic and that

$$(D + D^*)^2 = DD^* + D^*D = H - n + 2\mathcal{N},$$

where $\mathcal{N} = k$ on Λ^k. Deduce that the null space of $D + D^*$ on $\mathcal{S}'(\mathbb{R}^n; \Lambda^*)$ is spanned by $e^{-|x|^2/2} \in \mathcal{S}(\mathbb{R}^n; \Lambda^0)$, and that

$$\{u \in \mathcal{S}'(\mathbb{R}^n);\ Du = 0\} = \mathbb{C}e^{-|x|^2/2},$$
$$\{u \in \mathcal{S}'(\mathbb{R}^n; \Lambda^k);\ Du = 0\} = D\,\mathcal{S}'(\mathbb{R}^n; \Lambda^{k-1}),\ k \geq 1.$$

Conclude that $D + D^*$, as an elliptic element of $\Psi^1_{\mathrm{iso,ph}}(\mathbb{R}^n; \Lambda^{\mathrm{even}}, \Lambda^{\mathrm{odd}})$, has index 1 (and as an operator from odd to even forms, index -1).

Exercise 8.12 (Isotropic essential support)**.** For $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ with left symbol a, define the isotropic essential support (operator wavefront set) $\mathrm{WF}'_{\mathrm{iso}}(A) \subset \mathbb{R}^{2n} \setminus \{0\}$ as the complement of the set of $Z_0 \neq 0$ having a conic neighbourhood Γ in which a is of order $-\infty$: $|D^\alpha a(Z)| \leq C_{\alpha,k}\langle Z\rangle^{-k}$ for $Z \in \Gamma$ and all α, k. Show that $\mathrm{WF}'_{\mathrm{iso}}(A)$ is closed and conic, that it is unchanged if the left symbol is replaced by the right or Weyl symbol, and that

$$\mathrm{WF}'_{\mathrm{iso}}(A+B) \subset \mathrm{WF}'_{\mathrm{iso}}(A) \cup \mathrm{WF}'_{\mathrm{iso}}(B), \qquad \mathrm{WF}'_{\mathrm{iso}}(AB) \subset \mathrm{WF}'_{\mathrm{iso}}(A) \cap \mathrm{WF}'_{\mathrm{iso}}(B),$$
$$\mathrm{WF}'_{\mathrm{iso}}(A) = \emptyset \iff A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n).$$

Exercise 8.13 (Isotropic partition of unity)**.** Let U_i be a finite open cover of $\mathbb{S}^{2n-1}$ and $\tilde{U}_i = \{Z \in \mathbb{R}^{2n} \setminus \{0\};\ Z/|Z| \in U_i\}$ the corresponding conic cover of $\mathbb{R}^{2n} \setminus \{0\}$. Show that there are $A_i \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n)$ with $\mathrm{WF}'_{\mathrm{iso}}(A_i) \subset \tilde{U}_i$ and $\mathrm{Id} - \sum_i A_i \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$.

Exercise 8.14. Suppose $A \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$ is elliptic and has index zero. Show that the operator $A + F$ of Proposition 8.4.2 is an isomorphism of $\mathcal{S}(\mathbb{R}^n)$, of $\mathcal{S}'(\mathbb{R}^n)$, and from $H^s_{\mathrm{iso}}(\mathbb{R}^n)$ to $H^{s-m}_{\mathrm{iso}}(\mathbb{R}^n)$ for every s.

Exercise 8.15 (Isotropic wavefront set)**.** For $A \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n)$ let $\mathrm{Char}_{\mathrm{iso}}(A) = \{Z \neq 0;\ \sigma_0(A)(Z/|Z|) = 0\}$, where $\sigma_0(A)$ is the principal symbol. For $u \in \mathcal{S}'(\mathbb{R}^n)$ define

$$\mathrm{WF}_{\mathrm{iso}}(u) = \bigcap \{\mathrm{Char}_{\mathrm{iso}}(A);\ A \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}^n),\ Au \in \mathcal{S}(\mathbb{R}^n)\}.$$

Show that $\mathrm{WF}_{\mathrm{iso}}(u)$ is a closed cone, that $\mathrm{WF}_{\mathrm{iso}}(u) = \emptyset$ if and only if $u \in \mathcal{S}(\mathbb{R}^n)$, and that $\mathrm{WF}_{\mathrm{iso}}(Bu) \subset \mathrm{WF}_{\mathrm{iso}}(u) \cap \mathrm{WF}'_{\mathrm{iso}}(B)$ for $B \in \Psi^m_{\mathrm{iso,ph}}(\mathbb{R}^n)$. (Use Exercises 8.12 and 8.13.)

Exercise 8.16 (Traces and determinants of lower regularity)**.** Let $m < -2n$. Show that Tr is continuous on $\Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$, that $\mathrm{Id} + E$ with $E \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ invertible on L^2 has inverse $\mathrm{Id} + F$ with $F \in \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$, and that the definition of $\det_{\mathrm{Fr}}$ by integration of $\mathrm{Tr}(A^{-1}dA)$ along paths extends to $\mathrm{Id} + \Psi^m_{\mathrm{iso}}(\mathbb{R}^n)$ with the properties of Proposition 8.3.4 and Theorem 8.3.6. (Hint: approximate E by $\mathrm{Op}(\chi(Z/k)e)$ as in the proof of Lemma 8.2.2, which converges in $\Psi^{m+\epsilon}_{\mathrm{iso}}$, and show that the corresponding determinants converge.)

Notes

The main departures from the original notes are as follows. The Fredholm determinant is constructed by integrating $\mathrm{Tr}(A^{-1}dA)$ along paths and comparing with finite dimensional determinants via the harmonic oscillator basis (Lemma 8.3.2); this gives path independence and the limit formula (8.3.3) at once, and the holomorphy across non-invertible operators is reduced, via Lemma 8.3.5, to a finite dimensional determinant. The residue trace is normalized as $-\mathrm{Res}_{z=0}$, so that (8.7.2) holds with a positive sign; this agrees with the usual convention for Wodzicki's residue [13], since z here corresponds to $-s$ in the zeta-function regularization $\mathrm{Tr}(AQ^{-s})$. The exterior derivation is defined directly as commutation with $\dot{A}$, which avoids inverting the regularizing family away from $z = 0$. The determinant line bundle is constructed over finite dimensional families, where it is an honest smooth line bundle, and we have added the connection of Theorem 8.12.3, whose curvature is the obstruction form of Proposition 8.11.1.

For the residue trace on compact manifolds see [13, 5]; for the isotropic (Shubin) calculus and its spectral theory see [15]; the determinant bundle originates in [10]. The Chern–Weil description of the first Chern class used at the end of Section 8.12 can be found, for instance, in Appendix C of J. Milnor and J. Stasheff, *Characteristic classes*.

Part III

Boundary problems and propagation

Chapter 9

Elliptic boundary problems

An elliptic differential operator on a compact manifold *with boundary* has, in general, an infinite-dimensional null space. The basic problem of the theory of elliptic boundary problems is to parametrize this null space by boundary data, and then to decide which boundary conditions single out a Fredholm problem. In this chapter we carry this out, following the approach of Calderón, for first-order operators of Dirac type. The analysis rests on three ingredients: a careful description of function spaces and distributions on a manifold with boundary, culminating in the *jump formula* (an invariant form of Green's formula); the observation that a manifold with boundary can always be regarded as a closed subset of a manifold without boundary, so that the calculus of Chapter 5 is available; and an explicit analysis of the boundary behaviour of $Q(w\,\delta(x))$, where Q is a parametrix for a Dirac-type operator and $w\,\delta(x)$ is a layer supported on a hypersurface. For Dirac-type operators the symbol of Q is rational in the fibre variables, and contour deformation shows that these layer potentials are smooth up to the hypersurface from each side, with boundary values given by pseudodifferential operators.

We treat first the model situation of a hypersurface H dividing a compact manifold M without boundary, on which D is assumed invertible; here the Calderón projector $\Pi_{\mathcal{C}}$, a pseudodifferential projection on H whose range is the space of boundary values of solutions on one side, is obtained by an explicit formula. We then treat a compact manifold with boundary, where the same construction works modulo smoothing errors, and where some abstract functional analysis of smoothing operators and generalized inverses is needed to correct them; together with Green's formula and a right inverse for the adjoint this yields a Calderón projector, unique up to the choice of its null space. Finally we show that a pseudodifferential projection Π on the boundary defines a Fredholm boundary problem $\{u;\ \Pi(u|_{\partial X}) = 0\}$ precisely when Π restricted to the range of $\Pi_{\mathcal{C}}$ is Fredholm, with index the relative index of the two projections, and give a symbolic criterion for this which is the analogue of the Lopatinski–Shapiro condition. Figure 9.1 shows how these steps depend on each other.

Discussion. Why Calderón's approach? For an elliptic operator on a closed manifold the symbol calculus produces a parametrix, and everything follows. On a manifold with boundary the interior symbol knows nothing about the boundary, so a boundary condition has to be imposed by hand. Calderón's idea is to move the whole problem to the boundary: first describe exactly which boundary values solutions can have (the range of a projection $\Pi_{\mathcal{C}}$), and then ask how a proposed boundary condition interacts with that range. Both questions are answered by pseudodifferential operators on the closed manifold ∂X, where the calculus of Chapter 5 applies.

Throughout, $\Psi^m(Y; E, F)$ for a compact manifold Y without boundary is as in Chapter 5;

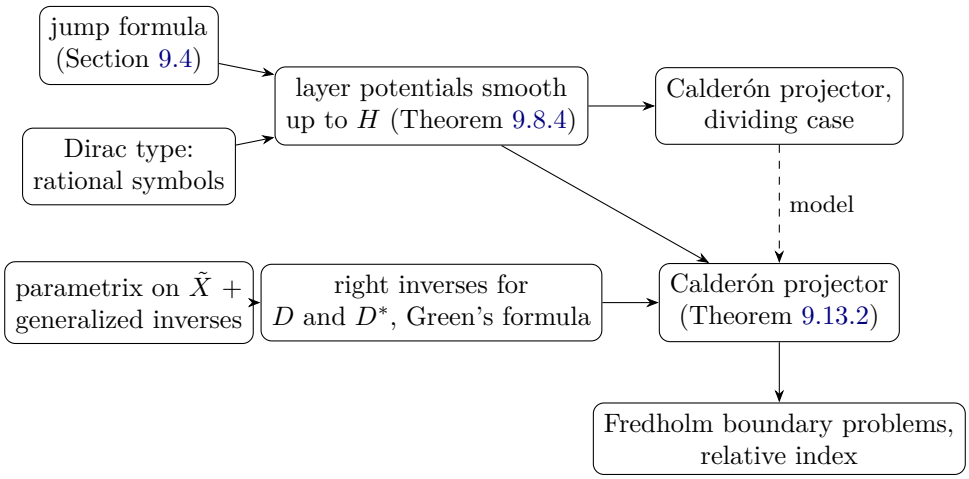


Figure 9.1: Logic of the chapter. The upper row is the model case of a dividing hypersurface, where D is invertible; the lower row repeats the construction on a manifold with boundary, where only a parametrix is available and finite-rank smoothing errors must be removed by functional analysis.

"classical" means that the local full symbols are polyhomogeneous (in S^m_{ph}), and then the principal symbol σ_m is a homogeneous function on $T^*Y \setminus 0$. A manifold is always assumed paracompact and Hausdorff, and vector bundles are complex.

9.1 Manifolds with boundary

We begin with the basic definitions. The one idea to retain from this section is that a compact manifold with boundary always sits inside a compact manifold without boundary, its double, so that functions and operators on X can be obtained by restriction from a closed manifold.

The local model for a manifold with boundary is the half-space

$$\mathbb{R}^{n,1} = [0,\infty) \times \mathbb{R}^{n-1}.$$

An *open set* in $\mathbb{R}^{n,1}$ means a relatively open set, $O = O' \cap \mathbb{R}^{n,1}$ with $O' \subset \mathbb{R}^n$ open. A map $F : O_1 \to O_2$ between open subsets of $\mathbb{R}^{n,1}$ is *smooth* if it is the restriction of a smooth map: there are open sets $O_i' \subset \mathbb{R}^n$ with $O_i = O_i' \cap \mathbb{R}^{n,1}$ and a $\mathcal{C}^\infty$ map $F' : O_1' \to O_2'$ with $F = F'|_{O_1}$. This is much stronger than smoothness of F on $O_1 \cap \{x_1 > 0\}$. A *diffeomorphism* is a smooth bijection with smooth inverse.

Definition 9.1.1. A *coordinate system* on a topological space X is a homeomorphism $\Phi : U \to O$ from an open set $U \subset X$ onto an open set $O \subset \mathbb{R}^{n,1}$. Two coordinate systems (U_i, Φ_i, O_i) are *compatible* if $U_1 \cap U_2 = \emptyset$ or $\Phi_2 \circ \Phi_1^{-1} : \Phi_1(U_1 \cap U_2) \to \Phi_2(U_1 \cap U_2)$ is a diffeomorphism. A $\mathcal{C}^\infty$ *manifold with boundary* is a paracompact Hausdorff space with a maximal covering by mutually compatible coordinate systems.

Equivalently, X is a manifold with boundary if there is a manifold $\tilde{X}$ without boundary and $f \in \mathcal{C}^\infty(\tilde{X})$, real-valued, with $df \neq 0$ on $\{f = 0\}$, such that $X = \{p \in \tilde{X};\ f(p) \geq 0\}$ with the coordinate systems obtained by restriction. One direction is clear; the doubling construction below gives the other.

The *boundary* ∂X is the union of the sets $\Phi^{-1}(O \cap (\{0\} \times \mathbb{R}^{n-1}))$ over coordinate systems; it is well defined because a diffeomorphism of open subsets of $\mathbb{R}^{n,1}$ maps boundary points

to boundary points (its extension is a local diffeomorphism of $\mathbb{R}^n$, hence an open map, so it cannot map a boundary point to an interior point of $\mathbb{R}^{n,1}$ and still map a neighbourhood into $\mathbb{R}^{n,1}$, and symmetrically for the inverse). ∂X is a manifold without boundary, compact if X is, and $X^\circ = X \setminus \partial X$ is a manifold without boundary.

A *boundary defining function* is a real $\rho \in \mathcal{C}^\infty(X)$ (see the next section) with $\rho \geq 0$, $\partial X = \{\rho = 0\}$ and $d\rho \neq 0$ on ∂X. Such functions exist: locally the first coordinate is one, and these may be patched together with a partition of unity, since positive combinations of local defining functions are again defining functions.

Lemma 9.1.2 (Collar neighbourhoods)**.** *Let X be a manifold with compact boundary and ρ a boundary defining function. Then there exist $\epsilon > 0$, an open neighbourhood C of ∂X and a diffeomorphism $\varphi : C \to [0, \epsilon) \times \partial X$ whose first component is ρ.*

Proof. Choose a Riemannian metric on X (by patching local ones) and let V be a smooth vector field equal to $\nabla\rho/|\nabla\rho|^2$ near ∂X; it is inward pointing and $V\rho = 1$ near ∂X. For $p \in \partial X$ the integral curve $t \mapsto \exp(tV)p$ exists for $0 \leq t < \epsilon$, for some $\epsilon > 0$ uniform by compactness of ∂X, and $\rho(\exp(tV)p) = t$ for small t. The map $\psi : [0, \epsilon) \times \partial X \ni (t, p) \mapsto \exp(tV)p$ is smooth, its differential at $t = 0$ is an isomorphism (it maps ∂_t to V, transversal to ∂X), and it is injective since $\rho(\psi(t, p)) = t$ and the curves starting at distinct boundary points are distinct. Shrinking ϵ, ψ is a diffeomorphism onto an open neighbourhood C; take $\varphi = \psi^{-1}$. □

Using a collar we can *double* X. Let

$$\tilde{X} = (X \sqcup X)/\partial X$$

be two copies of X with corresponding boundary points identified, and define $\mathcal{C}^\infty(\tilde{X})$ to consist of the pairs $(u_1, u_2) \in \mathcal{C}^\infty(X) \oplus \mathcal{C}^\infty(X)$ for which there is $f \in \mathcal{C}^\infty((-\epsilon, \epsilon) \times \partial X)$ with $(\varphi^{-1})^*(u_1|_C) = f(\rho, \cdot)$ and $(\varphi^{-1})^*(u_2|_C) = f(-\rho, \cdot)$. This is a $\mathcal{C}^\infty$ structure on $\tilde{X}$, a manifold without boundary which is compact if X is, the first copy embeds X as $\{x \geq 0\}$ where $x \in \mathcal{C}^\infty(\tilde{X})$ is equal to ρ on the first copy and $-\rho$ on the second near ∂X, and $\mathcal{C}^\infty(X) = \mathcal{C}^\infty(\tilde{X})|_X$. Vector bundles over X double in the same way (using the collar to identify E near ∂X with the pull-back of $E|_{\partial X}$). Thus all the usual objects — tensor bundles, densities, differential operators — may be obtained by restriction from $\tilde{X}$. Figure 9.2 shows the construction.

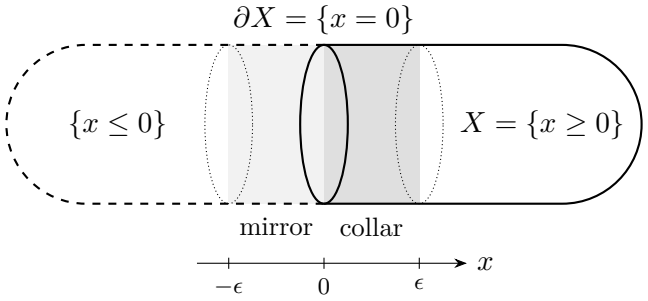


Figure 9.2: The double $\tilde{X}$ of X. The collar $C \cong [0, \epsilon) \times \partial X$ (dark) is reflected to the second copy (light, dashed outline); the function x equals ρ on the first copy and $-\rho$ on the second, and $X = \{x \geq 0\}$.

Discussion. The double is a device, not an object of interest: any closed manifold $\tilde{X}$ containing X as $\{x \geq 0\}$ would do, and nothing below depends on the choice. Its purpose is

to make the calculus of Chapter 5 available. The price is that we must keep track of *which side* of ∂X an object lives on. Much of the next three sections is bookkeeping of exactly this kind: a distribution on $\tilde{X}$ may be restricted to X, or it may be supported in X, and the two notions differ by terms concentrated on ∂X.

9.2 Smooth functions

For $O \subset \mathbb{R}^{n,1}$ open there are two natural definitions of smooth functions:

$$\mathcal{C}^\infty(O) = \{u : O \to \mathbb{C};\ u = \tilde{u}|_O \text{ for some } \tilde{u} \in \mathcal{C}^\infty(O'),\ O' \subset \mathbb{R}^n \text{ open},\ O = O' \cap \mathbb{R}^{n,1}\} \tag{9.2.1}$$

and the intrinsic one: $u \in \mathcal{C}^\infty(O^\circ)$, $O^\circ = O \cap ((0,\infty) \times \mathbb{R}^{n-1})$, with

$$\sup_{K \cap O^\circ} |D^\alpha u| < \infty \quad \text{for all } K \Subset O \text{ and all } \alpha. \tag{9.2.2}$$

Certainly (9.2.1) implies (9.2.2). Conversely, (9.2.2) implies that each $D^\alpha u$ is uniformly continuous on $K \cap O^\circ$, hence extends continuously to O, and these extensions are the derivatives of the extension of u in the closed half-space.

Proposition 9.2.1. *If u satisfies (9.2.2), then u extends to a smooth function on an open set $O' \subset \mathbb{R}^n$ with $O' \cap \mathbb{R}^{n,1} = O$; hence the two definitions agree. Moreover there is a linear extension map $E : \mathcal{C}^\infty(O) \to \mathcal{C}^\infty(O')$, continuous in the sense that for every compact $K' \subset O'$ and k there is a compact $K \subset O$ with $\sup_{K'} |D^\alpha Eu| \le C \sum_{|\beta| \le k} \sup_K |D^\beta u|$ for $|\alpha| \le k$.*

Proof. We prove the existence of extensions, which is all that is used below. By a partition of unity we may assume u has compact support in O. Let $u_k(y) = \partial_x^k u(0,y)$, which are in $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{n-1})$. By Borel's lemma with parameters (the usual proof, $\tilde{u}(x,y) = \sum_k \chi(x/\epsilon_k) x^k u_k(y)/k!$ with $\epsilon_k \downarrow 0$ chosen so that the kth term is bounded by 2^{-k} in $\mathcal{C}^{k-1}$, works verbatim; see [7, Section 1.2]) there is $\tilde{u} \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ with $\partial_x^k \tilde{u}(0,y) = u_k(y)$ for all k. Set $Eu = u$ for $x \ge 0$ and $Eu = \tilde{u}$ for $x < 0$. All derivatives of Eu from the two sides are continuous and agree on $x = 0$, so $Eu \in \mathcal{C}^k$ for every k (a function which is $\mathcal{C}^1$ on each closed half-space with matching derivatives is $\mathcal{C}^1$, by the mean value theorem, and one iterates). A linear continuous extension with the stated bounds is Seeley's extension operator $Eu(x,y) = \sum_k a_k \psi(-b_k x) u(-b_k x, y)$ for $x < 0$, with $b_k = 2^k$, ψ a cut-off and a_k chosen with $\sum_k a_k (-b_k)^j = 1$ for all j and $\sum_k |a_k| b_k^j < \infty$; see the Notes. □

For a manifold X with boundary, $\mathcal{C}^\infty(X)$ is the space of functions whose coordinate representations are smooth in this sense; it suffices to check this for one covering by coordinate systems. As for manifolds without boundary, $\mathcal{C}^\infty(X)$ admits partitions of unity, and its topology is given by the seminorms $\sup_K |D^\alpha (\Phi^{-1})^*(u|_U)|$, $K \Subset O$, of which a countable subset suffices; for X compact $\mathcal{C}^\infty(X)$ is a Fréchet space. The restriction map $\mathcal{C}^\infty(X) \ni u \mapsto u|_{\partial X} \in \mathcal{C}^\infty(\partial X)$ is surjective: using a collar, extend $v \in \mathcal{C}^\infty(\partial X)$ to be independent of ρ and cut off.

We think of $\mathcal{C}^\infty(X) \subset \mathcal{C}^\infty(X^\circ)$; a differential operator $P \in \mathrm{Diff}^m(X)$ is a differential operator on X° with coefficients smooth up to the boundary, and $P : \mathcal{C}^\infty(X) \to \mathcal{C}^\infty(X)$. The space of functions vanishing to infinite order at the boundary is

$$\dot{\mathcal{C}}^\infty(X) = \{u \in \mathcal{C}^\infty(X);\ (Pu)|_{\partial X} = 0\ \ \forall\, P \in \mathrm{Diff}^*(X)\}.$$

Extension by zero identifies $\dot{\mathcal{C}}^\infty(X)$ with the closed subspace of $\mathcal{C}^\infty(\tilde{X})$ of functions supported in X. Given a collar with normal variable $x = \rho$, Taylor's theorem and Borel's lemma give an isomorphism

$$\mathcal{C}^\infty(X)/\dot{\mathcal{C}}^\infty(X) \cong \mathcal{C}^\infty(\partial X)[[x]], \quad u \mapsto \sum_k \frac{x^k}{k!}\,(\partial_x^k u)|_{\partial X},$$

onto formal power series with coefficients in $\mathcal{C}^\infty(\partial X)$. The same applies to sections of vector bundles.

9.3 Distributions

Let X be a compact manifold with boundary and $\tilde{X}$ a compact manifold without boundary containing X as $\{x \geq 0\}$ (for example the double). There are three spaces of distributions associated with X. The coarsest is $\mathcal{C}^{-\infty}(X^\circ)$, the dual of $\mathcal{C}_c^\infty(X^\circ;\Omega)$; its elements may behave arbitrarily badly at ∂X and play only a conceptual role. The direct analogue of the boundaryless case is the space of *supported distributions*

$$\dot{\mathcal{C}}^{-\infty}(X) = \bigl(\mathcal{C}^\infty(X;\Omega)\bigr)'.$$

Since the restriction map $\mathcal{C}^\infty(\tilde{X};\Omega) \to \mathcal{C}^\infty(X;\Omega)$ is continuous and surjective, duality gives an injective map

$$\dot{\mathcal{C}}^{-\infty}(X) \ni u \longmapsto \tilde{u} \in \mathcal{C}^{-\infty}(\tilde{X}), \quad \tilde{u}(\varphi) = u(\varphi|_X). \tag{9.3.1}$$

Proposition 9.3.1. *The range of* (9.3.1) *is* $\{\tilde{u} \in \mathcal{C}^{-\infty}(\tilde{X});\ \operatorname{supp}\tilde{u} \subset X\}$.

The proof is given after Proposition 9.3.2. We use (9.3.1) to identify $\dot{\mathcal{C}}^{-\infty}(X)$ with the distributions on $\tilde{X}$ supported in X; this is consistent with the identification of $\dot{\mathcal{C}}^\infty(X)$ with smooth functions on $\tilde{X}$ supported in X. The closed subspace $\dot{\mathcal{C}}^\infty(X;\Omega) \subset \mathcal{C}^\infty(\tilde{X};\Omega)$ has dual the space of *extendible distributions*

$$\mathcal{C}^{-\infty}(X) = \bigl(\dot{\mathcal{C}}^\infty(X;\Omega)\bigr)',$$

and transposing the inclusion gives the restriction map

$$R_X : \mathcal{C}^{-\infty}(\tilde{X}) \to \mathcal{C}^{-\infty}(X), \quad R_X u(\varphi) = u(\varphi),\ \varphi \in \dot{\mathcal{C}}^\infty(X;\Omega). \tag{9.3.2}$$

This is not an identification: its null space is $\{v \in \mathcal{C}^{-\infty}(\tilde{X});\ \operatorname{supp} v \cap X^\circ = \emptyset\} = \dot{\mathcal{C}}^{-\infty}(\tilde{X}\backslash X^\circ)$, the distributions supported "on the other side of the boundary".

Proposition 9.3.2. *$\mathcal{C}_c^\infty(X^\circ;\Omega)$ is dense in $\dot{\mathcal{C}}^\infty(X;\Omega)$. Hence restriction gives an injective map $\mathcal{C}^{-\infty}(X) \hookrightarrow \mathcal{C}^{-\infty}(X^\circ)$, while $R_X : \dot{\mathcal{C}}^{-\infty}(X) \to \mathcal{C}^{-\infty}(X)$ (and a fortiori R_X on $\mathcal{C}^{-\infty}(\tilde{X})$) is surjective.*

Proof. Let V be a real vector field on $\tilde{X}$ which is inward pointing across ∂X, and $F_s = \exp(sV)$, a one-parameter group of diffeomorphisms of $\tilde{X}$ with $F_s(X) \subset X^\circ$ for $s > 0$. If $\varphi \in \dot{\mathcal{C}}^\infty(X)$ then $F_s^*\varphi \to \varphi$ in $\mathcal{C}^\infty(\tilde{X})$ as $s \to 0$, and for $s < 0$, $F_s^*\varphi = \varphi \circ F_s$ is supported in $F_{-s}(X) \Subset X^\circ$, so lies in $\mathcal{C}_c^\infty(X^\circ)$. The same argument applies to densities. Injectivity of $\mathcal{C}^{-\infty}(X) \to \mathcal{C}^{-\infty}(X^\circ)$ follows by duality. Surjectivity of R_X on $\dot{\mathcal{C}}^{-\infty}(X)$ follows from the Hahn–Banach theorem: a continuous linear functional on the closed subspace $\dot{\mathcal{C}}^\infty(X;\Omega) \subset \mathcal{C}^\infty(X;\Omega)$ extends to $\mathcal{C}^\infty(X;\Omega)$. □

Proof of Proposition 9.3.1. For $\tilde{u} \in \mathcal{C}^{-\infty}(\tilde{X})$ the condition $\operatorname{supp}\tilde{u} \subset X$ means $\tilde{u}(\varphi) = 0$ for all $\varphi \in \mathcal{C}^\infty_c(\tilde{X} \setminus X; \Omega)$. This certainly holds if $\tilde{u}$ comes from $\dot{\mathcal{C}}^{-\infty}(X)$, since then $\varphi|_X = 0$. Conversely, if it holds, then by continuity and the density of $\mathcal{C}^\infty_c(\tilde{X} \setminus X; \Omega)$ in $\dot{\mathcal{C}}^\infty(\tilde{X} \setminus X^\circ; \Omega)$ (Proposition 9.3.2 applied to the manifold with boundary $\tilde{X} \setminus X^\circ$), $\tilde{u}$ vanishes on the null space of the restriction $\mathcal{C}^\infty(\tilde{X}; \Omega) \to \mathcal{C}^\infty(X; \Omega)$, so it factors through this (open, by the open mapping theorem) surjection. □

On $\dot{\mathcal{C}}^{-\infty}(X)$ and $\mathcal{C}^{-\infty}(X)$ we may use the weak topologies, given by the seminorms $u \mapsto |u(\varphi)|$ for test functions φ.

Lemma 9.3.3. *$\mathcal{C}^\infty_c(X^\circ)$ is weakly dense in both $\dot{\mathcal{C}}^{-\infty}(X)$ and $\mathcal{C}^{-\infty}(X)$.*

Proof. For $u \in \dot{\mathcal{C}}^{-\infty}(X)$, with F_s as above and $s > 0$, the push-forward $u_s = (F_s)_* u$, $u_s(\varphi) = u(F_s^* \varphi)$, has compact support in X°, and $u_s \to u$ weakly since $F_s^* \varphi \to \varphi$ in $\mathcal{C}^\infty$. A distribution with compact support in X° is the weak limit of elements of $\mathcal{C}^\infty_c(X^\circ)$ (mollify in local coordinates after a partition of unity). Weak density being transitive, the first claim follows. For $u \in \mathcal{C}^{-\infty}(X)$ choose by Proposition 9.3.2 $\tilde{u} \in \mathcal{C}^{-\infty}(\tilde{X})$ with $R_X \tilde{u} = u$; $\tilde{u}$ is a weak limit of $\psi_k \in \mathcal{C}^\infty(\tilde{X})$ (on a compact manifold without boundary), so u is the weak limit of $\psi_k|_X \in \mathcal{C}^\infty(X)$. Finally $\psi \in \mathcal{C}^\infty(X)$ is the weak limit in $\mathcal{C}^{-\infty}(X)$ of $\chi_k \psi$ with $\chi_k \in \mathcal{C}^\infty_c(X^\circ)$, $0 \le \chi_k \le 1$, $\chi_k \to 1$ pointwise on X°, by dominated convergence. □

9.4 Boundary terms and the jump formula

To relate supported and extendible distributions we use the space of *boundary terms*

$$\dot{\mathcal{C}}^{-\infty}_{\partial X}(X) = \{u \in \dot{\mathcal{C}}^{-\infty}(X);\ \operatorname{supp} u \subset \partial X\},$$

the support being computed in $\tilde{X}$, and the map "cutting off at the boundary"

$$\mathcal{C}^\infty(X) \ni u \longmapsto u_c \in \dot{\mathcal{C}}^{-\infty}(X), \quad u_c(\varphi) = \int_X u\varphi \ \ \forall\, \varphi \in \mathcal{C}^\infty(X; \Omega),$$

which in terms of $\tilde{X}$ is extension by zero. The following is immediate from the definitions and Proposition 9.3.2.

Proposition 9.4.1. *For a compact manifold with boundary X there is a commutative diagram*

$$\begin{array}{ccccccccc} & & & & & & \dot{\mathcal{C}}^\infty(X) & & \\ & & & & & & \big\downarrow & & \\ & & & & & & \mathcal{C}^\infty(X) & & \\ & & & & & \overset{(\cdot)_c}{\swarrow} & \big\downarrow & & \\ 0 & \longrightarrow & \dot{\mathcal{C}}^{-\infty}_{\partial X}(X) & \longrightarrow & \dot{\mathcal{C}}^{-\infty}(X) & \xrightarrow{R_X} & \mathcal{C}^{-\infty}(X) & \longrightarrow & 0 \end{array}$$

with exact bottom row.

Proof. $R_X(u_c) = u$ for $u \in \mathcal{C}^\infty(X)$, which is the commutativity. Surjectivity of R_X is in Proposition 9.3.2. If $v \in \dot{\mathcal{C}}^{-\infty}_{\partial X}(X)$ then $v(\varphi) = 0$ for $\varphi \in \mathcal{C}^\infty_c(X^\circ; \Omega)$, hence by density for $\varphi \in \dot{\mathcal{C}}^\infty(X; \Omega)$, so $R_X v = 0$. Conversely if $u \in \dot{\mathcal{C}}^{-\infty}(X)$ and $R_X u = 0$ then $u(\varphi) = 0$ for $\varphi \in \mathcal{C}^\infty_c(X^\circ; \Omega)$, so $\operatorname{supp} u \cap X^\circ = \emptyset$ and $\operatorname{supp} u \subset \partial X$. □

Both maps from $\mathcal{C}^\infty(X)$ into distributions are injective; we regard the map into $\mathcal{C}^{-\infty}(X)$ as an identification, but *not* the map $u \mapsto u_c$.

Discussion. The model to keep in mind is $X = [0,\infty) \subset \mathbb{R}$ and the constant function 1. As an extendible distribution it is just 1: it only sees test functions vanishing to infinite order at 0, and its derivative is 0. As a supported distribution it is the Heaviside function $1_c = H(x)$, whose derivative is $\delta(x)$. The bottom row of Proposition 9.4.1 says that this is the only kind of discrepancy: two supported distributions with the same restriction to X differ by a distribution supported on ∂X. Green's formula, in the form of the jump formula below, computes the discrepancy for derivatives of smooth functions.

Actions of differential operators

Let E, F be vector bundles over X and $P \in \mathrm{Diff}^m(X; E, F)$. Its transpose $P^t \in \mathrm{Diff}^m(X; F^* \otimes \Omega, E^* \otimes \Omega)$ is defined by $\int_X \langle P^t\varphi, u\rangle = \int_X \langle \varphi, Pu\rangle$ for $u \in \mathcal{C}_c^\infty(X^\circ; E)$, $\varphi \in \mathcal{C}_c^\infty(X^\circ; F^* \otimes \Omega)$. Integration by parts shows that this identity continues to hold if $u \in \mathcal{C}^\infty(X; E)$ and $\varphi \in \dot{\mathcal{C}}^\infty(X; F^* \otimes \Omega)$, or vice versa, but *not* for u, φ both merely smooth up to the boundary. Accordingly P acts in two ways on distributions, by duality:

$$\begin{aligned} P &: \mathcal{C}^{-\infty}(X; E) \to \mathcal{C}^{-\infty}(X; F), && (Pu)(\varphi) = u(P^t\varphi),\ \varphi \in \dot{\mathcal{C}}^\infty(X; F^* \otimes \Omega), \\ \dot{P} &: \dot{\mathcal{C}}^{-\infty}(X; E) \to \dot{\mathcal{C}}^{-\infty}(X; F), && (\dot{P}u)(\varphi) = u(P^t\varphi),\ \varphi \in \mathcal{C}^\infty(X; F^* \otimes \Omega). \end{aligned}$$

The first extends the action on $\mathcal{C}^\infty(X; E)$. If $\tilde{P}$ is any extension of P to a differential operator on $\tilde{X}$, then $\dot{P}$ is the restriction of $\tilde{P}$ to distributions supported in X, since $\tilde{P}^t\tilde{\varphi}$ restricted to X depends only on $\tilde{\varphi}|_X$. The failure of $\dot{P}(u_c) = (Pu)_c$ for $u \in \mathcal{C}^\infty(X; E)$ is measured by a boundary term, $\dot{P}(u_c) - (Pu)_c \in \dot{\mathcal{C}}^{-\infty}_{\partial X}(X; F)$, which we now compute: this is Green's formula in distributional form.

Fix a collar $C \cong [0,\epsilon)_x \times \partial X$ as in Lemma 9.1.2, and identify E, F over C with the pull-backs of $E|_{\partial X}$, $F|_{\partial X}$ (possible since C retracts onto ∂X). On the double, x extends to $(-\epsilon, \epsilon) \times \partial X$ and we have the distributions

$$v\, D_x^j\delta(x) \in \dot{\mathcal{C}}^{-\infty}_{\partial X}(X; F), \quad v \in \mathcal{C}^\infty(\partial X; F),\ j \geq 0,$$

defined in local coordinates (x, y) as $v(y) \otimes D_x^j\delta(x)$, i.e. by $(v\, D_x^j\delta(x))(\varphi\,|dx\,dy|) = \int v(y) \cdot ((-D_x)^j\varphi)(0, y)\,dy$; this is independent of the choice of y. With $H(x)$ the Heaviside function, $u_c = H(x)u$ near ∂X, and $\delta(x) = \partial_x H(x) = iD_x H(x)$.

Proposition 9.4.2 (Jump formula)**.** *For each $P \in \mathrm{Diff}^m(X; E, F)$ there are differential operators $P_{jk} \in \mathrm{Diff}^{m-1-j-k}(\partial X; E, F)$, $j, k \geq 0$, $j + k \leq m - 1$, such that for all $u \in \mathcal{C}^\infty(X; E)$*

$$\dot{P}(u_c) - (Pu)_c = \sum_{j+k\leq m-1} (P_{jk}\,(D_x^k u)|_{\partial X})\, D_x^j\delta(x), \tag{9.4.1}$$

and $P_{0,m-1} = -i\,\sigma_m(P)(dx)|_{\partial X}$.

Proof. Both sides vanish if $u \in \dot{\mathcal{C}}^\infty(X; E)$ (then u_c is smooth on $\tilde{X}$), so only the Taylor series of u at ∂X matters and we may work in C. There

$$P = \sum_{l=0}^m P_l(x) D_x^l,$$

where $P_l(x) \in \mathrm{Diff}^{m-l}(\partial X; E, F)$ depends smoothly on x, and $P_m(0) = \sigma_m(P)(dx)$ (with the convention that D_x has symbol ξ). From $D_x(Hf) = HD_x f - i\delta(x)f(0,\cdot)$ one finds by induction

$$D_x^l(Hu) = HD_x^l u - i\sum_{k=0}^{l-1} ((D_x^k u)|_{\partial X})\, D_x^{l-1-k}\delta(x).$$

The tangential operators $P_l(x)$ commute with multiplication by H. Expanding $P_l(x) = \sum_{r<m} x^r P_{l,r} + x^m P'_l(x)$ by Taylor's formula and using $x^r D_x^p \delta(x) = c_{r,p} D_x^{p-r}\delta(x)$, where $c_{r,p} = i^r p!/(p-r)!$ for $r \le p$ and $c_{r,p} = 0$ for $r > p$ (in particular $c_{0,p} = 1$), we obtain (9.4.1) with

$$P_{jk} = -i \sum_{l,r:\ l-1-k-r=j} c_{r,l-1-k} P_{l,r}.$$

Here $P_{l,r}$ has order at most $m - l \le m - 1 - j - k$. The only contribution to $P_{0,m-1}$ comes from $l = m$, $r = 0$, giving $-iP_m(0) = -i\sigma_m(P)(dx)$. □

For first-order operators there is no choice involved:

Corollary 9.4.3. *If $D \in \mathrm{Diff}^1(X; E, F)$ and $u \in \mathcal{C}^\infty(X; E)$ then*

$$\dot{D}(u_c) = (Du)_c + \frac{1}{i}\,\sigma_1(D)(dx)(u|_{\partial X})\,\delta(x). \tag{9.4.2}$$

The right side does not depend on the choice of x: if $x' = ax$ with $a > 0$ then on ∂X, $\sigma_1(D)(dx') = a\sigma_1(D)(dx)$ while $\delta(x') = a^{-1}\delta(x)$. In the sequel we write b for the restriction map, $bu = u|_{\partial X}$.

Question. Why does only the boundary value bu appear in (9.4.2), and not the normal derivatives of u?

Answer. A first-order operator differentiates the Heaviside factor in $u_c = H(x)u$ at most once, producing $\delta(x)$ times the value of u at $x = 0$. Higher normal derivatives of u enter only through $D_x^l(Hu)$ with $l \ge 2$, as in the proof of Proposition 9.4.2, and they come with derivatives $D_x^j\delta(x)$ of the delta function. This is why first-order operators, and in particular Dirac operators, need only one piece of boundary data. □

The same discussion applies to an embedded hypersurface H with a defining function x in a manifold without boundary: if u is smooth on each of the two sides $\{\pm x \ge 0\}$ near H (i.e. $u|_{\{\pm x>0\}}$ extends smoothly to $\{\pm x \ge 0\}$), with boundary values $b_\pm u$, then applying (9.4.2) on both sides,

$$Du = (Du)_{+,c} + (Du)_{-,c} + \frac{1}{i}\sigma_1(D)(dx)(b_+u - b_-u)\delta(x), \tag{9.4.3}$$

where $(Du)_{\pm,c}$ is Du computed in $\{\pm x > 0\}$, extended by zero.

Formal solutions at the boundary

For an elliptic first-order operator, the Cauchy problem can always be solved formally at the boundary.

Lemma 9.4.4. *Let $D \in \mathrm{Diff}^1(X; V_1, V_2)$ with $\sigma_1(D)(dx)$ invertible at ∂X (this holds if D is elliptic).*

(i) For $f \in \mathcal{C}^\infty(X; V_2)$ and $v \in \mathcal{C}^\infty(\partial X; V_1)$ there is $u \in \mathcal{C}^\infty(X; V_1)$ with $bu = v$ and $Du - f \in \dot{\mathcal{C}}^\infty(X; V_2)$.

(ii) If $u \in \mathcal{C}^\infty(X; V_1)$, $bu = 0$ and $Du \in \dot{\mathcal{C}}^\infty(X; V_2)$, then $u \in \dot{\mathcal{C}}^\infty(X; V_1)$.

Proof. In the collar, shrinking it if necessary, $D = A(x)\big(D_x + E(x)\big)$ with $A(x)$ an invertible bundle map ($A(0) = \sigma_1(D)(dx)$) and $E(x) \in \mathrm{Diff}^1(\partial X; V_1)$ depending smoothly on x. Write $u_k = (D_x^k u)|_{\partial X}$. The condition that $D_x^k((D_x + E)u - A^{-1}f)$ vanish at $x = 0$ for all k is

$$u_{k+1} = (D_x^k(A^{-1}f))|_{x=0} - \sum_{j\le k} \binom{k}{j} (D_x^{k-j}E)(0)\, u_j, \quad k \ge 0.$$

For (i) set $u_0 = v$, define u_k recursively and use Borel's lemma to find u with these Taylor coefficients. For (ii), the recursion with f replaced by Du (all of whose Taylor coefficients vanish) and $u_0 = 0$ gives $u_k = 0$ for all k. □

9.5 Sobolev spaces

Boundary values of solutions will in general only lie in a Sobolev space, typically $H^{-\frac{1}{2}}(\partial X)$ for L^2 solutions, so we need Sobolev spaces on X and the regularity of layers $v\,\delta(x)$ on a hypersurface.

As for smooth functions, the (extendible) Sobolev spaces on a compact manifold with boundary $X \subset \tilde X$ are defined by restriction,

$$H^m(X) = R_X\big(H^m(\tilde X)\big) \subset \mathcal{C}^{-\infty}(X), \quad \|u\|_{H^m(X)} = \inf\{\|\tilde u\|_{H^m(\tilde X)};\ R_X\tilde u = u\}, \quad m \in \mathbb{R},$$

the quotient norm; independence of the choice of $\tilde X$ follows from the locality of Sobolev spaces and their invariance under diffeomorphisms. The *supported* Sobolev spaces are

$$\dot H^m(X) = \{u \in H^m(\tilde X);\ \operatorname{supp} u \subset X\} \subset \dot{\mathcal{C}}^{-\infty}(X),$$

closed subspaces of $H^m(\tilde X)$. Both are $\mathcal{C}^\infty(X)$-modules, and for a vector bundle E one sets $H^m(X;E) = H^m(X) \otimes_{\mathcal{C}^\infty(X)} \mathcal{C}^\infty(X;E)$, and similarly for $\dot H^m$.

Lemma 9.5.1. *$\mathcal{C}_c^\infty(X^\circ)$ is dense in $\dot H^m(X)$ for every $m \in \mathbb{R}$.*

Proof. With F_s as in the proof of Proposition 9.3.2, the operators $u \mapsto (F_s)_* u$ are uniformly bounded on $H^m(\tilde X)$ for $|s| \le 1$ (their norms depend on finitely many derivatives of F_s) and $(F_s)_* u \to u$ for $u \in \mathcal{C}^\infty(\tilde X)$, hence for all $u \in H^m(\tilde X)$ by density. For $u \in \dot H^m(X)$ and $s > 0$, $(F_s)_* u \in H^m(\tilde X)$ has compact support in X°, and is approximated in H^m by mollification in local coordinates. □

Proposition 9.5.2. *For $m \in \mathbb{R}$ the pairing $(u, v) \mapsto \int_X uv$, $u \in \mathcal{C}^\infty(X)$, $v \in \dot{\mathcal{C}}^\infty(X;\Omega)$, extends by continuity to a non-degenerate pairing identifying $H^m(X)$ with the dual of $\dot H^{-m}(X;\Omega)$, and $\dot H^m(X)$ with the dual of $H^{-m}(X;\Omega)$.*

Proof. The dual of the closed subspace $\dot H^{-m}(X;\Omega) \subset H^{-m}(\tilde X;\Omega)$ is the quotient of $H^m(\tilde X) = (H^{-m}(\tilde X;\Omega))'$ by the annihilator $\{v \in H^m(\tilde X);\ v(w) = 0\ \forall\, w \in \dot H^{-m}(X;\Omega)\}$. By Lemma 9.5.1 it suffices to test with $w \in \mathcal{C}_c^\infty(X^\circ;\Omega)$, so the annihilator is $\{v;\ v|_{X^\circ} = 0\}$, the null space of R_X on $H^m(\tilde X)$. Hence the dual is $H^m(\tilde X)/\operatorname{Nul}(R_X) \cong H^m(X)$ with the quotient norm. The second statement follows since Hilbert spaces are reflexive. □

A differential operator $P \in \mathrm{Diff}^k(X; E, F)$ defines bounded maps

$$P : H^m(X;E) \to H^{m-k}(X;F), \qquad \dot P : \dot H^m(X;E) \to \dot H^{m-k}(X;F),$$

by restriction from an extension $\tilde{P}$, consistently with the actions on distributions. Moreover

$$\mathcal{C}^{-\infty}(X;E) = \bigcup_m H^m(X;E), \qquad \mathcal{C}^{\infty}(X;E) = \bigcap_m H^m(X;E),$$
$$\dot{\mathcal{C}}^{-\infty}(X;E) = \bigcup_m \dot{H}^m(X;E), \qquad \dot{\mathcal{C}}^{\infty}(X;E) = \bigcap_m \dot{H}^m(X;E).$$

For the second: if $u \in H^m(X)$ for all m, then for each k, u is the restriction of a $\mathcal{C}^k$ function on $\tilde{X}$ (Sobolev embedding), so all derivatives of u are bounded on X° and $u \in \mathcal{C}^\infty(X)$ by Proposition 9.2.1. The others follow from the corresponding statements on $\tilde{X}$.

Layers on a hypersurface

We shall need the Sobolev regularity of distributions supported on a hypersurface. In $\mathbb{R}^n = \mathbb{R}_x \times \mathbb{R}^{n-1}_y$, for $v \in \mathcal{C}^{-\infty}_{\mathrm{c}}(\mathbb{R}^{n-1})$ the distribution $v \otimes \delta(x)$ has Fourier transform $\hat{v}(\eta)$, so, since $\int_{\mathbb{R}}(1+|\eta|^2+\xi^2)^t\,d\xi = c_t\langle\eta\rangle^{2t+1}$ with $c_t = \int(1+s^2)^t ds < \infty$ precisely for $t < -\frac{1}{2}$,

$$\|v \otimes \delta(x)\|^2_{H^t(\mathbb{R}^n)} = \frac{c_t}{2\pi}\|v\|^2_{H^{t+\frac{1}{2}}(\mathbb{R}^{n-1})}, \quad t < -\tfrac{1}{2}. \tag{9.5.1}$$

Lemma 9.5.3. *Let $H \subset M$ be a compact embedded hypersurface, with defining function x near H, in a manifold M without boundary, and E a vector bundle over M.*

(i) For $t < -\frac{1}{2}$, $v \mapsto v\,\delta(x)$ is bounded and injective $H^{t+\frac{1}{2}}(H;E) \to H^t_{\mathrm{c}}(M;E)$, with closed range; in particular for $t = -1$ it is an isomorphism of $H^{-\frac{1}{2}}(H;E)$ onto a closed subspace of $H^{-1}_{\mathrm{c}}(M;E)$.

(ii) Every $T \in H^{-1}_{\mathrm{c}}(M;E)$ with $\operatorname{supp} T \subset H$ is of the form $T = v\,\delta(x)$ with $v \in H^{-\frac{1}{2}}(H;E)$.

Proof. (i) follows from (9.5.1) by a partition of unity, which also shows that the norms of v and $v\,\delta(x)$ are equivalent. For (ii), localize: we may assume $T \in H^{-1}(\mathbb{R}^n)$ has compact support in $\{x=0\}$. If $\varphi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ vanishes on $\{x=0\}$, let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$ equal 1 near 0 and $\varphi_\epsilon = (1-\chi(x/\epsilon))\varphi$. Then $T(\varphi_\epsilon) = 0$, and $\varphi_\epsilon \to \varphi$ in H^1: indeed $\chi(x/\epsilon)\varphi$ is supported in $|x| \le C\epsilon$ where $|\varphi| \le C|x|$, so its L^2 norm is $O(\epsilon^{3/2})$ and that of its derivative is $O(\epsilon^{1/2})$ (the term $\epsilon^{-1}\chi'(x/\epsilon)\varphi$ being bounded). Hence $T(\varphi) = 0$, i.e. $T(\varphi)$ depends only on $\varphi(0,\cdot)$, and $T = v \otimes \delta(x)$ where $v(\psi) = T(\chi(x)\psi(y))$ is a compactly supported distribution. Then $\hat{T}(\xi,\eta) = \hat{v}(\eta)$ and $\|T\|^2_{H^{-1}} = c\int|\hat{v}(\eta)|^2\langle\eta\rangle^{-1}d\eta < \infty$, so $v \in H^{-\frac{1}{2}}$. □

9.6 Generalized Dirac operators

From now on we restrict to first-order operators whose symbol squares to the metric. The reason is analytic: for such operators the symbol of a parametrix is a *rational* function of the fibre variables (Lemma 9.8.2), and this is what makes the boundary analysis of Section 9.8 possible by a residue computation.

For $D \in \mathrm{Diff}^1(M;V_1,V_2)$, in local coordinates and trivializations $D = \sum_j A_j(z)D_{z_j} + A'(z)$, and the principal symbol is $\sigma_1(D)(z,\zeta) = \sum_j A_j(z)\zeta_j$, linear in $\zeta \in T^*_zM$.

Definition 9.6.1. Let M be a manifold with or without boundary, with a Riemannian metric g, and let V_1, V_2 be Hermitian vector bundles over M. An operator $D \in \mathrm{Diff}^1(M;V_1,V_2)$ is a *generalized Dirac operator* (or *of Dirac type*) if

$$\sigma_1(D)(z,\zeta)^*\,\sigma_1(D)(z,\zeta) = |\zeta|^2_g\,\mathrm{Id}_{V_1} \quad \text{for all } (z,\zeta) \in T^*M, \tag{9.6.1}$$

and $\operatorname{rank} V_1 = \operatorname{rank} V_2$. If $M_+ \subset M$ is a subset, D is of Dirac type on M_+ if (9.6.1) holds for $z \in M_+$.

We abbreviate $\sigma(\zeta) = \sigma_1(D)(z,\zeta)$ and write $\tau(\zeta) = \sigma(\zeta)^*$ for real ζ, extended complex-linearly to complex covectors, and $g(\zeta) = |\zeta|_g^2$, extended as a complex quadratic form. Then $\sigma(\zeta)$ is invertible for $\zeta \neq 0$ with

$$\sigma(\zeta)^{-1} = \frac{\tau(\zeta)}{g(\zeta)}, \qquad \sigma(\zeta)\tau(\zeta) = g(\zeta)\,\mathrm{Id}_{V_2},$$

since a left inverse of a square matrix is a right inverse. Thus D is elliptic, its formal adjoint D^* (with respect to the Riemannian density and the Hermitian structures) is again of Dirac type, with $\sigma_1(D^*) = \sigma_1(D)^*$, and D^*D, DD^* are second order with principal symbol $|\zeta|_g^2\,\mathrm{Id}$. Polarizing (9.6.1) and its counterpart on V_2 gives the *Clifford relations*

$$\sigma(\zeta)^*\sigma(\zeta') + \sigma(\zeta')^*\sigma(\zeta) = 2g(\zeta,\zeta')\,\mathrm{Id}_{V_1}, \quad \sigma(\zeta)\sigma(\zeta')^* + \sigma(\zeta')\sigma(\zeta)^* = 2g(\zeta,\zeta')\,\mathrm{Id}_{V_2} \tag{9.6.2}$$

for real covectors ζ, ζ'.

Example 9.6.2. (1) The Cauchy–Riemann operator $D = D_x + iD_y$ on $\mathbb{R}^2$ (or on a Riemann surface with a compatible metric, acting on functions with values in $(0,1)$-forms) has $\sigma(\xi,\eta) = \xi + i\eta$, $\sigma^*\sigma = \xi^2 + \eta^2$. (2) On a Riemannian manifold, $d + d^*$ acting from even to odd forms has symbol $i(\zeta \wedge - \iota_\zeta)$, and $(\zeta \wedge - \iota_\zeta)^2 = -|\zeta|^2$; so it is of Dirac type, as is $d + d^*$ on all forms. (3) The Dirac operator on the spinor bundle of a spin manifold, possibly twisted by a Hermitian bundle with connection, is of Dirac type; this is the origin of the name.

9.7 Dividing hypersurfaces

To isolate the essential idea of Calderón we first consider the following situation, in which no boundary is present at all: a hypersurface H cuts a closed manifold M into two pieces, and we compare solutions of $Du = 0$ on the two sides (Figure 9.3). Because D is assumed invertible on M, all smoothing errors disappear and the construction is completely explicit.

Standing hypotheses (dividing case). M is a compact Riemannian manifold without boundary, $D \in \mathrm{Diff}^1(M; V_1, V_2)$ is a generalized Dirac operator, and $x \in \mathcal{C}^\infty(M)$ is real with $dx \neq 0$ on $H = \{x = 0\}$; thus H is a compact hypersurface with trivial normal bundle, and

$$M = M_+ \cup M_-, \quad M_\pm = \{\pm x \geq 0\}, \quad M_+ \cap M_- = H,$$

where $M_\pm$ are compact manifolds with boundary H. In addition we assume

$$D : \mathcal{C}^\infty(M; V_1) \to \mathcal{C}^\infty(M; V_2) \text{ is an isomorphism.} \tag{9.7.1}$$

By Chapter 5, D is Fredholm, so (9.7.1) forces the index to vanish; it is made to simplify the discussion. By the results there, $D^{-1} \in \Psi^{-1}(M; V_2, V_1)$, and D^{-1} extends to an isomorphism $\mathcal{C}^{-\infty}(M; V_2) \to \mathcal{C}^{-\infty}(M; V_1)$ inverting D on distributions.

Our object is to parametrize the spaces of solutions

$$\mathcal{N}_\pm = \{u \in \mathcal{C}^\infty(M_\pm; V_1);\ Du = 0 \text{ in } M_\pm^\circ\}$$

by their boundary values. Let $V = V_1|_H$ and $b_\pm : \mathcal{C}^\infty(M_\pm; V_1) \to \mathcal{C}^\infty(H; V)$ the restriction maps.

Lemma 9.7.1. *Under the standing hypotheses $b_\pm : \mathcal{N}_\pm \to \mathcal{C}^\infty(H; V)$ are injective.*

Proof. If $u \in \mathcal{N}_+$ and $b_+ u = 0$, then by Lemma 9.4.4(ii) u vanishes to infinite order at H. Hence $\tilde{u}$, equal to u in M_+ and to 0 in M_-, is smooth on M, and $D\tilde{u} = 0$. By (9.7.1), $\tilde{u} = 0$. □

The main result in this setting is the following.

Theorem 9.7.2 (Calderón projector, dividing case)**.** *Under the standing hypotheses there is a unique $\Pi_{\mathcal{C}} \in \Psi^0(H; V)$ with $\Pi_{\mathcal{C}}^2 = \Pi_{\mathcal{C}}$ such that*

$$b_+ : \mathcal{N}_+ \to \Pi_{\mathcal{C}}\, \mathcal{C}^\infty(H; V) \quad \text{and} \quad b_- : \mathcal{N}_- \to (\mathrm{Id} - \Pi_{\mathcal{C}})\, \mathcal{C}^\infty(H; V)$$

are isomorphisms. In particular $\mathcal{C}^\infty(H; V) = b_+(\mathcal{N}_+) \oplus b_-(\mathcal{N}_-)$. $\Pi_{\mathcal{C}}$ is classical and, if $\nu = dx/|dx|$ is the unit conormal and T^*H is identified with the covectors orthogonal to ν,

$$\sigma_0(\Pi_{\mathcal{C}})(y, \eta) = \frac{1}{2}\Big(\mathrm{Id} + \frac{i\,\sigma(\nu)^{-1}\sigma(\eta)}{|\eta|}\Big), \quad (y, \eta) \in T^*H \setminus 0, \tag{9.7.2}$$

the orthogonal projection onto the $+1$ eigenspace of the self-adjoint involution $i\sigma(\nu)^{-1}\sigma(\eta)/|\eta|$.

The construction uses the jump formula "in reverse". Consider the space of piecewise solutions

$$\mathcal{N}_H = \{u \in L^1(M; V_1);\ u|_{M_\pm^\circ} \in \mathcal{N}_\pm\} \cong \mathcal{N}_+ \oplus \mathcal{N}_-.$$

By (9.4.3), for $u \in \mathcal{N}_H$,

$$Du = \frac{1}{i}\sigma(dx)(b_+ u - b_- u)\, \delta(x) \in \mathcal{C}^\infty(H; V_2)\, \delta(x). \tag{9.7.3}$$

So a piecewise solution is a global solution of $Du = 0$ except for a layer on H, whose density is (up to the invertible factor $\frac{1}{i}\sigma(dx)$) the jump of u across H. The idea is to run this backwards: given a layer, apply D^{-1}.

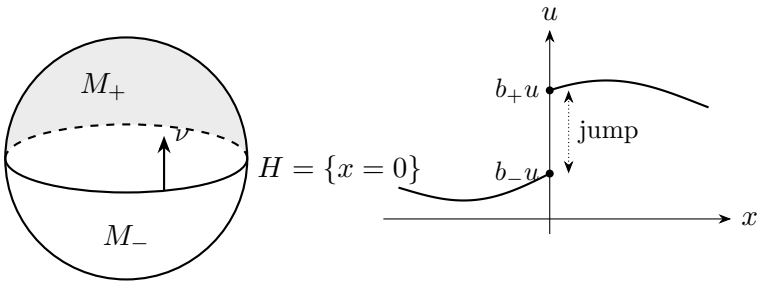


Figure 9.3: Left: a hypersurface H dividing a closed manifold M into M_+ and M_-, with unit conormal $\nu = dx/|dx|$. Right: a piecewise solution $u \in \mathcal{N}_H$ along a curve crossing H. It is smooth up to H from each side, and by (9.7.3) Du is a layer on H whose density is determined by the jump $b_+u - b_-u$.

Proposition 9.7.3. *Under the standing hypotheses, $D : \mathcal{N}_H \to \mathcal{C}^\infty(H; V_2)\, \delta(x)$ is an isomorphism with inverse $w\, \delta(x) \mapsto D^{-1}(w\, \delta(x))$.*

Proof. Injectivity: if $u \in \mathcal{N}_H$ and $Du = 0$ on M then $u = D^{-1}Du = 0$. For surjectivity we must show that for $w \in \mathcal{C}^\infty(H; V_2)$,

$$u = D^{-1}(w\, \delta(x)) \text{ satisfies } u|_{M_\pm^\circ} \in \mathcal{C}^\infty(M_\pm; V_1); \tag{9.7.4}$$

then $Du = 0$ in $M_\pm^\circ$ and $u \in \mathcal{N}_H$ (note $u \in H^{1/2-\epsilon} \subset L^1$ by Lemma 9.5.3). This is the case $Q = D^{-1}$ of Theorem 9.8.4 below, applied on both sides of H. □

The regularity statement (9.7.4) is the analytic heart of the matter. It is *false* for a general pseudodifferential operator of order -1 in place of D^{-1} (for instance for $\mathrm{Op}(\langle\zeta\rangle^{-1})$ in $\mathbb{R}^n$, $n \geq 2$, the result is not smooth up to H); it depends on the special form of the symbol of D^{-1}, which we now examine.

Discussion. The phenomenon is already visible in $\mathbb{R}^n = \mathbb{R}_x \times \mathbb{R}^{n-1}_y$. If $Q = \mathrm{Op}(q)$ with q independent of z, then $Q(w\,\delta(x))$ has partial Fourier transform in y equal to $\hat{w}(\eta)$ times the inverse Fourier transform in ξ of $\xi \mapsto q(\xi,\eta)$. For the Dirac-type symbol $q = 1/(1+\xi^2+|\eta|^2)$ (that of $(1+\Delta)^{-1}$; a Dirac parametrix has $\tau(\zeta)$ in the numerator) this is $e^{-\langle\eta\rangle|x|}/(2\langle\eta\rangle)$: a function which is smooth on each side of $x=0$, with a corner at $x=0$. The reason is that q is a rational function of ξ, and the inverse Fourier transform for $x>0$ is a residue at the pole $\xi_+ = i\langle\eta\rangle$ in the upper half plane (Figure 9.4). For $q = (1+\xi^2+|\eta|^2)^{-1/2}$, on the other hand, the inverse Fourier transform is $\pi^{-1}K_0(\langle\eta\rangle|x|)$, with K_0 the modified Bessel function, which has a logarithmic singularity at $x=0$: the branch cut of the square root replaces the pole. Section 9.8 carries out the residue argument with variable coefficients.

9.8 Rational symbols and layer potentials

This section proves the regularity of layer potentials, Theorem 9.8.4. The proof has three steps. First, the symbol of a parametrix of a Dirac-type operator agrees, to any finite order, with a rational function of ζ (Lemma 9.8.2). Second, the inverse Fourier transform of a rational function in the normal variable ξ is a residue, which is smooth up to $x=0$ and whose value at $x=0$ is a classical symbol in η (Lemma 9.8.3). Third, the remainder, of very negative order, is handled by absolute convergence.

Symbols over a closed set

Let $W \subset \mathbb{R}^n$ be open and $K \subset W$ closed in $\mathbb{R}^n$ (we shall take K a compact subset of a closed half-space). We say $a \in \mathcal{C}^\infty(W\times\mathbb{R}^n)$ is a symbol of order m *over* K, $a \in S^m(K)$, if for all α,β, $\sup_{z\in K}|D_z^\alpha D_\zeta^\beta a(z,\zeta)| \leq C_{\alpha\beta}\langle\zeta\rangle^{m-|\beta|}$. These classes are closed under products, D_z and D_ζ in the usual way. For a symbol $q \in S^m(\mathbb{R}^n;\mathbb{R}^n)$, $\mathrm{Op}(q)f(z)$ depends only on $q(z,\cdot)$.

Let $D = \sum_j A_j(z)D_{z_j} + A'(z)$ be a first-order operator on W (with matrix coefficients), with full symbol $d(z,\zeta) = \sigma(z,\zeta) + A'(z)$. Since d is linear in ζ, for $q \in S^m(\mathbb{R}^n;\mathbb{R}^n)$ with z-support in W,

$$D\ \mathrm{Op}(q) = \mathrm{Op}(d\circ q), \qquad d\circ q = dq + \sum_j A_j D_{z_j} q, \tag{9.8.1}$$

exactly, as follows by differentiating under the integral sign.

Rational symbols

Suppose now that W carries a Riemannian metric with dual metric $g(z,\zeta) = \sum g^{jk}(z)\zeta_j\zeta_k$ and Hermitian structures, and that $\tau(z,\zeta)$ is linear in ζ, with $\sigma\tau = g$ and $\tau\sigma = g$ for $z \in K$. (If D is of Dirac type on K, take $\tau = \sigma^*$ computed with respect to the Hermitian structures; outside K the identities need not hold.)

Definition 9.8.1. For $r \in \mathbb{Z}$, $\mathcal{R}^r$ is the space of finite sums of matrix-valued functions $p(z,\zeta)/(1+g(z,\zeta))^j$, where $j \geq 0$ and p is a polynomial in ζ of degree at most $2j+r$ with coefficients in $\mathcal{C}^\infty(W)$.

Clearly $\mathcal{R}^r\cdot\mathcal{R}^s\subset\mathcal{R}^{r+s}$, $\mathcal{R}^r\subset\mathcal{R}^{r+1}$, and $\mathcal{R}^r$ is preserved by D_{z_j} and multiplication by $\mathcal{C}^\infty(W)$; moreover $\partial_{\zeta_j}\mathcal{R}^r\subset\mathcal{R}^{r-1}$, since $\partial_{\zeta_j}(p(1+g)^{-j})=(\partial_{\zeta_j}p)(1+g)^{-j}-jp(\partial_{\zeta_j}g)(1+g)^{-j-1}$. Since $1+g\geq c\langle\zeta\rangle^2$ on compact sets, it follows that $\mathcal{R}^r\subset S^r(L)$ for every compact $L\subset W$.

Lemma 9.8.2 (Rational symbols)**.** *In this setting there are $\rho_N\in\mathcal{R}^{-1}$, $N\geq 1$, with $\rho_1=\tau/(1+g)$ and $\rho_{N+1}-\rho_N\in\mathcal{R}^{-N-1}$, such that the following holds. If $L\subset K$ is compact and $q\in S^{-1}(\mathbb{R}^n;\mathbb{R}^n)$ satisfies $d\circ q-1\in S^{-\infty}(L')$ for a compact neighbourhood L' of L in K, then*

$$q-\rho_N\in S^{-N-1}(L)\quad\text{for all } N.$$

Similarly, with $\phi\in\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n)$ equal to 1 near 0, $q-\sum_{l=1}^{N}(1-\phi)p_l g^{1-2l}\in S^{-N-1}(L)$, where $p_1=\tau$ and

$$p_{l+1}=-\tau\Big(gA'p_l+\sum_j A_j\big(gD_{z_j}p_l-(2l-1)(D_{z_j}g)p_l\big)\Big)$$

are polynomials in ζ, homogeneous of degree $3l-2$. Thus the homogeneous terms of the symbol of a parametrix of a generalized Dirac operator are of the form $g^{1-2l}p_l$.

Proof. Over K, $d\circ\rho_1=g/(1+g)+A'\tau/(1+g)+\sum_j A_jD_{z_j}(\tau/(1+g))=1-e_1$ with $e_1\in\mathcal{R}^{-1}$. Inductively, if $d\circ\rho_k=1-e_k$ over K with $e_k\in\mathcal{R}^{-k}$, set $\rho_{k+1}=\rho_k+\tau e_k/(1+g)$. Then $\tau e_k/(1+g)\in\mathcal{R}^{-k-1}$ and over K

$$d\circ\frac{\tau e_k}{1+g}=e_k-\frac{e_k}{1+g}+A'\frac{\tau e_k}{1+g}+\sum_j A_jD_{z_j}\frac{\tau e_k}{1+g},$$

so $d\circ\rho_{k+1}=1-e_{k+1}$ with $e_{k+1}\in\mathcal{R}^{-k-1}$. (The identity $\sigma\tau=g$ is used only pointwise; the derivatives fall on the smooth functions τ,g.)

Now let $e=q-\rho_N\in S^{-1}(L')$. Then $d\circ e=(d\circ q-1)+e_N\in S^{-N}(L')$. Suppose $e\in S^{-k}(L'')$ for a compact neighbourhood $L''\subset L'$ of L, $k\leq N$. Then $\sigma e=d\circ e-A'e-\sum_j A_jD_{z_j}e\in S^{-k}(L'')$ and $e=(1-\phi)\tau g^{-1}\sigma e+\phi e\in S^{-k-1}(L'')$. Starting from $k=1$ we reach $e\in S^{-N-1}(L)$ (the neighbourhoods may be fixed in advance, since only derivatives at points of L'' are involved). The homogeneous version is proved in the same way: writing $q_{-l}=p_lg^{1-2l}$, the terms of order $-l$ in $d\circ\sum_l q_{-l}=1$ give $\sigma q_{-1}=1$ and $q_{-l-1}=-g^{-1}\tau(A'q_{-l}+\sum_j A_jD_{z_j}q_{-l})$, and substituting $D_{z_j}q_{-l}=g^{-2l}(gD_{z_j}p_l-(2l-1)(D_{z_j}g)p_l)$ gives the recursion for p_l; the degree of p_{l+1} is $1+2+(3l-2)=3(l+1)-2$. □

The basic contour integral

Let $W=(-1,1)_x\times O$, $O\subset\mathbb{R}^{n-1}$ open, so $\zeta=(\xi,\eta)$, and write

$$g(z,\xi,\eta)=g^{11}(z)\xi^2+2\beta(z,\eta)\xi+g'(z,\eta),\quad \beta=\sum_{k\geq 2}g^{1k}\eta_k.$$

By positive-definiteness $g^{11}>0$ and $g^{11}g'(\eta)-\beta(\eta)^2\geq c|\eta|^2$ locally uniformly. Hence for each (z,η) the equation $1+g(z,\xi,\eta)=0$ has the two roots

$$\xi_\pm(z,\eta)=\frac{-\beta(z,\eta)\pm i\,s(z,\eta)}{g^{11}(z)},\quad s=\big(g^{11}(1+g')-\beta^2\big)^{1/2}\geq c\langle\eta\rangle,$$

which lie in the upper and lower half planes, at distance $\geq c\langle\eta\rangle$ from the real axis and from each other, with $|\xi_\pm|\leq C\langle\eta\rangle$. For $\rho=p/(1+g)^j\in\mathcal{R}^r$ with $r\leq -1$ and $x\geq 0$ set

$$G^+_\rho(z',x,\eta)=\frac{1}{2\pi}\oint_\gamma e^{ix\xi}\rho(z',\xi,\eta)\,d\xi=i\operatorname*{Res}_{\xi=\xi_+(z',\eta)}\big(e^{ix\xi}\rho(z',\xi,\eta)\big),\tag{9.8.2}$$

where γ is a positively oriented circle in $\{\operatorname{Im}\xi>0\}$ enclosing $\xi_+(z',\eta)$ but not $\xi_-(z',\eta)$; similarly $G^-_\rho(z',x,\eta)=-i\operatorname{Res}_{\xi=\xi_-}(e^{ix\xi}\rho)$ for $x\leq 0$.

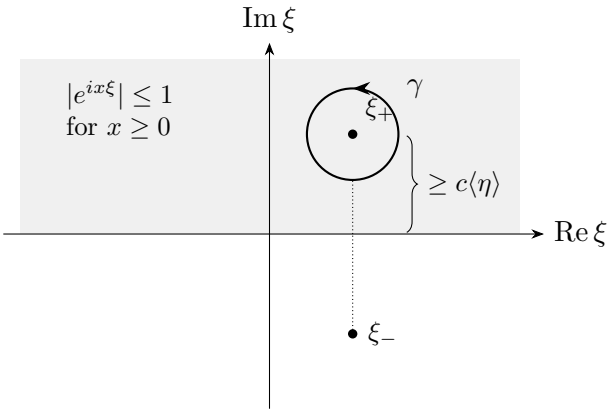


Figure 9.4: The two roots $\xi_\pm$ of $1+g(z',\xi,\eta)=0$ lie at distance $\geq c\langle\eta\rangle$ from the real axis, one in each half plane. For $x>0$ the factor $e^{ix\xi}$ is bounded in the upper half plane (shaded), so the inverse Fourier transform of a rational symbol is the residue at ξ_+, computed by integrating over the small circle γ; this is G^+_ρ, which is visibly smooth in $x\geq 0$ up to $x=0$.

Lemma 9.8.3. *Let $\rho\in\mathcal{R}^r$, $r\leq -1$, and let $L\subset W$ be compact.*

(i) For fixed (z',η) the inverse Fourier transform of $\xi\mapsto\rho(z',\xi,\eta)$ (an L^2 function) is $H(x)G^+_\rho(z',x,\eta)-H(-x)G^-_\rho(z',x,\eta)$.

(ii) For $z'\in L$, $x\geq 0$ and all α,k,β, $|D^\alpha_{z'}D^k_xD^\beta_\eta G^+_\rho(z',x,\eta)|\leq C\langle\eta\rangle^{r+1+k-|\beta|}$. In particular $G^+_\rho(z',0,\eta)\in S^{r+1}(L)$ (as a symbol in η).

(iii) For $z'\in L$, $G^+_\rho(z',0,\eta)$ is classical: for $|\eta|$ large it has an asymptotic expansion in functions homogeneous in η of degrees $r+1-l$, $l\geq 0$.

(iv) If $\rho=\tau/(1+g)$, the term of degree 0 in (iii) is $i\,\tau(z',\xi^0_+,\eta)/(g^{11}(\xi^0_+-\xi^0_-))$, where $\xi^0_\pm=(-\beta\pm i(g^{11}g'-\beta^2)^{1/2})/g^{11}$ are the roots of $g(z',\xi,\eta)=0$.

Proof. (i) Since $r<0$, ρ is a proper rational function of ξ and has a partial fraction decomposition $\sum_{k=1}^j\left(a_k(\xi-\xi_+)^{-k}+b_k(\xi-\xi_-)^{-k}\right)$. From $\int_0^\infty e^{-ix(\xi-\xi_+)}dx=-i(\xi-\xi_+)^{-1}$ (convergent as $\operatorname{Im}\xi_+>0$) we get $\mathcal{F}^{-1}[(\xi-\xi_+)^{-1}]=iH(x)e^{ix\xi_+}$, and differentiating in ξ_+, $\mathcal{F}^{-1}[(\xi-\xi_+)^{-k}]=iH(x)(ix)^{k-1}e^{ix\xi_+}/(k-1)!=iH(x)\operatorname{Res}_{\xi_+}(e^{ix\xi}(\xi-\xi_+)^{-k})$; similarly $\mathcal{F}^{-1}[(\xi-\xi_-)^{-k}]=-iH(-x)\operatorname{Res}_{\xi_-}(e^{ix\xi}(\xi-\xi_-)^{-k})$. Summing gives (i).

(ii) Near a given (z_0,η_0) we may fix γ to be the circle with centre $\xi_+(z_0,\eta_0)$ and radius $c'\langle\eta_0\rangle$, c' small; it encloses $\xi_+(z',\eta)$ and lies in $\operatorname{Im}\xi\geq c''\langle\eta\rangle$ for (z',η) in a neighbourhood of size comparable to $\langle\eta_0\rangle$ in η (by the estimates on $\xi_\pm$ and $|\partial_{z,\eta}\xi_\pm|\leq C$, $|\partial_z\xi_\pm|\leq C\langle\eta\rangle$). On γ, $|\xi|\leq C\langle\eta\rangle$ and $|1+g|=g^{11}|\xi-\xi_+||\xi-\xi_-|\geq c\langle\eta\rangle^2$, so $|\rho|\leq C\langle\eta\rangle^r$, and likewise $|D^\alpha_{z'}D^\beta_\eta\rho|\leq C\langle\eta\rangle^{r-|\beta|}$ because $D^\alpha_zD^\beta_\eta\rho\in\mathcal{R}^{r-|\beta|}$. Moreover $|e^{ix\xi}|\leq 1$ and $|D^k_xe^{ix\xi}|\leq C\langle\eta\rangle^k$ for $x\geq 0$. As γ has length $O(\langle\eta\rangle)$, differentiation under the integral gives (ii).

(iii) For $|\eta|\geq 1$ write $\eta=|\eta|\theta$, $\xi=|\eta|t$ and $\epsilon=|\eta|^{-2}$; then $1+g=|\eta|^2(\epsilon+g(z',t,\theta))$ and, decomposing p into homogeneous parts $p_{(k)}$,

$$G^+_\rho(z',0,\eta)=\sum_k|\eta|^{k+1-2j}\,\frac{1}{2\pi}\oint_{\gamma'}\frac{p_{(k)}(z',t,\theta)}{(\epsilon+g(z',t,\theta))^j}\,dt,$$

where γ' is a fixed circle around the root $t^0_+(z',\theta)$ of $g(z',t,\theta)=0$ in the upper half plane, which for $0\leq\epsilon\leq\epsilon_0$ still encloses exactly one root of $\epsilon+g=0$. The integrals are smooth in

(z', θ, ϵ) for $0 \le \epsilon \le \epsilon_0$; Taylor expansion in $\epsilon = |\eta|^{-2}$ gives an expansion in homogeneous functions of degrees $k + 1 - 2j - 2l \le r + 1 - 2l$.

(iv) This is the term $k = 1$, $l = 0$, i.e. $\epsilon = 0$: the residue of τ/g at the simple pole ξ_+^0. □

Boundary behaviour of layer potentials

Theorem 9.8.4. *Let M be a compact manifold without boundary, $x \in \mathcal{C}^\infty(M)$ real with $dx \neq 0$ on $H = \{x = 0\}$, $M_+ = \{x \ge 0\}$, and let $D \in \mathrm{Diff}^1(M; V_1, V_2)$ be elliptic on an open neighbourhood U of M_+ and of Dirac type on M_+ (for some metric and Hermitian structures on M). Let $Q \in \Psi^{-1}(M; V_2, V_1)$ satisfy $\chi(DQ - \mathrm{Id}) \in \Psi^{-\infty}$ for some $\chi \in \mathcal{C}^\infty(M)$ supported in U and equal to 1 near M_+. Then for $w \in \mathcal{C}^\infty(H; V_2)$:*

(i) $Q(w\,\delta(x))$ restricted to M_+° extends to an element of $\mathcal{C}^\infty(M_+; V_1)$;

(ii) the boundary value map $Q_+ w = b_+(Q(w\,\delta(x)))$ is a classical pseudodifferential operator $Q_+ \in \Psi^0(H; V_2, V_1)$;

(iii) with $\nu = dx/|dx|$ and T^*H identified with the covectors orthogonal to ν,

$$\sigma_0(Q_+)(y, \eta) = \frac{i|\eta|\,\sigma(\nu)^* + \sigma(\eta)^*}{2|dx|\,|\eta|}.$$

Proof. Since $\operatorname{sing\,supp} Q(w\,\delta(x)) \subset \operatorname{supp} w \subset H$, only points of H need attention, and by a partition of unity we may assume w is supported in a coordinate patch. Choose coordinates (x, y) on an open set $W = (-\epsilon, \epsilon) \times O \subset \{\chi = 1\}$ containing the patch, with x the given function, trivializing V_1, V_2 (away from H the analysis is not needed, by pseudolocality). Let $\phi, \psi \in \mathcal{C}_c^\infty(W)$ with $\psi = 1$ on $\operatorname{supp}\phi$ and near $\operatorname{supp} w \times \{0\}$, and $\phi = 1$ on a neighbourhood of a given compact $L_0 \subset W$. By Chapter 5, in these coordinates $\phi Q \psi = \mathrm{Op}(q) + R$ with $q \in S^{-1}(\mathbb{R}^n; \mathbb{R}^n)$ and R with smooth compactly supported kernel, and $\phi Q \psi(w\delta(x)) = \phi Q(w\delta(x))$. Moreover, by (9.8.1), $\mathrm{Op}(d \circ q) = D\phi Q\psi - DR = \phi\psi + \phi(DQ - \mathrm{Id})\psi + [D, \phi]Q\psi - DR$; the commutator is multiplication by a matrix function vanishing on $\{\phi = 1\}^\circ$, so over the compact set $K' = L' \cap \{x \ge 0\}$, for $L' \subset \{\phi = 1\}^\circ$ a compact neighbourhood of L_0, we have $d \circ q - 1 \in S^{-\infty}(K')$. Lemma 9.8.2 applies with $K = \{x \ge 0\}$ and gives $q = \rho_N + e$ with $e \in S^{-N-1}(L)$, $L = L_0 \cap \{x \ge 0\}$.

Let $\phi_1 \in \mathcal{C}_c^\infty(W)$ equal 1 near L_0. For $z \in L$, $\mathrm{Op}(q) f(z) = \mathrm{Op}(\phi_1 \rho_N) f(z) + \mathrm{Op}(e') f(z)$ with $e' = q - \phi_1 \rho_N \in S^{-1}$, $e' = e$ near L. We compute both terms on $f = w \otimes \delta(x)$, $\hat{f}(\xi, \eta) = \hat{w}(\eta)$, using the approximations $f_\epsilon = w \otimes \varphi_\epsilon(x)$, $\varphi_\epsilon = \epsilon^{-1}\varphi(x/\epsilon)$, which converge to f in $\mathcal{S}'$, so that $\mathrm{Op}(a) f_\epsilon \to \mathrm{Op}(a) f$ in $\mathcal{S}'$ for any symbol a.

The remainder. For $z \in L$, $|D_z^\alpha (e^{iz\cdot\zeta} e'(z, \zeta))| \le C\langle\zeta\rangle^{|\alpha|-N-1}$ and $\int \langle\zeta\rangle^{|\alpha|-N-1} d\xi \le C\langle\eta\rangle^{|\alpha|-N}$ if $|\alpha| \le N - 1$. Since $\hat{w}$ is rapidly decreasing, dominated convergence shows that $\mathrm{Op}(e') f_\epsilon(z) \to \iint e^{iz\cdot\zeta} e'(z, \zeta) \hat{w}(\eta)\, đ\xi đ\eta$ uniformly on L, together with z-derivatives of order $\le N - 1$. Thus on the interior of L (relative to $\mathbb{R}^n$), $\mathrm{Op}(e') f$ is this absolutely convergent integral, which is $\mathcal{C}^{N-1}$ on L, including up to $x = 0$. Its value at $x = 0$ is $\int e^{iy\cdot\eta} t_N(y, \eta) \hat{w}(\eta)\, đ\eta$ with $t_N(y, \eta) = \int e((0, y), \xi, \eta)\, đ\xi$, and since $D_\eta^\beta e \in S^{-N-1-|\beta|}(L)$, $t_N \in S^{-N}$.

The main term. $\phi_1 \rho_N$ is a finite sum of terms $\phi_1 \rho$, $\rho \in \mathcal{R}^{-1}$. By Lemma 9.8.3(i), $\int e^{ix\xi} \rho(z, \xi, \eta) \hat{\varphi}(\epsilon\xi)\, đ\xi$ is the convolution in x of $H(x) G_\rho^+ - H(-x) G_\rho^-$ with φ_ϵ; for $x > 0$ and ϵ small it converges to $G_\rho^+(z, x, \eta)$, with error at most $C\epsilon\langle\eta\rangle^{r+2}$ uniformly for $x \ge 2\epsilon$ by (ii) of that lemma. Hence for $z \in L$ with $x > 0$,

$$\mathrm{Op}(\phi_1 \rho) f(z) = \phi_1(z) \int e^{iy\cdot\eta} G_\rho^+(z, x, \eta) \hat{w}(\eta)\, đ\eta,$$

and by Lemma 9.8.3(ii) and the rapid decrease of $\hat{w}$ the right side is smooth in $z \in L$ up to $x = 0$, with boundary value $\int e^{iy\cdot\eta} G_\rho^+((0,y),0,\eta)\hat{w}(\eta)\,d\!\!\!^-\eta$.

Combining, $Q(w\delta(x))$ is $\mathcal{C}^{N-1}$ up to $x = 0$ near L for every N, which proves (i) since L_0 was arbitrary. Moreover, near L,

$$Q_+ w = \mathrm{Op}_H\,(\sigma^{(N)} + t_N)w + R_+ w, \qquad \sigma^{(N)}(y,\eta) = \sum G^+_{\phi_1\rho}((0,y),0,\eta),$$

with R_+ smoothing, the sum over the terms of ρ_N. Here $\sigma^{(N)} \in S^0$ is classical, $t_N \in S^{-N}$, and $\sigma^{(N+1)} - \sigma^{(N)} \in S^{-N}$ since $\rho_{N+1} - \rho_N \in \mathcal{R}^{-N-1}$. Hence locally Q_+ is a pseudodifferential operator of order 0 whose symbol agrees modulo S^{-N} with a classical symbol for every N, i.e. it is classical. If $\psi_1, \phi_2 \in \mathcal{C}^\infty(H)$ have disjoint supports, $\psi_1 Q_+ \phi_2$ has smooth kernel, since $Q(\phi_2 w\delta(x))$ is given near $\operatorname{supp}\psi_1$ by integrating the kernel of Q, smooth off the diagonal, against $\phi_2 w$ over H. Choosing a cover of H by coordinate patches such that any two that meet lie in a common patch, it follows that $Q_+ \in \Psi^0(H; V_2, V_1)$, which is (ii).

For (iii) we compute at a point of H. Changing the coordinates y (which does not affect x, $\delta(x)$ or Q_+) we may assume $dy_k \perp dx$ at that point, i.e. $\beta = 0$; then $g^{11} = |dx|^2$, $g'(\eta) = |\eta|^2$ and $\xi^0_\pm = \pm i|\eta|/|dx|$. The principal symbol of Q_+ comes from $\rho_1 = \tau/(1+g)$ only, and by Lemma 9.8.3(iv) equals $i\tau(\xi^0_+ dx + \eta)/(2i|dx||\eta|)$; since $\xi^0_+ dx = i|\eta|\nu$ and τ is complex linear, this is the stated formula. □

Remark 9.8.5. If D is of Dirac type on $M_- = \{x \le 0\}$ as well, the same argument gives smoothness of $Q(w\delta(x))$ up to H from M_-, with boundary value $Q_- \in \Psi^0(H; V_2, V_1)$, $\sigma_0(Q_-) = (-i|\eta|\sigma(\nu)^* + \sigma(\eta)^*)/(2|dx||\eta|)$, obtained from the residue at ξ_-.

9.9 The Calderón projector for a dividing hypersurface

We return to the standing hypotheses of Section 9.7. By Theorem 9.8.4 and Remark 9.8.5 with $Q = D^{-1}$, for $w \in \mathcal{C}^\infty(H; V_2)$ the distribution $D^{-1}(w\delta(x))$ is smooth up to H from both sides, which completes the proof of Proposition 9.7.3, and the boundary values

$$Q_\pm w = \lim_{\pm x \downarrow 0} D^{-1}(w\,\delta(x))$$

define $Q_\pm \in \Psi^0(H; V_2, V_1)$. Applying (9.7.3) to $u = D^{-1}(w\delta(x)) \in \mathcal{N}_H$ gives $\frac{1}{i}\sigma(dx)(Q_+ - Q_-)w = w$, i.e.

$$Q_+ - Q_- = i\,\sigma(dx)^{-1}. \tag{9.9.1}$$

Proof of Theorem 9.7.2. Define

$$\Pi_{\mathcal{C}} v = Q_+\Big(\frac{1}{i}\sigma(dx)v\Big) = b_+ D^{-1}\Big(\frac{1}{i}\sigma(dx)v\,\delta(x)\Big), \quad v \in \mathcal{C}^\infty(H; V), \tag{9.9.2}$$

so $\Pi_{\mathcal{C}} \in \Psi^0(H; V)$ is classical, and by (9.9.1) $\mathrm{Id} - \Pi_{\mathcal{C}} = -Q_- \frac{1}{i}\sigma(dx)$.

If $u \in \mathcal{N}_+$ then $u_c \in \mathcal{N}_H$ (zero on M_-) and by (9.7.3) $Du_c = \frac{1}{i}\sigma(dx)(b_+u)\delta(x)$, so $u_c = D^{-1}(\frac{1}{i}\sigma(dx)(b_+u)\delta(x))$ and, taking boundary values from M_+,

$$\Pi_{\mathcal{C}}\, b_+ u = b_+ u, \quad u \in \mathcal{N}_+. \tag{9.9.3}$$

Conversely for any $v \in \mathcal{C}^\infty(H; V)$, $u = D^{-1}(\frac{1}{i}\sigma(dx)v\delta(x))|_{M_+}$ lies in $\mathcal{N}_+$ and $b_+u = \Pi_{\mathcal{C}} v$. Hence $\operatorname{Ran}\Pi_{\mathcal{C}} = b_+(\mathcal{N}_+)$, and by (9.9.3) $\Pi^2_{\mathcal{C}} v = \Pi_{\mathcal{C}} v$. Together with Lemma 9.7.1, $b_+ : \mathcal{N}_+ \to \Pi_{\mathcal{C}}\mathcal{C}^\infty(H; V)$ is an isomorphism.

Similarly, for $u \in \mathcal{N}_-$, $Du_c = -\frac{1}{i}\sigma(dx)(b_-u)\delta(x)$, so $b_-u = -Q_-\frac{1}{i}\sigma(dx)b_-u = (\mathrm{Id} - \Pi_{\mathcal{C}})b_-u$; and for any v, $-D^{-1}(\frac{1}{i}\sigma(dx)v\delta(x))|_{M_-} \in \mathcal{N}_-$ has boundary value $(\mathrm{Id} - \Pi_{\mathcal{C}})v$. Hence $b_- : \mathcal{N}_- \to (\mathrm{Id} - \Pi_{\mathcal{C}})\mathcal{C}^\infty(H; V)$ is an isomorphism.

Uniqueness: a projection on $\mathcal{C}^\infty(H;V)$ is determined by its range and null space, and these are forced to be $b_+(\mathcal{N}_+)$ and $b_-(\mathcal{N}_-)$; a pseudodifferential operator is determined by its action on $\mathcal{C}^\infty$.

Finally, by Theorem 9.8.4(iii),

$$\sigma_0(\Pi_{\mathcal{C}}) = \sigma_0(Q_+)\frac{1}{i}|dx|\sigma(\nu) = \frac{1}{2}\Big(\sigma(\nu)^*\sigma(\nu) + \frac{\sigma(\eta)^*\sigma(\nu)}{i|\eta|}\Big).$$

By (9.6.1) $\sigma(\nu)^*\sigma(\nu) = \mathrm{Id}$, and by (9.6.2), since $g(\eta,\nu) = 0$, $\sigma(\eta)^*\sigma(\nu) = -\sigma(\nu)^*\sigma(\eta) = -\sigma(\nu)^{-1}\sigma(\eta)$, which gives (9.7.2). With $A = i\sigma(\nu)^{-1}\sigma(\eta)/|\eta| = i\sigma(\nu)^*\sigma(\eta)/|\eta|$, the second Clifford relation gives $\sigma(\eta)\sigma(\nu)^* = -\sigma(\nu)\sigma(\eta)^*$, so $A^2 = -\sigma(\nu)^*\sigma(\eta)\sigma(\nu)^*\sigma(\eta)/|\eta|^2 = \sigma(\nu)^*\sigma(\nu)\sigma(\eta)^*\sigma(\eta)/|\eta|^2 = \mathrm{Id}$, and $A^* = -i\sigma(\eta)^*\sigma(\nu)/|\eta| = A$. □

Example 9.9.1. For $D = D_x + iD_y$ (Example 9.6.2) we have $\sigma(\nu) = 1$, $\sigma(\eta) = i\eta$, so $\sigma_0(\Pi_{\mathcal{C}}) = \frac{1}{2}(1 - \eta/|\eta|)$, the projection onto negative frequencies. This is as it should be: $Du = 0$ means u is holomorphic in $x + iy$, and $e^{\eta(x+iy)}$ is bounded in $x > 0$ precisely when $\eta < 0$.

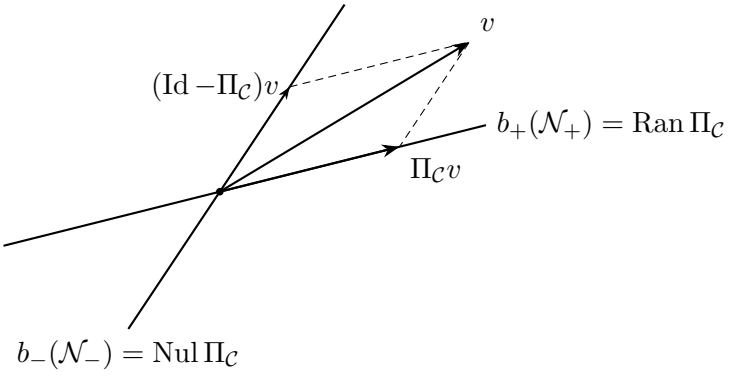


Figure 9.5: The Calderón projector in the dividing case. Every smooth section v on H splits uniquely as the boundary value of a solution on M_+ plus the boundary value of a solution on M_-; $\Pi_{\mathcal{C}}$ projects onto the first summand *along* the second. The two subspaces are in general not orthogonal, so $\Pi_{\mathcal{C}}$ is an oblique projection.

Question. Why is $\Pi_{\mathcal{C}}$ not simply the orthogonal projection onto $b_+(\mathcal{N}_+)$?

Answer. In the dividing case both the range and the null space of $\Pi_{\mathcal{C}}$ are forced on us (Figure 9.5), and nothing makes $b_+(\mathcal{N}_+)$ and $b_-(\mathcal{N}_-)$ orthogonal in $L^2(H;V)$. They are orthogonal "to leading order": $\sigma_0(\Pi_{\mathcal{C}})$ is the orthogonal projection onto an eigenspace of a self-adjoint involution, so $\Pi_{\mathcal{C}} - \Pi_{\mathcal{C}}^* \in \Psi^{-1}(H;V)$. For a manifold with boundary only the range is forced, and one can choose the projection to be self-adjoint (Exercise 9.4). □

The Poisson operator and a right inverse

The proof shows that

$$I_D : \Pi_{\mathcal{C}}\mathcal{C}^\infty(H;V) \to \mathcal{N}_+, \quad I_D v = D^{-1}\Big(\frac{1}{i}\sigma(dx)v\,\delta(x)\Big)\Big|_{M_+},$$

is the inverse of b_+: $u = I_D v$ is the unique solution of $Du = 0$ in M_+°, $u \in \mathcal{C}^\infty(M_+;V_1)$, $b_+u = v$. This is the *Poisson operator* for the Calderón boundary condition. As well as I_D we have a natural right inverse for D on M_+.

Lemma 9.9.2. *If $f \in \mathcal{C}^\infty(M_+;V_2)$ then $R_D f = D^{-1}(f_c)|_{M_+} \in \mathcal{C}^\infty(M_+;V_1)$, and $D \circ R_D =$ Id on $\mathcal{C}^\infty(M_+;V_2)$.*

Proof. Certainly $D(D^{-1}f_c) = f_c$, so $DR_Df = f$ in M_+°; the point is the regularity. By Lemma 9.4.4(i) there is $u' \in \mathcal{C}^\infty(M_+;V_1)$ with $b_+u' = 0$ and $h = Du' - f \in \dot{\mathcal{C}}^\infty(M_+;V_2)$. By the jump formula $D(u'_c) = (Du')_c = f_c + h_c$, where $h_c \in \mathcal{C}^\infty(M;V_2)$. Hence $D^{-1}f_c = u'_c - D^{-1}h_c$ and $R_Df = u' - (D^{-1}h_c)|_{M_+} \in \mathcal{C}^\infty(M_+;V_1)$. □

R_D is not a two-sided inverse: D has the large null space $\mathcal{N}_+ = I_D(\Pi_C\mathcal{C}^\infty(H;V))$ on M_+. Combining the two operators solves the boundary problem with Calderón boundary condition, see Exercise 9.3.

9.10 Smoothing operators and generalized inverses

On a manifold with boundary there is no invertible extension of D at hand, and we must work with parametrices; to correct their errors we use some abstract functional analysis. Let $\mathcal{H}$ be a separable Hilbert space, $\mathcal{B} = \mathcal{B}(\mathcal{H})$, $\mathcal{K} = \mathcal{K}(\mathcal{H})$ the compact operators and $\mathcal{F} = \mathcal{F}(\mathcal{H})$ the finite-rank operators, so $\mathcal{K}$ is the norm closure of $\mathcal{F}$.

Discussion. Why not simply work modulo compact operators? Because we want to keep regularity. A parametrix leaves an error which is smoothing, and when we correct it by a finite-rank term (to get an exact inverse on a complement of the null space, say) we want the correction to be smoothing as well, so that it maps L^2 into $\mathcal{C}^\infty$. A bi-ideal is an abstract class of "small" operators with exactly the closure properties needed for this: Proposition 9.10.4 shows that a generalized inverse of Id $+A$, A in the bi-ideal, differs from Id by an element of the same bi-ideal, and the null-space projections are in it too.

Definition 9.10.1. A subspace $\mathcal{J} \subset \mathcal{B}$ with a Fréchet topology given by norms $\|\cdot\|_k$, $k \in \mathbb{N}_0$, is a *bi-ideal* if:

(a) $\mathcal{J}$ is a subalgebra and for each k there are k' and C_k with $\|AB\|_k \leq C_k\|A\|_{k'}\|B\|_{k'}$, $A, B \in \mathcal{J}$;

(b) $\|A\|_{\mathcal{B}} \leq C\|A\|_0$ and $A^* \in \mathcal{J}$ with $\|A^*\|_k \leq C_k\|A\|_{k'}$ for $A \in \mathcal{J}$;

(c) $\mathcal{F} \cap \mathcal{J}$ is dense in $\mathcal{J}$;

(d) if $A_1, A_2 \in \mathcal{J}$ and $B \in \mathcal{B}$ then $A_1BA_2 \in \mathcal{J}$, with $\|A_1BA_2\|_k \leq C_k\|A_1\|_{k'}\|B\|_{\mathcal{B}}\|A_2\|_{k'}$.

By (b) and (c), $\mathcal{J} \subset \mathcal{K}$. Note that $\mathcal{J}$ need not be an ideal in $\mathcal{B}$; (d) replaces this.

Proposition 9.10.2. *Let X be a compact manifold, with or without boundary, with a smooth positive density, and E a Hermitian bundle. The operators on $\mathcal{H} = L^2(X;E)$ with kernels in $\mathcal{C}^\infty(X \times X; \mathrm{Hom}(E))$, $Au(z) = \int_X A(z,z')u(z')$, with the norms $\|A\|_k = \|A(\cdot,\cdot)\|_{\mathcal{C}^k(X\times X)}$ (defined using a fixed finite atlas), form a bi-ideal.*

Proof. The kernel of AB is obtained by integrating $A(z,z'')B(z'',z')$ over $z'' \in X$; its $\mathcal{C}^k$ norm is bounded by $\mathrm{vol}(X)\|A\|_k\|B\|_k$; the kernel of A^* is $A(z',z)^*$; and $\|A\|_{\mathcal{B}} \leq \|A(\cdot,\cdot)\|_{L^2(X\times X)} \leq C\|A\|_0$. For (d), the kernel of A_1BA_2 is $(BA_2(\cdot,z'), A_1(z,\cdot)^*)_{L^2}$, and derivatives in z, z' fall on A_1, A_2, giving the estimate. For (c), embed X in its double, extend the kernel to a smooth kernel on $\tilde{X} \times \tilde{X}$, and by a partition of unity reduce to a kernel in $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^n \times \mathbb{R}^n)$; regarding it as a smooth function on a large torus, its Fourier partial sums $\sum c_{jk}e^{ij\cdot z}e^{ik\cdot z'}$ converge in $\mathcal{C}^\infty$, and after multiplying by $\chi(z)\chi(z')$, $\chi = 1$ on the support, they give finite sums of products $\varphi(z)\psi(z')$, i.e. finite-rank operators in $\mathcal{J}$. □

The elements of this bi-ideal are the *smoothing operators*; they have the following characterization (in the boundaryless case $\dot{\mathcal{C}}^{-\infty} = \mathcal{C}^{-\infty}$).

Proposition 9.10.3. *A linear operator $A : \dot{\mathcal{C}}^\infty(X;E) \to \mathcal{C}^{-\infty}(X;F)$ has kernel in $\mathcal{C}^\infty(X \times X; \operatorname{Hom}(E,F) \otimes \Omega_R)$ if and only if it extends to a continuous map $\dot{\mathcal{C}}^{-\infty}(X;E) \to \mathcal{C}^\infty(X;F)$, i.e. to continuous maps $\dot{H}^{-m}(X;E) \to \mathcal{C}^\infty(X;F)$ for all m.*

Proof. If A has smooth kernel, $Au(z) = u(A(z,\cdot))$ makes sense for supported distributions u (the test functions being smooth up to the boundary) and is smooth and continuous in u. Conversely, for $z' \in X$ and $e \in E_{z'}$, $\delta_{z'}e \in \dot{H}^{-m}(X;E)$ when $m > n/2$, and $z' \mapsto \delta_{z'}$ is $\mathcal{C}^k$ into $\dot{H}^{-m}$ for $m > n/2 + k$ (as seen from the Fourier transform $e^{-iz'\cdot\zeta}$ in local coordinates). Thus $A(z,z')e = (A\delta_{z'}e)(z)$ is $\mathcal{C}^k$ in z' with values in $\mathcal{C}^\infty(X;F)$ for every k, hence smooth on $X \times X$. For $u \in \dot{\mathcal{C}}^\infty(X;E)$, $u = \int u(z')\delta_{z'}dz'$ as a Riemann integral in $\dot{H}^{-m}$, so by continuity $Au = \int A(\cdot,z')u(z')dz'$. □

Proposition 9.10.4. *If $\mathcal{J}$ is a bi-ideal and $A \in \mathcal{J}$, then $\operatorname{Id}+A$ has a generalized inverse $\operatorname{Id}+B$, $B \in \mathcal{J}$:*

$$(\operatorname{Id}+B)(\operatorname{Id}+A) = \operatorname{Id}-\pi_L, \quad (\operatorname{Id}+A)(\operatorname{Id}+B) = \operatorname{Id}-\pi_R,$$

where $\pi_L, \pi_R \in \mathcal{J} \cap \mathcal{F}$ are the orthogonal projections onto $\operatorname{Nul}(\operatorname{Id}+A)$ and onto $\operatorname{Ran}(\operatorname{Id}+A)^\perp = \operatorname{Nul}(\operatorname{Id}+A^*)$.

Proof. As A is compact, $\operatorname{Id}+A$ is Fredholm. If $u \in \operatorname{Nul}(\operatorname{Id}+A)$ then $u = -Au$, so $\pi_L = -A\pi_L$, and taking adjoints $\pi_L = -\pi_L A^*$; hence $\pi_L = A\pi_L A^* \in \mathcal{J}$ by (d). Similarly $\pi_R = A^*\pi_R A \in \mathcal{J}$. Let $G \in \mathcal{B}$ be the inverse of $\operatorname{Id}+A : \operatorname{Nul}(\operatorname{Id}+A)^\perp \to \operatorname{Ran}(\operatorname{Id}+A)$, extended by 0 on $\operatorname{Ran}(\operatorname{Id}+A)^\perp$; then $G(\operatorname{Id}+A) = \operatorname{Id}-\pi_L$ and $(\operatorname{Id}+A)G = \operatorname{Id}-\pi_R$. Setting $B = G - \operatorname{Id}$, these identities give $B = -\pi_L - A - BA$ and $B = -\pi_R - A - AB$; substituting the second into the first,

$$B = -\pi_L - A + \pi_R A + A^2 + ABA,$$

and each term lies in $\mathcal{J}$ by (a) and (d). □

Now let $\mathcal{H}_1, \mathcal{H}_2$ be Hilbert spaces, $\mathcal{J}_i \subset \mathcal{B}(\mathcal{H}_i)$ bi-ideals and $A \in \mathcal{B}(\mathcal{H}_1, \mathcal{H}_2)$. A *left parametrix* for A modulo $\mathcal{J}_1$ is $B_L \in \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$ with $B_L A = \operatorname{Id}+J_L$, $J_L \in \mathcal{J}_1$; a *right parametrix* modulo $\mathcal{J}_2$ is B_R with $AB_R = \operatorname{Id}+J_R$, $J_R \in \mathcal{J}_2$.

Proposition 9.10.5. *If A has a left parametrix B_L modulo $\mathcal{J}_1$, then A has closed range and finite-dimensional null space, the orthogonal projection π_L onto $\operatorname{Nul} A$ lies in $\mathcal{J}_1 \cap \mathcal{F}$, and there are $J \in \mathcal{J}_1$, $\pi'_L \in \mathcal{J}_1 \cap \mathcal{F}$ a self-adjoint projection with $\pi_L \le \pi'_L$, and $G \in \mathcal{B}(\mathcal{H}_2, \mathcal{H}_1)$ such that*

$$\begin{aligned} B'_L &= (\operatorname{Id}+J)B_L && \text{satisfies } B'_L A = \operatorname{Id}-\pi'_L, \quad \text{and} \\ B''_L &= B'_L + (\pi'_L - \pi_L)G && \text{satisfies } B''_L A = \operatorname{Id}-\pi_L, \end{aligned}$$

with $\pi'_L - \pi_L \in \mathcal{J}_1 \cap \mathcal{F}$.

Proof. By Proposition 9.10.4 there is $J \in \mathcal{J}_1$ with $(\operatorname{Id}+J)(\operatorname{Id}+J_L) = \operatorname{Id}-\pi'_L$, which gives B'_L. Then $\operatorname{Nul} A \subset \operatorname{Ran}\pi'_L$ is finite dimensional and $\pi_L = \pi'_L\pi_L\pi'_L \in \mathcal{J}_1$. On the closed subspace $\operatorname{Ran}(\operatorname{Id}-\pi'_L)$, $u = B'_L Au$, so $\|u\| \le \|B'_L\|\|Au\|$ and $A(\operatorname{Ran}(\operatorname{Id}-\pi'_L))$ is closed; adding the finite-dimensional space $A(\operatorname{Ran}\pi'_L)$, $\operatorname{Ran} A$ is closed. Let G be the generalized inverse of A ($GA = \operatorname{Id}-\pi_L$, $G = 0$ on $(\operatorname{Ran} A)^\perp$). Since $\pi'_L\pi_L = \pi_L$, $B''_L A = \operatorname{Id}-\pi'_L + (\pi'_L - \pi_L)(\operatorname{Id}-\pi_L) = \operatorname{Id}-\pi_L$, and $\pi'_L - \pi_L = \pi'_L(\operatorname{Id}-\pi_L)\pi'_L \in \mathcal{J}_1$. □

Proposition 9.10.6. *If A has a right parametrix B_R modulo $\mathcal{J}_2$, then A has closed range of finite codimension, the orthogonal projection π_R onto $(\operatorname{Ran} A)^\perp$ lies in $\mathcal{J}_2 \cap \mathcal{F}$, and there are $J \in \mathcal{J}_2$ and a self-adjoint projection $\pi'_R \in \mathcal{J}_2 \cap \mathcal{F}$ with $\pi_R \le \pi'_R$ such that $B'_R = B_R(\mathrm{Id}+J)$ satisfies $AB'_R = \mathrm{Id} - \pi'_R$; moreover $B''_R = B'_R + G(\pi'_R - \pi_R)$, G the generalized inverse of A, satisfies $AB''_R = \mathrm{Id} - \pi_R$.*

Proof. Choose J by Proposition 9.10.4 with $(\mathrm{Id}+J_R)(\mathrm{Id}+J) = \mathrm{Id} - \pi'_R$. Then $\operatorname{Ran} A \supset \operatorname{Ran}(\mathrm{Id} - \pi'_R)$, a closed subspace of finite codimension, so $\operatorname{Ran} A$ is closed of finite codimension, $(\operatorname{Ran} A)^\perp \subset \operatorname{Ran} \pi'_R$ and $\pi_R = \pi'_R \pi_R \pi'_R \in \mathcal{J}_2$. Finally $AG = \mathrm{Id} - \pi_R$ and $\pi_R \pi'_R = \pi_R$ give $AB''_R = \mathrm{Id} - \pi'_R + (\mathrm{Id} - \pi_R)(\pi'_R - \pi_R) = \mathrm{Id} - \pi_R$. □

An operator with closed range and either a null space of finite dimension or a range of finite codimension is called *semi-Fredholm*. The existence of a left or right parametrix modulo compact operators is necessary and sufficient for these two kinds of semi-Fredholmness.

9.11 Manifolds with boundary: a right inverse

Standing hypotheses (boundary case). From now on X is a compact Riemannian manifold with boundary, V_1, V_2 are Hermitian bundles over X, and $D \in \operatorname{Diff}^1(X; V_1, V_2)$ is a generalized Dirac operator. We embed X in its double $\tilde{X}$, a compact manifold without boundary, with $x \in \mathcal{C}^\infty(\tilde{X})$ a function such that $X = \{x \ge 0\}$ and $dx \ne 0$ on $\partial X = \{x = 0\}$, and extend the metric, the Hermitian bundles and D (by extending coefficients across ∂X in a collar and cutting off) to $\tilde{D} \in \operatorname{Diff}^1(\tilde{X}; \tilde{V}_1, \tilde{V}_2)$. Since ellipticity is an open condition, $\tilde{D}$ is elliptic on an open neighbourhood U of X; it is of Dirac type on X, but need not be elsewhere. By the parametrix construction of Chapter 5 (which is local), there is $\tilde{Q} \in \Psi^{-1}(\tilde{X}; \tilde{V}_2, \tilde{V}_1)$ and $\chi \in \mathcal{C}^\infty(\tilde{X})$ with $\operatorname{supp}\chi \subset U$, $\chi = 1$ near X, such that

$$E_R = \chi(\tilde{D}\tilde{Q} - \mathrm{Id}), \quad E_L = (\tilde{Q}\tilde{D} - \mathrm{Id})\chi \quad \text{are smoothing.}$$

We also use a smooth positive density on X (the Riemannian one) and the corresponding L^2 spaces. Finally we assume *unique continuation*:

$$\text{if } u \in \dot{\mathcal{C}}^\infty(X; V_1),\ Du = 0 \text{ then } u = 0; \quad \text{if } f \in \dot{\mathcal{C}}^\infty(X; V_2),\ D^*f = 0 \text{ then } f = 0. \tag{UC}$$

Remark 9.11.1. (UC) fails if X has a component without boundary on which D or D^* has null space. When every component of X meets ∂X it holds for operators of Dirac type: since D^*D and DD^* have scalar principal symbol $|\zeta|^2$, this follows from the unique continuation theorem of Aronszajn for such second-order operators, applied after extending the Dirac structure a little across the boundary near a boundary point; see the Notes. We simply assume it.

Discussion. Unique continuation plays the role that invertibility of D on M played in the dividing case. A section vanishing to infinite order at ∂X is "invisible" from the boundary: it has zero boundary values, and its extension by zero is smooth on $\tilde{X}$. (UC) says that no such section solves $Du = 0$ or $D^*f = 0$. The first half makes a solution determined by its boundary value (Lemma 9.12.1); the second half, via Proposition 9.11.3, makes D surjective, so that a right inverse exists.

For $f \in L^2(X; V_2)$ set

$$Q'f = R_X \tilde{Q}(f_c) \in H^1(X; V_1),$$

where $f_c \in L^2(\tilde{X}; \tilde{V}_2)$ is the extension by zero; $Q' : L^2(X; V_2) \to H^1(X; V_1)$ is bounded.

Lemma 9.11.2. *Q' maps $\mathcal{C}^\infty(X;V_2)$ into $\mathcal{C}^\infty(X;V_1)$, and $DQ' = \mathrm{Id} + E'$ where $E'f = R_X E_R(f_c)$ has kernel in $\mathcal{C}^\infty(X\times X;\mathrm{Hom}(V_2))$.*

Proof. The second statement holds since $\chi = 1$ on X: $DQ'f = R_X\tilde{D}\tilde{Q}f_c = f + R_X E_R f_c$. For the first, as in Lemma 9.9.2, choose by Lemma 9.4.4 $u' \in \mathcal{C}^\infty(X;V_1)$ with $bu' = 0$ and $h = Du' - f \in \dot{\mathcal{C}}^\infty(X;V_2)$. By the jump formula $\tilde{D}(u'_c) = f_c + h_c$ with h_c smooth, so $u'_c + E_L u'_c = \tilde{Q}\tilde{D}(u'_c) = \tilde{Q}f_c + \tilde{Q}h_c$, and $Q'f = u' + R_X(E_L u'_c - \tilde{Q}h_c) \in \mathcal{C}^\infty(X;V_1)$. □

By Proposition 9.10.2, E' lies in the bi-ideal of smoothing operators on $L^2(X;V_2)$, so Q' is a right parametrix for $D : H^1(X;V_1) \to L^2(X;V_2)$ and Proposition 9.10.6 shows that this map has closed range of finite codimension.

Proposition 9.11.3. *$D : \mathcal{C}^\infty(X;V_1) \to \mathcal{C}^\infty(X;V_2)$ is surjective if and only if there is no non-trivial $f \in \dot{\mathcal{C}}^\infty(X;V_2)$ with $D^*f = 0$. In that case $D : H^1(X;V_1) \to L^2(X;V_2)$ is surjective as well.*

Proof. If $f \in \dot{\mathcal{C}}^\infty(X;V_2)$ and $D^*f = 0$ then, integrating by parts (no boundary term arises as f vanishes at ∂X), $(Du, f) = (u, D^*f) = 0$ for all $u \in \mathcal{C}^\infty(X;V_1)$, so D is not surjective unless $f = 0$.

Conversely, assume there is no such f, and let $f \in L^2(X;V_2)$ be orthogonal to $D(H^1(X;V_1))$. For $\varphi \in \mathcal{C}^\infty_c(U;\tilde{V}_1)$, $(\tilde{D}\varphi, f_c) = (D(\varphi|_X), f) = 0$, so $\tilde{D}^*f_c = 0$ on U. By elliptic regularity f_c is smooth on U; as it vanishes outside X, $f \in \dot{\mathcal{C}}^\infty(X;V_2)$ and $D^*f = 0$, so $f = 0$. Thus $D : H^1 \to L^2$ is surjective.

For the smooth statement, apply Proposition 9.10.6: there are a smoothing J and a finite-rank smoothing self-adjoint projection π'_R, with range in $\mathcal{C}^\infty(X;V_2)$, such that $DQ'' = \mathrm{Id} - \pi'_R$, $Q'' = Q'(\mathrm{Id} + J)$. Since J maps L^2 into $\mathcal{C}^\infty$, Q'' maps $\mathcal{C}^\infty(X;V_2)$ into $\mathcal{C}^\infty(X;V_1)$, so the image W of $\mathcal{C}^\infty(X;V_1)$ under D contains $(\mathrm{Id} - \pi'_R)\mathcal{C}^\infty(X;V_2)$. Hence $f \in W$ if and only if $\pi'_R f \in W \cap \mathrm{Ran}\,\pi'_R$. If $W \neq \mathcal{C}^\infty(X;V_2)$, choose a non-zero linear functional λ on the finite-dimensional space $\mathrm{Ran}\,\pi'_R$ vanishing on $W \cap \mathrm{Ran}\,\pi'_R$; then $\ell = \lambda \circ \pi'_R$ is L^2-continuous, vanishes on W, but not on $\mathrm{Ran}\,\pi'_R \subset \mathcal{C}^\infty$. But W is dense in $D(H^1) = L^2$, since $\mathcal{C}^\infty$ is dense in H^1, so $\ell = 0$, a contradiction. □

Theorem 9.11.4. *Under the standing hypotheses (including (UC)) D has a right inverse*

$$Q : \mathcal{C}^\infty(X;V_2) \to \mathcal{C}^\infty(X;V_1), \quad DQ = \mathrm{Id}, \quad Q = Q' + E'',$$

with E'' smoothing; Q extends to a bounded map $L^2(X;V_2) \to H^1(X;V_1)$ and is a right inverse there too.

Proof. With Q'' and $\pi'_R = \sum_j e_j(\cdot, e_j)$ ($e_j \in \mathcal{C}^\infty$ orthonormal) as in the previous proof, choose by Proposition 9.11.3 $v_j \in \mathcal{C}^\infty(X;V_1)$ with $Dv_j = e_j$, and set $Q = Q'' + \sum_j v_j(\cdot, e_j)$. Then $DQ = \mathrm{Id} - \pi'_R + \pi'_R = \mathrm{Id}$. $Q - Q' = Q'J + \sum_j v_j(\cdot, e_j)$ has smooth kernel, since Q' is continuous on $\mathcal{C}^\infty(X)$ (closed graph theorem) and J has smooth kernel. □

9.12 The boundary map on solutions

For $s \in \mathbb{R}$ let

$$\mathcal{N}^s = \{u \in H^s(X;V_1);\ Du = 0 \text{ in } X^\circ\}, \qquad \mathcal{N}^\infty = \{u \in \mathcal{C}^\infty(X;V_1);\ Du = 0\} = \bigcap_s \mathcal{N}^s.$$

By Lemma 9.4.4(ii) and (UC), $b : \mathcal{N}^\infty \to \mathcal{C}^\infty(\partial X;V_1)$ is injective; its range

$$\mathfrak{C} = b(\mathcal{N}^\infty) \subset \mathcal{C}^\infty(\partial X;V_1)$$

is the space of *Cauchy data*. To show that it is the range of a pseudodifferential projection we extend b to L^2 solutions. Let

$$\mathcal{D} = \{u \in L^2(X; V_1);\ Du \in L^2(X; V_2)\}$$

with the graph norm, where Du is computed in $\mathcal{C}^{-\infty}(X; V_2)$.

Lemma 9.12.1. *For $u \in \mathcal{D}$ there is a unique $\beta(u) \in H^{-\frac{1}{2}}(\partial X; V_1)$ such that*

$$\tilde{D}(u_c) = (Du)_c + \frac{1}{i}\sigma(dx)\beta(u)\,\delta(x), \tag{9.12.1}$$

and $\beta : \mathcal{D} \to H^{-\frac{1}{2}}(\partial X; V_1)$ is continuous with $\beta(u) = bu$ for $u \in \mathcal{C}^\infty(X; V_1)$. In particular $\tilde{b} = \beta|_{\mathcal{N}^0} : \mathcal{N}^0 \to H^{-\frac{1}{2}}(\partial X; V_1)$ is a continuous injective extension of b.

Proof. $T = \tilde{D}(u_c) - (Du)_c \in H^{-1}(\tilde{X}; \tilde{V}_2)$ depends continuously on $u \in \mathcal{D}$ and is supported in ∂X: it vanishes outside X and equals $Du - Du = 0$ in X°. By Lemma 9.5.3, $T = w\,\delta(x)$ with $w \in H^{-\frac{1}{2}}$ depending continuously on T; set $\beta(u) = i\sigma(dx)^{-1}w$. For smooth u this is bu by Corollary 9.4.3. If $u \in \mathcal{N}^0$ and $\tilde{b}u = 0$, then $\tilde{D}(u_c) = 0$ on U, so u_c is smooth by elliptic regularity, whence $u \in \dot{\mathcal{C}}^\infty(X; V_1)$, $Du = 0$, and $u = 0$ by (UC). □

For $v \in H^{-\frac{1}{2}}(\partial X; V_1)$ the layer $T_v = \frac{1}{i}\sigma(dx)v\,\delta(x)$ lies in $H^{-1}(\tilde{X}; \tilde{V}_2)$, and we set

$$\tilde{I}_D v = R_X \tilde{Q}(T_v) \in L^2(X; V_1).$$

Since $R_X T_v = 0$ and $\chi = 1$ on X,

$$D\tilde{I}_D v = R_X \tilde{D}\tilde{Q}T_v = R_X E_R T_v =: R'v, \tag{9.12.2}$$

where R' has smooth kernel on $X \times \partial X$, and so extends to a continuous map $\mathcal{C}^{-\infty}(\partial X; V_1) \to \mathcal{C}^\infty(X; V_2)$. In particular $\tilde{I}_D : H^{-\frac{1}{2}}(\partial X; V_1) \to \mathcal{D}$ is continuous. Applying Theorem 9.8.4 to $\tilde{D}$ on $\tilde{X}$, with $M_+ = X$ and $Q = \tilde{Q}$, we obtain the classical operator

$$P = Q_+ \circ \frac{1}{i}\sigma(dx) \in \Psi^0(\partial X; V_1), \quad Pv = b\,\tilde{I}_D v \ \ (v \in \mathcal{C}^\infty(\partial X; V_1)), \tag{9.12.3}$$

with $\tilde{I}_D v \in \mathcal{C}^\infty(X; V_1)$ for smooth v, and $\sigma_0(P)$ given by the right side of (9.7.2) (the computation in the proof of Theorem 9.7.2 only uses the Dirac identities at ∂X).

Lemma 9.12.2. *(i) $\beta(\tilde{I}_D v) = Pv$ for all $v \in H^{-\frac{1}{2}}(\partial X; V_1)$.*

(ii) (Representation formula) For $u \in \mathcal{D}$,

$$u = Q'(Du) + \tilde{I}_D\beta(u) + Gu, \qquad Gu = -R_X E_L(u_c),$$

where G has smooth kernel on $X \times X$; in particular $u = \tilde{I}_D\tilde{b}u + Gu$ for $u \in \mathcal{N}^0$.

(iii) There is C such that $\|u\|_{L^2} \le C\|\tilde{b}u\|_{H^{-\frac{1}{2}}}$ for $u \in \mathcal{N}^0$. Hence $\overline{\mathfrak{C}} := \tilde{b}(\mathcal{N}^0)$ is closed in $H^{-\frac{1}{2}}(\partial X; V_1)$ and $\tilde{b} : \mathcal{N}^0 \to \overline{\mathfrak{C}}$ is an isomorphism.

Proof. (i) holds for smooth v by (9.12.3), and both sides are continuous on $H^{-\frac{1}{2}}$. (ii) By (9.12.1), $\tilde{D}(u_c) = (Du)_c + T_{\beta(u)}$; applying $\tilde{Q}$ and using $\tilde{Q}\tilde{D}(u_c) = u_c + E_L u_c$ gives the formula on restricting to X. (iii) If not, there are $u_k \in \mathcal{N}^0$ with $\|u_k\| = 1$ and $\tilde{b}u_k \to 0$. As G is compact on L^2 we may assume Gu_k converges; then $u_k = \tilde{I}_D\tilde{b}u_k + Gu_k$ converges in L^2 to some u, which lies in the closed subspace $\mathcal{N}^0$, has norm 1 and satisfies $\tilde{b}u = 0$ by continuity of $\tilde{b}$ (on $\mathcal{N}^0$ the graph norm is the L^2 norm), contradicting injectivity. The estimate gives closedness of the range and continuity of the inverse. □

Thus $\tilde{I}_D$ is a left parametrix for $\tilde{b}$, in the sense that $\tilde{I}_D\tilde{b} = \mathrm{Id} - G$ on $\mathcal{N}^0$ (where $\tilde{I}_D$ maps into $L^2(X;V_1) \supset \mathcal{N}^0$). To obtain a pseudodifferential projection onto $\overline{\mathfrak{C}}$ we need more: an *exact* formula expressing a solution in terms of its boundary values, with an error which is smoothing as a function of the boundary values. This comes from Green's formula, combined with a right inverse for the adjoint D^*.

Green's formula and the adjoint

We use the sesquilinear pairings $(f,g)_X = \int_X (f,g)\,d\mathrm{vol}$, given by the Riemannian density and the Hermitian structures, and similarly on $\tilde{X}$ for the extended structures; they extend to pairs (distribution, smooth section), and then $(\tilde{D}f, g)_{\tilde{X}} = (f, \tilde{D}^*g)_{\tilde{X}}$ for $f \in \mathcal{C}^{-\infty}(\tilde{X};\tilde{V}_1)$, $g \in \mathcal{C}^\infty(\tilde{X};\tilde{V}_2)$, where $\tilde{D}^*$ is the formal adjoint. On $Y = \partial X$ we use the smooth positive density ν_Y for which

$$(w\,\delta(x), g)_{\tilde{X}} = (w, g|_Y)_Y = \int_Y (w, g|_Y)\,\nu_Y, \quad w \in \mathcal{C}^\infty(Y;\tilde{V}_2|_Y),\ g \in \mathcal{C}^\infty(\tilde{X};\tilde{V}_2);$$

if $d\mathrm{vol} = \mu\,|dx\,dy|$ in local coordinates then $\nu_Y = \mu(0,y)\,|dy|$. By Lemma 9.5.3 both sides extend continuously to $w \in H^{-\frac{1}{2}}(Y;\tilde{V}_2|_Y)$, the right side being the extension of the $L^2(Y)$ pairing to $H^{-\frac{1}{2}} \times \mathcal{C}^\infty$. Smoothing operators on Y, and the spaces $L^2(Y)$, are defined with respect to ν_Y.

Lemma 9.12.3 (Green's formula)**.** *For $u \in \mathcal{D}$ and $\psi \in \mathcal{C}^\infty(X;V_2)$,*

$$(u, D^*\psi)_X = (Du, \psi)_X + \Big\langle \frac{1}{i}\sigma(dx)\beta(u), \psi|_Y \Big\rangle_Y.$$

Proof. Let $\tilde{\psi} \in \mathcal{C}^\infty(\tilde{X};\tilde{V}_2)$ be an extension of ψ. Pairing (9.12.1) with $\tilde{\psi}$ gives $(u_c, \tilde{D}^*\tilde{\psi})_{\tilde{X}} = ((Du)_c, \tilde{\psi})_{\tilde{X}} + (T_{\beta(u)}, \tilde{\psi})_{\tilde{X}}$, and the three terms are those of the formula, since $\tilde{D}^*\tilde{\psi} = D^*\psi$ on X. □

We need one more property of layers: for $v \in H^{-\frac{1}{2}}(Y;V)$ the pairing of $T_v \in H^{-1}(\tilde{X};\tilde{V}_2)$ with $w \in H^1(\tilde{X};\tilde{V}_2)$ only involves $R_X w$, provided the latter is smooth up to the boundary:

$$w \in H^1(\tilde{X};\tilde{V}_2),\ R_X w \in \mathcal{C}^\infty(X;V_2) \Longrightarrow (T_v, w)_{\tilde{X}} = \Big\langle \frac{1}{i}\sigma(dx)v, (R_X w)|_Y \Big\rangle_Y. \tag{9.12.4}$$

Indeed, let $w' \in \mathcal{C}^\infty(\tilde{X};\tilde{V}_2)$ extend $R_X w$; the formula holds for w' by definition of ν_Y. The difference $w - w' \in H^1(\tilde{X};\tilde{V}_2)$ is supported in the manifold with boundary $X' = \{x \le 0\}$, so by Lemma 9.5.1 (applied to X') it is the H^1-limit of sections in $\mathcal{C}^\infty_c(\{x<0\};\tilde{V}_2)$, all of which pair to zero with T_v. Hence $(T_v, w - w') = 0$.

The formal adjoint $D^* \in \mathrm{Diff}^1(X;V_2,V_1)$ satisfies the standing hypotheses as well: it is of Dirac type, $\tilde{D}^*$ is an extension elliptic on U, and $\tilde{Q}^* \in \Psi^{-1}(\tilde{X};\tilde{V}_1,\tilde{V}_2)$ is a parametrix for it near X in the required sense, since $\chi(\tilde{D}^*\tilde{Q}^* - \mathrm{Id}) = E_L^*$ and $(\tilde{Q}^*\tilde{D}^* - \mathrm{Id})\chi = E_R^*$ are smoothing; finally (UC) is symmetric in D and D^*. Applying Lemma 9.11.2 and Theorem 9.11.4 to D^* we obtain

$$Q_* : \mathcal{C}^\infty(X;V_1) \to \mathcal{C}^\infty(X;V_2), \quad D^*Q_* = \mathrm{Id}, \quad Q_* = Q_*' + E_*'', \quad Q_*'\varphi = R_X\tilde{Q}^*(\varphi_c), \tag{9.12.5}$$

where Q_*' maps $\mathcal{C}^\infty(X;V_1)$ into $\mathcal{C}^\infty(X;V_2)$ and E_*'' has smooth kernel on $X \times X$.

Proposition 9.12.4 (Exact representation of solutions)**.** *There is an operator S with smooth kernel on $X \times Y$, so $S : \mathcal{C}^{-\infty}(Y;V) \to \mathcal{C}^\infty(X;V_1)$, such that*

$$u = (\tilde{I}_D + S)\,\tilde{b}u \quad \textit{for all } u \in \mathcal{N}^0. \tag{9.12.6}$$

Proof. Let $u \in \mathcal{N}^0$, $v = \tilde{b}u$, and $\varphi \in \mathcal{C}^\infty(X;V_1)$. Applying Lemma 9.12.3 with $\psi = Q_*\varphi \in \mathcal{C}^\infty(X;V_2)$, so that $D^*\psi = \varphi$, and using $Du = 0$,

$$(u,\varphi)_X = \Big\langle \frac{1}{i}\sigma(dx)v, (Q'_*\varphi)|_Y \Big\rangle_Y + \Big\langle \frac{1}{i}\sigma(dx)v, (E''_*\varphi)|_Y \Big\rangle_Y.$$

For the first term, $w = \tilde{Q}^*(\varphi_c) \in H^1(\tilde{X};\tilde{V}_2)$ has $R_X w = Q'_*\varphi \in \mathcal{C}^\infty(X;V_2)$, so by (9.12.4)

$$\Big\langle \frac{1}{i}\sigma(dx)v, (Q'_*\varphi)|_Y \Big\rangle_Y = (T_v, \tilde{Q}^*(\varphi_c))_{\tilde{X}} = (\tilde{Q}T_v, \varphi_c)_{\tilde{X}} = (\tilde{I}_D v, \varphi)_X;$$

the middle identity holds for smooth sections in place of T_v, hence by continuity for $T_v \in H^{-1}$, as $\tilde{Q} : H^{-1} \to L^2$ and $\tilde{Q}^* : L^2 \to H^1$ are bounded. For the second, if $K(z,z')$ is the kernel of E''_* (with respect to $d\mathrm{vol}$), then $(\frac{1}{i}\sigma(dx)v, (E''_*\varphi)|_Y)_Y = (Sv,\varphi)_X$ where

$$Sv(z) = \int_Y K(y,z)^* \, \frac{1}{i}\sigma(dx)(y)\, v(y)\, \nu_Y(y),$$

the integral being interpreted as a distributional pairing in y. The kernel of S is smooth on $X \times Y$. Thus $(u - \tilde{I}_D v - Sv, \varphi)_X = 0$ for all $\varphi \in \mathcal{C}^\infty(X;V_1)$, which is dense in $L^2(X;V_1)$, so (9.12.6) holds. □

Next we produce many solutions from boundary data, correcting $\tilde{I}_D$ by means of the right inverse Q of Theorem 9.11.4. With R' as in (9.12.2), set

$$Iv = \tilde{I}_D v - QR'v, \quad v \in H^{-\frac{1}{2}}(Y;V).$$

Lemma 9.12.5. *I maps $H^{-\frac{1}{2}}(Y;V)$ continuously into $\mathcal{N}^0$, and $\mathcal{C}^\infty(Y;V)$ into $\mathcal{N}^\infty$. Moreover $\tilde{b} \circ I = P + B$ with $B \in \Psi^{-\infty}(Y;V)$.*

Proof. R' has smooth kernel $R'(z,y)$ on $X \times Y$, so $y \mapsto R'(\cdot,y)$ is a smooth map from Y into the Fréchet space $\mathcal{C}^\infty(X;\mathrm{Hom}(V,V_2))$. As Q is continuous on $\mathcal{C}^\infty$ (closed graph theorem), $y \mapsto QR'(\cdot,y)$ is smooth as well, i.e. QR' has smooth kernel on $X \times Y$; it maps $\mathcal{C}^{-\infty}(Y;V)$ continuously into $\mathcal{C}^\infty(X;V_1)$. Hence $Iv \in L^2(X;V_1)$ depends continuously on $v \in H^{-\frac{1}{2}}$, and $DIv = R'v - DQR'v = 0$, so $Iv \in \mathcal{N}^0$. For smooth v, $\tilde{I}_D v$ and hence Iv is smooth. Finally, by Lemma 9.12.2(i), $\beta(Iv) = Pv - (QR'v)|_Y$, so $\tilde{b} \circ I = P + B$ where B has kernel $-QR'(y',y)$, $(y',y) \in Y \times Y$, which is smooth. □

9.13 The Calderón projector

We now assemble the pieces. The argument has three steps. (1) The representation formula and Green's formula with a right inverse for D^* express every L^2 solution exactly through its boundary value (Proposition 9.12.4). (2) Applying the boundary value map gives a classical operator P on ∂X which is the identity on Cauchy data up to a smoothing error and maps onto Cauchy data up to a smoothing error (Lemma 9.13.1). (3) An abstract result of Chapter 5 corrects such an "almost projection" to an honest pseudodifferential projection with the right range.

We continue with the standing hypotheses of Section 9.11, including (UC). Recall that $P \in \Psi^0(Y;V)$, $Y = \partial X$, $V = V_1|_Y$, is the classical operator defined in (9.12.3), and $\overline{\mathfrak{C}} = \tilde{b}(\mathcal{N}^0) \subset H^{-\frac{1}{2}}(Y;V)$ is closed by Lemma 9.12.2.

Lemma 9.13.1. *There are $A, B \in \Psi^{-\infty}(Y;V)$ such that*

$$(\mathrm{Id} - P)v = Av \ \ \forall\, v \in \overline{\mathfrak{C}}, \qquad (P+B)H^{-\frac{1}{2}}(Y;V) \subset \overline{\mathfrak{C}}.$$

Consequently $P^2 - P \in \Psi^{-\infty}(Y;V)$.

Proof. The second statement, with B as in Lemma 9.12.5, is that lemma, since $(P+B)v = \tilde{b}(Iv)$. For the first, let $v = \tilde{b}u$, $u \in \mathcal{N}^0$. Applying β to (9.12.6) and using Lemma 9.12.2(i) and the smoothness of Sv,

$$v = \beta(\tilde{I}_D v) + \beta(Sv) = Pv + (Sv)|_Y,$$

so $(\mathrm{Id} - P)v = Av$ where A has kernel $S(y', y)$, the restriction of the kernel of S to $Y \times Y$. Finally, applying the first identity to $(P+B)v \in \overline{\mathfrak{C}}$ gives $(\mathrm{Id} - P)(P+B) = A(P+B)$ on $H^{-\frac{1}{2}}$, so

$$P - P^2 = A(P+B) - (\mathrm{Id} - P)B \in \Psi^{-\infty}(Y; V). \qquad \square$$

These are exactly the hypotheses of the characterization of ranges of pseudodifferential projections in Chapter 5 (Proposition 5.12.4, with $s = -\frac{1}{2}$ and $F = \overline{\mathfrak{C}}$).

Theorem 9.13.2 (Calderón projector)**.** *Let X be a compact Riemannian manifold with boundary and $D \in \mathrm{Diff}^1(X; V_1, V_2)$ a generalized Dirac operator satisfying* (UC)*. Then there is a classical $\Pi_{\mathcal{C}} \in \Psi^0(\partial X; V)$, $V = V_1|_{\partial X}$, with $\Pi_{\mathcal{C}}^2 = \Pi_{\mathcal{C}}$, such that:*

(i) $\Pi_{\mathcal{C}} H^{-\frac{1}{2}}(\partial X; V) = \tilde{b}(\mathcal{N}^0)$ *and* $\Pi_{\mathcal{C}} \mathcal{C}^\infty(\partial X; V) = \mathfrak{C} = b(\mathcal{N}^\infty)$*;*

(ii) $\Pi_{\mathcal{C}} - P \in \Psi^{-\infty}(\partial X; V)$*; in particular* $\sigma_0(\Pi_{\mathcal{C}})$ *is given by* (9.7.2)*, with* $\nu = dx/|dx|$ *the inward unit conormal;*

(iii) the Poisson operator $I_{\mathcal{C}} = (\tilde{I}_D + S)\Pi_{\mathcal{C}}$ *maps* $H^{-\frac{1}{2}}(\partial X; V)$ *continuously into* $\mathcal{N}^0$ *and* $\mathcal{C}^\infty(\partial X; V)$ *into* $\mathcal{N}^\infty$*, and*

$$\tilde{b} \circ I_{\mathcal{C}} = \Pi_{\mathcal{C}}, \qquad I_{\mathcal{C}} \circ \tilde{b} = \mathrm{Id} \ \text{ on } \mathcal{N}^0;$$

(iv) $\mathcal{N}^\infty$ *is dense in* $\mathcal{N}^0$*, and* $\tilde{b}$ *is the continuous extension of* $b : \mathcal{N}^\infty \to \mathfrak{C}$*.*

Proof. By Lemma 9.13.1 and Proposition 5.12.4 there is $B' \in \Psi^{-\infty}(Y; V)$ such that $\Pi_{\mathcal{C}} = P + B'$ satisfies $\Pi_{\mathcal{C}}^2 = \Pi_{\mathcal{C}}$ and $\Pi_{\mathcal{C}} H^{-\frac{1}{2}}(Y; V) = \overline{\mathfrak{C}}$. It is classical since P is, and (ii) holds; the symbol of P was identified after (9.12.3).

(iii) Let $v \in H^{-\frac{1}{2}}(Y; V)$. Then $\Pi_{\mathcal{C}} v \in \overline{\mathfrak{C}}$, so $\Pi_{\mathcal{C}} v = \tilde{b}u$ for a unique $u \in \mathcal{N}^0$, and by (9.12.6) $u = (\tilde{I}_D + S)\Pi_{\mathcal{C}} v = I_{\mathcal{C}} v$. Hence $I_{\mathcal{C}} v \in \mathcal{N}^0$ and $\tilde{b} I_{\mathcal{C}} v = \Pi_{\mathcal{C}} v$. If $u \in \mathcal{N}^0$ then $\Pi_{\mathcal{C}} \tilde{b} u = \tilde{b} u$, since $\Pi_{\mathcal{C}}$ is the identity on its range, and (9.12.6) gives $I_{\mathcal{C}} \tilde{b} u = u$. Continuity follows from that of $\tilde{I}_D : H^{-\frac{1}{2}} \to \mathcal{D}$ and of S. If v is smooth, then $\Pi_{\mathcal{C}} v$ is smooth, so $\tilde{I}_D \Pi_{\mathcal{C}} v$ is smooth by Theorem 9.8.4, and $S\Pi_{\mathcal{C}} v$ is smooth; thus $I_{\mathcal{C}} v \in \mathcal{N}^\infty$.

(i) The first statement is the choice of B'. If $v \in \mathfrak{C}$, $v = bu$ with $u \in \mathcal{N}^\infty$, then $v \in \overline{\mathfrak{C}}$ is smooth and $v = \Pi_{\mathcal{C}} v \in \Pi_{\mathcal{C}} \mathcal{C}^\infty(Y; V)$. Conversely if $v = \Pi_{\mathcal{C}} w$ with w smooth, then $u = I_{\mathcal{C}} w \in \mathcal{N}^\infty$ and $bu = \tilde{b}u = \Pi_{\mathcal{C}} w = v$, so $v \in \mathfrak{C}$.

(iv) $\Pi_{\mathcal{C}} \mathcal{C}^\infty(Y; V)$ is dense in $\Pi_{\mathcal{C}} H^{-\frac{1}{2}}(Y; V)$, as $\Pi_{\mathcal{C}}$ is continuous and $\mathcal{C}^\infty$ is dense in $H^{-\frac{1}{2}}$. Applying the continuous map $I_{\mathcal{C}}$, and using (iii), $\mathcal{N}^\infty = I_{\mathcal{C}}(\Pi_{\mathcal{C}} \mathcal{C}^\infty)$ is dense in $I_{\mathcal{C}}(\Pi_{\mathcal{C}} H^{-\frac{1}{2}}) = \mathcal{N}^0$. $\square$

Any $\Pi_{\mathcal{C}}$ with these properties is called a *Calderón projector* for D. In contrast to the dividing case (Theorem 9.7.2) it is not unique: only its range is determined. If $\Pi'_{\mathcal{C}}$ is another one then $\Pi_{\mathcal{C}} \Pi'_{\mathcal{C}} = \Pi'_{\mathcal{C}}$ and $\Pi'_{\mathcal{C}} \Pi_{\mathcal{C}} = \Pi_{\mathcal{C}}$, since each is the identity on the common range, and $\Pi'_{\mathcal{C}} - \Pi_{\mathcal{C}}$ is smoothing by (ii), but in general $\Pi'_{\mathcal{C}} \neq \Pi_{\mathcal{C}}$. A Calderón projector can always be chosen to be self-adjoint on $L^2(Y; V)$, see Exercise 9.4, but the orthogonal projection onto $\overline{\mathfrak{C}}$ in $H^{-\frac{1}{2}}$ is usually not pseudodifferential. From now on we no longer distinguish between $\tilde{b}$ and b.

Remark 9.13.3. (1) In the dividing case of Section 9.7, (UC) holds on $X = M_+$ (a solution vanishing to infinite order at H extends by zero to a smooth solution on M, which vanishes as D and D^* are invertible). The construction above works with M in place of the double $\tilde{X}$, and we may take $\tilde{Q} = D^{-1}$; then $E_R = E_L = 0$, $R' = 0$, and P is exactly the projection (9.9.2). By (iv) and continuity its range on $H^{-\frac{1}{2}}$ is the closure of $\mathfrak{C}$, which is $\tilde{b}(\mathcal{N}^0)$. So the projector of Theorem 9.7.2 is a Calderón projector in the present sense.

(2) Since $\sigma_0(\Pi_{\mathcal{C}})$ is the orthogonal projection onto the $+1$ eigenspace of the involution $A(y,\eta) = i\sigma(\nu)^{-1}\sigma(\eta)/|\eta|$, and $A(y,-\eta) = -A(y,\eta)$,

$$\sigma_0(\Pi_{\mathcal{C}})(y,-\eta) = \mathrm{Id} - \sigma_0(\Pi_{\mathcal{C}})(y,\eta), \qquad \operatorname{rank}\sigma_0(\Pi_{\mathcal{C}})(y,\eta) + \operatorname{rank}\sigma_0(\Pi_{\mathcal{C}})(y,-\eta) = \operatorname{rank} V.$$

In particular the range of $\Pi_{\mathcal{C}}$ and its null space are both infinite-dimensional (if $\dim X \geq 2$): the boundary values of solutions form "half" of all sections.

Question. Where exactly was the assumption that D is of Dirac type used?

Answer. In three places. Most importantly in Theorem 9.8.4: the symbol of a parametrix is rational in the normal variable, so layer potentials are smooth up to the boundary and their boundary values are classical pseudodifferential operators. Second, in the symbol formula (9.7.2), through the Clifford relations. Third, in the justification of (UC). For general elliptic operators Seeley proved that the Calderón projector is still pseudodifferential; the rational structure is then replaced by the *transmission property* of the symbol of the parametrix (see the Notes). □

9.14 Pseudodifferential boundary conditions

We continue under the hypotheses of Theorem 9.13.2, and fix a Calderón projector $\Pi_{\mathcal{C}}$. By that theorem a solution $u \in \mathcal{N}^\infty$ is uniquely determined by its boundary value, but bu ranges over the infinite-dimensional space $\mathfrak{C} = \Pi_{\mathcal{C}}\mathcal{C}^\infty(Y;V)$; a *boundary condition* should cut this down to a finite-dimensional space, without restricting the right-hand side Du too much.

Definition 9.14.1. For a projection $\Pi \in \Psi^0(Y;V)$, $\Pi^2 = \Pi$, the *realization* of D with boundary condition Π is

$$D_\Pi : \mathcal{D}_\Pi \to \mathcal{C}^\infty(X;V_2), \quad \mathcal{D}_\Pi = \{u \in \mathcal{C}^\infty(X;V_1);\ \Pi(bu) = 0\}, \quad D_\Pi u = Du.$$

A linear map between vector spaces is *Fredholm* if its null space and the quotient of its target by its range (its *cokernel*) are finite-dimensional; its index is then the difference of these dimensions.

For the Fréchet spaces involved here this algebraic notion agrees with the topological one: $\mathcal{D}_\Pi$ is a closed subspace of $\mathcal{C}^\infty(X;V_1)$, D_Π is continuous, and a continuous linear map between Fréchet spaces whose range has finite codimension automatically has closed range (apply the open mapping theorem to the map $\mathcal{D}_\Pi \oplus \mathbb{C}^k \to \mathcal{C}^\infty(X;V_2)$ obtained by adding a complement of the range).

The Fredholm property of D_Π is equivalent to that of an operator on the boundary, namely the restriction of Π to the boundary values of solutions,

$$T_\Pi : \mathfrak{C} = \Pi_{\mathcal{C}}\mathcal{C}^\infty(Y;V) \longrightarrow \Pi\mathcal{C}^\infty(Y;V), \quad T_\Pi v = \Pi v. \tag{9.14.1}$$

It depends only on Π and the space $\mathfrak{C}$, not on the choice of $\Pi_{\mathcal{C}}$; as an operator on $\mathcal{C}^\infty(Y;V)$ it is $\Pi\Pi_{\mathcal{C}}$. When T_Π is Fredholm its index is called the *relative index* of the pair $(\Pi_{\mathcal{C}}, \Pi)$.

Theorem 9.14.2. *Let $\Pi \in \Psi^0(Y;V)$ be a projection. Then b induces isomorphisms*

$$\mathrm{Nul}(D_\Pi) \cong \mathrm{Nul}(T_\Pi), \qquad \mathcal{C}^\infty(X;V_2)/D(\mathcal{D}_\Pi) \cong \Pi\mathcal{C}^\infty(Y;V)/\Pi(\mathfrak{C}).$$

In particular D_Π is Fredholm if and only if T_Π is Fredholm, and then $\mathrm{ind}(D_\Pi) = \mathrm{ind}(T_\Pi)$.

Proof. Null space. If $u \in \mathcal{D}_\Pi$ and $Du = 0$ then $u \in \mathcal{N}^\infty$, so $bu \in \mathfrak{C}$, and $\Pi bu = 0$. Conversely each $v \in \mathfrak{C}$ with $\Pi v = 0$ is bu for a unique $u \in \mathcal{N}^\infty$, by Theorem 9.13.2, and then $u \in \mathcal{D}_\Pi$. Since b is injective on $\mathcal{N}^\infty$, $b : \mathrm{Nul}(D_\Pi) \to \mathrm{Nul}(T_\Pi)$ is an isomorphism.

Cokernel. Define

$$\tau : \mathcal{C}^\infty(Y;V) \to \mathcal{C}^\infty(X;V_2)/D(\mathcal{D}_\Pi), \quad \tau(v) = [Du] \text{ where } u \in \mathcal{C}^\infty(X;V_1),\ bu = v.$$

This is well defined: such u exist since b is surjective, and if $bu = bu'$ then $\Pi b(u-u') = 0$, so $D(u-u') \in D(\mathcal{D}_\Pi)$. It is surjective: for $f \in \mathcal{C}^\infty(X;V_2)$ let $u = Qf$, with Q the right inverse of Theorem 9.11.4; then $\tau(bu) = [f]$. Its null space is

$$\mathrm{Nul}\,\tau = \mathfrak{C} + \mathrm{Nul}(\Pi|_{\mathcal{C}^\infty}), \qquad \mathrm{Nul}(\Pi|_{\mathcal{C}^\infty}) = (\mathrm{Id} - \Pi)\mathcal{C}^\infty(Y;V).$$

Indeed, if $v = w + v'$ with $w \in \mathfrak{C}$ and $\Pi v' = 0$, write $w = bu_0$ with $u_0 \in \mathcal{N}^\infty$ and $v' = bu'$ with $u' \in \mathcal{C}^\infty$; then $u' \in \mathcal{D}_\Pi$ and $\tau(v) = [Du_0 + Du'] = [Du'] = 0$. Conversely, if $\tau(v) = 0$, $v = bu$, then $Du = Du'$ for some $u' \in \mathcal{D}_\Pi$, so $u - u' \in \mathcal{N}^\infty$ and $v = b(u-u') + bu' \in \mathfrak{C} + \mathrm{Nul}(\Pi)$. Hence τ induces an isomorphism

$$\mathcal{C}^\infty(Y;V)/\big(\mathfrak{C} + (\mathrm{Id} - \Pi)\mathcal{C}^\infty(Y;V)\big) \cong \mathcal{C}^\infty(X;V_2)/D(\mathcal{D}_\Pi).$$

Finally $v \mapsto \Pi v$ induces an isomorphism of the left side onto $\Pi\mathcal{C}^\infty(Y;V)/\Pi(\mathfrak{C})$: it is surjective, and $\Pi v \in \Pi(\mathfrak{C})$ if and only if $\Pi(v-w) = 0$ for some $w \in \mathfrak{C}$, i.e. $v \in \mathfrak{C} + (\mathrm{Id} - \Pi)\mathcal{C}^\infty$. Since $\Pi(\mathfrak{C}) = \mathrm{Ran}(T_\Pi)$, this is the second isomorphism. □

Example 9.14.3. For $\Pi = \Pi_\mathcal{C}$, T_Π is the identity of $\mathfrak{C}$, so $D_{\Pi_\mathcal{C}} : \{u \in \mathcal{C}^\infty(X;V_1);\ \Pi_\mathcal{C}(bu) = 0\} \to \mathcal{C}^\infty(X;V_2)$ is an isomorphism; its inverse is $f \mapsto Qf - I_\mathcal{C} bQf$. At the other extreme, for $\Pi = \mathrm{Id} - \Pi_\mathcal{C}$ we have $T_\Pi = 0$ with infinite-dimensional null space $\mathfrak{C}$: this "boundary condition" is satisfied by every solution. For $\Pi = 0$ (no boundary condition) T_Π is surjective with null space $\mathfrak{C}$, while for $\Pi = \mathrm{Id}$ (the Dirichlet condition $bu = 0$) T_Π is injective with cokernel $(\mathrm{Id} - \Pi_\mathcal{C})\mathcal{C}^\infty$; neither is Fredholm (if $\dim X \geq 2$), in accordance with Remark 9.13.3(2).

Discussion. The unit disc (Exercises 9.5 and 9.6) shows the whole picture in Fourier modes $e^{ik\theta}$ on the boundary circle. Holomorphic functions have boundary values with only $k \geq 0$, so $\mathfrak{C}$ is "half" of $\mathcal{C}^\infty(\partial X)$. The boundary condition Π_N asks that the modes $k \geq N$ vanish. For $N > 0$ the modes $0 \leq k < N$ survive (null space of dimension N); for $N < 0$ the condition also asks that the modes $N \leq k < 0$ vanish, which solutions satisfy automatically, so the condition is $|N|$ conditions too many (cokernel of dimension $|N|$). The Dirichlet condition $\Pi = \mathrm{Id}$ asks infinitely many conditions too many, and $\Pi = 0$ infinitely many too few. A good boundary condition must therefore be "as large as" $\mathrm{Id} - \Pi_\mathcal{C}$, at least at the level of principal symbols; this is the symbolic criterion below.

A symbolic criterion

The operator T_Π is a "Toeplitz-type" operator between the ranges of two pseudodifferential projections, and its Fredholm property can be decided from principal symbols.

Theorem 9.14.4. *Let $\Pi \in \Psi^0(Y;V)$ be a classical projection, $\pi = \sigma_0(\Pi)$, and $p = \sigma_0(\Pi_{\mathcal{C}})$, given by (9.7.2). Suppose that for every $(y,\eta) \in T^*Y \setminus 0$*

$$\pi(y,\eta) : \operatorname{Ran} p(y,\eta) \longrightarrow \operatorname{Ran} \pi(y,\eta) \text{ is an isomorphism,} \tag{9.14.2}$$

equivalently $V_y = \operatorname{Ran} p(y,\eta) \oplus \operatorname{Nul} \pi(y,\eta)$. Then:

(i) for every $s \in \mathbb{R}$, $\Pi : \Pi_{\mathcal{C}} H^s(Y;V) \to \Pi H^s(Y;V)$ is Fredholm;

(ii) if $v \in \Pi_{\mathcal{C}} H^s(Y;V)$ and $\Pi v \in \mathcal{C}^\infty$ then $v \in \mathcal{C}^\infty$;

(iii) T_Π is Fredholm, with the same null space and index as the operators in (i).

Consequently D_Π is Fredholm and $\operatorname{ind}(D_\Pi) = \operatorname{ind}(T_\Pi)$ is the relative index of $(\Pi_{\mathcal{C}}, \Pi)$.

Proof. The equivalence of the two forms of (9.14.2): if π is an isomorphism $\operatorname{Ran} p \to \operatorname{Ran} \pi$, then $\operatorname{Ran} p \cap \operatorname{Nul} \pi = 0$ and $\dim \operatorname{Ran} p = \dim \operatorname{Ran} \pi = \dim V_y - \dim \operatorname{Nul} \pi$; conversely, if $V_y = \operatorname{Ran} p \oplus \operatorname{Nul} \pi$ then $\pi|_{\operatorname{Ran} p}$ is injective and $\pi(\operatorname{Ran} p) = \pi(V_y) = \operatorname{Ran} \pi$.

Since p and π are smooth projection-valued functions, homogeneous of degree 0, $\operatorname{Ran} p$ and $\operatorname{Ran} \pi$ are smooth vector bundles over $T^*Y \setminus 0$ and $\pi|_{\operatorname{Ran} p}$ is a smooth bundle isomorphism between them. Thus

$$q = \iota \circ (\pi|_{\operatorname{Ran} p})^{-1} \circ \pi \in \mathcal{C}^\infty(T^*Y \setminus 0; \hom(V)),$$

with ι the inclusion of $\operatorname{Ran} p$, is homogeneous of degree 0, and by construction

$$pq = q, \qquad q\pi p = p, \qquad \pi q = \pi.$$

(For the second, $q(\pi p w) = pw$ because $pw \in \operatorname{Ran} p$; for the third, $\pi q w = \pi z$ where $z \in \operatorname{Ran} p$ is the element with $\pi z = \pi w$.) Let $B_0 \in \Psi^0(Y;V)$ be classical with $\sigma_0(B_0) = q$ (Proposition 5.5.7) and set

$$B = \Pi_{\mathcal{C}} B_0 \Pi, \qquad T = \Pi \Pi_{\mathcal{C}}.$$

Then $\sigma_0(BT) = pq\pi\pi p = pq\pi p = p$ and $\sigma_0(TB) = \pi p p q \pi = \pi q \pi = \pi$, so

$$BT = \Pi_{\mathcal{C}} + R_1, \qquad TB = \Pi + R_2, \qquad R_1, R_2 \in \Psi^{-1}(Y;V).$$

Now T maps $\Pi_{\mathcal{C}} H^s$ into ΠH^s (where it acts as Π) and B maps ΠH^s into $\Pi_{\mathcal{C}} H^s$; on these closed subspaces $BT = \mathrm{Id} + R_1$ and $TB = \mathrm{Id} + R_2$, and R_1, R_2 map the respective subspaces into themselves (e.g. $R_1 v = BTv - v \in \Pi_{\mathcal{C}} H^s$ for $v \in \Pi_{\mathcal{C}} H^s$) and are compact there, being of negative order. By Atkinson's theorem (Appendix A), (i) follows.

For (ii), if $v = \Pi_{\mathcal{C}} v \in H^s$ and $\Pi v = Tv \in \mathcal{C}^\infty$, then $v = BTv - R_1 v \in H^{s+1}$; iterating, $v \in \mathcal{C}^\infty$.

(iii) By (ii) the null space of the operator in (i) consists of smooth sections, so it is $\operatorname{Nul}(T_\Pi)$, independent of s. Also by (ii), $\operatorname{Ran}(T_\Pi) = \Pi(\Pi_{\mathcal{C}} H^s) \cap \mathcal{C}^\infty$. The map $\Pi\mathcal{C}^\infty / \operatorname{Ran}(T_\Pi) \to \Pi H^s / \Pi(\Pi_{\mathcal{C}} H^s)$ induced by inclusion is therefore injective; it is also surjective, since the target is finite-dimensional and the image of the dense subspace $\Pi\mathcal{C}^\infty \subset \Pi H^s$ under the continuous quotient map (the range in (i) being closed) is dense, hence everything. So the cokernels have the same dimension. The last statement follows from Theorem 9.14.2. □

The condition (9.14.2) is the analogue for pseudodifferential boundary conditions of the Lopatinski–Shapiro ("covering") condition for local boundary conditions; it says that the boundary condition determines, at the symbolic level, exactly the half of the boundary data which is free for solutions. With some more work (boundary regularity for D) one shows that under (9.14.2) the realization is also Fredholm as a map $\{u \in H^1(X;V_1);\ \Pi b u = 0\} \to L^2(X;V_2)$, and that, conversely, the condition is necessary for the Fredholm property in these spaces; we do not pursue this here, see the Notes.

Remark 9.14.5. (1) If $\sigma_0(\Pi) = \sigma_0(\Pi_{\mathcal{C}})$ then (9.14.2) holds trivially. This is the case for $\Pi = \Pi_{\mathcal{C}}$, and for the boundary condition of Atiyah, Patodi and Singer when D has product form near the boundary: there Π is the spectral projection onto the non-negative eigenspaces of a self-adjoint elliptic operator on Y, whose symbol is the projection appearing in (9.7.2); see [16].

(2) *Local boundary conditions.* Suppose Π is a bundle projection, i.e. multiplication by a smooth $\pi \in \mathcal{C}^\infty(Y; \mathrm{End}(V))$ with $\pi^2 = \pi$; then $\sigma_0(\Pi)(y,\eta) = \pi(y)$ does not depend on η. If (9.14.2) holds then $\operatorname{rank} p(y,\eta) = \operatorname{rank} V - \dim \operatorname{Nul} \pi(y) = \operatorname{rank} \pi(y)$ for all η, so by Remark 9.13.3(2)

$$\operatorname{rank} V = \operatorname{rank} p(y,\eta) + \operatorname{rank} p(y,-\eta) = 2 \operatorname{rank} \pi(y).$$

Thus a local elliptic boundary condition can only exist if $\operatorname{rank} V_1$ is even. For instance the Cauchy–Riemann operator $D_x + iD_y$ acting on scalar functions on a surface with boundary has no local elliptic boundary condition, although it has many pseudodifferential ones; see Exercises 9.5 and 9.7.

9.15 Exercises

Exercise 9.1. Let X be a manifold with boundary, ∂X not necessarily compact. Show that there is a boundary defining function ρ, an open neighbourhood C of ∂X and a diffeomorphism $C \to [0,1) \times \partial X$ whose first component is ρ. (Hint: with V as in the proof of Lemma 9.1.2, find a smooth $\epsilon : \partial X \to (0,1]$ such that $\psi(t,p) = \exp(tV)p$ is defined, satisfies $\rho(\psi(t,p)) = t$, and is a diffeomorphism from $\{(t,p);\ 0 \le t < \epsilon(p)\}$ onto an open set; then use $(s,p) \mapsto \psi(s\epsilon(p),p)$, and modify ρ away from ∂X.) Deduce that the doubling construction of Section 9.1 works for all manifolds with boundary.

Exercise 9.2. Let X be a compact manifold with boundary and $k \in \mathbb{N}_0$. Show that

$$H^k(X) = \{u \in L^2(X);\ Pu \in L^2(X) \text{ for all } P \in \mathrm{Diff}^k(X)\},$$

where Pu is computed in $\mathcal{C}^{-\infty}(X)$, and that an equivalent norm is $\sum_j \|P_j u\|_{L^2}$ for any finite set of P_j generating $\mathrm{Diff}^k(X)$ as a $\mathcal{C}^\infty(X)$-module. (Hint: in a collar, use a finite reflection $Eu(x,y) = \sum_{j=0}^{k-1} a_j u(-2^j x, y)$ for $x < 0$, with $\sum_j a_j (-2^j)^l = 1$ for $0 \le l \le k-1$, to extend u across the boundary; check that E is bounded from the right side into H^k by approximating u with smooth functions, for which the derivatives of order $< k$ of Eu are continuous across $x = 0$.)

Exercise 9.3. Under the standing hypotheses of Section 9.7, show that for $f \in \mathcal{C}^\infty(M_+; V_2)$ and $v \in \mathcal{C}^\infty(H; V)$ there is a unique $u \in \mathcal{C}^\infty(M_+; V_1)$ with

$$Du = f \text{ in } M_+, \qquad \Pi_{\mathcal{C}}(b_+ u) = \Pi_{\mathcal{C}} v,$$

namely $u = R_D f + I_D \Pi_{\mathcal{C}} v$. (Hint: show that $D^{-1}(f_c)$ is smooth up to H from both sides, with equal boundary values by (9.4.3), and that its restriction to M_- lies in $\mathcal{N}_-$; deduce $\Pi_{\mathcal{C}} b_+ R_D f = 0$. Note that the condition $b_+ u = \Pi_{\mathcal{C}} v$ is not the right one: $(\mathrm{Id} - \Pi_{\mathcal{C}}) b_+ u = b_+ R_D f$ is determined by f.)

Exercise 9.4. Let $\mathcal{H}$ be a Hilbert space and $\Pi \in \mathcal{B}(\mathcal{H})$ with $\Pi^2 = \Pi$.

(a) Show that $T = \Pi + \Pi^* - \mathrm{Id}$ is self-adjoint with $T^2 = \mathrm{Id} + (\Pi - \Pi^*)(\Pi - \Pi^*)^* \ge \mathrm{Id}$, so T is invertible, and that $\Pi T = \Pi\Pi^* = T\Pi^*$, $T\Pi = \Pi^*\Pi = \Pi^* T$.

(b) Deduce that $\Pi_o = \Pi T^{-1}$ is the orthogonal projection onto $\operatorname{Ran}\Pi$.

(c) If $\Pi \in \Psi^0(Y;V)$ is a projection on a compact manifold, show that T is elliptic and that $\Pi_o \in \Psi^0(Y;V)$ (use Chapter 5 for the inverse of an invertible elliptic operator). Conclude that the Calderón projector may be chosen self-adjoint on $L^2(Y;V)$, and that its symbol is still (9.7.2).

Exercise 9.5. Let $X = \{z \in \mathbb{C};\ |z| \le 1\}$ with the Euclidean metric and $D = D_x + iD_y = -2i\partial_{\bar z}$ acting on functions.

(a) Show that D is of Dirac type and that (UC) holds.

(b) Show that $\mathfrak{C}$ consists of the $v \in \mathcal{C}^\infty(\partial X)$, $v(\theta) = \sum_k v_k e^{ik\theta}$, with $v_k = 0$ for $k < 0$, and that the Szegő projector S, $Sv = \sum_{k\ge 0} v_k e^{ik\theta}$, is a Calderón projector.

(c) Using local coordinates $x = 1 - r$, $y = \theta$ near $z = 1$, verify that (9.7.2) gives $\sigma_0(S)(\theta,\eta) = \frac12(1 + \eta/|\eta|)$, in agreement with (b).

Exercise 9.6. In the setting of Exercise 9.5, for $N \in \mathbb{Z}$ let Π_N be the projection onto $\operatorname{span}\{e^{ik\theta};\ k \ge N\}$. Show that $\Pi_N \in \Psi^0(\partial X)$, that (9.14.2) holds, and that the realization D_{Π_N} has index N; describe its null space and cokernel explicitly for $N > 0$ and $N < 0$.

Exercise 9.7. On the unit disc, let $D = \begin{pmatrix} D_x + iD_y & 0 \\ 0 & D_x - iD_y \end{pmatrix}$ act on $\mathbb{C}^2$-valued functions.

(a) Show that D is of Dirac type and compute $\mathfrak{C}$ and $\sigma_0(\Pi_{\mathcal{C}})$.

(b) Show that the local boundary condition $u_1|_{\partial X} = u_2|_{\partial X}$ is of the form $\Pi(bu) = 0$ for a bundle projection Π satisfying (9.14.2).

(c) Show directly that the null space of D_Π consists of the constant functions (c,c) and that D_Π is surjective, so $\operatorname{ind} D_\Pi = 1$; check this against Theorem 9.14.2.

Notes

The treatment of boundary problems by means of the Calderón projector goes back to A. P. Calderón (*Boundary value problems for elliptic equations*, Outlines of the joint Soviet–American symposium on partial differential equations, Novosibirsk, 1963) and R. T. Seeley (*Singular integrals and boundary value problems*, Amer. J. Math. **88** (1966)), who showed that the projector is pseudodifferential for general elliptic operators; the extension operator in Proposition 9.2.1 is from R. T. Seeley, *Extension of $\mathcal{C}^\infty$ functions defined in a half space*, Proc. Amer. Math. Soc. **15** (1964). The unique continuation theorem used to justify (UC) is that of N. Aronszajn (J. Math. Pures Appl. **36** (1957)) for second-order operators with scalar principal symbol. For operators of Dirac type a thorough account, including the invertible double, the gluing formula, local boundary conditions and the Atiyah–Patodi–Singer condition, is B. Booß-Bavnbek and K. P. Wojciechowski, *Elliptic boundary problems for Dirac operators* (Birkhäuser, 1993); for the classical theory of elliptic boundary problems with local boundary conditions see [8, Chapter 20], and for the APS condition [16].

Compared with the original notes we restrict to operators of Dirac type throughout and supply complete arguments in several places: the regularity of layer potentials is proved by the contour-deformation argument for rational symbols; the construction of the Calderón projector on a manifold with boundary, sketched in the original, is completed by the exact

representation formula Proposition 9.12.4, which uses Green's formula together with a right inverse for D^* and shows directly that $\mathrm{Id} - P$ is smoothing on the boundary values of solutions; and the reduction of a pseudodifferential boundary problem to the relative index of two projections (Theorem 9.14.2) is complemented by the symbolic criterion Theorem 9.14.4. The boundary condition in Exercise 9.3 corrects that of the original. Several sections announced in the original (the Poisson operator in Sobolev spaces, unique continuation, boundary regularity, local boundary conditions, absolute and relative Hodge cohomology, the transmission condition) were never written and are omitted, apart from the brief remarks above. The same applies to the gluing formula: if D is of Dirac type on a compact manifold M without boundary, divided by a hypersurface H into M_1 and M_2, then $\operatorname{ind} D$ equals the relative index of the pair $(\Pi_{1,\mathcal{C}}, \mathrm{Id} - \Pi_{2,\mathcal{C}})$ of Calderón projectors of the two sides; in the invertible case of Section 9.7 this reduces to $\Pi_{2,\mathcal{C}} = \mathrm{Id} - \Pi_{1,\mathcal{C}}$ (Theorem 9.7.2), and in general it is proved in the book of Booß-Bavnbek and Wojciechowski cited above.

Chapter 10

Lagrangian distributions and Hamilton–Jacobi theory

Elliptic operators have no singular solutions: if P is elliptic and $Pu \in \mathcal{C}^\infty$ then $u \in \mathcal{C}^\infty$ (Chapter 5). For non-elliptic operators, such as the wave operator, singular solutions do exist, and the simplest of them are conormal distributions. In this chapter we find out when, and how, one can solve $Pu \in \mathcal{C}^\infty$ with $u \in I^m(X, Y; \Omega^{\frac{1}{2}})$ conormal to a hypersurface Y. The principal symbol of Pu vanishes exactly when Y is *characteristic*; characteristic hypersurfaces are found by solving the eikonal equation $p(x, dh) = 0$, which we do by the Hamilton–Jacobi method, i.e. by symplectic geometry on T^*X. The next term in the symbol of Pu is then given by a first-order differential operator along the Hamilton vector field of p on N^*Y, and requiring it to vanish leads to the *transport equations*. Solving these iteratively produces solutions of $Pu \in \mathcal{C}^\infty$ with prescribed restriction to a transversal hypersurface ("travelling waves"); this is used in Chapter 11 to construct a parametrix for the wave equation.

The second half of the chapter removes the restriction to conormal distributions. When a characteristic hypersurface develops singularities, the union of the bicharacteristics making up its conormal bundle is still a smooth conic Lagrangian submanifold Λ of T^*X, but no longer a conormal bundle. We show that every conic Lagrangian is locally parametrized by a non-degenerate phase function, prove that any two parametrizations are related by stationary phase and a change of fibre variables, and define the space $I^m(X, \Lambda; \Omega^{\frac{1}{2}})$ of Lagrangian distributions: distributions which are microlocally oscillatory integrals $\int e^{i\varphi(x,\theta)} a(x, \theta)\, d\theta$ with phases parametrizing Λ, or, equivalently, push-forwards of conormal distributions. Conormal distributions are the case $\Lambda = N^*Y$, and the wave kernel of Chapter 11, an oscillatory integral whose phase solves the eikonal equation, is another example. Keller's example of a caustic for the wave equation in two space dimensions shows that the principal symbol of a Lagrangian distribution is not simply a half-density on Λ, but involves the Keller–Maslov line bundle; we define it and prove its invariance, and finally show how the symbol can be recovered by pairing with oscillatory test functions.

We use the conormal distributions of Chapter 4 (whose notation and normalization of orders we follow) and the pseudodifferential operators on manifolds of Chapter 5. Throughout, X is a manifold of dimension n, $\pi : T^*X \to X$ is the projection, and operators act on half-densities.

10.1 Pseudodifferential operators on conormal distributions

We first record what a pseudodifferential operator does to a conormal distribution. To leading order it simply multiplies the symbol by the principal symbol of the operator restricted to the conormal bundle; this already tells us where singular solutions can live.

Theorem 10.1.1. *Let $Y \subset X$ be a closed embedded submanifold and $A \in \Psi^{m'}(X;\Omega^{\frac12})$ properly supported. Then*

$$A : I^m(X,Y;\Omega^{\frac12}) \longrightarrow I^{m+m'}(X,Y;\Omega^{\frac12}), \qquad \sigma_{m+m'}(Au) = \sigma_{m'}(A)|_{N^*Y}\,\sigma_m(u),$$

and A preserves the classical subspaces if it has a polyhomogeneous symbol. The product on the right makes sense since $\sigma_{m'}(A) \in S^{m'-[1]}(T^*X)$ restricts to a class in $S^{m'-[1]}(N^*Y)$.

Proof. By a partition of unity we may assume u has compact support in an adapted chart. Since A is properly supported and its kernel is smooth off the diagonal, ϕAu is smooth if $\phi \in \mathcal{C}^\infty$ vanishes near $\operatorname{supp} u$, and for $\phi \in \mathcal{C}^\infty_{\mathrm{c}}$ of the chart, $\phi A\chi$ (with $\chi = 1$ near $\operatorname{supp} u$) is, in the chart, $\mathrm{Op}(b)$ plus a smoothing operator, with $b \in S^{m'}(\mathbb{R}^n;\mathbb{R}^n)$ compactly supported in z (Chapter 5). The result therefore follows from the local computation of Chapter 4: if $u = (\mathcal{I}[a]+f)|dz|^{1/2}$ then $\mathrm{Op}(b)u$ has full symbol congruent to $b(0,y,\xi,0)a(y,\xi)$ modulo $S^{M+m'-1}$, and $b(0,y,\xi,0)$ represents $\sigma_{m'}(A)$ at $(0,y;\xi,0) \in N^*Y$. Here and below $\mathcal{I}[a](x,y) = \int e^{ix\cdot\xi}a(y,\xi)\bar{d}\xi$ in adapted coordinates $(x,y) \in \mathbb{R}^k \times \mathbb{R}^d$, and $M = m + \frac{n}{4} - \frac{k}{2}$. □

In particular, if A is elliptic and $\sigma_m(u)$ is not in $S^{m+\frac{n}{4}-1}$, then $Au \notin \mathcal{C}^\infty$: there are no conormal solutions of $Au \in \mathcal{C}^\infty$. Suppose instead that $P \in \Psi^{m'}(X;\Omega^{\frac12})$ is not elliptic, with homogeneous principal symbol p, and that we look for $u \in I^m(X,Y;\Omega^{\frac12})$ which is genuinely singular, say with $\sigma_m(u)$ elliptic (bounded below by $c\langle\xi\rangle^{m+\frac{n}{4}}|dy\,d\xi|^{1/2}$ for large ξ), and $Pu \in \mathcal{C}^\infty$. By Theorem 10.1.1, $p|_{N^*Y}\sigma_m(u) = 0$ modulo lower order, so p must vanish on N^*Y.

Definition 10.1.2. Let $P \in \Psi^{m'}(X;\Omega^{\frac12})$ have principal symbol p, homogeneous of degree m' on $T^*X \setminus 0$, and characteristic variety $\Sigma(P) = \{p = 0\} \subset T^*X \setminus 0$. A closed embedded submanifold Y is *characteristic* for P if $N^*Y \setminus 0 \subset \Sigma(P)$, and characteristic on an open cone $\Gamma \subset N^*Y \setminus 0$ if $\Gamma \subset \Sigma(P)$.

Discussion. A conormal distribution to Y oscillates only in the directions conormal to Y: locally it is a superposition of plane waves $e^{ix\cdot\xi}$ with $\xi \in N^*Y$. For Pu to be smoother than expected, P must annihilate these plane waves to leading order at high frequency, i.e. p must vanish on N^*Y. For the wave operator this says that a singular wave front must move at the speed of light; for an elliptic operator it never happens.

Corollary 10.1.3. *If Y is characteristic for P on the open cone Γ, and $u \in I^m(X,Y;\Omega^{\frac12})$ has $\mathrm{WF}(u) \subset \Gamma$, then $Pu \in I^{m+m'-1}(X,Y;\Omega^{\frac12})$.*

Proof. By Chapter 4 (the characterization of the wavefront set of a conormal distribution by its full symbol), locally the full symbol a of u is of order $-\infty$ outside any conic neighbourhood of the closed cone $\mathrm{WF}(u)$; choosing a cut-off ψ homogeneous of degree 0, equal to one near $\mathrm{WF}(u)$ and supported in Γ, $a = \psi a$ modulo $S^{-\infty}$, and $p\psi = 0$ for large ξ. Hence $\sigma_{m+m'}(Pu) = p|_{N^*Y}\sigma_m(u) = 0$ and $Pu \in I^{m+m'-1}$ by the exact sequence of Chapter 4. □

So for characteristic Y we only gain one order; we return to this in Section 10.4. First we note that characteristic hypersurfaces and singular solutions do exist.

Example 10.1.4 (Plane waves)**.** Let $P = D_t^2 - \sum_{j=1}^n D_{x_j}^2 = D_t^2 - \Delta$ be the wave operator on $\mathbb{R}^{n+1} = \mathbb{R}_t \times \mathbb{R}_x^n$, acting on half-densities via the trivialization $|dt\,dx|^{1/2}$, so $p(t,x;\tau,\xi) = \tau^2 - |\xi|^2$. A hypersurface $H = \{h(t,x) = 0\}$ with $dh \neq 0$ on H has $N^*H = \{(t,x;\lambda\,dh(t,x));\ h(t,x) = 0,\ \lambda \neq 0\}$, so H is characteristic if and only if h satisfies the *eikonal equation*

$$\Big(\frac{\partial h}{\partial t}\Big)^2 = |d_x h|^2 \quad \text{on } \{h = 0\}.$$

Linear solutions are $h = at + b\cdot x$ with $a^2 = |b|^2$; up to scaling these are the planes $t = \omega\cdot x$, $\omega \in \mathbb{S}^{n-1}$. For $v \in I^{m_0}(\mathbb{R},\{0\})$ (a conormal distribution at a point, in one variable),

$$u = v(t-\omega\cdot x)\,|dt\,dx|^{1/2} \in I^{m_0-\frac{n}{4}}(\mathbb{R}^{n+1}, H; \Omega^{\frac12}), \qquad Pu = (1-|\omega|^2)v''(t-\omega\cdot x)|dt\,dx|^{1/2} = 0.$$

(The order follows from the linear change of coordinates $s = t - \omega\cdot x$, $x' = x$, of determinant one, and the relation $M = m + \frac{n+1}{4} - \frac12$ between the order of u and that of its symbol, which is $m_0 - \frac14$.) For instance $\delta(t - \omega\cdot x)$ is a travelling wave. We shall find all characteristic hypersurfaces locally, by the Hamilton–Jacobi method.

10.2 Symplectic geometry

The eikonal equation $p(x, dh) = 0$ is a non-linear first-order equation for h. The classical way to solve it is to stop thinking of h and to think of the graph of dh in T^*X instead: this is an n-dimensional submanifold on which the canonical two-form vanishes, and it must lie in $\{p = 0\}$. Constructing such submanifolds is a question about ordinary differential equations (the flow of the Hamilton vector field), which is what the symplectic language below makes systematic.

10.2.1 The canonical forms

The cotangent bundle carries a canonical one-form α, defined by

$$\alpha_{(x,\zeta)}(v) = \zeta(\pi_* v), \qquad v \in T_{(x,\zeta)}(T^*X).$$

If $x_1,\dots,x_n$ are local coordinates on X and $\zeta = \sum_j \xi_j dx_j$, then (x,ξ) are coordinates on T^*X and $\alpha = \sum_j \xi_j\,dx_j$. The *symplectic form* is

$$\omega = d\alpha = \sum_{j=1}^n d\xi_j \wedge dx_j.$$

A section $\theta : X \to T^*X$ is the same thing as a one-form on X, and the defining property of α gives $\theta^*\alpha = \theta$, hence $\theta^*\omega = d\theta$.

A *symplectic manifold* is a manifold S with a closed two-form ω which is non-degenerate at each point; then $\dim S = 2n$ is even and $\omega^{\wedge n} \neq 0$. A *symplectic vector space* is a real vector space E with a non-degenerate antisymmetric bilinear form ω; non-degeneracy means that $\tilde\omega : E \to E'$, $\tilde\omega(v)(w) = \omega(v,w)$, is an isomorphism. Each tangent space T_zS of a symplectic manifold is a symplectic vector space. For a subspace $W \subset E$ set $W^\omega = \{v \in E;\ \omega(v,w) = 0\ \forall\ w \in W\}$. A subspace is *Lagrangian* if $\omega|_{W\times W} = 0$ and $2\dim W = \dim E$.

Lemma 10.2.1. *If W is a subspace of a symplectic vector space E then* $\dim W + \dim W^\omega = \dim E$*, and W is Lagrangian if and only if $W^\omega = W$.*

Proof. Let $W^0 \subset E'$ be the annihilator of W, of dimension $\dim E - \dim W$. By definition $v \in W^\omega$ if and only if $\tilde\omega(v) \in W^0$, so $\tilde\omega$ restricts to an isomorphism $W^\omega \to W^0$ and the dimension formula follows. If W is Lagrangian then $W \subset W^\omega$ and the two have the same dimension, so they are equal. Conversely if $W = W^\omega$ then ω vanishes on W, and $2\dim W = \dim E$ by the dimension formula. □

A submanifold Λ of a symplectic manifold is *Lagrangian* if $T_\lambda\Lambda$ is a Lagrangian subspace of $T_\lambda S$ for every $\lambda \in \Lambda$, i.e. $\dim\Lambda = \frac{1}{2}\dim S$ and the pull-back of ω to Λ vanishes.

Lemma 10.2.2. *The fibres $T^*_x X$ and, for a submanifold $Y \subset X$, the conormal bundle N^*Y are Lagrangian submanifolds of T^*X.*

Proof. On a fibre $\pi_* v = 0$ for tangent vectors, so α pulls back to zero, hence so does $\omega = d\alpha$. On N^*Y, if v is tangent at (y,ζ) then $\pi_* v \in T_y Y$ and ζ annihilates $T_y Y$, so again α pulls back to zero. Both have dimension n. □

Lemma 10.2.3. *Let $\Lambda \subset T^*X$ be a submanifold of dimension n such that $\pi|_\Lambda$ is a local diffeomorphism near $\bar\lambda \in \Lambda$. Then near $\bar\lambda$, $\Lambda = \{(x, dh(x))\}$ for some h defined near $\pi(\bar\lambda)$, unique up to an additive constant, if and only if Λ is Lagrangian near $\bar\lambda$.*

Proof. Near $\bar\lambda$, Λ is the image of a section θ of T^*X over a neighbourhood U of $\pi(\bar\lambda)$, and $\theta : U \to \Lambda$ is a diffeomorphism with $\theta^*\omega = d\theta$. So Λ is Lagrangian if and only if the one-form θ is closed, which on a ball is equivalent to $\theta = dh$. □

10.2.2 Hamilton vector fields

On a symplectic manifold the isomorphism $\tilde\omega$ associates to each function f its *Hamilton vector field* H_f, defined by

$$\omega(v, H_f) = df(v) \quad \forall\ v \in T_z S.$$

On T^*X, in canonical coordinates,

$$H_f = \sum_{j=1}^{n} \Big(\frac{\partial f}{\partial \xi_j}\frac{\partial}{\partial x_j} - \frac{\partial f}{\partial x_j}\frac{\partial}{\partial \xi_j} \Big). \tag{10.2.1}$$

By antisymmetry $H_f f = df(H_f) = \omega(H_f, H_f) = 0$: f is constant along the integral curves of H_f. The flow $\exp(sH_f)$ preserves ω, since by Cartan's formula $\mathcal{L}_{H_f}\omega = d\iota_{H_f}\omega + \iota_{H_f} d\omega = -d\,df = 0$. If f is real, H_f is a real vector field; if f is homogeneous of degree μ in the fibre variables then by (10.2.1) H_f is homogeneous of degree $\mu - 1$ (its ∂_x-components are homogeneous of degree $\mu - 1$ and its ∂_ξ-components of degree μ), and its flow commutes with the dilations of the fibres up to reparametrization. The condition $d_{\mathrm{fib}} f \neq 0$, i.e. $(\partial f/\partial\xi_1, \dots, \partial f/\partial\xi_n) \neq 0$, means precisely that $\pi_* H_f \neq 0$, i.e. H_f is transversal to the fibres of T^*X.

Lemma 10.2.4. *If $\Lambda \subset S$ is Lagrangian and $f \in \mathcal{C}^\infty(S)$ vanishes on Λ, then H_f is tangent to Λ.*

Proof. For $v \in T_\lambda\Lambda$, $\omega(v, H_f) = df(v) = 0$; so $H_f(\lambda) \in (T_\lambda\Lambda)^\omega = T_\lambda\Lambda$ by Lemma 10.2.1. □

In particular, if Y is characteristic for P on $\Gamma \subset N^*Y$, then H_p is tangent to Γ: a characteristic conormal bundle is a union of integral curves of H_p (bicharacteristics). We use this to construct characteristic hypersurfaces.

10.3 Hamilton–Jacobi theory

Let $p \in \mathcal{C}^\infty(W)$ be a real-valued function on an open set $W \subset T^*X$. We seek solutions of the *eikonal equation*

$$p(x, dh(x)) = 0 \tag{10.3.1}$$

in a neighbourhood of a point $\bar{x}$, with h prescribed on a hypersurface through $\bar{x}$. Note that (10.3.1) is required on an open set, not only on $\{h = 0\}$. By Lemma 10.2.3, solving it amounts to finding a Lagrangian submanifold $\Lambda \subset p^{-1}(0)$ which projects diffeomorphically to a neighbourhood of $\bar{x}$; by Lemma 10.2.4 such a Λ must be a union of integral curves of H_p.

Theorem 10.3.1 (Hamilton–Jacobi)**.** *Let $p \in \mathcal{C}^\infty(W)$ be real, $\bar{z} \in W$ with $p(\bar{z}) = 0$, $\bar{x} = \pi(\bar{z})$, and let $H = \{t = 0\}$ be a hypersurface through $\bar{x}$, $t \in \mathcal{C}^\infty(X)$ real with $dt \neq 0$, such that*

$$H_p(\pi^* t)(\bar{z}) \neq 0. \tag{10.3.2}$$

Let $f \in \mathcal{C}^\infty(H)$ satisfy $df(\bar{x}) = \bar{z}|_{T_{\bar{x}}H}$. Then there are a neighbourhood O of $\bar{x}$ and $h \in \mathcal{C}^\infty(O)$ such that

$$p(x, dh(x)) = 0 \quad \text{in } O, \qquad h|_{H\cap O} = f|_{H\cap O}, \qquad dh(\bar{x}) = \bar{z},$$

and any other such solution coincides with h near $\bar{x}$. Moreover the graph of dh is the union of the integral curves of H_p through the submanifold Λ_0 defined in (10.3.3) below.

Proof. Choose coordinates (t, x') near $\bar{x} = 0$, $x' \in \mathbb{R}^{n-1}$, with dual coordinates (τ, ξ'); then $H = \{t = 0\}$, $\bar{z} = (0, 0; \bar{\tau}, \bar{\xi}')$ with $\bar{\xi}' = d_{x'} f(0)$, and (10.3.2) says $\partial_\tau p(\bar{z}) \neq 0$.

Initial manifold. By the implicit function theorem there is a unique smooth function $\tau(x')$, defined near 0, with $\tau(0) = \bar{\tau}$ and $p(0, x'; \tau(x'), d_{x'} f(x')) = 0$. Set

$$\Lambda_0 = \{(0, x'; \tau(x'), d_{x'} f(x'))\}, \tag{10.3.3}$$

an $(n-1)$-dimensional submanifold of $p^{-1}(0)$. It consists exactly of the points of $p^{-1}(0)$ near $\bar{z}$ lying over H whose restriction to TH is df. The pull-back of ω to Λ_0 is $d\tau \wedge dt + \sum d\xi'_j \wedge dx'_j$ restricted, which is $0 + d(d_{x'} f) = 0$ since $dt = 0$ on Λ_0.

Flow-out. Let $\Phi_s = \exp(sH_p)$ and $\Psi(s, x') = \Phi_s(0, x'; \tau(x'), d_{x'} f(x'))$ for $|s| < \epsilon$ and x' near 0. At $(0, 0)$, $\partial_s \Psi = H_p(\bar{z})$, whose projection to X has ∂_t-component $\partial_\tau p(\bar{z}) \neq 0$, while $\partial_{x'_j} \Psi$ projects to $\partial_{x'_j}$. Hence $\pi \circ \Psi$ has invertible differential at $(0, 0)$, so, shrinking the domain, Ψ is an embedding onto an n-dimensional submanifold Λ and $\pi|_\Lambda$ is a diffeomorphism onto a neighbourhood O of $\bar{x}$. Since $H_p p = 0$, $\Lambda \subset p^{-1}(0)$.

Λ is Lagrangian. At a point $z \in \Lambda_0$, $T_z\Lambda = T_z\Lambda_0 \oplus \mathbb{R} H_p(z)$; ω vanishes on $T_z\Lambda_0 \times T_z\Lambda_0$, $\omega(H_p, H_p) = 0$, and for $v \in T_z\Lambda_0$, $\omega(v, H_p) = dp(v) = 0$ because $\Lambda_0 \subset p^{-1}(0)$. So $T_z\Lambda$ is Lagrangian. Since $\Psi(s + s', x') = \Phi_s \Psi(s', x')$, the differential of Φ_s maps $T_z\Lambda$ onto $T_{\Phi_s(z)}\Lambda$, and Φ_s preserves ω; so every tangent space of Λ is Lagrangian.

Conclusion. By Lemma 10.2.3, Λ is the graph of dh for some $h \in \mathcal{C}^\infty(O)$, and $p(x, dh) = 0$. The points of Λ over H are those of Λ_0 (as $\pi|_\Lambda$ is injective and $\pi(\Lambda_0) = H \cap O$), so $d(h|_H) = d_{x'} f$; adjusting the constant, $h|_H = f$, and $dh(\bar{x}) = \bar{z}$ since $\bar{z} \in \Lambda_0$.

Uniqueness. If h' is another solution, its graph Λ' is Lagrangian and contained in $p^{-1}(0)$, so H_p is tangent to Λ' by Lemma 10.2.4. The points $(0, x'; \partial_t h'(0, x'), d_{x'} f(x'))$ of Λ' over H satisfy $p = 0$ and are close to $\bar{z}$, so by the uniqueness in the implicit function theorem $\partial_t h'(0, x') = \tau(x')$, i.e. $\Lambda_0 \subset \Lambda'$. The integral curves of H_p starting on Λ_0 therefore stay in Λ', so $\Lambda \subset \Lambda'$ near $\bar{z}$. Both are graphs over a neighbourhood of $\bar{x}$, hence $dh = dh'$ there, and $h = h'$ since they agree on H. □

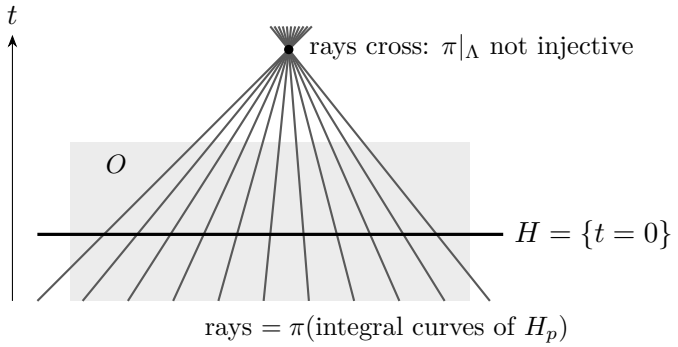


Figure 10.1: The Hamilton–Jacobi construction, projected to X. The data f on H and the equation $p = 0$ determine the initial manifold Λ_0 over H; the integral curves of H_p through Λ_0 project to "rays" crossing H transversally, by (10.3.2). In a neighbourhood O of $\bar{x}$ (shaded) the rays do not meet, the flow-out Λ is the graph of dh, and h solves the eikonal equation. Further away rays may cross; there Λ is no longer a graph, which is the starting point of Section 10.5.

Question. What is the role of the transversality condition (10.3.2)?

Answer. It says that the rays leave H instead of running along it (Figure 10.1). This is what makes the projection of the flow-out Λ a local diffeomorphism near $\bar{x}$, so that Λ is a graph. It is the analogue of the requirement, for the Cauchy problem of a linear first-order equation, that the initial surface be non-characteristic. If H_p were tangent to the fibres of π over H, the flow-out would not be determined by the data on H, and the eikonal equation could have no solution or many. □

Note that (10.3.2) implies $d_{\mathrm{fib}}p(\bar{z}) \neq 0$; conversely, if $d_{\mathrm{fib}}p(\bar{z}) \neq 0$ then $\pi_* H_p(\bar{z}) \neq 0$ and hypersurfaces H satisfying (10.3.2) exist.

For homogeneous p we obtain characteristic hypersurfaces with prescribed intersection with H. If $G \subset H$ is a hypersurface in H and $\bar{x} \in G$, write $N^*_{\bar{x}}G \subset T^*_{\bar{x}}H$ for its conormal space in H; and for $\bar{z} \in T^*_{\bar{x}}X$ write $\bar{z}_H = \bar{z}|_{T_{\bar{x}}H} \in T^*_{\bar{x}}H$.

Corollary 10.3.2. *Let $p \in \mathcal{C}^\infty(T^*X \setminus 0)$ be real and homogeneous of degree m', $\bar{z} \in T^*X \setminus 0$ with $p(\bar{z}) = 0$, $H = \{t = 0\}$ a hypersurface through $\bar{x} = \pi(\bar{z})$ satisfying (10.3.2), and $G \subset H$ a hypersurface of H through $\bar{x}$ with $\bar{z}_H \in N^*_{\bar{x}}G \setminus 0$. Then there is a hypersurface $\tilde{G}$ of X, defined near $\bar{x}$, with $\tilde{G} \cap H = G$ near $\bar{x}$, $\bar{z} \in N^*\tilde{G}$, and*

$$\Gamma = \{(x, \lambda\, d\tilde{g}(x));\ x \in \tilde{G},\ \lambda > 0\} \subset \Sigma(p),$$

where $\tilde{g}$ is a defining function of $\tilde{G}$ with $d\tilde{g}(\bar{x}) = \bar{z}$; that is, $\tilde{G}$ is characteristic on the half Γ of $N^*\tilde{G} \setminus 0$ containing $\bar{z}$. Moreover $\tilde{G}$ is unique near $\bar{x}$.

Proof. Write $G = \{g_0 = 0\}$ near $\bar{x}$ with $g_0 \in \mathcal{C}^\infty(H)$, $dg_0 \neq 0$; since $\bar{z}_H$ is a nonzero multiple of $dg_0(\bar{x})$ we may rescale g_0 so that $dg_0(\bar{x}) = \bar{z}_H$. Theorem 10.3.1 with $f = g_0$ gives $\tilde{g}$ with $p(x, d\tilde{g}) = 0$, $\tilde{g}|_H = g_0$ and $d\tilde{g}(\bar{x}) = \bar{z} \neq 0$. Then $\tilde{G} = \{\tilde{g} = 0\}$ is a hypersurface near $\bar{x}$, $\tilde{G} \cap H = \{g_0 = 0\} = G$, and $p(x, \lambda d\tilde{g}(x)) = \lambda^{m'} p(x, d\tilde{g}(x)) = 0$ for $\lambda > 0$.

For uniqueness, let $\tilde{G}'$ be another such hypersurface, with half-conormal bundle $\Gamma' \subset \Sigma(p)$ containing $\bar{z}$. Over G, the points of Γ and of Γ' near $\bar{z}$ are covectors in $\Sigma(p)$ whose restrictions to TH lie in N^*G; by (10.3.2) and the implicit function theorem (as for Λ_0 above, now with the extra dilation parameter) these sets coincide near $\bar{z}$; call this $(n-1)$-dimensional manifold Γ_0. Since Γ, Γ' are Lagrangian (Lemma 10.2.2) and contained in $p^{-1}(0)$, H_p is

tangent to both, so both contain the flow-out of Γ_0 by H_p. This flow-out is n-dimensional near $\bar z$: $H_p(\bar z)$ is not tangent to Γ_0 since $\pi_* H_p(\bar z) \notin T_{\bar x}H \supset \pi_* T\Gamma_0$. So it is an open subset of both Γ and Γ', and $\tilde G = \pi(\Gamma) = \pi(\Gamma') = \tilde G'$ near $\bar x$. □

10.4 The transport equation

10.4.1 The symbol of Pu for characteristic Y

We now compute the next term in the symbol of Pu when Y is characteristic. We assume that $P \in \Psi^{m'}(X;\Omega^{\frac12})$ is properly supported and has a polyhomogeneous symbol: in local coordinates, acting on coefficients with respect to $|dz|^{1/2}$, $P = \mathrm{Op}(b)$ modulo smoothing operators, with

$$b \sim b_{m'} + b_{m'-1} + \cdots, \qquad b_{m'-j} \text{ homogeneous of degree } m'-j \text{ in } |\zeta| \geq 1.$$

Differential operators are the main examples. The principal symbol is $p = b_{m'}$, and the *subprincipal symbol* is

$$p_s = b_{m'-1} - \frac{1}{2i}\sum_{j=1}^{n} \frac{\partial^2 p}{\partial z_j \partial \zeta_j}. \tag{10.4.1}$$

For operators on half-densities p_s is an invariantly defined function on $T^*X \setminus 0$, homogeneous of degree $m'-1$ [8, Theorem 18.1.33]; we shall only use its restriction to N^*Y, whose invariance follows from the next theorem. Recall that the Lie derivative acts on half-densities on a manifold with coordinates w by $\mathcal{L}_V(\beta|dw|^{1/2}) = (V\beta + \frac12(\operatorname{div} V)\beta)|dw|^{1/2}$, $\operatorname{div} V = \sum \partial_{w_j} V_j$.

Theorem 10.4.1. *Let P be as above, $Y \subset X$ a closed embedded submanifold and $\Gamma \subset N^*Y\setminus 0$ an open cone on which $p = 0$, so that H_p is tangent to Γ. If $u \in I^m(X,Y;\Omega^{\frac12})$ and $\mathrm{WF}(u) \subset \Gamma$, then $Pu \in I^{m+m'-1}(X,Y;\Omega^{\frac12})$ and*

$$\sigma_{m+m'-1}(Pu) = \big(-i\mathcal{L}_{H_p} + p_s\big)\sigma_m(u). \tag{10.4.2}$$

Here $\sigma_m(u) \in S^{m+\frac{n}{4}}(N^*Y;\Omega^{\frac12})$ is, modulo $S^{-\infty}$, supported in a closed cone contained in Γ, and $\mathcal{L}_{H_p}$ (with H_p restricted to Γ) and multiplication by p_s map $S^r(\Gamma;\Omega^{\frac12})$ to $S^{r+m'-1}(\Gamma;\Omega^{\frac12})$, since H_p and p_s are homogeneous of degree $m'-1$; so the right side of (10.4.2) is well defined in $S^{m+\frac{n}{4}+m'-1-[1]}(N^*Y;\Omega^{\frac12})$.

Proof. That $Pu \in I^{m+m'-1}$ is Corollary 10.1.3. The formula is local; in adapted coordinates (x,y) with $u = (\mathcal{I}[a]+f)|dz|^{1/2}$, the full symbol a' of Pu is given by the expansion of Chapter 4,

$$a' \sim \sum_{\alpha,\gamma} \frac{1}{\alpha!\gamma!}(-D_\xi)^\alpha \partial_x^\alpha \Big(\partial_\eta^\gamma b(x,y,\xi,0) D_y^\gamma a(y,\xi)\Big)\Big|_{x=0}.$$

Each α or γ lowers the order by one, so modulo $S^{M+m'-2}$ only the terms with $|\alpha|+|\gamma| \leq 1$ and the two leading terms of b contribute. Since $p(0,y,\xi,0) = 0$ where a is not of order $-\infty$,

$$\begin{aligned} a' &\equiv b_{m'-1}a + \sum_l \partial_{\eta_l} p\, D_{y_l} a - \sum_j D_{\xi_j}(\partial_{x_j} p\, a) \\ &= -i\Big(\sum_l \partial_{\eta_l} p\, \partial_{y_l} a - \sum_j \partial_{x_j} p\, \partial_{\xi_j} a\Big) + \Big(b_{m'-1} + i\sum_j \partial_{x_j}\partial_{\xi_j} p\Big)a, \end{aligned}$$

all functions of (y,ξ) evaluated at $x=0$, $\eta=0$. On $N^*Y=\{x=0,\ \eta=0\}$ we have $\partial_\xi p=0$ and $\partial_y p=0$ (as p vanishes there), so by (10.2.1)

$$H_p|_{N^*Y}=\sum_l \partial_{\eta_l}p\,\partial_{y_l}-\sum_j \partial_{x_j}p\,\partial_{\xi_j},$$

which is the first-order operator above. With respect to the half-density $|dy\,d\xi|^{1/2}$ its divergence is $\sum_l \partial_{y_l}\partial_{\eta_l}p-\sum_j \partial_{\xi_j}\partial_{x_j}p$, so

$$-iH_pa=-i\mathcal{L}_{H_p}a+\frac{i}{2}\Big(\sum_l \partial_{y_l}\partial_{\eta_l}p-\sum_j \partial_{x_j}\partial_{\xi_j}p\Big)a,$$

where $\mathcal{L}_{H_p}a$ denotes the coefficient of $\mathcal{L}_{H_p}(a|dy\,d\xi|^{1/2})$. Substituting,

$$a'\equiv -i\mathcal{L}_{H_p}a+\Big(b_{m'-1}+\frac{i}{2}\sum_{j=1}^{n}\frac{\partial^2 p}{\partial z_j\partial\zeta_j}\Big)a=-i\mathcal{L}_{H_p}a+p_sa$$

modulo $S^{M+m'-2}$, since $\frac{i}{2}=-\frac{1}{2i}$. This is (10.4.2). □

Thus, to obtain $Pu\in I^{m+m'-2}$ we must solve the *transport equation* $(-i\mathcal{L}_{H_p}+p_s)\sigma_m(u)=0$, a first-order ordinary differential equation along the integral curves of H_p in Γ; the lower order terms are then removed by solving inhomogeneous transport equations. To set up the initial value problem we need restriction to a transversal hypersurface.

Discussion. This is the mathematical form of geometrical optics. The eikonal equation ($p=0$ on N^*Y) says where the wave front goes; the transport equation says how the amplitude changes along each ray. The Lie derivative of a half-density measures how a bundle of nearby rays spreads or focuses, so the amplitude decreases where rays spread and increases where they focus, and the subprincipal symbol adds a damping or phase term. Each successive order of the symbol is found by integrating the same ODE with a source term, so the whole construction reduces to linear ODEs along rays.

10.4.2 Restriction to a transversal hypersurface

Let $t\in\mathcal{C}^\infty(X)$ be real with $dt\neq 0$ on $H=\{t=0\}$. Restriction of half-densities to H is defined using $|dt|^{1/2}$: in coordinates (t,w),

$$(\phi(t,w)|dt\,dw|^{1/2})\big|_H=\phi(0,w)|dw|^{1/2};\tag{10.4.3}$$

this is independent of the coordinates w (Exercise 10.2). Let $\tilde G\subset X$ be a hypersurface transversal to H, i.e. $dt|_{T\tilde G}\neq 0$ at each point of $G=\tilde G\cap H$; then G is a hypersurface in H (Exercise 10.1). Near any point of G there are coordinates (x_1,t,y'), $y'\in\mathbb{R}^{n-2}$, with $\tilde G=\{x_1=0\}$ and $H=\{t=0\}$. The restriction map $j:N^*\tilde G|_G\to N^*G$, $\zeta\mapsto\zeta|_{TH}$, is an isomorphism, in coordinates $(t=0,y';\xi_1)\mapsto(y';\xi_1)$. The set $N^*\tilde G|_G=N^*\tilde G\cap\pi^{-1}(H)$ is the hypersurface $\{\pi^*t=0\}$ of $N^*\tilde G$, so half-densities on $N^*\tilde G$ restrict to it as in (10.4.3), using $|d\pi^*t|^{1/2}$.

Proposition 10.4.2. *With H, $\tilde G$, G as above, restriction to H defines a map*

$$I^m(X,\tilde G;\Omega^{\frac12})\ni u\longmapsto u|_H\in I^{m+\frac14}(H,G;\Omega^{\frac12}),\qquad \sigma_{m+\frac14}(u|_H)=(j^{-1})^*(\sigma_m(u)|_{\{\pi^*t=0\}}),$$

mapping I^{m-l} to $I^{m-l+\frac14}$, and $\mathrm{WF}(u|_H)\subset j(\mathrm{WF}(u)\cap\pi^{-1}(H))$.

Proof. Since $\mathrm{WF}(u) \subset N^*\tilde G$ and $N^*\tilde G \cap N^*H = 0$ by transversality, the restriction (pull-back by the inclusion) of u to H is well defined [7, Theorem 8.2.4]; for half-densities we combine it with (10.4.3). In coordinates as above, $u = (\mathcal{I}[a] + f)|dx_1\, dt\, dy'|^{1/2}$ with $a = a(t, y', \xi_1) \in S^M$, $M = m + \frac{n}{4} - \frac{1}{2}$, and $\mathcal{I}[a]$ is a smooth function of t with values in distributions in (x_1, y'); so

$$u|_H = (\mathcal{I}[a(0,\cdot)] + f(\cdot, 0, \cdot))|dx_1\, dy'|^{1/2}.$$

Now H has dimension $n-1$ and G codimension one in it, so this is an element of $I^{m'}(H, G; \Omega^{\frac12})$ with $M = m' + \frac{n-1}{4} - \frac12$, i.e. $m' = m + \frac14$. Its symbol is $a(0, y', \xi_1)|dy'\, d\xi_1|^{1/2}$, which is the stated formula since $\sigma_m(u) = a|dt\, dy'\, d\xi_1|^{1/2}$. The wavefront set statement follows from the characterization of WF by the full symbol (Chapter 4). □

10.4.3 Travelling waves

We now construct conormal solutions of $Pu \in \mathcal{C}^\infty$ with prescribed restriction to a transversal hypersurface. We first solve the transport equations.

Lemma 10.4.3. *Under the hypotheses of Corollary 10.3.2, use coordinates (x_1, t, y') near $\bar x$ with $\tilde G = \{x_1 = 0\}$, $H = \{t = 0\}$, so that (t, y', ξ_1) are coordinates on $N^*\tilde G$ and $\Gamma = \{\xi_1 > 0\}$. Let q be a smooth function on Γ homogeneous of degree $m' - 1$. Then there are $\epsilon > 0$ and a neighbourhood U of $\bar y' = 0$ such that for every $c \in S^{r+m'-1}$ and $\beta_0 \in S^r$ (classes of half-densities on Γ, respectively on $\Gamma \cap \{t = 0\}$, of the stated orders, supported in $\xi_1 \geq 1$) the problem*

$$(-i\mathcal{L}_{H_p} + q)\beta = c \quad \text{in } \{|t| < \epsilon,\ y' \in U\}, \qquad \beta|_{t=0} = \beta_0$$

has a unique solution $\beta \in S^r$ there (restriction to $t = 0$ taken as in (10.4.3)). If β_0 and c are of order $-\infty$ outside closed cones K_0, K, then β is of order $-\infty$ outside the union of K and the H_p-flow-out of K_0.

Proof. Since H_p is tangent to Γ and homogeneous of degree $m' - 1$, in the coordinates (t, y', ξ_1)

$$H_p = \xi_1^{m'-1}\Big(e_0(t, y')\partial_t + \sum_l e_l(t, y')\partial_{y'_l} + e_n(t, y')\xi_1\partial_{\xi_1}\Big),$$

and $e_0(0, 0) \neq 0$ by (10.3.2). Writing $\beta = \tilde\beta|dt\, dy'\, d\xi_1|^{1/2}$ and dividing by $-i\xi_1^{m'-1}e_0$, the equation becomes

$$\Big(\partial_t + \sum_l \frac{e_l}{e_0}\partial_{y'_l} + \frac{e_n}{e_0}\xi_1\partial_{\xi_1}\Big)\tilde\beta + \kappa(t, y')\tilde\beta = \tilde c,$$

with κ smooth (the divergence of H_p and q both being $\xi_1^{m'-1}$ times functions of (t, y')) and $\tilde c = i\xi_1^{1-m'}e_0^{-1}c'$, c' the coefficient of c, of the same order as $\tilde\beta$. The vector field in parentheses has integral curves $s \mapsto (s, Y(s, y'_0), \xi_1 E(s, y'_0))$ with $E > 0$ smooth, the ξ_1-component being linear in ξ_1 because its coefficient is. Hence the solution is given explicitly by

$$\tilde\beta(s, Y(s, y'_0), \xi_1 E(s, y'_0)) = e^{-k(s,y'_0)}\Big(\tilde\beta_0(y'_0, \xi_1) + \int_0^s e^{k(\sigma,y'_0)}\tilde c(\sigma, Y(\sigma, y'_0), \xi_1 E(\sigma, y'_0))\, d\sigma\Big),$$

$k(s, y'_0) = \int_0^s \kappa(\sigma, Y(\sigma, y'_0))\, d\sigma$, for $|s| < \epsilon$ and y'_0 near 0. The change of variables $(s, y'_0, \xi_1) \mapsto (s, Y, \xi_1 E)$ is a diffeomorphism near $\{s = 0,\ y'_0 = 0\}$ which is linear and positive in the fibre variable, so it preserves symbol classes; hence β has order r, and the support statement is clear from the formula. □

Theorem 10.4.4 (Travelling waves)**.** *Let $P \in \Psi^{m'}(X;\Omega^{\frac12})$ be properly supported with polyhomogeneous symbol and real principal symbol p. Let $\bar z \in T^*X \setminus 0$ with $p(\bar z) = 0$, $\bar x = \pi(\bar z)$, let $H = \{t = 0\}$ be a hypersurface through $\bar x$ with $H_p(\pi^*t)(\bar z) \neq 0$, let $G \subset H$ be a hypersurface of H with $\bar z_H \in N^*_{\bar x}G \setminus 0$, and let $\tilde G$ be the characteristic hypersurface of Corollary 10.3.2. Then there are a neighbourhood O of $\bar x$ and an open conic neighbourhood γ of $\bar z_H$ in $T^*H \setminus 0$ such that for every $u_0 \in I^{m+\frac14}(H,G;\Omega^{\frac12})$ with $\mathrm{WF}(u_0) \subset \gamma$ there exists $u \in I^m(X,\tilde G;\Omega^{\frac12})$ with*

$$u|_{H\cap O} = u_0|_{H\cap O}, \qquad Pu \in \mathcal{C}^\infty(O;\Omega^{\frac12}),$$

and $\mathrm{WF}(u) \cap T^*O \subset \Gamma$. If P and u_0 are classical, u can be taken classical.

Proof. Use the coordinates of Lemma 10.4.3, and choose ϵ, U as there and γ so small that, in these coordinates, $j^{-1}(\gamma)$ lies over $\{y' \in U\}$ and within a cone whose H_p-flow-out for $|t| < \epsilon$ stays in Γ. Let $O = \{|x_1| < \epsilon', |t| < \epsilon,\ y' \in U'\}$ with $U' \Subset U$, and fix $\phi \in \mathcal{C}^\infty_{\mathrm c}(\{|t| < \epsilon, y' \in U\})$ equal to one near $\bar O$. We construct $u_j \in I^{m-j}(X,\tilde G;\Omega^{\frac12})$, with wavefront sets in Γ and supports in $\operatorname{supp}\phi$, such that for $v_N = u_0' + \cdots + u_N$ (we write u_0' for the first term to avoid confusion with the data u_0)

$$Pv_N \in I^{m+m'-N-2} \text{ in } O, \qquad u_0 - v_N|_H \in I^{m+\frac14-N-1}(H,G;\Omega^{\frac12}) \text{ in } H\cap O. \tag{10.4.4}$$

For $N = -1$ (with $v_{-1} = 0$) the second condition holds and the first is vacuous with the convention $Pv_{-1} = 0$. Given v_{N-1}, let $g = -Pv_{N-1} \in I^{m+m'-N-1}$ in O and $e = u_0 - v_{N-1}|_H \in I^{m+\frac14-N}(H,G;\Omega^{\frac12})$, both with wavefront sets in the appropriate cones. By Lemma 10.4.3 (with $q = p_s|_\Gamma$) solve

$$(-i\mathcal{L}_{H_p} + p_s)\beta_N = \sigma_{m+m'-N-1}(g), \qquad \beta_N|_{t=0} = (j)^*\sigma_{m+\frac14-N}(e),$$

with $\beta_N \in S^{m+\frac n4-N}$, of order $-\infty$ outside the flow-out cone, and let $u_N \in I^{m-N}$ be ϕ times a distribution with principal symbol β_N (Chapter 4). By Theorem 10.4.1, $\sigma_{m+m'-N-1}(Pu_N) = \sigma_{m+m'-N-1}(g)$ in O (where $\phi = 1$ and, P being pseudolocal, $P(\phi w) - \phi Pw$ is smooth), so $Pv_N = -g + Pu_N \in I^{m+m'-N-2}$ in O; and by Proposition 10.4.2, $\sigma_{m+\frac14-N}(u_N|_H) = \sigma_{m+\frac14-N}(e)$, so $e - u_N|_H \in I^{m+\frac14-N-1}$. This proves (10.4.4).

Let $\tilde u \sim \sum_N u_N$ be an asymptotic sum in $I^m(X,\tilde G;\Omega^{\frac12})$, with support in $\operatorname{supp}\phi$. Then $P\tilde u \in \bigcap_N I^{m+m'-N} = \mathcal{C}^\infty$ in O, and $u_0 - \tilde u|_H \in \bigcap_N I^{m+\frac14-N}(H,G) = \mathcal{C}^\infty$ in $H\cap O$, by Proposition 10.4.2. Finally choose $w \in \mathcal{C}^\infty(X;\Omega^{\frac12})$ with $w|_H = u_0 - \tilde u|_H$ near $H \cap \bar O$ (in coordinates, $w = \chi(t)\times$ the smooth error, extended constantly in t), and set $u = \tilde u + w$. Then $u|_{H\cap O} = u_0$, $Pu = P\tilde u + Pw \in \mathcal{C}^\infty(O)$, and $\mathrm{WF}(u) \cap T^*O \subset \Gamma$. In the classical case all symbols above are classical. □

The transport equation (10.4.2) for the leading symbol, and its inhomogeneous versions for the lower order terms, are thus solved by integration along the bicharacteristics (Duhamel's principle). In Chapter 11 this construction, applied to the wave operator with $H = \{t = 0\}$, gives a parametrix for the Cauchy problem.

10.5 Conic Lagrangian submanifolds and phase functions

The construction of Theorem 10.4.4 works only as long as the characteristic hypersurface $\tilde G$ is smooth. In general the integral curves of H_p which make up $N^*\tilde G$ continue to exist, and their union is still a smooth Lagrangian submanifold of T^*X, but it need not be the

conormal bundle of anything: its projection to X may become singular (a *caustic*, see Section 10.7). To describe the corresponding solutions we need a class of distributions associated with an arbitrary conic Lagrangian submanifold. In this section we develop the necessary symplectic geometry: every conic Lagrangian is locally *parametrized* by a non-degenerate phase function, and any two parametrizations are related in a precise way. Figure 10.2 shows how the results of this section and the next fit together.

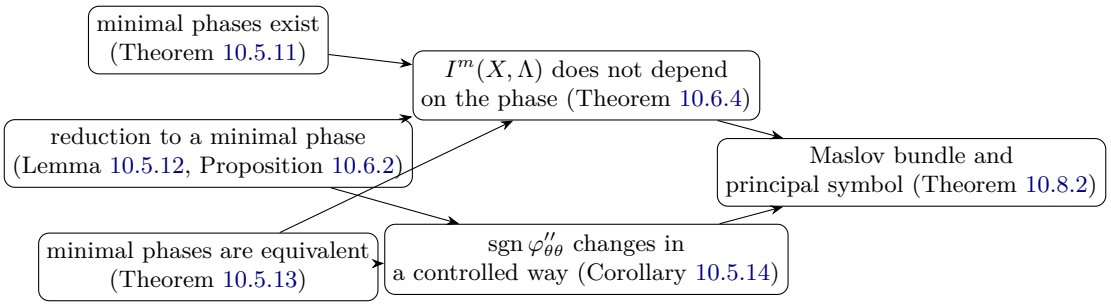


Figure 10.2: Logic of the theory of Lagrangian distributions. Geometric results on phase functions (left) feed into the analytic definitions (right).

10.5.1 Conic Lagrangian submanifolds

For $t > 0$ let $M_t : T^*X \to T^*X$, $M_t(x,\xi) = (x, t\xi)$, be the fibre dilation, and let $\varrho = \sum_j \xi_j \partial_{\xi_j}$ be the radial vector field, the generator of $s \mapsto M_{e^s}$. A subset $\Lambda \subset T^*X \setminus 0$ is *conic* if $M_t(\Lambda) = \Lambda$ for all $t > 0$.

Lemma 10.5.1. *A conic submanifold $\Lambda \subset T^*X \setminus 0$ of dimension n is Lagrangian if and only if the canonical one-form α pulls back to zero on Λ.*

Proof. If α pulls back to zero, so does $\omega = d\alpha$. Conversely, in canonical coordinates $\omega(\varrho, v) = \sum_j \left(d\xi_j(\varrho)dx_j(v) - dx_j(\varrho)d\xi_j(v)\right) = \sum_j \xi_j dx_j(v) = \alpha(v)$, i.e. $\iota_\varrho \omega = \alpha$. Since Λ is conic, ϱ is tangent to Λ, so if Λ is Lagrangian then $\alpha(v) = \omega(\varrho, v) = 0$ for $v \in T\Lambda$. □

The basic examples are the conormal bundles $N^*Y \setminus 0$ (Lemma 10.2.2). For $\lambda \in \Lambda$ let

$$K(\lambda) = \dim\left(T_\lambda\Lambda \cap T_\lambda(T^*_{\pi(\lambda)}X)\right) = n - \operatorname{rank}\left(d\pi|_{T_\lambda\Lambda}\right),$$

the dimension of the space of "vertical" tangent vectors. Since $0 \neq \varrho(\lambda)$ is vertical and tangent to Λ, $K(\lambda) \geq 1$ for a conic Lagrangian: unlike the Lagrangians of Lemma 10.2.3, a conic Lagrangian is never locally a graph over X. For $\Lambda = N^*Y \setminus 0$, $K \equiv \operatorname{codim} Y$. Points where $K(\lambda)$ jumps up project to the *caustic* of Λ.

Discussion. Why phase functions? Over a region where $\pi|_\Lambda$ is a diffeomorphism, Λ is the graph of dh and the natural distributions are $e^{ih}a$; over a fold this description fails (Figure 10.3). The remedy, as in the Fourier representation $\delta(x) = \int e^{ix\cdot\xi} d\xi$ of a distribution whose Lagrangian $T^*_0\mathbb{R}^n$ is not a graph at all, is to add auxiliary variables θ: Λ becomes the image of the critical set of $\varphi(x,\theta)$ in θ, and that critical set can be smooth even where the projection to X is not. The homogeneity in θ is the conic version of the same idea.

Conic Lagrangians in the characteristic variety arise as flow-outs, just as in the proof of Theorem 10.3.1; this does not require the projection to X to stay non-singular.

Lemma 10.5.2. *Let $p \in \mathcal{C}^\infty(T^*X \setminus 0)$ be real and homogeneous of degree m', and let $\Lambda_0 \subset p^{-1}(0)$ be a conic submanifold of dimension $n-1$ on which α pulls back to zero and*

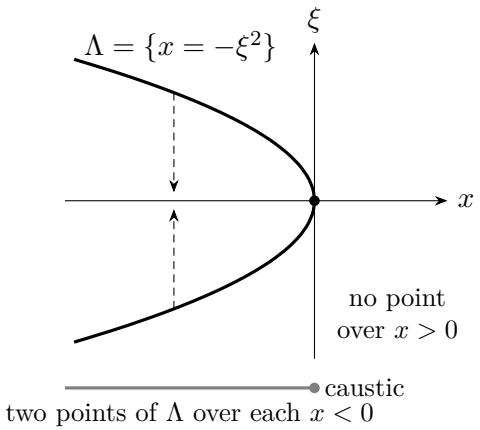


Figure 10.3: A Lagrangian curve with a fold (a non-homogeneous model, the Airy case). Away from $\xi = 0$, Λ is the union of the graphs of $dh_\pm$, $h_\pm(x) = \mp\frac{2}{3}(-x)^{3/2}$, with $dh_\pm = \pm(-x)^{1/2}\,dx$; at the fold point the projection to x is singular, so K jumps from 0 to 1, and its image is the caustic. The phase $\varphi(x,\theta) = x\theta + \frac{1}{3}\theta^3$ parametrizes all of Λ: $\partial_\theta\varphi = 0$ gives $x = -\theta^2$, $\xi = \theta$, and $\varphi''_{\theta\theta} = 2\theta$ changes sign at the fold.

to which H_p is nowhere tangent. Let Λ be the union of the maximal integral curves of H_p through Λ_0, and $\Psi(s,\lambda_0) = \exp(sH_p)(\lambda_0)$ on its (open) domain of definition in $\mathbb{R}\times\Lambda_0$. Then Ψ is an immersion, its image Λ is conic and contained in $p^{-1}(0)$, and Λ is a Lagrangian immersed submanifold: wherever Ψ is injective (in particular locally) its image is a conic Lagrangian submanifold.

Proof. At $s = 0$ the differential of Ψ has image $T\Lambda_0 \oplus \mathbb{R}H_p$, of dimension n; since $\Psi(s + s', \lambda_0) = \exp(sH_p)\Psi(s', \lambda_0)$ and $\exp(sH_p)$ is a diffeomorphism mapping H_p to H_p, Ψ is an immersion everywhere. That the tangent spaces are Lagrangian is proved exactly as in the proof of Theorem 10.3.1, using $dp = 0$ on $T\Lambda_0$ and the invariance of ω under the flow. Since $H_p p = 0$, $\Lambda \subset p^{-1}(0)$. Finally, homogeneity of p gives $(M_t)_* H_p = t^{1-m'} H_p \circ M_t$, so if γ is an integral curve of H_p then so is $s \mapsto M_t\gamma(t^{m'-1}s)$; as Λ_0 is conic, so is Λ. □

10.5.2 Non-degenerate phase functions

Let $X \subset \mathbb{R}^n$ be open (we work in local coordinates; see Lemma 10.6.1 for the change of coordinates) and let $\Gamma \subset X \times (\mathbb{R}^N \setminus 0)$ be an open cone, i.e. invariant under $(x,\theta) \mapsto (x, t\theta)$, $t > 0$.

Definition 10.5.3. A *phase function* on Γ is a real $\varphi \in \mathcal{C}^\infty(\Gamma)$, homogeneous of degree one in θ, with $d_{x,\theta}\varphi \neq 0$ on Γ. Its *critical set* is $C_\varphi = \{(x,\theta) \in \Gamma;\ d_\theta\varphi(x,\theta) = 0\}$, and φ is *non-degenerate* if the differentials $d(\partial\varphi/\partial\theta_1), \dots, d(\partial\varphi/\partial\theta_N)$ are linearly independent at each point of C_φ.

Proposition 10.5.4. *If φ is a non-degenerate phase function on Γ then C_φ is a conic submanifold of Γ of dimension n, and*

$$j_\varphi : C_\varphi \ni (x,\theta) \longmapsto (x, d_x\varphi(x,\theta)) \in T^*X \setminus 0$$

is a homogeneous immersion whose image is (locally) a conic Lagrangian submanifold Λ_φ.

Proof. C_φ is the common zero set of the N functions $\partial_{\theta_j}\varphi$, homogeneous of degree 0, with independent differentials; so it is a conic submanifold of dimension $n + N - N = n$. On C_φ,

$d_\theta\varphi = 0$ and $d\varphi \neq 0$, so $d_x\varphi \neq 0$ and j_φ maps into $T^*X \setminus 0$; it is homogeneous of degree one in the fibres since $d_x\varphi$ is. A tangent vector $(\delta x, \delta\theta)$ to C_φ satisfies $\varphi''_{\theta x}\delta x + \varphi''_{\theta\theta}\delta\theta = 0$; if moreover $dj_\varphi(\delta x, \delta\theta) = 0$ then $\delta x = 0$ and $\varphi''_{x\theta}\delta\theta = 0$, so $\varphi''_{\theta\theta}\delta\theta = 0$ as well. The $N \times (n+N)$ matrix $(\varphi''_{\theta x}\ \varphi''_{\theta\theta})$ has rank N by non-degeneracy, so its transpose is injective and $\delta\theta = 0$: j_φ is an immersion. Finally

$$j_\varphi^*\alpha = \sum_i \frac{\partial\varphi}{\partial x_i}dx_i\Big|_{C_\varphi} = \Big(d\varphi - \sum_j \frac{\partial\varphi}{\partial\theta_j}d\theta_j\Big)\Big|_{C_\varphi} = d(\varphi|_{C_\varphi}) = 0,$$

since $\varphi = \sum_j \theta_j\partial_{\theta_j}\varphi$ vanishes on C_φ by Euler's relation. By Lemma 10.5.1 the image is Lagrangian. □

Definition 10.5.5. Let $\Lambda \subset T^*X \setminus 0$ be a conic Lagrangian submanifold and $\bar\lambda \in \Lambda$. A non-degenerate phase function φ on Γ *parametrizes* Λ *near* $\bar\lambda$ if j_φ is a diffeomorphism of C_φ onto an open conic subset Λ_φ of Λ containing $\bar\lambda$.

If $j_\varphi(\bar x, \bar\theta) = \bar\lambda$ and $j_\varphi(C_\varphi) \subset \Lambda$ near $(\bar x, \bar\theta)$, then replacing Γ by a small conic neighbourhood of $(\bar x, \bar\theta)$ gives a parametrization near $\bar\lambda$, since an immersion between manifolds of the same dimension is a local diffeomorphism and both sides are conic.

Example 10.5.6. (i) In adapted coordinates $(x, y) \in \mathbb{R}^k \times \mathbb{R}^d$ for $Y = \{x = 0\}$, the function $\varphi(x, y, \xi) = x \cdot \xi$, $\xi \in \mathbb{R}^k \setminus 0$, is a non-degenerate phase function: $C_\varphi = \{x = 0\}$, $d(\partial_{\xi_j}\varphi) = dx_j$, and $j_\varphi(0, y, \xi) = (0, y; \xi, 0)$ is a diffeomorphism onto $N^*Y \setminus 0$. The conormal distributions $\mathcal{I}[a]$ of Chapter 4 are exactly oscillatory integrals with this phase.

(ii) The parametrix for the wave kernel constructed in Chapter 11 is a sum of oscillatory integrals with phases $\varphi_\pm(t, x, \eta) - y \cdot \eta$ on $\mathbb{R}_t \times X_x \times X_y \times (\mathbb{R}^n_\eta \setminus 0)$, where $\varphi_\pm$ solves the eikonal equation $\partial_t\varphi_\pm = \pm|d_x\varphi_\pm|_g$ with $\varphi_\pm(0, x, \eta) = x \cdot \eta$. These are non-degenerate phase functions, since $d(\partial_{\eta_j}(\varphi_\pm - y \cdot \eta)) = d(\partial_{\eta_j}\varphi_\pm) - dy_j$, and $d_y(\varphi_\pm - y \cdot \eta) = -\eta \neq 0$. The Lagrangian they parametrize is the flow-out $\{(t, x, y; \pm|\eta|_{g(y)}, \xi, -\eta);\ (x, \xi) = G_{\mp t}(y, \eta)\}$ by the geodesic flow G, the wavefront set computed in Chapter 11.

The number N of phase variables is not determined by Λ, but it is bounded below by K.

Lemma 10.5.7. *If φ is a non-degenerate phase function, $p \in C_\varphi$ and $\lambda = j_\varphi(p)$, then $K(\lambda) = N - \operatorname{rank}\varphi''_{\theta\theta}(p)$ for the (local) Lagrangian Λ_φ. In particular $N \geq K(\lambda)$, with equality if and only if $\varphi''_{\theta\theta}(p) = 0$.*

Proof. Since j_φ is an immersion and $\pi \circ j_\varphi(x, \theta) = x$, the vertical tangent vectors of Λ_φ at λ correspond to the vectors $(0, \delta\theta)$ tangent to C_φ, i.e. to the solutions of $\varphi''_{\theta\theta}(p)\delta\theta = 0$. □

We call φ *minimal* at p if $\varphi''_{\theta\theta}(p) = 0$. Note that $\varphi''_{\theta\theta}(x, \theta)\theta = 0$ by Euler's relation for the function $\partial_\theta\varphi$, homogeneous of degree zero, so θ itself is always in the null space.

10.5.3 Parametrization by hypersurfaces

Melrose's original approach parametrizes Λ by hypersurfaces in $X \times \mathbb{R}^k$; it is equivalent to the use of phase functions, as follows. Let $H = \{h = 0\} \subset X \times \mathbb{R}^k_z$ be a hypersurface, h real with $dh \neq 0$ on H, and let

$$C_H = \{(x, z) \in H;\ \partial_z h(x, z) = 0\}$$

be the set of points where the fibre $\{x\} \times \mathbb{R}^k$ is tangent to H. This depends only on H: if $h' = bh$ with $b \neq 0$ then $\partial_z h' = b\,\partial_z h$ on H. Call H *non-degenerate* at $(\bar x, \bar z) \in C_H$ if dh and $d(\partial_{z_1}h), \dots, d(\partial_{z_k}h)$ are linearly independent there; then C_H is a submanifold of dimension $n - 1$ near $(\bar x, \bar z)$.

Proposition 10.5.8. *Let $H = \{h = 0\}$ be non-degenerate along C_H and set, for $\theta = (\theta_0, \theta') \in \mathbb{R} \times \mathbb{R}^k$ with $\theta_0 > 0$,*

$$\varphi(x, \theta) = \theta_0\, h(x, \theta'/\theta_0).$$

Then φ is a non-degenerate phase function near C_φ, $C_\varphi = \{(x, \theta_0, \theta_0 z);\ (x, z) \in C_H,\ \theta_0 > 0\}$, and the Lagrangian it parametrizes is

$$\Lambda_H = \{(x, \tau\, d_x h(x, z));\ (x, z) \in C_H,\ \tau > 0\}.$$

Conversely, every phase function on a cone Γ on which, after a linear change of the θ variables, $\theta_0 > 0$, is of this form with $h(x, z) = \varphi(x, 1, z)$.

Proof. With $z = \theta'/\theta_0$ we have $\partial_{\theta_j}\varphi = (\partial_{z_j} h)(x, z)$ for $j \geq 1$ and $\partial_{\theta_0}\varphi = h - \sum_j z_j \partial_{z_j} h$, so C_φ is as stated, and $d_x\varphi = \theta_0 d_x h$ there. On C_φ, $d(\partial_{\theta_0}\varphi) = dh - \sum_j z_j d(\partial_{z_j} h)$ (all forms pulled back by the submersion $(x, \theta) \mapsto (x, \theta'/\theta_0)$), so the span of the $d(\partial_{\theta_j}\varphi)$ is the pull-back of the span of $dh, d(\partial_z h)$; pull-back by a submersion is injective, so the non-degeneracy conditions coincide. Also $d\varphi \neq 0$ near C_φ since $d_x\varphi = \theta_0 d_x h$ and $d_x h \neq 0$ on C_H ($dh \neq 0$ and $\partial_z h = 0$). The converse is homogeneity. □

Remark 10.5.9 (Lagrangian distributions as push-forwards). If $v \in I^{m'}(X \times \mathbb{R}^k, H)$ has compact support, then by Chapter 4 (any defining function may be used, and amplitudes may depend on all variables) $v = \frac{1}{2\pi}\int e^{i\theta_0 h(x,z)} c(x, z, \theta_0)\, d\theta_0$ modulo $\mathcal{C}^\infty$, with $c \in S^{\mu'}$, $\mu' = m' + \frac{n+k}{4} - \frac{1}{2}$. The substitution $\theta' = \theta_0 z$ in $\theta_0 > 0$ gives

$$\int_{\mathbb{R}^k} v(x, z)\, dz = (2\pi)^k \int_{\theta_0 > 0} e^{i\varphi(x,\theta)} c(x, \theta'/\theta_0, \theta_0)\, \theta_0^{-k}\, \bar{d}\theta + (\text{same with } \theta_0 < 0) + \mathcal{C}^\infty,$$

and $c(x, \theta'/\theta_0, \theta_0)\theta_0^{-k}$ is a symbol of order $\mu' - k$ in $\theta \in \mathbb{R}^{k+1}$ on the cone $\theta_0 > 0$ (as c is compactly supported in z). In the notation of Definition 10.6.3 below, the push-forward is therefore microlocally on Λ_H a Lagrangian distribution of order $m = m' - \frac{k}{4}$ (the part with $\theta_0 < 0$ lives on $-\Lambda_H$). This is Melrose's definition of Lagrangian distributions: they are, microlocally, push-forwards of conormal distributions under the projection $X \times \mathbb{R}^k \to X$, and the singularities of v that survive the integration are those at points where the fibres are tangent to H.

10.5.4 Existence of parametrizations

Lemma 10.5.10. *Let $\Lambda \subset T^*X \setminus 0$ be a conic Lagrangian submanifold, $\bar\lambda = (\bar x, \bar\xi) \in \Lambda$ and $K = K(\bar\lambda)$. There are local coordinates $x = (x', x'') \in \mathbb{R}^K \times \mathbb{R}^{n-K}$ near $\bar x$, with $\bar x = 0$ and $\bar\xi = dx_1$, such that, with $\xi = (\xi', \xi'')$ the dual coordinates, (ξ', x'') restrict to coordinates on Λ near $\bar\lambda$. Moreover dx' vanishes on $T_{\bar\lambda}\Lambda$.*

Proof. Let $S = d\pi(T_{\bar\lambda}\Lambda) \subset T_{\bar x}X$, of dimension $n - K$. For $v \in T_{\bar\lambda}\Lambda$, $\bar\xi(d\pi\, v) = \alpha(v) = 0$ by Lemma 10.5.1, so $\bar\xi$ annihilates S. Choose linear coordinates with $\bar x = 0$, $S = \operatorname{span}\{\partial_{x_{K+1}}, \dots, \partial_{x_n}\}$ and $\bar\xi = dx_1$ (possible since $\bar\xi \neq 0$ lies in the annihilator of S, which is spanned by $dx_1, \dots, dx_K$). Then dx' vanishes on $T_{\bar\lambda}\Lambda$. Let $V = T_{\bar\lambda}\Lambda \cap \ker d\pi$, of dimension K; its elements are $v = \sum \delta\xi_j \partial_{\xi_j}$, and for $w \in T_{\bar\lambda}\Lambda$, $0 = \omega(v, w) = \sum_j \delta\xi_j\, dx_j(w)$, i.e. $\delta\xi$ annihilates S, so $\delta\xi'' = 0$. Now if $w \in T_{\bar\lambda}\Lambda$ is annihilated by $d\xi'$ and dx'', then $d\pi\, w \in S$ has vanishing x''-components, so $d\pi\, w = 0$, $w \in V$, and then $\delta\xi' = 0 = \delta\xi''$, so $w = 0$. Hence the n functions (ξ', x'') have independent differentials on $T_{\bar\lambda}\Lambda$. □

Theorem 10.5.11. *Let Λ, $\bar\lambda$, K and the coordinates be as in Lemma 10.5.10. There is a smooth function $G(\theta, x'')$, defined for x'' near 0 and θ in a conic neighbourhood of $\bar\theta = (1, 0, \dots, 0) \in \mathbb{R}^K$, homogeneous of degree one in θ, such that*

$$\varphi(x, \theta) = x' \cdot \theta - G(\theta, x'') \tag{10.5.1}$$

is a non-degenerate phase function parametrizing Λ near $\bar\lambda$, minimal at the point $(0, \bar\theta)$ over $\bar\lambda$. Near $\bar\lambda$,

$$\Lambda = \{(x', x''; \xi', \xi'');\ x' = \partial_{\xi'} G(\xi', x''),\ \xi'' = -\partial_{x''} G(\xi', x'')\}.$$

Proof. By Lemma 10.5.10, near $\bar\lambda$, $\Lambda = \{x' = X'(\xi', x''),\ \xi'' = \Xi''(\xi', x'')\}$ for smooth X', Ξ''. Since M_t maps the point of Λ with coordinates (ξ', x'') to the point with coordinates $(t\xi', x'')$, X' is homogeneous of degree 0 and Ξ'' of degree 1 in ξ'; they are thus defined on a conic neighbourhood $\{\xi'/|\xi'| \text{ near } \bar\theta,\ x'' \text{ near } 0\}$, and the coordinate map is injective there (two points with the same coordinates can be rescaled by the same factor to lie near $\bar\lambda$). Let $G(\xi', x'') = X'(\xi', x'') \cdot \xi'$, homogeneous of degree one. On Λ, $0 = \alpha = \xi' \cdot dx' + \xi'' \cdot dx'' = d(x' \cdot \xi') - x' \cdot d\xi' + \xi'' \cdot dx''$, so in the coordinates (ξ', x'')

$$dG = X' \cdot d\xi' - \Xi'' \cdot dx'', \quad \text{i.e.} \quad \partial_{\xi'} G = X',\ \partial_{x''} G = -\Xi''.$$

For φ as in (10.5.1), $d_{x'}\varphi = \theta \neq 0$, $\partial_\theta \varphi = x' - \partial_\theta G$, whose differentials $dx'_j - d(\partial_{\theta_j} G)$ are independent, and $C_\varphi = \{x' = X'(\theta, x'')\}$ with $j_\varphi(x, \theta) = (x; \theta, -\partial_{x''} G) = (x; \theta, \Xi''(\theta, x''))$. So φ parametrizes Λ near $\bar\lambda$. Finally $\varphi''_{\theta\theta} = -\partial_\theta^2 G = -\partial_\theta X'$, which vanishes at $(\bar\theta, 0)$ because $dx' = 0$ on $T_{\bar\lambda}\Lambda$. □

10.5.5 Reduction and equivalence of phase functions

Lemma 10.5.12. *Let φ be a non-degenerate phase function on Γ and $\bar p = (\bar x, \bar\theta) \in C_\varphi$. After a linear change of the θ variables, $\theta = (\theta', \theta'') \in \mathbb{R}^K \times \mathbb{R}^r$, $r = N - K$, $K = K(j_\varphi(\bar p))$, where $\bar\theta'' = 0$, $\varphi''_{\theta''\theta''}(\bar p)$ is invertible and $\varphi''_{\theta'\theta}(\bar p) = 0$. Then near $(\bar x, \bar\theta')$ there is a unique smooth $\Theta(x, \theta')$, homogeneous of degree one, with $\Theta(\bar x, \bar\theta') = 0$ and $\partial_{\theta''}\varphi(x, \theta', \Theta(x, \theta')) = 0$, and*

$$\varphi_0(x, \theta') = \varphi(x, \theta', \Theta(x, \theta'))$$

is a non-degenerate phase function, minimal at $(\bar x, \bar\theta')$. The map $s(x, \theta') = (x, \theta', \Theta(x, \theta'))$ is a diffeomorphism of C_{φ_0} onto a neighbourhood of $\bar p$ in C_φ with $j_{\varphi_0} = j_\varphi \circ s$. At points of C_{φ_0},

$$\operatorname{sgn} \varphi''_{\theta\theta} \circ s = \operatorname{sgn} \varphi''_{0\theta'\theta'} + \operatorname{sgn} \varphi''_{\theta''\theta''} \circ s. \tag{10.5.2}$$

Proof. The symmetric matrix $\varphi''_{\theta\theta}(\bar p)$ has rank $N - K$ by Lemma 10.5.7; choose linear coordinates in which its null space is $\mathbb{R}^K \times 0$ and its range $0 \times \mathbb{R}^r$. By Euler's relation $\bar\theta$ is in the null space, so $\bar\theta'' = 0$, $\bar\theta' \neq 0$. The implicit function theorem gives Θ near $(\bar x, \bar\theta')$; since $\partial_{\theta''}\varphi$ is homogeneous of degree 0, $t\Theta(x, \theta')$ solves the equation at $(x, t\theta')$, so by uniqueness Θ is homogeneous of degree one and extends to a conic neighbourhood. Then $\partial_{\theta'}\varphi_0 = (\partial_{\theta'}\varphi) \circ s$ and $d_x\varphi_0 = (d_x\varphi) \circ s$, because $\partial_{\theta''}\varphi \circ s = 0$; hence $C_{\varphi_0} = s^{-1}(C_\varphi)$, s maps C_{φ_0} onto $C_\varphi \cap \{\theta'' = \Theta\}$, which is a neighbourhood of $\bar p$ in C_φ, and $j_{\varphi_0} = j_\varphi \circ s$. For non-degeneracy, suppose $\sum_j c_j d(\partial_{\theta'_j}\varphi_0) = 0$ at a point of C_{φ_0}, i.e. $s^*\beta = 0$ with $\beta = \sum_j c_j d(\partial_{\theta'_j}\varphi)$. The map s is a diffeomorphism onto $W = \{\partial_{\theta''}\varphi = 0\}$, so β vanishes on TW, i.e. is a combination of the $d(\partial_{\theta''_i}\varphi)$; by non-degeneracy of φ, $c = 0$. Also $d\varphi_0 \neq 0$ near C_{φ_0} since $d_x\varphi_0 \neq 0$ there. Finally, at a point of C_{φ_0} write $\varphi''_{\theta\theta} = \left(\begin{smallmatrix} A & B \\ B^t & D \end{smallmatrix}\right)$ with $D = \varphi''_{\theta''\theta''}$ invertible; differentiating $\partial_{\theta'}\varphi_0 = \partial_{\theta'}\varphi \circ s$ and using $\partial_{\theta'}\Theta = -D^{-1}B^t$ gives $\varphi''_{0\theta'\theta'} = A - BD^{-1}B^t$, and $\varphi''_{\theta\theta}$ is congruent to $\operatorname{diag}(A - BD^{-1}B^t, D)$; this gives (10.5.2). Minimality holds by Lemma 10.5.7, since φ_0 has K phase variables. □

Theorem 10.5.13 (Equivalence of minimal phase functions)**.** *Let φ_1, φ_2 be non-degenerate phase functions in the same number K of variables, both parametrizing Λ near $\bar\lambda$ and minimal at the points $(\bar x,\bar\theta_i) = j_{\varphi_i}^{-1}(\bar\lambda)$. Then there is a diffeomorphism $\kappa(x,\theta) = (x,\Theta(x,\theta))$ from a conic neighbourhood of $(\bar x,\bar\theta_1)$ onto a conic neighbourhood of $(\bar x,\bar\theta_2)$, with Θ homogeneous of degree one in θ, such that $\varphi_2\circ\kappa = \varphi_1$.*

Proof. Step 1: matching the critical sets. Since $\pi\circ j_{\varphi_i}$ is the projection $(x,\theta)\mapsto x$, the subspace $S = d\pi(T_{\bar\lambda}\Lambda)$ is the projection of TC_{φ_i} at $(\bar x,\bar\theta_i)$ for $i=1,2$. Choose linear coordinates $x=(x',x'')\in\mathbb{R}^K\times\mathbb{R}^{n-K}$ such that $dx''|_S$ is an isomorphism. By minimality every vector $(0,\delta\theta)$ is tangent to C_{φ_i} at $(\bar x,\bar\theta_i)$, and it follows (as in Lemma 10.5.10) that (θ,x'') are coordinates on C_{φ_i} near $(\bar x,\bar\theta_i)$: $C_{\varphi_i} = \{x' = X_i(\theta,x'')\}$, with X_i homogeneous of degree 0. The diffeomorphism $c = j_{\varphi_2}^{-1}\circ j_{\varphi_1} : C_{\varphi_1}\to C_{\varphi_2}$ preserves x, so in these coordinates $c(\theta,x'') = (\Theta_c(\theta,x''),x'')$ with $\partial_\theta\Theta_c$ invertible, Θ_c homogeneous of degree one and $X_2(\Theta_c(\theta,x''),x'') = X_1(\theta,x'')$. The map $\kappa_0(x,\theta) = (x,\Theta_c(\theta,x''))$ is a homogeneous fibre-preserving local diffeomorphism mapping C_{φ_1} onto C_{φ_2} and equal to c there. Replacing φ_2 by $\varphi_2\circ\kappa_0$ (again a minimal non-degenerate phase function parametrizing Λ) we may assume that $C_{\varphi_1} = C_{\varphi_2} = C$ and $j_{\varphi_1} = j_{\varphi_2}$ on C, i.e. $d_x\varphi_1 = d_x\varphi_2$ on C. As also $\varphi_i = 0$ and $d_\theta\varphi_i = 0$ on C, the difference $\psi = \varphi_2 - \varphi_1$ vanishes with its differential on C.

Step 2: Moser's argument. The functions $F_j = \partial_{\theta_j}\varphi_1$, homogeneous of degree 0, define C with independent differentials. Using homogeneous coordinates (ρ,c',F) near $(\bar x,\bar\theta_1)$, with $\rho = |\theta|$ and c' homogeneous of degree 0, Taylor's formula in F gives

$$\psi = \sum_{i,j} G_{ij}F_iF_j, \qquad G_{ij} = G_{ji} = \int_0^1 (1-\tau)\frac{\partial^2\psi}{\partial F_i\partial F_j}(\rho,c',\tau F)\,d\tau,$$

with G_{ij} homogeneous of degree one. Let $\varphi_t = \varphi_1 + t\psi$, $t\in[0,1]$. Then

$$\partial_{\theta_l}\varphi_t = \sum_j A_{lj}(t)F_j, \qquad A_{lj}(t) = \delta_{lj} + t\sum_i\Big(\frac{\partial G_{ij}}{\partial\theta_l}F_i + 2G_{ij}\frac{\partial^2\varphi_1}{\partial\theta_l\partial\theta_i}\Big),$$

and A is homogeneous of degree 0 and equal to the identity at $(\bar x,\bar\theta_1)$, because $F=0$ there and φ_1 is minimal. So $A(t)$ is invertible on a conic neighbourhood of $(\bar x,\bar\theta_1)$ for all $t\in[0,1]$; consequently $C_{\varphi_t} = C$, each φ_t is a non-degenerate phase function, and $d_x\varphi_t = d_x\varphi_1$ on C. We seek homogeneous fibre-preserving maps κ_t, $\kappa_0 = \mathrm{Id}$, with $\varphi_t\circ\kappa_t = \varphi_1$, generated by vertical vector fields $V_t = \sum_l v_l(t,x,\theta)\partial_{\theta_l}$, $\frac{d}{dt}\kappa_t = V_t\circ\kappa_t$. Differentiating, the condition becomes $(\psi + V_t\varphi_t)\circ\kappa_t = 0$, which holds if

$$v_l = -\sum_{i,j} G_{ij}F_i\,(A(t)^{-1})_{jl},$$

since then $V_t\varphi_t = \sum_l v_l\sum_j A_{lj}F_j = -\sum_{i,j}G_{ij}F_iF_j = -\psi$. The coefficients v_l are homogeneous of degree one, so V_t commutes with the dilations, and V_t vanishes on C. Since V_t vanishes at $(\bar x,\bar\theta_1)$, its flow is defined for $t\in[0,1]$ on a neighbourhood of this point, hence by homogeneity on a conic neighbourhood, and $\kappa = \kappa_1$ satisfies $\varphi_2\circ\kappa = \varphi_1$. Composing with κ_0 from Step 1 proves the theorem. □

Corollary 10.5.14. *Let φ_1, φ_2 be non-degenerate phase functions, with N_1 and N_2 variables, parametrizing Λ on conic open sets Λ_{φ_1}, Λ_{φ_2}. For $\lambda\in\Lambda_{\varphi_i}$ let $s_i(\lambda) = \operatorname{sgn}\varphi_i''{}_{\theta\theta}(j_{\varphi_i}^{-1}(\lambda))$. Then $s_1 - s_2$ is locally constant on $\Lambda_{\varphi_1}\cap\Lambda_{\varphi_2}$, and $s_1 - s_2 \equiv N_1 - N_2 \pmod 2$.*

Proof. Fix $\bar\lambda$ in the intersection and reduce both phases at the points over $\bar\lambda$ by Lemma 10.5.12, to minimal phases $\varphi_{1,0}$, $\varphi_{2,0}$ in $K = K(\bar\lambda)$ variables. By Theorem 10.5.13, $\varphi_{2,0}\circ\kappa = \varphi_{1,0}$, and

differentiating twice at points of $C_{\varphi_{1,0}}$, where $\partial_\theta\varphi_{2,0}\circ\kappa = 0$, $\varphi''_{1,0\theta\theta} = (\partial_\theta\Theta)^t\,(\varphi''_{2,0\theta\theta}\circ\kappa)\,\partial_\theta\Theta$; moreover $j_{\varphi_{2,0}}\circ\kappa = j_{\varphi_{1,0}}$ on $C_{\varphi_{1,0}}$. So the minimal phases have equal signatures at points over the same λ, and by (10.5.2) $s_1 - s_2 = \operatorname{sgn} D_1 - \operatorname{sgn} D_2$ near $\bar\lambda$, where D_i are the invertible $r_i\times r_i$ matrices $\varphi''_{i,\theta''\theta''}$. These signatures are locally constant, and $\operatorname{sgn} D_i \equiv r_i = N_i - K \pmod 2$. □

Question. The number N of phase variables is not determined by Λ. Why does this not matter?

Answer. Extra phase variables in which the phase is a non-degenerate quadratic form can be added (Exercise 10.6) or removed by stationary phase (Proposition 10.6.2). Removing r such variables multiplies the amplitude by $(2\pi)^{-\frac{r}{2}}e^{\frac{i\pi}{4}\operatorname{sgn} D}|\det D|^{-\frac12}$ and raises its order by $\frac{r}{2}$. The order convention $\mu_N = m + \frac{n}{4} - \frac{N}{2}$ absorbs the change of order, the half-density d_{C_φ} absorbs $|\det D|$, and the factor $e^{\frac{i\pi}{4}\operatorname{sgn} D}$ is what the Maslov bundle of Section 10.8 keeps track of. □

10.6 Lagrangian distributions

10.6.1 Oscillatory integrals

Let $X\subset\mathbb{R}^n$ be open and φ a phase function on the open cone $\Gamma\subset X\times(\mathbb{R}^N\setminus 0)$. For $\mu\in\mathbb{R}$ let S^μ_Γ be the space of $a\in S^\mu(\mathbb{R}^n;\mathbb{R}^N)$ which vanish for $|\theta|\le 1$ and whose support is contained in $L\times\mathbb{R}^N$, $L\subset X$ compact, and in a closed cone (in $X\times(\mathbb{R}^N\setminus 0)$) contained in Γ. For such a the oscillatory integral

$$I_\varphi(a)(x) = \int_{\mathbb{R}^N} e^{i\varphi(x,\theta)}a(x,\theta)đ\theta, \qquad đ\theta = (2\pi)^{-N}d\theta,$$

is defined as the distribution $\phi\mapsto\lim_{\epsilon\downarrow 0}\iint e^{i\varphi}a\,\chi_0(\epsilon\theta)\phi\,dxđ\theta$, $\chi_0\in\mathcal{C}^\infty_{\mathrm{c}}$, $\chi_0(0)=1$; the limit exists and is independent of χ_0, because $d_{x,\theta}\varphi\neq 0$ allows repeated integration by parts [7, Section 7.8]. If $\mu < -N - r$ the integral converges absolutely and $I_\varphi(a)\in\mathcal{C}^r$; in particular $I_\varphi(a)\in\mathcal{C}^\infty$ if $a\in S^{-\infty}$. Moreover [7, Section 8.1], as recalled in Chapter 11,

$$\mathrm{WF}(I_\varphi(a))\subset j_\varphi(C_\varphi\cap\operatorname{cone}\operatorname{supp} a), \tag{10.6.1}$$

where $\operatorname{cone}\operatorname{supp} a$ is the smallest closed cone outside which a is of order $-\infty$; and if $d_\theta\varphi\neq 0$ on $\operatorname{cone}\operatorname{supp} a$ then $I_\varphi(a)\in\mathcal{C}^\infty$ (integrate by parts with $L = |d_\theta\varphi|^{-2}\sum_j\partial_{\theta_j}\varphi\,D_{\theta_j}$, $Le^{i\varphi} = e^{i\varphi}$). On a manifold we work with half-densities: in a chart, $I_\varphi(a)|dx|^{1/2}$. We write

$$\mu_N = m + \frac{n}{4} - \frac{N}{2}$$

for the amplitude order corresponding to order m; for the conormal phase of Example 10.5.6(i), $N=k$ and $\mu_k = M$ is the order of Chapter 4.

Lemma 10.6.1. *(i) If $\kappa(x,\theta) = (x,\Theta(x,\theta))$ is a homogeneous fibre-preserving diffeomorphism between open cones and $\varphi_2\circ\kappa = \varphi_1$, then $I_{\varphi_2}(a) = I_{\varphi_1}(a\circ\kappa\,|\det\partial_\theta\Theta|)$ for $a\in S^\mu_{\Gamma_2}$, and $a\circ\kappa|\det\partial_\theta\Theta|\in S^\mu_{\Gamma_1}$. (ii) If $F: X'\to X$ is a diffeomorphism of open sets then $F^*(I_\varphi(a)|dx|^{1/2}) = I_{\varphi\circ F}(a\circ F\,|\det F'|^{1/2})|dx'|^{1/2}$, where $\varphi\circ F(x',\theta) = \varphi(F(x'),\theta)$ parametrizes $(F^\sharp)^{-1}(\Lambda_\varphi)$, $F^\sharp$ being the induced map $T^*X'\to T^*X$. (iii) If $a\in S^\mu_\Gamma$ and $a|_{C_\varphi}\in S^{\mu-1}$ (that is, a coincides on C_φ with an element of $S^{\mu-1}_\Gamma$) then $I_\varphi(a) = I_\varphi(a') + \mathcal{C}^\infty$ with $a'\in S^{\mu-1}_\Gamma$.*

Proof. (i) and (ii) are changes of variables (first for a of very negative order, then by continuity); composition with a homogeneous diffeomorphism preserves symbol classes, and $\det \partial_\theta \Theta$ is homogeneous of degree 0. (iii) Subtract an element of $S^{\mu-1}_\Gamma$ equal to a on C_φ; so we may assume $a = 0$ on C_φ. Let χ be homogeneous of degree zero for $|\theta| \geq 1$, equal to one near C_φ and supported in a conic neighbourhood of C_φ covered by finitely many conic sets in which homogeneous coordinates (ρ, c', F), $F = \partial_\theta \varphi$, exist (we use a subordinate partition of unity homogeneous of degree 0 and treat each piece separately). Then $d_\theta \varphi \neq 0$ on the support of $(1-\chi)a$, so $I_\varphi((1-\chi)a) \in \mathcal{C}^\infty$; and as in the proof of Theorem 10.5.13, $\chi a = \sum_j F_j b_j$ with $b_j = \chi \int_0^1 \partial_{F_j} a(\rho, c', \tau F)\, d\tau \in S^\mu_\Gamma$. Since $F_j e^{i\varphi} = D_{\theta_j} e^{i\varphi}$, integration by parts gives $I_\varphi(\chi a) = -\sum_j I_\varphi(D_{\theta_j} b_j)$, and $D_{\theta_j} b_j \in S^{\mu-1}_\Gamma$. □

The next result is the analytic counterpart of Lemma 10.5.12: phase variables in which the phase is non-degenerate can be eliminated by stationary phase. Recall that for an invertible real symmetric $r \times r$ matrix D,

$$\int_{\mathbb{R}^r} e^{\frac{i}{2}\langle Dw, w\rangle}\, dw = (2\pi)^{\frac{r}{2}} |\det D|^{-\frac{1}{2}} e^{\frac{i\pi}{4}\operatorname{sgn} D} \tag{10.6.2}$$

as an oscillatory integral [7, Section 7.6].

Proposition 10.6.2. *Let φ, $\bar p$, Θ, φ_0 be as in Lemma 10.5.12, and write $D(x,\theta') = \varphi''_{\theta''\theta''}(x,\theta',\Theta(x,\theta'))$, homogeneous of degree -1, with constant signature $\operatorname{sgn} D$ near $(\bar x, \bar\theta')$. There are conic neighbourhoods Γ_1 of $\bar p$ and Γ_0 of $(\bar x, \bar\theta')$ such that:*

(i) for every $a \in S^\mu_{\Gamma_1}$ there is $b \in S^{\mu+\frac{r}{2}}_{\Gamma_0}$ with $I_\varphi(a) - I_{\varphi_0}(b) \in \mathcal{C}^\infty$ and

$$b(x,\theta') \equiv (2\pi)^{-\frac{r}{2}} e^{\frac{i\pi}{4}\operatorname{sgn} D} |\det D(x,\theta')|^{-\frac{1}{2}} a(x,\theta',\Theta(x,\theta')) \mod S^{\mu+\frac{r}{2}-1}; \tag{10.6.3}$$

(ii) conversely, for every $b \in S^{\mu+\frac{r}{2}}_{\Gamma_0}$ there is $a \in S^\mu_{\Gamma_1}$ with $I_\varphi(a) - I_{\varphi_0}(b) \in \mathcal{C}^\infty$.

Classical amplitudes correspond to classical amplitudes.

Sketch of proof. (i) Choose $\chi(x,\theta)$, homogeneous of degree 0, supported where $|\theta'' - \Theta(x,\theta')| < \epsilon|\theta'|$ and equal to one where $|\theta'' - \Theta| < \epsilon|\theta'|/2$. On the support of $(1-\chi)a$ (for Γ_1 small) $\partial_{\theta''}\varphi \neq 0$, so $I_\varphi((1-\chi)a) \in \mathcal{C}^\infty$. For χa substitute $\theta'' = \Theta(x,\theta') + |\theta'| w$; then $\varphi = |\theta'|\Phi(x,\hat\theta', w)$, $\hat\theta' = \theta'/|\theta'|$, where Φ has, in w, a unique non-degenerate critical point at $w = 0$ with value $\varphi_0(x,\hat\theta')$ and Hessian $|\theta'| D(x,\theta')$. The method of stationary phase with the large parameter $|\theta'|$ [7, Theorem 7.7.5], applied with symbol amplitudes (the derivatives in w of $a(x,\theta',\Theta + |\theta'|w)$ are $O(|\theta'|^\mu)$ and each term of the expansion gains a factor $|\theta'|^{-1}$; see [8, Section 25.1]), shows that

$$\int e^{i\varphi(x,\theta',\theta'')} \chi a(x,\theta',\theta'')\, đ\theta'' = e^{i\varphi_0(x,\theta')} b(x,\theta')$$

with $b \in S^{\mu+\frac{r}{2}}$ having an asymptotic expansion whose leading term, by (10.6.2) and the factor $(2\pi)^{-r}$ in $đ\theta''$, is (10.6.3). (ii) Given b, put $a_0 = (2\pi)^{\frac{r}{2}} e^{-\frac{i\pi}{4}\operatorname{sgn} D} |\det D|^{\frac{1}{2}} b(x,\theta')\, \chi(x,\theta)$, of order μ; by (i), $I_\varphi(a_0) = I_{\varphi_0}(b - b_1)$ modulo $\mathcal{C}^\infty$ with $b_1 \in S^{\mu+\frac{r}{2}-1}$. Repeating with b_1 and so on, and summing the a_j asymptotically, gives a. □

10.6.2 Definition and independence of the parametrization

Definition 10.6.3. Let X be a manifold of dimension n and $\Lambda \subset T^*X \setminus 0$ a closed conic Lagrangian submanifold. The space $I^m(X, \Lambda; \Omega^{\frac{1}{2}})$ of *Lagrangian distributions* of order m consists of those $u \in \mathcal{C}^{-\infty}(X; \Omega^{\frac{1}{2}})$ such that $\mathrm{WF}(u) \subset \Lambda$ and, for every $\bar\lambda \in \Lambda$, there are a chart near $\pi(\bar\lambda)$, a non-degenerate phase function φ in N variables parametrizing Λ near $\bar\lambda$, and $a \in S^{\mu_N}_\Gamma$, such that

$$\bar\lambda \notin \mathrm{WF}\left(u - I_\varphi(a)|dx|^{\frac{1}{2}}\right).$$

The classical subspace $I^m_{\mathrm{ph}}(X, \Lambda; \Omega^{\frac{1}{2}})$ is defined by requiring a to be classical.

Thus a Lagrangian distribution is smooth away from $\pi(\Lambda)$ and is, microlocally near each point of Λ, an oscillatory integral with a phase function parametrizing Λ. By (10.6.1), $I_\varphi(a)|dx|^{1/2}$ itself lies in $I^m(X, \Lambda; \Omega^{\frac{1}{2}})$ when $a \in S^{\mu_N}_\Gamma$ and φ parametrizes Λ (with Γ small, so that $j_\varphi(C_\varphi) \subset \Lambda$). Since the definition only asks for *some* parametrization, the real content of the following theorem is that *every* parametrization can be used.

Theorem 10.6.4. *Let $u \in I^m(X, \Lambda; \Omega^{\frac{1}{2}})$, $\bar\lambda \in \Lambda$, and let φ be any non-degenerate phase function, in N variables, parametrizing Λ near $\bar\lambda$ (in some chart). Then there is $a \in S^{\mu_N}_\Gamma$ with $\bar\lambda \notin \mathrm{WF}(u - I_\varphi(a)|dx|^{1/2})$; a is classical if u is.*

Proof. By Lemma 10.6.1(ii) we may use a single chart. Let φ_1, $a_1 \in S^{\mu_{N_1}}$ be as in the definition. By Lemma 10.5.12 and Proposition 10.6.2(i) (after a linear change of the θ variables, which changes a_1 by a constant factor), and after cutting a_1 off to a small conic neighbourhood of the point over $\bar\lambda$ (which by (10.6.1) changes $I_{\varphi_1}(a_1)$ only away from a conic neighbourhood of $\bar\lambda$), $I_{\varphi_1}(a_1) \equiv I_{\varphi_{1,0}}(b_1)$ modulo $\mathcal{C}^\infty$, with $\varphi_{1,0}$ minimal in K variables and $b_1 \in S^{\mu_K}$, since $\mu_{N_1} + \frac{N_1 - K}{2} = \mu_K$. Reduce φ in the same way to a minimal φ_0. By Theorem 10.5.13 and Lemma 10.6.1(i), $I_{\varphi_{1,0}}(b_1) = I_{\varphi_0}(b)$ with $b \in S^{\mu_K}$, and by Proposition 10.6.2(ii), $I_{\varphi_0}(b) \equiv I_\varphi(a)$ modulo $\mathcal{C}^\infty$ with $a \in S^{\mu_K - \frac{N-K}{2}} = S^{\mu_N}$. All steps preserve classical amplitudes. □

Remark 10.6.5. Using a microlocal partition of unity one can show that every $u \in I^m(X, \Lambda; \Omega^{\frac{1}{2}})$ is a locally finite sum of oscillatory integrals $I_{\varphi_j}(a_j)|dx|^{1/2}$ plus a smooth half-density, and that properly supported pseudodifferential operators of order m' map $I^m(X, \Lambda; \Omega^{\frac{1}{2}})$ into $I^{m+m'}(X, \Lambda; \Omega^{\frac{1}{2}})$; see [8, Chapter 25], [6]. We shall only need the special case in the proof of the next theorem.

10.6.3 Conormal distributions

Conormal distributions are the Lagrangian distributions associated with conormal bundles, with the same orders.

Theorem 10.6.6. *Let $Y \subset X$ be a closed embedded submanifold. Then*

$$I^m(X, N^*Y \setminus 0; \Omega^{\frac{1}{2}}) = I^m(X, Y; \Omega^{\frac{1}{2}}),$$

and similarly for the classical subspaces.

Proof. Let $u \in I^m(X, Y; \Omega^{\frac{1}{2}})$. Then $\mathrm{WF}(u) \subset N^*Y \setminus 0$ (Chapter 4). In an adapted chart, $\phi u = (\mathcal{I}[a] + f)|dz|^{1/2}$ with $a \in S^M$, $M = m + \frac{n}{4} - \frac{k}{2} = \mu_k$, which we may cut off to vanish for $|\xi| \le 1$ (changing $\mathcal{I}[a]$ by a smooth function); so ϕu is, modulo $\mathcal{C}^\infty$, an oscillatory integral with the phase $x \cdot \xi$ of Example 10.5.6(i), and $u \in I^m(X, N^*Y \setminus 0; \Omega^{\frac{1}{2}})$.

Conversely let $u \in I^m(X, N^*Y \setminus 0; \Omega^{\frac12})$. Then u is smooth outside Y. Fix $\bar y \in Y$ and an adapted chart near $\bar y$. By Theorem 10.6.4, applied with the phase $x\cdot\xi$, each $\bar\lambda \in N^*_{\bar y}Y \setminus 0$ has an open conic neighbourhood $\Gamma_{\bar\lambda}$ in $T^*X\setminus 0$ and $a_{\bar\lambda} \in S^M$ with $\mathrm{WF}(u - \mathcal{I}[a_{\bar\lambda}]|dz|^{1/2}) \cap \Gamma_{\bar\lambda} = \emptyset$. By compactness of the sphere in $N^*_{\bar y}Y$, finitely many $\Gamma_j = \Gamma_{\bar\lambda_j}$ cover $N^*_yY \setminus 0$ for y in a neighbourhood U of $\bar y$. Choose properly supported $B_j \in \Psi^0(X)$ with $\mathrm{WF}'(B_j) \subset \Gamma_j$ and $\sum_j B_j = \mathrm{Id}$ microlocally near $N^*Y \cap T^*U$, i.e. $\sum_j \sigma_0(B_j) = 1$ on a conic neighbourhood of $N^*Y \cap T^*U$ (Chapter 3). For $\phi \in \mathcal{C}^\infty_{\mathrm c}(U)$, since $\mathrm{WF}(\phi u) \subset N^*Y$,

$$\phi u \equiv \sum_j B_j(\phi u) \equiv \sum_j B_j(\phi \mathcal{I}[a_j]|dz|^{\frac12}) \mod \mathcal{C}^\infty,$$

because $\mathrm{WF}(B_j w) \subset \mathrm{WF}'(B_j) \cap \mathrm{WF}(w)$. The right side is conormal, since pseudodifferential operators preserve conormal distributions (Theorem 10.1.1). So $u \in I^m(X, Y; \Omega^{\frac12})$. The classical case is the same. □

Remark 10.6.7 (The wave kernel). By Example 10.5.6(ii), the short-time parametrix for the wave kernel constructed in Chapter 11 is a sum of oscillatory integrals with non-degenerate phases $\varphi_\pm(t,x,\eta) - y\cdot\eta$, in $N = n$ variables, on the manifold $\mathbb{R}\times X\times X$ of dimension $2n+1$, with amplitudes of order 0 (for the cosine kernel). Since $0 = m + \frac{2n+1}{4} - \frac{n}{2}$, the cosine kernel is (for small $|t|$) a Lagrangian distribution of order $-\frac14$ associated with the flow-out of the diagonal described there; the sine kernel has order $-\frac54$. The conormal superposition used in Chapter 11 is the hypersurface parametrization of Proposition 10.5.8 in polar coordinates.

10.7 Keller's example of a caustic

J. B. Keller was the first to compute with Lagrangian distributions in a situation, that of a caustic, where the factors which are now organized into the Keller–Maslov line bundle cannot be avoided. The example motivates the definition of the principal symbol in the next section.

Let $P = D_t^2 - D_x^2 - D_y^2$ on $\mathbb{R}^3$, with $p = \tau^2 - \xi^2 - \eta^2$ and $\Sigma(P) = \{\tau^2 = \xi^2 + \eta^2\}$. Its null bicharacteristics are the *light rays*: (τ,ξ,η) is constant and (t,x,y) moves on the straight line $t = t_0 + s$, $x = x_0 - \xi s/\tau$, $y = y_0 - \eta s/\tau$ (after reparametrization). Characteristic planes such as $\{t = \omega\cdot(x,y)\}$, $|\omega| = 1$, carry the travelling waves of Example 10.1.4, and by Corollary 10.3.2 any smooth curve C in the plane $\{t=0\}$ is the intersection with $\{t=0\}$ of two characteristic surfaces, defined for $|t|$ small (one for each choice of the sign of τ relative to the normal). These surfaces in general develop singularities, but by Lemma 10.5.2 the corresponding Lagrangian, the union of the light rays through the conormal covectors of C, stays smooth. We consider the parabola

$$C = \{y = \tfrac12 x^2\}, \qquad N^*C = \{(x_0, \tfrac12 x_0^2; -x_0\eta, \eta);\ x_0, \eta \in \mathbb{R}\}.$$

Proposition 10.7.1. *Let* $\Lambda_0 = \{(0, x_0, \frac12 x_0^2; \tau, -x_0\eta, \eta);\ \eta \neq 0,\ \tau = -(1+x_0^2)^{1/2}\eta\}$ *and let* Λ *be the flow-out of* Λ_0 *by* H_p. *Then, with* $w = w(x_0) = (1+x_0^2)^{-1/2}$,

$$\Lambda = \Big\{(t,\ x_0(1-tw),\ \tfrac12 x_0^2 + tw;\ -\eta/w,\ -x_0\eta,\ \eta);\ x_0, t \in \mathbb{R},\ \eta \neq 0\Big\}, \tag{10.7.1}$$

a closed conic Lagrangian submanifold of $T^*\mathbb{R}^3 \setminus 0$ *contained in* $\Sigma(P)$, *invariant under* $\zeta \mapsto -\zeta$. *Moreover:*

(i) $K(\lambda) = 1$ *at all points of* Λ *except those with* $t = (1+x_0^2)^{3/2}$*, where* $K = 2$*; these project to the curve*

$$\mathcal{K} = \{(t,x,y) = ((1+x_0^2)^{\frac{3}{2}},\ -x_0^3,\ 1+\tfrac{3}{2}x_0^2);\ x_0 \in \mathbb{R}\},$$

the caustic, which is singular only at $(1,0,1)$*.*

(ii) In $\{t<1\}$*,* $\Lambda = N^*G \setminus 0$ *where* $G = \{(t, x_0(1-tw), \tfrac{1}{2}x_0^2 + tw);\ t<1,\ x_0 \in \mathbb{R}\}$ *is a smooth embedded characteristic surface.*

Proof. Λ_0 is conic, of dimension 2, contained in $\Sigma(P)$ (as $\tau^2 = (1+x_0^2)\eta^2 = \xi^2+\eta^2$), $\alpha = \tau\,dt + \xi\,dx + \eta\,dy$ vanishes on it ($dt = 0$ and $-x_0\eta\,dx_0 + \eta\,x_0 dx_0 = 0$), and $H_p = 2\tau\partial_t - 2\xi\partial_x - 2\eta\partial_y$ is transversal to it since $\tau \neq 0$. The light ray through a point of Λ_0 is $t \mapsto (t, x_0 - \xi t/\tau, \tfrac{1}{2}x_0^2 - \eta t/\tau)$ with $\xi/\tau = x_0 w$, $\eta/\tau = -w$, which gives (10.7.1); the parametrization by (x_0, t, η) is injective (the covector determines η and x_0), so by Lemma 10.5.2 Λ is a conic Lagrangian submanifold. It is closed in $T^*\mathbb{R}^3 \setminus 0$: if $\lambda_k \in \Lambda$ converge to a point of $T^*\mathbb{R}^3 \setminus 0$ then η_k stays away from 0 (otherwise, the covectors being bounded below, $|x_{0,k}| \to \infty$ and then $|x| = |x_{0,k}|\,|1-tw| \to \infty$), so $(x_{0,k}, t_k, \eta_k)$ converges.

(i) The projection is $(x_0, t, \eta) \mapsto (t, x, y)$; its differential has rank one in the (t)-direction plus the rank of $\partial_{x_0}(x,y)$, which is

$$\partial_{x_0}(x_0(1-tw)) = 1 - t(1+x_0^2)^{-\frac{3}{2}}, \qquad \partial_{x_0}(\tfrac{1}{2}x_0^2 + tw) = x_0(1 - t(1+x_0^2)^{-\frac{3}{2}}),$$

so the rank of $d\pi|_{T\Lambda}$ is 2 (i.e. $K = 1$) unless $t = (1+x_0^2)^{3/2}$, where it is 1. Substituting this value of t gives $\mathcal{K}$. All three components of the derivative of the parametrization of $\mathcal{K}$, $(3x_0(1+x_0^2)^{1/2}, -3x_0^2, 3x_0)$, vanish only at $x_0 = 0$.

(ii) For $t<1$, $\partial_{x_0}x = 1 - t(1+x_0^2)^{-3/2} > 0$, so for fixed t the map $x_0 \mapsto (x,y)$ is an injective immersion, and G is a smooth embedded surface; $\Lambda \cap \{t<1\}$ is a conic Lagrangian projecting onto G with $K = 1$, contained in N^*G (as $\alpha = 0$ on it) and of the same dimension. It is characteristic since $\Lambda \subset \Sigma(P)$. □

For fixed t, the curve $C_t = G \cap \{t\}$ is the parallel curve to C at distance t, on the concave side: $(x,y) = (x_0, \tfrac{1}{2}x_0^2) + t(-x_0, 1)w$, the unit normal being $(-x_0, 1)w$. The projection of $\mathcal{K}$ to the (x,y) plane is the evolute of the parabola, the locus of its centres of curvature (the radius of curvature of C at x_0 is $(1+x_0^2)^{3/2}$), the semicubical parabola $(y-1)^3 = \frac{27}{8}x^2$. For $t<1$, C_t is smooth; at $t=1$ it has a single singular point $(0,1)$; for $t>1$ it has two cusps, at $x_0 = \pm(t^{2/3}-1)^{1/2}$, on the evolute, and a self-intersection on the axis $x=0$, at $y = \frac{1}{2}(t^2+1)$ (Figure 10.4). The arc of C_t between the cusps, which contains the vertex $(0,t)$, is curved the opposite way from the rest (Exercise 10.8).

10.7.1 A parametrization near the cusp

We follow the proof of Theorem 10.5.11 near the point of Λ over $(t,x,y) = (1,0,1)$ with $x_0 = 0$, $\eta = 1$, i.e. $\bar\lambda = (1,0,1;-1,0,1)$, where $K = 2$. The tangent space of Λ projects onto $\operatorname{span}\{\partial_t + \partial_y\}$, and $\bar\lambda = dy - dt$. So we use the linear coordinates and their dual variables

$$S = y - t, \quad x, \quad R = \tfrac{1}{2}(y+t) - 1; \qquad \sigma = \tfrac{1}{2}(\eta - \tau), \quad \xi, \quad \rho = \eta + \tau,$$

so that $\tau\,dt + \xi\,dx + \eta\,dy = \sigma\,dS + \xi\,dx + \rho\,dR$, the base point is $(S,x,R) = 0$, $\bar\lambda = dS$, and $x' = (S,x)$, $x'' = R$, $\theta = (\sigma,\xi)$ in the notation of Theorem 10.5.11. On Λ, with $q = (1+x_0^2)^{1/2}$,

$$S = \tfrac{1}{2}x_0^2 + t(w-1), \quad R = \tfrac{1}{4}x_0^2 + \tfrac{1}{2}t(1+w) - 1, \quad \sigma = \tfrac{1}{2}\eta(1+q), \quad \xi = -x_0\eta, \quad \rho = \eta(1-q).$$

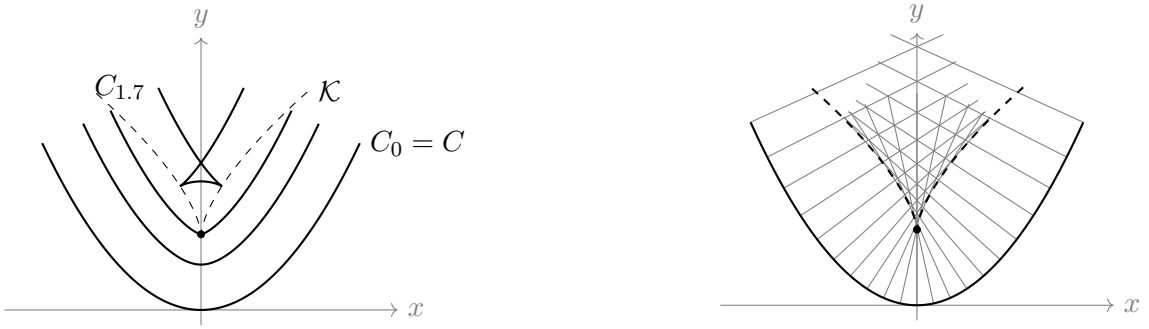


Figure 10.4: Keller's example, projected to the (x, y)-plane. Left: the parallel curves C_t of the parabola for $t = 0$, 0.6, 1, 1.7 and the caustic $\mathcal{K}$ (dashed). Right: the light rays, i.e. the projections of the null bicharacteristics making up Λ, are the normal lines of C; their envelope is the caustic. The dot is the cusp $(0, 1)$ of $\mathcal{K}$, reached at $t = 1$.

It is convenient to use the homogeneous coordinate $z = \xi/\sigma = -2x_0/(1+q)$; then

$$x_0 = -\frac{4z}{4-z^2}, \qquad q = \frac{4+z^2}{4-z^2}, \qquad w = \frac{4-z^2}{4+z^2}, \qquad |z| < 2,$$

and solving for t from (R, z) and substituting one finds, on Λ,

$$x = z(1+R) - \frac{16z}{(4-z^2)^2}, \qquad S = \frac{2z^2(4+z^2)}{(4-z^2)^2} - \frac{z^2}{2}(1+R), \qquad \frac{\rho}{\sigma} = -\frac{z^2}{2}.$$

These are $x = \partial_z F$, $S = F - z\partial_z F$ and $\rho/\sigma = -\partial_R F$ for

$$F(R, z) = \frac{z^2}{2}(1+R) - \frac{2z^2}{4-z^2} = \frac{z^2}{2}R - \frac{z^4}{8} - \frac{z^6}{32} - \cdots,$$

and $G(\sigma, \xi, R) = \sigma F(R, \xi/\sigma)$ is the generating function of Theorem 10.5.11. Thus Λ is parametrized near $\bar\lambda$ by the minimal phase function

$$\varphi(S, x, R; \sigma, \xi) = \sigma S + \xi x - \sigma F(R, \xi/\sigma) = \sigma\, h(S, x, R, \xi/\sigma), \qquad h = S + zx - F(R, z), \tag{10.7.2}$$

on $\sigma > 0$, $|\xi| < 2\sigma$, the hypersurface $\{h = 0\} \subset \mathbb{R}^3 \times \mathbb{R}_z$ being a parametrization in the sense of Proposition 10.5.8. (For $\eta < 0$ one uses $\sigma < 0$ in the same formula.) Since $\partial_z^2 F(R, 0) = R$,

$$\varphi''_{\xi\xi} = \frac{1}{\sigma}\partial_z^2 h = -\frac{R}{\sigma} \quad \text{at } z = 0.$$

The points of C_φ with $\xi = 0$ are those with $x = \partial_z F(R, 0) = 0$ and $S = F(R, 0) = 0$, i.e. they lie over the points $(t, 0, t)$ of the axis, the vertices of the fronts C_t, and there $R = t - 1$. So $\varphi''_{\theta\theta}$ has rank one (and $K = 1$) at these points for $t \neq 1$, with

$$\operatorname{sgn} \varphi''_{\theta\theta} = \operatorname{sgn} \varphi''_{\xi\xi} = \operatorname{sgn}\big((1-t)\sigma\big). \tag{10.7.3}$$

10.7.2 The phase shift across the caustic

Let $u = I_\varphi(a)$ with $a \in S^\mu$, $\mu = m + \frac{3}{4} - 1$, classical, supported in a small conic neighbourhood of $\{(0, \sigma, 0);\ \sigma > 0\}$, so that $u \in I^m(\mathbb{R}^3, \Lambda)$ near $(1, 0, 1)$. Near a point $(t, 0, t)$ with $0 < |t-1|$ small, u is microlocally conormal to the smooth sheet of the front through that point (the

part of G near its vertex for $t<1$, and the reversed arc of the front for $t>1$). Indeed Proposition 10.6.2, applied with $\theta'=\sigma$, $\theta''=\xi$ (at these points $\varphi''_{\sigma\sigma}=\varphi''_{\sigma\xi}=0$ by Euler's relation), shows that $u\equiv I_{\varphi_0}(b)$ with the phase $\varphi_0(S,x,R,\sigma)=\sigma g(S,x,R)$, where $g=h(\cdot,Z)$, $\partial_z h(\cdot,Z)=0$, defines the front, and

$$b\equiv(2\pi)^{-\frac12}e^{\frac{i\pi}{4}\operatorname{sgn}((1-t)\sigma)}\,\frac{|\sigma|^{\frac12}}{|t-1|^{\frac12}}\,a(S,x,R,\sigma,0)\quad\text{at the points over }(t,0,t).$$

Two features appear. The factor $|t-1|^{-1/2}$ shows that the conormal amplitude blows up as the vertex approaches the caustic, the intensity predicted by geometrical optics becoming infinite; this factor is the ratio $(d_{C_\varphi}/d_{C_{\varphi_0}})^{1/2}$ of Section 10.8 and is accounted for by using half-densities on Λ. The second is *Keller's observation*: after removing the factor $|t-1|^{-1/2}$ and the smooth amplitude a, the conormal symbol on the sheet beyond the caustic differs from that before it by

$$e^{\frac{i\pi}{4}(\operatorname{sgn}(1-t)-\operatorname{sgn}(1-t'))\operatorname{sgn}\sigma}\Big|_{t>1>t'}=e^{-\frac{i\pi}{2}\operatorname{sgn}\sigma}=\mp i\quad\text{for }\pm\sigma>0.$$

So one must expect factors $e^{\pm i\pi/2}$ in passing from one conormal description of a Lagrangian distribution to another across a caustic. These are the transition functions of the Keller–Maslov bundle. (Solutions of $Pu\in\mathcal{C}^\infty$ of this form exist: the travelling waves with initial data conormal to C constructed in Theorem 10.4.4 continue, by the theory of Fourier integral operators [6], as Lagrangian distributions associated with Λ through the caustic, with principal symbol satisfying a transport equation along Λ.)

10.8 The principal symbol and the Maslov bundle

For conormal distributions the principal symbol is a half-density on N^*Y. For a general Lagrangian the amplitude a restricted to C_φ depends on the phase φ, and Keller's example shows the two ways in which it does: through a Jacobian, absorbed by working with half-densities, and through the factor $e^{\frac{i\pi}{4}\operatorname{sgn}\varphi''_{\theta\theta}}$, absorbed by a flat line bundle.

Discussion. In the model of Figure 10.3, $\varphi''_{\theta\theta}=2\theta$, so $\operatorname{sgn}\varphi''_{\theta\theta}=\pm1$ on the two sheets. Stationary phase in θ turns the Airy integral $\int e^{i(x\theta+\theta^3/3)}\,d\theta$, for $x<0$, into a sum of two terms $e^{ih_\pm(x)}$ with amplitudes containing $e^{\pm\frac{i\pi}{4}}$: the two sheets differ by the phase $e^{\frac{i\pi}{2}}$, exactly as in Keller's example. The Maslov bundle records these factors once and for all, so that the principal symbol becomes an invariant object on Λ.

10.8.1 The density on the critical set

Let φ be a non-degenerate phase function on $\Gamma\subset X\times(\mathbb{R}^N\setminus0)$, $X\subset\mathbb{R}^n$ open, and $F_j=\partial_{\theta_j}\varphi$. Near C_φ there are local coordinates (c,F) with c restricting to coordinates on C_φ; if $|dx\,d\theta|=J(c,F)|dc\,dF|$, we set $d_{C_\varphi}=J(c,0)|dc|$. This density on C_φ is independent of the choice of c (if $\tilde c=\tilde c(c,F)$ then $|dc\,dF|=|\det\partial_c\tilde c|^{-1}|d\tilde c\,dF|$), and it is homogeneous of degree N under $\theta\mapsto t\theta$, since $|dx\,d\theta|$ is homogeneous of degree N and the F_j of degree 0. Thus for $a\in S^{\mu_N}_\Gamma$, $(j_\varphi)_*(a|_{C_\varphi}\sqrt{d_{C_\varphi}})$ is a half-density on Λ_φ of "order" $\mu_N+\frac N2=m+\frac n4$. For the conormal phase $x\cdot\xi$ of Example 10.5.6(i), $F=x$ and $|dx\,dy\,d\xi|=|dy\,d\xi|\,|dF|$, so $d_{C_\varphi}=|dy\,d\xi|$.

10.8.2 Symbol spaces

Let Λ be a conic manifold (with the action of M_t, $t > 0$, free and proper). A function $g \in \mathcal{C}^\infty(\Lambda)$ is a *symbol of order* 0 if $V_1 \cdots V_k g$ is bounded on $\{\lambda \in U;\ |\lambda| \geq 1\}$ for every k, every conic set U with compact image in $\Lambda/\mathbb{R}_+$, and all smooth vector fields V_j commuting with the dilations; here $|\lambda|$ is any positive function homogeneous of degree one. The space $S^r(\Lambda; \Omega^{\frac12})$ consists of the half-densities $g\beta_0$, locally, with g a symbol of order 0 and β_0 a smooth non-vanishing half-density homogeneous of degree r. For $\Lambda = N^*Y$, in coordinates (y, ξ) the vector fields ∂_y and $\xi_i \partial_{\xi_j}$ generate those commuting with dilations and $|dy\, d\xi|^{1/2}$ is homogeneous of degree $\frac{k}{2}$, so this is the space $S^r(N^*Y; \Omega^{\frac12})$ of Chapter 4 (restricted to $N^*Y \setminus 0$). The quotient $S^{r-[1]} = S^r/S^{r-1}$ and, for flat line bundles L, the spaces $S^r(\Lambda; \Omega^{\frac12} \otimes L)$ are defined in the same way using local flat trivializations.

10.8.3 The Maslov bundle

Let $\Lambda \subset T^*X \setminus 0$ be a conic Lagrangian submanifold and cover it by open conic sets Λ_{φ_i} parametrized by non-degenerate phase functions φ_i in N_i variables (in charts of X). Let $s_i(\lambda) = \operatorname{sgn} \varphi_i''{}_{\theta\theta}(j_{\varphi_i}^{-1}(\lambda))$. By Corollary 10.5.14, the functions

$$g_{ij} = e^{\frac{i\pi}{4}(s_j - s_i)} \quad \text{on } \Lambda_{\varphi_i} \cap \Lambda_{\varphi_j}$$

are locally constant (with values in the eighth roots of unity, in fact in $\{\pm 1, \pm i\}$ times $e^{\frac{i\pi}{4}(N_j - N_i)}$), and they trivially satisfy $g_{ij} g_{jk} = g_{ik}$.

Definition 10.8.1. The *Keller–Maslov bundle* M_Λ is the flat complex line bundle over Λ defined by the cocycle (g_{ij}): a section is a family of functions c_i on Λ_{φ_i} with $c_i = g_{ij} c_j$ on the overlaps, i.e. such that the (in general discontinuous) function $e^{\frac{i\pi}{4} s_i} c_i$ is independent of i.

Adding more parametrizations to the cover does not change the bundle (up to canonical isomorphism), by Corollary 10.5.14 again. Over a conormal bundle $N^*Y \setminus 0$ the phases $x \cdot \xi$ in adapted coordinates all have $\varphi''_{\theta\theta} = 0$, so they define a canonical trivialization of M_{N^*Y}. In Keller's example, (10.7.3) shows that the trivializations of M_Λ defined by conormal phases for the sheets of the front before and after the caustic differ by the factor $e^{\mp i\pi/2}$ found in Section 10.7. Each component of that Λ is contractible, so M_Λ is trivial there, but no trivialization is compatible with all conormal descriptions. In general M_Λ need not be trivial: since $\operatorname{sgn} \varphi''_{\theta\theta}$ jumps by ± 2 when a curve in Λ crosses the caustic transversally (in the generic situation), the holonomy of M_Λ around a closed curve γ is $i^{\pm \operatorname{ind}(\gamma)}$, where $\operatorname{ind}(\gamma)$ is the Maslov index of γ, an integer counting these crossings with signs; see [8, Chapter 21, Section 25.1].

10.8.4 The principal symbol

Theorem 10.8.2. *Let $\Lambda \subset T^*X \setminus 0$ be a closed conic Lagrangian submanifold. There is a unique linear map*

$$\sigma_m : I^m(X, \Lambda; \Omega^{\frac12}) \longrightarrow S^{m + \frac{n}{4} - [1]}(\Lambda; \Omega^{\frac12} \otimes M_\Lambda)$$

such that, whenever φ parametrizes Λ on Λ_φ and $\bar\lambda \notin \mathrm{WF}(u - I_\varphi(a)|dx|^{1/2})$ with $a \in S^{\mu_N}_\Gamma$, then near $\bar\lambda$, in the trivialization of M_Λ given by φ,

$$\sigma_m(u) = (2\pi)^{\frac{n}{4} - \frac{N}{2}} (j_\varphi)_* \big(a|_{C_\varphi} \sqrt{d_{C_\varphi}}\big) \mod S^{m + \frac{n}{4} - 1}. \tag{10.8.1}$$

Equivalently: if $\bar\lambda \notin \mathrm{WF}(u - I_{\varphi_i}(a_i)|dx|^{1/2})$ *for* $i = 1, 2$, *then near* $\bar\lambda$

$$(2\pi)^{-\frac{N_1}{2}} e^{\frac{i\pi}{4}s_1}(j_{\varphi_1})_*(a_1\sqrt{d_{C_{\varphi_1}}}) \equiv (2\pi)^{-\frac{N_2}{2}} e^{\frac{i\pi}{4}s_2}(j_{\varphi_2})_*(a_2\sqrt{d_{C_{\varphi_2}}}) \mod S^{m+\frac{n}{4}-1}. \quad (10.8.2)$$

There is a short exact sequence

$$0 \longrightarrow I^{m-1}(X, \Lambda; \Omega^{\frac12}) \longrightarrow I^m(X, \Lambda; \Omega^{\frac12}) \xrightarrow{\sigma_m} S^{m+\frac{n}{4}-[1]}(\Lambda; \Omega^{\frac12} \otimes M_\Lambda) \longrightarrow 0,$$

and similarly for the classical spaces, with values in the smooth sections of $\Omega^{\frac12} \otimes M_\Lambda$ *homogeneous of degree* $m + \frac{n}{4}$. *For* $\Lambda = N^*Y \setminus 0$, *with the canonical trivialization of* M_Λ,

$$\sigma_m(u) = (2\pi)^{\frac{n}{4}-\frac{k}{2}}\, \sigma_m^Y(u), \qquad k = \operatorname{codim} Y, \quad (10.8.3)$$

where σ_m^Y *is the principal symbol of Chapter 4.*

The normalizing constant $(2\pi)^{\frac{n}{4}-\frac{N}{2}}$ is chosen so that (10.8.1) is the symbol of [6] and [8, Section 25.1], where oscillatory integrals are normalized by the factor $(2\pi)^{-\frac{n+2N}{4}}$ instead of our $(2\pi)^{-N}$. Since the relation between the two symbols in (10.8.3) depends on k, no normalization makes it the identity for all submanifolds at once.

Proof (with the analytic steps sketched). (a) Equivalent phases. Let $\varphi_2 \circ \kappa = \varphi_1$, $\kappa(x, \theta) = (x, \Theta(x, \theta))$, and $a_1 = (a_2 \circ \kappa)|\det \partial_\theta \Theta|$ as in Lemma 10.6.1(i). On C_{φ_1}, $\partial_\theta \varphi_1 = (\partial_\theta \Theta)^t (\partial_\theta \varphi_2 \circ \kappa)$, so $d(\partial_\theta \varphi_1) = (\partial_\theta \Theta)^t\, \kappa^* d(\partial_\theta \varphi_2)$ there, and $\kappa^*|dx\, d\theta| = |\det \partial_\theta \Theta|\, |dx\, d\theta|$; hence $\kappa^* d_{C_{\varphi_2}} = |\det \partial_\theta \Theta|^2 d_{C_{\varphi_1}}$ and $a_1\sqrt{d_{C_{\varphi_1}}} = \kappa^*(a_2\sqrt{d_{C_{\varphi_2}}})$. Also $j_{\varphi_2} \circ \kappa = j_{\varphi_1}$ on C_{φ_1}, $N_1 = N_2$ and $s_1 = s_2$ (proof of Corollary 10.5.14). So both sides of (10.8.2) agree exactly.

(b) Reduction. In the situation of Lemma 10.5.12, use near C_φ the coordinates (x, θ', w), $w = \partial_{\theta''}\varphi$, for which $|dx\, d\theta| = |\det \varphi''_{\theta''\theta''}|^{-1} |dx\, d\theta'\, dw|$; on $\{w = 0\} = s(\{(x, \theta')\})$ the remaining defining functions $\partial_{\theta'}\varphi$ restrict to $\partial_{\theta'}\varphi_0$. Hence $s^* d_{C_\varphi} = |\det D|^{-1} d_{C_{\varphi_0}}$, and with (10.6.3) and (10.5.2)

$$\begin{aligned}(2\pi)^{-\frac{K}{2}} e^{\frac{i\pi}{4}\operatorname{sgn}\varphi_0''} b\sqrt{d_{C_{\varphi_0}}} &\equiv (2\pi)^{-\frac{K}{2}-\frac{r}{2}} e^{\frac{i\pi}{4}(\operatorname{sgn}\varphi_0'' + \operatorname{sgn} D)} |\det D|^{-\frac12} a\, |\det D|^{\frac12} s^*\sqrt{d_{C_\varphi}} \\ &= (2\pi)^{-\frac{N}{2}} e^{\frac{i\pi}{4}\operatorname{sgn}\varphi''} s^*(a\sqrt{d_{C_\varphi}}),\end{aligned}$$

where a is evaluated at $\theta'' = \Theta$ and $\varphi'' = \varphi''_{\theta\theta} \circ s$. So (10.8.2) holds for a phase and its reduction.

(c) Independence of the amplitude. By the proof of Theorem 10.6.4, (a) and (b) reduce (10.8.2) to the following statement: if φ is minimal at the point over $\bar\lambda$ and $\bar\lambda \notin \mathrm{WF}(I_\varphi(a) - I_\varphi(a'))$ with $a \in S^\mu$, $a' \in S^{\mu-1}$, then $a|_{C_\varphi} \in S^{\mu-1}$ near the point over $\bar\lambda$. By (a), Theorem 10.5.13 and Lemma 10.6.1(ii) we may take $\varphi = x' \cdot \theta - G(\theta, x'')$ as in Theorem 10.5.11. Since $C_\varphi = \{x' = \partial_\theta G\}$, Lemma 10.6.1(iii) allows us to replace a and a' by their restrictions to C_φ, regarded as functions of (x'', θ) independent of x' (a cut-off in x' is unnecessary: for $|x'|$ large $|d_\theta \varphi| \geq |x'|/2$, and integration by parts shows that the resulting integrals are rapidly decreasing in x' with all derivatives). Then, for $c = a - a'$ cut off to a small conic neighbourhood of $(\bar x'', \bar\theta)$,

$$v(x', x'') = \int e^{ix'\cdot\theta}(e^{-iG(\theta, x'')} c(x'', \theta))\, \bar d\theta$$

is, for each x'', the inverse Fourier transform in x' of $e^{-iG}c$. By (10.6.1), $\mathrm{WF}(v)$ lies in Λ, where $|\xi''| \leq C|\xi'|$, and by hypothesis it avoids a conic neighbourhood of $\bar\lambda$; so for $\psi \in \mathcal{C}_c^\infty$ supported near $\bar x''$, the Fourier transform of $\psi(x'')v$ in all variables is rapidly decreasing

in the region where ξ' lies in a small cone around $\bar\theta$ (both where $|\xi''| \le 2C|\xi'|$, by the hypothesis, and where $|\xi''| \ge 2C|\xi'|$, which meets no point of Λ). Integrating out ξ'', the partial Fourier transform $\psi(x'')e^{-iG}c(x'',\theta)$ is rapidly decreasing, with all derivatives, for θ in that cone. Since the derivatives of e^{iG} grow only polynomially, $c \in S^{-\infty}$ near $(\bar x'', \bar\theta)$; hence $a|_{C_\varphi} \in S^{\mu-1}$ there.

(d) Conclusion. By (a)–(c) the local expressions (10.8.1) are well defined modulo $S^{m+\frac{n}{4}-1}$ and are compatible on overlaps with the transition functions of M_Λ, so they define $\sigma_m(u)$. If $u \in I^{m-1}$ its local amplitudes have order $\mu_N - 1$ and $\sigma_m(u) = 0$. Conversely if $\sigma_m(u) = 0$, the local amplitudes satisfy $a|_{C_\varphi} \in S^{\mu_N-1}$, and Lemma 10.6.1(iii) shows that u is microlocally an oscillatory integral of order $m-1$ near each point, i.e. $u \in I^{m-1}$. For surjectivity, take a locally finite cover of Λ by conic sets Λ_{φ_j} with a subordinate partition of unity ψ_j by functions homogeneous of degree 0; given β, extend the coefficient $(2\pi)^{\frac{N}{2}-\frac{n}{4}}(\psi_j\beta)/\sqrt{d_{C_{\varphi_j}}}$ (pulled back to C_{φ_j}) to an amplitude $a_j \in S^{\mu_{N_j}}_{\Gamma_j}$, and let $u = \sum_j I_{\varphi_j}(a_j)|dx|^{1/2}$ (with cut-offs in x to make the sum locally finite). By (10.6.1), $u \in I^m(X,\Lambda;\Omega^{\frac12})$, and $\sigma_m(u) = \sum_j \psi_j\beta = \beta$. The classical case is the same. Finally, (10.8.3) follows from (10.8.1) with $N = k$, $d_{C_\varphi} = |dy\,d\xi|$ and the definition of σ^Y_m in Chapter 4. □

Remark 10.8.3. The results of Sections 10.1 and 10.4 extend to Lagrangian distributions [8, Chapter 25], [6]: if $A \in \Psi^{m'}(X;\Omega^{\frac12})$ is properly supported then $\sigma_{m+m'}(Au) = \sigma_{m'}(A)|_\Lambda\,\sigma_m(u)$; and if $\Lambda \subset \Sigma(P)$, so that H_p is tangent to Λ (Lemma 10.2.4), then $Pu \in I^{m+m'-1}(X,\Lambda;\Omega^{\frac12})$ and $\sigma_{m+m'-1}(Pu) = (-i\mathcal{L}_{H_p} + p_s)\sigma_m(u)$, where the Lie derivative acts on $\Omega^{\frac12}\otimes M_\Lambda$ using the flat structure of M_Λ. For $\Lambda = N^*Y$ these are Theorem 10.1.1 and Theorem 10.4.1. Since the transport equation is an ordinary differential equation along the bicharacteristics in Λ, it can be solved through caustics, where the methods of Section 10.4 break down; this is how the travelling waves of Theorem 10.4.4 are continued for all time, and in Keller's example it produces the phase shift $e^{\mp i\pi/2}$ of Section 10.7.

10.9 Oscillatory testing

Discussion. The idea is that of the wavefront set itself. To decide whether u is singular at (x,ξ) one localizes near x and takes the Fourier transform in the direction ξ. Here we localize with ν and replace the linear phase $x\cdot\xi$ by a non-linear phase f with $df(\bar x) = \bar\xi$, chosen so that its graph meets Λ transversally at $\bar\lambda$. Then stationary phase picks out the single point $\bar\lambda$, and the leading coefficient of the resulting symbol in s is the value of $\sigma_m(u)$ there.

The symbol of a conormal distribution is obtained by a Fourier transform across the submanifold. For a Lagrangian distribution there is an analogous procedure: pair u with an oscillatory test function $e^{-isf}\nu$ whose phase f has differential at a point equal to a given $\lambda \in \Lambda$, and study the behaviour as $s \to \infty$.

Lemma 10.9.1. *Let $\Lambda \subset T^*X \setminus 0$ be a conic Lagrangian and $\bar\lambda = (\bar x, \bar\xi) \in \Lambda$.*

(i) There is a real $f \in \mathcal{C}^\infty(X)$ with $f(\bar x) = 0$, $df(\bar x) = \bar\xi$ and graph$(df) = \{(x, df(x))\}$ *transversal to Λ at $\bar\lambda$; then $\bar\lambda$ is an isolated point of* graph$(df) \cap \Lambda$.

(ii) If φ parametrizes Λ near $\bar\lambda$ and $j_\varphi(\bar x,\bar\theta) = \bar\lambda$, then $\Phi(x,\theta) = \varphi(x,\theta) - f(x)$ has a critical point at $(\bar x,\bar\theta)$, with $\Phi(\bar x,\bar\theta) = 0$, and its Hessian $\Phi''(\bar x,\bar\theta)$ in all variables (x,θ) is non-degenerate if and only if graph(df) *is transversal to Λ at $\bar\lambda$.*

Proof. (i) In the coordinates of Theorem 10.5.11, $\Lambda = \{x' = \partial_\theta G,\ \xi'' = -\partial_{x''}G\}$ near $\bar\lambda$, and since $dx' = 0$ on $T_{\bar\lambda}\Lambda$, $\partial^2_{\theta\theta}G = 0$ and $\partial^2_{\theta x''}G = 0$ at $\bar\lambda$. So $T_{\bar\lambda}\Lambda = \{\delta x' = 0,\ \delta\xi'' = -G_{x''x''}\delta x''\}$, with $G_{x''x''} = \partial^2_{x''}G(\bar\theta, 0)$. For $f = x_1 + \frac12\langle Bx'', x''\rangle$ the tangent space of the graph of df at $\bar\lambda$ is $\{\delta\xi' = 0,\ \delta\xi'' = B\delta x''\}$, and the intersection is $\{\delta x' = 0,\ \delta\xi' = 0,\ (B + G_{x''x''})\delta x'' = 0\}$, which is zero if $B + G_{x''x''}$ is invertible. Both submanifolds have dimension n, so this is transversality, which implies that the intersection is a discrete set near $\bar\lambda$. (ii) At points of C_φ, $d_\theta\Phi = 0$, and $d_x\Phi = d_x\varphi - df = 0$ exactly when $j_\varphi(x,\theta) \in \operatorname{graph}(df)$; and $\varphi = 0$ on C_φ. A vector $(\delta x, \delta\theta)$ is in the null space of the Hessian if and only if $\varphi''_{\theta x}\delta x + \varphi''_{\theta\theta}\delta\theta = 0$, i.e. it is tangent to C_φ, and $\varphi''_{xx}\delta x + \varphi''_{x\theta}\delta\theta = f''\delta x$, i.e. $dj_\varphi(\delta x, \delta\theta) = (\delta x, f''\delta x)$ is tangent to $\operatorname{graph}(df)$. Since dj_φ is injective, the null space is isomorphic to $T_{\bar\lambda}\Lambda \cap T_{\bar\lambda}\operatorname{graph}(df)$. □

Proposition 10.9.2. *Let $u \in I^m(X, \Lambda; \Omega^{\frac12})$, $\bar\lambda \in \Lambda$ and f as in Lemma 10.9.1(i). If $\nu \in \mathcal{C}^\infty_{\mathrm{c}}(X; \Omega^{\frac12})$ has support in a sufficiently small neighbourhood of $\bar x$, then*

$$A(s) = \langle u, e^{-isf}\nu\rangle, \qquad s \geq 1,$$

is a symbol of order $m - \frac{n}{4}$ on $[1,\infty)$, classical if u is. If $\bar\lambda \notin \mathrm{WF}(u - I_\varphi(a)|dx|^{1/2})$ and $\nu = \nu_0|dx|^{1/2}$ in the chart, then, with Φ as in Lemma 10.9.1(ii),

$$A(s) \equiv (2\pi)^{\frac{n-N}{2}} s^{\frac{N-n}{2}} e^{\frac{i\pi}{4}\operatorname{sgn}\Phi''} |\det\Phi''|^{-\frac12}\nu_0(\bar x)\, a(\bar x, s\bar\theta) \mod S^{m-\frac{n}{4}-1}, \tag{10.9.1}$$

$\Phi'' = \Phi''(\bar x, \bar\theta)$. In particular, for classical u and $\nu_0(\bar x) \neq 0$, $A \in S^{m-\frac{n}{4}-1}$ if and only if the principal symbol $\sigma_m(u)$ vanishes at $\bar\lambda$.

Sketch of proof. Localization. If $w \in \mathcal{C}^{-\infty}(X; \Omega^{\frac12})$ and $\mathrm{WF}(w)$ contains no point $(x, t\,df(x))$ with $x \in \operatorname{supp}\nu$, $t > 0$, then $\langle w, e^{-isf}\nu\rangle = O(s^{-\infty})$: in local coordinates in which $f = y_1$ (possible near $\bar x$ as $df \neq 0$) the pairing is the Fourier transform of the compactly supported distribution νw at $s\,dy_1$, which is rapidly decreasing by the definition of the wavefront set (Chapter 3), using a partition of unity on $\operatorname{supp}\nu$ if necessary. Since Λ is closed and conic and $\bar\lambda$ is isolated in $\operatorname{graph}(df) \cap \Lambda$, for $\operatorname{supp}\nu$ small the only points of Λ of the form $(x, t\,df(x))$, $x \in \operatorname{supp}\nu$, are those of the ray through $\bar\lambda$. Hence $A(s) \equiv \langle I_\varphi(a)|dx|^{1/2}, e^{-isf}\nu\rangle$ modulo $O(s^{-\infty})$, and we may cut a off to a small conic neighbourhood of $(\bar x, \bar\theta)$.

Stationary phase. Substituting $\theta = s\vartheta$,

$$A(s) \equiv (2\pi)^{-N} s^N \iint e^{is\Phi(x,\vartheta)} a(x, s\vartheta)\nu_0(x)\, dx\, d\vartheta.$$

Where $|\vartheta|$ is small, $|d_x\Phi| \geq |df| - C|\vartheta| > 0$; where $|\vartheta|$ is large, $|d_x\varphi| + |\vartheta||d_\vartheta\varphi| \geq c|\vartheta|$ on the support forces $d\Phi \neq 0$; in both regions integration by parts gives $O(s^{-\infty})$. On the remaining compact region Φ has the single critical point $(\bar x, \bar\theta)$ (by Lemma 10.9.1), non-degenerate, with $\Phi = 0$ there. The method of stationary phase in the $n + N$ variables (x, ϑ) [7, Theorem 7.7.5] gives (10.9.1), the factor being $(2\pi)^{-N}s^N(2\pi/s)^{\frac{n+N}{2}}$; the order is $\mu_N + \frac{N-n}{2} = m - \frac{n}{4}$. For classical a, expanding $a(x, s\vartheta) \sim \sum_j s^{\mu_N - j}a_{\mu_N - j}(x, \vartheta)$ gives a full expansion of A in powers $s^{m-\frac{n}{4}-j}$. The symbol estimates for $\partial_s^k A$ follow in the same way: ∂_s produces the factors $i\Phi$, which vanishes to second order at the critical point and hence lowers the order by one, and $\vartheta \cdot \partial_\theta a(x, s\vartheta)$, of order $\mu_N - 1$ in s. The last statement follows since, for classical a, the principal symbol vanishes at $\bar\lambda$ if and only if $a_{\mu_N}(\bar x, \bar\theta) = 0$, by (10.8.1). □

Comparing the leading terms in (10.9.1) for two different representations of the same u shows directly that the vanishing of the principal symbol at a point does not depend on the parametrization; this is the injectivity statement in Theorem 10.8.2 for classical

distributions. Conversely, oscillatory testing can be used to *define* the symbol: the leading coefficient of $A(s)$ is a non-zero multiple of the value of $\sigma_m(u)$ on the ray through $\bar\lambda$, the multiple depending on f and ν (through Φ'') and on the parametrization only through the Maslov factor $e^{\frac{i\pi}{4}\operatorname{sgn}\varphi''_{\theta\theta}}$; compare the transformation law (10.8.2).

10.10 Exercises

Exercise 10.1. Let $\tilde G\subset X$ be a hypersurface and $t\in\mathcal{C}^\infty(X)$ real with $dt|_{T_q\tilde G}\neq 0$ for all $q\in G=\tilde G\cap\{t=0\}$. Show that $H=\{t=0\}$ is a hypersurface near G and that $G=H\cap\tilde G$ is a submanifold of codimension one in both H and $\tilde G$.

Exercise 10.2. Assuming $dt\neq 0$ on $H=\{t=0\}$, show that dt gives an isomorphism of line bundles $\Omega^{\frac12}(H)\cong\Omega^{\frac12}_H(X)\otimes|N^*H|^{-1/2}$, where $\Omega^{\frac12}_H(X)$ is the restriction to H of $\Omega^{\frac12}(X)$ and $|N^*H|^{-1/2}$ is trivialized by $|dt|^{-1/2}$, and hence that (10.4.3) defines a coordinate-independent restriction map $\mathcal{C}^\infty(X;\Omega^{\frac12})\to\mathcal{C}^\infty(H;\Omega^{\frac12})$.

Exercise 10.3. Prove Proposition 10.4.2 directly from the coordinate formula: show that the restriction defined in adapted coordinates is independent of the choice of coordinates (x_1,t,y') with t fixed.

Exercise 10.4. For the wave operator $P=D_t^2-\Delta$ on $\mathbb{R}^{n+1}$, $H=\{t=0\}$ and $f(x)=\omega\cdot x$, $|\omega|=1$, find the two solutions $h_\pm$ of the eikonal equation $p(x,dh)=0$ with $h|_H=f$ given by Theorem 10.3.1 (one for each choice of $\bar z$ over 0). Do the same with $f(x)=|x|$ near a point $x\neq 0$, and describe the characteristic hypersurfaces through $\{t=0,\ |x|=1\}$. Show that the light cone $\{t^2=|x|^2\}$ is characteristic away from its vertex.

Exercise 10.5. Show that the subprincipal symbol of $D_t^2-\Delta$ on $\mathbb{R}^{n+1}$, acting on half-densities via $|dt\,dx|^{1/2}$, vanishes, and solve the transport equation for the plane waves of Example 10.1.4. For a Riemannian metric g on X compute the subprincipal symbol of $D_t^2-\Delta_g$ acting on half-densities by $u\mapsto(D_t^2-\Delta_g)(u\,|dg|^{-1/2})\,|dg|^{1/2}$, $|dg|$ the Riemannian density.

Exercise 10.6. Let φ be a non-degenerate phase function in N variables parametrizing Λ, and B a real invertible symmetric $q\times q$ matrix. Show that $\tilde\varphi(x,\theta,\omega)=\varphi(x,\theta)+\langle B\omega,\omega\rangle/(2|\theta|)$, defined for $|\omega|<c|\theta|$, is a non-degenerate phase function in $N+q$ variables parametrizing the same Lagrangian, with $C_{\tilde\varphi}=C_\varphi\times\{0\}$ and $\operatorname{sgn}\tilde\varphi''=\operatorname{sgn}\varphi''+\operatorname{sgn}B$. Conclude that Λ can be parametrized near $\bar\lambda$ with any number $N\geq K(\bar\lambda)$ of phase variables, and check (10.8.2) directly for φ and $\tilde\varphi$ using (10.6.2).

Exercise 10.7. Let φ be a non-degenerate phase function with $\varphi''_{\theta\theta}=0$ at every point of C_φ. Show that Λ_φ is locally the conormal bundle of a submanifold of codimension N. (Hint: by Lemma 10.5.7, $\pi|_{\Lambda_\varphi}$ has constant rank $n-N$, so its image is locally a submanifold Y; use Lemma 10.5.1 to show $\Lambda_\varphi\subset N^*Y$.) Conversely, if $Y=\{g_1=\cdots=g_k=0\}$ with independent dg_j, show that $\varphi(x,\theta)=\sum_j\theta_jg_j(x)$ parametrizes $N^*Y\setminus 0$ and is minimal everywhere.

Exercise 10.8. In Section 10.7: (a) Show that the curvature of the parallel curve C_t at the point coming from x_0, measured with respect to the normal $(-x_0,1)w$, is $\kappa/(1-t\kappa)$ with $\kappa=(1+x_0^2)^{-3/2}$, and deduce that for $t>1$ the arc $|x_0|<(t^{2/3}-1)^{1/2}$ between the two cusps is curved the opposite way from the rest of C_t. (b) Verify the formulas for x_0,q,w in terms of z and for x, S, ρ/σ and F on Λ. (c) Show that over points of the caustic other than $(1,0,1)$ the phase (10.7.2) (or its analogue at other base points) has $\varphi''_{\theta\theta}$ of rank one, and that the jump of $\operatorname{sgn}\varphi''_{\theta\theta}$ across $\mathcal{K}$ along Λ is again ±2.

Exercise 10.9. Let $\bar\lambda \in \Lambda$, $\bar x = \pi(\bar\lambda)$, and let $S \subset X$ be a submanifold through $\bar x$ with $\bar\lambda \in N^*S$ and $T_{\bar\lambda}N^*S = T_{\bar\lambda}\Lambda$. Show that such S exist (in the coordinates of Theorem 10.5.11 take $S = \{x' = Q(x'')\}$ with Q quadratic) and that, for real f with $df(\bar x) = \bar\lambda$, $\operatorname{graph}(df)$ is transversal to Λ at $\bar\lambda$ if and only if $f|_S$ has a non-degenerate critical point at $\bar x$. Deduce that transversality is an open condition: it persists at $\lambda' = (x', df(x'))$ for small perturbations $f + \epsilon g$ with $\lambda' \in \Lambda$, even though $K(\lambda')$ may differ from $K(\bar\lambda)$.

Notes

This chapter is a rewriting of the second half of the conormal/wave kernel chapter of the 1998 notes. The section on Riemann metrics and quantization there was only a heading and has been omitted. The Hamilton–Jacobi theorem is proved here in full (the original only sketched the strategy), the transport equation is derived for submanifolds of any codimension with the zeroth order term identified as the subprincipal symbol, the restriction theorem (an exercise in the original) is proved, and the construction of travelling waves is made precise. For the general theory see [8, Sections 18.1, 25.2] and [6], where the analogue of (10.4.2) is proved on any Lagrangian submanifold contained in $\Sigma(P)$.

The sections on Lagrangian parametrization, Lagrangian distributions, Keller's example and oscillatory testing are based on Sections 9.2–9.5 of Melrose's revised notes *From microlocal to global analysis* (2008), which are themselves only partly written. We work throughout with Hörmander's homogeneous phase functions and explain how Melrose's parametrization by hypersurfaces in $X \times \mathbb{R}^k$ (and his definition of Lagrangian distributions as push-forwards of conormal distributions) fits in (Proposition 10.5.8 and Remark 10.5.9), keeping track of orders, which the source does not do. The equivalence of minimal parametrizations is proved in full by Moser's method in the homogeneous setting, where minimality is exactly what makes the interpolating phases non-degenerate; the reduction of non-minimal phases is done by stationary phase, which we only sketch. The principal symbol, its Maslov factor and the exact sequence, left by the source for a later section, are stated precisely, with the analytic steps of the proof sketched; for complete proofs see [6], [8, Sections 21.2, 25.1]. In Keller's example we have corrected and completed the computations: the caustic is $\{x = -x_0^3,\ y = 1 + \frac{3}{2}x_0^2\}$ (the same set as in the source), the generating function is $F(R, z) = \frac{z^2}{2}(1 + R) - \frac{2z^2}{4 - z^2}$ (the source has $F = x\xi$, which omits the term coming from S), and the Hessian of the phase at the vertices is $(1 - t)/\sigma$; accordingly the phase shift across the caustic is $e^{-i\pi/2}$ on the half $\sigma > 0$ and $e^{i\pi/2}$ on $\sigma < 0$, rather than "multiplication by i". The source's proof that the oscillatory test is a classical symbol (by exhibiting it as the Fourier transform of a conormal distribution at a point) is replaced by a stationary phase argument, which also gives the leading term. The "problems" referred to in the source are Exercises 10.6 to 10.9.

Chapter 11

The wave equation and Weyl asymptotics

In this chapter the theory of conormal distributions (Chapter 4) and the Hamilton–Jacobi and transport equations (Chapter 10) are applied to the wave operator $P = D_t^2 - \Delta$ on $\mathbb{R} \times X$, where X is a compact Riemannian manifold. We first solve the Cauchy problem and construct the forward fundamental solution, with energy estimates and finite speed of propagation. Then we construct, for short times, a parametrix for the wave kernel as a superposition of conormal distributions associated with characteristic hypersurfaces (equivalently, as an oscillatory integral whose phase solves the eikonal equation). Restricting this parametrix to the diagonal and integrating gives the singularity at $t = 0$ of the trace of the wave group, $T(t) = \sum_j \cos(t\lambda_j)$; a Tauberian argument then yields Weyl's asymptotic formula for the eigenvalues of Δ with remainder $O(\lambda^{n-1})$. Finally we explain how the propagation of singularities leads to the Poisson relation: the singularities of T lie at 0 and at the lengths of closed geodesics. Figure 11.1 summarizes the logic.

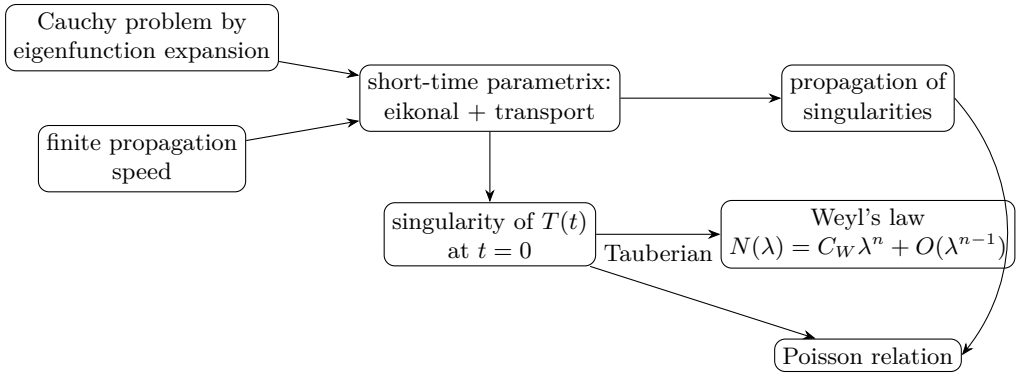


Figure 11.1: Logic of the chapter. The analysis of the wave kernel for short times gives the leading singularity of the wave trace, and hence Weyl's law; propagation of singularities controls the wave trace for all times.

Discussion. Why should the wave equation say anything about eigenvalues? The wave trace $T(t) = \sum_j \cos(t\lambda_j)$ is, up to a factor, the Fourier transform of the counting measure $\sum_j \delta_{\lambda_j}$. Large-$\lambda$ behaviour of the counting measure corresponds to the singularities of T in t, and the strongest singularity sits at $t = 0$. On the other hand, by finite propagation speed the wave kernel for small $|t|$ only depends on the geometry of X near the diagonal, where it can be computed by a parametrix. Weyl's law comes from matching these two descriptions of T near $t = 0$.

11.1 Setup

11.1.1 The Laplacian on a compact manifold

Throughout, X is a compact $\mathcal{C}^\infty$ manifold without boundary, of dimension $n \geq 1$, with a Riemannian metric g. In local coordinates $g = \sum g_{ij}(x)\, dx^i dx^j$, $(g^{ij}) = (g_{ij})^{-1}$, $|g| = \det(g_{ij})$, the Riemannian measure is $d\mathrm{vol} = |g|^{1/2} dx$ and the (non-negative) Laplacian is

$$\Delta u = -|g|^{-1/2} \sum_{i,j=1}^{n} \partial_i (|g|^{1/2} g^{ij} \partial_j u), \qquad \int_X \Delta u\, \bar{v}\, d\mathrm{vol} = \int_X \langle du, dv \rangle_g \, d\mathrm{vol}.$$

For a covector we write $|\xi|_{g(x)} = (\sum g^{ij}(x)\xi_i\xi_j)^{1/2}$ and set $q(x,\xi) = |\xi|_{g(x)}$. We use $d\mathrm{vol}$ to identify densities with functions: $\mathcal{C}^{-\infty}(X)$ is the dual of $\mathcal{C}^\infty(X)$ under $\langle u, v\rangle = \int_X uv\, d\mathrm{vol}$, and the Schwartz kernel K_A of an operator A is taken with respect to $d\mathrm{vol}$, $Au(x) = \int_X K_A(x,y)u(y)\, d\mathrm{vol}(y)$. In particular $K_{\mathrm{Id}} = |g(y)|^{-1/2}\delta(x-y)$ in local coordinates.

We use the following consequences of the elliptic theory on compact manifolds (Chapter 5; see also [15]).

(F1) There is an orthonormal basis $\{\phi_j\}_{j\geq 0}$ of $L^2(X)$ of real eigenfunctions $\phi_j \in \mathcal{C}^\infty(X)$, $\Delta\phi_j = \lambda_j^2 \phi_j$, with $0 = \lambda_0 \leq \lambda_1 \leq \lambda_2 \leq \cdots \to \infty$ (repeated with multiplicity).

(F2) For $s \in \mathbb{R}$, writing $u_j = \langle u, \phi_j\rangle$ and $\langle\lambda\rangle = (1+\lambda^2)^{1/2}$,

$$H^s(X) = \Big\{ u \in \mathcal{C}^{-\infty}(X);\ \|u\|_s^2 = \sum_j \langle\lambda_j\rangle^{2s} |u_j|^2 < \infty \Big\}$$

with a norm equivalent to the usual one. Thus $\mathcal{C}^\infty(X) = \bigcap_s H^s(X)$, $\mathcal{C}^{-\infty}(X) = \bigcup_s H^s(X)$, every $u \in \mathcal{C}^{-\infty}(X)$ is the sum of its expansion $\sum_j u_j\phi_j$ (in some H^s), and for $u \in \mathcal{C}^\infty(X)$ the expansion converges in $\mathcal{C}^\infty(X)$. For $s > n/2 + k$ there is the Sobolev embedding $\|u\|_{\mathcal{C}^k} \leq C\|u\|_s$.

For a bounded Borel function F on $[0,\infty)$ we set $F(\Lambda)u = \sum_j F(\lambda_j) u_j \phi_j$; here $\Lambda = \Delta^{1/2}$. Such an operator is bounded on every $H^s(X)$ with norm $\leq \sup|F|$. We shall use $\cos(t\Lambda)$ and $\Lambda^{-1}\sin(t\Lambda)$, the latter defined by the function $\lambda \mapsto \sin(t\lambda)/\lambda$ (equal to t at $\lambda = 0$). The counting function is

$$N(\lambda) = \#\{j;\ \lambda_j \leq \lambda\} = \text{number of eigenvalues of } \Delta \text{ in } [0, \lambda^2], \text{ with multiplicity.}$$

Lemma 11.1.1. *For every $s > n/2$ there is C_s with $\sum_j \langle\lambda_j\rangle^{-2s} |\phi_j(x)|^2 \leq C_s^2$ for all $x \in X$; hence $\sum_j \langle\lambda_j\rangle^{-2s} \leq C_s^2\, \mathrm{vol}(X)$ and $N(\lambda) \leq C_s^2\, \mathrm{vol}(X) \langle\lambda\rangle^{2s}$.*

Proof. By the Sobolev embedding, $u \mapsto u(x)$ is a continuous linear functional on $H^s(X)$ of norm at most C_s, uniformly in x. In the spectral description (F2), $u(x) = \sum_j u_j\phi_j(x)$, and the norm of this functional is exactly $(\sum_j \langle\lambda_j\rangle^{-2s} |\phi_j(x)|^2)^{1/2}$. Integrating over X gives the second statement, and $N(\lambda)\langle\lambda\rangle^{-2s} \leq \sum_{\lambda_j \leq \lambda} \langle\lambda_j\rangle^{-2s}$ gives the third. □

In particular $\sum_j \langle\lambda_j\rangle^{-M} < \infty$ for $M > n$, and the sup norms of ϕ_j and their derivatives grow at most polynomially in λ_j.

11.1.2 The wave operator and its bicharacteristics

The wave operator is $P = D_t^2 - \Delta$ on $\mathbb{R} \times X$, $D_t = -i\partial_t$. In the canonical coordinates (t, x, τ, ξ) of $T^*(\mathbb{R} \times X) = T^*\mathbb{R} \times T^*X$ its principal symbol is

$$p(t, x, \tau, \xi) = \tau^2 - |\xi|^2_{g(x)},$$

and the characteristic set is $\Sigma(P) = \mathrm{Char}(P) = \{p = 0\} \setminus 0$. On $\Sigma(P)$ we have $\tau = \pm|\xi|_g$ with $\tau \neq 0$, $\xi \neq 0$, so $d_{(\tau,\xi)}p = (2\tau, -2g^{-1}\xi) \neq 0$: P is of real principal type and $\Sigma(P)$ is a smooth conic hypersurface. We use the Hamilton vector field $H_f = \sum_j (\partial_{\xi_j} f\, \partial_{x_j} - \partial_{x_j} f\, \partial_{\xi_j})$ (with the corresponding τ, t terms on $T^*(\mathbb{R} \times X)$); a *null bicharacteristic* of P is an integral curve of H_p in $\Sigma(P)$. Let

$$G_s = \exp(sH_q) \colon T^*X \setminus 0 \to T^*X \setminus 0, \qquad q = |\xi|_g,$$

be the (homogeneous) geodesic flow. It is complete since X is compact, commutes with dilations in ξ and preserves q, and the projection to X of $s \mapsto G_s(x, \xi)$ is the unit speed geodesic with initial velocity $g^{-1}\xi/|\xi|_g$: indeed $H_q = q^{-1}H_{q^2/2}$ on $T^*X \setminus 0$, and Hamilton's equations for $\frac{1}{2}q^2$ are the geodesic equations ($\dot{x} = g^{-1}\xi$ is the Legendre transform).

Lemma 11.1.2. *Along a null bicharacteristic, τ is constant and t is strictly monotone. Parametrised by t, the null bicharacteristic through $(\bar{t}, \bar{x}, \tau, \bar{\xi})$ with $\tau = \mp|\bar{\xi}|_g$ is*

$$t \longmapsto \bigl(t, G_{\pm(t-\bar{t})}(\bar{x}, \bar{\xi}), \tau\bigr), \qquad t \in \mathbb{R}.$$

In particular its projection to X is a unit speed geodesic.

Proof. $H_p = 2\tau\partial_t - H_{q^2} = 2\tau\partial_t - 2qH_q$, so $\dot{\tau} = 0$, $\dot{t} = 2\tau \neq 0$ and $d(x, \xi)/dt = -(q/\tau)H_q = \pm H_q$ when $\tau = \mp q$. □

11.1.3 Conormal distributions and oscillatory integrals

We recall from Chapter 4 the local description of conormal distributions associated with a hypersurface, which is all we need. Let M be a manifold of dimension N and $H \subset M$ a closed embedded hypersurface. Then $u \in I^m(M, H)$ if $u \in \mathcal{C}^{-\infty}(M)$ is smooth on $M \setminus H$ and near each point of H, in local coordinates $(s, z) \in \mathbb{R} \times \mathbb{R}^{N-1}$ with $H = \{s = 0\}$,

$$u = \frac{1}{2\pi} \int_{\mathbb{R}} e^{is\rho} a(z, \rho)\, d\rho + \mathcal{C}^\infty, \qquad a \in S^{m+\frac{N}{4}-\frac{1}{2}}(\mathbb{R}^{N-1}_z; \mathbb{R}_\rho). \tag{11.1.1}$$

(We work with functions rather than half-densities; in fixed coordinates the two differ by a smooth non-vanishing factor.) The class of a modulo symbols of one order lower is the local form of the principal symbol. Any defining function h of H (with $dh \neq 0$ on H) may be used in place of s, since it can be taken as a coordinate; and amplitudes $a(s, z, \rho)$ depending also on s give elements of the same space (Chapter 4; a special case is proved in Lemma 11.5.3 below). For $M = \mathbb{R}$, $H = \{0\}$: a compactly supported $u \in \mathcal{C}^{-\infty}(\mathbb{R})$ lies in $I^m(\mathbb{R}, \{0\})$ iff $\hat{u} \in S^{m-\frac{1}{4}}(\mathbb{R})$. We always have $\mathrm{WF}(u) \subset N^*H$.

A hypersurface $H = \{h = 0\}$ is *characteristic* for P if $N^*H \subset \Sigma(P)$, i.e. $p(dh) = 0$ on H; for P this is the eikonal equation $(\partial_t h)^2 = |d_x h|^2_g$. Characteristic hypersurfaces are obtained by Hamilton–Jacobi theory, and the symbols of conormal solutions of $Pu \in \mathcal{C}^\infty$ satisfy transport equations along H_p (Chapter 10). For P we carry this out explicitly in Section 11.4.

Finally we use standard facts about oscillatory integrals [7, Sections 8.1, 8.2]. Let $\Omega \subset \mathbb{R}^d$ be open and $\Phi \in \mathcal{C}^\infty(\Omega \times (\mathbb{R}^N \setminus 0))$ real, homogeneous of degree one in θ, with

$d_{z,\theta}\Phi \neq 0$, and let $a \in S^{\mu}(\Omega;\mathbb{R}^N)$ vanish near $\theta = 0$. Then $I(z) = \int e^{i\Phi(z,\theta)}a(z,\theta)\,d\theta$, defined as $\lim_{\epsilon\downarrow 0}\int e^{i\Phi}a(z,\theta)\chi_0(\epsilon\theta)\,d\theta$ for any $\chi_0 \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^N)$ with $\chi_0(0) = 1$, is a distribution on Ω, and

$$\mathrm{WF}(I) \subset \{(z, d_z\Phi(z,\theta));\ d_\theta\Phi(z,\theta) = 0,\ \theta \in \operatorname{cone\,supp} a\}. \tag{11.1.2}$$

In particular $I \in \mathcal{C}^\infty$ if $a \in S^{-\infty}$, or near points where $d_\theta\Phi \neq 0$ on the support of a. If some of the variables z are parameters t and $d_{z',\theta}\Phi \neq 0$ in the remaining variables, I is a smooth function of t with values in distributions.

11.2 The Cauchy problem

Theorem 11.2.1. *Let $s \in \mathbb{R}$, let $I \subset \mathbb{R}$ be an open interval containing 0, and let $u_0 \in H^s(X)$, $u_1 \in H^{s-1}(X)$, $f \in \mathcal{C}^0(I;H^{s-1}(X))$. There is a unique $u \in \mathcal{C}^0(I;H^s)\cap\mathcal{C}^1(I;H^{s-1})$ with*

$$Pu = f \quad \text{in } I\times X, \qquad u(0) = u_0, \qquad D_t u(0) = u_1, \tag{11.2.1}$$

namely

$$u(t) = \cos(t\Lambda)u_0 + i\Lambda^{-1}\sin(t\Lambda)u_1 - \int_0^t \Lambda^{-1}\sin((t-r)\Lambda)f(r)\,dr. \tag{11.2.2}$$

It satisfies, for $t \in I$, the energy estimate

$$\|u(t)\|_s + \|D_t u(t)\|_{s-1} \leq 4(1+|t|)\Big(\|u_0\|_s + \|u_1\|_{s-1} + \Big|\int_0^t \|f(r)\|_{s-1}dr\Big|\Big), \tag{11.2.3}$$

and if $f = 0$ the energy $\|\partial_t u(t)\|^2_{L^2} + \|du(t)\|^2_{L^2}$ is independent of t (for $s \geq 1$). Uniqueness holds in the larger class $\mathcal{C}^2(I;H^{-M}(X))$, any M. If $u_0, u_1 \in \mathcal{C}^\infty(X)$ and $f \in \mathcal{C}^\infty(I\times X)$ then $u \in \mathcal{C}^\infty(I\times X)$.

Proof. Expanding in eigenfunctions, u solves (11.2.1) iff its coefficients $u_j(t) = \langle u(t), \phi_j\rangle$ solve $u_j'' + \lambda_j^2 u_j = -f_j$, $u_j(0) = u_{0,j}$, $u_j'(0) = iu_{1,j}$ (recall $D_t^2 = -\partial_t^2$ and $\langle \Delta u, \phi_j\rangle = \lambda_j^2 u_j$), whose unique solution is the j-th coefficient of (11.2.2). This proves uniqueness in $\mathcal{C}^2(I;H^{-M})$: if the data vanish, every u_j vanishes, hence $u = 0$ by (F2). For the estimate use $|\sin(t\lambda)/\lambda| \leq \min(|t|,\lambda^{-1}) \leq 2(1+|t|)\langle\lambda\rangle^{-1}$:

$$\langle\lambda_j\rangle^s|u_j(t)| \leq \langle\lambda_j\rangle^s|u_{0,j}| + 2(1+|t|)\Big(\langle\lambda_j\rangle^{s-1}|u_{1,j}| + \Big|\int_0^t \langle\lambda_j\rangle^{s-1}|f_j(r)|\,dr\Big|\Big),$$

$$\langle\lambda_j\rangle^{s-1}|D_t u_j(t)| \leq \langle\lambda_j\rangle^s|u_{0,j}| + \langle\lambda_j\rangle^{s-1}|u_{1,j}| + \Big|\int_0^t \langle\lambda_j\rangle^{s-1}|f_j(r)|\,dr\Big|,$$

the second because $D_t u_j = i\lambda_j\sin(t\lambda_j)u_{0,j} + \cos(t\lambda_j)u_{1,j} + i\int_0^t \cos((t-r)\lambda_j)f_j(r)\,dr$. Taking ℓ^2 norms in j (and Minkowski's inequality for the integral) gives (11.2.3); the same bounds and dominated convergence show that (11.2.2) lies in $\mathcal{C}^0(I;H^s)\cap\mathcal{C}^1(I;H^{s-1})$. If $f = 0$ then $|u_j'|^2 + \lambda_j^2|u_j|^2$ is constant for each j, and summing gives the conservation of $\|\partial_t u\|^2 + \langle\Delta u, \bar u\rangle = \|\partial_t u\|^2 + \|du\|^2$. Smoothness is the case $Y = \{\mathrm{pt}\}$ of Proposition 11.2.2. □

The *wave group* is the map $U(t)\colon (u_0,u_1) \mapsto (u(t), D_t u(t))$ for $f = 0$; explicitly

$$U(t) = \begin{pmatrix} \cos t\Lambda & i\Lambda^{-1}\sin t\Lambda \\ i\Lambda\sin t\Lambda & \cos t\Lambda \end{pmatrix}.$$

By Theorem 11.2.1 it is a strongly continuous group of bounded operators on $H^s(X)\oplus H^{s-1}(X)$ for each s: $U(t+t') = U(t)U(t')$ by uniqueness. Its (distributional) trace is $2\sum_j \cos(t\lambda_j)$, see Section 11.6.

We need a version with parameters. Let Y be a compact manifold (possibly a point) and let Δ_X act on the first factor of $X\times Y$.

Proposition 11.2.2. *Let $I \ni 0$ be an open interval. If $w \in \mathcal{C}^2(I; H^{-M}(X \times Y))$ satisfies $(D_t^2 - \Delta_X)w = 0$ and $w(0) = D_t w(0) = 0$, then $w = 0$. If $F \in \mathcal{C}^\infty(I \times X \times Y)$ and $w_0, w_1 \in \mathcal{C}^\infty(X \times Y)$, the solution of $(D_t^2 - \Delta_X)w = F$, $w(0) = w_0$, $D_t w(0) = w_1$, given by (11.2.2) with $\Lambda = \Lambda_X$, lies in $\mathcal{C}^\infty(I \times X \times Y)$.*

Proof. Choose a metric on Y with Laplacian Δ_Y and eigenbasis ψ_l, $\Delta_Y \psi_l = \mu_l^2 \psi_l$. Then $\{\phi_j \otimes \psi_l\}$ is an orthonormal basis of $L^2(X \times Y)$ of eigenfunctions of the Laplacian $\Delta_X + \Delta_Y$ of the product metric, so by (F2) on $X \times Y$, $H^s(X \times Y)$ is characterised by $\sum_{j,l}(1 + \lambda_j^2 + \mu_l^2)^s |w_{jl}|^2 < \infty$. Uniqueness follows as before from the ODEs $w_{jl}'' + \lambda_j^2 w_{jl} = 0$. The operators $\cos(t\Lambda_X)$ and $\Lambda_X^{-1}\sin(t\Lambda_X)$ act on the coefficients by multiplication by functions bounded by $1 + |t|$, so they are bounded on every $H^s(X \times Y)$, locally uniformly and strongly continuously in t. Hence $w \in \mathcal{C}^0(I; H^s(X \times Y))$ for all s; the equation gives $\partial_t^2 w = -\Delta_X w - F \in \mathcal{C}^0(I; H^{s-2})$, and inductively $w \in \mathcal{C}^k(I; H^s(X \times Y))$ for all k, s. By the Sobolev embedding, $w \in \mathcal{C}^\infty(I \times X \times Y)$. □

Before turning to singularities we show that waves travel at speed at most one. The proof is an energy estimate on a shrinking ball (Figure 11.2): the energy inside the ball $\{d(x_0, x) < R - t\}$ can only decrease with t, because energy can leave the ball through its moving boundary but, travelling at unit speed, cannot enter.

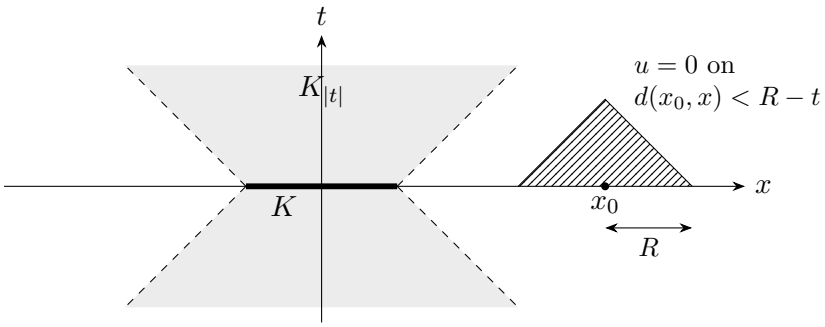


Figure 11.2: Finite propagation speed. If the Cauchy data vanish outside K, then $u(t)$ vanishes outside $K_{|t|}$, the region between the dashed lines of slope ± 1. The proof shows that u vanishes on each cone $\{d(x_0, x) < R - t\}$ (hatched) with base a ball disjoint from K.

Theorem 11.2.3 (Finite propagation speed)**.** *Let $K \subset X$ be closed, $K_r = \{x \in X;\ d(x, K) \le r\}$ for $r \ge 0$, where d is the Riemannian distance. If u is the solution of (11.2.1) with $f = 0$ and $\operatorname{supp} u_0 \cup \operatorname{supp} u_1 \subset K$ (any s), then $\operatorname{supp} u(t) \subset K_{|t|}$ for all t.*

Proof. First let $u_0, u_1 \in \mathcal{C}^\infty(X)$, so $u \in \mathcal{C}^\infty(\mathbb{R} \times X)$. Fix $x_0 \in X$ and $0 < R < \operatorname{inj}(X)$ (the injectivity radius, positive as X is compact) with $R < d(x_0, K)$. On the ball $B(x_0, \operatorname{inj} X)$ the function $d(x_0, \cdot)^2$ is smooth, so $\rho_\epsilon = (d(x_0, \cdot)^2 + \epsilon^2)^{1/2}$ is smooth there and $|d\rho_\epsilon|_g = d(x_0, \cdot)/\rho_\epsilon < 1$. Let $\theta_\delta \in \mathcal{C}^\infty(\mathbb{R})$ be non-decreasing, $\theta_\delta = 0$ on $(-\infty, 0]$, $\theta_\delta = 1$ on $[\delta, \infty)$, and $\psi(t, x) = \theta_\delta(R - t - \rho_\epsilon(x))$, a smooth function supported in $\{\rho_\epsilon \le R - t\} \subset B(x_0, R)$ for $t \ge 0$. With the energy density $e = |\partial_t u|^2 + |du|_g^2$ and $\partial_t^2 u = -\Delta u$ we have $\partial_t e = 2\operatorname{Re}(\overline{\partial_t u}\, \partial_t^2 u + \langle d\partial_t u, du\rangle_g)$ and, integrating by parts, the terms with two derivatives of u cancel:

$$\frac{d}{dt}\int_X \psi e \, d\mathrm{vol} = \int_X (\partial_t \psi) e \, d\mathrm{vol} - 2\operatorname{Re}\int_X \overline{\partial_t u}\, \langle d\psi, du\rangle_g \, d\mathrm{vol} \le \int_X (\partial_t \psi + |d\psi|_g) e \, d\mathrm{vol} \le 0,$$

because $\partial_t \psi = -\theta_\delta'$ and $|d\psi|_g = \theta_\delta' |d\rho_\epsilon|_g \le \theta_\delta'$. At $t = 0$ the integral vanishes since u_0, u_1 vanish on $\bar{B}(x_0, R)$. Letting $\delta, \epsilon \to 0$ we get $e(t, x) = 0$ for $0 \le t < R$, $d(x_0, x) < R - t$, hence

$\partial_t u = 0$ there, and since $u(0) = 0$ on $B(x_0, R)$, $u = 0$ on this cone. Given $0 \le t < \mathrm{inj}(X)$ and $x_0 \notin K_t$ we may choose $R \in (t, \min(\mathrm{inj}\, X, d(x_0, K)))$, so $u(t, x_0) = 0$: $\operatorname{supp} u(t) \subset K_t$ for $0 \le t < \mathrm{inj}\, X$. Since $(K_a)_b \subset K_{a+b}$ by the triangle inequality, applying this repeatedly (to the solution with data at time t_1, using uniqueness and time-translation invariance) gives all $t \ge 0$; $t \le 0$ follows by time reversal $t \mapsto -t$.

For distributional data use duality: $\cos(t\Lambda)$ and $\Lambda^{-1}\sin(t\Lambda)$ are symmetric for the bilinear pairing (the ϕ_j are real). If $\phi \in \mathcal{C}^\infty(X)$ has $\operatorname{supp}\phi \cap K_{|t|} = \emptyset$, then $d(\operatorname{supp}\phi, K) > |t|$, so by the smooth case $\cos(t\Lambda)\phi$ and $\Lambda^{-1}\sin(t\Lambda)\phi$ vanish near K and $\langle u(t), \phi\rangle = \langle u_0, \cos(t\Lambda)\phi\rangle + i\langle u_1, \Lambda^{-1}\sin(t\Lambda)\phi\rangle = 0$. □

11.3 The forward fundamental solution

For $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R} \times X)$ set

$$E\phi(t) = -\int_{-\infty}^{t} \Lambda^{-1}\sin((t-r)\Lambda)\phi(r)\, dr, \qquad E^\dagger\phi(t) = -\int_{t}^{\infty} \Lambda^{-1}\sin((r-t)\Lambda)\phi(r)\, dr.$$

By Theorem 11.2.1 and Proposition 11.2.2 (with data at a time before, resp. after, $\operatorname{supp}\phi$) these are the unique solutions in $\mathcal{C}^\infty(\mathbb{R}\times X)$ of $Pu = \phi$ vanishing for $t \ll 0$, resp. $t \gg 0$, and they depend continuously on ϕ. The Schwartz kernel of E is

$$E(t, x; t', x') = -H(t-t')\, K_{\sin}(t-t', x, x'), \qquad K_{\sin}(t, x, y) = \sum_j \frac{\sin(t\lambda_j)}{\lambda_j}\phi_j(x)\phi_j(y), \tag{11.3.1}$$

(H the Heaviside function), and $E^\dagger$ is its transpose for the pairing $\langle u, v\rangle = \int uv\, dt\, d\mathrm{vol}$, since $K_{\sin}$ is symmetric in (x, y). For $A \subset \mathbb{R}\times X$ let $J^\pm(A) = \{(t, x);\ \exists (t', x') \in A,\ d(x, x') \le \pm(t - t')\}$ be the causal future and past.

Theorem 11.3.1. *Let $f \in \mathcal{C}^{-\infty}(\mathbb{R}\times X)$ have $\operatorname{supp} f \subset [T, \infty)\times X$ for some T. Then there is a unique $u \in \mathcal{C}^{-\infty}(\mathbb{R}\times X)$ with support bounded below in t such that $Pu = f$. It satisfies $\operatorname{supp} u \subset J^+(\operatorname{supp} f) \subset [T, \infty)\times X$, it is given by $\langle u, \phi\rangle = \langle f, E^\dagger\phi\rangle$ for $\phi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}\times X)$ (so $u = Ef$, extending E), and $u \in \mathcal{C}^\infty$ if $f \in \mathcal{C}^\infty$. In particular E is the unique forward fundamental solution of P: $PE = EP = \mathrm{Id}$ on distributions with support bounded below, and $\operatorname{supp} E \subset \{(t, x, t', x');\ d(x, x') \le t - t'\}$.*

Proof. By Theorem 11.2.3 (applied with initial time $\sup\{t; (t, x) \in \operatorname{supp}\phi\}$), $\operatorname{supp} E^\dagger\phi \subset J^-(\operatorname{supp}\phi)$, so $\operatorname{supp} f \cap \operatorname{supp} E^\dagger\phi$ is compact and $\langle f, E^\dagger\phi\rangle$ is defined; it is continuous in ϕ, so it defines $u \in \mathcal{C}^{-\infty}(\mathbb{R}\times X)$. Since P is formally self-adjoint for the bilinear pairing ($P^t = P$), $\langle Pu, \phi\rangle = \langle f, E^\dagger P\phi\rangle = \langle f, \phi\rangle$, because $E^\dagger P\phi - \phi$ solves the homogeneous equation and vanishes for $t \gg 0$, hence is zero by Theorem 11.2.1. If $\operatorname{supp}\phi \cap J^+(\operatorname{supp} f) = \emptyset$ then $J^-(\operatorname{supp}\phi) \cap \operatorname{supp} f = \emptyset$, so $\langle u, \phi\rangle = 0$; this gives the support statement. For uniqueness let v have support bounded below and $Pv = 0$. For $\phi \in \mathcal{C}^\infty_{\mathrm{c}}$, $\operatorname{supp} v \cap \operatorname{supp} E^\dagger\phi$ is compact, so $\langle v, \phi\rangle = \langle v, PE^\dagger\phi\rangle = \langle Pv, E^\dagger\phi\rangle = 0$. If $f \in \mathcal{C}^\infty$, the Duhamel integral $-\int_{-\infty}^t \Lambda^{-1}\sin((t-r)\Lambda)f(r)\, dr$ is a smooth solution (Proposition 11.2.2) vanishing for $t < T$, hence equals u. The statements about E are a reformulation, the support of the kernel coming from Theorem 11.2.3. □

Remark 11.3.2. On non-compact manifolds (for instance $\mathbb{R}^n$ with a metric that is Euclidean outside a compact set) the spectral formula is not available, and one instead constructs E from a local parametrix E' with $PE' = \mathrm{Id} + F$, where F has a smooth kernel supported in $t \ge t'$; $\mathrm{Id} + F$ is then inverted by the Volterra (Neumann) series $\sum_k (-F)^{\circ k}$ for the t-convolution, which converges because the k-fold composition gains a factor $|t-t'|^{k-1}/(k-1)!$ (see Exercise 11.5).

11.4 The short-time structure of the wave kernel

We now describe the Schwartz kernels

$$K_{\cos}(t,x,y)=\sum_j \cos(t\lambda_j)\phi_j(x)\phi_j(y), \qquad K_{\sin}(t,x,y)=\sum_j \frac{\sin(t\lambda_j)}{\lambda_j}\phi_j(x)\phi_j(y)$$

of $\cos(t\Lambda)$ and $\Lambda^{-1}\sin(t\Lambda)$ for small $|t|$. For fixed y, $K_{\cos}$ solves $PK_{\cos}=0$ with $K_{\cos}(0)=\delta_y$, $D_tK_{\cos}(0)=0$, and $K_{\sin}$ solves $PK_{\sin}=0$ with $K_{\sin}(0)=0$, $\partial_tK_{\sin}(0)=\delta_y$. Both series converge in $\mathcal{C}^\infty(\mathbb{R};H^{-M}(X\times X))$ for M large, by Lemma 11.1.1.

11.4.1 Plane waves

To construct the wave kernel we must solve the Cauchy problem with the singular datum δ_y. The method of Chapter 10 produces solutions conormal to characteristic hypersurfaces, so we first write δ_y as a superposition of distributions conormal to hyperplanes.

The initial datum δ_y is conormal to the point $\{y\}$, but the only characteristic submanifolds for P are hypersurfaces: if $Z\subset\mathbb{R}\times X$ has codimension k and $N^*Z\subset\Sigma(P)\cup 0$, then each fibre N^*_zZ is a k-dimensional linear space on which the Lorentzian quadratic form p of signature $(1,n)$ vanishes, so $k\le 1$. The way around this is to write δ as a superposition of distributions conormal to hyperplanes. In $\mathbb{R}^n$, using polar coordinates $\xi=r\omega$ and symmetrising $(r,\omega)\mapsto(-r,-\omega)$,

$$\delta(x)=(2\pi)^{-n}\int_{\mathbb{S}^{n-1}}\int_0^\infty e^{irx\cdot\omega}r^{n-1}\,dr\,d\omega=\frac{1}{2(2\pi)^{n-1}}\int_{\mathbb{S}^{n-1}}f_n(x\cdot\omega)\,d\omega, \tag{11.4.1}$$

where

$$f_n(s)=\frac{1}{2\pi}\int_{\mathbb{R}}e^{isr}|r|^{n-1}\,dr=|D_s|^{n-1}\delta(s).$$

For n odd, $f_n=D_s^{n-1}\delta$; for n even, $|D_s|^{n-1}$ is a pseudodifferential operator and f_n is not supported at 0.

Lemma 11.4.1. *$f_n\in I^{n-\frac34}(\mathbb{R},\{0\})$ (modulo a smooth function), and the pull-back $f_n(x\cdot\omega)$ under the submersion $\mathbb{R}^n\times\mathbb{S}^{n-1}\ni(x,\omega)\mapsto x\cdot\omega$ lies in $I^{\frac n2-\frac14}(\mathbb{R}^n\times\mathbb{S}^{n-1},R)$, $R=\{x\cdot\omega=0\}$.*

Proof. Let $\rho\in\mathcal{C}^\infty(\mathbb{R})$ vanish near 0 and equal 1 for $|r|\ge1$. Then $f_n=\frac{1}{2\pi}\int e^{isr}\rho(r)|r|^{n-1}dr+\frac{1}{2\pi}\int e^{isr}(1-\rho(r))|r|^{n-1}dr$; the second term is smooth (the inverse Fourier transform of a compactly supported function) and the first is of the form (11.1.1) with symbol $\rho|r|^{n-1}\in S^{n-1}$, so of order m with $m+\frac14-\frac12=n-1$. The map $(x,\omega)\mapsto x\cdot\omega$ has differential $\omega\cdot dx+x\cdot d\omega\ne0$, so it is a submersion, R is a smooth hypersurface and $x\cdot\omega$ is a defining function. The pull-back is $\frac{1}{2\pi}\int e^{i(x\cdot\omega)r}\rho(r)|r|^{n-1}dr+\mathcal{C}^\infty$, with the same symbol, now on a manifold of dimension $N=2n-1$; thus $m+\frac{2n-1}{4}-\frac12=n-1$, i.e. $m=\frac n2-\frac14$. □

Proposition 11.4.2 (Radon inversion)**.** *If $\pi_1\colon\mathbb{R}^n\times\mathbb{S}^{n-1}\to\mathbb{R}^n$ is the projection, then $(\pi_1)_*(f_n(x\cdot\omega)\,d\omega)=2(2\pi)^{n-1}\delta$; this is (11.4.1).*

Proof. For $\phi\in\mathcal{S}(\mathbb{R}^n)$ and fixed ω, $\int f_n(x\cdot\omega)\phi(x)\,dx=\langle f_n,\mathcal{R}_\omega\phi\rangle$, where $\mathcal{R}_\omega\phi(s)=\int_{x\cdot\omega=s}\phi$ satisfies $\widehat{\mathcal{R}_\omega\phi}(r)=\hat\phi(r\omega)$. Hence $\langle f_n,\mathcal{R}_\omega\phi\rangle=\frac{1}{2\pi}\int_{\mathbb{R}}|r|^{n-1}\hat\phi(-r\omega)\,dr$ and, by polar coordinates and Fourier inversion, $\int_{\mathbb{S}^{n-1}}\langle f_n,\mathcal{R}_\omega\phi\rangle\,d\omega=\frac{2}{2\pi}\int_{\mathbb{R}^n}\hat\phi(\xi)\,d\xi=2(2\pi)^{n-1}\phi(0)$. □

Thus, to solve the Cauchy problem with datum δ_y we may solve it, for each ω, with data conormal to the hyperplane $\{(x-y)\cdot\omega=0\}$, using the two characteristic hypersurfaces through it, and then integrate over ω. It is convenient to keep the radial variable as well: this amounts to an oscillatory integral whose phase solves the eikonal equation.

Question. Why are there two characteristic hypersurfaces through each hyperplane, and why do we need both?

Answer. Over a conormal $\omega\,dx$ to the hyperplane the characteristic set contains the two points $\tau = \pm|\omega|_g$, giving two families of bicharacteristics and hence two characteristic hypersurfaces: a plane wave moving in the direction ω and one moving in the direction $-\omega$. The wave equation is of second order in t, so we must match two pieces of Cauchy data, $u(0)$ and $D_t u(0)$; a single travelling wave has its time derivative determined by its value, and only the sum of the two can match arbitrary data (Exercise 11.4). In one dimension this is d'Alembert's formula $\cos(tD_x)\delta = \frac{1}{2}(\delta(x-t)+\delta(x+t))$. □

11.4.2 The eikonal equation

Work in a coordinate chart $U \subset \mathbb{R}^n$ of X, and let $q(x,\eta) = |\eta|_{g(x)}$.

Lemma 11.4.3. *Let $W \Subset U$ be open. There exist $t_0 > 0$ and unique real $\varphi_\pm \in \mathcal{C}^\infty((-t_0,t_0)\times W \times (\mathbb{R}^n\setminus 0))$, homogeneous of degree one in η, with*

$$\partial_t \varphi_\pm = \pm q(x, d_x\varphi_\pm), \qquad \varphi_\pm(0,x,\eta) = x\cdot\eta. \tag{11.4.2}$$

Moreover $\varphi_-(t,x,\eta) = \varphi_+(-t,x,\eta)$, $d_x\varphi_\pm \neq 0$, and for $y \in W$:

$$d_\eta(\varphi_\pm(t,x,\eta) - y\cdot\eta) = 0 \iff (x, d_x\varphi_\pm(t,x,\eta)) = G_{\mp t}(y,\eta), \tag{11.4.3}$$

in which case $(x, d_x\varphi_\pm)$ lies on the geodesic flow-out and $d(x,y) \le |t|$.

Proof. This is the method of characteristics (the Hamilton–Jacobi theorem of Chapter 10) for $F_\pm = \tau \mp q(x,\xi)$ with initial surface $\{t=0\}$, which is non-characteristic since $\partial_\tau F_\pm = 1$. Explicitly, consider φ_+ and let $(x(t),\xi(t)) = G_{-t}(x_0,\eta)$, i.e. $\dot x = -\partial_\xi q$, $\dot\xi = \partial_x q$, which are the x,ξ components of the bicharacteristics of F_+. For $|t| < t_0$ small, uniformly for $|\eta| = 1$ and x_0 near $\bar W$, the map $x_0 \mapsto x(t;x_0,\eta)$ is a diffeomorphism onto a neighbourhood of $\bar W$, and φ_+ is defined by $\varphi_+(t,x(t;x_0,\eta),\eta) = x_0\cdot\eta$, extended homogeneously; the Hamilton–Jacobi theorem shows that it solves (11.4.2) and that $d_x\varphi_+(t,x(t;x_0,\eta),\eta) = \xi(t) \neq 0$ (consistently, $\frac{d}{dt}\varphi_+(t,x(t),\eta) = q - \xi\cdot\partial_\xi q = 0$ by Euler's relation). Uniqueness also follows from the characteristics. Since $\psi(t) = \varphi_+(-t)$ solves the equation with the sign $-$, $\varphi_-(t) = \varphi_+(-t)$.

For (11.4.3), differentiate $\varphi_+(t,x(t;x_0,\eta),\eta) = x_0\cdot\eta$ in η:

$$d_\eta\varphi_+(t,x(t),\eta) + \xi(t)\cdot\partial_\eta x(t;x_0,\eta) = x_0.$$

The second term vanishes: with $w = \partial_\eta x\cdot\delta\eta$, $v = \partial_\eta\xi\cdot\delta\eta$, $\frac{d}{dt}(\xi\cdot w) = \partial_x q\cdot w - \xi\cdot(\partial_x\partial_\xi q\, w + \partial_\xi^2 q\, v) = 0$, since $\xi\cdot\partial_\xi\partial_x q = \partial_x q$ and $\xi\cdot\partial_\xi^2 q = 0$ by homogeneity, and $w = 0$ at $t=0$. (This is the invariance of the Liouville form $\xi\cdot dx$ under the homogeneous flow G_s.) Hence $d_\eta\varphi_+(t,x,\eta) = y$ iff $x = x(t;y,\eta)$, i.e. iff $(x, d_x\varphi_+(t,x,\eta)) = G_{-t}(y,\eta)$. The case of φ_- follows from $\varphi_-(t) = \varphi_+(-t)$. Since G_s moves base points along unit speed geodesics, $d(x,y) \le |t|$. □

11.4.3 Transport equations

For a real phase $\phi(t,x)$ and an amplitude $a(t,x)$, a direct computation (using $\Delta e^{i\phi} = e^{i\phi}(|d_x\phi|_g^2 + i\Delta\phi)$) gives

$$P(e^{i\phi}a) = e^{i\phi}\Big[((\partial_t\phi)^2 - |d_x\phi|_g^2)a - iL_\phi a + Pa\Big], \tag{11.4.4}$$

$$L_\phi a = 2(\partial_t\phi\,\partial_t a - \langle d_x\phi, d_x a\rangle_g) + (\partial_t^2\phi + \Delta\phi)a.$$

If ϕ solves the eikonal equation the first term vanishes, and L_ϕ is a first order operator whose principal part is the vector field $V_\phi = 2(\partial_t\phi\,\partial_t - \sum g^{ij}\partial_i\phi\,\partial_j)$, the projection to $\mathbb{R}\times X$ of H_p restricted to the Lagrangian manifold $\{(t,x,d\phi)\}$. For $\phi = \varphi_\pm(t,x,\eta) - y\cdot\eta$ the ∂_t-coefficient is $\pm 2q(x,d_x\varphi_\pm) \neq 0$, so V_ϕ is transversal to $\{t=0\}$ and its integral curves are (reparametrised) unit speed geodesics. If $a \sim \sum_{k\geq 0} a_k$ with a_k homogeneous of degree $\mu - k$ in η, then $P(e^{i\phi}a) = e^{i\phi}r$ with $r \in S^{-\infty}$ in η provided the *transport equations*

$$L_\phi a_0 = 0, \qquad L_\phi a_k = -iPa_{k-1} \quad (k \geq 1) \tag{11.4.5}$$

hold; both sides of the k-th equation are homogeneous of degree $\mu + 1 - k$. Each is a linear ODE along the integral curves of V_ϕ, uniquely solvable for small $|t|$ given $a_k(0,\cdot)$, and the solution is homogeneous of the right degree.

11.4.4 The parametrix

Fix $\chi \in \mathcal{C}^\infty(\mathbb{R}^n)$ with $\chi(\eta) = 0$ for $|\eta| \leq \frac{1}{2}$ and $\chi(\eta) = 1$ for $|\eta| \geq 1$. Choose a finite cover of X by coordinate charts U_k, open sets $V_k \Subset U_k$ still covering X, a partition of unity χ_k with $\operatorname{supp}\chi_k \subset V_k$, and $\delta_0 > 0$ such that the $3\delta_0$-neighbourhood of $\bar V_k$ is relatively compact in U_k for all k. Let W_k be the $2\delta_0$-neighbourhood of $\bar V_k$ and $\zeta_k \in \mathcal{C}^\infty_{\mathrm{c}}(W_k)$ with $\zeta_k = 1$ on the δ_0-neighbourhood of $\bar V_k$.

Theorem 11.4.4. *There are $t_0 \in (0,\delta_0)$, phase functions $\varphi_{k,\pm}$ on $(-t_0,t_0)\times W_k\times(\mathbb{R}^n\setminus 0)$ as in Lemma 11.4.3, and polyhomogeneous symbols $a_{k,\pm} \in S^0_{\mathrm{ph}}$, $b_{k,\pm} \in S^{-1}_{\mathrm{ph}}$ in $\eta\in\mathbb{R}^n$, depending smoothly on $(t,x,y) \in (-t_0,t_0)\times W_k\times V_k$, such that*

$$K_{\cos}(t,x,y) \equiv \sum_{k,\pm} \zeta_k(x)\chi_k(y)(2\pi)^{-n}\int e^{i(\varphi_{k,\pm}(t,x,\eta)-y\cdot\eta)}a_{k,\pm}(t,x,y,\eta)\chi(\eta)\,d\eta,$$
$$K_{\sin}(t,x,y) \equiv \sum_{k,\pm} \zeta_k(x)\chi_k(y)(2\pi)^{-n}\int e^{i(\varphi_{k,\pm}(t,x,\eta)-y\cdot\eta)}b_{k,\pm}(t,x,y,\eta)\chi(\eta)\,d\eta,$$

modulo $\mathcal{C}^\infty((-t_0,t_0)\times X\times X)$. At $t=0$,

$$a_{k,\pm}(0,x,y,\eta) - \tfrac{1}{2}|g(y)|^{-1/2} \in S^{-1}, \qquad b_{k,\pm}(0,x,y,\eta) \mp \frac{|g(y)|^{-1/2}}{2i\,q(x,\eta)} \in S^{-2}. \tag{11.4.6}$$

Proof. Apply Lemma 11.4.3 on W_k (with $W_k \Subset U_k$), shrinking t_0 so that also all integral curves of the transport fields starting in W_k stay in the $3\delta_0$-neighbourhood for $|t| < t_0$. We construct the cosine term; write $\phi_\pm = \varphi_{k,\pm} - y\cdot\eta$ and $S = \sum_\pm (2\pi)^{-n}\int e^{i\phi_\pm}a_\pm\chi\,d\eta$.

Amplitudes. We seek $a_\pm \sim \sum_{j\geq 0} a_{\pm,j}$, $a_{\pm,j}$ homogeneous of degree $-j$, solving (11.4.5) for $\phi_\pm$, with initial values fixed by the requirements $S(0) \equiv |g(y)|^{-1/2}\delta(x-y)$, $D_tS(0)\equiv 0$. Since $\phi_\pm(0) = (x-y)\cdot\eta$ and $\partial_t\phi_\pm(0) = \pm q(x,\eta)$, these amount to

$$a_{+,0} + a_{-,0} = |g(y)|^{-1/2}, \qquad a_{+,j} + a_{-,j} = 0 \ (j\geq 1),$$
$$q(x,\eta)(a_{+,j} - a_{-,j}) + D_t a_{+,j-1} + D_t a_{-,j-1} = 0 \ (j \geq 0)$$

at $t=0$ (with $a_{\pm,-1} = 0$), the last being the part homogeneous of degree $1-j$ of $\sum_\pm(\partial_t\phi_\pm a_\pm + D_t a_\pm)|_{t=0}$. Inductively in j: the initial values of $a_{\pm,j}$ are determined by these equations, given $a_{\pm,j-1}$ for all t, and then $a_{\pm,j}$ is determined for $|t| < t_0$ by the transport equation. In particular $a_{\pm,0}(0) = \frac{1}{2}|g(y)|^{-1/2}$. Let $a_\pm$ be a Borel sum of the series, multiplied by χ. Then $PS = (2\pi)^{-n}\sum_\pm\int e^{i\phi_\pm}r_\pm\,d\eta$ with $r_\pm \in S^{-\infty}$, so $PS \in \mathcal{C}^\infty$;

similarly $S(0) - |g(y)|^{-1/2}\delta(x-y)$ and $D_tS(0)$ are oscillatory integrals with amplitudes in $S^{-\infty}$, hence smooth. For the sine term the conditions are $\sum_\pm b_\pm(0) \in S^{-\infty}$ and $\sum_\pm (i\partial_t\phi_\pm b_\pm + \partial_t b_\pm)(0) - |g(y)|^{-1/2} \in S^{-\infty}$, giving $b_{\pm,0}(0) = \pm|g(y)|^{-1/2}/(2iq(x,\eta))$ and then the same induction.

Singular support. The phase $\phi_\pm$ is non-degenerate in (y,η) ($d_y\phi_\pm = -\eta \neq 0$), so S is a smooth function of t with values in distributions on $W_k \times V_k$, and by (11.1.2) and (11.4.3) it is smooth on $\{d(x,y) > |t|\}$.

Comparison. Let $w = \chi_k(y)K_{\cos} - \zeta_k(x)\chi_k(y)S$ on $(-t_0,t_0)\times X \times X$. Then

$$Pw = -\zeta_k\chi_k PS - [P,\zeta_k]\chi_k S \in \mathcal{C}^\infty,$$

the commutator term because $d\zeta_k = 0$ on the δ_0-neighbourhood of $\operatorname{supp}\chi_k$, while S is smooth where $d(x,y) > |t|$ and $|t| < t_0 < \delta_0$. Also $w(0) = \chi_k(y)(K_{\mathrm{Id}} - \zeta_k(x)S(0))$ and $D_tw(0)$ are smooth, since $\zeta_k(x) = 1$ near the support of $\chi_k(y)\delta(x-y)$. Both $K_{\cos}$ and S lie in $\mathcal{C}^\infty((-t_0,t_0); H^{-M}(X\times X))$ for M large, so by Proposition 11.2.2 (with $Y = X$ the y-factor) $w \in \mathcal{C}^\infty$. Summing over k proves the claim for $K_{\cos}$, and the same argument applies to $K_{\sin}$. □

Remark 11.4.5 (Conormal superposition). In polar coordinates $\eta = r\omega$, homogeneity gives $\varphi_\pm(t,x,r\omega) - ry\cdot\omega = rh_\pm(t,x,y,\omega)$, so each term of the parametrix is

$$(2\pi)^{1-n}\int_{\mathbb{S}^{n-1}} u_\pm(t,x;y,\omega)\,d\omega, \qquad u_\pm = \frac{1}{2\pi}\int_0^\infty e^{irh_\pm(t,x,y,\omega)}a_\pm(t,x,y,r\omega)\chi(r\omega)r^{n-1}\,dr.$$

For fixed (y,ω), $u_\pm$ is a conormal distribution, of order $\frac{3n-3}{4}$, associated with the hypersurface $H_\pm(y,\omega) = \{h_\pm = 0\} \subset (-t_0,t_0)\times W_k$, which is characteristic for P ($(\partial_t h_\pm)^2 = |d_x h_\pm|_g^2$ by (11.4.2)) and meets $\{t=0\}$ in the hyperplane $\{(x-y)\cdot\omega = 0\}$; there are exactly two such characteristic hypersurfaces, corresponding to the two points $\tau = \pm|\omega|_g$ of $\Sigma(P)$ over the conormal $\omega\,dx$. The amplitude of $u_\pm$ solves the transport equation, and at $t=0$ the sum $u_+ + u_-$ reproduces the plane wave decomposition (11.4.1) of δ_y. As a function of all variables (t,x,y,ω), $u_\pm$ is conormal with respect to the hypersurface $\{h_\pm = 0\}$, smooth since $d_y h_\pm = -\omega \neq 0$, and the kernel is its push-forward along $\mathbb{S}^{n-1}$.

Corollary 11.4.6. *For $|t| < t_0$ the wavefront sets of $K_{\cos}$ and $K_{\sin}$ are contained in*

$$\Gamma = \{(t,x,y;\tau,\xi,-\eta);\ \eta\neq 0,\ \tau = \pm|\eta|_{g(y)},\ (x,\xi) = G_{\mp t}(y,\eta)\}.$$

In particular their singular supports are contained in $\{d(x,y) = |t|\}$ for $|t| < \min(t_0, \operatorname{inj} X)$.

Proof. By (11.1.2), the wavefront set of each term is contained in the set of points

$$(t,x,y;\partial_t\phi_\pm, d_x\phi_\pm, d_y\phi_\pm) \quad\text{with}\quad d_\eta\phi_\pm = 0, \qquad \phi_\pm = \varphi_{k,\pm}(t,x,\eta) - y\cdot\eta.$$

By (11.4.3) there $(x, d_x\varphi_\pm) = G_{\mp t}(y,\eta)$, while $d_y\phi_\pm = -\eta$ and $\partial_t\phi_\pm = \pm q(x,d_x\varphi_\pm) = \pm q(y,\eta)$ since G preserves q. Within the injectivity radius the geodesic of length $|t|$ is minimising, so $d(x,y) = |t|$. □

By Lemma 11.1.2, Γ is the union of the null bicharacteristics of $D_t^2 - \Delta_x$ (acting in (t,x), with (y,η) fixed) through the points $(0,y,y;\pm|\eta|,\eta,-\eta)$, i.e. the flow-out of the wavefront set of the initial datum $\delta(x-y)$. In words: the singularity of δ_y travels out along all unit speed geodesics from y, and at time t it sits on the geodesic sphere of radius $|t|$ about y (Figure 11.3).

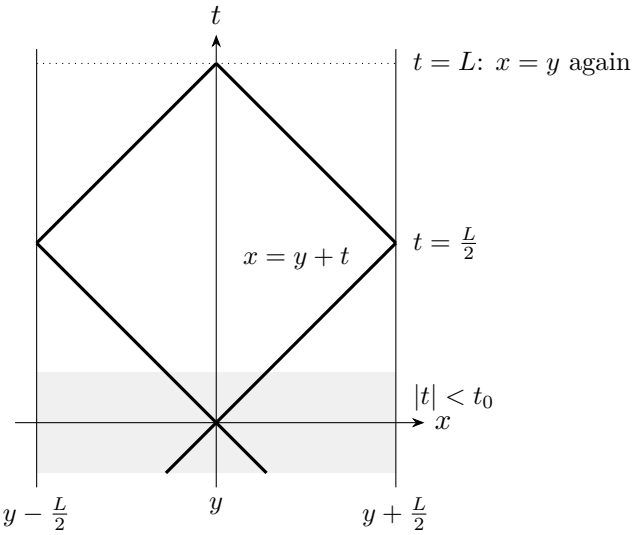


Figure 11.3: The singular support of $K_{\cos}(t, \cdot, y)$ on the circle $X = \mathbb{R}/L\mathbb{Z}$, drawn on a fundamental domain $|x - y| \le \frac{L}{2}$ whose ends are identified. For small $|t|$ (shaded) it is the geodesic sphere $\{d(x, y) = |t|\}$, as in Corollary 11.4.6. For larger t the singularity keeps moving along the geodesics and returns to $x = y$ at $t = L$, the length of the closed geodesic; this is why the wave trace is singular at $t = L$ (Section 11.8).

11.5 Operations on conormal distributions

We record the behaviour of conormal distributions under push-forward along fibres and restriction to transversal submanifolds, and the reduction of amplitudes depending on the normal variable. Together these compute the trace of the wave group in Section 11.6.

Proposition 11.5.1 (Push-forward)**.** *Let M_1 be a manifold of dimension N_1, Z a compact manifold of dimension d with a smooth positive density ν, $\pi\colon M_1 \times Z \to M_1$ the projection, and $H_1 \subset M_1$ a hypersurface. If $u \in I^m(M_1 \times Z, H_1 \times Z)$ then $\pi_* u = \int_Z u\,\nu \in I^{m+\frac{d}{4}}(M_1, H_1)$; in local coordinates, with $u = \frac{1}{2\pi}\int e^{is\rho} a(z', w, \rho)\,d\rho$, the push-forward has amplitude $\int_Z a(z', w, \rho)\,\nu(w)$.*

Proof. π_* maps $\mathcal{C}^\infty$ to $\mathcal{C}^\infty$ and distributions smooth off $H_1 \times Z$ to distributions smooth off H_1. Near H_1 choose coordinates (s, z') on M_1 with $H_1 = \{s = 0\}$; then $s \circ \pi$ defines $H_1 \times Z$ and, using a partition of unity on Z and coordinates w there, u is locally as in (11.1.1) with $a(z', w, \rho) \in S^\mu$, $\mu = m + \frac{N_1 + d}{4} - \frac{1}{2}$. Integrating in w gives an amplitude in $S^\mu(\mathbb{R}^{N_1 - 1}; \mathbb{R})$, and $\mu = m' + \frac{N_1}{4} - \frac{1}{2}$ with $m' = m + \frac{d}{4}$. □

Proposition 11.5.2 (Restriction)**.** *Let $H \subset M$ be a hypersurface, $\dim M = N$, and $Y \subset M$ a submanifold of codimension k transversal to H ($T_z H + T_z Y = T_z M$ for $z \in H \cap Y$). Then restriction extends from $\mathcal{C}^\infty$ to a map $I^m(M, H) \to I^{m+\frac{k}{4}}(Y, H \cap Y)$, and the local amplitude restricts.*

Proof. By transversality there are local coordinates (s, z', z'') with $H = \{s = 0\}$ and $Y = \{z'' = 0\}$, $z'' \in \mathbb{R}^k$. Then $u = \frac{1}{2\pi}\int e^{is\rho} a(z', z'', \rho)\,d\rho$ depends smoothly on z'' as a distribution in (s, z'), and $u|_Y = \frac{1}{2\pi}\int e^{is\rho} a(z', 0, \rho)\,d\rho$, consistent with the restriction of smooth functions (and with $N^*Y \cap N^*H = 0$). The symbol order $\mu = m + \frac{N}{4} - \frac{1}{2}$ is unchanged, and $\mu = m' + \frac{N-k}{4} - \frac{1}{2}$ gives $m' = m + \frac{k}{4}$. □

Lemma 11.5.3. *Let $B \in \mathcal{C}^\infty(\mathbb{R}\times\mathbb{R})$ satisfy $B(t,\rho)=0$ for $|t|\geq t_1$ and $|\partial_t^k\partial_\rho^l B(t,\rho)|\leq C_{kl}\langle\rho\rangle^{m-l}$. Then*

$$J_B(\lambda)=\int_{\mathbb{R}}\Big(\int_{\mathbb{R}} e^{it(\rho-\lambda)}B(t,\rho)\,dt\Big)d\rho$$

(an absolutely convergent iterated integral) lies in $S^m(\mathbb{R})$, and $J_B(\lambda)\sim 2\pi\sum_{k\geq 0}\frac{i^k}{k!}\partial_t^k\partial_\rho^k B(0,\lambda)$; in particular $J_B(\lambda)-2\pi B(0,\lambda)\in S^{m-1}(\mathbb{R})$. If B_ϵ is bounded in this class for $\epsilon\in(0,1]$ and $B_\epsilon\to B$ pointwise, then $J_{B_\epsilon}\to J_B$ pointwise. Consequently $v=\frac{1}{2\pi}\int e^{it\rho}B(t,\rho)\,d\rho\in I^{m+\frac14}(\mathbb{R},\{0\})$, with $\hat v=J_B/(2\pi)$.

Proof. Let $\hat B(\sigma,\rho)=\int e^{-it\sigma}B(t,\rho)\,dt$; then $|\partial_\sigma^j\partial_\rho^l\hat B|\leq C_{Njl}\langle\sigma\rangle^{-N}\langle\rho\rangle^{m-l}$, and $J_B(\lambda)=\int\hat B(\lambda-\rho,\rho)\,d\rho=\int\hat B(\sigma,\lambda-\sigma)\,d\sigma$. Taylor's formula in the second variable gives $\hat B(\sigma,\lambda-\sigma)=\sum_{k<K}\frac{(-\sigma)^k}{k!}\partial_\rho^k\hat B(\sigma,\lambda)+R_K$ with, by Peetre's inequality $\langle\lambda-s\sigma\rangle^\mu\leq 2^{|\mu|}\langle\lambda\rangle^\mu\langle\sigma\rangle^{|\mu|}$, $|R_K|\leq C_N\langle\sigma\rangle^{K+|m-K|-N}\langle\lambda\rangle^{m-K}$, which integrates to $O(\langle\lambda\rangle^{m-K})$. Since $\sigma^k\hat B=\widehat{D_t^kB}$ and $\int\hat f\,d\sigma=2\pi f(0)$, the k-th term is $2\pi\frac{(-1)^k}{k!}D_t^k\partial_\rho^kB(0,\lambda)=2\pi\frac{i^k}{k!}\partial_t^k\partial_\rho^kB(0,\lambda)$. Finally $\partial_\lambda J_B=J_{\partial_\rho B}$ and $\partial_\rho B$ is of the same type with m replaced by $m-1$, so the bound $|J_B|\leq C\langle\lambda\rangle^m$ (case $K=0$) gives the symbol estimates. The convergence statement follows by dominated convergence from the uniform bound $|\hat B_\epsilon(\lambda-\rho,\rho)|\leq C\langle\lambda-\rho\rangle^{-N}\langle\rho\rangle^m$. For the last statement, $\hat v(\lambda)=\frac{1}{2\pi}J_B(\lambda)\in S^m$ and v has compact support, so $v\in I^{m+\frac14}(\mathbb{R},\{0\})$. □

In Lemma 11.5.3 the amplitude $B(t,\rho)$ is replaced by the amplitude $\hat v(\rho)$ independent of t; this is the one-dimensional case of the statement that amplitudes may depend on the normal variable.

11.6 The trace of the wave group

We now integrate the parametrix over the diagonal. Only the restriction of the kernel to $\{x=y\}$ matters, and for small $|t|\neq 0$ this restriction is smooth, because the singularities sit on $\{d(x,y)=|t|\}$; all the singularity of the trace near $t=0$ therefore comes from $t=0$ itself.

For $\psi\in\mathcal{S}(\mathbb{R})$ let $C_\psi=\int\psi(t)\cos(t\Lambda)\,dt=\tilde\psi(\Lambda)$, where $\tilde\psi(\lambda)=\int\psi(t)\cos(t\lambda)\,dt=\frac12(\hat\psi(\lambda)+\hat\psi(-\lambda))$ is rapidly decreasing. By Lemma 11.1.1 and the polynomial bounds on $\|\phi_j\|_{\mathcal{C}^k}$, the kernel $K_{C_\psi}(x,y)=\sum_j\tilde\psi(\lambda_j)\phi_j(x)\phi_j(y)$ converges in $\mathcal{C}^\infty(X\times X)$; C_ψ is smoothing and of trace class, and

$$\operatorname{Tr}C_\psi=\sum_j\tilde\psi(\lambda_j)=\int_X K_{C_\psi}(x,x)\,d\mathrm{vol}(x).$$

Since $\sum_j\langle\lambda_j\rangle^{-M}<\infty$ for $M>n$, this is a continuous linear functional on $\mathcal{S}(\mathbb{R})$:

Definition 11.6.1. The *(even) wave trace* is the tempered distribution

$$T(t)=\operatorname{Tr}\cos(t\Lambda)=\sum_{j\geq 0}\cos(t\lambda_j),\qquad T(\psi)=\operatorname{Tr}C_\psi.$$

The trace of the wave group $U(t)$ is $2T$. With the measure $\mu=\sum_j\delta_{\lambda_j}$ on $[0,\infty)$ and its even part $\mu_e=\frac12\sum_j(\delta_{\lambda_j}+\delta_{-\lambda_j})$, $T(t)=\int e^{it\sigma}d\mu_e(\sigma)$, i.e. $\hat T=2\pi\mu_e$, and $N(\lambda)=\mu([0,\lambda])$. For $\psi\in\mathcal{C}_c^\infty(\mathbb{R})$,

$$\widehat{\psi T}(\lambda)=T(\psi e^{-i\lambda\cdot})=\frac12\sum_j\big(\hat\psi(\lambda-\lambda_j)+\hat\psi(\lambda+\lambda_j)\big).\tag{11.6.1}$$

Let ω_n denote the volume of the unit ball in $\mathbb{R}^n$ (so $\mathrm{vol}(\mathbb{S}^{n-1})=n\omega_n$).

Theorem 11.6.2. *Let $t_0 > 0$ be as in Theorem 11.4.4. For every $\psi \in \mathcal{C}_c^\infty((-t_0,t_0))$, $\psi T \in I^{n-\frac34}(\mathbb{R},\{0\})$, i.e. $\widehat{\psi T} \in S^{n-1}(\mathbb{R})$; more precisely, $\widehat{\psi T}$ is polyhomogeneous as $\lambda \to \pm\infty$ and*

$$\widehat{\psi T}(\lambda) = \frac{n\,\omega_n}{2(2\pi)^{n-1}}\,\psi(0)\,\mathrm{vol}(X)\,|\lambda|^{n-1} + O(|\lambda|^{n-2}), \qquad |\lambda| \to \infty. \tag{11.6.2}$$

Proof. Reduction to the parametrix. With $\psi_\lambda = \psi e^{-i\lambda\cdot}$, $\widehat{\psi T}(\lambda) = \int_X K_{C_{\psi_\lambda}}(x,x)\,d\mathrm{vol}$ (the definition of C_ψ extends to complex ψ), and $K_{C_{\psi_\lambda}} = \int \psi_\lambda(t) K_{\cos}(t)\,dt$. By Theorem 11.4.4, $K_{\cos} = \sum_{k,\pm} S_{k,\pm} + R$, with $S_{k,\pm}$ the terms of the parametrix, on $(-t_0,t_0)\times X\times X$ with R smooth; the contribution $\iint \psi(t) e^{-it\lambda} R(t,x,x)\,dt\,d\mathrm{vol}$ of R is $O(|\lambda|^{-\infty})$. Fix k and the sign $+$ (the sign $-$ is analogous) and write $\varphi = \varphi_{k,+}$, $a = a_{k,+}$. Then

$$\int \psi_\lambda(t) S_{k,+}(t,x,y)\,dt = (2\pi)^{-n} \int e^{-iy\cdot\eta}\alpha(x,y,\eta)\,d\eta,$$
$$\alpha = \zeta_k(x)\chi_k(y)\chi(\eta)\int e^{i(\varphi(t,x,\eta)-t\lambda)}\psi(t)a\,dt.$$

For fixed λ, $|\partial_t\varphi - \lambda| = |q(x,d_x\varphi) - \lambda| \geq c|\eta|$ for $|\eta| \geq C(1+|\lambda|)$, so repeated integration by parts with $(\partial_t\varphi - \lambda)^{-1} D_t$ shows that α and all its (x,y)-derivatives are rapidly decreasing in η. Hence this kernel is smooth, given by an absolutely convergent integral, and its value at $x = y$ is the absolutely convergent iterated integral $(2\pi)^{-n}\int(\int\cdots dt)\,d\eta$. Thus

$$I_{k,+}(\lambda) = (2\pi)^{-n}\int_X\int_{\mathbb{R}^n}\Big(\int_{\mathbb{R}} e^{i(\varphi(t,x,\eta)-x\cdot\eta-t\lambda)} b(t,x,\eta)\,dt\Big)d\eta\,d\mathrm{vol}(x),$$

with $b(t,x,\eta) = \psi(t)\chi_k(x)a(t,x,x,\eta)\chi(\eta)$ (using $\zeta_k\chi_k = \chi_k$). By dominated convergence we may insert $\chi_0(\epsilon\eta)$, $\chi_0 \in \mathcal{C}_c^\infty(\mathbb{R}^n)$, $\chi_0(0) = 1$, and let $\epsilon \to 0$; for $\epsilon > 0$ the integral is absolutely convergent and Fubini's theorem allows the t-integration to be done last.

Restriction to the diagonal. Since $\varphi(0,x,\eta) = x\cdot\eta$,

$$\varphi(t,x,\eta) - x\cdot\eta = tQ(t,x,\eta), \qquad Q(t,x,\eta) = \int_0^1 q\big(x, d_x\varphi(st,x,\eta)\big)\,ds,$$

with Q smooth, positive, homogeneous of degree one in η and $Q(0,x,\eta) = |\eta|_{g(x)}$. (Thus the restriction of the characteristic hypersurface $\{h_+ = 0\}$ of Remark 11.4.5 to the diagonal is exactly $\{t = 0\}$, the transversal situation of Proposition 11.5.2.) In polar coordinates $\eta = r\omega$ and then $\rho = rQ(t,x,\omega)$ for fixed (t,x,ω), $d\eta = \rho^{n-1}Q(t,x,\omega)^{-n}d\rho\,d\omega$, and we obtain $I_{k,+}(\lambda) = \lim_{\epsilon\to0}\iint e^{it(\rho-\lambda)}B_\epsilon(t,\rho)\,dt\,d\rho$ with

$$B_\epsilon(t,\rho) = (2\pi)^{-n}\int_X\int_{\mathbb{S}^{n-1}} b\Big(t,x,\frac{\rho\omega}{Q}\Big)\chi_0\Big(\frac{\epsilon\rho\omega}{Q}\Big)\frac{\rho^{n-1}}{Q(t,x,\omega)^n}\,d\omega\,d\mathrm{vol}(x)$$

for $\rho > 0$, and $B_\epsilon = 0$ for $\rho \leq 0$ (as $\chi = 0$ near 0). This is the push-forward of Proposition 11.5.1 along $X\times\mathbb{S}^{n-1}$. The B_ϵ are bounded in the class of Lemma 11.5.3 with $m = n-1$ and converge to $B = B_0$, whose amplitude is polyhomogeneous in ρ. By Lemma 11.5.3, $I_{k,+} = J_B \in S^{n-1}$, polyhomogeneous, and $I_{k,+}(\lambda) = 2\pi B(0,\lambda) + O(\langle\lambda\rangle^{n-2})$. In particular $I_{k,+}(\lambda) = O(|\lambda|^{-\infty})$ as $\lambda \to -\infty$.

Leading term. At $t = 0$, $b(0,x,\eta) = \psi(0)\chi_k(x)a(0,x,x,\eta)$ for $|\eta| \geq 1$, and by (11.4.6) $a(0,x,x,\eta) = \frac12|g(x)|^{-1/2} + S^{-1}$. Hence for $\rho \to +\infty$

$$B(0,\rho) = \frac{\psi(0)}{2(2\pi)^n}\rho^{n-1}\int_X \chi_k(x)|g(x)|^{-1/2}\int_{\mathbb{S}^{n-1}}|\omega|_{g(x)}^{-n}\,d\omega\,d\mathrm{vol}(x) + O(\rho^{n-2}).$$

In polar coordinates the ellipsoid $\{\eta;\ |\eta|_{g(x)} \leq 1\}$ has volume $\frac1n\int_{\mathbb{S}^{n-1}}|\omega|_{g(x)}^{-n}d\omega$, and it is the image of the unit ball under a linear map of determinant $\det(g^{ij})^{-1/2} = |g(x)|^{1/2}$; so

the inner integral is $n\omega_n|g(x)|^{1/2}$ and $B(0,\rho) = \frac{n\omega_n}{2(2\pi)^n}\psi(0)\rho^{n-1}\int_X \chi_k\,d\mathrm{vol} + O(\rho^{n-2})$. The terms with the sign $-$ have phase $\varphi_{k,-} - x\cdot\eta = -tQ_-$ and give the same expression with λ replaced by $-\lambda$, rapidly decreasing as $\lambda \to +\infty$. Summing over k ($\sum \chi_k = 1$) and multiplying by 2π gives (11.6.2). □

Remark 11.6.3. For the circle $X = \mathbb{R}/L\mathbb{Z}$, $\lambda = 2\pi|k|/L$ for $k \in \mathbb{Z}$ and $T(t) = \sum_{k\in\mathbb{Z}} \cos(2\pi kt/L)$, while by Poisson summation $\sum_k \hat\psi(\lambda - 2\pi k/L) = L\sum_m \psi(mL)e^{-imL\lambda} = L\psi(0)$ if $\operatorname{supp}\psi \subset (-L, L)$. Thus $\widehat{\psi T} = L\psi(0)$, in agreement with (11.6.2) ($n = 1$, $\omega_1 = 2$). The restriction on the support of ψ is necessary: T is singular at every $t \in L\mathbb{Z}$, the lengths of the closed geodesics (Section 11.8).

11.7 Weyl asymptotics

Discussion. Weyl's law has a simple heuristic. By the uncertainty principle each eigenfunction occupies a cell of volume $(2\pi)^n$ in phase space, and an eigenfunction with $\lambda_j \le \lambda$ "lives" in the region $\{(x,\xi);\ |\xi|_{g(x)} \le \lambda\}$ of T^*X. That region has volume $\omega_n \operatorname{vol}(X)\lambda^n$, so one expects $N(\lambda) \approx (2\pi)^{-n}\omega_n \operatorname{vol}(X)\lambda^n$. The proof below makes this precise with the trace formula of Theorem 11.6.2, whose leading coefficient is exactly this phase-space volume, written as a density in λ.

Theorem 11.7.1 (Weyl's law with remainder)**.** *Let X be a compact Riemannian manifold of dimension $n \ge 2$. Then*

$$N(\lambda) = (2\pi)^{-n}\omega_n \operatorname{vol}(X)\lambda^n + O(\lambda^{n-1}), \qquad \lambda \to \infty.$$

Figure 11.4 compares $N(\lambda)$ with $C_W\lambda^n$ for the flat torus $\mathbb{R}^2/2\pi\mathbb{Z}^2$, where $N(\lambda)$ counts lattice points in a disc (Exercise 11.7).

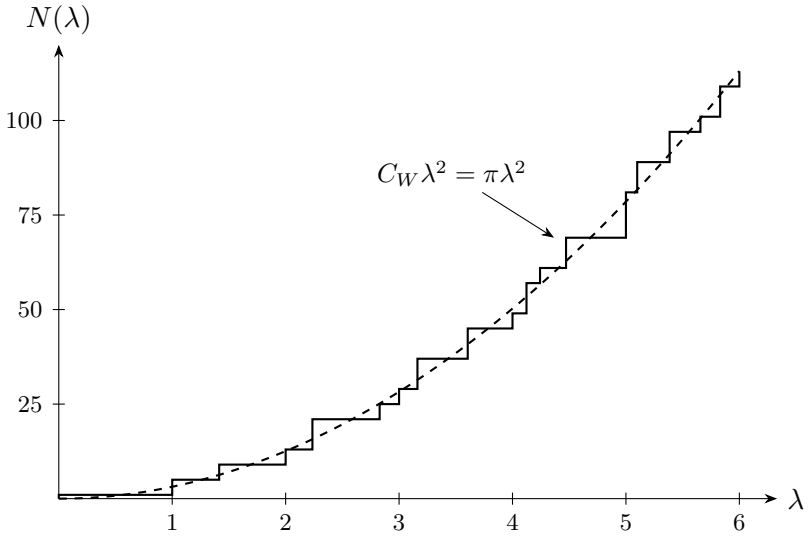


Figure 11.4: The counting function $N(\lambda) = \#\{k \in \mathbb{Z}^2;\ |k| \le \lambda\}$ of the flat torus $\mathbb{R}^2/2\pi\mathbb{Z}^2$ (staircase) and the Weyl term $C_W\lambda^2 = \pi\lambda^2$ (dashed). The jumps of N are the multiplicities, here at most of order λ^ϵ; on the round sphere they are of order λ^{n-1}, which is why the remainder $O(\lambda^{n-1})$ cannot be improved in general.

The proof has three steps. The trace formula controls a *smoothed* spectral density $\rho * \mu$, where ρ is a fixed positive bump. Positivity of ρ gives an a priori bound on the number

of eigenvalues in any window of fixed length (Step 1). Integrating $\rho * \mu$ gives a smoothed counting function with the correct asymptotics (Step 2), and the a priori bound shows that smoothing changes N by at most $O(\lambda^{n-1})$ (Step 3). Figure 11.5 shows the smoothing.

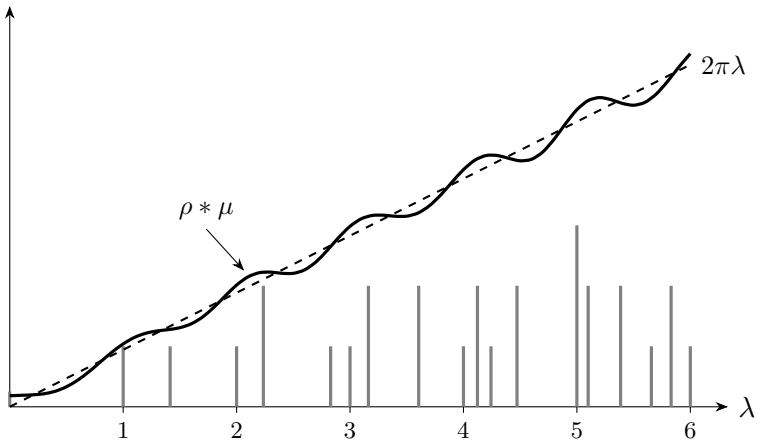


Figure 11.5: The Tauberian step for the torus of Figure 11.4. The spectral measure $\mu = \sum_j \delta_{\lambda_j}$ (gray spikes, heights proportional to multiplicities) is too irregular to have asymptotics, but its convolution with a fixed positive bump ρ of width about $\frac{1}{3}$ (thick curve) follows the smooth density $nC_W\lambda^{n-1} = 2\pi\lambda$ (dashed) up to an error of lower order.

Proof. Write $C_W = (2\pi)^{-n}\omega_n \operatorname{vol}(X)$.

Choice of a test function. Let $\theta \in \mathcal{C}_c^\infty((-t_0/2, t_0/2))$ be real, even, non-negative and not identically zero, normalised by $\int \theta^2 = 1$, and let $\psi = \theta * \theta$. Then $\psi \in \mathcal{C}_c^\infty((-t_0, t_0))$ is even, $\psi(0) = 1$ and $\hat\psi = \hat\theta^2 = |\hat\theta|^2 \geq 0$. Set $\rho = \hat\psi/(2\pi)$: $\rho \in \mathcal{S}(\mathbb{R})$ is even and non-negative, $\int \rho = \psi(0) = 1$, and $\rho(0) = (2\pi)^{-1}(\int \theta)^2 > 0$, so there are $a, c_0 > 0$ with $\rho \geq c_0$ on $[-a, a]$.

Smoothed density. By (11.6.1), $(\rho * \mu)(\lambda) = \sum_j \rho(\lambda - \lambda_j) = \pi^{-1}\widehat{\psi T}(\lambda) - \sum_j \rho(\lambda + \lambda_j)$. For $\lambda \geq 0$, $\sum_j \rho(\lambda + \lambda_j) \leq C_M \sum_j (1 + \lambda + \lambda_j)^{-M} = O(\lambda^{-\infty})$, by Lemma 11.1.1. Hence by Theorem 11.6.2

$$(\rho * \mu)(\lambda) = nC_W\lambda^{n-1} + O(\lambda^{n-2}), \quad \lambda \to +\infty; \qquad (\rho * \mu)(\lambda) = O(\langle\lambda\rangle^{-\infty}), \quad \lambda \to -\infty, \tag{11.7.1}$$

the second because $\lambda - \lambda_j \leq \lambda < 0$. Note $\pi^{-1}\frac{n\omega_n}{2(2\pi)^{n-1}} = n(2\pi)^{-n}\omega_n$.

Step 1: a priori bound. Since $\rho \geq 0$ and $\rho \geq c_0$ on $[-a, a]$, $\mu([\lambda - a, \lambda + a]) \leq c_0^{-1}(\rho * \mu)(\lambda) \leq C(1 + |\lambda|)^{n-1}$ for all $\lambda \in \mathbb{R}$ (the function $\rho * \mu$ is continuous). Covering an interval of length $2r$ by at most $1 + r/a$ intervals of length $2a$,

$$\mu([\lambda - r, \lambda + r]) \leq C'(1 + r)^n(1 + |\lambda|)^{n-1}, \qquad \lambda \in \mathbb{R},\ r \geq 0. \tag{11.7.2}$$

Step 2: the smoothed counting function. Let $N_\rho(\lambda) = \int N(\lambda - \sigma)\rho(\sigma)\,d\sigma = \sum_j R(\lambda - \lambda_j)$, $R(s) = \int_{-\infty}^s \rho$. Termwise differentiation (justified by the locally uniform convergence of $\sum_j \rho(\lambda - \lambda_j)$) gives $N_\rho' = \rho * \mu$, and $N_\rho(\lambda) \to 0$ as $\lambda \to -\infty$. Thus, by (11.7.1) and $n \geq 2$,

$$N_\rho(\lambda) = \int_{-\infty}^1 (\rho * \mu)\,ds + \int_1^\lambda \left(nC_W s^{n-1} + O(s^{n-2})\right) ds = C_W\lambda^n + O(\lambda^{n-1}).$$

Step 3: removing the smoothing. Since $\int \rho = 1$, $\rho \geq 0$ and N is non-decreasing, $|N(\lambda) - N(\lambda - \sigma)| \leq \mu([\lambda - |\sigma|, \lambda + |\sigma|])$, so by (11.7.2)

$$|N(\lambda) - N_\rho(\lambda)| \leq \int |N(\lambda) - N(\lambda - \sigma)|\rho(\sigma)\,d\sigma \leq C'(1 + |\lambda|)^{n-1} \int (1 + |\sigma|)^n \rho(\sigma)\,d\sigma.$$

Combining with Step 2 proves the theorem. □

Remark 11.7.2. (i) The remainder cannot be improved in general: on the round sphere $\mathbb{S}^n$ the eigenvalues $k(k+n-1)$ of Δ have multiplicities of order k^{n-1}, so N has jumps of order λ^{n-1} (Exercise 11.8). The remainder is $o(\lambda^{n-1})$ if the set of periodic points of the geodesic flow on the unit cosphere bundle has measure zero (Duistermaat–Guillemin, Ivrii); this uses the structure of T at the non-zero periods. The weaker formula $N(\lambda) \sim C_W \lambda^n$ can also be obtained from the heat kernel or, as in [4], from residues of the zeta function of Δ.

(ii) For $n = 1$ each component of X is a circle and everything is explicit (Remark 11.6.3); in that case $N(\lambda) = C_W \lambda + O(1)$, but the argument of Step 2 only gives $O(\log \lambda)$ unless one uses that the lower order terms in (11.6.2) vanish.

(iii) The proof only used: the trace formula (11.6.2) for *one* positive test function with small support, the positivity of μ, and a polynomial a priori bound. See [8, Section 17.5] for the same method based on the Hadamard parametrix.

Question. Why must the test function be chosen with $\hat{\psi} \geq 0$?

Answer. Because Step 1 bounds the number of eigenvalues in a window $[\lambda - a, \lambda + a]$ by $c_0^{-1}(\rho * \mu)(\lambda)$, which needs $\rho = \hat{\psi}/(2\pi) \geq 0$ everywhere and $\rho \geq c_0 > 0$ on $[-a, a]$. With a kernel taking negative values, a large cluster of eigenvalues near λ could be cancelled in $\rho * \mu$ by the contributions of eigenvalues elsewhere, and the trace formula would give no control on N. Since ψ must also have small support (the parametrix is only valid for $|t| < t_0$), $\psi = \theta * \theta$ is the natural choice: compact support and $\hat{\psi} = \hat{\theta}^2 \geq 0$. □

11.8 Propagation of singularities and the Poisson relation

In this section we explain, with some steps only sketched, how the global behaviour of T is governed by closed geodesics.

Discussion. The trace T is obtained by restricting the wave kernel to the diagonal $x = y$ and integrating. A singularity of $K_{\cos}(t, \cdot, y)$ starting at y travels along a geodesic, so it can be on the diagonal at time $t \neq 0$ only if that geodesic returns to its starting point at time $|t|$ with the same direction, i.e. is closed of length dividing $|t|$ (Figure 11.3). After integration in x only these returns can produce singularities of T. Propagation of singularities makes this argument rigorous for all times.

The full theory requires Fourier integral operators [6] and the propagation theorem for operators of real principal type, which is proved in Chapter 3; for the strictly hyperbolic case see [8, Chapter XXIII].

Let $C_+ \subset \Sigma(P) \times \Sigma(P)$ be the forward bicharacteristic relation: $(\zeta, \zeta') \in C_+$ if ζ lies on the null bicharacteristic through ζ' and $t(\zeta) \geq t(\zeta')$. Write Δ^* for the diagonal of $(T^*(\mathbb{R} \times X) \setminus 0)^2$ and $\mathrm{WF}'(E) = \{(\zeta, \zeta');\ (\zeta, -\zeta') \in \mathrm{WF}(E)\}$.

Lemma 11.8.1. $\mathrm{WF}'(E) \cap \{|t - t'| < t_0\} \subset \Delta^* \cup C_+$, *and* $\mathrm{WF}(E)$ *contains no points with zero covector in either factor.*

Sketch of proof. By (11.3.1), $E = -H(t - t')K_{\sin}(t - t', x, x')$. For $0 < |t - t'| < t_0$, Corollary 11.4.6 gives $\mathrm{WF}(E) \subset$ pull-back of Γ, and by Lemma 11.1.2 a point of Γ at time $t - t' > 0$ corresponds exactly to a pair in C_+ (the point (t', x', τ, η) is carried to $(t, G_{\mp(t-t')}(x', \eta), \tau)$ for $\tau = \pm|\eta|$). At $t = t'$, $\mathrm{WF}(H(t - t'))$ consists of covectors $(\sigma, 0; -\sigma, 0)$ and $\mathrm{WF}(K_{\sin}(t - t', x, x'))$ of covectors $(\tau, \xi; -\tau, -\xi)$ with $x = x'$, $\tau = \pm|\xi|$; their sum is never

zero, so the product is defined [7, Section 8.2], and all resulting points over $\{t = t', x = x'\}$ have the form $(\zeta, -\zeta)$ with $\zeta \neq 0$, i.e. lie in Δ^* in the primed notation. The last claim follows from the same description. □

Theorem 11.8.2 (Propagation of singularities)**.** *Let $u \in \mathcal{C}^{-\infty}(\mathbb{R} \times X)$. Then $\mathrm{WF}(u) \setminus \mathrm{WF}(Pu) \subset \Sigma(P)$, and if $\gamma\colon J \to \Sigma(P) \setminus \mathrm{WF}(Pu)$ is a null bicharacteristic on an interval J with $\gamma(s_0) \in \mathrm{WF}(u)$ for some $s_0 \in J$, then $\gamma(J) \subset \mathrm{WF}(u)$.*

Proof. The first statement is microlocal elliptic regularity, since $\Sigma(P)$ is the characteristic set of P. The second follows from the propagation theorem for operators with real homogeneous principal symbol on Euclidean space proved in Chapter 3, applied in local coordinates. Let $A = \{s \in J;\ \gamma(s) \in \mathrm{WF}(u)\}$; it is non-empty, and closed in J since $\mathrm{WF}(u)$ is closed. Let $s_1 \in A$. Choose a coordinate chart $\kappa\colon U \to U' \subset \mathbb{R}^{n+1}$ of $\mathbb{R} \times X$ containing the projection of $\gamma(s_1)$, and a compact interval $I \subset J$ which is a neighbourhood of s_1 in J and such that the projection of $\gamma(I)$ lies in an open set V with $\overline{V} \subset U$ compact. Let $\phi, \psi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$ with $\phi = 1$ near $\overline{V}$ and $\psi = 1$ near $\operatorname{supp}\phi$. In the coordinates, $P = \sum_{|\alpha| \leq 2} a_\alpha(y) D_y^\alpha$ with smooth coefficients on U', and $\tilde{P} = \sum_{|\alpha| \leq 2} \psi a_\alpha D_y^\alpha$ (with ψ written in the coordinates) is a differential operator with coefficients in $\mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^{n+1})$, hence lies in $\Psi^2(\mathbb{R}^{n+1})$ and has real homogeneous principal symbol ψp. Let $\tilde{u} = \phi u$, a distribution with compact support in U'. Since P is a differential operator and $\phi = \psi = 1$ on V, $\tilde{P}\tilde{u} = Pu$ on V, so over V the wavefront sets of $\tilde{u}$ and $\tilde{P}\tilde{u}$ coincide with those of u and Pu (the wavefront set is local, and invariantly defined on $T^*(\mathbb{R} \times X)$). Over V, $\psi p = p$, so the Hamilton vector field of ψp there is the coordinate expression of H_p, and $\gamma|_I$ is a null bicharacteristic of $\tilde{P}$ which does not meet $\mathrm{WF}(\tilde{P}\tilde{u})$ and meets $\mathrm{WF}(\tilde{u})$ at $\gamma(s_1)$. By the propagation theorem of Chapter 3, $\gamma(I) \subset \mathrm{WF}(\tilde{u})$, i.e. $\gamma(I) \subset \mathrm{WF}(u)$. Hence A is open in J, and since J is connected, $A = J$. □

The same argument applies to the differential operator $D_t^2 - \Delta_X$ on $\mathbb{R} \times X \times Y$, whose principal symbol $\tau^2 - |\xi|_g^2$ is real: its null bicharacteristics through points with $(\tau, \xi) \neq 0$ are those of Lemma 11.1.2, with the (y, η) components fixed, while points with $\tau = 0$, $\xi = 0$ are stationary (there $H_p = 0$).

Proposition 11.8.3. *For all $t \in \mathbb{R}$, $\mathrm{WF}(K_{\cos}) \subset \Gamma$, with Γ as in Corollary 11.4.6.*

Sketch of proof. $K_{\cos}$ is annihilated by $D_t^2 - \Delta_x$ and by $D_t^2 - \Delta_y$ (as $\cos(t\Lambda)$ commutes with Δ), so $\mathrm{WF}(K_{\cos}) \subset \{\tau^2 = |\xi|^2 = |\eta|^2\}$, which does not meet $\{\tau = 0, \xi = 0\}$. By the extension of Theorem 11.8.2, $\mathrm{WF}(K_{\cos})$ is a union of complete bicharacteristics of $D_t^2 - \Delta_x$; each of these meets $\{|t| < t_0\}$, where $\mathrm{WF}(K_{\cos}) \subset \Gamma$ by Corollary 11.4.6. Since Γ is a union of such bicharacteristics, $\mathrm{WF}(K_{\cos}) \subset \Gamma$. □

Let $\mathcal{L}$ be the set of lengths of closed (non-constant, unit speed) geodesics of X. It is closed, and $\inf \mathcal{L} \geq 2\,\mathrm{inj}(X) > 0$, since a geodesic loop at x cannot be shorter than $2\,\mathrm{inj}(X)$.

Theorem 11.8.4 (Poisson relation)**.** $\operatorname{sing\,supp} T \subset \{0\} \cup \mathcal{L} \cup (-\mathcal{L})$.

Sketch of proof. Let $\iota\colon \mathbb{R} \times X \to \mathbb{R} \times X \times X$, $(t, x) \mapsto (t, x, x)$, and $\pi\colon \mathbb{R} \times X \to \mathbb{R}$. The conormal bundle of $\iota(\mathbb{R} \times X)$ consists of covectors $(0, \xi, -\xi)$, which are not in Γ (there $\tau \neq 0$), so $\iota^* K_{\cos}$ is defined and $\mathrm{WF}(\iota^* K_{\cos}) \subset \{(t, x; \tau, \xi - \eta);\ (t, x, x; \tau, \xi, -\eta) \in \Gamma\}$ [7, Section 8.2]. Since integration over t against ψ commutes with ι^* and turns $K_{\cos}$ into the smooth kernel K_{C_ψ}, $T = \pi_* \iota^* K_{\cos}$. For the push-forward, $\mathrm{WF}(\pi_* v) \subset \{(t, \tau);\ \exists x,\ (t, x; \tau, 0) \in \mathrm{WF}(v)\}$. Hence if $(t, \tau) \in \mathrm{WF}(T)$ there are x and $\eta \neq 0$ with $(x, \eta) = G_{\mp t}(x, \eta)$, i.e. either $t = 0$, or the geodesic through x in the direction $g^{-1}\eta$ is closed of length dividing $|t|$ — in particular $|t| \in \mathcal{L}$ (a multiple of a closed geodesic is a closed geodesic). □

Corollary 11.8.5. *The conclusions of Theorem 11.6.2 hold for every* $\psi \in \mathcal{C}^\infty_{\mathrm{c}}((-\inf \mathcal{L}, \inf \mathcal{L}))$.

Proof. Take $\psi_0 \in \mathcal{C}^\infty_{\mathrm{c}}((-t_0, t_0))$ with $\psi_0 = 1$ near 0. Then $(1-\psi_0)\psi T$ is smooth with compact support by Theorem 11.8.4, so its Fourier transform is rapidly decreasing, and Theorem 11.6.2 applies to $\psi_0\psi$, with $(\psi_0\psi)(0) = \psi(0)$. □

11.9 Exercises

Exercise 11.1. Let E be a real vector space with a non-degenerate antisymmetric bilinear form ω. Show that $\dim E = 2n$ is even and that there is a basis $v_1, \dots, v_n, w_1, \dots, w_n$ with dual basis v_j^*, w_j^* such that $\omega = \sum_j v_j^* \wedge w_j^*$. *Hint:* choose $v_1 \neq 0$, then w_1 with $\omega(v_1, w_1) = 1$; then v_2 in the ω-orthogonal complement of $\operatorname{span}(v_1, w_1)$ (why is this complement a complement, and why is ω non-degenerate on it?), and so on. Deduce that there is a linear isomorphism $L\colon E \to \mathbb{R}^{2n}$ with $\omega = L^*\omega_0$, where $\omega_0 = \sum_j d\xi_j \wedge dx_j$ is the standard symplectic form.

Exercise 11.2. Extend Exercise 11.1: if $V \subset E$ is a Lagrangian subspace ($\dim V = n$, $\omega|_V = 0$), then L can be chosen to map V onto $\{x = 0\} \subset \mathbb{R}^{2n}$. *Hint:* construct the basis with all $v_j \in V$ (or all w_j, depending on the sign convention).

Exercise 11.3. Let (S, ω) be a symplectic manifold and H_f the Hamilton vector field of $f \in \mathcal{C}^\infty(S)$, defined by $\iota_{H_f}\omega = -df$ (or with the opposite sign). Show that the Poisson bracket $\{f, g\} = H_f g$ makes $\mathcal{C}^\infty(S)$ into a Lie algebra, and that $[H_f, H_g] = H_{\{f,g\}}$. *Hint:* the Jacobi identity is equivalent to $d\omega = 0$.

Exercise 11.4. Let $H_0 = \{(x-y)\cdot\omega = 0\} \subset \mathbb{R}^n$ and let $H_\pm$ be the two characteristic hypersurfaces of Remark 11.4.5 through $\{0\} \times H_0$. Show that for $v_0 \in I^m(\mathbb{R}^n, H_0)$ and $v_1 \in I^{m+1}(\mathbb{R}^n, H_0)$ there are conormal $u_\pm$, associated with $H_\pm$ and solving $Pu_\pm \in \mathcal{C}^\infty$ near $t = 0$, with $u_+ + u_- = v_0$ and $D_t(u_+ + u_-) = v_1$ at $t = 0$ modulo $\mathcal{C}^\infty$, and that their full symbols at $t = 0$ are uniquely determined. (Write the conditions on the symbols as a triangular system, as in the proof of Theorem 11.4.4.)

Exercise 11.5. Let $F \in \mathcal{C}^0([0, t_1] \times X \times X)$ with $|F| \leq M$, extended by 0 for $t < 0$, and define the t-convolution $(F \circ G)(t, x, y) = \int_0^t \int_X F(t-s, x, z) G(s, z, y)\, d\mathrm{vol}(z)\, ds$. Show that the iterates $F^{(1)} = F$, $F^{(k)} = F^{(k-1)} \circ F$ satisfy $|F^{(k)}(t, x, y)| \leq M^k \operatorname{vol}(X)^{k-1} t^{k-1}/(k-1)!$, and deduce that $\mathrm{Id} + F$ (with Id the identity for t-convolution, kernel $\delta(t)\delta_y(x)$) has inverse $\mathrm{Id} + \tilde{F}$, $\tilde{F} = \sum_{k\geq 1} (-1)^k F^{(k)}$, with $\tilde{F}$ continuous. Show that if F is smooth and vanishes to infinite order at $t = 0$ then so does $\tilde{F}$, and use this, together with a local parametrix, to construct a forward fundamental solution as in Remark 11.3.2.

Exercise 11.6. For $X = \mathbb{R}/L\mathbb{Z}$ compute $N(\lambda)$ exactly and verify $N(\lambda) = \frac{L}{\pi}\lambda + O(1)$. Show that $T = \sum_{k\in\mathbb{Z}} \cos(2\pi k t/L) = L \sum_{m\in\mathbb{Z}} \delta(t - mL)$ and compare with Theorem 11.8.4.

Exercise 11.7. For the flat torus $X = \mathbb{R}^n/2\pi\mathbb{Z}^n$ show that the eigenvalues of Δ are $|k|^2$, $k \in \mathbb{Z}^n$, and deduce from Theorem 11.7.1 the lattice point estimate $\#\{k \in \mathbb{Z}^n;\ |k| \leq \lambda\} = \omega_n \lambda^n + O(\lambda^{n-1})$ $(n \geq 2)$.

Exercise 11.8. On the round sphere $\mathbb{S}^2$ the eigenvalues of Δ are $k(k+1)$, $k \in \mathbb{N}_0$, with multiplicity $2k+1$. Show that $C_W = 1$ and that $N(\lambda) - \lambda^2$ takes values $\geq \frac{1}{2}\lambda$ and $\leq -\frac{1}{2}\lambda$ for arbitrarily large λ, so the remainder in Theorem 11.7.1 is not $o(\lambda^{n-1})$ in general. Compute T for $\mathbb{S}^2$ and locate its singularities.

Exercise 11.9. Show that f_n is homogeneous of degree $-n$, that $f_n = D_s^{n-1}\delta$ for n odd, and that for n even $f_n(s) = c_n s^{-n}$ for $s \neq 0$, with a constant $c_n \neq 0$ (compute it). Deduce from Proposition 11.4.2 that, for n odd, δ is a superposition of distributions supported on hyperplanes, whereas for n even it is not.

Notes

Compared with the original notes, we have restricted attention to compact X (dropping the case of $\mathbb{R}^n$ with a metric that is Euclidean at infinity), and solved the Cauchy problem and constructed the forward fundamental solution by spectral decomposition, keeping the Volterra-series construction as a remark and an exercise. The short-time parametrix is written as an oscillatory integral whose phase solves the eikonal equation; in polar coordinates it is exactly the superposition of conormal solutions associated with characteristic hypersurfaces through hyperplanes that the original uses (Remark 11.4.5). The order of $f_n(x\cdot\omega)$ in Lemma 11.4.1 is $\frac{n}{2}-\frac{1}{4}$ (not $\frac{n}{4}-\frac{1}{4}$), and Weyl's formula with the remainder $O(\lambda^{n-1})$ is what the Tauberian argument proves; the original stated $o(\lambda^{n-1})$, which is false in general (Exercise 11.8).

The remainder $O(\lambda^{n-1})$ is due to Avakumović, Levitan and Hörmander (L. Hörmander, *The spectral function of an elliptic operator*, Acta Math. 121 (1968)); see also [8, Section 17.5]. The Poisson relation and the improvement to $o(\lambda^{n-1})$ under a condition on periodic geodesics are due to Chazarain, Colin de Verdière and Duistermaat–Guillemin (J. J. Duistermaat and V. W. Guillemin, *The spectrum of positive elliptic operators and periodic bicharacteristics*, Invent. Math. 29 (1975)), and Ivrii. For Fourier integral operators and Lagrangian distributions, of which the wave kernel is the basic example, see [6], [2]; for a different proof of the leading term see [4].

Part IV

Index theory

Chapter 12

Homotopy invariance of the index and Toeplitz operators

An elliptic operator on a compact manifold is Fredholm: its null space and its cokernel are finite dimensional. Neither dimension is stable. A small perturbation can create or destroy solutions of $Au = 0$. Their difference, the index, does not move. This stability is the reason the index can be computed from topological data, and this chapter explains where it comes from. It also builds the first tools for the computation: K-theory, Chern forms and the simplest index formula, for Toeplitz operators on the circle. Figure 12.1 shows how the sections fit together.

This chapter opens the part of the book devoted to index theory. Let X be a compact manifold without boundary and E, F complex vector bundles over X. By Chapter 5 an elliptic operator $A \in \Psi^m(X; E, F)$ has a parametrix and is Fredholm, so its index

$$\operatorname{ind}(A) = \dim \operatorname{Nul}(A) - \dim\bigl(\mathcal{C}^\infty(X; F)/A\,\mathcal{C}^\infty(X; E)\bigr)$$

is defined. The first two sections show that $\operatorname{ind}(A)$ can be written as a difference of traces of smoothing operators, and deduce from this that the index is constant along smooth families of elliptic operators. It therefore depends only on the homotopy class of the principal symbol. The index theorem computes this integer from topological data. To set that up we take *odd K-theory* to be the homotopy classes of maps into the group $G^{-\infty}(Y; E)$ of invertible smoothing perturbations of the identity. We show that this is an abelian group and introduce its even counterpart, which is classified by a loop group $G^{-\infty}_{(1)}$ (Section 12.3). The bi-invariant *Chern forms* $\operatorname{Tr}((A^{-1}dA)^{2k-1})$ on $G^{-\infty}$ give de Rham classes on these K-groups. The simplest of them, $\operatorname{Tr}(A^{-1}dA)$, has integral periods, and this gives a direct construction of the Fredholm determinant (Section 12.4). The last three sections turn to the Toeplitz algebra on the circle. It is the model for the "symbol sequence" that leads to Bott periodicity in Chapter 13. We compute the index of Toeplitz operators. We show that families of elliptic operators with vanishing "families index" can be perturbed to be invertible, and that the matrix groups can be deformed away inside the invertible Toeplitz operators.

12.1 The index as a trace

The index is the difference of two dimensions. Dimensions of finite rank projections are traces (Lemma 12.1.1), and traces of smoothing operators depend continuously on the operator. So the first step is to write the index as a difference of traces of smoothing operators built from a parametrix.

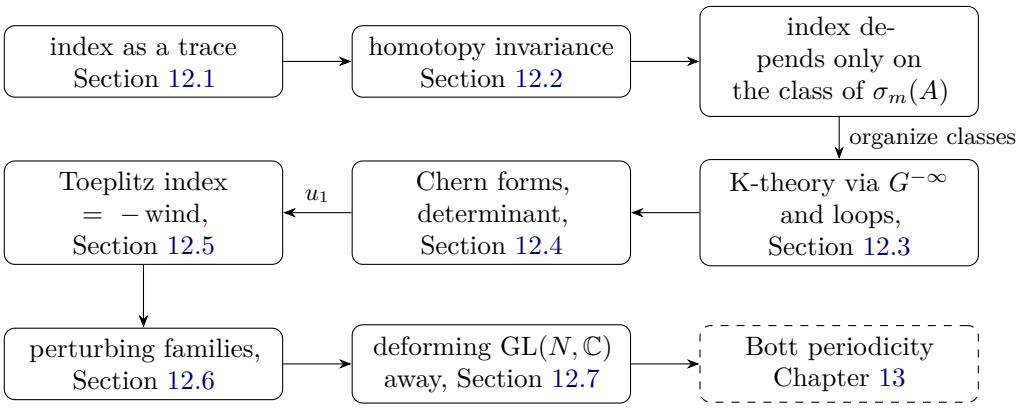


Figure 12.1: The logic of Chapter 12. The trace formula gives homotopy invariance; homotopy invariance leads to K-theory; the first Chern form u_1 computes the Toeplitz index as a winding number; the last two sections prepare the proof of Bott periodicity in Chapter 13.

12.1.1 What we use from earlier chapters

Throughout, X is a compact manifold without boundary, of positive dimension, and E, F are complex vector bundles over it. Operators in $\Psi^m(X;E,F)$ are classical (one-step polyhomogeneous). As in Chapters 4 and 5 their Schwartz kernels are conormal distributions at the diagonal. The principal symbol $\sigma_m(A)$ is a section of $\hom(\pi^*E,\pi^*F)$ over $T^*X\setminus 0$, homogeneous of degree m. After fixing a Riemannian metric we identify it with its restriction to the cosphere bundle S^*X. We use the following facts.

(i) The smoothing operators are $\Psi^{-\infty}(X;E,F) = \mathcal{C}^\infty(X^2;\mathrm{Hom}(E,F)\otimes\Omega_R)$, with $\mathrm{Hom}(E,F)=\pi_L^*F\otimes\pi_R^*E'$ and Ω_R the density bundle from the right factor. They carry the $\mathcal{C}^\infty$ (Fréchet) topology. A smoothing operator maps $\mathcal{C}^{-\infty}(X;E)$ continuously into $\mathcal{C}^\infty(X;F)$. In particular it is bounded on L^2 for any choice of Hermitian metrics and of a smooth positive density, and the operator norm is a continuous seminorm on $\Psi^{-\infty}$.

(ii) If $A\in\Psi^m$ and $R\in\Psi^{-\infty}$ then AR and RA are smoothing (when defined), and depend continuously on R. The formal adjoint and the transpose of A lie in Ψ^m, and $\sigma_m(A^*)=\sigma_m(A)^*$.

(iii) $A\in\Psi^m(X;E,F)$ is bounded $H^s(X;E)\to H^{s-m}(X;F)$ for every s. If A is elliptic, i.e. $\sigma_m(A)$ is invertible on S^*X, it has a parametrix $B_0\in\Psi^{-m}(X;F,E)$:

$$B_0A=\mathrm{Id}_E-R,\quad AB_0=\mathrm{Id}_F-R',\quad R\in\Psi^{-\infty}(X;E),\ R'\in\Psi^{-\infty}(X;F). \tag{12.1.1}$$

(iv) Finite rank smoothing operators are dense in $\Psi^{-\infty}(X;E,F)$. A smoothing operator of finite rank has kernel $\sum_{i=1}^N f_i(x)e_i(y)$ with $f_i\in\mathcal{C}^\infty(X;F)$ and $e_i\in\mathcal{C}^\infty(X;E'\otimes\Omega)$. The approximation can be made uniform for families: if $a\in\mathcal{C}^\infty(Z;\Psi^{-\infty}(X;E,F))$ with Z compact, the approximating kernels may be taken as $\sum_{i,j\le N}c_{ij}(z)f_i(x)e_j(y)$ with fixed f_i, e_j.

(v) The trace $\mathrm{Tr}(A)=\int_X \mathrm{tr}\,A(x,x)$ is a continuous linear functional on $\Psi^{-\infty}(X;E)$. It equals the matrix trace on finite rank operators, and $\mathrm{Tr}(AB)=\mathrm{Tr}(BA)$ for $A\in\Psi^{-\infty}(X;E,F)$, $B\in\Psi^{-\infty}(X;F,E)$. The proof in Chapter 5 for $E=F$ applies verbatim.

(vi) If $A \in \Psi^{-\infty}(X;E)$ and $\mathrm{Id}+A$ is invertible on $L^2(X;E)$ (equivalently on $\mathcal{C}^\infty(X;E)$), then $(\mathrm{Id}+A)^{-1} = \mathrm{Id}+B$ with $B \in \Psi^{-\infty}(X;E)$. The group

$$G^{-\infty}(X;E) = \{\mathrm{Id}+A;\ A \in \Psi^{-\infty}(X;E),\ \mathrm{Id}+A \text{ invertible}\}$$

is an open subset of $\mathrm{Id}+\Psi^{-\infty}(X;E)$, a topological group, and connected.

We identify $G^{-\infty}(X;E)$ with the open subset $\{A;\ \mathrm{Id}+A \text{ invertible}\} \subset \Psi^{-\infty}(X;E)$ whenever we speak of smooth maps into it. We also need two simple facts about finite rank operators.

Lemma 12.1.1. *If $\pi \in \Psi^{-\infty}(X;E)$ is a projection ($\pi^2 = \pi$) of finite rank, then* $\mathrm{Tr}(\pi) = \dim \mathrm{Ran}(\pi)$.

Proof. Let $f_1, \dots, f_N \in \mathcal{C}^\infty(X;E)$ be a basis of $\mathrm{Ran}(\pi)$. Then $\pi u = \sum_i \ell_i(u) f_i$, where the ℓ_i are continuous linear functionals given by smooth sections e_i of $E' \otimes \Omega$. Since $\pi f_j = f_j$ we get $\ell_i(f_j) = \delta_{ij}$, so $\mathrm{Tr}(\pi) = \sum_i \ell_i(f_i) = N$. □

Lemma 12.1.2. *If $A \in \Psi^m(X;E,F)$ and $C \in \Psi^{-\infty}(X;F,E)$ then* $\mathrm{Tr}(AC) = \mathrm{Tr}(CA)$, *the traces being taken on $\Psi^{-\infty}(X;F)$ and $\Psi^{-\infty}(X;E)$ respectively.*

Proof. Both sides are continuous in C by (ii) and (v). By (iv) it suffices to treat C of rank one, $Cf = \ell(f)w$ with $w \in \mathcal{C}^\infty(X;E)$ and $\ell(f) = \int_X v \cdot f$, $v \in \mathcal{C}^\infty(X;F' \otimes \Omega)$. Then AC is the rank one operator $f \mapsto \ell(f)\, Aw$ and CA is $g \mapsto \ell(Ag)\, w$. The functional $g \mapsto \ell(Ag)$ is given by the smooth section $A^t v$, where $A^t \in \Psi^m$ is the transpose. Thus

$$\mathrm{Tr}(AC) = \int_X v \cdot Aw = \int_X (A^t v) \cdot w = \mathrm{Tr}(CA). \qquad \square$$

12.1.2 The generalized inverse

The parametrix of an elliptic operator can be improved to an exact inverse up to finite rank projections.

Proposition 12.1.3. *Let $A \in \Psi^m(X;E,F)$ be elliptic. Choose Hermitian metrics on E, F and a smooth positive density on X, and let $A^* \in \Psi^m(X;F,E)$ be the formal adjoint. Then* $\mathrm{Nul}(A) \subset \mathcal{C}^\infty(X;E)$ *and* $\mathrm{Nul}(A^*) \subset \mathcal{C}^\infty(X;F)$ *are finite dimensional and*

$$\mathcal{C}^\infty(X;F) = A\big(\mathcal{C}^\infty(X;E)\big) \oplus \mathrm{Nul}(A^*), \tag{12.1.2}$$

an L^2-orthogonal decomposition. Let π and π' be the orthogonal projections onto $\mathrm{Nul}(A)$ *and* $\mathrm{Nul}(A^*)$. *There is a unique $B \in \Psi^{-m}(X;F,E)$, the* generalized inverse *of A, such that*

$$BA = \mathrm{Id}_E - \pi, \quad AB = \mathrm{Id}_F - \pi', \quad \pi B = 0, \quad B\pi' = 0. \tag{12.1.3}$$

The projections π, π' are self-adjoint finite rank smoothing operators, and

$$\mathrm{ind}(A) = \dim \mathrm{Nul}(A) - \dim \mathrm{Nul}(A^*) = \mathrm{Tr}(\pi) - \mathrm{Tr}(\pi').$$

Proof. Let B_0, R, R' be as in (12.1.1). As a bounded map $A : H^m(X;E) \to L^2(X;F)$, A satisfies $B_0 A = \mathrm{Id} - R$ and $AB_0 = \mathrm{Id} - R'$, where R is compact on H^m and R' is compact on L^2. Indeed both map into $\mathcal{C}^\infty$ continuously from H^{-N} for every N, and $H^N \hookrightarrow H^m$ is compact for $N > m$ (Rellich). By Appendix A $A : H^m \to L^2$ is Fredholm. In particular its range is closed. A distribution $u \in H^m$ with $Au = 0$ satisfies $u = Ru \in \mathcal{C}^\infty$, so $\mathrm{Nul}(A)$ is the same on H^m and on $\mathcal{C}^\infty$, and is finite dimensional. The orthogonal complement of $A(H^m)$ in L^2 consists of the $v \in L^2$ with $A^* v = 0$ in the sense of distributions. Since

$B_0^* A^* = \mathrm{Id} - (R')^*$, such v satisfy $v = (R')^* v \in \mathcal{C}^\infty$, so this complement is $\mathrm{Nul}(A^*)$, a finite dimensional subspace of $\mathcal{C}^\infty(X;F)$. If $f \in \mathcal{C}^\infty(X;F)$ is orthogonal to $\mathrm{Nul}(A^*)$ then $f = Au$ with $u \in H^m$, and $u = B_0 f + Ru \in \mathcal{C}^\infty$. This proves (12.1.2). The orthogonal projection onto a finite dimensional space $W \subset \mathcal{C}^\infty$ with orthonormal basis w_i has kernel $\sum_i w_i(x)\overline{w_i(y)}\nu(y)$, so π and π' are self-adjoint finite rank smoothing operators.

The map $A : \mathrm{Nul}(A)^\perp \cap H^m \to A(H^m)$ is a continuous bijection between Hilbert spaces. Let $B : L^2(X;F) \to H^m(X;E)$ be its inverse on $A(H^m)$, extended by zero on $\mathrm{Nul}(A^*)$. By the open mapping theorem B is bounded, and (12.1.3) holds on L^2. These four identities determine B. Substituting $\mathrm{Id} = AB_0 + R'$ on the right and $\mathrm{Id} = B_0 A + R$ on the left,

$$\begin{aligned} B &= B(AB_0 + R') = (\mathrm{Id} - \pi)B_0 + BR' = (\mathrm{Id} - \pi)B_0 + (B_0 A + R)BR' \\ &= (\mathrm{Id} - \pi)B_0 + B_0(\mathrm{Id} - \pi')R' + RBR'. \end{aligned}$$

The first two terms lie in $\Psi^{-m}(X;F,E)$. The kernel of RBR' is smooth. The map $y \mapsto R'(\cdot, y)$ is smooth from X into $\mathcal{C}^\infty(X;F)$, hence into L^2. Then B is bounded on L^2, and $R : L^2 \to \mathcal{C}^\infty$ is continuous. So $y \mapsto RBR'(\cdot, y)$ is a smooth map into $\mathcal{C}^\infty(X;E)$. Thus $B \in \Psi^{-m}(X;F,E)$. Finally $\mathrm{Tr}(\pi) = \dim \mathrm{Nul}(A)$ and $\mathrm{Tr}(\pi') = \dim \mathrm{Nul}(A^*) = \dim \mathcal{C}^\infty(X;F)/A\mathcal{C}^\infty(X;E)$ by Lemma 12.1.1 and (12.1.2). □

Any two parametrices of A differ by a smoothing operator. Indeed, if $B_1 A - \mathrm{Id}$ and $AB_2 - \mathrm{Id}$ are smoothing then $B_1 - B_2 = B_1(\mathrm{Id} - AB_2) - (\mathrm{Id} - B_1 A)B_2 \in \Psi^{-\infty}$. In particular the generalized inverse is a parametrix in the sense of (12.1.1).

Theorem 12.1.4. *Let $A \in \Psi^m(X;E,F)$ be elliptic and let $B \in \Psi^{-m}(X;F,E)$ be any parametrix, so that $\mathrm{Id}_E - BA$ and $\mathrm{Id}_F - AB$ are smoothing. Then*

$$\mathrm{ind}(A) = \mathrm{Tr}(\mathrm{Id}_E - BA) - \mathrm{Tr}(\mathrm{Id}_F - AB). \tag{12.1.4}$$

Proof. For the generalized inverse $\tilde{B}$ of Proposition 12.1.3 this is the identity $\mathrm{ind}(A) = \mathrm{Tr}(\pi) - \mathrm{Tr}(\pi')$. For a general parametrix, $C = B - \tilde{B} \in \Psi^{-\infty}(X;F,E)$ and

$$(\mathrm{Tr}(\mathrm{Id} - BA) - \mathrm{Tr}(\mathrm{Id} - AB)) - (\mathrm{Tr}(\mathrm{Id} - \tilde{B}A) - \mathrm{Tr}(\mathrm{Id} - A\tilde{B})) = \mathrm{Tr}(AC) - \mathrm{Tr}(CA) = 0$$

by Lemma 12.1.2. □

Question. In finite dimensions, how does (12.1.4) reproduce the familiar formula for the index?

Answer. Let $A : \mathbb{C}^n \to \mathbb{C}^m$ and let $B : \mathbb{C}^m \to \mathbb{C}^n$ be any linear map. Then $\mathrm{Tr}(\mathrm{Id}_n - BA) - \mathrm{Tr}(\mathrm{Id}_m - AB) = n - m$, because $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$. By rank–nullity, $n - m = \dim \mathrm{Nul}\, A - \dim(\mathbb{C}^m / A\mathbb{C}^n)$. In infinite dimensions $n - m$ becomes "$\infty - \infty$". The parametrix makes $\mathrm{Id} - BA$ and $\mathrm{Id} - AB$ smoothing, hence trace class, so the two infinite traces of the identity cancel before any trace is taken. □

Remark 12.1.5. The right side of (12.1.4) is $\mathrm{Tr}([A,B])$ in a formal sense: $\mathrm{Id} - BA$ and $\mathrm{Id} - AB$ are individually trace class, but AB and BA are not. The formula shows that the index measures how far the trace identity $\mathrm{Tr}(AB) = \mathrm{Tr}(BA)$ fails when the operators are not smoothing.

12.2 Homotopy invariance of the index

Theorem 12.1.4 writes the index as a continuous function of the operator, as long as the parametrix can be chosen continuously. A continuous integer-valued function on an interval is constant. So the only work is to construct parametrices that depend smoothly on a parameter.

12.2.1 Smooth families

Let Z be a manifold, possibly with boundary or corners (we mostly use $Z = [0,1]$ and $Z = [0,1] \times X'$ with X' compact). A *smooth family* $A \in \mathcal{C}^\infty(Z; \Psi^m(X; E, F))$ is a family $A_z \in \Psi^m(X; E, F)$ whose kernels have the following properties. Away from $Z \times \mathrm{Diag}$ they are smooth on $Z \times (X^2 \setminus \mathrm{Diag})$. In local coordinates and trivializations near the diagonal they are of the form

$$\int e^{i(x-y)\cdot\xi} a(z, x, \xi) đ\xi + K(z, x, y),$$

where K is smooth in all variables and a is a classical symbol of order m depending smoothly on z. That is, every z-derivative of a satisfies the symbol estimates locally uniformly in z, and the homogeneous terms of the expansion of a are smooth in z. For $m = -\infty$ this simply means $A \in \mathcal{C}^\infty(Z \times X^2; \mathrm{Hom}(E, F) \otimes \Omega_R)$.

All the constructions of Chapters 4 and 5 (composition, adjoints, quantization of symbols, asymptotic summation, parametrices) are given by formulas that are continuous in the symbol seminorms, so they apply to smooth families with the result again smooth. The only point needing a comment is asymptotic summation (Borel's lemma). The cut-offs $\chi(\epsilon_j \xi)$ can be chosen with ϵ_j decreasing so fast that the estimates hold uniformly on an exhaustion of Z by compact sets. We record the consequences we need.

Lemma 12.2.1. *Let $A \in \mathcal{C}^\infty(Z; \Psi^m(X; E, F))$ be a smooth family with A_z elliptic for every $z \in Z$.*

(a) There is a smooth family $B \in \mathcal{C}^\infty(Z; \Psi^{-m}(X; F, E))$ with $\mathrm{Id} - B_z A_z$ and $\mathrm{Id} - A_z B_z$ in $\mathcal{C}^\infty(Z; \Psi^{-\infty})$.

(b) If $m = 0$, $E = F$ and every A_z is invertible, then $z \mapsto A_z^{-1}$ is a smooth family in $\mathcal{C}^\infty(Z; \Psi^0(X; E))$.

Moreover, given $a \in \mathcal{C}^\infty(Z \times S^*X; \hom(\pi^*E, \pi^*F))$ there is a smooth family $Q(a) \in \mathcal{C}^\infty(Z; \Psi^m(X; E, F))$ with $\sigma_m(Q(a)_z) = a_z$, depending linearly on a.

Proof. The quantization Q is obtained by extending a homogeneously of degree m, cutting off near the zero section, quantizing in coordinate charts and summing with a partition of unity. Each step preserves smooth dependence on z. For (a), let $B' = Q(\sigma_m(A)^{-1})$ (with $-m$ in place of m). Then $\mathrm{Id} - B'A \in \mathcal{C}^\infty(Z; \Psi^{-1})$, and the left parametrix $B_L \sim \sum_{j \geq 0} (\mathrm{Id} - B'A)^j B'$ is an asymptotic sum of smooth families, so can be chosen smooth in z. Similarly for a right parametrix B_R, and $B_L - B_R = B_L(\mathrm{Id} - AB_R) - (\mathrm{Id} - B_L A)B_R$ is a smooth family of smoothing operators.

For (b), let $G_z = A_z^{-1}$. It lies in Ψ^0 by Proposition 12.1.3 with $\pi = \pi' = 0$. A smooth family in Ψ^0 is a smooth family of bounded operators on L^2, because the L^2 norm of an operator of order 0 is bounded by a symbol seminorm. So $z \mapsto G_z$ is smooth into the bounded operators, being the inverse of a smooth family of invertible bounded operators. With B from (a) and $R = \mathrm{Id} - BA$, $R' = \mathrm{Id} - AB$, the argument in the proof of Proposition 12.1.3 (with $\pi = \pi' = 0$) gives

$$G = G(AB + R') = B + GR' = B + (BA + R)GR' = B + BR' + RGR'.$$

Since R_z, R'_z are smooth families of smoothing operators and G_z is smooth in operator norm, the kernel of RGR' is smooth in all variables. Hence G is a smooth family. □

12.2.2 Invariance

Theorem 12.2.2. *If $A \in \mathcal{C}^\infty([0,1]; \Psi^m(X;E,F))$ is a smooth family with A_t elliptic for every $t \in [0,1]$, then $\operatorname{ind}(A_t)$ is independent of t.*

Proof. Choose a smooth family of parametrices B_t as in Lemma 12.2.1. By Theorem 12.1.4

$$\operatorname{ind}(A_t) = \operatorname{Tr}(\mathrm{Id} - B_t A_t) - \operatorname{Tr}(\mathrm{Id} - A_t B_t).$$

The arguments of the traces are continuous (indeed smooth) maps $[0,1] \to \Psi^{-\infty}$ and the trace is continuous, so $\operatorname{ind}(A_t)$ is a continuous integer-valued function on $[0,1]$. Hence it is constant. □

Discussion. Where was ellipticity used? Only in producing the smooth family of parametrices B_t. If ellipticity fails at one time t_0, then $\mathrm{Id} - B_t A_t$ cannot be smoothing near t_0, the traces are undefined, and the index can jump. Figure 12.2 shows the typical picture for the Toeplitz operators of Section 12.5: along the family of symbols $a_t(\theta) = (1-t) + te^{i\theta}$ the index is 0 for $t < \frac{1}{2}$ and -1 for $t > \frac{1}{2}$, and at $t = \frac{1}{2}$ the symbol vanishes at one point.

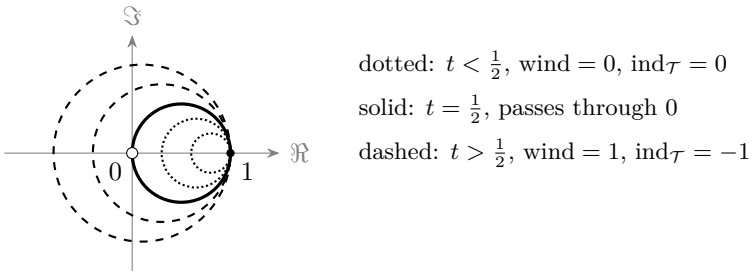


Figure 12.2: The images in $\mathbb{C}$ of the symbols $a_t(\theta) = (1-t) + te^{i\theta}$, circles of centre $1-t$ and radius t. As long as the curve avoids 0 (the symbol is elliptic) its winding number about 0, and with it the index of T_{a_t} (Theorem 12.5.4), cannot change. The only way to change the index is to pass through a non-elliptic symbol, here at $t = \frac{1}{2}$.

Remark 12.2.3. The same conclusion holds for families that are only continuous in t (in the topology of Ψ^m). One way to see this is to note that such a family is norm-continuous as a family of Fredholm operators $H^m \to L^2$. Its index there equals the index on $\mathcal{C}^\infty$, by the proof of Proposition 12.1.3. The index of Fredholm operators is locally constant in the norm topology (Appendix A).

Two elliptic operators $A, A' \in \Psi^m(X;E,F)$ with the same principal symbol are joined by the smooth family $(1-t)A + tA'$, which has constant principal symbol and so is elliptic. Hence $\operatorname{ind}(A) = \operatorname{ind}(A')$. More precisely, we have the following.

Corollary 12.2.4. *Let $A_0, A_1 \in \Psi^m(X;E,F)$ be elliptic. Suppose that*

$$a \in \mathcal{C}^\infty([0,1] \times S^*X; \hom(\pi^*E, \pi^*F))$$

is invertible at every point, with $a_0 = \sigma_m(A_0)$ and $a_1 = \sigma_m(A_1)$. Then $\operatorname{ind}(A_0) = \operatorname{ind}(A_1)$. Thus the index of an elliptic operator depends only on the homotopy class of its principal symbol in $\mathcal{C}^\infty(S^*X; \operatorname{Iso}(\pi^*E, \pi^*F))$.

Proof. With Q as in Lemma 12.2.1,

$$A_t = Q(a)_t + (1-t)(A_0 - Q(a)_0) + t(A_1 - Q(a)_1)$$

is a smooth family with $\sigma_m(A_t) = a_t$ invertible, and A_0, A_1 are its end points. Apply Theorem 12.2.2. □

Remark 12.2.5. (1) A continuous homotopy of invertible symbols between smooth end points can be approximated, with the same end points, by a smooth one, since the invertible elements form an open set. So continuous homotopies suffice in Corollary 12.2.4.

(2) The order is also irrelevant. For each $s \in \mathbb{R}$ there is an invertible elliptic $\Lambda_s \in \Psi^s(X;E)$. Take any elliptic $\Lambda \in \Psi^{s/2}(X;E)$ and set $\Lambda_s = \Lambda^*\Lambda + \pi$, with π the orthogonal projection onto $\mathrm{Nul}(\Lambda)$. If $\Lambda_s u = 0$ then $\|\Lambda u\|^2 + \|\pi u\|^2 = 0$, so $u = 0$. Since Λ_s is self-adjoint, (12.1.2) shows it is also surjective. For $A \in \Psi^m(X;E,F)$ elliptic, $A\Lambda_{-m} \in \Psi^0(X;E,F)$ has the same index, since $\mathrm{Nul}(A\Lambda_{-m}) = \Lambda_{-m}^{-1}\,\mathrm{Nul}(A)$ and the ranges coincide. So the index is a function on the homotopy classes of $\mathcal{C}^\infty(S^*X; \mathrm{Iso}(\pi^*E, \pi^*F))$. Even this contains more information than the index sees. The index theorem eliminates the extra data by passing to K-theory, which we now introduce.

12.3 Smoothing groups and K-theory

Corollary 12.2.4 reduces the index to a function on homotopy classes of invertible symbols. To compute it we need a good way of organizing such classes. K-theory does this by ignoring the size of the matrices: a map g into $\mathrm{GL}(N,\mathbb{C})$ is identified with its *stabilization* $g \oplus \mathrm{Id} \in \mathrm{GL}(N+1,\mathbb{C})$. The group $G^{-\infty}(Y;E)$ of invertible smoothing perturbations of the identity contains all the groups $\mathrm{GL}(N,\mathbb{C})$, nested inside each other, so it performs this stabilization automatically.

12.3.1 Odd K-theory

Let Y be a compact manifold of positive dimension and E a vector bundle over Y. Let X be a compact manifold, possibly with boundary or corners. A *smooth map* $f : X \to G^{-\infty}(Y;E)$ is a map such that $f - \mathrm{Id} \in \mathcal{C}^\infty(X; \Psi^{-\infty}(Y;E)) = \mathcal{C}^\infty(X \times Y^2; \mathrm{Hom}(E) \otimes \Omega_R)$ and $f(x)$ is invertible for each x. Then $x \mapsto f(x)^{-1} - \mathrm{Id}$ is again smooth. If $f = \mathrm{Id} + a$ and $f^{-1} = \mathrm{Id} + b$, the identity $b = -a + a^2 + aba$ shows that b is smooth into $\Psi^{-\infty}$ once it is continuous in L^2 norm, and the latter holds because inversion is continuous on bounded operators. A smooth homotopy is a smooth map on $[0,1] \times X$. As in Chapter 5 we set

$$K^{-1}(X) = [X; G^{-\infty}(Y;E)], \tag{12.3.1}$$

the set of smooth homotopy classes of smooth maps. We recall the result of Chapter 5: (12.3.1) is independent of (Y,E). The natural bijections are induced by finite rank approximation, which deforms any smooth f to a map of the form

$$f(x) = \mathrm{Id} + \sum_{i,j=1}^{N} (g_{ij}(x) - \delta_{ij})\phi_i \otimes \phi_j^*, \quad g \in \mathcal{C}^\infty(X; \mathrm{GL}(N,\mathbb{C})), \tag{12.3.2}$$

for an orthonormal family $\phi_1, \dots, \phi_N \in \mathcal{C}^\infty(Y;E)$, where $\phi_i \otimes \phi_j^*$ denotes $u \mapsto (u, \phi_j)\phi_i$. Such an f acts as the matrix $g(x)$ on $W_N = \mathrm{span}\{\phi_i\}$ and as the identity on $W_N^\perp$. It is invertible precisely when $g(x)$ is. To see that a smooth $f = \mathrm{Id} + a$ is homotopic to a map of this form, approximate a uniformly on X in $\mathcal{C}^\infty(X;\Psi^{-\infty})$ by $a^{(N)} = \sum_{i,j \le N} c_{ij}(x)\phi_i \otimes \phi_j^*$, as in (iv). Since $G^{-\infty}$ is open and X compact, $\mathrm{Id} + (1-s)a + sa^{(N)}$ is invertible for $s \in [0,1]$ once N is large.

Proposition 12.3.1. *Pointwise multiplication induces a product on $K^{-1}(X)$ which makes it an abelian group, with identity the class of the constant map* Id *and inverse $[f]^{-1} = [f^{-1}]$. The product is also induced by the "direct sum": if f and f' are of the form (12.3.2) with matrices g, g' acting on orthogonal spaces W, W', then $[f][f']$ is represented by the map acting as $g \oplus g'$ on $W \oplus W'$ and as the identity on $(W \oplus W')^\perp$.*

Proof. If f_s, f'_s are homotopies then so is $f_s f'_s$. Hence the product is well defined on homotopy classes. Associativity, the identity and the inverse are inherited from $G^{-\infty}$. For commutativity, deform f and f' to the form (12.3.2) with matrices g, g' with respect to the same orthonormal family $\phi_1, \dots, \phi_N$. Enlarge the family to $\phi_1, \dots, \phi_{2N}$ (possible since $\dim Y > 0$, so $\mathcal{C}^\infty(Y; E)$ is infinite dimensional). Let W_1 and W_2 be spanned by the first and last N elements. For $\theta \in [0, \pi/2]$ let R_θ be the unitary operator acting on $W_1 \oplus W_2 = \mathbb{C}^N \oplus \mathbb{C}^N$ by the block rotation

$$R_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

and as the identity on $(W_1 \oplus W_2)^\perp$. Then $R_\theta - \mathrm{Id}$ is a finite rank smoothing operator and $R_\theta f' R_\theta^{-1}$ is a smooth homotopy in $G^{-\infty}$. At $\theta = \pi/2$ it acts as $\mathrm{Id}_{W_1} \oplus g'$, with g' now acting on W_2; this is a direct computation with $R_{\pi/2} = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. Thus f' is homotopic to $f'' = \mathrm{Id}_{W_1} \oplus g'$, which commutes with $f = g \oplus \mathrm{Id}_{W_2}$, and

$$[f][f'] = [f f''] = [g \oplus g'] = [f'' f] = [f'][f].$$

The middle term is the direct sum description. □

Question. The groups $\mathrm{GL}(N, \mathbb{C})$ and $G^{-\infty}(Y; E)$ are not commutative. Why is $K^{-1}(X)$?

Answer. Because commutativity is only needed up to homotopy, and there is room to move. The rotation R_θ slides g' into a block W_2 orthogonal to the block W_1 on which g acts, and maps acting on orthogonal blocks commute. The room comes from $\dim \mathcal{C}^\infty(Y; E) = \infty$, which is why $\dim Y > 0$ is assumed. Inside a fixed $\mathrm{GL}(N, \mathbb{C})$, with g and g' using all N dimensions, there is no such room and the argument fails. □

Remark 12.3.2. In terms of homotopy groups, the assertion that (12.3.1) is odd K-theory, i.e. that $G^{-\infty}$ is a classifying space for K^{-1}, amounts to Bott periodicity in the form

$$\pi_k(G^{-\infty}(Y; E)) = \begin{cases} 0 & k \text{ even}, \\ \mathbb{Z} & k \text{ odd}. \end{cases} \tag{12.3.3}$$

This is proved in Chapter 13. In the standard topological treatment $K^{-1}(X)$ is defined by stabilizing maps into $\mathrm{GL}(N, \mathbb{C})$ as $N \to \infty$ [17]. The group $G^{-\infty}$ makes the stabilization automatic: by (12.3.2) every map is stably a map into some $\mathrm{GL}(N, \mathbb{C})$, and the rotation argument above is the usual proof that the stabilized product is commutative.

12.3.2 Even K-theory

Odd K-theory is built from invertible maps. Even K-theory, K^0, is usually built from vector bundles (Chapter 13). The bridge is the *clutching construction*: a loop $g : \mathbb{S}^1 \to \mathrm{GL}(N, \mathbb{C})$ glues the trivial bundles $\mathbb{C}^N$ over the two hemispheres of $\mathbb{S}^2$ along the equator, and every bundle on $\mathbb{S}^2$ arises in this way (Figure 13.5, Lemma 13.9.7). So loops of invertible maps are transition functions of bundles, and they give a model of even K-theory.

Let $\mathbb{S}^1 = \{e^{i\theta}\}$ with base point 1.

Definition 12.3.3. $K^{-2}(X)$ is the set of smooth homotopy classes of *pointed* smooth maps $f : X \times \mathbb{S}^1 \to G^{-\infty}(Y; E)$, meaning $f(x, 1) = \mathrm{Id}$ for all $x \in X$, where homotopies are also required to be pointed at every time.

Pointwise product makes $K^{-2}(X)$ a group. To see that it is abelian, and how it is related to K^{-1}, we compare it with $K^{-1}(X \times \mathbb{S}^1)$.

Lemma 12.3.4. *The map $\Phi : K^{-1}(X) \times K^{-2}(X) \to K^{-1}(X \times \mathbb{S}^1)$, $\Phi([g],[h]) = [gh]$, where g is regarded as a map on $X \times \mathbb{S}^1$ constant in θ, is an isomorphism of groups. Its inverse is*

$$[f] \mapsto \big([f_1],[f_2]\big), \quad f_1(x) = f(x,1), \quad f_2(x,\theta) = f(x,1)^{-1} f(x,\theta). \tag{12.3.4}$$

In particular $K^{-2}(X)$ is abelian and $K^{-1}(X \times \mathbb{S}^1) \cong K^{-1}(X) \oplus K^{-2}(X)$.

Proof. Φ is well defined, since homotopies of g and h multiply to a homotopy of gh. The map (12.3.4) is also well defined. If F is a homotopy from f to f', then $F(\cdot,1)$ is a homotopy from f_1 to f_1' and $F(\cdot,1)^{-1}F$ is a pointed homotopy from f_2 to f_2'. The two maps are mutually inverse at the level of maps, since $f = f_1 f_2$ and $(gh)_1 = g$, $(gh)_2 = h$. Finally Φ is a homomorphism because $K^{-1}(X \times \mathbb{S}^1)$ is abelian by Proposition 12.3.1:

$$\Phi([g][g'],[h][h']) = [g][g'][h][h'] = [g][h][g'][h'] = \Phi([g],[h])\Phi([g'],[h']).$$

Since Φ restricted to $\{1\} \times K^{-2}(X)$ is an injective homomorphism into an abelian group, $K^{-2}(X)$ is abelian. □

12.3.3 The loop group

Pointed maps may be further normalized to be *flat* at the base point, i.e. to differ from the identity by a map vanishing to infinite order at $X \times \{1\}$. Such maps are naturally Schwartz functions on $X \times \mathbb{R}$.

Definition 12.3.5. The (smooth, flat, pointed) loop group of $G^{-\infty}(Y;E)$ is

$$G^{-\infty}_{(1)}(Y;E) = \big\{a \in \mathcal{S}(\mathbb{R};\Psi^{-\infty}(Y;E));\ \mathrm{Id}+a(t) \in G^{-\infty}(Y;E)\ \forall\, t \in \mathbb{R}\big\},$$

where $\mathcal{S}(\mathbb{R};\Psi^{-\infty}(Y;E)) = \mathcal{S}(\mathbb{R}\times Y^2;\mathrm{Hom}(E)\otimes\Omega_R)$ is the space of kernels that are Schwartz in t, uniformly with all derivatives in Y^2. The product is $(\mathrm{Id}+a)(\mathrm{Id}+b) = \mathrm{Id}+a+b+ab$, computed pointwise in t.

Lemma 12.3.6. *$G^{-\infty}_{(1)}(Y;E)$ is an open subset of $\mathcal{S}(\mathbb{R};\Psi^{-\infty}(Y;E))$ and a topological group.*

Proof. If $a \in G^{-\infty}_{(1)}$ then $\|a(t)\|_{L^2} \to 0$ as $|t| \to \infty$. So $\sup_t \|(\mathrm{Id}+a(t))^{-1}\| < \infty$ by continuity of inversion, and every a' with $\sup_t \|a'(t)-a(t)\|_{L^2}$ small also has $\mathrm{Id}+a'(t)$ invertible for all t. The supremum is bounded by a Schwartz seminorm, so $G^{-\infty}_{(1)}$ is open. Products of Schwartz families are Schwartz. For the inverse $\mathrm{Id}+b(t)$ we have $b = -a + a^2 + aba$ pointwise. Here $b(t)$ is bounded on L^2 uniformly in t, as are its t-derivatives, by repeated differentiation of $\partial_t b = -(\mathrm{Id}+b)(\partial_t a)(\mathrm{Id}+b)$. The kernel of aba is then Schwartz in t, because each factor a contributes Schwartz decay and $\mathcal{C}^\infty$ regularity in the outer variable. So $b \in G^{-\infty}_{(1)}$, and composition and inversion are continuous for the same reasons. □

Proposition 12.3.7. *Let $\gamma(t) = \exp\big(i\pi(1 + t/\langle t\rangle)\big)$, a diffeomorphism of $\mathbb{R}$ onto $\mathbb{S}^1 \setminus \{1\}$. Pulling back by γ gives a bijection*

$$K^{-2}(X) \cong [X; G^{-\infty}_{(1)}(Y;E)]$$

onto the smooth homotopy classes of smooth maps $X \to G^{-\infty}_{(1)}(Y;E)$.

Proof. First, pulling back by γ identifies the functions on $\mathbb{S}^1$ vanishing to infinite order at 1 with $\mathcal{S}(\mathbb{R})$, with parameters. Indeed, near $t = \pm\infty$ the angle $\theta = \pi(1 + t/\langle t\rangle)$ satisfies $\theta \sim \pi/(2t^2)$ and $2\pi - \theta \sim \pi/(2t^2)$. Moreover $d\theta/dt = \pi\langle t\rangle^{-3}$, so ∂_t and ∂_θ differ by factors that are polynomially bounded in t or in θ^{-1}. Infinite order vanishing in θ is therefore equivalent to rapid decay of all t-derivatives. Hence smooth maps $X \to G^{-\infty}_{(1)}$ correspond exactly to smooth pointed maps $X \times \mathbb{S}^1 \to G^{-\infty}$ that are flat at $X \times \{1\}$, and the same holds for homotopies.

It remains to show that every pointed map is homotopic, through pointed maps, to a flat one, and that two flat maps which are pointed homotopic are homotopic through flat maps. Let $f = \mathrm{Id} + F$ be pointed. Then $F(x, 1) = 0$, so by compactness there is $\epsilon > 0$ with $\|F(x, e^{i\theta})\|_{L^2} < 1$ for $|\theta| \leq 2\epsilon$. Choose $\rho \in \mathcal{C}^\infty(\mathbb{S}^1)$ with $0 \leq \rho \leq 1$, $\rho = 1$ for $|\theta| \leq \epsilon$ and $\rho = 0$ for $|\theta| \geq 2\epsilon$. Then

$$f_s = \mathrm{Id} + (1 - s\rho)F, \quad s \in [0, 1],$$

is a pointed homotopy. Where $\rho \neq 0$ we have $\|(1 - s\rho)F\|_{L^2} < 1$, so f_s is invertible. The end point f_1 equals Id near $X \times \{1\}$, so it is flat. If f is already flat, every f_s is flat. Applying the same construction to a pointed homotopy between flat maps, with X replaced by $[0, 1] \times X$, gives a homotopy through flat maps from f_1 to f_1'. Combined with the flat homotopies f_s and f_s' at the ends, this connects f to f' through flat maps. □

Remark 12.3.8. Thus, essentially by definition, $G^{-\infty}(Y; E)$ and $G^{-\infty}_{(1)}(Y; E)$ are classifying spaces for odd and even K-theory respectively. The second is the space of based loops in the first, so $\pi_k(G^{-\infty}_{(1)}) = \pi_{k+1}(G^{-\infty})$, and (12.3.3) becomes $\pi_0(G^{-\infty}_{(1)}) = \mathbb{Z}$, $\pi_1(G^{-\infty}_{(1)}) = 0$ and so on. In Chapter 13 $G^{-\infty}_{(1)}$ is reinterpreted as the "symbol group" for elliptic Toeplitz operators on the circle with values in smoothing operators on Y. This leads to a *classifying sequence*

$$G^{-\infty}(\mathbb{S}^1 \times Y; E) \longrightarrow G \longrightarrow G^{-\infty}_{(1),0}(Y; E), \tag{12.3.5}$$

in which G is a weakly contractible group of invertible Toeplitz perturbations of the identity, and the subscript 0 denotes the identity component of the loop group, on which the index vanishes. The last three sections of this chapter prepare for this.

12.4 Chern forms and the Fredholm determinant

For a loop g in $\mathrm{GL}(N, \mathbb{C})$ the basic invariant is the winding number of $\det g$, which equals $\frac{1}{2\pi i}\int \mathrm{tr}(g^{-1}dg)$. This section builds the analogue on $G^{-\infty}$, with the trace of smoothing operators in place of the matrix trace. The forms $\mathrm{Tr}((A^{-1}dA)^{2k-1})$ give de Rham classes on the K-groups, and the first of them, integrated along paths, gives the Fredholm determinant.

12.4.1 Forms with values in smoothing operators

Since $G^{-\infty} = G^{-\infty}(Y; E)$ is an open subset of $\mathrm{Id} + \Psi^{-\infty}(Y; E)$, we may treat it as a (Fréchet) manifold whose tangent space at every point is $\Psi^{-\infty}(Y; E)$. The tangent vector to a smooth curve $\mathrm{Id} + a_t$ at $A = \mathrm{Id} + a_0$ is $dA = \frac{d}{dt}a_t|_{t=0}$. The *Maurer–Cartan form* $A^{-1}dA$, the map $T_A G^{-\infty} = \Psi^{-\infty} \ni b \mapsto A^{-1}b$, is a $\Psi^{-\infty}$-valued 1-form. It is invariant under left multiplication: if $L_B(A) = BA$ then $L_B^*(A^{-1}dA) = A^{-1}B^{-1}B\,dA = A^{-1}dA$. Products of such forms are defined by composition of the values combined with antisymmetrization, and Tr is applied to the values. Explicitly, the *(unnormalized) odd Chern forms* are

$$\begin{aligned} u_{2k-1} &= \mathrm{Tr}\big((A^{-1}dA)^{2k-1}\big), \\ u_{2k-1}(b_1, \dots, b_{2k-1}) &= \sum_{\sigma \in \Sigma_{2k-1}} \mathrm{sgn}(\sigma)\,\mathrm{Tr}(A^{-1}b_{\sigma(1)}A^{-1}b_{\sigma(2)} \cdots A^{-1}b_{\sigma(2k-1)}), \end{aligned} \tag{12.4.1}$$

for $k = 1, 2, \dots$. They are smooth by the smoothness of composition, inversion and the trace.

To avoid differential calculus on Fréchet manifolds we work with the pull-backs of these forms under smooth maps $f : X \to G^{-\infty}(Y; E)$, where X is a finite dimensional manifold. Let

$$\Lambda^p(X; \Psi^{-\infty}(Y; E)) = \mathcal{C}^\infty\big(X \times Y^2; \Lambda^p T^*X \boxtimes (\operatorname{Hom}(E) \otimes \Omega_R)\big)$$

be the p-forms on X with values in smoothing operators on Y. We also allow 0-forms of the form $c\operatorname{Id} + a$. The wedge product of $\alpha \otimes a$ and $\beta \otimes b$ is $\alpha \wedge \beta \otimes ab$, extended by linearity and locally finite sums. The differential d acts in the X variables, and Tr acts on the values, producing ordinary forms. The following are immediate from the corresponding identities for scalar forms and from $\operatorname{Tr}(ab) = \operatorname{Tr}(ba)$:

$$\begin{aligned} d(\omega \wedge \eta) &= d\omega \wedge \eta + (-1)^p \omega \wedge d\eta, \qquad d\operatorname{Tr}(\omega) = \operatorname{Tr}(d\omega), \\ \operatorname{Tr}(\omega \wedge \eta) &= (-1)^{pq} \operatorname{Tr}(\eta \wedge \omega), \end{aligned} \tag{12.4.2}$$

for ω of degree p, η of degree q, and at least one of them smoothing-valued. For the last identity, write locally $\omega = \sum \alpha_I \otimes a_I$ and $\eta = \sum \beta_J \otimes b_J$. Then $\operatorname{Tr}(\omega \wedge \eta) = \sum \alpha_I \wedge \beta_J \operatorname{Tr}(a_I b_J) = (-1)^{pq} \sum \beta_J \wedge \alpha_I \operatorname{Tr}(b_J a_I)$. With the antisymmetrization convention of (12.4.1), the pull-back of u_{2k-1} by f is

$$f^* u_{2k-1} = \operatorname{Tr}(\theta^{2k-1}), \quad \theta = f^{-1}df \in \Lambda^1(X; \Psi^{-\infty}(Y; E)).$$

Lemma 12.4.1. *For every smooth $f : X \to G^{-\infty}(Y; E)$, with $\theta = f^{-1}df$:*

(a) $d\theta = -\theta \wedge \theta$;

(b) $\operatorname{Tr}(\theta^{2k}) = 0$ *for* $k \geq 1$;

(c) $f^* u_{2k-1} = \operatorname{Tr}(\theta^{2k-1})$ *is closed;*

(d) $f^* u_{2k-1}$ *is unchanged if f is replaced by Bf or fB with $B \in G^{-\infty}(Y; E)$ fixed. That is, u_{2k-1} is bi-invariant.*

Proof. Differentiating $f^{-1}f = \operatorname{Id}$ gives $d(f^{-1}) = -f^{-1}(df)f^{-1}$, so $d\theta = d(f^{-1}) \wedge df = -\theta \wedge \theta$. By (12.4.2) with $\omega = \theta$ and $\eta = \theta^{2k-1}$, $\operatorname{Tr}(\theta^{2k}) = (-1)^{2k-1} \operatorname{Tr}(\theta^{2k})$, which is (b). For (c), the Leibniz rule and (a) give

$$d(\theta^n) = \sum_{j=0}^{n-1} (-1)^j \theta^j \wedge d\theta \wedge \theta^{n-1-j} = -\Big(\sum_{j=0}^{n-1} (-1)^j\Big)\theta^{n+1}.$$

For $n = 2k-1$ odd the sum is 1, so $d\operatorname{Tr}(\theta^{2k-1}) = -\operatorname{Tr}(\theta^{2k}) = 0$ by (b). For (d), $(Bf)^{-1}d(Bf) = \theta$, while $(fB)^{-1}d(fB) = B^{-1}\theta B$ and $\operatorname{Tr}((B^{-1}\theta B)^{2k-1}) = \operatorname{Tr}(B^{-1}\theta^{2k-1}B) = \operatorname{Tr}(\theta^{2k-1})$. □

Lemma 12.4.2. *Let $F : [0,1] \times X \to G^{-\infty}(Y; E)$ be smooth and $f_t = F(t, \cdot)$. Then*

$$f_1^* u_{2k-1} - f_0^* u_{2k-1} = d \int_0^1 v_t \, dt, \quad v_t = (2k-1) \operatorname{Tr}(f_t^{-1} \partial_t f_t \, (f_t^{-1} d_X f_t)^{2k-2}).$$

Proof. On $[0,1] \times X$ write $\Theta = F^{-1}dF = dt\,\alpha + \theta_t$ with $\alpha = f_t^{-1}\partial_t f_t$ and $\theta_t = f_t^{-1} d_X f_t$. Expanding $\operatorname{Tr}(\Theta^{2k-1})$, the terms containing dt are

$$\sum_{j=0}^{2k-2} \operatorname{Tr}(\theta_t^j \wedge dt\,\alpha\,\theta_t^{2k-2-j}) = dt \wedge \sum_j (-1)^j \operatorname{Tr}(\theta_t^j \alpha \theta_t^{2k-2-j}) = dt \wedge v_t.$$

The last step uses (12.4.2) to move θ_t^j to the end at the cost of $(-1)^{j(2k-2-j)} = (-1)^j$. So $\operatorname{Tr}(\Theta^{2k-1}) = dt \wedge v_t + f_t^* u_{2k-1}$, and this is closed by Lemma 12.4.1. Its dt-component gives $\partial_t(f_t^* u_{2k-1}) - d_X v_t = 0$. Integrating in t proves the lemma. □

Proposition 12.4.3. *For each $k \in \mathbb{N}$ the de Rham class of $f^* u_{2k-1}$ defines a group homomorphism*

$$U_{2k-1} : K^{-1}(X) \longrightarrow H^{2k-1}_{\mathrm{dR}}(X;\mathbb{C}), \quad [f] \mapsto [f^* u_{2k-1}].$$

Proof. The class is well defined on homotopy classes by Lemma 12.4.2. For additivity, deform f and f' as in the proof of Proposition 12.3.1 so that $f = g \oplus \mathrm{Id}$ and $f' = \mathrm{Id} \oplus g'$ act on orthogonal finite dimensional spaces W_1, W_2. Then $\theta = f^{-1}df$ and $\theta' = (f')^{-1}df'$ satisfy $\theta \wedge \theta' = 0 = \theta' \wedge \theta$, and $(ff')^{-1}d(ff') = \theta + \theta'$. Hence $(\theta + \theta')^{2k-1} = \theta^{2k-1} + (\theta')^{2k-1}$ and $(ff')^* u_{2k-1} = f^* u_{2k-1} + (f')^* u_{2k-1}$. □

Remark 12.4.4. For a map of the form (12.3.2), $f^* u_{2k-1} = \operatorname{tr}((g^{-1}dg)^{2k-1})$ is the familiar form built from the matrix-valued map g, since the trace of a finite rank operator is the matrix trace. With suitable normalizing constants c_k the combination $\sum_k c_k U_{2k-1} : K^{-1}(X) \to H^{\mathrm{odd}}_{\mathrm{dR}}(X;\mathbb{C})$ is the *odd Chern character*; the constants are fixed in Chapter 15. It is a standard result of topology that its range spans $H^{\mathrm{odd}}(X;\mathbb{C})$ and that its null space is the torsion subgroup of $K^{-1}(X)$ (see [17]). We do not use this.

12.4.2 Even Chern forms

Even forms are defined in the same way on the loop group. A smooth map $f : X \to G^{-\infty}_{(1)}(Y;E)$ is a map $F : X \times \mathbb{R} \to G^{-\infty}(Y;E)$ with $F - \mathrm{Id}$ Schwartz in t, smooth in x. Set

$$f^* u_{2k} = \int_{\mathbb{R}} \operatorname{Tr}\big((F^{-1}d_X F)^{2k}\, F^{-1}\partial_t F\big)\, dt, \quad k = 0, 1, 2, \ldots. \tag{12.4.3}$$

The integrand is Schwartz in t. By the computation in the proof of Lemma 12.4.2, with $(2k+1)$ in place of $(2k-1)$, we have $\operatorname{Tr}((F^{-1}dF)^{2k+1}) = dt \wedge v + w$ on $X \times \mathbb{R}$, where $v = (2k+1)\operatorname{Tr}(F^{-1}\partial_t F (F^{-1}d_X F)^{2k})$. Since $\operatorname{Tr}(\alpha\theta^{2k}) = \operatorname{Tr}(\theta^{2k}\alpha)$, this gives

$$(2k+1) f^* u_{2k} = \int_{\mathbb{R}} v\, dt.$$

This is the integral over the fibres of $X \times \mathbb{R} \to X$ of the closed form $F^* u_{2k+1}$. Because $\partial_t w = d_X v$ and $w \to 0$ rapidly as $|t| \to \infty$, we get $d_X \int_{\mathbb{R}} v\, dt = \int_{\mathbb{R}} \partial_t w\, dt = 0$. So $f^* u_{2k}$ is closed. The argument of Lemma 12.4.2, applied on $[0,1] \times X \times \mathbb{R}$, shows that its class is a homotopy invariant. Exactly as in Proposition 12.4.3, using Proposition 12.3.7, we obtain homomorphisms

$$U_{2k} : K^{-2}(X) \longrightarrow H^{2k}_{\mathrm{dR}}(X;\mathbb{C}), \quad k \in \mathbb{N}_0.$$

A suitable combination of them is the (ordinary) Chern character.

The simplest and most fundamental cases are the first odd form and its even counterpart:

$$u_1 = \operatorname{Tr}(A^{-1}dA) \text{ on } G^{-\infty}, \qquad u_0(A) = \int_{\mathbb{R}} \operatorname{Tr}\big(A(t)^{-1}\partial_t A(t)\big)\, dt, \ A \in G^{-\infty}_{(1)}. \tag{12.4.4}$$

Proposition 12.4.5. *For every piecewise smooth closed curve $\gamma : \mathbb{S}^1 \to G^{-\infty}(Y;E)$,*

$$\int_\gamma u_1 \in 2\pi i\mathbb{Z}.$$

Consequently $u_0/2\pi i$ is an integer-valued, locally constant function on $G^{-\infty}_{(1)}(Y;E)$.

Proof. The integral is unchanged by reparametrization, so we may assume γ smooth (round off corners by a reparametrization that is flat at them). By Lemma 12.4.2 with $X = \mathbb{S}^1$ and $k = 1$, homotopic loops have equal integrals. By finite rank approximation we may therefore replace γ by a loop of the form (12.3.2) with $g : \mathbb{S}^1 \to \mathrm{GL}(N, \mathbb{C})$. Then $\gamma^* u_1 = \mathrm{tr}(g^{-1} dg)$. By the matrix identity $d \det g = \det g \ \mathrm{tr}(g^{-1} dg)$,

$$\int_\gamma u_1 = \int_{\mathbb{S}^1} \frac{d \det g}{\det g} = 2\pi i \ \mathrm{wind}(\det g),$$

where wind is the winding number about 0. If $A \in G^{-\infty}_{(1)}$, then $u_0(A)$ is the integral of u_1 over the closed loop $A \circ \gamma^{-1}$ in the notation of Proposition 12.3.7, extended by Id at 1, so it lies in $2\pi i \mathbb{Z}$. It is continuous on $G^{-\infty}_{(1)}$, hence locally constant. □

Remark 12.4.6. We shall see in Chapter 13 that $u_0/2\pi i : \pi_0(G^{-\infty}_{(1)}) \to \mathbb{Z}$ is a bijection, so it labels the components of the loop group. It is the simplest case of the index formula: under the identification of Remark 12.3.8 it computes the index of the corresponding Toeplitz operator, up to sign.

12.4.3 The Fredholm determinant

Proposition 12.4.5 yields a direct construction of the Fredholm determinant on $G^{-\infty}(Y; E)$. Any two points of $G^{-\infty}$ are joined by a piecewise linear path. Indeed $G^{-\infty}$ is connected, and each point $A \in G^{-\infty}$ has a convex neighbourhood $\{A + b;\ \|b\|_{L^2} < \|A^{-1}\|^{-1}\}$ inside $G^{-\infty}$. After reparametrization such a path is smooth.

Discussion. In finite dimensions $d \log \det A = \mathrm{tr}(A^{-1} dA)$, so $\log \det A$ is the integral of $\mathrm{tr}(A^{-1} dA)$ along a path from Id to A. The logarithm is multivalued: two paths differ by a loop, and the integral over a loop is $2\pi i$ times a winding number. The exponential does not see this ambiguity. The same argument works on $G^{-\infty}$ because Proposition 12.4.5 says that the periods of u_1 lie in $2\pi i \mathbb{Z}$. We never need a trace of $\log A$ or a product of eigenvalues.

Theorem 12.4.7. *For $A \in G^{-\infty}(Y; E)$ let $\gamma : [0, 1] \to G^{-\infty}(Y; E)$ be any piecewise smooth path with $\gamma(0) = \mathrm{Id}$ and $\gamma(1) = A$, and set*

$$\det(A) = \exp\Big(\int_\gamma u_1\Big) = \exp\Big(\int_0^1 \mathrm{Tr}(\gamma(t)^{-1} \gamma'(t))\, dt\Big). \tag{12.4.5}$$

This is independent of γ. It defines a smooth homomorphism $\det : G^{-\infty}(Y; E) \to \mathbb{C}^*$ with $d \log \det = u_1$, i.e. $\frac{d}{ds} \det(A_s) = \det(A_s) \mathrm{Tr}(A_s^{-1} \frac{dA_s}{ds})$ along smooth curves. If $A = \mathrm{Id} + a$ with a of finite rank and range in a finite dimensional space $W \subset \mathcal{C}^\infty(Y; E)$, then $\det(A)$ is the determinant of the restriction $A|_W : W \to W$.

Proof. If γ_1, γ_2 are two such paths, then γ_1 followed by γ_2 reversed is a closed piecewise smooth loop. By Proposition 12.4.5 the two integrals differ by an element of $2\pi i \mathbb{Z}$. For multiplicativity, if γ_A and γ_B join Id to A and B, then $\gamma_A \gamma_B$ joins Id to AB and

$$(\gamma_A \gamma_B)^{-1} (\gamma_A \gamma_B)' = \gamma_B^{-1} (\gamma_A^{-1} \gamma_A') \gamma_B + \gamma_B^{-1} \gamma_B'.$$

Taking traces gives $\det(AB) = \det(A) \det(B)$. For smoothness near $A_0 \in G^{-\infty}$, choose the path to pass through A_0 and then follow the segment to A. This gives

$$\det(A) = \det(A_0) \exp\Big(\int_0^1 \mathrm{Tr}\big((A_0 + s(A - A_0))^{-1} (A - A_0)\big)\, ds\Big),$$

which is smooth in A near A_0. Differentiating along a curve gives $d\log\det = u_1$. Finally, suppose a has range in W. Then A preserves W and induces the identity on $\mathcal{C}^\infty(Y;E)/W$. In terms of a decomposition $\mathcal{C}^\infty = W\oplus W'$ it is upper triangular, $\left(\begin{smallmatrix} g & c \\ 0 & \mathrm{Id}\end{smallmatrix}\right)$ with $g = A|_W \in \mathrm{GL}(W)$. Since $\mathrm{GL}(W)$ is connected we can join A to Id by $\gamma(t) = \left(\begin{smallmatrix} g_t & tc \\ 0 & \mathrm{Id}\end{smallmatrix}\right)$ with g_t a path in $\mathrm{GL}(W)$ from Id to g. Then $\gamma^{-1}\gamma'$ has range in W, and its trace is $\mathrm{tr}(g_t^{-1}g_t')$. Integrating gives $\det(A) = \det(g)$. □

By Theorem 12.4.7 the determinant agrees with the matrix determinant under any finite rank embedding $\mathrm{GL}(N,\mathbb{C}) \hookrightarrow G^{-\infty}$. It is the same function as the Fredholm determinant constructed for the residual isotropic algebra in Chapter 8; the characterizing properties are identical. It extends to all smoothing perturbations of the identity, vanishing exactly on the non-invertible ones.

Theorem 12.4.8. *The function $a \mapsto \det(\mathrm{Id}+a)$, defined on $\{a \in \Psi^{-\infty}(Y;E);\ \mathrm{Id}+a \in G^{-\infty}\}$, extends uniquely to a continuous function on $\Psi^{-\infty}(Y;E)$. The extension is smooth, entire along complex lines (i.e. $z \mapsto \det(\mathrm{Id}+a+zb)$ is entire), and vanishes exactly where $\mathrm{Id}+a$ is not invertible. Moreover $G^{-\infty}$ is dense in $\mathrm{Id}+\Psi^{-\infty}(Y;E)$, and $\frac{d}{ds}\det(\mathrm{Id}+sa)|_{s=0} = \mathrm{Tr}(a)$.*

Proof. Fix $a_0 \in \Psi^{-\infty}(Y;E)$. Using (iv) write $a_0 = p_0 + r_0$ with p_0 of finite rank, with range in a finite dimensional $W \subset \mathcal{C}^\infty(Y;E)$, and $\|r_0\|_{L^2} < 1/2$. For a in the neighbourhood $\mathcal{U} = \{\|a-a_0\|_{L^2} < 1/2\}$ put $r = a - p_0$. Then $\|r\| < 1$, so $\mathrm{Id}+r \in G^{-\infty}$, and

$$\mathrm{Id}+a = \big(\mathrm{Id}+p_0(\mathrm{Id}+r)^{-1}\big)(\mathrm{Id}+r).$$

The first factor is Id plus an operator with range in W. As in the proof of Theorem 12.4.7 it is invertible exactly when its restriction $g(a)$ to W is. Define on $\mathcal{U}$

$$D(a) = \det\big(g(a)\big)\det(\mathrm{Id}+r), \quad g(a) = \big(\mathrm{Id}+p_0(\mathrm{Id}+a-p_0)^{-1}\big)\big|_W.$$

This is smooth and holomorphic along complex lines, because $a \mapsto (\mathrm{Id}+a-p_0)^{-1}$ is. It vanishes exactly where $\mathrm{Id}+a$ is not invertible. Where $\mathrm{Id}+a \in G^{-\infty}$ it equals $\det(\mathrm{Id}+a)$ by Theorem 12.4.7, which gives both multiplicativity and the value on finite rank perturbations.

Next we show that $\{a \in \mathcal{U};\ \mathrm{Id}+a \in G^{-\infty}\}$ is dense in $\mathcal{U}$. Let π_W be the orthogonal projection onto W, a finite rank smoothing operator, and for $a \in \mathcal{U}$ set $a_\epsilon = a + \epsilon\pi_W(\mathrm{Id}+r)$, $r = a - p_0$. Then

$$\mathrm{Id}+a_\epsilon = \big(\mathrm{Id}+p_0(\mathrm{Id}+r)^{-1} + \epsilon\pi_W\big)(\mathrm{Id}+r),$$

and the first factor is Id plus an operator with range in W, whose restriction to W is $g(a) + \epsilon\,\mathrm{Id}_W$. This is invertible for all small $\epsilon \neq 0$, and $a_\epsilon \to a$. Two continuous functions agreeing on a dense set agree. So the local definitions agree on overlaps and define a global extension, which is unique for the same reason. Finally $\det(\mathrm{Id}+sa) = \exp\int_0^s \mathrm{Tr}((\mathrm{Id}+\sigma a)^{-1}a)\,d\sigma$ for small s, whose derivative at 0 is $\mathrm{Tr}(a)$. □

12.5 The Szegő projector and Toeplitz operators

12.5.1 Motivation

The next step towards the index theorem is a group-level version of the symbol sequence

$$\Psi^{-1}(Z;\mathbb{C}^N) \longrightarrow \Psi^0(Z;\mathbb{C}^N) \xrightarrow{\sigma_0} \mathcal{C}^\infty(S^*Z; M(N,\mathbb{C})).$$

Passing to invertible elements gives

$$G^{-1}(Z;\mathbb{C}^N) \longrightarrow G^0(Z;\mathbb{C}^N) \longrightarrow \mathcal{C}^\infty(S^*Z;\mathrm{GL}(N,\mathbb{C})), \tag{12.5.1}$$

where $G^0(Z;\mathbb{C}^N)$ is the group of invertible elliptic elements of $\Psi^0(Z;\mathbb{C}^N)$ (their inverses lie in Ψ^0 by Proposition 12.1.3) and $G^{-1}(Z;\mathbb{C}^N) = G^0(Z;\mathbb{C}^N) \cap (\mathrm{Id} + \Psi^{-1}(Z;\mathbb{C}^N))$. The second map is in general not surjective. An elliptic symbol is the symbol of an invertible operator if and only if the index of some (hence every) operator with that symbol vanishes. So the index is the obstruction to exactness of (12.5.1).

We would like to arrange that the middle group is contractible and the group on the right is essentially a loop group $G^{-\infty}_{(1)}$. This requires two things. First, S^*Z should be a circle. Second, $\mathbb{C}^N$ should be replaced by an infinite dimensional space, so that $\mathrm{GL}(N,\mathbb{C})$ becomes one of the groups $G^{-\infty}$ (this is done in Chapter 13). A compact manifold with $S^*Z = \mathbb{S}^1$ does not exist, since Z would be a compact 1-dimensional manifold and $S^*\mathbb{S}^1 = \mathbb{S}^1 \sqcup \mathbb{S}^1$. There are two ways round this:

(A) replace the circle by the line $\mathbb{R}$ and use the isotropic calculus of Chapter 7, for which the "cosphere" is the boundary circle of the radial compactification of $T^*\mathbb{R} = \mathbb{R}^2$;

(B) keep the circle but discard half of $S^*\mathbb{S}^1$, which leads to the Toeplitz algebra.

We follow (B). It is less transparent geometrically but has a long history. A third route uses the Heisenberg calculus [22].

12.5.2 The Hardy space and the Szegő projector

Expanding in Fourier series, $\mathcal{C}^\infty(\mathbb{S}^1)$ is the direct sum of

$$\mathcal{C}^\infty_+(\mathbb{S}^1) = \Big\{\sum_{k\geq 0} c_k e^{ik\theta}\Big\}, \qquad \mathcal{C}^\infty_-(\mathbb{S}^1) = \Big\{\sum_{k<0} c_k e^{ik\theta}\Big\},$$

the coefficients being rapidly decreasing. The *Szegő projector* is

$$S : \mathcal{C}^\infty(\mathbb{S}^1) \longrightarrow \mathcal{C}^\infty_+(\mathbb{S}^1), \quad S\Big(\sum_{k\in\mathbb{Z}} c_k e^{ik\theta}\Big) = \sum_{k\geq 0} c_k e^{ik\theta}. \tag{12.5.2}$$

It satisfies $S^2 = S = S^*$ (with respect to $d\theta$), its null space is $\mathcal{C}^\infty_-$ and its range is $\mathcal{C}^\infty_+$. The product of two elements of $\mathcal{C}^\infty_+$ is again in $\mathcal{C}^\infty_+$, so the Hardy space $\mathcal{C}^\infty_+(\mathbb{S}^1)$ is a ring. It consists of the boundary values of holomorphic functions on the unit disc $\mathbb{D} \subset \mathbb{C}$ (here $\mathbb{D}$ is the open disc and $\overline{\mathbb{D}}$ its closure), so it should be thought of as the functions on "half" of $\mathbb{S}^1$. The cotangent bundle is $T^*\mathbb{S}^1 = \mathbb{S}^1 \times \mathbb{R}_\xi$, and the Fourier variable k corresponds to ξ. This suggests that S is microlocally the identity on the positive half $S^*_+\mathbb{S}^1 = \{\xi > 0\}$ and zero on the negative half $S^*_-\mathbb{S}^1$ (Figure 12.3).

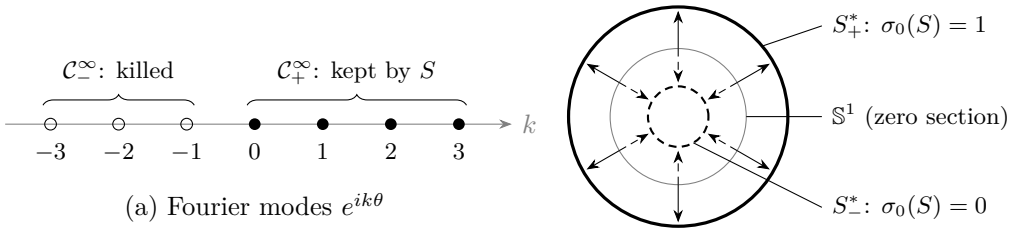
Proposition 12.5.1. *Restriction to the boundary is an isomorphism*

$$\mathcal{C}^\infty_{\mathrm{hol}}(\overline{\mathbb{D}}) = \{u \in \mathcal{C}^\infty(\overline{\mathbb{D}});\ \bar\partial u = 0 \text{ in } \mathbb{D}\} \longrightarrow \mathcal{C}^\infty_+(\mathbb{S}^1).$$

For $f \in \mathcal{C}^\infty(\mathbb{S}^1)$ the Cauchy integral

$$\tilde S(f)(z) = \frac{1}{2\pi i}\int_{|\tau|=1} \frac{f(\tau)}{\tau - z}\, d\tau, \quad |z| < 1,$$

is the holomorphic extension of Sf, and $Sf(e^{i\theta}) = \lim_{r\uparrow 1} \tilde S f(re^{i\theta})$.



(a) Fourier modes $e^{ik\theta}$

(b) $T^*\mathbb{S}^1$, $\xi > 0$ outward

Figure 12.3: The Szegő projector. (a) On Fourier series, S keeps the modes $k \geq 0$ and kills the modes $k < 0$. (b) The cotangent bundle $T^*\mathbb{S}^1 = \mathbb{S}^1 \times \mathbb{R}_\xi$, drawn with $\xi > 0$ pointing outwards. The cosphere bundle has two components, S^*_+ (outer circle) and S^*_- (inner circle); the principal symbol of S is 1 on the first and 0 on the second (Theorem 12.5.2).

Proof. If $a = \sum_{k\geq 0} a_k e^{ik\theta} \in \mathcal{C}^\infty_+$ then the coefficients are rapidly decreasing. Hence $u_a(z) = \sum_{k\geq 0} a_k z^k$ converges on $\overline{\mathbb{D}}$ with all derivatives, is holomorphic in $\mathbb{D}$, and restricts to a. Conversely, if $u \in \mathcal{C}^\infty_{\mathrm{hol}}(\overline{\mathbb{D}})$ has the Taylor series $\sum_{k\geq 0} b_k z^k$ in $\mathbb{D}$, then for $r < 1$ and $k \in \mathbb{Z}$

$$\frac{1}{2\pi}\int_0^{2\pi} e^{-ik\theta} u(re^{i\theta})\, d\theta = \begin{cases} b_k r^k & k \geq 0, \\ 0 & k < 0. \end{cases}$$

Letting $r \uparrow 1$, the Fourier coefficients of $u|_{\mathbb{S}^1}$ are b_k for $k \geq 0$ and 0 for $k < 0$. So the boundary value lies in $\mathcal{C}^\infty_+$, and it vanishes only if $u = 0$. Thus restriction is an isomorphism. For the Cauchy integral, by rapid convergence of the Fourier series we can integrate term by term. For $|z| < 1$,

$$\frac{1}{2\pi i}\int_{|\tau|=1} \frac{\tau^k}{\tau - z}\, d\tau = \begin{cases} z^k & k \geq 0, \\ 0 & k < 0, \end{cases}$$

by the residue theorem. For $k < 0$ the integrand $\tau^k/(\tau - z)$ is holomorphic outside the disc and $O(|\tau|^{-2})$ at infinity, so we deform the contour to infinity. So $\tilde{S}(f) = u_{Sf}$, and the boundary values follow from the first part. □

Theorem 12.5.2. *$S \in \Psi^0(\mathbb{S}^1)$, and its principal symbol is*

$$\sigma_0(S) = \begin{cases} 1 & \text{on } S^*_+\mathbb{S}^1 = \{\xi > 0\}, \\ 0 & \text{on } S^*_-\mathbb{S}^1 = \{\xi < 0\}. \end{cases}$$

More precisely, near the diagonal the kernel of S is $\int e^{i(\theta-\theta')\xi}\chi(\xi)đ\xi$ modulo a smooth function, where $\chi \in \mathcal{C}^\infty(\mathbb{R})$, $\chi(\xi) = 1$ for $\xi \geq 1$ and $\chi(\xi) = 0$ for $\xi \leq -1$. Thus the amplitude vanishes identically in the negative direction.

Proof. The kernel of S with respect to $d\theta'$ is the distribution

$$K_S(\theta, \theta') = \frac{1}{2\pi}\sum_{k\geq 0} e^{ik(\theta-\theta')} = \lim_{r\uparrow 1} \frac{1}{2\pi}\frac{1}{1 - re^{i\phi}}, \quad \phi = \theta - \theta'.$$

The limit is taken in distributions, since the Fourier coefficients $r^k \to 1$ with uniform polynomial bounds. For $\phi \notin 2\pi\mathbb{Z}$ the limit is the smooth function $(2\pi)^{-1}(1 - e^{i\phi})^{-1}$, locally uniformly with all derivatives. So K_S is smooth off the diagonal. Near $\phi = 0$, write $r = e^{-\epsilon}$, so $re^{i\phi} = e^{i(\phi + i\epsilon)}$, and use the Laurent expansion

$$\frac{1}{1 - e^{iz}} = \frac{i}{z} + h(z), \quad h \text{ holomorphic in } |z| < 2\pi \quad (h(0) = \tfrac{1}{2}).$$

Hence for $|\phi| < \pi$

$$K_S = \frac{1}{2\pi}\Big(\frac{i}{\phi + i0} + h(\phi)\Big), \qquad \frac{i}{\phi + i0} = \lim_{\epsilon \downarrow 0} \frac{i}{\phi + i\epsilon} = \lim_{\epsilon \downarrow 0} \int_0^\infty e^{i\phi\xi - \epsilon\xi}\, d\xi.$$

So $(2\pi)^{-1} i/(\phi + i0) = \int e^{i\phi\xi} H(\xi) d\xi$ with H the Heaviside function. Since $H - \chi$ has compact support, $\int e^{i\phi\xi}(H - \chi) d\xi$ is entire in ϕ. Thus near the diagonal K_S is the kernel of $\mathrm{Op}(\chi)$ modulo a smooth function. As χ is a classical symbol of order 0 (it is homogeneous of degree 0 for $|\xi| \geq 1$), $S \in \Psi^0(\mathbb{S}^1)$ with principal symbol χ restricted to $\{\xi = \pm 1\}$. □

12.5.3 The Toeplitz algebra

On $\mathbb{C}^N$-valued functions S acts componentwise. Set

$$\Psi^m_{\mathcal{T}}(\mathbb{S}^1; \mathbb{C}^N) = \{A \in \Psi^m(\mathbb{S}^1; \mathbb{C}^N);\ A = SAS\}, \quad m \in \{0, -\infty\},$$

regarded as operators on $\mathcal{C}^\infty_+(\mathbb{S}^1; \mathbb{C}^N)$. This is a subalgebra with identity S: if $A_i = SA_iS$ then $S(A_1A_2)S = (SA_1S)(SA_2S) = A_1A_2$. Its elements map $\mathcal{C}^\infty_-$ to zero and $\mathcal{C}^\infty$ into $\mathcal{C}^\infty_+$, and $\Psi^{-\infty}_{\mathcal{T}} = S\Psi^{-\infty}S$. Since $\sigma_0(A) = \sigma_0(S)\sigma_0(A)\sigma_0(S)$, the principal symbol of $A \in \Psi^0_{\mathcal{T}}$ vanishes on $S^*_-\mathbb{S}^1$. We call A *elliptic* (as a Toeplitz operator) if $\sigma_0(A)|_{S^*_+\mathbb{S}^1}$, a smooth function on $\mathbb{S}^1$ with values in $M(N, \mathbb{C})$, is invertible. The basic examples are the *Toeplitz operators* $T_a = SaS$ with multiplier $a \in \mathcal{C}^\infty(\mathbb{S}^1; M(N, \mathbb{C}))$. They are elliptic exactly when a takes values in $\mathrm{GL}(N, \mathbb{C})$.

Proposition 12.5.3. *If $A \in \Psi^0_{\mathcal{T}}(\mathbb{S}^1; \mathbb{C}^N)$ is elliptic, then $A + (\mathrm{Id} - S) \in \Psi^0(\mathbb{S}^1; \mathbb{C}^N)$ is elliptic, and A is Fredholm on $\mathcal{C}^\infty_+(\mathbb{S}^1; \mathbb{C}^N)$ with*

$$\mathrm{ind}_{\mathcal{T}}(A) = \dim \mathrm{Nul}(A|_{\mathcal{C}^\infty_+}) - \dim(\mathcal{C}^\infty_+ / A\mathcal{C}^\infty_+) = \mathrm{ind}(A + \mathrm{Id} - S).$$

There is $B \in \Psi^0_{\mathcal{T}}$ with $AB - S$ and $BA - S$ in $\Psi^{-\infty}_{\mathcal{T}}$. If A is invertible on $\mathcal{C}^\infty_+$, its inverse lies in $\Psi^0_{\mathcal{T}}$. The Toeplitz index is constant along smooth families of elliptic elements of $\Psi^0_{\mathcal{T}}$.

Proof. The symbol of $P = A + \mathrm{Id} - S$ is $\sigma_0(A)$ on S^*_+ and Id on S^*_-, so P is elliptic. With respect to $\mathcal{C}^\infty = \mathcal{C}^\infty_+ \oplus \mathcal{C}^\infty_-$, P is diagonal, acting as A on $\mathcal{C}^\infty_+$ and as Id on $\mathcal{C}^\infty_-$. So the null space and cokernel of P on $\mathcal{C}^\infty$ are those of A on $\mathcal{C}^\infty_+$. If Q is a parametrix (or the inverse) of P, then $B = SQS$ satisfies $BA = SQSAS = SQPS = S - S(\mathrm{Id} - QP)S$. The other identity is similar. The last statement follows from Theorem 12.2.2 applied to the smooth family $A_t + \mathrm{Id} - S$. □

12.5.4 The index on the circle

The *shifts* on $\mathcal{C}^\infty_+(\mathbb{S}^1)$ are

$$U = Se^{i\theta}S : \sum_{k \geq 0} c_k e^{ik\theta} \mapsto \sum_{k \geq 0} c_k e^{i(k+1)\theta}, \quad L = Se^{-i\theta}S : \sum_{k \geq 0} c_k e^{ik\theta} \mapsto \sum_{k \geq 1} c_k e^{i(k-1)\theta}.$$

Both are elliptic elements of $\Psi^0_{\mathcal{T}}(\mathbb{S}^1)$, and

$$LU = \mathrm{Id}, \quad UL = \mathrm{Id} - \pi_0, \tag{12.5.3}$$

where Id means S, the identity on $\mathcal{C}^\infty_+$, and π_0 is the orthogonal projection onto the constants. Thus U is injective with range of codimension one (complemented by the constants), and L is surjective with null space the constants:

$$\mathrm{ind}_{\mathcal{T}}(U) = -1, \qquad \mathrm{ind}_{\mathcal{T}}(L) = 1.$$

More generally $U^k = T_{e^{ik\theta}}$ for $k \geq 0$ has index $-k$ and $L^k = T_{e^{-ik\theta}}$ has index k. Figure 12.4 shows the mechanism.

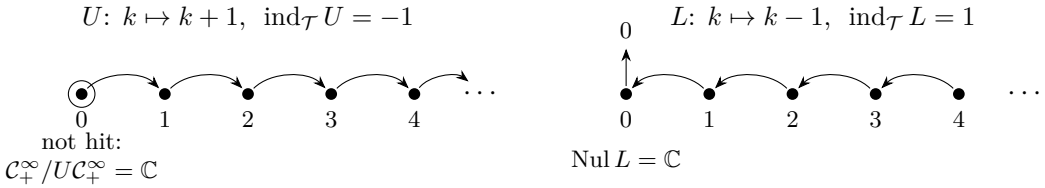


Figure 12.4: The shifts on the Hardy space, acting on the Fourier modes $e^{ik\theta}$, $k \geq 0$. The forward shift U is injective but misses the constants. The backward shift L is surjective but kills the constants. On the full space $\mathcal{C}^\infty(\mathbb{S}^1)$, where k runs over $\mathbb{Z}$, both are invertible.

Question. The shift U is an isometry and its symbol $e^{i\theta}$ never vanishes. How can it fail to be invertible?

Answer. An injective linear map of a finite dimensional space to itself is surjective, but this is false in infinite dimensions ("Hilbert's hotel"). Ellipticity only says that U is invertible modulo smoothing operators, and the index measures the remaining defect. Multiplication by $e^{i\theta}$ on all of $\mathcal{C}^\infty(\mathbb{S}^1)$ is invertible; it is the restriction to half of the frequencies, i.e. to half of $S^*\mathbb{S}^1$, that creates the index. □

Theorem 12.5.4. *If $a \in \mathcal{C}^\infty(\mathbb{S}^1; \mathrm{GL}(N,\mathbb{C}))$ then the Toeplitz operator $T_a = SaS$ on $\mathcal{C}^\infty_+(\mathbb{S}^1;\mathbb{C}^N)$ is Fredholm and*

$$\operatorname{ind}_{\mathcal{T}}(T_a) = -\operatorname{wind}(\det a) = -\frac{1}{2\pi i}\int_{\mathbb{S}^1} \operatorname{tr}(a^{-1}da).$$

Proof. The second equality is the computation in the proof of Proposition 12.4.5. Let $k = \operatorname{wind}(\det a)$. We claim that a is smoothly homotopic in $\mathcal{C}^\infty(\mathbb{S}^1;\mathrm{GL}(N,\mathbb{C}))$ to $a_k = \operatorname{diag}(e^{ik\theta},1,\dots,1)$. Write $a = \operatorname{diag}(\det a,1,\dots,1)\, b$ with b a loop in $\mathrm{SL}(N,\mathbb{C})$. The scalar loop $\det a$ is homotopic in $\mathbb{C}^*$ to $e^{ik\theta}$: write $\det a = e^{ik\theta}e^{\phi(\theta)}$ with ϕ a smooth function on the circle, and deform ϕ to 0. Since $\mathrm{SL}(N,\mathbb{C})$ is connected and simply connected (a standard fact; it retracts onto $\mathrm{SU}(N)$), b is smoothly homotopic to the constant loop Id. The family T_{a_t} along the homotopy is a smooth family of elliptic Toeplitz operators, so by Proposition 12.5.3 $\operatorname{ind}_{\mathcal{T}}(T_a) = \operatorname{ind}_{\mathcal{T}}(T_{a_k})$. Now $T_{a_k} = U^k \oplus \mathrm{Id} \oplus \cdots \oplus \mathrm{Id}$ if $k \geq 0$ and $L^{-k} \oplus \mathrm{Id} \oplus \cdots \oplus \mathrm{Id}$ if $k < 0$. Its index is $-k$. □

Discussion. The proof follows a pattern that recurs in the rest of the book. First, homotopy invariance reduces the problem to a symbol in a *normal form*, here $\operatorname{diag}(e^{ik\theta},1,\dots,1)$. Second, the index of an explicit operator with that symbol is computed by hand, here by counting Fourier modes. In Chapter 13 the same two steps are carried out for families, where the simple homotopy above is replaced by Atiyah's linearization of loops.

Remark 12.5.5. Via finite rank embeddings $\mathrm{GL}(N,\mathbb{C}) \hookrightarrow G^{-\infty}$, Theorem 12.5.4 says that the Toeplitz index is the homomorphism $-U_1/2\pi i : K^{-1}(\mathbb{S}^1) \to \mathbb{Z}$. It vanishes on the classes of constant maps, so by Lemma 12.3.4 it factors through $K^{-2}(\mathrm{pt})$. By Proposition 12.3.7 it is there given by $-u_0/2\pi i$ on $\pi_0(G^{-\infty}_{(1)})$. That it is an isomorphism is the case $k = 1$ of (12.3.3). This is the prototype of the index formula.

12.6 Smoothing perturbations of elliptic families

An elliptic operator of index zero becomes invertible after adding a suitable finite rank smoothing operator. For a family $A(x)$ we would like to do this smoothly in x. Index zero

at each point is necessary but not sufficient: the cokernels can twist into a non-trivial vector bundle as x varies. This section shows that this twisting, measured by the families index, is the only obstruction.

Throughout this section X is a compact manifold (the parameter space, possibly with boundary). We use the standard L^2 structure on $\mathbb{S}^1$ and the projections onto finite Fourier series

$$\pi_r\Big(\sum_k c_k e^{ik\theta}\Big) = \sum_{|k|\le r} c_k e^{ik\theta}, \quad r \in \mathbb{N}_0,$$

acting componentwise on $\mathbb{C}^N$-valued functions. They are self-adjoint finite rank smoothing operators of rank $(2r+1)N$.

Lemma 12.6.1. *If $R \in \mathcal{C}^\infty(X; \Psi^{-\infty}(\mathbb{S}^1; \mathbb{C}^N))$ then $R\pi_r \to R$ and $\pi_r R \to R$ in $\mathcal{C}^\infty(X; \Psi^{-\infty})$, uniformly on X, as $r \to \infty$.*

Proof. The kernel of $R_x\pi_r$ is

$$\int R_x(\theta,\theta'')\pi_r(\theta'',\theta')\,d\theta'' = \sum_{|k|\le r} b_k(x,\theta)e^{-ik\theta'}, \quad b_k(x,\theta) = \frac{1}{2\pi}\int R_x(\theta,\theta'')e^{ik\theta''}\,d\theta'',$$

i.e. the partial sums of the Fourier series of $R_x(\theta,\theta')$ in the second variable. Integration by parts in θ'' shows that every (x,θ)-derivative of b_k is $O(\langle k\rangle^{-q})$ for all q, uniformly in (x,θ). So the series converges with all derivatives. The case of $\pi_r R$ follows by taking adjoints. □

Lemma 12.6.2. *Let $A \in \mathcal{C}^\infty(X; \Psi^0(\mathbb{S}^1; \mathbb{C}^N))$ be a smooth family of elliptic operators and $A'_r(x) = A(x)(\mathrm{Id} - \pi_r)$. There is r_0 such that for all $r \ge r_0$ and $x \in X$:*

(a) $\mathrm{Nul}(A'_r(x)) = \mathrm{Ran}(\pi_r)$, of dimension $(2r+1)N$;

(b) the range of $A'_r(x)$ is closed, with complement

$$F_r(x) = \mathrm{Nul}\big((\mathrm{Id} - \pi_r)A(x)^*\big) = \mathrm{Ran}(A'_r(x))^\perp$$

of dimension $(2r+1)N - \operatorname{ind} A(x)$;

(c) the $F_r(x)$ form a smooth vector bundle $F_r \to X$.

Proof. Let Q be a smooth family of parametrices, $Q(x)A(x) = \mathrm{Id} - R(x)$ with $R \in \mathcal{C}^\infty(X; \Psi^{-\infty})$ (Lemma 12.2.1). Then

$$Q(x)A'_r(x) = (\mathrm{Id} - R(x))(\mathrm{Id} - \pi_r) = \big(\mathrm{Id} - R(x)(\mathrm{Id} - \pi_r)\big)(\mathrm{Id} - \pi_r).$$

By Lemma 12.6.1, $R(x)(\mathrm{Id} - \pi_r) \to 0$ uniformly in $\mathcal{C}^\infty(X; \Psi^{-\infty})$, hence in L^2 norm. So for $r \ge r_0$ the operator $\mathrm{Id} - R(x)(\mathrm{Id} - \pi_r)$ is invertible, with inverse $\mathrm{Id} - S_r(x)$ depending smoothly on x in $\Psi^{-\infty}$ by (vi). With $Q'_r = (\mathrm{Id} - S_r)Q$ this gives

$$Q'_r(x)A'_r(x) = \mathrm{Id} - \pi_r. \tag{12.6.1}$$

So $\mathrm{Nul}(A'_r) \subset \mathrm{Ran}(\pi_r)$, and the reverse inclusion is clear, which proves (a). $A'_r(x)$ is elliptic and differs from $A(x)$ by a smoothing operator, so it has the same index. By Proposition 12.1.3 its range has the orthogonal complement $\mathrm{Nul}((A'_r)^*) = F_r(x)$, of dimension $(2r+1)N - \operatorname{ind} A(x)$. This proves (b).

For (c), the dimension is locally constant by Theorem 12.2.2. Let $T(x) = A'_r(x)^* A'_r(x) + \pi_r$, a smooth family of elliptic self-adjoint operators of order 0. It is injective, since $T(x)u = 0$ implies $\|A'_r u\|^2 + \|\pi_r u\|^2 = 0$, so $u \in \mathrm{Ran}(\pi_r) \cap \mathrm{Nul}(\pi_r) = \{0\}$. It is therefore invertible by

(12.1.2), and $G(x) = T(x)^{-1}$ is a smooth family by Lemma 12.2.1(b). Set $P(x) = A'_r G A'^*_r$. Since $A'_r \pi_r = 0$ we have $T\pi_r = \pi_r$, hence $G\pi_r = \pi_r$ and $GA'^*_r A'_r = G(T - \pi_r) = \mathrm{Id} - \pi_r$. Therefore $P^2 = A'_r(\mathrm{Id} - \pi_r)GA'^*_r = P$, $P^* = P$ and $PA'_r = A'_r(\mathrm{Id} - \pi_r) = A'_r$. Since also $\operatorname{Ran} P \subset \operatorname{Ran} A'_r$, $P(x)$ is the orthogonal projection onto $\operatorname{Ran}(A'_r(x))$. Thus $\mathrm{Id} - P(x)$ is the orthogonal projection onto $F_r(x)$. It is a smooth family in Ψ^0 whose values are smoothing (finite rank with range in $\mathcal{C}^\infty$). Its full symbol is therefore residual for each x, which for a smooth family of classical symbols means the family lies in $\mathcal{C}^\infty(X; \Psi^{-\infty})$. If $v_1, \dots, v_d$ is a basis of $F_r(x_0)$, then $(\mathrm{Id} - P(x))v_i$ are smooth sections of F_r which remain a basis near x_0. So F_r is a smooth vector bundle. □

The formal difference "$\operatorname{Ran}(\pi_r) \ominus F_r$" of the trivial bundle of null spaces and the bundle F_r complementary to the range is the *families index* of A, an element of $K^0(X)$. It is discussed in Chapter 15.

Discussion. Why cut off the first $2r+1$ Fourier modes? The null spaces $\operatorname{Nul} A(x)$ may jump in dimension as x varies, so they do not form a bundle. Composing with $\mathrm{Id} - \pi_r$ forces the null space to be the *fixed* space $\operatorname{Ran}(\pi_r)$ for every x (Lemma 12.6.2(a)). All the variation is then pushed into the complement $F_r(x)$ of the range, whose dimension is constant because the index is. The trivial bundle $\operatorname{Ran}(\pi_r)$ and the bundle F_r together carry exactly the information that a single number, the index, carries for one operator.

Theorem 12.6.3. *Let $A \in \mathcal{C}^\infty(X; \Psi^0(\mathbb{S}^1; \mathbb{C}^N))$ be a smooth family of elliptic operators. There is a smooth family $B \in \mathcal{C}^\infty(X; \Psi^{-\infty}(\mathbb{S}^1; \mathbb{C}^N))$ such that $A(x) + B(x)$ is invertible for every $x \in X$ if and only if, for some (equivalently every) $r \geq r_0$, the bundle F_r of Lemma 12.6.2 is isomorphic to the trivial bundle $X \times \mathbb{C}^{(2r+1)N}$.*

Proof. Suppose F_r is trivial for some $r \geq r_0$. Since $\operatorname{Ran}(\pi_r)$ has dimension $(2r+1)N$, there is a smooth bundle isomorphism $\Phi(x) : \operatorname{Ran}(\pi_r) \to F_r(x)$. Then $B_r(x) = \Phi(x)\pi_r$ is a smooth family of finite rank smoothing operators: its kernel is $\sum_j \Phi(x)e_j \otimes e_j^*$ for an orthonormal basis e_j of $\operatorname{Ran}(\pi_r)$. The operator $A'_r(x) + B_r(x)$ maps $\operatorname{Ran}(\mathrm{Id} - \pi_r)$ injectively onto $\operatorname{Ran}(A'_r(x))$ and $\operatorname{Ran}(\pi_r)$ isomorphically onto $F_r(x)$. As $L^2 = \operatorname{Ran}(A'_r) \oplus F_r$, it is invertible. Thus $B = B_r - A\pi_r$ has the required property, since $A + B = A'_r + B_r$.

Conversely, suppose $A + B$ is invertible on X. By Lemma 12.6.1, $B\pi_r \to B$ in L^2 norm, uniformly on X. Since the inverses $(A(x) + B(x))^{-1}$ are bounded uniformly on the compact set X, $A + B\pi_r$ is invertible for all large r. Write

$$A + B\pi_r = A'_r + B_r, \quad B_r = (A + B)\pi_r.$$

Invertibility of $A'_r + B_r$ means that $L^2 = A'_r(\operatorname{Ran}(\mathrm{Id} - \pi_r)) \oplus B_r(\operatorname{Ran}(\pi_r))$. So $V(x) = B_r(x) \operatorname{Ran}(\pi_r)$ is a smooth bundle complementary to $\operatorname{Ran}(A'_r(x))$, trivialized by B_r (which is injective on $\operatorname{Ran}(\pi_r)$). The orthogonal projection $\mathrm{Id} - P(x)$ onto $F_r(x)$ restricts to an isomorphism $V(x) \to F_r(x)$, since its null space is $\operatorname{Ran}(A'_r(x))$. Hence F_r is trivial for all large r. Together with the first part, triviality for one $r \geq r_0$ implies triviality for all large r. □

In particular a necessary condition is that $\operatorname{ind} A(x) = 0$, and when X is a point the theorem reduces to the statement of Chapter 5 that an elliptic operator has an invertible smoothing perturbation if and only if its index vanishes. The proofs extend to any compact manifold in place of the circle (and to fibrations by compact manifolds) once a sequence of finite rank smoothing projections with the property of Lemma 12.6.1 is available. They also pass unchanged to the Toeplitz algebra if π_r is replaced by the projection π_r^+ onto $\operatorname{span}\{e^{ik\theta};\ 0 \leq k \leq r\}$. This projection commutes with S, lies in $\Psi_{\mathcal{T}}^{-\infty}$, and has rank $(r+1)N$.

Corollary 12.6.4. *Let $A \in \mathcal{C}^\infty(X; \Psi^0_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^N))$ be a smooth family of elliptic Toeplitz operators. There is $B \in \mathcal{C}^\infty(X; \Psi^{-\infty}_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^N))$ with $A(x)+B(x)$ invertible on $\mathcal{C}^\infty_+$ for all x if and only if, for large r, the bundle $F^+_r(x) = \mathrm{Nul}((S-\pi^+_r)A(x)^*) \cap \mathcal{C}^\infty_+$ is trivial of rank $(r+1)N$. The inverses then lie in $\Psi^0_{\mathcal{T}}$.*

Corollary 12.6.5. *Let $A \in \mathcal{C}^\infty([0,1]\times X; \Psi^0_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^N))$ (or $\mathcal{C}^\infty([0,1]\times X; \Psi^0(\mathbb{S}^1;\mathbb{C}^N))$) be elliptic at each point, with $A_0(x)$ invertible for all $x \in X$. Then there is a smooth family $B \in \mathcal{C}^\infty([0,1]\times X; \Psi^{-\infty}_{\mathcal{T}})$ (respectively $\Psi^{-\infty}$) with $B_0 = 0$ and $A_t(x)+B_t(x)$ invertible for all (t,x).*

Proof. We give the proof in the Ψ^0 case; the Toeplitz case is identical. Let r be large. Applying the converse half of Theorem 12.6.3 on X with $B=0$ shows that the restriction of F_r to $\{0\}\times X$ is trivial. Since $[0,1]\times X$ retracts onto $\{0\}\times X$, the bundle F_r over $[0,1]\times X$ is isomorphic to the pull-back of its restriction to $\{0\}\times X$ (homotopy invariance of vector bundles [17]), so it is trivial. The first half of the theorem gives a smooth family B' of smoothing operators with $C_t = A_t + B'_t$ invertible everywhere. To make the perturbation vanish at $t=0$, set

$$D_t = C_t C_0^{-1} A_0 = A_t + B_t, \quad B_t = -A_t(A_0+B'_0)^{-1}B'_0 + B'_t C_0^{-1} A_0.$$

Here we used $C_0^{-1}A_0 = \mathrm{Id} - C_0^{-1}B'_0$. D_t is invertible as a product of invertibles, $D_0 = A_0$, and B_t is a smooth family of smoothing operators, since $C_0^{-1} \in \Psi^0$ by Lemma 12.2.1(b). It vanishes at $t=0$ because $D_0 = A_0$. □

This allows homotopies of elliptic families to be lifted to homotopies of invertible families. It is used in Chapter 13.

12.7 Deforming matrices away in the Toeplitz group

In Chapter 13 we show that the group of invertible Toeplitz operators is (weakly) contractible. A first obstacle is the copy of $\mathrm{GL}(N,\mathbb{C})$ inside it: the constant matrices, which are not contractible in $\mathrm{GL}(N,\mathbb{C})$ itself. The point of this section is that, after doubling $\mathbb{C}^N$ to $\mathbb{C}^{2N}$, the shifts U and L provide enough room to contract them.

Let

$$\tilde{G}^0_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^N) = \{A \in \Psi^0_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^N);\ A \text{ is elliptic and invertible on } \mathcal{C}^\infty_+,\ A^{-1} \in \Psi^0_{\mathcal{T}}\}$$

(the tilde distinguishes it from a smaller group used in Chapter 13). By Proposition 12.5.3 the last condition is automatic. The smoothing elements $G^{-\infty}_{\mathcal{T}} = \tilde{G}^0_{\mathcal{T}} \cap (S + \Psi^{-\infty}_{\mathcal{T}})$ form a subgroup containing a copy of $\mathrm{GL}(N,\mathbb{C})$:

$$\mathrm{GL}(N,\mathbb{C}) \ni g \longmapsto \mathrm{Id} - \pi_0 + \pi_0 g \pi_0 \in G^{-\infty}_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^N), \tag{12.7.1}$$

which acts as g on the constants and as the identity on higher modes; here $\mathrm{Id} = S$. We also use the standard stabilization $\mathrm{GL}(N,\mathbb{C}) \hookrightarrow \mathrm{GL}(2N,\mathbb{C})$, $g \mapsto \left(\begin{smallmatrix} g & 0 \\ 0 & \mathrm{Id}_N \end{smallmatrix}\right)$.

Proposition 12.7.1. *Embed $\mathrm{GL}(N,\mathbb{C})$ in $\tilde{G}^0_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^{2N})$ by combining the stabilization with (12.7.1) (for $2N$ in place of N). Then there is a continuous map $M : [0,1]\times \mathrm{GL}(N,\mathbb{C}) \to \tilde{G}^0_{\mathcal{T}}(\mathbb{S}^1;\mathbb{C}^{2N})$, smooth in all variables, such that $M(0,g)$ is the image of g and $M(1,g) = \mathrm{Id}$ for all g.*

Proof. Write $\mathbb{C}^{2N} = \mathbb{C}^N \oplus \mathbb{C}^N$ and operators as 2×2 block matrices of $N \times N$ Toeplitz operators. A constant matrix g commutes with S, so it defines the Toeplitz operator gS, which we write as g. The image of g is

$$M_0 = \begin{pmatrix} \mathrm{Id} - \pi_0 + \pi_0 g \pi_0 & 0 \\ 0 & \mathrm{Id} \end{pmatrix}.$$

The homotopy has three stages, each depending smoothly on g:

$$M_0 \xrightarrow{\text{Stage 1}} M_{\pi/2} \xrightarrow{\text{Stage 2}} \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix} \xrightarrow{\text{Stage 3}} \mathrm{Id}\,.$$

In Stages 1 and 2 the Toeplitz structure is essential: the shifts move the matrix g off the constants, so that it can be traded for g^{-1} in the second block. Stage 3 is the finite dimensional fact that $\mathrm{diag}(g, g^{-1})$ is connected to Id uniformly in g.

Stage 1. For $\tau \in [0, \pi/2]$ let

$$M_\tau = \begin{pmatrix} \cos\tau(\mathrm{Id} - \pi_0) + \pi_0 g \pi_0 & \sin\tau\, gU \\ -\sin\tau\, g^{-1} L & \cos\tau\ \mathrm{Id} \end{pmatrix}.$$

Its symbol on S^*_+ is $\left(\begin{smallmatrix} \cos\tau & \sin\tau\, g e^{i\theta} \\ -\sin\tau\, g^{-1} e^{-i\theta} & \cos\tau \end{smallmatrix}\right)$, of determinant 1, so M_τ is elliptic. To see that it is invertible, write $u = \sum_{k\geq 0} u_k e^{ik\theta}$, $v = \sum_{k\geq 0} v_k e^{ik\theta}$ with $u_k, v_k \in \mathbb{C}^N$. Then M_τ maps $(u_0, 0)$ to $(gu_0, 0)$. For $k \geq 1$ it maps the pair of modes $(u_k e^{ik\theta}, v_{k-1} e^{i(k-1)\theta})$ to $(f_k e^{ik\theta}, h_{k-1} e^{i(k-1)\theta})$ with

$$\begin{pmatrix} f_k \\ h_{k-1} \end{pmatrix} = \begin{pmatrix} \cos\tau & \sin\tau\, g \\ -\sin\tau\, g^{-1} & \cos\tau \end{pmatrix} \begin{pmatrix} u_k \\ v_{k-1} \end{pmatrix},$$

and

$$\begin{pmatrix} \cos\tau & \sin\tau\, g \\ -\sin\tau\, g^{-1} & \cos\tau \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & g^{-1} \end{pmatrix} R_\tau \begin{pmatrix} 1 & 0 \\ 0 & g \end{pmatrix},$$

where $R_\tau = \left(\begin{smallmatrix} \cos\tau & \sin\tau \\ -\sin\tau & \cos\tau \end{smallmatrix}\right)$. These matrices are invertible, with inverses of the same form with τ replaced by $-\tau$. So M_τ is invertible, and its inverse is the Toeplitz operator obtained by replacing τ by $-\tau$ and g by g^{-1} in the $(1,1)$ entry. Thus $M_\tau \in \tilde{G}^0_{\mathcal{T}}$. At $\tau = \pi/2$,

$$M_{\pi/2} = \begin{pmatrix} \pi_0 g \pi_0 & gU \\ -g^{-1} L & 0 \end{pmatrix}.$$

Stage 2. For τ decreasing from $\pi/2$ to 0 let

$$M'_\tau = \begin{pmatrix} \cos\tau\, g(\mathrm{Id} - \pi_0) + \pi_0 g \pi_0 & \sin\tau\, gU \\ -\sin\tau\, g^{-1} L & \cos\tau\, g^{-1} \end{pmatrix},$$

which equals $M_{\pi/2}$ at $\tau = \pi/2$. Its symbol is $\mathrm{diag}(g, g^{-1}) \left(\begin{smallmatrix} \cos\tau & \sin\tau\, e^{i\theta} \\ -\sin\tau\, e^{-i\theta} & \cos\tau \end{smallmatrix}\right)$, which is invertible. On modes it acts as g on u_0 and by the matrix $\left(\begin{smallmatrix} \cos\tau\, g & \sin\tau\, g \\ -\sin\tau\, g^{-1} & \cos\tau\, g^{-1} \end{smallmatrix}\right) = \mathrm{diag}(g, g^{-1}) R_\tau$ on the pairs (u_k, v_{k-1}). So it is again in $\tilde{G}^0_{\mathcal{T}}$. At $\tau = 0$ we reach the constant matrix

$$M'_0 = \begin{pmatrix} g & 0 \\ 0 & g^{-1} \end{pmatrix}.$$

Stage 3. It remains to deform $\operatorname{diag}(g, g^{-1})$ to the identity in $\mathrm{GL}(2N, \mathbb{C})$, uniformly in g. First take $\left(\begin{smallmatrix} \cos\tau\, g & \sin\tau \\ -\sin\tau & \cos\tau\, g^{-1} \end{smallmatrix}\right)$, $\tau \in [0, \pi/2]$. For $\cos\tau \neq 0$ the Schur complement of the invertible $(1,1)$ entry is $\cos\tau\, g^{-1} + \sin^2\tau (\cos\tau)^{-1} g^{-1} = (\cos\tau)^{-1} g^{-1}$, so the matrix is invertible. At $\tau = \pi/2$ it is $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$. Then follow the rotations R_τ from $\tau = \pi/2$ back to $\tau = 0$. Constant invertible matrices are elements of $\tilde{G}^0_{\mathcal{T}}$. After smoothing the corners by reparametrization, the concatenation of the three stages is the required homotopy. □

Remark 12.7.2. Proposition 12.7.1 shows that the "$G^{-\infty}_{\mathcal{T}}$ part" of the Toeplitz group can be (weakly) deformed away after stabilization. In Chapter 13 this is combined with Atiyah's linearization of loops in $\mathrm{GL}(N, \mathbb{C})$, which shows how invertible elliptic Toeplitz operators can be deformed into the smoothing subgroup. Together these give the weak contractibility of the middle group in (12.3.5).

12.8 Exercises

Exercise 12.1. Let $A_1 \in \Psi^{m_1}(X; F, G)$ and $A_2 \in \Psi^{m_2}(X; E, F)$ be elliptic. Using Theorem 12.1.4 with the parametrix $B_2 B_1$ of $A_1 A_2$, show that $\operatorname{ind}(A_1 A_2) = \operatorname{ind}(A_1) + \operatorname{ind}(A_2)$. (Write $\mathrm{Id} - B_2 B_1 A_1 A_2 = (\mathrm{Id} - B_2 A_2) + B_2 R_1 A_2$ with $R_1 = \mathrm{Id} - B_1 A_1$, and use Lemma 12.1.2.)

Exercise 12.2. Show that $\operatorname{ind}(A^*) = -\operatorname{ind}(A)$ for elliptic A, and that every elliptic self-adjoint $A \in \Psi^m(X; E)$ has index zero.

Exercise 12.3. Let $A \in \mathcal{C}^\infty(X'; \Psi^m(X; E, F))$ be a smooth family of elliptic operators parametrized by a connected compact manifold X'. Show directly from Theorem 12.1.4 and Lemma 12.2.1 that $x \mapsto \operatorname{ind} A(x)$ is constant.

Exercise 12.4. Show that for $a \in \Psi^{-\infty}(Y; E)$ the function $z \mapsto \det(\mathrm{Id} + za)$ is entire, and that its zeros are the points $z = -1/\lambda$ with $\lambda \neq 0$ an eigenvalue of a. Show further that the order of the zero at $-1/\lambda$ is the algebraic multiplicity of λ. (Use the local formula in the proof of Theorem 12.4.8 with W containing the generalized eigenspace.)

Exercise 12.5. Let $f : \mathbb{S}^1 \to G^{-\infty}(\mathbb{S}^1)$, $f(e^{is}) = \mathrm{Id} + (e^{is} - 1)\pi_0$, where π_0 is the projection onto the constants. Compute $\int_{\mathbb{S}^1} f^* u_1$ and $\det f(e^{is})$. Deduce that $U_1 : K^{-1}(\mathbb{S}^1) \to H^1(\mathbb{S}^1; \mathbb{C}) \cong \mathbb{C}$ has image $2\pi i\mathbb{Z}$.

Exercise 12.6. Show that the forms $\operatorname{Tr}((A^{-1}dA)^{2k})$ on $G^{-\infty}$ vanish identically, directly from the antisymmetrized formula (12.4.1) with $2k$ entries. Show also that u_{2k-1} is invariant under right multiplication, using $R_B^*(A^{-1}dA) = B^{-1}(A^{-1}dA)B$.

Exercise 12.7. Show that $[S, f] \in \Psi^{-\infty}(\mathbb{S}^1)$ for every $f \in \mathcal{C}^\infty(\mathbb{S}^1)$. Use the symbol χ of Theorem 12.5.2 and the composition formula, noting that all ξ-derivatives of χ have compact support. Deduce that $T_a T_b - T_{ab} \in \Psi^{-\infty}_{\mathcal{T}}$ for $a, b \in \mathcal{C}^\infty(\mathbb{S}^1; M(N, \mathbb{C}))$. Also show that, if a and b are trigonometric polynomials, $T_a T_b - T_{ab}$ has finite rank.

Exercise 12.8. A compact manifold Z is recovered from the ring $\mathcal{C}^\infty(Z)$ as the set of its multiplicative linear functionals ("valuations") p, with $p(fg) = p(f)p(g)$ and $p(1) = 1$. Show that the continuous multiplicative linear functionals on the ring $\mathcal{C}^\infty_+(\mathbb{S}^1)$ are exactly $f \mapsto u_f(z)$, $z \in \overline{\mathbb{D}}$, where u_f is the holomorphic extension of Proposition 12.5.1. So the "space" of the Hardy ring is the closed disc, not half of the circle.

Exercise 12.9. Using Theorem 12.5.4, show that $\operatorname{ind}_{\mathcal{T}}(T_a T_b) = \operatorname{ind}_{\mathcal{T}}(T_a) + \operatorname{ind}_{\mathcal{T}}(T_b)$ for $a, b \in \mathcal{C}^\infty(\mathbb{S}^1; \mathrm{GL}(N, \mathbb{C}))$. Check this against Exercise 12.1 applied to $T_a + \mathrm{Id} - S$ and $T_b + \mathrm{Id} - S$.

Notes

The source lectures stated the existence of smooth families of parametrices, the smoothness of the bundles F_r, and the extension of the Fredholm determinant to all of $\Psi^{-\infty}$ without proof. Here they are supplied in Lemma 12.2.1, Lemma 12.6.2 and Theorem 12.4.8. The second stage of the homotopy in Proposition 12.7.1 has been corrected so that it starts where the first stage ends. The parametrization of $\mathbb{S}^1 \setminus \{1\}$ in Proposition 12.3.7 has been adjusted to omit the base point. The classical index theorem for Toeplitz operators on the circle (Theorem 12.5.4) is due to F. Noether and to Gohberg and Krein; it has been added as the simplest instance of the index formula. Loop groups and their relation with Toeplitz operators are treated at length in [21]. The approach via the isotropic algebra on $\mathbb{R}$ is carried out in the setting of manifolds with boundary in [20].

The Szegő projector also exists in higher dimensions. For the boundary $\mathbb{S}^{2n-1}$ of the unit ball in $\mathbb{C}^n$ (or of a strictly pseudoconvex domain) the orthogonal projection onto boundary values of holomorphic functions is a Fourier integral operator of Hermite type. The index of the Toeplitz operator $T_a = SaS$ with $a \in \mathcal{C}^\infty(\mathbb{S}^{2n-1}; \mathrm{GL}(N, \mathbb{C}))$ is a universal multiple of $\int_{\mathbb{S}^{2n-1}} \mathrm{tr}((a^{-1}da)^{2n-1})$, i.e. of the pairing of the odd Chern form u_{2n-1} with the fundamental class. This is a theorem of L. Boutet de Monvel (*On the index of Toeplitz operators of several complex variables*, Invent. Math. 50 (1979)). Theorem 12.5.4 is the case $n = 1$.

Chapter 13

Linearization of symbols, K-theory and Bott periodicity

The index of an elliptic operator depends only on the homotopy class of its symbol. To compute it we need to understand such homotopy classes, and the simplest non-trivial case already contains the key idea: loops of invertible matrices, which are the symbols of Toeplitz operators on the circle. This chapter shows that every such loop can be deformed into an explicit *normal form*, reads off the index from the normal form, and deduces Bott periodicity, the basic theorem of K-theory, from the resulting index formula.

In Chapter 12 the index of an elliptic operator was shown to be a homotopy invariant, the group $G^{-\infty}(Y;E)$ of invertible perturbations of the identity by smoothing operators was introduced as a model for odd K-theory, and the Toeplitz algebra on the circle was set up. This chapter combines these ingredients. The central technical result is a normal form, due to Atiyah, for loops of invertible matrices: after stabilization and deformation every (pointed) loop can be brought to the form $\pi_- e^{-i\theta} + \pi_0 + \pi_+ e^{i\theta}$ with commuting idempotents $\pi_\pm, \pi_0$. Applied to the symbols of Toeplitz operators this *linearization* reduces index computations to an explicit model. As consequences we obtain:

- the index formula for Toeplitz and pseudodifferential operators on the circle;
- the families index of elliptic Toeplitz families, with values in K^0, and the fact that it gives an isomorphism $K^{-2}(X) \cong K^0(X)$ (Bott periodicity in its first form);
- the weak contractibility of the stabilized group $G^0_{\mathcal{T}}$ of invertible Toeplitz operators, hence a *classifying sequence* $G^{-\infty}_{\mathcal{T}} \to G^0_{\mathcal{T}} \to G^0_{\mathcal{T}}/G^{-\infty}_{\mathcal{T}}$ for K-theory;
- Bott periodicity in its usual form, $K^0_{\mathrm{c}}(\mathbb{R}^2 \times X) \cong K^0(X)$, and the homotopy groups of $G^{-\infty}$: $\pi_k(G^{-\infty}) = \mathbb{Z}$ for k odd and 0 for k even;
- a second, independent, realization of the index isomorphism through the isotropic calculus of Chapter 7: the circle at infinity of the phase plane $\mathbb{R}^2$ replaces the circle, the eigenprojections of the harmonic oscillator replace the projections onto Fourier modes, and the annihilation operator replaces the Toeplitz shift;
- the *isotropic–semiclassical quantization maps* $K^{-1}_{\mathrm{c}}(\mathbb{R}^{2n}\times X) \to K^{-1}_{\mathrm{c}}(X)$ and $K^0_{\mathrm{c}}(\mathbb{R}^{2n}\times X) \to K^0_{\mathrm{c}}(X)$, obtained by quantizing compactly supported symbols with values in invertible matrices or in projections, which realize Bott periodicity and are used in Chapter 16;
- complex and symplectic vector bundles and the Thom isomorphism;

- the analytic index map $K^0_c(T^*Z) \to \mathbb{Z}$, the starting point of the index theorem.

K-theory is defined from scratch in Section 13.2; we follow [17] for the topological background. The main line uses Toeplitz operators on the circle; compare [21] for the relation between loop groups and Toeplitz operators. The isotropic version (Section 13.7) runs parallel to it and the two index maps are compared in Proposition 13.7.7.

13.1 Conventions and background

Throughout, X denotes a compact manifold, possibly with boundary (we need $[0,1] \times X$), and all maps and homotopies are smooth. The circle is $\mathbb{S} = \mathbb{R}/2\pi\mathbb{Z}$ with variable θ; we write $z = e^{i\theta}$, so that $\mathbb{S}$ is also the unit circle in $\mathbb{C}$, and we use the point $1 \in \mathbb{S}$ ($\theta = 0$) as base point. $M(N,\mathbb{C})$ is the algebra of $N \times N$ matrices. For maps from a product we use the usual notion of smoothness; a smooth map into a Fréchet space F (such as $\Psi^0(\mathbb{S})$ with its standard topology, see Chapters 4 and 5) is one all of whose derivatives exist and are continuous.

Two smooth homotopies $h, h' \colon [0,1] \times X \to W$ with $h_1 = h'_0$ can be concatenated to a smooth homotopy after reparametrizing each of them by a smooth non-decreasing surjection $[0,1] \to [0,1]$ which is constant near 0 and 1. We use this tacitly whenever homotopies are composed.

13.1.1 Facts recalled from Chapter 12

Let Y be a compact manifold of positive dimension, $E \to Y$ a Hermitian vector bundle, and fix a smooth positive density on Y; write $L^2(Y;E)$ for the corresponding Hilbert space.

(B1) $\Psi^{-\infty}(Y;E)$, the smoothing operators (kernels in $\mathcal{C}^\infty(Y^2; \mathrm{Hom}(E) \otimes \Omega_R)$), is a Fréchet algebra and

$$G^{-\infty}(Y;E) = \{\mathrm{Id} + A;\ A \in \Psi^{-\infty}(Y;E),\ \mathrm{Id} + A \text{ invertible on } L^2(Y;E)\}.$$

The inverse of $\mathrm{Id} + A \in G^{-\infty}(Y;E)$ is of the form $\mathrm{Id} + B$ with $B \in \Psi^{-\infty}(Y;E)$; $G^{-\infty}(Y;E)$ is an open subset of $\mathrm{Id} + \Psi^{-\infty}(Y;E)$, a topological group with smooth multiplication and inversion, and it is connected.

(B2) (Finite rank approximation.) Given a finite dimensional subspace $V_0 \subset \mathcal{C}^\infty(Y;E)$ there is an increasing sequence of finite dimensional subspaces $V_0 \subset V_1 \subset V_2 \subset \cdots \subset \mathcal{C}^\infty(Y;E)$ (for instance V_0 plus the span of the first j eigensections of a Laplacian) whose orthogonal projections Π_j satisfy $\Pi_j A \Pi_j \to A$ in $\Psi^{-\infty}(Y;E)$, uniformly with all derivatives on smooth families A parametrized by a compact manifold. For a finite dimensional subspace V with orthogonal projection Π_V we identify $\mathrm{GL}(V)$, and so $\mathrm{GL}(N,\mathbb{C})$ after choosing an orthonormal basis, with the subgroup $\{\mathrm{Id} - \Pi_V + g\Pi_V;\ g \in \mathrm{GL}(V)\} \subset G^{-\infty}(Y;E)$. Enlarging V to $V' \supset V$ corresponds to the stabilization $g \mapsto g \oplus \mathrm{Id}$.

(B3) $K^{-1}(X) = [X; G^{-\infty}(Y;E)]$, the set of smooth homotopy classes of smooth maps, is an abelian group under pointwise multiplication, independent of the choice of (Y,E).

(B4) For a smooth loop $b \colon \mathbb{S} \to G^{-\infty}(Y;E)$ the number

$$\mathrm{wind}(b) = \frac{1}{2\pi i} \int_0^{2\pi} \mathrm{Tr}\left(b^{-1} \frac{db}{d\theta}\right) d\theta$$

is an integer, the winding number of the Fredholm determinant along b. For b with values in $\mathrm{GL}(N,\mathbb{C})$ it is the winding number of $\det b$, since $d\log\det b = \mathrm{Tr}(b^{-1}db)$. It is invariant under homotopy of loops.

(B5) The Szegő projector S, which removes the negative Fourier modes, is a self-adjoint projection in $\Psi^0(\mathbb{S})$ with principal symbol equal to 1 on $\mathbb{S}_+ = \{\xi > 0\}$ and 0 on $\mathbb{S}_- = \{\xi < 0\}$ in $S^*\mathbb{S} = \mathbb{S}_+ \sqcup \mathbb{S}_-$. Its range on $\mathcal{C}^\infty(\mathbb{S})$ is $\mathcal{C}^\infty_+(\mathbb{S})$, the smooth functions with only non-negative Fourier modes, and on L^2 it is $L^2_+(\mathbb{S})$. The Toeplitz algebra is

$$\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N) = \{A \in \Psi^0(\mathbb{S};\mathbb{C}^N);\ A = SAS\},$$

acting on $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$; $\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N) = S\Psi^{-\infty}(\mathbb{S};\mathbb{C}^N)S$. The (Toeplitz) symbol $\sigma_0(A) \in \mathcal{C}^\infty(\mathbb{S}; M(N,\mathbb{C}))$ is the restriction of the principal symbol to $\mathbb{S}_+ \cong \mathbb{S}$; it is multiplicative and for $a \in \mathcal{C}^\infty(\mathbb{S}; M(N,\mathbb{C}))$ the operator $T_a = SaS$ (multiplication by a, then projection) lies in $\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ with $\sigma_0(T_a) = a$. A constant matrix commutes with S, so $T_c = c$. A is *elliptic* if $\sigma_0(A)$ is invertible; then A has a parametrix in $\Psi^0_{\mathcal{T}}$ modulo $\Psi^{-\infty}_{\mathcal{T}}$, is Fredholm on $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$ and on $L^2_+(\mathbb{S};\mathbb{C}^N)$ with null spaces in $\mathcal{C}^\infty_+$, and its index depends only on the homotopy class of $\sigma_0(A)$ in $\mathcal{C}^\infty(\mathbb{S};\mathrm{GL}(N,\mathbb{C}))$. If an elliptic A is invertible, then $A^{-1} \in \Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$.

(B6) The shifts $U = T_z$ and $L = T_{z^{-1}}$ act by $U(e^{ik\theta}) = e^{i(k+1)\theta}$, $L(e^{ik\theta}) = e^{i(k-1)\theta}$ for $k \geq 1$ and $L(1) = 0$. Thus $LU = \mathrm{Id}$, $UL = \mathrm{Id} - P_0$, $\mathrm{ind}\, U = -1$ and $\mathrm{ind}\, L = 1$. Here, and below, P_k is the orthogonal projection onto the span of $e^{ij\theta}$, $0 \leq j \leq k$ (acting componentwise on vector valued functions). For a smooth family K of elements of $\Psi^{-\infty}_{\mathcal{T}}$ parametrized by a compact manifold, $KP_k \to K$ and $P_kK \to K$ in $\Psi^{-\infty}_{\mathcal{T}}$ uniformly as $k \to \infty$.

(B7) (Lifting homotopies.) If $A\colon [0,1] \times X \to \Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ is a smooth family of elliptic operators and $A_0(x)$ is invertible for all $x \in X$, then there is a smooth family $B\colon [0,1] \times X \to \Psi^{-\infty}_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ with $B_0 = 0$ such that $A_t(x) + B_t(x)$ is invertible for all (t,x).

A smooth family of invertible elliptic Toeplitz operators has smoothly varying inverses; we prove this in Lemma 13.5.1 below.

13.1.2 Elementary deformations of loops

A *pointed loop* (over X) is a smooth map $a\colon X \times \mathbb{S} \to \mathrm{GL}(N,\mathbb{C})$ with $a(x,1) = \mathrm{Id}$ for all x; a pointed homotopy is a smooth family of pointed loops. Loops are *stabilized* by $a \mapsto a \oplus \mathrm{Id}_{M-N}$, and a, b are *stably pointed homotopic*, $a \sim b$, if $a \oplus \mathrm{Id}$ and $b \oplus \mathrm{Id}$ are pointed homotopic in some $\mathrm{GL}(M,\mathbb{C})$. The following three lemmas are used constantly.

Discussion. Stabilization is what makes loops manageable. In a fixed $\mathrm{GL}(N,\mathbb{C})$ two loops a, b need not commute, and there is no direct reason for ab and ba to be homotopic. After adding a block of identity matrices there is room to rotate one loop into the new block (Lemma 13.1.2), where it commutes with the other. All the deformations below use this freedom to add blocks.

Lemma 13.1.1. *Let $h\colon [0,1] \times X \times \mathbb{S} \to \mathrm{GL}(N,\mathbb{C})$ be smooth and set*

$$\hat{h}_t(x,\theta) = h_t(x,1)^{-1} h_t(x,\theta).$$

Then $\hat{h}$ is a pointed homotopy; $\hat{h}_t = h_t$ whenever h_t is pointed, and $\hat{h}_t = \mathrm{Id}$ whenever $h_t(x,\theta)$ is independent of θ.

Proof. Immediate from the definition. □

Lemma 13.1.2 (Rotation)**.** *For pointed loops a, b with values in $\mathrm{GL}(N,\mathbb{C})$, $\mathrm{diag}(ab, \mathrm{Id}_N)$ and $\mathrm{diag}(a,b)$ are pointed homotopic in $\mathrm{GL}(2N,\mathbb{C})$. Consequently $ab \sim \mathrm{diag}(a,b) \sim ba$, $\mathrm{diag}(a_1 \cdots a_r, \mathrm{Id}, \dots, \mathrm{Id}) \sim \mathrm{diag}(a_1, \dots, a_r)$, and block sums of pointed loops are commutative up to $\sim$.*

Proof. Let $R_\tau = \begin{pmatrix} \cos\tau\,\mathrm{Id}_N & -\sin\tau\,\mathrm{Id}_N \\ \sin\tau\,\mathrm{Id}_N & \cos\tau\,\mathrm{Id}_N \end{pmatrix}$ and $h_\tau = \mathrm{diag}(a,\mathrm{Id}) R_\tau \,\mathrm{diag}(b,\mathrm{Id}) R_\tau^{-1}$, $\tau \in [0,\pi/2]$. Then $h_0 = \mathrm{diag}(ab,\mathrm{Id})$, $h_{\pi/2} = \mathrm{diag}(a,\mathrm{Id})\,\mathrm{diag}(\mathrm{Id},b) = \mathrm{diag}(a,b)$, and $h_\tau(x,1) = \mathrm{Id}$. The remaining statements follow by iteration and by applying the same argument with the roles of the blocks exchanged (the constant permutation matrix exchanging two blocks is joined to Id by R_τ). □

Lemma 13.1.3 (Conjugation)**.** *Let a be a pointed loop in $\mathrm{GL}(N,\mathbb{C})$ and $c\colon X \to \mathrm{GL}(N,\mathbb{C})$ smooth (independent of θ). Then $cac^{-1} \sim a$; if c is constant then cac^{-1} and a are pointed homotopic without stabilization.*

Proof. If c is constant, join it to Id by a path c_t in the connected group $\mathrm{GL}(N,\mathbb{C})$; then $c_t a c_t^{-1}$ is a pointed homotopy. In general, $h_\tau = \mathrm{diag}(c,\mathrm{Id}) R_\tau \,\mathrm{diag}(c^{-1},\mathrm{Id}) R_\tau^{-1}$ is a θ-independent path from Id to $\mathrm{diag}(c,c^{-1})$, and $h_\tau\,\mathrm{diag}(a,\mathrm{Id})h_\tau^{-1}$ is a pointed homotopy from $\mathrm{diag}(a,\mathrm{Id})$ to $\mathrm{diag}(c,c^{-1})\,\mathrm{diag}(a,\mathrm{Id})\,\mathrm{diag}(c^{-1},c) = \mathrm{diag}(cac^{-1},\mathrm{Id})$. □

13.2 Vector bundles and K-theory

All vector bundles are smooth complex vector bundles of finite rank; $\mathbb{C}^N$ also denotes the trivial bundle. For bundles E, F over the same base we write $E \cong F$ if there is a smooth bundle isomorphism, i.e. a smooth map of total spaces which is a linear isomorphism $E_x \to F_x$ on each fibre.

13.2.1 Bundles

Lemma 13.2.1. *Let X be compact.*

(i) Every vector bundle E over X is isomorphic to the range of a smooth family of orthogonal projections $p\colon X \to M(N,\mathbb{C})$; in particular there is a bundle F with $E \oplus F \cong \mathbb{C}^N$.

(ii) If $\pi\colon X \to M(N,\mathbb{C})$ is a smooth family of idempotents, $\pi^2 = \pi$, then $\operatorname{Ran}\pi = \{(x,v);\ v \in \operatorname{Ran}\pi(x)\}$ is a smooth vector bundle.

Proof. (ii) The rank of $\pi(x)$ equals $\operatorname{tr}\pi(x)$, so is locally constant. If $v_1, \dots, v_r$ is a basis of $\operatorname{Ran}\pi(x_0)$ then $\pi(x)v_1, \dots, \pi(x)v_r$ are linearly independent for x near x_0, hence a smooth local frame of $\operatorname{Ran}\pi$.

(i) Choose open sets $U_1, \dots, U_s$ covering X over which E has trivializations $\tau_i\colon E|_{U_i} \to U_i \times \mathbb{C}^n$, and smooth $\psi_i \geq 0$ with $\operatorname{supp}\psi_i \subset U_i$ and $\sum_i \psi_i > 0$. Then

$$\iota(v) = (\psi_1(x)\tau_1(v), \dots, \psi_s(x)\tau_s(v)), \quad v \in E_x,$$

is a smooth fibrewise injective bundle map $E \to \mathbb{C}^N$, $N = sn$. The orthogonal projection $p(x)$ onto $\iota(E_x)$ depends smoothly on x (locally it is given by the Gram–Schmidt process applied to a smooth frame), so $E \cong \operatorname{Ran} p$, and $F = \operatorname{Ran}(\mathrm{Id} - p)$ is a complement. □

Lemma 13.2.2 (Homotopy invariance of bundles)**.** *Let W be a manifold.*

(i) If E is a vector bundle over $[0,1]\times W$, then $E|_{\{0\}\times W}\cong E|_{\{1\}\times W}$ as bundles over W.

(ii) If $\varpi\colon V\to W$ is a real vector bundle (for instance $V=\mathbb{R}^n\times W$ or $V=T^*Z$) and E is a vector bundle over the total space V, then $E\cong\varpi^*(E|_W)$, where W is identified with the zero section.

Proof. Choose a connection on E (patching local trivial connections with a partition of unity). In (i), parallel transport along the curves $t\mapsto(t,w)$ gives linear isomorphisms $E_{(0,w)}\to E_{(1,w)}$ which depend smoothly on w by smooth dependence of solutions of linear ordinary differential equations on parameters. In (ii), parallel transport along $t\mapsto t\zeta$, $t\in[0,1]$, from the zero section to $\zeta\in V$ gives a smooth isomorphism $\varpi^*(E|_W)\to E$. □

13.2.2 The group K^0

Definition 13.2.3. For compact X, $K^0(X)$ is the set of equivalence classes of pairs (E_+,E_-) of vector bundles over X, where $(E_+,E_-)\sim(F_+,F_-)$ if there is a bundle H with

$$E_+\oplus F_-\oplus H\cong F_+\oplus E_-\oplus H.$$

The class of (E_+,E_-) is written $[E_+]-[E_-]$, and $[E]=[E]-[0]$. Addition is $([E_+]-[E_-])+([F_+]-[F_-])=[E_+\oplus F_+]-[E_-\oplus F_-]$.

The relation is reflexive and symmetric; it is transitive because if $E_+\oplus F_-\oplus H\cong F_+\oplus E_-\oplus H$ and $F_+\oplus G_-\oplus H'\cong G_+\oplus F_-\oplus H'$ then $E_+\oplus G_-\oplus(F_+\oplus F_-\oplus H\oplus H')\cong G_+\oplus E_-\oplus(F_+\oplus F_-\oplus H\oplus H')$. Addition is well defined, associative and commutative; $[E]-[E]$ is the zero element and $[E_-]-[E_+]$ is the inverse of $[E_+]-[E_-]$. Thus $K^0(X)$ is an abelian group. By Lemma 13.2.1 every element can be written $[E]-[\mathbb{C}^N]$ (add a complement of E_- to both bundles), and $[E_+]-[E_-]=0$ if and only if $E_+\oplus\mathbb{C}^L\cong E_-\oplus\mathbb{C}^L$ for some L (replace H by $H\oplus H^\perp\cong\mathbb{C}^L$). A smooth map $f\colon X'\to X$ induces a homomorphism $f^*\colon K^0(X)\to K^0(X')$ by pull-back of bundles. By Lemma 13.2.2, $K^0([0,1]\times X)\to K^0(X)$, restriction to $\{t\}\times X$, is independent of t. The difference of ranks at a point identifies $K^0(\mathrm{pt})=\mathbb{Z}$; the *rank* of $[E_+]-[E_-]$ at x is $\dim E_{+,x}-\dim E_{-,x}$. If $x_0\in X$ is a base point, $\tilde K^0(X)=\ker\bigl(K^0(X)\to K^0(\{x_0\})\bigr)$ is the *reduced* group, and $K^0(X)=\mathbb{Z}\oplus\tilde K^0(X)$, the splitting being given by pull-back along $X\to\{x_0\}$.

Question. Why use formal differences $[E_+]-[E_-]$ rather than bundles themselves?

Answer. Because the index is a difference. For one operator, $\dim\operatorname{Nul}A-\dim\operatorname{Nul}A^*$ can be negative, so it lives in $\mathbb{Z}$, not in $\mathbb{N}_0$. For a family the null spaces and cokernels become bundles, and the index becomes "null bundle minus cokernel bundle". $K^0(X)$ is built from bundles exactly as $\mathbb{Z}$ is built from $\mathbb{N}_0$. The extra bundle H in Definition 13.2.3 is needed because cancellation can fail for bundles: $E\oplus H\cong F\oplus H$ does not imply $E\cong F$. □

13.2.3 Odd K-theory and pointed loops

Odd K-theory is $K^{-1}(X)=[X;G^{-\infty}(Y;E)]$, see (B3). The loops in $G^{-\infty}$ give a second model of even K-theory.

Definition 13.2.4. For compact X, $K^{-2}(X)$ is the set of smooth homotopy classes of smooth maps $f\colon X\times\mathbb{S}\to G^{-\infty}(Y;E)$ with $f(x,1)=\mathrm{Id}$ for all x, where homotopies are required to satisfy the same condition. It is a group under pointwise multiplication.

Equivalently, $K^{-2}(X)$ is the set of homotopy classes of smooth maps from X into the pointed loop group

$$\Omega G^{-\infty}(Y;E) = \{b \in \mathcal{C}^\infty(\mathbb{S}; G^{-\infty}(Y;E));\ b(1) = \mathrm{Id}\}$$

(this is homotopy equivalent to the flat loop group $G^{-\infty}_{(1)}(Y;E)$ of Chapter 12, by the flattening argument used there). The number $\mathrm{wind}(f(x,\cdot))$ of (B4) is locally constant in x; we denote by $\Omega_0 G^{-\infty}(Y;E)$ the loops with wind $= 0$.

Lemma 13.2.5. *Let $X' \subset X$ be closed (e.g. $X' = X \times \{1\} \subset X \times \mathbb{S}$).*

(i) Every smooth $f\colon X \to G^{-\infty}(Y;E)$ with $f = \mathrm{Id}$ on X' is homotopic, through maps equal to Id *on X', to a map into a subgroup $\mathrm{GL}(V) \cong \mathrm{GL}(N,\mathbb{C})$ as in (B2).*

(ii) If two such maps into $\mathrm{GL}(V)$ are homotopic in $G^{-\infty}(Y;E)$ through maps equal to Id *on X', they are homotopic through such maps in $\mathrm{GL}(V')$ for some finite dimensional $V' \supset V$.*

In particular $K^{-2}(X)$ is the set of stable pointed homotopy classes of pointed loops in the groups $\mathrm{GL}(N,\mathbb{C})$; in it $[a]+[b] = [ab] = [\mathrm{diag}(a,b)]$, and $K^{-2}(X)$ is abelian. The same holds for $K^{-1}(X)$, with free maps $X \to \mathrm{GL}(N,\mathbb{C})$.

Proof. (i) Write $f = \mathrm{Id} + A$. With Π_j as in (B2), the maps $f_t = \mathrm{Id} + (1-t)A + t\Pi_j A \Pi_j$ converge to f uniformly on $[0,1]\times X$ as $j \to \infty$; since $f(X)$ is compact and $G^{-\infty}$ is open, $f_t \in G^{-\infty}$ for j large. Where $A = 0$ also $f_t = \mathrm{Id}$, and $f_1 = \mathrm{Id} - \Pi_j + (\mathrm{Id} + \Pi_j A \Pi_j)\Pi_j$ has values in $\mathrm{GL}(V_j)$. (ii) Apply (i) to the homotopy, a map on $[0,1]\times X$, with $V_0 = V$ and closed set $[0,1]\times X' \cup \{0,1\}\times X$; on $\{0,1\}\times X$ the operator A satisfies $A = \Pi_V A \Pi_V = \Pi_j A \Pi_j$, so the approximation does not change the endpoints. The final statements follow, using Lemma 13.1.2. □

Remark 13.2.6 (The isotropic model). The proof of Lemma 13.2.5 uses only the finite rank approximations of (B2). The residual isotropic group $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N) = \{\mathrm{Id} + A;\ A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N),\ \mathrm{Id}+A \text{ invertible}\}$ of Chapter 7 has the same property, with Π_k the orthogonal projection onto the span of the Hermite functions u_α, $|\alpha| \le k$ (tensored with $\mathbb{C}^N$): for every $A \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$, $\Pi_k A \Pi_k \to A$ in $\Psi^{-\infty}_{\mathrm{iso}}$, uniformly on compact families, because the Hermite expansion of a Schwartz kernel converges in $\mathcal{S}$. Hence smooth homotopy classes of maps into $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ are again the stable classes of maps into the groups $\mathrm{GL}(M,\mathbb{C})$, and $K^{-1}(X) = [X; G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)]$, $K^{-2}(X)$ can equally be defined with $G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$ in place of $G^{-\infty}(Y;E)$. We use this model freely below.

Lemma 13.2.7. *For compact X let $\varpi\colon \mathbb{S}\times X \to X$ be the projection and $\iota_1\colon X \to \mathbb{S}\times X$, $x \mapsto (1,x)$. Then $\iota_1^*\varpi^* = \mathrm{Id}$ and forgetting the base point condition identifies $K^{-2}(X)$ with $\ker \iota_1^*$, so*

$$K^{-1}(\mathbb{S}\times X) \cong K^{-2}(X) \oplus K^{-1}(X), \qquad [f] \longmapsto \left([f\cdot \varpi^*\iota_1^* f^{-1}], [\iota_1^* f]\right).$$

Proof. If $f\colon \mathbb{S}\times X \to G^{-\infty}$, then $f\cdot\varpi^*\iota_1^* f^{-1}$, i.e. $(\theta,x) \mapsto f(\theta,x)f(1,x)^{-1}$, is pointed, and it is pointed homotopic to f when f is pointed. A free homotopy h_t between pointed maps gives the pointed homotopy $h_t \cdot \varpi^*\iota_1^* h_t^{-1}$, so the forgetful map $K^{-2}(X) \to \ker \iota_1^*$ is injective, and by the first remark it is surjective. Since $K^{-1}(\mathbb{S}\times X)$ is abelian, $[f] = [f\cdot\varpi^*\iota_1^* f^{-1}] + \varpi^*[\iota_1^* f]$, which gives the splitting. □

Via Lemma 13.9.3 below, $K^{-2}(X) \cong K_c^{-1}(\mathbb{R}\times X)$, so the lemma says that $K^{-1}(\mathbb{S}\times X)$ combines the odd K-theory of X and the odd K-theory of $\mathbb{R}\times X$; the latter will be identified with $K^0(X)$.

13.3 Linearization of loops

A loop of invertible matrices can be complicated. We now show that, up to stabilization and pointed homotopy, it can be replaced by the simplest possible loop: on each of three complementary subspaces it is z^{-1}, 1 or z. For such a loop the index of the Toeplitz operator can be read off by counting dimensions. The following normal form is due to Atiyah [17].

Theorem 13.3.1. *Let $a\colon X\times\mathbb{S}\to \mathrm{GL}(N,\mathbb{C})$ be a pointed loop, X compact. Then for some $M\geq N$ the stabilized loop $a\oplus \mathrm{Id}_{M-N}$ is pointed homotopic in $\mathrm{GL}(M,\mathbb{C})$ to a loop of the form*

$$\pi_-(x)z^{-1}+\pi_0(x)+\pi_+(x)z, \tag{13.3.1}$$

where $\pi_-,\pi_0,\pi_+\colon X\to M(M,\mathbb{C})$ are smooth families of idempotents which commute with each other and satisfy $\pi_-+\pi_0+\pi_+=\mathrm{Id}$.

Note that (13.3.1) is invertible, with inverse $\pi_- z+\pi_0+\pi_+ z^{-1}$, and is automatically pointed. We call it a loop in *normal form*. The proof occupies the rest of this section and proceeds through Laurent polynomials, polynomials and linear loops (Figure 13.1).

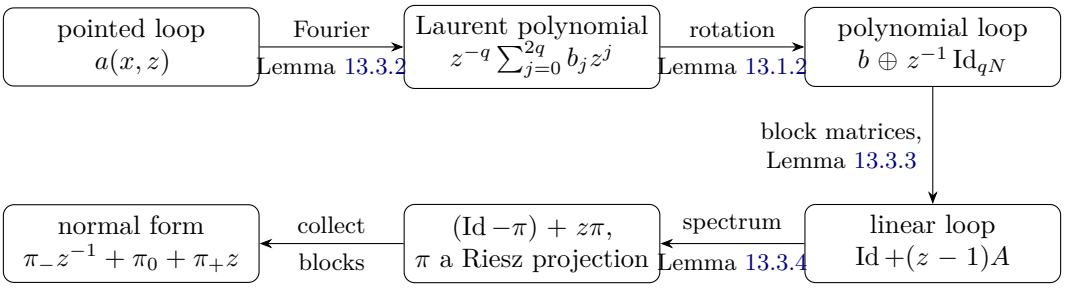


Figure 13.1: The steps in the linearization of loops (Theorem 13.3.1). Each arrow is a stable pointed homotopy. The degree in z is first made finite, then reduced to one by enlarging the matrices, and finally the linear loop is straightened by its spectral projection.

Lemma 13.3.2. *A pointed loop a is pointed homotopic to a pointed Laurent polynomial $\sum_{|j|\leq q} a'_j(x)z^j$ with $a'_j\in\mathcal{C}^\infty(X;M(N,\mathbb{C}))$.*

Proof. The Fourier coefficients $a_j(x)=\frac{1}{2\pi}\int_0^{2\pi} e^{-ij\theta}a(x,\theta)\,d\theta$ satisfy $\sup_X|a_j|\leq C_m(1+|j|)^{-m}$ for every m, by integration by parts in θ, so $a_{(q)}=\sum_{|j|\leq q}a_j z^j\to a$ uniformly on $X\times\mathbb{S}$. Let $\delta=\min_{X\times\mathbb{S}}\|a^{-1}\|^{-1}>0$ and choose q with $\sup\|a_{(q)}-a\|<\delta$. Then $h_t=(1-t)a+ta_{(q)}=a\big(\mathrm{Id}+ta^{-1}(a_{(q)}-a)\big)$ is invertible for $t\in[0,1]$, and by Lemma 13.1.1 $\hat{h}_t$ is a pointed homotopy from a to $a_{(q)}(x,1)^{-1}a_{(q)}$, which is a pointed Laurent polynomial. □

So we may assume $a=z^{-q}b$ where $b(x,z)=\sum_{j=0}^{p}b_j(x)z^j$, $p=2q$, is a polynomial in z which is invertible for $|z|=1$ and satisfies $b(x,1)=\mathrm{Id}$. Since z^{-q} is a scalar, Lemma 13.1.2 gives

$$\mathrm{diag}(a,\mathrm{Id}_N)=\mathrm{diag}(b\,z^{-q}\,\mathrm{Id}_N,\mathrm{Id}_N)\sim\mathrm{diag}(b,z^{-q}\,\mathrm{Id}_N),$$
$$\mathrm{diag}(z^{-q}\,\mathrm{Id}_N,\mathrm{Id}_{(q-1)N})\sim z^{-1}\,\mathrm{Id}_{qN},$$

the second by writing $z^{-q}=z^{-1}\cdots z^{-1}$. It therefore suffices to deal with the polynomial loop b.

Lemma 13.3.3. *Let $b=\sum_{j=0}^{p}b_j z^j$ be a pointed polynomial loop in $\mathrm{GL}(N,\mathbb{C})$. Then $\mathrm{diag}(b,\mathrm{Id}_{pN})$ is pointed homotopic in $\mathrm{GL}((p+1)N,\mathbb{C})$ to a loop $\mathrm{Id}+(z-1)A(x)$, with $A\in\mathcal{C}^\infty(X;M((p+1)N,\mathbb{C}))$.*

Proof. Set $c_j(x,z) = \sum_{l=j}^{p} b_l(x) z^{l-j}$ for $0 \le j \le p$, so $c_0 = b$, $c_p = b_p$ and $c_j - z c_{j+1} = b_j$. In $(p+1)\times(p+1)$ block form with $N \times N$ blocks consider

$$T(t) = \begin{pmatrix} b & tc_1 & \cdots & tc_p \\ 0 & \mathrm{Id} & & \\ \vdots & & \ddots & \\ 0 & & & \mathrm{Id} \end{pmatrix}, \quad \Lambda(s) = \begin{pmatrix} \mathrm{Id} & & & \\ -sz & \mathrm{Id} & & \\ & \ddots & \ddots & \\ & & -sz & \mathrm{Id} \end{pmatrix}, \quad t, s \in [0,1].$$

$T(t)$ is block upper triangular with invertible diagonal blocks when $|z| = 1$, and $\Lambda(s)$ is block lower triangular with identity diagonal, so both are invertible. The homotopy $T(t)$, $t \in [0,1]$, followed by $T(1)\Lambda(s)$, $s \in [0,1]$, starts at $\mathrm{diag}(b, \mathrm{Id})$ and ends at

$$T(1)\Lambda(1) = \begin{pmatrix} b_0 & b_1 & \cdots & b_{p-1} & b_p \\ -z & \mathrm{Id} & & & \\ & -z & \mathrm{Id} & & \\ & & \ddots & \ddots & \\ & & & -z & \mathrm{Id} \end{pmatrix} = B_0(x) + zB_1(x),$$

since the $(1,j)$ entry of the product is $c_j - zc_{j+1} = b_j$ for $j < p$ and $c_p = b_p$ for $j = p$. Applying Lemma 13.1.1 gives a pointed homotopy (the initial loop is pointed) ending at $(B_0+B_1)^{-1}(B_0+zB_1) = \mathrm{Id} + (z-1)A$ with $A = (B_0+B_1)^{-1}B_1$. □

Lemma 13.3.4. *Let $A \in \mathcal{C}^\infty(X; M(M,\mathbb{C}))$. The loop $\ell(x,z) = \mathrm{Id} + (z-1)A(x)$ is invertible for all $|z| = 1$ if and only if no eigenvalue of any $A(x)$ lies on the line $\{\Re\lambda = \frac{1}{2}\}$. In that case ℓ is pointed homotopic, through loops of the same form, to $(\mathrm{Id} - \pi) + z\pi$ where $\pi(x)$ is the spectral projection of $A(x)$ onto the generalized eigenspaces with $\Re\lambda > \frac{1}{2}$.*

Proof. For $z \neq 1$, $\ell(z) = (z-1)(A-w)$ with $w = (1-z)^{-1}$, and as $z = e^{i\theta}$ runs over $\mathbb{S} \setminus \{1\}$,

$$\frac{1}{1-e^{i\theta}} = \frac{1}{2} + \frac{i}{2}\cot\frac{\theta}{2}$$

runs over the whole line $\Re w = \frac{1}{2}$; at $z = 1$, $\ell = \mathrm{Id}$. This proves the first statement (see Figure 13.2). Now the set $\{(x,\lambda);\ \det(\lambda - A(x)) = 0\}$ is compact and disjoint from $X \times \{\Re\lambda = \frac{1}{2}\}$, so there is $\delta > 0$ such that all eigenvalues satisfy $|\Re\lambda - \frac{1}{2}| \ge \delta$. Let Γ be the positively oriented boundary of the rectangle $\{\frac{1}{2} + \frac{\delta}{2} \le \Re\lambda \le R,\ |\Im\lambda| \le R\}$ with $R > 1 + \sup_X \|A\|$. Then

$$\pi(x) = \frac{1}{2\pi i}\int_\Gamma (\lambda - A(x))^{-1}\, d\lambda$$

is a smooth family of idempotents commuting with $A(x)$, with range the sum of the generalized eigenspaces of $A(x)$ for eigenvalues with $\Re\lambda > \frac{1}{2}$ and null space the sum of the others (the Riesz projection). Set $A_t = (1-t)A + t\pi$. On $\mathrm{Ran}\,\pi$, which is A-invariant, $A_t = (1-t)A + t$ has eigenvalues $(1-t)\lambda + t$ with $\Re\lambda > \frac{1}{2}$, so real part $> \frac{1}{2}$; on $\mathrm{Nul}\,\pi$, $A_t = (1-t)A$ has eigenvalues $(1-t)\lambda$ with $\Re\lambda < \frac{1}{2}$, so real part $< \frac{1}{2}$. By the first part $\mathrm{Id} + (z-1)A_t$ is a pointed homotopy, and it ends at $\mathrm{Id} + (z-1)\pi = (\mathrm{Id} - \pi) + z\pi$. □

Proof of Theorem 13.3.1. By Lemma 13.3.2 and the reduction following it, $a \oplus \mathrm{Id}$ is stably pointed homotopic to $\mathrm{diag}(b, z^{-1}\,\mathrm{Id}_{qN})$, where b is a pointed polynomial loop of degree at most $p = 2q$. By Lemmas 13.3.3 and 13.3.4, $\mathrm{diag}(b, \mathrm{Id}_{pN}) \sim (\mathrm{Id} - \pi) + z\pi$. Hence, with $M = (3q+1)N$, $a \oplus \mathrm{Id}_{M-N}$ is pointed homotopic (after a constant permutation of blocks, Lemma 13.1.3) to (13.3.1) with $\pi_+ = \pi \oplus 0$, $\pi_0 = (\mathrm{Id} - \pi) \oplus 0$, $\pi_- = 0 \oplus \mathrm{Id}_{qN}$. □

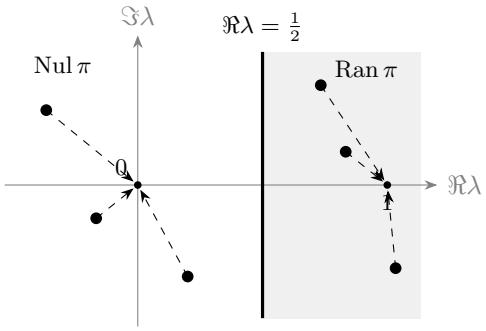


Figure 13.2: The proof of Lemma 13.3.4. As z runs over the circle, the loop $\mathrm{Id} + (z-1)A$ fails to be invertible exactly when an eigenvalue of A lies on the line $\Re\lambda = \frac{1}{2}$. The homotopy $A_t = (1-t)A + t\pi$ moves the eigenvalues on the right of the line to 1 and those on the left to 0, without crossing the line.

Remark 13.3.5. The proof gives more: π_- may be taken constant. We shall not use this. The spectral projection in Lemma 13.3.4 can also be written as a contour integral of the logarithmic derivative of the linear loop over $|z| = 1$; the formulation above does not require the coefficient of z to be invertible.

The decisive point for K-theory is that a loop in normal form is trivial as soon as $\operatorname{Ran}\pi_-$ and $\operatorname{Ran}\pi_+$ are isomorphic.

Lemma 13.3.6. *Let π_-, π_0, π_+ be commuting smooth idempotents on $\mathbb{C}^M$ over X with sum* Id*, and suppose there is a smooth bundle isomorphism $F\colon \operatorname{Ran}\pi_- \to \operatorname{Ran}\pi_+$. Then $\pi_- z^{-1} + \pi_0 + \pi_+ z$ is pointed homotopic to* Id *in* $\mathrm{GL}(M, \mathbb{C})$.

Proof. The fibres split as $\mathbb{C}^M = V_- \oplus V_0 \oplus V_+$, $V_\bullet = \operatorname{Ran}\pi_\bullet$. Regard $F\pi_-$ and $F^{-1}\pi_+$ as smooth maps into $M(M, \mathbb{C})$ and set, for $\tau \in [0, \pi/2]$,

$$h_\tau = \pi_0 + \cos\tau\,(z^{-1}\pi_- + z\pi_+) + \sin\tau\,(F^{-1}\pi_+ - F\pi_-).$$

h_τ is the identity on V_0 and preserves $V_- \oplus V_+$; identifying V_+ with V_- by F^{-1} it acts there as $\begin{pmatrix} \cos\tau\, z^{-1} & \sin\tau \\ -\sin\tau & \cos\tau\, z \end{pmatrix} \otimes \mathrm{Id}_{V_-}$, which has determinant 1. So h_τ is a loop in $\mathrm{GL}(M, \mathbb{C})$, h_0 is the given loop, and $h_{\pi/2} = \pi_0 + F^{-1}\pi_+ - F\pi_-$ is independent of z. By Lemma 13.1.1, $\hat{h}_\tau$ is a pointed homotopy from h_0 to Id. □

Discussion. The scalar case explains the lemma. The loop $\operatorname{diag}(z^{-1}, z)$ has $\det = 1$, so winding number zero, and the rotation in the proof joins it to a constant matrix. In general the factor z^{-1} on $\operatorname{Ran}\pi_-$ and the factor z on $\operatorname{Ran}\pi_+$ cancel as soon as the two bundles can be matched by an isomorphism F. So a normal form loop carries exactly one piece of information: the pair of bundles $(\operatorname{Ran}\pi_-, \operatorname{Ran}\pi_+)$, up to isomorphism. This is why the index of loops will take values in K^0.

13.4 The index of Toeplitz operators on the circle

We now quantize the normal form: replace z^{-1} by the backward shift L and z by the forward shift U. The resulting operator acts on each Fourier mode by an explicit matrix, so its null space and range can be computed directly. Together with the linearization this gives a second proof of Theorem 12.5.4, and one that extends to families.

For commuting smooth idempotents π_-, π_0, π_+ on $\mathbb{C}^M$ with sum Id consider the family of Toeplitz operators on $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^M)$

$$Q(x) = \pi_-(x)L + \pi_0(x) + \pi_+(x)U, \qquad \sigma_0(Q) = \pi_- z^{-1} + \pi_0 + \pi_+ z. \tag{13.4.1}$$

The matrices act pointwise and commute with S, U, L and the P_k. For $c \in \mathbb{C}^M$ write $c^\bullet = \pi_\bullet c$.

Lemma 13.4.1. *For every $k \geq 0$ the null space of $Q(x)(\mathrm{Id} - P_k)$ is $\operatorname{Ran} P_k$, and*

$$W_k(x) = \operatorname{span}\{e^{ij\theta}\mathbb{C}^M\}_{0\leq j\leq k-1} \oplus e^{ik\theta}\operatorname{Ran}(\pi_0+\pi_+)(x) \oplus e^{i(k+1)\theta}\operatorname{Ran}\pi_+(x)$$

is a complement to the range of $Q(x)(\mathrm{Id} - P_k)$ in $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^M)$. In particular $\operatorname{Nul} Q(x) = \operatorname{Ran}\pi_-(x)$ (constant functions), $\operatorname{Nul} Q(x)^* = \operatorname{Ran}\pi_+(x)^*$ and $\operatorname{ind} Q(x) = \operatorname{rank}\pi_- - \operatorname{rank}\pi_+$.

Proof. For $u = \sum_{j\geq k+1} c_j e^{ij\theta}$ (rapidly decreasing coefficients),

$$Q(\mathrm{Id} - P_k)u = \sum_{j\geq k+1} (c_j^- e^{i(j-1)\theta} + c_j^0 e^{ij\theta} + c_j^+ e^{i(j+1)\theta}) = \sum_m d_m e^{im\theta}$$

with $d_m^- = c_{m+1}^-$, $d_m^0 = c_m^0$, $d_m^+ = c_{m-1}^+$. These relations determine u from the image, so $Q(\mathrm{Id} - P_k)$ is injective on $\operatorname{Ran}(\mathrm{Id} - P_k)$, and its range consists exactly of the $\sum d_m e^{im\theta} \in \mathcal{C}^\infty_+$ with $d_m^- = 0$ for $m < k$, $d_m^0 = 0$ for $m < k+1$ and $d_m^+ = 0$ for $m < k+2$. The complement W_k follows. For a constant c_0, $Qc_0 = c_0^0 + c_0^+ e^{i\theta}$, so $\operatorname{Nul} Q = \operatorname{Ran}\pi_-$. The index is $\dim\operatorname{Ran} P_0 - \dim W_0 = M - \operatorname{rank}\pi_0 - 2\operatorname{rank}\pi_+ = \operatorname{rank}\pi_- - \operatorname{rank}\pi_+$. Finally $Q^* = \pi_-^* U + \pi_0^* + \pi_+^* L$ is of the same form, with the commuting idempotents $\pi_\pm^*$ in exchanged roles, so $\operatorname{Nul} Q^* = \operatorname{Ran}\pi_+^*$. □

Corollary 13.4.2 (Index formula for Toeplitz operators)**.** *If $P \in \Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ is elliptic, then*

$$\operatorname{ind} P = \dim\operatorname{Nul} P - \dim\operatorname{Nul} P^* = \frac{i}{2\pi}\int_0^{2\pi} \operatorname{Tr}\Big(\sigma_0(P)^{-1}\frac{d\sigma_0(P)}{d\theta}\Big)d\theta = -\operatorname{wind}\sigma_0(P). \tag{13.4.2}$$

Proof. The last two expressions agree by (B4). Let $c = \sigma_0(P)(1) \in \mathrm{GL}(N,\mathbb{C})$. Then $c^{-1}P$ is elliptic with pointed symbol $c^{-1}\sigma_0(P)$; it has the same index as P since c is an invertible operator, and the same winding number since $\det c$ is constant. So we may assume $\sigma_0(P)$ pointed. Both sides of (13.4.2) depend only on the stable pointed homotopy class of the symbol: the index by (B5), and because stabilizing P by Id adds nothing to either side; the winding number by (B4). By Theorem 13.3.1 (with X a point) we may thus assume $\sigma_0(P) = \pi_- z^{-1} + \pi_0 + \pi_+ z$, and then P may be replaced by Q of (13.4.1). Then $\operatorname{ind} Q = \operatorname{rank}\pi_- - \operatorname{rank}\pi_+$ by Lemma 13.4.1, while $\det\sigma_0(Q) = z^{\operatorname{rank}\pi_+ - \operatorname{rank}\pi_-}$, since $\sigma_0(Q)$ acts as z^{-1}, 1, z on the summands of $\mathbb{C}^M = \operatorname{Ran}\pi_- \oplus \operatorname{Ran}\pi_0 \oplus \operatorname{Ran}\pi_+$, so $\operatorname{wind}\sigma_0(Q) = \operatorname{rank}\pi_+ - \operatorname{rank}\pi_-$. □

The same holds for pseudodifferential operators on the circle, except that now both halves $\mathbb{S}_\pm$ of $S^*\mathbb{S}$ contribute. For $P \in \Psi^0(\mathbb{S};\mathbb{C}^N)$ write $\sigma_\pm(\theta) = \sigma_0(P)(\theta,\pm 1)$.

Proposition 13.4.3. *If $P \in \Psi^0(\mathbb{S};\mathbb{C}^N)$ is elliptic, then*

$$\operatorname{ind} P = \frac{i}{2\pi}\int_0^{2\pi}\operatorname{Tr}(\sigma_+^{-1}\partial_\theta\sigma_+)d\theta - \frac{i}{2\pi}\int_0^{2\pi}\operatorname{Tr}(\sigma_-^{-1}\partial_\theta\sigma_-)d\theta = \operatorname{wind}\sigma_- - \operatorname{wind}\sigma_+.$$

Equivalently $\operatorname{ind} P = \frac{i}{2\pi}\int_{S^*\mathbb{S}}\operatorname{Tr}(\sigma_0(P)^{-1}d\sigma_0(P))$ *with $\mathbb{S}_+$ oriented by $d\theta$ and $\mathbb{S}_-$ by $-d\theta$.*

Proof. Let $S' = \mathrm{Id} - S$, the projection onto $\mathcal{C}^\infty_-(\mathbb{S})$, the span of the negative Fourier modes. By (B5), SPS' and $S'PS$ have vanishing principal symbol, so $P_t = P - t(SPS' + S'PS)$ is a family of elliptic operators with constant principal symbol, and $\operatorname{ind} P = \operatorname{ind} P_1$. Now $P_1 = SPS + S'PS'$ preserves the decomposition $\mathcal{C}^\infty(\mathbb{S};\mathbb{C}^N) = \mathcal{C}^\infty_+ \oplus \mathcal{C}^\infty_-$, so $\operatorname{ind} P = \operatorname{ind}(SPS|_{\mathcal{C}^\infty_+}) + \operatorname{ind}(S'PS'|_{\mathcal{C}^\infty_-})$. The first is an elliptic Toeplitz operator with symbol σ_+, of index $-\operatorname{wind}\sigma_+$ by Corollary 13.4.2. For the second, let $Ru(\theta) = e^{-i\theta}u(-\theta)$. Then $R(e^{ik\theta}) = e^{-i(k+1)\theta}$, so R is an isomorphism of $\mathcal{C}^\infty(\mathbb{S};\mathbb{C}^N)$ exchanging $\mathcal{C}^\infty_-$ and $\mathcal{C}^\infty_+$, and $RS'R^{-1} = S$. Thus $R(S'PS')R^{-1} = STS$ with $T = RPR^{-1} \in \Psi^0(\mathbb{S};\mathbb{C}^N)$; as R is the composite of the pull-back by $\theta \mapsto -\theta$ and a multiplication, $\sigma_0(T)(\theta,\xi) = \sigma_0(P)(-\theta,-\xi)$. So $S'PS'|_{\mathcal{C}^\infty_-}$ is conjugate to the elliptic Toeplitz operator with symbol $\sigma_-(-\theta)$, whose index is $-\operatorname{wind}(\sigma_-(-\cdot)) = \operatorname{wind}\sigma_-$. □

Remark 13.4.4. The index of an elliptic *differential* operator $P \in \mathrm{Diff}^m(\mathbb{S};\mathbb{C}^N)$ vanishes. Indeed $\sigma_m(P) = p(\theta)\xi^m$ with p invertible, and $P(\mathrm{Id} + D_\theta^2)^{-m/2} \in \Psi^0$ has the same index as P (the second factor is invertible) and symbol $\sigma_\pm = (\pm 1)^m p$. Since $\operatorname{wind}((-1)^m p) = \operatorname{wind} p$, the two terms in Proposition 13.4.3 cancel.

13.5 The families index

We first record that inverses of smooth families are smooth. The same argument applies to the operators with values in smoothing operators of Section 13.8.

Lemma 13.5.1. *Let Z be a manifold and $A \in \mathcal{C}^\infty(Z;\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$ a smooth family of elliptic operators, each invertible on $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$ (equivalently on $L^2_+(\mathbb{S};\mathbb{C}^N)$). Then $A^{-1} \in \mathcal{C}^\infty(Z;\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$.*

Proof. The symbolic parametrix construction (Chapter 5) works with smooth dependence on parameters: it uses only the multiplicativity of the symbol, the inversion of the symbol and asymptotic summation. This gives a smooth family B with $BA = \mathrm{Id} - R$, $AB = \mathrm{Id} - R'$, where $R, R' \in \mathcal{C}^\infty(Z;\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$. The inclusion of $\Psi^0_{\mathcal{T}}$ into the bounded operators on L^2_+ is continuous, so $z \mapsto A(z)$ is smooth in operator norm, and hence so is $V = A^{-1}$. From $V = B + RV$ and $V = B + VR'$ we get $V = B + RB + RVR'$. The bilinear map $T \mapsto RTR'$ is continuous from bounded operators to smoothing operators (its kernel is $(w,w') \mapsto (TR'(\cdot,w'), \overline{R(w,\cdot)})$ and all derivatives fall on R and R'), so RVR', and hence V, is a smooth family in $\Psi^0_{\mathcal{T}}$. □

Now let Z be a compact manifold (possibly with boundary) and $A \in \mathcal{C}^\infty(Z;\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$ a smooth family of elliptic Toeplitz operators. Since $A(z)$ is Fredholm, the range of $A(z)(\mathrm{Id} - P_k) = A(z) - A(z)P_k$ is closed and of finite codimension, equal to $(k+1)N - \operatorname{ind} A(z)$ if its null space is $\operatorname{Ran} P_k$; the same holds on $\mathcal{C}^\infty_+$ and on L^2_+.

Definition 13.5.2. An integer $k \geq 0$ is *admissible* for A if $\operatorname{Nul}(A(z)(\mathrm{Id} - P_k)) = \operatorname{Ran} P_k$ for all $z \in Z$. A *complement* for (A,k) is a smooth vector bundle W over Z whose fibres $W(z) \subset \mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$ satisfy $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N) = A(z)(\mathrm{Id} - P_k)\mathcal{C}^\infty_+ \oplus W(z)$; smoothness means that W has local frames given by smooth maps into $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$.

If k is admissible then so is every $k' \geq k$, since $\operatorname{Ran}(\mathrm{Id} - P_{k'}) \subset \operatorname{Ran}(\mathrm{Id} - P_k)$.

Lemma 13.5.3. *Let A be a smooth elliptic family as above.*

(i) There is k_0 such that every $k \geq k_0$ is admissible.

(ii) If k is admissible, the orthocomplements $F_k(z)$ of the ranges of $A'(z) = A(z)(\mathrm{Id} - P_k)$ in L^2_+ form a complement for (A, k), and the orthogonal projections onto $F_k(z)$ form a smooth family. Every complement W for (A, k) is isomorphic to F_k.

(iii) If k is admissible with complement W, then $W \oplus A(e^{i(k+1)\theta}\mathbb{C}^N) \cong W \oplus \mathbb{C}^N$ is a complement for $(A, k+1)$.

Consequently the class

$$\operatorname{ind}(A) = [\operatorname{Ran} P_k] - [W] = [\mathbb{C}^{(k+1)N}] - [W] \in K^0(Z)$$

is independent of the admissible k and of the complement W for (A, k).

Proof. (i) Choose a smooth family of parametrices B, so $BA = \mathrm{Id} - R$ with $R \in \mathcal{C}^\infty(Z; \Psi^{-\infty}_{\mathcal{T}})$. Then $BA(\mathrm{Id} - P_k) = (\mathrm{Id} - R(\mathrm{Id} - P_k))(\mathrm{Id} - P_k)$. By (B6), $R(\mathrm{Id} - P_k) \to 0$ in $\Psi^{-\infty}_{\mathcal{T}}$, hence in L^2 operator norm, uniformly on Z, so for $k \geq k_0$ the operators $\mathrm{Id} - R(\mathrm{Id} - P_k)$ are invertible and $(\mathrm{Id} - P_k)(\mathrm{Id} - R(\mathrm{Id} - P_k))^{-1} BA(\mathrm{Id} - P_k) = \mathrm{Id} - P_k$. Hence $\operatorname{Nul} A(\mathrm{Id} - P_k) = \operatorname{Ran} P_k$.

(ii) The family $C = A'^* A' + P_k$ is elliptic (its symbol is $\sigma_0(A)^*\sigma_0(A)$) and injective: $(Cu, u) = \|A'u\|^2 + \|P_k u\|^2 = 0$ forces $P_k u = 0$ and $A'u = 0$, so $u \in \operatorname{Ran}(\mathrm{Id} - P_k) \cap \operatorname{Ran} P_k = \{0\}$. Being self-adjoint and Fredholm it is invertible, and C^{-1} is a smooth family by Lemma 13.5.1. Since $CP_k = P_k$ and $C(\mathrm{Id} - P_k) = A'^* A'$, the family $G = C^{-1} A'^*$ satisfies $GA' = \mathrm{Id} - P_k$. Then $A'G$ is the orthogonal projection onto $\operatorname{Ran} A'$ (it is idempotent, self-adjoint as $A'G = A'C^{-1}A'^*$, and has range $\operatorname{Ran} A'$), so $\Pi = \mathrm{Id} - A'G$ is a smooth family of orthogonal projections onto F_k. Its range consists of smooth functions (it is the null space of the elliptic operator A'^*) and its rank $(k+1)N - \operatorname{ind} A(z)$ is locally constant. If $v_1, \dots, v_r$ is a basis of $F_k(z_0)$, then $\Pi(z)v_1, \dots, \Pi(z)v_r$ are smooth maps into $\mathcal{C}^\infty_+$, linearly independent near z_0, so form a local frame: F_k is a smooth bundle. As $\operatorname{Ran} A'$ is closed and $A'(\mathcal{C}^\infty_+) = \operatorname{Ran} A' \cap \mathcal{C}^\infty_+$ (if $A'v \in \mathcal{C}^\infty_+$ then $GA'v = (\mathrm{Id} - P_k)v \in \mathcal{C}^\infty_+$), F_k is a complement. Finally, if W is a complement then $\Pi|_W \colon W \to F_k$ is a smooth bundle map; it is injective because $\operatorname{Nul}\Pi \cap \mathcal{C}^\infty_+ = A'(\mathcal{C}^\infty_+)$ meets W only in 0, and $\dim W = \dim F_k$, both being the codimension of $A'(\mathcal{C}^\infty_+)$. So $W \cong F_k$.

(iii) Since A is injective on $\operatorname{Ran}(\mathrm{Id} - P_k)$, we have $A(\mathrm{Id} - P_k)\mathcal{C}^\infty_+ = A(\mathrm{Id} - P_{k+1})\mathcal{C}^\infty_+ \oplus A(e^{i(k+1)\theta}\mathbb{C}^N)$. Hence $W \oplus A(e^{i(k+1)\theta}\mathbb{C}^N)$ is a complement for $(A, k+1)$, and $v \mapsto A(e^{i(k+1)\theta}v)$ trivializes the second summand.

For the consequence: given admissible $k \leq k'$ with complements W, W', (iii) applied $k' - k$ times gives a complement $W \oplus \mathbb{C}^{(k'-k)N}$ for k', isomorphic to W' by (ii); since $\operatorname{Ran} P_{k'} = \operatorname{Ran} P_k \oplus \mathbb{C}^{(k'-k)N}$ the classes agree. □

Definition 13.5.4. The class $\operatorname{ind}(A) \in K^0(Z)$ of Lemma 13.5.3 is the *families index* of the elliptic family A.

Question. Why is the families index a class in $K^0(Z)$ and not just an integer?

Answer. The integer $\operatorname{ind} A(z)$ is the rank of $\operatorname{ind}(A)$ at z, and it is constant on each component of Z. But the null spaces and the cokernels can twist as z moves. For the normal form operator Q of (13.4.1), the null space is the bundle $\operatorname{Ran}\pi_-$ and the cokernel is $\operatorname{Ran}\pi_+$ (Lemma 13.4.1). If $\operatorname{Ran}\pi_-$ is a non-trivial line bundle and $\operatorname{Ran}\pi_+ = 0$, the integer index is 1 at every point, yet $\operatorname{ind}(Q) = [\operatorname{Ran}\pi_-]$ differs from $[\mathbb{C}]$ whenever this line bundle is not stably trivial, as for the tautological line bundle over $\mathbb{S}^2$. The class in $K^0(Z)$ records this twisting. □

For Z a point this is the numerical index, since $(k+1)N - \dim W = \operatorname{ind} A$. In the notation of Chapter 12, $F_k = \operatorname{Nul}((\mathrm{Id} - P_k)A^*)$, and Lemma 13.5.3 shows that the bundle

considered there represents, together with the trivial bundle $\operatorname{Ran} P_k$, a well-defined element of $K^0(Z)$. The families index has the expected properties.

Proposition 13.5.5. *Let A, A' be smooth elliptic families of Toeplitz operators over Z.*

(i) (Homotopy) If A_t is an elliptic family over $[0,1]\times Z$, then $\operatorname{ind}(A_0)=\operatorname{ind}(A_1)$. In particular $\operatorname{ind}(A)$ depends only on the homotopy class of $\sigma_0(A)$ in $\mathcal{C}^\infty(Z\times\mathbb{S};\mathrm{GL}(N,\mathbb{C}))$.

(ii) (Additivity) $\operatorname{ind}(A\oplus A')=\operatorname{ind}(A)+\operatorname{ind}(A')$; $\operatorname{ind}(A\oplus \mathrm{Id})=\operatorname{ind}(A)$.

(iii) (Naturality) $\operatorname{ind}(f^*A)=f^*\operatorname{ind}(A)$ for smooth $f\colon Z'\to Z$.

(iv) If each $A(z)$ is invertible then $\operatorname{ind}(A)=0$. If $b,c\colon Z\to\mathrm{GL}(N,\mathbb{C})$ then $\operatorname{ind}(bAc)=\operatorname{ind}(A)$.

(v) (Normal form) For Q as in (13.4.1), $\operatorname{ind}(Q)=[\operatorname{Ran}\pi_-]-[\operatorname{Ran}\pi_+]$.

Proof. (i) Choose k admissible for the family over $[0,1]\times Z$; the restrictions of its complement F_k to $t=0,1$ are complements for A_0, A_1 and are isomorphic by Lemma 13.2.2. Two families with homotopic symbols are joined by an elliptic family: first the linear homotopy between A and $T_{\sigma_0(A)}$, which has constant symbol, then T_{a_t} for a homotopy a_t of symbols. (ii) and (iii) are clear from the definition, using P_k diagonally; a complement for (Id,k) is $\operatorname{Ran}P_k$. (iv) If A is invertible every k is admissible and $A(\operatorname{Ran}P_k)\cong\mathbb{C}^{(k+1)N}$ is a complement. Multiplication by $b(z)$ and $c(z)$ commutes with P_k, so $(bAc)(\mathrm{Id}-P_k)$ has null space $c^{-1}\operatorname{Ran}P_k=\operatorname{Ran}P_k$, range $bA(\mathrm{Id}-P_k)\mathcal{C}^\infty_+$ and complement $bW\cong W$. (v) By Lemma 13.4.1 every k is admissible, with complement $W_k\cong\mathbb{C}^{kM}\oplus\operatorname{Ran}(\pi_0+\pi_+)\oplus\operatorname{Ran}\pi_+$ (a smooth bundle by Lemma 13.2.1(ii)); since $\mathbb{C}^M=\operatorname{Ran}\pi_-\oplus\operatorname{Ran}(\pi_0+\pi_+)$, $[\mathbb{C}^{(k+1)M}]-[W_k]=[\operatorname{Ran}\pi_-]-[\operatorname{Ran}\pi_+]$. □

The families index is the only obstruction to invertibility after perturbation; this is the criterion of Chapter 12 (there phrased as triviality of the bundle F_k), up to stabilization.

Proposition 13.5.6. *Let A be a smooth elliptic family of Toeplitz operators on $\mathbb{C}^N$ over compact Z. Then $\operatorname{ind}(A)=0$ in $K^0(Z)$ if and only if for some L there is a smooth family $B\colon Z\to\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S};\mathbb{C}^{N+L})$ such that $A\oplus\mathrm{Id}_L+B$ is invertible for all $z\in Z$.*

Proof. If $A\oplus\mathrm{Id}_L+B$ is invertible, then $\operatorname{ind}(A)=\operatorname{ind}(A\oplus\mathrm{Id}_L+B)=0$ by Proposition 13.5.5. Conversely, let k be admissible with complement W. If $\operatorname{ind}(A)=0$ then $W\oplus\mathbb{C}^J\cong\mathbb{C}^{(k+1)N+J}$ for some J. Choose L with $(k+1)L\geq J$; then $\tilde A=A\oplus\mathrm{Id}_L$ has complement $\tilde W=W\oplus\mathbb{C}^{(k+1)L}\cong\mathbb{C}^{(k+1)(N+L)}=\operatorname{Ran}P_k$, so there is a smooth bundle isomorphism $\Theta\colon\operatorname{Ran}P_k\to\tilde W$. Put $B=\Theta P_k-\tilde A P_k$; this is a smooth family of finite rank operators with smooth kernels, so $B\in\Psi^{-\infty}_{\mathcal{T}}$, and $\tilde A+B=\tilde A(\mathrm{Id}-P_k)+\Theta P_k$ maps $\operatorname{Ran}(\mathrm{Id}-P_k)$ injectively onto $\tilde A(\mathrm{Id}-P_k)\mathcal{C}^\infty_+$ and $\operatorname{Ran}P_k$ onto the complement $\tilde W$. So it is a bijection of $\mathcal{C}^\infty_+$, hence invertible, and the inverses form a smooth family by Lemma 13.5.1. □

13.6 The index isomorphism and Bott periodicity, first form

We can now assemble the pieces. Quantizing a loop gives a family of Toeplitz operators, and its families index is a class in $K^0(X)$. The main theorem says that this is a bijection between loops (up to stable pointed homotopy) and $K^0(X)$. Injectivity comes from the linearization and Lemma 13.3.6; surjectivity from an explicit inverse, the Bott map, which turns a pair of bundles into a normal form loop.

For a pointed loop $a\colon X\times\mathbb{S}\to\mathrm{GL}(N,\mathbb{C})$, $T_a(x)=Sa(x,\cdot)S$ is a smooth elliptic family of Toeplitz operators with symbol a; set

$$\operatorname{ind}(a)=\operatorname{ind}(T_a)\in K^0(X).$$

By Proposition 13.5.5(i) any elliptic family with symbol a has the same index. By Proposition 13.5.5(i),(ii), $\operatorname{ind}(a)$ depends only on the stable pointed homotopy class of a, so by Lemma 13.2.5 it defines a map

$$\operatorname{ind}\colon K^{-2}(X) \longrightarrow K^0(X),$$

which is a homomorphism since $[a]+[b] = [\operatorname{diag}(a,b)]$ and $T_{\operatorname{diag}(a,b)} = T_a \oplus T_b$. For X a point, $\operatorname{ind}(a) = -\operatorname{wind}(a)$ by Corollary 13.4.2; in general the rank of $\operatorname{ind}(a)$ at x is $-\operatorname{wind} a(x,\cdot)$.

Proposition 13.6.1. *If a is a pointed loop over X with $\operatorname{ind}(a) = 0$ in $K^0(X)$, then a is stably pointed homotopic to* Id.

Proof. By Theorem 13.3.1 and homotopy invariance we may assume $a = \pi_- z^{-1} + \pi_0 + \pi_+ z$, and then $[\operatorname{Ran}\pi_-] - [\operatorname{Ran}\pi_+] = \operatorname{ind}(a) = 0$ by Proposition 13.5.5(v), i.e. $\operatorname{Ran}\pi_- \oplus H \cong \operatorname{Ran}\pi_+ \oplus H$ for some bundle H. By Lemma 13.2.1 we may take $H = \operatorname{Ran} q$ for a smooth family of idempotents q on some $\mathbb{C}^K$. The loop

$$\beta_q = \operatorname{diag}\left(qz^{-1} + (\operatorname{Id} - q),\ (\operatorname{Id} - q) + qz\right)$$

on $\mathbb{C}^{2K}$ is in normal form with $\operatorname{Ran}\pi_-^{\beta} = \operatorname{Ran} q \oplus 0 \cong 0 \oplus \operatorname{Ran} q = \operatorname{Ran}\pi_+^{\beta}$, so β_q is pointed homotopic to Id by Lemma 13.3.6. Hence $a \sim \operatorname{diag}(a, \beta_q)$, and $\operatorname{diag}(a,\beta_q)$ is in normal form with idempotents $\pi_- \oplus q \oplus 0$, $\pi_0 \oplus (\operatorname{Id} - q) \oplus (\operatorname{Id} - q)$, $\pi_+ \oplus 0 \oplus q$, whose ranges satisfy $\operatorname{Ran}\pi_- \oplus H \cong \operatorname{Ran}\pi_+ \oplus H$. By Lemma 13.3.6 again, $\operatorname{diag}(a,\beta_q)$ is pointed homotopic to Id. □

Theorem 13.6.2 (Bott periodicity, first form)**.** *For every compact manifold X the index map* $\operatorname{ind}\colon K^{-2}(X) \to K^0(X)$ *is an isomorphism of abelian groups, natural with respect to pull-back. Its inverse, the* Bott map β*, is given as follows: if $E_\pm \cong \operatorname{Ran} p_\pm$ with $p_\pm\colon X \to M(N_\pm, \mathbb{C})$ smooth families of projections, then*

$$\beta([E_+] - [E_-]) = \left[(p_+ z^{-1} + (\operatorname{Id} - p_+)) \oplus ((\operatorname{Id} - p_-) + p_- z)\right]. \tag{13.6.1}$$

Proof. Injectivity is Proposition 13.6.1, and naturality follows from Proposition 13.5.5(iii). The loop in (13.6.1) is in normal form with $\pi_- = p_+ \oplus 0$ and $\pi_+ = 0 \oplus p_-$, so by Proposition 13.5.5(v) its index is $[\operatorname{Ran} p_+] - [\operatorname{Ran} p_-] = [E_+] - [E_-]$. As every element of $K^0(X)$ is of the form $[E_+] - [E_-]$ and every bundle is the range of a family of projections (Lemma 13.2.1), ind is surjective, and β is its inverse. □

Concretely, for the operator with symbol (13.6.1), namely $(p_+ L + \operatorname{Id} - p_+) \oplus (\operatorname{Id} - p_- + p_- U)$, the null space is the bundle E_+ of constant sections and the null space of the adjoint is E_-. The Toeplitz operator L, of index $1 = [\mathbb{C}] \in K^0(\mathrm{pt})$, is the *Bott element*: $\beta[\mathbb{C}] = [z^{-1}]$. Figure 13.3 summarizes the construction.

Discussion. Why "periodicity"? Each step from K^0 to K^{-1} to K^{-2} adds one real variable: by clutching, $K^{-1}(X) \cong K_{\mathrm{c}}^0(\mathbb{R} \times X)$ (Lemma 13.9.4), and $K^{-2}(X) \cong K_{\mathrm{c}}^{-1}(\mathbb{R} \times X)$ (Lemma 13.9.3). Theorem 13.6.2 says that after two such steps we are back at $K^0(X)$. So the K-groups repeat with period two; in terms of homotopy groups, $\pi_{k+2}(G^{-\infty}) \cong \pi_k(G^{-\infty})$, which is proved in Corollary 13.9.9.

Remark 13.6.3. The signs matter here: the loop $\pi_- z^{-1} + \pi_0 + \pi_+ z$ with $\operatorname{Ran}\pi_\pm \cong E_\pm$ represents $-\beta([E_+] - [E_-])$, because $\operatorname{ind} U = -1$.

Corollary 13.6.4. *If $x_0 \in X$ is a base point and $\tilde{K}^{-2}(X)$ denotes the kernel of restriction $K^{-2}(X) \to K^{-2}(\{x_0\})$, then* ind *restricts to an isomorphism $\tilde{K}^{-2}(X) \cong \tilde{K}^0(X)$. Moreover $K^{-2}(\mathrm{pt}) \cong \mathbb{Z}$, i.e. pointed loops in $G^{-\infty}(Y;E)$ (equivalently in $\mathrm{GL}(N,\mathbb{C})$, stably) are classified up to pointed homotopy by their winding number, and every winding number occurs.*

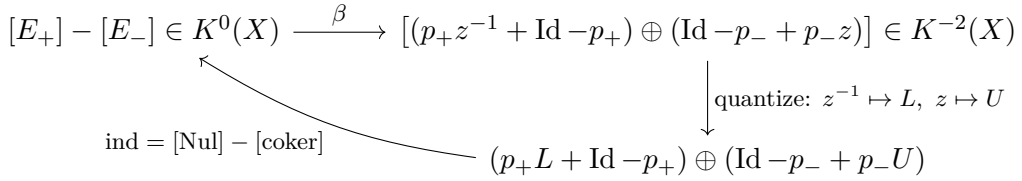


Figure 13.3: The Bott map β and the index. A pair of bundles $E_\pm = \operatorname{Ran} p_\pm$ is turned into a loop in normal form, with z^{-1} on E_+ and z on E_-. The corresponding Toeplitz family has null bundle E_+ (the constant sections killed by L) and cokernel E_- (the constants missed by U), so its families index is $[E_+]-[E_-]$.

Proof. Restriction and pull-back along $X \to \{x_0\}$ commute with ind, and ind is an isomorphism for X and for $\{x_0\}$, so it maps the kernel of one restriction bijectively onto the kernel of the other. The second statement is the case $X = \mathrm{pt}$, where $\mathrm{ind}(a) = -\operatorname{wind}(a)$. □

13.7 The isotropic index map

Toeplitz operators are one way to 'quantize' loops of matrices. A second way uses the isotropic calculus on $\mathbb{R}$ of Chapter 7: the principal symbol of an isotropic operator of order 0 on $\mathbb{R}$ is a function on the circle at infinity of the phase plane $\mathbb{R}^2$, so loops are again symbols. The role of the projections P_k onto the first Fourier modes is taken by the eigenprojections of the harmonic oscillator, and the role of the shift L by the annihilation operator. In this section we carry this out and obtain a second proof of Theorem 13.6.2; the two index maps are compared in Proposition 13.7.7. Unlike the Toeplitz construction, the isotropic one extends directly to $\mathbb{R}^n$, which is exploited in Section 13.10.

13.7.1 Conventions

The phase plane is $\mathbb{R}^2$ with coordinates $w = (x, \xi)$, oriented by $dx \wedge d\xi$. We identify $\mathbb{S}$ with the unit circle by $\theta \mapsto (\cos\theta, \sin\theta)$, so that $\mathbb{S}$ is traversed counterclockwise, and on it

$$z = e^{i\theta} = \frac{x + i\xi}{|w|};$$

the base point $\theta = 0$ is $(x, \xi) = (1, 0)$. For $A \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R}; \mathbb{C}^N)$, a polyhomogeneous isotropic operator of order 0, the principal symbol is $\sigma_0(A) \in \mathcal{C}^\infty(\mathbb{S}; M(N, \mathbb{C}))$, and A is elliptic if $\sigma_0(A)$ is invertible. Conversely, for $a \in \mathcal{C}^\infty(Z \times \mathbb{S}; M(N, \mathbb{C}))$ the family

$$\mathrm{Op}_\chi(a)(y) = \mathrm{Op}\big(\chi(|w|)\, a(y, w/|w|)\big), \qquad y \in Z, \tag{13.7.1}$$

where $\chi \in \mathcal{C}^\infty(\mathbb{R})$, $\chi = 0$ near 0 and $\chi = 1$ on $[1, \infty)$, is a smooth family in $\Psi^0_{\mathrm{iso,ph}}(\mathbb{R}; \mathbb{C}^N)$ with principal symbol $a(y, \cdot)$, the symbol being homogeneous of degree 0 outside the unit ball. We write u_k, $k \in \mathbb{N}_0$, for the Hermite functions on $\mathbb{R}$ (the normalized eigenfunctions of $H = D_x^2 + x^2$, $Hu_k = (2k+1)u_k$) and Π_k for the orthogonal projection onto $\operatorname{span}\{u_0, \dots, u_k\} \otimes \mathbb{C}^N$.

13.7.2 The dictionary

The constructions of Section 13.5 used the following properties of the Toeplitz algebra: the existence of parametrices depending smoothly on parameters, the continuity of the inclusion into the bounded operators on L^2_+, elliptic regularity, the fact that RTR' is a smoothing operator, continuously in T, when R, R' are smoothing and T is bounded, and (B6). Each of these holds for the isotropic calculus by the results of Chapter 7:

(I1) An elliptic $A \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R};\mathbb{C}^N)$ has a parametrix $B \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R};\mathbb{C}^N)$, $AB-\mathrm{Id}$, $BA-\mathrm{Id} \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R};\mathbb{C}^N)$; the construction is symbolic (inversion of the symbol and asymptotic summation), so for a smooth family A the parametrices may be chosen to form a smooth family.

(I2) An elliptic A is Fredholm on $L^2(\mathbb{R};\mathbb{C}^N)$ and on $\mathcal{S}(\mathbb{R};\mathbb{C}^N)$; the null spaces of A and A^* lie in $\mathcal{S}$, they are the same on $\mathcal{S}$, L^2 and $\mathcal{S}'$, and the range on each of these spaces is the annihilator of $\operatorname{Nul} A^*$. The inclusion $\Psi^0_{\mathrm{iso,ph}}(\mathbb{R};\mathbb{C}^N) \to \mathcal{B}(L^2)$ is continuous.

(I3) If $R, R' \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R};\mathbb{C}^N)$ and T is bounded on L^2 then $RTR' \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R};\mathbb{C}^N)$, with seminorms bounded by seminorms of R, R' times $\|T\|$.

(I4) For a smooth family K in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R};\mathbb{C}^N)$ parametrized by a compact manifold, $K\Pi_k \to K$ and $\Pi_k K \to K$ in $\Psi^{-\infty}_{\mathrm{iso}}$, uniformly, as $k \to \infty$ (the Hermite expansion of a Schwartz kernel converges in $\mathcal{S}(\mathbb{R}^2)$). Moreover $\operatorname{Ran}\Pi_{k+1} = \operatorname{Ran}\Pi_k \oplus u_{k+1}\mathbb{C}^N$.

The dictionary is therefore

$$\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N) \rightsquigarrow \mathcal{S}(\mathbb{R};\mathbb{C}^N), \quad L^2_+(\mathbb{S};\mathbb{C}^N) \rightsquigarrow L^2(\mathbb{R};\mathbb{C}^N), \quad \Psi^0_{\mathcal{T}} \rightsquigarrow \Psi^0_{\mathrm{iso,ph}},$$
$$\Psi^{-\infty}_{\mathcal{T}} \rightsquigarrow \Psi^{-\infty}_{\mathrm{iso}}, \quad P_k \rightsquigarrow \Pi_k, \quad e^{ij\theta} \rightsquigarrow u_j,$$

with T_a replaced by $\operatorname{Op}_\chi(a)$. Making these replacements in Lemmas 13.5.1 and 13.5.3 and Proposition 13.5.6 and in Proposition 13.5.5(i)–(iv) we obtain, with the same proofs:

Proposition 13.7.1. *Let Z be a compact manifold and $A \in \mathcal{C}^\infty(Z;\Psi^0_{\mathrm{iso,ph}}(\mathbb{R};\mathbb{C}^N))$ a smooth family of elliptic operators.*

(i) If each $A(z)$ is invertible then $A^{-1} \in \mathcal{C}^\infty(Z;\Psi^0_{\mathrm{iso,ph}}(\mathbb{R};\mathbb{C}^N))$.

(ii) For k large, $\operatorname{Nul}(A(z)(\mathrm{Id}-\Pi_k)) = \operatorname{Ran}\Pi_k$ for all z; the orthocomplements $F_k(z) \subset \mathcal{S}(\mathbb{R};\mathbb{C}^N)$ of the ranges of $A(z)(\mathrm{Id}-\Pi_k)$ form a smooth vector bundle, and

$$\operatorname{ind}_{\mathrm{iso}}(A) = [\operatorname{Ran}\Pi_k] - [F_k] = [\mathbb{C}^{(k+1)N}] - [F_k] \in K^0(Z)$$

is independent of k; F_k may be replaced by any smooth complement of the range of $A(\mathrm{Id}-\Pi_k)$ in $\mathcal{S}(\mathbb{R};\mathbb{C}^N)$. For Z a point, $\operatorname{ind}_{\mathrm{iso}}(A) = \operatorname{ind} A$.

(iii) $\operatorname{ind}_{\mathrm{iso}}$ is invariant under homotopies of elliptic families, depends only on the homotopy class of $\sigma_0(A)$ in $\mathcal{C}^\infty(Z\times\mathbb{S};\mathrm{GL}(N,\mathbb{C}))$, is additive under direct sums, unchanged by stabilization $A \mapsto A\oplus\mathrm{Id}$, natural under pull-back, vanishes for invertible families, and $\operatorname{ind}_{\mathrm{iso}}(bAc) = \operatorname{ind}_{\mathrm{iso}}(A)$ for $b, c \in \mathcal{C}^\infty(Z;\mathrm{GL}(N,\mathbb{C}))$.

(iv) $\operatorname{ind}_{\mathrm{iso}}(A) = 0$ if and only if $A\oplus\mathrm{Id}_L + B$ is invertible for some L and some smooth family B in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R};\mathbb{C}^{N+L})$.

In (iii), two families with homotopic symbols are joined by the linear homotopy from A to $\operatorname{Op}_\chi(\sigma_0(A))$, which has constant symbol, followed by $\operatorname{Op}_\chi(a_t)$ for a homotopy a_t of symbols. The following lemma, valid in both the Toeplitz and the isotropic setting, is the most convenient way to compute families indices.

Lemma 13.7.2. *Let A be a smooth elliptic family over a compact manifold Z, either of Toeplitz operators on $V = \mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$ or of isotropic operators on $V = \mathcal{S}(\mathbb{R};\mathbb{C}^N)$. Suppose that the null spaces $\operatorname{Nul} A(z)$ form a smooth vector bundle $\mathcal{N}$ (local frames given by smooth maps into V) and that W is a smooth vector bundle with $V = A(z)V \oplus W(z)$ for all z. Then $\operatorname{ind}(A) = [\mathcal{N}] - [W]$.*

Proof. Let $P = P_k$, resp. $P = \Pi_k$, with k admissible, and $A' = A(\mathrm{Id} - P)$. If $n \in \mathcal{N}(z)$ and $Pn = 0$, then $A'n = An = 0$, so $n \in \operatorname{Nul} A' = \operatorname{Ran} P$ and $n = 0$. Hence P maps $\mathcal{N}$ isomorphically onto a smooth subbundle $P\mathcal{N}$ of the trivial bundle $\operatorname{Ran} P$; let Y be its orthocomplement in $\operatorname{Ran} P$, again a smooth bundle. We claim that $A(V) = A(Y) \oplus A'(V)$ with A injective on Y. Indeed, if $y \in Y$ and $Ay = A'v$, then $y - (\mathrm{Id} - P)v \in \mathcal{N}$, and applying P, $y \in P\mathcal{N} \cap Y = \{0\}$. Conversely, since $V = \operatorname{Ran} P + \operatorname{Ran}(\mathrm{Id} - P)$, every element of $A(V)$ is $Ar + A'v$ with $r \in \operatorname{Ran} P$; writing $r = y + Pn$ with $y \in Y$, $n \in \mathcal{N}$, we get $Ar = Ay - A'n$. Thus $W \oplus A(Y)$ is a complement for (A, k); it is smooth since A maps local frames of Y to local frames of $A(Y)$. By Lemma 13.5.3 (or Proposition 13.7.1) $\operatorname{ind}(A) = [\operatorname{Ran} P] - [W] - [Y] = [P\mathcal{N}] - [W] = [\mathcal{N}] - [W]$. □

13.7.3 The annihilation operator

The operator $x + \partial_x = i(D_x - ix)$ is i times the annihilation operator of the harmonic oscillator. Its symbol is $x + i\xi$, of order one; we normalize it to order 0.

Lemma 13.7.3. *Let $B = \frac{1}{2}(\mathrm{Op}(\langle w\rangle^{-1}) + \mathrm{Op}(\langle w\rangle^{-1})^*)$ and $\mathfrak{a} = (x + \partial_x)B$. Then $\mathfrak{a} \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R})$ is elliptic, $\sigma_0(\mathfrak{a}) = z$, and $\operatorname{ind}\mathfrak{a} = 1$. Moreover there is a finite rank $K \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R})$ such that $\mathfrak{a}' = \mathfrak{a} + K$ is surjective on $\mathcal{S}(\mathbb{R})$ and on $L^2(\mathbb{R})$, with $\operatorname{Nul}\mathfrak{a}' = \mathbb{C}\phi_0$ for some $\phi_0 \in \mathcal{S}(\mathbb{R})$ with $\|\phi_0\| = 1$; then $\mathfrak{a}'^*$ is injective, with range $\{f;\ (f, \phi_0) = 0\}$.*

Proof. $B \in \Psi^{-1}_{\mathrm{iso,ph}}(\mathbb{R})$ is formally self-adjoint with principal symbol $|w|^{-1}$, so it is elliptic of order -1, and $\mathfrak{a}$ is polyhomogeneous of order 0 with principal symbol $(x + i\xi)/|w| = z$. By (I2) (which holds for elliptic operators of any order on $\mathcal{S}$), B is Fredholm on $\mathcal{S}(\mathbb{R})$ with index $\dim\operatorname{Nul} B - \dim\operatorname{Nul} B^* = 0$. The operator $x + \partial_x$ is elliptic of order one; $(x + \partial_x)u = 0$ means $\partial_x(e^{x^2/2}u) = 0$, so $\operatorname{Nul}(x + \partial_x) = \mathbb{C}e^{-x^2/2}$, while the adjoint $x - \partial_x$ has no tempered null space ($e^{x^2/2}$ is not tempered), so $x + \partial_x$ is surjective on $\mathcal{S}$ and its index is 1. For linear maps with finite dimensional null spaces and cokernels the index of a composite is the sum of the indices, so $\operatorname{ind}\mathfrak{a} = 1$ on $\mathcal{S}$, and also on L^2 by (I2).

Let $N_0 = \operatorname{Nul}\mathfrak{a}$, $N_1 = \operatorname{Nul}\mathfrak{a}^*$, both in $\mathcal{S}$, with $\dim N_0 = \dim N_1 + 1$. Choose a unit vector $\phi_0 \in N_0$, let $N_0' = N_0 \ominus \mathbb{C}\phi_0$, and let K map N_0' isomorphically onto N_1 and vanish on $(N_0')^\perp$; K has finite rank with range and co-range in $\mathcal{S}$, so $K \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R})$. If $(\mathfrak{a}+K)u = 0$, write $u = u' + u''$ with $u' \in N_0'$, $u'' \perp N_0'$; then $\mathfrak{a}u'' \in \operatorname{Ran}\mathfrak{a} \perp N_1 \ni Ku'$, so both vanish, $u' = 0$ and $u'' \in N_0 \ominus N_0' = \mathbb{C}\phi_0$. The range of $\mathfrak{a} + K$ contains $\operatorname{Ran}\mathfrak{a}$ and N_1, hence everything. The statements about $\mathfrak{a}'^*$ follow from (I2). □

13.7.4 Normal forms and Bott periodicity

Proposition 13.7.4. *Let $\pi_-, \pi_0, \pi_+ \colon Z \to M(M, \mathbb{C})$ be smooth commuting idempotents with $\pi_- + \pi_0 + \pi_+ = \mathrm{Id}$, $V_\bullet = \operatorname{Ran}\pi_\bullet$. The family*

$$Q = \pi_-\mathfrak{a}'^* + \pi_0 + \pi_+\mathfrak{a}', \qquad \sigma_0(Q) = \pi_- z^{-1} + \pi_0 + \pi_+ z,$$

has $\operatorname{Nul} Q = \phi_0 \otimes V_+$, the bundle $\phi_0 \otimes V_-$ is a complement of its range, and

$$\operatorname{ind}_{\mathrm{iso}}(Q) = [V_+] - [V_-] \in K^0(Z).$$

Proof. The matrices act on the factor $\mathbb{C}^M$ of $\mathcal{S}(\mathbb{R}; \mathbb{C}^M) = \mathcal{S}(\mathbb{R}) \otimes \mathbb{C}^M$ and commute with $\mathfrak{a}'$, $\mathfrak{a}'^*$, so with respect to $\mathcal{S}(\mathbb{R}; \mathbb{C}^M) = \mathcal{S}(\mathbb{R}) \otimes V_- \oplus \mathcal{S}(\mathbb{R}) \otimes V_0 \oplus \mathcal{S}(\mathbb{R}) \otimes V_+$ the operator Q is $\mathfrak{a}'^* \otimes \mathrm{Id} \oplus \mathrm{Id} \oplus \mathfrak{a}' \otimes \mathrm{Id}$. By Lemma 13.7.3 the null space is $\phi_0 \otimes V_+$ and the range is $\{u \in \mathcal{S}(\mathbb{R}) \otimes V_-;\ (u, \phi_0) = 0\} \oplus \mathcal{S}(\mathbb{R}) \otimes V_0 \oplus \mathcal{S}(\mathbb{R}) \otimes V_+$, where $(u, \phi_0) \in V_-$ denotes the pairing in the $\mathbb{R}$-variable; so $\phi_0 \otimes V_-$ is a complement. Both are smooth bundles, with local frames $\phi_0 \otimes \pi_\pm e_j$ (Lemma 13.2.1). Apply Lemma 13.7.2. □

Corollary 13.7.5 (Index formula for isotropic operators on $\mathbb{R}$)**.** *If $A \in \Psi^0_{\mathrm{iso,ph}}(\mathbb{R};\mathbb{C}^N)$ is elliptic, then*

$$\operatorname{ind} A = \operatorname{wind}\sigma_0(A) = \frac{1}{2\pi i}\int_0^{2\pi} \operatorname{Tr}\Big(\sigma_0(A)^{-1}\frac{d\sigma_0(A)}{d\theta}\Big)d\theta,$$

the circle at infinity being traversed counterclockwise in the (x,ξ)-plane.

Proof. As in the proof of Corollary 13.4.2 we may assume $\sigma_0(A)$ pointed and, by Theorem 13.3.1 and Proposition 13.7.1(iii), in normal form $\pi_- z^{-1} + \pi_0 + \pi_+ z$ with constant idempotents. Then $\operatorname{ind} A = \operatorname{rank}\pi_+ - \operatorname{rank}\pi_-$ by Proposition 13.7.4, and this is the winding number of $\det(\pi_- z^{-1} + \pi_0 + \pi_+ z) = z^{\operatorname{rank}\pi_+ - \operatorname{rank}\pi_-}$. □

For example $x + \partial_x$, the generator of the index, has symbol of winding number one, while the creation operator $x - \partial_x$ has symbol $x - i\xi$ of winding number -1, in accordance with $\operatorname{Nul}(x-\partial_x) = 0$ and $\operatorname{Nul}(x+\partial_x) = \mathbb{C}e^{-x^2/2}$.

For a pointed loop $a\colon X \times \mathbb{S} \to \mathrm{GL}(N,\mathbb{C})$, X compact, set $\operatorname{ind}_{\mathrm{iso}}(a) = \operatorname{ind}_{\mathrm{iso}}(\mathrm{Op}_\chi(a)) \in K^0(X)$. By Proposition 13.7.1(iii) and Lemma 13.2.5 this defines a homomorphism

$$\operatorname{ind}_{\mathrm{iso}}\colon K^{-2}(X) \longrightarrow K^0(X),$$

natural under pull-back.

Theorem 13.7.6 (Bott periodicity, isotropic form)**.** *For every compact manifold X the isotropic index map $\operatorname{ind}_{\mathrm{iso}}\colon K^{-2}(X) \to K^0(X)$ is an isomorphism. Its inverse is*

$$\beta_{\mathrm{iso}}([E_+]-[E_-]) = \big[(p_+ z + (\mathrm{Id} - p_+)) \oplus ((\mathrm{Id} - p_-) + p_- z^{-1})\big], \tag{13.7.2}$$

where $E_\pm \cong \operatorname{Ran} p_\pm$ for smooth families of projections $p_\pm$. The element $[z] \in K^{-2}(\mathrm{pt})$, the class of the symbol of $\mathfrak{a}$, corresponds to $[\mathbb{C}] \in K^0(\mathrm{pt})$.

Proof. The loop in (13.7.2) is in normal form with $\pi_+ = p_+ \oplus 0$ and $\pi_- = 0 \oplus p_-$, so by Proposition 13.7.4 its isotropic index is $[E_+] - [E_-]$; thus $\operatorname{ind}_{\mathrm{iso}} \circ \beta_{\mathrm{iso}} = \mathrm{Id}$ and $\operatorname{ind}_{\mathrm{iso}}$ is surjective. For injectivity let $\operatorname{ind}_{\mathrm{iso}}(a) = 0$. By Theorem 13.3.1 we may assume $a = \pi_- z^{-1} + \pi_0 + \pi_+ z$, and then $[V_+] = [V_-]$ by Proposition 13.7.4. The argument of Proposition 13.6.1, which only involves loops (and Lemma 13.3.6, which is symmetric in $\pi_\pm$), shows that a is stably pointed homotopic to Id. □

The proof shows that the only analytic input needed for Bott periodicity, besides the linearization of loops, is the computation of the index of one operator of index one, here $\mathfrak{a}$, and of its twisted versions in Proposition 13.7.4.

Proposition 13.7.7. *Let $\rho\colon \mathbb{S} \to \mathbb{S}$, $\theta \mapsto -\theta$. For every pointed loop a over a compact manifold X,*

$$\operatorname{ind}_{\mathrm{iso}}(a) = \operatorname{ind}(a \circ \rho) = -\operatorname{ind}(a),$$

where ind *is the Toeplitz index map of Section 13.6. In other words, after the orientation reversing identification of the circle at infinity of the phase plane with the circle $\mathbb{S}_+$ of the Toeplitz calculus, the isotropic and the Toeplitz index maps coincide.*

Proof. All three expressions define homomorphisms $K^{-2}(X) \to K^0(X)$, so by Theorem 13.3.1 it suffices to evaluate them on $a = \pi_- z^{-1} + \pi_0 + \pi_+ z$. Then $a \circ \rho = \pi_- z + \pi_0 + \pi_+ z^{-1}$, and by Proposition 13.5.5(v), Proposition 13.7.4, $\operatorname{ind}(a\circ\rho) = [V_+] - [V_-] = \operatorname{ind}_{\mathrm{iso}}(a) = -\operatorname{ind}(a)$. □

Remark 13.7.8. The comparison can be made at the level of operators. The unitary map $L^2(\mathbb{R}) \to L^2_+(\mathbb{S})$, $u_k \mapsto e^{ik\theta}$, carries H to $2D_\theta + 1$ and $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R})$ onto $\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S})$ (both correspond to the matrices $(c_{jk})_{j,k\geq 0}$ with rapidly decreasing entries), and $\mathfrak{a}'$ to an operator which, like L, is surjective with one-dimensional null space. The reversal of orientation reflects the fact that the annihilation operator lowers the eigenvalue of H, while L lowers the Fourier mode, although their symbols are z and z^{-1}.

Remark 13.7.9 (The rotation argument). Injectivity of the index map can also be proved without the linearization of loops, by an argument due to Atiyah. In the present setting it runs as follows. Let a be a loop with $\mathrm{ind}_{\mathrm{iso}}(a) = 0$; by Proposition 13.7.1(iv) there is (after stabilization) an *invertible* family $A(x)$ with symbol $a(x,\cdot)$. Clutch a to a family of projections Π_a on $\mathbb{R}^2 \times X$, given by a matrix whose entries are affine in a and a^{-1} (as in Lemma 13.10.7 below). Using a loop β of winding number one in a third variable r, the family $e = \beta(r)\Pi_a + \mathrm{Id} - \Pi_a$ of loops in r has isotropic index the class of Π_a, i.e. the clutching class of a. Rotating the plane of r and the variable of a by a right angle does not change the index, but after the rotation the loop variable is the variable of a, and replacing a, a^{-1} by A, A^{-1} in the entries of Π_a gives an invertible quantization; so the index, hence the clutching class of a, vanishes. Making this precise requires the families index over the non-compact parameter space $\mathbb{R}^2 \times X$ with compact supports, and careful normalization at infinity; we do not pursue it.

13.8 The stabilized Toeplitz group and the classifying sequence

We now pass from matrices to operators with values in the smoothing operators on Y, to obtain groups which contain all the $\mathrm{GL}(N,\mathbb{C})$ at once.

The aim is a short exact sequence of groups $G^{-\infty}_{\mathcal{T}} \to G^0_{\mathcal{T}} \to G^0_{\mathcal{T}}/G^{-\infty}_{\mathcal{T}}$ in which the middle group is weakly contractible. In topology, a contractible space EG with a free action of a group G has quotient BG with $\pi_k(BG) = \pi_{k-1}(G)$, a "delooping" of G. Here the subgroup classifies odd K-theory, so the quotient, which is a space of symbols, classifies even K-theory (Figure 13.4). This is the analytic form of the passage from K^{-1} to K^0.

13.8.1 Toeplitz operators with values in smoothing operators

Let Y, E be as in Section 13.1. On $\Psi^0(\mathbb{S})$ we use the Fréchet topology of Chapters 4 and 5: an element is a conormal distribution at the diagonal, i.e. the sum of a smooth kernel and, near the diagonal, the inverse Fourier transform of a classical symbol, and the seminorms are those of uniform convergence of all derivatives of these two pieces. Composition is continuous, and so is the inclusion into the bounded operators on L^2.

Definition 13.8.1. $\Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S}, Y; E)$ is the space of smooth sections $A \in \mathcal{C}^\infty(Y^2; \Psi^0_{\mathcal{T}}(\mathbb{S}) \otimes \mathrm{Hom}(E) \otimes \Omega_R)$, i.e. smooth maps $(y,y') \mapsto A(y,y')$ with values in Toeplitz operators tensored with $\mathrm{Hom}(E_{y'}, E_y)$ and a density in y'. It acts on $\mathcal{C}^\infty(\mathbb{S}\times Y; E)$, and on $\mathcal{C}^\infty_+(\mathbb{S}\times Y; E) = S\mathcal{C}^\infty(\mathbb{S}\times Y; E)$, by $(Au)(\cdot, y) = \int_Y A(y,y')u(\cdot,y')$, and is an algebra with $(AB)(y,y') = \int_Y A(y,y'') \circ B(y'',y')$. The symbol

$$\sigma_0 \colon \Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S}, Y; E) \longrightarrow \mathcal{C}^\infty(\mathbb{S}; \Psi^{-\infty}(Y; E)), \quad \sigma_0(A)(\theta)(y,y') = \sigma_0(A(y,y'))(\theta),$$

is multiplicative. The elements with $A(y,y') \in \Psi^{-\infty}_{\mathcal{T}}(\mathbb{S})$ form the ideal $\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S}\times Y; E) = S\Psi^{-\infty}(\mathbb{S}\times Y; E)S$, the Toeplitz smoothing operators on $\mathbb{S}\times Y$.

If $V \subset \mathcal{C}^\infty(Y;E)$ has orthonormal basis $\phi_1,\dots,\phi_N$, then $(A_{lm}) \mapsto \sum_{l,m} A_{lm}\,\phi_l(y)\phi_m(y')^*$ embeds $\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ as a subalgebra (without unit); correspondingly an operator A on $\mathcal{C}^\infty_+(\mathbb{S};\mathbb{C}^N)$ is identified with $(\mathrm{Id}-\Pi_V)+A$ on $\mathcal{C}^\infty_+(\mathbb{S}\times Y;E)$. Different choices of V give embeddings which are conjugate by unitaries joined to the identity, and enlarging V corresponds to stabilization $A \mapsto A \oplus \mathrm{Id}$.

Definition 13.8.2. The *stabilized normalized Toeplitz group* is

$$G^0_{\mathcal{T}}(\mathbb{S},Y;E) = \{\mathrm{Id}+A;\ A \in \Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S},Y;E),\ \sigma_0(A)(1)=0,\\ (\mathrm{Id}+A)^{-1} = \mathrm{Id}+B \text{ with } B \in \Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S},Y;E)\},$$

and $G^{-\infty}_{\mathcal{T}}(\mathbb{S}\times Y;E)$ is its subgroup of elements with $A \in \Psi^{-\infty}_{\mathcal{T}}(\mathbb{S}\times Y;E)$. Similarly $G^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ denotes the group of invertible $A \in \Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ with $\sigma_0(A)(1)=\mathrm{Id}$, and $G^{-\infty}_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ the subgroup with $A-\mathrm{Id} \in \Psi^{-\infty}_{\mathcal{T}}$.

Elements of $G^0_{\mathcal{T}}$ are elliptic: $\mathrm{Id}+\sigma_0(A)(\theta)$ has inverse $\mathrm{Id}+\sigma_0(B)(\theta)$, so $\mathrm{Id}+\sigma_0(A) \in \Omega G^{-\infty}(Y;E)$. The normalization $\sigma_0(A)(1)=0$ removes the evaluation map $G^0_{\mathcal{T}} \to G^{-\infty}(Y;E)$ at $1 \in \mathbb{S}$, which would otherwise carry odd K-theory. A map $f\colon X \to G^0_{\mathcal{T}}(\mathbb{S},Y;E)$ is smooth if $f-\mathrm{Id}$ is smooth into the Fréchet space $\Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S},Y;E)$; by Lemma 13.8.3 below, f^{-1} is then smooth too, and products of smooth maps are smooth.

Lemma 13.8.3. *Let $A \in \mathcal{C}^\infty(Z;\Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S},Y;E))$ be a smooth family such that, for all $z \in Z$ and $\theta \in \mathbb{S}$, $\mathrm{Id}+\sigma_0(A(z))(\theta) \in G^{-\infty}(Y;E)$ and $\mathrm{Id}+A(z)$ is invertible on $L^2_+(\mathbb{S}\times Y;E)$. Then $(\mathrm{Id}+A)^{-1} = \mathrm{Id}+B$ with $B \in \mathcal{C}^\infty(Z;\Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S},Y;E))$.*

Proof. The proof of Lemma 13.5.1 applies verbatim. The parametrix construction goes through for operators with values in $\Psi^{-\infty}(Y;E)$; the symbol of a parametrix is $(\mathrm{Id}+\sigma_0(A))^{-1}-\mathrm{Id} \in \mathcal{C}^\infty(\mathbb{S};\Psi^{-\infty}(Y;E))$, inversion in $G^{-\infty}(Y;E)$ being smooth, and the lower order corrections are obtained by the same asymptotic summation. This gives $\mathrm{Id}+B_0$ with errors R, R' in $\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S}\times Y;E)$, and then

$$V = (\mathrm{Id}+A)^{-1} = (\mathrm{Id}+B_0) + R(\mathrm{Id}+B_0) + RVR'.$$

□

Lemma 13.8.4. *Let $A \in \mathcal{C}^\infty(X;\Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S},Y;E))$ with $\mathrm{Id}+A(x) \in G^0_{\mathcal{T}}$ for all x, X compact. If $A' \in \mathcal{C}^\infty(X;\Psi^{0,-\infty}_{\mathcal{T}})$ satisfies $\sigma_0(A')(1)=0$ and is sufficiently close to A uniformly on X (in a suitable seminorm), then $\mathrm{Id}+A'(x) \in G^0_{\mathcal{T}}$ for all x. The same holds for $G^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ and for the groups $G^{-\infty}_{\mathcal{T}}$.*

Proof. The symbols $\mathrm{Id}+\sigma_0(A')(\theta)$ are uniformly close to the compact set of values of $\mathrm{Id}+\sigma_0(A)$ in the open set $G^{-\infty}(Y;E)$, so $\mathrm{Id}+A'$ is elliptic; the operators are close in L^2 operator norm to invertible ones, so are invertible on $L^2_+(\mathbb{S}\times Y;E)$. By Lemma 13.8.3 (or Lemma 13.5.1) the inverses lie in the algebra. □

13.8.2 Weak contractibility

Proposition 13.8.5. *Let $g\colon X \to G^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ be smooth. Then for some $M \geq N$, $g\oplus \mathrm{Id}_{M-N}$ is smoothly homotopic in $G^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^M)$ to a map into $G^{-\infty}_{\mathcal{T}}(\mathbb{S};\mathbb{C}^M)$.*

Proof. The symbol $a=\sigma_0(g)$ is a pointed loop, and $\mathrm{ind}(a)=\mathrm{ind}(g)=0$ since g is invertible (Proposition 13.5.5(iv)). By Proposition 13.6.1 there is a pointed homotopy a_t, $t \in [0,1]$, in some $\mathrm{GL}(M,\mathbb{C})$ from $a\oplus\mathrm{Id}$ to Id. The family $A_t = g \oplus \mathrm{Id} + T_{a_t a_0^{-1}}$ is elliptic with symbol

a_t, and $A_0 = g \oplus \mathrm{Id}$ is invertible. By (B7) there is a smooth family B_t of Toeplitz smoothing operators with $B_0 = 0$ such that $A_t + B_t$ is invertible; its inverse is smooth by Lemma 13.5.1, and its symbol a_t is pointed. So $A_t + B_t$ is a homotopy in $G^0_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^M)$, and $A_1 + B_1$ has symbol Id, i.e. lies in $G^{-\infty}_{\mathcal{T}}$. □

Theorem 13.8.6. *The group $G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$ is weakly contractible: every smooth map $f\colon X \to G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$ from a compact manifold is smoothly homotopic in $G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$ to the constant map* Id.

Proof. Step 1: reduction to matrices. Write $f = \mathrm{Id} + A$. Let Π_j be the projections of (B2). The expansion of the smooth map $(x, y, y') \mapsto A(x)(y, y')$ with values in the Fréchet space $\Psi^0_{\mathcal{T}}(\mathbb{S})$ (tensored with $\mathrm{Hom}(E)$) shows, exactly as for smoothing operators, that $\Pi_j A \Pi_j \to A$ uniformly on X in $\Psi^{0,-\infty}_{\mathcal{T}}$. Since $\sigma_0(\Pi_j A \Pi_j)(1) = \Pi_j \sigma_0(A)(1) \Pi_j = 0$, Lemma 13.8.4 shows that $\mathrm{Id} + (1-t)A + t\Pi_j A \Pi_j$, $t \in [0,1]$, is a homotopy in $G^0_{\mathcal{T}}$ for j large. Its end point is $(\mathrm{Id} - \Pi_j) + g$, with $g\colon X \to G^0_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^N)$, $N = \dim V_j$.

Step 2: reduction to smoothing operators. By Proposition 13.8.5, after enlarging V_j (which does not change the element of $G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$), g is homotopic in $G^0_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^M)$ to a map $g' = \mathrm{Id} + K$, $K \in \mathcal{C}^\infty(X; \Psi^{-\infty}_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^M))$.

Step 3: reduction to the zero Fourier mode. By (B6), $P_k K P_k \to K$ uniformly, so by Lemma 13.8.4 $\mathrm{Id} + (1-t)K + tP_kKP_k$ is a homotopy in $G^{-\infty}_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^M)$ for k large. Its end point acts on $\operatorname{Ran} P_k$, spanned by $e^{ij\theta} e_l$, $0 \le j \le k$, $1 \le l \le M$, and is the identity on $\operatorname{Ran}(\mathrm{Id} - P_k)$. Stabilize once more, to $\mathbb{C}^{(k+1)M}$ with orthonormal basis $e_{l,j}$, $1 \le l \le M$, $0 \le j \le k$, where $e_{l,0} = e_l$. For $1 \le j \le k$ the pairs $(e^{ij\theta} e_{l,0}, e_{l,j})$, the second being a constant function, are orthonormal and span mutually orthogonal planes. Let $\mathcal{R}_\tau$ be the unitary rotating each of these planes by the angle τ and fixing their orthocomplement. Then $\mathcal{R}_\tau - \mathrm{Id}$ has finite rank and smooth kernel, so $\mathcal{R}_\tau \in G^{-\infty}_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^{(k+1)M})$, and conjugation by $\mathcal{R}_\tau$, $\tau \in [0, \pi/2]$, is a homotopy in $G^{-\infty}_{\mathcal{T}}$ ending at an operator of the form

$$g'' = \mathrm{Id} - P_0 + P_0 h P_0, \qquad h\colon X \to \mathrm{GL}((k+1)M, \mathbb{C}),$$

since $\mathcal{R}_{\pi/2}$ maps $\operatorname{Ran} P_k \otimes e_{\cdot,0}$ onto the constants $\operatorname{Ran} P_0$.

Step 4: rotating away the matrix. For $h \in \mathrm{GL}(N', \mathbb{C})$ consider on $\mathcal{C}^\infty_+(\mathbb{S}; \mathbb{C}^{N'})^2$ the two curves of Toeplitz operators, $c = \cos\tau$, $s = \sin\tau$,

$$M_\tau = \begin{pmatrix} c(\mathrm{Id} - P_0) + P_0 h P_0 & s\,hU \\ -s\,h^{-1}L & c \end{pmatrix}, \quad M'_\tau = \begin{pmatrix} c\,h(\mathrm{Id} - P_0) + P_0 h P_0 & s\,hU \\ -s\,h^{-1}L & c\,h^{-1} \end{pmatrix},$$

$\tau \in [0, \pi/2]$; they agree at $\tau = \pi/2$, $M_0 = \mathrm{diag}(g'', \mathrm{Id})$ and $M'_0 = \mathrm{diag}(h, h^{-1})$. Both are invertible: on $(u_k e^{ik\theta}, v_k e^{i(k-1)\theta})$, $k \ge 1$, they act by the matrices $\begin{pmatrix} c & sh \\ -sh^{-1} & c \end{pmatrix} = \mathrm{diag}(h, 1) R'_\tau \,\mathrm{diag}(h^{-1}, 1)$ and $\mathrm{diag}(h, h^{-1}) R'_\tau$, $R'_\tau = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}$, and they map $(u_0, 0)$ to $(hu_0, 0)$. Let $C_\tau = \sigma_0(M_\tau)(1)$, a constant invertible matrix (it commutes with S). Then $C_\tau^{-1} M_\tau$ followed by $(C'_\tau)^{-1} M'_\tau$ (run from $\tau = \pi/2$ to 0, $C'_\tau = \sigma_0(M'_\tau)(1)$) is a homotopy in $G^0_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^{2N'})$, smooth in $h = h(x)$, from $\mathrm{diag}(g'', \mathrm{Id})$ to $\mathrm{diag}(h, h^{-1})^{-1} \mathrm{diag}(h, h^{-1}) = \mathrm{Id}$. (This is the deformation of Chapter 12, normalized at $1 \in \mathbb{S}$.)

Composing the homotopies of Steps 1–4, all of which take place in $G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$ via the embeddings above, gives a homotopy from f to Id. □

Remark 13.8.7. Continuous maps from compact manifolds into $G^0_{\mathcal{T}}$ can be approximated by smooth ones, and nearby maps are homotopic (linear homotopy and Lemma 13.8.4), so all homotopy groups of $G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$ vanish. Whether $G^0_{\mathcal{T}}$ is actually contractible is discussed in the Notes.

13.8.3 The classifying sequence

To describe the quotient $G^0_{\mathcal{T}}/G^{-\infty}_{\mathcal{T}}$ we use full symbols. On the circle, with the normal fibration of the diagonal given by $\theta - \theta'$, an operator $A \in \Psi^0(\mathbb{S})$ has, modulo $\Psi^{-\infty}$, a kernel given near the diagonal by the inverse Fourier transform of a classical symbol $a(\theta, \xi)$; its Taylor series at $\xi = +\infty$ in $\rho = 1/\xi$ is the *full symbol* at $\mathbb{S}_+$, $\Phi(A) = \sum_{j\geq 0} \rho^j a_j(\theta)$, with $a_0 = \sigma_0(A)|_{\mathbb{S}_+}$. For Toeplitz operators the full symbol at $\mathbb{S}_-$ vanishes, so by Borel's lemma (asymptotic completeness, Chapter 4) there is a short exact sequence

$$\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S}) \longrightarrow \Psi^0_{\mathcal{T}}(\mathbb{S}) \xrightarrow{\Phi} \mathcal{C}^\infty(\mathbb{S})[[\rho]],$$

and by the composition formula $\Phi(AB) = \Phi(A) * \Phi(B)$ for a bilinear product $*$ on formal series whose leading term is the pointwise product. Applying Φ to the kernels $A(y, y')$ gives, for $\Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S}, Y; E)$, the exact sequence

$$\Psi^{-\infty}_{\mathcal{T}}(\mathbb{S} \times Y; E) \longrightarrow \Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S}, Y; E) \xrightarrow{\Phi} \mathcal{C}^\infty(\mathbb{S}; \Psi^{-\infty}(Y; E))[[\rho]]. \tag{13.8.1}$$

Lemma 13.8.8. *If $A \in \Psi^{0,-\infty}_{\mathcal{T}}(\mathbb{S}, Y; E)$ and $\mathrm{Id} + \sigma_0(A)$ is a loop in $G^{-\infty}(Y; E)$, then $\mathrm{Id} + A$ is Fredholm on $L^2_+(\mathbb{S} \times Y; E)$ and $\operatorname{ind}(\mathrm{Id} + A) = -\operatorname{wind}(\mathrm{Id} + \sigma_0(A))$. If this index vanishes there is $K \in \Psi^{-\infty}_{\mathcal{T}}(\mathbb{S} \times Y; E)$ of finite rank with $\mathrm{Id} + A + K$ invertible; if moreover $\sigma_0(A)(1) = 0$ then $\mathrm{Id} + A + K \in G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$.*

Proof. A parametrix (Lemma 13.8.3) inverts $\mathrm{Id} + A$ modulo compact operators, so it is Fredholm, with null space and orthocomplement of the range (the null space of the adjoint, which is of the same type) consisting of smooth sections. For the index, the family $\mathrm{Id} + (1-t)A + t\Pi_j A \Pi_j$ consists of elliptic, hence Fredholm, operators for j large and is continuous in norm, so the index is constant; the winding number of the symbol is constant as well. At $t = 1$ the operator is $\mathrm{Id} - \Pi_j$ plus an elliptic Toeplitz operator on $\mathbb{C}^{\dim V_j}$, and Corollary 13.4.2 applies. If the index vanishes, let K map an orthonormal basis of the null space to one of the orthocomplement of the range and vanish on the orthocomplement of the null space; it has finite rank and smooth kernel, $\mathrm{Id} + A + K$ is bijective on L^2_+, and Lemma 13.8.3 applies. □

Theorem 13.8.9 (Classifying sequence)**.** *For Y, E as above there is a short exact sequence of groups*

$$G^{-\infty}_{\mathcal{T}}(\mathbb{S} \times Y; E) \longrightarrow G^0_{\mathcal{T}}(\mathbb{S}, Y; E) \xrightarrow{\Phi} \mathcal{Q}, \tag{13.8.2}$$

where $\mathcal{Q}$ is the set of formal series $\sum_{j\geq 0} \rho^j b_j$ with $b_j \in \mathcal{C}^\infty(\mathbb{S}; \Psi^{-\infty}(Y; E))$ and $\mathrm{Id} + b_0 \in \Omega_0 G^{-\infty}(Y; E)$ (pointed loops of winding number zero), with the group structure induced by $*$. Moreover, for every compact manifold X:

(i) $[X; G^{-\infty}_{\mathcal{T}}(\mathbb{S} \times Y; E)] \cong K^{-1}(X)$;

(ii) $G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$ *is weakly contractible;*

(iii) if X is connected, $[X; \mathcal{Q}] \cong [X; \Omega_0 G^{-\infty}(Y; E)] \cong \tilde{K}^0(X)$, where $\mathcal{Q}$ has the product topology of the coefficients.

Thus $G^{-\infty}_{\mathcal{T}}$ classifies odd K-theory, $G^0_{\mathcal{T}}$ is weakly contractible, and $\mathcal{Q}$ classifies reduced even K-theory.

Question. Why is a weakly contractible middle group useful?

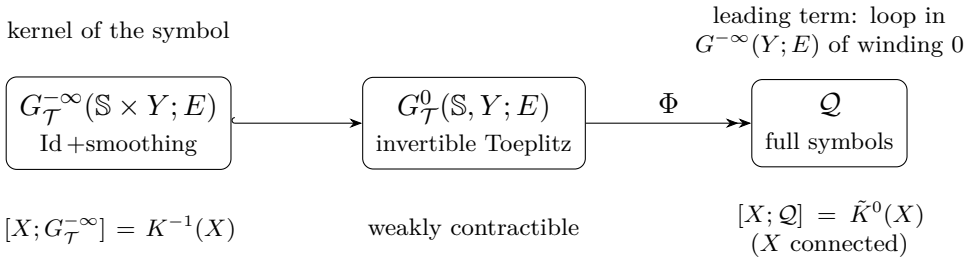


Figure 13.4: The classifying sequence of Theorem 13.8.9, a group-level version of the symbol sequence $\Psi^{-\infty} \to \Psi^0 \to$ symbols. Ellipticity and invertibility are built into the middle group, so the index obstruction of (12.5.1) has been removed by restricting the quotient to loops of winding number zero.

Answer. It converts operators into topology. A map $X \to \mathcal{Q}$ is a family of symbols; it lifts to a family of invertible operators in $G_{\mathcal{T}}^0$, and any two lifts, or any lift of a homotopy, are related through the contractible group $G_{\mathcal{T}}^0$. The ambiguity in the lift is a map into the subgroup $G_{\mathcal{T}}^{-\infty}$, which carries odd K-theory. If the sequence were a fibration this would give $\pi_k(\mathcal{Q}) \cong \pi_{k-1}(G_{\mathcal{T}}^{-\infty})$, i.e. the delooping relation between even and odd K-theory, as in the remark below. □

Proof. By (13.8.1) the kernel of Φ on $G_{\mathcal{T}}^0$ is $G_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)$ (an element of $G_{\mathcal{T}}^0$ with A smoothing has inverse $\mathrm{Id}+B$ with $B=-A-AB\in\Psi_{\mathcal{T}}^{-\infty}$). If $\mathrm{Id}+A\in G_{\mathcal{T}}^0$, then $\mathrm{Id}+\sigma_0(A)$ is a pointed loop in $G^{-\infty}(Y;E)$, and its winding number is $-\operatorname{ind}(\mathrm{Id}+A)=0$ by Lemma 13.8.8; so $\Phi(G_{\mathcal{T}}^0)\subset\mathcal{Q}$. Conversely, given $\sum\rho^j b_j\in\mathcal{Q}$, Borel's lemma gives A with $\Phi(A)=\sum\rho^j b_j$, and by Lemma 13.8.8 there is $K\in\Psi_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)$ with $\mathrm{Id}+A+K\in G_{\mathcal{T}}^0$; since $\Phi(K)=0$, Φ is surjective. This proves exactness, and Φ is a homomorphism for the product $*$.

(i) $G_{\mathcal{T}}^{-\infty}(\mathbb{S}\times Y;E)$ is the group of invertible perturbations of the identity by smoothing operators on the Hilbert space $L^2_+(\mathbb{S}\times Y;E)$. The orthonormal system $e^{il\theta}\phi_m$, $l\geq 0$, with ϕ_m the eigensections of (B2), gives finite rank approximations exactly as in (B2), so the proof of Lemma 13.2.5 identifies $[X;G_{\mathcal{T}}^{-\infty}]$ with the stable homotopy classes of maps $X\to \mathrm{GL}(N,\mathbb{C})$, which is $K^{-1}(X)$.

(ii) is Theorem 13.8.6.

(iii) As a space, $\mathcal{Q}=\Omega_0 G^{-\infty}(Y;E)\times\prod_{j\geq 1}\mathcal{C}^\infty(\mathbb{S};\Psi^{-\infty}(Y;E))$, and the second factor is a vector space; scaling it by $t\in[0,1]$ shows that projection onto b_0 induces a bijection $[X;\mathcal{Q}]\to[X;\Omega_0 G^{-\infty}]$. A smooth map $X\to\Omega G^{-\infty}$ is the same as a pointed map $X\times\mathbb{S}\to G^{-\infty}$; for connected X its winding number is constant, and it is zero exactly when the rank of the index class is zero. By Theorem 13.6.2 the maps into $\Omega_0 G^{-\infty}$ thus correspond to $\{\xi\in K^0(X);\ \operatorname{rank}\xi=0\}=\tilde{K}^0(X)$. □

Remark 13.8.10. If (13.8.2) were known to be a fibration, the long exact sequence of homotopy groups and (ii) would give $\pi_k(\mathcal{Q})\cong\pi_{k-1}(G_{\mathcal{T}}^{-\infty})$. This is consistent with the results above: by (iii) and Corollary 13.9.9 below, $\pi_k(\mathcal{Q})=\pi_k(\Omega_0 G^{-\infty})=\pi_{k+1}(G^{-\infty})$, which by Bott periodicity equals $\pi_{k-1}(G^{-\infty})=\pi_{k-1}(G_{\mathcal{T}}^{-\infty})$.

13.9 Compactly supported K-theory and Bott periodicity

The symbol of an elliptic operator on Z lives on T^*Z, which is not compact, and it is invertible only outside a compact set (the zero section). The natural K-group for such data is K-theory with compact supports: pairs of bundles together with an isomorphism between

them away from a compact set. This section defines these groups, relates them to the loop models by clutching, and proves Bott periodicity in its usual form, $K_{\mathrm{c}}^0(\mathbb{R}^2 \times X) \cong K^0(X)$.

13.9.1 Definitions

Let M be a manifold, not necessarily compact. We write $K \Subset M$ for a compact subset.

Definition 13.9.1. (i) $K_{\mathrm{c}}^{-1}(M)$ is the set of smooth maps $f\colon M \to G^{-\infty}(Y;E)$ with $f = \mathrm{Id}$ on $M \setminus K$ for some $K \Subset M$, modulo homotopies f_t with $f_t = \mathrm{Id}$ on $M \setminus K$ for one $K \Subset M$ and all t.

(ii) $K_{\mathrm{c}}^{-2}(M)$ is defined in the same way, using maps $f\colon M \times \mathbb{S} \to G^{-\infty}(Y;E)$ with $f = \mathrm{Id}$ on $(M \setminus K) \times \mathbb{S} \cup M \times \{1\}$.

(iii) $K_{\mathrm{c}}^0(M)$ is the set of equivalence classes of triples (E_+, E_-, a), where $E_\pm$ are vector bundles over M and $a\colon E_+ \to E_-$ is a smooth bundle isomorphism over $M \setminus K$ for some $K \Subset M$. The equivalence relation is generated by:

(a) $(E_+, E_-, a) \sim (E'_+, E'_-, \phi_- a \phi_+^{-1})$ for bundle isomorphisms $\phi_\pm\colon E_\pm \to E'_\pm$ over M;

(b) $(E_+, E_-, a) \sim (E_+, E_-, a')$ if $a = a'$ outside a compact set;

(c) $(E_+, E_-, a_0) \sim (E_+, E_-, a_1)$ if a_t, $t \in [0,1]$, is a smooth family of isomorphisms over $M \setminus K$ for one $K \Subset M$;

(d) $(E_+, E_-, a) \sim (E_+ \oplus H, E_- \oplus H, a \oplus \mathrm{Id}_H)$ for any bundle H.

The sum is $[(E_+, E_-, a)] + [(F_+, F_-, b)] = [(E_+ \oplus F_+, E_- \oplus F_-, a \oplus b)]$.

The groups K_{c}^{-1}, K_{c}^{-2} are abelian groups by pointwise multiplication (as in Lemma 13.2.5). $K_{\mathrm{c}}^0(M)$ is an abelian monoid with zero $[(0,0,\cdot)]$; a triple (F,F,c) with c an isomorphism over all of M represents 0 (use (a) with $\phi_- = c^{-1}$ and then (d)). It is a group: $(E_+ \oplus E_-, E_- \oplus E_+, a \oplus a^{-1})$ is equivalent by (a), composing with the exchange of the summands of $E_- \oplus E_+$, to (F,F,b) with $F = E_+ \oplus E_-$ and $b = \begin{pmatrix} 0 & a^{-1} \\ a & 0 \end{pmatrix}$, and $b_\tau = \begin{pmatrix} \cos\tau & \sin\tau\, a^{-1} \\ \sin\tau\, a & -\cos\tau \end{pmatrix}$ satisfies $b_\tau^2 = \mathrm{Id}$ and joins b to the globally defined isomorphism $\mathrm{diag}(\mathrm{Id}, -\mathrm{Id})$.

For compact M the conditions at infinity are void, and $K_{\mathrm{c}}^{-1}(M) = K^{-1}(M)$, $K_{\mathrm{c}}^{-2}(M) = K^{-2}(M)$; also $K_{\mathrm{c}}^0(M) = K^0(M)$, since (E_+, E_-, a) may be taken with a defined on the empty set and then (a), (d) are exactly the relation of Definition 13.2.3.

13.9.2 Flattening and suspension

Lemma 13.9.2 (Flattening)**.** *Let $W_1 \subset M_1$ and $W_2 \subset M_2$ be points (or empty) in compact manifolds and $W = W_1 \times M_2 \cup M_1 \times W_2$. Every smooth $f\colon M_1 \times M_2 \to G^{-\infty}$ with $f = \mathrm{Id}$ on W is homotopic, through such maps, to one which is* Id *in a neighbourhood of W; and two maps which are* Id *near W and homotopic through maps equal to* Id *on W are homotopic through maps equal to* Id *near W.*

Proof. In a coordinate ball centred at $W_1 = \{w\}$, let $\psi_t(y) = ((1-t) + t\chi(|y|))y$, where $\chi \in \mathcal{C}^\infty([0,\infty))$ vanishes on $[0,\delta]$ and equals 1 on $[2\delta,\infty)$, and extend ψ_t by the identity. Then ψ_t fixes w, $\psi_0 = \mathrm{Id}$ and ψ_1 maps a neighbourhood of w to w. With the analogous map on M_2, $\Psi_t = \psi_t \times \psi_t$ maps W into W for all t, and $f \circ \Psi_t$ is the required homotopy.

For the second statement apply Ψ_1 to the homotopy; $h \circ \Psi_t$ joins a map h which is Id near W to $h \circ \Psi_1$ through maps equal to Id near W, provided δ is small (by compactness Ψ_t maps a small neighbourhood of W into the set where $h = \mathrm{Id}$). □

Applied with $M_1 = \mathrm{pt}$ or with the stereographic identification $\mathbb{S}^k \setminus \{e\} \cong \mathbb{R}^k$ this shows that the homotopy groups $\pi_k(G^{-\infty}) = [\mathbb{S}^k, e; G^{-\infty}, \mathrm{Id}]$ (pointed smooth homotopy classes) satisfy $\pi_k(G^{-\infty}) \cong K_{\mathrm{c}}^{-1}(\mathbb{R}^k)$. Similarly, identifying $\mathbb{S} \setminus \{1\}$ with $\mathbb{R}$:

Lemma 13.9.3. *For every manifold M, $K_{\mathrm{c}}^{-2}(M) \cong K_{\mathrm{c}}^{-1}(\mathbb{R} \times M)$.*

Proof. By (the proof of) Lemma 13.9.2 in the circle factor, representatives and homotopies in $K_{\mathrm{c}}^{-2}(M)$ may be taken equal to Id near $M \times \{1\}$; restricted to $M \times (\mathbb{S} \setminus \{1\}) \cong M \times \mathbb{R}$ they are exactly the representatives and homotopies of $K_{\mathrm{c}}^{-1}(\mathbb{R} \times M)$, and conversely such maps extend by Id. □

13.9.3 Clutching

For $n \geq 1$ let $\mathbb{S}^{n-1} \subset \mathbb{R}^n$ be the unit sphere with base point e, and write $K^{-1}(\mathbb{S}^{n-1} \times X, X)$ for the set of homotopy classes of maps $g\colon \mathbb{S}^{n-1} \times X \to G^{-\infty}(Y; E)$ with $g(e, x) = \mathrm{Id}$, through such maps. Thus $K^{-1}(\mathbb{S}^0 \times X, X) = K^{-1}(X)$ and $K^{-1}(\mathbb{S} \times X, X) = K^{-2}(X)$.

Lemma 13.9.4 (Clutching)**.** *For compact X and $n \geq 1$ there is a natural isomorphism*

$$K_{\mathrm{c}}^0(\mathbb{R}^n \times X) \cong K^{-1}(\mathbb{S}^{n-1} \times X, X),$$

whose inverse sends the class of a pointed g with values in $\mathrm{GL}(N, \mathbb{C})$ to the class of $(\mathbb{C}^N, \mathbb{C}^N, \tilde{g})$, $\tilde{g}(r\omega, x) = g(\omega, x)$ for $r \geq 1$, $\omega \in \mathbb{S}^{n-1}$.

Proof. Let $\varpi\colon \mathbb{R}^n \times X \to X$. By Lemma 13.2.2(ii) every bundle on $\mathbb{R}^n \times X$ is isomorphic to a pull-back, so by (a) every class is represented by $(\varpi^* F_+, \varpi^* F_-, a)$ with a an isomorphism on $\{|v| \geq R\}$. Choose a bundle G and an isomorphism $\tau\colon F_- \oplus G \to \mathbb{C}^N$, and set

$$g(\omega, x) = \tau\,(a_R \oplus \mathrm{Id}_G)(\omega, x)\,(a_R \oplus \mathrm{Id}_G)(e, x)^{-1}\tau^{-1}, \qquad a_R(\omega, x) = a(R\omega, x),$$

where $a_R(\omega, x)\colon F_{+,x} \to F_{-,x}$. This is a pointed map into $\mathrm{GL}(N, \mathbb{C})$; let $\kappa[(E_+, E_-, a)]$ be its class.

Well-definedness. Changing R, or a by a homotopy (c), changes g by a pointed homotopy, and (b) is covered by changing R. Under (d), or a change of (G, τ), g is changed by stabilization and conjugation by a map $X \to \mathrm{GL}$ independent of ω, which does not change the class by Lemma 13.1.3. Under (a) with isomorphisms $\phi_\pm$ of the pull-back bundles, $\phi_\pm(sR\omega, x)$, $s \in [0, 1]$, joins $\phi_\pm|_{\{|v|=R\}}$ to $\phi_\pm(0, x)$, which is independent of ω; along this homotopy g is changed by a pointed homotopy, and at $s = 0$ it is replaced by its conjugate by $\tau(\phi_-(0, \cdot) \oplus \mathrm{Id})\tau^{-1}$ and again the class does not change. κ is additive, since direct sums of triples give block sums of the g's.

Surjectivity. For pointed g with values in $\mathrm{GL}(N, \mathbb{C})$ (Lemma 13.2.5) the triple $(\mathbb{C}^N, \mathbb{C}^N, \tilde{g})$ of the statement is mapped to $[g]$ (with $G = 0$, $\tau = \mathrm{Id}$, $R = 1$).

Injectivity. Using (d) with $\varpi^* G$ and then (a) with τ we may assume $F_- = \mathbb{C}^N$; since $a_R(e, x)\colon F_+ \to \mathbb{C}^N$ is an isomorphism, (a) allows us to assume $F_+ = \mathbb{C}^N$ and $a_R(e, x) = \mathrm{Id}$. The homotopy $a_s(r\omega, x) = a\big(((1 - s)r + sR)\omega, x\big)$, $r \geq R$, shows that the triple is equivalent to $(\mathbb{C}^N, \mathbb{C}^N, \tilde{g})$ with $\tilde{g}(r\omega, x) = g(\omega, x)$ for $r \geq R$. If $[g] = 0$, then by Lemma 13.2.5 $g \oplus \mathrm{Id}$ is joined to Id by a pointed homotopy g_t in some $\mathrm{GL}(N', \mathbb{C})$, and (d), (c) give $(\mathbb{C}^N, \mathbb{C}^N, \tilde{g}) \sim (\mathbb{C}^{N'}, \mathbb{C}^{N'}, \tilde{g}_1) = (\mathbb{C}^{N'}, \mathbb{C}^{N'}, \mathrm{Id})$, which represents 0. □

13.9.4 Bott periodicity

Theorem 13.9.5 (Bott periodicity)**.** *For every compact manifold X there is a natural isomorphism*

$$K^0(X) \longrightarrow K^0_{\mathrm{c}}(\mathbb{R}^2 \times X).$$

Identifying $\mathbb{R}^2 = \mathbb{C}$ with variable w, it maps $[E]$ to the class of $(\varpi^*E, \varpi^*E, \bar{w}/|w|)$; that is, it is multiplication by the Bott class $b = [(\mathbb{C}, \mathbb{C}, \bar{w}/|w|)] \in K^0_{\mathrm{c}}(\mathbb{R}^2)$.

Proof. The composite of the inverse of the index isomorphism (Theorem 13.6.2) and of the inverse of the clutching isomorphism (Lemma 13.9.4, $n = 2$), $K^0(X) \to K^{-2}(X) = K^{-1}(\mathbb{S} \times X, X) \to K^0_{\mathrm{c}}(\mathbb{R}^2 \times X)$, is an isomorphism. By (13.6.1), if $E = \operatorname{Ran} p$, $p\colon X \to M(N, \mathbb{C})$, then $\beta[E]$ is represented by $pz^{-1} + (\mathrm{Id} - p)$, and the clutching construction gives $(\mathbb{C}^N, \mathbb{C}^N, p\,\bar{w}/|w| + \mathrm{Id} - p)$ for $|w| \geq 1$, since $z^{-1} = \bar{w}/|w|$ on the circle $\omega = w/|w|$. Writing $\mathbb{C}^N = E \oplus \operatorname{Ran}(\mathrm{Id} - p)$ and using (d) this is equivalent to $(\varpi^*E, \varpi^*E, \bar{w}/|w|)$. □

Discussion. The Bott class b is the simplest compactly supported class on $\mathbb{R}^2$: the trivial line bundle, compared with itself by multiplication by $\bar{w}/|w|$ outside the unit disc. Restricted to the circle at infinity, this is a loop of winding number -1, the symbol of the Toeplitz operator L of index 1. The theorem says that every compactly supported class on $\mathbb{R}^2 \times X$ is obtained from a class on X by "twisting" with this one loop, so the plane $\mathbb{R}^2$ contributes nothing but a shift.

Remark 13.9.6. Periodicity can also be deduced from the chain $K^0_{\mathrm{c}}(\mathbb{R}^2 \times X) \cong K^{-1}_{\mathrm{c}}(\mathbb{R} \times X) \cong K^{-2}_{\mathrm{c}}(X) \cong K^0_{\mathrm{c}}(X)$, which uses the clutching isomorphism for the non-compact manifold $\mathbb{R} \times X$ and the index isomorphism for non-compact X (Exercise 13.1). For compact X the clutching lemma with $n = 1$ gives $K^0_{\mathrm{c}}(\mathbb{R} \times X) \cong K^{-1}(X)$, and Lemma 13.9.3 gives $K^{-2}(X) \cong K^{-1}_{\mathrm{c}}(\mathbb{R} \times X)$, so the chain is $K^0_{\mathrm{c}}(\mathbb{R}^2 \times X) \cong K^{-2}(X) \cong K^{-1}_{\mathrm{c}}(\mathbb{R} \times X)$ and $K^0_{\mathrm{c}}(\mathbb{R} \times X) \cong K^{-1}(X)$.

13.9.5 The homotopy groups of $G^{-\infty}$

Spheres need one more form of the clutching construction.

Lemma 13.9.7. *For $m \geq 1$, $\tilde{K}^0(\mathbb{S}^m) \cong \pi_{m-1}(G^{-\infty}(Y; E))$.*

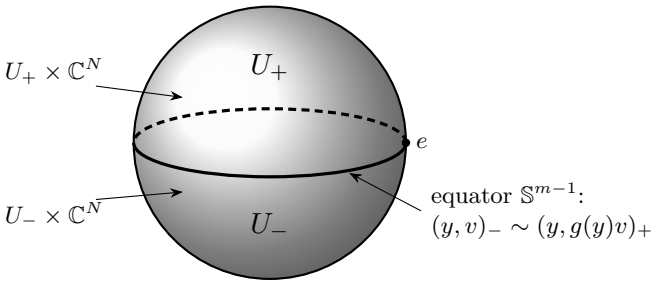


Figure 13.5: The clutching construction. Over the two hemispheres $U_\pm$ of $\mathbb{S}^m$ (both contractible) a bundle is trivial; it is determined by how the two trivializations are glued along the equator, i.e. by a map $g : \mathbb{S}^{m-1} \to \mathrm{GL}(N, \mathbb{C})$. For $m = 2$ the equator is a circle and g is a loop; the loop $g(z) = z$ gives, up to conjugation, the tautological line bundle over $\mathbb{S}^2 = \mathbb{CP}^1$.

Proof. Let $U_\pm = \{\pm x_{m+1} > -\epsilon\} \subset \mathbb{S}^m \subset \mathbb{R}^{m+1}$, so that $U_\pm$ are diffeomorphic to $\mathbb{R}^m$ and $U_+ \cap U_-$ retracts, by a map r, onto the equator $\mathbb{S}^{m-1}$, which carries the base point e. For pointed $g\colon \mathbb{S}^{m-1} \to \mathrm{GL}(N,\mathbb{C})$ let E_g be the bundle obtained from $U_+ \times \mathbb{C}^N$ and $U_- \times \mathbb{C}^N$ by identifying $(y,v)_-$ with $(y, g(r(y))v)_+$ over $U_+ \cap U_-$ (Figure 13.5), and set $\Psi(g) = [E_g] - [\mathbb{C}^N]$, an element of $\tilde K^0(\mathbb{S}^m)$. A pointed homotopy g_t defines a bundle over $[0,1] \times \mathbb{S}^m$, so $E_{g_0} \cong E_{g_1}$ by Lemma 13.2.2; $E_{g\oplus \mathrm{Id}} = E_g \oplus \mathbb{C}$ and $E_{\mathrm{diag}(g,h)} = E_g \oplus E_h$. By Lemma 13.2.5 and Lemma 13.1.2, Ψ induces a homomorphism $\pi_{m-1}(G^{-\infty}) \to \tilde K^0(\mathbb{S}^m)$.

It is surjective: an element of $\tilde K^0(\mathbb{S}^m)$ has the form $[E] - [\mathbb{C}^N]$ with $\operatorname{rank} E = N$ ($\mathbb{S}^m$ is connected). E is trivial over $U_\pm$ (Lemma 13.2.2(ii)); for trivializations $\tau_\pm$ the transition map $\tilde g = \tau_+\tau_-^{-1}$ on $U_+ \cap U_-$ is homotopic to $\tilde g \circ r$ (deform along the retraction), so $E \cong E_g$ with $g = \tilde g|_{\mathbb{S}^{m-1}}$; replacing τ_- by $g(e)\tau_-$ makes g pointed.

It is injective: if $[E_g] = [\mathbb{C}^N]$, then $E_g \oplus \mathbb{C}^L \cong \mathbb{C}^{N+L}$, i.e. $E_{g'}$, $g' = g \oplus \mathrm{Id}_L$, has a global trivialization. In the local trivializations over $U_\pm$ it is given by maps $\tau_\pm\colon U_\pm \to \mathrm{GL}(N+L,\mathbb{C})$ with $\tau_- = \tau_+ g'$, i.e. $g' = \tau_+^{-1}\tau_-$, on the equator. As $U_\pm$ are star-shaped in suitable coordinates, $\tau_\pm$ are homotopic to constants, so g' is homotopic to a constant, and then (as GL is connected) to Id; normalizing by the value at e (Lemma 13.1.1) makes the homotopy pointed. So $[g] = 0$. □

Lemma 13.9.8. *For $m \geq 0$, with base point $e \in \mathbb{S}^m$, $\tilde K^{-2}(\mathbb{S}^m) \cong \pi_{m+1}(G^{-\infty}(Y;E))$.*

Proof. An element of $\tilde K^{-2}(\mathbb{S}^m)$ is represented by $f\colon \mathbb{S}^m \times \mathbb{S} \to G^{-\infty}$ with $f = \mathrm{Id}$ on $\mathbb{S}^m \times \{1\}$ and such that the loop $f(e,\cdot)$ is null-homotopic through pointed loops h_s, $h_0 = f(e,\cdot)$, $h_1 = \mathrm{Id}$. Choose $\chi \in \mathcal{C}^\infty(\mathbb{S}^m)$ with values in $[0,1]$ and $\chi(e) = 1$. Then $F_s(x,\theta) = f(x,\theta)f(e,\theta)^{-1}h_{s\chi(x)}(\theta)$ is a homotopy in K^{-2} from f to a map equal to Id on $W = \mathbb{S}^m \times \{1\} \cup \{e\} \times \mathbb{S}$. Similarly a homotopy H_t between two maps equal to Id on W is replaced by $H_t(x,\theta)H_t(e,\theta)^{-1}$, which is Id on W and has the same ends. Hence $\tilde K^{-2}(\mathbb{S}^m)$ is the set of homotopy classes, relative to W, of maps equal to Id on W. By Lemma 13.9.2 these are the classes of maps which are Id near W, and since $(\mathbb{S}^m \setminus \{e\}) \times (\mathbb{S} \setminus \{1\}) \cong \mathbb{R}^{m+1}$ this is $K_c^{-1}(\mathbb{R}^{m+1}) \cong \pi_{m+1}(G^{-\infty})$. (For $m = 0$, $\mathbb{S}^0 = \{e,-e\}$ and the statement is $K^{-2}(\{-e\}) = \pi_1(G^{-\infty})$.) □

Corollary 13.9.9. *For every compact Y of positive dimension and every bundle E,*

$$\pi_k(G^{-\infty}(Y;E)) \cong \begin{cases} \mathbb{Z} & k \text{ odd}, \\ 0 & k \text{ even}. \end{cases}$$

The isomorphism $\pi_1(G^{-\infty}) \cong \mathbb{Z}$ is given by the winding number. Consequently also $K^{-1}(\mathbb{S}^k) = [\mathbb{S}^k; G^{-\infty}] = \pi_k(G^{-\infty})$, and $\tilde K^0(\mathbb{S}^k)$ is $\mathbb{Z}$ for k even and 0 for k odd, $k \geq 1$.

Proof. $\pi_0(G^{-\infty}) = 0$ since $G^{-\infty}$ is connected. For $m \geq 0$, $\pi_{m+1}(G^{-\infty}) \cong \tilde K^{-2}(\mathbb{S}^m) \cong \tilde K^0(\mathbb{S}^m)$ by Lemma 13.9.8 and Corollary 13.6.4. For $m = 0$, $\tilde K^0(\mathbb{S}^0) = K^0(\mathrm{pt}) = \mathbb{Z}$, so $\pi_1(G^{-\infty}) = \mathbb{Z}$, the isomorphism being minus the index, i.e. the winding number. For $m \geq 1$, $\tilde K^0(\mathbb{S}^m) \cong \pi_{m-1}(G^{-\infty})$ by Lemma 13.9.7, so $\pi_{m+1}(G^{-\infty}) \cong \pi_{m-1}(G^{-\infty})$ and the result follows by induction. Free and pointed homotopy classes of maps $\mathbb{S}^k \to G^{-\infty}$ agree, since $f \mapsto f(e)^{-1}f$ is a homotopy inverse to the inclusion of pointed maps ($G^{-\infty}$ is a connected group). □

This justifies calling $G^{-\infty}$ a classifying space for odd K-theory: its homotopy groups are those of the stable unitary group (Bott's original theorem).

13.10 Isotropic–semiclassical quantization maps

The isotropic index map quantizes a loop of matrices, viewed as a function on the circle at infinity of $\mathbb{R}^2$, to a Fredholm operator. There is a second, 'odd', way to pass from symbols to operators, which works on every $\mathbb{R}^{2n}$: a compactly supported map $g\colon \mathbb{R}^{2n} \to \mathrm{GL}(N,\mathbb{C})$ (equal to Id outside a compact set) is quantized to an invertible perturbation of the identity by an isotropic smoothing operator on $\mathbb{R}^n$, after rescaling the symbol by a small parameter ϵ. The same procedure applied to projections gives an even version. These maps are used in Chapter 16; here we define them, show that they commute with clutching, and show that they realize Bott periodicity.

13.10.1 Weyl operators and semiclassical scaling

On $\mathbb{R}^{2n}$ we use the variables $w = (x,\xi)$ and dual variables $\zeta = (p,q)$, with $\langle \zeta, w\rangle = p\cdot x + q\cdot\xi$, and the Fourier transform $\hat a(\zeta) = \int e^{-i\langle\zeta,w\rangle} a(w)\,dw$. The *Weyl operators*

$$\mathcal{W}(\zeta)u(x) = e^{ip\cdot x + ip\cdot q/2}\,u(x+q), \qquad \zeta = (p,q) \in \mathbb{R}^{2n},$$

are unitary on $L^2(\mathbb{R}^n)$, strongly continuous in ζ, and by direct computation

$$\mathcal{W}(\zeta)\mathcal{W}(\zeta') = e^{i\sigma(\zeta,\zeta')/2}\,\mathcal{W}(\zeta+\zeta'), \qquad \sigma((p,q),(p',q')) = p'\cdot q - p\cdot q'. \tag{13.10.1}$$

From the kernel of the Weyl quantization Op^{W} of Chapter 7 one finds $\mathrm{Op}^{\mathrm{W}}(e^{i\langle\zeta,\cdot\rangle}) = \mathcal{W}(\zeta)$: its kernel is $e^{ip\cdot(x+y)/2}\delta(x-y+q)$. Since $u,v \mapsto \langle \mathrm{Op}^{\mathrm{W}}(a)u, v\rangle$ is the pairing of a with a Schwartz function of w (the kernel formula of the Weyl quantization), we may insert the Fourier inversion formula $a = (2\pi)^{-2n}\int \hat a(\zeta) e^{i\langle\zeta,\cdot\rangle} d\zeta$ and get, for $a \in \mathcal{S}(\mathbb{R}^{2n})$,

$$\mathrm{Op}^{\mathrm{W}}(a) = (2\pi)^{-2n}\int \hat a(\zeta)\,\mathcal{W}(\zeta)\,d\zeta, \qquad \|\mathrm{Op}^{\mathrm{W}}(a)\|_{\mathcal{B}(L^2)} \le (2\pi)^{-2n}\|\hat a\|_{L^1}, \tag{13.10.2}$$

the integral converging strongly. The same holds for matrix-valued symbols, with $\|\hat a\|_{L^1} = \int |\hat a(\zeta)|\,d\zeta$ and the operator norm on matrices.

Definition 13.10.1. For $0 < \epsilon \le 1$ and $a \in \mathcal{S}(\mathbb{R}^{2n}; M(N,\mathbb{C}))$ set

$$q_\epsilon(a) = \mathrm{Op}^{\mathrm{W}}(a_\epsilon) \in \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N), \qquad a_\epsilon(w) = a(\sqrt{\epsilon}\,w).$$

Conjugating by the unitary dilation $u \mapsto \epsilon^{-n/4}u(\epsilon^{-1/2}\cdot)$ transforms $q_\epsilon(a)$ into the semiclassical Weyl quantization of a with parameter ϵ, $(2\pi\epsilon)^{-n}\int e^{i(x-y)\cdot\xi/\epsilon} a(\frac{x+y}{2},\xi)u(y)\,dy\,d\xi$, so q_ϵ is the isotropic version of the semiclassical quantization of Chapter 6. The dilations form a smooth path of unitaries joining this conjugation to the identity and preserving $\Psi^{-\infty}_{\mathrm{iso}}$, so for the K-theoretic constructions below it makes no difference which of the two is used.

Lemma 13.10.2. *For $a, b \in \mathcal{S}(\mathbb{R}^{2n}; M(N,\mathbb{C}))$ and $0 < \epsilon \le 1$:*

(i) $q_\epsilon(a)^* = q_\epsilon(a^*)$ *and* $\|q_\epsilon(a)\| \le (2\pi)^{-2n}\|\hat a\|_{L^1}$;

(ii) $\|q_\epsilon(a)q_\epsilon(b) - q_\epsilon(ab)\| \le \epsilon\, 2^{-1}(2\pi)^{-4n}\,\||\zeta|\hat a\|_{L^1}\||\zeta|\hat b\|_{L^1}$;

(iii) $q_\epsilon(a) \to a(0)$ *strongly on* $L^2(\mathbb{R}^n;\mathbb{C}^N)$ *as* $\epsilon \downarrow 0$.

If a depends smoothly on parameters, with $a = 0$ outside a compact subset of $\mathbb{R}^{2n}\times X$, the bounds are locally uniform in the parameters and $q_\epsilon(a)$ is a smooth function of $(\epsilon,x) \in (0,1]\times X$ with values in $\Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$.

Proof. The adjoint formula is that of the Weyl quantization. Since $\widehat{a_\epsilon}(\zeta) = \epsilon^{-n}\hat{a}(\zeta/\sqrt{\epsilon})$, (13.10.2) gives $q_\epsilon(a) = (2\pi)^{-2n}\int \hat{a}(\zeta)\mathcal{W}(\sqrt{\epsilon}\zeta)\,d\zeta$, which implies the norm bound and, by dominated convergence and $\mathcal{W}(\sqrt{\epsilon}\zeta)u \to u$, (iii). By (13.10.1),

$$q_\epsilon(a)q_\epsilon(b) = (2\pi)^{-4n}\iint \hat{a}(\zeta)\hat{b}(\zeta')\,e^{i\epsilon\sigma(\zeta,\zeta')/2}\,\mathcal{W}(\sqrt{\epsilon}(\zeta+\zeta'))\,d\zeta\,d\zeta',$$

while $\widehat{ab} = (2\pi)^{-2n}\hat{a} * \hat{b}$ gives the same formula for $q_\epsilon(ab)$ without the factor $e^{i\epsilon\sigma/2}$. Since $|e^{is}-1| \le |s|$ and $|\sigma(\zeta,\zeta')| \le |\zeta||\zeta'|$, (ii) follows. The last statement is clear from the kernel of $\mathrm{Op}^{\mathrm{W}}(a_\epsilon)$, a Schwartz function depending smoothly on (ϵ, x). □

Discussion. The parameter ϵ is a semiclassical parameter. The rescaled symbol $a(\sqrt{\epsilon}\,w)$ varies on the scale $\epsilon^{-1/2}$, which is large compared with the scale 1 of the uncertainty principle. On this scale quantization is almost multiplicative, $q_\epsilon(a)q_\epsilon(b) = q_\epsilon(ab) + O(\epsilon)$. Hence an invertible symbol is quantized to an almost invertible operator, which can be corrected to an invertible one, and an idempotent symbol to an almost idempotent operator, whose spectral projection is an honest projection. This is all that the two quantization maps below use.

13.10.2 The odd quantization map

Proposition 13.10.3. *Let X be a manifold and $g \in \mathcal{C}^\infty(\mathbb{R}^{2n}\times X; \mathrm{GL}(N,\mathbb{C}))$ with $g = \mathrm{Id}$ outside a compact set. There is $\epsilon_0 > 0$ such that for $0 < \epsilon \le \epsilon_0$*

$$G_\epsilon(x) = \mathrm{Id} + q_\epsilon\big(g(\cdot,x) - \mathrm{Id}\big) \in G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N) \quad \textit{for all } x \in X;$$

G_ϵ is a smooth map, equal to Id *outside a compact subset of X. Its class $\mathrm{Ind}^{odd}_{\mathrm{sl}}[g] = [G_\epsilon] \in K_{\mathrm{c}}^{-1}(X)$ is independent of $\epsilon \in (0,\epsilon_0]$, depends only on $[g] \in K_{\mathrm{c}}^{-1}(\mathbb{R}^{2n}\times X)$, and defines a homomorphism*

$$\mathrm{Ind}^{odd}_{\mathrm{sl}} \colon K_{\mathrm{c}}^{-1}(\mathbb{R}^{2n}\times X) \longrightarrow K_{\mathrm{c}}^{-1}(X),$$

natural under proper maps and under extension from open subsets of X.

Proof. Let $\alpha = g - \mathrm{Id}$ and $\beta = g^{-1} - \mathrm{Id}$, both compactly supported, with $\alpha + \beta + \alpha\beta = 0 = \alpha + \beta + \beta\alpha$. Then

$$\big(\mathrm{Id} + q_\epsilon(\alpha)\big)\big(\mathrm{Id} + q_\epsilon(\beta)\big) = \mathrm{Id} + q_\epsilon(\alpha)q_\epsilon(\beta) - q_\epsilon(\alpha\beta) = \mathrm{Id} + O(\epsilon),$$

uniformly in x by Lemma 13.10.2(ii), and similarly in the other order. So for small ϵ, $G_\epsilon(x)$ has a left and a right inverse on L^2, i.e. is invertible, and since it lies in $\mathrm{Id} + \Psi^{-\infty}_{\mathrm{iso}}$ its inverse does too (Chapter 7), so $G_\epsilon(x) \in G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n;\mathbb{C}^N)$, with $\|G_\epsilon^{-1}\|$ uniformly bounded. Smoothness and the support property follow from Lemma 13.10.2. A homotopy g_t as in Definition 13.9.1, or the path $\epsilon \in (0,\epsilon_0]$, gives a homotopy of the G_ϵ, and stabilization of g stabilizes G_ϵ. For two maps g, h, $G_\epsilon(g)G_\epsilon(h) - G_\epsilon(gh) = q_\epsilon(\alpha_g)q_\epsilon(\alpha_h) - q_\epsilon(\alpha_g\alpha_h) = O(\epsilon)$, so for small ϵ the linear path between $G_\epsilon(g)G_\epsilon(h)$ and $G_\epsilon(gh)$ stays in $G^{-\infty}_{\mathrm{iso}}$; thus $\mathrm{Ind}^{odd}_{\mathrm{sl}}$ is a homomorphism. Naturality is clear. □

13.10.3 Projections

Even classes are conveniently represented by projections. By *projection data* on a manifold M we mean a smooth family of orthogonal projections $p \in \mathcal{C}^\infty(M; M(N,\mathbb{C}))$ which is equal to a constant projection p_∞ outside a compact set. It defines

$$[p] - [p_\infty] = \big[(\mathrm{Ran}\,p,\ M\times \mathrm{Ran}\,p_\infty,\ p_\infty|_{\mathrm{Ran}\,p})\big] \in K^0_{\mathrm{c}}(M); \tag{13.10.3}$$

$p_\infty|_{\mathrm{Ran}\,p}$ is the identity outside a compact set. For compact M this is the difference of the classes of the bundles $\mathrm{Ran}\,p$ and $\mathrm{Ran}\,p_\infty$.

Lemma 13.10.4. *(i) The class* (13.10.3) *is unchanged under homotopies p_t of projection data with $p_t = p_\infty$ outside one compact set, under stabilization $(p, p_\infty) \mapsto (p \oplus r, p_\infty \oplus r)$ by a constant projection r, and under conjugation by a constant unitary. The same holds for smooth families of idempotents (not necessarily self-adjoint).*

(ii) If M is compact, or is the total space of a real vector bundle over a compact manifold (for instance $\mathbb{R}^k \times X$ or T^*Z with X, Z compact), every element of $K^0_{\mathrm{c}}(M)$ is of the form (13.10.3).

Proof. (i) For a homotopy, $\operatorname{Ran} p_t$ is a bundle over $[0,1] \times M$; parallel transport in t for the connection $p_t\,\partial_t$ (Lemma 13.2.2) is an isomorphism $\operatorname{Ran} p_0 \to \operatorname{Ran} p_1$ which is the identity where p_t does not depend on t, hence outside a compact set, and relations (a), (b) of Definition 13.9.1 apply. Stabilization is (d) and conjugation is (a). For idempotents use the connection $p_t\,\partial_t$ in the same way.

(ii) Let (E_+, E_-, a) be a triple, a invertible outside a compact set K. In the second case bundles on M are pulled back from the compact base (Lemma 13.2.2); in either case there is a bundle F with $E_- \oplus F \cong M \times \mathbb{C}^N$ and an embedding $E_+ \oplus F \subset M \times \mathbb{C}^{N'}$ (Lemma 13.2.1). By (d) and (a) we may assume $E_- = \mathbb{C}^N$ and $E_+ \subset \mathbb{C}^{N'}$. Choose $\chi \in \mathcal{C}^\infty(M; [0, \pi/2])$ with $\chi = 0$ near K and $\chi = \pi/2$ outside a compact set, and let p be the orthogonal projection onto $\{(\cos\chi\, e, \sin\chi\, a(e));\ e \in E_+\} \subset \mathbb{C}^{N'} \oplus \mathbb{C}^N$, a bundle isomorphic to E_+, and p_∞ the projection onto $0 \oplus \mathbb{C}^N$. Outside a compact set $p = p_\infty$, and in terms of the isomorphism with E_+ the map $p_\infty|_{\operatorname{Ran} p}$ is $e \mapsto \sin\chi\, a(e)$, which equals a outside a compact set. By (a) and (b) the class (13.10.3) is that of $(E_+, \mathbb{C}^N, a)$. □

Conversely, two projection data define the same class if and only if they are related by the moves in (i) ([17], Ch. II); we use this only to regard the even quantization map below as defined on K^0_{c}.

In the quantized setting the projections differ from a constant one by a smoothing operator. For such families the relative class is defined by finite rank approximation. Here $p_\infty \in M(N, \mathbb{C})$ acts on $L^2(\mathbb{R}^n; \mathbb{C}^N)$ pointwise, and commutes with the Hermite projections Π_k of $L^2(\mathbb{R}^n; \mathbb{C}^N)$ (onto the span of the $u_\alpha \otimes \mathbb{C}^N$, $|\alpha| \le k$).

Lemma 13.10.5. *Let X be a manifold and $Q \in \mathcal{C}^\infty(X; p_\infty + \Psi^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n; \mathbb{C}^N))$ a family of idempotents with $Q = p_\infty$ outside a compact subset of X. There is a well-defined class $\delta(Q) \in K^0_{\mathrm{c}}(X)$ with the following properties.*

(i) If $V \subset \operatorname{Ran} \Pi_k$ is a p_∞-invariant subspace such that Q commutes with the orthogonal projection Π_V onto V and $Q = p_\infty$ on $V^\perp$, then $\delta(Q)$ is the class of the triple $(\operatorname{Ran}(Q\Pi_V), X \times \operatorname{Ran}(p_\infty \Pi_V), p_\infty)$. In particular, for X a point, $\delta(Q) = \operatorname{rank}(Q|_V) - \operatorname{rank}(p_\infty|_V)$.

(ii) $\delta(Q_0) = \delta(Q_1)$ if Q_t, $t \in [0,1]$, is a smooth family of such idempotents with $Q_t = p_\infty$ outside one compact set; δ is additive under direct sums and natural under proper maps.

Proof. On the circle $\Gamma = \{|\lambda - 1| = \frac{1}{2}\}$ the resolvent of an idempotent Q is $(\lambda - Q)^{-1} = \lambda^{-1}(\mathrm{Id} - Q) + (\lambda - 1)^{-1} Q$, of norm at most $2(1 + 2\|Q\|)$. Hence if T is an operator with $\|T - Q\| < (4 + 8\|Q\|)^{-1}$, then Γ lies in the resolvent set of T and the Riesz projection $\mathcal{R}(T) = (2\pi i)^{-1} \oint_\Gamma (\lambda - T)^{-1} d\lambda$ is an idempotent, equal to Q if $T = Q$. If moreover $T = p_\infty + R$ with $R \in \Psi^{-\infty}_{\mathrm{iso}}$, two applications of the resolvent identity give

$$(\lambda - T)^{-1} = (\lambda - p_\infty)^{-1} + (\lambda - p_\infty)^{-1} R (\lambda - p_\infty)^{-1} + (\lambda - p_\infty)^{-1} R (\lambda - T)^{-1} R (\lambda - p_\infty)^{-1},$$

and $(\lambda - p_\infty)^{-1}$ is a matrix, so by (I3) (in n dimensions, Chapter 7) $\mathcal{R}(T) - p_\infty \in \Psi^{-\infty}_{\mathrm{iso}}$, depending smoothly on parameters on which R depends smoothly.

Now write $Q = p_\infty + R$, $R = 0$ outside the compact set $C \subset X$. By Remark 13.2.6, $\Pi_k R \Pi_k \to R$ uniformly on C, so for k large the operators $T_s = p_\infty + (1-s)R + s\Pi_k R \Pi_k$ are within the bound above of Q, and $Q_s = \mathcal{R}(T_s)$ is a homotopy of idempotents of the stated kind from Q to $Q^{(k)} = \mathcal{R}(p_\infty + \Pi_k R \Pi_k)$. Since $p_\infty + \Pi_k R \Pi_k$ commutes with Π_k and equals p_∞ on $\mathrm{Ran}(\mathrm{Id} - \Pi_k)$, so does $Q^{(k)}$, and we define $\delta(Q)$ by the triple in (i) for $Q^{(k)}$ and $V = \operatorname{Ran} \Pi_k$. Here $\mathrm{Ran}(Q^{(k)}\Pi_k)$ is a smooth subbundle of $X \times \operatorname{Ran} \Pi_k$ (Lemma 13.2.1) equal to $\mathrm{Ran}(p_\infty \Pi_k)$ outside C, where p_∞ restricts to the identity.

We first show: (*) if Q_t is a smooth family as in (ii), and all Q_t commute with Π_V and equal p_∞ on $V^\perp$, then the triples of (i) for Q_0 and Q_1 are equivalent. Indeed, transport along t for the connection $Q_t \partial_t$ on the bundle $\mathrm{Ran}(Q_t \Pi_V)$ over $[0,1] \times X$ is an isomorphism which is the identity outside a compact set, and (a), (b) of Definition 13.9.1 apply. Moreover, if $V \subset V'$ with both p_∞-invariant, then the triple for V' is the triple for V plus $(\mathrm{Ran}(p_\infty \Pi_{V' \ominus V}), \mathrm{Ran}(p_\infty \Pi_{V' \ominus V}), \mathrm{Id})$, a stabilization.

Independence of k: for $k' \geq k$ both $\Pi_k R \Pi_k$ and $\Pi_{k'} R \Pi_{k'}$ are close to R, so the Riesz projections of the operators on the segment between $p_\infty + \Pi_k R \Pi_k$ and $p_\infty + \Pi_{k'} R \Pi_{k'}$ form a family as in (*) with $V = \operatorname{Ran} \Pi_{k'}$, joining $Q^{(k)}$ to $Q^{(k')}$. With the stabilization remark this shows that $\delta(Q)$ does not depend on k. The same argument, with one k chosen uniformly on $[0,1] \times C$, gives (ii). For (i), $V \subset \operatorname{Ran} \Pi_k$, and Q commutes with Π_k and equals p_∞ on $\operatorname{Ran} \Pi_k \ominus V$; so $Q = Q^{(k)}$ and the triples for V and $\operatorname{Ran} \Pi_k$ differ by a stabilization. Additivity and naturality are clear. □

For X a point, $\delta(Q)$ is the relative dimension of the two projections, i.e. the index of $p_\infty \colon \operatorname{Ran} Q \to \operatorname{Ran} p_\infty$ when Q is self-adjoint.

13.10.4 The even quantization map

Proposition 13.10.6. *Let X be a manifold and p projection data on $\mathbb{R}^{2n} \times X$ with values in $M(N, \mathbb{C})$ and constant value p_∞. There is $\epsilon_0 > 0$ such that for $0 < \epsilon \leq \epsilon_0$ the self-adjoint operators*

$$P_\epsilon(x) = p_\infty + q_\epsilon(p(\cdot, x) - p_\infty)$$

satisfy $\|P_\epsilon^2 - P_\epsilon\| \leq \frac{1}{8}$; their spectral projections $Q_\epsilon = \mathcal{R}(P_\epsilon) = \chi_{(1/2,\infty)}(P_\epsilon)$ form a family as in Lemma 13.10.5, and

$$\mathrm{Ind}^{even}_{\mathrm{sl}}([p] - [p_\infty]) = \delta(Q_\epsilon) \in K^0_{\mathrm{c}}(X)$$

is independent of $\epsilon \in (0, \epsilon_0]$ and invariant under the moves of Lemma 13.10.4(i). It is additive under direct sums and natural. Via Lemma 13.10.4 it defines a homomorphism $\mathrm{Ind}^{even}_{\mathrm{sl}} \colon K^0_{\mathrm{c}}(\mathbb{R}^{2n} \times X) \to K^0_{\mathrm{c}}(X)$.

Proof. With $r = p - p_\infty$, $p^2 = p$ means $p_\infty r + r p_\infty + r^2 = r$, so $P_\epsilon^2 - P_\epsilon = q_\epsilon(r) q_\epsilon(r) - q_\epsilon(r^2) = O(\epsilon)$ uniformly in x (Lemma 13.10.2). For a self-adjoint P with $\|P^2 - P\| \leq \frac{1}{8}$ the spectrum lies in $\{|\lambda^2 - \lambda| \leq \frac{1}{8}\}$, which does not meet $\{|\lambda - \frac{1}{2}| \leq \frac{1}{2}\sqrt{1/2}\}$, so Γ is in the resolvent set and $\mathcal{R}(P)$ is the spectral projection of $(1/2, \infty)$. The proof of Lemma 13.10.5 shows that $Q_\epsilon - p_\infty \in \Psi^{-\infty}_{\mathrm{iso}}$ depends smoothly on (ϵ, x), and $Q_\epsilon = p_\infty$ where $r(\cdot, x) = 0$. Homotopies of p, and the parameter ϵ, give homotopies of Q_ϵ, so (ii) of Lemma 13.10.5 gives the invariance; stabilization and constant unitary conjugation pass through the construction. □

13.10.5 Clutching

Fix $\vartheta \in \mathcal{C}^\infty(\mathbb{R}; [0, \pi/2])$ with $\vartheta = 0$ on $(-\infty, -1]$ and $\vartheta = \pi/2$ on $[1, \infty)$. For an invertible element g of a unital algebra and $\tau \in \mathbb{R}$ let

$$S(\tau, g) = \begin{pmatrix} \cos\tau & -\sin\tau\, g^{-1} \\ \sin\tau\, g & \cos\tau \end{pmatrix}, \qquad S(\tau, g)^{-1} = S(-\tau, g),$$

and, with $\Pi_\infty = \operatorname{diag}(\mathrm{Id}, 0)$,

$$\Pi_g(t) = S(\vartheta(t), \mathrm{Id})^{-1} S(\vartheta(t), g)\, \Pi_\infty\, S(\vartheta(t), g)^{-1} S(\vartheta(t), \mathrm{Id}), \qquad t \in \mathbb{R}. \tag{13.10.4}$$

This is an idempotent; since $S(\tau, g)\Pi_\infty S(\tau, g)^{-1} = \begin{pmatrix} \cos^2\tau & \cos\tau\sin\tau\, g^{-1} \\ \cos\tau\sin\tau\, g & \sin^2\tau \end{pmatrix}$, its entries are affine functions of g and g^{-1} with coefficients depending on t, and a direct computation gives $\Pi_g(t) = \Pi_\infty$ for $|t| \geq 1$ and for $g = \mathrm{Id}$. If g is unitary, $S(\tau, g)$ is unitary and $\Pi_g(t)$ is an orthogonal projection.

Lemma 13.10.7. *(i) For a manifold M and $g \in \mathcal{C}^\infty(M; \mathrm{U}(N))$ equal to* Id *outside a compact set, Π_g is projection data on $\mathbb{R} \times M$, and $[\Pi_g] - [\Pi_\infty]$ is the class of the triple $(\mathbb{C}^N, \mathbb{C}^N, \tilde{g})$, $\tilde{g}(t, m) = \mathrm{Id}$ for $t \leq -1$, $\tilde{g}(t, m) = g(m)$ for $t \geq 1$. For compact M the map* clu: $K^{-1}(M) \to K_{\mathrm{c}}^0(\mathbb{R} \times M)$, $[g] \mapsto [\Pi_g] - [\Pi_\infty]$, *is the isomorphism of Lemma 13.9.4 ($n = 1$, base point $e = -1$).*

(ii) For $G \in \mathcal{C}^\infty(M; G_{\mathrm{iso}}^{-\infty}(\mathbb{R}^n; \mathbb{C}^N))$ equal to Id *outside a compact set,* (13.10.4) *with G in place of g is a family of idempotents as in Lemma 13.10.5 over $\mathbb{R} \times M$, and* $\mathrm{clu}[G] = \delta(\Pi_G)$ *extends* clu *to the isotropic model of $K_{\mathrm{c}}^{-1}(M)$.*

(iii) Quantization commutes with clutching: for $g \in \mathcal{C}^\infty(\mathbb{R}^{2n} \times M; \mathrm{U}(N))$ equal to Id *outside a compact set,*

$$\mathrm{Ind}_{\mathrm{sl}}^{even}([\Pi_g] - [\Pi_\infty]) = \mathrm{clu}\big(\mathrm{Ind}_{\mathrm{sl}}^{odd}[g]\big) \in K_{\mathrm{c}}^0(\mathbb{R} \times M),$$

where on the left Π_g is regarded as projection data on $\mathbb{R}^{2n} \times (\mathbb{R} \times M)$.

Proof. (i) The map $F(t)v = S(\vartheta, \mathrm{Id})^{-1} S(\vartheta, g)(v, 0)$ is a global isomorphism $\mathbb{C}^N \to \operatorname{Ran}\Pi_g$, and $\Pi_\infty F(t)v$ is the first component of $S(-\vartheta, \mathrm{Id})(\cos\vartheta\, v, \sin\vartheta\, gv)$, i.e. $(\cos^2\vartheta + \sin^2\vartheta\, g)v$. This is v for $t \leq -1$ and gv for $t \geq 1$; by (a), (b) of Definition 13.9.1 the triple of (13.10.3) is equivalent to $(\mathbb{C}^N, \mathbb{C}^N, \tilde{g})$, which is the image of $[g]$ under the inverse of the isomorphism of Lemma 13.9.4 (every class of $K^{-1}(M)$ has a unitary representative, by the polar decomposition $g \mapsto g(g^*g)^{-1/2}$ and the homotopy $g(g^*g)^{-s/2}$).

(ii) $\Pi_G - \Pi_\infty$ is affine in $G - \mathrm{Id}$ and $G^{-1} - \mathrm{Id}$, so smoothing. If G takes values in a subgroup $\mathrm{GL}(V)$ as in Remark 13.2.6, Π_G is reduced by $V \oplus V$ and Lemma 13.10.5(i) identifies $\delta(\Pi_G)$ with the class in (i); as every class has such a representative and δ is homotopy invariant, $\delta(\Pi_G)$ depends only on $[G]$.

(iii) Write $\Pi_g = \Pi_\infty + L_t(g - \mathrm{Id}) + L_t'(g^{-1} - \mathrm{Id})$, where L_t, L_t' place their argument, multiplied by functions of t, into blocks of a 2×2 matrix and then conjugate by the constant (in w) matrix $S(\vartheta(t), \mathrm{Id})$. The quantization q_ϵ acts in the variable $w \in \mathbb{R}^{2n}$ and commutes with L_t, L_t'. With $\alpha = g - \mathrm{Id}$, $\beta = g^{-1} - \mathrm{Id}$ and $G_\epsilon = \mathrm{Id} + q_\epsilon(\alpha)$,

$$P_\epsilon = \Pi_\infty + L_t(q_\epsilon(\alpha)) + L_t'(q_\epsilon(\beta)), \qquad \Pi_{G_\epsilon} = \Pi_\infty + L_t(q_\epsilon(\alpha)) + L_t'(G_\epsilon^{-1} - \mathrm{Id}).$$

As in the proof of Proposition 13.10.3, $G_\epsilon^{-1} - \mathrm{Id} - q_\epsilon(\beta) = -G_\epsilon^{-1}(q_\epsilon(\alpha)q_\epsilon(\beta) - q_\epsilon(\alpha\beta)) = O(\epsilon)$, so $\|P_\epsilon - \Pi_{G_\epsilon}\| = O(\epsilon)$ uniformly. For small ϵ the Riesz projections of the segment from P_ϵ to the idempotent Π_{G_ϵ} form a homotopy as in Lemma 13.10.5(ii) from Q_ϵ to Π_{G_ϵ}. Hence $\mathrm{Ind}_{\mathrm{sl}}^{even}([\Pi_g] - [\Pi_\infty]) = \delta(Q_\epsilon) = \delta(\Pi_{G_\epsilon}) = \mathrm{clu}[G_\epsilon]$. □

13.10.6 Bott periodicity

We identify the phase plane $\mathbb{R}^2$ (for $n=1$) with $\mathbb{C}$ by $w=(x,\xi)\mapsto x+i\xi$, and write $z=(x+i\xi)/|w|$ as in Section 13.7. Let

$$b_{\mathrm{iso}}=[(\mathbb{C},\mathbb{C},z)]\in K^0_{\mathrm{c}}(\mathbb{R}^2),\qquad b_{\mathrm{iso}}\cdot[E]=[(\varpi^*E,\varpi^*E,z)]\in K^0_{\mathrm{c}}(\mathbb{R}^2\times X)\tag{13.10.5}$$

for a bundle E over X. Since $(\varpi^*E,\varpi^*E,z)\oplus(\varpi^*E,\varpi^*E,\bar z)$ is equivalent to 0 (by the rotation homotopy of Lemma 13.1.2 joining $\mathrm{diag}(z,\bar z)$ to Id), $b_{\mathrm{iso}}\cdot[E]$ is minus the image of $[E]$ under the isomorphism of Theorem 13.9.5; so $[E]\mapsto b_{\mathrm{iso}}\cdot[E]$ extends to an isomorphism $K^0(X)\to K^0_{\mathrm{c}}(\mathbb{R}^2\times X)$. Note that b_{iso} is the clutching of the symbol z of the operator $\mathfrak{a}$ of index one.

Proposition 13.10.8. *Let X be compact and $n=1$. For every vector bundle E over X, $\mathrm{Ind}_{\mathrm{sl}}^{even}(b_{\mathrm{iso}}\cdot[E])=[E]$. Consequently $\mathrm{Ind}_{\mathrm{sl}}^{even}\colon K^0_{\mathrm{c}}(\mathbb{R}^2\times X)\to K^0(X)$ is the inverse of the isomorphism $[E]\mapsto b_{\mathrm{iso}}\cdot[E]$.*

Proof. Projection data for b_{iso}. Let $\varphi\in\mathcal{C}^\infty([0,\infty);[0,\pi/2])$ with $\varphi=0$ on $[0,1]$ and $\varphi=\pi/2$ on $[2,\infty)$, and let $P_B(w)$ be the orthogonal projection onto the line spanned by the unit vector $v(w)=(\cos\varphi(|w|),\sin\varphi(|w|)\,z)\in\mathbb{C}^2$, smooth since $\varphi=0$ near 0; $P_B=p_\infty=\mathrm{diag}(0,1)$ for $|w|\ge 2$. The section v trivializes $\operatorname{Ran}P_B$, and $p_\infty v=(0,\sin\varphi\, z)$, which is z times the basis vector of $\operatorname{Ran}p_\infty$ for $|w|\ge 2$. So $[P_B]-[p_\infty]=b_{\mathrm{iso}}$. If $E=\operatorname{Ran}e$ for a smooth family of orthogonal projections e on $\mathbb{C}^L$, then $\tilde P=P_B\otimes e+p_\infty\otimes(\mathrm{Id}-e)$ represents $b_{\mathrm{iso}}\cdot[E]$, as one sees by splitting $\mathbb{C}^L=E\oplus E^\perp$.

Charge. Let $H=D_x^2+x^2$ on $\mathbb{R}$. For $\zeta=(p,q)$ one computes $[D_x,\mathcal{W}(\zeta)]=p\mathcal{W}(\zeta)$, $[x,\mathcal{W}(\zeta)]=-q\mathcal{W}(\zeta)$, and from the kernels $\mathrm{Op^W}(\xi e^{i\langle\zeta,\cdot\rangle})=\frac12(D_x\mathcal{W}+\mathcal{W}D_x)$, $\mathrm{Op^W}(xe^{i\langle\zeta,\cdot\rangle})=\frac12(x\mathcal{W}+\mathcal{W}x)$. Hence $[H,\mathcal{W}(\zeta)]=\mathrm{Op^W}(2(p\xi-qx)e^{i\langle\zeta,\cdot\rangle})$, and by superposition

$$[H,\mathrm{Op^W}(c)]=\mathrm{Op^W}(2i\,\partial_\theta c),\qquad \partial_\theta=x\partial_\xi-\xi\partial_x,\qquad c\in\mathcal{S}(\mathbb{R}^2).$$

If $c=f(|w|)z^m$ (with f vanishing near 0 if $m\neq 0$), then $2i\partial_\theta c=-2mc$, so $H\,\mathrm{Op^W}(c)=\mathrm{Op^W}(c)(H-2m)$ and $\mathrm{Op^W}(c)$ maps u_k into $\mathbb{C}u_{k-m}$ ($u_j=0$ for $j<0$). The entries of P_B-p_∞ are of this form, with $m=0$ on the diagonal, $m=1$ in the lower left and $m=-1$ in the upper right entry, and dilation does not change m. Hence P_ϵ, p_∞ and therefore Q_ϵ are reduced by the subspaces

$$B_0=\mathbb{C}\,u_0\otimes e_1,\qquad B_k=\operatorname{span}\{u_k\otimes e_1,u_{k-1}\otimes e_2\},\ k\ge 1,$$

whose orthogonal sum is $L^2(\mathbb{R};\mathbb{C}^2)$.

Counting. For $\epsilon\le\epsilon_0$ the Riesz projection Q_ϵ depends continuously on ϵ in norm, so $\operatorname{rank}(Q_\epsilon|_{B_k})$ is independent of ϵ. By Lemma 13.10.2(iii), $P_\epsilon\to p_\infty+(P_B(0)-p_\infty)=\mathrm{diag}(1,0)$ strongly, hence in norm on each B_k, so $Q_\epsilon|_{B_k}$ tends to the projection onto $u_k\otimes e_1$, of rank one, for every $k\ge 0$. On the other hand $\operatorname{rank}(p_\infty|_{B_0})=0$ and $\operatorname{rank}(p_\infty|_{B_k})=1$ for $k\ge 1$. Since $Q_\epsilon-p_\infty$ is compact, $\|(Q_\epsilon-p_\infty)|_{V_K^\perp}\|<\frac12$ for $V_K=B_0\oplus\cdots\oplus B_K$ and K large; the Riesz projections of $p_\infty+(Q_\epsilon-p_\infty)(\Pi_{V_K}+s(\mathrm{Id}-\Pi_{V_K}))$, $s\in[0,1]$, join Q_ϵ to an idempotent which equals Q_ϵ on V_K and p_∞ on $V_K^\perp$. As $V_K\subset\operatorname{Ran}\Pi_K$ is p_∞-invariant, Lemma 13.10.5 gives $\delta(Q_\epsilon)=(K+1)-K=1$.

Twisting. $\tilde P_\epsilon=P_\epsilon\otimes e+p_\infty\otimes(\mathrm{Id}-e)$ commutes with $\mathrm{Id}\otimes e$, so its Riesz projection is $Q_\epsilon\otimes e+p_\infty\otimes(\mathrm{Id}-e)$. Applying the same homotopy in the first factor and Lemma 13.10.5(i) with $V=V_K\otimes\mathbb{C}^L$, δ is the class of the bundle $\operatorname{Ran}(Q_\epsilon|_{V_K})\otimes E\oplus\operatorname{Ran}(p_\infty|_{V_K})\otimes E^\perp$ minus that of $\operatorname{Ran}(p_\infty|_{V_K})\otimes\mathbb{C}^L$, that is $(K+1)[E]-K[E]=[E]$.

The last statement follows from Theorem 13.9.5, by which every class of $K^0_{\mathrm{c}}(\mathbb{R}^2\times X)$ is $b_{\mathrm{iso}}\cdot y$ for a unique $y\in K^0(X)$, and additivity. □

Theorem 13.10.9. *For every manifold X and every $n \geq 1$ the maps*

$$\mathrm{Ind}_{\mathrm{sl}}^{odd} \colon K_{\mathrm{c}}^{-1}(\mathbb{R}^{2n} \times X) \to K_{\mathrm{c}}^{-1}(X), \qquad \mathrm{Ind}_{\mathrm{sl}}^{even} \colon K_{\mathrm{c}}^{0}(\mathbb{R}^{2n} \times X) \to K_{\mathrm{c}}^{0}(X)$$

are isomorphisms. The inverse of $\mathrm{Ind}_{\mathrm{sl}}^{even}$ is exterior multiplication (as in Chapter 15) by $b_{\mathrm{iso}}^{\boxtimes n} \in K_{\mathrm{c}}^{0}(\mathbb{R}^{2n})$, the coordinates of $\mathbb{R}^{2n}$ being paired as (x_j, ξ_j); the maps commute with clutching (Lemma 13.10.7), and quantizing in stages, $\mathbb{R}^{2n} = \mathbb{R}^{2n'} \times \mathbb{R}^{2n''}$, gives the same maps.

Sketch of proof. For $n = 1$ and compact X the even statement is Proposition 13.10.8. The extension to non-compact X uses the index isomorphism with compact supports (Exercise 13.1) and the same computation, which is local in X. Given the even statement for all X, Lemma 13.10.7(iii) and the clutching isomorphism $K_{\mathrm{c}}^{-1}(M) \cong K_{\mathrm{c}}^{0}(\mathbb{R} \times M)$ (Lemma 13.9.4, extended to non-compact M in the same way) give the odd statement. For the stages, the Weyl quantization on $\mathbb{R}^{n'+n''}$ of a product $a'(w')a''(w'')$ is $\mathrm{Op^W}(a') \otimes \mathrm{Op^W}(a'')$, and the estimates of Lemma 13.10.2 show that quantizing first in w'' (with values in matrices) and then, after a finite rank approximation of the result as in Lemma 13.10.5, in w', gives a family homotopic to the one obtained by quantizing in w at once. Iterating the case $n = 1$ gives the general case. Complete proofs of the topological ingredients are in [17]. □

13.11 Complex and symplectic bundles; the Thom isomorphism

The quantization maps of Section 13.10 were defined for the trivial bundle $\mathbb{R}^{2n} \times X$ with its constant symplectic form. In the index theorem they are needed for a bundle of symplectic vector spaces over X, such as $N \oplus N^*$ for a real vector bundle N, and in the classical formulation for complex vector bundles. We first recall that the two notions are equivalent up to homotopy.

Lemma 13.11.1. *Let $V \to X$ be a real vector bundle of even rank. Hermitian forms are taken linear in the first and antilinear in the second variable.*

(i) If J is a complex structure on V ($J^2 = -\mathrm{Id}$) and h is a Hermitian metric for J, then $\omega = \Im h$ is a symplectic form on V, $\omega(Jv, Jw) = \omega(v, w)$, and $h(v, w) = \omega(Jv, w) + i\,\omega(v, w)$.

(ii) If ω is a symplectic form on V and g is a Euclidean metric, define J' by $\omega(J'v, w) = g(v, w)$. Then J' is g-skew-adjoint, $-J'^2$ is positive definite and commutes with J', and $J = (-J'^2)^{-1/2}J'$ is a complex structure with $\omega(Jv, Jw) = \omega(v, w)$ for which $h(v, w) = \omega(Jv, w) + i\,\omega(v, w)$ is a Hermitian metric.

Both constructions depend smoothly on parameters, and each followed by the other (with $g = \Re h$ in (ii)) returns the structure one started with. Since the choices of h and of g range over convex sets, homotopy classes of complex structures and of symplectic structures on V correspond bijectively; if the structures are constant outside a compact set (in a trivialization), so is the correspondence.

Proof. (i) From $h(Jv, w) = ih(v, w)$ we get $\omega(Jv, w) = \Im(ih(v, w)) = \Re h(v, w)$, which is symmetric and positive definite; so $h = \omega(J\cdot, \cdot) + i\omega$. The form ω is real and antisymmetric because $h(w, v) = \overline{h(v, w)}$, non-degenerate because $\omega(Jv, v) = h(v, v) > 0$ for $v \neq 0$, and J-invariant because $h(Jv, Jw) = h(v, w)$.

(ii) J' is invertible since ω and g are non-degenerate. Using the definition twice, $g(v,J'w)=\omega(J'v,J'w)=-\omega(J'w,J'v)=-g(w,J'v)=-g(J'v,w)$, so J' is skew-adjoint and $g(-J'^2v,v)=g(J'v,J'v)>0$. Let $A=(-J'^2)^{1/2}$, the positive square root, which commutes with J' and is self-adjoint; then $J=A^{-1}J'$ satisfies $J^2=A^{-2}J'^2=-\mathrm{Id}$, and $\omega(Jv,w)=\omega(J'A^{-1}v,w)=g(A^{-1}v,w)$ is symmetric and positive definite. Replacing w by Jw and using the skew-adjointness of J' one checks $\omega(Jv,Jw)=\omega(v,w)$, so h is Hermitian with $h(Jv,w)=ih(v,w)$. The square root of a positive definite matrix is a smooth (indeed real analytic) function of the matrix, which gives smoothness in parameters. If (J,h) is given and $\omega=\Im h$, $g=\Re h$, then $\omega(J'v,w)=g(v,w)=\omega(Jv,w)$ forces $J'=J$, $A=\mathrm{Id}$; conversely $\Im h=\omega$ in (ii). □

On $\mathbb{C}$ with $h(v,w)=v\bar w$ and $v=x+i\xi$, $\omega(v,w)=\Im(v\bar w)$ is the form $d\xi\wedge dx$ used for the phase plane in Chapter 7; thus the complex structure compatible with the standard symplectic form of $\mathbb{R}^{2n}=\{(x,\xi)\}$ is the identification $(x,\xi)\mapsto x+i\xi$ used in Section 13.10.

13.11.1 The Bott class of a complex bundle

Let $V\to X$ be a complex vector bundle with Hermitian metric and $\varpi\colon V\to X$ the projection. On Λ^*V let ι_v be the adjoint of exterior multiplication by v, so that $v\wedge\iota_v+\iota_v v\wedge=|v|^2$. Then $c(v)=v\wedge\cdot-\iota_v$ maps $\Lambda^{\mathrm{ev}}V_x$ to $\Lambda^{\mathrm{odd}}V_x$ and $c(v)^*c(v)=|v|^2$.

Definition 13.11.2. The *Bott class* (or Thom class) of V is

$$\beta_V=\left[(\varpi^*\Lambda^{\mathrm{ev}}V,\ \varpi^*\Lambda^{\mathrm{odd}}V,\ c)\right]\in K^0_{\mathrm{c}}(V),$$

for compact X; for a bundle E over X we write $\beta_V\cdot[E]=[(\varpi^*(\Lambda^{\mathrm{ev}}V\otimes E),\varpi^*(\Lambda^{\mathrm{odd}}V\otimes E),c\otimes\mathrm{Id})]$.

For $V=X\times\mathbb{C}$, $c(v)1=v$ and $\beta_V\cdot[E]=[(\varpi^*E,\varpi^*E,v)]=b_{\mathrm{iso}}\cdot[E]$, since v and $v/|v|$ are homotopic outside the zero section. For $V=V_1\oplus V_2$, $\Lambda V=\Lambda V_1\otimes\Lambda V_2$ as graded bundles and $c(v_1+v_2)$ is the graded (Clifford) combination of $c(v_1)$ and $c(v_2)$; so $\beta_{V_1\oplus V_2}$ is the product of $\varpi_1^*\beta_{V_1}$ and $\varpi_2^*\beta_{V_2}$ in the sense of Chapter 15, and $\beta_{X\times\mathbb{C}^n}=b_{\mathrm{iso}}^{\boxtimes n}$. The class $\lambda_{\mathcal{V}}$ used in Chapter 15 for a real bundle $\mathcal{V}$ is $\beta_{\mathcal{V}\otimes\mathbb{C}}$.

Theorem 13.11.3 (Thom isomorphism)**.** *Let X be compact and $V\to X$ a Hermitian complex vector bundle of rank n. Then*

$$K^0(X)\ni[E]-[F]\longmapsto\beta_V\cdot[E]-\beta_V\cdot[F]\in K^0_{\mathrm{c}}(V)$$

is an isomorphism; denote its inverse by $\mathrm{Thom}_V\colon K^0_{\mathrm{c}}(V)\to K^0(X)$. *If $V=X\times\mathbb{C}^n$ with the standard structure, then* $\mathrm{Thom}_V=\mathrm{Ind}_{\mathrm{sl}}^{even}$, *the even quantization map of Proposition 13.10.6 for $\mathbb{C}^n=\mathbb{R}^{2n}$.*

Proof and references. For $V=X\times\mathbb{C}$ this is Proposition 13.10.8 together with Theorem 13.9.5. For trivial V of rank n it follows from $\beta_V=b_{\mathrm{iso}}^{\boxtimes n}$ and Theorem 13.10.9. For general V choose a complement V', $V\oplus V'\cong X\times\mathbb{C}^m$ (Lemma 13.2.1); multiplicativity of the Bott class and the trivial case, applied over the total space of V (a non-compact base, as in Theorem 13.10.9), show that $\beta_V\cdot$ has a left and a right inverse. We do not give the details, which are standard; see [17], §2.7. □

Remark 13.11.4 (Thom isomorphism by quantization). For a symplectic bundle (V,ω) over X, Thom_V is again given by fibrewise quantization. Choose a compatible complex structure and Hermitian metric (Lemma 13.11.1). Unitary frames identify the fibres

symplectically with $\mathbb{R}^{2n}$, and two such frames differ by an element U of the unitary group $\mathrm{U}(n) \subset \mathrm{Sp}(2n,\mathbb{R})$. By the symplectic invariance of the Weyl quantization (Chapter 7), $\mathrm{Op}^{\mathrm{W}}(a\circ U) = M_U^{-1}\,\mathrm{Op}^{\mathrm{W}}(a)M_U$ for a unitary operator M_U determined up to a constant factor; since $|w|^2\circ U = |w|^2$, M_U commutes with the harmonic oscillator and preserves its eigenspaces. Hence the constructions of Lemma 13.10.5 and Proposition 13.10.6, carried out in local unitary frames, are compatible on overlaps up to conjugation by such operators, which does not change the finite rank bundles up to isomorphism. They define $\mathrm{Ind}^{\mathrm{even}}_{\mathrm{sl},V}\colon K^0_{\mathrm{c}}(V)\to K^0(X)$ (and similarly an odd map $K^{-1}_{\mathrm{c}}(V)\to K^{-1}(X)$), and $\mathrm{Ind}^{\mathrm{even}}_{\mathrm{sl},V}(\beta_V\cdot[E]) = [E]$ by the computation of Proposition 13.10.8 in each factor; thus $\mathrm{Ind}^{\mathrm{even}}_{\mathrm{sl},V} = \mathrm{Thom}_V$. To make the patching canonical one uses the Fock space of each fibre, on which $\mathrm{U}(n)$ acts by a genuine representation; we omit the details.

13.12 The analytic index map

Let Z be a compact manifold and $\varpi\colon T^*Z\to Z$. An elliptic operator $P\in\Psi^0(Z;E_+,E_-)$ is Fredholm, and its index

$$\operatorname{ind} P = \dim\operatorname{Nul} P - \dim\left(\mathcal{C}^\infty(Z;E_-)/P\mathcal{C}^\infty(Z;E_+)\right)$$

is a homotopy invariant (Chapter 12), hence depends only on $E_\pm$ and the homotopy class of $\sigma_0(P)\in\mathcal{C}^\infty(S^*Z;\mathrm{Iso}(E_+,E_-))$. The principal symbol, extended homogeneously of degree 0 to $T^*Z\setminus 0$, defines a triple $(\varpi^*E_+,\varpi^*E_-,\sigma_0(P))$ and so a class in $K^0_{\mathrm{c}}(T^*Z)$.

Proposition 13.12.1. *There is a unique homomorphism* $\mathrm{ind}_a\colon K^0_{\mathrm{c}}(T^*Z)\to\mathbb{Z}$*, the* analytic index*, with* $\operatorname{ind}P = \mathrm{ind}_a[(\varpi^*E_+,\varpi^*E_-,\sigma_0(P))]$ *for every elliptic* $P\in\Psi^0(Z;E_+,E_-)$.

Proof. Every class is of this form. By Lemma 13.2.2(ii) and (a) a class is represented by $(\varpi^*E_+,\varpi^*E_-,a)$ with $E_\pm$ bundles over Z and a an isomorphism for $|\zeta|\geq R$ (with respect to some metric). The homotopy (c) $a_s(\zeta) = a(((1-s)|\zeta|+sR)\zeta/|\zeta|)$, $|\zeta|\geq R$, ends at a map homogeneous of degree 0, i.e. at the symbol σ of some elliptic $P\in\Psi^0(Z;E_+,E_-)$. Define ind_a of the class to be $\operatorname{ind}P$.

This does not depend on the choices: by homotopy invariance $\operatorname{ind}P$ depends only on the homotopy class of σ, which takes care of R, of (b) and of (c). Under (d), P is replaced by $P\oplus\mathrm{Id}_H$, with the same index. Under (a), bundle isomorphisms $\phi_\pm$ of $\varpi^*E_\pm$ are homotopic, through $\phi_\pm(t\zeta)$, to the pull-backs of their restrictions $\phi^0_\pm$ to the zero section; so σ is replaced by a symbol homotopic to $\phi^0_-\sigma(\phi^0_+)^{-1}$, which is the symbol of $\phi^0_-P(\phi^0_+)^{-1}$, an operator with the same index since $\phi^0_\pm$ are invertible. Additivity follows from $\operatorname{ind}(P\oplus P') = \operatorname{ind}P+\operatorname{ind}P'$, and uniqueness from the fact that every class arises from an elliptic operator. □

The problem posed by Gelfand around 1960 was to compute this map topologically. The answer of Atiyah and Singer is to define a second, *topological*, index map by embedding Z into a Euclidean space (or a sphere), constructing on the larger space an operator with the same index and with symbol derived from that of P, and using the fact, essentially Bott periodicity, that for suitable operators on Euclidean space the index map is an isomorphism; the two maps are then shown to coincide. This is taken up in Chapter 15.

13.13 Exercises

Exercise 13.1. Show that the index isomorphism of Theorem 13.6.2 carries over to non-compact manifolds: for every manifold M there is an isomorphism $K^{-2}_{\mathrm{c}}(M)\cong K^0_{\mathrm{c}}(M)$. (Hint: represent elements of $K^{-2}_{\mathrm{c}}(M)$ by loops equal to Id outside $K\times\mathbb{S}$, choose a compact

manifold with boundary $M' \Subset M$ containing a neighbourhood of K, and apply the families index over M', observing that the index class comes with a canonical trivialization over $M \setminus K$.)

Exercise 13.2. Show directly that the Toeplitz operator T_{z^k} has index $-k$. For a smooth family of projections p on $\mathbb{C}^N$ over X show that $pL + (\mathrm{Id} - p)$ is surjective with null space the constant sections of $\operatorname{Ran} p$, and compute $\operatorname{ind}(pL + \mathrm{Id} - p) = [\operatorname{Ran} p]$ directly from Definition 13.5.4 with $k = 0$.

Exercise 13.3. Show that for pointed loops a, b over X the families index satisfies $\operatorname{ind}(T_a T_b) = \operatorname{ind}(T_{ab}) = \operatorname{ind}(a) + \operatorname{ind}(b)$. (Use Lemma 13.1.2 and the fact that $T_a T_b - T_{ab}$ has vanishing symbol.)

Exercise 13.4. Let $a(\theta) = \operatorname{diag}(e^{2i\theta}, e^{-i\theta})$. Carry out the steps of the proof of Theorem 13.3.1 for a and find the resulting normal form; check that its index is -1 in accordance with Corollary 13.4.2.

Exercise 13.5. Show that $K^0_{\mathrm{c}}(\mathbb{R}) = 0$ and $K^0_{\mathrm{c}}(\mathbb{R}^2) \cong \mathbb{Z}$, generated by the Bott class $[(\mathbb{C}, \mathbb{C}, \bar{w}/|w|)]$.

Exercise 13.6. Let $H \to \mathbb{S}^2 = \mathbb{C} \cup \{\infty\}$ be the line bundle obtained by clutching the trivial bundles over the two hemispheres with the map $z \mapsto z$ on the equator (this is the tautological bundle over $\mathbb{CP}^1$, up to conjugation). Show that $[H] - [\mathbb{C}]$ generates $\tilde{K}^0(\mathbb{S}^2) \cong \mathbb{Z}$, and prove that $(H \otimes H) \oplus \mathbb{C} \cong H \oplus H$ by comparing clutching functions and using Lemma 13.1.2.

Exercise 13.7. Show that if $A \colon X \to \Psi^0_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^N)$ is a smooth family of elliptic Toeplitz operators whose null spaces have constant dimension, then $\operatorname{Nul} A$ and $\operatorname{Nul} A^*$ are smooth vector bundles and $\operatorname{ind}(A) = [\operatorname{Nul} A] - [\operatorname{Nul} A^*]$. (Use the generalized inverse to construct local frames, then Lemma 13.7.2.) Prove the same for isotropic families.

Exercise 13.8. Let $\mathrm{U}^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n) = \{\mathrm{Id} + A \in G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n);\ (\mathrm{Id} + A)^{-1} = \mathrm{Id} + A^*\}$. Show that for $g = \mathrm{Id} + A \in G^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$ the operator $(g^*g)^{-1/2} - \mathrm{Id}$ is smoothing (write $g^*g = \mathrm{Id} + B$ with B smoothing and self-adjoint, and use the Cauchy integral for $\lambda^{-1/2}$ together with the resolvent identities, as in the proof of Lemma 13.10.5), and deduce that $g \mapsto g(g^*g)^{-s/2}$, $s \in [0, 1]$, deforms compactly supported maps into $G^{-\infty}_{\mathrm{iso}}$ to maps into $\mathrm{U}^{-\infty}_{\mathrm{iso}}$. Conclude that $K_{\mathrm{c}}^{-1}(X)$ is the set of homotopy classes of compactly supported maps into $\mathrm{U}^{-\infty}_{\mathrm{iso}}(\mathbb{R}^n)$.

Exercise 13.9. Prove directly that $\mathrm{GL}(N, \mathbb{C})$ is connected, by induction on N: an invertible a is joined to one with $ae_1 = e_1$ by moving ae_1 to e_1 in $\mathbb{C}^N \setminus \{0\}$ while keeping the other columns linearly independent of it. Show also that a loop a in $\mathrm{GL}(N, \mathbb{C})$ with $\operatorname{wind} \det a = 0$ is null-homotopic: approximate a by a trigonometric polynomial, deform it so that $e_1^* a(\theta) e_1$ never vanishes, and reduce to $\mathrm{GL}(N-1, \mathbb{C})$. Compare with Corollary 13.6.4, where the stable version is deduced from the index.

Exercise 13.10. Use the commutation relation $[H, \mathrm{Op^W}(c)] = \mathrm{Op^W}(2i\partial_\theta c)$ from the proof of Proposition 13.10.8 to show that for $f \in \mathcal{S}(\mathbb{R})$ the operator $\mathrm{Op^W}(f(x^2 + \xi^2))$ on $L^2(\mathbb{R})$ is diagonal in the Hermite basis. Compute its eigenvalues for $f(s) = e^{-s}$, and show that the projection onto $\mathbb{C}u_0$ has Weyl symbol $2e^{-(x^2+\xi^2)}$.

Exercise 13.11. Let Q, P be orthogonal projections on a Hilbert space with $Q - P$ compact. Show that $P|_{\operatorname{Ran} Q} \colon \operatorname{Ran} Q \to \operatorname{Ran} P$ is Fredholm, that its index is unchanged under norm continuous deformations of Q with $Q - P$ compact, and that it equals $\operatorname{rank}(Q|_V) - \operatorname{rank}(P|_V)$ whenever V is a finite dimensional subspace reducing both P and Q with $Q = P$ on $V^\perp$. Deduce that $\delta(Q)$ in Lemma 13.10.5 is this index when X is a point.

Exercise 13.12. For $a \in \mathcal{C}^\infty(X; \mathrm{GL}(N,\mathbb{C}))$, X compact, let E_a be the bundle over $\mathbb{S} \times X$ whose smooth sections are the $s \in \mathcal{C}^\infty(\mathbb{R}\times X; \mathbb{C}^N)$ with $s(t+2\pi, x) = a(x)s(t,x)$. Show that E_a is a smooth vector bundle, that $E_a \oplus E_{a^{-1}}$ is trivial, and that $[E_a] - [\mathbb{C}^N] \in K^0(\mathbb{S}\times X)$ is the image of $\mathrm{clu}[a] \in K_c^0(\mathbb{R}\times X)$ (Lemma 13.10.7) under extension from $(0,2\pi)\times X$, up to sign.

Notes

The linearization of loops (Section 13.3) is the heart of Atiyah's proof of Bott periodicity via clutching functions [17]; the use of the families index of Toeplitz operators to invert the Bott map follows the source lectures and is closely related to the treatment of Bott periodicity through the Grassmannian of a polarized Hilbert space in [21]. Compared to the source we have replaced the residue argument for linear loops by the Riesz spectral projection (Lemma 13.3.4), proved that a normal form loop is trivial when the ranges of $\pi_\pm$ are isomorphic in a single lemma (Lemma 13.3.6), defined the families index through arbitrary smooth complements of the range, corrected the sign of the Bott map (which, with $\operatorname{ind} U = -1$, sends $[E_+] - [E_-]$ to the class with $p_+ z^{-1}$), corrected the second curve in the deformation of $\mathrm{Id} - P_0 + P_0 h P_0$ and normalized it at $1 \in \mathbb{S}$, and derived the homotopy groups of $G^{-\infty}$ from the index isomorphism for compact spheres together with the clutching construction, instead of from the (unproved) non-compact index isomorphism.

Sections 13.7, 13.10 and 13.11 and Lemma 13.2.7 come from the K-theory chapter of the 2008 revision of the lectures (*From microlocal to global analysis*), which starts from the residual isotropic group as the model for odd K-theory (Remark 13.2.6). There the injectivity of the isotropic index map is proved by Atiyah's rotation argument, which we only sketch (Remark 13.7.9); instead we combine the isotropic index with the linearization of loops, so that the only analytic input is the normalized annihilation operator (Lemma 13.7.3). We orient the circle at infinity counterclockwise in the (x,ξ)-plane, so that $\operatorname{ind} A = +\operatorname{wind}\sigma_0(A)$; the source writes $-$ wind, with the opposite orientation. The semiclassical quantization maps are only stated in the source; we have defined them through the rescaled Weyl quantization, proved the basic estimate with the Weyl operators (Lemma 13.10.2), given the relative class of projections (Lemma 13.10.5), proved that quantization commutes with clutching using a simplified clutching projection (13.10.4), and computed the even map for $n = 1$ explicitly by means of the rotation invariance of the harmonic oscillator (Proposition 13.10.8); the general case and the Thom isomorphism for non-trivial bundles are only sketched, with reference to [17]. The source's sections on Chern forms, the Chern character and the Todd class are incorporated in Chapter 15; its sections on stabilization, delooping and looping sequences, C^*-algebras and the norm closure of Ψ^0 are empty and have been omitted. The isotropic proof of Bott periodicity via the index of the harmonic oscillator is the approach of M. F. Atiyah, *Bott periodicity and the index of elliptic operators*, Quart. J. Math. Oxford **19** (1968), 113–140.

The source conjectures that $G^0_{\mathcal{T}}(\mathbb{S}, Y; E)$ is actually contractible. Since it is an open subset of an affine Fréchet space (Lemma 13.8.4), it has the homotopy type of a CW complex by a theorem of R. Palais (Homotopy theory of infinite dimensional manifolds, Topology 5 (1966)), and a weakly contractible CW complex is contractible by Whitehead's theorem; we have not checked the details of this argument in the present setting.

Chapter 14

Fibrations and product-type operators

Many problems in index theory involve two kinds of operators at once: operators acting along the fibres of a fibration, and operators acting on its total space. The proof of the index theorem in Chapter 15 deforms an operator built from both kinds, and needs a single algebra in which such deformations make sense. Neither the ordinary pseudodifferential operators on M nor the fibrewise families are large enough. This chapter builds the algebra and its two symbols.

The families version of the Atiyah–Singer index theorem concerns elliptic operators on the fibres of a fibration $\phi : M \longrightarrow B$; rather than a family $A_b \in \Psi^m(Z; E, F)$ simply parametrized by $b \in B$, the manifold on which A_b acts is allowed to be 'twisted' by diffeomorphisms of the fibre Z as b varies. This chapter sets up the analytic machinery needed for that theorem and for the index formulae of Chapter 15 and Chapter 17.

In the first three sections we define fibrations, the space $\Psi^m(M/B; E, F)$ of pseudodifferential operators acting on the fibres, and the index bundle $\mathrm{ind}_a(A) \in K^0(B)$ of an elliptic family. The rest of the chapter constructs a larger algebra, the *product-type* pseudodifferential operators $\Psi^{m,m'}_{\phi\text{-pt}}(M; E, F)$, which contains both the ordinary pseudodifferential operators on the total space M and the fibrewise families. Their kernels are conormal, in a 'product' sense, to the nested pair of submanifolds $\mathrm{Diag}_M \subset M^2_\phi \subset M^2$, where M^2_ϕ is the fibre product. Following the pattern of Chapter 4, these kernels are defined as inverse Fourier transforms of symbols, now smooth (up to weights) on a compactification of a vector space W relative to a subspace V; this *relative compactification* turns out to be a product of two radial compactifications. A product-type operator has two symbols: the usual principal symbol, defined on a blown-up cosphere bundle, and a *base symbol*, which is a family of pseudodifferential operators on the fibres parametrized by $T^*B \setminus 0$. If both are invertible the operator is *fully elliptic*, has a parametrix modulo smoothing operators, and is Fredholm.

We use freely the radial compactification of a vector space (Chapter 1), conormal distributions and their symbols (Chapter 4), pseudodifferential operators on compact manifolds and their Fredholm properties (Chapter 5), and the elementary K-theory of Chapter 12 and Chapter 13. All manifolds are without boundary unless otherwise stated, and all vector bundles are complex.

14.1 Fibrations

Definition 14.1.1. A *fibration* (of compact manifolds) is a smooth surjective map $\phi : M \longrightarrow B$ between compact manifolds which is a *submersion*, i.e. $\phi_* : T_mM \longrightarrow T_{\phi(m)}B$ is

surjective for every $m \in M$. The *fibre* over $b \in B$ is $Z_b = \phi^{-1}(b)$ (Figure 14.1). We indicate a fibration by the diagram $Z — M \xrightarrow{\phi} B$, where Z is the typical fibre (see below).

If M is compact, B is connected and ϕ is a submersion, then ϕ is automatically surjective (if $M \neq \emptyset$): its range is open, since ϕ is a submersion, and closed, since M is compact.

Theorem 14.1.2 (Local triviality)**.** *Let $\phi : M \longrightarrow B$ be a fibration.*

(1) *Each fibre Z_b is a compact embedded submanifold of M of dimension $\dim M - \dim B$; if B is connected all fibres are diffeomorphic to a fixed compact manifold Z.*

(2) *Each $b \in B$ has an open neighbourhood U_b with a diffeomorphism $f_b : \phi^{-1}(U_b) \longrightarrow Z \times U_b$ such that $\pi_U \circ f_b = \phi$, where $\pi_U : Z \times U_b \longrightarrow U_b$ is the projection.*

(3) *For two such trivializations, $f_{b'} \circ f_b^{-1}$ is a diffeomorphism of $Z \times (U_b \cap U_{b'})$ of the form $(z, c) \longmapsto (f_{b'b}(c)(z), c)$ where $f_{b'b} : U_b \cap U_{b'} \longrightarrow \mathrm{Dfm}(Z)$ is a smooth family of diffeomorphisms of Z (meaning that $(z, c) \mapsto f_{b'b}(c)(z)$ is smooth), and the cocycle condition $f_{b''b'} f_{b'b} = f_{b''b}$ holds on $U_b \cap U_{b'} \cap U_{b''}$.*

Proof. If $t_1, \dots, t_p$ are local coordinates on B centred at b, the functions $\phi^* t_i$ have linearly independent differentials at each point of Z_b because ϕ is a submersion, so by the implicit function theorem $Z_b = \{\phi^* t_1 = \cdots = \phi^* t_p = 0\}$ is an embedded submanifold of codimension $p = \dim B$; it is compact, being closed in M.

Choose a Riemannian metric on M and let $H_m \subset T_m M$ be the orthogonal complement of $\ker \phi_* = T_m Z_{\phi(m)}$. Then $\phi_* : H_m \longrightarrow T_{\phi(m)} B$ is an isomorphism and depends smoothly on m, so each vector field v on an open set $U \subset B$ has a unique smooth 'horizontal lift' $\tilde{v}$ to $\phi^{-1}(U)$, with $\tilde{v}(m) \in H_m$ and $\phi_* \tilde{v} = v$. Let U_b be a coordinate ball, $\{|t| < r\}$, centred at b. For $c \in U_b$ with coordinates $t(c)$ consider the constant-coefficient vector field $v_c = \sum_i t_i(c) \partial_{t_i}$ and its horizontal lift $\tilde{v}_c$. The integral curve of v_c starting at b is $s \mapsto st(c)$, which stays in the compact set $\{|t| \leq |t(c)|\} \subset U_b$ for $s \in [0, 1]$; hence the integral curves of $\tilde{v}_c$ starting on Z_b stay in the compact set $\phi^{-1}(\{|t| \leq |t(c)|\})$ and exist up to time 1. Let $F(z, c) = \exp(\tilde{v}_c)(z)$ be the time-one flow. By smooth dependence of solutions of ordinary differential equations on parameters, $F : Z_b \times U_b \longrightarrow \phi^{-1}(U_b)$ is smooth, and $\phi(F(z, c)) = c$. It is a bijection with smooth inverse $m \longmapsto (\exp(-\tilde{v}_{\phi(m)})(m), \phi(m))$, so $f_b = F^{-1}$ is a diffeomorphism as in (2), with Z replaced by Z_b. In particular the fibres over U_b are all diffeomorphic to Z_b; since B is connected, the set of $c \in B$ with Z_c diffeomorphic to a given fibre is open and closed, which proves (1) and (2) with a fixed Z. Part (3) is immediate from (2): $f_{b'} \circ f_b^{-1}$ commutes with the projections to $U_b \cap U_{b'}$, so has the stated form, and the cocycle condition is just associativity of composition. □

Conversely, M is recovered from the trivializations and the transition maps $f_{b'b}$; thus a fibration is a fibre bundle with fibre Z and structure group $\mathrm{Dfm}(Z)$. If the transition maps can be chosen to take values in a subgroup $G \subset \mathrm{Dfm}(Z)$, the structure group is said to be *reduced* to G.

Proposition 14.1.3 (Pull-back)**.** *Let $\phi : M \longrightarrow B$ be a fibration and $F : \tilde{B} \longrightarrow B$ a smooth map from a compact manifold. Then $\tilde{M} = \{(m, \tilde{b}) \in M \times \tilde{B}; \phi(m) = F(\tilde{b})\}$ is an embedded submanifold of $M \times \tilde{B}$ and $\tilde{\phi} : \tilde{M} \longrightarrow \tilde{B}$, $\tilde{\phi}(m, \tilde{b}) = \tilde{b}$, is a fibration with fibre $\tilde{\phi}^{-1}(\tilde{b}) = Z_{F(\tilde{b})}$.*

Proof. $\tilde{M}$ is the preimage of Diag_B under $\Phi = \phi \times F : M \times \tilde{B} \longrightarrow B \times B$. Since ϕ is a submersion, the range of Φ_* contains $TB \times \{0\}$, and $TB \times \{0\}$ together with $T\,\mathrm{Diag}_B$ spans $T(B \times B)$; so Φ is transversal to Diag_B and $\tilde{M}$ is an embedded submanifold of codimension

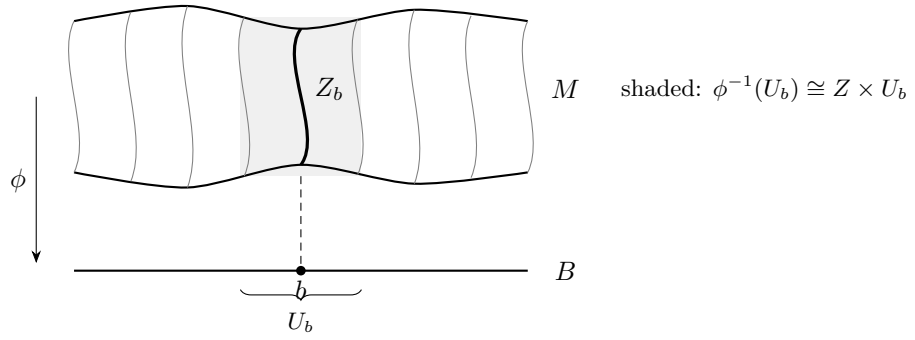


Figure 14.1: A fibration $\phi : M \to B$. Each fibre $Z_b = \phi^{-1}(b)$ is a compact manifold diffeomorphic to a fixed Z, and over a small open set U_b the total space is a product (Theorem 14.1.2). Globally the fibres may be glued together by diffeomorphisms, so M need not be $Z \times B$.

$\dim B$, compact since it is closed. Its tangent space is $\{(v, w); \phi_* v = F_* w\}$; given $w \in T_{\tilde{b}}\tilde{B}$ we can solve $\phi_* v = F_* w$, so $\tilde{\phi}$ is a submersion, and it is surjective because F is defined on all of $\tilde{B}$ and ϕ is surjective. □

Proposition 14.1.4 (Composites and products)**.** *If $\phi' : M' \longrightarrow M$ and $\phi : M \longrightarrow B$ are fibrations then $\phi \circ \phi' : M' \longrightarrow B$ is a fibration; its fibre over b is $(\phi')^{-1}(Z_b)$, which is itself the total space of the fibration $\phi'| : (\phi')^{-1}(Z_b) \longrightarrow Z_b$ obtained by restriction. If $\phi_i : M_i \longrightarrow B_i$, $i = 1, 2$, are fibrations then so is $\phi_1 \times \phi_2 : M_1 \times M_2 \longrightarrow B_1 \times B_2$.*

Proof. Composites and products of surjective submersions are surjective submersions. The restriction of ϕ' to $(\phi')^{-1}(Z_b)$ is the pull-back of ϕ' under the inclusion $Z_b \hookrightarrow M$, so is a fibration by Proposition 14.1.3. □

Note that the fibre of $\phi \circ \phi'$ need not be diffeomorphic to the product $Z \times Z'$ of the fibres; it is in general a non-trivial fibration over Z.

14.1.1 Vertical bundles and connections

The *vertical tangent bundle* $T(M/B) \subset TM$ is the null bundle of ϕ_*, so $T_m(M/B) = T_m Z_{\phi(m)}$, and there are short exact sequences of vector bundles over M

$$0 \longrightarrow T(M/B) \longrightarrow TM \xrightarrow{\phi_*} \phi^* TB \longrightarrow 0, \qquad 0 \longrightarrow \phi^* T^* B \xrightarrow{\phi^*} T^* M \longrightarrow T^*(M/B) \longrightarrow 0, \tag{14.1.1}$$

where $T^*(M/B)$ is the dual of $T(M/B)$. The second sequence identifies $\phi^* T^* B$ with the subbundle of covectors annihilating the fibres. We denote by $S^*(M/B)$ and $\phi^* S^* B$ the corresponding sphere bundles (quotients of the complements of the zero sections by $\mathbb{R}_+$).

Definition 14.1.5. A *connection* on the fibration ϕ is a choice of a smooth subbundle $H \subset TM$ complementary to $T(M/B)$. Then $\phi_* : H \longrightarrow \phi^* TB$ is an isomorphism, so a connection is the same as a splitting of the first sequence in (14.1.1), and dually of the second.

The proof of Theorem 14.1.2 used the connection given by the orthogonal complement for a Riemannian metric, so connections always exist.

The *fibre density bundle* $\Omega(M/B) = \Omega(T(M/B))$ is the bundle of densities on the fibres. From (14.1.1) there is a canonical isomorphism $\Omega(M) \cong \Omega(M/B) \otimes \phi^* \Omega(B)$; in particular a smooth positive section of $\Omega(M/B)$ restricts to a smooth positive density on each fibre.

14.1.2 The fibre product

Applying Proposition 14.1.3 with $F = \phi$ gives the *fibre product*

$$M^2_\phi = M \times_\phi M = \{(m, m') \in M^2; \phi(m) = \phi(m')\},$$

an embedded submanifold of M^2 of codimension $b = \dim B$. It fibres over M (by either projection) with fibre Z, and over B with fibre Z^2. It contains the diagonal, so we have a nested pair of embedded submanifolds

$$\mathrm{Diag}_M \subset M^2_\phi \subset M^2. \tag{14.1.2}$$

If (z, y) are local coordinates on M in which $\phi(z, y) = y$ (*fibred coordinates*), and (z', y') is a second copy, then near a point of M^2_ϕ

$$\mathrm{Diag}_M = \{z = z',\ y = y'\} \subset M^2_\phi = \{y = y'\} \subset M^2. \tag{14.1.3}$$

We write $\pi_L, \pi_R : M^2 \longrightarrow M$ for the projections and also for their restrictions to M^2_ϕ, and set $\Omega_R = \pi_R^* \Omega(M/B)$ over M^2_ϕ; for bundles E, F over M we write $\mathrm{Hom}(E, F) = \pi_L^* F \otimes \pi_R^* E^*$ over M^2 or M^2_ϕ.

Lemma 14.1.6. *For a fibration $\phi : M \longrightarrow B$:*

(i) *the conormal bundle of the fibre diagonal is $N^*_{(m,m')} M^2_\phi = \{(\phi^*\eta, -\phi^*\eta); \eta \in T^*_{\phi(m)} B\}$, so $N^* M^2_\phi \cong (\phi \circ \pi_L)^* T^* B$ and its sphere bundle is the fibre product $SN^* M^2_\phi \cong M^2_\phi \times_B S^* B$, which is the fibre product of the pulled-back fibration $\pi^* M = M \times_B S^* B \longrightarrow S^* B$, $\pi : S^* B \longrightarrow B$;*

(ii) *$N^* \mathrm{Diag}_M \cong T^* M$, by $(\xi, -\xi) \longmapsto \xi$;*

(iii) *under* (ii)*, the restriction $N^* M^2_\phi|_{\mathrm{Diag}_M}$ is identified with $\phi^* T^* B \subset T^* M$, and the quotient*

$$N^* \mathrm{Diag}_M / N^* M^2_\phi|_{\mathrm{Diag}_M}$$

with $T^*(M/B)$, the conormal bundle of Diag_M as a submanifold of M^2_ϕ.

Proof. M^2_ϕ is the preimage of Diag_B under $\phi \times \phi$, a submersion, so its conormal space at (m, m') is the pull-back of $N^* \mathrm{Diag}_B = \{(\eta, -\eta)\}$. This gives (i); the description of the sphere bundle follows. Part (ii) is standard, and (iii) follows by comparing (i) and (ii) with (14.1.1); in the coordinates (14.1.3) it says that $N^* M^2_\phi$ is spanned by the $dy_j - dy'_j$. □

14.2 Families of pseudodifferential operators

Throughout, $\phi : M \longrightarrow B$ is a fibration with typical fibre Z, and we set $f = \dim Z$, $b = \dim B$. One could define operators on the fibres by patching together, using Theorem 14.1.2, smooth families of operators on Z, relying on the invariance of pseudodifferential operators under diffeomorphisms. It is more direct to use conormal distributions on the fibre product. Recall from Chapter 4 and Chapter 5 that $\Psi^m(X; E, F) = I^m(X^2, \mathrm{Diag}; \mathrm{Hom}(E, F) \otimes \Omega_R)$, with the order normalization of Chapter 4 (for which a conormal distribution of order m at a submanifold of codimension c in a manifold of dimension n has a symbol of order $m + \frac{n}{4} - \frac{c}{2}$).

Definition 14.2.1. The space of *pseudodifferential operators on the fibres* of ϕ, acting from sections of E to sections of F, is

$$\Psi^m(M/B; E, F) = I^{m - \frac{b}{4}}(M^2_\phi, \mathrm{Diag}_M; \mathrm{Hom}(E, F) \otimes \Omega_R).$$

The shift in order is such that the symbols have order m: $\dim M_\phi^2 = 2f + b$ and Diag_M has codimension f in M_ϕ^2, so the symbol order is $m - \frac{b}{4} + \frac{2f+b}{4} - \frac{f}{2} = m$.

Discussion. Why the fibre product? An operator acting on each fibre separately never moves information from one fibre to another, so its Schwartz kernel on M^2 vanishes unless the two points lie in the same fibre, i.e. it is supported in M_ϕ^2. On M_ϕ^2 it is an ordinary pseudodifferential kernel in the fibre variables, conormal to the diagonal. Viewed on M^2 it is a δ-function in the base directions $y - y'$ times such a kernel; this picture is the starting point for the product-type calculus later in the chapter.

Lemma 14.2.2. *Let $U \subset B$ be an open ball over which there is a trivialization $\phi^{-1}(U) \cong Z \times U$ as in Theorem 14.1.2, and bundle isomorphisms $E|_{\phi^{-1}(U)} \cong \pi_Z^* E_0$, $F|_{\phi^{-1}(U)} \cong \pi_Z^* F_0$ for bundles E_0, F_0 over Z (these exist since U is contractible). Then restriction of kernels gives an identification*

$$\Psi^m(M/B; E, F)\big|_U \cong \mathcal{C}^\infty(U; \Psi^m(Z; E_0, F_0)),$$

where $\Psi^m(Z; E_0, F_0)$ carries its Fréchet topology and the right side consists of the smooth maps into it.

Proof. Over U the trivialization identifies M_ϕ^2 with $Z^2 \times U$, Diag_M with $\mathrm{Diag}_Z \times U$ and Ω_R with the right density bundle of Z. The submanifold $\mathrm{Diag}_Z \times U$ is a product, the variables in U being variables along it, so by the definition of conormal distributions in local coordinates (Chapter 4) an element of $I^{m''}(Z^2 \times U, \mathrm{Diag}_Z \times U)$ is locally the inverse Fourier transform in the normal variables of a symbol depending smoothly on the variables in Z and U; that is, it is a smooth family, parametrized by U, of elements of $I^{m''+\frac{b}{4}}(Z^2, \mathrm{Diag}_Z)$ (the shift by $\frac{b}{4}$ comes from the change of dimension). With $m'' = m - \frac{b}{4}$ this is $\mathcal{C}^\infty(U; \Psi^m(Z; E_0, F_0))$. □

In particular $\Psi^m(M/B; E, F)$ does not depend on choices, and its elements are the families which are, locally in B, smooth families of pseudodifferential operators on a fixed fibre, glued by the transition maps of Theorem 14.1.2(3).

Note that the elements of $\Psi^m(M/B; E, F)$, as operators on M, have Schwartz kernels supported in M_ϕ^2, with a δ-singularity in the normal directions, whereas elements of $\Psi^m(M; E, F)$ have kernels which are smooth off the diagonal. For $\dim B > 0$ the only operators in both spaces are fibrewise differential operators (Exercise 14.2).

Proposition 14.2.3. *Let $\phi : M \longrightarrow B$ be a fibration and E, F, G vector bundles over M.*

(1) (Action) *Each $A \in \Psi^m(M/B; E, F)$ defines a continuous linear map $A : \mathcal{C}^\infty(M; E) \longrightarrow \mathcal{C}^\infty(M; F)$. It acts on the fibres: if $u \in \mathcal{C}^\infty(M; E)$ and $u|_{Z_b} = 0$ then $Au|_{Z_b} = 0$, so*

$$A_b v = (Au)|_{Z_b}, \quad u \in \mathcal{C}^\infty(M; E),\ u|_{Z_b} = v,$$

defines $A_b \in \Psi^m(Z_b; E_b, F_b)$, where $E_b = E|_{Z_b}$. Moreover $A(\phi^*\chi\, u) = \phi^*\chi\, Au$ for all $\chi \in \mathcal{C}^\infty(B)$.

(2) (Smoothing families) $\Psi^{-\infty}(M/B; E, F) = \bigcap_m \Psi^m(M/B; E, F) = \mathcal{C}^\infty(M_\phi^2; \mathrm{Hom}(E, F) \otimes \Omega_R)$.

(3) (Symbol) *There is a principal symbol map giving a short exact sequence*

$$\Psi^{m-1}(M/B; E, F) \hookrightarrow \Psi^m(M/B; E, F) \xrightarrow{\sigma_m} \mathcal{C}^\infty_{[m]}(T^*(M/B) \setminus 0; \mathrm{hom}(E, F)),$$

where $\mathcal{C}^\infty_{[m]}$ denotes smooth sections, homogeneous of degree m, of the pull-back of $\mathrm{hom}(E, F)$ to $T^*(M/B) \setminus 0$; $\sigma_m(A)$ restricts on $T^*Z_b \setminus 0$ to $\sigma_m(A_b)$.

(4) (Composition) *If* $A \in \Psi^m(M/B;F,G)$ *and* $B' \in \Psi^{m'}(M/B;E,F)$ *then* $AB' \in \Psi^{m+m'}(M/B;E,G)$, $(AB')_b = A_bB'_b$ *and* $\sigma_{m+m'}(AB') = \sigma_m(A)\sigma_{m'}(B')$.

(5) (Adjoints) *Given Hermitian metrics on* E, F *and a positive fibre density, the fibrewise adjoints* A_b^* *form an element* $A^* \in \Psi^m(M/B;F,E)$, *with* $\sigma_m(A^*) = \sigma_m(A)^*$.

(6) (Asymptotic completeness) *If* $A_j \in \Psi^{m-j}(M/B;E,F)$ *for* $j \in \mathbb{N}_0$, *then there exists an element* $A \in \Psi^m(M/B;E,F)$, *unique modulo* $\Psi^{-\infty}(M/B;E,F)$, *such that* $A - \sum_{j<N} A_j \in \Psi^{m-N}(M/B;E,F)$ *for all* N.

Proof. All statements are local in B, since the kernels are supported in M_ϕ^2. Choose a finite cover of B by balls U_i as in Lemma 14.2.2 and a partition of unity $\chi_i \in \mathcal{C}_c^\infty(U_i)$. Multiplication by $\phi^*\chi_i$ commutes with A because the kernel of A is supported in M_ϕ^2, on which $\phi^*\chi_i(m) = \phi^*\chi_i(m')$. So $A = \sum_i \phi^*\chi_i\, A$ and each term is, by Lemma 14.2.2, a smooth family over U_i of pseudodifferential operators on Z, with compact support in U_i.

Every construction of the calculus on the compact manifold Z is continuous in the Fréchet topologies involved: the action $\Psi^m(Z) \times \mathcal{C}^\infty(Z) \longrightarrow \mathcal{C}^\infty(Z)$, composition, the symbol map, adjoints, and the quantization of symbols. Hence each of them maps smooth families to smooth families, and (1), (3), (4), (5) follow from the corresponding statements for a single compact manifold in Chapter 5. For (1) note also that a smooth family $u \in \mathcal{C}^\infty(U \times Z; E_0)$ is mapped to a smooth family, i.e. to an element of $\mathcal{C}^\infty(U \times Z; F_0)$. In (3) the homogeneous symbols of the local families patch together because the symbol is invariant under the diffeomorphisms $f_{b'b}(c)$, and its restriction to T^*Z_b is the symbol of A_b; the null space of σ_m is $\Psi^{m-1}(M/B)$ by the corresponding property of conormal distributions, and surjectivity follows by quantizing $\chi_i\sigma$ in each U_i and summing. Part (2) is the definition of conormal distributions of order $-\infty$, and (6) follows from asymptotic completeness of conormal distributions on M_ϕ^2 (or locally from the parameter version of Borel's lemma) together with the partition of unity. □

Definition 14.2.4. $A \in \Psi^m(M/B;E,F)$ is *elliptic* if $\sigma_m(A)$ is invertible at each point of $T^*(M/B) \setminus 0$, i.e. if each A_b is elliptic.

Proposition 14.2.5. *If* $A \in \Psi^m(M/B;E,F)$ *is elliptic there exists* $Q \in \Psi^{-m}(M/B;F,E)$ *with*

$$QA - \mathrm{Id}_E \in \Psi^{-\infty}(M/B;E), \qquad AQ - \mathrm{Id}_F \in \Psi^{-\infty}(M/B;F).$$

Proof. Over each U_i as above, A is a smooth family of elliptic operators on Z. The construction of a parametrix in Chapter 5 — quantize the inverse of the principal symbol to get Q_0 with $Q_0A = \mathrm{Id} - R$, R of order -1, then take $Q \sim \sum_j R^jQ_0$ — uses only operations continuous in the Fréchet topologies together with asymptotic summation, which can be carried out with smooth dependence on parameters. So there are $Q_i \in \mathcal{C}^\infty(U_i;\Psi^{-m}(Z))$ with $Q_iA - \mathrm{Id}$ and $AQ_i - \mathrm{Id}$ smoothing families over U_i. Set $Q = \sum_i \phi^*\chi_i\, Q_i \in \Psi^{-m}(M/B;F,E)$. Since multiplication by $\phi^*\chi_i$ commutes with fibre operators,

$$QA = \sum_i \phi^*\chi_i\, Q_iA = \mathrm{Id}_E - \sum_i \phi^*\chi_i(\mathrm{Id} - Q_iA),$$
$$AQ = \sum_i \phi^*\chi_i\, AQ_i = \mathrm{Id}_F - \sum_i \phi^*\chi_i(\mathrm{Id} - AQ_i),$$

and the error terms are smoothing families. □

Unlike the case of a single operator, the partition of unity on B introduces no error at all, because it commutes with the operators. The errors could also be improved directly: if $Q_0A = \mathrm{Id} - R$ with $R \in \Psi^{-1}(M/B;E)$, the Neumann series $\sum_j R^jQ_0$ is asymptotically summable by Proposition 14.2.3(6).

14.3 The index bundle

We now associate to an elliptic family $A \in \Psi^m(M/B; E, F)$ an element of $K^0(B)$ generalizing the index of a single elliptic operator. The null spaces $\mathrm{Nul}(A_b)$ need not have constant dimension, so need not form a vector bundle; following [18] we first stabilize the family by a finite rank perturbation so that it becomes surjective.

Question. Why not define the index bundle as $[\mathrm{Nul}\, A] - [\mathrm{Nul}\, A^*]$ directly?

Answer. Because $\dim \mathrm{Nul}\, A_b$ can jump. For the elliptic family $A_b = D_\theta + b$ on the circle, $b \in [-\frac{1}{2}, \frac{1}{2}]$, the eigenvalues are $k + b$, $k \in \mathbb{Z}$, so the null space is the constants at $b = 0$ and zero elsewhere; the same happens for $A_b^* = A_b$. Only the difference $\dim \mathrm{Nul}\, A_b - \dim \mathrm{Nul}\, A_b^*$ is stable. Adding N fixed sections makes every $\tilde{A}_b$ surjective, so the cokernel vanishes and the null spaces have the constant dimension $N + \operatorname{ind} A_b$. This is the same idea as cutting off Fourier modes in Section 12.6. □

For each $b \in B$, A_b is an elliptic operator on the compact manifold Z_b, so by Chapter 5 it is Fredholm as a map $H^{s+m}(Z_b; E_b) \longrightarrow H^s(Z_b; F_b)$ for every s, its null space is a finite-dimensional subspace of $\mathcal{C}^\infty(Z_b; E_b)$ independent of s, its range is closed and $H^s(Z_b; F_b) = \mathrm{Ran}(A_b) \oplus \mathrm{Nul}(A_b^*)$ (orthogonal decomposition when $s = 0$; in general $\mathrm{Nul}(A_b^*) \subset \mathcal{C}^\infty$ is a complement); the index does not depend on s.

Lemma 14.3.1. *In a local trivialization as in Lemma 14.2.2, if $A \in \mathcal{C}^\infty(U; \Psi^m(Z; E_0, F_0))$ then for each s the map $U \ni c \longmapsto A(c) \in \mathcal{B}(H^{s+m}(Z; E_0), H^s(Z; F_0))$ is smooth in the norm topology.*

Proof. The inclusion $\Psi^m(Z; E_0, F_0) \longrightarrow \mathcal{B}(H^{s+m}, H^s)$ is linear and has closed graph (convergence in either space implies convergence of the Schwartz kernels as distributions), so is continuous by the closed graph theorem. The composite of a smooth map into a Fréchet space with a continuous linear map into a Banach space is smooth. □

Lemma 14.3.2 (Stabilization)**.** *If $A \in \Psi^m(M/B; E, F)$ is elliptic there are $N \in \mathbb{N}_0$ and sections $s_1, \dots, s_N \in \mathcal{C}^\infty(M; F)$ such that for every $b \in B$ the map*

$$\tilde{A}_b : \mathcal{C}^\infty(Z_b; E_b) \oplus \mathbb{C}^N \longrightarrow \mathcal{C}^\infty(Z_b; F_b), \quad (u, c) \longmapsto A_b u + \sum_{j=1}^{N} c_j s_j|_{Z_b} \tag{14.3.1}$$

is surjective.

Proof. Fix $b_0 \in B$. Since $\mathrm{Ran}(A_{b_0})$ has finite codimension in $\mathcal{C}^\infty(Z_{b_0}; F_{b_0})$ we may choose $v_1, \dots, v_p \in \mathcal{C}^\infty(Z_{b_0}; F_{b_0})$ spanning a complement, and extend them to sections $s_j \in \mathcal{C}^\infty(M; F)$. In a trivialization over a ball $U \ni b_0$ consider

$$T(c) : H^m(Z; E_0) \oplus \mathbb{C}^p \longrightarrow L^2(Z; F_0), \quad T(c)(u, a) = A(c)u + \sum_j a_j s_j|_{Z_c}.$$

The v_j also span a complement to the range of A_{b_0} on H^m: this range is $\mathrm{Nul}(A_{b_0}^*)^\perp$, of codimension $\dim \mathrm{Nul}(A_{b_0}^*) = p$, and if $\sum_j c_j v_j = A_{b_0} u$ with $u \in H^m$ then $u \in \mathcal{C}^\infty$ by elliptic regularity, so $c = 0$. Thus $T(b_0)$ is surjective. By Lemma 14.3.1, $T(c)$ depends continuously on c in norm, and surjectivity is an open condition (a surjective bounded operator has a bounded right inverse, and small perturbations of an operator with a bounded right inverse still have one), so $T(c)$ is surjective for c in a neighbourhood U' of b_0. If $g \in \mathcal{C}^\infty(Z_c; F_c)$ and $g = T(c)(u, a)$ with $u \in H^m$, then $A_c u = g - \sum a_j s_j \in \mathcal{C}^\infty$, so $u \in \mathcal{C}^\infty$ by elliptic regularity; hence (14.3.1), with these p sections, is surjective for $c \in U'$. Covering B by finitely many such neighbourhoods and taking all the sections so obtained gives the lemma, since adding sections preserves surjectivity. □

Proposition 14.3.3. *If $\tilde{A}_b$ in (14.3.1) is surjective for all b then the null spaces $\mathrm{Nul}(\tilde{A}_b) \subset \mathcal{C}^\infty(Z_b; E_b) \oplus \mathbb{C}^N$ form a smooth vector bundle $\mathrm{Nul}(\tilde{A})$ over B, of rank $N + \mathrm{ind}(A_b)$, in the sense that each point has a neighbourhood U over which there are $e_1, \dots, e_r \in \mathcal{C}^\infty(\phi^{-1}(U); E) \oplus \mathcal{C}^\infty(U; \mathbb{C}^N)$ restricting to a basis of $\mathrm{Nul}(\tilde{A}_c)$ for each $c \in U$.*

Proof. As a map $H^{s+m} \oplus \mathbb{C}^N \longrightarrow H^s$, $\tilde{A}_c$ is Fredholm with index $\mathrm{ind}(A_c) + N$, and surjective (its range is closed and contains the dense subspace $\mathcal{C}^\infty$); its null space lies in $\mathcal{C}^\infty \oplus \mathbb{C}^N$ by elliptic regularity. So $\dim \mathrm{Nul}(\tilde{A}_c) = \mathrm{ind}(A_c) + N$, which is locally constant since the index is (Chapter 12).

Fix b_0 and work in a trivialization over a ball $U \ni b_0$. Let $K_0 = \mathrm{Nul}(\tilde{A}(b_0))$ and let P be the orthogonal projection onto K_0 in $H^m \oplus \mathbb{C}^N$ (for some choice of inner product). Then

$$T(c) = (\tilde{A}(c), P) : H^{s+m}(Z; E_0) \oplus \mathbb{C}^N \longrightarrow H^s(Z; F_0) \oplus K_0$$

is invertible for $c = b_0$ and every $s \geq 0$. By Lemma 14.3.1 it is invertible, with inverse depending smoothly on c, for c in a neighbourhood U_0 of b_0 when $s = 0$. For $s > 0$ and $c \in U_0$, $T(c)$ is Fredholm of index 0 on the H^{s+m} level and injective, since its null space is contained in that on the H^m level; so it is invertible for all $s \geq 0$ and all $c \in U_0$, and the inverse is smooth in c in each of these norms. If $\kappa_1, \dots, \kappa_r$ is a basis of K_0, the elements $e_i(c) = T(c)^{-1}(0, \kappa_i)$ lie in $\mathrm{Nul}(\tilde{A}(c))$ and are linearly independent, hence form a basis by the dimension count. Since $c \mapsto e_i(c)$ is smooth into $H^{s+m}(Z; E_0) \oplus \mathbb{C}^N$ for every s, Sobolev embedding shows that $e_i \in \mathcal{C}^\infty(U_0 \times Z; E_0) \oplus \mathcal{C}^\infty(U_0; \mathbb{C}^N)$. □

Definition 14.3.4. The *(analytic) index* of an elliptic family $A \in \Psi^m(M/B; E, F)$ is

$$\mathrm{ind}_a(A) = [\mathrm{Nul}(\tilde{A})] - [\mathbb{C}^N] \in K^0(B),$$

where $\tilde{A}$ is any stabilization as in Lemma 14.3.2.

Theorem 14.3.5. *Let $A \in \Psi^m(M/B; E, F)$ be elliptic.*

(i) $\mathrm{ind}_a(A)$ *is independent of the choice of stabilization.*

(ii) *If A_t, $t \in [0,1]$, is a smooth family of elliptic elements of $\Psi^m(M/B; E, F)$ then $\mathrm{ind}_a(A_0) = \mathrm{ind}_a(A_1)$. Consequently $\mathrm{ind}_a(A)$ depends only on the homotopy class of $\sigma_m(A)$ through elliptic symbols.*

(iii) *If $\dim \mathrm{Nul}(A_b)$ is independent of b, then $\mathrm{Nul}(A)$ and $\mathrm{Nul}(A^*)$ are smooth vector bundles and $\mathrm{ind}_a(A) = [\mathrm{Nul}(A)] - [\mathrm{Nul}(A^*)]$. In particular for B a point, $\mathrm{ind}_a(A) = \mathrm{ind}(A) \in \mathbb{Z} = K^0(\mathrm{pt})$.*

(iv) $\mathrm{ind}_a(A_1 \oplus A_2) = \mathrm{ind}_a(A_1) + \mathrm{ind}_a(A_2)$.

Proof. (i) Let S and S' be stabilizations, by N and N' sections. Then the combination $S'' = (S, S')$, by all $N + N'$ sections, is again a stabilization. The map $\mathrm{Nul}(\tilde{A}'') \ni (u, c, c') \longmapsto c' \in \mathbb{C}^{N'}$ is a smooth bundle map, surjective because $\tilde{A}$ is (given c' solve $A_b u + S_b c = -S'_b c'$), and its null space is $\{(u, c, 0); A_b u + S_b c = 0\} = \mathrm{Nul}(\tilde{A})$. So $0 \to \mathrm{Nul}(\tilde{A}) \to \mathrm{Nul}(\tilde{A}'') \to \mathbb{C}^{N'} \to 0$ is a short exact sequence of vector bundles, which splits, and $[\mathrm{Nul}(\tilde{A}'')] - [\mathbb{C}^{N+N'}] = [\mathrm{Nul}(\tilde{A})] - [\mathbb{C}^N]$. By symmetry the same holds with S' in place of S.

(ii) A smooth family A_t is an elliptic family for the fibration $M \times [0,1] \longrightarrow B \times [0,1]$, and the proofs above apply verbatim over the compact base $B \times [0,1]$ (manifold with boundary); thus a single stabilization works for all t and $\mathrm{Nul}(\tilde{A})$ is a vector bundle over $B \times [0,1]$. Its restrictions to $B \times \{0\}$ and $B \times \{1\}$ are isomorphic [17]. If σ_t is a smooth

homotopy of elliptic symbols, quantizing it with smooth dependence on t gives a family A'_t with $\sigma_m(A'_t) = \sigma_t$; if $\sigma_m(A) = \sigma_m(A')$ then $(1-t)A + tA'$ is elliptic for all t. Together these show the second statement.

(iii) Suppose $\dim \mathrm{Nul}(A_b) = r$ for all b. Fix b_0, work in a trivialization near b_0, let P be a projection of $H^m(Z;E_0)$ onto $K_0 = \mathrm{Nul}(A(b_0))$ and Q the projection of $L^2(Z;F_0)$ onto $\mathrm{Ran}(A(b_0))$ along $\mathrm{Nul}(A(b_0)^*)$. Then $T(c) = (QA(c), P) : H^m \longrightarrow \mathrm{Ran}(A(b_0)) \oplus K_0$ is invertible at b_0, hence for c near b_0, and $\mathrm{Nul}(QA(c)) = T(c)^{-1}(\{0\} \oplus K_0)$ has dimension r. Since $\mathrm{Nul}(A(c)) \subset \mathrm{Nul}(QA(c))$ and both have dimension r they are equal, so as in the proof of Proposition 14.3.3 $\mathrm{Nul}(A)$ is a smooth bundle. The same applies to A^*, since $\dim \mathrm{Nul}(A_b^*) = r - \mathrm{ind}(A_b)$ is also constant. Now choose a bundle G with $\mathrm{Nul}(A^*) \oplus G \cong \mathbb{C}^N$ and let $S_b : \mathbb{C}^N \longrightarrow \mathcal{C}^\infty(Z_b;F_b)$ be the inclusion on $\mathrm{Nul}(A_b^*)$ and zero on G_b; this is given by N smooth sections of F. Since $\mathcal{C}^\infty(Z_b;F_b) = \mathrm{Ran}(A_b) \oplus \mathrm{Nul}(A_b^*)$, $\tilde{A}$ is surjective, and $\mathrm{Nul}(\tilde{A}) = \mathrm{Nul}(A) \oplus G$. Hence $\mathrm{ind}_a(A) = [\mathrm{Nul}\, A] + [G] - [\mathrm{Nul}(A^*) \oplus G] = [\mathrm{Nul}\, A] - [\mathrm{Nul}\, A^*]$.

(iv) If $\tilde{A}_1$, $\tilde{A}_2$ are stabilizations then $\tilde{A}_1 \oplus \tilde{A}_2$ is a stabilization of $A_1 \oplus A_2$ with null bundle $\mathrm{Nul}(\tilde{A}_1) \oplus \mathrm{Nul}(\tilde{A}_2)$. □

Remark 14.3.6. The principal symbol of an elliptic family, $\sigma_m(A)$, is an isomorphism $\pi^*E \longrightarrow \pi^*F$ off the zero section of $\pi : T^*(M/B) \longrightarrow M$, so defines a class $[\sigma_m(A)] \in K_c^0(T^*(M/B))$. By Theorem 14.3.5(ii) the index depends only on the homotopy class of the symbol; arguing exactly as for a single operator in Chapter 12 (stabilization by adding trivial bundles, invariance under bundle isomorphisms, and the fact that every compactly supported class is represented by the symbol of an elliptic family of order 0) one obtains a well-defined homomorphism

$$\mathrm{ind}_a : K_c^0(T^*(M/B)) \longrightarrow K^0(B),$$

the *analytic index map* of the fibration; see [18]. Its identification with a topological index is the families index theorem of Chapter 15.

14.4 The relative compactification

We now turn to the product-type calculus. Its kernels will be conormal to a nested pair of submanifolds $Z \subset Y \subset X$; at a point of Z the symbols are functions on the conormal space $W = N_p^*Z$, which contains the distinguished subspace $V = N_p^*Y$, and the relevant compactification of W is one which 'remembers' V.

Question. Why not use the radial compactification $\overline{W}$, as for ordinary symbols?

Answer. Because the symbols we need are not smooth on it. A fibrewise operator has a symbol that depends only on the fibre covariable $\zeta \in U = W/V$, for instance $a = \langle\zeta\rangle^\mu$. Near a point of $SV \subset \partial\overline{W}$ use the coordinates $x = 1/|\eta|$ and $v = \zeta/|\eta|$, in which $SV = \{x = 0,\ v = 0\}$. Then $\langle w\rangle^{-\mu} a$ is $(x^2 + |v|^2)^{\mu/2}$ times a smooth positive function, which is not smooth in (x,v) unless $\mu \in 2\mathbb{N}_0$. It is, however, smooth in polar coordinates around $\{x = v = 0\}$. The relative compactification introduces exactly these polar coordinates: it inserts a whole face H_V over SV (Figure 14.2 and Exercise 14.5), and on the result such symbols are smooth. □

Recall from Chapter 1 that the radial compactification of a real vector space U with a Euclidean norm is the closed half-sphere

$$\overline{U} = \{(t,u') \in \mathbb{R} \times U;\ t \geq 0,\ t^2 + |u'|^2 = 1\}, \quad U \ni u \longmapsto \Big(\frac{1}{\langle u\rangle}, \frac{u}{\langle u\rangle}\Big) \in \overline{U},$$

which embeds U as the interior of a compact manifold with boundary $\partial\overline{U} = SU$, the unit sphere; $\rho_U = \langle u\rangle^{-1}$ extends to a boundary defining function, near the boundary $1/|u|$ and $u/|u|$ give smooth coordinates, and $\rho_U^{-\mu}\mathcal{C}^\infty(\overline{U})$ is the space $S^\mu_{\mathrm{ph}}(U)$ of classical symbols of order μ. We shall use the following parameter version of the invariance of $\overline{U}$ under affine maps.

Lemma 14.4.1. *Let P be a manifold, possibly with corners, and $L \in \mathcal{C}^\infty(P;\mathrm{GL}(U))$, $c \in \mathcal{C}^\infty(P;U)$. Then $P \times U \ni (p,u) \longmapsto (p, L(p)u + c(p))$ extends to a diffeomorphism of $P \times \overline{U}$.*

Proof. Away from the boundary there is nothing to prove. Near a boundary point use the coordinates $x = 1/|u|$, $\theta = u/|u|$. Then $Lu + c = x^{-1}(L\theta + xc)$ and $L\theta \neq 0$, so for x small

$$\frac{1}{|Lu+c|} = \frac{x}{|L\theta + xc|}, \qquad \frac{Lu+c}{|Lu+c|} = \frac{L\theta + xc}{|L\theta + xc|}$$

are smooth functions of (p,x,θ) up to $x = 0$. The inverse map is of the same form, with L^{-1} and $-L^{-1}c$. □

Now let W be a real vector space of dimension $k+l$ and $V \subset W$ a subspace of dimension l; we assume $k, l > 0$ (the degenerate cases are discussed in Exercise 14.3). Choose a complement U, so $W = U \oplus V$, and Euclidean norms on U and V; we write $w = (\zeta, \eta)$, $\zeta \in U$, $\eta \in V$ and $\langle \zeta, \eta\rangle = (1 + |\zeta|^2 + |\eta|^2)^{1/2}$. Note that U is identified with W/V, so the variable ζ is the class of w in W/V.

Definition 14.4.2. The *relative compactification* of W with respect to V is the compact manifold with corners

$$^V\overline{W} = \overline{U} \times \overline{V}, \quad \text{with} \quad \iota_V : W \ni (\zeta,\eta) \longmapsto \Big(\zeta, \frac{\eta}{\langle\zeta\rangle}\Big) \in U \times V \subset \overline{U} \times \overline{V}.$$

Its two boundary hypersurfaces are

$$H_W = \partial\overline{U} \times \overline{V} = SU \times \overline{V} \quad (\text{the } \textit{main face}), \qquad H_V = \overline{U} \times \partial\overline{V} = \overline{U} \times SV \quad (\text{the } \textit{front face}),$$

meeting in the corner $H_W \cap H_V = SU \times SV$. The functions

$$\rho_W = \frac{1}{\langle\zeta\rangle}, \qquad \rho_V = \frac{1}{\langle \eta/\langle\zeta\rangle\rangle} = \frac{\langle\zeta\rangle}{\langle\zeta,\eta\rangle}$$

on W extend to boundary defining functions of H_W and H_V respectively, and $\rho_W\rho_V = \langle\zeta,\eta\rangle^{-1}$.

Clearly ι_V is a diffeomorphism of W onto the interior, with inverse $(\zeta, q) \mapsto (\zeta, \langle\zeta\rangle q)$. Composing with the embeddings of $\overline{U}$ and $\overline{V}$ as half-spheres gives Melrose's original description: $^V\overline{W}$ is the closure of the image of

$$W \ni (\zeta,\eta) \longmapsto (t, s, \zeta', \eta') = \Big(\frac{1}{\langle\zeta\rangle}, \frac{\langle\zeta\rangle}{\langle\zeta,\eta\rangle}, \frac{\zeta}{\langle\zeta\rangle}, \frac{\eta}{\langle\zeta,\eta\rangle}\Big) \in \mathbb{R}^2 \times W,$$

which is $\{t, s \geq 0,\ t^2 + |\zeta'|^2 = 1 = s^2 + |\eta'|^2\}$, the product of two half-spheres. The point is that the identification of the interior with W is *not* the obvious product one: the second factor is compactified in the rescaled variable $q = \eta/\langle\zeta\rangle$.

Proposition 14.4.3. *Smooth coordinates (more precisely, smooth functions among which a coordinate system can be chosen) near the boundary of $^V\overline{W}$ are given by*

(i) *near interior points of* H_W*:* $1/|\zeta|$, $\zeta/|\zeta|$ *and* $\eta/|\zeta|$*;*

(ii) *near interior points of* H_V*:* ζ, $1/|\eta|$ *and* $\eta/|\eta|$*;*

(iii) *near the corner:* $1/|\zeta|$, $|\zeta|/|\eta|$, $\zeta/|\zeta|$ *and* $\eta/|\eta|$,

where the first one, respectively the first two in (iii)*, are local boundary defining functions.*

Proof. Near H_W, $|\zeta|$ is large and $\rho_W = \langle\zeta\rangle^{-1}$ is a smooth function of $1/|\zeta|$ (since $x(1{+}x^2)^{-1/2}$ is a smooth function of x near 0 with nonvanishing derivative), while $\zeta/\langle\zeta\rangle = (\zeta/|\zeta|)\,|\zeta|/\langle\zeta\rangle$. In the interior of H_W, $q = \eta/\langle\zeta\rangle$ is a smooth coordinate on V and $\eta/|\zeta| = q\langle\zeta\rangle/|\zeta|$. This gives (i). In (ii), ζ is a coordinate on U and $q = \eta/\langle\zeta\rangle$ is large, so $1/|q| = \langle\zeta\rangle/|\eta|$ and $q/|q| = \eta/|\eta|$ are coordinates on $\overline{V}$ near its boundary; multiplying by the positive smooth function $1/\langle\zeta\rangle$ gives (ii). Near the corner both $|\zeta|$ and $|q|$ are large and $1/|q| = \langle\zeta\rangle/|\eta|$ is a smooth positive multiple of $|\zeta|/|\eta|$, giving (iii). □

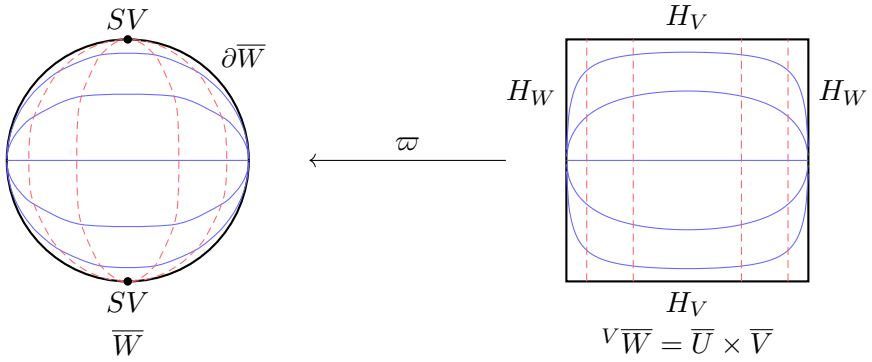


Figure 14.2: The relative compactification for $U = \mathbb{R}_\zeta$ (horizontal) and $V = \mathbb{R}_\eta$ (vertical). The lines $\eta = \text{const}$ (solid) all end at the two points $\zeta = \pm\infty$, $\eta/\langle\zeta\rangle = 0$ of H_W; the lines $\zeta = \text{const}$ (dashed), which in $\overline{W}$ all end at the two points of SV, end at distinct points of the front faces H_V. The blow-down map ϖ collapses each component of H_V to a point of SV.

Proposition 14.4.4 (Invariance)**.** *Let* $\mathrm{GL}(W,V) = \{L \in \mathrm{GL}(W); L(V) = V\}$*. The action of* $\mathrm{GL}(W,V)$ *on* W *and the action of* W *on itself by translations extend to smooth actions*

$$\mathrm{GL}(W,V) \times {}^V\overline{W} \longrightarrow {}^V\overline{W}, \qquad W \times {}^V\overline{W} \longrightarrow {}^V\overline{W}$$

by diffeomorphisms preserving H_W *and* H_V*. Consequently* ${}^V\overline{W}$*, with the embedding* ι_V*, is independent of the choice of* U *and of the norms: two choices give compactifications related by a unique diffeomorphism extending the identity of* W*.*

Proof. In terms of $W = U \oplus V$ an element $L \in \mathrm{GL}(W,V)$ is block lower triangular, $L(\zeta,\eta) = (A\zeta, C\zeta + D\eta)$ with $A \in \mathrm{GL}(U)$, $D \in \mathrm{GL}(V)$, $C \in \mathrm{Hom}(U,V)$. In the coordinates (ζ, q) of the interior, L becomes

$$(\zeta, q) \longmapsto \Big(A\zeta,\ g(\zeta)\Big(Dq + C\frac{\zeta}{\langle\zeta\rangle}\Big)\Big), \qquad g(\zeta) = \frac{\langle\zeta\rangle}{\langle A\zeta\rangle}. \tag{14.4.1}$$

Here $\zeta \mapsto A\zeta$ extends to a diffeomorphism of $\overline{U}$ by Lemma 14.4.1, and g and $\zeta/\langle\zeta\rangle$ are smooth on $\overline{U}$ (they are classical symbols of order 0), with $g > 0$. So for fixed $\zeta \in \overline{U}$ the second component is an invertible affine map of V whose coefficients are smooth on $\overline{U}$,

and Lemma 14.4.1 with $P = \overline{U}$ shows that (14.4.1) extends to a diffeomorphism of $\overline{U} \times \overline{V}$; the same argument with $P = \mathrm{GL}(W,V) \times \overline{U}$ gives joint smoothness of the action. For a translation by (a,c) the lifted map is $(\zeta, q) \mapsto (\zeta + a,\ \tilde{g}(\zeta)(q + c/\langle\zeta\rangle))$ with $\tilde{g} = \langle\zeta\rangle/\langle\zeta + a\rangle$, and the same argument applies. Finally, changing the complement U and the norms amounts to composing ι_V with an element of $\mathrm{GL}(W,V)$. □

From (14.4.1) one sees that the boundary faces have some natural structure. As $|q| \to \infty$ the direction of the second component tends to the direction of Dq, so on H_V the lifted action is $(\zeta, \theta) \mapsto (A\zeta, D\theta/|D\theta|)$; thus there is a natural product decomposition

$$H_V = \overline{W/V} \times SV. \tag{14.4.2}$$

On the other hand on H_W the action is $(\omega, q) \mapsto (A\omega/|A\omega|, |\omega|_A^{-1}(Dq + C\omega))$, with $|\omega|_A = |A\omega|$; it covers the action of $\mathrm{GL}(W/V)$ on $S(W/V)$, but the off-diagonal part C acts on the fibres by translations. Thus H_W is naturally a fibration

$$\overline{V}\text{—}H_W \longrightarrow S(W/V), \tag{14.4.3}$$

trivial but not naturally a product, whose fibres are radial compactifications of affine spaces modelled on V.

Proposition 14.4.5. (i) *The identity of W extends to a smooth surjective map $\varpi : {}^V\overline{W} \longrightarrow \overline{W}$, the* blow-down map*, with $\varpi^*\rho = \rho_W\rho_V$ for $\rho = \langle\zeta,\eta\rangle^{-1}$. It restricts to a diffeomorphism of $H_W \setminus H_V$ onto $SW \setminus SV$ and maps H_V onto $SV = \partial\overline{V} \subset \partial\overline{W}$, by the projection $\overline{W/V} \times SV \longrightarrow SV$.*

(ii) *The projection $W \longrightarrow W/V$ extends to the fibration ${}^V\overline{W} \longrightarrow \overline{W/V}$, with fibre $\overline{V}$, given in the product description by projection onto the first factor; the inclusion $V \hookrightarrow W$ extends to the embedding $\overline{V} \ni \eta \longmapsto (0,\eta)$ as the fibre over 0.*

Proof. In the half-sphere description of $\overline{W}$,

$$\varpi(\zeta, q) = \Big(\frac{1}{\langle\zeta,\eta\rangle}, \frac{\zeta}{\langle\zeta,\eta\rangle}, \frac{\eta}{\langle\zeta,\eta\rangle}\Big) = \Big(\rho_W\rho_V,\ \rho_V\frac{\zeta}{\langle\zeta\rangle},\ \frac{q}{\langle q\rangle}\Big),$$

which is smooth on $\overline{U} \times \overline{V}$. On H_V ($\rho_V = 0$) the image is $(0,0,\theta)$, $\theta = \lim q/|q| \in SV$. On $H_W \setminus H_V$ ($\rho_W = 0$, $\rho_V > 0$) the image is $(0, \rho_V\omega, q/\langle q\rangle)$ with $\omega = \zeta/|\zeta|$, $\rho_V = \langle q\rangle^{-1}$, a point of SW with non-zero component in U; conversely a point $(\hat\zeta, \hat\eta) \in SW$ with $\hat\zeta \neq 0$ is the image of $\omega = \hat\zeta/|\hat\zeta|$, $q = \hat\eta/|\hat\zeta|$, and this inverse is smooth. Part (ii) is immediate from the definition. □

In fact ϖ identifies ${}^V\overline{W}$ with the manifold obtained from $\overline{W}$ by introducing polar coordinates around the submanifold SV of its boundary, the *blow-up* $[\overline{W}; SV]$ (Exercise 14.5); H_V is the 'front face' produced by the blow-up and H_W is the lift of the original boundary.

Remark 14.4.6. By Proposition 14.4.4 the construction applies fibrewise to a real vector bundle $W \longrightarrow X$ with a subbundle V: the transition maps of local trivializations in which V is trivialized as a subbundle take values in $\mathrm{GL}(W,V)$, so the fibrewise relative compactifications form a compact (if X is compact) manifold with corners ${}^V\overline{W}$, fibred over X, with boundary faces H_W and H_V and defining functions ρ_W, ρ_V well defined up to smooth positive factors. Moreover $H_V = \overline{W/V} \times_X SV$, and H_W fibres over $S(W/V)$.

14.5 Product-type symbols

We keep the notation $W = U \oplus V$, $w = (\zeta, \eta)$, $q = \eta/\langle\zeta\rangle$ of the previous section, and let $\Omega \subset \mathbb{R}^p$ be an open set of parameters. Estimates are required to hold uniformly for x in compact subsets of Ω. For $M, M' \in \mathbb{R}$ define the weight

$$w_{M,M'}(\zeta,\eta) = \langle q\rangle^{M}\langle\zeta\rangle^{M'} = \langle\zeta,\eta\rangle^{M}\langle\zeta\rangle^{M'-M} = \rho_V^{-M}\rho_W^{-M'}.$$

The first order M is the order at the front face H_V and the second, M', the order at the main face H_W.

Definition 14.5.1. The space $S^{M,M'}(\Omega; W, V)$ of *product-type symbols* consists of the $a \in \mathcal{C}^\infty(\Omega \times W)$ such that for all multi-indices

$$|\partial_x^\gamma \partial_\zeta^\alpha \partial_\eta^\beta a(x,\zeta,\eta)| \le C\langle\zeta\rangle^{M'-M-|\alpha|}\langle\zeta,\eta\rangle^{M-|\beta|} = C\, w_{M,M'}(\zeta,\eta)\langle\zeta\rangle^{-|\alpha|}\langle\zeta,\eta\rangle^{-|\beta|}. \quad (14.5.1)$$

The subspace of *classical* (polyhomogeneous) product-type symbols is

$$S^{M,M'}_{\mathrm{ph}}(\Omega; W, V) = \{a;\ a = \rho_V^{-M}\rho_W^{-M'} b \text{ on } \Omega \times W,\ b \in \mathcal{C}^\infty(\Omega \times {}^V\overline{W})\}.$$

By Proposition 14.4.4, $S^{M,M'}_{\mathrm{ph}}$ does not depend on the choices made, since the defining functions change by smooth positive factors; we shall see in Proposition 14.5.3(vii) that the same is true of $S^{M,M'}$. Thus $\langle\zeta,\eta\rangle^{\mu} \in S^{\mu,\mu}_{\mathrm{ph}}$, while $\langle\zeta\rangle^{\mu} \in S^{0,\mu}_{\mathrm{ph}}$ is constant along V. Each ∂_ζ-derivative improves the estimate by $\langle\zeta\rangle^{-1}$, each ∂_η-derivative by $\langle\zeta,\eta\rangle^{-1}$.

Discussion. The two orders measure growth in two different regimes. Near the main face H_W, $|\zeta|$ is large and η is at most comparable to ζ: this is where an ordinary symbol of order M' lives. Near the front face H_V, η is much larger than ζ: the symbol grows like $|\eta|^M$ and, for fixed direction of η, is a symbol of order $M' - M$ in ζ. The estimate (14.5.1) is the standard symbol estimate with these two regimes separated: ∂_ζ gains only $\langle\zeta\rangle^{-1}$, not $\langle\zeta,\eta\rangle^{-1}$.

Lemma 14.5.2 (Peetre's inequality)**.** *For all M, M' there is C such that for all $w, \theta \in W$*

$$w_{M,M'}(w+\theta) \le C\, w_{M,M'}(w)\langle\theta\rangle^{|M|+|M'-M|}.$$

Proof. Apply Peetre's inequality $\langle x+y\rangle^s \le 2^{|s|/2}\langle x\rangle^s\langle y\rangle^{|s|}$ to each factor of $\langle\zeta,\eta\rangle^{M}\langle\zeta\rangle^{M'-M}$. □

Proposition 14.5.3. (i) *$a \in S^{M,M'}(\Omega; W, V)$ if and only if $c(x,\zeta,q) = a(x,\zeta,\langle\zeta\rangle q)$ satisfies the product estimates*

$$|\partial_x^\gamma \partial_\zeta^\alpha \partial_q^\beta c| \le C\langle\zeta\rangle^{M'-|\alpha|}\langle q\rangle^{M-|\beta|}.$$

(ii) *$S^{M,M'}_{\mathrm{ph}} \subset S^{M,M'}$.*

(iii) *$S^{M_1,M_1'} \cdot S^{M_2,M_2'} \subset S^{M_1+M_2,M_1'+M_2'}$, and the same for classical symbols.*

(iv) *$\partial_\zeta : S^{M,M'} \longrightarrow S^{M,M'-1}$ and $\partial_\eta : S^{M,M'} \longrightarrow S^{M-1,M'-1}$, and the same for classical symbols. Moreover $S^{M,M'} \subset S^{M+1,M'} \cap S^{M,M'+1}$ and $S^{M-1,M'-1} \subset S^{M,M'-1}$.*

(v) *The standard symbols on W are included: $S^{\mu}(\Omega; W) \subset S^{\mu,\mu}(\Omega; W, V)$ and $S^{\mu}_{\mathrm{ph}}(\Omega; W) \subset S^{\mu,\mu}_{\mathrm{ph}}(\Omega; W, V)$; more precisely $S^{\mu}_{\mathrm{ph}}(\Omega; W) = \varpi^*\big(\rho^{-\mu}\mathcal{C}^\infty(\Omega \times \overline{W})\big)$.*

(vi) *Symbols on W/V lift: if $a \in S^\mu(\Omega; W/V)$ then $a(x,\zeta) \in S^{0,\mu}(\Omega; W, V)$, and similarly for classical symbols.*

(vii) *If $L \in \mathcal{C}^\infty(\Omega; \mathrm{GL}(W,V))$ and $a \in S^{M,M'}$ then $a(x, L(x)w) \in S^{M,M'}$, and the same for classical symbols.*

(viii) *The residual spaces are*

$$S^{M,-\infty} = \bigcap_{M'} S^{M,M'} = \mathcal{S}(U; S^M(\Omega; V)), \quad S^{M,-\infty}_{\mathrm{ph}} = \bigcap_{M'} S^{M,M'}_{\mathrm{ph}} = \mathcal{S}(U; S^M_{\mathrm{ph}}(\Omega; V)),$$
$$S^{-\infty,-\infty} = \bigcap_{M,M'} S^{M,M'} = \bigcap_{M,M'} S^{M,M'}_{\mathrm{ph}} = \mathcal{C}^\infty(\Omega; \mathcal{S}(W)),$$

where $\mathcal{S}(U; S^M(\Omega; V))$ consists of the a with $|\partial_x^\gamma \partial_\zeta^\alpha \partial_\eta^\beta a| \le C_N \langle\zeta\rangle^{-N} \langle\eta\rangle^{M-|\beta|}$ for all N, and the classical version requires in addition that $a(x,\zeta,\cdot)$ be classical, with all terms of the expansion rapidly decreasing in ζ.

Proof. (i) Consider the vector fields $L_j = \langle\zeta\rangle\partial_{\zeta_j}$ and $Q_i = \langle\zeta,\eta\rangle\partial_{\eta_i}$ in the coordinates (ζ,η), and $L'_j = \langle\zeta\rangle\partial_{\zeta_j}$, $Q'_i = \langle q\rangle\partial_{q_i}$ in the coordinates (ζ, q). Since $\partial_q = \langle\zeta\rangle\partial_\eta$ and $\langle q\rangle\langle\zeta\rangle = \langle\zeta,\eta\rangle$,

$$Q'_i = Q_i, \qquad L'_j = L_j + \frac{\zeta_j}{\langle\zeta\rangle} \sum_i \frac{q_i}{\langle q\rangle} Q_i.$$

The functions $\zeta_j/\langle\zeta\rangle$ and $q_i/\langle q\rangle$ are annihilated by Q', respectively L', and any number of applications of L', respectively Q', to them gives bounded functions of the same type. Hence the products of L's and Q's, and the products of L''s and Q''s, are linear combinations of each other with bounded coefficients all of whose L', Q' derivatives are bounded. By the usual induction, the estimates (14.5.1) are equivalent to $|Pa| \le C w_{M,M'}$ for all products P of the L_j, Q_i and ∂_x (using $\partial_\zeta^\alpha = \langle\zeta\rangle^{-|\alpha|} L^\alpha +$ lower order terms with coefficients which are bounded symbols of negative order, and similarly for η), and the product estimates are equivalent to $|P'c| \le C w_{M,M'}$ for all products P' of the L'_j, Q'_i and ∂_x. This proves (i).

(ii) If $b \in \mathcal{C}^\infty(\Omega \times \overline{U} \times \overline{V})$ then L'_j and Q'_i lift to smooth vector fields on $\overline{U} \times \overline{V}$ (they are tangent to the boundary), so $P'b$ is bounded for every product P'. The weight $\langle\zeta\rangle^{M'}\langle q\rangle^M$ is itself a product-type symbol in the sense of (i), so (ii) follows from (i) and Leibniz' formula.

(iii) and (iv) follow directly from (14.5.1) and Leibniz' formula, and for classical symbols from the fact that L'_j, Q'_i, and hence $\rho_W^{-1}\partial_{\zeta_j} = L_j$ and $(\rho_W\rho_V)^{-1}\partial_{\eta_i} = Q_i$ (in the original coordinates), are smooth vector fields on ${}^V\overline{W}$ tangent to the boundary; the inclusions in (iv) hold because $\langle q\rangle \ge 1$ and $\langle\zeta\rangle \ge 1$.

(v) For $a \in S^\mu(W)$, $|\partial_\zeta^\alpha \partial_\eta^\beta a| \le C\langle\zeta,\eta\rangle^{\mu-|\alpha|-|\beta|} \le C\langle\zeta\rangle^{-|\alpha|}\langle\zeta,\eta\rangle^{\mu-|\beta|}$, which is (14.5.1) with $M = M' = \mu$. The classical statement follows from Proposition 14.4.5(i) since $\varpi^*\rho = \rho_V\rho_W$.

(vi) is clear from (i), since c is independent of q; for classical symbols use Proposition 14.4.5(ii).

(vii) If $L(x)(\zeta,\eta) = (A\zeta, C\zeta + D\eta)$ then by the chain rule $\partial_\eta(a \circ L) = D^t(\partial_\eta a) \circ L$ and $\partial_\zeta(a \circ L) = A^t(\partial_\zeta a) \circ L + C^t(\partial_\eta a) \circ L$, and x-derivatives produce similar terms with coefficients $\partial_x A$, etc. Using (iv) and the fact that $w_{M,M'} \circ L$ is bounded above and below by multiples of $w_{M,M'}$ (since $\langle A\zeta\rangle \simeq \langle\zeta\rangle$ and $\langle L(\zeta,\eta)\rangle \simeq \langle\zeta,\eta\rangle$, locally uniformly in x), the estimates follow by induction. The classical case follows from the smoothness of the action in Proposition 14.4.4.

(viii) Since $\langle\zeta\rangle^{-N-M}\langle\zeta,\eta\rangle^{M-|\beta|}$ lies between $\langle\zeta\rangle^{-N-M\mp|M-|\beta||}\langle\eta\rangle^{M-|\beta|}$ (by Peetre's inequality in the form $\langle\eta\rangle^s\langle\zeta\rangle^{-|s|} \le \langle\zeta,\eta\rangle^s \le \langle\eta\rangle^s\langle\zeta\rangle^{|s|}$), the first identity is clear. For

the classical case, $a \in S^{M,M'}_{\mathrm{ph}}$ for all M' means that $a = \langle q\rangle^M b(\zeta, q)$ where b is smooth on $\overline{U} \times \overline{V}$ and vanishes to infinite order at $H_W = SU \times \overline{V}$, i.e. b is a Schwartz function of ζ with values in the Fréchet space $\mathcal{C}^\infty(\overline{V})$. For fixed ζ, $\eta \mapsto a(\zeta, \eta)$ is obtained from $\langle q\rangle^M b(\zeta, q) \in \rho_V^{-M}\mathcal{C}^\infty(\overline{V}) = S^M_{\mathrm{ph}}(V)$ by the dilation $q = \eta/\langle\zeta\rangle$. Dilations by $\lambda \in (0, 1]$ act on $S^M_{\mathrm{ph}}(V)$ with each seminorm bounded by $C\lambda^{-K}$ times a seminorm, so $a \in \mathcal{S}(U; S^M_{\mathrm{ph}}(V))$; the converse is proved in the same way using the inverse dilation. Finally, taking $M = M' = -N$ gives $w_{M,M'} = \langle\zeta, \eta\rangle^{-N}$, so $S^{-\infty,-\infty}$ consists of Schwartz functions; conversely $w_{M,M'} \geq \langle\zeta, \eta\rangle^{-|M|-|M'-M|}$, so Schwartz functions lie in every $S^{M,M'}$, and they are classical, being $\rho^\infty\mathcal{C}^\infty(\overline{W})$ lifted by ϖ. □

Lemma 14.5.4 (Translations)**.** *For $\theta \in W$ the translation $a \longmapsto a(\cdot+\theta)$ maps $S^{M,M'}_{\mathrm{ph}}(\Omega; W, V)$ and $S^{M,M'}(\Omega; W, V)$ into themselves; each seminorm of $a(\cdot + \theta)$ is bounded by $C\langle\theta\rangle^K$ times a seminorm of a.*

Proof. For $S^{M,M'}$ this is Lemma 14.5.2 applied to (14.5.1). For classical symbols, the translation lifts to a diffeomorphism T_θ of ${}^V\overline{W}$ depending smoothly on θ (Proposition 14.4.4) and preserving the boundary faces, so $T_\theta^*(\rho_V^{-M}\rho_W^{-M'})/(\rho_V^{-M}\rho_W^{-M'})$ is smooth and positive. From the explicit form $(\zeta, q) \mapsto (\zeta + \theta_1, \langle\zeta\rangle\langle\zeta + \theta_1\rangle^{-1}(q + \theta_2/\langle\zeta\rangle))$ one checks, using the coordinates of Proposition 14.4.3, that all derivatives of T_θ and of this quotient are bounded by powers of $\langle\theta\rangle$. □

Proposition 14.5.5 (Asymptotic completeness)**.** (i) *If $a_j \in S^{M,M'-j}_{\mathrm{ph}}(\Omega; W, V)$ for $j \in \mathbb{N}_0$, there is $a \in S^{M,M'}_{\mathrm{ph}}$ with $a - \sum_{j<N} a_j \in S^{M,M'-N}_{\mathrm{ph}}$ for all N; a is unique modulo $S^{M,-\infty}_{\mathrm{ph}}$.*

(ii) *If $a_j \in S^{M-j,M'}_{\mathrm{ph}}$, there is $a \in S^{M,M'}_{\mathrm{ph}}$ with $a - \sum_{j<N} a_j \in S^{M-N,M'}_{\mathrm{ph}}$ for all N; a is unique modulo $S^{-\infty,M'}_{\mathrm{ph}} = \bigcap_j S^{M-j,M'}_{\mathrm{ph}}$.*

(iii) *If $a_j \in S^{M-j,M'-j}_{\mathrm{ph}}$, there is $a \in S^{M,M'}_{\mathrm{ph}}$ with $a - \sum_{j<N} a_j \in S^{M-N,M'-N}_{\mathrm{ph}}$ for all N; a is unique modulo $S^{-\infty,-\infty}$.*

The same statements hold for the spaces $S^{M,M'}$.

Proof. Write $a_j = \rho_V^{-M}\rho_W^{-M'}\rho_W^j b_j$ in case (i). Near H_W the factor $\overline{U}$ has a collar $[0, 1)_{\rho_W} \times SU$, so ${}^V\overline{W}$ has a collar $[0, 1) \times H_W$, and Borel's lemma (with parameters in $\Omega \times H_W$) gives $b \in \mathcal{C}^\infty(\Omega \times {}^V\overline{W})$ with $b - \sum_{j<N} \rho_W^j b_j \in \rho_W^N\mathcal{C}^\infty$ for all N: take $b = \sum_j \chi(\rho_W/\epsilon_j)\rho_W^j b_j$ with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}([0, 1))$ equal to 1 near 0 and $\epsilon_j \downarrow 0$ so fast that the j-th term has C^j norm on the j-th set of an exhaustion of Ω at most 2^{-j}. Then $a = \rho_V^{-M}\rho_W^{-M'}b$. Uniqueness holds since the difference of two choices vanishes to infinite order at H_W. Case (ii) is the same with ρ_V in place of ρ_W, and case (iii) the same with $\rho = \rho_V\rho_W$: since $|\partial^\alpha(\chi(\rho/\epsilon))| \leq C_\alpha\epsilon^{-|\alpha|}$ on $\{\rho \leq \epsilon\}$, the C^k norm of $\rho^j\chi(\rho/\epsilon)b_j$ is $O(\epsilon^{j-k})$, and the construction goes through. For non-classical symbols one uses instead the usual cut-offs $1 - \chi(\langle\zeta\rangle/R_j)$, $1 - \chi(\langle q\rangle/R_j)$, respectively $1 - \chi(\langle\zeta, \eta\rangle/R_j)$, with $R_j \to \infty$ rapidly, exactly as for standard symbols. □

14.5.1 Symbol maps in the model case

For $a \in S^{M,M'}_{\mathrm{ph}}(\Omega; W, V)$, $a = \rho_V^{-M}\rho_W^{-M'}b$, the restrictions of b to the two boundary faces are given by limits. For $w = (\zeta, \eta)$ with $\zeta \neq 0$ (that is, $w \notin V$) and for $\eta \in V \setminus 0$, put

$$\sigma_W(a)(x, w) = \lim_{t\to\infty} t^{-M'}a(x, tw), \qquad \sigma_V(a)(x, \zeta, \eta) = \lim_{t\to\infty} t^{-M}a(x, \zeta, t\eta). \tag{14.5.2}$$

Using $\langle tw\rangle \simeq t|w|$ and $tq \to \infty$ one finds

$$\sigma_W(a)(x,w) = |\zeta|^{M'-M}|w|^M b(x, \tfrac{\zeta}{|\zeta|}, \tfrac{\eta}{|\zeta|}), \qquad \sigma_V(a)(x,\zeta,\eta) = \langle\zeta\rangle^{M'-M}|\eta|^M b(x,\zeta,\tfrac{\eta}{|\eta|}),$$

where $(\zeta/|\zeta|, \eta/|\zeta|) \in SU \times V$ is a point of the interior of H_W and $(\zeta, \eta/|\eta|)$ a point of $H_V = \overline{U} \times SV$, and $|w|$ is the Euclidean norm on $W = U \oplus V$. Thus $\sigma_W(a)$ is homogeneous of degree M' on $W \setminus V$, while for each $\eta \neq 0$, $\sigma_V(a)(x,\cdot,\eta)$ is a classical symbol of order $M'-M$ on $U = W/V$, and $\sigma_V(a)$ is homogeneous of degree M in η. Let $\Sigma_W^{M,M'}$ and $\Sigma_V^{M,M'}$ denote the spaces of such functions, i.e. of the functions s of this form with b a smooth function on $\Omega \times H_W$, respectively $\Omega \times H_V$.

Lemma 14.5.6. *The maps σ_W and σ_V give short exact sequences*

$$S^{M,M'-1}_{\mathrm{ph}} \hookrightarrow S^{M,M'}_{\mathrm{ph}} \xrightarrow{\sigma_W} \Sigma_W^{M,M'}, \qquad S^{M-1,M'}_{\mathrm{ph}} \hookrightarrow S^{M,M'}_{\mathrm{ph}} \xrightarrow{\sigma_V} \Sigma_V^{M,M'},$$
$$S^{M-1,M'-1}_{\mathrm{ph}} \hookrightarrow S^{M,M'}_{\mathrm{ph}} \xrightarrow{(\sigma_W,\sigma_V)} \{(s_W,s_V) \in \Sigma_W^{M,M'} \times \Sigma_V^{M,M'};\ (14.5.3)\ \textit{holds}\},$$

where the compatibility condition, which expresses the equality of the restrictions of $b|_{H_W}$ and $b|_{H_V}$ to the corner, is

$$\lim_{t\to\infty} t^{-M} s_W(x,\zeta+t\eta) = \lim_{r\to\infty} r^{M-M'} s_V(x,r\zeta,\eta), \qquad \zeta \in U\setminus 0,\ \eta \in V \setminus 0; \tag{14.5.3}$$

that is, the leading homogeneous term of s_W at V is the principal symbol of the classical symbol $s_V(x,\cdot,\eta)$ on U.

Proof. $b|_{H_W} = 0$ if and only if $b \in \rho_W \mathcal{C}^\infty$, which gives the null space of σ_W, and similarly for σ_V. If b vanishes on both faces then $b = \rho_V b'$ with b' vanishing on $H_W \setminus H_V$, hence by continuity on H_W, so $b = \rho_V\rho_W b''$. For surjectivity use the product structure: if $c_V \in \mathcal{C}^\infty(\Omega \times \overline{U} \times SV)$ and $c_W \in \mathcal{C}^\infty(\Omega \times SU \times \overline{V})$ agree on $SU \times SV$, let $\chi \in \mathcal{C}^\infty_{\mathrm{c}}([0,1))$ be 1 near 0, set $\tilde{c}_V(x,\zeta,q) = \chi(\rho_V)c_V(x,\zeta,q/|q|)$ and

$$b = \tilde{c}_V + \chi(\rho_W)(c_W - \tilde{c}_V|_{H_W})(x,\zeta/|\zeta|,q).$$

Then $b|_{H_V} = c_V$, since $c_W - \tilde{c}_V|_{H_W}$ vanishes at the corner, and $b|_{H_W} = c_W$. Taking $c_W = 0$ or $c_V = 0$ (and dropping the compatibility) gives surjectivity of the separate maps. The computation of the limits above shows that the compatibility of $b|_{H_W}$ and $b|_{H_V}$ at the corner is (14.5.3). □

14.6 Product-type conormal distributions

Let $Z \subset Y \subset X$ be closed embedded submanifolds of a manifold X of dimension n, with $\dim Z = d$, $\dim Y = d+k$, so that Z has codimension k in Y and Y has codimension $l = n-d-k$ in X. (In this section Z denotes a submanifold, not the fibre of a fibration.) We define spaces $I^{m,m'}(X,Y,Z;E)$ of distributions which are conormal to Y of order m and, in a product sense, conormal to Z of order m'. Thus the *first* index refers to the larger submanifold Y.

14.6.1 The model case

Let $X = \mathbb{R}^n = \mathbb{R}^d_w \times \mathbb{R}^k_z \times \mathbb{R}^l_y$, $Y = \{y=0\}$, $Z = \{z=0, y=0\}$. The conormal space to Z at each point is $W = \mathbb{R}^k_\zeta \times \mathbb{R}^l_\eta$, the dual of the normal variables (z,y), and the conormal space to Y is its subspace $V = \{\zeta = 0\}$, the annihilator of the z-directions; the quotient

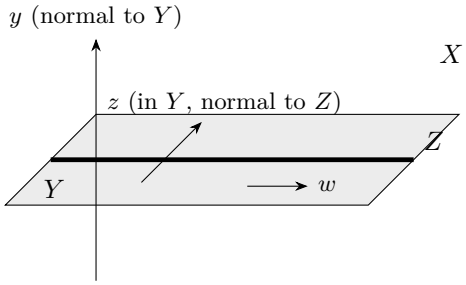


Figure 14.3: The nested pair $Z \subset Y \subset X$ in adapted coordinates (w, z, y): $Y = \{y = 0\}$ (shaded) and $Z = \{z = y = 0\}$ (thick line). The dual variables are ζ (to z) and η (to y). A product-type conormal distribution is conormal to Y away from Z (for instance a δ-function in the directions normal to Y) and has an additional conormal singularity along Z. For the fibrewise and product-type operators, $X = M^2$, $Y = M^2_\phi$ and $Z = \mathrm{Diag}_M$.

$W/V \cong \mathbb{R}^k_\zeta$ is the conormal space of Z in Y. For a compactly supported distribution u we write $\hat{u}(w, \zeta, \eta) = \int e^{-i(z\cdot\zeta + y\cdot\eta)} u(w, z, y)\, dz\, dy$ for the partial Fourier transform in the normal variables. Set

$$M = m + \frac{n}{4} - \frac{l}{2}, \qquad M' = m' + \frac{n}{4} - \frac{k+l}{2}; \tag{14.6.1}$$

these are the symbol orders corresponding, with the normalization of Chapter 4, to conormal order m at Y and m' at Z. Recall from Chapter 4 that, in these coordinates, $u \in I^m(\Omega, Y)$ if and only if for each $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ the partial Fourier transform of χu in y alone is a classical symbol of order M in η, depending smoothly on (w, z), and $u \in I^{m'}(\Omega, Z)$ if and only if each $\widehat{\chi u}$ lies in $S^{M'}_{\mathrm{ph}}(\mathbb{R}^d; W)$.

Definition 14.6.1. For $\Omega \subset \mathbb{R}^n$ open, $I^{m,m'}(\Omega, Y, Z)$ is the space of $u \in \mathcal{C}^{-\infty}(\Omega)$ such that for every $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$, $\widehat{\chi u} \in S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^d; W, V)$, with M, M' given by (14.6.1).

Thus, for $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$, $\chi u = \int e^{i(z\cdot\zeta + y\cdot\eta)} a(w, \zeta, \eta) đ\zeta đ\eta$ (an oscillatory integral) with $a = \widehat{\chi u}$ a product-type symbol, compactly supported in w. The definition is local: if $u \in I^{m,m'}(\Omega, Y, Z)$ and $\psi \in \mathcal{C}^\infty(\Omega)$ then $\psi u \in I^{m,m'}(\Omega, Y, Z)$, and membership can be checked using χ's from a partition of unity. The basic tool for handling these distributions is the following reduction of amplitudes.

Lemma 14.6.2 (Amplitudes)**.** *Let $r \in S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^n; W, V)$ have support in $K \times W$ for a compact $K \subset \mathbb{R}^n$, and let*

$$v(x) = \int e^{i(z\cdot\zeta + y\cdot\eta)} r(x, \zeta, \eta) đ\zeta đ\eta, \quad x = (w, z, y),$$

as an oscillatory integral. Then $\hat{v} \in S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^d; W, V)$ and, with $\xi = (\zeta, \eta)$,

$$\hat{v}(w, \xi) \sim \sum_\gamma \frac{1}{\gamma!} (-D_\xi)^\gamma \partial^\gamma_{(z,y)} r(w, 0, 0, \xi), \tag{14.6.2}$$

in the sense that the difference between $\hat{v}$ and the sum over $|\gamma| < N$ is in $S^{M,M'-N}$ for every N. The terms with $\gamma = (\alpha, 0)$ lie in $S^{M,M'-|\alpha|}_{\mathrm{ph}}$, those with $\gamma = (\alpha, \beta)$, $\beta \neq 0$, in $S^{M-1,M'-|\gamma|}_{\mathrm{ph}}$. In particular $v \in I^{m,m'}(\mathbb{R}^n, Y, Z)$.

Proof. Let $\tilde r(w,\theta;\xi) = \int e^{-i(z,y)\cdot\theta} r(w,z,y,\xi)\,dz\,dy$ be the Fourier transform of r in (z,y); since r has compact support in x and satisfies the estimates (14.5.1) with all x-derivatives, integration by parts shows that for each L the family $\langle\theta\rangle^L\tilde r(\cdot,\theta;\cdot)$, $\theta\in\mathbb{R}^{k+l}$, is bounded in $S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^d;W,V)$; that is, $\tilde r$ is rapidly decreasing in θ with values in $S^{M,M'}_{\mathrm{ph}}$. Inserting the definition of v and integrating first in (z,y) (which is justified by inserting a cut-off $\chi(\epsilon\xi)$ and letting $\epsilon\to 0$),

$$\hat v(w,\xi) = \int \tilde r(w,\xi-\xi';\xi')đ\xi' = \int \tilde r(w,\theta;\xi-\theta)đ\theta.$$

By Lemma 14.5.4 the integrand is, for each θ, in $S^{M,M'}_{\mathrm{ph}}$ with seminorms bounded by $C_L\langle\theta\rangle^{K-L}$, so the integral converges in $S^{M,M'}_{\mathrm{ph}}$ and $\hat v$ is a classical product-type symbol. For the expansion, apply Taylor's formula in the last variable:

$$\tilde r(w,\theta;\xi-\theta) = \sum_{|\gamma|<N}\frac{(-\theta)^\gamma}{\gamma!}\partial_\xi^\gamma\tilde r(w,\theta;\xi) + \sum_{|\gamma|=N}\frac{N(-\theta)^\gamma}{\gamma!}\int_0^1(1-s)^{N-1}\partial_\xi^\gamma\tilde r(w,\theta;\xi-s\theta)\,ds.$$

Since $\int\theta^\gamma\tilde r(w,\theta;\xi)đ\theta = D^\gamma_{(z,y)}r(w,0,0,\xi)$, the terms with $|\gamma|<N$ give the sum in (14.6.2), noting $(-\partial_\xi)^\gamma D^\gamma = (-D_\xi)^\gamma\partial^\gamma$. By Proposition 14.5.3(iv), $\partial_\xi^\gamma r\in S^{M,M'-N}$ when $|\gamma|=N$, so by Lemma 14.5.2 the remainder is bounded by $C\int\langle\theta\rangle^{-L}w_{M,M'-N}(\xi-s\theta)\,d\theta \le C'w_{M,M'-N}(\xi)$, and the same argument applies to its derivatives. The orders of the individual terms follow from Proposition 14.5.3(iv), since $(-D_\xi)^\gamma$ contains at least one η-derivative if $\beta\neq 0$. □

Proposition 14.6.3. *Let $\Omega\subset\mathbb{R}^n$ be open.*

(i) *$I^m(\Omega,Y)\subset I^{m,-\infty}(\Omega,Y,Z)=\bigcap_{m'}I^{m,m'}(\Omega,Y,Z)$, with equality, and $I^{m'}(\Omega,Z)\subset I^{m'-k/2,m'}(\Omega,Y,Z)$.*

(ii) *$\bigcap_j I^{m-j,m'-j}(\Omega,Y,Z)=\mathcal{C}^\infty(\Omega)$.*

(iii) *If $u\in I^{m,m'}(\Omega,Y,Z)$ then $u|_{\Omega\setminus Z}\in I^m(\Omega\setminus Z,Y\setminus Z)$; in particular u is smooth off Y.*

(iv) *Multiplication by y_j maps $I^{m,m'}$ into $I^{m-1,m'-1}$, multiplication by z_j maps it into $I^{m,m'-1}$, and smooth vector fields tangent to both Y and Z map $I^{m,m'}$ into itself.*

Proof. (i) If $u\in I^m(\Omega,Y)$ and $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\Omega)$, the partial Fourier transform of χu in y is a classical symbol of order M in η which is smooth and compactly supported in (w,z); its Fourier transform in z is therefore in $\mathcal{S}(\mathbb{R}^k_\zeta;S^M_{\mathrm{ph}}(\mathbb{R}^d;\mathbb{R}^l))=S^{M,-\infty}_{\mathrm{ph}}$ by Proposition 14.5.3(viii). Conversely, if $\widehat{\chi u}\in S^{M,-\infty}_{\mathrm{ph}}$ then its inverse Fourier transform in ζ is smooth in (w,z) and a classical symbol in η, so $\chi u\in I^m(\Omega,Y)$. The second inclusion is Proposition 14.5.3(v): the symbol order M' of $I^{m'}(\Omega,Z)$ is the second order in (14.6.1), and the first order $M=M'$ corresponds to $m=m'-k/2$.

(ii) By Proposition 14.5.3(viii), $\widehat{\chi u}$ is Schwartz in ξ, so χu is smooth.

(iii) Points of $\Omega\setminus Z$ either lie off Y or have $z\neq 0$. Let $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ and $a=\widehat{\chi u}$. Multiplication by y^β corresponds to applying $(i\partial_\eta)^\beta$ to a, which by Proposition 14.5.3(iv) gives an element of $S^{M-|\beta|,M'-|\beta|}$; since $w_{M-N,M'-N}\le\langle\xi\rangle^{-N+|M|+|M'-M|}$, $y^\beta\chi u\in C^j$ when $|\beta|$ is large compared to j, so χu is smooth where $y\neq 0$.

Now consider $\{z\neq 0\}$. The partial inverse Fourier transform $A(w,z,\eta)=\int e^{iz\cdot\zeta}a(w,\zeta,\eta)đ\zeta$ satisfies, for $z\neq 0$ and any N, $A=|z|^{-2N}\int e^{iz\cdot\zeta}(-\Delta_\zeta)^N a đ\zeta$, and $(-\Delta_\zeta)^N a\in S^{M,M'-2N}$ satisfies $|\partial_\eta^\beta\partial_\zeta^\alpha\Delta_\zeta^N a|\le C\langle\zeta\rangle^{M'-M-2N-|\alpha|+|M-|\beta||}\langle\eta\rangle^{M-|\beta|}$ by Peetre's inequality. Taking N large, it follows that on $|z|>\epsilon$, A is smooth in (w,z) and satisfies symbol estimates

of order M in η. To see that it is classical, write $a = \langle\zeta\rangle^{M'}\langle q\rangle^{M} b$ and expand b at H_V: $b = \sum_{j<J} \rho_V^j b_j(w,\zeta,\eta/|\eta|) + \rho_V^J b'_J$. Up to a term in $S^{M-J,M'}$, whose contribution to A on $|z|>\epsilon$ is $O(\langle\eta\rangle^{M-J})$ with all derivatives by the estimate just used, a is a sum of terms $(1-\chi(q))|\eta|^{M-j}\langle\zeta\rangle^{M'-M+j} b_j(w,\zeta,\hat\eta)$, $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(V)$ equal to 1 near 0 (absorbing smooth factors into the b_j). The cut-off $\chi(q)=\chi(\eta/\langle\zeta\rangle)$ is supported where $\langle\eta\rangle\le C\langle\zeta\rangle$, so after integrating by parts in ζ its contribution on $|z|>\epsilon$ is $O(\langle\eta\rangle^{-\infty})$; the remaining term contributes $|\eta|^{M-j}g_j(w,z,\hat\eta)$ with $g_j=\int e^{iz\cdot\zeta}\langle\zeta\rangle^{M'-M+j}b_j(w,\zeta,\hat\eta)đ\zeta$, which is smooth for $z\neq 0$. Hence A is a classical symbol of order M in η on $|z|>\epsilon$, and $\chi u=\int e^{iy\cdot\eta}A đ\eta$ is conormal to Y there.

(iv) Multiplication by y_j and z_j corresponds to $i\partial_{\eta_j}$ and $i\partial_{\zeta_j}$ on the symbol, so this follows from Proposition 14.5.3(iv). The vector fields tangent to Y and Z are spanned over $\mathcal{C}^\infty$ by ∂_w, $z_i\partial_{z_j}$, $y_i\partial_{z_j}$ and $y_i\partial_{y_j}$, which correspond to ∂_w and to $-\partial_{\zeta_i}\zeta_j$, $-\partial_{\eta_i}\zeta_j$ and $-\partial_{\eta_i}\eta_j$; by Proposition 14.5.3 these preserve $S^{M,M'}_{\mathrm{ph}}$ ($\zeta_j\in S^{0,1}$, $\eta_j\in S^{1,1}$). Multiplication by smooth functions preserves the class by the definition. □

Proposition 14.6.4 (Invariance)**.** *Let $F:\Omega'\longrightarrow\Omega$ be a diffeomorphism between open subsets of $\mathbb{R}^n$ with $F(Y\cap\Omega')=Y\cap\Omega$ and $F(Z\cap\Omega')=Z\cap\Omega$. Then $F^*: I^{m,m'}(\Omega,Y,Z)\longrightarrow I^{m,m'}(\Omega',Y,Z)$. Moreover, near Z the symbol of F^*u is given, modulo $S^{M,M'-1}_{\mathrm{ph}}$, by*

$$\widehat{\chi F^*u}(w,\xi)\equiv\chi(w,0,0)\,a\big(F_w(w),(\Phi(w)^t)^{-1}\xi\big)\,|\det\Phi(w)|^{-1},\tag{14.6.3}$$

where $a=\widehat{\psi u}$ for $\psi\in\mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ equal to 1 on $F(\operatorname{supp}\chi)$, $F_w(w)$ is the w-component of $F(w,0,0)$, and $\Phi(w)$ is the differential of F at $(w,0,0)$ acting on the normal variables (z,y).

Proof. By the localizability of the definition and Proposition 14.6.3(iii) together with the invariance of $I^m(\Omega,Y)$ under diffeomorphisms preserving Y (Chapter 4) and Proposition 14.6.3(i), it suffices to consider χF^*u where $\chi\in\mathcal{C}^\infty_{\mathrm{c}}(\Omega')$ has support in a small neighbourhood of a point of Z.

Write $F=(F_w,F_z,F_y)$. Since F_y vanishes on $Y=\{y=0\}$ and F_z vanishes on $Z=\{z=y=0\}$, Hadamard's lemma gives

$$F_z(x)=A(x)z+B(x)y,\qquad F_y(x)=C(x)y,$$

with A, B, C smooth matrices. At points of Z the differential of F is block triangular with diagonal blocks $\partial_w F_w$, A, C, so A and C are invertible near Z. Set $\Phi(x)=\begin{pmatrix}A & B\\ 0 & C\end{pmatrix}$, so that $(F_z,F_y)=\Phi(x)(z,y)$. Then, with $a=\widehat{\psi u}$,

$$\begin{aligned}\chi F^*u(x)&=\chi(x)\int e^{i(F_z(x)\cdot\zeta+F_y(x)\cdot\eta)}a(F_w(x),\zeta,\eta)đ\zeta đ\eta\\&=\int e^{i(z\cdot\zeta'+y\cdot\eta')}\chi(x)\,a(F_w(x),L(x)(\zeta',\eta'))|\det\Phi(x)|^{-1}đ\zeta' đ\eta',\end{aligned}$$

where $(\zeta',\eta')=\Phi(x)^t(\zeta,\eta)=(A^t\zeta,B^t\zeta+C^t\eta)$ and $L(x)=(\Phi(x)^t)^{-1}$; the change of variables is justified by inserting a cut-off $\chi(\epsilon\xi)$. The crucial point is that Φ^t, and hence $L(x)$, preserves $V=\{\zeta=0\}$, so $L\in\mathcal{C}^\infty(\operatorname{supp}\chi;\mathrm{GL}(W,V))$. By Proposition 14.5.3(vii) the new amplitude $r(x,\xi')=\chi(x)a(F_w(x),L(x)\xi')|\det\Phi(x)|^{-1}$ lies in $S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^n;W,V)$ with compact support in x, so Lemma 14.6.2 shows that $\chi F^*u\in I^{m,m'}(\Omega',Y,Z)$; (14.6.3) is the leading term of (14.6.2). □

14.6.2 The general case

An *adapted chart* for $Z \subset Y \subset X$ near $p \in Z$ is a coordinate chart (w, z, y) in which $Y = \{y = 0\}$ and $Z = \{z = y = 0\}$; such charts exist near every point of Z (take coordinates on Y adapted to Z and extend them using a tubular neighbourhood of Y).

Definition 14.6.5. For a vector bundle E over X, $I^{m,m'}(X, Y, Z; E)$ is the space of $u \in \mathcal{C}^{-\infty}(X; E)$ such that $u|_{X\setminus Z} \in I^m(X \setminus Z, Y \setminus Z; E)$ and such that each point of Z has an adapted chart Ω, over which E is trivial, with $u|_\Omega \in I^{m,m'}(\Omega, Y, Z) \otimes \mathbb{C}^{\operatorname{rank} E}$.

By Proposition 14.6.4, if $u \in I^{m,m'}(X, Y, Z; E)$ then $u|_\Omega$ is of model form in *every* adapted chart; by Proposition 14.6.3(iii) the first condition is only a condition away from a neighbourhood of Z.

Proposition 14.6.6. *Let $Z \subset Y \subset X$ be as above, $k = \dim Y - \dim Z$, $l = \operatorname{codim} Y$.*

(i) *$I^{m,m'}(X, Y, Z; E)$ is a $\mathcal{C}^\infty(X)$-module, invariant under diffeomorphisms preserving Y and Z, and under differential operators generated by vector fields tangent to Y and Z; also $I^{m,m'} \subset I^{m+1,m'} \cap I^{m,m'+1}$.*

(ii) *$I^m(X, Y; E) = \bigcap_{m'} I^{m,m'}(X, Y, Z; E)$ and $I^{m'}(X, Z; E) \subset I^{m'-k/2,m'}(X, Y, Z; E)$.*

(iii) *$\bigcap_j I^{m-j,m'-j}(X, Y, Z; E) = \mathcal{C}^\infty(X; E)$.*

(iv) *Let $\Omega(N^*Y)$ be the density bundle of the conormal bundle. If $v \in I^{m''}(Y, Z; E|_Y \otimes \Omega(N^*Y))$ then $v\,\delta_Y$, defined by $\langle v\delta_Y, \psi\rangle = \langle v, \psi|_Y\rangle$ for test densities ψ on X, lies in $I^{(l-d-k)/4,\, m''+l/4}(X, Y, Z; E)$.*

(v) *Asymptotic completeness holds in the three forms of Proposition 14.5.5, modulo $I^{m,-\infty} = I^m(X, Y)$, modulo $I^{-\infty,m'} = \bigcap_j I^{m-j,m'}$, and modulo $\mathcal{C}^\infty(X; E)$ respectively.*

Proof. Parts (i)–(iii) follow from Proposition 14.6.3, Proposition 14.6.4 and the corresponding properties of $I^m(X, Y)$ away from Z. For (iv), in an adapted chart in which $\psi|_Y = \psi(w, z, 0)|dw\,dz|$ (absorbing $|dy|$ into the trivialization of $\Omega(N^*Y)$), $v\delta_Y = v(w, z)\delta(y)$, so $\widehat{\chi v\delta_Y}(w, \zeta, \eta) = \widehat{\chi v}(w, \zeta)$ is independent of η and is a classical symbol of order $\mu = m'' + \frac{d+k}{4} - \frac{k}{2}$ in ζ. By Proposition 14.5.3(vi) it is in $S^{0,\mu}_{\mathrm{ph}}$; the orders in (14.6.1) with $M = 0$, $M' = \mu$ are $m = \frac{l}{2} - \frac{n}{4} = \frac{l-d-k}{4}$ and $m' = m'' + \frac{l}{4}$. Part (v) follows from Proposition 14.5.5 in adapted charts, patched with a partition of unity, together with asymptotic completeness of $I^m(X, Y)$. □

14.6.3 Symbol maps

At a point $p \in Z$ the symbol of $u \in I^{m,m'}(X, Y, Z; E)$ is a function on $W = N^*_p Z$ which is a product-type symbol with respect to $V = N^*_p Y$. By Remark 14.4.6 the fibrewise relative compactifications form a manifold with corners ${}^{N^*Y}\overline{N^*Z}$ fibred over Z, with main face H_W, whose interior is identified by the blow-down map with $SN^*Z \setminus S(N^*Y|_Z)$ and which fibres over the sphere bundle of $N^*Z/N^*Y|_Z$, the conormal bundle of Z in Y, and with front face $H_V = \overline{N^*Z/N^*Y|_Z} \times_Z S(N^*Y|_Z)$.

Proposition 14.6.7 (Main symbol)**.** *For $u \in I^{m,m'}(X, Y, Z; E)$ the function $\sigma_W(\hat{u})$ of (14.5.2), computed in adapted charts, defines a homogeneous function of degree M' on $N^*Z \setminus N^*Y|_Z$ with values in $E \otimes \Omega_{\mathrm{fib}}(N^*Z)$, smooth up to H_W after multiplication by the weight as in Lemma 14.5.6; this gives a short exact sequence*

$$I^{m,m'-1}(X, Y, Z; E) \hookrightarrow I^{m,m'}(X, Y, Z; E) \xrightarrow{\sigma_W} \Sigma^{M,M'}_W(N^*Z, N^*Y|_Z; E \otimes \Omega_{\mathrm{fib}}(N^*Z)).$$

Proof. By (14.6.3), under a change of adapted coordinates the symbol changes, modulo $S^{M,M'-1}_{\mathrm{ph}}$, by composition with the transpose inverse of the normal differential and multiplication by the inverse of its determinant. This is precisely the transformation law of a section of $\Omega_{\mathrm{fib}}(N^*Z)$ over N^*Z (as for the principal symbol of an element of $I^{m'}(X,Z)$ in Chapter 4), and the linear action extends smoothly to the relative compactification by Proposition 14.4.4. So $\sigma_W(u)$ is well defined. Exactness follows from Lemma 14.5.6 in adapted charts, using a partition of unity for surjectivity, and from the fact that away from Z the spaces $I^{m,m'-1}$ and $I^{m,m'}$ coincide. □

The other symbol is associated with Y. Away from Z, u is conormal to Y and has its usual principal symbol, a homogeneous function of degree M on $N^*(Y\setminus Z)\setminus 0$. At Z this symbol develops a conormal singularity: in an adapted chart, for $\eta\in\mathbb{R}^l\setminus 0$, set

$$\sigma_V(u)(w,z,\eta)=\int e^{iz\cdot\zeta}\sigma_V(\hat{u})(w,\zeta,\eta)đ\zeta. \tag{14.6.4}$$

Since $\sigma_V(\hat{u})(w,\cdot,\eta)$ is a classical symbol of order $M'-M$ on $\mathbb{R}^k$, $\sigma_V(u)$ is a distribution on $N^*Y\setminus 0$ (with coordinates (w,z,η)), homogeneous of degree M in η and conormal to $N^*Y|_Z$, with symbol of order $M'-M$ in the conormal variables ζ of Z in Y.

Proposition 14.6.8 (Symbol at Y)**.** *For $u\in I^{m,m'}(X,Y,Z;E)$, (14.6.4) defines invariantly a distribution $\sigma_V(u)$ on $N^*Y\setminus 0$ with values in $E\otimes\Omega_{\mathrm{fib}}(N^*Y)$, homogeneous of degree M and conormal to $N^*Y|_Z\setminus 0$ with symbol of order $M'-M$, which coincides with the usual symbol of u as a conormal distribution to Y over $Y\setminus Z$. There is a short exact sequence*

$$I^{m-1,m'}(X,Y,Z;E)\hookrightarrow I^{m,m'}(X,Y,Z;E)\xrightarrow{\sigma_V}\Sigma_V^{M,M'}(N^*Y,N^*Y|_Z;E\otimes\Omega_{\mathrm{fib}}(N^*Y)),$$

where the last space consists of all such homogeneous conormal distributions. The joint symbol (σ_W,σ_V) is surjective onto the pairs satisfying the compatibility condition (14.5.3) at the corner (the principal symbol of $\sigma_V(u)$ at $N^*Y|_Z$ is the leading part of $\sigma_W(u)$ at $N^*Y|_Z$), with null space $I^{m-1,m'-1}(X,Y,Z;E)$.

Proof. Away from Z, (14.6.4) is the usual principal symbol, by the computation in the proof of Proposition 14.6.3(iii) (the term $j=0$ there). Near Z consider a change of adapted coordinates as in the proof of Proposition 14.6.4. In the expansion (14.6.2) of the new symbol, the terms involving ∂_y lie in $S^{M-1,M'}$ and do not contribute to σ_V; the terms involving only ∂_z are exactly the terms of the expansion of Lemma 14.6.2 applied, on Y, to the amplitude $\lim_{t\to\infty}t^{-M}r(w,z,0,\zeta,t\eta)$. Now

$$L(\zeta,t\eta)=(A^{-t}\zeta,\ tC^{-t}\eta-C^{-t}B^tA^{-t}\zeta),$$

and the second term is a bounded translation in η for fixed ζ, which does not affect the limit in (14.5.2) (it is invisible at H_V). Hence

$$\sigma_V(F^*u)(\cdot,\cdot,\eta)=|\det C|^{-1}\,(F|_Y)^*\big[\sigma_V(u)(\cdot,\cdot,C^{-t}\eta)\big],$$

where the pull-back of the distribution in (w,z) by the diffeomorphism $F|_Y:(w,z)\mapsto(F_w(w,z,0),A(w,z,0)z)$ is computed, as in the standard invariance proof for $I^{\cdot}(Y,Z)$, by the same Kuranishi argument ($F_z|_Y=Az$, so $z\cdot\zeta$ becomes $z\cdot A^t\zeta$). This is the transformation law of a distribution on N^*Y with values in $\Omega_{\mathrm{fib}}(N^*Y)$ under the map induced by F, so $\sigma_V(u)$ is invariantly defined. The exact sequences and surjectivity of the joint symbol follow from Lemma 14.5.6, via a partition of unity, together with the usual symbol sequence for $I^m(X\setminus Z,Y\setminus Z)$; on the corner $H_V\cap H_W$ the restrictions of the two symbols correspond to the leading term of $\sigma_V(\hat{u})$ as a symbol in ζ, which is the principal symbol of the conormal distribution $\sigma_V(u)$. □

14.7 Product-type pseudodifferential operators

Let $\phi : M \longrightarrow B$ be a fibration with fibre Z, $f = \dim Z$, $b = \dim B$, and let E, F be vector bundles over M. We apply the theory of the previous section to the nested pair (14.1.2), $\mathrm{Diag}_M \subset M^2_\phi \subset M^2$, for which $d = f + b$, $k = f$ and $l = b$.

Definition 14.7.1. The space of *product-type pseudodifferential operators* associated with ϕ is

$$\Psi^{m,m'}_{\phi\text{-pt}}(M; E, F) = I^{m-\frac{f}{2},m'}(M^2, M^2_\phi, \mathrm{Diag}_M; \mathrm{Hom}(E, F) \otimes \Omega_R),$$

where $\Omega_R = \pi_R^*\Omega(M)$, identified with a space of operators through the Schwartz kernel theorem.

With this normalization the symbol orders (14.6.1) are exactly (m, m'): $M = m - \frac{f}{2} + \frac{f+b}{2} - \frac{b}{2} = m$ and $M' = m'$. The first order m is the order at the fibre diagonal M^2_ϕ; we call it the *base order*. The second, m', is the order at the diagonal, the *fibre order*. As we shall see, $\Psi^m(M) \subset \Psi^{m,m}_{\phi\text{-pt}}(M)$, $\Psi^{m'}(M/B) \subset \Psi^{0,m'}_{\phi\text{-pt}}(M)$, and compressions of operators of order m on the base lie in $\Psi^{m,-\infty}_{\phi\text{-pt}}(M)$.

14.7.1 Local structure

Let $x = (z, y)$ be fibred coordinates on an open set $\Omega \subset M$, so $\phi(z, y) = y$, and let $\xi = (\zeta, \eta)$ be the dual variables. Then $T^*_x M = \mathbb{R}^f_\zeta \times \mathbb{R}^b_\eta$ and $\phi^*T^*B = \{\zeta = 0\}$; we write $S^{m,m'}_{\mathrm{ph}}(\Omega)$ for $S^{m,m'}_{\mathrm{ph}}(\Omega; \mathbb{R}^{f+b}, \{\zeta = 0\})$. For a symbol $a(x, \xi)$, $\mathrm{Op}(a)$ is the left quantization.

Proposition 14.7.2. *If $A \in \Psi^{m,m'}_{\phi\text{-pt}}(M; E, F)$ and $\chi, \chi' \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega)$ then, after trivializing E and F over Ω and $\Omega(M)$ by $|dx|$, $\chi A\chi' = \mathrm{Op}(a)$ with $a \in S^{m,m'}_{\mathrm{ph}}(\Omega)$, of compact support in x. Conversely $\mathrm{Op}(a)\chi' \in \Psi^{m,m'}_{\phi\text{-pt}}(M; E, F)$ for every such a. If $\operatorname{supp}\chi \cap \operatorname{supp}\chi' = \emptyset$ then the kernel of $\chi A\chi'$ is conormal to M^2_ϕ, of order $m - \frac{f}{2}$ there, and smooth off M^2_ϕ.*

Proof. The coordinates $(w, \tilde z, \tilde y) = (x, z - z', y - y')$ on $\Omega \times \Omega$ are adapted to (14.1.2), by (14.1.3). The partial Fourier transform of $\chi(x)K_A(x, x')\chi'(x')$ in $(\tilde z, \tilde y)$ is $a(x, \xi) = \int e^{-i\nu\cdot\xi}\chi(x)K_A(x, x-\nu)\chi'(x-\nu)\,d\nu$, and Fourier inversion gives $\chi K_A\chi' = \int e^{i(x-x')\cdot\xi}a(x, \xi)\text{đ}\xi$, i.e. $\chi A\chi' = \mathrm{Op}(a)$. By Definition 14.6.1 and the computation of the orders above, $a \in S^{m,m'}_{\mathrm{ph}}(\Omega)$. Conversely the kernel of $\mathrm{Op}(a)\chi'$ is of model form near Diag and, by Proposition 14.6.3(iii), conormal to M^2_ϕ away from it; it is smooth and rapidly decreasing away from $\{y = y'\}$, so extends by zero. The last statement is Proposition 14.6.3(iii). □

Away from the diagonal the kernels are conormal to M^2_ϕ but extend across the whole fibre, so it is natural to describe A over a trivializing open set of the base as an operator on B with values in operators on Z. Let $O \subset B$ be open, with coordinates y, a trivialization $\phi^{-1}(O) \cong Z \times O$ and bundle isomorphisms $E \cong \pi_Z^*E_0$, $F \cong \pi_Z^*F_0$ there.

Definition 14.7.3. $S^{m,m'}_{\mathrm{ph}}(O; Z; E_0, F_0)$ is the space of families $k(y, \eta)$, $(y, \eta) \in O \times \mathbb{R}^b$, of distributional kernels on Z^2 (sections of $\mathrm{Hom}(E_0, F_0) \otimes \Omega_R$) such that

(a) if $\chi \in \mathcal{C}^\infty(Z^2)$ vanishes near Diag_Z then χk is a classical symbol of order m in η with values in the Fréchet space $\mathcal{C}^\infty(Z^2; \mathrm{Hom}(E_0, F_0) \otimes \Omega_R)$;

(b) if $\psi, \psi' \in \mathcal{C}^\infty_{\mathrm{c}}(\Omega_Z)$ for a coordinate chart $\Omega_Z \subset Z$ then $\psi k(y, \eta)\psi' = \mathrm{Op}_z(a(\cdot, y, \cdot, \eta))$ with $a \in S^{m,m'}_{\mathrm{ph}}(\Omega_Z \times O)$, compactly supported in z.

For k with compact support in y we set $\mathrm{Op}_y(k)u(y) = \int e^{iy\cdot\eta} k(y,\eta)\hat{u}(\eta)đ\eta$, i.e. the kernel of $\mathrm{Op}_y(k)$ is $\int e^{i(y-y')\cdot\eta}k(y,\eta)đ\eta$. The non-classical spaces $S^{m,m'}(O;Z;E_0,F_0)$ are defined in the same way using non-classical symbols.

Proposition 14.7.4. *If $A \in \Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ and $\chi,\chi' \in \mathcal{C}^\infty_{\mathrm{c}}(O)$ then $\phi^*\chi\, A\, \phi^*\chi' = \mathrm{Op}_y(k)$ with $k \in S^{m,m'}_{\mathrm{ph}}(O;Z;E_0,F_0)$ compactly supported in y; conversely $\mathrm{Op}_y(k)\phi^*\chi' \in \Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ for each such k.*

Proof. Set $k(y,\eta) = \int e^{-i(y-y')\cdot\eta}K(\cdot,y,\cdot,y')\,dy'$, where K is the kernel of $\phi^*\chi A\phi^*\chi'$. Condition (a) holds since away from Diag_Z the kernel is conormal to $\{y=y'\}$ of symbol order m (the variables (z,z',y) are coordinates along M^2_ϕ, and the characterization of $I^{\cdot}(M^2,M^2_\phi)$ recalled in Section 14.6 applies, with $y-y'$ as normal variable). For (b) choose a partition of unity ψ_p on the chart with small supports. If $\operatorname{supp}\psi_p$ and $\operatorname{supp}\psi_q$ are close, then $\psi_p k\psi_q = \mathrm{Op}_z(a_{pq})$ with $a_{pq}\in S^{m,m'}_{\mathrm{ph}}$ by Proposition 14.7.2, since $\mathrm{Op}(a) = \mathrm{Op}_y(\mathrm{Op}_z(a))$. If they are disjoint then by (a) $\psi_p k\psi_q$ is a smooth kernel, compactly supported in the chart, which is a classical symbol in η; it equals $\mathrm{Op}_z(c)$ with $c(z,y,\zeta,\eta) = \int e^{-i(z-z')\cdot\zeta}(\psi_p k\psi_q)(z,z')\,dz'$, and c is Schwartz in ζ with values in classical symbols in η, so lies in $S^{m,-\infty}_{\mathrm{ph}} \subset S^{m,m'}_{\mathrm{ph}}$ by Proposition 14.5.3(viii). Summing gives (b). The converse follows by reversing these arguments. □

Lemma 14.7.5. *Let $k \in S^{m,m'}(O;Z;E_0,F_0)$ and $K \subset O$ compact.*

(i) *For all multi-indices γ, β the family*

$$\langle\eta\rangle^{-m+|\beta|}\partial_y^\gamma\partial_\eta^\beta k(y,\eta), \quad y\in K,\ \eta\in\mathbb{R}^b,$$

is bounded in $\Psi^{m'-m+|m-|\beta||}(Z;E_0,F_0)$.

(ii) *If k is classical, then for $\hat\eta\in\mathbb{S}^{b-1}$ the family $r^m k(y,\hat\eta/r)$ is smooth in $r\in[0,1)$ with values in $\Psi^\infty(Z;E_0,F_0)$, its j-th derivative in r being bounded in $\Psi^{m'-m+|m|+j}(Z;E_0,F_0)$. In particular*

$$\beta(k)(y,\eta) = \lim_{t\to\infty} t^{-m}k(y,t\eta) \tag{14.7.1}$$

exists in $\Psi^{m'-m+|m|+1}(Z;E_0,F_0)$, and lies in $\Psi^{m'-m}(Z;E_0,F_0)$; its local symbols are $\sigma_V(a)$ of (14.5.2).

Proof. (i) For the local symbols, $\partial^\beta_\eta a \in S^{m-|\beta|,m'-|\beta|}$, and by Peetre's inequality

$$\langle\zeta\rangle^{m'-m-|\alpha|}\langle\zeta,\eta\rangle^{m-|\beta|} \le C\langle\eta\rangle^{m-|\beta|}\langle\zeta\rangle^{m'-m+|m-|\beta||-|\alpha|};$$

the smooth parts are handled by (a). (ii) In terms of $a = \langle\zeta\rangle^{m'}\langle q\rangle^m b(\zeta,q)$ with $q=\hat\eta/(r\langle\zeta\rangle)$ we have $r^m a = \langle\zeta\rangle^{m'-m}(1+r^2\langle\zeta\rangle^2)^{m/2}b(\zeta,q)$. Near H_V, b is a smooth function of $\zeta\in\overline{U}$, $1/|q| = r\langle\zeta\rangle$ and $q/|q| = \hat\eta$, while where $|q|$ is bounded we have $r\langle\zeta\rangle \ge c > 0$ and $|\partial_r q| = |q|/r \le C\langle\zeta\rangle$. In either case each r-derivative produces at most a factor $\langle\zeta\rangle$ (also when it falls on $(1+r^2\langle\zeta\rangle^2)^{m/2}$), which gives the stated bounds; setting $r=0$ gives $\langle\zeta\rangle^{m'-m}b(\zeta,\infty\hat\eta)$, which is the symbol $\sigma_V(a)(\cdot,\hat\eta)$ of order $m'-m$. The smooth parts are classical in η by (a). □

Proposition 14.7.6. *Each $A\in\Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ defines continuous linear maps $\mathcal{C}^\infty(M;E)\longrightarrow\mathcal{C}^\infty(M;F)$ and $\mathcal{C}^{-\infty}(M;E)\longrightarrow\mathcal{C}^{-\infty}(M;F)$. The transpose and, given Hermitian metrics and a positive density, the adjoint of A are product-type operators of the same orders.*

Proof. The transpose has kernel obtained by pulling back under the involution $(m, m') \mapsto (m', m)$ of M^2, which preserves M^2_ϕ and Diag_M; so it is in the class by Proposition 14.6.6(i), and similarly for the adjoint. The action on distributions follows by duality from that on smooth sections. Choose a partition of unity χ_i on B with supports so small that whenever $\operatorname{supp}\chi_i \cap \operatorname{supp}\chi_j \neq \emptyset$, both lie in a trivializing open set. Then $A = \sum_{i,j} \phi^*\chi_i A \phi^*\chi_j$; if the supports are disjoint the kernel is supported off M^2_ϕ, so is smooth, and the term is smoothing. The other terms are of the form $\mathrm{Op}_y(k)$, and $\mathrm{Op}_y(k)u(\cdot, y) = \int e^{iy\cdot\eta} k(y,\eta)\hat{u}(\cdot,\eta)đ\eta$, where $\hat{u}(\cdot,\eta)$ is a Schwartz function of η with values in $\mathcal{C}^\infty(Z; E_0)$. By Lemma 14.7.5(i) and the continuity of the action $\Psi^\mu(Z) \times \mathcal{C}^\infty(Z) \longrightarrow \mathcal{C}^\infty(Z)$, the integrand and all its y-derivatives are bounded in $\mathcal{C}^\infty(Z; F_0)$ by $C\langle\eta\rangle^{-N}$, so the integral defines an element of $\mathcal{C}^\infty(O \times Z; F_0)$ depending continuously on u. □

14.7.2 Symbols

By Lemma 14.1.6, the conormal bundle $W = N^*\,\mathrm{Diag}_M$ is T^*M, with $V = N^*M^2_\phi|_{\mathrm{Diag}} = \phi^*T^*B$ and $W/V = T^*(M/B)$. The relative compactification (Remark 14.4.6) is a manifold with corners ${}^{\phi^*T^*B}\overline{T^*M}$ over M. Its main face

$$H_W = [S^*M; \phi^*S^*B]$$

has interior $S^*M \setminus \phi^*S^*B$ and fibres over $S^*(M/B)$ with fibre over a point above $b \in B$ a radially compactified affine space modelled on T^*_bB, by (14.4.3); its boundary, the corner, is $S^*(M/B) \times_M \phi^*S^*B$, which is the vertical cosphere bundle $S^*(\pi^*M/S^*B)$ of the pulled-back fibration $\pi^*M = M \times_B S^*B \longrightarrow S^*B$. The front face is $H_V = \overline{T^*(M/B)} \times_M \phi^*S^*B$. On the other hand $N^*M^2_\phi \setminus 0 = M^2_\phi \times_B (T^*B \setminus 0)$ is the fibre product for the fibration $\pi^*M \longrightarrow T^*B \setminus 0$, and its intersection with the conormal bundle of the diagonal is the diagonal of that fibre product. Since $\Omega(M) \cong \Omega(M/B) \otimes \phi^*\Omega(B)$ and $\Omega(B) \otimes \Omega(T^*_bB)$ is canonically trivial, the density factors in Proposition 14.6.7 and Proposition 14.6.8 are absorbed.

Accordingly we define the symbol spaces as follows. Choose a metric and let $|\zeta|$ denote the length of the image of $\xi \in T^*_xM$ in $T^*_x(M/B)$.

- $\Sigma^{m,m'}_\phi(M; E, F)$ is the space of $s \in \mathcal{C}^\infty(T^*M \setminus \phi^*T^*B; \hom(E,F))$, homogeneous of degree m', such that $|\zeta|^{m-m'}|\xi|^{-m}s$, restricted to $S^*M \setminus \phi^*S^*B$, extends to a smooth function on $[S^*M; \phi^*S^*B]$.
- $\mathcal{B}^{m,m'}_\phi(M; E, F)$ is the space of smooth families $P(b,\eta) \in \Psi^{m'-m}(Z_b; E_b, F_b)$, $(b,\eta) \in T^*B \setminus 0$, which are homogeneous of degree m, $P(b, t\eta) = t^mP(b,\eta)$ for $t > 0$; that is, the homogeneous elements of $\Psi^{m'-m}(\pi^*M/(T^*B \setminus 0); E, F)$.

Theorem 14.7.7 (Symbol maps)**.** *There are two symbol maps on* $\Psi^{m,m'}_{\phi\text{-pt}}(M; E, F)$*:*

(i) *the* principal symbol $\sigma_{m,m'} : \Psi^{m,m'}_{\phi\text{-pt}}(M; E, F) \longrightarrow \Sigma^{m,m'}_\phi(M; E, F)$*, given in local coordinates by* $\sigma_{m,m'}(A)(x,\xi) = \lim_{t\to\infty} t^{-m'}a(x, t\xi)$ *for* $\xi \notin \phi^*T^*B$*;*

(ii) *the* base symbol $\beta_{m,m'} : \Psi^{m,m'}_{\phi\text{-pt}}(M; E, F) \longrightarrow \mathcal{B}^{m,m'}_\phi(M; E, F)$*, given in a local trivialization by* (14.7.1)*.*

They give short exact sequences

$$\Psi^{m,m'-1}_{\phi\text{-pt}} \hookrightarrow \Psi^{m,m'}_{\phi\text{-pt}} \xrightarrow{\sigma_{m,m'}} \Sigma^{m,m'}_\phi, \qquad \Psi^{m-1,m'}_{\phi\text{-pt}} \hookrightarrow \Psi^{m,m'}_{\phi\text{-pt}} \xrightarrow{\beta_{m,m'}} \mathcal{B}^{m,m'}_\phi,$$

and the joint symbol $(\sigma_{m,m'}, \beta_{m,m'})$ *has null space* $\Psi_{\phi\text{-pt}}^{m-1,m'-1}(M;E,F)$ *and range the compatible pairs, i.e. those satisfying*

$$\lim_{t\to\infty} t^{-m} s(x, \tilde{\zeta} + t\,\phi^*\eta) = \sigma_{m'-m}(P(\phi(x),\eta))(x,\zeta) \tag{14.7.2}$$

for all $x \in M$, $\eta \in T^*_{\phi(x)}B \setminus 0$, $\zeta \in T^*_x(M/B) \setminus 0$ *and any lift* $\tilde{\zeta} \in T^*_x M$ *of* ζ*; here* $\sigma_{m'-m}(P(b,\eta))$ *is the principal symbol of the operator* $P(b,\eta)$ *on* Z_b*.*

Proof. The principal symbol is $\sigma_W(K_A)$ of Proposition 14.6.7 and the base symbol is $\sigma_V(K_A)$ of Proposition 14.6.8, under the identifications above; in particular both are invariantly defined. That $\sigma_V(K_A)$ is given by (14.7.1) is Lemma 14.7.5(ii) near the diagonal and the definition of the principal symbol of a conormal distribution away from it. The exact sequences and the range of the joint symbol are Proposition 14.6.7 and Proposition 14.6.8, and (14.7.2) is (14.5.3): indeed in local coordinates $\tilde{\zeta} + t\eta$ corresponds to $\zeta + t\eta$, and the right side of (14.5.3) is the principal symbol of the classical symbol $\sigma_V(a)(x,\cdot,\eta)$ of order $m' - m$, which is the local symbol of $\beta_{m,m'}(A)$. □

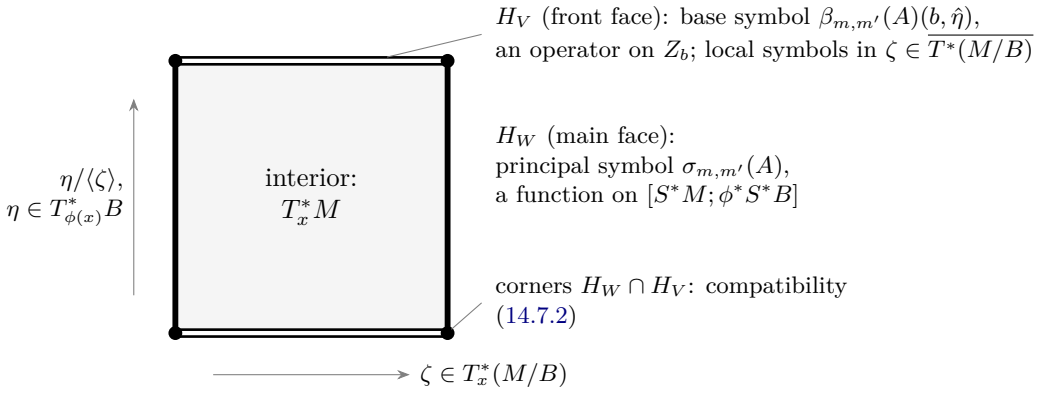


Figure 14.4: Which symbol lives where. The fibre of the relative compactification ${}^{\phi^*T^*B}\overline{T^*M}$ over $x \in M$ is a square $\overline{T^*_x(M/B)} \times \overline{\mathbb{R}^b}$ (Figure 14.2). The principal symbol is the leading behaviour at the main face H_W (sides, thick), where the fibre covariable ζ is large. The base symbol is the leading behaviour at the front face H_V (top and bottom, double), where the base covariable η is much larger than ζ; for each direction $\hat{\eta}$ it is a whole pseudodifferential operator on the fibre. The two must agree at the corners. Full ellipticity (Definition 14.7.13) asks for invertibility on both faces.

Discussion. Why is the base symbol an operator and not a function? On the front face the fibre variable ζ is not sent to infinity: it stays a finite point of $\overline{T^*(M/B)}$. So the leading behaviour in η is a whole classical symbol in ζ, of order $m' - m$, not just its leading part. Together with the part of the kernel away from the diagonal (which is conormal to M^2_ϕ and also has a leading term in η) this is an operator on the fibre Z_b. Composition in the calculus therefore composes base symbols as operators (Theorem 14.7.8), and their invertibility is a global question on each fibre, not a pointwise one.

Concretely, the surjectivity of the base symbol amounts to the following: if $P \in \mathcal{B}^{m,m'}_\phi$ has local symbols $p(z,y,\zeta,\hat{\eta})$ (of order $m' - m$ in ζ) near the fibre diagonal, then $a(z,y,\zeta,\eta) = (1 - \chi(\eta/\langle\zeta\rangle))|\eta|^m p(z,y,\zeta,\eta/|\eta|)$, with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^b)$ equal to 1 near 0, lies in $S^{m,m'}_{\mathrm{ph}}$ and has $\sigma_V(a) = |\eta|^m p$. Note that the cut-off must be a function of $q = \eta/\langle\zeta\rangle$, not of η: $(1 - \chi(\eta))|\eta|^m p$ is not smooth on the relative compactification at $H_W \cap \{q = 0\}$.

14.7.3 Composition

Theorem 14.7.8. *For any fibration $\phi : M \longrightarrow B$ of compact manifolds and bundles E, F, G over M,*

$$\Psi^{m_1,m_1'}_{\phi\text{-pt}}(M;F,G)\circ\Psi^{m_2,m_2'}_{\phi\text{-pt}}(M;E,F)\subset\Psi^{m_1+m_2,m_1'+m_2'}_{\phi\text{-pt}}(M;E,G),$$

and both symbol maps are multiplicative:

$$\sigma_{m_1+m_2,m_1'+m_2'}(AA') = \sigma_{m_1,m_1'}(A)\,\sigma_{m_2,m_2'}(A'),$$
$$\beta_{m_1+m_2,m_1'+m_2'}(AA')(b,\eta) = \beta_{m_1,m_1'}(A)(b,\eta)\circ\beta_{m_2,m_2'}(A')(b,\eta),$$

the latter being composition of pseudodifferential operators on Z_b.

Proof. Step 1: reduction to a trivialization. Choose a finite cover of B by trivializing open sets, let $\delta>0$ be a Lebesgue number for it (for some metric on B) and χ_i a partition of unity with $\operatorname{diam}\operatorname{supp}\chi_i<\delta/4$. Write $A=\sum A_{ij}$, $A_{ij}=\phi^*\chi_i A\phi^*\chi_j$, and similarly $A'=\sum A'_{kl}$. As in the proof of Proposition 14.7.6, the terms with $\operatorname{supp}\chi_i\cap\operatorname{supp}\chi_j=\emptyset$ are smoothing. If R is smoothing and A'' is any product-type operator then RA'' and $A''R$ are smoothing, since for instance the kernel of RA'' is $(x,x')\mapsto((A'')^tK_R(x,\cdot))(x')$, which is smooth by Proposition 14.7.6 applied with the parameter x. Moreover $A_{ij}A'_{kl}=0$ unless $\operatorname{supp}\chi_j\cap\operatorname{supp}\chi_k\neq\emptyset$. The remaining products involve four sets whose union has diameter less than δ, so lies in a trivializing set O. By Proposition 14.7.4 it therefore suffices to show that $\operatorname{Op}_y(k_1)\operatorname{Op}_y(k_2)=\operatorname{Op}_y(k)$ with $k\in S^{m_1+m_2,m_1'+m_2'}_{\text{ph}}(O;Z;E_0,G_0)$ when $k_i\in S^{m_i,m_i'}_{\text{ph}}(O;Z)$ have compact support in y.

Step 2: products on the fibre. We first show that the pointwise product $k_1(y,\eta)k_2(y,\eta)$ (composition of operators on Z) lies in $S^{m_1+m_2,m_1'+m_2'}_{\text{ph}}(O;Z)$, with local symbol having leading part a_1a_2 modulo $S^{m_1+m_2,m_1'+m_2'-1}$. Decompose each k_i using a partition of unity ψ_p on Z whose supports are small compared to a Lebesgue number of a cover of Z by charts. A product of a piece $\psi_pk_1\psi_q$ and a piece $\psi_rk_2\psi_s$ vanishes unless q and r overlap. If both pieces are near the diagonal, all four supports lie in one chart and the product is $\operatorname{Op}_z(a_1)\operatorname{Op}_z(a_2)=\operatorname{Op}_z(c)$ with

$$c(z,y,\zeta,\eta)=\iint e^{-iz'\cdot\theta}a_1(z,y,\zeta+\theta,\eta)a_2(z+z',y,\zeta,\eta)\,dz'đ\theta.$$

This is the standard composition formula with (y,η) as parameters. The standard proof (Taylor expansion of a_1 in θ and integration by parts in z', Chapter 2) goes through verbatim with the weights $w_{M,M'}$ in place of $\langle\xi\rangle^\mu$, using Lemma 14.5.2 for translations in ζ alone and the fact that ∂_ζ lowers the second order; it gives

$$c\sim\sum_\alpha\frac{1}{\alpha!}\partial_\zeta^\alpha a_1\,D_z^\alpha a_2,\qquad \partial_\zeta^\alpha a_1\,D_z^\alpha a_2\in S^{m_1+m_2,m_1'+m_2'-|\alpha|}_{\text{ph}},$$

modulo $S^{m_1+m_2,-\infty}$. Moreover c is classical: after integrating by parts to make the integral absolutely convergent, the integrand is, for fixed (z',θ), a classical product-type symbol with seminorms $O(\langle z'\rangle^{-L}\langle\theta\rangle^{K-L})$ by Lemma 14.5.4. If one of the pieces, say $\psi_pk_1\psi_q$, has a smooth kernel, then the product has kernel $(z,z')\mapsto(k_2(y,\eta)^t(\psi_pk_1\psi_q)(z,\cdot))(z')$, which by Lemma 14.7.5 is a classical symbol of order m_1+m_2 in η with values in $\mathcal{C}^\infty(Z^2)$: the factor $\langle\eta\rangle^{m_2}$ comes from (i), and classicality from (ii), since the smooth kernel absorbs the loss of orders in the r-derivatives. If both pieces are near the diagonal but $\operatorname{supp}\psi_p$ and $\operatorname{supp}\psi_s$ are

far apart then the product vanishes by the choice of supports. Summing all pieces proves the claim.

Step 3: composition in the base. Exactly as for operators on $\mathbb{R}^b$,

$$\mathrm{Op}_y(k_1)\,\mathrm{Op}_y(k_2) = \mathrm{Op}_y(k), \quad k(y,\eta) = \iint e^{-iy'\cdot\theta} k_1(y,\eta+\theta) k_2(y+y',\eta)\, dy' đ\theta,$$

an oscillatory integral of products of operators on Z (the order of the factors matters). Taylor expansion of $k_1(y,\eta+\theta)$ in θ and integration by parts in y' give

$$k \sim \sum_\alpha \frac{1}{\alpha!} \partial_\eta^\alpha k_1 \cdot D_y^\alpha k_2, \qquad \partial_\eta^\alpha k_1 \cdot D_y^\alpha k_2 \in S^{m_1+m_2-|\alpha|, m_1'+m_2'-|\alpha|}_{\mathrm{ph}}(O;Z), \tag{14.7.3}$$

by Step 2 and Proposition 14.5.3(iv); the remainder after N terms is an integral of products $\partial_\eta^\alpha k_1(y,\eta+s\theta)\,\partial_y^\alpha k_2(y+y',\eta)$, $|\alpha| = N$, at different values of η. The argument of Step 2 applies to such products, and Peetre's inequality $w_{M,M'}(\zeta,\eta+s\theta) \le C w_{M,M'}(\zeta,\eta)\langle\theta\rangle^K$ shows, as in the scalar case, that the remainder lies in the non-classical space $S^{m_1+m_2-N, m_1'+m_2'-N}(O;Z)$. By Proposition 14.5.5(iii) there is a classical $\tilde{k}$ with the expansion (14.7.3); then $k-\tilde{k} \in \bigcap_N S^{-N,-N}(O;Z)$, and this space consists of Schwartz functions of η with values in smoothing operators on Z, which are classical. Hence $k \in S^{m_1+m_2, m_1'+m_2'}_{\mathrm{ph}}(O;Z)$.

Step 4: symbols. The local symbol of k is $a_1 a_2$ modulo $S^{m_1+m_2, m_1'+m_2'-1}$, which gives multiplicativity of σ. All terms of (14.7.3) except the first have base order at most m_1+m_2-1, so $\beta(AA') = \lim t^{-m_1-m_2} k_1(y,t\eta) k_2(y,t\eta)$; by Lemma 14.7.5(ii) the factors converge in spaces of pseudodifferential operators of fixed order, and composition is continuous, so the limit is $\beta(A)\beta(A')$. □

Proposition 14.7.9. (i) $\bigcap_j \Psi^{m-j,m'-j}_{\phi\text{-pt}}(M;E,F) = \Psi^{-\infty}(M;E,F)$, *the smoothing operators.*

(ii) $\Psi^{m,-\infty}_{\phi\text{-pt}}(M;E,F) = \bigcap_{m'} \Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ *consists of the operators whose kernels are conormal to* M^2_ϕ *of order* $m - \frac{f}{2}$*; locally in* B *these are the* $\mathrm{Op}_y(k)$ *with* k *a classical symbol of order* m *with values in smoothing operators on* Z*.*

(iii) *Asymptotic completeness holds for* $\Psi^{m,m'}_{\phi\text{-pt}}$ *in the three forms of Proposition 14.5.5, modulo* $\Psi^{m,-\infty}_{\phi\text{-pt}}$*, modulo* $\Psi^{-\infty,m'}_{\phi\text{-pt}}$ *and modulo* $\Psi^{-\infty}(M)$ *respectively.*

Proof. This is Proposition 14.6.6(ii), (iii) and (v) in the present setting, together with Proposition 14.7.4. □

14.7.4 Subalgebras

Proposition 14.7.10. (i) (Fibrewise operators) $\Psi^{m'}(M/B;E,F) \subset \Psi^{0,m'}_{\phi\text{-pt}}(M;E,F)$*; for* $A \in \Psi^{m'}(M/B;E,F)$*,* $\sigma_{0,m'}(A) = \gamma^*\sigma_{m'}(A)$*, where* $\gamma : T^*M\backslash\phi^*T^*B \longrightarrow T^*(M/B)\backslash 0$ *is the quotient map, and* $\beta_{0,m'}(A)(b,\eta) = A_b$ *for all* $\eta \neq 0$*.*

(ii) (Ordinary operators) $\Psi^m(M;E,F) \subset \Psi^{m,m}_{\phi\text{-pt}}(M;E,F)$*;* $\sigma_{m,m}(A)$ *is the restriction of* $\sigma_m(A)$ *to* $T^*M \setminus \phi^*T^*B$*, and* $\beta_{m,m}(A)(b,\eta)$ *is the operator of multiplication on* Z_b *by* $x \longmapsto \sigma_m(A)(x,\phi^*\eta)$*.*

(iii) (Compressed base operators) *Let* $\pi_1 \in \Psi^{-\infty}(M/B;E)$ *and* $\pi_2 \in \Psi^{-\infty}(M/B;F)$ *be smooth families of projections. Their ranges are finite-dimensional and form vector*

bundles $\tilde E$, $\tilde F$ over B, subbundles of $\mathcal{C}^\infty(M/B;E)$, $\mathcal{C}^\infty(M/B;F)$. If $P\in\Psi^m(B;\tilde E,\tilde F)$ then $\tilde P=\iota_2 P\pi_1$, where $\iota_2:\mathcal{C}^\infty(B;\tilde F)\hookrightarrow\mathcal{C}^\infty(M;F)$, lies in $\Psi^{m,-\infty}_{\phi\text{-pt}}(M;E,F)$, with $\beta_{m,-\infty}(\tilde P)(b,\eta)=\sigma_m(P)(b,\eta)\circ\pi_1(b)$ (followed by the inclusion $\tilde F_b\subset\mathcal{C}^\infty(Z_b;F_b)$).

Proof. (i) By Proposition 14.6.6(iv) with $Y=M_\phi^2$, $Z=\mathrm{Diag}_M$, $d=f+b$, $k=f$, $l=b$: an element of $I^{m'-b/4}(M_\phi^2,\mathrm{Diag})$ with values in $\mathrm{Hom}(E,F)\otimes\Omega_R(M/B)$ times $\delta_{M_\phi^2}$ (the factor $\Omega(N^*M_\phi^2)\cong\phi^*\Omega(T^*B)$ combining with $\Omega_R(M/B)$ to give $\Omega_R(M)$) lies in $I^{(b-2f-b)/4,\,m'}=I^{-f/2,m'}$, i.e. in $\Psi^{0,m'}_{\phi\text{-pt}}$. In local coordinates the symbol $a(z,y,\zeta)$ is independent of η, so $\sigma_W(a)$ is the pull-back of $\sigma_{m'}(A)$ and $\sigma_V(a)=a$; in the operator-valued description $k(y,\eta)=A_y$ is independent of η.

(ii) The local symbols of A lie in $S^m_{\mathrm{ph}}\subset S^{m,m}_{\mathrm{ph}}$ by Proposition 14.5.3(v), and the kernel is smooth off the diagonal; the limit (14.5.2) of $t^{-m}a(z,y,\zeta,t\eta)$ is the principal symbol at $(0,\eta)$, independent of ζ, i.e. a multiplication operator.

(iii) The rank of $\pi_1(b)$ is $\mathrm{Tr}\,\pi_1(b)$, a continuous integer-valued function, so locally constant, and near b_0 the sections $\pi_1(b)e_j$, for $e_j\in\mathcal{C}^\infty(M;E)$ restricting to a basis of $\mathrm{Ran}\,\pi_1(b_0)$, form a smooth local frame of $\tilde E$. Over a trivializing open set O choose smooth frames $e_1,\dots,e_r\in\mathcal{C}^\infty(\phi^{-1}(O);E)$ of $\tilde E$ and $f_1,\dots,f_s$ of $\tilde F$, and $e_j^*\in\mathcal{C}^\infty(\phi^{-1}(O);E^*\otimes\Omega(M/B))$ with $\pi_1 u=\sum_j e_j\int_{Z_b}e_j^*u$. If P has matrix (P_{ij}) in these frames, the kernel of $\tilde P$ over O is $\sum_{i,j}f_i(z,y)K_{P_{ij}}(y,y')e_j^*(z',y')$, i.e. $\tilde P=\mathrm{Op}_y(k)$ with $k(y,\eta)=\sum_{ij}f_i\,p_{ij}(y,\eta)\,e_j^*$, a classical symbol of order m with values in smoothing operators of finite rank on Z (up to a smoothing term coming from the left reduction in y). The base symbol is the limit (14.7.1). □

Remark 14.7.11. For a product $M=Z\times B$ one might expect the lift $\mathrm{Id}_Z\otimes P$ of $P\in\Psi^m(B)$ to be a product-type operator, of base order m and fibre order 0; this is not so, which is why the compression in (iii) is needed. Its local symbol $p(y,\eta)$ is independent of ζ, and near H_W, where $|\zeta|\to\infty$ with $q=\eta/\langle\zeta\rangle$ bounded, $p(y,\langle\zeta\rangle q)$ behaves like $\langle\zeta\rangle^m|q|^m p_m(y,q/|q|)$, which is not smooth at $q=0$ unless p_m is a polynomial. In fact $\mathrm{Id}_Z\otimes P$ is product-type (and then lies in $\Psi^m(M)\subset\Psi^{m,m}_{\phi\text{-pt}}(M)$) only when P is a differential operator (Exercise 14.7).

14.7.5 Boundedness, ellipticity and the Fredholm property

Proposition 14.7.12. *Each $A\in\Psi^{0,0}_{\phi\text{-pt}}(M;E,F)$ extends to a bounded operator $L^2(M;E)\longrightarrow L^2(M;F)$ (for any choice of Hermitian metrics and smooth positive density).*

Proof. As in the proof of Proposition 14.7.6, up to smoothing terms (which are bounded) A is a finite sum of operators $\mathrm{Op}_y(k)$ with $k\in S^{0,0}_{\mathrm{ph}}(O;Z)$ compactly supported in $y\in O\subset\mathbb{R}^b$. By Lemma 14.7.5(i) with $\beta=0$, the operators $\partial_y^\gamma k(y,\eta)$ are bounded in $\Psi^0(Z)$, hence uniformly bounded on $L^2(Z;E_0)$ (the inclusion $\Psi^0(Z)\hookrightarrow\mathcal{B}(L^2(Z))$ is continuous). Write $k(y,\eta)=\int e^{iy\cdot\theta}\hat k(\theta,\eta)đ\theta$, where

$$\|\hat k(\theta,\eta)\|_{\mathcal{B}(L^2(Z))}\le C_N\langle\theta\rangle^{-2N}\sup_y\|(1-\Delta_y)^N k(y,\eta)\|\le C'_N\langle\theta\rangle^{-2N}.$$

Then

$$\mathrm{Op}_y(k)u=\int e^{iy\cdot\theta}\hat k(\theta,D_y)u đ\theta,$$

where $\hat k(\theta,D_y)$ is the Fourier multiplier, acting on the space $L^2(\mathbb{R}^b;L^2(Z;E_0))$, with operator-valued multiplier $\hat k(\theta,\eta)$; by Plancherel's theorem its norm is at most $\sup_\eta\|\hat k(\theta,\eta)\|$. Multiplication by $e^{iy\cdot\theta}$ is unitary, so

$$\|\mathrm{Op}_y(k)\|\le\int C'_N\langle\theta\rangle^{-2N}đ\theta<\infty.$$

□

Definition 14.7.13. $A \in \Psi^{m,m'}_{\phi\text{-pt}}(M; E, F)$ is *fully elliptic* if

(a) $\sigma_{m,m'}(A)$ is invertible, in the sense that the normalized symbol $|\zeta|^{m-m'}|\xi|^{-m}\sigma_{m,m'}(A)$ is invertible at every point of $[S^*M; \phi^*S^*B]$, including its boundary, and

(b) $\beta_{m,m'}(A)(b,\eta) : \mathcal{C}^\infty(Z_b; E_b) \longrightarrow \mathcal{C}^\infty(Z_b; F_b)$ is invertible for every $(b,\eta) \in T^*B \setminus 0$.

By (14.7.2), (a) implies that each $\beta(A)(b,\eta)$ is an elliptic operator of order $m' - m$ on Z_b; (b) then requires these elliptic operators to be invertible, which in particular forces their index to vanish.

Question. Why is an invertible principal symbol not enough for the Fredholm property?

Answer. Take $M = Z \times B$ with $\dim B > 0$ and the constant family $P_b = P_0$, where $P_0 \in \Psi^0(Z; E_0, F_0)$ is elliptic with $\operatorname{Nul} P_0 \neq 0$, regarded as an element of $\Psi^{0,0}_{\phi\text{-pt}}(M; E, F)$. Its principal symbol is invertible, since P_0 is elliptic. But if $P_0 v = 0$, then $u(z,y) = \chi(y)v(z)$ satisfies $Pu = 0$ for every $\chi \in \mathcal{C}^\infty(B)$, so the null space on M is infinite dimensional. The base symbol, here P_0 itself (Proposition 14.7.10(i)), detects this: it is not invertible. □

Lemma 14.7.14. *If $P(p) \in \Psi^\mu(Z; E_0, F_0)$ is a smooth family of elliptic operators, depending on parameters p in a manifold, and each $P(p)$ is invertible on $\mathcal{C}^\infty$, then $P(p)^{-1} \in \Psi^{-\mu}(Z; F_0, E_0)$ is again a smooth family.*

Proof. Let $Q(p)$ be a smooth family of parametrices, $QP = \mathrm{Id} - R$, $PQ = \mathrm{Id} - R'$, with R, R' smooth families of smoothing operators. Then $P^{-1} = Q + RP^{-1} = Q + RQ + RP^{-1}R'$. The inverses $P(p)^{-1} : H^s \longrightarrow H^{s+\mu}$ depend smoothly on p in norm, by Lemma 14.3.1, for every s, and since R' maps H^{-N} to H^N and R maps H^{-N} to H^N smoothly in p for all N, $RP^{-1}R'$ is a smooth family of smoothing operators. □

Theorem 14.7.15. *If $A \in \Psi^{m,m'}_{\phi\text{-pt}}(M; E, F)$ is fully elliptic there exists $Q \in \Psi^{-m,-m'}_{\phi\text{-pt}}(M; F, E)$ such that*

$$QA = \mathrm{Id}_E - R, \quad AQ = \mathrm{Id}_F - R', \qquad R \in \Psi^{-\infty}(M; E),\ R' \in \Psi^{-\infty}(M; F).$$

Conversely, if such a Q exists then A is fully elliptic.

Proof. The inverse of the normalized principal symbol is smooth on $[S^*M; \phi^*S^*B]$, so $\sigma_{m,m'}(A)^{-1} \in \Sigma^{-m,-m'}_\phi(M; F, E)$. By Lemma 14.7.14 applied to the fibration $\pi^*M \longrightarrow T^*B \setminus 0$, $\beta_{m,m'}(A)^{-1} \in \mathcal{B}^{-m,-m'}_\phi(M; F, E)$; it is homogeneous of degree $-m$, and its principal symbol is the inverse of that of $\beta(A)$, so by (14.7.2) the pair $(\sigma(A)^{-1}, \beta(A)^{-1})$ is compatible. By Theorem 14.7.7 there is $Q_0 \in \Psi^{-m,-m'}_{\phi\text{-pt}}(M; F, E)$ with $\sigma(Q_0) = \sigma(A)^{-1}$ and $\beta(Q_0) = \beta(A)^{-1}$. By Theorem 14.7.8, $Q_0 A = \mathrm{Id} - R_0$ with $\sigma_{0,0}(R_0) = 0$ and $\beta_{0,0}(R_0) = 0$, so $R_0 \in \Psi^{-1,-1}_{\phi\text{-pt}}(M; E)$. By Proposition 14.7.9(iii) the Neumann series $\sum_j R_0^j$ is asymptotically summable, to $S \in \Psi^{0,0}_{\phi\text{-pt}}$ with $S(\mathrm{Id} - R_0) - \mathrm{Id} \in \Psi^{-\infty}$, and $Q = SQ_0$ is a left parametrix. A right parametrix Q' is constructed in the same way, and then $Q - Q' = Q(\mathrm{Id} - AQ') - (\mathrm{Id} - QA)Q'$ is smoothing, so Q is a two-sided parametrix. Conversely, if Q exists then by multiplicativity $\sigma(Q)\sigma(A) = \mathrm{Id}$ and $\beta(Q)\beta(A) = \mathrm{Id}$, and similarly on the other side, since smoothing operators have vanishing symbols. □

Discussion. The construction is the usual one, done on two faces at once. For an ordinary elliptic operator one inverts the principal symbol and removes the error, of order -1, by a Neumann series. Here one inverts both symbols, which is possible exactly when A is fully elliptic. The error then vanishes at both faces, so it gains one order in each index ($\Psi^{-1,-1}_{\phi\text{-pt}}$),

and the Neumann series converges modulo $\bigcap_j \Psi_{\phi\text{-pt}}^{-j,-j}$, which consists of smoothing operators. Inverting only the principal symbol would leave an error in $\Psi_{\phi\text{-pt}}^{0,-1}$, which in general is not compact on L^2.

Corollary 14.7.16. *A fully elliptic $A \in \Psi_{\phi\text{-pt}}^{m,m'}(M;E,F)$ is Fredholm as a map $\mathcal{C}^\infty(M;E) \longrightarrow \mathcal{C}^\infty(M;F)$, its null space and that of its adjoint consist of smooth sections, and*

$$\operatorname{ind}(A) = \operatorname{Tr}(\mathrm{Id}_E - QA) - \operatorname{Tr}(\mathrm{Id}_F - AQ)$$

for any parametrix Q as in Theorem 14.7.15. If A_t, $t \in [0,1]$, is a smooth family of fully elliptic operators then $\operatorname{ind}(A_t)$ *is constant. For $m = m' = 0$, A is also Fredholm on L^2, with the same index.*

Proof. With smoothing errors, the arguments of Chapter 5 and Chapter 12 for elliptic operators apply verbatim: $\operatorname{Nul}(A) \subset \operatorname{Nul}(\mathrm{Id} - R)$ is a finite-dimensional space of smooth sections, the range is closed of finite codimension, and the trace formula holds. The construction of the parametrix can be carried out with smooth dependence on t, so the right side of the trace formula is continuous in t, and integer valued. For $m = m' = 0$, A, Q are bounded on L^2 by Proposition 14.7.12 and R, R' are compact, so A is Fredholm on L^2; its null space, and that of its adjoint, are the same as on $\mathcal{C}^\infty$ by the parametrix identities. □

Proposition 14.7.17. *An operator $A \in \Psi_{\phi\text{-pt}}^{0,0}(M;E,F)$ is Fredholm on L^2 if and only if it is fully elliptic.*

Proof. Sufficiency is Corollary 14.7.16. Conversely suppose A is Fredholm on L^2; then there are a bounded Q and a compact K with $QA = \mathrm{Id} + K$, and the same holds for A^*. We sketch the construction of approximate null sequences. Suppose first that $(x_0,\xi_0) \in T^*M$, $\xi_0 \notin \phi^*T^*B$, and $e \in E_{x_0}$. In local coordinates let $u_t(x) = t^{n/4}\chi(t^{1/2}(x - x_0))e^{it x\cdot\xi_0}e$, $n = \dim M$, $\chi \in \mathcal{C}_{\mathrm{c}}^\infty$. Then $\|u_t\|$ is bounded below, $u_t \to 0$ weakly, and a direct computation with the local form $\operatorname{Op}(a)$ (Taylor expanding a at $(x_0, t\xi_0)$) shows that $Au_t - \sigma_{0,0}(A)(x_0,\xi_0)u_t \to 0$ in L^2. Since $Ku_t \to 0$, $\|u_t\| \le \|Q\|\|Au_t\| + o(1)$, so $|\sigma_{0,0}(A)(x_0,\xi_0)e| \ge \|Q\|^{-1}|e|$. By continuity this lower bound extends to all of $[S^*M;\phi^*S^*B]$, and applying the same argument to A^* shows that $\sigma_{0,0}(A)$ is invertible. Next, for $(b_0,\eta_0) \in T^*B\backslash 0$ and $v \in \mathcal{C}^\infty(Z_{b_0};E_{b_0})$, in a trivialization let $u_t(z,y) = t^{b/4}\chi(t^{1/2}(y - y_0))e^{ity\cdot\eta_0}v(z)$. The same computation, now using the operator-valued form $\operatorname{Op}_y(k)$ and Lemma 14.7.5, shows $Au_t - (\beta_{0,0}(A)(b_0,\eta_0)v) \otimes t^{b/4}\chi(t^{1/2}(y - y_0))e^{ity\cdot\eta_0} \to 0$, using the homogeneity of degree 0 of $\beta_{0,0}(A)$. Hence $\|\beta_{0,0}(A)(b_0,\eta_0)v\| \ge \|Q\|^{-1}\|v\|$; the same for A^*, with $\beta(A^*) = \beta(A)^*$, shows that $\beta_{0,0}(A)(b_0,\eta_0)$ is invertible on L^2, hence, being elliptic, on $\mathcal{C}^\infty$. □

14.7.6 The tensor product construction

The following construction, which is the starting point for the proof of the Atiyah–Singer index theorem in Chapter 15, shows how product-type operators connect the index of an operator on the base with operators on the total space of a fibration.

Proposition 14.7.18. *Let $\phi : M \longrightarrow B$ be a fibration, with Hermitian metrics and a fibre density chosen. Let $P \in \Psi^0(M/B;E_+,E_-)$ be an elliptic family such that each P_b is surjective and* $\operatorname{Nul}(P)$ *is a trivial line bundle, spanned by $e \in \mathcal{C}^\infty(M;E_+)$ with $\|e|_{Z_b}\| = 1$, and let $\pi_\nu = e \otimes e^* \in \Psi^{-\infty}(M/B;E_+)$ be the orthogonal projection onto* $\operatorname{Nul}(P)$. *Let $A \in \Psi^0(B;F_+,F_-)$ be elliptic, and let $A_\nu \in \Psi_{\phi\text{-pt}}^{0,-\infty}(M;E_+ \otimes F_+, E_+ \otimes F_-)$ be the*

compression $A_\nu u = e \otimes A(\int_{Z_b}(u,e)_{E_+})$ *of Proposition 14.7.10(iii) (here and below* $F_\pm$ *also denotes* $\phi^* F_\pm$*). Set*

$$H_+ = E_+ \otimes F_+ \oplus E_- \otimes F_-, \quad H_- = E_- \otimes F_+ \oplus E_+ \otimes F_-,$$
$$P_A = \begin{pmatrix} P & 0 \\ A_\nu & P^* \end{pmatrix} : \mathcal{C}^\infty(M; H_+) \longrightarrow \mathcal{C}^\infty(M; H_-),$$

where P *and* P^* *act on the first factors of the tensor products. Then* $P_A \in \Psi^{0,0}_{\phi\text{-pt}}(M; H_+, H_-)$ *is fully elliptic and* $\operatorname{ind}(P_A) = \operatorname{ind}(A)$.

Proof. By Proposition 14.7.10, P, $P^* \in \Psi^{0,0}_{\phi\text{-pt}}$ and $A_\nu \in \Psi^{0,-\infty}_{\phi\text{-pt}} \subset \Psi^{0,0}_{\phi\text{-pt}}$, with

$$\sigma_{0,0}(P_A) = \begin{pmatrix} \gamma^*\sigma_0(P) & 0 \\ 0 & \gamma^*\sigma_0(P)^* \end{pmatrix}, \qquad \beta_{0,0}(P_A)(b,\eta) = \begin{pmatrix} P_b & 0 \\ \pi_\nu a & P_b^* \end{pmatrix}, \quad a = \sigma_0(A)(b,\eta),$$

where $a : F_{+,b} \longrightarrow F_{-,b}$ is invertible. The principal symbol is invertible, P being elliptic. To see that $\beta(P_A)(b,\eta)$ is invertible, solve $P_b u = g$, $\pi_\nu a u + P_b^* v = h$. Since P_b is surjective with null space $\mathbb{C}e_b$, $u = u_0 + e_b \otimes c$ where u_0 is the solution orthogonal to e_b and $c \in F_{+,b}$ is free. Since P_b^* is injective with range $(\mathbb{C}e_b)^\perp$, the second equation splits into $\pi_\nu a u = \pi_\nu h$, i.e. $e_b \otimes ac = \pi_\nu h$, which determines c uniquely because a is invertible, and $P_b^* v = (\mathrm{Id} - \pi_\nu)h$, which determines v. So $\beta(P_A)(b,\eta)$ is bijective, and P_A is fully elliptic.

For the index, let $(u,v) \in \operatorname{Nul}(P_A)$. Then $Pu = 0$, so $u = e \otimes g$ with $g \in \mathcal{C}^\infty(B; F_+)$; and $A_\nu u + P^* v = e \otimes Ag + P^* v = 0$, where the first term lies in $\operatorname{Nul}(P) \otimes F_-$ and the second in its orthogonal complement, so $Ag = 0$ and $P^* v = 0$, i.e. $v = 0$. Conversely $(e \otimes g, 0) \in \operatorname{Nul}(P_A)$ if $Ag = 0$, so $\operatorname{Nul}(P_A) \cong \operatorname{Nul}(A)$. The adjoint is $P_A^* = \begin{pmatrix} P^* & A_\nu^* \\ 0 & P \end{pmatrix}$ with $A_\nu^* u = e \otimes A^* \int_{Z_b}(u,e)$, and the same argument gives $\operatorname{Nul}(P_A^*) \cong \operatorname{Nul}(A^*)$. Hence $\operatorname{ind}(P_A) = \operatorname{ind}(A)$. □

The operator P_A can be regarded as a 'Clifford' tensor product of P and A. In Proposition 14.7.18 its full ellipticity is guaranteed by the fibre operator P, while its index comes from the base operator A. In Chapter 15 P_A is deformed, through fully elliptic product-type operators (so without changing the index, by Corollary 14.7.16), to an ordinary pseudodifferential operator on M whose symbol, given a connection on ϕ, is of the form

$$\begin{pmatrix} \chi_1\sigma_0(P) & -\chi_2\sigma_0(A)^* \\ \chi_2\sigma_0(A) & \chi_1\sigma_0(P)^* \end{pmatrix},$$

with χ_1, χ_2 a partition of unity on S^*M subordinate to the complements of $\phi^* S^* B$ and of the vertical cosphere bundle. Applied to the fibration of the radially compactified normal bundle of an embedding $B \hookrightarrow \mathbb{S}^N$, this reduces the computation of the index of A to a problem on the sphere, which can then be solved using Bott periodicity (Chapter 13).

14.8 Exercises

Exercise 14.1. Let $\phi : M \longrightarrow B$ be a fibration. Show that there is an embedding $e : M \longrightarrow \mathbb{S}^N \times B$, with range in $(\mathbb{S}^N \setminus \{p\}) \times B$ for a fixed point p, such that $\pi_B \circ e = \phi$. *Hint:* embed M in a sphere avoiding one point (Whitney), and take e to be the product of this embedding with ϕ. Such an embedding identifies M with a fibration over B whose fibres are submanifolds of $\mathbb{R}^N$; it is used in the proof of the families index theorem.

Exercise 14.2. Let $\phi : M \longrightarrow B$ be a fibration with $\dim B > 0$. Show that $\Psi^m(M;E,F) \cap \Psi^m(M/B;E,F)$ is the space of fibrewise differential operators of order at most m (so it is $\{0\}$ for $m < 0$). *Hint:* the kernel of an element of the intersection is smooth off Diag_M and supported in M^2_ϕ, which has positive codimension; so it is supported on the diagonal.

Exercise 14.3. Show that the definition of ${}^V\overline{W}$ extends to the degenerate cases, with ${}^{\{0\}}\overline{W} = \overline{W}$ and ${}^W\overline{W} = \overline{W}$, and that then $S^{M,M'}_{\mathrm{ph}}(W,\{0\}) = S^{M'}_{\mathrm{ph}}(W)$ and $S^{M,M'}_{\mathrm{ph}}(W,W) = S^M_{\mathrm{ph}}(W)$.

Exercise 14.4. In the notation of Section 14.4, show that the vector fields ∂_{ζ_j}, ∂_{η_i}, $\zeta_j\partial_{\zeta_{j'}}$, $\zeta_j\partial_{\eta_i}$ and $\eta_i\partial_{\eta_{i'}}$ on W lift to smooth vector fields on ${}^V\overline{W}$, tangent to its boundary, whereas $\eta_i\partial_{\zeta_j}$ does not. Relate this to Proposition 14.4.4 by observing that the first set spans the Lie algebra of the affine group of W preserving the direction V.

Exercise 14.5. Near a point of $SV \subset \partial\overline{W}$ use the coordinates $x = 1/|\eta|$, $v = \zeta/|\eta| \in \mathbb{R}^k$ and $\theta = \eta/|\eta|$ on $\overline{W}$, in which $SV = \{x = 0, v = 0\}$. Introduce polar coordinates $R = (x^2+|v|^2)^{1/2}$, $(x,v)/R$ in the half-space $x \geq 0$. Show that the map from a neighbourhood of H_V in ${}^V\overline{W}$ given in the coordinates of Proposition 14.4.3(ii),(iii) to $(R,(x,v)/R,\theta)$ is a diffeomorphism onto a neighbourhood of $\{R = 0\} \times \mathbb{S}^k_+ \times \mathbb{S}^{l-1}$. Conclude that ${}^V\overline{W}$ is the blow-up $[\overline{W};SV]$ of $\overline{W}$ along SV, with H_V the front face.

Exercise 14.6. Using (14.4.1), show that the identification (14.4.2) of H_V with $\overline{W/V} \times SV$ is natural, whereas H_W is only naturally a fibration over $S(W/V)$. In the setting of a nested pair $Z \subset Y \subset X$, show that H_W is naturally identified with the blow-up $[SN^*Z; S(N^*Y|_Z)]$, fibred over the cosphere bundle of Z in Y.

Exercise 14.7. Let $p \in S^m_{\mathrm{ph}}(\mathbb{R}^b)$ and regard $a(\zeta,\eta) = p(\eta)$ as a function on $\mathbb{R}^f \times \mathbb{R}^b$. Show that if $a \in S^{M,M'}_{\mathrm{ph}}(\mathbb{R}^{f+b},\{\zeta = 0\})$ for some M, M' then p is a polynomial. *Hint:* near the interior of H_W, ρ_V is bounded below and $a = p(q/\rho_W)$; differentiate $\rho_W^{M'}p(q/\rho_W)$ in q along the curves $q = \rho_W v$. Deduce the statement in Remark 14.7.11.

Exercise 14.8. For $A \in \Psi^{m,m'}_{\phi\text{-pt}}(M;E,F)$ show that $\sigma_{m,m'}(A^*) = \sigma_{m,m'}(A)^*$ and that $\beta_{m,m'}(A^*)(b,\eta)$ is the adjoint $\beta_{m,m'}(A)(b,\eta)^*$ (the adjoint on Z_b with respect to the induced metrics and fibre density).

Exercise 14.9. Let $P \in \Psi^{m'}(M/B;E,F)$ be an elliptic family, regarded as an element of $\Psi^{0,m'}_{\phi\text{-pt}}(M;E,F)$. Show that P is fully elliptic if and only if each P_b is invertible, and that then P is invertible with inverse in $\Psi^{-m'}(M/B;F,E)$. In particular, for $M = Z \times B$ with product metric, the fibre Laplacian $\Delta_Z + 1$ is a fully elliptic element of $\Psi^{0,2}_{\phi\text{-pt}}(M)$ which is not elliptic as an operator on M if $\dim B > 0$; while $\Delta_Z + \Delta_B$ and $\Delta_M + 1$ lie in $\Psi^{2,2}_{\phi\text{-pt}}(M)$ and are fully elliptic. Compute their base symbols.

Exercise 14.10. Let $A \in \Psi^m(M/B;E,F)$ be an elliptic family and $b \in B$. Show that the restriction of $\mathrm{ind}_a(A)$ to $\{b\}$, i.e. its image in $K^0(\{b\}) = \mathbb{Z}$, is $\mathrm{ind}(A_b)$, and that $\mathrm{ind}_a(A^*) = -\mathrm{ind}_a(A)$ when $m = 0$.

Notes

This chapter combines four lectures, with their (largely empty) addenda, of Melrose's 2005 course. The main departures from the original are as follows. The construction of the index bundle via finite rank projections π_N approximating the identity, which was stated without

proof in the original, has been replaced by the stabilization procedure of Atiyah and Singer [18], which the original adopted later in the course. The relative compactification has been presented as the product $\overline{U} \times \overline{V}$ in the rescaled variables $(\zeta, \eta/\langle\zeta\rangle)$; this makes its linear invariance and the structure of product-type symbols transparent, and the classical estimate characterization (14.5.1) is a by-product. The order conventions of the original are not consistent between lectures; here the first order always refers to the larger submanifold (for operators, the base order at M_ϕ^2) and the second to the smaller (the fibre order at the diagonal), so that $\Psi^m(M) \subset \Psi^{m,m}_{\phi\text{-pt}}(M)$, $\Psi^{m'}(M/B) \subset \Psi^{0,m'}_{\phi\text{-pt}}(M)$ and compressed base operators lie in $\Psi^{m,-\infty}_{\phi\text{-pt}}(M)$. The base symbol of an element of $\Psi^{m,m'}_{\phi\text{-pt}}(M)$ has fibre order $m' - m$ (not m'). The original asserted that operators on the base lift to $\Psi^{m,0}_{\phi\text{-pt}}(M)$; this is false unless they are differential operators (Remark 14.7.11), and the original's later arguments only use the compressed operators of Proposition 14.7.10(iii). The proofs of coordinate invariance, of the symbol maps and of the composition formula, which were left to unwritten addenda in the original, have been supplied, as has the proof that full ellipticity is equivalent to the Fredholm property.

Pseudodifferential operators with symbols of product type in two groups of variables go back to the 'bisingular' operators of Rodino; the calculus adapted to a fibration used here is that of Melrose's lectures, and is closely related to the calculi used for families of operators on manifolds with boundary in [20] and to the adiabatic limits of [19]. For the families index and its relation to K-theory see [18] and [17]; for the conormal distributions underlying the definitions see [8], Chapter 18.

Chapter 15

Multiplicativity, the Chern character and the families index theorem

An elliptic operator on a compact manifold has an index, the dimension of its null space minus the codimension of its range. For a family of elliptic operators parametrized by a base B, the null spaces jump, but their formal difference is a well-defined K-class on B, the index bundle. The index theorem says that this analytic object can be computed from the principal symbol alone, by a sequence of purely topological operations. This chapter proves that statement and then turns it into an integral formula in cohomology.

In Chapter 14 the analytic index of a family of elliptic pseudodifferential operators on the fibres of a fibration $\phi : M \to B$ of compact manifolds was defined as a homomorphism $\mathrm{ind_a} : K_{\mathrm{c}}(T^*(M/B)) \to K^0(B)$, and the calculus of pseudodifferential operators of product type was set up. This chapter puts these together to prove the families index theorem of Atiyah and Singer in K-theory, and then passes to cohomology through the Chern character.

The proof rests on two properties of the analytic index. The first, *multiplicativity* (Section 15.2), says that for a tower of fibrations $\tilde{M} \to M \to B$ the index is unchanged when a family on M/B is "multiplied" by a family on the fibres of $\tilde{M} \to M$ whose index is the trivial line bundle. The product is first formed in the product-type calculus, where its index is computed directly, and is then deformed, through fully elliptic product-type families, into an ordinary family on $\tilde{M}/B$. The second property, *excision* (Section 15.3), says that the index of a symbol that is trivial outside an open set only depends on that open set. Embedding the fibration into a trivial one, $M \hookrightarrow \mathbb{R}^N \times B$, these two properties reduce the index to the case of the torus fibration $\mathbb{T}^N \times B \to B$. There the index is computed by multiplicativity and Bott periodicity. This gives the equality of the analytic and topological index maps (Theorem 15.4.4). One ingredient, the index of the Thom class of a non-trivial normal bundle, is not proved here; we say exactly what is used. (It is proved in Chapter 16.) The logic of the proof is summarized in Figure 15.1.

The second half of the chapter reviews Chern–Weil theory and defines the Chern character on K^0 (Section 15.5). It then defines the odd Chern character and computes the Chern character of the index bundle of a family of Toeplitz operators (Section 15.6). The relative Chern character of a compactly supported K-class is given by an explicit formula of Fedosov type (Section 15.7); its boundary term is an odd Chern form of the symbol, and the odd Chern character itself is the fibre integral of the Chern character of the clutched class (Proposition 15.7.5), which explains its normalization independently of Toeplitz operators. Finally the cohomological form of the families index theorem is stated

(Section 15.8), and the Todd class is related to the Chern character of the Bott class through the Thom isomorphism (Remark 15.8.3).

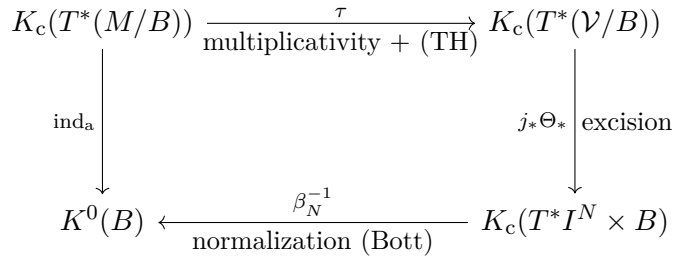


Figure 15.1: Roadmap of the proof of Theorem 15.4.4. The composite of the top, right and bottom arrows is the topological index $\mathrm{ind}_{\mathrm{t}}$. Each arrow preserves the analytic index, for the reason written next to it: multiplicativity (Theorem 15.2.4) together with the index of the Thom class (TH); excision (Theorem 15.3.1); and the normalization on the torus (Proposition 15.4.2). Hence the square commutes, which is the statement $\mathrm{ind}_{\mathrm{a}} = \mathrm{ind}_{\mathrm{t}}$.

15.1 Conventions and the index bundle

15.1.1 Superbundles and compactly supported K-theory

A *superbundle* over a manifold X is a pair $\mathbb{E} = (E_+, E_-)$ of complex vector bundles, always with Hermitian metrics. The *graded tensor product* of two superbundles is

$$(\mathbb{E} \otimes \mathbb{F})_+ = E_+ \otimes F_+ \oplus E_- \otimes F_-, \qquad (\mathbb{E} \otimes \mathbb{F})_- = E_- \otimes F_+ \oplus E_+ \otimes F_-. \tag{15.1.1}$$

Matrices of maps $(\mathbb{E} \otimes \mathbb{F})_+ \to (\mathbb{E} \otimes \mathbb{F})_-$ below are always written with respect to these ordered decompositions.

We recall from Chapter 13 (see also [17]) the compactly supported K-group of a manifold X. A *K_{c}-triple* (E_+, E_-, a) consists of complex vector bundles $E_\pm$ over X and $a \in \mathcal{C}^\infty(X \setminus K; \mathrm{Hom}(E_+, E_-))$ invertible, for some compact $K \subset X$. The group $K_{\mathrm{c}}(X)$ is the set of equivalence classes of such triples under the equivalence relation generated by bundle isomorphisms, by changing a on a compact set, by homotopies of triples (triples on $[0,1] \times X$ with a invertible outside $[0,1] \times K$, K compact), and by stabilization $(E_+, E_-, a) \sim (E_+ \oplus H, E_- \oplus H, a \oplus \mathrm{Id}_H)$. The group operation is direct sum. We use the following standard facts.

(K1) If a is invertible on all of X the triple represents 0.

(K2) If X is compact, $[(E_+, E_-, a)] \mapsto [E_+] - [E_-]$ identifies $K_{\mathrm{c}}(X)$ with $K^0(X)$.

(K3) An open embedding $j : U \hookrightarrow X$ induces a homomorphism $j_* : K_{\mathrm{c}}(U) \to K_{\mathrm{c}}(X)$, *extension by the identity*. Every class in $K_{\mathrm{c}}(U)$ has a representative (E_+, E_-, a) with $E_\pm$ trivialized outside a compact set $C \subset U$ and $a = \mathrm{Id}$ outside C in these trivializations; j_* extends $E_\pm$ trivially and a by Id. This is independent of choices. ([17], §2.4; in the cases needed below the normalization is carried out in the proof of Theorem 15.3.1.)

(K4) A proper smooth map $f : X' \to X$ induces $f^* : K_{\mathrm{c}}(X) \to K_{\mathrm{c}}(X')$ by pulling back triples.

If $V \to Y$ is a real vector bundle over a compact manifold Y and SV its sphere bundle, a homogeneous map $a \in \mathcal{C}^\infty(V \setminus 0; \mathrm{Hom}(\pi^*E_+, \pi^*E_-))$ of degree 0 which is invertible defines the class $[(\pi^*E_+, \pi^*E_-, a)] \in K_{\mathrm{c}}(V)$; here a is only used outside the compact set $\{|v| \leq 1\}$. Every class in $K_{\mathrm{c}}(V)$ arises in this way (Chapter 14): bundles over V are pulled back from Y, and a may be deformed to be homogeneous.

15.1.2 The Clifford product

The following elementary computation is used repeatedly. Let $p : E_+ \to E_-$ and $\alpha : F_+ \to F_-$ be linear maps between finite dimensional Hermitian spaces (or bounded operators between Hilbert spaces) and $c_1, c_2 \geq 0$. Set

$$m(c_1, c_2) = \begin{pmatrix} c_1\, p \otimes 1 & -c_2\, 1 \otimes \alpha^* \\ c_2\, 1 \otimes \alpha & c_1\, p^* \otimes 1 \end{pmatrix} : (\mathbb{E} \otimes \mathbb{F})_+ \longrightarrow (\mathbb{E} \otimes \mathbb{F})_-. \tag{15.1.2}$$

Lemma 15.1.1. *With the notation of* (15.1.2),

$$m^*m = \begin{pmatrix} c_1^2\, p^*p \otimes 1 + c_2^2\, 1 \otimes \alpha^*\alpha & 0 \\ 0 & c_1^2\, pp^* \otimes 1 + c_2^2\, 1 \otimes \alpha\alpha^* \end{pmatrix},$$

and mm^* is given by the same formula with (p, α) replaced by $(p^*, -\alpha)$ (and the factors in the other order). Hence m is invertible if either $c_1 > 0$ and p is invertible, or $c_2 > 0$ and α is invertible.

Proof. Since $m^* = \begin{pmatrix} c_1 p^* \otimes 1 & c_2\, 1 \otimes \alpha^* \\ -c_2\, 1 \otimes \alpha & c_1 p \otimes 1 \end{pmatrix}$, the off-diagonal entries of m^*m are $\pm c_1c_2(p^* \otimes \alpha^* - p^* \otimes \alpha^*) = 0$, because maps acting on different tensor factors commute. The diagonal entries are as stated. Note that m^* has the form (15.1.2) with (p, α) replaced by $(p^*, -\alpha)$ and the roles of the two summands interchanged, which gives mm^*. Under either hypothesis both m^*m and mm^* are sums of non-negative self-adjoint operators one of which is bounded below by a positive constant, so they are invertible, and then m is injective with closed range and surjective. □

Discussion. The matrix $m(c_1, c_2)$ is the familiar trick for "adding" two elliptic problems. For the Dirac-type operator $\left(\begin{smallmatrix} 0 & p^* \\ p & 0 \end{smallmatrix}\right)$ the square is $p^*p \oplus pp^*$; the graded tensor product of two such operators squares to the *sum* of the two squares, because the cross terms anticommute and cancel. This is exactly what Lemma 15.1.1 says, and it is why one invertible factor is enough to make the product invertible. Every product of K-classes and every deformation in the multiplicativity proof below is built from this single identity.

Definition 15.1.2. Let $x_i = [(E_+^i, E_-^i, a_i)] \in K_{\mathrm{c}}(X_i)$, $i = 1, 2$, and extend a_i (after multiplying by a cut-off) to $a_i \in \mathcal{C}^\infty(X_i; \mathrm{Hom}(E_+^i, E_-^i))$, invertible outside a compact set K_i. The *exterior product* $x_1 \boxtimes x_2 \in K_{\mathrm{c}}(X_1 \times X_2)$ is the class of $((\mathbb{E}^1 \otimes \mathbb{E}^2)_+, (\mathbb{E}^1 \otimes \mathbb{E}^2)_-, a_1 \# a_2)$, $a_1 \# a_2 = m(1, 1)$ with $p = a_1$, $\alpha = a_2$. If V_1, V_2 are vector bundles over the same compact manifold Y and $x_i \in K_{\mathrm{c}}(V_i)$, the *product* $x_1 \cdot x_2 \in K_{\mathrm{c}}(V_1 \oplus V_2)$ is the pull-back of $x_1 \boxtimes x_2$ under the proper embedding $V_1 \oplus V_2 \hookrightarrow V_1 \times V_2$.

By Lemma 15.1.1, $a_1 \# a_2$ is invertible outside $K_1 \times K_2$. (Where it is invertible the ranks of the two bundles agree.) Changing the extension of a_1 inside K_1, the straight line homotopy between the two extensions gives a homotopy of triples, since $a_1 \# a_2$ stays invertible wherever a_2 is. Homotopies, bundle isomorphisms and direct sums pass through the construction. If x_2 is the stabilizing triple (H, H, Id) then $a_1 \#\, \mathrm{Id}$ is invertible everywhere, so the product vanishes by (K1). Thus the products are well defined and bilinear. This is the standard product in K-theory ([17], §2.6).

15.1.3 Fibrewise families and the analytic index

Let $\phi : M \to B$ be a fibration of compact manifolds with typical fibre Z, $Z_b = \phi^{-1}(b)$. From Chapter 14 we use:

(F1) The algebra $\Psi^m(M/B; E_+, E_-)$ of smooth families of pseudodifferential operators on the fibres, with principal symbol σ_m on $T^*(M/B) \setminus 0$, composition, adjoints (with respect to Hermitian metrics and a smooth positive fibre density), parametrices for elliptic families, and the fact that in a local trivialization $\phi^{-1}(U) \cong Z \times U$ an element of $\Psi^0(M/B)$ is a smooth family in $\Psi^0(Z)$, hence a norm-smooth family of bounded operators on $L^2(Z)$.

(F2) An elliptic $A \in \Psi^0(M/B; E_+, E_-)$ defines $[\sigma_0(A)] \in K_{\mathrm{c}}(T^*(M/B))$, and every class arises in this way.

(F3) The analytic index of A, defined by stabilizing with finite rank projections, depends only on $[\sigma_0(A)]$ and gives a homomorphism $\mathrm{ind}_{\mathrm{a}} : K_{\mathrm{c}}(T^*(M/B)) \to K^0(B)$.

The degenerate case $M = B$, $\phi = \mathrm{Id}$, is allowed. Then $T^*(B/B) = B$, the "fibrewise operators" are bundle maps $a \in \mathcal{C}^\infty(B; \mathrm{Hom}(E_+, E_-))$, there is no ellipticity condition, and $K_{\mathrm{c}}(T^*(B/B)) = K^0(B)$.

15.1.4 Product-type families

Consider a tower of fibrations of compact manifolds

$$\begin{array}{ccc} Y & \longrightarrow & \tilde{M} \\ & & \downarrow{\scriptstyle \tilde{\phi}} \\ Z & \longrightarrow & M \\ & & \downarrow{\scriptstyle \phi} \\ & & B \end{array} \tag{15.1.3}$$

so $\tilde{M} \to B$ is a fibration with fibre $\tilde{M}_b = (\phi\tilde{\phi})^{-1}(b)$, itself fibred over $M_b = Z_b$ by $\tilde{\phi}$ with fibre Y. The space $\Psi^{m,m'}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B; E_+, E_-)$ of *product-type families* consists of smooth families, parametrized by $b \in B$, of product-type operators on $\tilde{M}_b$ with respect to $\tilde{\phi} : \tilde{M}_b \to M_b$, as defined in Chapter 14; their kernels are product-type conormal distributions on the fibre product $\tilde{M} \times_B \tilde{M}$ with respect to $\tilde{M} \times_M \tilde{M} \supset \mathrm{Diag}$. Since all constructions in Chapter 14 are local in the base, they hold with parameters in B. Let

$$\Sigma = S(\tilde{\phi}^* T^*(M/B)) \subset S^*(\tilde{M}/B)$$

be the sphere bundle of the subbundle $\tilde{\phi}^* T^*(M/B) \subset T^*(\tilde{M}/B)$ (the annihilator of $T(\tilde{M}/M)$). We use the following from Chapter 14, in the families version and for order $(0, 0)$.

(P1) There are two symbol maps. The *usual symbol* $\sigma(A)$ is a smooth function on the blow-up $[S^*(\tilde{M}/B); \Sigma]$. The *base symbol* $\beta(A)$ is a family, parametrized by $\eta \in S^*(M/B)$, of operators in $\Psi^0(Y_m)$, $Y_m = \tilde{\phi}^{-1}(m)$, m the base point of η. Both are multiplicative and compatible with adjoints. The two symbols determine A modulo $\Psi^{-\infty}(\tilde{M}/B)$, and any compatible pair arises.

(P2) A is *fully elliptic* if $\sigma(A)$ and $\beta(A)$ are invertible. A fully elliptic family has a parametrix modulo smoothing families, so it is Fredholm on $L^2(\tilde{M}_b)$ for each b; its null space consists of smooth sections. In a local trivialization of $\tilde{M} \to B$ it is a norm-smooth family of bounded operators.

(P3) A family $P \in \Psi^0(\tilde{M}/M; E_+, E_-)$ on the fibres of $\tilde{\phi}$ lies in $\Psi^{0,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B)$, with $\sigma(P) = \gamma^*\sigma_0(P)$ and $\beta(P)(\eta) = P_m$. Here $\gamma : [S^*(\tilde{M}/B); \Sigma] \to S^*(\tilde{M}/M)$ is the extension to the blow-up of the radial projection of the restriction map $T^*(\tilde{M}/B) \to T^*(\tilde{M}/M)$, which vanishes exactly on $\tilde{\phi}^*T^*(M/B)$.

(P4) $\Psi^0(\tilde{M}/B) \subset \Psi^{0,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B)$; $\sigma(A)$ is the pull-back of $\sigma_0(A)$ under the blow-down map, and $\beta(A)(\eta)$ is multiplication by the restriction of $\sigma_0(A)$ to Σ, i.e. to $\tilde{\phi}^*S^*(M/B)$, at η. For such A full ellipticity is ellipticity.

(P5) If $e \in \mathcal{C}^\infty(\tilde{M}; E_+)$ has L^2 norm one on each fibre Y_m, Π is the orthogonal projection onto $\mathbb{C}e$ on each fibre, and $A \in \Psi^0(M/B; F_+, F_-)$, then

$$A_\Pi : u \longmapsto e \otimes A\big(\langle u, e\rangle\big), \quad \langle u, e\rangle(m) = \int_{Y_m} (u, e)\, \nu,$$

lies in $\Psi^{-\infty,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B; E_+ \otimes F_+, E_+ \otimes F_-)$, with $\sigma(A_\Pi) = 0$ and $\beta(A_\Pi)(\eta) = \Pi \otimes \sigma_0(A)(\eta)$.

Here ν is the fibre density on $\tilde{M}/M$ used to define adjoints, and the bundles $F_\pm$ over M are lifted to $\tilde{M}$ without change of notation. A fibrewise family P on $\tilde{M}/M$ acts on sections of $E_\pm \otimes F$ as $P \otimes \mathrm{Id}_F$, since F is constant on the fibres of $\tilde{\phi}$; we keep writing P.

15.1.5 Stabilization and the index bundle

Discussion. Why is a construction needed at all? For a single operator the index is $\dim \mathrm{Nul}\, A - \dim \mathrm{Nul}\, A^*$. In a family, $\dim \mathrm{Nul}\, A_b$ can jump up at special values of b, so the null spaces do not form a vector bundle. The jumps in $\mathrm{Nul}\, A_b$ and $\mathrm{Nul}\, A_b^*$ are equal, and the remedy is to add a fixed finite-dimensional space to one side so that the operators become surjective for every b. Then the null spaces have constant dimension and form a bundle, and subtracting the added trivial bundle gives a K-class that does not depend on the choices.

The index bundle of a fibrewise elliptic family was constructed in Chapter 14 using finite rank projections. For the comparison of different families below, and for product-type families, an equivalent and more flexible construction is convenient: one stabilizes the *range* rather than the domain. In the next lemma, A is either an elliptic family in $\Psi^0(M/B; E_+, E_-)$ or a fully elliptic family in $\Psi^{0,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B; E_+, E_-)$; to treat both cases at once we write M for the total space and M_b for its fibres. Sections $v_1, \dots, v_N \in \mathcal{C}^\infty(M; E_-)$ define $S \in \mathcal{C}^\infty(M; \mathrm{Hom}(\mathbb{C}^N, E_-))$ and the operators

$$A_b \oplus S_b : L^2(M_b; E_+) \oplus \mathbb{C}^N \longrightarrow L^2(M_b; E_-), \quad (u, c) \longmapsto A_b u + \sum_j c_j v_j|_{M_b}.$$

Lemma 15.1.3 (Stabilization)**.** *(i) There exist N and S such that $A_b \oplus S_b$ is surjective for all $b \in B$.*

(ii) For such S, the null spaces $\mathrm{Nul}(A_b \oplus S_b) \subset \mathcal{C}^\infty(M_b; E_+) \oplus \mathbb{C}^N$ form a smooth vector bundle $\mathcal{N}(A, S)$ over B.

(iii) The class

$$\operatorname{ind}(A) = [\mathcal{N}(A,S)] - [\mathbb{C}^N] \in K^0(B)$$

is independent of S, and is unchanged under smooth homotopies of A through families of the same kind (elliptic, resp. fully elliptic).

(iv) For an elliptic $A \in \Psi^0(M/B; E_+, E_-)$, $\operatorname{ind}(A) = \operatorname{ind}_{\mathrm{a}}([\sigma_0(A)])$.

Proof. (i) Fix b_0 and a trivialization over a neighbourhood U of b_0. The operator A_{b_0} is Fredholm, so its range is closed of finite codimension. Since smooth sections are dense in L^2, there are smooth sections $w_1, \dots, w_r$ of E_- over M_{b_0} whose classes span $L^2/\operatorname{Ran}(A_{b_0})$. Extend them, using the trivialization and a cut-off on the base equal to 1 near b_0, to smooth sections over M supported in $\phi^{-1}(U)$. In the trivialization $b \mapsto A_b \oplus S_b$ is norm continuous and surjective bounded operators form an open set, so $A_b \oplus S_b$ is surjective for b near b_0. Cover B by finitely many such neighbourhoods and use all the sections so obtained; adding columns preserves surjectivity.

(ii) Let $\mathcal{N}_b = \operatorname{Nul}(A_b \oplus S_b)$. Fix b_0, let Π_0 be the orthogonal projection onto the finite dimensional space $\mathcal{N}_{b_0}$ in $L^2 \oplus \mathbb{C}^N$, and set

$$T_b(u,c) = \big(A_b u + S_b c,\ \Pi_0(u,c)\big) \in L^2(M_b; E_-) \oplus \mathcal{N}_{b_0}.$$

T_{b_0} is an isomorphism: it is injective since $\mathcal{N}_{b_0} \cap \mathcal{N}_{b_0}^{\perp} = 0$, and given (g,n) choose w with $(A_{b_0} \oplus S_{b_0})w = g$; then $w - \Pi_0 w + n$ is a preimage. Hence T_b is invertible for b near b_0, depending smoothly on b in the trivialization, and $n \mapsto T_b^{-1}(0,n)$ is an isomorphism $\mathcal{N}_{b_0} \to \mathcal{N}_b$ (if $w \in \mathcal{N}_b$ then $w = T_b^{-1}(0, \Pi_0 w)$). This gives local frames. The elements of $\mathcal{N}_b$ are smooth sections, by elliptic regularity: if Q is a parametrix, smooth in b, with $QA = \mathrm{Id} - R$, R smoothing, then $u = -QSc + Ru$. Applying this to the frame elements, which depend smoothly on b in L^2, shows that they are smooth on M. Smoothness of $\mathcal{N}$ does not depend on the trivialization, since its frames are smooth sections of E_+ over open subsets of M, plus vectors in $\mathbb{C}^N$.

(iii) Let S, S' (with N, N' columns) both satisfy (i) and let $S'' = (S, S')$. The sequence of bundles

$$0 \longrightarrow \mathcal{N}(A,S) \longrightarrow \mathcal{N}(A,S'') \longrightarrow \mathbb{C}^{N'} \longrightarrow 0, \quad (u,c) \mapsto (u,c,0), \quad (u,c,c') \mapsto c',$$

is exact: the last map is surjective because, given c', we may solve $Au + Sc = -S'c'$. Exact sequences of vector bundles split, so $[\mathcal{N}(A,S'')] - [\mathbb{C}^{N+N'}] = [\mathcal{N}(A,S)] - [\mathbb{C}^N]$. By symmetry the same holds for S'. If A_t, $t \in [0,1]$, is a smooth homotopy, apply (i) and (ii) on the compact manifold (with boundary) $B \times [0,1]$. The resulting bundle over $B \times [0,1]$ has isomorphic restrictions to $B \times \{0\}$ and $B \times \{1\}$ ([17], Lemma 1.4.3), and the restrictions of S stabilize A_0 and A_1. So $\operatorname{ind}(A_0) = \operatorname{ind}(A_1)$.

(iv) In Chapter 14 the index is $[\operatorname{Ran}\pi] - [W]$, where $\pi \in \Psi^{-\infty}(M/B; E_+)$ is a family of finite rank projections such that $\operatorname{Nul}(A(\mathrm{Id} - \pi)) = \operatorname{Ran}\pi$ and $W = \operatorname{Nul}((\mathrm{Id} - \pi)A^*)$ are smooth bundles, W_b being the orthocomplement of the range of $A_b(\mathrm{Id} - \pi_b)$. Choose finitely many smooth sections $w_1, \dots, w_K$ of W (sections of E_- over M) spanning W_b for every b, and let S_W be the corresponding map from $\mathbb{C}^K$. Then $A(\mathrm{Id} - \pi) \oplus S_W$ is surjective. Its null space is $\operatorname{Ran}\pi \oplus L$, where L is the kernel of the surjective bundle map $\mathbb{C}^K \to W$, since the two terms of $A(\mathrm{Id} - \pi)u + S_W c$ lie in complementary subspaces. As $L \oplus W \cong \mathbb{C}^K$, the stabilized class of $A(\mathrm{Id} - \pi)$ is $[\operatorname{Ran}\pi] + [L] - [\mathbb{C}^K] = [\operatorname{Ran}\pi] - [W]$. Finally $A(\mathrm{Id} - t\pi)$, $t \in [0,1]$, is a smooth homotopy of elliptic families, so by (iii) $\operatorname{ind}(A) = \operatorname{ind}(A(\mathrm{Id} - \pi))$. □

Remark 15.1.4. (a) If B is a point, $\dim \mathcal{N}(A,S) = \dim \operatorname{Nul} A + N - \operatorname{codim} \operatorname{Ran} A$, so $\operatorname{ind}(A)$ is the numerical index. (b) In the degenerate case $M = B$, for a bundle map

$a : E_+ \to E_-$ the sequence $0 \to \mathcal{N}(a,S) \to E_+ \oplus \mathbb{C}^N \to E_- \to 0$ is exact, so $\operatorname{ind}(a) = [E_+] - [E_-]$; that is, $\operatorname{ind}_{\mathrm{a}}$ is the identification (K2). (c) The index bundle is often written as $[\operatorname{Nul}(A+A')] - [\operatorname{Nul}((A+A')^*)]$ for a smoothing perturbation A' after which the null spaces form bundles; this is the construction of Chapter 14 in another guise. (d) The construction of (i)–(ii) is compatible with pull-back: for a smooth map $f : B' \to B$ and the pulled-back fibration, $\operatorname{ind}(f^*A) = f^* \operatorname{ind}(A)$, since f^*S stabilizes f^*A and $\mathcal{N}(f^*A, f^*S) = f^*\mathcal{N}(A,S)$.

By Lemma 15.1.3, the index of a fully elliptic product-type family is a well-defined element of $K^0(B)$, invariant under homotopies through fully elliptic families. For ordinary families it agrees with $\operatorname{ind}_{\mathrm{a}}$.

15.2 Multiplicativity

Multiplicativity is the analytic heart of the proof. Suppose that on the fibres Y of $\tilde{M} \to M$ we have a family P whose index is the trivial line bundle, and on the fibres Z of $M \to B$ an arbitrary elliptic family A. Heuristically, an operator on $\tilde{M}_b$ built from P and A has null space "$\operatorname{Nul} P \otimes \operatorname{Nul} A$", and since $\operatorname{Nul} P$ is one-dimensional this should have the same index as A. The difficulty is to make this into an honest statement about an ordinary elliptic family on $\tilde{M}/B$.

Discussion. The plan has three steps. (1) Form the *Clifford tensor product* $P \otimes A$ ((15.2.5)). It is not an ordinary pseudodifferential family, because A acts only in the base directions; but it is a fully elliptic product-type family, and its index is computed directly to be $\operatorname{ind}(A)$ (Proposition 15.2.3). (2) Deform $P \otimes A$, through fully elliptic product-type families, to an ordinary elliptic family $\tilde{A}_P$ (Theorem 15.2.4). The index does not change along the deformation. (3) Identify the symbol class of $\tilde{A}_P$ with the K-theory product of the symbol classes of P and A (Lemma 15.2.1). The product-type calculus of Chapter 14 is used only to have a common home for both ends of the deformation.

Throughout this section we consider the tower (15.1.3), a superbundle $\mathbb{E} = (E_+, E_-)$ over $\tilde{M}$, a superbundle $\mathbb{F} = (F_+, F_-)$ over M (also lifted to $\tilde{M}$), and $\mathbb{G} = \mathbb{E} \otimes \mathbb{F}$. We fix a fibre density ν on $\tilde{M}/M$ and Hermitian metrics.

15.2.1 The splitting of the symbol space

The vertical cotangent bundle $T^*(\tilde{M}/B)$ contains the subbundle $\tilde{\phi}^*T^*(M/B)$, the annihilator of $T(\tilde{M}/M)$, with quotient $T^*(\tilde{M}/M)$. A connection for $\tilde{\phi}$, i.e. a choice of complement H to $T(\tilde{M}/M)$ in $T(\tilde{M}/B)$, splits this sequence:

$$T^*(\tilde{M}/B) \cong T^*(\tilde{M}/M) \oplus \tilde{\phi}^*T^*(M/B), \tag{15.2.1}$$

where $T^*(\tilde{M}/M)$ is identified with the annihilator H° of H. Any two connections are homotopic. Let

$$\Sigma_1 = S(\tilde{\phi}^*T^*(M/B)), \qquad \Sigma_2 = S(H^\circ),$$

two disjoint closed subsets of $S^*(\tilde{M}/B)$. For elliptic families $P \in \Psi^0(\tilde{M}/M; E_+, E_-)$ and $A \in \Psi^0(M/B; F_+, F_-)$ write

$$p = \sigma_0(P), \qquad \alpha = \sigma_0(A),$$

homogeneous of degree 0 and invertible. The function p, composed with the restriction map $T^*(\tilde{M}/B) \to T^*(\tilde{M}/M)$, is defined on $S^*(\tilde{M}/B) \setminus \Sigma_1$, and α, composed with the

projection onto the second summand in (15.2.1), is defined on $S^*(\tilde{M}/B) \setminus \Sigma_2$. Choose $\chi_1, \chi_2 \in \mathcal{C}^\infty(S^*(\tilde{M}/B))$ with

$$\chi_1, \chi_2 \geq 0, \quad \chi_1 + \chi_2 = 1, \quad \chi_1 = 0 \text{ near } \Sigma_1, \quad \chi_2 = 0 \text{ near } \Sigma_2, \tag{15.2.2}$$

and define, with m as in (15.1.2),

$$a_P = m(\chi_1, \chi_2) = \begin{pmatrix} \chi_1\, p & -\chi_2\, \alpha^* \\ \chi_2\, \alpha & \chi_1\, p^* \end{pmatrix} \in \mathcal{C}^\infty\big(S^*(\tilde{M}/B); \operatorname{Hom}(G_+, G_-)\big). \tag{15.2.3}$$

It is smooth since each coefficient vanishes near the set where its factor is undefined. By Lemma 15.1.1 it is invertible at every point, since $\chi_1 > 0$ or $\chi_2 > 0$ everywhere. So a_P, extended as a homogeneous function of degree 0, is an elliptic symbol on $\tilde{M}/B$ (Figure 15.2).

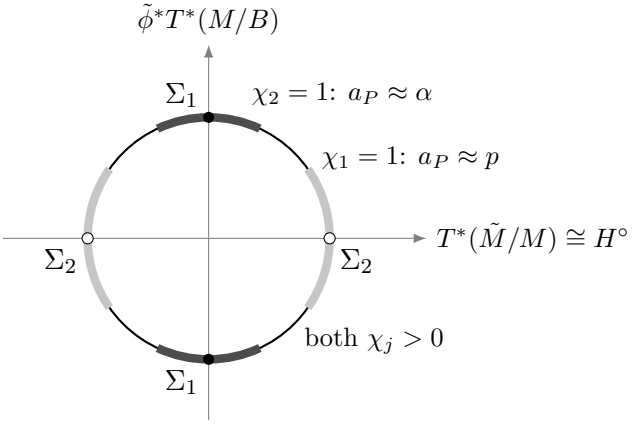


Figure 15.2: One fibre of $S^*(\tilde{M}/B)$ when the fibre and base directions are one-dimensional. The fibre symbol p is undefined at the points Σ_1 of pure base covectors, and the base symbol α is undefined at the points Σ_2 of pure fibre covectors. The product symbol $a_P = m(\chi_1, \chi_2)$ uses p near Σ_2 (light arcs), α near Σ_1 (dark arcs), and a Clifford combination of both in between; by Lemma 15.1.1 it is invertible everywhere.

Lemma 15.2.1. *The class of a_P in $K_{\mathrm{c}}(T^*(\tilde{M}/B))$ is the product $[p] \cdot \tilde{\phi}^*[\alpha]$ of Definition 15.1.2, for the splitting (15.2.1). In particular it is independent of the choice of χ_1, χ_2 and of the connection.*

Proof. Write $\xi = (\xi_1, \xi_2)$ according to (15.2.1), and let p', α' be smooth extensions to all of $T^*(\tilde{M}/M)$, resp. $\tilde{\phi}^*T^*(M/B)$, equal to p, α for $|\xi_1| \geq 1$, resp. $|\xi_2| \geq 1$. These represent $[p]$ and $\tilde{\phi}^*[\alpha]$, and $m(1,1)$ formed from them represents the product. By compactness there is $\delta \in (0,1)$ with $|\xi_1| \geq \delta|\xi|$ on $\operatorname{supp}\chi_1$ and $|\xi_2| \geq \delta|\xi|$ on $\operatorname{supp}\chi_2$ (both extended homogeneously). Consider, for $t \in [0,1]$,

$$m_t = m\big((1-t)\chi_1 + t, (1-t)\chi_2 + t\big) \quad \text{formed with } p',\ \alpha'.$$

For $|\xi| \geq 2/\delta$ this is invertible by Lemma 15.1.1. If $t > 0$ both coefficients are positive, and $|\xi_1| \geq 1$ or $|\xi_2| \geq 1$. If $t = 0$, at a point where $\chi_1 > 0$ we have $|\xi_1| \geq 2$, and where $\chi_2 > 0$ we have $|\xi_2| \geq 2$. So m_t is a homotopy of triples. Moreover m_0 coincides with a_P for $|\xi| \geq 2/\delta$, and $m_1 = m(1,1)$. Different choices of cut-offs or connections are joined by homotopies, through which the same argument applies. □

15.2.2 The Clifford tensor product of families

We now make the following assumption on P:

$$\begin{aligned} P \in \Psi^0(\tilde{M}/M; E_+, E_-) \text{ is elliptic, each } P_m \text{ is surjective, and } \operatorname{Nul}(P_m) = \mathbb{C}e(m), \\ e \in \mathcal{C}^\infty(\tilde{M}; E_+),\ \textstyle\int_{Y_m} |e|^2 \nu = 1 \text{ for all } m \in M. \end{aligned} \tag{15.2.4}$$

So $\operatorname{ind}(P) = 1$ is the trivial line bundle, and the null bundle itself is trivialized. Let Π be the orthogonal projection onto $\mathbb{C}e$. The next lemma shows that any P with $\operatorname{ind}(P) = 1 \in K^0(M)$ can be brought to this form without changing its symbol class.

Lemma 15.2.2. *Let $P \in \Psi^0(\tilde{M}/M; E_+, E_-)$ be elliptic with $\operatorname{ind}(P) = [\mathbb{C}] \in K^0(M)$. Then for some N there is $P' \in \Psi^0(\tilde{M}/M; E_+ \oplus \mathbb{C}^N, E_- \oplus \mathbb{C}^N)$ with $P' - P \oplus \operatorname{Id} \in \Psi^{-\infty}$ satisfying (15.2.4). In particular $[\sigma_0(P')] = [\sigma_0(P)]$.*

Proof. Choose S with N columns as in Lemma 15.1.3 and let $L = \mathcal{N}(P, S)$, a bundle over M with $[L] - [\mathbb{C}^N] = [\mathbb{C}]$. Thus $L \oplus \mathbb{C}^k \cong \mathbb{C}^{N+1+k}$ for some k; adding k zero columns to S replaces L by $L \oplus \mathbb{C}^k$, so we may assume that $L \cong \mathbb{C}^{N+1}$. Choose a smooth frame $\ell_1, \dots, \ell_{N+1}$ of L and let $q : L \to \mathbb{C}^N$ be the first N coordinates, extended to $\mathcal{C}^\infty(Y_m; E_+) \oplus \mathbb{C}^N$ by composing with the orthogonal projection onto L. Normalize ν to have mass one on each fibre Y_m and let $\Pi_0 w = \int_{Y_m} w\,\nu$ be the projection of $\mathcal{C}^\infty(Y_m; \mathbb{C}^N)$ onto constants. Define

$$P'(u, w) = \big(Pu + S\Pi_0 w,\ (\operatorname{Id} - \Pi_0)w + q(u, \Pi_0 w)\big),$$

with $q(\cdot) \in \mathbb{C}^N$ regarded as a constant function. All the terms except P and Id have smooth kernels, so $P' - P \oplus \operatorname{Id}$ is smoothing. If $P'(u, w) = 0$ then the mean-zero part $(\operatorname{Id} - \Pi_0)w$ and the constant $q(u, \Pi_0 w)$ both vanish, so $w = c$ is constant, $Pu + Sc = 0$ and $q(u, c) = 0$; thus $(u, c) \in \mathbb{C}\ell_{N+1}$. Conversely every multiple of ℓ_{N+1} gives an element of the null space. To see surjectivity, given (g, h) write $h = h_0 + h_1$ with $h_0 = \Pi_0 h$. Take $w = h_1 + c$, and choose (u, c) with $Pu + Sc = g$ and $q(u, c) = h_0$; this is possible since solutions of $Pu + Sc = g$ form an affine space modelled on L and $q : L \to \mathbb{C}^N$ is onto. Normalizing ℓ_{N+1} gives e. □

Under (15.2.4), and for any elliptic $A \in \Psi^0(M/B; F_+, F_-)$, define the *Clifford tensor product*

$$P \otimes A = \begin{pmatrix} P & 0 \\ A_\Pi & P^* \end{pmatrix} \in \Psi^{0,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B; G_+, G_-), \tag{15.2.5}$$

acting from $G_+ = E_+ \otimes F_+ \oplus E_- \otimes F_-$ to $G_- = E_- \otimes F_+ \oplus E_+ \otimes F_-$, with A_Π as in (P5).

Proposition 15.2.3. *Under (15.2.4), $P \otimes A$ is fully elliptic and $\operatorname{ind}(P \otimes A) = \operatorname{ind}(A)$ in $K^0(B)$.*

Proof. By (P3) and (P5),

$$\sigma(P \otimes A) = \begin{pmatrix} p & 0 \\ 0 & p^* \end{pmatrix}, \qquad \beta(P \otimes A)(\eta) = \begin{pmatrix} P_m & 0 \\ \Pi \otimes \alpha(\eta) & P_m^* \end{pmatrix}.$$

The first is invertible. For the second, recall that P_m is surjective, so P_m^* is injective with closed range $(\mathbb{C}e)^\perp = \operatorname{Ran}(\operatorname{Id} - \Pi)$, while $\Pi \otimes \alpha$ takes values in $\operatorname{Ran}\Pi$. If $P_m u = 0$ and $(\Pi \otimes \alpha)u + P_m^* v = 0$ then $u = e \otimes f$, both terms vanish separately, $v = 0$, and $\alpha f = 0$ forces $f = 0$. Given (g, h), choose u_0 with $P_m u_0 = g$, write $h - (\Pi \otimes \alpha)u_0 = e \otimes k + P_m^* v$, and set $u = u_0 + e \otimes \alpha^{-1} k$; then (u, v) is a preimage. So the base symbol is invertible (its inverse is again a pseudodifferential operator on Y_m), and $P \otimes A$ is fully elliptic.

Choose $S \in \mathcal{C}^\infty(M; \mathrm{Hom}(\mathbb{C}^N, F_-))$ with $A \oplus S$ surjective, and let $\tilde{S} = e \otimes S$, mapping $\mathbb{C}^N$ into the second summand $E_+ \otimes F_-$ of G_-. We claim that $(P \otimes A) \oplus \tilde{S}$ is surjective on each fibre, with null space

$$\mathrm{Nul}\big((P \otimes A) \oplus \tilde{S}\big) = \{(e \otimes f, 0, c);\ (f, c) \in \mathrm{Nul}(A \oplus S)\}. \tag{15.2.6}$$

Indeed, $(P \otimes A)(u, v) + \tilde{S}c = (Pu,\ e \otimes (A\langle u, e\rangle + Sc) + P^*v)$. The second component is the sum of a term in $\mathrm{Ran}\,\Pi$ and a term in $\mathrm{Ran}(\mathrm{Id} - \Pi)$. If the whole vanishes then $u = e \otimes f$, $v = 0$ and $Af + Sc = 0$. Conversely, given (g, h) choose u_0 with $Pu_0 = g$, decompose $h - e \otimes A\langle u_0, e\rangle = e \otimes k + P^*v$, solve $Af + Sc = k$ and put $u = u_0 + e \otimes f$. The map $(f, c) \mapsto (e \otimes f, 0, c)$ is a smooth bundle isomorphism $\mathcal{N}(A, S) \to \mathcal{N}(P \otimes A, \tilde{S})$, so the two stabilized indices coincide. □

Question. Since $\mathrm{ind}(P \otimes A) = \mathrm{ind}(A)$ is so easy, why not stop here?

Answer. Because $P \otimes A$ is not an ordinary elliptic family on $\tilde{M}/B$, so it does not define a class in $K_{\mathrm{c}}(T^*(\tilde{M}/B))$. Its usual symbol is $\mathrm{diag}(p, p^*)$, which is undefined on Σ_1; all the information about A sits in the base symbol β. The two homotopies in the proof of Theorem 15.2.4 move this information from β into the usual symbol, until the base symbol is the multiplication operator determined by an ordinary symbol, a_P. □

15.2.3 Deformation to an ordinary family

Now choose families

$$\tilde{A}_\pm \in \Psi^0(\tilde{M}/B; E_\pm \otimes F_+, E_\pm \otimes F_-), \quad \tilde{P}_\pm \in \Psi^0(\tilde{M}/B; E_+ \otimes F_\pm, E_- \otimes F_\pm),$$

with principal symbols $\sigma_0(\tilde{A}_\pm) = \chi_2\ \mathrm{Id}_{E_\pm} \otimes \alpha$ and $\sigma_0(\tilde{P}_\pm) = \chi_1\, p \otimes \mathrm{Id}_{F_\pm}$, extended homogeneously.

Theorem 15.2.4 (Multiplicativity)**.** *Assume* (15.2.4) *and let* $A \in \Psi^0(M/B; F_+, F_-)$ *be elliptic. The family*

$$\tilde{A}_P = \begin{pmatrix} \tilde{P}_+ & -\tilde{A}_-^* \\ \tilde{A}_+ & \tilde{P}_-^* \end{pmatrix} \in \Psi^0(\tilde{M}/B; G_+, G_-)$$

is elliptic with principal symbol a_P*, and*

$$\mathrm{ind}(\tilde{A}_P) = \mathrm{ind}(A) \in K^0(B).$$

Proof. That $\sigma_0(\tilde{A}_P) = a_P$ is immediate, so $\tilde{A}_P$ is elliptic. We connect $P \otimes A$ to $\tilde{A}_P$ by two homotopies of fully elliptic product-type families,

$$P \otimes A = Q_0 \xrightarrow{Q_t} Q_1 = R_0 \xrightarrow{R_t} R_1 = \tilde{A}_P;$$

the first switches on the lifted base family $\tilde{A}_\pm$ and switches off A_Π, the second replaces P by $\tilde{P}_\pm$. By Proposition 15.2.3 and Lemma 15.1.3(iii) this proves the theorem. In each step we check that both symbols stay invertible, always by Lemma 15.1.1.

First homotopy. For $t \in [0, 1]$ let

$$Q_t = \begin{pmatrix} P & -t\tilde{A}_-^* \\ (1-t)A_\Pi + t\tilde{A}_+ & P^* \end{pmatrix} \in \Psi^{0,0}_{\tilde{\phi}\text{-pt}}(\tilde{M}/B; G_+, G_-),$$

so $Q_0 = P \otimes A$. On the blow-up of $\Sigma = \Sigma_1$ the functions γ^*p and $\chi_2\alpha$ are smooth, and by (P3)–(P5)

$$\sigma(Q_t) = \begin{pmatrix} \gamma^*p & -t\chi_2\alpha^* \\ t\chi_2\alpha & \gamma^*p^* \end{pmatrix} = m(1, t\chi_2),$$

invertible by Lemma 15.1.1 since γ^*p is invertible. For the base symbol note that $\chi_2 = 1$ near Σ_1, and on Σ_1 the projection onto the second summand of (15.2.1) is the identity, so $\beta(\tilde{A}_\pm)(\eta)$ is multiplication by $\mathrm{Id}_{E_\pm} \otimes \alpha(\eta)$. Hence, writing $\alpha = \alpha(\eta)$,

$$\beta(Q_t)(\eta) = \begin{pmatrix} P_m & -t\alpha^* \\ (1-t)\Pi\alpha + t\alpha & P_m^* \end{pmatrix}.$$

Here α acts on the F factor and commutes with P_m, P_m^* and Π. Decompose the domain as $\operatorname{Ran}\Pi \otimes F_+ \oplus \operatorname{Ran}(\mathrm{Id}-\Pi) \otimes F_+ \oplus E_- \otimes F_-$ and the range as $E_- \otimes F_+ \oplus \operatorname{Ran}\Pi \otimes F_- \oplus \operatorname{Ran}(\mathrm{Id}-\Pi) \otimes F_-$. Then $\beta(Q_t)(\eta)$ becomes

$$\begin{pmatrix} 0 & P_1 & -t\alpha^* \\ \alpha & 0 & 0 \\ 0 & t\alpha & P_1^* \end{pmatrix},$$

where $P_1 : \operatorname{Ran}(\mathrm{Id}-\Pi) \to L^2(Y_m; E_-)$ is the restriction of P_m, an isomorphism. The middle row is invertible on the first summand, and the remaining 2×2 block is $m(1,t)$ formed with P_1 and α, which is invertible by Lemma 15.1.1. So Q_t is fully elliptic for all $t \in [0,1]$, and $Q_1 = \left(\begin{smallmatrix} P & -\tilde{A}_-^* \\ \tilde{A}_+ & P^* \end{smallmatrix}\right)$.

Second homotopy. For $t \in [0,1]$ let

$$R_t = \begin{pmatrix} (1-t)P + t\tilde{P}_+ & -\tilde{A}_-^* \\ \tilde{A}_+ & (1-t)P^* + t\tilde{P}_-^* \end{pmatrix},$$

so $R_0 = Q_1$ and $R_1 = \tilde{A}_P$. Its usual symbol is $m(1-t\chi_2, \chi_2)$ formed with γ^*p and $\chi_2\alpha$ (recall $1-t+t\chi_1 = 1-t\chi_2$). Where $\chi_2 < 1$ the first coefficient is positive and p is invertible; where $\chi_2 = 1$ the second coefficient is positive and α is invertible. So $\sigma(R_t)$ is invertible. Since $\chi_1 = 0$ near Σ_1, $\beta(\tilde{P}_\pm) = 0$, and

$$\beta(R_t)(\eta) = \begin{pmatrix} (1-t)P_m & -\alpha^* \\ \alpha & (1-t)P_m^* \end{pmatrix} = m(1-t, 1),$$

which is invertible by Lemma 15.1.1 because α is invertible. Thus R_t is fully elliptic for all t. By (P4) the index of R_1 as a product-type family is its index as an ordinary family. □

Corollary 15.2.5. *For the tower (15.1.3), let $b \in K_{\mathrm{c}}(T^*(\tilde{M}/M))$ satisfy $\mathrm{ind}_{\mathrm{a}}(b) = 1 \in K^0(M)$. Then for every $x \in K_{\mathrm{c}}(T^*(M/B))$,*

$$\mathrm{ind}_{\mathrm{a}}(b \cdot \tilde{\phi}^* x) = \mathrm{ind}_{\mathrm{a}}(x) \in K^0(B),$$

where the product is formed using (15.2.1). That is, the diagram

$$\begin{array}{ccc} K_{\mathrm{c}}(T^*(M/B)) & \xrightarrow{\;b\cdot\tilde{\phi}^*(\cdot)\;} & K_{\mathrm{c}}(T^*(\tilde{M}/B)) \\ & {\scriptstyle \mathrm{ind}_{\mathrm{a}}} \searrow \quad \swarrow {\scriptstyle \mathrm{ind}_{\mathrm{a}}} & \\ & K^0(B) & \end{array}$$

commutes.

Proof. By (F2) write $b = [\sigma_0(P)]$ and $x = [\sigma_0(A)]$. By Lemma 15.2.2 we may assume (15.2.4). Then $b \cdot \tilde{\phi}^* x = [a_P] = [\sigma_0(\tilde{A}_P)]$ by Lemma 15.2.1, and Theorem 15.2.4 gives the result. □

Remark 15.2.6. The degenerate case $M = B$ is included: then A is a bundle map a over B, $\Sigma_1 = \emptyset$, and the corollary says $\mathrm{ind_a}(b \cdot x) = x$ for $x \in K^0(B) = K_\mathrm{c}(B)$.

The product is also natural with respect to open embeddings, in the following sense.

Lemma 15.2.7. *Let V_1, V_2 be vector bundles over a compact manifold Y, $O \subset Y$ open and $k : V_1|_O \hookrightarrow V_1$, $k' : (V_1 \oplus V_2)|_O \hookrightarrow V_1 \oplus V_2$ the inclusions. For $x_1 \in K_\mathrm{c}(V_1|_O)$ and $x_2 \in K_\mathrm{c}(V_2)$,*

$$k_*(x_1) \cdot x_2 = k'_*(x_1 \cdot x_2|_O) \in K_\mathrm{c}(V_1 \oplus V_2).$$

Proof. The product $x_1 \cdot x_2|_O$ is compactly supported since x_1 is supported over a compact subset of O. By (K3) represent x_1 by (E_+, E_-, a_1) with $E_\pm$ trivialized and $a_1 = \mathrm{Id}$ outside a compact set $C \subset V_1|_O$, and let a_2 (extended) represent x_2. Then $k_* x_1 \cdot x_2$ is represented by $m(1,1)$ formed with the extension of a_1 by Id. Choose $\psi \in \mathcal{C}^\infty_\mathrm{c}(V_1|_O)$, $0 \le \psi \le 1$, equal to 1 near C, pulled back to $V_1 \oplus V_2$, and consider $m(1, (1-t) + t\psi)$. Where a_1 is not invertible, $\psi = 1$ and this equals $m(1,1)$, invertible where a_2 is; elsewhere the first coefficient is 1 and a_1 invertible. So it is a homotopy of triples. At $t = 1$, outside $\operatorname{supp}\psi$ the map is $\left(\begin{smallmatrix} \mathrm{Id} & 0 \\ 0 & \mathrm{Id} \end{smallmatrix}\right)$ in the trivializations, so the class is the extension by the identity of the triple $m(1, \psi)$ over $(V_1 \oplus V_2)|_O$, which in turn is homotopic there to $m(1,1)$ by the same homotopy. □

15.3 Excision

The index is a global quantity: elements of the null space are sections over the whole fibre. Excision says that it is nevertheless *local in the symbol*: if the symbol class is trivial outside an open set U, the index can be computed on any compact manifold containing U. This is what allows us to move the problem from M into a torus.

Let $\phi_j : M_j \to B$, $j = 1, 2$, be fibrations of compact manifolds and U a manifold with open embeddings $i_j : U \hookrightarrow M_j$ such that

$$\begin{array}{ccccc} M_1 & \xleftarrow{i_1} & U & \xrightarrow{i_2} & M_2 \\ & {\scriptstyle\phi_1}\searrow & \downarrow{\scriptstyle\phi_U} & \swarrow{\scriptstyle\phi_2} & \\ & & B & & \end{array} \tag{15.3.1}$$

commutes. Then ϕ_U is a submersion (with non-compact fibres), $T^*(U/B)$ is well defined, and i_j induce open embeddings $T^*(U/B) \hookrightarrow T^*(M_j/B)$, hence extension maps $(i_j)_*$ by (K3).

Theorem 15.3.1 (Excision)**.** *In the situation* (15.3.1)*,*

$$\mathrm{ind_a} \circ (i_1)_* = \mathrm{ind_a} \circ (i_2)_* : K_\mathrm{c}(T^*(U/B)) \longrightarrow K^0(B).$$

Discussion. The idea of the proof is simple. Choose the operators P_1 on M_1 and P_2 on M_2 so that they are a bundle isomorphism τ outside a compact subset of U, and agree on U. Then an equation $P_j u = g$ can be solved outside U pointwise, by $u = \tau^{-1} g$, and inside U the two equations are the same. So the stabilized null spaces live inside U and can be transplanted from M_1 to M_2. The proof has three steps: a normal form for the symbol (Step 1), operators with the locality properties (a), (b) (Step 2), and a stabilization supported in U that works for both manifolds (Step 3).

Proof. Step 1: normal form. Let $\varpi : T^*(U/B) \to U$ be the projection. Let $x \in K_\mathrm{c}(T^*(U/B))$ be represented by (E_+, E_-, a) with a invertible outside a compact set $K \subset T^*(U/B)$. Since $T^*(U/B)$ retracts onto U we may assume $E_\pm = \varpi^* E_\pm$ for bundles $E_\pm$ over U. The image

$K' = \varpi(K)$ is compact, and over $U \setminus K'$ the map a is defined and invertible on the whole fibres of $T^*(U/B)$. Choose $\rho \in \mathcal{C}^\infty_{\mathrm{c}}(U)$, $0 \le \rho \le 1$, with $\rho = 1$ near K', and set $a_s(u,\xi) = a(u,(1-s+s\rho(u))\xi)$. Where $\rho(u) = 1$, $a_s = a$; elsewhere $u \notin K'$ and a_s is invertible. So a_s is a homotopy of triples, and $a_1 = \varpi^* h$ over $U \setminus \operatorname{supp}\rho$, where $h(u) = a(u,0)$ is a bundle isomorphism $E_+ \to E_-$ there. Using h to identify E_+ with E_- outside a compact set, and after stabilization and the construction of (K3), we arrive at a representative of the following form: bundles $E_\pm$ over U with an isomorphism $\tau : E_+ \to E_-$ over $U \setminus C$, $C \subset U$ compact, both trivialized there, and a symbol a, homogeneous of degree 0 for $|\xi| \ge 1$, with $a = \tau$ over $U \setminus C$. Then $(i_j)_* x$ is represented by the bundles $\tilde{E}^j_\pm$ over M_j obtained by extending $E_\pm$ trivially, with the symbol $\tilde{a}^j$ extended by $\tau = \mathrm{Id}$.

Step 2: operators that agree. Choose compact sets $C \subset K_1 \subset \operatorname{int} K_2 \subset K_2 \subset \operatorname{int} K_3 \subset K_3 \subset U$ and $\psi, \psi' \in \mathcal{C}^\infty_{\mathrm{c}}(U)$ with $\psi = 1$ near C, $\operatorname{supp}\psi \subset \operatorname{int} K_1$, $\psi' = 1$ on K_1, $\operatorname{supp}\psi' \subset \operatorname{int} K_2$. The bundle map $J = (1-\psi)\tau$ is defined on all of M_j (as τ outside $i_j(U)$). The symbol $\tilde{a}^j - J$ vanishes outside $\varpi^{-1}(\operatorname{supp}\psi)$. Choose $T_U \in \Psi^0(U/B; E_+, E_-)$ with $\sigma_0(T_U) = a - (1-\psi)\tau$ and replace it by $\psi' T_U \psi'$, which has the same symbol and Schwartz kernel supported in $K_2 \times_B K_2$. Set

$$P_j = J + (i_j)_* T_U \in \Psi^0(M_j/B; \tilde{E}^j_+, \tilde{E}^j_-),$$

where $(i_j)_* T_U$ is the operator whose kernel is the push-forward of that of T_U. Then $\sigma_0(P_j) = \tilde{a}^j$, so $[\sigma_0(P_j)] = (i_j)_* x$, and P_1, P_2 have the following locality properties:

(a) for every v and every point $y \notin i_j(K_2)$, $(P_j v)(y) = \tau(y) v(y)$; if moreover v vanishes near $i_j(K_2)$ then $P_j v = \tau v$ (both sides vanishing on $i_j(K_2)$);

(b) if v is supported in $i_j(L)$, $L \subset U$ compact, then $P_j v$ is supported in $i_j(L \cup K_2)$, and $i_1^*(P_1 v_1) = i_2^*(P_2 v_2)$ whenever v_1, v_2 are supported in $i_1(L)$, $i_2(L)$ and $i_1^* v_1 = i_2^* v_2$.

(In (b) only the restriction of the bundles to U, where they are the same, is involved.)

Step 3: common stabilization. Fix b. Sections supported in $i_j(\operatorname{int} K_3)$ span $L^2(M_{j,b}; \tilde{E}_-)$ modulo $\operatorname{Ran} P_{j,b}$: given g, choose $\psi_3 \in \mathcal{C}^\infty_{\mathrm{c}}(\operatorname{int} K_3)$ equal to 1 near K_2; then by (a) $P_j(\tau^{-1}(1-\psi_3)g) = (1-\psi_3)g$. Since the cokernel is finite dimensional and smooth sections are dense, it is spanned by finitely many smooth sections supported in $i_j(\operatorname{int} K_3)$. Running the proof of Lemma 15.1.3(i) for P_1 with this choice (and cut-offs in the base) gives sections $v_1, \dots, v_N$ of E_- supported in the interior of a compact subset of U, which we use for both j, so that $P_1 \oplus S$ is surjective.

Then $P_2 \oplus S$ is surjective too. Given g on $M_{2,b}$, write $g = \psi_3 g + (1-\psi_3)g$; the second term is $P_2(\tau^{-1}(1-\psi_3)g)$. For the first, transplant it to M_1 and solve $P_1 u + Sc = \psi_3 g$. Outside $i_1(K_2 \cup \operatorname{supp}\psi_3 \cup \operatorname{supp} S)$, (a) gives $\tau u = 0$ pointwise, so u is supported in $i_1(U)$; by (b) its transplant u_2 to M_2 satisfies $P_2 u_2 + Sc = \psi_3 g$. The same argument shows that the elements of $\operatorname{Nul}(P_j \oplus S)$ are supported in a fixed compact subset of $i_j(U)$, and that transplanting gives an isomorphism of bundles $\mathcal{N}(P_1, S) \cong \mathcal{N}(P_2, S)$ over B. Hence $\operatorname{ind_a}((i_1)_* x) = \operatorname{ind}(P_1) = \operatorname{ind}(P_2) = \operatorname{ind_a}((i_2)_* x)$ by Lemma 15.1.3. □

15.4 The families index theorem in K-theory

We now assemble the proof. The strategy, due to Atiyah and Singer, is to transport the symbol class from $T^*(M/B)$ to the cotangent bundle of a *trivial* fibration, where the index can be computed by hand. The transport has three stages (Figures 15.1 and 15.3): embed M in $\mathbb{R}^N \times B$ and push the symbol into a tubular neighbourhood by the Thom homomorphism; extend it by the identity to $I^N \times B \subset \mathbb{T}^N \times B$; and read off the index on the torus by Bott periodicity.

15.4.1 Embedding the fibration

Lemma 15.4.1. *Let $\phi : M \to B$ be a fibration of compact manifolds with fibres of dimension n. For N large there is an embedding $e : M \to \mathbb{R}^N \times B$ with $\pi_B \circ e = \phi$. If $\mathcal{V} \to M$ is the normal bundle, of rank $k = N - n$, there is a diffeomorphism Θ from the total space of $\mathcal{V}$ onto an open neighbourhood Ω of $e(M)$, with $\Theta(0_m) = e(m)$ and $\pi_B \circ \Theta = \phi \circ \varpi$, $\varpi : \mathcal{V} \to M$ the projection. Moreover $e(M)$ and Ω may be taken inside $(0, 2\pi)^N \times B$.*

Proof. Let $\iota : M \to \mathbb{R}^N$ be an embedding (Whitney) and $e = (\iota, \phi)$; it is an injective immersion of a compact manifold, hence an embedding, and $\pi_B \circ e = \phi$. Define $\mathcal{V}_m = (d\iota(T_m(M/B)))^\perp \subset \mathbb{R}^N$ and $\Theta(m, v) = (\iota(m) + v, \phi(m))$. At the zero section $d\Theta(w, v) = (d\iota(w) + v, d\phi(w))$; if this vanishes then w is vertical, and $d\iota(w) + v = 0$ with $v \perp d\iota(w)$ forces $w = 0$, $v = 0$. Since $\dim M + k = N + \dim B$, Θ is a local diffeomorphism near the zero section, injective on it, hence (by the usual compactness argument) a diffeomorphism from $\{|v| < \delta\}$ onto an open set. Composing with $v \mapsto \delta v/(1 + |v|^2)^{1/2}$ gives Θ. Translating and scaling ι puts everything inside $(0, 2\pi)^N \times B$. □

Let $I = (0, 2\pi)$. Both Θ and the inclusion $\Omega \subset I^N \times B$ commute with the projections to B, so they induce maps of vertical cotangent bundles: $\Theta_* : K_c(T^*(\mathcal{V}/B)) \to K_c(T^*(\Omega/B))$ and the extension $j_* : K_c(T^*(\Omega/B)) \to K_c(T^*(I^N \times B/B)) = K_c(T^*I^N \times B)$; note $T^*I^N \cong \mathbb{R}^{2N}$.

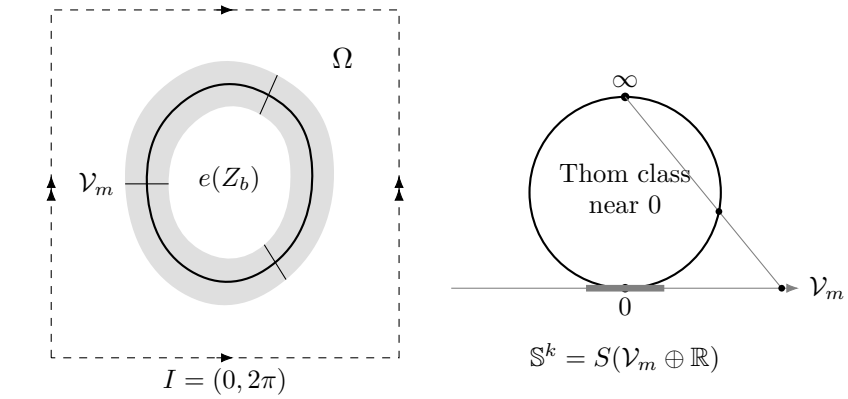


Figure 15.3: Left: the fibre Z_b embedded in the box I^N ($N = 2$), with its tubular neighbourhood $\Omega \cong \mathcal{V}$; identifying opposite sides of the box gives the torus $\mathbb{T}^N$. Right: a normal fibre $\mathcal{V}_m \cong \mathbb{R}^k$ ($k = 1$) compactified to the sphere $\mathbb{S}^k$ by inverse stereographic projection. The Thom class $\lambda_\mathcal{V}$ is concentrated near the zero section, so it can be extended by the identity both to the sphere bundle $\tilde{M}$ (where its index is computed by (TH)) and, via Θ, to the torus (where it is computed by Bott periodicity). Excision says the two computations agree.

15.4.2 The Thom class

Consider $\mathcal{V}$ as a fibration over M. Its vertical cotangent bundle is $T^*(\mathcal{V}/M) = \varpi^*\mathcal{V}^*$, whose total space is $\mathcal{V} \oplus \mathcal{V}^* \cong \mathcal{V} \otimes \mathbb{C}$ via $(v, \eta) \mapsto v + i\eta^\sharp$. On the complex vector bundle $W = \varpi^*(\mathcal{V} \otimes \mathbb{C})$ over this total space, with the Hermitian metric induced by a metric on $\mathcal{V}$, Clifford multiplication $c(w) = w\wedge\cdot - \iota_w$ maps $\Lambda^{\mathrm{ev}}W \to \Lambda^{\mathrm{odd}}W$ and satisfies $c(w)^*c(w) = |w|^2$. The *Thom class* is

$$\lambda_\mathcal{V} = [(\Lambda^{\mathrm{ev}}W, \Lambda^{\mathrm{odd}}W, c)] \in K_c(T^*(\mathcal{V}/M)). \tag{15.4.1}$$

By (15.2.1) for the tower $\mathcal{V} \to M \to B$ (with non-compact fibres, which does not affect the linear algebra), $T^*(\mathcal{V}/B) \cong T^*(\mathcal{V}/M) \oplus \varpi^* T^*(M/B)$, and we define the *Thom homomorphism*

$$\tau : K_{\mathrm{c}}(T^*(M/B)) \longrightarrow K_{\mathrm{c}}(T^*(\mathcal{V}/B)), \quad \tau(x) = \lambda_{\mathcal{V}} \cdot \varpi^* x.$$

(The product is compactly supported because $\lambda_{\mathcal{V}}$ is supported in the zero section, i.e. over a compact set.) Since $T^*(\mathcal{V}/M) \cong \mathcal{V} \otimes \mathbb{C}$ is a complex (equivalently, by Chapter 13, symplectic) vector bundle, $\lambda_{\mathcal{V}}$ is its Bott class and, by the Thom isomorphism theorem of Chapter 13, τ is an isomorphism; we shall not need this.

Discussion. On a single fibre $\mathcal{V}_m \cong \mathbb{R}^k$, with coordinates (v, η) on its cotangent bundle, the Thom symbol is Clifford multiplication by $v + i\eta$. For $k = 1$ this is the scalar symbol $x + i\xi$, the symbol of the annihilation operator of the harmonic oscillator (Remark 15.4.3). It is invertible except at the origin, so it defines a compactly supported class concentrated at the zero section. Multiplying a symbol x on M/B by it does not change where x lives on M; it only adds the normal directions in the most economical way. That this costs nothing at the level of the index is the content of (TH) below.

15.4.3 The Bott element and the trivial fibration

Let $\mathbb{S} = \mathbb{R}/2\pi\mathbb{Z}$, so that $I = (0, 2\pi) \subset \mathbb{S}$. Let $\Pi_>$ be the orthogonal projection of $L^2(\mathbb{S})$ onto $\overline{\mathrm{span}}\{e^{ik\theta}; k \geq 1\}$; it lies in $\Psi^0(\mathbb{S})$ with symbol 1 for $\xi > 0$ and 0 for $\xi < 0$ (it differs from the Szegő projector of Chapter 12 by the projection onto constants). Define

$$P_1 = e^{-i\theta}\Pi_> + (\mathrm{Id} - \Pi_>) \in \Psi^0(\mathbb{S}), \quad \sigma_0(P_1)(\theta, \xi) = \begin{cases} e^{-i\theta} & \xi > 0, \\ 1 & \xi < 0. \end{cases} \tag{15.4.2}$$

On Fourier coefficients, $(P_1 u)_j = u_{j+1}$ for $j \geq 1$, $(P_1 u)_0 = u_0 + u_1$ and $(P_1 u)_j = u_j$ for $j < 0$. So P_1 is surjective and $\mathrm{Nul}(P_1) = \mathbb{C}(e^{i\theta} - 1)$; in particular $\mathrm{ind}(P_1) = 1$, and the constant family P_1 satisfies (15.2.4) for any fibration with fibre $\mathbb{S}$ of product form.

Let $f \in \mathcal{C}^\infty([0, 2\pi])$ be non-decreasing with $f = 0$ near 0, $f = 2\pi$ near 2π. The symbols with $e^{-i\theta}$ replaced by $e^{-i((1-s)\theta + s f(\theta))}$ are a homotopy of elliptic symbols, and at $s = 1$ the symbol is 1 for θ near 0. Hence there is a class

$$b_1 \in K_{\mathrm{c}}(T^*I) = K_{\mathrm{c}}(\mathbb{R}^2), \qquad (k_1)_* b_1 = [\sigma_0(P_1)] \in K_{\mathrm{c}}(T^*\mathbb{S}),$$

where $k_1 : T^*I \hookrightarrow T^*\mathbb{S}$; b_1 is the *Bott element*. Let $\mathbb{T}^N = \mathbb{S}^N$ and $k_N : T^*I^N \times B \hookrightarrow T^*(\mathbb{T}^N \times B/B)$. Set

$$\beta_N : K^0(B) \longrightarrow K_{\mathrm{c}}(T^*I^N \times B), \quad \beta_N(y) = b_1 \boxtimes \cdots \boxtimes b_1 \boxtimes y.$$

We use Bott periodicity in the following form.

(BP) For every manifold X, the map $K_{\mathrm{c}}(X) \ni x \mapsto b_1 \boxtimes x \in K_{\mathrm{c}}(T^*I \times X)$ is an isomorphism. Consequently each β_N is an isomorphism.

This is proved in Chapter 13 through the Toeplitz index; in the stated form (exterior multiplication by the generator) it is [17], Theorem 2.4.9. For compact X the two forms are equivalent by the following proposition, which shows that the index on the circle is a left inverse of $b_1 \boxtimes \cdot$.

Proposition 15.4.2 (Normalization)**.** *For every $N \geq 0$,*

$$\mathrm{ind}_{\mathrm{a}} \circ (k_N)_* \circ \beta_N = \mathrm{Id}_{K^0(B)}.$$

Hence, by (BP), $\mathrm{ind}_{\mathrm{a}} \circ (k_N)_* = \beta_N^{-1} : K_{\mathrm{c}}(T^*I^N \times B) \to K^0(B)$.

Proof. Induction on N; for $N = 0$ this is Remark 15.1.4(b). Consider the tower $\mathbb{T}^N \times B \to \mathbb{T}^{N-1} \times B \to B$, the first map forgetting the last circle factor, and the constant family P_1 on its fibres, which satisfies (15.2.4). Here the splitting (15.2.1) is the product decomposition. For $y \in K^0(B)$, Lemma 15.2.7 (applied in both factors) gives

$$[\sigma_0(P_1)] \cdot (k_{N-1})_* \beta_{N-1}(y) = (k_1)_* b_1 \cdot (k_{N-1})_* \beta_{N-1}(y) = (k_N)_* \beta_N(y).$$

So Corollary 15.2.5 and the inductive hypothesis give

$$\mathrm{ind_a}\big((k_N)_* \beta_N(y)\big) = \mathrm{ind_a}\big((k_{N-1})_* \beta_{N-1}(y)\big) = y.$$

□

Remark 15.4.3. The Thom class of the trivial line bundle, $\lambda_{\mathbb{R}} \in K_{\mathrm{c}}(T^*\mathbb{R})$, is represented by the symbol $x + i\xi$ on $\Lambda^0 \to \Lambda^1$, which has winding number 1 around the circle $|(x,\xi)| = R$ traversed counterclockwise in the (x,ξ)-plane. The symbol of b_1 has the same winding number (it is 1 on the lower edge $\xi = -R$ and on the vertical edges of a large square, and $e^{-if(\theta)}$ on the upper edge, traversed in the direction of decreasing θ). Since classes in $K_{\mathrm{c}}(\mathbb{R}^2)$ are determined by this winding number (Chapter 13), $\lambda_{\mathbb{R}} = b_1$ under $I \cong \mathbb{R}$. For a trivial bundle $\mathcal{V} = M \times \mathbb{R}^k$ it follows that $\lambda_{\mathcal{V}}$ is the pull-back of $b_1 \boxtimes \cdots \boxtimes b_1$ (k factors), since $\Lambda(W_1 \oplus W_2) = \Lambda W_1 \otimes \Lambda W_2$ as graded bundles and Clifford multiplication is the Clifford product (15.1.2).

15.4.4 The theorem

Using Lemma 15.4.1 and (BP) we define the *topological index*

$$\mathrm{ind_t} = \beta_N^{-1} \circ j_* \circ \Theta_* \circ \tau : K_{\mathrm{c}}(T^*(M/B)) \longrightarrow K^0(B). \tag{15.4.3}$$

The definition involves the choice of the embedding (and of the tubular neighbourhood and the metrics); the following theorem shows a posteriori that $\mathrm{ind_t}$ does not depend on these choices.

Theorem 15.4.4 (Atiyah–Singer, K-theory form)**.** *For every fibration $\phi : M \to B$ of compact manifolds,* $\mathrm{ind_a} = \mathrm{ind_t}$*, i.e. the diagram*

$$\begin{array}{ccccc} K_{\mathrm{c}}(T^*(M/B)) & \xrightarrow{\ \tau\ } & K_{\mathrm{c}}(T^*(\mathcal{V}/B)) & \xrightarrow{\ j_*\Theta_*\ } & K_{\mathrm{c}}(T^*I^N \times B) \\ & \searrow^{\mathrm{ind_a}} & & & \downarrow{\scriptstyle \beta_N^{-1}} \\ & & & & K^0(B) \end{array}$$

commutes. In particular $\mathrm{ind_t}$ *is independent of the embedding.*

Besides what was recalled in Section 15.1, the proof uses one result which we do not prove.

(TH) Let $\tilde{M} = S(\mathcal{V} \oplus \mathbb{R}) \to M$ be the fibrewise one-point compactification of $\mathcal{V}$ (the unit sphere bundle of $\mathcal{V} \oplus \mathbb{R}$), with fibre $\mathbb{S}^k$, and $k_{\mathcal{V}} : \mathcal{V} \hookrightarrow \tilde{M}$ the inverse stereographic projection from the section at infinity. Then

$$\mathrm{ind_a}\big((k_{\mathcal{V}})_* \lambda_{\mathcal{V}}\big) = 1 \in K^0(M).$$

Restricted to any point of M (using Remark 15.1.4(d)) this holds by Remark 15.4.3, Proposition 15.4.2 (with B a point) and excision (comparing the sphere $\mathbb{S}^k$ with the torus $\mathbb{T}^k$, both of which contain $\mathbb{R}^k$). By the same argument it holds in general when $\mathcal{V}$ is trivial. In general, however, a class in $K^0(M)$ is not determined by its restrictions to points. The standard proof shows that the index of the Bott operator on $\mathbb{S}^k$ is the trivial one-dimensional representation of $O(k)$, and then constructs the family over M from the orthonormal frame bundle of $\mathcal{V}$; see the Notes and [18]. A short proof, based on the harmonic oscillator, is given in Corollary 16.6.4.

Question. Each fibre of $\tilde{M} \to M$ is a sphere on which the index is 1. Why is it not automatic that the index bundle is the trivial line bundle?

Answer. A line bundle over M whose restriction to every point is $\mathbb{C}$ can still be non-trivial; and a class in $K^0(M)$ is not even determined by its rank. What is needed is a *canonical* identification of the one-dimensional index spaces of the operators on the fibres $\mathbb{S}^k$, compatible with the structure group $O(k)$ of $\mathcal{V}$. The classical argument checks that $O(k)$ acts trivially on the index; in Chapter 16 the index space is spanned by an $O(k)$-invariant Gaussian, which makes the triviality visible. □

The proof follows Figure 15.1 arrow by arrow: multiplicativity and (TH) for the Thom step, excision to move from the sphere bundle $\tilde{M}$ to the torus, and the normalization to evaluate the index there.

Proof of Theorem 15.4.4. Let $x \in K_{\mathrm{c}}(T^*(M/B))$. Consider the tower $\tilde{M} \xrightarrow{\tilde{\phi}} M \xrightarrow{\phi} B$ with $\tilde{M} = S(\mathcal{V} \oplus \mathbb{R})$ as in (TH), and let $\hat{k} : T^*(\mathcal{V}/B) \hookrightarrow T^*(\tilde{M}/B)$ be the open embedding induced by $k_{\mathcal{V}}$. By (TH) the class $b = (k_{\mathcal{V}})_*\lambda_{\mathcal{V}}$ has $\mathrm{ind}_{\mathrm{a}}(b) = 1$, so by Corollary 15.2.5 and Lemma 15.2.7 (with $Y = \tilde{M}$, $O = \mathcal{V}$; a splitting of $T^*(\tilde{M}/B)$ restricting to the one used for τ over $\mathcal{V}$ is chosen)

$$\mathrm{ind}_{\mathrm{a}}(x) = \mathrm{ind}_{\mathrm{a}}(b \cdot \tilde{\phi}^* x) = \mathrm{ind}_{\mathrm{a}}(\hat{k}_* \tau(x)).$$

Now apply Theorem 15.3.1 with $U = \mathcal{V}$, $M_1 = \tilde{M}$, $i_1 = k_{\mathcal{V}}$, and $M_2 = \mathbb{T}^N \times B$, $i_2 = \Theta$ followed by the inclusion $\Omega \subset I^N \times B \subset \mathbb{T}^N \times B$ (all commuting with the maps to B):

$$\mathrm{ind}_{\mathrm{a}}(\hat{k}_* \tau(x)) = \mathrm{ind}_{\mathrm{a}}((k_N)_* j_* \Theta_* \tau(x)).$$

Finally Proposition 15.4.2 identifies the right side with $\beta_N^{-1}(j_*\Theta_*\tau(x)) = \mathrm{ind}_{\mathrm{t}}(x)$. □

Remark 15.4.5. To summarize the logical structure: multiplicativity (Theorem 15.2.4), excision (Theorem 15.3.1) and the normalization (Proposition 15.4.2) are proved in this chapter from the calculus of Chapter 14; Bott periodicity (BP) comes from Chapter 13; the index of the Thom class (TH) is imported, except when the normal bundle is trivial. Traditionally (15.4.3) is the *definition* of the topological index. Read the other way, Theorem 15.4.4 is an effective method for computing the analytic index: everything has been reduced to the index on the trivial fibration, which is given by Bott periodicity, i.e. by repeated use of the Toeplitz index isomorphism of Chapter 13. In practice one may replace $\mathbb{R}^N \times B$ by $\mathbb{S}^N \times B$, as excision shows.

15.5 Chern–Weil theory and the Chern character

The K-theoretic index theorem identifies two maps but does not, by itself, produce a number one can compute. To obtain an integral formula we map K-theory to de Rham cohomology by the Chern character. It is built from the curvature of a connection, so it is given by

explicit differential forms, and it is a ring homomorphism, so it turns the products of K-classes used above into wedge products of forms. We work with de Rham cohomology with complex coefficients; $H^{\mathrm{ev}}(X;\mathbb{C})$ denotes the cohomology in even degrees.

15.5.1 Connections and curvature

Let $E \to X$ be a complex vector bundle over a manifold. A *connection* on E is a differential operator $\nabla \in \mathrm{Diff}^1(X; E, \Lambda^1 \otimes E)$ with

$$\nabla(fu) = df \otimes u + f\nabla u, \quad f \in \mathcal{C}^\infty(X),\ u \in \mathcal{C}^\infty(X;E).$$

If $E = X \times \mathbb{C}^N$ then d, acting on the components, is a connection, and every connection is then $d + \gamma$ with γ a 1-form with values in $N \times N$ matrices. If ∇_i are connections on E over the sets of an open cover and ρ_i is a subordinate partition of unity then $\sum_i \rho_i \nabla_i$ is a connection. Hence every vector bundle admits connections, and the connections form an affine space over $\mathcal{C}^\infty(X; \Lambda^1 \otimes \mathrm{End}\, E)$. If $a : E \to \tilde{E}$ is a bundle isomorphism then $a\nabla a^{-1}$ is a connection on $\tilde{E}$.

A connection extends uniquely to $\nabla : \mathcal{C}^\infty(X; \Lambda^k \otimes E) \to \mathcal{C}^\infty(X; \Lambda^{k+1} \otimes E)$ for all k, so that

$$\nabla(\alpha \wedge u) = d\alpha \wedge u + (-1)^j \alpha \wedge \nabla u, \quad \alpha \in \mathcal{C}^\infty(X; \Lambda^j),\ u \in \mathcal{C}^\infty(X; \Lambda^* \otimes E).$$

(Locally $\nabla = d + \gamma$ with γ acting by exterior multiplication; the sign records the passage of ∇ past j factors.) The *curvature* $\omega_\nabla = \nabla^2$ commutes with multiplication by functions:

$$\nabla^2(fu) = \nabla(df \wedge u + f\nabla u) = -df \wedge \nabla u + df \wedge \nabla u + f\nabla^2 u = f\nabla^2 u,$$

so $\omega_\nabla \in \mathcal{C}^\infty(X; \Lambda^2 \otimes \mathrm{End}\, E)$; locally $\omega = d\gamma + \gamma \wedge \gamma$. The bundle $\Lambda^* \otimes \mathrm{End}\, E$ is a bundle of $\mathbb{Z}_2$-graded algebras, with product $(\alpha \otimes a)(\beta \otimes b) = \alpha \wedge \beta \otimes ab$, and $\mathrm{End}(E)$-valued forms act on $\Lambda^* \otimes E$. For $a \in \mathcal{C}^\infty(X; \Lambda^k \otimes \mathrm{End}\, E)$ set

$$\nabla a = [\nabla, a] = \nabla \circ a - (-1)^k a \circ \nabla \in \mathcal{C}^\infty(X; \Lambda^{k+1} \otimes \mathrm{End}\, E),$$

locally $da + \gamma \wedge a - (-1)^k a \wedge \gamma$. It satisfies $\nabla(ab) = (\nabla a)b + (-1)^k a \nabla b$, and the *Bianchi identity*

$$\nabla\omega_\nabla = [\nabla, \nabla^2] = 0.$$

The usual constructions are compatible with curvature. A connection on E induces one on E^* by $d(u^*(u)) = (\nabla^* u^*)(u) + u^*(\nabla u)$, with curvature $-\omega^t$. On $E \oplus F$, $\nabla_E \oplus \nabla_F$ has curvature $\omega_E \oplus \omega_F$. On $E \otimes F$, $\nabla(u \otimes v) = \nabla_E u \otimes v + u \otimes \nabla_F v$ has curvature

$$\omega_{E\otimes F} = \omega_E \otimes \mathrm{Id}_F + \mathrm{Id}_E \otimes \omega_F.$$

This gives connections on $\mathrm{Hom}(E,F) = F \otimes E^*$; the induced action on $a \in \mathcal{C}^\infty(X; \mathrm{Hom}(E,F))$ is $\nabla a = \nabla_F \circ a - a \circ \nabla_E$. For a smooth map $f : X' \to X$ the pull-back $f^*\nabla$ has curvature $f^*\omega_\nabla$. If $a : E \to \tilde{E}$ is an isomorphism, $a\nabla a^{-1}$ has curvature $a\omega_\nabla a^{-1}$.

15.5.2 The Chern character form

The fibre trace extends to $\mathrm{tr} : \mathcal{C}^\infty(X; \Lambda^j \otimes \mathrm{End}\, E) \to \mathcal{C}^\infty(X; \Lambda^j)$, $\mathrm{tr}(\alpha \otimes a) = \alpha\, \mathrm{tr}(a)$, and is graded-cyclic:

$$\mathrm{tr}(ab) = (-1)^{jk}\, \mathrm{tr}(ba), \quad a \in \Lambda^j \otimes \mathrm{End}\, E,\ b \in \Lambda^k \otimes \mathrm{End}\, E. \tag{15.5.1}$$

Lemma 15.5.1. *For any connection ∇ on E and $a \in \mathcal{C}^\infty(X; \Lambda^k \otimes \operatorname{End} E)$, $d \operatorname{tr}(a) = \operatorname{tr}(\nabla a)$.*

Proof. Both sides are local. In a trivialization $\nabla a = da + [\gamma, a]$, the bracket being graded, and $\operatorname{tr}[\gamma, a] = 0$ by (15.5.1); also $\operatorname{tr}(da) = d \operatorname{tr}(a)$. □

Definition 15.5.2. The *Chern character form* of (E, ∇) is

$$\operatorname{Ch}(\nabla) = \operatorname{tr} \exp\Big(\frac{i}{2\pi}\omega_\nabla\Big) = \operatorname{rank} E + \sum_{k \geq 1} \frac{i^k}{(2\pi)^k k!} \operatorname{tr}(\omega_\nabla^k) \in \mathcal{C}^\infty(X; \Lambda^{\mathrm{ev}}).$$

The sum is finite since $\omega^k = 0$ for $2k > \dim X$. Even forms commute, so exp has its usual properties. The normalization $i/2\pi$ makes the classes rational, as recalled below.

Discussion. Why $\operatorname{tr} \exp$? The Chern character must be additive for $\oplus$ and multiplicative for $\otimes$. The curvature of $E \oplus F$ is block diagonal, and the trace of a block diagonal matrix is the sum of the traces; the curvature of $E \otimes F$ is $\omega_E \otimes \mathrm{Id} + \mathrm{Id} \otimes \omega_F$, and the exponential turns this sum of commuting terms into a tensor product, whose trace is the product of the traces. Closedness of $\operatorname{Ch}(\nabla)$ and independence of ∇ both come from the Bianchi identity and the graded cyclicity of tr.

Proposition 15.5.3. *(i)* $\operatorname{Ch}(\nabla)$ *is closed.*

(ii) If $\nabla_t = \nabla + t\theta$, $\theta = \nabla' - \nabla$, *then*

$$\operatorname{Ch}(\nabla') - \operatorname{Ch}(\nabla) = d \operatorname{cs}(\nabla, \nabla'), \quad \operatorname{cs}(\nabla, \nabla') = \frac{i}{2\pi} \int_0^1 \operatorname{tr}\Big(\theta \exp\Big(\frac{i}{2\pi}\omega_{\nabla_t}\Big)\Big) dt.$$

So the class $\operatorname{Ch}(E) \in H^{\mathrm{ev}}(X; \mathbb{C})$ *of* $\operatorname{Ch}(\nabla)$ *is independent of* ∇.

(iii) $\operatorname{Ch}(E \oplus F) = \operatorname{Ch}(E) + \operatorname{Ch}(F)$, $\operatorname{Ch}(E \otimes F) = \operatorname{Ch}(E) \wedge \operatorname{Ch}(F)$, $\operatorname{Ch}(f^*E) = f^* \operatorname{Ch}(E)$, *and* $\operatorname{Ch}(E) = \operatorname{Ch}(\tilde{E})$ *if* $E \cong \tilde{E}$.

Proof. Write $c = i/2\pi$. (i) By Lemma 15.5.1, $d \operatorname{Ch}(\nabla) = \operatorname{tr}(\nabla e^{c\omega})$. Since ω is even, $\nabla(\omega^k) = \sum_j \omega^j (\nabla\omega) \omega^{k-1-j} = 0$ by Bianchi.

(ii) The curvature of ∇_t is $\omega_t = \omega + t\nabla\theta + t^2\theta^2$, so $\partial_t \omega_t = \nabla\theta + 2t\theta^2 = \nabla_t \theta$ (graded commutator, θ odd). By cyclicity, since ω_t is even,

$$\partial_t \operatorname{tr} e^{c\omega_t} = c \operatorname{tr}((\nabla_t \theta) e^{c\omega_t}) = c \operatorname{tr}(\nabla_t(\theta e^{c\omega_t})) = c\, d \operatorname{tr}(\theta e^{c\omega_t}),$$

using $\nabla_t e^{c\omega_t} = 0$ and Lemma 15.5.1. Integrate in t. (This is the transgression; equivalently, apply (i) to the connection $(1-t)\nabla + t\nabla' + dt\, \partial_t$ on the pull-back of E to $[0,1] \times X$ and integrate the dt-component.)

(iii) The curvature of a direct sum is block diagonal, so the traces add. For the tensor product, $\omega_E \otimes \mathrm{Id}$ and $\mathrm{Id} \otimes \omega_F$ commute, so $e^{c\omega_{E\otimes F}} = e^{c\omega_E} \otimes e^{c\omega_F}$ and $\operatorname{tr}_{E\otimes F} = \operatorname{tr}_E \otimes \operatorname{tr}_F$. Naturality follows from $\omega_{f^*\nabla} = f^*\omega_\nabla$, and invariance under isomorphism from $\operatorname{tr}(a\omega^k a^{-1}) = \operatorname{tr}(\omega^k)$. □

Definition 15.5.4. The *Chern character* $\operatorname{Ch} : K^0(X) \to H^{\mathrm{ev}}(X; \mathbb{C})$ is $\operatorname{Ch}([E_+] - [E_-]) = \operatorname{Ch}(E_+) - \operatorname{Ch}(E_-)$.

By Proposition 15.5.3(iii) this is well defined (it is unchanged by adding the same bundle to E_+ and E_- and by isomorphisms) and additive. For superbundles, with the graded tensor product (15.1.1), $\operatorname{Ch}(G_+) - \operatorname{Ch}(G_-) = (\operatorname{Ch}(E_+) - \operatorname{Ch}(E_-)) \wedge (\operatorname{Ch}(F_+) - \operatorname{Ch}(F_-))$. The product on $K^0(X)$ is induced by the graded tensor product, so Ch is a ring homomorphism,

$$\operatorname{Ch}(x \cdot y) = \operatorname{Ch}(x) \wedge \operatorname{Ch}(y), \quad x, y \in K^0(X).$$

With more care one shows that Ch takes values in rational cohomology. We will not prove, nor use, the theorem of Atiyah and Hirzebruch that $\mathrm{Ch} : K^0(X) \otimes \mathbb{C} \to H^{\mathrm{ev}}(X;\mathbb{C})$ is an isomorphism for compact X.

Example 15.5.5. Let $\pi \in \mathcal{C}^\infty(X; M(N,\mathbb{C}))$ be a smooth family of projections and $E = \pi\mathbb{C}^N$. The *Grassmann connection* is $\nabla u = \pi\, du$. From $\pi^2 = \pi$ we get $\pi\, d\pi + d\pi\, \pi = d\pi$, hence $\pi\, d\pi\, \pi = 0$ and $\pi(d\pi)^2 = (d\pi)^2\pi$. For $u = \pi u$,

$$\nabla^2 u = \pi\, d(\pi\, du) = \pi\, d\pi \wedge du = \pi\, d\pi \wedge (d\pi\, u + \pi\, du) = \pi(d\pi)^2 u,$$

so $\omega = \pi(d\pi)^2\pi$ and

$$\mathrm{Ch}(\nabla) = \sum_{k\geq 0} \frac{1}{k!}\Big(\frac{i}{2\pi}\Big)^k \operatorname{tr}(\pi(d\pi)^{2k}). \tag{15.5.2}$$

For the tautological line bundle over $\mathbb{C}P^1$, near $w = 0$ in the affine chart, $\pi = vv^*/|v|^2$ with $v = (1, w)$, and at $w = 0$ one finds $\operatorname{tr}(\pi(d\pi)^2) = d\bar{w} \wedge dw = 2i\, dx \wedge dy$. So $\frac{i}{2\pi}\operatorname{tr}(\pi(d\pi)^2)$ is negative at $w = 0$ (and, by $U(2)$-invariance, at every point), and the degree two part of Ch integrates to -1 (Exercise 15.2).

15.6 The odd Chern character and Toeplitz families

15.6.1 The odd Chern character

For a smooth map $a : X \to \mathrm{GL}(N,\mathbb{C})$ set $\theta = a^{-1}da$. In Chapter 12 it was shown that $\operatorname{tr}(\theta^{2k+1})$ is closed, that its class depends only on the homotopy class of a, and that it is additive. We fix the normalization:

$$\mathrm{Ch}_{\mathrm{odd}}(a) = \sum_{k\geq 0} c_k \operatorname{tr}((a^{-1}da)^{2k+1}), \qquad c_k = \frac{i}{2\pi}\Big(\frac{-i}{2\pi}\Big)^k \frac{k!}{(2k+1)!}. \tag{15.6.1}$$

The class of $\mathrm{Ch}_{\mathrm{odd}}(a)$ in $H^{\mathrm{odd}}(X;\mathbb{C})$ depends only on the class of a in $K^{-1}(X)$, since stabilization by Id does not change the forms, and by Chapter 12 the forms extend to maps into $G^{-\infty}$ (with Tr in place of tr), every class having a finite rank representative. For later use we recall the transgression formula from Chapter 12: if a_t is a smooth family and $\varphi = a_t^{-1}\partial_t a_t$ then

$$\partial_t \operatorname{tr}(\theta_t^{2k+1}) = (2k+1)\, d \operatorname{tr}(\varphi\, \theta_t^{2k}). \tag{15.6.2}$$

The normalization (15.6.1) is chosen so that the Chern character of the index of a Toeplitz family is the push-forward of the odd Chern character of its symbol, as we now show.

Question. The constants c_k look arbitrary. Are they?

Answer. No. Each form $\operatorname{tr}((a^{-1}da)^{2k+1})$ is closed on its own, so closedness says nothing about the relative weights. They are fixed by requiring that the odd character be compatible with the even one. There are two independent ways to express this requirement, and both give the same c_k: the Toeplitz index (Proposition 15.6.1), and clutching (Proposition 15.7.5), where $\mathrm{Ch}_{\mathrm{odd}}(g)$ is the integral over $\mathbb{R}$ of the even Chern character of the bundle clutched by g. The factorials come from the Beta integral $\int_0^1 s^k(1-s)^k ds$. □

15.6.2 The Chern character of the Toeplitz index

Recall from Chapters 12 and 13 that a smooth family of elliptic Toeplitz operators $A \in \mathcal{C}^\infty(B; \Psi^0_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^N))$, acting on the Hardy space $\mathcal{C}^\infty_+(\mathbb{S}; \mathbb{C}^N)$, has symbol $a = \sigma(A) \in \mathcal{C}^\infty(B \times \mathbb{S}; \mathrm{GL}(N, \mathbb{C}))$, and that the families index gives an isomorphism

$$\mathrm{ind}_{\mathcal{T}} : K^{-2}(B) \longrightarrow K^0(B)$$

on homotopy classes of such symbols normalized by $a(b, 0) = \mathrm{Id}$. Our sign convention is that of Chapter 12: the lowering operator $L = T_{e^{-i\theta}}$ has index 1. For a form $\mu = d\theta \wedge \mu_1 + \mu_2$ on $B \times \mathbb{S}$, with μ_1, μ_2 not containing $d\theta$, set $\int_{\mathbb{S}} \mu = \int_0^{2\pi} \mu_1 \, d\theta$; this commutes with d.

Proposition 15.6.1. *For any $a \in \mathcal{C}^\infty(B \times \mathbb{S}; \mathrm{GL}(N, \mathbb{C}))$ with $a(\cdot, 0) = \mathrm{Id}$,*

$$\mathrm{Ch}(\mathrm{ind}_{\mathcal{T}}[a]) = \int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(a) = \sum_{k \geq 0} \frac{(-1)^k i^{k+1} k!}{(2\pi)^{k+1}(2k+1)!} \int_{\mathbb{S}} \mathrm{tr}((a^{-1}da)^{2k+1}) \in H^{\mathrm{ev}}(B; \mathbb{C}).$$

Proof. Both sides depend only on $[a] \in K^{-2}(B)$ and are additive: the right side by (15.6.2) and since $\int_{\mathbb{S}}$ commutes with d, the left side because $\mathrm{ind}_{\mathcal{T}}$ is a homomorphism and Ch is additive. Since $\mathrm{ind}_{\mathcal{T}}$ is injective (Chapter 13) and every class in $K^0(B)$ has the form $[E] - [F]$ with $E = \pi_E \mathbb{C}^{N_1}$, $F = \pi_F \mathbb{C}^{N_2}$ for families of projections, it suffices to treat

$$a_E = \pi_E e^{-i\theta} + (\mathrm{Id} - \pi_E), \qquad a_F = \pi_F e^{i\theta} + (\mathrm{Id} - \pi_F),$$

whose Toeplitz families have null bundle E (constants), resp. cokernel bundle F, so $\mathrm{ind}_{\mathcal{T}}[a_E] = [E]$, $\mathrm{ind}_{\mathcal{T}}[a_F] = -[F]$.

For $a = a_E$, write $\pi = \pi_E$, $z = e^{i\theta}$ and $Q = \mathrm{Id} + (z-1)\pi$, so that $a^{-1} = Q$ and

$$a^{-1}da = -i\pi \, d\theta + \beta, \qquad \beta = (\bar{z} - 1) Q \, d\pi.$$

The form β is off-diagonal with respect to $\mathbb{C}^{N_1} = \mathrm{Ran}\,\pi \oplus \mathrm{Ran}(\mathrm{Id} - \pi)$, because $\pi \, d\pi \, \pi = 0 = (\mathrm{Id} - \pi) d\pi (\mathrm{Id} - \pi)$, and $\pi Q = z\pi$, $Q(\mathrm{Id} - \pi) = \mathrm{Id} - \pi$. Hence

$$\pi \beta^2 \pi = (\bar{z} - 1)^2 z \, \pi (d\pi)^2 \pi = -(2 - z - \bar{z}) \, \pi (d\pi)^2 \pi, \qquad \pi \beta^{2k} = (-1)^k (2 - 2\cos\theta)^k \pi (d\pi)^{2k}.$$

In $\mathrm{tr}((a^{-1}da)^{2k+1})$ only terms with exactly one factor $-i\pi \, d\theta$ contribute to $\int_{\mathbb{S}}$. By (15.5.1) each of the $2k+1$ such terms can be cycled to the front without change of sign (the sign is $(-1)^{j(2k+1-j)} = 1$), so the relevant part is

$$-i(2k+1) \, d\theta \wedge \mathrm{tr}(\pi \beta^{2k}) = -i(2k+1)(-1)^k (2 - 2\cos\theta)^k \, d\theta \wedge \mathrm{tr}\big(\pi (d\pi)^{2k}\big).$$

Since $\int_0^{2\pi} (2 - 2\cos\theta)^k d\theta = 4^k \int_0^{2\pi} \sin^{2k}(\theta/2) \, d\theta = 2\pi \frac{(2k)!}{(k!)^2}$,

$$c_k \int_{\mathbb{S}} \mathrm{tr}((a^{-1}da)^{2k+1}) = c_k (-i)(-1)^k 2\pi \frac{(2k+1)!}{(k!)^2} \, \mathrm{tr}(\pi (d\pi)^{2k}) = \frac{1}{k!} \Big(\frac{i}{2\pi}\Big)^k \mathrm{tr}(\pi (d\pi)^{2k}),$$

which is the degree $2k$ part of $\mathrm{Ch}(E)$ by (15.5.2). Finally $a_F(\theta) = a_{E'}(-\theta)$ with $E' = F$, and the reflection $\theta \mapsto -\theta$ reverses the orientation of $\mathbb{S}$, so $\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(a_F) = -\mathrm{Ch}(F)$. □

Corollary 15.6.2. *If $A \in \mathcal{C}^\infty(B; \Psi^0_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^N))$ is an elliptic family of Toeplitz operators with symbol a, the Chern character of its index bundle is $\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(a)$.*

Proof. Let $g(b) = a(b,0)^{-1}$, independent of θ. Multiplying A by g, an invertible bundle map over B, does not change the index bundle, and ga is normalized. By the additivity of the odd Chern forms under pointwise products (Chapter 12; see also Exercise 15.4), $\mathrm{Ch}_{\mathrm{odd}}(ga) - \mathrm{Ch}_{\mathrm{odd}}(g) - \mathrm{Ch}_{\mathrm{odd}}(a)$ is exact, and $\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(g) = 0$ because $\mathrm{Ch}_{\mathrm{odd}}(g)$ contains no $d\theta$. Since the families index depends only on the symbol, Proposition 15.6.1 applies to ga. □

Remark 15.6.3. For $k = 0$ the formula reads $\operatorname{rank} \operatorname{ind}_{\mathcal{T}}(a) = \frac{i}{2\pi} \int_{\mathbb{S}} \operatorname{tr}(a^{-1}da) = -\frac{1}{2\pi i} \int \operatorname{tr}(a^{-1}da)$, which is Chapter 12's formula for the index of a single Toeplitz operator.

Remark 15.6.4. In Chapter 13 the same index map is also realized by isotropic operators on $\mathbb{R}$, whose symbols are functions on the circle at infinity of the phase plane, traversed counterclockwise in the (x,ξ)-plane. The isotropic index of a loop a is the Toeplitz index of $a \circ \rho$, $\rho(\theta) = -\theta$, and ρ reverses the orientation of $\mathbb{S}$, so

$$\mathrm{Ch}(\mathrm{ind}_{\mathrm{iso}}[a]) = -\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(a),$$

whose degree zero part is $-\frac{i}{2\pi} \int \operatorname{tr}(a^{-1}\partial_\theta a)\, d\theta$, the winding number of $\det a$: this is the index formula for isotropic operators on $\mathbb{R}$; for the symbol $z = e^{i\theta}$ of the normalized annihilation operator it gives 1.

15.7 The relative Chern character

The cohomological index formula involves the Chern character of the symbol, a compactly supported class on $T^*(M/B)$. We define the Chern character on $K_{\mathrm{c}}(\operatorname{int} X)$, for X a compact manifold with boundary, by explicit forms. This covers $T^*(M/B)$, the interior of its radial compactification, and $\mathbb{R}^{2N} \times B$.

15.7.1 Relative de Rham cohomology

Let $\iota : \partial X \hookrightarrow X$ be the inclusion. The compactly supported de Rham cohomology $H^*_{\mathrm{c}}(\operatorname{int} X)$ is isomorphic to the relative cohomology $H^*(X, \partial X)$. The latter is the cohomology of the complex of pairs

$$(\mu,\nu) \in \mathcal{C}^\infty(X;\Lambda^k) \oplus \mathcal{C}^\infty(\partial X;\Lambda^{k-1}), \quad D(\mu,\nu) = (d\mu, \iota^*\mu - d\nu),$$

and the isomorphism is induced by $\mu \mapsto (\mu, 0)$ for compactly supported μ. A closed pair is (μ,ν) with $d\mu = 0$ and $\iota^*\mu = d\nu$. The pairing with absolute classes,

$$\langle [(\mu,\nu)], [\varphi] \rangle = \int_X \mu \wedge \varphi - \int_{\partial X} \nu \wedge \iota^*\varphi, \quad d\varphi = 0, \tag{15.7.1}$$

is well defined (by Stokes' theorem, ∂X carrying the boundary orientation), and for $\nu = 0$ it is $\int_X \mu \wedge \varphi$.

15.7.2 The Chern form of a triple

Let (E_+, E_-, a) be a K_{c}-triple on $\operatorname{int} X$. As in the proof of Theorem 15.3.1 we may assume that $E_\pm$ and a extend smoothly to X and that a is invertible on a neighbourhood of ∂X. Choose connections $\nabla_\pm$ on $E_\pm$, with curvatures $\omega_\pm$, and let

$$\nabla a = \nabla_- \circ a - a \circ \nabla_+, \qquad \gamma = a^{-1}\nabla a,$$

an $\mathrm{End}(E_+)$-valued 1-form near ∂X. Choose $\rho \in \mathcal{C}^\infty(X)$ with $1-\rho$ supported in the set where a is invertible and $\rho = 0$ near ∂X. Since $a^{-1}\nabla_- a = \nabla_+ + \gamma$,

$$\nabla^\rho = \rho\nabla_+ + (1-\rho)a^{-1}\nabla_- a = \nabla_+ + (1-\rho)\gamma \tag{15.7.2}$$

is a connection on E_+ which equals $a^{-1}\nabla_- a$ near ∂X. Set $c = i/2\pi$ and

$$\lambda_\rho = \operatorname{tr} e^{c\,\omega_{\nabla^\rho}} - \operatorname{tr} e^{c\,\omega_-}. \tag{15.7.3}$$

It is closed, and vanishes near ∂X since there $\omega_{\nabla^\rho} = a^{-1}\omega_- a$.

Proposition 15.7.1. *The class* $\mathrm{Ch}(E_+, E_-, a) = [\lambda_\rho] \in H^{\mathrm{ev}}_{\mathrm{c}}(\operatorname{int} X)$ *depends only on the class of the triple in* $K_{\mathrm{c}}(\operatorname{int} X)$*, and defines a homomorphism*

$$\mathrm{Ch} : K_{\mathrm{c}}(\operatorname{int} X) \longrightarrow H^{\mathrm{ev}}_{\mathrm{c}}(\operatorname{int} X) = H^{\mathrm{ev}}(X, \partial X).$$

If X is closed this is the Chern character of Definition 15.5.4.

Proof. All variations are handled by Proposition 15.5.3(ii). Changing ρ (or ∇_+ with ∇_-, a fixed) changes the connection on E_+ by a θ supported away from ∂X, so the transgression form is compactly supported. Changing ∇_- to ∇_-^t changes λ_ρ by $d\big(\mathrm{cs}(\nabla^\rho, \nabla_t^\rho) - \mathrm{cs}(\nabla_-, \nabla_-^t)\big)$; near ∂X the two Chern–Simons forms are equal because the connections on E_+ there are $a^{-1}\nabla_-^t a$ and the trace is conjugation invariant. For a homotopy a_t the transgression form of ∇_t^ρ near ∂X is $c\int \operatorname{tr}(\theta_t e^{c\omega_t})dt$ with $\theta_t = \partial_t(a_t^{-1}\nabla_- a_t) = \nabla_t^\rho(\varphi_t)$, $\varphi_t = a_t^{-1}\partial_t a_t$; so there it equals $d\big(c\int \operatorname{tr}(\varphi_t e^{c\omega_t})dt\big)$, and subtracting d of χ times this, with $\chi = 1$ near ∂X and supported where $\rho = 0$, gives a compactly supported transgression. Bundle isomorphisms and stabilization by (H, H, Id) obviously do not change the class, and direct sums add. If $\partial X = \emptyset$ we may take $\rho = 1$. □

15.7.3 An explicit representative

The class can be represented without the cut-off ρ. On E_+ consider the linear path of connections from ∇_+ to $a^{-1}\nabla_- a$ near ∂X:

$$\nabla_+ + s\gamma, \quad s \in [0,1], \qquad \text{with curvature} \qquad \Omega(s) = (1-s)\omega_+ + s\,a^{-1}\omega_- a - s(1-s)\gamma^2.$$

To check the curvature, $(\nabla_+ + s\gamma)^2 = \omega_+ + s\nabla_+\gamma + s^2\gamma^2$, and at $s=1$ this is $a^{-1}\omega_- a$, so $\nabla_+\gamma = a^{-1}\omega_- a - \omega_+ - \gamma^2$.

Theorem 15.7.2 (Fedosov-type formula)**.** *The relative class* $\mathrm{Ch}(E_+, E_-, a) \in H^{\mathrm{ev}}(X, \partial X)$ *is represented by the closed pair*

$$\Big(\operatorname{tr} e^{c\omega_+} - \operatorname{tr} e^{c\omega_-},\ -c\,\iota^* \int_0^1 \operatorname{tr}\big(\gamma\, e^{c\Omega(s)}\big)ds\Big), \qquad c = \frac{i}{2\pi}.$$

Equivalently, for every closed form φ on X,

$$\int_X \lambda_\rho \wedge \varphi = \int_X \big(\operatorname{tr} e^{c\omega_+} - \operatorname{tr} e^{c\omega_-}\big) \wedge \varphi + c\int_{\partial X} \iota^*\Big(\int_0^1 \operatorname{tr}\big(a^{-1}(\nabla a)e^{c\Omega(s)}\big)ds \wedge \varphi\Big).$$

Proof. Let $\mu = c\int_0^1 \operatorname{tr}\big((1-\rho)\gamma\, e^{c\omega_t}\big)dt$, where ω_t is the curvature of $\nabla_+ + t(1-\rho)\gamma$. By Proposition 15.5.3(ii), $d\mu = \operatorname{tr} e^{c\omega_{\nabla^\rho}} - \operatorname{tr} e^{c\omega_+}$, which is λ_ρ minus the first entry of the pair. Near ∂X, $\rho = 0$ and $\omega_t = \Omega(t)$, so $\iota^*\mu$ is minus the second entry. Hence

$$(\lambda_\rho, 0) - \Big(\operatorname{tr} e^{c\omega_+} - \operatorname{tr} e^{c\omega_-}, -\iota^*\mu\Big) = (d\mu, \iota^*\mu) = D(\mu, 0).$$

Thus the pair is closed and cohomologous to $(\lambda_\rho, 0)$. The second statement is (15.7.1). □

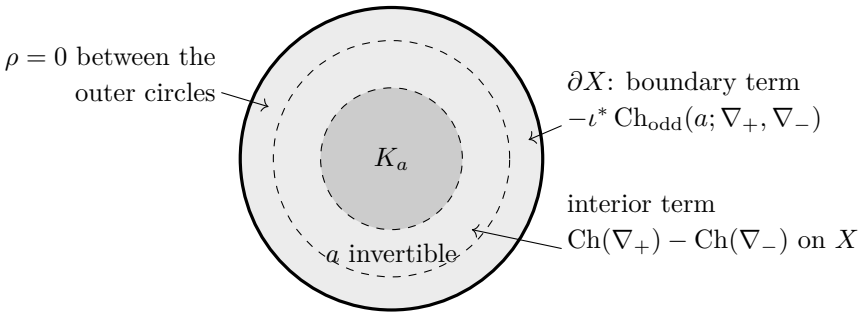


Figure 15.4: A fibre of the radially compactified cotangent bundle X. The symbol a fails to be invertible only in the compact set K_a. The representative λ_ρ of (15.7.3) is a compactly supported form, obtained by bending the connection on E_+ into $a^{-1}\nabla_- a$ before reaching ∂X. The Fedosov-type formula (Theorem 15.7.2) instead keeps the two given connections up to the boundary and pays for it with an odd Chern–Simons form of the symbol on $\partial X = S^*(M/B)$.

Discussion. The two representatives in Figure 15.4 correspond to the two descriptions of $H^*_{\mathrm{c}}(\operatorname{int} X) = H^*(X, \partial X)$: a compactly supported closed form, or a closed form on X together with a primitive of its restriction to ∂X. For the symbol of an elliptic family the second is more natural, because the symbol is given at infinity, that is on ∂X, and the boundary term is then built from $a^{-1}\nabla a$ alone. When the bundles are trivial with trivial connections, the interior term vanishes and the whole class is carried by the odd Chern form of the symbol on the cosphere bundle; this is the form of the index formula used in Chapter 17.

Remark 15.7.3. The formula can be obtained by letting ρ tend to the characteristic function of X. Let x be a boundary defining function and $\rho = \chi(x/\epsilon)$ with $\chi \in \mathcal{C}^\infty(\mathbb{R})$, $\chi = 0$ on $(-\infty, \frac{1}{2}]$, $\chi = 1$ on $[1, \infty)$. From (15.7.2),

$$\omega_{\nabla^\rho} = \Omega(1-\rho) - d\rho \wedge \gamma,$$

and since $(d\rho)^2 = 0$, $\operatorname{tr} e^{c\omega_{\nabla^\rho}} = \operatorname{tr} e^{c\Omega(1-\rho)} - c\, d\rho \wedge \operatorname{tr}(\gamma e^{c\Omega(1-\rho)})$. As $\epsilon \to 0$ the first term tends to $\operatorname{tr} e^{c\omega_+}$ boundedly almost everywhere. In the second, $d\rho = \epsilon^{-1}\chi'(x/\epsilon)dx$; replacing the coefficients by their boundary values (with an error that tends to zero) and substituting $s = 1 - \chi(x/\epsilon)$ shows that it converges, as a current, to a multiple of $\delta(x)dx$ times $\iota^* \int_0^1 \operatorname{tr}(\gamma e^{c\Omega(s)})ds$. The limit is the "conormal" representative $\mu + \delta(x)dx \wedge \nu$ of the relative class; such a current is closed (on forms smooth up to the boundary) exactly when $d\mu = 0$ and $\iota^*\mu = \pm d\nu$, which is the closedness of the pair in Theorem 15.7.2. The proof given above avoids distributions. Note the analogy with Proposition 15.6.1, where the index class is also given by an integral of the odd forms of the symbol over a boundary circle.

For an elliptic family $A \in \Psi^0(M/B; E_+, E_-)$ we apply this with X the radial compactification of $T^*(M/B)$ (a ball bundle), $\partial X = S^*(M/B)$, the lifted bundles with pulled-back connections, and $a = \sigma_0(A)$ homogeneous. This defines

$$\operatorname{Ch}(\sigma_0(A)) \in H^{\mathrm{ev}}_{\mathrm{c}}(T^*(M/B); \mathbb{C}),$$

represented by the pair of Theorem 15.7.2, whose second entry lives on $S^*(M/B)$ and is built from $\sigma_0(A)^{-1}\nabla\sigma_0(A)$.

15.7.4 The odd Chern character as a suspension

The constants c_k in (15.6.1) were fixed by the Toeplitz index. They are also forced by the even Chern character, through the clutching construction: the odd Chern character of g is the fibre integral of the Chern character of the compactly supported class on $\mathbb{R}\times X$ obtained by clutching with g. We use the integral form of $\mathrm{Ch}_{\mathrm{odd}}$.

Lemma 15.7.4. *For $g\in\mathcal{C}^\infty(X;\mathrm{GL}(N,\mathbb{C}))$, with $\theta=g^{-1}dg$ and $c=i/2\pi$,*

$$\mathrm{Ch}_{\mathrm{odd}}(g)=c\int_0^1\mathrm{tr}\Big(\theta\,\exp\big(c\,s(s-1)\,\theta^2\big)\Big)ds.$$

Proof. The term of degree $2k+1$ on the right is $\frac{c^{k+1}}{k!}\int_0^1(s^2-s)^k ds\ \mathrm{tr}(\theta^{2k+1})$, and $\int_0^1(s^2-s)^k ds=(-1)^k\int_0^1 s^k(1-s)^k ds=(-1)^k\frac{(k!)^2}{(2k+1)!}$. So the coefficient is $c(-c)^k\frac{k!}{(2k+1)!}=\frac{i}{2\pi}\big(\frac{-i}{2\pi}\big)^k\frac{k!}{(2k+1)!}=c_k$. □

For compact X and $g\in\mathcal{C}^\infty(X;\mathrm{GL}(N,\mathbb{C}))$ let

$$\mathrm{clu}[g]=[(\mathbb{C}^N,\mathbb{C}^N,\tilde g)]\in K_{\mathrm{c}}(\mathbb{R}\times X),\qquad \tilde g(t,x)=\mathrm{Id}\ (t\le -1),\quad \tilde g(t,x)=g(x)\ (t\ge 1),$$

the clutching isomorphism $K^{-1}(X)\cong K_{\mathrm{c}}(\mathbb{R}\times X)$ of Chapter 13. We regard $\mathbb{R}\times X$ as the interior of a compact manifold with boundary by compactifying $\mathbb{R}$ to a closed interval, so that Proposition 15.7.1 applies, and for a compactly supported form $\mu=dt\wedge\mu_1+\mu_2$ on $\mathbb{R}\times X$ (μ_1, μ_2 without dt) set $\int_{\mathbb{R}}\mu=\int_{\mathbb{R}}\mu_1\,dt$; this commutes with d, as does $\int_{\mathbb{S}}$ in Section 15.6.

Proposition 15.7.5. *For compact X and $g\in\mathcal{C}^\infty(X;\mathrm{GL}(N,\mathbb{C}))$,*

$$\int_{\mathbb{R}}\mathrm{Ch}(\mathrm{clu}[g])=\mathrm{Ch}_{\mathrm{odd}}(g)\in H^{\mathrm{odd}}(X;\mathbb{C}).$$

In fact, for the trivial connections and a suitable cut-off, the representative λ_ρ of (15.7.3) satisfies $\int_{\mathbb{R}}\lambda_\rho=\mathrm{Ch}_{\mathrm{odd}}(g)$ as forms.

Proof. Take $\nabla_\pm=d$ on $E_\pm=\mathbb{C}^N$. Where $\tilde g$ is invertible, $\gamma=\tilde g^{-1}d\tilde g$ is 0 for $t\le -1$ and $\theta=g^{-1}dg$ for $t\ge 1$. Choose $\chi\in\mathcal{C}^\infty(\mathbb{R})$ with $\chi=0$ on $(-\infty,1]$, $\chi=1$ on $[2,\infty)$, and ρ with $\rho=1-\chi$ on $[0,\infty)$, $\rho=1$ on $[-1,0]$ and $\rho=0$ near $-\infty$; since $\gamma=0$ for $t\le -1$, (15.7.2) is $\nabla^\rho=d+\chi(t)\theta$. Its curvature is, using $d\theta=-\theta^2$,

$$\omega=d(\chi\theta)+\chi^2\theta^2=d\chi\wedge\theta+(\chi^2-\chi)\theta^2.$$

The two terms commute (the second is an even form) and $(d\chi\wedge\theta)^2=0$, so $\mathrm{tr}\,e^{c\omega}=\mathrm{tr}\,e^{c(\chi^2-\chi)\theta^2}+c\,d\chi\wedge\mathrm{tr}\big(\theta e^{c(\chi^2-\chi)\theta^2}\big)$. Hence $\lambda_\rho=\mathrm{tr}\,e^{c\omega}-N$ is supported in $1\le t\le 2$, and

$$\int_{\mathbb{R}}\lambda_\rho=c\int_{\mathbb{R}}\chi'(t)\,\mathrm{tr}\big(\theta e^{c(\chi^2-\chi)\theta^2}\big)dt=c\int_0^1\mathrm{tr}\big(\theta e^{c(s^2-s)\theta^2}\big)ds=\mathrm{Ch}_{\mathrm{odd}}(g)$$

by the substitution $s=\chi(t)$ and Lemma 15.7.4. □

Remark 15.7.6. More generally, for a bundle isomorphism $a\colon F\to F'$ and connections on F, F', the form

$$\mathrm{Ch}_{\mathrm{odd}}(a;\nabla_F,\nabla_{F'})=c\int_0^1\mathrm{tr}\big(\gamma\,e^{c\Omega(s)}\big)ds,$$
$$\gamma=a^{-1}\nabla a,\qquad \Omega(s)=(1-s)\omega_F+s\,a^{-1}\omega_{F'}a+s(s-1)\gamma^2,$$

reduces to Lemma 15.7.4 for trivial bundles and connections. It is the Chern–Simons form of the path $\nabla_F + s\gamma$ from ∇_F to $a^{-1}\nabla_{F'}a$, so $d\,\mathrm{Ch}_{\mathrm{odd}}(a;\nabla_F,\nabla_{F'}) = \mathrm{Ch}(\nabla_{F'}) - \mathrm{Ch}(\nabla_F)$, and Theorem 15.7.2 says that the relative Chern character of (E_+, E_-, a) is represented by the pair $(\mathrm{Ch}(\nabla_+) - \mathrm{Ch}(\nabla_-), -\iota^*\,\mathrm{Ch}_{\mathrm{odd}}(a;\nabla_+,\nabla_-))$: the boundary term is the odd Chern character of the symbol. The proof of Proposition 15.7.5 is the special case in which the second entry is moved into the interior.

15.8 The cohomological form of the index theorem

Orient the fibres $T^*_b(Z)$ of $\pi : T^*(M/B) \to B$, where π is the composite of the bundle projection and ϕ, by $dz_1 \wedge d\zeta_1 \wedge \cdots \wedge dz_n \wedge d\zeta_n$ in canonical coordinates (z coordinates on the fibre Z_b, ζ the dual coordinates). This is the n-th power of the canonical symplectic form (up to a universal sign), so it is well defined even if Z is not orientable. Integration over the fibres is then a map $\pi_* : H^*_{\mathrm{c}}(T^*(M/B)) \to H^{*-2n}(B)$. Let Td denote the Todd class, the multiplicative characteristic class with $\mathrm{Td}(L) = c_1(L)/(1 - e^{-c_1(L)})$ for a line bundle L.

Theorem 15.8.1 (Atiyah–Singer, cohomological form)**.** *For every $x \in K_{\mathrm{c}}(T^*(M/B))$, with n the dimension of the fibres of ϕ,*

$$\mathrm{Ch}(\mathrm{ind}_{\mathrm{a}}(x)) = (-1)^n \pi_*\big(\mathrm{Ch}(x) \wedge \pi_0^*\,\mathrm{Td}(T(M/B)\otimes\mathbb{C})\big) \in H^{\mathrm{ev}}(B;\mathbb{C}),$$

$\pi_0 : T^*(M/B) \to M$ the projection. In particular, for an elliptic family A, $\mathrm{Ch}(\mathrm{ind}\,A) = (-1)^n \int_{T^*(M/B)/B} \mathrm{Ch}(\sigma_0(A))\,\mathrm{Td}(T(M/B)\otimes\mathbb{C})$.

This is [18], Theorem 5.1. Since the sign depends on the orientation convention, it is often absorbed into it, as in the form $\mathrm{Ch}(\mathrm{ind}\,A) = \int \mathrm{Td}\cdot\mathrm{Ch}(\sigma(A))$. We explain how the theorem follows from Theorem 15.4.4; the topological inputs (C1) and (C4) below are not proved here.

Discussion. Where does the Todd class come from? Apply Ch to each arrow of Figure 15.1. The extension $j_*\Theta_*$ is harmless, and β_N^{-1} becomes integration over $\mathbb{R}^{2N}$ up to a sign. The Thom map τ multiplies by $\lambda_{\mathcal{V}}$, and in cohomology this becomes "integrate over the normal directions" only up to a correction factor: the fibre integral of $\mathrm{Ch}(\lambda_{\mathcal{V}})$ is not ±1 but $\pm\,\mathrm{Td}(\mathcal{V}\otimes\mathbb{C})^{-1}$. Since $\mathcal{V}$ is a complement of $T(M/B)$, this correction is $\mathrm{Td}(T(M/B)\otimes\mathbb{C})$. In other words, the Todd class measures the failure of the Chern character to commute with the Thom isomorphism (Remark 15.8.3).

Derivation from Theorem 15.4.4. We use the following properties of the Chern character of Proposition 15.7.1.

(C1) Ch is multiplicative for the products of Definition 15.1.2: $\mathrm{Ch}(x_1\cdot x_2) = \mathrm{Ch}(x_1)\wedge\mathrm{Ch}(x_2)$. (This is standard; on a compact space it is Proposition 15.5.3(iii), and the compactly supported case follows by passing to one-point compactifications.)

(C2) Ch commutes with extensions j_* for open embeddings (extension of compactly supported forms by zero) and with pull-backs, since the forms λ_ρ are compactly supported and natural.

(C3) $\int_{T^*I} \mathrm{Ch}(b_1) = -1$ for the orientation $d\theta \wedge d\xi$ (Exercise 15.3). Hence by (C1), (C2), $\mathrm{Ch}(\beta_N^{-1}z) = (-1)^N\pi_*\,\mathrm{Ch}(z)$ for $z \in K_{\mathrm{c}}(T^*I^N \times B)$, with the orientation above on T^*I^N.

(C4) For a real vector bundle $\mathcal{V}$ of rank k over M, integration over the fibres of $T^*(\mathcal{V}/M) \cong \mathcal{V}\otimes\mathbb{C}$ gives $\varpi_*\operatorname{Ch}(\lambda_{\mathcal{V}}) = (-1)^k\operatorname{Td}(\mathcal{V}\otimes\mathbb{C})^{-1}$. For trivial $\mathcal{V}$ this is (C3) with $B = M$ and Remark 15.4.3; in general it is the classical computation of Atiyah and Hirzebruch (see the Notes).

(C5) Td is multiplicative and $T(M/B)\oplus\mathcal{V}$ is trivial, so $\operatorname{Td}(\mathcal{V}\otimes\mathbb{C})^{-1} = \operatorname{Td}(T(M/B)\otimes\mathbb{C})$.

Then, with $x \in K_{\mathrm{c}}(T^*(M/B))$ and the orientations multiplying as they should (all of them are products of oriented two-planes),

$$\begin{aligned}\operatorname{Ch}(\operatorname{ind}_{\mathrm{a}} x) &= \operatorname{Ch}(\beta_N^{-1} j_*\Theta_*\tau(x)) = (-1)^N\pi_*\operatorname{Ch}(\lambda_{\mathcal{V}}\cdot\varpi^*x)\\ &= (-1)^N\pi_*(\varpi_*\operatorname{Ch}(\lambda_{\mathcal{V}})\wedge\operatorname{Ch}(x)) = (-1)^{N+k}\pi_*(\operatorname{Td}(T(M/B)\otimes\mathbb{C})\wedge\operatorname{Ch}(x)),\end{aligned}$$

and $(-1)^{N+k} = (-1)^n$ since $N = n+k$. □

Remark 15.8.2. Even without (C4) the derivation shows that $\operatorname{Ch}(\operatorname{ind}_{\mathrm{a}} x) = \pi_*(\operatorname{Ch}(x)\wedge\pi_0^*\mathcal{T})$ for a class $\mathcal{T}\in H^{\mathrm{ev}}(M;\mathbb{C})$ which depends only on the fibration (through its normal bundle), not on x. The content of (C4) is the identification of $\mathcal{T}$ with $(-1)^n\operatorname{Td}(T(M/B)\otimes\mathbb{C})$. For B a point and $n = 1$, $M = \mathbb{S}$, the theorem reduces to $\operatorname{ind}(P_1) = 1 = -\int_{T^*\mathbb{S}}\operatorname{Ch}(\sigma_0(P_1))$, which is Exercise 15.3.

Remark 15.8.3 (The Todd class and the Thom isomorphism). The class $\mathcal{T}$ can be described through the Thom isomorphism of Chapter 13. Let $V\to M$ be a Hermitian complex vector bundle of rank n, $\varpi\colon V\to M$, with Bott class $\beta_V\in K_{\mathrm{c}}(V)$ (for $V = \mathcal{V}\otimes\mathbb{C}$ this is $\lambda_{\mathcal{V}}$), and orient the fibres as complex vector spaces. By Chapter 13, $x\mapsto\beta_V\cdot\varpi^*x$ is an isomorphism $K^0(M)\to K_{\mathrm{c}}(V)$, with inverse Thom_V. By (C1) and the projection formula $\varpi_*(\varpi^*\alpha\wedge\mu) = \alpha\wedge\varpi_*\mu$ for even α,

$$\varpi_*\operatorname{Ch}(\beta_V\cdot\varpi^*x) = \operatorname{Ch}(x)\wedge\varpi_*\operatorname{Ch}(\beta_V).$$

Restricted to a point, β_V is the n-fold product of the class $b_1 = \lambda_{\mathbb{R}}$ of Remark 15.4.3, so by (C1), (C3) the degree zero part of $\varpi_*\operatorname{Ch}(\beta_V)$ is $(-1)^n$, and $\mathcal{T}(V) = (-1)^n(\varpi_*\operatorname{Ch}(\beta_V))^{-1}\in H^{\mathrm{ev}}(M;\mathbb{C})$ is defined. Thus

$$\operatorname{Ch}(\operatorname{Thom}_V(y)) = (-1)^n\varpi_*\operatorname{Ch}(y)\wedge\mathcal{T}(V),\qquad y\in K_{\mathrm{c}}(V).$$

Since $\beta_{V\oplus W} = \beta_V\cdot\beta_W$ and β is natural, $\mathcal{T}$ is a multiplicative characteristic class with $\mathcal{T}(\mathbb{C}^n) = 1$; if $V\oplus W$ is trivial of rank m, then $\mathcal{T}(V) = (-1)^{\operatorname{rank} W}\varpi_*\operatorname{Ch}(\beta_W)$, which is the definition of the Todd class as the fibre integral of the Chern character of the Bott element of a complement used in the source. The identification of $\mathcal{T}(\mathcal{V}\otimes\mathbb{C})$ with $\operatorname{Td}(\mathcal{V}\otimes\mathbb{C})$ is (C4).

15.9 Exercises

Exercise 15.1. Show directly that the exterior product of Definition 15.1.2 is well defined on K_{c}-classes and bilinear, and that $x\boxtimes y$ corresponds to $y\boxtimes x$ under the flip $X_1\times X_2\cong X_2\times X_1$. (For the latter, find a homotopy between $a_1\#a_2$ and the conjugate of $a_2\#a_1$ by the obvious bundle isomorphisms.)

Exercise 15.2. Complete Example 15.5.5: show that for the tautological line bundle over $\mathbb{C}P^1$, $\frac{i}{2\pi}\operatorname{tr}(\pi(d\pi)^2) = -\frac{1}{\pi}(1+|w|^2)^{-2}dx\wedge dy$ in the affine coordinate $w = x+iy$, so that its integral is -1.

Exercise 15.3. Let $X \subset T^*I$ be a large square $[\epsilon, 2\pi-\epsilon]\times[-R,R]$ in the coordinates (θ,ξ), oriented by $d\theta\wedge d\xi$, and represent b_1 by the trivial bundles with the trivial connection and the symbol of (15.4.2) modified as in the definition of b_1. Use Theorem 15.7.2 with $\varphi=1$ to show that $\int_{T^*I}\mathrm{Ch}(b_1)=\frac{i}{2\pi}\int_{\partial X}a^{-1}da=-1$.

Exercise 15.4. Show that for maps $a,a' : X\to \mathrm{GL}(N,\mathbb{C})$ the forms $\mathrm{Ch}_{\mathrm{odd}}(aa')$ and $\mathrm{Ch}_{\mathrm{odd}}(a)+\mathrm{Ch}_{\mathrm{odd}}(a')$ differ by an exact form. (Use the homotopy $\left(\begin{smallmatrix}a&0\\0&1\end{smallmatrix}\right)R_t\left(\begin{smallmatrix}a'&0\\0&1\end{smallmatrix}\right)R_t^{-1}$ with R_t a rotation from Id to $\left(\begin{smallmatrix}0&1\\-1&0\end{smallmatrix}\right)$, and (15.6.2).)

Exercise 15.5. Prove the base change property Remark 15.1.4(d) in detail, and deduce that if $\mathrm{ind_a}(b)=1$ in $K^0(M)$ for $b\in K_{\mathrm{c}}(T^*(\tilde{M}/M))$ then the restriction of b to the fibre over any $m\in M$ has numerical index 1.

Exercise 15.6. Show that $K_{\mathrm{c}}(\mathbb{R}^2)\to\mathbb{Z}$, taking the class of a triple $(\mathbb{C}^r,\mathbb{C}^r,a)$ with a invertible outside a disc to the winding number of $\det a$ on a large circle, is well defined, and that it maps b_1 to 1. Deduce from Proposition 15.4.2 with $N=1$ and B a point that the index of a symbol in $K_{\mathrm{c}}(T^*I)$, extended to the circle, equals this winding number.

Notes

This chapter follows Lectures 19–21 of the 2005 course, with several changes. The proofs of multiplicativity and excision were completed: the invertibility of the symbols along the deformation is now deduced from the single identity of Lemma 15.1.1; the families index of product-type families and the comparison of the stabilization by sections with the construction of Chapter 14 (left to the addenda in the original) are in Lemma 15.1.3; the hypothesis of a trivialized null line bundle is reduced to $\mathrm{ind_a}(b)=1$ in Lemma 15.2.2. The normalization on the trivial fibration is proved via multiplicativity on the torus. In the original the curvature of the Grassmann connection carries a sign error, which propagated to the constants in the formula for the Chern character of a Toeplitz index; the corrected constants are in (15.6.1) (we checked the case $k=1$ also numerically). The original section on the Bott element was empty; here its role is played by P_1, the normalization proposition and the imported statement (TH).

The families index theorem is due to Atiyah and Singer [18]; the K-theoretic method of proof, by embedding, excision and multiplicativity, goes back to M. F. Atiyah and I. M. Singer, *The index of elliptic operators I*, Ann. of Math. **87** (1968), 484–530, where the equivariant index of the Bott operator needed for (TH) is computed. For K-theory, products, the Thom isomorphism and Bott periodicity see [17]. The Riemann–Roch identity (C4) and the theorem that the Chern character is a rational isomorphism are due to M. F. Atiyah and F. Hirzebruch, *Vector bundles and homogeneous spaces*, Proc. Sympos. Pure Math. **3** (1961), 7–38. The explicit formula of Theorem 15.7.2 is of the type used by B. V. Fedosov in his work on index formulas.

The material on Chern forms, the odd Chern character, and the Todd class in the K-theory chapter of the 2008 revision of the lectures (*From microlocal to global analysis*) has been merged here. There the odd Chern character is derived from a connection on the bundle over $\mathbb{S}\times X$ clutched by g, which yields the integral formula of Lemma 15.7.4; the source leaves its constants open (and, with an unnormalized even Chern character $\mathrm{tr}\,e^{\omega}$, warns of possible sign errors). With the normalization $\mathrm{tr}\,e^{i\omega/2\pi}$ the clutching computation (Proposition 15.7.5) reproduces exactly the constants c_k of (15.6.1) that were fixed by the Toeplitz index, a non-trivial consistency check, and the source's general formula for an isomorphism of bundles with connections is the boundary term of Theorem 15.7.2 (Remark 15.7.6). The source states the Chern character of the isotropic index as $-\frac{1}{2\pi i}\int\mathrm{Ch}_{\mathrm{odd}}$ for unnormalized forms

and its own orientation of the circle; in our conventions it is Remark 15.6.4. The source defines the Todd class of E as the fibre integral of the Chern character of the Bott element of a complement; Remark 15.8.3 shows how this follows from the Thom isomorphism, with the signs made explicit.

Chapter 16

The index theorem via semiclassical and adiabatic limits

Quantum mechanics suggests a direct way to go from a symbol to an operator without losing information: put a Planck constant ϵ in front of the momentum. As $\epsilon \to 0$ the operator $\mathrm{Op}_\epsilon(q)$ "becomes" its symbol q: products of operators become products of symbols, and the Schwartz kernel collapses onto the diagonal. This chapter uses that limit to build an index map straight from compactly supported symbols, shows that it agrees with the analytic index, and uses the harmonic oscillator to compute the one ingredient of the families index theorem that Chapter 15 had to import.

In Chapter 15 the families index theorem of Atiyah and Singer was proved in K-theory by embedding a fibration $\phi : M \to B$ into a trivial fibration $\mathbb{R}^N \times B \to B$ and using three properties of the analytic index: multiplicativity for towers of fibrations, excision, and a normalization on the trivial fibration. One input, the index of the Thom class on a sphere bundle (statement (TH) there), was imported. This chapter gives a second proof, based on a different realization of the index map: the *semiclassical index*.

The idea is simple. A compactly supported K-class on $T^*(M/B)$ can be represented by data which are trivial outside a compact set; for idempotents, say, by a smooth family e of idempotent matrices which is equal to a constant idempotent p_0 outside a compact subset of $T^*(M/B)$. Such a symbol can be quantized, with a small parameter $\epsilon > 0$ (Planck's constant), to a family of idempotent operators on the fibres which differs from p_0 by a *semiclassical family of smoothing operators*. A pair of idempotents differing by a smoothing operator has a relative index, an element of $K^0(B)$, and this is the semiclassical index $\mathrm{Ind}_{\mathrm{sl}}$. As $\epsilon \downarrow 0$ the Schwartz kernels concentrate at the fibre diagonal, so the construction is local: excision for $\mathrm{Ind}_{\mathrm{sl}}$ is immediate. The main analytic result of the chapter (Theorem 16.5.4) is that $\mathrm{Ind}_{\mathrm{sl}}$ *coincides* with the analytic index of Chapters 14 and 15. The proof is short and algebraic: from an elliptic family and a parametrix one forms an explicit idempotent ((16.3.1)) whose relative index is the index bundle; if the family and the parametrix are themselves semiclassical quantizations, this idempotent is a semiclassical quantization of the corresponding symbol-level idempotent.

The second ingredient is the index of the Thom class. In the semiclassical picture this becomes a statement about the harmonic oscillator on the fibres of the normal bundle: after rescaling by $\epsilon^{1/2}$, the semiclassical quantization of the Thom symbol is the Koszul complex of the harmonic oscillator, whose null space is spanned by the Gaussian. Since the Gaussian is invariant under the orthogonal group, its span is a *canonically trivial* line bundle; no equivariant index computation is needed (Theorem 16.6.1). As a consequence, statement (TH) of Chapter 15 ((TH)) is proved, and the semiclassical normalization on the trivial

fibration follows as well.

Putting these together (Section 16.7) gives the families index theorem in the following form: excision and the normalization are replaced by their (essentially trivial) semiclassical versions together with Theorem 16.5.4, and (TH) by Theorem 16.6.1; the only property of the analytic index from Chapter 15 which is still used is multiplicativity (Theorem 15.2.4 and Corollary 15.2.5). In R. Melrose's treatment, on which this chapter is based, the multiplicativity of the semiclassical index is also proved semiclassically, by an adiabatic limit; we describe that argument in Section 16.8, without giving the full details of the underlying two-parameter calculus.

The odd version of the semiclassical index, $K_{\mathrm{c}}^{-1}(T^*(M/B)) \to K^{-1}(B)$, obtained by quantizing compactly supported perturbations of the identity, is the most elementary of these maps; we show in Section 16.4 that the even index is the odd one applied to loops, followed by the Toeplitz index isomorphism of Chapter 13 (Theorem 13.6.2). Throughout, the K-groups of the base are realized, as in Chapters 12 and 13, through the group $G^{-\infty}$ of invertible perturbations of the identity by smoothing operators. The required passage from families of operators on the varying fibres Z_b to maps into a fixed group $G^{-\infty}(Y;F)$ is made in Section 16.2 by an explicit isometric embedding of the fibres into a torus.

Discussion. The construction of the semiclassical index has four steps, each of which is elementary:

$$x \xmapsto{\text{Milnor}} e_{a,b} \xmapsto{\text{quantize}} \mathfrak{E}(\epsilon) \xmapsto{\text{compare}} d(\mathfrak{E}(\epsilon), p_0) \in K^0(B).$$

Here $x = [(E_+, E_-, a)]$, $e_{a,b}$ is an idempotent-valued symbol and $\mathfrak{E}(\epsilon)$ an idempotent family of operators. The first arrow is pure linear algebra (Lemma 16.3.1), the second is semiclassical quantization followed by a correction which makes the operator an exact idempotent (Proposition 16.3.2), and the third compares two infinite rank idempotents which differ by a smoothing operator (Definition 16.2.6). Only the second step involves analysis, and for small ϵ it is essentially local.

16.1 Setting and recollections

16.1.1 Fibrations

Throughout, B is a compact manifold, possibly with boundary (we need $B \times [0,1]$ for homotopies). A *fibration over* B in this chapter is a submersion $\phi : M \to B$ such that M is an open subset of the total space of a fibration $\bar\phi : \bar M \to B$ of compact manifolds in the sense of Chapter 14 (Definition 14.1.1), with $\phi = \bar\phi|_M$. The fibres $Z_b = \phi^{-1}(b)$ may thus be non-compact. The cases we need are:

- fibrations of compact manifolds ($M = \bar M$);
- open subsets $U \subset \bar M$;
- the total space $\mathcal{V}$ of a real vector bundle $\varpi : \mathcal{V} \to M$ over the total space of a fibration of compact manifolds, fibred over B by $\phi \circ \varpi$; it is an open subset of the sphere bundle $S(\mathcal{V} \oplus \mathbb{R})$, via inverse stereographic projection;
- the trivial fibrations $\mathbb{R}^N \times B$ and $I^N \times B$, $I = (0, 2\pi)$, inside $\mathbb{T}^N \times B$, $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$;
- the degenerate fibration $B \to B$, $\phi = \mathrm{Id}$, whose fibres are points.

We write $M^2_\phi = M \times_B M$ for the fibre product, $\mathrm{Diag} \subset M^2_\phi$ for the (fibre) diagonal, $T^*(M/B)$ for the vertical cotangent bundle and $\varpi : T^*(M/B) \to M$ for its projection. A Hermitian bundle $E \to M$ and a smooth positive fibre density ν are always fixed; they define the Hilbert spaces $L^2(Z_b; E_b)$. We write $\Psi_{\mathrm{c}}^{-\infty}(M/B; E)$ for the smoothing families with compactly supported kernels, $\mathcal{C}^\infty_{\mathrm{c}}(M^2_\phi; \mathrm{Hom}(E) \otimes \Omega_R)$, where Ω_R is the fibre density bundle lifted from the right factor (Chapter 14); for M compact this is $\Psi^{-\infty}(M/B; E)$.

Remark 16.1.1 (Adjoining a trivial bundle). We shall often need operators on $L^2(Z_b; E_b) \oplus \mathbb{C}^L$, for instance after stabilizing an elliptic family by finitely many sections as in Chapter 15. These are covered by the formalism: $M \sqcup B \to B$ is a fibration whose fibres are $Z_b \sqcup \{b\}$, and $\mathbb{C}^L$ is the space of sections of the trivial bundle of rank L over the point fibre $\{b\}$. Smoothing families on $M \sqcup B$ are block matrices whose off-diagonal entries are given by finitely many smooth sections of E and $E^* \otimes \Omega$, and whose lower right entry is a smooth map $B \to M(L, \mathbb{C})$. Pseudodifferential families on $(M \sqcup B)/B$ of order m are matrices $\left(\begin{smallmatrix} P & S \\ T & c \end{smallmatrix}\right)$ with $P \in \Psi^m(M/B; E)$ and S, T, c as just described; they compose as matrices and the smoothing families form an ideal. We use this without further comment.

16.1.2 K-theory

We use the conventions of Chapters 13 and 15. For a manifold X, $K_{\mathrm{c}}(X) = K^0_{\mathrm{c}}(X)$ is the group of equivalence classes of triples (E_+, E_-, a), with $a : E_+ \to E_-$ an isomorphism outside a compact set, under bundle isomorphisms, change of a on compact sets, homotopies and stabilization. If X is the total space of a vector bundle over M, bundles over X are pulled back from M (Lemma 13.2.2), so every class is represented by a triple $(\varpi^* E_+, \varpi^* E_-, a)$ with $E_\pm$ bundles over M and $a \in \mathcal{C}^\infty(X; \mathrm{Hom}(\varpi^* E_+, \varpi^* E_-))$ defined everywhere and invertible outside a compact set; we usually drop ϖ^*. We also use:

(K1) For a compact manifold Y of positive dimension (not necessarily connected) and a Hermitian bundle $F \to Y$, $G^{-\infty}(Y; F) \subset \mathrm{Id} + \Psi^{-\infty}(Y; F)$ is the group of invertible perturbations of the identity by smoothing operators, and for compact X

$$K^{-1}(X) = [X; G^{-\infty}(Y; F)], \qquad K^{-2}(X) = [X \times \mathbb{S}, X \times \{1\}; G^{-\infty}(Y; F), \mathrm{Id}],$$

smooth homotopy classes of smooth maps, respectively of pointed maps ((B3) and Definition 13.2.4). By the finite rank approximation (B2) and Lemma 13.2.5, every class is represented by a map into a subgroup $\mathrm{GL}(V) = \{\mathrm{Id} - \Pi_V + g\Pi_V\}$, $V \subset \mathcal{C}^\infty(Y; F)$ finite dimensional, and two such maps define the same class if and only if they are stably homotopic as maps into $\mathrm{GL}(N, \mathbb{C})$ (after choosing orthonormal bases). In this sense the groups do not depend on (Y, F).

(K2) (Bott periodicity, first form; Theorem 13.6.2.) The Toeplitz families index (Definition 13.5.4) is a natural isomorphism $\mathrm{ind}_{\mathcal{T}} : K^{-2}(X) \to K^0(X)$ for compact X. If $p : X \to M(N, \mathbb{C})$ is a smooth family of projections then, by (13.6.1),

$$\mathrm{ind}_{\mathcal{T}}[pz^{-1} + (\mathrm{Id} - p)] = [\mathrm{Ran}\, p], \qquad \mathrm{ind}_{\mathcal{T}}[pz + (\mathrm{Id} - p)] = -[\mathrm{Ran}\, p], \tag{16.1.1}$$

where $z = e^{i\theta}$ is the variable on $\mathbb{S}$.

(K3) From Chapter 15: the exterior product $\boxtimes$ and the product $x \cdot y$ for bundles over the same base, defined by the Clifford product of symbols (Definition 15.1.2); the extension maps j_* for open embeddings ((K3)); the Thom class $\lambda_{\mathcal{V}} \in K_{\mathrm{c}}(T^*(\mathcal{V}/M))$ of a real vector bundle (15.4.1), represented on ΛW, $W = \varpi^*(\mathcal{V} \otimes \mathbb{C})$, by Clifford multiplication $c(w) = w \wedge \cdot - \iota_w$, $w = v + i\eta$ (in the notation of Definition 13.11.2, $\lambda_{\mathcal{V}} = \beta_{\mathcal{V} \otimes \mathbb{C}}$);

the Thom homomorphism $\tau(x) = \lambda_{\mathcal{V}} \cdot \varpi^* x$; the Bott element $b_1 \in K_{\mathrm{c}}(T^*I)$, the maps $\beta_N(y) = b_1 \boxtimes \cdots \boxtimes b_1 \boxtimes y$, which are isomorphisms by (BP); the identification $\lambda_{\mathbb{R}} = b_1$ under $I \cong \mathbb{R}$ (Remark 15.4.3) and hence $\lambda_{\mathbb{R}^N} = b_1^{\boxtimes N}$; and the topological index (15.4.3), $\mathrm{ind}_{\mathrm{t}} = \beta_N^{-1} \circ j_* \circ \Theta_* \circ \tau$, defined by an embedding $M \hookrightarrow \mathbb{R}^N \times B$ with tubular neighbourhood $\Theta : \mathcal{V} \cong \Omega \subset I^N \times B$ (Lemma 15.4.1).

(K4) The analytic index $\mathrm{ind}_{\mathrm{a}} : K_{\mathrm{c}}(T^*(M/B)) \to K^0(B)$ for a fibration of compact manifolds (Definition 14.3.4, Lemma 15.1.3): if $A \in \Psi^0(M/B; E_+, E_-)$ is elliptic and $S : \mathbb{C}^L \to \mathcal{C}^\infty(M; E_-)$ is given by L sections such that $A \oplus S$ is surjective on every fibre, then the null spaces of $A \oplus S$ form a smooth bundle $\mathcal{N}(A, S)$ over B and $\mathrm{ind}_{\mathrm{a}}[\sigma_0(A)] = [\mathcal{N}(A, S)] - [\mathbb{C}^L]$.

16.1.3 The semiclassical calculus

We use the semiclassical calculus of Chapter 6, in the version for families on the fibres of a fibration (Remark 6.5.11). For $q \in \mathcal{C}^\infty([0,1]_\epsilon; S^m(\mathbb{R}^n; \mathbb{R}^n))$ the semiclassical (left) quantization is (Definition 6.2.1)

$$\mathrm{Op}_\epsilon(q)u(z) = (2\pi\epsilon)^{-n} \int e^{i(z-z')\cdot\zeta/\epsilon} q(\epsilon, z, \zeta) u(z')\, dz' d\zeta = \int e^{i(z-z')\cdot\xi} q(\epsilon, z, \epsilon\xi) u(z')\, dz' đ\xi, \tag{16.1.2}$$

$\epsilon \in (0,1]$. Its kernel is $\epsilon^{-n} \check{q}(\epsilon, z, (z-z')/\epsilon)$, where $\check{q}$ is the inverse Fourier transform in the last variable ((6.2.2)). If q is Schwartz in ζ the kernel concentrates, as $\epsilon \downarrow 0$, in an ϵ-neighbourhood of the diagonal (Figure 16.1).

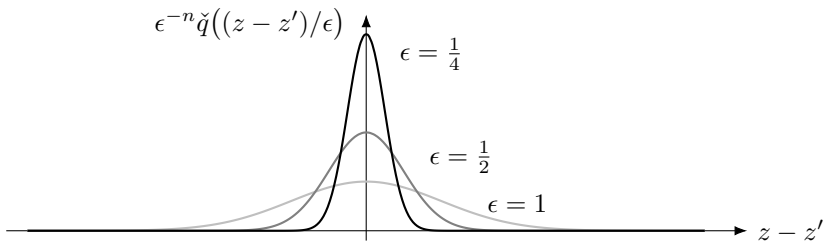


Figure 16.1: The Schwartz kernel of $\mathrm{Op}_\epsilon(q)$ for a symbol q which is Schwartz in ζ, plotted across the diagonal for $n = 1$ and three values of ϵ. Its integral is fixed, while its width is $O(\epsilon)$. This is why constructions with semiclassical families are local for small ϵ, and why excision is trivial for the semiclassical index (Remark 16.3.6).

The results of Chapter 6 are stated for compact manifolds; since all constructions are local (Lemma 6.5.2), they hold for families with kernels supported over a fixed compact subset of a non-compact fibred manifold, which is the form in which we state them.

(S1) (Semiclassical smoothing families; Proposition 6.5.6.) For a fibration $\phi : M \to B$ and $E \to M$, $\Psi_{\mathrm{sl}}^{-\infty}(M/B; E)$ is the space of families $A = A(\epsilon)$, $\epsilon \in (0,1]$, of elements of $\Psi_{\mathrm{c}}^{-\infty}(M/B; E)$ whose kernels are supported in $(0,1] \times K \times_B K$ for some compact $K \subset M$, and which are semiclassical: away from Diag the kernel lies in $\epsilon^\infty \mathcal{C}^\infty([0,1] \times M_\phi^2)$, and in local coordinates (y, z), y in the base and z in the fibre, near Diag it is $\epsilon^{-n} K(\epsilon, y, z, (z-z')/\epsilon)$ (times the coordinate density) with K smooth and rapidly decreasing, with all derivatives, in the last variable, $n = \dim Z$. The families whose kernels are smooth on $[0,1] \times M_\phi^2$ and vanish to infinite order at $\epsilon = 0$ form the ideal $\Psi_{\epsilon^\infty}^{-\infty}(M/B; E)$, denoted $\epsilon^\infty \Psi_{\mathrm{sl}}^{-\infty} = \dot{\mathcal{C}}^\infty([0,1]; \Psi^{-\infty})$ in Chapter 6.

(S2) (The semiclassical symbol; Proposition 6.5.6 and Theorem 6.5.3.) There is a surjective linear map

$$\sigma_{\mathrm{sl}} : \Psi_{\mathrm{sl}}^{-\infty}(M/B; E) \longrightarrow \mathcal{S}_{\mathrm{c}}(T^*(M/B); \operatorname{End} E),$$

locally $\sigma_{\mathrm{sl}}(A)(y, z, \zeta) = \hat{K}(0, y, z, \zeta)$, onto the smooth sections which are Schwartz in the fibres of $T^*(M/B)$ and supported over a compact subset of M. It is multiplicative, $\sigma_{\mathrm{sl}}(AB) = \sigma_{\mathrm{sl}}(A)\sigma_{\mathrm{sl}}(B)$, and $\sigma_{\mathrm{sl}}(A^*) = \sigma_{\mathrm{sl}}(A)^*$. Its null space is $\epsilon\Psi_{\mathrm{sl}}^{-\infty}(M/B; E)$, and $\bigcap_k \epsilon^k \Psi_{\mathrm{sl}}^{-\infty} = \Psi_{\epsilon^\infty}^{-\infty}$. Every sequence $A_k \in \epsilon^k \Psi_{\mathrm{sl}}^{-\infty}$, with kernels supported in a fixed compact set, has an asymptotic sum $A \in \Psi_{\mathrm{sl}}^{-\infty}$, $A - \sum_{k<K} A_k \in \epsilon^K \Psi_{\mathrm{sl}}^{-\infty}$ for all K. A smooth bundle map $e \in \mathcal{C}^\infty(M; \operatorname{End} E)$, regarded as a multiplication operator, satisfies $eA, Ae \in \Psi_{\mathrm{sl}}^{-\infty}$ with symbols $e\sigma_{\mathrm{sl}}(A)$, $\sigma_{\mathrm{sl}}(A)e$. If $\sigma_{\mathrm{sl}}(A)$ is supported in $\varpi^{-1}(K)$, $K \subset M$ compact, then A may be chosen (modifying it by an element of $\Psi_{\epsilon^\infty}^{-\infty}$) with kernel supported in any given neighbourhood of $\mathrm{Diag} \cap K \times_B K$ (Corollary 6.3.3). All of this holds with smooth dependence on additional parameters (Remark 6.5.11).

(S3) (Naturality; Proposition 6.4.10.) A diffeomorphism $F : M \to M'$ of fibrations over B, covered by a unitary bundle isomorphism $E \to E'$, conjugates $\Psi_{\mathrm{sl}}^{-\infty}(M/B; E)$ to $\Psi_{\mathrm{sl}}^{-\infty}(M'/B; E')$ (with the obvious density factors), and the symbol transforms by the induced bundle isomorphism $T^*(M/B) \to T^*(M'/B)$.

(S4) (Finite order; compact fibres; Theorem 6.5.3.) For a fibration of compact manifolds, $\Psi_{\mathrm{sl}}^m(M/B; E_+, E_-)$ is the space of families which are, locally, $\mathrm{Op}_\epsilon(q)$ with q in $S_{\mathrm{sl}}^m = \mathcal{C}^\infty([0,1]; S^m)$, plus elements of $\Psi_{\epsilon^\infty}^{-\infty}$. Then $\Psi_{\mathrm{sl}}^{-\infty} = \bigcap_k \Psi_{\mathrm{sl}}^{-k}$, the space Ψ_{sl}^0 is an algebra containing $\Psi_{\mathrm{sl}}^{-\infty}$ as an ideal, and for fixed $\epsilon > 0$, $A(\epsilon) \in \Psi^0(M/B)$. There are two multiplicative symbols: the semiclassical symbol $\sigma_{\mathrm{sl}}(A)$, a symbol of order 0 on $T^*(M/B)$ with values in $\operatorname{Hom}(E_+, E_-)$, locally $q(0, \cdot)$, and the uniform principal symbol $\tilde{\sigma}_0(A)$, the class of q modulo S_{sl}^{-1}; for $\epsilon > 0$ the principal symbol of $A(\epsilon)$ is represented by $q(\epsilon, z, \epsilon\xi)$, by part (f) of Theorem 6.5.3. For classical symbols which are homogeneous of degree 0 for $|\zeta| \geq 1$ the rescaling does not change the homogeneous part, so $\sigma_0(A(\epsilon))$ is the boundary value of $q(\epsilon, \cdot)$ at infinity. The sequences

$$\epsilon\Psi_{\mathrm{sl}}^m \hookrightarrow \Psi_{\mathrm{sl}}^m \xrightarrow{\sigma_{\mathrm{sl}}} S^m, \qquad \Psi_{\mathrm{sl}}^{m-1} \hookrightarrow \Psi_{\mathrm{sl}}^m \xrightarrow{\tilde{\sigma}_m} S_{\mathrm{sl}}^m / S_{\mathrm{sl}}^{m-1}$$

are exact (Proposition 6.4.2); if $T_k \in \epsilon^k \Psi_{\mathrm{sl}}^{-k}$ then there is $T \in \Psi_{\mathrm{sl}}^0$ with $T - \sum_{k<K} T_k \in \epsilon^K \Psi_{\mathrm{sl}}^{-K}$ for all K, and $\bigcap_K \epsilon^K \Psi_{\mathrm{sl}}^{-K} = \Psi_{\epsilon^\infty}^{-\infty}$; and every symbol a of order 0 on $T^*(M/B)$ with values in $\operatorname{Hom}(E_+, E_-)$ is $\sigma_{\mathrm{sl}}(A)$ for some A, which may be taken with $\tilde{\sigma}_0(A) = [a]$ independent of ϵ (the quantization map q_ϵ of Theorem 6.5.3(d) applied to a, regarded as constant in ϵ).

(S5) (Euclidean estimates; Theorems 6.2.7 and 6.4.7.) On $\mathbb{R}^k$, for symbols $p, q \in S^0(\mathbb{R}^k; \mathbb{R}^k)$ (bounded with all derivatives in the base variable), $\sup_{0<\epsilon\leq 1} \|\mathrm{Op}_\epsilon(p)\|_{L^2} \leq C$, and

$$\|\mathrm{Op}_\epsilon(p)\,\mathrm{Op}_\epsilon(q) - \mathrm{Op}_\epsilon(pq)\|_{L^2} \leq C\epsilon, \quad 0 < \epsilon \leq 1,$$

since $p\#_\epsilon q - pq \in \epsilon S_{\mathrm{sl}}^{-1}$. Moreover $\mathrm{Op}_\epsilon(p)\,\mathrm{Op}_\epsilon(q) = \mathrm{Op}_\epsilon(p\#_\epsilon q)$ with

$$p\#_\epsilon q \sim \sum_\alpha \frac{\epsilon^{|\alpha|}}{\alpha!} \partial_\zeta^\alpha p\, D_z^\alpha q$$

by (6.2.9); the expansion is *exact* (a finite sum) if q is a polynomial in z, or if p is a polynomial in ζ, and in these cases it holds for polynomial symbols of any order.

We add two consequences. For compact fibres they are Theorem 6.5.10 and Corollary 6.5.9; we include the short proofs, which apply verbatim to compactly supported families on non-compact fibres.

Lemma 16.1.2. *If* $A \in \Psi_{\mathrm{sl}}^{-\infty}(M/B;E)$ *then* $\sup_{\epsilon,b}\|A_b(\epsilon)\|_{L^2} < \infty$, *and if* $A \in \Psi_{\epsilon^\infty}^{-\infty}(M/B;E)$ *then* $\sup_b\|A_b(\epsilon)\| = O(\epsilon^\infty)$.

Proof. The second statement is clear, since the kernels are smooth with compact support and all their sup norms are $O(\epsilon^\infty)$. For the first, by a partition of unity we may assume the kernel is $\epsilon^{-n}K(\epsilon,y,z,(z-z')/\epsilon)$ with compact support in z. Since $\sup_{\epsilon,y,z}|K(\epsilon,y,z,Z)| \leq C\langle Z\rangle^{-n-1}$,

$$\sup_z \int \epsilon^{-n}|K(\epsilon,y,z,(z-z')/\epsilon)|dz' \leq C\int \langle Z\rangle^{-n-1}dZ,$$

and the same holds with the roles of z and z' exchanged. Schur's test gives the bound. □

Lemma 16.1.3 (Invertibility)**.** *Let* $A \in \Psi_{\mathrm{sl}}^{-\infty}(M/B;E)$, *possibly depending smoothly on parameters in a compact manifold* P, *and suppose that* $\mathrm{Id}+\sigma_{\mathrm{sl}}(A)$ *is invertible at every point. Then there is* $\epsilon_0 > 0$ *such that* $\mathrm{Id}+A_b(\epsilon)$ *is invertible on* $L^2(Z_b;E_b)$ *for all* $b \in B$, *all parameters and all* $\epsilon \in (0,\epsilon_0]$, *and* $(\mathrm{Id}+A)^{-1} = \mathrm{Id}+A'$ *with* $A' \in \Psi_{\mathrm{sl}}^{-\infty}(M/B;E)$ *(restricted to* $\epsilon \leq \epsilon_0$*), depending smoothly on the parameters. If* P *is a complex manifold and* A *depends holomorphically on the parameters, so does* A'.

Proof. The symbol $a' = (\mathrm{Id}+\sigma_{\mathrm{sl}}(A))^{-1} - \mathrm{Id}$ is again Schwartz and supported over a compact set. Choose A'_0 with $\sigma_{\mathrm{sl}}(A'_0) = a'$. Then $(\mathrm{Id}+A)(\mathrm{Id}+A'_0) = \mathrm{Id}+\epsilon R_1$, $R_1 \in \Psi_{\mathrm{sl}}^{-\infty}$. Inductively, if $(\mathrm{Id}+A)(\mathrm{Id}+A'_{(k)}) = \mathrm{Id}+\epsilon^k R_k$, set $A'_{(k+1)} = A'_{(k)} - \epsilon^k Q_k$ with $\sigma_{\mathrm{sl}}(Q_k) = (\mathrm{Id}+\sigma_{\mathrm{sl}}(A))^{-1}\sigma_{\mathrm{sl}}(R_k)$; then the error is $O(\epsilon^{k+1})$. Taking an asymptotic sum gives A'' with $(\mathrm{Id}+A)(\mathrm{Id}+A'') = \mathrm{Id}+R$, $R \in \Psi_{\epsilon^\infty}^{-\infty}$, and similarly a left inverse modulo $\Psi_{\epsilon^\infty}^{-\infty}$. By Lemma 16.1.2, $\|R\| = O(\epsilon^\infty)$, so for small ϵ, $\mathrm{Id}+R$ is invertible and

$$(\mathrm{Id}+R)^{-1} = \mathrm{Id}-R+R(\mathrm{Id}+R)^{-1}R.$$

The last term has kernel $(w,w') \mapsto ((\mathrm{Id}+R)^{-1}R(\cdot,w'),R^*(w,\cdot))$, which is smooth, with all derivatives (in w, w', ϵ and the parameters) $O(\epsilon^\infty)$, because derivatives of $(\mathrm{Id}+R)^{-1}$ are products of $(\mathrm{Id}+R)^{-1}$ with derivatives of R. So $(\mathrm{Id}+R)^{-1} \in \mathrm{Id}+\Psi_{\epsilon^\infty}^{-\infty}$, $\mathrm{Id}+A$ has a right inverse in $\mathrm{Id}+\Psi_{\mathrm{sl}}^{-\infty}$, and in the same way a left inverse; they coincide. Holomorphy is preserved at each step (the asymptotic sum may be taken with coefficients holomorphic in the parameters, and inversion is holomorphic). □

Remark 16.1.4 (The semiclassical cotangent bundle). The symbol $\sigma_{\mathrm{sl}}(A)$ depends on the parameter ϵ, not only on the family of operators: under a reparametrization $\epsilon' = \epsilon T(\epsilon)$ with $T > 0$ smooth, the symbol is composed with the fibre dilation by $T(0)$. Invariantly, $\sigma_{\mathrm{sl}}(A)$ is a function on a bundle ${}^{\mathrm{sl}}T^*(M/B)$ which is isomorphic, but not canonically, to $T^*(M/B)$; a covector of ${}^{\mathrm{sl}}T^*$ at z is the class of df/ϵ for a function f with $f(z) = 0$. Once ϵ is fixed, as it is throughout this chapter, the identification is fixed. Since the dilations are homotopic to the identity, none of the K-theoretic statements below depends on this choice; see Remark 6.5.4.

16.2 Smoothing families and the K-theory of the base

Let $\phi : M \to B$ be a fibration over the compact manifold B and $E \to M$ a Hermitian bundle. Set

$$G_{\mathrm{c}}^{-\infty}(M/B;E) = \{g = \mathrm{Id}+A;\ A \in \Psi_{\mathrm{c}}^{-\infty}(M/B;E),\\ \mathrm{Id}+A_b \text{ invertible on } L^2(Z_b;E_b) \text{ for every } b \in B\}.$$

If $g = \mathrm{Id} + A \in G_{\mathrm{c}}^{-\infty}$, then $g_b^{-1} = \mathrm{Id} + A_b'$ with A' again in $\Psi_{\mathrm{c}}^{-\infty}(M/B; E)$: indeed $A' = -A + A(\mathrm{Id}+A)^{-1}A$, and as in the proof of Lemma 16.1.3 the last term has a smooth kernel depending smoothly on b, with support in that of A. Thus $G_{\mathrm{c}}^{-\infty}(M/B; E)$ is a group, the group of smooth sections of a bundle of groups over B with fibre $G_{\mathrm{c}}^{-\infty}(Z_b; E_b)$. We assign to each of its elements a class in $K^{-1}(B)$. (Melrose does this by showing that the Hilbert bundle with fibres $L^2(Z_b; E_b)$ has a smooth orthonormal frame; see Exercise 16.2. The following embedding is more elementary.)

Discussion. The difficulty addressed in this section is bookkeeping, not analysis. In Chapters 12 and 13, classes in $K^{-1}(B)$ are homotopy classes of maps from B into one fixed group $G^{-\infty}(Y; F)$. A family g_b of invertible operators on the fibres Z_b is not such a map, because the Hilbert spaces $L^2(Z_b; E_b)$ change with b. We fix this by embedding all of them isometrically and smoothly into one space $L^2(\mathbb{T}^N; \mathbb{C}^R)$, using the embedding of the fibration into a torus already used in Chapter 15, and then check that nothing depends on the embedding.

16.2.1 Admissible embeddings

Definition 16.2.1. Let $K \subset M$ be compact and $W \supset K$ open. An *admissible embedding* of $(M/B, E)$ over W is a family of linear isometries

$$J_b : L^2(W_b; E_b) \longrightarrow L^2(Y; F), \quad W_b = W \cap Z_b,$$

where Y is a compact manifold and $F \to Y$ a Hermitian bundle, such that J_b maps $\mathcal{C}_{\mathrm{c}}^\infty(W_b; E_b)$ into $\mathcal{C}^\infty(Y; F)$ and, for every $A \in \Psi_{\mathrm{c}}^{-\infty}(M/B; E)$ with kernel supported in $K \times_B K$, the map $b \mapsto J_b A_b J_b^*$ is a smooth map $B \to \Psi^{-\infty}(Y; F)$ (depending continuously on A).

Lemma 16.2.2. *For every compact $K \subset M$ there is an admissible embedding of $(M/B, E)$ over some $W \supset K$, with $Y = \mathbb{T}^N$ and $F = \mathbb{C}^R$ for some N, R.*

Proof. Let $\bar{M} \supset M$ be the compact fibration of which M is an open subset. As in Lemma 15.4.1, choose an embedding $\iota : \bar{M} \to \mathbb{R}^N$ such that $(\iota, \bar{\phi})$ embeds $\bar{M}$ into $\mathbb{R}^N \times B$, with image in $(0, 2\pi)^N \times B$, and let $\mathcal{V}_m = d\iota(T_m(\bar{M}/B))^\perp$. For some $\delta > 0$ the map $\Theta(m, v) = \iota(m) + v$ is, for each b, a diffeomorphism from $\{(m, v);\ m \in \bar{Z}_b,\ v \in \mathcal{V}_m,\ |v| < \delta\}$ onto an open set $\Omega_b \subset (0, 2\pi)^N \subset \mathbb{T}^N$, depending smoothly on b; let $p_b : \Omega_b \to \bar{Z}_b$ be the corresponding projection. Choose $h \in \mathcal{C}_{\mathrm{c}}^\infty([0, \delta))$ real, not identically zero near 0, and define $\gamma'(y) = h(|v|)$ for $y = \Theta(m, v)$, extended by 0. By the coarea formula

$$\int_{\Omega_b} |\gamma'(y)|^2 |u(p_b(y))|^2 dy = \int_{\bar{Z}_b} f(m) |u(m)|^2 \nu(m)$$

for a smooth positive function f on $\bar{M}$ (the integral of $|\gamma'|^2$ over $p_b^{-1}(m)$, with respect to the measure induced by dy and ν). Set $\gamma = \gamma' \cdot (f \circ p)^{-1/2}$. Next let $W \supset K$ be open with compact closure in M, and $W' \supset \bar{W}$ open with compact closure; as in the proof of Lemma 13.2.1, a finite cover of $\bar{W}'$ by trivializing open sets and a partition of unity give a fibrewise injective bundle map $E|_{W'} \to \mathbb{C}^R$, which after Gram–Schmidt we may take isometric, $\kappa : E|_{W'} \to \mathbb{C}^R$. Define

$$(J_b u)(y) = \gamma(y)\, \kappa(u(p_b(y))) \ \text{ if } p_b(y) \in W, \qquad (J_b u)(y) = 0 \ \text{ otherwise.}$$

By construction $\|J_b u\| = \|u\|$. If $u \in \mathcal{C}_{\mathrm{c}}^\infty(W_b)$ then $J_b u$ is smooth with compact support in Ω_b. For A with kernel $A(m, m')$ supported in $K \times_B K$, $J_b A_b J_b^*$ has the kernel

$$\gamma(y)\, \kappa\, A_b(p_b(y), p_b(y'))\, \kappa^*\, \gamma(y')\, j_b(y'),$$

where j_b is the smooth positive Jacobian factor relating ν, dy and the measures on the fibres of p_b; it is smooth in (b, y, y') with compact support in $\Omega_b \times \Omega_b$. □

For a non-compact vector bundle fibre we shall also need kernels which are Schwartz, rather than compactly supported, along the fibres.

Remark 16.2.3. Let $\varpi : \mathcal{V} \to M$ be a real vector bundle with a metric over a fibration of compact manifolds $M \to B$. Let $\Psi^{-\infty}_{\mathcal{S}}(\mathcal{V}/B; E)$ denote the families whose kernels are smooth on $\mathcal{V} \times_B \mathcal{V}$ and Schwartz in the fibre variables, uniformly with all derivatives (i.e. they extend smoothly, vanishing to infinite order at the boundary, to the fibrewise radial compactification; see Chapter 1). Composing the construction above with the fibre compression $v \mapsto v(1 + |v|^2)^{-1/2}$, which identifies $\mathcal{V}$ with the open unit ball bundle, and using a function γ which is Gaussian in the original variable, one obtains isometries $J_b : L^2(\mathcal{V}_b) \to L^2(\mathbb{T}^N; \mathbb{C}^R)$ with the properties of Definition 16.2.1 for all elements of $\Psi^{-\infty}_{\mathcal{S}}$; the Jacobian factors grow at most polynomially in $|v|$ and are absorbed by the rapid decay. Everything below applies to such families without change.

16.2.2 The class of a section

Definition 16.2.4. Let $g = \mathrm{Id} + A \in G_c^{-\infty}(M/B; E)$, with A supported in $K \times_B K$, and let J be an admissible embedding over $W \supset K$. Then

$$\rho_J(g)(b) = \mathrm{Id} + J_b A_b J_b^* \in G^{-\infty}(Y; F),$$

and we set $[g] = [\rho_J(g)] \in [B; G^{-\infty}(Y; F)] = K^{-1}(B)$.

Indeed $\rho_J(g)(b)$ is invertible: with respect to $L^2(Y; F) = \operatorname{Ran} J_b \oplus (\operatorname{Ran} J_b)^\perp$ it is $J_b g_b J_b^* \oplus \mathrm{Id}$, where g_b acts on $L^2(W_b)$ (it preserves this space and its complement, being the identity on $L^2(Z_b \setminus K)$), and its inverse is $\mathrm{Id} + J_b A_b' J_b^*$. Note that $J^*J = \mathrm{Id}$ implies $\rho_J(gh) = \rho_J(g)\rho_J(h)$.

Lemma 16.2.5. *(i) If $I : L^2(Y; F) \to L^2(Y'; F')$ is an isometry which maps $\mathcal{C}^\infty(Y; F)$ into $\mathcal{C}^\infty(Y'; F')$ and $\Psi^{-\infty}(Y; F)$ continuously into $\Psi^{-\infty}(Y'; F')$ by $T \mapsto ITI^*$, then the map $h \mapsto IhI^* + \mathrm{Id} - II^*$ induces the canonical identification $[B; G^{-\infty}(Y; F)] \cong [B; G^{-\infty}(Y'; F')]$ of (K1).*

(ii) The class $[g] \in K^{-1}(B)$ does not depend on K, W or J.

(iii) $[gh] = [g] + [h]$; smoothly homotopic sections define the same class; and if $U \subset M$ is open and $g \in G_c^{-\infty}(U/B; E)$, then the class of g is the same as that of its extension (by the identity) to an element of $G_c^{-\infty}(M/B; E)$.

(iv) If $\Phi : E \to E'$ is a unitary isomorphism covering a diffeomorphism of fibrations over B (in particular Φ may be a unitary bundle map on M), then $[\Phi g \Phi^{-1}] = [g]$.

Proof. (i) By (K1) a class is represented by h with values in a subgroup $\mathrm{GL}(V)$, $V \subset \mathcal{C}^\infty(Y; F)$ finite dimensional. Then $IhI^* + \mathrm{Id} - II^*$ takes values in $\mathrm{GL}(IV)$, $IV \subset \mathcal{C}^\infty(Y'; F')$, and in the orthonormal bases $\{e_j\}$ of V and $\{Ie_j\}$ of IV the two maps are given by the same matrices. This is the identification of (K1).

(ii) Enlarging K does not change $\rho_J(g)$. If J, J' are admissible over W, W', both restrict to admissible embeddings over $W \cap W' \supset K$, and for A supported in $K \times_B K$ the restriction does not change JAJ^*. So we may assume $W = W'$. Consider $Y \sqcup Y'$ with the bundle $F \sqcup F'$, so $L^2(Y \sqcup Y') = L^2(Y) \oplus L^2(Y')$, and

$$R_{t,b} = (\cos t\, J_b,\ \sin t\, J_b') : L^2(W_b; E_b) \longrightarrow L^2(Y; F) \oplus L^2(Y'; F'), \quad t \in [0, \tfrac{\pi}{2}].$$

Since the ranges of $(J_b, 0)$ and $(0, J'_b)$ are orthogonal, $R_t^* R_t = \cos^2 t\, J^*J + \sin^2 t\, J'^* J' = \mathrm{Id}$, so each R_t is admissible, and $\rho_{R_t}(g)$ is a homotopy in $G^{-\infty}(Y \sqcup Y')$ from $\rho_J(g) \oplus \mathrm{Id}$ to $\mathrm{Id} \oplus \rho_{J'}(g)$. By (i), applied to the inclusions of $L^2(Y)$ and $L^2(Y')$ as summands, these represent $[\rho_J(g)]$ and $[\rho_{J'}(g)]$.

(iii) The first statement follows from $\rho_J(gh) = \rho_J(g)\rho_J(h)$. A homotopy is a section over $B \times [0,1]$, and one admissible embedding (Lemma 16.2.2 over $B \times [0,1]$) serves for all t. An admissible embedding of M over $W \subset U$ is one of U.

(iv) If J is an admissible embedding for $(M'/B, E')$ over W', then $J\Phi$ is one for $(M/B, E)$ over the preimage of W', and since Φ is unitary, $J(\Phi A \Phi^{-1})J^* = (J\Phi)A(J\Phi)^*$. Thus $\rho_J(\Phi g \Phi^{-1}) = \rho_{J\Phi}(g)$, and (ii) applies. □

Applying this to the fibration $M \times \mathbb{S} \to B \times \mathbb{S}$, a smooth *pointed loop* $g : \mathbb{S} \to G_{\mathrm{c}}^{-\infty}(M/B; E)$, i.e. an element of $G_{\mathrm{c}}^{-\infty}((M \times \mathbb{S})/(B \times \mathbb{S}); E)$ equal to Id over $B \times \{1\}$, has a class $[g] \in K^{-2}(B)$ (take J independent of z).

16.2.3 The relative index of idempotents

Let $p_0 \in \mathcal{C}^\infty(M; \mathrm{End}\, E)$ be a smooth family of idempotents ($p_0^2 = p_0$), acting as a multiplication operator.

Definition 16.2.6. A *compactly supported idempotent perturbation* of p_0 is a smooth family $e = p_0 + Q$, $Q \in \Psi_{\mathrm{c}}^{-\infty}(M/B; E)$ (or, in the situation of Remark 16.2.3, $Q \in \Psi_{\mathcal{S}}^{-\infty}$), with $e_b^2 = e_b$ for all b. Its *relative index* is

$$d(e, p_0) = \mathrm{ind}_{\mathcal{T}}[g_{e,p_0}] \in K^0(B), \qquad g_{e,p_0}(z) = \big(ez^{-1} + \mathrm{Id} - e\big)\big(p_0 z + \mathrm{Id} - p_0\big),\ z \in \mathbb{S}.$$

For an idempotent f, $z \mapsto fz^{-1} + \mathrm{Id} - f$ is a loop of invertible operators with inverse $fz + \mathrm{Id} - f$. Expanding, with $e = p_0 + Q$, and using $p_0(\mathrm{Id} - p_0) = 0$,

$$g_{e,p_0}(z) = \mathrm{Id} + Q(2p_0 - \mathrm{Id}) + z^{-1}Q(\mathrm{Id} - p_0) - zQp_0, \tag{16.2.1}$$

so g_{e,p_0} is a pointed loop in $G_{\mathrm{c}}^{-\infty}(M/B; E)$, and $d(e, p_0)$ is defined. (The construction is that of the Bott map (16.1.1), applied to e, followed by subtraction of the same construction for p_0; the two infinite rank contributions cancel.)

Question. Why not define the relative index simply as $[\mathrm{Ran}\, e] - [\mathrm{Ran}\, p_0]$?

Answer. Both ranges are infinite-dimensional, so their difference is meaningless as it stands. What is finite is the *discrepancy* between them, and the loop g_{e,p_0} packages it: it is the identity plus a smoothing family, so it defines a class in $K^{-2}(B)$, and the Toeplitz index converts that class into an honest element of $K^0(B)$. The test case is (d4): if $e = p_0 + \Pi$ with Π a finite rank idempotent orthogonal to p_0, then $d(e, p_0)$ is $[\mathrm{Ran}\, \Pi]$, as the naive formula suggests. □

Proposition 16.2.7. *(d1) If e_t, $t \in [0,1]$, is a smooth family of compactly supported idempotent perturbations of p_0 (supports in a fixed compact set), then $d(e_0, p_0) = d(e_1, p_0)$.*

(d2) $d(e \oplus e', p_0 \oplus p'_0) = d(e, p_0) + d(e', p'_0)$, and $d(p_0, p_0) = 0$.

(d3) If Φ is as in Lemma 16.2.5(iv) and commutes with p_0 (resp. carries p_0 to p'_0), then $d(\Phi e \Phi^{-1}, p'_0) = d(e, p_0)$. This also holds for Φ a smooth family of invertible (not necessarily unitary) bundle maps on M.

(d4) *If* $\Pi \in \Psi_c^{-\infty}(M/B;E)$ *is a smooth family of finite rank idempotents with* $p_0\Pi = \Pi p_0 = 0$*, then the ranges of* Π_b *form a smooth vector bundle* $\operatorname{Ran}\Pi$ *over* B *and* $d(p_0+\Pi,p_0) = [\operatorname{Ran}\Pi]$*. If instead* $p_0\Pi = \Pi p_0 = \Pi$ *then* $d(p_0-\Pi,p_0) = -[\operatorname{Ran}\Pi]$*.*

(d5) *If* $F \to B$ *is a vector bundle, lifted to* M*, then* $d(e\otimes \mathrm{Id}_F, p_0\otimes \mathrm{Id}_F) = d(e,p_0)\cdot[F]$*.*

(d6) *If* $U \subset M$ *is open and* $e-p_0$ *is supported in* $K\times_B K$*,* $K\subset U$ *compact, then* $d(e,p_0)$ *is the same whether computed on* U *or on* M*.*

Proof. (d1), (d2) and (d6) follow from Lemma 16.2.5(iii), using that $\operatorname{ind}_{\mathcal{T}}$ is a homomorphism, that block sums of loops represent sums in $K^{-2}(B)$, and that $g_{p_0,p_0} = \mathrm{Id}$. For (d3), $g_{\Phi e\Phi^{-1},p_0'} = \Phi g_{e,p_0}\Phi^{-1}$; if Φ is not unitary, the polar decomposition $\Phi_s = \Phi(\Phi^*\Phi)^{-s/2}$, $s\in[0,1]$, joins it to a unitary through invertible bundle maps, each of which commutes with p_0 if Φ and Φ^* do; in general conjugation by Φ_s is a homotopy of loops in $G_c^{-\infty}$, and we use Lemma 16.2.5(iii),(iv).

(d4). The rank of Π_b is $\operatorname{Tr}\Pi_b$, which is locally constant, and if $v_1,\dots,v_r$ span $\operatorname{Ran}\Pi_{b_0}$ then $\Pi_b v_1,\dots,\Pi_b v_r$ are smooth local frames near b_0; so $\operatorname{Ran}\Pi$ is a smooth bundle, of smooth sections. If $p_0\Pi = \Pi p_0 = 0$ then p_0 and Π are commuting idempotents with product zero, so $(p_0+\Pi)z^{-1}+\mathrm{Id}-p_0-\Pi = (p_0z^{-1}+\mathrm{Id}-p_0)(\Pi z^{-1}+\mathrm{Id}-\Pi)$, the factors commuting, and therefore $g_{p_0+\Pi,p_0} = \Pi z^{-1}+\mathrm{Id}-\Pi$. It remains to show that the class of this loop is the Bott class of the bundle $N=\operatorname{Ran}\Pi$, so that (16.1.1) applies. Transport by an admissible embedding J; then $J\Pi J^*$ is a smooth family of finite rank idempotents in $\Psi^{-\infty}(Y;F)$ with range $JN\cong N$, and we write Π for it. Let P_N be the orthogonal projection onto N (given locally by orthonormal frames, so a smooth family of finite rank smoothing operators). Since $\Pi P_N = P_N$ and $P_N\Pi = \Pi$, $\Pi_s = P_N + (1-s)P_N\Pi(\mathrm{Id}-P_N)$ is an idempotent with range N for $s\in[0,1]$; this deforms Π to P_N. By the finite rank approximation (B2) there is a finite dimensional $V\subset\mathcal{C}^\infty(Y;F)$ with orthogonal projection P_V such that $\|P_N - P_VP_NP_V\|\le\frac12$, hence $\|(\mathrm{Id}-P_V)P_N\|\le\frac12$, for all b. Then $T_s = \mathrm{Id}-s(\mathrm{Id}-P_V)$ is injective on N_b, so $N_s = T_sN$ is a smooth bundle isomorphic to N, and the orthogonal projections P_{N_s} deform P_N to the projection onto $N_1 = P_VN\subset V$. The loop $P_{N_1}z^{-1}+\mathrm{Id}-P_{N_1}$ lies in $\mathrm{GL}(V)$, where it is the loop $pz^{-1}+\mathrm{Id}-p$ of (16.1.1) with $\operatorname{Ran}p = N_1\cong N$. Hence $d(p_0+\Pi,p_0) = [N]$. If $p_0\Pi = \Pi p_0 = \Pi$, then similarly $(p_0-\Pi)z^{-1}+\mathrm{Id}-p_0+\Pi = (p_0z^{-1}+\mathrm{Id}-p_0)(\Pi z+\mathrm{Id}-\Pi)$, so $g_{p_0-\Pi,p_0} = \Pi z+\mathrm{Id}-\Pi$, and the second formula in (16.1.1) gives $-[\operatorname{Ran}\Pi]$.

(d5). Write $F=\operatorname{Ran}q$ for a family of projections $q: B\to M(r,\mathbb{C})$. By (d2) and (d3), $d(e\otimes\mathrm{Id}_F, p_0\otimes\mathrm{Id}_F)$ is the Toeplitz index of the loop $h = g\otimes q+\mathrm{Id}\otimes(\mathrm{Id}-q)$ on $L^2\otimes\mathbb{C}^r$, $g=g_{e,p_0}$. By (K1) we may assume that g takes values in $\mathrm{GL}(n,\mathbb{C})$, and then h takes values in $\mathrm{GL}(nr,\mathbb{C})$. Let T_g be the Toeplitz family of g, k admissible for it with complement W (in the sense of Definition 13.5.2, Lemma 13.5.3). Since q is constant on the circle it commutes with the Szegő projector and with P_k, so $T_h = T_g\otimes q+\mathrm{Id}\otimes(\mathrm{Id}-q)$, k is admissible for T_h, and $W\otimes F\oplus\operatorname{Ran}P_k\otimes\operatorname{Ran}(\mathrm{Id}-q)$ is a complement. Hence $\operatorname{ind}_{\mathcal{T}}(h) = [\operatorname{Ran}P_k\otimes\mathbb{C}^r]-[W\otimes F]-[\operatorname{Ran}P_k\otimes\operatorname{Ran}(\mathrm{Id}-q)] = ([\operatorname{Ran}P_k]-[W])[F] = \operatorname{ind}_{\mathcal{T}}(g)[F]$. □

Remark 16.2.8. If $e=P$, $p_0=Q$ are orthogonal projections, then $QP:\operatorname{Ran}P\to\operatorname{Ran}Q$ is a Fredholm family (it differs from the identity of $\operatorname{Ran}Q$ by a compact operator after identifying $\operatorname{Ran}P$ with $\operatorname{Ran}Q$), and $d(P,Q)$ is its index bundle, the *relative index* of the pair. This is Melrose's description of the even semiclassical index ("the formal difference of the pair $P\ominus\pi_\infty$"); we do not need it, but see Exercise 16.3.

16.3 The semiclassical index

16.3.1 An idempotent attached to an invertible-at-infinity map

The following purely algebraic construction is used twice: for symbols, to represent a compactly supported K-class by idempotents which are trivial outside a compact set, and for operators, to express the index bundle of an elliptic family as a relative index.

Lemma 16.3.1. *Let $A : H_+ \to H_-$ and $B : H_- \to H_+$ be linear maps (between vector spaces, modules, or elements of a matrix algebra), and set $R_+ = \mathrm{Id} - BA$, $R_- = \mathrm{Id} - AB$. Then*

$$W = \begin{pmatrix} A + R_- A & -R_- \\ R_+ & B \end{pmatrix} : H_+ \oplus H_- \to H_- \oplus H_+,$$

$$W' = \begin{pmatrix} B & R_+ \\ -R_- & A + R_- A \end{pmatrix} : H_- \oplus H_+ \to H_+ \oplus H_-$$

are inverse to each other, and

$$e_{A,B} = W \begin{pmatrix} \mathrm{Id} & 0 \\ 0 & 0 \end{pmatrix} W' = \begin{pmatrix} \mathrm{Id} - R_-^2 & (\mathrm{Id} + R_-) A R_+ \\ R_+ B & R_+^2 \end{pmatrix} \tag{16.3.1}$$

is an idempotent on $H_- \oplus H_+$. With $p_0 = \left(\begin{smallmatrix} \mathrm{Id} & 0 \\ 0 & 0 \end{smallmatrix}\right)$ (the projection onto H_-), $e_{A,B} - p_0$ is a matrix of polynomials in A, B, $R_\pm$ each term of which contains a factor R_+ or R_-. If $B = A^{-1}$ then $e_{A,B} = p_0$.

Proof. Direct multiplication, using $AR_+ = R_- A$ and $R_+ B = B R_-$. For instance the upper left entry of $W'W$ is

$$B(A + R_- A) + R_+^2 = BA + R_+ BA + R_+^2 = BA + R_+ = \mathrm{Id},$$

and the lower left entry is

$$-R_-(A + R_- A) + (A + R_- A) R_+ = -R_-^2 A + R_- A R_+ = 0;$$

the other entries, and those of WW', are checked in the same way. Then

$$e_{A,B}^2 = W \,\mathrm{diag}(\mathrm{Id}, 0) W' W \,\mathrm{diag}(\mathrm{Id}, 0) W' = e_{A,B},$$

and (16.3.1) follows from $(A + R_- A)B = (\mathrm{Id} + R_-)(\mathrm{Id} - R_-)$. □

The idempotent $e_{A,B}$ is the image of the class of A under the boundary map in algebraic K-theory, in the explicit form due to Milnor. For an invertible A it is trivial; in general it measures the failure of B to be an inverse.

Discussion. To see why $e_{A,B}$ carries the index, take A a Fredholm operator and B a parametrix with $R_+ = \Pi_{\mathrm{Nul}\, A}$ and $R_- = \Pi_{\mathrm{Nul}\, A^*}$, the orthogonal projections onto the null space and the cokernel. Then $e_{A,B}$ differs from p_0 by adding $\mathrm{Nul}\, A$ in the H_+ slot and removing $\mathrm{Nul}\, A^*$ from the H_- slot, so its "relative rank" is $\dim \mathrm{Nul}\, A - \dim \mathrm{Nul}\, A^*$. The point of Milnor's formula is that it is a polynomial in A and B. A polynomial can be applied equally well to symbols and to operators, and semiclassical quantization is multiplicative to leading order; this is what makes the comparison with the analytic index in Section 16.5 short.

16.3.2 Quantization of idempotents

Let $\phi : M \to B$ be a fibration, $E \to M$, and $p_0 \in \mathcal{C}^\infty(M; \operatorname{End} E)$ an idempotent.

Proposition 16.3.2. *Let $e \in \mathcal{C}^\infty(T^*(M/B); \operatorname{End} E)$ be a family of idempotents such that $e - p_0 \in \mathcal{S}_{\mathrm{c}}(T^*(M/B); \operatorname{End} E)$. Then there are $\epsilon_0 > 0$ and $\mathfrak{E} = p_0 + Q$, $Q \in \Psi_{\mathrm{sl}}^{-\infty}(M/B; E)$, with $\sigma_{\mathrm{sl}}(Q) = e - p_0$ and $\mathfrak{E}(\epsilon)^2 = \mathfrak{E}(\epsilon)$ for $0 < \epsilon \le \epsilon_0$; we call $\mathfrak{E}$ a* semiclassical quantization *of e. If $\mathfrak{E}'$ is another one, then for ϵ small $\mathfrak{E}(\epsilon)$ and $\mathfrak{E}'(\epsilon)$ are joined by a smooth family of compactly supported idempotent perturbations of p_0. The same holds with smooth dependence on parameters; in particular a smooth homotopy e_t, with $e_t - p_0$ supported in a fixed compact set, has quantizations $\mathfrak{E}_t$ depending smoothly on t.*

Proof. Choose $Q_0 \in \Psi_{\mathrm{sl}}^{-\infty}$ with $\sigma_{\mathrm{sl}}(Q_0) = e - p_0$ and set $P = p_0 + Q_0$. Then $P^2 - P = p_0Q_0 + Q_0p_0 + Q_0^2 - Q_0 \in \Psi_{\mathrm{sl}}^{-\infty}$ has symbol $(p_0+q)^2 - (p_0+q) = e^2 - e = 0$, $q = e - p_0$, so $P^2 - P = \epsilon E$ with $E \in \Psi_{\mathrm{sl}}^{-\infty}$. For s on the circle $\Gamma = \{|s-1| = \frac12\}$, $|s(s-1)| \ge \frac14$ and

$$(s - P)(s - 1 + P) = s(s-1) - (P^2 - P) = s(s-1)\Big(\mathrm{Id} - \frac{\epsilon E}{s(s-1)}\Big).$$

The last factor is Id plus an element of $\Psi_{\mathrm{sl}}^{-\infty}$, depending holomorphically on s near Γ, with symbol 0; by Lemma 16.1.3 it is invertible for small ϵ, with inverse $\mathrm{Id} + \epsilon E'_s$, $E'_s \in \Psi_{\mathrm{sl}}^{-\infty}$ holomorphic in s. Since $s - P$ and $s - 1 + P$ commute, $s \notin \operatorname{Spec}(P)$ and

$$(s - P)^{-1} = \frac{s - 1 + P}{s(s-1)}(\mathrm{Id} + \epsilon E'_s), \quad s \in \Gamma.$$

(For compact fibres the existence statement is Theorem 6.6.3, proved there by a Newton iteration, and the uniqueness up to homotopy is Proposition 6.6.5; we give a direct argument.) Let $\mathfrak{E} = \frac{1}{2\pi i}\oint_\Gamma (s - P)^{-1} ds$ be the Riesz idempotent of P for the part of the spectrum inside Γ; that it is an idempotent is the standard consequence of the resolvent identity. The residue of $(s - 1 + P)/(s(s-1))$ at $s = 1$, the only pole inside Γ, is P, so

$$\mathfrak{E} = P + \frac{\epsilon}{2\pi i}\oint_\Gamma \frac{s - 1 + P}{s(s-1)} E'_s \, ds \in p_0 + Q_0 + \epsilon\Psi_{\mathrm{sl}}^{-\infty},$$

which has the required properties (the kernels of E'_s are supported in the same compact set as those of E).

If $\mathfrak{E}'$ is a second quantization, then $P_t = (1-t)\mathfrak{E} + t\mathfrak{E}'$ is of the same form as P above, uniformly in $t \in [0,1]$, so for small ϵ its Riesz idempotents $\mathfrak{E}_t$ are defined and depend smoothly on t; $\mathfrak{E}_0 = \mathfrak{E}$ and $\mathfrak{E}_1 = \mathfrak{E}'$, since the Riesz idempotent of an idempotent is itself. Parameters are handled in the same way. □

16.3.3 Definition of the semiclassical index

Let $x \in K_{\mathrm{c}}(T^*(M/B))$ be represented by a triple (E_+, E_-, a), where $E_\pm$ are bundles over M and a is a smooth section of $\operatorname{Hom}(E_+, E_-)$ over $T^*(M/B)$, invertible outside a compact set K_a. Choose a smooth section b of $\operatorname{Hom}(E_-, E_+)$ such that

$$r_+ = \mathrm{Id} - ba \quad \text{and} \quad r_- = \mathrm{Id} - ab \quad \text{have compact support;} \tag{16.3.2}$$

for example $b = (1 - \chi)a^{-1}$ with $\chi \in \mathcal{C}_{\mathrm{c}}^\infty(T^*(M/B))$ equal to 1 near K_a. On $E = E_- \oplus E_+$ let p_0 be the projection onto E_- and $e_{a,b}$ the idempotent (16.3.1), formed pointwise. Then $e_{a,b} - p_0$ is compactly supported.

Definition 16.3.3. The *semiclassical index* of x is

$$\mathrm{Ind}_{\mathrm{sl}}(x) = d(\mathfrak{E}(\epsilon), p_0) \in K^0(B),$$

where $\mathfrak{E}$ is a semiclassical quantization of $e_{a,b}$ and $\epsilon > 0$ is sufficiently small.

Theorem 16.3.4. $\mathrm{Ind}_{\mathrm{sl}}$ *is a well-defined homomorphism* $K_{\mathrm{c}}(T^*(M/B)) \to K^0(B)$*, and it has the following properties.*

(i) (Excision) If $U \subset M$ *is open (a fibration over* B*) and* $j : T^*(U/B) \hookrightarrow T^*(M/B)$ *is the inclusion, then* $\mathrm{Ind}_{\mathrm{sl}}^{M/B} \circ j_* = \mathrm{Ind}_{\mathrm{sl}}^{U/B}$.

(ii) (Naturality) If $F : M \to M'$ *is a diffeomorphism of fibrations over* B*, then* $\mathrm{Ind}_{\mathrm{sl}}^{M'/B} \circ F_* = \mathrm{Ind}_{\mathrm{sl}}^{M/B}$. *If* $f : B' \to B$ *is smooth, then* $\mathrm{Ind}_{\mathrm{sl}}(f^*x) = f^* \mathrm{Ind}_{\mathrm{sl}}(x)$ *for the pulled back fibration.*

(iii) (Degenerate fibration) For $\phi = \mathrm{Id} : B \to B$*, the map* $\mathrm{Ind}_{\mathrm{sl}}$ *on* $K_{\mathrm{c}}(T^*(B/B)) = K^0(B)$ *is the identity.*

(iv) (Module property) $\mathrm{Ind}_{\mathrm{sl}}(x \cdot \phi^* y) = \mathrm{Ind}_{\mathrm{sl}}(x)\, y$ *for* $y \in K^0(B)$*, where* $\phi^* y$ *is pulled back to* $T^*(M/B)$.

Proof. Independence of the choices. By Proposition 16.3.2 and (d1), $d(\mathfrak{E}(\epsilon), p_0)$ does not depend on the quantization or on $\epsilon \le \epsilon_0$. If b_0, b_1 both satisfy (16.3.2), so does $b_t = (1-t)b_0 + tb_1$, with supports in a fixed compact set; the quantizations of e_{a,b_t} (Proposition 16.3.2 with parameter t) give a homotopy, so $\mathrm{Ind}_{\mathrm{sl}}$ does not depend on b.

Independence of the representative. The equivalence relation on triples is generated by the following moves, which we check in turn. (b) Changing a on a compact set: if $a' = a$ outside a compact set, then $a_t = (1-t)a + ta'$ is invertible outside a fixed compact set, and the same b satisfies (16.3.2) for all t. (c) A homotopy a_t, invertible outside a fixed compact set: take $b_t = (1-\chi)a_t^{-1}$ with χ fixed. (a) Bundle isomorphisms $\phi_\pm$ of the lifted bundles: the dilation $\phi_\pm(t\zeta)$, $t \in [0,1]$, joins them to the lifts of their restrictions $\phi_\pm^0$ to the zero section, and along it the triple changes by a homotopy (c). For bundle isomorphisms $\phi_\pm^0$ over M, put $\Phi = \mathrm{diag}(\phi_-^0, \phi_+^0)$; then $e_{\phi_-^0 a(\phi_+^0)^{-1}, \phi_+^0 b(\phi_-^0)^{-1}} = \Phi e_{a,b} \Phi^{-1}$, $\Phi \mathfrak{E} \Phi^{-1}$ is a quantization of it, Φ commutes with p_0, and (d3) applies. (The same argument shows that the choice of the bundles $E_\pm$ over M in a triple on $T^*(M/B)$ is immaterial.) (d) Stabilization by (H, H, Id): with $b = \mathrm{Id}$ on H, the extra block of $e_{a,b}$ is p_0, so by (d2) nothing changes. Direct sums of triples give direct sums of idempotents, so $\mathrm{Ind}_{\mathrm{sl}}$ is additive.

(i) As in Step 1 of the proof of Theorem 15.3.1, $x \in K_{\mathrm{c}}(T^*(U/B))$ has a representative (E_+, E_-, a) with $E_\pm$ bundles over U, an isomorphism $\tau : E_+ \to E_-$ over $U \setminus C$, $C \subset U$ compact, with both bundles trivialized there and $\tau = \mathrm{Id}$ in these trivializations, and $a = \tau$ over $U \setminus C$; $j_* x$ is represented by the extended bundles $\tilde{E}_\pm$ over M and the extension $\tilde{a}$ of a by Id. The set where a is not invertible is compact in $T^*(U/B)$; choose $\chi \in \mathcal{C}_{\mathrm{c}}^\infty(T^*(U/B))$ equal to 1 near it and $b = (1-\chi)\tilde{a}^{-1}$. Then $e_{\tilde{a},b} = p_0$ outside $\operatorname{supp}\chi$. By (S2) a quantization on U/B can be chosen with kernels supported in $K \times_B K$, $K \subset U$ compact; extended by p_0 it is a quantization on M/B. Now (d6) gives the claim.

(ii) The first statement follows from (S3) and Lemma 16.2.5(iv): the push-forward of a quantization is a quantization of the push-forward symbol, and the loops are conjugated by a unitary. For the second, pull back the quantization and use the naturality of $\mathrm{ind}_{\mathcal{T}}$.

(iii) The fibres are points, $T^*(B/B) = B$, $\Psi_{\mathrm{sl}}^{-\infty}(B/B; E)$ consists of families of bundle maps, and the quantization of a symbol is the symbol itself. Since B is compact, any a and

b satisfy (16.3.2); take $a = 0$, $b = 0$, so $r_\pm = \mathrm{Id}$ and $e_{0,0} = \mathrm{diag}(0, \mathrm{Id}_{E_+})$. By (d2) and (d4) (with $\Pi = \mathrm{Id}_{E_-}$ in the first block and $\Pi = \mathrm{Id}_{E_+}$ in the second),

$$\mathrm{Ind}_{\mathrm{sl}}([E_+] - [E_-]) = d(0, \mathrm{Id}_{E_-}) + d(\mathrm{Id}_{E_+}, 0) = -[E_-] + [E_+].$$

(iv) By additivity it suffices to take $y = [F]$, F a bundle over B. The product $x \cdot \phi^*[F]$ is represented by $(E_+ \otimes F, E_- \otimes F, a \otimes \mathrm{Id}_F)$ (the Clifford product with the triple $(F, 0, 0)$), so we may take $b \otimes \mathrm{Id}_F$, $e_{a,b} \otimes \mathrm{Id}_F$ and the quantization $\mathfrak{E} \otimes \mathrm{Id}_F$, which is in the calculus since F is constant on the fibres. Now apply (d5). □

Remark 16.3.5 (Comparison with the quantization maps of Chapter 13). For the trivial fibration $\mathbb{R}^n \times X \to X$, X compact, $T^*((\mathbb{R}^n \times X)/X) = \mathbb{R}^{2n} \times X$ and $\mathrm{Ind}_{\mathrm{sl}}^{(\mathbb{R}^n \times X)/X}$ coincides with the even isotropic–semiclassical quantization map $\mathrm{Ind}_{\mathrm{sl}}^{even} : K_{\mathrm{c}}^0(\mathbb{R}^{2n} \times X) \to K^0(X)$ of Proposition 13.10.6. Indeed, both quantize projection-valued (or idempotent-valued) symbols equal to a constant p_∞ outside a compact set; the quantizations differ only in the use of the Weyl instead of the left quantization and of the isotropic rescaling instead of the semiclassical one, and Definition 13.10.1 and the remark following it, together with (S5), show that the two families of idempotents have the same semiclassical symbol and are joined by a homotopy of idempotent perturbations of p_∞ for small ϵ. Finally the relative class $\delta(Q)$ of Lemma 13.10.5 equals $d(Q, p_\infty)$: both are invariant under homotopies, and on the finite rank model of Lemma 13.10.5(i) both are given by the range of the finite rank part, by (d4). (Projection data p give idempotents e through Lemma 13.10.4(ii), whose construction is the Clifford rotation used in Lemma 16.3.1; the classes agree by (d1).) Thus $\mathrm{Ind}_{\mathrm{sl}}$ extends $\mathrm{Ind}_{\mathrm{sl}}^{even}$ from trivial bundles of phase planes to arbitrary fibrations, and its odd counterpart $\mathrm{Ind}_{\mathrm{sl}}^{odd}$ of Section 16.4 extends the map of Proposition 13.10.3.

Remark 16.3.6. Excision, which required some care for the analytic index (Theorem 15.3.1), is immediate here because nothing outside the support of the symbol enters the construction: for ϵ small the operators are p_0 plus families with kernels in an arbitrarily small neighbourhood of the fibre diagonal over the support. Note also that the fibres need not be compact.

16.4 The odd semiclassical index

In Melrose's presentation the basic object is the odd index, defined by quantizing compactly supported perturbations of the identity. Recall from Definition 13.9.1 that $K_{\mathrm{c}}^{-1}(X)$ is the group of homotopy classes of smooth maps $a : X \to \mathrm{GL}(N, \mathbb{C})$ (stably in N) with $a = \mathrm{Id}$ outside a compact set, through such maps, and $K_{\mathrm{c}}^{-2}(X)$ the analogous group of pointed loops.

Proposition 16.4.1. *Let $a \in \mathcal{C}^\infty(T^*(M/B); \mathrm{GL}(N, \mathbb{C}))$ with $a - \mathrm{Id}$ compactly supported, and let $Q \in \Psi_{\mathrm{sl}}^{-\infty}(M/B; \mathbb{C}^N)$ have $\sigma_{\mathrm{sl}}(Q) = a - \mathrm{Id}$. For ϵ small, $\mathrm{Id} + Q(\epsilon) \in G_{\mathrm{c}}^{-\infty}(M/B; \mathbb{C}^N)$, and its class*

$$\mathrm{Ind}_{\mathrm{sl}}^{odd}[a] = [\mathrm{Id} + Q(\epsilon)] \in K^{-1}(B)$$

depends only on $[a] \in K_{\mathrm{c}}^{-1}(T^*(M/B))$. This defines a homomorphism

$$\mathrm{Ind}_{\mathrm{sl}}^{odd} : K_{\mathrm{c}}^{-1}(T^*(M/B)) \to K^{-1}(B)$$

with the excision and naturality properties of Theorem 16.3.4. Applied to pointed loops (the fibration $M \times \mathbb{S} \to B \times \mathbb{S}$) it gives $\mathrm{Ind}_{\mathrm{sl}}^{odd,\mathbb{S}} : K_{\mathrm{c}}^{-2}(T^*(M/B)) \to K^{-2}(B)$.

Proof. Invertibility for small ϵ is Lemma 16.1.3. If Q' is another choice, then $Q'-Q=\epsilon R$ with $R\in\Psi_{\mathrm{sl}}^{-\infty}$, and $\mathrm{Id}+Q+t\epsilon R=(\mathrm{Id}+Q)(\mathrm{Id}+t\epsilon(\mathrm{Id}+Q)^{-1}R)$ is invertible for $t\in[0,1]$ and ϵ small, by Lemma 16.1.2; by Lemma 16.2.5 the class is unchanged. Homotopies of a are quantized with a parameter, stabilization $a\oplus\mathrm{Id}$ adds an identity block, and $\mathrm{Ind}_{\mathrm{sl}}^{odd}[ab]=\mathrm{Ind}_{\mathrm{sl}}^{odd}[a]+\mathrm{Ind}_{\mathrm{sl}}^{odd}[b]$ because a quantization of ab is $(\mathrm{Id}+Q_a)(\mathrm{Id}+Q_b)-\mathrm{Id}$. Excision and naturality are proved as for the even index. □

The even index is the odd one applied to loops. For $x=[(E_+,E_-,a)]$ as above, the symbol-level loop

$$\ell_x(z)=\big(e_{a,b}z^{-1}+\mathrm{Id}-e_{a,b}\big)\big(p_0z+\mathrm{Id}-p_0\big) \tag{16.4.1}$$

is a pointed loop of invertible maps which is the identity outside a compact set, by the symbol-level version of (16.2.1). (After embedding E in a trivial bundle and extending by Id it defines a class in $K_{\mathrm{c}}^{-2}(T^*(M/B))$; this is the inverse of the Toeplitz index for compactly supported classes.)

Proposition 16.4.2. $\mathrm{Ind}_{\mathrm{sl}}(x)=\mathrm{ind}_{\mathcal{T}}\big(\mathrm{Ind}_{\mathrm{sl}}^{odd,\mathbb{S}}[\ell_x]\big)$.

Proof. If $\mathfrak{E}=p_0+Q$ is a quantization of $e_{a,b}$, then by (16.2.1) $g_{\mathfrak{E},p_0}-\mathrm{Id}\in\Psi_{\mathrm{sl}}^{-\infty}$ with symbol $\ell_x-\mathrm{Id}$, so it is a valid quantization in Proposition 16.4.1. Hence $[g_{\mathfrak{E},p_0}]=\mathrm{Ind}_{\mathrm{sl}}^{odd,\mathbb{S}}[\ell_x]$, and $\mathrm{Ind}_{\mathrm{sl}}(x)=\mathrm{ind}_{\mathcal{T}}[g_{\mathfrak{E},p_0}]$ by definition. □

Remark 16.4.3. Melrose proves the compatibility of the odd and even semiclassical index maps with Bott periodicity (his Proposition 12.7, the commutativity of a square with vertical maps given by isotropic quantization) by introducing a joint isotropic–semiclassical calculus. With the definitions adopted here the even index is *defined* through loops, and Proposition 16.4.2 replaces that square; the isotropic quantization enters only in Section 16.6.

16.5 The analytic and semiclassical indices agree

In this section $\phi:M\to B$ is a fibration of compact manifolds. We first express the analytic index bundle as a relative index of idempotents, using Lemma 16.3.1 at the operator level.

Discussion. The idea of the comparison is shown in Figure 16.2. Quantize the symbol a semiclassically to an elliptic family $A(\epsilon)$, with a semiclassical parametrix $B(\epsilon)$. For each fixed $\epsilon>0$ the Milnor idempotent $e_{A(\epsilon),B(\epsilon)}$ computes the analytic index (Proposition 16.5.1). As $\epsilon\downarrow 0$ the same idempotent is a semiclassical quantization of the symbol-level idempotent $e_{a,b}$, so for small ϵ it computes the semiclassical index. The relative index cannot change along the path, so the two indices agree.

Proposition 16.5.1. *Let $A\in\Psi^0(M/B;E_+,E_-)$ be elliptic and let $B\in\Psi^0(M/B;E_-,E_+)$ satisfy $R_+=\mathrm{Id}-BA\in\Psi^{-\infty}(M/B;E_+)$ and $R_-=\mathrm{Id}-AB\in\Psi^{-\infty}(M/B;E_-)$. Then $e_{A,B}$ of (16.3.1) is a compactly supported idempotent perturbation of p_0 on $E_-\oplus E_+$, and*

$$d(e_{A,B},p_0)=\mathrm{ind}_{\mathrm{a}}([\sigma_0(A)]).$$

The same holds for the operators of Remark 16.1.1, i.e. with $\mathbb{C}^L$ adjoined to E_+ or E_-.

Proof. The entries of $e_{A,B}-p_0$ are products of elements of $\Psi^0(M/B)$ with at least one factor $R_\pm$, so they are smoothing families; $e_{A,B}$ is an idempotent by Lemma 16.3.1. Write $I(A,B)=d(e_{A,B},p_0)$.

Step 1: I depends only on A, and is homotopy invariant. If B' is a second such parametrix, then $B_t=(1-t)B+tB'$ has $\mathrm{Id}-B_tA=(1-t)R_++tR'_+$ and similarly on the

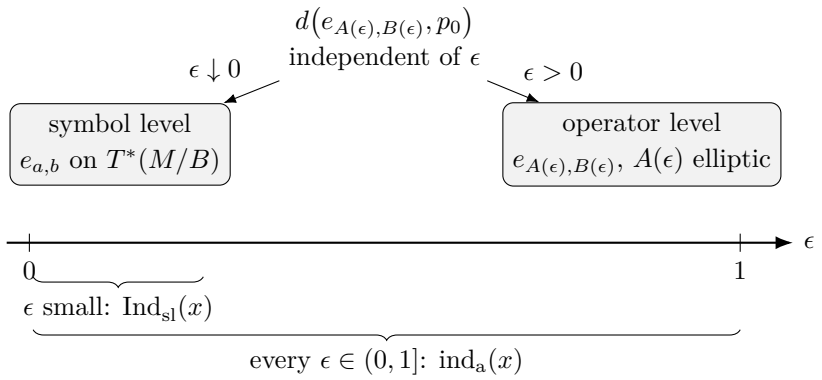


Figure 16.2: The semiclassical limit deforms an elliptic family into its symbol. For every $\epsilon \in (0,1]$ the relative index of the Milnor idempotent is the analytic index (right); for small ϵ the idempotent is a semiclassical quantization of its symbol, and the relative index is the semiclassical index (left). This is Theorem 16.5.4.

other side, so e_{A,B_t} is a smooth family of compactly supported idempotent perturbations and (d1) applies. If (A_t, B_t) is a smooth family of such pairs, the same argument applies. We write $I(A)$.

Step 2: stabilization. Let $S : \mathbb{C}^L \to \mathcal{C}^\infty(M; E_-)$ be given by L smooth sections, such that $A \oplus S : L^2(Z_b; E_+) \oplus \mathbb{C}^L \to L^2(Z_b; E_-)$ is surjective for every b. Consider the operators $A_t = (A, tS)$ from $E_+ \oplus \mathbb{C}^L$ to E_-, in the sense of Remark 16.1.1, with the fixed $B' = \left(\begin{smallmatrix} B \\ 0 \end{smallmatrix}\right)$. Then

$$\mathrm{Id} - B' A_t = \begin{pmatrix} R_+ & -tBS \\ 0 & \mathrm{Id}_{\mathbb{C}^L} \end{pmatrix}, \qquad \mathrm{Id} - A_t B' = R_-,$$

both smoothing in the extended sense, so $I(A_0) = I(A_1)$. For $t = 0$, A_0 is the direct sum of A and the zero map $\mathbb{C}^L \to \{0\}$, and B' is the direct sum of B and 0; for the second summand $R_+ = \mathrm{Id}_{\mathbb{C}^L}$, $R_- = 0$ and the idempotent is $\mathrm{Id}_{\mathbb{C}^L}$, with $p_0 = 0$ there. By (d2), (d3) (to reorder the summands) and (d4),

$$I(A, S) = I(A_1) = I(A_0) = I(A) + [\mathbb{C}^L].$$

Step 3: the surjective case. Write $A' = (A, S)$ and let $\mathcal{N} = \mathcal{N}(A, S)$ be the null bundle, a smooth bundle of rank r whose fibres consist of smooth sections ((K4)); the orthogonal projections $\Pi_{\mathcal{N}}$ onto its fibres form a smooth family of finite rank smoothing operators, being given locally by orthonormal frames. The family $C = A'^* A' + \Pi_{\mathcal{N}}$ is elliptic in the extended sense, and injective, since $(Cu, u) = \|A'u\|^2 + \|\Pi_{\mathcal{N}} u\|^2$; being self-adjoint and Fredholm of index 0, it is invertible. Its inverse is a smooth family in the extended calculus: in a local trivialization C is norm-smooth (Lemma 14.3.1), hence so is $V = C^{-1}$, and if Q is a parametrix with $QC = \mathrm{Id} - R$, $CQ = \mathrm{Id} - R'$, then $V = Q + RQ + RVR'$, where RVR' has a smooth kernel depending smoothly on b (as in the proof of Lemma 13.5.1). Put $G = C^{-1} A'^*$. Since $C\Pi_{\mathcal{N}} = \Pi_{\mathcal{N}}$ we have $C^{-1}\Pi_{\mathcal{N}} = \Pi_{\mathcal{N}} = \Pi_{\mathcal{N}} C^{-1}$, hence

$$GA' = C^{-1}(C - \Pi_{\mathcal{N}}) = \mathrm{Id} - \Pi_{\mathcal{N}}, \qquad A'GA' = A'(\mathrm{Id} - \Pi_{\mathcal{N}}) = A',$$

and, A' being surjective, $A'G = \mathrm{Id}$. So G is a parametrix of A' with $R_+ = \Pi_{\mathcal{N}}$, $R_- = 0$. Since $A'\Pi_{\mathcal{N}} = 0$ and $\Pi_{\mathcal{N}} G = \Pi_{\mathcal{N}} C^{-1} A'^* = (A'\Pi_{\mathcal{N}})^* = 0$, (16.3.1) gives

$$e_{A',G} = \begin{pmatrix} \mathrm{Id} & 0 \\ 0 & \Pi_{\mathcal{N}} \end{pmatrix} = p_0 + \Pi, \qquad \Pi = \mathrm{diag}(0, \Pi_{\mathcal{N}}),$$

with $p_0\Pi = \Pi p_0 = 0$. By (d4), $I(A,S) = [\mathcal{N}]$.

Combining the steps, $I(A) = [\mathcal{N}(A,S)] - [\mathbb{C}^L] = \mathrm{ind_a}[\sigma_0(A)]$. □

Remark 16.5.2. For B a point, the rank of $d(e_{A,B}, p_0)$ is $\mathrm{Tr}(e_{A,B} - p_0) = \mathrm{Tr}(R_+^2) - \mathrm{Tr}(R_-^2)$, which is the formula $\mathrm{ind}\, A = \mathrm{Tr}(R_+^k) - \mathrm{Tr}(R_-^k)$ of Theorem 12.1.4 for $k = 2$. Indeed, Tr of the relative index of two idempotents differing by a trace class operator is the difference of their ranks.

Next we need parametrices in the semiclassical calculus.

Lemma 16.5.3. *Let $a \in S^0_{\mathrm{ph}}(T^*(M/B); \mathrm{Hom}(E_+, E_-))$ be invertible outside a compact set, and $b \in S^0_{\mathrm{ph}}(T^*(M/B); \mathrm{Hom}(E_-, E_+))$ satisfy (16.3.2). Let $A \in \Psi^0_{\mathrm{sl}}(M/B; E_+, E_-)$ be a quantization of a with symbols independent of ϵ, as in (S4). Then there is $B \in \Psi^0_{\mathrm{sl}}(M/B; E_-, E_+)$ with $\sigma_{\mathrm{sl}}(B) = b$ and*

$$\mathrm{Id} - BA \in \Psi^{-\infty}_{\mathrm{sl}}(M/B; E_+), \qquad \mathrm{Id} - AB \in \Psi^{-\infty}_{\mathrm{sl}}(M/B; E_-).$$

Proof. Since $ba = \mathrm{Id}$ near infinity, the boundary values of a and b are inverse to each other. Let B_0 be a quantization of b with symbols independent of ϵ; by (S4), $\sigma_0(B_0(\epsilon)) = \sigma_0(A(\epsilon))^{-1}$ for all $\epsilon \in [0,1]$. Then $R = \mathrm{Id} - B_0 A \in \Psi^0_{\mathrm{sl}}$ has vanishing principal symbols, so $R \in \Psi^{-1}_{\mathrm{sl}}$, and $\sigma_{\mathrm{sl}}(R) = r_+$ is compactly supported. Choose $R_1 \in \Psi^{-\infty}_{\mathrm{sl}}$ with $\sigma_{\mathrm{sl}}(R_1) = r_+$. Then $R - R_1 \in \Psi^{-1}_{\mathrm{sl}}$ has vanishing semiclassical symbol, so $R - R_1 = \epsilon T$ with $T \in \Psi^{-1}_{\mathrm{sl}}$. Let $F \in \Psi^0_{\mathrm{sl}}$ be an asymptotic sum of $\sum_{k\geq 0} \epsilon^k T^k$. Then $F(\mathrm{Id} - \epsilon T) - \mathrm{Id} \in \bigcap_K \epsilon^K \Psi^{-K}_{\mathrm{sl}} = \Psi^{-\infty}_{\epsilon^\infty}$, $\sigma_{\mathrm{sl}}(F) = \mathrm{Id}$, and with $B_1 = FB_0$,

$$\mathrm{Id} - B_1 A = \mathrm{Id} - F(\mathrm{Id} - R_1 - \epsilon T) = \big(\mathrm{Id} - F(\mathrm{Id} - \epsilon T)\big) + FR_1 \in \Psi^{-\infty}_{\mathrm{sl}}.$$

In the same way there is B_2 with $\sigma_{\mathrm{sl}}(B_2) = b$ and $\mathrm{Id} - AB_2 \in \Psi^{-\infty}_{\mathrm{sl}}$. Then $B_1 - B_2 = B_1(\mathrm{Id} - AB_2) - (\mathrm{Id} - B_1 A)B_2 \in \Psi^{-\infty}_{\mathrm{sl}}$, so $B = B_1$ satisfies $\mathrm{Id} - AB = (\mathrm{Id} - AB_2) - A(B_1 - B_2) \in \Psi^{-\infty}_{\mathrm{sl}}$. □

Theorem 16.5.4. *For every fibration of compact manifolds $\phi : M \to B$,*

$$\mathrm{Ind_{sl}} = \mathrm{ind_a} : K_{\mathrm{c}}(T^*(M/B)) \longrightarrow K^0(B).$$

Proof. Let $x \in K_{\mathrm{c}}(T^*(M/B))$. As in Chapters 13 and 15 we may represent x by (E_+, E_-, a) with $E_\pm$ bundles over M and a homogeneous of degree 0 and invertible for $|\zeta| \geq 1$ (for some metric), extended smoothly to $T^*(M/B)$; then $a \in S^0_{\mathrm{ph}}$. Let $b = (1 - \chi(|\zeta|))a^{-1}$ with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}([0,\infty))$ equal to 1 on $[0,1]$; it is a classical symbol of order 0 satisfying (16.3.2). Choose $A \in \Psi^0_{\mathrm{sl}}(M/B; E_+, E_-)$ quantizing a with symbols independent of ϵ, and B as in Lemma 16.5.3.

For each fixed $\epsilon \in (0,1]$, $A(\epsilon)$ is an elliptic family in $\Psi^0(M/B; E_+, E_-)$ with $\sigma_0(A(\epsilon)) = a|_{|\zeta|\geq 1}$, so $[\sigma_0(A(\epsilon))] = x$, and $B(\epsilon)$ is a parametrix modulo smoothing families. By Proposition 16.5.1,

$$d(e_{A(\epsilon),B(\epsilon)}, p_0) = \mathrm{ind_a}(x) \quad \text{for every } \epsilon \in (0,1].$$

On the other hand, $e_{A,B} - p_0$ is a matrix of products of elements of Ψ^0_{sl}, each containing a factor $R_\pm \in \Psi^{-\infty}_{\mathrm{sl}}$, so $e_{A,B} - p_0 \in \Psi^{-\infty}_{\mathrm{sl}}(M/B; E_- \oplus E_+)$; by multiplicativity of σ_{sl} its symbol is $e_{a,b} - p_0$, where $e_{a,b}$ is formed from $a, b, r_\pm = \sigma_{\mathrm{sl}}(R_\pm)$; and $e_{A,B}$ is an idempotent for every ϵ. So $e_{A,B}$ is a semiclassical quantization of $e_{a,b}$ in the sense of Proposition 16.3.2, and by definition $\mathrm{Ind_{sl}}(x) = d(e_{A(\epsilon),B(\epsilon)}, p_0)$ for small ϵ. □

Corollary 16.5.5 (Excision for the analytic index)**.** *In the situation of Theorem 15.3.1 (two fibrations of compact manifolds M_1, M_2 over B containing a common open subset U),* $\mathrm{ind_a} \circ (i_1)_* = \mathrm{ind_a} \circ (i_2)_*$ *on* $K_{\mathrm{c}}(T^*(U/B))$.

Proof. By Theorem 16.5.4 and Theorem 16.3.4(i),(ii), $\mathrm{ind_a} \circ (i_j)_* = \mathrm{Ind}_{\mathrm{sl}}^{M_j/B} \circ (i_j)_* = \mathrm{Ind}_{\mathrm{sl}}^{U/B}$ for $j = 1, 2$. □

Remark 16.5.6. Melrose's proof of Theorem 16.5.4 (his Theorem 12.1) is organized around a common generalization of the two index maps: triples (π^+, π^-, a) of families of projections on the radially compactified cotangent bundle with an isomorphism a of their ranges at infinity, quantized as Toeplitz-type operators $\Pi^- A' \Pi^+$ with Π^- a semiclassical quantization of π^-. The analytic index corresponds to constant $\pi^\pm$, the semiclassical index to $a = \mathrm{Id}$, and the equality follows from homotopy invariance of the general index and the fact that both kinds of data are retracts of the general one. The idempotent $e_{A,B}$ plays the role of these triples here, and reduces the argument to the multiplicativity of the semiclassical symbol.

16.6 The Thom class and the harmonic oscillator

Let $\varpi : \mathcal{V} \to M$ be a real vector bundle of rank k with a metric, over a fibration of compact manifolds $M \to B$ (or over M itself, which is the case $B = M$). The fibration $\mathcal{V} \to M$ has non-compact fibres; it is an open subset of the sphere bundle $\tilde{M} = S(\mathcal{V} \oplus \mathbb{R})$ via inverse stereographic projection $k_{\mathcal{V}}$, so its semiclassical index is defined.

Theorem 16.6.1. *The Thom class has semiclassical index one:*

$$\mathrm{Ind}_{\mathrm{sl}}^{\mathcal{V}/M}(\lambda_{\mathcal{V}}) = 1 = [\mathbb{C}] \in K^0(M).$$

The proof occupies the rest of this subsection. We first describe the model on a single fibre, then compare the semiclassical quantization with the harmonic oscillator.

Discussion. Here is the whole argument in heuristic form. On one fibre $\mathbb{R}^k$ the Thom symbol is Clifford multiplication by $w + i\xi$. Its quantization is, up to a factor, the operator $D + D^*$ of the Koszul complex of the harmonic oscillator, whose square is $H - k + 2\deg$. On odd forms this is invertible; on even forms its null space is spanned by the Gaussian ground state $u_0 = \pi^{-k/4} e^{-|w|^2/2}$. So the index on each fibre is the line $\mathbb{C}u_0$. Semiclassical quantization at scale ϵ gives the same operator after the dilation $w \mapsto \epsilon^{-1/2} w$, with ground state a Gaussian of width $\epsilon^{1/2}$. Finally, the Gaussian depends only on $|w|$, so it is invariant under the structure group $O(k)$ and defines a nowhere vanishing section of the line bundle of ground states over M (Figure 16.3). That line bundle is therefore trivial.

16.6.1 The model

On $\mathbb{R}^k$, with variables $X = (w, \xi) \in \mathbb{R}^{2k}$, let $\Lambda = \Lambda\mathbb{C}^k = \Lambda^{\mathrm{ev}} \oplus \Lambda^{\mathrm{odd}}$ with its standard Hermitian structure, $\{e_j\}$ the standard basis, $e_j\wedge$ exterior multiplication and ι_j its adjoint. Let

$$c(X) = \sum_j (w_j + i\xi_j)\, e_j \wedge - \sum_j (w_j - i\xi_j)\, \iota_j : \Lambda^{\mathrm{ev}} \longrightarrow \Lambda^{\mathrm{odd}},$$

which is Clifford multiplication $c(u) = u\wedge\cdot - \iota_u$ by $u = w + i\xi$ as in Chapter 15; $c(X)^* = -c(X)$ and $c(X)^* c(X) = |X|^2$. The group $O(k)$ acts on $\mathbb{R}^{2k}$ by $g(w, \xi) = (gw, g\xi)$ (the cotangent lift of its action on $\mathbb{R}^k$) and on Λ by Λg, and $c(gX) = \Lambda g\, c(X)\, (\Lambda g)^{-1}$.

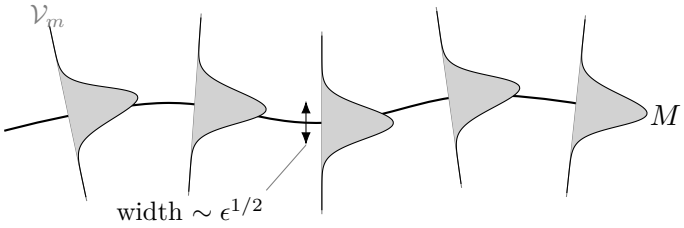


Figure 16.3: The rescaled Gaussian ground state $U_\epsilon^{-1}u_0$ on each normal fibre $\mathcal{V}_m$ (drawn as a profile over the fibre). Because u_0 is invariant under $O(k)$, these profiles fit together into a smooth nowhere vanishing section over M, so the index bundle of the Thom class is canonically the trivial line bundle.

Let $C_j = D_{w_j} + iw_j$, $A_j = D_{w_j} - iw_j$ be the creation and annihilation operators of Section 7.8, and $D = \sum_j A_j\, e_j\wedge$ the Koszul operator, $D^* = \sum_j C_j \iota_j$. The left symbol of A_j is $\xi_j - iw_j = -i(w_j + i\xi_j)$ and that of C_j is $\xi_j + iw_j = i(w_j - i\xi_j)$; since c is a sum of terms each of which is linear in w or in ξ, its quantization involves no ordering ambiguity, and

$$\mathrm{Op}_1(c) = i(D + D^*)\big|_{\Lambda^{\mathrm{ev}}} =: i\mathcal{D}. \tag{16.6.1}$$

Using $[A_j, C_l] = 2\delta_{jl}$, $H = \sum_j C_j A_j + k$ and $e_j \wedge \iota_l + \iota_l e_j\wedge = \delta_{jl}$,

$$(D + D^*)^2 = DD^* + D^*D = \sum_{j,l} C_l A_j (e_j \wedge \iota_l + \iota_l e_j\wedge) + 2\sum_j e_j \wedge \iota_j = H - k + 2\deg$$

on Λ, since $D^2 = 0$ (compare Exercise 7.5). By Propositions 7.8.2 and 7.8.3, $H \geq k$ with $\mathrm{Nul}(H-k) = \mathbb{C}u_0$, $u_0 = \pi^{-k/4}e^{-|w|^2/2}$. Hence $\mathrm{Nul}(\mathcal{D}) = \mathrm{Nul}(D+D^*) = \mathbb{C}u_0 \subset \Lambda^0$, and on Λ^{odd}, $\mathcal{D}\mathcal{D}^* = H - k + 2\deg \geq 2$ is invertible, with inverse in $\Psi^{-2}_{\mathrm{iso}}(\mathbb{R}^k; \Lambda^{\mathrm{odd}})$ (Proposition 7.6.3). Put $G_0 = \mathcal{D}^*(\mathcal{D}\mathcal{D}^*)^{-1} \in \Psi^{-1}_{\mathrm{iso}}$. Then

$$\mathcal{D}G_0 = \mathrm{Id}, \qquad G_0\mathcal{D} = \mathrm{Id} - \Pi_0, \tag{16.6.2}$$

where Π_0 is the orthogonal projection onto $\mathbb{C}u_0$ ($G_0\mathcal{D}$ is the self-adjoint idempotent with range $\mathrm{Ran}\,\mathcal{D}^* = (\mathrm{Nul}\,\mathcal{D})^\perp$). All these operators commute with the action $(g\cdot u)(w) = \Lambda g\, u(g^{-1}w)$ of $O(k)$, and u_0 is invariant.

Now let

$$a = c, \qquad b = -(1 - \chi(|X|))\frac{c(X)}{|X|^2}, \qquad r_\pm = \chi(|X|)\,\mathrm{Id},$$

with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}([0,\infty))$ equal to 1 near $[0,1]$. Since $c^{-1} = c^*/|X|^2 = -c/|X|^2$, b satisfies (16.3.2) with these $r_\pm$, it is a symbol of order -1, and $e_{a,b}$ is an $O(k)$-equivariant family of idempotents on $\Lambda^{\mathrm{odd}} \oplus \Lambda^{\mathrm{ev}}$ equal to p_0 (the projection onto Λ^{odd}) for $|X|$ large.

16.6.2 Transport to the bundle

Let $P \to M$ be the orthonormal frame bundle of $\mathcal{V}$, so that a point $f \in P_m$ is an isometry $f : \mathbb{R}^k \to \mathcal{V}_m$, inducing $\Lambda\mathbb{C}^k \cong \Lambda W_m$ and $L^2(\mathbb{R}^k;\Lambda) \cong L^2(\mathcal{V}_m;\Lambda W_m)$. An $O(k)$-equivariant operator T on $L^2(\mathbb{R}^k;\Lambda)$ defines operators $T_m = f_* T f_*^{-1}$ on $L^2(\mathcal{V}_m;\Lambda W_m)$, independent of $f \in P_m$, and using smooth local sections of P one sees that $(T_m)_{m\in M}$ is a smooth family whenever the kernel of T is smooth (or Schwartz). The cotangent lift of f identifies $\mathbb{R}^{2k}$ with $T^*_{(m,\cdot)}(\mathcal{V}_m) = \mathcal{V}_m \oplus \mathcal{V}_m^*$, and under this identification $e_{a,b}$ becomes the idempotent formed, as in Definition 16.3.3, from the triple $(\Lambda^{\mathrm{ev}}W, \Lambda^{\mathrm{odd}}W, c)$ representing $\lambda_{\mathcal{V}}$.

Let Q_0 be the operator with kernel $\epsilon^{-k}\check{q}(w,(w-w')/\epsilon)\psi(|w'|)$, $q = e_{a,b} - p_0$, where $\psi \in \mathcal{C}^\infty_c([0,\infty))$ is 1 on a large interval; it is $O(k)$-equivariant, and $\mathrm{Op}_\epsilon(q) - Q_0$ has a kernel which is $O(\epsilon^\infty)$ in $\mathcal{S}$. Running the construction of Proposition 16.3.2 with this Q_0 gives an $O(k)$-equivariant semiclassical quantization $\mathfrak{F} = p_0 + Q$ of $e_{a,b}$ on $\mathbb{R}^k$ (the Riesz integral preserves equivariance). Transported, $\mathfrak{F}_m$ is a semiclassical quantization of the idempotent of $\lambda_\mathcal{V}$, by (S3). Hence

$$\mathrm{Ind}_{\mathrm{sl}}^{\mathcal{V}/M}(\lambda_\mathcal{V}) = d(\mathfrak{F}_\mathcal{V}, p_0), \tag{16.6.3}$$

where the subscript denotes the transported family.

16.6.3 Comparison with the harmonic oscillator

Let $U_\epsilon u(w) = \epsilon^{k/4}u(\epsilon^{1/2}w)$, a unitary operator on $L^2(\mathbb{R}^k;\Lambda)$ commuting with $O(k)$. Changing variables in (16.1.2),

$$U_\epsilon \,\mathrm{Op}_\epsilon(q) U_\epsilon^{-1} = \mathrm{Op}_1(q_\epsilon), \qquad q_\epsilon(w,\xi) = q(\epsilon^{1/2}w, \epsilon^{1/2}\xi). \tag{16.6.4}$$

In particular $U_\epsilon \,\mathrm{Op}_\epsilon(c) U_\epsilon^{-1} = \epsilon^{1/2}\,\mathrm{Op}_1(c) = i\epsilon^{1/2}\mathcal{D}$ by (16.6.1): the semiclassical quantization of the Thom symbol is, up to a unitary rescaling and a factor, the Koszul complex of the harmonic oscillator. Conjugation by U_ϵ maps $\Psi^m_{\mathrm{iso}}(\mathbb{R}^k)$ to itself, and an operator has a Schwartz kernel if and only if its conjugate does. We write $\Psi^{-\infty}_{\mathrm{iso}}$ for the operators with Schwartz kernels, an ideal in $\bigcup_m \Psi^m_{\mathrm{iso}}$ (Lemma 7.2.2 and Theorem 7.2.7).

Question. Why is the natural length scale $\epsilon^{1/2}$ rather than ϵ?

Answer. The operator $\mathrm{Op}_\epsilon(c)$ is built from multiplication by w_j and from ϵD_{w_j}. On a function varying on the length scale L, the first is of size L and the second of size ϵ/L. They balance when $L^2 = \epsilon$, and this is the width of the ground state, just as for the semiclassical harmonic oscillator $|w|^2 + \epsilon^2\Delta$. After the dilation U_ϵ the two terms have equal weight and the operator becomes the ϵ-independent oscillator, multiplied by $\epsilon^{1/2}$. □

Lemma 16.6.2. *Let $A = \mathrm{Op}_\epsilon(c)$ and $B_0 = \mathrm{Op}_\epsilon(b)$. For ϵ small there is $B \in U_\epsilon^{-1}\Psi^{-1}_{\mathrm{iso}}U_\epsilon$, $O(k)$-equivariant, with $R_+ = \mathrm{Id} - BA$ and $R_- = \mathrm{Id} - AB$ in $\Psi^{-\infty}_{\mathrm{iso}}$ and*

$$\|e_{A,B} - p_0 - \mathrm{Op}_\epsilon(e_{a,b} - p_0)\|_{L^2} \le C\epsilon.$$

Proof. Since c is linear in ξ and in w, the expansions of (S5) are exact:

$$B_0 A = \mathrm{Op}_\epsilon\Big(ba + \epsilon\sum_j \partial_{\xi_j} b\, D_{w_j} c\Big) = \mathrm{Id} - \mathrm{Op}_\epsilon(r_+) + \epsilon\,\mathrm{Op}_\epsilon(s), \qquad s = -i\sum_j \partial_{\xi_j} b\, c(e_j),$$

where $c(e_j) = e_j \wedge - \iota_j$ is constant, and similarly $AB_0 = \mathrm{Id} - \mathrm{Op}_\epsilon(r_-) + \epsilon\,\mathrm{Op}_\epsilon(s')$ with $s' = \sum_j \partial_{\xi_j} c\, D_{w_j} b$. Both s, s' are symbols of order -2 in X, so they lie in $S^0(\mathbb{R}^k;\mathbb{R}^k)$ and $T_1 = \epsilon\,\mathrm{Op}_\epsilon(s)$ has norm $\le C\epsilon$ by (S5). For small ϵ, $\mathrm{Id} + T_1$ is invertible on L^2; after conjugation by U_ϵ it is an elliptic element of Ψ^0_{iso} (its symbol is $1 + \epsilon s_\epsilon$, with $s_\epsilon \in \Gamma^{-2}$), so its inverse is in Ψ^0_{iso} (an elliptic isotropic operator invertible on L^2 has trivial null spaces on $\mathcal{S}'$, since these lie in $\mathcal{S}$, and its generalized inverse in Proposition 7.6.3 is then an inverse). Set $B = (\mathrm{Id} + T_1)^{-1}B_0$. Then

$$R_+ = \mathrm{Id} - BA = (\mathrm{Id} + T_1)^{-1}\,\mathrm{Op}_\epsilon(r_+) \in \Psi^{-\infty}_{\mathrm{iso}}, \qquad \|R_+ - \mathrm{Op}_\epsilon(r_+)\| + \|B - B_0\| \le C\epsilon.$$

Next, $R_- A = AR_+ \in \Psi^{-\infty}_{\mathrm{iso}}$, and A has the exact right inverse $G = U_\epsilon^{-1}(i\epsilon^{1/2})^{-1}G_0U_\epsilon$ by (16.6.2), so $R_- = R_- AG = AR_+G \in \Psi^{-\infty}_{\mathrm{iso}}$. For the norm, write

$$R_- = \mathrm{Op}_\epsilon(r_-) - \epsilon\,\mathrm{Op}_\epsilon(s') + AT_1(\mathrm{Id} + T_1)^{-1}B_0.$$

Here $AT_1 = \epsilon \operatorname{Op}_\epsilon(c \#_\epsilon s)$ and $c\#_\epsilon s = cs + \epsilon \sum_j \partial_{\xi_j} c\, D_{w_j} s$ (exact, c being linear in ξ) is a symbol of order -1, so $\|AT_1\| \le C\epsilon$ and $\|R_- - \operatorname{Op}_\epsilon(r_-)\| \le C\epsilon$.

Finally, the entries of $e_{A,B} - p_0$ are $-R_-^2$, $(\mathrm{Id}+R_-)AR_+$, R_+B and R_+^2; all of them are in $\Psi_{\mathrm{iso}}^{-\infty}$, since $R_\pm$ are. By the estimates just proved and (S5), R_-^2, R_+B and R_+^2 are within $C\epsilon$ of $\operatorname{Op}_\epsilon(r_-^2)$, $\operatorname{Op}_\epsilon(r_+b)$, $\operatorname{Op}_\epsilon(r_+^2)$. For the remaining entry, $AR_+ = A\operatorname{Op}_\epsilon(r_+) - AT_1(\mathrm{Id}+T_1)^{-1}\operatorname{Op}_\epsilon(r_+)$ and $A\operatorname{Op}_\epsilon(r_+) = \operatorname{Op}_\epsilon(cr_+ + \epsilon\sum_j \partial_{\xi_j} c\, D_{w_j} r_+)$ exactly, with r_+ compactly supported; so AR_+ is within $C\epsilon$ of $\operatorname{Op}_\epsilon(cr_+)$, and $(\mathrm{Id}+R_-)AR_+$ within $C\epsilon$ of $\operatorname{Op}_\epsilon((1+r_-)cr_+)$. These symbols are the entries of $e_{a,b} - p_0$. Equivariance holds at each step. □

Proof of Theorem 16.6.1. By Proposition 16.3.2 and Lemma 16.1.2, $\|\mathfrak{F} - p_0 - \operatorname{Op}_\epsilon(e_{a,b} - p_0)\| \le C\epsilon$, so with $e = e_{A,B}$ from Lemma 16.6.2, $\|e - \mathfrak{F}\| \le C\epsilon$. Both $e - p_0$ and $\mathfrak{F} - p_0$ have Schwartz kernels. For $t \in [0,1]$ put $e_t = (1-t)e + t\mathfrak{F}$; a short computation gives

$$e_t^2 - e_t = -t(1-t)(e-\mathfrak{F})^2,$$

which has norm $\le C\epsilon^2$ and a Schwartz kernel. As in the proof of Proposition 16.3.2, for small ϵ the Riesz idempotents $\mathfrak{e}_t = \frac{1}{2\pi i}\oint_{|s-1|=1/2}(s-e_t)^{-1}ds$ are defined, $\mathfrak{e}_0 = e$, $\mathfrak{e}_1 = \mathfrak{F}$, and

$$(s-e_t)^{-1} = \frac{s-1+e_t}{s(s-1)}\Big(\mathrm{Id} - \frac{e_t^2 - e_t}{s(s-1)}\Big)^{-1}.$$

The inverse on the right is Id plus an operator with Schwartz kernel (Corollary 7.4.8: if K has a Schwartz kernel and $\mathrm{Id}+K$ is invertible, $(\mathrm{Id}+K)^{-1} = \mathrm{Id} - K + K(\mathrm{Id}+K)^{-1}K$, and the last term again has a Schwartz kernel, depending smoothly on parameters). Hence $\mathfrak{e}_t - p_0$ has a Schwartz kernel depending smoothly on t, and $\mathfrak{e}_t$ is $O(k)$-equivariant. Transporting to $\mathcal{V}$ and using Remark 16.2.3 and (d1),

$$d(\mathfrak{F}\mathcal{V}, p_0) = d(e\mathcal{V}, p_0), \qquad e = e_{A,B}.$$

Now G (the exact right inverse of A above) satisfies $\mathrm{Id} - GA = \Pi$, the orthogonal projection onto $\mathbb{C}U_\epsilon^{-1}u_0$, and $\mathrm{Id} - AG = 0$. The linear path $B_t = (1-t)B + tG$ has $\mathrm{Id} - B_tA = (1-t)R_+ + t\Pi$ and $\mathrm{Id} - AB_t = (1-t)R_-$, both with Schwartz kernels, so as in Step 1 of the proof of Proposition 16.5.1, $d(e\mathcal{V}, p_0) = d((e_{A,G})\mathcal{V}, p_0)$. Since $A\Pi = 0$ and $\Pi G = 0$ (as $\Pi\mathcal{D}^* = 0$), (16.3.1) gives $e_{A,G} = p_0 + \operatorname{diag}(0,\Pi)$. The transported family $\Pi_\mathcal{V}$ is the projection onto the span of the section $m \mapsto (U_\epsilon^{-1}u_0)_m$, which is well defined because u_0 is $O(k)$-invariant and nowhere zero; so $\operatorname{Ran}\Pi_\mathcal{V}$ is the trivial line bundle and, by (d4), $d((e_{A,G})\mathcal{V}, p_0) = [\mathbb{C}]$. With (16.6.3) this proves the theorem. □

Remark 16.6.3. The analogous statement for the analytic index, (TH), is traditionally proved by computing the index of the Bott operator on the sphere $\mathbb{S}^k$ as a representation of $O(k)$ and then twisting by the frame bundle. Here the corresponding representation is visible directly: it is the span of the Gaussian ground state, on which $O(k)$ acts trivially. Since $\lambda_\mathcal{V} = \beta_{\mathcal{V}\otimes\mathbb{C}}$, Theorem 16.6.1 is the statement that the fibrewise quantization map of Remark 13.11.4 takes the Bott class of the complex bundle $\mathcal{V}\otimes\mathbb{C} = \mathcal{V}\oplus\mathcal{V}^*$ to 1; for a complexified real bundle the structure group reduces to $O(k) \subset \mathrm{U}(k)$, which acts honestly (not projectively) on $L^2(\mathbb{R}^k)$, so the patching issue mentioned there does not arise. This is Melrose's observation that "the bundle of ground states of the harmonic oscillators is canonically trivial". The analysis needed is the comparison Lemma 16.6.2 between semiclassical quantization at scale ϵ and isotropic quantization after rescaling by $\epsilon^{1/2}$.

Corollary 16.6.4 (The index of the Thom class)**.** *Let $\tilde{M} = S(\mathcal{V}\oplus\mathbb{R}) \to M$ and $k_{\mathcal{V}} : \mathcal{V} \hookrightarrow \tilde{M}$ be as in statement (TH) of Chapter 15. Then*

$$\operatorname{ind}_{\mathrm{a}}^{\tilde{M}/M}((k_{\mathcal{V}})_*\lambda_{\mathcal{V}}) = 1 \in K^0(M).$$

Thus (TH) holds, and the proof of Theorem 15.4.4 given in Chapter 15 is complete.

Proof. By Theorem 16.5.4 for the fibration $\tilde{M} \to M$ of compact manifolds, excision (Theorem 16.3.4(i)) and Theorem 16.6.1, $\operatorname{ind}_{\mathrm{a}}((k_{\mathcal{V}})_*\lambda_{\mathcal{V}}) = \operatorname{Ind}_{\mathrm{sl}}^{\tilde{M}/M}((k_{\mathcal{V}})_*\lambda_{\mathcal{V}}) = \operatorname{Ind}_{\mathrm{sl}}^{\mathcal{V}/M}(\lambda_{\mathcal{V}}) = 1$. □

Corollary 16.6.5 (Semiclassical normalization)**.** *For every $N \geq 0$ and every compact B,*

$$\operatorname{Ind}_{\mathrm{sl}}^{(I^N\times B)/B} \circ \beta_N = \mathrm{Id}_{K^0(B)}, \qquad \textit{hence} \qquad \operatorname{ind}_{\mathrm{a}} \circ (k_N)_* \circ \beta_N = \mathrm{Id}_{K^0(B)},$$

where $k_N : T^*I^N \times B \hookrightarrow T^*(\mathbb{T}^N \times B/B)$.

Proof. Apply Theorem 16.6.1 to the trivial bundle $\mathcal{V} = \mathbb{R}^N \times B$ over $M = B$ (the degenerate fibration $B \to B$): $\operatorname{Ind}_{\mathrm{sl}}^{(\mathbb{R}^N\times B)/B}(\lambda_{\mathbb{R}^N}) = 1$. By the module property Theorem 16.3.4(iv), $\operatorname{Ind}_{\mathrm{sl}}(\lambda_{\mathbb{R}^N} \cdot y) = y$ for $y \in K^0(B)$, and $\lambda_{\mathbb{R}^N} \cdot y = \lambda_{\mathbb{R}^N} \boxtimes y$. Under a diffeomorphism $I \cong \mathbb{R}$, $\lambda_{\mathbb{R}^N} = b_1^{\boxtimes N}$ ((K3)), so by naturality $\operatorname{Ind}_{\mathrm{sl}}^{(I^N\times B)/B}(\beta_N(y)) = y$. The second statement follows by excision ($I^N \times B \subset \mathbb{T}^N \times B$) and Theorem 16.5.4. □

For $N = 1$ and with $b_1 = b_{\mathrm{iso}}$ ((13.10.5); both are represented by $x + i\xi$ up to homotopy), the first identity is also Proposition 13.10.8, via Remark 16.3.5. The second identity is Proposition 15.4.2, which was proved there by the multiplicativity of the analytic index on the torus; here it is obtained from the harmonic oscillator alone.

16.7 The families index theorem

We now give the second proof of the K-theoretic form of the families index theorem of Chapter 15. Recall the data defining $\operatorname{ind}_{\mathrm{t}}$ ((K3)): an embedding $M \hookrightarrow \mathbb{R}^N \times B$ over B with normal bundle $\mathcal{V} \to M$, a tubular neighbourhood $\Theta : \mathcal{V} \cong \Omega \subset I^N \times B$, the Thom homomorphism τ, the extension j_* from Ω to $I^N \times B$, and the Bott isomorphism β_N.

Theorem 16.7.1 (Atiyah–Singer)**.** *For every fibration of compact manifolds $\phi : M \to B$,*

$$\operatorname{ind}_{\mathrm{a}} = \operatorname{Ind}_{\mathrm{sl}} = \operatorname{ind}_{\mathrm{t}} : K_{\mathrm{c}}(T^*(M/B)) \longrightarrow K^0(B).$$

Moreover $\operatorname{Ind}_{\mathrm{sl}}^{(I^N\times B)/B} = \beta_N^{-1}$ and, for the fibration $\mathcal{V} \to B$,

$$\operatorname{Ind}_{\mathrm{sl}}^{\mathcal{V}/B} \circ \tau = \operatorname{Ind}_{\mathrm{sl}}^{M/B}. \tag{16.7.1}$$

Proof. The first equality is Theorem 16.5.4.

The trivial fibration. By Corollary 16.6.5, $\operatorname{Ind}_{\mathrm{sl}}^{(I^N\times B)/B} \circ \beta_N = \mathrm{Id}$, and β_N is an isomorphism by Bott periodicity (Chapters 13 and 15); hence $\operatorname{Ind}_{\mathrm{sl}}^{(I^N\times B)/B} = \beta_N^{-1}$.

The Thom step (16.7.1). Let $\tilde{M} = S(\mathcal{V} \oplus \mathbb{R}) \xrightarrow{\tilde{\phi}} M$ and $\hat{k} : T^*(\mathcal{V}/B) \hookrightarrow T^*(\tilde{M}/B)$ the open embedding induced by $k_{\mathcal{V}}$. The class $b = (k_{\mathcal{V}})_*\lambda_{\mathcal{V}} \in K_{\mathrm{c}}(T^*(\tilde{M}/M))$ has $\operatorname{ind}_{\mathrm{a}}(b) = 1$ by Corollary 16.6.4. By Corollary 15.2.5 and Lemma 15.2.7 (with a splitting of $T^*(\tilde{M}/B)$ extending the one used to define τ),

$$\operatorname{ind}_{\mathrm{a}}^{M/B}(x) = \operatorname{ind}_{\mathrm{a}}^{\tilde{M}/B}(b \cdot \tilde{\phi}^* x) = \operatorname{ind}_{\mathrm{a}}^{\tilde{M}/B}(\hat{k}_*\tau(x)).$$

By Theorem 16.5.4 for $\tilde{M} \to B$ and $M \to B$, and excision for $\mathrm{Ind}_{\mathrm{sl}}$, the outer terms are $\mathrm{Ind}_{\mathrm{sl}}^{M/B}(x)$ and $\mathrm{Ind}_{\mathrm{sl}}^{\mathcal{V}/B}(\tau(x))$.

Conclusion. By naturality and excision for $\mathrm{Ind}_{\mathrm{sl}}$ (Theorem 16.3.4),

$$\mathrm{Ind}_{\mathrm{sl}}^{\mathcal{V}/B}(\tau x) = \mathrm{Ind}_{\mathrm{sl}}^{\Omega/B}(\Theta_*\tau x) = \mathrm{Ind}_{\mathrm{sl}}^{(I^N\times B)/B}(j_*\Theta_*\tau x) = \beta_N^{-1}(j_*\Theta_*\tau x) = \mathrm{ind}_{\mathrm{t}}(x).$$

Together with (16.7.1) this gives $\mathrm{Ind}_{\mathrm{sl}} = \mathrm{ind}_{\mathrm{t}}$. □

Remark 16.7.2 (Comparison of the two proofs). See Figure 16.4. Both proofs pass through the same diagram, the embedding of M into $I^N \times B$ followed by the Thom homomorphism. In Theorem 15.4.4 the analytic index was carried along it using (1) multiplicativity, proved with the product-type calculus of Chapter 14, (2) excision, proved by a direct comparison of stabilized null spaces, (3) the normalization on the torus, proved by multiplicativity, and (4) the index of the Thom class, (TH), which was imported. Here (2) and (3) are replaced by excision for $\mathrm{Ind}_{\mathrm{sl}}$ (which is trivial) together with Theorem 16.5.4, and by Corollary 16.6.5; and (4) is proved (Corollary 16.6.4). Only (1) is still taken from Chapter 15. In Melrose's treatment (1) is replaced as well, by multiplicativity of the semiclassical index under iterated fibrations, proved by an adiabatic limit; we explain this in the next section. With it, (16.7.1) is proved directly and the proof becomes entirely semiclassical: $\mathrm{ind}_{\mathrm{a}} = \mathrm{Ind}_{\mathrm{sl}}^{M/B} = \mathrm{Ind}_{\mathrm{sl}}^{\mathcal{V}/B} \circ \tau = \beta_N^{-1} \circ j_* \circ \Theta_* \circ \tau$.

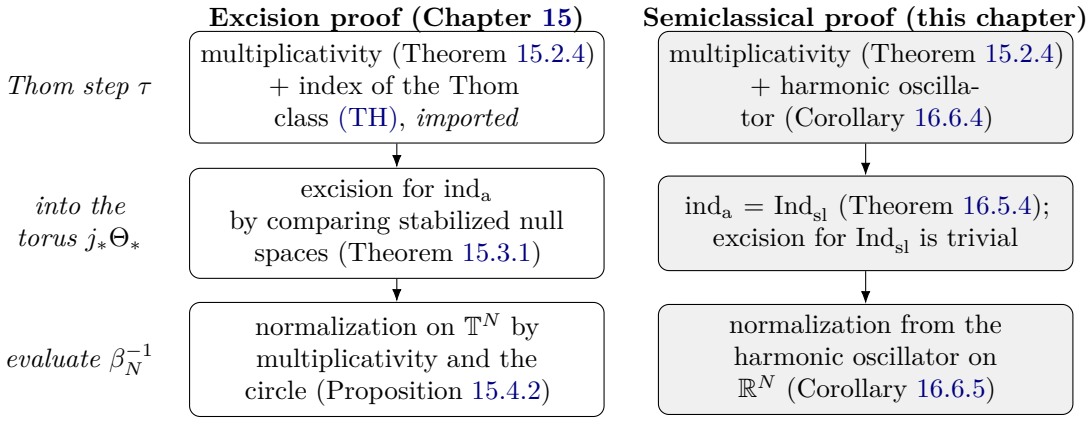


Figure 16.4: The two proofs of $\mathrm{ind}_{\mathrm{a}} = \mathrm{ind}_{\mathrm{t}}$ side by side. Both follow the arrows of Figure 15.1; they differ in how each arrow is shown to preserve the index. The shaded steps are those where the semiclassical index replaces an analytic argument, and in particular the imported statement (TH) is proved.

16.8 Multiplicativity and the adiabatic limit

This section describes, without complete proofs, how the remaining ingredient, (16.7.1), and more generally the multiplicativity of the semiclassical index, can be obtained by semiclassical methods.

Discussion. In an *adiabatic limit* one makes an operator semiclassical in some directions only. For a tower $\tilde{M} \to M \to B$ one lets the kernels concentrate at the diagonal in the directions of M/B while keeping them general along the fibres of $\tilde{M} \to M$. The limiting symbol is then not a matrix-valued function but a family of operators on the fibres of $\tilde{M} \to M$, parametrized by covectors of M/B. Quantizing first in the fibres and then in the

base, or all at once, are two faces of one family, and the index cannot change between them; this is the multiplicativity (16.8.1).

16.8.1 Iterated fibrations

Consider a tower $\tilde{M} \xrightarrow{\tilde{\phi}} M \xrightarrow{\phi} B$ of fibrations, $\psi = \phi \circ \tilde{\phi}$. A class $\tilde{x} \in K_{\mathrm{c}}(T^*(\tilde{M}/B))$ can be quantized in two steps. First, in the fibres of $\tilde{\phi}$ only: this requires the fibres of $\tilde{\phi}$ to carry, as parameters, the covectors in $T^*(M/B)$, i.e. one considers the pulled back fibration $\tilde{\phi}^* : {}^{\mathrm{sl}}T^*(M/B) \times_M \tilde{M} \to {}^{\mathrm{sl}}T^*(M/B)$, and quantizes $\tilde{x}$ semiclassically along its fibres. This gives a class $\mathrm{Ind}_{\mathrm{sl}}^{\tilde{\phi}^*}(\tilde{x}) \in K_{\mathrm{c}}(T^*(M/B))$ (compactly supported in the parameters). Second, one applies $\mathrm{Ind}_{\mathrm{sl}}^{M/B}$. Melrose's multiplicativity theorem (his Proposition 12.15) states that

$$\mathrm{Ind}_{\mathrm{sl}}^{\psi} = \mathrm{Ind}_{\mathrm{sl}}^{\phi} \circ \mathrm{Ind}_{\mathrm{sl}}^{\tilde{\phi}^*} . \tag{16.8.1}$$

For the tower $\mathcal{V} \to M \to B$ and $\tilde{x} = \lambda_{\mathcal{V}} \cdot \varpi^* x$ the inner index is $\mathrm{Ind}_{\mathrm{sl}}^{\mathcal{V}/M}(\lambda_{\mathcal{V}}) \cdot x = x$ by Theorem 16.6.1 and the module property (applied with parameters in $T^*(M/B)$), so (16.8.1) contains (16.7.1).

16.8.2 The adiabatic calculus

To compare the two quantizations one uses two parameters: ϵ, the semiclassical parameter in the directions of M/B, and δ, an adiabatic parameter in the fibres of $\tilde{\phi}$. For a single fibration $Y \to Z$ with fibre $\tilde{Z}$ (the case $B = \mathrm{pt}$), an *adiabatic family* is a family $A(\delta)$ of smoothing operators on Y whose kernels vanish to infinite order as $\delta \downarrow 0$ away from the fibre diagonal $Y \times_Z Y$ and near it have the form

$$\delta^{-m} A'\Big(\delta, z, \frac{z - z'}{\delta}, \tilde{z}, \tilde{z}'\Big), \quad m = \dim Z,$$

in local product coordinates, with A' smooth, compactly supported in z and Schwartz in $(z - z')/\delta$, and smooth in the fibre variables $\tilde{z}$, $\tilde{z}'$. They form an algebra with a multiplicative symbol sequence

$$\delta\Psi_{\mathrm{ad}}^{-\infty}(Y; E) \hookrightarrow \Psi_{\mathrm{ad}}^{-\infty}(Y; E) \xrightarrow{\sigma_{\mathrm{ad}}} \mathcal{S}\big({}^{\mathrm{sl}}T^*Z; \Psi^{-\infty}(\pi^*Y/{}^{\mathrm{sl}}T^*Z; E)\big),$$

the adiabatic symbol being a Schwartz family, parametrized by the semiclassical cotangent bundle of the base, of smoothing operators on the fibres $\tilde{Z}$. If a is an invertible-at-infinity adiabatic symbol, its adiabatic quantizations are invertible for small δ, and this defines an index $K_{\mathrm{c}}^{-1}(T^*Z) \to K^{-1}(\mathrm{pt})$ with *operator-valued* symbols; by deforming a to finite rank (Exercise 16.2) one sees that it agrees with the semiclassical index (Melrose's Proposition 12.14). Combining the two parameters gives the adiabatic–semiclassical algebra, whose kernels are of the form $\delta^{-n-k} A(\delta, \epsilon, (\tilde{z} - \tilde{z}')/\epsilon, z, (z - z')/(\epsilon\delta))$ near the fibre diagonal, with two symbol maps (at $\epsilon = 0$ and at $\delta = 0$) and a joint symbol; the relevant cotangent bundle ${}^{\mathrm{sl\,ad}}T^*\tilde{M}$ is spanned by df/ϵ and $dg/(\epsilon\delta)$, $f \in \mathcal{C}^\infty(\tilde{M})$, $g \in \mathcal{C}^\infty(Z)$, and restricts to ${}^{\mathrm{sl}}T^*\tilde{M}$, to the adiabatic cotangent bundle and to $T^*\tilde{M}$ on the three faces. The multiplicativity (16.8.1) is then the statement that a family in this algebra quantizing $\tilde{x}$ restricts, at the two faces, to the two composite quantizations, and that the index is constant along the family.

For the Thom tower the same idea can be made concrete in the language of this chapter. Let A be a semiclassical quantization of the symbol of x on M/B, lifted to $\mathcal{V}$ by a connection, and let $\mathcal{D}_\epsilon$ be the transported, rescaled Koszul operator of Section 16.6 on the fibres of $\mathcal{V}$, whose null space is spanned by the transported Gaussian at scale $\epsilon^{1/2}$ and which is

surjective. This is the hypothesis (H) of the Section 15.2 ((15.2.4)), and the Clifford tensor product $\mathcal{D}_\epsilon \otimes A$ formed there has relative index equal to that of A, by the same linear algebra. The remaining step is to show that the idempotent of Lemma 16.3.1 built from $\mathcal{D}_\epsilon \otimes A$ is homotopic, through compactly supported (or Schwartz) idempotent perturbations of p_0, to a semiclassical quantization on $\mathcal{V}/B$ of the idempotent of $\lambda_{\mathcal{V}} \cdot \varpi^* x$. In the analytic proof this step required the product-type calculus; here it requires the estimate that the commutators of the lifted operators with $\mathcal{D}_\epsilon$ are $O(\epsilon^{1/2})$ relative to $\mathcal{D}_\epsilon$, which is what the joint semiclassical–isotropic calculus provides. We do not develop that calculus here.

16.9 The Chern character of the index bundle

Since $\mathrm{ind_a} = \mathrm{Ind_{sl}} = \mathrm{ind_t}$, the cohomological form of the index theorem,

$$\mathrm{Ch}(\mathrm{ind_a}\, x) = (-1)^n \pi_* \big(\mathrm{Ch}(x) \wedge \pi_0^* \,\mathrm{Td}(T(M/B) \otimes \mathbb{C})\big),$$

follows from the K-theoretic form exactly as in Theorem 15.8.1, with the same inputs and the same conventions (multiplicativity of Ch for products, compatibility with extension and pull-back, the integral of the Chern character of b_1, and the Riemann–Roch identity for the Thom class). We add two remarks on the semiclassical point of view.

Remark 16.9.1 (The odd form). Melrose states the result for the odd index: for $a \in K_{\mathrm{c}}^{-1}(T^*(M/B))$, $\mathrm{Ch_{odd}}(\mathrm{Ind}_{\mathrm{sl}}^{odd}(a))$ is the fibre integral of $\mathrm{Ch_{odd}}(a)$ against the Todd class of the fibration. The two forms are equivalent. Indeed, by Proposition 16.4.2, $\mathrm{Ind_{sl}}(x) = \mathrm{ind}_{\mathcal{T}}(\mathrm{Ind}_{\mathrm{sl}}^{odd,\mathbb{S}}[\ell_x])$, and Proposition 15.6.1 shows that the Chern character of a Toeplitz index is the integral over the circle of the odd Chern character of the loop; the same computation, now with compact supports, identifies the integral over the circle of the odd Chern character of ℓ_x with the Chern character of x (we do not carry this out). The signs depend on the orientation conventions, which are fixed in Chapter 15.

Remark 16.9.2 (The Todd class). The derivation shows that $\mathrm{Ch}(\mathrm{ind_a}\, x)$ is the fibre integral of $\mathrm{Ch}(x)$ against a class $\mathcal{T} \in H^{\mathrm{ev}}(M)$ obtained by integrating $\mathrm{Ch}(\lambda_{\mathcal{V}})$ over the fibres of $T^*(\mathcal{V}/M)$, where $\mathcal{V}$ is the normal bundle of an embedding. That $\mathcal{T}$ does not depend on the embedding follows from Theorem 16.7.1 (for B a point and suitable x) or directly: the class is stable under adding trivial bundles, since $\mathrm{Ch}(\lambda_{\mathbb{R}})$ integrates to a constant, and any two normal bundles are stably isomorphic, being stable inverses of $T(M/B)$. Its identification with the Todd class of $T(M/B) \otimes \mathbb{C}$ is the Riemann–Roch computation of Atiyah and Hirzebruch recalled in Chapter 15.

Remark 16.9.3 (Dirac families and spectral sections). For families of Dirac operators the index formula can be refined to the level of differential forms, through the heat kernel and Bismut's superconnection; see [19] and Chapter 17, where eta forms appear in the index theorem for families with boundary. In that setting one also needs *spectral sections* (smooth families of projections which differ from the positive spectral projections of a self-adjoint family by smoothing operators; [16]): such a family exists if and only if the odd index of the boundary family vanishes in K^1 of the base, an obstruction which is the odd counterpart of the even index studied here. These topics are not pursued in this chapter.

16.10 Exercises

Exercise 16.1 (Strong contractibility of the unitary group)**.** Let $H = L^2([0,1])$ and $\mathrm{U}(H)$ its unitary group. For $U \in \mathrm{U}(H)$ and $t \in (0,1]$ let $\kappa_t : L^2([0,t]) \to L^2([0,1])$, $(\kappa_t v)(x) = t^{1/2} v(tx)$, and define $U_t = \kappa_t^{-1} U \kappa_t \oplus \mathrm{Id}_{L^2([t,1])}$, $U_0 = \mathrm{Id}$. Show that each U_t

is unitary, that $\|U_t v - v\| \le 2\|v\|_{L^2([0,t])}$, that $(UV)_t = U_t V_t$, and that $(U,t,v) \mapsto U_t v$ is continuous on $\mathrm{U}(H) \times [0,1] \times H$ when $\mathrm{U}(H)$ has the strong topology. Conclude that $\mathrm{U}(H)$ is contractible in the strong topology. (By Kuiper's theorem it is even contractible in the norm topology; this is not needed.)

Exercise 16.2 (Smooth orthonormal frames)**.** Let $\phi : M \to B$ be a fibration of compact manifolds and $E \to M$ Hermitian. (a) Using an admissible embedding J (Lemma 16.2.2) and the finite rank approximation (B2) on $\mathbb{T}^N$, construct smooth families of finite rank orthogonal projections $\pi_k \in \Psi^{-\infty}(M/B; E)$ with $\|\pi_k A \pi_k - A\| \to 0$ uniformly on B for every $A \in \Psi^{-\infty}(M/B; E)$. (Hint: compress $J^* P_V J$ and use the Riesz projection near 1 of $J^* P_V J$ for V large.) (b) Alternatively, following Melrose, use Exercise 16.1 and the fact that a fibre bundle with contractible fibre over a compact manifold has a section to find continuous sections e_i' of the Hilbert bundle forming an orthonormal basis in each fibre; approximate them by smooth sections e_i'' with $\sup_b \|e_i' - e_i''\| \le 2^{-i-4}$, and show that the operator $T_b u = \sum_i (u, e_i'(b)) e_i''(b)$ satisfies $\|T_b - \mathrm{Id}\| \le \frac{1}{4}$, so that Gram–Schmidt applied to the e_i'' gives a smooth orthonormal basis.

Exercise 16.3. Let P, Q be orthogonal projections on a Hilbert space with $P - Q$ compact. Show that $QP : \operatorname{Ran} P \to \operatorname{Ran} Q$ is Fredholm, and that for a pair of smooth families as in Definition 16.2.6 (with B a point) its index equals $d(P,Q)$. (Reduce, by a homotopy and finite rank approximation, to the case $P = Q + \Pi$ with Π of finite rank and $Q\Pi = 0$.)

Exercise 16.4. In Lemma 16.3.1, show that if $R_\pm$ are trace class then $\mathrm{Tr}(e_{A,B} - p_0) = \mathrm{Tr}(R_+^2) - \mathrm{Tr}(R_-^2)$, and that $\mathrm{Tr}(R_+^2) - \mathrm{Tr}(R_-^2) = \mathrm{Tr}(R_+) - \mathrm{Tr}(R_-)$ (use $AR_+ = R_- A$ and $\mathrm{Tr}[A, BR_-] = 0$ when the commutator is a commutator of a bounded and a trace class operator). Deduce Proposition 16.5.1 for B a point from the trace formula of Theorem 12.1.4.

Exercise 16.5. Let $A \in \Psi_{\mathrm{sl}}^{-\infty}(\mathbb{R}^n)$, $u \in \mathcal{C}_{\mathrm{c}}^\infty(\mathbb{R}^n)$ and $f \in \mathcal{C}^\infty(\mathbb{R}^n)$ real-valued. Show that

$$e^{-if/\epsilon} A(\epsilon)(e^{if/\epsilon} u) \longrightarrow \sigma_{\mathrm{sl}}(A)(z, df(z))\, u(z) \quad \text{in } \mathcal{C}^\infty(\mathbb{R}^n) \text{ as } \epsilon \downarrow 0,$$

and use this to show that σ_{sl} is invariant under changes of coordinates, in the sense of (S3). Discuss what happens if ϵ is replaced by 2ϵ (Remark 16.1.4).

Exercise 16.6. Let $\mathcal{V} \to Z$ be a real vector bundle with a metric over a compact manifold, and $\gamma \in \mathcal{S}(\mathcal{V})$ the fibrewise normalized Gaussian. Show that $A \mapsto \Gamma A \Gamma^*$, $\Gamma u = \gamma \cdot \varpi^* u$, embeds $\Psi^{-\infty}(Z; E)$ as a subalgebra of the operators on $\mathcal{V}$ with Schwartz kernels (Melrose's Proposition 12.3), that it maps $G^{-\infty}(Z; E)$ into $\mathrm{Id} + \Psi_{\mathcal{S}}^{-\infty}$, and that it preserves the classes in $K^{-1}(\mathrm{pt})$ of Definition 16.2.4.

Exercise 16.7. For $M = \mathbb{T}$, $B = \mathrm{pt}$, compute $\mathrm{Ind}_{\mathrm{sl}}$ of the class of the symbol of the operator $P_1 = e^{-i\theta}\Pi_> + (\mathrm{Id} - \Pi_>)$ of (15.4.2) directly: show that a symbol representing $[\sigma_0(P_1)]$ can be chosen equal to 1 for $\xi < -1$ and to $e^{-i\theta}$ for $\xi > 1$, construct $e_{a,b}$, and show that the relative index of its quantization is 1 by deforming it to the case $\epsilon = 1$, where the quantization is a Fourier multiplier composed with multiplication operators.

Notes

This chapter is based on Chapter 12 of R. Melrose's notes *From microlocal to global analysis* (2008), which outlines a proof of the Atiyah–Singer index theorem through semiclassical and adiabatic limits; many steps there are only sketched, and several sections are empty. We have reorganized the material around a definition of the even semiclassical index as a

relative index of idempotents, computed by the Toeplitz index of a loop (Definitions 16.2.6 and 16.3.3); with it, the comparison with the analytic index (Theorem 16.5.4) reduces to Milnor's explicit form of the index idempotent (Lemma 16.3.1) and the multiplicativity of the semiclassical symbol, in place of the combined 'Toeplitz' calculus sketched in the source. The passage from families of operators on the fibres to classes in $K^{-1}(B)$ is made through an explicit isometric embedding of the fibres (Lemma 16.2.2) instead of Melrose's smooth orthonormal frames, whose construction (via strong contractibility of the unitary group and the existence of sections of bundles with contractible fibres) is left to Exercises 16.1 and 16.2. The index of the Thom class (Theorem 16.6.1) is computed from the harmonic oscillator, as suggested by the source; this also completes the proof in Chapter 15. The multiplicativity of the semiclassical index under iterated fibrations, via the adiabatic calculus, is only outlined (Section 16.8); the final proof uses instead the analytic multiplicativity of Chapter 15. The source's sections on Dirac families and spectral sections are empty and are replaced by Remark 16.9.3.

The semiclassical approach to index theory has many antecedents; the use of the harmonic oscillator to produce the Thom isomorphism goes back to the proof of Bott periodicity by M. F. Atiyah, *Bott periodicity and the index of elliptic operators*, Quart. J. Math. **19** (1968), 113–140, and the K-theoretic setting is that of [17, 18]. For the semiclassical calculus see the references in Chapter 6; for the isotropic calculus and the harmonic oscillator, [15]. The idempotent (16.3.1) is the standard formula for the index map in algebraic K-theory, from J. Milnor, *Introduction to algebraic K-theory*, Ann. of Math. Studies **72** (1971).

Chapter 17

Eta forms and the index of product-type families

A fully elliptic product-type operator has two symbols: an ordinary principal symbol, and a base family of invertible operators on the fibres. The Atiyah–Singer formula integrates a characteristic form built from the principal symbol. One would like to add a similar contribution from the base family, namely its odd Chern character $\mathrm{Tr}((B^{-1}dB)^{2k+1})$. But $B^{-1}dB$ is a pseudodifferential operator of order 0, not a trace class operator, so this trace does not exist. This chapter shows how to regularize it, and that the regularized form, the *eta form*, is exactly the missing term in the index formula.

The index formula for product-type operators involves, besides the familiar Atiyah–Singer integrand, a 'regularized odd Chern character' of the base family, which we call an eta form. In this chapter we first construct the necessary regularization of the trace on a compact manifold, using holomorphic families of pseudodifferential operators: Seeley's theorem on the meromorphic continuation of traces, the residue trace, the regularized trace and the trace-defect formula (these are the analogues for compact manifolds of the constructions for the isotropic algebra in Chapter 8). We then specialize to the circle and define eta forms on the group of invertible Toeplitz operators, compute their exterior derivative and relate them to the odd Chern forms and the Fredholm determinant of Chapter 12. The main result, Theorem 17.8.1, is an index formula for fully elliptic operators of Toeplitz product type on $M \times \mathbb{S}$; the proof uses the product-type calculus of Chapter 14, the weak contractibility of the Toeplitz group from Chapter 13 and the Atiyah–Singer theorem of Chapter 15. A version for the full (non-Toeplitz) product-type calculus follows. The last section discusses, more briefly and partly conjecturally, the general product-type index problem, a corresponding cohomology theory and the determinant line bundle. The eta forms here are close relatives of the eta invariant of Atiyah–Patodi–Singer [16] and of the eta forms of Bismut and Cheeger [19]; see also [20].

Throughout, Z denotes a compact manifold without boundary of dimension $d \geq 1$ and $E \longrightarrow Z$ a complex vector bundle. We fix a smooth positive density on Z and a Hermitian structure on E, so that operators act on $L^2(Z;E)$; the trace does not depend on these choices.

17.1 The trace on operators of low order

The regularization is modelled on the zeta function: $\sum_{k\geq 1} k^{-s}$ diverges at $s=0$, but it converges for $\Re s > 1$ and has a meromorphic continuation, whose value at $s=0$ is a sensible substitute for the divergent sum. To imitate this we multiply an operator A by a family

$E(s)$ of order s, take the trace where it converges, and continue in s. For this we need operators of complex order.

17.1.1 Operators of complex order

Recall (Chapters 1 and 4) that the classical symbols of order m on $\mathbb{R}^d$ are exactly the functions of the form $\rho^{-m} b$ with $b \in \mathcal{C}^\infty(\overline{\mathbb{R}^d})$, where $\overline{\mathbb{R}^d}$ is the radial compactification and ρ is a boundary defining function, for instance $\rho = \langle \xi \rangle^{-1}$. This description makes sense for any $m \in \mathbb{C}$: we say that $a \in \mathcal{C}^\infty(U \times \mathbb{R}^d; M(N,\mathbb{C}))$ is a *classical symbol of order* $m \in \mathbb{C}$ if $a = \rho^{-m} b$ with $b \in \mathcal{C}^\infty(U \times \overline{\mathbb{R}^d}; M(N,\mathbb{C}))$. Such a symbol has an asymptotic expansion $a \sim \sum_{j \geq 0} a_{m-j}$ with a_{m-j} homogeneous of degree $m-j$ in $|\xi| \geq 1$, obtained from the Taylor series of b at the boundary. Correspondingly $\Psi^m(Z;E)$, $m \in \mathbb{C}$, denotes the operators whose kernels are smooth away from the diagonal and which in any coordinate chart U over which E is trivial, after cut-off, are of the form $\operatorname{Op}(a)$ with a a classical symbol of order m (left quantization), modulo smoothing operators. The composition theorem and the symbol calculus of Chapter 5 hold without change for complex orders: $\Psi^m \circ \Psi^{m'} \subset \Psi^{m+m'}$, principal symbols multiply and $\Psi^m \subset \Psi^{m'}$ if $\Re m' - \Re m \in \mathbb{N}_0$ and $m' - m \in \mathbb{N}_0$. We write

$$\Psi^{\mathbb{Z}}(Z;E) = \bigcup_{m \in \mathbb{Z}} \Psi^m(Z;E)$$

for the algebra of classical operators of integral order.

Lemma 17.1.1. *If $A \in \Psi^m(Z;E)$ with $\Re m < -d$ then A is of trace class on $L^2(Z;E)$, its kernel K_A is continuous and*

$$\operatorname{Tr}(A) = \int_Z \operatorname{tr}_E K_A(z,z).$$

If U is a coordinate chart over which E is trivialized, $\phi, \psi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$ and $A = \phi \operatorname{Op}(a) \psi$ with a a symbol of order m, then

$$\operatorname{Tr}(A) = (2\pi)^{-d} \int_U \int_{\mathbb{R}^d} \phi(z)\psi(z) \operatorname{tr} a(z,\xi)\, d\xi\, dz. \tag{17.1.1}$$

Proof. Using a partition of unity $\{\phi_i\}$ subordinate to coordinate charts and $\psi_i \in \mathcal{C}^\infty_{\mathrm{c}}$ equal to 1 near $\operatorname{supp}\phi_i$, write $A = \sum_i \phi_i A \psi_i + R$ with $R = \sum_i \phi_i A(1-\psi_i)$ smoothing. Smoothing operators on a compact manifold are of trace class with trace given by the integral of the kernel over the diagonal, so it suffices to consider $A = \phi \operatorname{Op}(a)\psi$ in a chart. Write $a = a' \langle \xi \rangle^{m/2}$ with $a' = a \langle \xi \rangle^{-m/2}$ of order $m/2$. Since the second factor is a Fourier multiplier, $\operatorname{Op}(a')\operatorname{Op}(\langle \xi \rangle^{m/2}) = \operatorname{Op}(a)$ exactly in the left quantization, so $A = T_1 T_2$ with $T_1 = \phi \operatorname{Op}(a')$, $T_2 = \operatorname{Op}(\langle \xi \rangle^{m/2})\psi$. The kernel of T_2 is $k(z-z')\psi(z')$ with $\hat{k} = \langle \xi \rangle^{m/2} \in L^2(\mathbb{R}^d)$ since $\Re m < -d$; hence $\int\int |k(z-z')\psi(z')|^2 dz\, dz' = \|k\|^2 \|\psi\|^2$ and T_2 is Hilbert–Schmidt. The kernel of T_1 is $\phi(z) \int e^{i(z-z')\cdot\xi} a'(z,\xi) d\!\!{}^-\xi$ which, for each z, is in L^2 in z' with norm $\|a'(z,\cdot)\|_{L^2}$, uniformly bounded; as ϕ has compact support T_1 is Hilbert–Schmidt. Thus A is of trace class and $\operatorname{Tr}(A) = \int\int K_{T_1}(z,z') K_{T_2}(z',z)\, dz'\, dz$. Since a is integrable in ξ, locally uniformly in z, the kernel $K_A(z,z') = \phi(z) \int e^{i(z-z')\cdot\xi} a(z,\xi) d\!\!{}^-\xi\, \psi(z')$ is continuous and the double integral equals $\int K_A(z,z)\, dz$, which is (17.1.1). □

Lemma 17.1.2. *If $A \in \Psi^{m_1}(Z;E)$, $B \in \Psi^{m_2}(Z;E)$ and $\Re(m_1+m_2) < -d$ then $\operatorname{Tr}(AB) = \operatorname{Tr}(BA)$.*

Proof. If A is smoothing then AB and BA are smoothing and both traces are the pairing of $K_A(z,z')$ with $K_B(z',z)$, so they agree. In general choose $\delta > 0$ with $\Re(m_1+m_2) + \delta < -d$

and smoothing operators $A_\epsilon \to A$ in $\Psi^{m_1+\delta}$ as $\epsilon \to 0$ (for instance by cutting off the local symbols with $\chi(\epsilon\xi)$, $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R}^d)$ equal to 1 near 0). Composition is continuous and, by (17.1.1), the trace is continuous on Ψ^m for $\Re m < -d$, so $\operatorname{Tr}(AB) = \lim \operatorname{Tr}(A_\epsilon B) = \lim \operatorname{Tr}(BA_\epsilon) = \operatorname{Tr}(BA)$. □

17.2 Holomorphic families

The space $\Psi^m(Z;E)$ of fixed order has a natural Fréchet topology and there is no difficulty in speaking of holomorphic maps $\Omega \longrightarrow \Psi^m(Z;E)$ for $\Omega \subset \mathbb{C}$ open. We need families in which the order varies holomorphically as well; these are not maps into a fixed space.

Definition 17.2.1. Let $\Omega \subset \mathbb{C}$ be open and $\mu : \Omega \longrightarrow \mathbb{C}$ holomorphic. A family $A(s) \in \Psi^{\mu(s)}(Z;E)$, $s \in \Omega$, is a *holomorphic family of order* μ if

(i) for every $\chi \in \mathcal{C}^\infty(Z^2)$ vanishing near the diagonal, $\chi K_{A(s)}$ is a holomorphic map from Ω to $\mathcal{C}^\infty(Z^2; \operatorname{Hom}(E) \otimes \Omega_R)$, and

(ii) for every coordinate chart U over which E is trivial and all $\phi, \psi \in \mathcal{C}^\infty_{\mathrm{c}}(U)$,

$$\phi A(s) \psi = \operatorname{Op}(\rho^{-\mu(s)} b(s)) + R(s),$$

where $b \in \mathcal{C}^\infty(\Omega \times U \times \overline{\mathbb{R}^d}; M(N,\mathbb{C}))$ is holomorphic in s with support in a fixed compact subset of U in z, and $R(s)$ is a holomorphic family of smoothing operators.

Changing the boundary defining function ρ to $\rho' = f\rho$, $f > 0$ smooth, replaces $b(s)$ by $f^{\mu(s)} b(s)$, which is again smooth and holomorphic in s; so the definition does not depend on ρ. Its independence of the coordinates follows, exactly as for a single operator in Chapter 5, from the fact that the coordinate-change formula for symbols is an asymptotic sum whose terms are obtained from the symbol by applying differential operators in (z,ξ) and multiplying by smooth functions; these operations preserve $\rho^{-\mu(s)}\mathcal{C}^\infty(\Omega \times U \times \overline{\mathbb{R}^d})$, with holomorphic dependence, and the asymptotic summation can be carried out holomorphically (Borel's lemma with cut-offs chosen independently of s, uniformly on compact subsets of Ω). For μ constant this is the usual notion of a holomorphic family. In the same way one obtains:

Lemma 17.2.2. *If $A(s) \in \Psi^{\mu(s)}(Z;F,G)$ and $B(s) \in \Psi^{\nu(s)}(Z;E,F)$ are holomorphic families on Ω then $A(s)B(s)$ is a holomorphic family of order $\mu+\nu$.*

Sketch. Off the diagonal this follows from the corresponding property of compositions with smoothing operators. Near the diagonal, in local coordinates, the symbol of the composite has the expansion $\sum_\alpha \frac{1}{\alpha!}\partial_\xi^\alpha a(s) D_z^\alpha b(s)$, each term lying in $\rho^{-\mu(s)-\nu(s)}\mathcal{C}^\infty$ holomorphically, and the remainder after J terms satisfies symbol estimates of order $\Re(\mu+\nu) - J$ locally uniformly in s; by Cauchy's formula the same holds for its s-derivatives, which gives holomorphy of the composite symbol. □

We also need a division property.

Lemma 17.2.3. *If $F(s)$ is a holomorphic family of order μ on Ω and $F(s_0) = 0$ for some $s_0 \in \Omega$, then $F(s) = (s - s_0)F_1(s)$ with F_1 a holomorphic family of order μ.*

Proof. Off the diagonal $\chi K_{F(s)}$ is holomorphic and vanishes at s_0, so is divisible by $s - s_0$. Near the diagonal $\phi F(s)\psi = \operatorname{Op}(\rho^{-\mu(s)} b(s)) + R(s)$, and $\rho^{-\mu(s_0)} b(s_0)$ is a symbol of order $-\infty$, so $b(s_0)$ vanishes to infinite order at the boundary; subtracting $\rho^{-\mu(s)}\rho^{\mu(s_0)} b(s_0)$

(which changes R by a holomorphic smoothing family vanishing at s_0) we may assume $b(s_0) = 0$ and then $b(s) = (s - s_0)b_1(s)$ with b_1 smooth and holomorphic. Similarly $R(s) = (s - s_0)R_1(s)$. □

Proposition 17.2.4. *There is an entire family $E(s) \in \Psi^s(Z; E)$, $s \in \mathbb{C}$, with*

$$E(0) = \mathrm{Id}$$

and principal symbol $|\xi|_g^s \mathrm{Id}_E$ for a Riemannian metric g; in particular $E(s)$ is elliptic for every s.

Proof. Choose a finite cover by coordinate charts U_i over which E is trivialized, a subordinate partition of unity ϕ_i, functions $\psi_i \in \mathcal{C}_{\mathrm{c}}^\infty(U_i)$ equal to 1 near $\operatorname{supp}\phi_i$, and a metric g. Set

$$E(s) = \sum_i \phi_i \operatorname{Op}\big((1 + |\xi|^2_{g(z)})^{s/2} \mathrm{Id}\big)\psi_i.$$

This is an entire family of order s with principal symbol $|\xi|_g^s \mathrm{Id}$, and since $\operatorname{Op}(1) = \mathrm{Id}$, $E(0) = \sum_i \phi_i\psi_i = \sum_i \phi_i = \mathrm{Id}$. □

Using Seeley's complex powers [11] one can also take $E(s) = (\Delta + 1)^{s/2}$, with Δ a Laplacian, which is in addition invertible for all s; we shall not need this. We call an entire family $E(s)$ of order s with $E(0) = \mathrm{Id}$ *admissible*.

Theorem 17.2.5 (Seeley)**.** *Let $A(s)$ be a holomorphic family of order $m + s$, $m \in \mathbb{C}$, on a connected open set $\Omega \subset \mathbb{C}$ which meets the half-plane $\{\Re s < -d - \Re m\}$. Then the function $\operatorname{Tr}(A(s))$, holomorphic on $\Omega \cap \{\Re s < -d - \Re m\}$, extends to a meromorphic function on Ω with at most simple poles, located in $-d - m + \mathbb{N}_0$. The operator $A(s_j)$ at $s_j = j - d - m$ has integral order $j - d$, and the residue there is*

$$\operatorname*{Res}_{s=s_j} \operatorname{Tr}(A(s)) = -\operatorname{res}(A(s_j)), \tag{17.2.1}$$

where for $B \in \Psi^k(Z; E)$, $k \in \mathbb{Z}$,

$$\operatorname{res}(B) = (2\pi)^{-d} \sum_i \int_{U_i} \phi_i(z) \int_{|\xi|=1} \operatorname{tr} b^{(i)}_{-d}(z, \xi)\, dS(\xi)\, dz,$$

$b^{(i)}_{-d}$ being the term of homogeneity $-d$ in the symbol of B in the chart U_i (and 0 if $k < -d$); here $\{\phi_i\}$ is a partition of unity subordinate to charts U_i over which E is trivial.

Proof. As in the proof of Lemma 17.1.1, $A(s) = \sum_i \phi_i A(s)\psi_i + R(s)$ where $R(s)$ is a holomorphic family of smoothing operators, so $\operatorname{Tr}(R(s))$ is entire. By the definition, $\phi_i A(s)\psi_i = \operatorname{Op}(\rho^{-m-s} b_i(s))$ modulo holomorphic smoothing families, and we may choose $\rho = |\xi|^{-1}$ in $|\xi| \geq 1$. By (17.1.1), for $\Re s < -d - \Re m$,

$$\begin{aligned}\operatorname{Tr}(\phi_i A(s)\psi_i) = (2\pi)^{-d} \int_U \int_{|\xi|\leq 1} \operatorname{tr}(\cdots)\, d\xi\, dz \\ + (2\pi)^{-d} \int_U \int_{\mathbb{S}^{d-1}} \int_0^1 r^{-m-s-d-1} \operatorname{tr} b_i(s, z, r, \omega)\, dr\, d\omega\, dz,\end{aligned}$$

where we have used polar coordinates $\xi = r^{-1}\omega$ in $|\xi| \geq 1$, so that $d\xi = r^{-d-1} dr\, d\omega$, and b_i is smooth in $r \in [0, 1]$ (this is the meaning of smoothness on $\overline{\mathbb{R}^d}$). The first term is entire.

Taylor expanding, $b_i(s,z,r,\omega)=\sum_{j<J} b_{i,j}(s,z,\omega)r^j + r^J b_{i,J}(s,z,r,\omega)$ with all coefficients smooth and holomorphic in s, so

$$\int_0^1 r^{-m-s-d-1}b_i\,dr = \sum_{j<J}\frac{b_{i,j}(s,z,\omega)}{j-d-m-s} + \int_0^1 r^{J-m-s-d-1}b_{i,J}\,dr,$$

and the last integral is holomorphic in $\Re s < J-d-\Re m$. Since J is arbitrary this gives the meromorphic continuation, with at most simple poles at the points s_j, and residue $-(2\pi)^{-d}\sum_i\int\int \operatorname{tr} b_{i,j}(s_j,z,\omega)\,d\omega\,dz$ there. Now $r^{-m-s_j+j}b_{i,j}(s_j,z,\omega)=|\xi|^{-d}b_{i,j}(s_j,z,\omega)$ is precisely the term of homogeneity $-d$ in the symbol of $\phi_i A(s_j)\psi_i$. Finally the full symbol of $\phi_i B\psi_i$ is ϕ_i times the local symbol of B modulo $S^{-\infty}$, because $\psi_i\equiv 1$ near $\operatorname{supp}\phi_i$; this gives (17.2.1). □

Discussion. The proof shows where the poles come from. In polar coordinates the trace is an integral $\int^\infty r^{m+s+d-1}(\cdots)\,dr$ over the size $r=|\xi|$ of the covector. Each homogeneous term of the symbol contributes a power of r, and $\int_1^\infty r^{\lambda-1}dr=-1/\lambda$ continues meromorphically with a simple pole at $\lambda=0$. The pole at $s=s_j$ comes from the one term of the symbol which, at $s=s_j$, has homogeneity exactly $-d$, the borderline case for integrability in $\xi\in\mathbb{R}^d$. Its residue is therefore a *local* quantity, an integral over the cosphere bundle of one term of the symbol.

17.3 The residue trace and the regularized trace

Seeley's theorem gives two numbers for each operator A of integral order: the residue of $\operatorname{Tr}(AE(s))$ at $s=0$ and its finite part. The residue is a genuine trace on the whole algebra, but it vanishes on trace class operators. The finite part extends the ordinary trace, but it is not a trace. The interplay between the two, the trace-defect formula (Lemma 17.3.7), is what produces the eta forms.

Proposition 17.3.1. *For $A\in\Psi^m(Z;E)$, $m\in\mathbb{Z}$, and any holomorphic family $A(s)$ of order $m+s$ on a connected open set containing 0 and meeting $\{\Re s<-d-m\}$, with $A(0)=A$,*

$$\operatorname*{Res}_{s=0}\operatorname{Tr}(A(s)) = -\operatorname{res}(A).$$

The residue trace $\operatorname{Tr}_{\mathrm{R}}(A):=\operatorname{Res}_{s=0}\operatorname{Tr}(AE(s))$, *$E$ admissible, therefore does not depend on E, and*

$$\operatorname{Tr}_{\mathrm{R}}(A) = -\operatorname{res}(A) = -(2\pi)^{-d}\int_{S^*Z}\operatorname{tr}\sigma_{-d}(A), \tag{17.3.1}$$

where $\sigma_{-d}(A)$ is the homogeneous term of degree $-d$ in the symbol of A and the integral means, in local coordinates, $\int\int_{|\xi|=1}\operatorname{tr}\sigma_{-d}(A)\,dS(\xi)\,dz$; the integrand is a well-defined density on S^*Z although $\sigma_{-d}(A)$ itself depends on the coordinates. Moreover $\operatorname{Tr}_{\mathrm{R}}$ vanishes on $\Psi^m(Z;E)$ for $m<-d$ and on $\Psi^{-\infty}(Z;E)$, and

$$\operatorname{Tr}_{\mathrm{R}}([A,B])=0 \quad \forall\ A,B\in\Psi^{\mathbb{Z}}(Z;E).$$

Proof. The first statement is (17.2.1) at $s_j=0$, since the expression for res only involves $A(0)=A$. By Lemma 17.2.2, $AE(s)$ is such a family. The formula (17.2.1) is a priori written with a partition of unity; applying it to an operator supported in the intersection of two charts shows that the density $\left(\int_{|\xi|=1}\operatorname{tr}a_{-d}(z,\xi)dS(\xi)\right)dz$ is independent of the coordinates (one can also check this directly, since for a function h homogeneous of degree $-d$ the form $\iota_R(h\,d\xi)$, R the radial vector field, is closed on $\mathbb{R}^d\setminus 0$). If $m<-d$, $AE(s)$ has order $m+s$ with $\Re(m+s)<-d$ near $s=0$, so there is no pole.

For the trace property, $\operatorname{Tr}(ABE(s)) = \operatorname{Tr}(BE(s)A)$ for $\Re s$ very negative by Lemma 17.1.2, hence for all s by meromorphic continuation, so

$$\operatorname{Tr}([A,B]E(s)) = \operatorname{Tr}(B[E(s),A]).$$

The family $[E(s),A]$ is holomorphic of order m_A+s and vanishes at $s=0$, so by Lemma 17.2.3 $[E(s),A] = sC(s)$ with C holomorphic. Since $\operatorname{Tr}(BC(s))$ has at most a simple pole at 0, $s\operatorname{Tr}(BC(s))$ is regular there, so $\operatorname{Tr_R}([A,B]) = 0$. □

Thus, with our convention that $E(s)$ has order $+s$, the residue trace is minus the Wodzicki–Guillemin residue [13, 5].

Definition 17.3.2. For an admissible family E the *regularized trace* of $A \in \Psi^{\mathbb{Z}}(Z;E)$ is the finite part

$$\overline{\operatorname{Tr}}_E(A) = \operatorname*{FP}_{s=0} \operatorname{Tr}(AE(s)) = \lim_{s\to 0}\Big(\operatorname{Tr}(AE(s)) - \frac{1}{s}\operatorname{Tr_R}(A)\Big).$$

Clearly $\overline{\operatorname{Tr}}_E$ is linear and $\overline{\operatorname{Tr}}_E(A) = \operatorname{Tr}(A)$ if $A \in \Psi^m$, $m < -d$, in particular on smoothing operators. From the proof of Theorem 17.2.5 the Laurent coefficients of $\operatorname{Tr}(AE(s))$ at $s=0$ are given by convergent integrals of the (Taylor coefficients of the) symbol of $AE(s)$, so $\overline{\operatorname{Tr}}_E$ is continuous on each $\Psi^m(Z;E)$. However $\overline{\operatorname{Tr}}_E$ is not a trace, and it depends on E.

Question. Is there a better choice: a trace on all of $\Psi^{\mathbb{Z}}(Z;E)$ which extends Tr?

Answer. No. On the circle, $[e^{i\theta}, e^{-i\theta}\Pi]$ is minus a rank one projection (Example 17.4.2), so any extension of Tr takes the value -1 on this commutator of two operators of order 0, and cannot vanish on all commutators. Hence one must choose between keeping the trace property, as $\operatorname{Tr_R}$ does at the cost of vanishing on trace class operators, and extending Tr, as $\overline{\operatorname{Tr}}_E$ does at the cost of the trace property. The defect is local in the symbols (Lemma 17.3.7), and this locality is the key to everything that follows. □

Lemma 17.3.3. *If E and E' are admissible then $E'(s) = E(s) + sF(s)$ with F an entire family of order s; setting $D(E,E') = F(0) \in \Psi^0(Z;E)$,*

$$\overline{\operatorname{Tr}}_{E'}(A) = \overline{\operatorname{Tr}}_E(A) + \operatorname{Tr_R}\big(AD(E,E')\big) \quad \forall\ A \in \Psi^{\mathbb{Z}}(Z;E).$$

Proof. The first statement is Lemma 17.2.3. Then $\operatorname{Tr}(AE'(s)) = \operatorname{Tr}(AE(s)) + s\operatorname{Tr}(AF(s))$; here $AF(s)$ is holomorphic of order $m+s$ with value $AD(E,E')$ at 0, so $s\operatorname{Tr}(AF(s))$ is regular at 0 with value $\operatorname{Tr_R}(AD(E,E'))$, while the residues of $\operatorname{Tr}(AE'(s))$ and $\operatorname{Tr}(AE(s))$ agree. □

Proposition 17.3.4. *The regularized traces $\overline{\operatorname{Tr}}_E$, as E ranges over admissible families, form an affine space modelled on $\Psi^0(Z;E)/\Psi^{-\infty}(Z;E)$: the map $E' \longmapsto [D(E,E')]$ identifies $\{\overline{\operatorname{Tr}}_{E'}\}$ with $\Psi^0/\Psi^{-\infty}$.*

Proof. Every $D \in \Psi^0$ occurs: $E'(s) = E(s) + sDE(s)$ is admissible with $D(E,E') = D$. By Lemma 17.3.3 it remains to see that the functional $A \mapsto \operatorname{Tr_R}(AD)$ on $\Psi^{\mathbb{Z}}$ vanishes if and only if $D \in \Psi^{-\infty}$. If D is smoothing so is AD. Conversely, if $D \notin \Psi^{-\infty}$ let $k \in \mathbb{N}_0$ be such that $D \in \Psi^{-k}$ has non-vanishing principal symbol $\sigma_{-k}(D)$. For $A \in \Psi^{k-d}$ the term of homogeneity $-d$ in the symbol of AD is $\sigma_{k-d}(A)\sigma_{-k}(D)$, so by (17.3.1)

$$\operatorname{Tr_R}(AD) = -(2\pi)^{-d}\int_{S^*Z} \operatorname{tr}(\sigma_{k-d}(A)\sigma_{-k}(D)).$$

Choosing $\sigma_{k-d}(A) = \chi\,\sigma_{-k}(D)^*$ (extended homogeneously) with $0 \le \chi$, $\chi > 0$ somewhere where $\sigma_{-k}(D) \neq 0$, makes this non-zero. □

17.3.1 The trace-defect formula

The failure of $\overline{\mathrm{Tr}}_E$ to be a trace is measured by a derivation.

Definition 17.3.5. For an admissible family E and $A \in \Psi^{\mathbb{Z}}(Z;E)$ let

$$D_E A = \frac{d}{ds}[E(s), A]\Big|_{s=0},$$

so that $[E(s), A] = sC(s)$ with C holomorphic of order $m+s$ and $C(0) = D_E A$.

Formally, if $E(s) = Q^s$ then $D_E A = [\log Q, A]$; $\log Q$ is not in the algebra, but the commutator is.

Proposition 17.3.6. *D_E is a derivation of $\Psi^{\mathbb{Z}}(Z;E)$ mapping Ψ^m into Ψ^m; if the principal symbol of $E(s)$ is $|\xi|_g^s \operatorname{Id}$ (as in Proposition 17.2.4) then $D_E(\Psi^m) \subset \Psi^{m-1}$ and*

$$\sigma_{m-1}(D_E A) = -i\{\log|\xi|_g, \sigma_m(A)\}. \tag{17.3.2}$$

For two admissible families, $D_{E'}A = D_E A + [D(E,E'), A]$. Moreover

$$\mathrm{Tr}_{\mathrm{R}}(D_E A) = 0 \quad \forall\ A \in \Psi^{\mathbb{Z}}(Z;E).$$

Proof. Differentiating $[E(s), AB] = [E(s),A]B + A[E(s),B]$ at $s=0$ gives the derivation property, and the difference formula by differentiating $[E(s)+sF(s), A]$. If the principal symbol e_s of $E(s)$ is scalar, the principal symbol of $[E(s),A]$ of order $m+s$ vanishes, and the term of order $m+s-1$ is

$$-i\{e_s, \sigma_m(A)\} = -is|\xi|_g^s\{\log|\xi|_g, \sigma_m(A)\};$$

dividing by s and setting $s=0$ gives (17.3.2). Finally, the family $C(s)$ has order $m+s$ and value $D_E A$ at 0, so $\mathrm{Tr}_{\mathrm{R}}(D_E A)$ is the residue of $\mathrm{Tr}(C(s)) = s^{-1}\mathrm{Tr}([E(s),A])$; but $\mathrm{Tr}([E(s),A]) = 0$ for $\Re s$ very negative by Lemma 17.1.2, hence identically. □

One can show that D_E is not an inner derivation (see Exercise 17.2 for the circle).

Lemma 17.3.7 (Trace-defect formula)**.** *For all $A, B \in \Psi^{\mathbb{Z}}(Z;E)$,*

$$\overline{\mathrm{Tr}}_E([A,B]) = \mathrm{Tr}_{\mathrm{R}}\big((D_E A)B\big) = \mathrm{Tr}_{\mathrm{R}}\big(BD_E A\big).$$

Proof. As in the proof of Proposition 17.3.1, $\mathrm{Tr}([A,B]E(s)) = \mathrm{Tr}(B[E(s),A]) = s\,\mathrm{Tr}(BC(s))$, which is regular at $s=0$ with value $\mathrm{Res}_{s=0}\mathrm{Tr}(BC(s)) = \mathrm{Tr}_{\mathrm{R}}(BD_E A)$. Since $\mathrm{Tr}_{\mathrm{R}}([A,B]) = 0$, this value is $\overline{\mathrm{Tr}}_E([A,B])$; the second equality is the trace property of Tr_{R}. □

We shall apply this to operator-valued differential forms. If X is a manifold and $\alpha = \sum_I A_I\, dx^I$, $\beta = \sum_J B_J\, dx^J$ are smooth forms of degrees p and q with values in $\Psi^{\mathbb{Z}}(Z;E)$, products are taken in the algebra and in the exterior algebra, $\overline{\mathrm{Tr}}_E$, Tr_{R} and D_E act on coefficients, and the graded commutator is $[\alpha,\beta] = \alpha\beta - (-1)^{pq}\beta\alpha = \sum_{I,J}[A_I, B_J]\, dx^I \wedge dx^J$. Hence:

Corollary 17.3.8. *For operator-valued forms α, β as above, $\overline{\mathrm{Tr}}_E([\alpha,\beta]) = \mathrm{Tr}_{\mathrm{R}}\big((D_E\alpha)\wedge\beta\big)$, and Tr_{R} vanishes on graded commutators. Moreover $d\,\overline{\mathrm{Tr}}_E(\alpha) = \overline{\mathrm{Tr}}_E(d\alpha)$, by continuity and linearity of $\overline{\mathrm{Tr}}_E$.*

17.4 The circle

On $\mathbb{S} = \mathbb{R}/2\pi\mathbb{Z}$, with variable θ and dual variable ξ, we use the admissible family

$$E(s)e^{ik\theta} = \lambda_k^s e^{ik\theta}, \qquad \lambda_k = (1+k^2)^{1/2}, \tag{17.4.1}$$

acting diagonally on $\mathcal{C}^\infty(\mathbb{S};\mathbb{C}^N)$. It is the Fourier multiplier with symbol $\langle\xi\rangle^s$, an entire family of invertible elliptic operators of order s with $E(0) = \mathrm{Id}$. From now on $\overline{\mathrm{Tr}} = \overline{\mathrm{Tr}}_E$ and D_E refer to this family. Here $S^*\mathbb{S} = \mathbb{S}\times\{\pm 1\}$.

Lemma 17.4.1. *(i) For $A \in \Psi^{-1}(\mathbb{S};\mathbb{C}^N)$ with symbol term a_{-1} of homogeneity -1,*

$$\mathrm{Tr}_{\mathrm{R}}(A) = -\frac{1}{2\pi}\int_0^{2\pi} \mathrm{tr}\big(a_{-1}(\theta,1) + a_{-1}(\theta,-1)\big)\,d\theta.$$

(ii) For $A \in \Psi^m(\mathbb{S};\mathbb{C}^N)$ with principal symbol a, $D_E A \in \Psi^{m-1}$ has principal symbol

$$\sigma_{m-1}(D_E A)(\theta,\xi) = -i\,\xi^{-1}\partial_\theta a(\theta,\xi);$$

in fact $D_E A$ has full symbol $\sum_{k\geq 1}\frac{1}{k!}\partial_\xi^k(\log\langle\xi\rangle)D_\theta^k a$.

(iii) If $f \in \mathcal{C}^\infty(\mathbb{S}; M(N,\mathbb{C}))$ acts by multiplication and Π is the Szegő projector then

$$\overline{\mathrm{Tr}}(f) = 0, \qquad \overline{\mathrm{Tr}}(\Pi f\Pi) = \frac{1}{4\pi}\int_0^{2\pi}\mathrm{tr}\, f\,d\theta;$$

in particular $\overline{\mathrm{Tr}}(\Pi) = \frac{1}{2}$, and $\mathrm{Tr}_{\mathrm{R}}(f) = \mathrm{Tr}_{\mathrm{R}}(\Pi f\Pi) = 0$.

Proof. (i) is (17.3.1) for $d = 1$. For (ii), since the symbol $e_s = \langle\xi\rangle^s$ depends only on ξ, $\mathrm{Op}(a)\,\mathrm{Op}(e_s) = \mathrm{Op}(ae_s)$ while $\mathrm{Op}(e_s)\,\mathrm{Op}(a)$ has symbol $\sum_k \frac{1}{k!}\partial_\xi^k e_s D_\theta^k a$; every term with $k \geq 1$ is divisible by s, with $\frac{d}{ds}\partial_\xi^k e_s|_{s=0} = \partial_\xi^k \log\langle\xi\rangle$. The leading term is $\partial_\xi\log\langle\xi\rangle\, D_\theta a = \xi\langle\xi\rangle^{-2}(-i\partial_\theta a)$. For (iii), the multiplication operator has diagonal matrix entries $\langle fe_k, e_k\rangle = \frac{1}{2\pi}\int f\,d\theta$ for every $k \in \mathbb{Z}$ (in $\Pi f\Pi$ only for $k \geq 0$), so for $\Re s < -1$

$$\mathrm{Tr}(fE(s)) = \frac{1}{2\pi}\int \mathrm{tr}\, f\,d\theta \sum_{k\in\mathbb{Z}}\lambda_k^s, \quad \mathrm{Tr}(\Pi f\Pi E(s)) = \frac{1}{2\pi}\int \mathrm{tr}\, f\,d\theta\sum_{k\geq 0}\lambda_k^s.$$

Now $\sum_{k\geq 0}\lambda_k^s = 1 + \zeta(-s) + h(s)$ with $h(s) = \sum_{k\geq 1}(\lambda_k^s - k^s)$; since $\lambda_k^s - k^s = O(|s|k^{\Re s - 2})$, h is holomorphic in $\Re s < 1$ with $h(0) = 0$. The Riemann zeta function is regular at 0 with $\zeta(0) = -\frac{1}{2}$, so the finite part (in fact the value) at $s = 0$ is $\frac{1}{2}$, and $\sum_{k\in\mathbb{Z}}\lambda_k^s = 2\sum_{k\geq 0}\lambda_k^s - 1$ has value 0 (Figure 17.1). □

Example 17.4.2. Let A be multiplication by $e^{i\theta}$ and $B = e^{-i\theta}\Pi$. Then $AB = \Pi$ and $BA = e^{-i\theta}\Pi e^{i\theta}$ is the projection onto the span of the $e^{ik\theta}$, $k \geq -1$, so $[A,B] = -\pi_{-1}$, minus the projection onto $e^{-i\theta}$, and $\overline{\mathrm{Tr}}([A,B]) = -1$. On the other hand, by Lemma 17.4.1(ii), $(D_E A)B$ has symbol ξ^{-1} in $\xi > 0$ and is rapidly decreasing in $\xi < 0$, so $\mathrm{Tr}_{\mathrm{R}}((D_E A)B) = -\frac{1}{2\pi}\int_0^{2\pi} 1\,d\theta = -1$, as it must be by Lemma 17.3.7. This confirms the signs in the trace-defect formula.

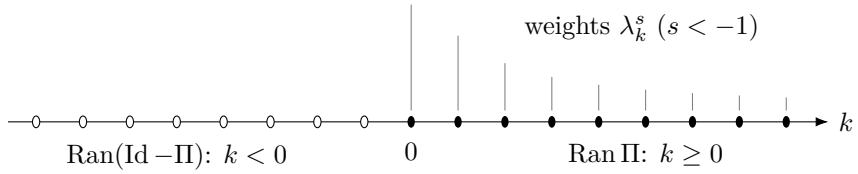


Figure 17.1: The Szegő projector Π keeps the Fourier modes $e^{ik\theta}$ with $k\geq 0$ (filled dots). The family $E(s)$ weights the k-th mode by λ_k^s, $\lambda_k=(1+k^2)^{1/2}$; for $\Re s<-1$ the weighted traces converge, and their values at $s=0$, obtained by analytic continuation, are $\frac{1}{2}$ for Π and 0 for Id. "Half of the modes" of the circle have regularized dimension $\frac{1}{2}$, and the full space has regularized dimension 0.

17.5 Odd Chern forms and the Toeplitz index

17.5.1 Toeplitz operators

We recall from Chapter 12. The Szegő projector $\Pi\in\Psi^0(\mathbb{S})$ projects onto the span $\mathcal{H}$ of the $e^{ik\theta}$, $k\geq 0$; its symbol is 1 on $\xi>0$ and its full symbol is rapidly decreasing as $\xi\to-\infty$. The Toeplitz algebra

$$\Psi_{\mathcal{T}}^m(\mathbb{S};\mathbb{C}^N)=\Pi\Psi^m(\mathbb{S};\mathbb{C}^N)\Pi$$

acts on $\mathcal{H}\otimes\mathbb{C}^N$, with identity Π; the full symbols of its elements are rapidly decreasing as $\xi\to-\infty$, and we write $\sigma_0(A)(\theta)$ for the principal symbol at $(\theta,+1)$. An element of $\Psi_{\mathcal{T}}^0$ is Fredholm on $\mathcal{H}\otimes\mathbb{C}^N$ if and only if $\sigma_0(A)$ is invertible, and then

$$\mathrm{ind}(A)=\frac{i}{2\pi}\int_0^{2\pi}\mathrm{tr}\big(\sigma_0(A)^{-1}\partial_\theta\sigma_0(A)\big)d\theta=-\operatorname{wind}\det\sigma_0(A). \tag{17.5.1}$$

For instance the shift $U=\Pi e^{i\theta}\Pi$ has index -1. Let

$$G_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)=\{B\in\Psi_{\mathcal{T}}^0(\mathbb{S};\mathbb{C}^N);\ B\text{ invertible on }\mathcal{H}\otimes\mathbb{C}^N,\ B^{-1}\in\Psi_{\mathcal{T}}^0(\mathbb{S};\mathbb{C}^N)\}$$

and $G_{\mathcal{T}}^{-\infty}=G_{\mathcal{T}}\cap(\Pi+\Psi_{\mathcal{T}}^{-\infty})$. Symbols of elements of $G_{\mathcal{T}}$ are loops in $\mathrm{GL}(N,\mathbb{C})$ with winding number 0. Products of operators on $\mathcal{H}\otimes\mathbb{C}^N$ are products in $\Psi^{\mathbb{Z}}(\mathbb{S};\mathbb{C}^N)$, so $\overline{\mathrm{Tr}}$ and Tr_{R} apply to elements of $\Psi_{\mathcal{T}}^{\mathbb{Z}}$.

17.5.2 Odd Chern forms

For a manifold W and $a\in\mathcal{C}^\infty(W;\mathrm{GL}(N,\mathbb{C}))$ set

$$\mathrm{Ch}_{\mathrm{odd}}(a)=\sum_{k\geq 0}c_k\,\mathrm{tr}\big((a^{-1}da)^{2k+1}\big),\qquad c_k=-\frac{k!}{(2\pi i)^{k+1}(2k+1)!}. \tag{17.5.2}$$

The same formula, with tr replaced by Tr, defines $\mathrm{Ch}_{\mathrm{odd}}$ for maps into the group $G^{-\infty}$ of Chapter 12. The forms are closed: with $\nu=a^{-1}da$ we have $d\nu=-\nu^2$, hence

$$d(\nu^{2k+1})=\sum_{j=0}^{2k}(-1)^j\nu^j d\nu\,\nu^{2k-j}=-\nu^{2k+2}, \tag{17.5.3}$$

and $\mathrm{tr}(\nu^{2k+2})=0$ because $\mathrm{tr}(\nu\wedge\nu^{2k+1})=-\mathrm{tr}(\nu^{2k+1}\wedge\nu)$. The constants c_k are fixed by Proposition 17.5.2 below; note $c_0=i/2\pi$, consistent with (17.5.1).

Conventions. On a product $W\times\mathbb{S}$ every form can be written uniquely as $\omega=\omega_0+\omega_1\wedge d\theta$ with ω_0,ω_1 not containing $d\theta$, and we set $\int_{\mathbb{S}}\omega=\int_0^{2\pi}\omega_1\,d\theta$, a form on W. With this ('fibre last') convention $d\int_{\mathbb{S}}=\int_{\mathbb{S}}d$, $\int_{\mathbb{S}}(\pi^*\alpha\wedge\omega)=\alpha\wedge\int_{\mathbb{S}}\omega$, and $\int_{W\times\mathbb{S}}=\int_W\int_{\mathbb{S}}$ when $W\times\mathbb{S}$ carries the product orientation (first W, then $d\theta$).

Lemma 17.5.1. *Let $a_t\in\mathcal{C}^\infty(W;\mathrm{GL}(N,\mathbb{C}))$ depend smoothly on $t\in[0,1]$ and regard a as a map on $W\times[0,1]$. Then*

$$\mathrm{Ch}_{\mathrm{odd}}(a)=\mathrm{Ch}_{\mathrm{odd}}(a_t)+\Theta(a_t)\wedge dt,\qquad \Theta(a_t)=\sum_{k\ge0}(2k+1)c_k\,\mathrm{tr}\big((a_t^{-1}da_t)^{2k}a_t^{-1}\partial_t a_t\big),$$

and $\partial_t\,\mathrm{Ch}_{\mathrm{odd}}(a_t)=d\Theta(a_t)$.

Proof. Write $a^{-1}Da=\nu+\lambda\,dt$, $D=d+dt\,\partial_t$, $\nu=a_t^{-1}da_t$, $\lambda=a_t^{-1}\partial_t a_t$. The dt-component of $(\nu+\lambda dt)^{2k+1}$ is

$$\sum_j \nu^j\lambda\,dt\,\nu^{2k-j}=\sum_j(-1)^{2k-j}\nu^j\lambda\nu^{2k-j}\wedge dt,$$

and $\mathrm{tr}(\nu^j\lambda\nu^{2k-j})=(-1)^{j(2k-j)}\,\mathrm{tr}(\lambda\nu^{2k})=(-1)^j\,\mathrm{tr}(\lambda\nu^{2k})$, giving $(2k+1)\,\mathrm{tr}(\nu^{2k}\lambda)\wedge dt$. The second identity is the dt-component of $D\,\mathrm{Ch}_{\mathrm{odd}}(a)=0$, using $dt\wedge\partial_t\omega=-\partial_t\omega\wedge dt$ for ω of odd degree. □

Applying this to the rotation homotopy from $g\oplus h$ to $gh\oplus\mathrm{Id}$ shows that $\mathrm{Ch}_{\mathrm{odd}}(gh)-\mathrm{Ch}_{\mathrm{odd}}(g)-\mathrm{Ch}_{\mathrm{odd}}(h)$ is exact (Exercise 17.4).

Proposition 17.5.2. *Let W be compact, $b\in\mathcal{C}^\infty(W\times\mathbb{S};\mathrm{GL}(N,\mathbb{C}))$ and let $T(w)\in\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)$ be a smooth family with symbol $b(w,\cdot)$. Then the Chern character of the index bundle $\mathrm{ind}(T)\in K^0(W)$ is*

$$\mathrm{Ch}(\mathrm{ind}\,T)=\Big[\int_{\mathbb{S}}\mathrm{Ch}_{\mathrm{odd}}(b)\Big]\in H^{\mathrm{even}}(W;\mathbb{C}),\tag{17.5.4}$$

where Ch *of a bundle with curvature Ω is represented by* $\mathrm{tr}\exp(i\Omega/2\pi)$.

Proof. This is the families index theorem for Toeplitz operators of Chapter 15; we recall the structure of the argument and verify the constants. Both sides are additive under direct sums, unchanged by stabilization and homotopy invariant (the right side by Lemma 17.5.1). Multiplying T by the θ-independent invertible matrix $b(w,1)^{-1}$ changes neither side (for the right side use Exercise 17.4 and the fact that $\int_{\mathbb{S}}\mathrm{Ch}_{\mathrm{odd}}(g)=0$ for θ-independent g). By the homotopy results of Chapter 13, after these reductions and stabilization b may be deformed to $\pi_-e^{-i\theta}+\pi_0+\pi_+e^{i\theta}$, with $\pi_\pm,\pi_0$ smooth families of commuting projections on some $\mathbb{C}^M$ summing to Id. This is the product of the commuting symbols $\pi_+e^{i\theta}+(\mathrm{Id}-\pi_+)$ and $\pi_-e^{-i\theta}+(\mathrm{Id}-\pi_-)$; the index is additive under composition (the Toeplitz operator of a product differs from the product of the Toeplitz operators by a compact family) and $\mathrm{Ch}_{\mathrm{odd}}$ is additive up to exact forms (Exercise 17.4), so it suffices to consider the two factors. The second is obtained from a symbol of the first type by $\theta\mapsto-\theta$, which reverses the sign of both sides. So it suffices to consider $b=\pi e^{i\theta}+\pi'$, with Toeplitz operator $\pi U+\pi'$: it is injective with cokernel $\pi\mathbb{C}^M$ (in the constants), so $\mathrm{ind}(T)=-[V]$, $V=\mathrm{Ran}\,\pi$. The connection πd on V has curvature $\Omega=\pi\,d\pi\wedge d\pi\,\pi$, so the left side of (17.5.4) is $-\sum_k\frac{1}{k!}(\frac{i}{2\pi})^k\,\mathrm{tr}\,\Omega^k$.

On the other hand, writing d' for the differential on $W\times\mathbb{S}$,

$$b^{-1}d'b=i\pi\,d\theta+\nu,\quad \nu=f\,\pi\,d\pi\,\pi'+g\,\pi'd\pi\,\pi,\quad f=1-e^{-i\theta},\ g=e^{i\theta}-1,$$

using $\pi\, d\pi\, \pi = 0$ and $\pi' d\pi\, \pi' = 0$. The $d\theta$-component of $(b^{-1}d'b)^{2k+1}$ has trace $(2k+1)i\,\mathrm{tr}(\pi\nu^{2k}\pi)\wedge d\theta$ (by cyclicity, as in Lemma 17.5.1) and $\pi\nu^{2k}\pi = (fg)^k(\pi\, d\pi\, \pi' d\pi\, \pi)^k = (-1)^k(2-2\cos\theta)^k\Omega^k$. Since $\int_0^{2\pi}(2-2\cos\theta)^k d\theta = 2\pi\frac{(2k)!}{(k!)^2}$,

$$\int_{\mathbb{S}} \mathrm{tr}((b^{-1}d'b)^{2k+1}) = 2\pi i(-1)^k\frac{(2k+1)!}{(k!)^2}\,\mathrm{tr}\,\Omega^k,$$

and multiplying by c_k gives exactly $-\frac{1}{k!}(\frac{i}{2\pi})^k\,\mathrm{tr}\,\Omega^k$. □

17.5.3 The Fredholm determinant

Recall from Chapter 12 that on $G^{-\infty}$ the Fredholm determinant satisfies $d\log\det A = \mathrm{Tr}(A^{-1}dA)$, so that the degree-one part $c_0\,\mathrm{Tr}(A^{-1}dA)$ of $\mathrm{Ch}_{\mathrm{odd}}$ is $\frac{i}{2\pi}d\log\det A$, and that $\frac{1}{2\pi i}\int_\gamma \mathrm{Tr}(A^{-1}dA)\in\mathbb{Z}$ for every loop γ; this integer is the case $W=\{\mathrm{pt}\}$ of (17.5.1)–(17.5.4).

17.6 Eta forms

Let X be a manifold and $B\in\mathcal{C}^\infty(X;G_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$ a smooth family of invertible Toeplitz operators. Then

$$\alpha = B^{-1}dB\in\mathcal{C}^\infty(X;\Lambda^1\otimes\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$$

and the symbol $b=\sigma_0(B)\in\mathcal{C}^\infty(X\times\mathbb{S};\mathrm{GL}(N,\mathbb{C}))$ satisfies $\mathrm{wind}\det b(x,\cdot)=0$. We write

$$\nu = b^{-1}d_X b,\qquad \mu = b^{-1}\partial_\theta b,$$

for the components of $b^{-1}d'b=\nu+\mu\,d\theta$ on $X\times\mathbb{S}$.

Definition 17.6.1. The *eta forms* of B are

$$\eta_{2k+1}(B) = \overline{\mathrm{Tr}}((B^{-1}dB)^{2k+1}),\qquad \eta(B)=\sum_{k\ge0}c_k\eta_{2k+1}(B),$$

and the associated *symbolic correction forms* are

$$\gamma_{2k+1}(b) = \frac{i}{4\pi}\int_0^{2\pi}\mathrm{tr}(\nu^{2k+1}\mu)d\theta,\qquad \gamma(b)=\sum_{k\ge0}c_k\gamma_{2k+1}(b).$$

The γ-forms are local in the symbol b; they vanish if b is independent of θ. If $B\in\mathcal{C}^\infty(X;G^{-\infty}_{\mathcal{T}})$ then $B^{-1}dB$ is smoothing, so $\overline{\mathrm{Tr}}=\mathrm{Tr}$ and $\eta(B)=\mathrm{Ch}_{\mathrm{odd}}(B)$, while $b=\mathrm{Id}$ and $\gamma(b)=0$. Thus η *extends the odd Chern character from the normal subgroup* $G^{-\infty}_{\mathcal{T}}$ *to the whole group* $G_{\mathcal{T}}$, at the expense of no longer being closed.

Discussion. How far from closed is η? The odd Chern forms are closed because $d(\alpha^{2k+1})=-\alpha^{2k+2}$ and $\mathrm{Tr}(\alpha^{2k+2})=\frac12\,\mathrm{Tr}[\alpha,\alpha^{2k+1}]=0$. With $\overline{\mathrm{Tr}}$ in place of Tr the first identity still holds, but the graded commutator no longer has trace zero: by the trace-defect formula its regularized trace is a residue trace, an integral of symbols. So $d\eta$ is a *local* expression in the principal symbol b of the family. Part of it is exact, $-d\gamma(b)$, and the rest is the fibre integral of the odd Chern form of b over the circle (Theorem 17.6.2 and Figure 17.2).

Question. The regularized trace depends on the choice of the family $E(s)$. Does the eta form?

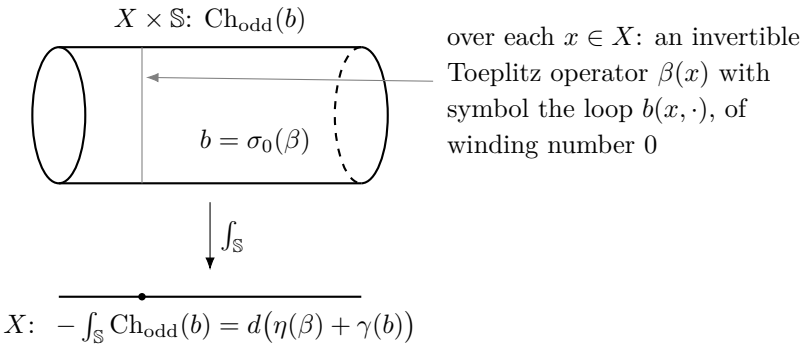


Figure 17.2: The variation formula (17.6.2). A family β of invertible Toeplitz operators over X has a symbol b on $X \times \mathbb{S}$. The odd Chern form of b, integrated over the circle, is exact on X, and the eta form of β, corrected by the local term $\gamma(b)$, is an explicit primitive. When β is a smoothing perturbation of Π, the symbol is trivial and $\eta(\beta)$ is the closed odd Chern form of β.

Answer. Yes, but only mildly. By Lemma 17.3.3 a change of E changes $\overline{\mathrm{Tr}}$ by a residue trace, which is an integral of symbols. So the eta forms defined with two admissible families differ by a form that is local in the symbol of B. When $D(E, E')$ has order -1, Exercise 17.3 shows that the change can be absorbed into the symbolic correction γ without affecting the index formula. □

Theorem 17.6.2. *For every smooth family $B \in \mathcal{C}^\infty(X; G_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^N))$ with symbol b,*

$$d\big(\eta_{2k+1}(B) + \gamma_{2k+1}(b)\big) = \frac{i}{4\pi(2k+3)} \int_{\mathbb{S}} \mathrm{tr}\big((b^{-1}d'b)^{2k+3}\big) \tag{17.6.1}$$

and consequently

$$d\big(\eta(B) + \gamma(b)\big) = -\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(b). \tag{17.6.2}$$

Proof. By Corollary 17.3.8 and (17.5.3) applied to α,

$$d\eta_{2k+1}(B) = \overline{\mathrm{Tr}}\big(d(\alpha^{2k+1})\big) = -\overline{\mathrm{Tr}}(\alpha^{2k+2}) = -\frac{1}{2}\overline{\mathrm{Tr}}([\alpha, \alpha^{2k+1}]) = -\frac{1}{2}\,\mathrm{Tr_R}\big((D_E\alpha)\wedge\alpha^{2k+1}\big),$$

since the graded commutator is $[\alpha, \alpha^{2k+1}] = 2\alpha^{2k+2}$. The operator $(D_E\alpha)\wedge\alpha^{2k+1}$ has order -1, and its full symbol is rapidly decreasing as $\xi \to -\infty$ because α is Toeplitz. By Lemma 17.4.1(i),(ii) only the principal symbol at $\xi = +1$, namely $-i(\partial_\theta\nu)\nu^{2k+1}$, contributes:

$$d\eta_{2k+1}(B) = -\frac{1}{2}\cdot\Big(-\frac{1}{2\pi}\Big)\int_0^{2\pi} \mathrm{tr}\big(-i\,\partial_\theta\nu\wedge\nu^{2k+1}\big)d\theta = -\frac{i}{4\pi}\int_0^{2\pi} \mathrm{tr}\big(\partial_\theta\nu\wedge\nu^{2k+1}\big)d\theta.$$

Differentiating $\nu = b^{-1}d_X b$ and $\mu = b^{-1}\partial_\theta b$ gives the Maurer–Cartan identity $\partial_\theta\nu = d_X\mu + \nu\mu - \mu\nu$. By cyclicity $\mathrm{tr}(\nu\mu\nu^{2k+1}) = -\,\mathrm{tr}(\mu\nu^{2k+2})$, and by (17.5.3) $\mathrm{tr}(d\mu\wedge\nu^{2k+1}) = d\,\mathrm{tr}(\mu\nu^{2k+1}) + \mathrm{tr}(\mu\nu^{2k+2})$. Hence

$$\mathrm{tr}(\partial_\theta\nu\wedge\nu^{2k+1}) = d\,\mathrm{tr}(\mu\nu^{2k+1}) - \mathrm{tr}(\mu\nu^{2k+2}),$$

and, μ being of degree 0,

$$d\eta_{2k+1}(B) = -d\gamma_{2k+1}(b) + \frac{i}{4\pi}\int_0^{2\pi} \mathrm{tr}(\nu^{2k+2}\mu)\,d\theta.$$

Finally, the $d\theta$-component of $(\nu + \mu\, d\theta)^{2k+3}$ is $\sum_{j=0}^{2k+2}(-1)^j \nu^j \mu \nu^{2k+2-j} \wedge d\theta$, whose trace is $(2k+3)\operatorname{tr}(\nu^{2k+2}\mu) \wedge d\theta$ since $\operatorname{tr}(\nu^j \mu \nu^{2k+2-j}) = (-1)^j \operatorname{tr}(\mu \nu^{2k+2})$. This proves (17.6.1). For (17.6.2) note that the constants satisfy

$$c_k \cdot \frac{i}{4\pi(2k+3)} = -c_{k+1},$$

so the sum over k of (17.6.1) gives $-\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(b)$ except for the term $c_0 \int_{\mathbb{S}} \operatorname{tr}(b^{-1}d'b) = c_0 \int_0^{2\pi} \operatorname{tr}\mu\, d\theta$ of degree 0. By (17.5.1) this is $\operatorname{ind}(B(x)) = 0$. □

Thus $-(\eta(B) + \gamma(b))$ is an explicit primitive of the form $\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(b)$, which by Proposition 17.5.2 represents the Chern character of the index bundle of any Toeplitz family with symbol b, a class that vanishes here because the family B is invertible. Note that the variation of the eta form depends only on the leading part of the symbol, not on the lower order terms or on the smoothing part of B. The situation can be summarized in the diagram

$$\mathrm{Ch}_{\mathrm{odd}} \xleftarrow{\text{restrict}} \eta \xrightarrow{d(\cdot+\gamma)} -\textstyle\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}$$

$$G_{\mathcal{T}}^{-\infty}(\mathbb{S};\mathbb{C}^N) \longrightarrow G_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N) \xrightarrow{\sigma_0} \mathcal{C}^\infty(\mathbb{S};\mathrm{GL}(N,\mathbb{C}))_0$$

where the right-hand group consists of loops of winding number 0. In the stabilized setting of Chapter 13 (where $\mathbb{C}^N$ is replaced by $\Psi^{-\infty}(Y;F)$ and tr by the trace on Y; all the arguments above go through verbatim) the bottom row is the classifying sequence for K-theory, with weakly contractible middle group, and the eta form is a transgression form for the even Chern character of the quotient. That the transgression only involves the principal symbol is a higher analogue of the 'miracle of the loop group' of Pressley and Segal [21].

Proposition 17.6.3. *Let $B \in \mathcal{C}^\infty(X; G_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$ and $g \in \mathcal{C}^\infty(X; G_{\mathcal{T}}^{-\infty})$. Then*

(i) $\eta_1(g) = \operatorname{Tr}(g^{-1}dg) = d\log\det g$;

(ii) $\eta_1(Bg) = \eta_1(B) + d\log\det g$ *and* $\eta_1(gB) = \eta_1(B) + d\log\det g$;

(iii) $d(\eta_1(B) + \gamma_1(b)) = \dfrac{i}{12\pi}\displaystyle\int_{\mathbb{S}} \operatorname{tr}((b^{-1}d'b)^3)$.

Proof. (i) was noted above and (iii) is (17.6.1) for $k = 0$. For (ii), $(Bg)^{-1}d(Bg) = g^{-1}\alpha g + g^{-1}dg$, and $g^{-1}\alpha g - \alpha = [g^{-1}\alpha, g]$, so by Lemma 17.3.7

$$\overline{\operatorname{Tr}}(g^{-1}\alpha g) - \overline{\operatorname{Tr}}(\alpha) = \operatorname{Tr}_{\mathrm{R}}(D_E(g^{-1}\alpha)\, g).$$

Since $g - \Pi$ is smoothing and $\sigma_0(g^{-1}\alpha) = \nu$, by Lemma 17.4.1 the right side is

$$\frac{i}{2\pi}\int_0^{2\pi} \partial_\theta \operatorname{tr}\nu\, d\theta = 0.$$

Also $\overline{\operatorname{Tr}}(g^{-1}dg) = \operatorname{Tr}(g^{-1}dg)$. For gB, we have $(gB)^{-1}d(gB) = B^{-1}(g^{-1}dg)B + \alpha$ and $\operatorname{Tr}(B^{-1}(g^{-1}dg)B) = \operatorname{Tr}(g^{-1}dg)$ because $g^{-1}dg$ is of trace class. □

Remark 17.6.4. Consider the principal bundle $G_{\mathcal{T}} \longrightarrow G_{\mathcal{T}}/G_{\mathcal{T}}^{-\infty}$ and the line bundle L associated to it by the character $\det : G_{\mathcal{T}}^{-\infty} \longrightarrow \mathbb{C}^*$. By (ii), and since $\gamma_1(\sigma_0(Bg)) = \gamma_1(\sigma_0(B))$, the 1-form $\eta_1 + \gamma_1$ on $G_{\mathcal{T}}$ is a connection form for L (its restriction to the fibres is $d\log\det$ and it transforms correctly); by (iii) its curvature is the 2-form $\frac{i}{12\pi}\int_{\mathbb{S}} \operatorname{tr}((b^{-1}d'b)^3)$,

which depends only on the symbol and so descends. The first Chern form $\frac{i}{2\pi}d(\eta_1+\gamma_1)$ of L is therefore $-\frac{1}{24\pi^2}\int_{\mathbb{S}}\operatorname{tr}((b^{-1}d'b)^3)$, which is minus the degree-two part of $\int_{\mathbb{S}}\mathrm{Ch}_{\mathrm{odd}}(b)$. This is consistent with the identification of L with the determinant line bundle $\Lambda^{\max}(\mathrm{Nul})^*\otimes\Lambda^{\max}(\mathrm{coker})$ of the index bundle, whose first Chern class is $-c_1(\mathrm{ind})$. Thus η_1 is a regularized form of $d\log\det$ on the Toeplitz group; see also Section 17.10.4.

17.7 Product-type Toeplitz operators

17.7.1 The product-type calculus on $M\times\mathbb{S}$

Let M be a compact manifold without boundary of dimension $n\geq 1$ and consider the trivial fibration $\pi: M\times\mathbb{S}\longrightarrow M$. We use the following facts from Chapter 14 about the product-type operators $\Psi^{0,0}_{\mathrm{pt}}(M\times\mathbb{S};\mathbb{C}^N)$ associated to this fibration.

(P1) They form an algebra, acting on $\mathcal{C}^\infty(M\times\mathbb{S};\mathbb{C}^N)$ and boundedly on L^2, closed under adjoints. There are two multiplicative symbol maps. The *principal symbol* $\sigma(A)$ is a smooth function on the blow-up of $S^*(M\times\mathbb{S})$ along $S^*M\times\mathbb{S}$ (the covectors annihilating the fibres, $\tau=0$ where τ is dual to θ); this blow-up is the disjoint union of the two closed half-sphere bundles

$$S^*_\pm=\{\pm\tau\geq 0\}\subset S^*(M\times\mathbb{S}),\qquad \partial S^*_\pm=S^*M\times\mathbb{S}.$$

The *base family* $\beta(A)\in\mathcal{C}^\infty(S^*M;\Psi^0(\mathbb{S};\mathbb{C}^N))$ is a family of operators on the fibre parametrized by S^*M.

(P2) The symbols are compatible: the principal symbol of $\beta(A)(x)$ on $S^*\mathbb{S}=\mathbb{S}\times\{\pm1\}$ is the restriction of $\sigma(A)$ to $\{x\}\times\mathbb{S}\subset\partial S^*_\pm$. Conversely any compatible pair, depending smoothly on parameters, is the joint symbol of a smooth family of operators.

(P3) If $\sigma(A)=0$ and $\beta(A)=0$ then A is compact on L^2.

(P4) If A is *fully elliptic*, i.e. $\sigma(A)$ and all $\beta(A)(x)$ are invertible, then A is Fredholm on L^2 and on $\mathcal{C}^\infty$, with the same index; the index depends only on the joint symbol and is constant along smooth families of fully elliptic operators.

(P5) The ordinary operators $\Psi^0(M\times\mathbb{S};\mathbb{C}^N)$ are product-type, with $\sigma(P)$ the lift of the usual principal symbol and $\beta(P)(x)$ the multiplication operator by $\sigma(P)|_{\{x\}\times\mathbb{S}}$. Fibrewise families $Q\in\mathcal{C}^\infty(M;\Psi^0(\mathbb{S};\mathbb{C}^N))$ are product-type, with $\sigma(Q)$ the lift of the fibre symbol and $\beta(Q)(m,\xi)=Q(m)$; in particular multiplication by $h\in\mathcal{C}^\infty(M\times\mathbb{S};M(N,\mathbb{C}))$ has symbol h and base family the multiplication by $h(m,\cdot)$.

Since the Szegő projector Π, acting in θ, is a fibrewise operator, it is in $\Psi^{0,0}_{\mathrm{pt}}$ with $\sigma(\Pi)=1$ on S^*_+, $\sigma(\Pi)=0$ on S^*_-, and $\beta(\Pi)=\Pi$. The *product-type Toeplitz algebra* is

$$\Psi^{0,0}_{\mathrm{pt},\mathcal{T}}(M\times\mathbb{S};\mathbb{C}^N)=\Pi\,\Psi^{0,0}_{\mathrm{pt}}(M\times\mathbb{S};\mathbb{C}^N)\,\Pi,$$

acting on $\mathcal{H}(M\times\mathbb{S};\mathbb{C}^N)=\Pi\mathcal{C}^\infty(M\times\mathbb{S};\mathbb{C}^N)$ (or its L^2 closure), with identity Π. For A in this algebra the principal symbol vanishes on S^*_-; we set

$$a=\sigma(A)|_{S^*_+}\in\mathcal{C}^\infty(S^*_+;M(N,\mathbb{C})),$$
$$\beta=\beta(A)\in\mathcal{C}^\infty(S^*M;\Psi^0_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N)),\quad b=a|_{\partial S^*_+}=\sigma_0(\beta).$$

A is *fully elliptic* if a takes values in $\mathrm{GL}(N,\mathbb{C})$ and $\beta\in\mathcal{C}^\infty(S^*M;G_{\mathcal{T}}(\mathbb{S};\mathbb{C}^N))$; then A is Fredholm on $\mathcal{H}$ (apply (P4) to $A+\mathrm{Id}-\Pi$) and its index has the properties in (P4). We write $\mathrm{ind}(A)$ for the index on $\mathcal{H}$.

Discussion. Why restrict to Toeplitz operators? Compressing by Π kills the half S^*_- of the cosphere bundle, so the symbol lives on a single ball bundle S^*_+ with boundary $S^*M \times \mathbb{S}$, and the base family takes values in the Toeplitz group $G_{\mathcal{T}}$, where the eta forms of Section 17.6 are defined. The general product-type case is reduced to this one in Section 17.9 by a unitary doubling. Full ellipticity has two parts, and each feeds one term of the index formula: invertibility of a on S^*_+ (the Chern term) and invertibility of the base family β (the eta term).

17.7.2 Geometry and orientations

Fix a metric on M and use the product metric on $M \times \mathbb{S}$. In each fibre of $S^*_+ \to M \times \mathbb{S}$, stereographic projection $(\xi, \tau) \mapsto \xi/(1+\tau)$ identifies the closed half-sphere $\{|\xi|^2 + \tau^2 = 1,\ \tau \geq 0\}$ diffeomorphically with the closed unit ball in $T^*_m M$; thus $S^*_+ \cong \overline{B^*M} \times \mathbb{S}$ is a ball bundle over $M \times \mathbb{S}$, with 'centre' the section $(\xi, \tau) = (0, 1)$, i.e. $d\theta$.

We orient cotangent bundles by the Liouville form: if $\omega = \sum_j d\xi_j \wedge dx_j$ then T^*M is oriented by $\omega^n/n!$. The ball bundle $\overline{B^*M}$ inherits this orientation, and $S^*M = \partial\overline{B^*M}$ is given the boundary orientation (outward normal first); we call this the *Liouville orientation* of S^*M, and similarly for $S^*(M \times \mathbb{S})$, whose open subset $\{\tau > 0\}$ orients S^*_+. For $n = 1$ the two components $\{\xi = \pm 1\}$ of $S^*M \cong M \times \{\pm 1\}$ receive opposite orientations ($\pm dx$ if M is oriented by dx).

Lemma 17.7.1. *With these orientations, $\partial S^*_+ = S^*M \times \mathbb{S}$ with the product orientation, S^*M first and $d\theta$ last.*

Proof. The Liouville form of $T^*(M \times \mathbb{S})$ is $\omega_M + d\tau \wedge d\theta$, and $(\omega_M + d\tau \wedge d\theta)^{n+1}/(n+1)! = \omega_M^n/n! \wedge d\tau \wedge d\theta$. At a point with $\tau = 0$ the outward radial vector R is tangent to the ξ directions, so $S^*(M \times \mathbb{S})$ is oriented there by $\iota_R(\omega_M^n/n!) \wedge d\tau \wedge d\theta = \sigma \wedge d\tau \wedge d\theta$, where σ is the Liouville orientation of S^*M. The outward normal of S^*_+ at $\tau = 0$ is $-\partial_\tau$, and $\iota_{-\partial_\tau}(\sigma \wedge d\tau \wedge d\theta) = (-1)^{2n}\sigma \wedge d\theta$ since σ has degree $2n - 1$. □

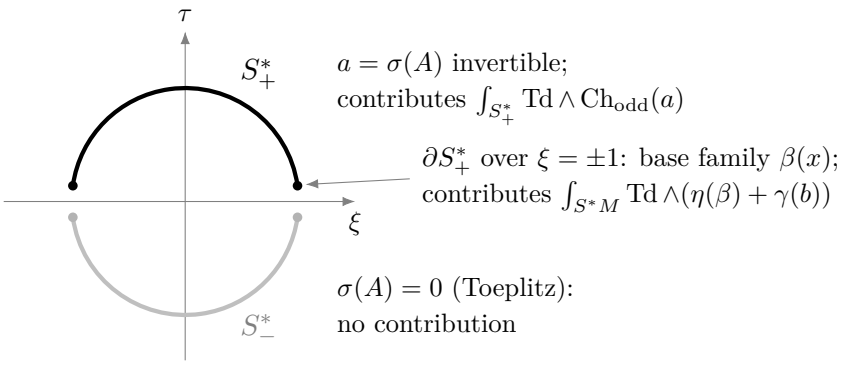


Figure 17.3: The fibre of $S^*(M \times \mathbb{S})$ over a point of $M \times \mathbb{S}$, for $n = 1$, in the coordinates (ξ, τ). The blow-up along $\tau = 0$ separates the two closed half-circles $S^*_\pm$. Each point $\tau = 0$, $\xi = \pm 1$, of S^*M is doubled, and over it the base family $\beta(x)$ acts on the circle fibre. For a Toeplitz operator the symbol vanishes on S^*_-; the index formula (17.8.1) integrates the odd Chern form of the symbol over S^*_+, and the eta form of the base family over its boundary.

We use the Atiyah–Singer theorem in the following form, with $\mathrm{Td}(M)$ a closed form representing the Todd class of $TM \otimes \mathbb{C}$ (by Stokes' theorem the right side does not depend on the choice of form).

Theorem 17.7.2 (Atiyah–Singer, cosphere form)**.** *If W is a compact manifold without boundary and $P \in \Psi^0(W;\mathbb{C}^N)$ is elliptic with principal symbol $p \in \mathcal{C}^\infty(S^*W;\mathrm{GL}(N,\mathbb{C}))$ then*

$$\operatorname{ind}(P) = \int_{S^*W} \mathrm{Td}(W) \wedge \mathrm{Ch}_{\mathrm{odd}}(p),$$

where S^*W carries the Liouville orientation and only the component of degree $2\dim W - 1$ is integrated.

This is the case of a point base of the families index theorem of Chapter 15, once the relative Chern character of the symbol class in $K_c(T^*W)$ is transgressed to the cosphere bundle; the normalization of $\mathrm{Ch}_{\mathrm{odd}}$ is the one forced by Bott periodicity, Proposition 17.5.2. For $W = \mathbb{S}$ it is the elementary formula $\operatorname{ind} P = -\operatorname{wind}\det p(\cdot,1) + \operatorname{wind}\det p(\cdot,-1)$, which follows from (17.5.1) by splitting P into its positive and negative Toeplitz parts (see Chapter 13); with the Liouville orientations ($d\theta$ on $\xi = 1$, $-d\theta$ on $\xi = -1$) this agrees with the theorem, which checks the orientation convention.

17.8 The index formula

For a fully elliptic $A \in \Psi^{0,0}_{\mathrm{pt},\mathcal{T}}(M \times \mathbb{S};\mathbb{C}^N)$ with symbols (a,β), $b = \sigma_0(\beta)$, set

$$I(A) = \int_{S^*M} \mathrm{Td}(M) \wedge \big(\eta(\beta) + \gamma(b)\big) + \int_{S^*_+} \mathrm{Td}(M) \wedge \mathrm{Ch}_{\mathrm{odd}}(a), \tag{17.8.1}$$

where $\mathrm{Td}(M)$ is pulled back to S^*M and to S^*_+, the orientations are those of the previous section, and only components of top degree are integrated. Note that $I(A)$ depends only on the joint symbol of A.

Theorem 17.8.1. *For every fully elliptic $A \in \Psi^{0,0}_{\mathrm{pt},\mathcal{T}}(M \times \mathbb{S};\mathbb{C}^N)$,*

$$\operatorname{ind}(A) = \int_{S^*M} \mathrm{Td}(M) \wedge \big(\eta(\beta) + \gamma(b)\big) + \int_{S^*_+(M\times\mathbb{S})} \mathrm{Td}(M) \wedge \mathrm{Ch}_{\mathrm{odd}}(a).$$

For M a point (formally $n = 0$) the formula reduces to (17.5.1), S^*_+ being the circle oriented by $d\theta$. The proof occupies the rest of this section. We write $\Delta(A) = \operatorname{ind}(A) - I(A)$.

Discussion. The formula has the shape of the Atiyah–Patodi–Singer theorem: an interior integral of a characteristic form, here over S^*_+, plus a boundary correction, here over $\partial S^*_+ = S^*M \times \mathbb{S}$, given by an eta form (Figure 17.3). Neither term is an integer by itself. The variation formula Theorem 17.6.2 says that when the symbols are deformed, the change of the interior term is exactly cancelled by the change of the boundary term; this is Proposition 17.8.2. So both sides of the formula are homotopy invariants, and the proof consists in deforming A to an operator for which the formula is already known.

Proposition 17.8.2. *If (a_t,β_t), $t \in [0,1]$, is a smooth family of compatible pairs of fully elliptic symbols then I is constant in t. Consequently $\Delta(A)$ depends only on the homotopy class of the joint symbol of A through fully elliptic symbols.*

Proof. Write $T = \mathrm{Td}(M)$, a closed even form, $X = S^*M$, and $I(t) = I_1(t) + I_2(t)$ for the two terms. By Lemma 17.5.1, $\partial_t \mathrm{Ch}_{\mathrm{odd}}(a_t) = d\Theta(a_t)$, so by Stokes' theorem and Lemma 17.7.1

$$I_2'(t) = \int_{S^*_+} T \wedge d\Theta(a_t) = \int_{X\times\mathbb{S}} T \wedge \Theta(b_t) = \int_X T \wedge \int_{\mathbb{S}} \Theta(b_t),$$

$$A \xrightarrow{\text{Step 1}} \beta = \Pi g \Pi \xrightarrow{\text{Step 2}} \beta = \Pi h \Pi \xrightarrow{\text{Step 3}} \beta = \Pi \xrightarrow{\text{Step 4}} \Pi P \Pi \xrightarrow{\text{Lemma 17.8.5}} \Delta(\Pi P \Pi) = 0$$

Figure 17.4: Logic of the proof of Theorem 17.8.1. Step 1 makes the base family constant in θ, using the weak contractibility of the Toeplitz group; Step 2 makes it pulled back from M; Step 3 removes it by composing with the inverse of a multiplication operator (Lemmas 17.8.3 and 17.8.4); Step 4 makes the symbol trivial near ∂S^*_+, so the operator is the compression of an ordinary operator P, for which $\operatorname{ind}(P) = I(\Pi P \Pi)$ is the Atiyah–Singer theorem on $M \times \mathbb{S}$. Every step changes neither ind nor I (Proposition 17.8.2).

where $\Theta(b_t)$ is the restriction of $\Theta(a_t)$ to the boundary, given by the same formula in terms of b_t. Now apply Theorem 17.6.2 to the family β on $X \times [0,1]$, with total differential $D = d + dt\,\partial_t$. Writing $\eta(\beta) + \gamma(b) = \zeta(t) + \zeta'(t) \wedge dt$ and, by Lemma 17.5.1 applied on $X \times \mathbb{S} \times [0,1]$,

$$\int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(b) = \int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(b_t) + \int_{\mathbb{S}} (\Theta(b_t) \wedge dt) = \int_{\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(b_t) - \Big(\int_{\mathbb{S}} \Theta(b_t)\Big) \wedge dt$$

(moving dt past $d\theta$), the dt-component of (17.6.2) reads $-\partial_t \zeta + d_X \zeta' = \int_{\mathbb{S}} \Theta(b_t)$, using $dt \wedge \partial_t \zeta = -\partial_t \zeta \wedge dt$ (ζ is odd). Since T is closed and X is compact without boundary,

$$I_1'(t) = \int_X T \wedge \partial_t \zeta = -\int_X T \wedge \int_{\mathbb{S}} \Theta(b_t) = -I_2'(t).$$

The second statement follows from (P2) and (P4): a homotopy of joint symbols is realized by a smooth family of fully elliptic operators, along which the index is constant. □

We use direct sums and stabilization freely: if A acts on $\mathbb{C}^N$-valued functions, $A \oplus \Pi$ acting on $\mathbb{C}^{N+K}$-valued functions has the same index and the same I, and both ind and I are additive under direct sums (all the forms involved are traces of block-diagonal expressions).

Lemma 17.8.3. *If $A, A' \in \Psi^{0,0}_{\mathrm{pt},\mathcal{T}}(M \times \mathbb{S}; \mathbb{C}^N)$ are fully elliptic then $\Delta(AA') = \Delta(A) + \Delta(A')$.*

Proof. The index is additive under composition of Fredholm operators. Let R_t be the constant unitary $\left(\begin{smallmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{smallmatrix}\right) \otimes \mathrm{Id}_N$ on $\mathbb{C}^{2N}$; it commutes with Π and is an invertible element of the algebra. Then

$$C_t = R_t (A \oplus \Pi) R_t^{-1} (\Pi \oplus A'), \qquad 0 \le t \le \pi/2,$$

is a smooth family of fully elliptic operators with $C_0 = A \oplus A'$ and $C_{\pi/2} = \Pi \oplus AA'$. By Proposition 17.8.2 and additivity, $I(A) + I(A') = I(\Pi \oplus AA') = I(AA')$, since $I(\Pi) = 0$ ($\beta(\Pi)$ is constant and $a = \mathrm{Id}$). □

Lemma 17.8.4. *If $h \in \mathcal{C}^\infty(M; \mathrm{GL}(N, \mathbb{C}))$ and H denotes multiplication by h on $\mathcal{H}(M \times \mathbb{S}; \mathbb{C}^N)$, then H is fully elliptic and $\operatorname{ind}(H) = I(H) = 0$.*

Proof. H is invertible, so $\operatorname{ind}(H) = 0$. Its symbols are $a = h$ (pulled back to S^*_+) and $\beta = \Pi h \Pi$, constant in ξ and θ. Hence $\gamma(b) = 0$ and, by Lemma 17.4.1(iii), $\eta(\beta) = \frac{1}{2}\,\mathrm{Ch}_{\mathrm{odd}}(h)$ pulled back from M. The form $\mathrm{Td}(M) \wedge \mathrm{Ch}_{\mathrm{odd}}(h)$ is pulled back from M, which has dimension n, whereas $\dim S^*_+ = 2n + 1 > n$ and $\dim S^*M = 2n - 1$; so both integrals in (17.8.1) vanish when $n \ge 2$. If $n = 1$ the second still vanishes, and the first vanishes because $S^*M \cong M \times \{\pm 1\}$ with opposite orientations on the two components. □

Lemma 17.8.5. *Let $P \in \Psi^0(M \times \mathbb{S}; \mathbb{C}^N)$ be elliptic with principal symbol p equal to* Id *outside a compact subset of $\{\tau > 0\} \subset S^*(M \times \mathbb{S})$. Then $\Pi P \Pi \in \Psi^{0,0}_{\mathrm{pt},\mathcal{T}}$ is fully elliptic with $a = p|_{S^*_+}$, $\beta = \Pi$, and*

$$\operatorname{ind}(\Pi P \Pi) = \operatorname{ind}(P) = I(\Pi P \Pi).$$

Proof. By (P5), $\beta(P)$ is multiplication by $p|_{\tau=0} = \mathrm{Id}$, so $\beta(\Pi P\Pi) = \Pi$ and $\sigma(\Pi P\Pi) = p$ on S^*_+. With respect to $L^2 = \mathcal{H} \oplus \mathcal{H}^\perp$ write $P = \Pi P \Pi + (\mathrm{Id} - \Pi)P(\mathrm{Id} - \Pi) + K$, $K = \Pi P(\mathrm{Id} - \Pi) + (\mathrm{Id} - \Pi)P\Pi$. Both symbols of K vanish ($\sigma(\Pi)\sigma(\mathrm{Id} - \Pi) = 0$ and $\Pi\,\mathrm{Id}\,(\mathrm{Id} - \Pi) = 0$), and both symbols of $(\mathrm{Id} - \Pi)P(\mathrm{Id} - \Pi) - (\mathrm{Id} - \Pi)$ vanish (on S^*_-, $p = \mathrm{Id}$). By (P3) these are compact, so $\operatorname{ind}(P) = \operatorname{ind}(\Pi P\Pi) + \operatorname{ind}((\mathrm{Id} - \Pi) + \text{compact}) = \operatorname{ind}(\Pi P\Pi)$. On the other hand $\eta(\Pi) = 0$ and $\gamma(\mathrm{Id}) = 0$, so $I(\Pi P\Pi) = \int_{S^*_+} \mathrm{Td}(M) \wedge \mathrm{Ch}_{\mathrm{odd}}(p)$. Since $p = \mathrm{Id}$ on S^*_-, $\mathrm{Ch}_{\mathrm{odd}}(p) = 0$ there, and $\mathrm{Td}(M \times \mathbb{S}) = \mathrm{Td}(M)$ (as $T(M \times \mathbb{S}) \otimes \mathbb{C} = \pi^*(TM \otimes \mathbb{C}) \oplus \mathbb{C}$); so Theorem 17.7.2 on $M \times \mathbb{S}$ gives $\operatorname{ind}(P) = I(\Pi P\Pi)$. □

We also need the weak contractibility of the normalized Toeplitz group from Chapter 13, in the following form. If $\beta' \in \mathcal{C}^\infty(X; G_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^N))$, X compact, satisfies the normalization $\sigma_0(\beta')(x, 1) = \mathrm{Id}$ (at $\theta = 0$), then after stabilization $\mathbb{C}^N \subset \mathbb{C}^K$, $\beta' \oplus \Pi$ is smoothly homotopic through $\mathcal{C}^\infty(X; G_{\mathcal{T}}(\mathbb{S}; \mathbb{C}^K))$ to the constant family Π. Indeed Chapter 13 gives such a homotopy in the group stabilized by the smoothing operators on an auxiliary manifold Y, with $\mathbb{C}^N$ embedded as the span of the first N eigenfunctions of a Laplacian on Y; compressing the homotopy by the spectral projection onto the first K eigenfunctions gives, for K large, a homotopy of the required kind, since the compressions converge in the $\mathcal{C}^\infty$ topology and invertibility is an open condition, uniformly on the compact parameter space.

Proof of Theorem 17.8.1. Let A be fully elliptic with symbols (a, β), $X = S^*M$. We show $\Delta(A) = 0$ in four steps (Figure 17.4); each step replaces A by an operator with the same Δ.

Homotopies with prescribed boundary values. Choose a collar $\partial S^*_+ \times [0, \epsilon)_r$ of the boundary of S^*_+ and $\phi \in \mathcal{C}^\infty([0, \epsilon))$ with $\phi = 0$ near 0 and $\phi = 1$ on $[\epsilon/2, \epsilon)$. If b_t is a smooth homotopy of $b_0 = a|_{\partial S^*_+}$ through maps into $\mathrm{GL}(N, \mathbb{C})$, then

$$a_t(q, r) = b_{t(1-\phi(r))}(q)\, b_0(q)^{-1} a(q, r) \text{ in the collar}, \qquad a_t = a \text{ elsewhere},$$

is a smooth homotopy of a through invertible symbols with $a_t|_{\partial S^*_+} = b_t$. So any smooth homotopy β_t of β through $\mathcal{C}^\infty(X; G_{\mathcal{T}})$ extends to a homotopy of fully elliptic compatible pairs (a_t, β_t), which does not change Δ (Proposition 17.8.2).

Step 1: β constant in θ. Let $g(x) = b(x, 1) \in \mathrm{GL}(N, \mathbb{C})$ and $\beta' = g^{-1}\beta$ (the matrix $g(x)^{-1}$ acting on $\mathcal{H} \otimes \mathbb{C}^N$, which commutes with Π). Then β' is normalized, so after stabilization there is a homotopy β'_t from β' to Π, and $\beta_t = g\beta'_t$ is a homotopy from β to $\Pi g \Pi$. After extending it as above we may assume that $\beta(x) = \Pi g(x) \Pi$ with $g \in \mathcal{C}^\infty(X; \mathrm{GL}(N, \mathbb{C}))$; then $b(x, \theta) = g(x)$.

Step 2: β pulled back from M. The submanifold $S^*_+ \cap \{\theta = 0\}$ is a closed ball bundle over M with boundary X, and a restricted to it extends g. Contracting the balls linearly to their centres (in the stereographic picture, $v \mapsto (1 - t)v$) gives a smooth homotopy g_t, $g_0 = g$, from g to $h \circ \pi_M$, where $h(m)$ is the value of a at the centre over $(m, 0)$. The family $\beta_t = \Pi g_t \Pi$ consists of invertible operators; extending as above we may assume $\beta = \Pi h \Pi$ with $h \in \mathcal{C}^\infty(M; \mathrm{GL}(N, \mathbb{C}))$.

Step 3: $\beta = \Pi$. Let H be multiplication by h and $A' = H^{-1}A$. Then $\beta(A') = \Pi$ and $a' = h^{-1}a$ satisfies $a'|_{\partial S^*_+} = \mathrm{Id}$. By Lemmas 17.8.3 and 17.8.4, $\Delta(A) = \Delta(H) + \Delta(A') = \Delta(A')$.

Step 4: reduction to an ordinary operator. Let $\kappa \in \mathcal{C}^\infty([0,\epsilon))$ be non-decreasing with $\kappa = 0$ on $[0,\epsilon/2]$ and $\kappa(r) = r$ near ϵ, and let $\Phi_t(q,r) = (q,(1-t)r + t\kappa(r))$ in the collar, $\Phi_t = \mathrm{Id}$ elsewhere. Then $a' \circ \Phi_t$ is a homotopy of invertible symbols with boundary value Id, and $a'' = a' \circ \Phi_1$ equals Id near ∂S^*_+. Keeping $\beta = \Pi$, this is a homotopy of fully elliptic pairs. Extend a'' by Id to a smooth function p on $S^*(M \times \mathbb{S})$ and choose $P \in \Psi^0(M \times \mathbb{S}; \mathbb{C}^N)$ with principal symbol p. Then $\Pi P \Pi$ has the joint symbol (a'', Π), so by Lemma 17.8.5 and Proposition 17.8.2, $\Delta(A') = \Delta(\Pi P \Pi) = 0$. □

Remark 17.8.6. The theorem contains the Atiyah–Singer formula on M. Let $\pi_{\le k}$ be the projection onto the span of $e^{ij\theta}$, $0 \le j \le k$, and let $Q \in \Psi^0(M; \mathbb{C}^N \otimes \mathbb{C}^{k+1})$ be elliptic, $\mathbb{C}^{k+1} \cong \operatorname{Ran} \pi_{\le k}$. The operator $A = Q\pi_{\le k} + (\Pi - \pi_{\le k})$ is in $\Psi^{0,0}_{\mathrm{pt},\mathcal{T}}$ (compare the compressions in Chapter 14) with $a = \mathrm{Id}$ and $\beta = \sigma(Q)\pi_{\le k} + \Pi - \pi_{\le k}$, a finite-rank perturbation of Π; so $b = \mathrm{Id}$, $\gamma(b) = 0$ and $\eta(\beta) = \mathrm{Ch}_{\mathrm{odd}}(\sigma(Q))$. Since $\mathrm{ind}(A) = \mathrm{ind}(Q)$, Theorem 17.8.1 reduces to Theorem 17.7.2 on M with the Liouville orientation of S^*M, consistently with Lemma 17.7.1. Conversely, one might try to prove the theorem by deforming every A to this form; but a fully elliptic A need not be deformable to one with $a = \mathrm{Id}$, because the obstruction, the index bundle over M of the Toeplitz family with symbol a restricted to the centre section, need not vanish (it only vanishes after pull-back to S^*M). This is why the proof above deforms in the opposite direction and uses the Atiyah–Singer theorem on $M \times \mathbb{S}$.

Remark 17.8.7. The symbolic term $\gamma(b)$ cannot be omitted. Take $n = 1$, $M = \mathbb{S}_x$, $N = 1$, and symbols $a_t = \exp(it\chi\sin(x+\theta))$ on S^*_+, where χ is a function on the fibres of S^*_+ (half-circles) equal to 1 at the end $\xi = +1$ and to 0 at the end $\xi = -1$. Thus $b_t = \exp(it\sin(x+\theta))$ on the component $\xi = +1$ of S^*M and $b_t = 1$ on $\xi = -1$. Since (a_t) starts at $a_0 = 1$, invertible base families β_t with symbols b_t and $\beta_0 = \Pi$ exist (lift the homotopy as in Chapter 12). In degree one, with $\mathrm{Td} = 1$ there,

$$\int_{S^*M} \gamma(b_t) = c_0 \int_0^{2\pi} \frac{i}{4\pi} \int_0^{2\pi} (it\cos(x+\theta))^2 \, d\theta \, dx = \frac{i}{2\pi} \cdot \frac{i}{4\pi} \cdot (-t^2) \cdot 2\pi^2 = \frac{t^2}{4},$$

which depends on t. Since I is constant along the family by Proposition 17.8.2, the remaining terms of I vary by $-t^2/4$, so the formula without γ cannot hold for all t.

17.9 Non-Toeplitz product-type operators

The formula extends to the full product-type calculus, where both halves $S^*_\pm$ of the blown-up cosphere bundle carry symbols. For a fully elliptic $P \in \Psi^{0,0}_{\mathrm{pt}}(M \times \mathbb{S}; \mathbb{C}^N)$ write $a_\pm = \sigma(P)|_{S^*_\pm}$, $\beta = \beta(P) \in \mathcal{C}^\infty(S^*M; \Psi^0(\mathbb{S}; \mathbb{C}^N))$ and $b_\pm = a_\pm|_{\partial S^*_\pm}$, regarded as functions on $S^*M \times \mathbb{S}$. The eta form $\eta(\beta)$ is defined exactly as before, with $\overline{\mathrm{Tr}}$ on $\Psi^0(\mathbb{S}; \mathbb{C}^N)$. For a map $c \in \mathcal{C}^\infty(S^*M \times \mathbb{S}; \mathrm{GL}(N,\mathbb{C}))$ let $\langle \mathrm{Ch}_{\mathrm{odd}}(c) \rangle = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{Ch}_{\mathrm{odd}}(c(\cdot,\theta))\, d\theta$ be the θ-average of the odd Chern forms of the maps $c(\cdot,\theta)$ on S^*M.

Proposition 17.9.1. *For every fully elliptic $P \in \Psi^{0,0}_{\mathrm{pt}}(M \times \mathbb{S}; \mathbb{C}^N)$,*

$$\mathrm{ind}(P) = \int_{S^*M} \mathrm{Td}(M) \wedge \big(\eta(\beta) + \langle \mathrm{Ch}_{\mathrm{odd}}(b_-) \rangle + \gamma(b_+) - \gamma(b_-)\big) + \int_{S^*_+} \mathrm{Td}(M) \wedge \mathrm{Ch}_{\mathrm{odd}}(a_+) + \int_{S^*_-} \mathrm{Td}(M) \wedge \mathrm{Ch}_{\mathrm{odd}}(a_-),$$

with $S^*_\pm$ oriented as subsets of $S^*(M \times \mathbb{S})$ and S^*M Liouville oriented.

Proof. Let $J : \mathcal{H}^{\perp} \longrightarrow \mathcal{H}$ be the unitary $(Ju)(\theta) = e^{-i\theta}u(-\theta)$, mapping $e^{-i(k+1)\theta}$ to $e^{ik\theta}$, and

$$\mathcal{U} : L^2(M \times \mathbb{S}; \mathbb{C}^N) \longrightarrow \mathcal{H}(M \times \mathbb{S}; \mathbb{C}^{2N}), \quad \mathcal{U}u = \big(\Pi u, J(\mathrm{Id} - \Pi)u\big).$$

The reflection $\theta \mapsto -\theta$ is a fibre-preserving diffeomorphism, so it preserves the product-type calculus, and the off-diagonal blocks $\Pi P(\mathrm{Id} - \Pi)$, $(\mathrm{Id} - \Pi)P\Pi$ have vanishing principal symbol. Hence $P^{\sharp} = \mathcal{U}P\mathcal{U}^{-1}$ is a fully elliptic element of $\Psi^{0,0}_{\mathrm{pt},\mathcal{T}}(M \times \mathbb{S}; \mathbb{C}^{2N})$ with

$$a^{\sharp} = a_+ \oplus R^* a_-, \qquad \beta^{\sharp} = \mathcal{U}\beta\mathcal{U}^{-1},$$

where $R : S^*_+ \longrightarrow S^*_-$ is induced by $(\theta, \tau) \mapsto (-\theta, -\tau)$, and $\mathrm{ind}(P) = \mathrm{ind}(P^{\sharp})$. We evaluate $I(P^{\sharp})$.

The map R preserves the Liouville form $d\tau \wedge d\theta$ and the radial structure, so it preserves orientations and $\int_{S^*_+} \mathrm{Td} \wedge \mathrm{Ch}_{\mathrm{odd}}(R^* a_-) = \int_{S^*_-} \mathrm{Td} \wedge \mathrm{Ch}_{\mathrm{odd}}(a_-)$. The boundary symbol of $P^{\sharp}$ is $b_+ \oplus b_-^R$ with $b_-^R(x, \theta) = b_-(x, -\theta)$; the substitution $\theta \mapsto -\theta$ changes the sign of μ and not of ν, so $\gamma(b_-^R) = -\gamma(b_-)$. Finally $\overline{\mathrm{Tr}}(\mathcal{U}Q\mathcal{U}^{-1}) = \overline{\mathrm{Tr}}_{E_1}(Q)$ where $E_1 = \mathcal{U}^{-1}E\mathcal{U}$ multiplies $e^{ij\theta}$ by λ_j^s for $j \geq 0$ and by $\lambda_{|j|-1}^s$ for $j < 0$. This is an admissible family and, by Lemma 17.3.3, $\overline{\mathrm{Tr}}_{E_1}(Q) = \overline{\mathrm{Tr}}(Q) + \mathrm{Tr}_{\mathrm{R}}(QD)$ where $D = \frac{d}{ds}(E_1 - E)|_{s=0}$ is the Fourier multiplier by $\log \lambda_{|j|-1} - \log \lambda_{|j|}$ for $j < 0$ and 0 for $j \geq 0$, with symbol ξ^{-1} (modulo lower order) in $\xi < 0$ and rapidly decreasing in $\xi > 0$. With $Q = (\beta^{-1}d\beta)^{2k+1}$, Lemma 17.4.1(i) gives

$$\mathrm{Tr}_{\mathrm{R}}(QD) = -\frac{1}{2\pi}\int_0^{2\pi} \mathrm{tr}(\nu_-^{2k+1} \cdot (-1))d\theta = \frac{1}{2\pi}\int_0^{2\pi} \mathrm{tr}\big((b_-^{-1}d_X b_-)^{2k+1}\big)d\theta,$$

so $\eta(\beta^{\sharp}) = \eta(\beta) + \langle \mathrm{Ch}_{\mathrm{odd}}(b_-)\rangle$. Now apply Theorem 17.8.1 to $P^{\sharp}$. □

As a check, let $P \in \Psi^0(M \times \mathbb{S}; \mathbb{C}^N)$ be an ordinary elliptic operator. Then β is multiplication by $b = b_+ = b_-$, so $\eta(\beta) = 0$ by Lemma 17.4.1(iii) and the γ-terms cancel. For fixed θ the map $b(\cdot, \theta)$ on S^*M extends over the ball bundle $S^*_+ \cap \{\theta\}$, hence is homotopic to a map pulled back from M, so $\int_{S^*M} \mathrm{Td} \wedge \mathrm{Ch}_{\mathrm{odd}}(b(\cdot, \theta)) = 0$ as in Lemma 17.8.4. The formula therefore reduces to Theorem 17.7.2 on $M \times \mathbb{S}$. The asymmetric term $\langle \mathrm{Ch}_{\mathrm{odd}}(b_-)\rangle$ reflects the fact that the regularization (17.4.1) is not invariant under J.

17.10 Towards general product-type index theorems

In this final section we indicate how the results above should fit into a general index theory for product-type families. Parts of it are conjectural, and are labelled as such.

17.10.1 Thom isomorphisms and the Todd class

For an oriented real vector bundle $V \longrightarrow M$ of rank r over a compact manifold, integration over the fibres gives the Thom isomorphism $H^*_c(V) \longrightarrow H^{*-r}(M)$. For a complex vector bundle there is also a Thom isomorphism $K_c(V) \longrightarrow K(M)$ in K-theory, extending Bott periodicity (the case of trivial V). Both sides carry Chern characters, giving a square

$$\begin{array}{ccc} K_c(V) & \xrightarrow{\ \mathrm{Thom}\ } & K(M) \\ {\scriptstyle \mathrm{Ch}}\downarrow & & \downarrow{\scriptstyle \mathrm{Ch}} \\ H_c^{\mathrm{even}}(V) & \xrightarrow[\ \mathrm{Thom}\ ]{} & H^{\mathrm{even}}(M) \end{array}$$

which does *not* commute: the two composites differ by multiplication by a characteristic class of V, which (with appropriate conventions) is the Todd class. This is the origin of the factor $\mathrm{Td}(M) = \mathrm{Td}(TM \otimes \mathbb{C})$ in the index formula. A direct computation of the Chern character of the index bundle in the families index theorem of Chapter 15 produces the Todd class in the same way.

17.10.2 Product-type K-theory

Recall that for a fibration $\phi : M \longrightarrow B$ of compact manifolds the families index theorem is the equality of two maps

$$\mathrm{ind}_a = \mathrm{ind}_t : K_c^0(T^*(M/B)) \longrightarrow K^0(B),$$

the analytic index being defined by representing a class by the symbol of an elliptic family $A \in \Psi^0(M/B; \mathbb{E})$ and taking the stabilized index bundle, the topological index by embedding the fibration in a product $\mathbb{S}^N \times B$.

Consider now a compound fibration of compact manifolds

$$M \xrightarrow{\psi} Y \xrightarrow{\Phi} B, \qquad \phi = \Phi \circ \psi,$$

and the families $A \in \Psi^{0,0}_{\mathrm{pt}}(M/B; \mathbb{E})$ of operators on the fibres of ϕ which are of product type with respect to the fibration (over $\Phi^{-1}(b)$) induced by ψ. Such a family has two symbols,

$$\sigma(A) \in \mathcal{C}^\infty\big([S^*(M/B), \psi^* S^*(Y/B)]; \hom(\mathbb{E})\big), \qquad \beta(A) \in \Psi^0\big(\pi^* M/S^*(Y/B); \mathbb{E}\big),$$

the second being a family of operators on the fibres of ψ parametrized by $S^*(Y/B)$, and full ellipticity means invertibility of both. Then the analytic index $\mathrm{ind}_a(A) \in K^0(B)$ is well defined by stabilization and depends only on $(\sigma(A), \beta(A))$. Considering all compatible invertible pairs modulo bundle isomorphism, stabilization and homotopy one expects an abelian group $K^0_{c,\psi\text{-pt}}(T^*(M/B))$ (and an odd version $K^{-1}_{c,\psi\text{-pt}}$) on which the analytic index is defined. Forgetting β gives a map to $K_c^0(T^*(M/Y))$ (evaluation of σ in the vertical directions of ψ), and the families index of ψ, pulled back to $S^*(Y/B)$, gives a map $K_c^0(T^*(M/Y)) \longrightarrow K^0(S^*(Y/B))$ which vanishes on the image, expressing the invertibility of β. Identifying $K^{-1}(S^*(Y/B))$ with homotopy classes of maps into $G^{-\infty}$, i.e. with invertible base families with symbol the identity, one is led to:

Remark 17.10.1 (Conjectural six-term sequence). There should be an exact sequence

$$\begin{array}{ccccc} K^{-1}(S^*(Y/B)) & \longrightarrow & K^0_{c,\psi\text{-pt}}(T^*(M/B)) & \xrightarrow{\sigma} & K_c^0(T^*(M/Y)) \\ {\scriptstyle \mathrm{ind}}\uparrow & & & & \downarrow{\scriptstyle \mathrm{ind}} \\ K_c^{-1}(T^*(M/Y)) & \xleftarrow{\sigma} & K^{-1}_{c,\psi\text{-pt}}(T^*(M/B)) & \longleftarrow & K^0(S^*(Y/B)) \end{array}$$

and the analytic index should extend to an equality $\mathrm{ind}_a = \mathrm{ind}_t : K^0_{c,\psi\text{-pt}}(T^*(M/B)) \longrightarrow K^0(B)$ with a topological index defined by embedding. A natural problem is to find a more traditional realization of the group $K^0_{c,\psi\text{-pt}}(T^*(M/B))$, for which the six-term sequence should serve as a guide. Groups of this type were subsequently constructed by Melrose and Rochon; see [20] for the closely related families index for manifolds with boundary.

17.10.3 Product-type cohomology

The cohomological form of the index theorem expresses the Chern character of the index in terms of a Chern character of the symbol. Theorem 17.8.1 suggests what the latter should be in the product-type case: a pair consisting of the Chern form of σ and the eta form of β, related by Theorem 17.6.2. We formulate the corresponding cohomology in an abstract setting (correcting the degree conventions of the original notes).

Let Z be a compact manifold with boundary and $\psi : \partial Z \longrightarrow Y$ a fibration over a compact manifold Y without boundary, with compact oriented fibres of dimension f, and let ψ_* denote integration over the fibres with the 'fibre last' convention, so that $\psi_* d = d\psi_*$ and $\int_{\partial Z} = \int_Y \psi_*$ when ∂Z is oriented by Y followed by the fibres. Let $\iota : \partial Z \hookrightarrow Z$. Set

$$C^k(Z,\psi) = \mathcal{C}^\infty(Z;\Lambda^k) \oplus \mathcal{C}^\infty(Y;\Lambda^{k-f-1}), \qquad \mathcal{D}(u,\tau) = (du,\ \psi_*\iota^* u - d\tau).$$

Then $\mathcal{D}^2 = 0$; a pair is closed if $du = 0$ and $d\tau = \psi_*\iota^* u$, and exact if it is of the form $(dv, \psi_*\iota^* v - d\varsigma)$. Denote the cohomology by $H^k(Z,\psi)$.

Proposition 17.10.2. *(i) There is a long exact sequence*

$$\cdots \longrightarrow H^{k-f-1}(Y) \longrightarrow H^k(Z,\psi) \longrightarrow H^k(Z) \xrightarrow{\psi_*\iota^*} H^{k-f}(Y) \longrightarrow H^{k+1}(Z,\psi) \longrightarrow \cdots .$$

(ii) If $\dim Z = n$, *the pairing* $\langle (u,\tau), [Z,\psi]\rangle = \int_Z u - \int_Y \tau$ *on* $C^n(Z,\psi)$ *vanishes on exact pairs, so defines a linear form on* $H^n(Z,\psi)$.

Proof. (i) The sequence of complexes $0 \longrightarrow \mathcal{C}^\infty(Y;\Lambda^{\bullet-f-1}) \longrightarrow C^\bullet(Z,\psi) \longrightarrow \mathcal{C}^\infty(Z;\Lambda^\bullet) \longrightarrow 0$, $\tau \mapsto (0,\tau)$ and $(u,\tau) \mapsto u$, is exact and compatible with the differentials (up to the sign of d on the first complex, which does not affect cohomology). The connecting map sends $[u]$ to the class of $\psi_*\iota^* u$, since $\mathcal{D}(u,0) = (0, \psi_*\iota^* u)$. (ii) $\int_Z dv - \int_Y (\psi_*\iota^* v - d\varsigma) = \int_{\partial Z} v - \int_{\partial Z} v = 0$ by Stokes' theorem. □

When ψ is the identity of ∂Z ($f = 0$), $C^\bullet(Z,\psi)$ is the mapping cone of ι^*, and $H^k(Z,\psi)$ is the relative cohomology $H^k(Z,\partial Z)$, i.e. the compactly supported cohomology of the interior (Exercise 17.7).

In the setting of Theorem 17.8.1, with $Z = S^*_+$, $Y = S^*M$ and ψ the projection $S^*M \times \mathbb{S} \longrightarrow S^*M$ ($f = 1$), Theorem 17.6.2 says precisely that

$$\mathrm{Ch}_{\mathrm{pt}}(A) = \big(\mathrm{Td}(M) \wedge \mathrm{Ch}_{\mathrm{odd}}(a),\ -\mathrm{Td}(M) \wedge (\eta(\beta) + \gamma(b))\big)$$

is closed, and Theorem 17.8.1 becomes

$$\mathrm{ind}(A) = \langle \mathrm{Ch}_{\mathrm{pt}}(A), [S^*_+, \psi]\rangle.$$

The proof of Proposition 17.8.2 amounts to showing that the class of $\mathrm{Ch}_{\mathrm{pt}}(A)$ is homotopy invariant. In general one expects a Chern character $K^0_{c,\psi\text{-pt}}(T^*(M/B)) \longrightarrow H^{\mathrm{even}}_{c,\psi\text{-pt}}(T^*(M/B))$, given by the pair (relative Chern form of σ, eta form of β) and defined using connections, mapping the K-theory six-term sequence of Remark 17.10.1 to the corresponding cohomological one, together with a push-forward $H^{\mathrm{even}}_{c,\psi\text{-pt}}(T^*(M/B)) \longrightarrow H^{\mathrm{even}}(B)$ when the fibres are oriented, so that $\mathrm{Ch}(\mathrm{ind}_a(A))$ is the push-forward of $\mathrm{Td} \cdot \mathrm{Ch}_{\mathrm{pt}}(A)$.

17.10.4 The determinant line bundle

The numerical index is the degree-zero part of the Chern character of the index bundle. In the odd case the degree-one part may be thought of as the 'spectral flow' of the phase of a determinant. In the even case the degree-two part corresponds to the determinant line bundle, which we now construct; the degree-three part in the odd case corresponds to the curvature of a gerbe, and the degree-four part in the even case to a 2-gerbe.

Let $\phi : M \longrightarrow B$ be a fibration with compact fibres Z_b and let $P \in \Psi^0(M/B; E_+, E_-)$ be an elliptic family of numerical index 0. Then each P_b can be perturbed by a smoothing operator to be invertible, so the sets

$$\mathcal{P}_b = \{P_b + Q_b;\ Q_b \in \Psi^{-\infty}(Z_b; E_+, E_-),\ (P_b + Q_b)^{-1} \in \Psi^0(Z_b; E_-, E_+)\}$$

are non-empty. Each $\mathcal{P}_b$ is a principal homogeneous space for the group $G^{-\infty}(Z_b; E_-)$ acting on the left: if $A, A' \in \mathcal{P}_b$ then $A' = (\mathrm{Id} + R)A$ with $\mathrm{Id} + R = A'A^{-1}$, $R = (A' - A)A^{-1}$ smoothing. Thus $\mathcal{P} = \bigcup_b \mathcal{P}_b \longrightarrow B$ behaves like a principal bundle, although the structure groups $G^{-\infty}(Z_b; E_-)$ form a bundle of groups rather than a fixed group. (Using exhausting families of smoothing projections, as in the stabilization of the index bundle in Chapter 12, one sees that the components of the space of sections of this bundle of groups are canonically $K^{-1}(B)$, so it is 'weakly trivial'.) The Fredholm determinant is a character of each $G^{-\infty}(Z_b; E_-)$, so we may form the associated line bundle.

Proposition 17.10.3. *Let* $\mathrm{Det}_b = (\mathcal{P}_b \times \mathbb{C})/\sim$, *where* $(A, z) \sim ((\mathrm{Id} + R)A, \det(\mathrm{Id} + R)^{-1} z)$ *for* $\mathrm{Id} + R \in G^{-\infty}(Z_b; E_-)$. *Then* $\mathrm{Det} = \bigcup_b \mathrm{Det}_b$ *is a smooth line bundle over* B, *and*

$$s(b) = [A, \det(A^{-1} P_b)], \qquad A \in \mathcal{P}_b,$$

is a well-defined smooth section of Det *which vanishes exactly at the points* b *where* P_b *is not invertible.*

Proof. The relation is an equivalence relation because det is multiplicative. For fixed $A_0 \in \mathcal{P}_b$, every element of $\mathcal{P}_b$ is uniquely $(\mathrm{Id} + R)A_0$, so $z \mapsto [A_0, z]$ is a bijection $\mathbb{C} \longrightarrow \mathrm{Det}_b$, and the linear structures induced by different choices of A_0 agree. Local triviality: if $A_0 = P_{b_0} + Q_0 \in \mathcal{P}_{b_0}$, choose a local trivialization of the fibration near b_0 and transport Q_0 to a smooth family Q_b; since invertibility is open, $P_b + Q_b \in \mathcal{P}_b$ for b near b_0, giving a local smooth section of $\mathcal{P}$, hence a local trivialization, and the transition functions are det of smooth families in $G^{-\infty}$, hence smooth. For s: $A^{-1}P_b = \mathrm{Id} - A^{-1}Q_b \in \mathrm{Id} + \Psi^{-\infty}$ has a Fredholm determinant, which is smooth in b for a smooth local section A. Replacing A by $(\mathrm{Id} + R)A$ gives $\det(A^{-1}(\mathrm{Id} + R)^{-1}P_b) = \det(\mathrm{Id} + R)^{-1} \det(A^{-1}P_b)$, using the multiplicativity of det and its invariance under conjugation by A; so s is well defined. Finally $\det(A^{-1}P_b) \neq 0$ if and only if $A^{-1}P_b$, equivalently P_b, is invertible. □

When P_b is invertible, $s(b) = [P_b, 1]$. The section s is the determinant of the family. The curvature of natural connections on Det (Quillen, Bismut–Freed) represents the degree-two part of $-\mathrm{Ch}(\mathrm{ind}\, P)$ (compare Remark 17.6.4); with the opposite convention for the equivalence relation one obtains the dual line bundle.

17.11 Exercises

Exercise 17.1. Let $U = \Pi e^{i\theta}\Pi$ and $L = \Pi e^{-i\theta}\Pi$ be the shifts on $\mathcal{H}$. Show that $[L, U]$ is the projection onto the constants, and verify the trace-defect formula $\overline{\mathrm{Tr}}([L, U]) = \mathrm{Tr}_{\mathrm{R}}((D_E L)U) = 1$ directly from Lemma 17.4.1.

Exercise 17.2. Show that D_E is not an inner derivation of $\Psi^{\mathbb{Z}}(\mathbb{S})$. (Hint: if $D_E = [C, \cdot]$ with $C \in \Psi^k(\mathbb{S})$, compare the principal symbols of $D_E f = -i\xi^{-1} f' + \dots$ and $[C, f]$, $f \in \mathcal{C}^\infty(\mathbb{S})$; show that $k \le 0$ is forced and that $k = 0$ would require $\partial_\xi \sigma_0(C) = \xi^{-1}$ with $\sigma_0(C)$ homogeneous of degree 0.)

Exercise 17.3. Let E' be another admissible family on $\mathbb{S}$ with $D = D(E, E') \in \Psi^{-1}(\mathbb{S})$ of principal symbol $\delta(\theta, \xi)$. Show that the eta forms defined using E' are

$$\eta'(B) = \eta(B) - \frac{1}{2\pi} \sum_k c_k \int_0^{2\pi} \operatorname{tr}(\nu^{2k+1})\, \delta(\theta, 1)\, d\theta,$$

so that they differ from η by a term local in the symbol of B. Deduce that Theorem 17.8.1 holds with η' in place of η provided γ is replaced by γ minus this local term.

Exercise 17.4. Let $g, h \in \mathcal{C}^\infty(W; \mathrm{GL}(N, \mathbb{C}))$. Using the homotopy $R_t(g \oplus \mathrm{Id}) R_t^{-1}(\mathrm{Id} \oplus h)$ and Lemma 17.5.1, show that $\mathrm{Ch}_{\mathrm{odd}}(gh) - \mathrm{Ch}_{\mathrm{odd}}(g) - \mathrm{Ch}_{\mathrm{odd}}(h)$ is exact, and that its degree-one part vanishes.

Exercise 17.5. Show that for elliptic $P \in \Psi^0(\mathbb{S}; \mathbb{C}^N)$, $\operatorname{ind}(P) = \int_{S^*\mathbb{S}} \mathrm{Ch}_{\mathrm{odd}}(\sigma_0(P))$ with the Liouville orientation, by writing P, up to compact operators, as $\Pi P_+ \Pi + (\mathrm{Id} - \Pi) P_- (\mathrm{Id} - \Pi)$ and applying (17.5.1) to both parts (for the negative part use J from the proof of Proposition 17.9.1).

Exercise 17.6. Verify directly in local coordinates that for $n = 1$ the Liouville orientation of $S^*M \cong M \times \{\pm 1\}$ is $\pm dx$, and that the orientation of ∂S^*_+ is $\pm dx \wedge d\theta$ on the two components.

Exercise 17.7. Check that when $\psi : \partial Z \longrightarrow \partial Z$ is the identity, $H^k(Z, \psi)$ is naturally isomorphic to $H^k(Z, \partial Z) = H^k_c(Z \setminus \partial Z)$. (Hint: given a closed pair (u, τ), use a collar and a cut-off χ equal to 1 near ∂Z to replace it by the cohomologous pair $(u - d(\chi\tau), 0)$, with $u - d(\chi\tau)$ vanishing near ∂Z.)

Exercise 17.8. Give the details of Proposition 17.10.2(i), and show that the connecting map $H^k(Z) \longrightarrow H^{k-f}(Y)$ is induced by $\psi_* \iota^*$.

Exercise 17.9. In the setting of Proposition 17.10.3, suppose $P_b = P_0 + K_b$ where P_0 is a fixed invertible family (in a local trivialization of the fibration) and $K_b \in \Psi^{-\infty}$. Show that in the trivialization $z \mapsto [P_0, z]$ of Det the determinant section is $s(b) = \det(\mathrm{Id} + P_0^{-1} K_b)$, and that Det is trivial if every P_b is invertible.

Notes

Regularized traces of this kind were studied by Kontsevich and Vishik, and the trace-defect formula appears in the treatment of the b-trace in [16]; the residue trace is due to Wodzicki and Guillemin [13, 5], and the meromorphic continuation to Seeley [11]. In the APS index theorem the eta invariant arises in the same way, as a regularized trace of the indicial family [16]; the eta forms here are the corresponding objects for the odd Chern character, analogous to the eta forms of Bismut and Cheeger [19], and appear in the families index theorem of Melrose and Rochon [20].

The main departures from the original lecture notes are as follows. We have fixed all normalizations: with $E(s)$ of order $+s$ the residue trace is minus the Wodzicki residue, the trace-defect formula reads $\overline{\mathrm{Tr}}_E([A, B]) = \mathrm{Tr}_{\mathrm{R}}(B D_E A)$ with $D_E A = \frac{d}{ds}[E(s), A]|_{s=0}$, and

the constants of the odd Chern character are determined by the Toeplitz index (Proposition 17.5.2). The variation formula for the eta forms, Theorem 17.6.2, has the structure of the original one (eta form plus a symbolic term γ), with the constants now determined. In the original proof that the right side of the index formula is homotopy invariant, the t-derivative of the symbolic term $\gamma(b_t)$ was dropped when the variation formula was pulled back to $X \times [0,1]$; in fact γ must appear in the index formula (Remark 17.8.7: along that homotopy the index and the Chern term are constant while $\int \gamma$ is not). The original coefficient $-\frac{1}{2}$ of the eta term (the original explicitly left its constants unchecked) is also not a matter of normalization: the eta form restricts to the odd Chern form on finite-rank perturbations of Π, so the reduction to the Atiyah–Singer theorem on M, which the original also uses (Remark 17.8.6), forces the coefficient ± 1, the sign depending only on the orientation of S^*M; with the Liouville orientation it is $+1$. The original proof of the index formula reduced to the Atiyah–Singer theorem on M by composing with the inverse of a bundle isomorphism L; in the Toeplitz algebra the compression $\Pi L^{-1}\Pi$ need not be invertible, nor can it in general be made invertible by a smooth family of smoothing perturbations, since the Toeplitz family with symbol L can have non-zero index bundle over M (Remark 17.8.6), and we have instead given a proof based on the weak contractibility of the Toeplitz group and the Atiyah–Singer theorem on $M \times \mathbb{S}$. The non-Toeplitz case is deduced by a unitary doubling. In the product-type cohomology the compatibility condition has been changed to $d\tau = \psi_* \iota^* u$ (push-forward along the fibres), which is what the variation formula requires; the suggested identification with the cohomology of the space obtained by collapsing the fibres of ψ refers to the pull-back condition and has been dropped. The original also asserts that D_E is the unique continuous outer derivation up to inner derivations and constant multiples; we have not proved this. For the Todd class in the comparison of Thom isomorphisms see Atiyah and Singer, *The index of elliptic operators III*, and for determinant line bundles Quillen and Bismut–Freed.

Part V

Algebraic structure

Chapter 18

Hochschild homology

In Chapter 8 we met the residue trace, a trace on the algebra of polyhomogeneous operators modulo smoothing operators, and we noted that it is essentially the only one. Why is there exactly one trace, and are there "higher" invariants of the same kind? Hochschild homology is the tool that answers such questions: its degree-zero part is dual to the space of traces, and its higher parts are the natural higher analogues. For symbol algebras it turns out to be computable, and the answer is purely topological.

The algebra $\Psi^{\mathbb{Z}}(X)$ of classical pseudodifferential operators of integral order on a compact manifold X, studied in Chapter 5, is a non-commutative deformation of a commutative algebra: modulo $\Psi^{-\infty}(X)$ it is, via the full symbol, the algebra of formal sums of homogeneous functions on $T^*X \setminus 0$, with a product whose leading term is pointwise multiplication and whose first correction is the Poisson bracket. Hochschild homology is the basic homological invariant of an associative algebra; in degree zero it is the dual of the space of traces. The aim of this chapter is to compute the Hochschild homology of the algebra $\Psi^{\mathbb{Z}}(X)/\Psi^{-\infty}(X)$ of formal symbols, following Brylinski–Getzler [1] and Wodzicki [13, 14]:

$$\mathrm{HH}_k(\Psi^{\mathbb{Z}}(X)/\Psi^{-\infty}(X)) \cong H^{2n-k}(S^*X) \oplus H^{2n-k-1}(S^*X), \qquad n = \dim X.$$

The computation proceeds in three stages. First (Sections 18.1 and 18.2) we set up the Hochschild complex and compute it for polynomial algebras by means of the Koszul resolution. Secondly (Sections 18.3 and 18.4) we prove the theorem of Hochschild–Kostant–Rosenberg and Connes that the Hochschild homology of $\mathcal{C}^\infty(X)$ is the space of differential forms, and deduce the analogous statement for the commutative algebra of homogeneous symbols. Finally (Sections 18.5, 18.6, 18.8 and 18.9) we filter the Hochschild complex of the symbol algebra by order; the resulting spectral sequence has E_1 term the homogeneous differential forms on $T^*X \setminus 0$, its first differential is the Poisson (Koszul–Brylinski) differential, a symplectic Hodge star identifies E_2 with de Rham cohomology, and a homogeneity argument shows that the spectral sequence degenerates and converges. Figure 18.1 summarizes the logic.

Throughout, vector spaces are complex, $\otimes$ without decoration is the algebraic tensor product over $\mathbb{C}$, and X is a smooth manifold of dimension n.

18.1 The Hochschild complex

18.1.1 Chains and differentials

Let $\mathcal{A}$ be an associative algebra over $\mathbb{C}$, not necessarily with identity. The space of (algebraic) Hochschild k-chains, $k \in \mathbb{N}_0$, is

$$C_k(\mathcal{A}) = \mathcal{A}^{\otimes(k+1)} = \mathcal{A} \otimes \mathcal{A} \otimes \cdots \otimes \mathcal{A}.$$

$$\begin{array}{ccc}
\mathrm{HH}_\bullet(\mathbb{C}[x]) \cong \Omega^\bullet_{\mathrm{pol}} & \xrightarrow{\text{Taylor, Borel}} & \mathrm{HH}_\bullet(\mathcal{C}^\infty(X)) \cong \Omega^\bullet(X) \\
 & & \Big\downarrow \text{Künneth} \\
E^1_{p,\bullet} = \Omega^\bullet_{(p)}(Y), \quad d^1 = -i\delta & \xleftarrow{\text{filter by order}} & \mathrm{HH}_\bullet(\mathcal{A}_0) \cong \Omega^\bullet_{(\cdot)}(Y) \\
\Big\downarrow \text{symplectic } * & & \\
E^2 \cong H^{2n-\bullet}(S^*X) \oplus H^{2n-\bullet-1}(S^*X) & \xrightarrow{\text{degeneration}} & \mathrm{HH}_\bullet(\mathcal{A})
\end{array}$$

Figure 18.1: Roadmap of the chapter, with $Y = T^*X \setminus 0$. The first three entries are Hochschild homologies of commutative algebras (Koszul resolution, Connes' theorem, homogeneous symbols). The last three treat the non-commutative symbol algebra $\mathcal{A}$ as a deformation of the commutative algebra $\mathcal{A}_0$; the E^2 term is concentrated in the bidegrees with $p = k - n$.

For $k \geq 1$ define the *face maps* $d_i : C_k(\mathcal{A}) \longrightarrow C_{k-1}(\mathcal{A})$, $0 \leq i \leq k$, by

$$\begin{aligned}
d_i(a_0 \otimes \cdots \otimes a_k) &= a_0 \otimes \cdots \otimes a_{i-1} \otimes a_i a_{i+1} \otimes a_{i+2} \otimes \cdots \otimes a_k, \quad 0 \leq i \leq k-1, \\
d_k(a_0 \otimes \cdots \otimes a_k) &= a_k a_0 \otimes a_1 \otimes \cdots \otimes a_{k-1},
\end{aligned} \tag{18.1.1}$$

and set

$$b' = \sum_{i=0}^{k-1} (-1)^i d_i, \qquad b = b' + (-1)^k d_k = \sum_{i=0}^{k} (-1)^i d_i \quad \text{on } C_k(\mathcal{A}), \tag{18.1.2}$$

with $b = b' = 0$ on $C_0(\mathcal{A}) = \mathcal{A}$.

Discussion. It helps to picture a chain $a_0 \otimes \cdots \otimes a_k$ as $k+1$ beads on a necklace (Figure 18.2). The face map d_i multiplies two neighbouring beads. The last face d_k multiplies a_k with a_0, closing the necklace; it is this cyclic face that distinguishes b from b'. In the lowest degree, $b(a_0 \otimes a_1) = a_0 a_1 - a_1 a_0$, so the cyclic face is what makes Hochschild homology see commutators, and hence traces.

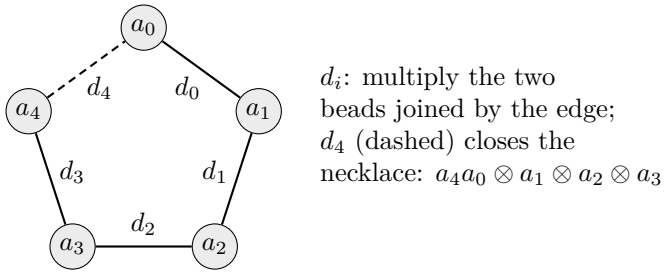


Figure 18.2: A Hochschild 4-chain $a_0 \otimes a_1 \otimes a_2 \otimes a_3 \otimes a_4$ as a necklace. Each edge corresponds to a face map d_i. The bar differential b' uses only the solid edges; the Hochschild differential b also uses the dashed cyclic edge.

Lemma 18.1.1. *For $k \geq 2$ and $0 \leq i < j \leq k$ the face maps satisfy $d_i d_j = d_{j-1} d_i$ as maps $C_k(\mathcal{A}) \longrightarrow C_{k-2}(\mathcal{A})$. Consequently $(b')^2 = 0$ and $b^2 = 0$.*

Proof. If $j \leq k-1$ both sides only multiply neighbouring factors and the identity is the associativity of the product (for $j = i+1$ both sides replace $a_i \otimes a_{i+1} \otimes a_{i+2}$ by $a_i a_{i+1} a_{i+2}$;

for $j > i+1$ the two multiplications commute). If $j = k$ and $i = 0$, both sides give $a_k a_0 a_1 \otimes a_2 \otimes \cdots \otimes a_{k-1}$; if $j = k$ and $0 < i < k-1$, both give $a_k a_0 \otimes \cdots \otimes a_i a_{i+1} \otimes \cdots \otimes a_{k-1}$; and if $j = k$, $i = k-1$, both give $a_{k-1} a_k a_0 \otimes a_1 \otimes \cdots \otimes a_{k-2}$.

Now $b^2 = \sum_{i=0}^{k-1} \sum_{j=0}^{k} (-1)^{i+j} d_i d_j$ on $C_k(\mathcal{A})$. The map $(i,j) \longmapsto (j-1, i)$ is a bijection from the pairs with $i < j \leq k$ onto the pairs (i', j') with $j' \leq i' \leq k-1$, and by the identity the corresponding terms are equal but carry the opposite signs $(-1)^{i+j}$ and $(-1)^{i+j-1}$; hence $b^2 = 0$. The same argument with k replaced by $k-1$ in the range of j gives $(b')^2 = 0$. □

Definition 18.1.2. The *Hochschild homology* of $\mathcal{A}$ is the homology of the complex $(C_\bullet(\mathcal{A}), b)$:

$$\mathrm{HH}_k(\mathcal{A}) = \frac{\mathrm{Nul}(b : C_k(\mathcal{A}) \to C_{k-1}(\mathcal{A}))}{\mathrm{Ran}(b : C_{k+1}(\mathcal{A}) \to C_k(\mathcal{A}))}, \qquad k \in \mathbb{N}_0.$$

In degree zero $b(a_0 \otimes a_1) = a_0 a_1 - a_1 a_0$, so $\mathrm{HH}_0(\mathcal{A}) = \mathcal{A}/[\mathcal{A}, \mathcal{A}]$, where $[\mathcal{A}, \mathcal{A}]$ is the span of the commutators. Thus the linear functionals on $\mathrm{HH}_0(\mathcal{A})$ are precisely the *traces* on $\mathcal{A}$, the linear maps τ with $\tau(ab) = \tau(ba)$. If $\mathcal{A}$ is commutative then $b = 0$ on $C_1(\mathcal{A})$ and $\mathrm{HH}_0(\mathcal{A}) = \mathcal{A}$.

18.1.2 The bar complex

The complex $(C_\bullet(\mathcal{A}), b')$ is the *bar complex*. It is, in contrast to the Hochschild complex, trivial for algebras with identity.

Lemma 18.1.3. *If $\mathcal{A}$ has an identity 1, then $s(a_0 \otimes \cdots \otimes a_k) = 1 \otimes a_0 \otimes \cdots \otimes a_k$ defines maps $s : C_k(\mathcal{A}) \to C_{k+1}(\mathcal{A})$ with*

$$b's + sb' = \mathrm{Id} \quad \text{on } C_k(\mathcal{A}),\ k \geq 0;$$

in particular $(C_\bullet(\mathcal{A}), b')$ is acyclic.

Proof. On $C_{k+1}(\mathcal{A})$, $d_0(1 \otimes a) = a$ for $a \in C_k(\mathcal{A})$, while for $1 \leq i \leq k$, $d_i(1 \otimes a) = 1 \otimes d_{i-1} a$. Hence $b'sa = a + \sum_{i=1}^{k} (-1)^i\, 1 \otimes d_{i-1} a = a - sb'a$. □

Definition 18.1.4. An associative algebra is *H-unital* if its bar complex $(C_\bullet(\mathcal{A}), b')$ is acyclic.

So every unital algebra is H-unital. For H-unital algebras Hochschild homology has good excision properties (Wodzicki); we shall not need these.

18.1.3 Hochschild homology as a derived functor

Suppose $\mathcal{A}$ has an identity and let $\mathcal{A}^e = \mathcal{A} \otimes \mathcal{A}^{\mathrm{op}}$, so that $\mathcal{A}$-bimodules are the same as left $\mathcal{A}^e$-modules; $\mathcal{A}$ itself is an $\mathcal{A}^e$-module by $(a \otimes a') \cdot c = aca'$. Set

$$P_k = \mathcal{A} \otimes \mathcal{A}^{\otimes k} \otimes \mathcal{A}\ (= \mathcal{A}^{\otimes (k+2)}), \qquad (a \otimes a') \cdot (a_0 \otimes \cdots \otimes a_{k+1}) = a a_0 \otimes a_1 \otimes \cdots \otimes a_{k+1} a'.$$

Each $P_k \cong \mathcal{A}^e \otimes \mathcal{A}^{\otimes k}$ is a free $\mathcal{A}^e$-module, the bar differential b' of (18.1.2) (on $\mathcal{A}^{\otimes (k+2)}$, i.e. $b' = \sum_{i=0}^{k} (-1)^i d_i$) is $\mathcal{A}^e$-linear, and by Lemma 18.1.3 the complex

$$\ldots \xrightarrow{b'} P_2 \xrightarrow{b'} P_1 \xrightarrow{b'} P_0 \xrightarrow{m} \mathcal{A} \longrightarrow 0, \qquad m(a_0 \otimes a_1) = a_0 a_1, \tag{18.1.3}$$

is exact: it is a free resolution of $\mathcal{A}$ as an $\mathcal{A}^e$-module. Applying $\mathcal{A} \otimes_{\mathcal{A}^e} (\cdot)$ and using the isomorphism

$$\mathcal{A} \otimes_{\mathcal{A}^e} P_k \longrightarrow C_k(\mathcal{A}), \quad c \otimes (a_0 \otimes a_1 \otimes \cdots \otimes a_k \otimes a_{k+1}) \longmapsto a_{k+1} c a_0 \otimes a_1 \otimes \cdots \otimes a_k,$$

the differential $\mathrm{Id}\otimes b'$ becomes b: the term d_0 of b' gives d_0, and the last term $(-1)^k(\cdots\otimes a_ka_{k+1})$ of b' on P_k (with $a_{k+1}=1$) gives $(-1)^k a_k c a_0\otimes\cdots=(-1)^k d_k$. Thus

$$\mathrm{HH}_k(\mathcal{A})=\mathrm{Tor}_k^{\mathcal{A}^e}(\mathcal{A},\mathcal{A}).$$

The practical content of this is the following standard fact (the *comparison theorem* of homological algebra): if $Q_\bullet\to\mathcal{A}\to 0$ is another resolution of $\mathcal{A}$ by free (or projective) $\mathcal{A}^e$-modules, then there are $\mathcal{A}^e$-linear chain maps $\Phi:Q_\bullet\to P_\bullet$ and $\Psi:P_\bullet\to Q_\bullet$ inducing the identity on $\mathcal{A}$, and $\Psi\Phi$, $\Phi\Psi$ are $\mathcal{A}^e$-linearly chain homotopic to the identity. The maps are constructed degree by degree: on a basis element e of Q_k one chooses $\Phi(e)$ in P_k with $b'\Phi(e)=\Phi(\partial e)$, which is possible because $\Phi(\partial e)$ is a cycle (for $k=0$, lies over the right element of $\mathcal{A}$) and $P_\bullet$ is exact; the homotopies are constructed in the same way. Consequently $\mathrm{HH}_\bullet(\mathcal{A})$ is also the homology of $\mathcal{A}\otimes_{\mathcal{A}^e}Q_\bullet$.

18.1.4 Topological algebras

For the 'large' algebras of analysis the algebraic tensor product is inappropriate and is replaced by a completion. If $\mathcal{A}$ is a Fréchet algebra (multiplication jointly continuous) one uses the completed projective tensor product $\mathcal{A}^{\hat\otimes(k+1)}$; the face maps extend by continuity and Lemmas 18.1.1 and 18.1.3 continue to hold, by density of the algebraic tensor product. The resulting homology is the *continuous* Hochschild homology.

For $\mathcal{A}=\mathcal{C}^\infty(X)$, X compact, a theorem of Grothendieck identifies $\mathcal{C}^\infty(X)\hat\otimes\mathcal{C}^\infty(Y)$ with $\mathcal{C}^\infty(X\times Y)$ (this rests on the nuclearity of $\mathcal{C}^\infty(X)$; the algebraic tensor product is in any case dense in $\mathcal{C}^\infty(X\times Y)$). We shall simply *define* the chain spaces for $\mathcal{C}^\infty(X)$ to be $\mathcal{C}^\infty(X^{k+1})$, see Section 18.3; for the symbol algebras the completion used is specified in Definitions 18.4.1 and 18.5.1.

18.2 Polynomial algebras

Let $\mathcal{A}=\mathbb{C}[x]=\mathbb{C}[x_1,\dots,x_n]$, regarded as the algebra of polynomial functions on $\mathbb{R}^n$. Then $C_k(\mathcal{A})=\mathbb{C}[z_0,z_1,\dots,z_k]$, $z_j\in\mathbb{R}^n$, is the space of polynomials on $(\mathbb{R}^n)^{k+1}$, the element $a_0\otimes\cdots\otimes a_k$ corresponding to $a_0(z_0)a_1(z_1)\cdots a_k(z_k)$. In these terms the face maps (18.1.1) are restrictions to diagonals:

$$\begin{aligned}(d_iu)(z_0,\dots,z_{k-1})&=u(z_0,\dots,z_{i-1},z_i,z_i,z_{i+1},\dots,z_{k-1}),\quad 0\le i\le k-1,\\ (d_ku)(z_0,\dots,z_{k-1})&=u(z_0,z_1,\dots,z_{k-1},z_0).\end{aligned}\tag{18.2.1}$$

Let $\Omega^k_{\mathrm{pol}}=\mathbb{C}[x]\otimes\Lambda^k(\mathbb{C}^n)$ be the space of k-forms on $\mathbb{R}^n$ with polynomial coefficients, dx^j corresponding to the j-th basis vector e_j of $\mathbb{C}^n$.

Discussion. For a commutative algebra of functions, a k-chain is a function of $k+1$ points $z_0,\dots,z_k$, and every face map restricts it to a diagonal where two neighbouring points coincide. A cycle is therefore a function whose alternating sum of diagonal restrictions vanishes, and only the behaviour of chains infinitely near the full diagonal $z_0=\cdots=z_k$ can survive in homology. Near the diagonal a function of $k+1$ points that vanishes when any two neighbours meet looks, to first order, like a k-linear expression in the differences z_j-z_{j-1}; the cycle condition makes it alternating. This is why Hochschild homology of a commutative algebra of functions consists of differential forms. The map ε below extracts exactly this first-order part.

18.2.1 The Hochschild–Kostant–Rosenberg map

For a function u of $(z_0, \dots, z_k)$ set

$$\varepsilon(u)(x) = \frac{1}{k!} \sum_{j_1,\dots,j_k=1}^{n} \frac{\partial^k u}{\partial z_1^{j_1} \cdots \partial z_k^{j_k}}(x, x, \dots, x)\, dx^{j_1} \wedge \cdots \wedge dx^{j_k}, \tag{18.2.2}$$

where z_l^j is the j-th coordinate of z_l. Thus $\varepsilon(a_0 \otimes a_1 \otimes \cdots \otimes a_k) = \frac{1}{k!} a_0\, da_1 \wedge \cdots \wedge da_k$. For $a_0, \dots, a_k \in \mathcal{A}$ we also set

$$\varphi(a_0; a_1, \dots, a_k) = \sum_{\sigma \in \mathfrak{S}_k} \operatorname{sgn}(\sigma)\, a_0 \otimes a_{\sigma(1)} \otimes \cdots \otimes a_{\sigma(k)} \in C_k(\mathcal{A}). \tag{18.2.3}$$

Lemma 18.2.1. *For any commutative algebra $\mathcal{A}$, define $\varepsilon : C_k(\mathcal{A}) \to \Omega^k_{\mathcal{A}}$ by $\varepsilon(a_0 \otimes \cdots \otimes a_k) = \frac{1}{k!} a_0\, da_1 \wedge \cdots \wedge da_k$, where $\Omega^\bullet_{\mathcal{A}}$ is any graded-commutative algebra containing $\mathcal{A}$ in degree zero, with a derivation d (for instance the de Rham complex of a manifold, with $\mathcal{A}$ a subalgebra of the smooth functions). Then $\varepsilon \circ b = 0$, $b\,\varphi(a_0; a_1, \dots, a_k) = 0$ and $\varepsilon(\varphi(a_0; a_1, \dots, a_k)) = a_0\, da_1 \wedge \cdots \wedge da_k$.*

Proof. Write $T_i = a_0 a_i\, da_1 \wedge \cdots \wedge \widehat{da_i} \wedge \cdots \wedge da_k$. By the Leibniz rule $d(a_i a_{i+1}) = a_{i+1} da_i + a_i da_{i+1}$, so $(k-1)!\,\varepsilon(b(a_0 \otimes \cdots \otimes a_k))$ equals

$$T_1 + \sum_{i=1}^{k-1} (-1)^i (T_{i+1} + T_i) + (-1)^k T_k = 0.$$

In $b\varphi$, the terms d_i with $1 \le i \le k-1$ cancel in pairs, since σ and $\sigma \circ (i\ i{+}1)$ have opposite signs and $\mathcal{A}$ is commutative. The term d_0 for σ is $\operatorname{sgn}(\sigma) a_0 a_{\sigma(1)} \otimes a_{\sigma(2)} \otimes \dots$, and the term $(-1)^k d_k$ for σ', where $\sigma'(j) = \sigma(j+1)$ for $j < k$ and $\sigma'(k) = \sigma(1)$, is $(-1)^k \operatorname{sgn}(\sigma') a_{\sigma(1)} a_0 \otimes a_{\sigma(2)} \otimes \dots$; since $\operatorname{sgn}(\sigma') = (-1)^{k-1} \operatorname{sgn}(\sigma)$ these cancel. Finally $\varepsilon\varphi = \frac{1}{k!} \sum_\sigma \operatorname{sgn}(\sigma) a_0\, da_{\sigma(1)} \wedge \cdots \wedge da_{\sigma(k)} = a_0\, da_1 \wedge \cdots \wedge da_k$. □

18.2.2 The Koszul resolution

We identify $\mathcal{A}^e = \mathbb{C}[x] \otimes \mathbb{C}[x]$ with $\mathbb{C}[x, y]$, polynomials in $(x, y) \in \mathbb{R}^n \times \mathbb{R}^n$, via $a \otimes a' \mapsto a(x)a'(y)$; then $\mathcal{A}$ is the $\mathcal{A}^e$-module $\mathbb{C}[x, y]/(x_1 - y_1, \dots, x_n - y_n)$. The *Koszul complex* is

$$K_k = \mathbb{C}[x, y] \otimes \Lambda^k(\mathbb{C}^n), \quad \partial(f\, e_{i_1} \wedge \cdots \wedge e_{i_k}) = \sum_{l=1}^{k} (-1)^{l-1} (x_{i_l} - y_{i_l}) f\, e_{i_1} \wedge \cdots \wedge \widehat{e_{i_l}} \wedge \cdots \wedge e_{i_k},$$

with augmentation $K_0 = \mathbb{C}[x, y] \to \mathcal{A}$, $f \mapsto f(x, x)$.

Lemma 18.2.2. *The augmented Koszul complex $\cdots \to K_1 \to K_0 \to \mathcal{A} \to 0$ is exact, so $K_\bullet$ is a free resolution of $\mathcal{A}$ as an $\mathcal{A}^e$-module.*

Proof. In the variables $u = x - y$, $v = y$ identify K_k with the k-forms in u with coefficients in $\mathbb{C}[u, v]$, $e_i \leftrightarrow du_i$. Then $\partial = \iota_E$ is contraction with the Euler field $E = \sum_i u_i \partial_{u_i}$ and the augmentation is $f \mapsto f|_{u=0}$. Let d_u be the exterior derivative in u (with v as a parameter). By Cartan's formula $d_u \iota_E + \iota_E d_u = \mathcal{L}_E$, and $\mathcal{L}_E$ multiplies $u^\alpha du_I$ (with coefficients in $\mathbb{C}[v]$) by $|\alpha| + |I|$. Hence if $\omega \in K_k$, $k \ge 1$, is homogeneous of total degree $N = |\alpha| + |I| \ge 1$ in u and $\partial\omega = 0$, then $\omega = \partial(d_u\omega/N)$; splitting a general cycle into homogeneous parts, $K_\bullet$ is exact in degrees $k \ge 1$. In degree 0 the kernel of the augmentation consists of the f with no term of degree 0 in u, and such an f equals $\partial(\sum_N d_u f_N/N)$, f_N its homogeneous parts. Finally the augmentation is surjective. □

Theorem 18.2.3 (Hochschild–Kostant–Rosenberg)**.** *Let* $\mathcal{A} = \mathbb{C}[x_1, \dots, x_n]$. *The map* (18.2.2) *induces an isomorphism*

$$\varepsilon_* : \mathrm{HH}_k(\mathbb{C}[x]) \longrightarrow \Omega^k_{\mathrm{pol}} = \mathbb{C}[x] \otimes \Lambda^k(\mathbb{C}^n), \qquad k \in \mathbb{N}_0,$$

with inverse induced by $a_0\, dx^{i_1} \wedge \cdots \wedge dx^{i_k} \longmapsto \varphi(a_0; x_{i_1}, \dots, x_{i_k})$. *In particular* $\mathrm{HH}_k(\mathbb{C}[x]) = 0$ *for* $k > n$.

Proof. Let $P_\bullet$ be the bar resolution (18.1.3) and define the $\mathcal{A}^e$-linear maps $\Phi : K_k \to P_k$ by

$$\Phi(e_{i_1} \wedge \cdots \wedge e_{i_k}) = \sum_{\sigma \in \mathfrak{S}_k} \mathrm{sgn}(\sigma)\, 1 \otimes x_{i_{\sigma(1)}} \otimes \cdots \otimes x_{i_{\sigma(k)}} \otimes 1.$$

This is a chain map. Indeed, applying b', the inner terms (multiplying two adjacent x's) cancel in pairs as in Lemma 18.2.1. The first term is $x_{i_{\sigma(1)}} \cdot (1 \otimes x_{i_{\sigma(2)}} \otimes \cdots \otimes 1)$ (left action, i.e. multiplication by the variable x) and, grouping the σ with $\sigma(1) = l$, contributes $\sum_l (-1)^{l-1} x_{i_l} \Phi(e_{i_1} \wedge \dots \widehat{e_{i_l}} \cdots \wedge e_{i_k})$; the last term is $(-1)^k (1 \otimes \cdots \otimes 1) \cdot x_{i_{\sigma(k)}}$ (right action, multiplication by y) and, grouping the σ with $\sigma(k) = l$ (sign $(-1)^{k-l}$), contributes $-\sum_l (-1)^{l-1} y_{i_l} \Phi(\dots \widehat{e_{i_l}} \dots)$. Together this is $\Phi(\partial e_{i_1} \wedge \cdots \wedge e_{i_k})$. In degree 0, Φ is the identity of $\mathcal{A}^e$, so Φ lies over the identity of $\mathcal{A}$.

By Lemma 18.2.2 both complexes are free resolutions of $\mathcal{A}$, so by the comparison theorem (Section 18.1.3) $\mathrm{Id} \otimes \Phi$ induces an isomorphism from the homology of $\mathcal{A} \otimes_{\mathcal{A}^e} K_\bullet$ to that of $\mathcal{A} \otimes_{\mathcal{A}^e} P_\bullet$, i.e. to $\mathrm{HH}_\bullet(\mathcal{A})$. Now $\mathcal{A} \otimes_{\mathcal{A}^e} K_k = \mathcal{A} \otimes \Lambda^k(\mathbb{C}^n) = \Omega^k_{\mathrm{pol}}$ and the induced differential vanishes, since $x_i - y_i$ acts on $\mathcal{A}$ as 0. Under the identification $\mathcal{A} \otimes_{\mathcal{A}^e} P_k = C_k(\mathcal{A})$, $\mathrm{Id} \otimes \Phi$ becomes $a_0\, dx^{i_1} \wedge \cdots \wedge dx^{i_k} \mapsto \varphi(a_0; x_{i_1}, \dots, x_{i_k})$, which therefore induces an isomorphism $\varphi_* : \Omega^k_{\mathrm{pol}} \to \mathrm{HH}_k(\mathcal{A})$. By Lemma 18.2.1, ε descends to homology and $\varepsilon_* \varphi_* = \mathrm{Id}$, so $\varepsilon_* = \varphi_*^{-1}$. □

Remark 18.2.4. The same argument applies to the algebra $\mathbb{C}[r, r^{-1}]$ of Laurent polynomials in one variable. Here $\mathcal{A}^e = \mathbb{C}[r^{\pm 1}, s^{\pm 1}]$ and the Koszul resolution is $0 \to \mathcal{A}^e \xrightarrow{r-s} \mathcal{A}^e \to \mathcal{A} \to 0$; it is exact because $r - s$ is not a zero divisor and $f(r,s) - f(r,r)$ is divisible by $r - s$ in $\mathcal{A}^e$ (write $f = (rs)^{-N} g$ with g a polynomial). Hence $\mathrm{HH}_0(\mathbb{C}[r^{\pm 1}]) = \mathbb{C}[r^{\pm 1}]$, $\mathrm{HH}_1(\mathbb{C}[r^{\pm 1}]) = \mathbb{C}[r^{\pm 1}]\, dr$ and $\mathrm{HH}_k(\mathbb{C}[r^{\pm 1}]) = 0$ for $k \geq 2$, the isomorphisms again being induced by ε.

Question. Why use the Koszul resolution at all, when the bar resolution already computes $\mathrm{HH}_\bullet(\mathcal{A})$?

Answer. The bar resolution is enormous: $P_k = \mathcal{A}^{\otimes(k+2)}$ in every degree, and the complex $C_\bullet(\mathcal{A})$ it produces has non-zero differential everywhere. The Koszul resolution has length n and finite rank over $\mathcal{A}^e$, and after tensoring with $\mathcal{A}$ its differential vanishes identically, because $x_i - y_i$ acts as zero. So the homology can be read off with no computation; the comparison theorem transfers the answer back to the Hochschild complex. □

A more elementary argument for Theorem 18.2.3 proceeds instead by successively restricting to chains vanishing on diagonals. This reduction is very useful for smooth functions, where no finite free resolution is at hand, and we now describe it.

18.3 Smooth functions

18.3.1 The complex

Let X be a compact manifold. We take as the space of Hochschild k-chains of $\mathcal{C}^\infty(X)$ the space $C_k = \mathcal{C}^\infty(X^{k+1})$, a completion of $\mathcal{C}^\infty(X)^{\otimes(k+1)}$ (see Section 18.1.4), with the

face maps (18.2.1); these are the continuous extensions of (18.1.1), and $b = \sum_i (-1)^i d_i$ as before. The map $\varepsilon : C_k \to \Omega^k(X) = \mathcal{C}^\infty(X; \Lambda^k T^*X)$ is defined by (18.2.2) in local coordinates. This is invariant: the mixed derivative $\partial^k u / \partial z_1^{j_1} \cdots \partial z_k^{j_k}$, in which each variable is differentiated once, restricted to the diagonal, transforms as a section of $(T^*X)^{\otimes k}$ under change of coordinates, and $\varepsilon(u)$ is $\frac{1}{k!}$ times its image under $v_1 \otimes \cdots \otimes v_k \mapsto v_1 \wedge \cdots \wedge v_k$.

Lemma 18.3.1. *$\varepsilon \circ b = 0$ on C_{k+1}, and $\varepsilon_* : \mathrm{HH}_k(\mathcal{C}^\infty(X)) \to \Omega^k(X)$ is surjective.*

Proof. The value of $\varepsilon(bu)$ at $x \in X$ only involves derivatives of u of order at most $k+1$ at $(x, \dots, x)$. Replacing u, in local coordinates, by its Taylor polynomial of that order, which is a sum of products of functions of the individual variables, the first statement follows from Lemma 18.2.1. Embedding X in some $\mathbb{R}^N$, every k-form is a finite sum of forms $a_0\, da_1 \wedge \cdots \wedge da_k$ with $a_j \in \mathcal{C}^\infty(X)$ (take $a_1, \dots, a_k$ among the restricted coordinate functions). By Lemma 18.2.1, $\varphi(a_0; a_1, \dots, a_k)$ is a cycle with ε-image $a_0\, da_1 \wedge \cdots \wedge da_k$. □

The main result of this section is the corresponding injectivity. Its proof has three steps. First, every class is represented by a *normalized* chain, one that vanishes on all the diagonals $\{z_i = z_{i+1}\}$ (Proposition 18.3.3). Second, a normalized cycle with $\varepsilon(u) = 0$ is *locally* a boundary: away from the full diagonal by a direct construction, and near it by Taylor expansion and the polynomial case (Lemma 18.3.8). Third, the boundaries form a module over $\mathcal{C}^\infty(X^{k+1})$ (Lemma 18.3.4), so local solutions can be patched together with a partition of unity.

Theorem 18.3.2 (Connes)**.** *For a compact manifold X, ε induces an isomorphism $\varepsilon_* : \mathrm{HH}_k(\mathcal{C}^\infty(X)) \longrightarrow \Omega^k(X)$ for each $k \in \mathbb{N}_0$.*

The bar complex $(C_\bullet, b')$ is acyclic by the continuous version of Lemma 18.1.3, with $(su)(z_0, \dots, z_{k+1}) = u(z_1, \dots, z_{k+1})$; so $\mathcal{C}^\infty(X)$ is (continuously) H-unital.

18.3.2 Reduction to normalized chains

The following discussion applies verbatim to $\mathcal{C}^\infty(X)$ and to the polynomial algebra (with $X = \mathbb{R}^n$ and C_k the polynomials on $(\mathbb{R}^n)^{k+1}$). For $u \in C_k$ consider the diagonals

$$D_i = \{z \in X^{k+1}; z_i = z_{i+1}\},\ 0 \le i \le k-1, \qquad D_k = \{z \in X^{k+1}; z_k = z_0\},$$

so that $d_i u$ is essentially the restriction of u to D_i, and set

$$N_k = \{u \in C_k;\ u|_{D_i} = 0,\ 0 \le i \le k-1\}, \qquad M_k = \{u \in N_k;\ u|_{D_k} = 0\},$$
$$B_k = d_{k+1}(N_{k+1}) = \{v(z_0, \dots, z_k, z_0);\ v \in N_{k+1}\}.$$

Note $N_0 = C_0$, $M_0 = C_0$, and $B_k \subset M_k$: a $v \in N_{k+1}$ vanishes on $\{z_k = z_{k+1}\}$, so $v(z_0, \dots, z_k, z_0)$ vanishes when $z_k = z_0$. If $u \in N_k$ then $d_i u = 0$ for $i < k$, so $bu = (-1)^k d_k u$; and $d_k u$ vanishes on the diagonals $\{z_i = z_{i+1}\}$, $i \le k-2$, of X^k. Thus $(N_\bullet, b)$ is a subcomplex of $(C_\bullet, b)$, and

$$H_k(N_\bullet, b) = M_k / B_k. \tag{18.3.1}$$

Proposition 18.3.3. *The inclusion $N_\bullet \hookrightarrow C_\bullet$ induces an isomorphism in homology. Hence $\mathrm{HH}_k = M_k/B_k$; in particular a chain in M_k which is a b-boundary in $C_\bullet$ lies in B_k.*

Proof. For $j \in \mathbb{N}_0$ let $J^j_m = \{u \in C_m;\ u|_{D_i} = 0 \text{ for } 0 \le i < \min(j, m)\}$. Thus $J^0 = C$, and $J^j_m = N_m$ for $m \le j$. Each $J^j_\bullet$ is a subcomplex: if $u \in J^j_m$ then $d_i u = 0$ for $i < \min(j, m)$, and for the other i the restriction $d_i u$ still vanishes on $\{z_l = z_{l+1}\}$ for $l < \min(j, m-1)$ (for $i \le m-1$ because $i \ge l+1$; for $i = m$ trivially).

Let $V_m^j = \{u \in J_m^j;\ u \text{ independent of } z_{j+1}\}$ for $m \geq j+1$, and $V_m^j = 0$ for $m \leq j$. For $m \geq j+1$ every $u \in J_m^j$ decomposes uniquely as

$$u = u' + u'', \qquad u'(z) = u(z_0, \dots, z_j, z_j, z_{j+2}, \dots, z_m) \in V_m^j, \quad u'' = u - u' \in J_m^{j+1},$$

(u'' vanishes on $\{z_j = z_{j+1}\}$; uniqueness because a function independent of z_{j+1} vanishing at $z_{j+1} = z_j$ vanishes). So $J_\bullet^j = J_\bullet^{j+1} \oplus V_\bullet^j$ as graded spaces. We show that $V_\bullet^j$ is a contractible subcomplex; then this is a direct sum of complexes and $J^{j+1} \hookrightarrow J^j$ is a quasi-isomorphism.

Let $u \in V_m^j$. In bu the terms $d_i u$, $i < j$, vanish, and $d_j u = d_{j+1} u$ (both delete the variable z_{j+1} on which u does not depend; if $j+1 = m$, d_{j+1} is the cyclic face d_m and the same holds), so these cancel. Hence

$$bu = \sum_{i=j+2}^{m} (-1)^i d_i u,$$

which is independent of z_{j+1} and lies in J_{m-1}^j; so $bu \in V_{m-1}^j$ (and $bu = 0$ if $m = j+1$). Define $H : V_m^j \to V_{m+1}^j$ by inserting a dummy variable in position $j+2$:

$$(Hu)(z_0, \dots, z_{m+1}) = u(z_0, \dots, z_{j+1}, z_{j+3}, \dots, z_{m+1}).$$

Then $d_{j+2} H u = u$ (if $m = j+1$ this is the cyclic face, which also just deletes the dummy variable), $d_i H u = H d_{i-1} u$ for $j+3 \leq i \leq m+1$ (for $i = m+1$ both sides equal $u(z_0, \dots, z_{j+1}, z_{j+3}, \dots, z_m, z_0)$ after renaming), hence

$$bHu = (-1)^j u + \sum_{i=j+3}^{m+1} (-1)^i H d_{i-1} u = (-1)^j u - Hbu,$$

i.e. $bH + Hb = (-1)^j \,\mathrm{Id}$ on $V_\bullet^j$. Thus $V_\bullet^j$ is contractible.

Consequently $H_k(C) = H_k(J^0) \cong H_k(J^1) \cong \dots \cong H_k(J^{k+1})$, and J^{k+1} coincides with N in degrees $\leq k+1$, so $H_k(J^{k+1}) = H_k(N)$. All isomorphisms are induced by inclusions. The last statement follows from (18.3.1). □

18.3.3 Structure of the normalized cycles

Lemma 18.3.4. (i) *B_k is a $\mathcal{C}^\infty(X^{k+1})$-submodule of M_k.*

(ii) *If $f \in \mathcal{C}^\infty(X)$ and $g \in M_k$, then $(f(z_j) - f(z_{j+1}))g \in B_k$ for $0 \leq j \leq k-1$, and $(f(z_k) - f(z_0))g \in B_k$.*

Proof. (i) If $\psi \in \mathcal{C}^\infty(X^{k+1})$ and $v \in N_{k+1}$, then $\tilde{\psi} v \in N_{k+1}$, where $\tilde{\psi}(z_0, \dots, z_{k+1}) = \psi(z_0, \dots, z_k)$, and $d_{k+1}(\tilde{\psi} v) = \psi\, d_{k+1} v$.

(ii) For $0 \leq j \leq k-1$ set

$$w(z_0, \dots, z_{k+1}) = (-1)^{j+1}(f(z_j) - f(z_{j+1}))\, g(z_0, \dots, z_j, z_{j+2}, \dots, z_{k+1}) \in C_{k+1}.$$

We compute bw. For $i < j$, $d_i w = 0$ since g vanishes on $\{z_i = z_{i+1}\}$; $d_j w = 0$ because of the first factor; for $j+2 \leq i \leq k$, $d_i w = 0$ because g vanishes on its diagonal D_{i-1}; and $d_{k+1} w = 0$ because g vanishes on its diagonal D_k (here $j \leq k-1$ is used, so that z_{k+1} is a variable of g). Finally $d_{j+1} w$ identifies z_{j+1} with z_{j+2} and gives $(-1)^{j+1}(f(z_j) - f(z_{j+1}))g(z_0, \dots, z_k)$. Hence $bw = (f(z_j) - f(z_{j+1}))g$. This lies in M_k and is a boundary in $C_\bullet$, so it lies in B_k by Proposition 18.3.3. The last case follows from $f(z_k) - f(z_0) = -\sum_{j=0}^{k-1}(f(z_j) - f(z_{j+1}))$. □

Near the full diagonal $\Delta = \{z_0 = z_1 = \cdots = z_k\}$ we use the following coordinates. Let $x_0 \in X$, choose coordinates near x_0 identifying a neighbourhood with an open set in $\mathbb{R}^n$, and on a neighbourhood $W = \{|x - x_0| < \delta,\ |y_j| < \delta\}$ of $(x_0, \dots, x_0)$ use

$$x = z_0, \qquad y_j = z_j - z_{j-1},\ j = 1, \dots, k. \tag{18.3.2}$$

Then $D_{j-1} = \{y_j = 0\}$ for $1 \le j \le k$, $D_k = \{y_1 + \cdots + y_k = 0\}$ and $\Delta = \{y = 0\}$.

Lemma 18.3.5. *Let q be a complex-valued k-linear form on $\mathbb{R}^n$. Then $q(y_1, \dots, y_{k-1}, -(y_1 + \cdots + y_{k-1})) \equiv 0$ if and only if q is alternating.*

Proof. If q is alternating the expression is $-\sum_{j<k} q(y_1, \dots, y_{k-1}, y_j) = 0$. Conversely each term $q(y_1, \dots, y_{k-1}, y_j)$, $j < k$, is of degree 2 in y_j and of degree 1 in the other y_i, $i < k$; these multidegrees are distinct, so each term vanishes separately. Thus q vanishes whenever the j-th and k-th arguments coincide, i.e. q is alternating in these two arguments. The transpositions $(j\ k)$ generate $\mathfrak{S}_k$, so q is alternating. □

Lemma 18.3.6. *Let $u \in N_k$. In the coordinates (18.3.2), $u = \sum_I y_{1,i_1} \cdots y_{k,i_k} w_I(x, y)$ on W, with w_I smooth, where $I = (i_1, \dots, i_k) \in \{1, \dots, n\}^k$ and $y_{j,i}$ is the i-th component of y_j. Hence the Taylor series of u at $y = 0$ (with coefficients smooth in x) starts with the k-linear term $u_{(k)}(x; y) = \sum_I w_I(x, 0) y_{1,i_1} \cdots y_{k,i_k}$, and*

$$\varepsilon(u)(x) = \frac{1}{k!} \sum_I w_I(x, 0)\, dx^{i_1} \wedge \cdots \wedge dx^{i_k}.$$

If $u \in M_k$, then $u_{(k)}(x; \cdot)$ is alternating and $\varepsilon(u)(x) = 0$ if and only if $u_{(k)}(x; \cdot) = 0$, i.e. if and only if u vanishes to order $k + 1$ at $(x, \dots, x)$.

Proof. By Hadamard's lemma in y_1, $u = \sum_i y_{1,i} u_i$ with $u_i = \int_0^1 (\partial_{y_{1,i}} u)(x, t y_1, y_2, \dots, y_k)\, dt$; the u_i still vanish on $\{y_j = 0\}$, $j \ge 2$. Repeating in $y_2, \dots, y_k$ gives the representation. Since $z_l = x + y_1 + \cdots + y_l$, the derivative ∂_{z_l} equals $\partial_{y_l} - \partial_{y_{l+1}}$ (∂_{y_l} for $l = k$). In $\partial_{z_1^{i_1}} \cdots \partial_{z_k^{i_k}} u$ at $y = 0$ only terms in which every factor $y_{l,\cdot}$ is differentiated survive; since ∂_{z_k} can only hit y_k, then $\partial_{z_{k-1}}$ only y_{k-1}, and so on, the result is $w_I(x, 0)$, which gives the formula for $\varepsilon(u)$. If $u \in M_k$, the homogeneous term $u_{(k)}$ vanishes on $\{y_1 + \cdots + y_k = 0\}$, so it is alternating by Lemma 18.3.5. An alternating tensor is determined by its image in Λ^k, whence the last statement. □

Lemma 18.3.7. *Let $k \ge 1$ and $m \ge k+1$. Let $\mathcal{M}_m$ be the space of homogeneous polynomials q of degree m in $y = (y_1, \dots, y_k) \in (\mathbb{R}^n)^k$ which vanish on each of the subspaces $\{y_j = 0\}$, $1 \le j \le k$, and $\{y_1 + \cdots + y_k = 0\}$, and let $\mathcal{V}_m$ be the space of homogeneous polynomials of degree m in $(y_1, \dots, y_{k+1}) \in (\mathbb{R}^n)^{k+1}$ vanishing on each $\{y_j = 0\}$, $1 \le j \le k + 1$. Then*

$$R : \mathcal{V}_m \longrightarrow \mathcal{M}_m, \quad (RV)(y_1, \dots, y_k) = V(y_1, \dots, y_k, -(y_1 + \cdots + y_k)),$$

is surjective.

Proof. Given $q \in \mathcal{M}_m$, the polynomial chain $u(z_0, \dots, z_k) = q(z_1 - z_0, \dots, z_k - z_{k-1})$ for $\mathbb{C}[x]$ lies in M_k, and $\varepsilon(u) = 0$ since u vanishes to order $m > k$ on Δ. By Theorem 18.2.3 and Proposition 18.3.3 (for polynomials) $u \in B_k$, i.e. $u = d_{k+1} v$ with $v \in N_{k+1}$. Writing v in the coordinates $x = z_0$, $y_j = z_j - z_{j-1}$, $1 \le j \le k + 1$, as $V(x; y_1, \dots, y_{k+1})$, V vanishes on the subspaces $\{y_j = 0\}$, and $d_{k+1} v = V(x; y_1, \dots, y_k, -(y_1 + \cdots + y_k))$ because $z_{k+1} = z_0$ means $y_{k+1} = -(y_1 + \cdots + y_k)$. The component of V of degree 0 in x and degree m in y lies in $\mathcal{V}_m$ and is mapped by R to the corresponding component of u, which is q. □

Lemma 18.3.8. *Let $u \in M_k$, $k \geq 1$, satisfy $\varepsilon(u) = 0$. Then every point $p \in X^{k+1}$ has a neighbourhood on which u coincides with an element of B_k.*

Proof. First let $p \notin \Delta$. Then $p_j \neq p_{j+1}$ for some $0 \leq j \leq k-1$ (else all p_j coincide), and we choose $f \in \mathcal{C}^\infty(X)$ with $f(p_j) \neq f(p_{j+1})$ and $\psi \in \mathcal{C}^\infty(X^{k+1})$ supported where $f(z_j) \neq f(z_{j+1})$, equal to 1 near p. Then $g = \psi u/(f(z_j) - f(z_{j+1})) \in M_k$ and $u = (f(z_j) - f(z_{j+1}))g$ near p; this lies in B_k by Lemma 18.3.4.

Now let $p = (x_0, \dots, x_0) \in \Delta$ and use the coordinates (18.3.2). The Taylor series of u in y at $y = 0$ is $\sum_m u_{(m)}(x; y)$, where $u_{(m)}(x; \cdot)$ is homogeneous of degree m and vanishes on the same linear subspaces as u, so $u_{(m)} \in \mathcal{C}^\infty(B) \otimes \mathcal{M}_m$, $B = \{|x - x_0| < \delta\}$. By Lemma 18.3.6, $u_{(m)} = 0$ for $m \leq k$. For $m \geq k+1$, choosing a linear right inverse of R in Lemma 18.3.7, we find $V_{(m)} \in \mathcal{C}^\infty(B) \otimes \mathcal{V}_m$ with $RV_{(m)} = u_{(m)}$. By Borel's lemma (see [7, Section 1.2]), shrinking δ, there is a smooth $V(x; y_1, \dots, y_{k+1}) = \sum_m \chi(\lambda_m |y|) V_{(m)}(x; y)$, with $\chi \in \mathcal{C}^\infty_{\mathrm{c}}(\mathbb{R})$ equal to 1 near 0 and $\lambda_m \to \infty$ fast enough, having Taylor series $\sum_m V_{(m)}$ at $y = 0$; each term, hence V, vanishes on the subspaces $\{y_j = 0\}$. Let $\Theta \in \mathcal{C}^\infty_{\mathrm{c}}$ be a cut-off near $(x_0, \dots, x_0) \in X^{k+2}$ and let $v = \Theta V \in N_{k+1}$, extended by zero (written in the variables $x = z_0$, $y_j = z_j - z_{j-1}$, $j \leq k+1$). Then $s = d_{k+1} v \in B_k$ has the same Taylor series as u at Δ near p. So $g = u - s \in M_k$ vanishes to infinite order at Δ near p. Near p,

$$g = \sum_{j=1}^{k} \sum_{i=1}^{n} y_{j,i}\, g_{j,i}, \qquad g_{j,i} = \frac{y_{j,i}\, g}{|y|^2},$$

where $g_{j,i}$, extended by 0 on Δ, is smooth since g is flat there, and vanishes wherever g does. Taking $f_i \in \mathcal{C}^\infty(X)$ equal to the i-th coordinate near x_0, and a cut-off ψ equal to 1 near p, we have $y_{j,i} = -(f_i(z_{j-1}) - f_i(z_j))$ on the support of ψ, so near p, $g = -\sum_{j,i} (f_i(z_{j-1}) - f_i(z_j)) \psi g_{j,i} \in B_k$ by Lemma 18.3.4. Hence $u = s + g$ coincides near p with an element of B_k. □

Proof of Theorem 18.3.2. For $k = 0$, $(bu)(z_0) = u(z_0, z_0) - u(z_0, z_0) = 0$ for $u \in C_1$, so $\mathrm{HH}_0 = \mathcal{C}^\infty(X) = \Omega^0(X)$. Let $k \geq 1$. By Lemma 18.3.1 it remains to show injectivity. By Proposition 18.3.3 every class in HH_k is represented by some $u \in M_k$, and u is a boundary if (and only if) $u \in B_k$. Suppose $\varepsilon(u) = 0$. By Lemma 18.3.8 and compactness there is a finite partition of unity $\sum_\beta \psi_\beta = 1$ on X^{k+1} and $s_\beta \in B_k$ with $u = s_\beta$ on $\operatorname{supp} \psi_\beta$. Then $u = \sum_\beta \psi_\beta s_\beta \in B_k$ by Lemma 18.3.4(i). □

Remark 18.3.9. Compactness was only used to make the partitions of unity finite. With some care about supports (using locally finite partitions of unity and cut-offs in all variables) the same argument applies to non-compact X.

18.4 The commutative symbol algebra

Before deforming, we compute the Hochschild homology of the commutative algebra whose product is the leading term of the symbol product: formal sums of homogeneous functions on $T^*X \setminus 0$ with pointwise multiplication. Polar coordinates in ξ split such a function into a function on the cosphere bundle and a power of the radial variable, and the answer is assembled from Connes' theorem for S^*X and the Laurent polynomial computation of Remark 18.2.4.

From now on X is compact, $n = \dim X \geq 1$, and

$$Y = T^*X \setminus 0, \qquad \rho_t(x, \xi) = (x, t\xi),\ t > 0, \qquad S^*X = Y/\mathbb{R}^+.$$

A function a on Y is homogeneous of degree $m \in \mathbb{Z}$ if $\rho_t^* a = t^m a$ for all $t > 0$, and similarly for differential forms; we write $\mathcal{C}^\infty_{(m)}(Y)$ and $\Omega^k_{(m)}(Y)$ for the smooth homogeneous functions and k-forms of degree m. (Equivalently, $\mathcal{C}^\infty_{(m)}(Y)$ is the space of sections of a line bundle over S^*X.) Choosing a Riemannian metric on X and setting $R = |\xi|$ identifies S^*X with $\{R = 1\}$ and

$$Y \cong S^*X \times (0,\infty), \qquad (x,\xi) \longmapsto \big((x,\xi/|\xi|),|\xi|\big); \tag{18.4.1}$$

we write $s \in S^*X$ and $r = R$ for the two factors.

The *commutative algebra of formal symbols* is

$$\mathcal{A}_0 = \Big\{ a = \sum_{m \le M} a_m;\ a_m \in \mathcal{C}^\infty_{(m)}(Y),\ M \in \mathbb{Z} \Big\}, \qquad (ab)_m = \sum_{l \in \mathbb{Z}} a_{m-l} b_l,$$

the sum being finite. It contains the algebra $\mathcal{A}_{\mathrm{fin}} = \bigoplus_m \mathcal{C}^\infty_{(m)}(Y)$ of finite sums.

Definition 18.4.1. For a multi-index $I = (I_0,\dots,I_k) \in \mathbb{Z}^{k+1}$ let $\mathcal{C}^\infty_{(I)}(Y^{k+1})$ be the space of smooth functions on Y^{k+1} homogeneous of degree I_j in the j-th factor, and for $m \in \mathbb{Z}$ let $C_{m,k} = \bigoplus_{|I|=m} \mathcal{C}^\infty_{(I)}(Y^{k+1})$ (finite sums), $|I| = I_0 + \cdots + I_k$. The space of Hochschild k-chains of the symbol algebra is

$$\mathcal{B}^{(k)} = \Big\{ u = \sum_{m \le M} u_m;\ u_m \in C_{m,k},\ M \in \mathbb{Z} \Big\}.$$

Each $\mathcal{C}^\infty_{(I)}(Y^{k+1})$ is the completed projective tensor product of the spaces $\mathcal{C}^\infty_{(I_j)}(Y)$, and $\mathcal{A}_0^{\otimes(k+1)} \subset \mathcal{B}^{(k)}$: a finite sum of products $a^0(\zeta_0)\cdots a^k(\zeta_k)$ of formal sums has, in each total degree m, only finitely many non-zero multihomogeneous components, because each a^j has degrees bounded above. The face maps (18.2.1), with $\zeta_j \in Y$ in place of z_j, map $\mathcal{C}^\infty_{(I)}(Y^{k+1})$ to $\mathcal{C}^\infty_{(I')}(Y^k)$ where I' is obtained by adding two adjacent (or the last and the first) entries; they preserve the total degree and extend the face maps of $\mathcal{A}_0$. So $(\mathcal{B}^{(\bullet)}, b)$ is a complex, the direct product over m (with degrees bounded above) of the complexes $(C_{m,\bullet}, b)$. Homology commutes with such products, so

$$\mathrm{HH}_k(\mathcal{A}_0) = \Big\{ \sum_{m \le M} h_m;\ h_m \in H_k(C_{m,\bullet}, b) \Big\}. \tag{18.4.2}$$

Theorem 18.4.2. *The map ε of (18.2.2) (applied on Y) induces isomorphisms $H_k(C_{m,\bullet},b) \cong \Omega^k_{(m)}(Y)$ for all $m \in \mathbb{Z}$, $k \in \mathbb{N}_0$. Hence*

$$\mathrm{HH}_k(\mathcal{A}_0) \cong \Big\{ \alpha = \sum_{m \le M} \alpha_m;\ \alpha_m \in \Omega^k_{(m)}(Y) \Big\},$$

the space of formal sums of homogeneous k-forms on $T^*X \setminus 0$.

Proof. Under (18.4.1), an element of the space $\mathcal{C}^\infty_{(I)}(Y^{k+1})$ is of the form $r_0^{I_0}\cdots r_k^{I_k} v(s_0,\dots,s_k)$ with $v \in \mathcal{C}^\infty((S^*X)^{k+1})$, so

$$\bigoplus_m C_{m,k} = \mathcal{C}^\infty\big((S^*X)^{k+1}\big) \otimes \mathbb{C}[r_0^{\pm1},\dots,r_k^{\pm1}].$$

The face maps of the left side are the tensor products of the face maps of the Hochschild complexes of $\mathcal{C}^\infty(S^*X)$ (Section 18.3) and of $\mathbb{C}[r^{\pm1}]$. With the degeneracies s_i (insertion of the identity in position $i+1$) the Hochschild complex of a unital algebra is a simplicial vector space, and the complex above is the diagonal of the tensor product of two simplicial

vector spaces. By the Eilenberg–Zilber theorem, the shuffle map sh from the tensor product complex $C_\bullet(\mathcal{C}^\infty(S^*X))\otimes C_\bullet(\mathbb{C}[r^{\pm1}])$ to the diagonal complex is a quasi-isomorphism, and by the Künneth formula over $\mathbb{C}$ the homology of the tensor product complex is $\mathrm{HH}_\bullet(\mathcal{C}^\infty(S^*X))\otimes \mathrm{HH}_\bullet(\mathbb{C}[r^{\pm1}])$. A direct computation from the definition of the shuffle map shows $\varepsilon(\mathrm{sh}(\alpha\otimes\beta))=\varepsilon(\alpha)\wedge\varepsilon(\beta)$ (the $\binom{p+q}{p}$ shuffles each contribute the same wedge product, and $\binom{p+q}{p}/(p+q)!=1/(p!q!)$). By Theorem 18.3.2 and Remark 18.2.4 it follows that ε induces an isomorphism of $H_k(\bigoplus_m C_{m,\bullet})$ onto

$$\Omega^k(S^*X)\otimes\mathbb{C}[r^{\pm1}]\ \oplus\ \Omega^{k-1}(S^*X)\otimes\mathbb{C}[r^{\pm1}]\,dr,$$

which by (18.4.1) is the space of finite sums of homogeneous k-forms on Y: a k-form homogeneous of degree m is exactly $r^m(\alpha'+\alpha''\wedge dr/r)$ with α', α'' forms on S^*X. Since ε maps $C_{m,k}$ to $\Omega^k_{(m)}(Y)$, the isomorphism respects the grading by m; (18.4.2) gives the last statement. □

18.5 The algebra of formal symbols

18.5.1 Full symbols and the star product

Let $\Psi^m(X)$ denote the classical pseudodifferential operators of order m on X (Chapter 5), $\Psi^{\mathbb{Z}}(X)=\bigcup_{m\in\mathbb{Z}}\Psi^m(X)$ and $\Psi^{-\infty}(X)=\bigcap_m\Psi^m(X)$. Then $\Psi^{-\infty}(X)$ is an ideal and we consider the quotient algebra

$$\mathcal{A}=\Psi^{\mathbb{Z}}(X)/\Psi^{-\infty}(X).$$

The principal symbol gives exact sequences $0\to\Psi^{m-1}(X)\to\Psi^m(X)\xrightarrow{\sigma_m}\mathcal{C}^\infty_{(m)}(Y)\to0$, with $\sigma_{m+m'}(AB)=\sigma_m(A)\sigma_{m'}(B)$ and

$$\sigma_{m+m'-1}([A,B])=-i\{\sigma_m(A),\sigma_{m'}(B)\},\qquad \{a,b\}=\sum_{j=1}^n\Big(\frac{\partial a}{\partial\xi_j}\frac{\partial b}{\partial x_j}-\frac{\partial a}{\partial x_j}\frac{\partial b}{\partial\xi_j}\Big)=H_a b,$$

the Poisson bracket, H_a being the Hamilton vector field of a (in canonical coordinates (x,ξ) on T^*X).

To compute Hochschild homology we need the full symbol. We use the following standard facts on global symbol calculus (see [1] and the references there). After the choice of a Riemannian metric, and hence of a global quantization map, there is a linear bijection

$$\sigma:\mathcal{A}\longrightarrow\mathcal{A}_0,\qquad \sigma\big(\Psi^m(X)\big)=\{a\in\mathcal{A}_0;\ a_j=0\text{ for }j>m\},$$

whose top-degree component on $\Psi^m(X)$ is the principal symbol, and there are bidifferential operators P_l, $l\in\mathbb{N}_0$, on Y with $P_l(\mathcal{C}^\infty_{(p)}(Y),\mathcal{C}^\infty_{(q)}(Y))\subset\mathcal{C}^\infty_{(p+q-l)}(Y)$, such that the product transported to $\mathcal{A}_0$ is

$$\begin{aligned} a\star b&=\sigma\big(\sigma^{-1}(a)\sigma^{-1}(b)\big)=\sum_{l\ge0}P_l(a,b),\\ P_0(a,b)&=ab,\qquad P_1(a,b)-P_1(b,a)=-i\{a,b\}. \end{aligned}\tag{18.5.1}$$

The sum is meaningful in $\mathcal{A}_0$: the component of degree m of $a\star b$ is the finite sum $\sum_{l\ge0}\sum_p P_l(a_p,b_{m+l-p})$. In local coordinates, with the left quantization, the composition formula of Chapter 2 gives $P_l(a,b)=\sum_{|\alpha|=l}(\alpha!)^{-1}\partial_\xi^\alpha a\,D_x^\alpha b$, and then indeed

$$P_1(a,b)-P_1(b,a)=-i\sum_j(\partial_{\xi_j}a\,\partial_{x_j}b-\partial_{\xi_j}b\,\partial_{x_j}a).$$

The relations (18.5.1) for P_0 and the antisymmetric part of P_1 are forced by the principal symbol calculus; these, together with associativity of $\star$, are all that will be used.

18.5.2 Hochschild chains

Definition 18.5.1. The Hochschild complex of $\mathcal{A} \cong (\mathcal{A}_0, \star)$ has the chain spaces $\mathcal{B}^{(k)}$ of Definition 18.4.1, with face maps $d_i^\star = \sum_{l\geq 0} d_i^{(l)}$, where, for $u \in \mathcal{C}^\infty_{(I)}(Y^{k+1})$, $d_i^{(l)}u$ is obtained by applying the bidifferential operator P_l in the variables (ζ_i, ζ_{i+1}) (in (ζ_k, ζ_0) for $i = k$, the first argument being ζ_k) and then restricting as in (18.2.1). We write

$$b = \sum_{i=0}^{k} (-1)^i d_i^\star = \sum_{l\geq 0} b_l, \qquad b_l = \sum_{i=0}^{k} (-1)^i d_i^{(l)} : C_{m,k} \longrightarrow C_{m-l,k-1}.$$

On $a_0 \otimes \cdots \otimes a_k$ the map $d_i^{(l)}$ replaces $a_i \otimes a_{i+1}$ by $P_l(a_i, a_{i+1})$ (and $d_k^{(l)}$ gives $P_l(a_k, a_0) \otimes a_1 \otimes \cdots \otimes a_{k-1}$), so b extends the Hochschild differential of $(\mathcal{A}_0, \star)$. For $u = \sum_{m\leq M} u_m$ the component of degree m of bu is the finite sum $\sum_{l\geq 0} b_l u_{m+l}$, so b is well defined on $\mathcal{B}^{(k)}$. Moreover $b^2 = 0$: on the algebraic tensor product of copies of $\mathcal{A}_{\mathrm{fin}}$ this is Lemma 18.1.1 (which only uses associativity), each component of b^2 is a finite sum of operators continuous in the $\mathcal{C}^\infty$ topology, and finite sums of products are dense in each $\mathcal{C}^\infty_{(I)}(Y^{k+1})$. The differential b_0 is the Hochschild differential of the commutative algebra $\mathcal{A}_0$ of Section 18.4. We define $\mathrm{HH}_k(\mathcal{A})$ to be the homology of $(\mathcal{B}^{(\bullet)}, b)$.

Remark 18.5.2 (Semiclassical interpretation). For $h > 0$ let $L_h : \mathcal{A}_0 \to \mathcal{A}_0$, $L_h(\sum_j a_j) = \sum_j h^j a_j$, and define $a \star_h b = L_h^{-1}(L_h a \star L_h b)$. Since $P_l(h^p a_p, h^q b_q) = h^{p+q} P_l(a_p, b_q)$ has degree $p + q - l$,

$$a \star_h b = \sum_{l\geq 0} h^l P_l(a, b),$$

a family of associative products, all isomorphic to $\star = \star_1$ via L_h, which at $h = 0$ reduces to the commutative product. The Hochschild differential of $\star_h$ is $b_h = \sum_l h^l b_l$. Thus the symbol algebra is a deformation of $\mathcal{A}_0$ in the direction of the Poisson bracket, and its Hochschild homology can be computed by expanding in powers of h, the 'semiclassical limit'. Expanding chains in powers of h is equivalent to filtering them by homogeneity, which is how we proceed: the spectral sequence of the next section is the one associated to the Taylor expansion of h-dependent chains at $h = 0$.

18.6 The filtration by order and its spectral sequence

18.6.1 The first terms

The complex $(\mathcal{B}^{(\bullet)}, b)$ is filtered by the subcomplexes $F_p\mathcal{B}^{(k)} = \{u;\ u_m = 0 \text{ for } m > p\}$ of chains of order at most p; $b(F_p) \subset F_p$ since each b_l lowers degrees by $l \geq 0$. The filtration is exhaustive ($\bigcup_p F_p\mathcal{B}^{(k)} = \mathcal{B}^{(k)}$), separated ($\bigcap_p F_p = 0$), and complete: a series $\sum_j e_j$ with $e_j \in F_{p_j}$, $p_j \to -\infty$ and $\sup p_j < \infty$ converges in the sense that each component is a finite sum, and b commutes with such sums.

Comparing components of degree $m - l$ in $b^2 = 0$ gives $\sum_{i+j=l} b_i b_j = 0$ for every l, in particular

$$b_0^2 = 0, \qquad b_0 b_1 + b_1 b_0 = 0, \qquad b_1^2 + b_0 b_2 + b_2 b_0 = 0. \tag{18.6.1}$$

Define

$$E^1_{p,k} = H_k(C_{p,\bullet}, b_0),$$

the homology of the associated graded complex. By (18.6.1), b_1 maps b_0-cycles to b_0-cycles and b_0-boundaries to b_0-boundaries, so induces $d^1 : E^1_{p,k} \to E^1_{p-1,k-1}$, and $(d^1)^2 = 0$. Set

$$E^2_{p,k} = \frac{\mathrm{Nul}(d^1 : E^1_{p,k} \to E^1_{p-1,k-1})}{\mathrm{Ran}(d^1 : E^1_{p+1,k+1} \to E^1_{p,k})}.$$

These are the first two terms of the spectral sequence of the filtered complex; the higher terms E^r, with differentials $d^r : E^r_{p,k} \to E^r_{p-r,k-1}$, will not be needed explicitly. By Theorem 18.4.2,

$$E^1_{p,k} \cong \Omega^k_{(p)}(Y), \tag{18.6.2}$$

the isomorphism being induced by ε.

Discussion. In plain words, we try to solve the cycle equation $bu = 0$ one order at a time. The top-order part of the equation is $b_0 u_p = 0$, a problem for the *commutative* algebra, whose solutions modulo b_0-boundaries form E^1, the homogeneous differential forms. The next order adds the first correction b_1 of the product, and passing to E^2 amounts to keeping only those leading parts that survive this correction. In Remark 18.5.2 this is the expansion in powers of h: E^1 is the classical limit, and d^1 is the first quantum correction, governed by the Poisson bracket.

18.6.2 Degeneration and convergence

The following purely algebraic statement reduces the computation of $\mathrm{HH}_\bullet(\mathcal{A})$ to that of E^2. It applies to any complex of the form $C_k = \{\sum_{p \le P} c_p;\ c_p \in C_{p,k}\}$ with $b = \sum_{l \ge 0} b_l$, $b_l : C_{p,k} \to C_{p-l,k-1}$, $b^2 = 0$, and E^1, E^2 defined as above.

Proposition 18.6.1. *Suppose there is an integer n such that $E^2_{p,k} = 0$ whenever $p \neq k - n$. Then every class in $H_k(C, b)$ has a representative $c \in F_{k-n}C_k$, and*

$$\Theta : H_k(C, b) \longrightarrow E^2_{k-n,k}, \qquad [c] \longmapsto [c_{k-n}],$$

is a well-defined isomorphism. Moreover all differentials d^r, $r \ge 2$, of the spectral sequence vanish.

Note that if $c \in F_{k-n}$ and $bc = 0$, then $b_0 c_{k-n} = (bc)_{k-n} = 0$ and $b_1 c_{k-n} = -b_0 c_{k-n-1}$, so c_{k-n} defines an element of $E^2_{k-n,k}$.

Proof. We use the following step.

(P) Let $c \in F_p C_k$ satisfy $b_0 c_p = 0$ and $b_1 c_p + b_0 c_{p-1} = 0$ (e.g. $bc = 0$, or more generally $bc \in F_{p-2}$), and suppose that the class of c_p in $E^2_{p,k}$ vanishes. Then there are $x \in C_{p+1,k+1}$ with $b_0 x = 0$ and $y \in C_{p,k+1}$ such that $c - b(x + y) \in F_{p-1}$.

Indeed, $[c_p] \in E^1_{p,k}$ is a d^1-cycle, and as it vanishes in E^2 it equals $d^1[x] = [b_1 x]$ for some b_0-cycle $x \in C_{p+1,k+1}$; so $c_p - b_1 x = b_0 y$ for some $y \in C_{p,k+1}$. The components of $b(x + y)$ of degree $p + 1$ and p are $b_0 x = 0$ and $b_1 x + b_0 y = c_p$, which proves (P).

Representatives. If $c \in F_P C_k$ is a cycle with $P > k - n$, then $E^2_{P,k} = 0$ and (P) replaces c by a homologous cycle in F_{P-1}. After finitely many steps $c \in F_{k-n}$.

Well-definedness. It suffices to show that if $c = be \in F_{k-n}C_k$, $e \in F_q C_{k+1}$, then $[c_{k-n}] = 0$ in E^2. If $q > k + 1 - n$, then $be \in F_{q-2}$ and $E^2_{q,k+1} = 0$, so by (P) applied to e we may replace e by $e - b(x + y) \in F_{q-1}$ without changing be. Repeating, we may assume $q \le k + 1 - n$. Then $b_0 e_{k+1-n} = (be)_{k+1-n} = 0$ and $c_{k-n} = b_1 e_{k+1-n} + b_0 e_{k-n}$, so the class of c_{k-n} in E^1 is $d^1[e_{k+1-n}]$, which vanishes in E^2.

Injectivity. Let $c \in F_{k-n}C_k$ be a cycle with $[c_{k-n}] = 0$ in E^2. By (P), $c - b(x_{k-n} + y_{k-n}) = c^{(1)} \in F_{k-n-1}$. Since $E^2_{p,k} = 0$ for $p < k - n$, (P) applies again to $c^{(1)}$, and so on: we obtain $e_p = x_p + y_p \in F_{p+1}C_{k+1}$, $p = k - n, k - n - 1, \dots$, with $c - b(\sum_{p \ge P} e_p) \in F_{P-1}$ for every P. By completeness $e = \sum_p e_p$ exists, and $c = be$.

Surjectivity. Let $x \in C_{k-n,k}$ with $b_0 x = 0$ represent a class in $E^2_{k-n,k}$. Then $[b_1 x] = d^1[x] = 0$ in E^1, i.e. $b_1 x = -b_0 y$ for some $y \in C_{k-n-1,k}$, and $c^{(1)} = x + y$ satisfies

$bc^{(1)} \in F_{k-n-2}$. Suppose $c^{(r)} \in F_{k-n}$, $r \geq 1$, has $c^{(r)}_{k-n} = x$ and $z = bc^{(r)} \in F_pC_{k-1}$, $p = k-n-r-1$. Then z is a cycle and $p \neq (k-1)-n$, so (P) gives $x' \in C_{p+1,k}$, $y' \in C_{p,k}$ with $z - b(x'+y') \in F_{p-1}$. Set $c^{(r+1)} = c^{(r)} - x' - y'$; since $p+1 < k-n$, its component of degree $k-n$ is still x, and $bc^{(r+1)} \in F_{p-1}$. The corrections have degrees tending to $-\infty$, so $c = \lim_r c^{(r)}$ exists, $bc = 0$ and $\Theta[c] = [x]$.

Degeneration. Each $E^r_{p,k}$, $r \geq 2$, is a subquotient of $E^2_{p,k}$, so vanishes unless $p = k-n$. The differential d^r maps $E^r_{k-n,k}$ to $E^r_{k-n-r,k-1}$, which vanishes unless $k-n-r = k-1-n$, i.e. $r = 1$. Hence $d^r = 0$ for $r \geq 2$. □

18.7 The first differential: the Poisson complex

The canonical symplectic form on T^*X is $\omega = \sum_j d\xi_j \wedge dx_j$ in canonical coordinates. Its inverse is the Poisson bivector Π; we regard it as the antisymmetric bilinear form on T^*Y with $\Pi(da, db) = \{a, b\}$, i.e. $\Pi(\theta, \eta) = \sum_j (\theta(\partial_{\xi_j})\eta(\partial_{x_j}) - \theta(\partial_{x_j})\eta(\partial_{\xi_j}))$. Let $\iota_\Pi : \Omega^k(Y) \to \Omega^{k-2}(Y)$ be contraction with Π:

$$\iota_\Pi(\theta_1 \wedge \cdots \wedge \theta_k) = \sum_{j<l} (-1)^{j+l+1} \Pi(\theta_j, \theta_l)\, \theta_1 \wedge \cdots \wedge \widehat{\theta_j} \wedge \cdots \wedge \widehat{\theta_l} \wedge \cdots \wedge \theta_k;$$

in canonical coordinates $\iota_\Pi = \sum_j \iota_{\partial_{x_j}} \iota_{\partial_{\xi_j}}$, as one checks by expanding the right side on decomposable forms. Following Koszul and Brylinski, set

$$\delta = \iota_\Pi d - d\iota_\Pi : \Omega^k(Y) \longrightarrow \Omega^{k-1}(Y).$$

Since Π lowers homogeneity by one, δ maps $\Omega^k_{(p)}(Y)$ to $\Omega^{k-1}_{(p-1)}(Y)$.

Lemma 18.7.1. *For functions $a_0, \ldots, a_k$,*

$$\begin{aligned}\delta(a_0\, da_1 \wedge \cdots \wedge da_k) = &\sum_{j=1}^{k} (-1)^{j+1} \{a_0, a_j\}\, da_1 \wedge \cdots \wedge \widehat{da_j} \wedge \cdots \wedge da_k \\ &+ \sum_{1 \leq j < l \leq k} (-1)^{j+l} a_0\, d\{a_j, a_l\} \wedge da_1 \wedge \cdots \wedge \widehat{da_j} \wedge \cdots \wedge \widehat{da_l} \wedge \cdots \wedge da_k.\end{aligned}$$

Moreover, if f is a function and β a form with constant coefficients in canonical coordinates, then $\delta(f\beta) = \iota_{H_f}\beta$.

Proof. Write D_{jl} for $da_1 \wedge \cdots \wedge da_k$ with da_j and da_l omitted (D_{0l}: only da_l omitted). Applying ι_Π to $da_0 \wedge da_1 \wedge \cdots \wedge da_k$, the pairs $(0, l)$ give $\sum_l (-1)^{l+1}\{a_0, a_l\} D_{0l}$ and the pairs $1 \leq j < l$ give $\sum (-1)^{j+l+1}\{a_j, a_l\} da_0 \wedge D_{jl}$. On the other hand $d\iota_\Pi(a_0\, da_1 \wedge \cdots \wedge da_k) = \sum_{j<l} (-1)^{j+l+1}(\{a_j, a_l\} da_0 + a_0\, d\{a_j, a_l\}) \wedge D_{jl}$. Subtracting gives the formula. For the second statement, $d\beta = 0$ and $d\iota_\Pi\beta = 0$, so $\delta(f\beta) = \iota_\Pi(df \wedge \beta) - df \wedge \iota_\Pi\beta$. For vector fields X, Z and a 1-form θ one has $\iota_X\iota_Z(\theta \wedge \beta) - \theta \wedge \iota_X\iota_Z\beta = \theta(Z)\iota_X\beta - \theta(X)\iota_Z\beta$; with $X = \partial_{x_j}$, $Z = \partial_{\xi_j}$, $\theta = df$, summed over j, this is $\iota_V\beta$ with $V = \sum_j (\partial_{\xi_j} f\, \partial_{x_j} - \partial_{x_j} f\, \partial_{\xi_j}) = H_f$. □

Proposition 18.7.2. *Under the identification* (18.6.2), *$d^1 = -i\,\delta$.*

Proof. Homogeneous forms of degree p are finite sums of forms $\alpha = a_0\, da_1 \wedge \cdots \wedge da_k$ with a_j homogeneous of total degree p (use (18.4.1) and an embedding of S^*X in some $\mathbb{R}^N$). By Lemma 18.2.1 the chain $F = \varphi(a_0; a_1, \ldots, a_k) \in C_{p,k}$ is a b_0-cycle with $\varepsilon(F) = \alpha$, so $d^1\alpha = \varepsilon(b_1 F)$.

Decompose $P_1 = S + A$ with $S(a,b) = \frac{1}{2}(P_1(a,b) + P_1(b,a))$ symmetric and $A(a,b) = -\frac{i}{2}\{a,b\}$ by (18.5.1). Replacing P_1 by S in b_1F, the inner terms cancel in pairs (as σ

and $\sigma \circ (i\ i{+}1)$ have opposite signs) and the first and last terms cancel as in the proof of Lemma 18.2.1, now using $S(a_0, a_{\sigma(1)}) = S(a_{\sigma(1)}, a_0)$. Thus $b_1 F$ may be computed with P_1 replaced by $c\{\cdot,\cdot\}$, $c = -\frac{i}{2}$. Then $(k-1)!\,\varepsilon(b_1 F)$ is the sum of the following, with the notation D_{jl} of the proof of Lemma 18.7.1:

(a) the first terms $c\sum_\sigma \operatorname{sgn}(\sigma)\{a_0, a_{\sigma(1)}\} da_{\sigma(2)} \wedge \cdots \wedge da_{\sigma(k)}$; grouping according to $\sigma(1) = j$ (sign $(-1)^{j-1}$) this is $c(k-1)!\sum_j (-1)^{j-1}\{a_0, a_j\} D_{0j}$;

(b) the last terms $(-1)^k c\sum_\sigma \operatorname{sgn}(\sigma)\{a_{\sigma(k)}, a_0\} da_{\sigma(1)} \wedge \cdots \wedge da_{\sigma(k-1)}$; grouping according to $\sigma(k) = j$ (sign $(-1)^{k-j}$) this equals the expression in (a);

(c) the inner terms $c\sum_\sigma \operatorname{sgn}(\sigma)\sum_{i=1}^{k-1}(-1)^i a_0\, da_{\sigma(1)} \wedge \cdots \wedge d\{a_{\sigma(i)}, a_{\sigma(i+1)}\} \wedge \cdots \wedge da_{\sigma(k)}$. Moving $d\{\cdot,\cdot\}$ to the front costs $(-1)^{i-1}$; for fixed i and $(\sigma(i), \sigma(i+1)) = (j, l)$, $j < l$, the sign of σ is $(-1)^{j+l+1}$ times the sign of the induced ordering of the remaining indices, and the pair (l, j) gives the same contribution since both $\{a_l, a_j\}$ and the sign change. Each i thus contributes $-2c(k-2)!\sum_{j<l}(-1)^{j+l+1} a_0\, d\{a_j, a_l\} \wedge D_{jl}$, and there are $k-1$ values of i.

Dividing by $(k-1)!$,

$$\varepsilon(b_1 F) = 2c\Big(\sum_j (-1)^{j+1}\{a_0, a_j\} D_{0j} + \sum_{j<l}(-1)^{j+l} a_0\, d\{a_j, a_l\} \wedge D_{jl}\Big) = -i\,\delta\alpha$$

by Lemma 18.7.1. □

In particular $\delta^2 = 0$, which can also be checked directly. The complex $(\Omega^\bullet(Y), \delta)$ is the *canonical* (or Koszul–Brylinski) complex of the Poisson manifold Y.

Discussion. Why does the Poisson bracket appear? The symmetric part of P_1, which depends on the choice of quantization, drops out of d^1, as the cancellation in the proof shows; only the antisymmetric part $-\frac{i}{2}\{a, b\}$, which is forced by the principal symbol calculus, survives. On forms, the operation built from a bracket of functions in the way d is built from the product is exactly the Koszul–Brylinski operator δ (Lemma 18.7.1).

18.8 The E^2 term

To compute the homology of δ we convert it into the de Rham differential. In Riemannian geometry the Hodge star turns d into (a multiple of) its adjoint d^*; on a symplectic manifold an analogous star operator, built from ω instead of a metric, turns δ into $\pm d$. The homology of δ is then de Rham cohomology, restricted to forms of a fixed homogeneity.

18.8.1 The symplectic star operator

Fix $\zeta \in Y$ and let $V = T_\zeta Y$, of dimension $2n$. The bilinear form Π on V^* is non-degenerate; let $\Pi^\sharp : V^* \to V$ be defined by $\eta(\Pi^\sharp\theta) = \Pi(\theta, \eta)$, so that $\Pi^\sharp(df) = H_f$. On $\Lambda^k V^*$ define the bilinear form

$$\Lambda(\theta_1 \wedge \cdots \wedge \theta_k, \eta_1 \wedge \cdots \wedge \eta_k) = \det\big(\Pi(\theta_i, \eta_j)\big)_{i,j=1}^k,$$

which is non-degenerate since $\Lambda(\alpha, \beta) = (-1)^k\langle \alpha, \Lambda^k\Pi^\sharp\,\beta\rangle$ and $\Pi^\sharp$ is invertible. Let $\mathrm{vol} = \omega^n/n!$. The *symplectic star operator* $* : \Lambda^k V^* \to \Lambda^{2n-k} V^*$ is defined by

$$\alpha \wedge *\beta = \Lambda(\alpha, \beta)\,\mathrm{vol} \quad \text{for all } \alpha \in \Lambda^k V^*;$$

it is an isomorphism because the wedge pairing $\Lambda^k \times \Lambda^{2n-k} \to \Lambda^{2n}$ is non-degenerate. Applied pointwise it gives a bundle isomorphism, hence isomorphisms $* : \Omega^k(Y) \to \Omega^{2n-k}(Y)$, and in canonical coordinates it maps forms with constant coefficients to forms with constant coefficients (as Π and vol have constant coefficients). This is the operator W_ω of the original notes, which describes it in the basis $dx_i, d\xi_i$.

Lemma 18.8.1. *For $\theta \in V^*$ and $\beta \in \Lambda^k V^*$, $k \geq 1$,*

$$*^{-1}(\theta \wedge *\beta) = (-1)^{k-1} \iota_{\Pi^\sharp \theta} \beta.$$

Proof. Expanding the determinant along the first row, $\Lambda(\theta \wedge \alpha, \beta) = \Lambda(\alpha, \iota_{\Pi^\sharp\theta}\beta)$ for $\alpha \in \Lambda^{k-1}V^*$, since $\iota_v(\eta_1 \wedge \cdots \wedge \eta_k) = \sum_j (-1)^{j+1} \eta_j(v)\, \eta_1 \wedge \ldots \widehat{\eta_j} \cdots \wedge \eta_k$ and $\eta_j(\Pi^\sharp\theta) = \Pi(\theta, \eta_j)$. Let $\mu = *^{-1}(\theta \wedge *\beta) \in \Lambda^{k-1}V^*$. For all $\alpha \in \Lambda^{k-1}V^*$,

$$\Lambda(\alpha, \mu)\mathrm{vol} = \alpha \wedge *\mu = \alpha \wedge \theta \wedge *\beta = (-1)^{k-1}\theta \wedge \alpha \wedge *\beta = (-1)^{k-1}\Lambda(\alpha, \iota_{\Pi^\sharp\theta}\beta)\mathrm{vol},$$

and Λ is non-degenerate. □

Proposition 18.8.2 (Brylinski)**.** *On k-forms, $\delta = (-1)^{k-1} *^{-1} d *$. Moreover $*$ maps $\Omega^k_{(p)}(Y)$ onto $\Omega^{2n-k}_{(p+n-k)}(Y)$.*

Proof. Both sides are first-order differential operators, so it suffices to compare them on $f\beta$ with f a function and β a form with constant coefficients in canonical coordinates. Then $*\beta$ has constant coefficients, $d(f*\beta) = df \wedge *\beta$ and, by Lemma 18.8.1, $*^{-1}d*(f\beta) = (-1)^{k-1}\iota_{H_f}\beta$, which is $(-1)^{k-1}\delta(f\beta)$ by Lemma 18.7.1.

Since $\{a, b\}$ is homogeneous of degree $p + q - 1$ if a, b are homogeneous of degrees p, q, the function $\Lambda(\alpha, \beta)$ is homogeneous of degree $p + q - k$ for $\alpha \in \Omega^k_{(p)}$, $\beta \in \Omega^k_{(q)}$, and vol is homogeneous of degree n. Hence for such α, β,

$$\alpha \wedge \rho_t^*(*\beta) = \rho_t^*\left(t^{-p}\alpha \wedge *\beta\right) = t^{-p}\rho_t^*\left(\Lambda(\alpha, \beta)\mathrm{vol}\right) = t^{q+n-k}\alpha \wedge *\beta.$$

As the values of the forms $\alpha \in \Omega^k_{(p)}(Y)$ at any point span Λ^k, $\rho_t^*(*\beta) = t^{q+n-k} * \beta$. The same argument applies to $*^{-1}$. □

18.8.2 Homogeneous de Rham cohomology

Proposition 18.8.3. *Let $c \in \mathbb{Z}$. The cohomology of the complex $(\Omega^\bullet_{(c)}(Y), d)$ of forms homogeneous of degree c vanishes if $c \neq 0$, while for $c = 0$*

$$H^j(\Omega^\bullet_{(0)}(Y), d) \cong H^j(S^*X) \oplus H^{j-1}(S^*X).$$

Proof. Let E be the generator of the dilations ρ_t (the radial vector field $\sum_j \xi_j \partial_{\xi_j}$). Since ρ_t commutes with the flow of E, the contraction ι_E preserves homogeneity, and $\mathcal{L}_E\alpha = c\alpha$ for $\alpha \in \Omega^\bullet_{(c)}(Y)$. By Cartan's formula $d\iota_E + \iota_E d = \mathcal{L}_E$, so if $c \neq 0$ and $d\alpha = 0$ then $\alpha = d(\iota_E\alpha/c)$.

For $c = 0$ use (18.4.1). Every form on $S^*X \times (0, \infty)$ can be written $\alpha' + \alpha'' \wedge dr/r$ with α', α'' not containing dr; since $\rho_t^*(dr/r) = dr/r$, the form is homogeneous of degree 0 if and only if α' and α'' are (pull-backs of) forms on S^*X. As $d(\alpha' + \alpha'' \wedge dr/r) = d\alpha' + d\alpha'' \wedge dr/r$, the complex $\Omega^\bullet_{(0)}(Y)$ is the direct sum of the de Rham complex of S^*X and of the same complex shifted by one, which gives the result. □

The splitting depends on the choice of the radial function R; replacing R by aR, $a \in \mathcal{C}^\infty(S^*X)$ positive, changes α' by $\alpha'' \wedge d\log a$, which does not change the induced map on cohomology from the second summand, nor the subspace given by the first.

Question. Why does the final answer involve only the cohomology of S^*X, and not that of the bigger space $Y = T^*X \setminus 0$?

Answer. As a manifold $Y \cong S^*X \times (0,\infty)$ has the same cohomology as S^*X. But we only see forms that are homogeneous, and the dilation ρ_t acts on each homogeneity class by a fixed power of t. Cartan's formula shows that a closed form of non-zero homogeneity is exact, while a form of homogeneity 0 is $\alpha' + \alpha'' \wedge dr/r$ with α', α'' forms on S^*X. The extra dr/r is responsible for the second summand $H^{j-1}(S^*X)$. □

Theorem 18.8.4. *For the spectral sequence of Section 18.6,*

$$E^2_{p,k} \cong \begin{cases} H^{2n-k}(S^*X) \oplus H^{2n-k-1}(S^*X), & p = k-n, \\ 0, & p \neq k-n. \end{cases}$$

Proof. By (18.6.2) and Proposition 18.7.2, $E^2_{p,k}$ is the homology at $\Omega^k_{(p)}(Y)$ of $\Omega^{k+1}_{(p+1)} \xrightarrow{\delta} \Omega^k_{(p)} \xrightarrow{\delta} \Omega^{k-1}_{(p-1)}$. By Proposition 18.8.2, $*$ maps this sequence isomorphically onto $\Omega^{2n-k-1}_{(c)} \to \Omega^{2n-k}_{(c)} \to \Omega^{2n-k+1}_{(c)}$, $c = p+n-k$, intertwining δ with $\pm d$ (the sign depending only on the degree). So $E^2_{p,k} \cong H^{2n-k}(\Omega^\bullet_{(p+n-k)}(Y), d)$, and Proposition 18.8.3 gives the result. □

18.9 The Hochschild homology of the symbol algebra

Theorem 18.9.1 (Brylinski–Getzler, Wodzicki)**.** *Let X be a compact manifold of dimension $n \geq 1$. The Hochschild homology of the algebra $\mathcal{A} = \Psi^{\mathbb{Z}}(X)/\Psi^{-\infty}(X)$ of formal symbols, with the chain spaces of Definition 18.5.1, is*

$$\mathrm{HH}_k(\mathcal{A}) \cong H^{2n-k}(S^*X) \oplus H^{2n-k-1}(S^*X), \qquad k \in \mathbb{N}_0.$$

More precisely, every class in $\mathrm{HH}_k(\mathcal{A})$ is represented by a cycle of order at most $k-n$, and the map assigning to such a cycle the class in $E^2_{k-n,k}$ of the homogeneous k-form $\varepsilon(c_{k-n})$ of degree $k-n$ is an isomorphism; composing with Theorem 18.8.4 gives the displayed isomorphism. In particular $\mathrm{HH}_k(\mathcal{A}) = 0$ for $k > 2n$ and $\mathrm{HH}_{2n}(\mathcal{A}) \cong H^0(S^*X)$.

Proof. By Theorem 18.8.4 the hypothesis of Proposition 18.6.1 holds, with $n = \dim X$. □

The degeneration of the spectral sequence at E^2, part of Proposition 18.6.1, thus comes down to the observation that the E^2 term is concentrated on the line $p = k-n$, where the order of a chain is tied to its degree; differentials d^r, $r \geq 2$, would have to connect different such lines (Figure 18.3).

Remark 18.9.2 (The residue trace). In degree 0, $\mathrm{HH}_0(\mathcal{A}) = \mathcal{A}/b(\mathcal{B}^{(1)})$, the quotient by (the closure of) the commutators, and its dual is the space of traces on $\mathcal{A}$. By Theorem 18.9.1, $\mathrm{HH}_0(\mathcal{A}) \cong H^{2n-1}(S^*X)$, and the isomorphism is computed as follows. Every $A \in \Psi^{\mathbb{Z}}(X)$ is congruent modulo $b(\mathcal{B}^{(1)})$ to some $A' \in \Psi^{-n}(X)$. The class of the function $a = \sigma_{-n}(A') \in \Omega^0_{(-n)}(Y)$ corresponds under $*$ to $a\,\mathrm{vol} \in \Omega^{2n}_{(0)}(Y)$, and writing $a\,\mathrm{vol} = \alpha'' \wedge dr/r$ as in Proposition 18.8.3, $\alpha'' = -\iota_E(a\,\mathrm{vol})|_{S^*X}$. Hence, if X is connected and $n \geq 2$, so that S^*X is connected (and it is always orientable) and $H^{2n-1}(S^*X) \cong \mathbb{C}$, the isomorphism $\mathrm{HH}_0(\mathcal{A}) \cong \mathbb{C}$ is, up to a constant factor,

$$A \longmapsto \int_{S^*X} \sigma_{-n}(A')\, \iota_E\Big(\frac{\omega^n}{n!}\Big).$$

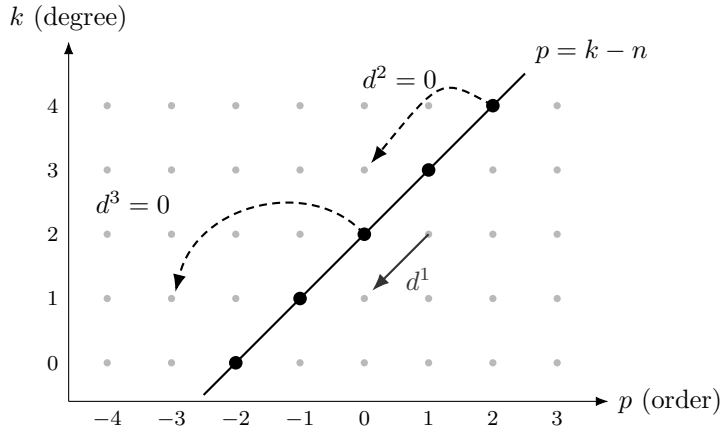


Figure 18.3: The spectral sequence of Section 18.6 for $n = \dim X = 2$. Gray dots are the positions (p,k) where $E^1_{p,k} = \Omega^k_{(p)}(Y)$ can be non-zero ($0 \le k \le 2n$). The differential $d^1 : E^1_{p,k} \to E^1_{p-1,k-1}$ moves along lines parallel to $p = k - n$. By Theorem 18.8.4 the E^2 term lives only on the line $p = k-n$ (black dots), while d^r, $r \ge 2$, maps (p,k) to $(p-r, k-1)$, off that line; hence $d^r = 0$.

This is the noncommutative residue of Wodzicki and Guillemin [13, 5]: up to normalization it is $\int_{S^*X} \sigma_{-n}(A)\, \iota_E(\omega^n/n!)$, computed for general A in local coordinates from the term of homogeneity $-n$ of the full symbol. Thus, for $n \ge 2$ and X connected, the residue is the unique trace on $\Psi^{\mathbb{Z}}(X)/\Psi^{-\infty}(X)$ up to a multiple. For $n = 1$, S^*X has two components over each component of X, and there are correspondingly two independent residue traces, one for each direction of ξ.

18.10 Exercises

Exercise 18.1. Let $\mathcal{A} = \mathbb{C}^N$ with componentwise multiplication. Show that $\mathcal{A}$ is a projective $\mathcal{A}^e$-module (the idempotent $\sum_j e_j \otimes e_j \in \mathcal{A}^e$ splits the multiplication map $\mathcal{A}^e \to \mathcal{A}$) and deduce $\mathrm{HH}_0(\mathcal{A}) = \mathbb{C}^N$, $\mathrm{HH}_k(\mathcal{A}) = 0$ for $k \ge 1$.

Exercise 18.2. Let $\mathcal{A} = M_N(\mathbb{C})$. Show that $\mathcal{A}$ is a projective $\mathcal{A}^e$-module, that $\mathrm{HH}_k(\mathcal{A}) = 0$ for $k \ge 1$, and that the trace induces an isomorphism $\mathrm{HH}_0(\mathcal{A}) \cong \mathbb{C}$.

Exercise 18.3. Using Lemma 18.7.1, compute δ on 1-forms and on 2-forms on Y, and verify directly that $\delta^2 = 0$ on 2-forms. (Hint: use the Jacobi identity for the Poisson bracket.)

Exercise 18.4. Let $X = \mathbb{S}^1$. Show that S^*X is the disjoint union of two circles and deduce from Theorem 18.9.1 that $\mathrm{HH}_0(\mathcal{A}) \cong \mathbb{C}^2$, $\mathrm{HH}_1(\mathcal{A}) \cong \mathbb{C}^4$, $\mathrm{HH}_2(\mathcal{A}) \cong \mathbb{C}^2$ and $\mathrm{HH}_k(\mathcal{A}) = 0$ for $k \ge 3$. Find explicit representatives of order -1 for the classes in $\mathrm{HH}_0(\mathcal{A})$, and the two traces.

Exercise 18.5. Show directly, using Theorem 18.8.4 and Proposition 18.6.1, that a class in $\mathrm{HH}_{2n}(\mathcal{A})$ is represented by a cycle of order at most n whose leading part corresponds to a locally constant multiple of ω^n on Y.

Notes

The polynomial case of the Hochschild–Kostant–Rosenberg theorem is proved here by the Koszul resolution instead of the direct reduction used in the original notes, which is kept, in

a sharpened form (Proposition 18.3.3), for the proof of Connes' theorem. That proof was incomplete in the original (the key exactness statement was left open); the argument above reduces it, via the module property of the boundaries (Lemma 18.3.4) and Borel's lemma, to the polynomial case. Connes' original proof (A. Connes, *Non-commutative differential geometry*, Publ. Math. IHÉS **62**, 1985) uses a topological Koszul resolution. The Eilenberg–Zilber and Künneth theorems used in Theorem 18.4.2 are standard; see J.-L. Loday, *Cyclic homology*, Springer, 1992, Chapters 1 and 4, which also covers H-unitality. The identification $\mathcal{C}^\infty(X)\hat{\otimes}\mathcal{C}^\infty(Y) = \mathcal{C}^\infty(X \times Y)$ is in F. Trèves, *Topological vector spaces, distributions and kernels*, Academic Press, 1967.

The semiclassical family of the original has been replaced by the equivalent filtration by order, and the degeneration and convergence of the spectral sequence are proved in Proposition 18.6.1 by the homogeneity argument. The complex $(\Omega^\bullet, \delta)$ and the symplectic star operator are due to Koszul and to J.-L. Brylinski (*A differential complex for Poisson manifolds*, J. Differential Geom. **28**, 1988). The main theorem is due to Brylinski–Getzler [1] and Wodzicki [13, 14]; these papers also treat cyclic homology and the algebras $\Psi^{-\infty}(X)$ and $\Psi^{\mathbb{Z}}(X)$, and they use a global symbol calculus (due to Widom) for which the product has the form (18.5.1).

Appendix A

Bounded operators on Hilbert space

This appendix collects the properties of bounded operators on complex Hilbert spaces which are used throughout the text: elementary facts about the algebra of bounded operators, compact operators, Fredholm operators and their index, and the Hilbert–Schmidt and trace-class ideals. Proofs are short and standard. Hilbert spaces H, H_1, $H_2, \dots$ are complex; where orthonormal bases are used they are assumed separable, so that bases are countable.

A.1 Bounded operators

Proposition A.1.1. *A linear map $A : H_1 \to H_2$ is continuous if and only if it is bounded, i.e. $\|A\| = \sup_{\|x\|\le 1} \|Ax\| < \infty$. The space $\mathcal{B}(H_1, H_2)$ of bounded operators is a Banach space with this norm, and $\|AB\| \le \|A\|\,\|B\|$ whenever the composite is defined. In particular $\mathcal{B}(H) = \mathcal{B}(H, H)$ is a Banach algebra.*

Proof. If A is continuous at 0 there is $\delta > 0$ with $\|Ax\| \le 1$ for $\|x\| \le \delta$, so $\|A\| \le \delta^{-1}$; conversely $\|Ax - Ay\| \le \|A\|\|x - y\|$. If (A_k) is Cauchy in norm then $A_k x$ is Cauchy for each x, and the limit Ax is linear with $\|A - A_k\| \to 0$. Finally $\|ABx\| \le \|A\|\|Bx\| \le \|A\|\|B\|\|x\|$. □

By the Riesz representation theorem each $A \in \mathcal{B}(H_1, H_2)$ has an adjoint $A^* \in \mathcal{B}(H_2, H_1)$, $(Ax, y) = (x, A^*y)$, with $(AB)^* = B^*A^*$, $A^{**} = A$ and $\|A^*\| = \|A\|$ (since $\|A\| = \sup_{\|x\|,\|y\|\le 1} |(Ax, y)|$). We also use the open mapping theorem: a bounded bijection between Banach spaces has bounded inverse.

Proposition A.1.2 (C^*-identity)**.** *For $A \in \mathcal{B}(H_1, H_2)$, $\|A^*A\| = \|AA^*\| = \|A\|^2$.*

Proof. $\|Ax\|^2 = (A^*Ax, x) \le \|A^*A\|\|x\|^2$, so $\|A\|^2 \le \|A^*A\| \le \|A^*\|\|A\| = \|A\|^2$. Apply this to A^* for the other identity. □

Proposition A.1.3 (Neumann series)**.** *If $A \in \mathcal{B}(H)$ and $\|A\| < 1$ then $\mathrm{Id} - A$ is invertible, with $(\mathrm{Id} - A)^{-1} = \sum_{k\ge 0} A^k$ and $\|(\mathrm{Id} - A)^{-1}\| \le (1 - \|A\|)^{-1}$. Consequently:*

(a) *$A - \lambda$ is invertible for $|\lambda| > \|A\|$, so the spectrum of A lies in $\{|\lambda| \le \|A\|\}$;*

(b) *the invertible elements of $\mathcal{B}(H_1, H_2)$ form an open set: if A is invertible and $\|E\| < \|A^{-1}\|^{-1}$ then $A + E = A(\mathrm{Id} + A^{-1}E)$ is invertible; and inversion is continuous.*

Proof. The series converges absolutely in the Banach space $\mathcal{B}(H)$ since $\|A^k\| \le \|A\|^k$, and $(\mathrm{Id}-A)\sum_{k<N} A^k = \sum_{k<N} A^k(\mathrm{Id}-A) = \mathrm{Id}-A^N \to \mathrm{Id}$. For (a) write $A-\lambda = -\lambda(\mathrm{Id}-A/\lambda)$. In (b) the inverse is $(A+E)^{-1} = \sum_k (-A^{-1}E)^k A^{-1}$, which depends continuously on E. □

The condition in (a) cannot be weakened to $|\lambda| \ge \|A\|$: for $A = \mathrm{Id}$, $\lambda = 1$.

Lemma A.1.4. *If $T \in \mathcal{B}(H)$ is self-adjoint then $\|T\| = \sup_{\|x\|=1} |(Tx, x)|$.*

Proof. Let M be the supremum; clearly $M \le \|T\|$. By polarization, for $\|x\| = \|y\| = 1$, $4\,\mathrm{Re}(Tx, y) = (T(x+y), x+y) - (T(x-y), x-y) \le M(\|x+y\|^2 + \|x-y\|^2) = 4M$. Taking $y = Tx/\|Tx\|$ gives $\|Tx\| \le M$. □

A.2 Compact operators

Definition A.2.1. An operator $K \in \mathcal{B}(H_1, H_2)$ is *compact* if the closure of the image of the unit ball is compact; equivalently, if every bounded sequence (x_n) has a subsequence along which Kx_n converges. The compact operators form a subspace $\mathcal{K}(H_1, H_2)$, and $\mathcal{K}(H) = \mathcal{K}(H, H)$.

Proposition A.2.2. (a) *If K is compact and A, B are bounded, then AKB is compact (when defined); $\mathcal{K}(H)$ is a two-sided ideal in $\mathcal{B}(H)$.*

(b) *K is compact if and only if $x_n \rightharpoonup x$ weakly implies $Kx_n \to Kx$ in norm.*

(c) *$\mathcal{K}(H_1, H_2)$ is the norm closure of the finite-rank operators.*

(d) *K is compact if and only if K^* is compact, if and only if K^*K is compact.*

(e) (Riesz) *If the closed unit ball of H is compact,* $\dim H < \infty$. *In particular if K is compact,* $\mathrm{Nul}(\mathrm{Id}-K)$ *is finite-dimensional.*

Proof. (a) Bounded operators map bounded sets to bounded sets and convergent sequences to convergent sequences.

(b) Suppose K compact and $x_n \rightharpoonup x$. Then (x_n) is bounded and $Kx_n \rightharpoonup Kx$, since $(Kx_n, z) = (x_n, K^*z)$. Every subsequence of (Kx_n) has a norm-convergent subsequence, whose limit must be Kx; hence $Kx_n \to Kx$. Conversely, every bounded sequence has a weakly convergent subsequence, which K maps to a convergent sequence.

(c) Finite-rank operators are compact, and a norm limit K of compact operators K_j is compact: given $\epsilon > 0$ choose j with $\|K - K_j\| < \epsilon$; then $K(B)$ lies within ϵ of the totally bounded set $K_j(B)$, so is totally bounded. Conversely, let K be compact and $\epsilon > 0$. Cover the compact set $\overline{K(B)}$ by finitely many balls $B(y_i, \epsilon)$, and let P be the orthogonal projection onto the span of the y_i. For $\|x\| \le 1$ choose i with $\|Kx - y_i\| < \epsilon$; then $\|Kx - PKx\| \le \|Kx - y_i\| + \|P(y_i - Kx)\| < 2\epsilon$. So $\|K - PK\| \le 2\epsilon$, and PK has finite rank.

(d) The adjoint of a finite-rank operator has finite rank and $\|K^* - F^*\| = \|K - F\|$, so the first equivalence follows from (c). If K is compact so is K^*K, by (a). Conversely, suppose K^*K is compact and $x_n \rightharpoonup x$, $\|x_n\| \le M$. Then $\|K(x_n - x)\|^2 = (K^*K(x_n - x), x_n - x) \le 2M\|K^*K(x_n - x)\| \to 0$ by (b), so K is compact by (b).

(e) If $\dim H = \infty$ an orthonormal sequence (e_n) has $\|e_n - e_m\| = \sqrt{2}$ for $n \ne m$, so has no convergent subsequence. On $\mathrm{Nul}(\mathrm{Id}-K)$, $x = Kx$, so the unit ball of this closed subspace is compact. □

Theorem A.2.3 (Spectral theorem)**.** *Let $T \in \mathcal{K}(H)$ be self-adjoint. Then there is an orthonormal sequence (e_k) (finite or infinite) of eigenvectors, $Te_k = \lambda_k e_k$, with real eigenvalues $|\lambda_1| \geq |\lambda_2| \geq \cdots > 0$ tending to zero if infinite, such that $H = \operatorname{Nul}(T) \oplus \overline{\operatorname{span}}\{e_k\}$ and $Tx = \sum_k \lambda_k (x, e_k) e_k$. Each non-zero eigenvalue has finite multiplicity.*

Proof. If $T \neq 0$, by Lemma A.1.4 there are unit vectors x_n with $(Tx_n, x_n) \to \lambda$, $|\lambda| = \|T\|$, λ real. Then $\|Tx_n - \lambda x_n\|^2 = \|Tx_n\|^2 - 2\lambda(Tx_n, x_n) + \lambda^2 \leq 2\lambda^2 - 2\lambda(Tx_n, x_n) \to 0$. Passing to a subsequence, $Tx_n \to y$, so $x_n \to y/\lambda$, and $e = y/\|y\|$ satisfies $Te = \lambda e$. The orthocomplement of e is invariant under T; repeating the argument there gives the sequence (e_k) with $|\lambda_k|$ non-increasing, the restriction of T to the orthocomplement of $e_1, \ldots, e_k$ having norm $|\lambda_{k+1}|$. If infinitely many $|\lambda_k| \geq \epsilon > 0$, then $\|Te_k - Te_j\|^2 = \lambda_k^2 + \lambda_j^2 \geq 2\epsilon^2$, contradicting compactness; so $\lambda_k \to 0$, and each eigenvalue occurs finitely often. On the orthocomplement W of all the e_k, which is invariant, $\|T|_W\| \leq |\lambda_k|$ for all k (or the process would have stopped), so $T|_W = 0$ and $W \subset \operatorname{Nul}(T)$; the reverse inclusion holds since $(x, e_k) = \lambda_k^{-1}(x, Te_k) = \lambda_k^{-1}(Tx, e_k)$. □

A.3 Fredholm operators

Definition A.3.1. An operator $A \in \mathcal{B}(H_1, H_2)$ is *Fredholm* if $\operatorname{Nul}(A)$ is finite-dimensional, $\operatorname{Ran}(A)$ is closed, and $\operatorname{Ran}(A)^\perp$ is finite-dimensional. Its *index* is $\operatorname{ind}(A) = \dim \operatorname{Nul}(A) - \dim \operatorname{Ran}(A)^\perp = \dim \operatorname{Nul}(A) - \dim \operatorname{Nul}(A^*)$.

Here $\operatorname{Ran}(A)^\perp = \operatorname{Nul}(A^*)$ for any bounded A, and if $\operatorname{Ran}(A)$ is closed it is the orthocomplement of $\operatorname{Nul}(A^*)$. Thus A is Fredholm if and only if A^* is, and $\operatorname{ind}(A^*) = -\operatorname{ind}(A)$.

Discussion. Fredholm operators are the operators that are "invertible up to finite dimensions". Their null space and cokernel can jump under perturbation, but only together, so that the difference of their dimensions, the index, is stable. Atkinson's theorem below makes this practical: to prove that an operator is Fredholm it suffices to invert it modulo compact operators, which is exactly what a parametrix does on a compact manifold.

Proposition A.3.2 (Generalized inverse)**.** *If $A \in \mathcal{B}(H_1, H_2)$ is Fredholm there is a unique $G \in \mathcal{B}(H_2, H_1)$ with*

$$GA = \operatorname{Id} - \pi_N, \qquad AG = \operatorname{Id} - \pi_C, \qquad G\pi_C = 0,$$

where π_N and π_C are the orthogonal projections onto $\operatorname{Nul}(A)$ and $\operatorname{Nul}(A^*)$. G is Fredholm with $\operatorname{ind}(G) = -\operatorname{ind}(A)$.

Proof. A restricts to a bounded bijection $\operatorname{Nul}(A)^\perp \to \operatorname{Ran}(A)$ between Hilbert spaces, which has bounded inverse A_0^{-1} by the open mapping theorem; set $G = A_0^{-1}(\operatorname{Id} - \pi_C)$. The identities are immediate and determine G. $\operatorname{Nul}(G) = \operatorname{Nul}(A^*)$ and $\operatorname{Ran}(G) = \operatorname{Nul}(A)^\perp$, so $\operatorname{ind}(G) = \dim \operatorname{Nul}(A^*) - \dim \operatorname{Nul}(A)$. □

Theorem A.3.3 (Atkinson)**.** *$A \in \mathcal{B}(H_1, H_2)$ is Fredholm if and only if there exists $B \in \mathcal{B}(H_2, H_1)$ such that $BA - \operatorname{Id}$ and $AB - \operatorname{Id}$ are compact. One may then take B such that both are of finite rank.*

Proof. If A is Fredholm, the generalized inverse is such a B, with finite-rank errors. Conversely suppose $BA = \operatorname{Id} - K$ and $AB = \operatorname{Id} - K'$ with K, K' compact. Then $\operatorname{Nul}(A) \subset \operatorname{Nul}(\operatorname{Id} - K)$, which is finite-dimensional by Proposition A.2.2(e). Let $L = \operatorname{Id} - K'$. On $\operatorname{Nul}(L)^\perp$ there is $c > 0$ with $\|Lx\| \geq c\|x\|$: otherwise there are unit vectors $x_n \in \operatorname{Nul}(L)^\perp$ with $Lx_n \to 0$; passing to a subsequence $K'x_n \to y$, so $x_n = Lx_n + K'x_n \to y$, with

$\|y\| = 1$, $y \in \mathrm{Nul}(L)^\perp$ and $Ly = 0$, a contradiction. Hence $\mathrm{Ran}(L)$ is closed, and $\mathrm{Ran}(L)^\perp = \mathrm{Nul}(\mathrm{Id} - K'^*)$ is finite-dimensional. Since $\mathrm{Ran}(A) \supset \mathrm{Ran}(AB) = \mathrm{Ran}(L)$, $\mathrm{Ran}(A)$ is the sum of the closed subspace $\mathrm{Ran}(L)$ and a finite-dimensional space, hence closed, and $\mathrm{Ran}(A)^\perp \subset \mathrm{Ran}(L)^\perp$ is finite-dimensional. □

Lemma A.3.4. *If $F \in \mathcal{B}(H)$ has finite rank then $\mathrm{Id} - F$ is Fredholm of index zero.*

Proof. Let $V = \mathrm{Ran}(F) + \mathrm{Ran}(F^*)$, which is finite-dimensional. Then F maps V into itself and vanishes on $V^\perp \subset \mathrm{Nul}(F)$, so $\mathrm{Id} - F = (\mathrm{Id}_V - F|_V) \oplus \mathrm{Id}_{V^\perp}$, and the index is that of an endomorphism of a finite-dimensional space, which is zero. □

Proposition A.3.5. *If $B \in \mathcal{B}(H_1, H_2)$ and $A \in \mathcal{B}(H_2, H_3)$ are Fredholm, then AB is Fredholm and* $\mathrm{ind}(AB) = \mathrm{ind}(A) + \mathrm{ind}(B)$.

Proof. If B', A' are parametrices as in Theorem A.3.3 then $B'A'$ is one for AB. Consider the sequence of finite-dimensional spaces and natural maps

$$0 \to \mathrm{Nul}\, B \hookrightarrow \mathrm{Nul}\, AB \xrightarrow{B} \mathrm{Nul}\, A \to H_2/\mathrm{Ran}\, B \xrightarrow{A} H_3/\mathrm{Ran}\, AB \to H_3/\mathrm{Ran}\, A \to 0,$$

where the third map is inclusion followed by the quotient map and the last is the quotient map ($\mathrm{Ran}\, AB \subset \mathrm{Ran}\, A$). It is exact: the kernel of B on $\mathrm{Nul}\, AB$ is $\mathrm{Nul}\, B$; the image of $\mathrm{Nul}\, AB$ under B is $\mathrm{Nul}\, A \cap \mathrm{Ran}\, B$, the kernel of the third map; the kernel of the map induced by A consists of the classes of v with $Av \in \mathrm{Ran}\, AB$, i.e. $v \in \mathrm{Nul}\, A + \mathrm{Ran}\, B$; and the kernel of the last map is $\mathrm{Ran}\, A/\mathrm{Ran}\, AB$. The alternating sum of the dimensions in an exact sequence vanishes, and $\dim H_2/\mathrm{Ran}\, B = \dim \mathrm{Nul}\, B^*$ etc. since the ranges are closed; this gives the formula. □

Question. Why must the range be closed in the definition? Is it not enough that $\mathrm{Nul}(A)$ and $\mathrm{Ran}(A)^\perp$ are finite-dimensional?

Answer. No. The diagonal operator $Ae_k = k^{-1}e_k$ on a space with orthonormal basis (e_k) is injective with dense range, so both spaces are zero; but it is compact, its range is not closed, and it has no bounded inverse on its range. Being compact, it is certainly not invertible modulo compact operators, so Atkinson's theorem fails for it. Closedness of the range is what the parametrix characterization and the stability of the index rely on. □

Theorem A.3.6 (Stability of the index)**.** *Let $A \in \mathcal{B}(H_1, H_2)$ be Fredholm.*

(a) *If G is a generalized inverse of A and $\|E\| < \|G\|^{-1}$, then $A + E$ is Fredholm and* $\mathrm{ind}(A + E) = \mathrm{ind}(A)$. *Thus the Fredholm operators form an open subset of $\mathcal{B}(H_1, H_2)$ on which the index is locally constant, hence constant along continuous paths.*

(b) *If $K \in \mathcal{K}(H_1, H_2)$ then $A + K$ is Fredholm and* $\mathrm{ind}(A + K) = \mathrm{ind}(A)$.

Proof. (a) Since $\|GE\| < 1$ and $\|EG\| < 1$, $\mathrm{Id} + GE$ and $\mathrm{Id} + EG$ are invertible, and

$$(\mathrm{Id} + GE)^{-1}G(A + E) = \mathrm{Id} - (\mathrm{Id} + GE)^{-1}\pi_N,$$
$$(A + E)G(\mathrm{Id} + EG)^{-1} = \mathrm{Id} - \pi_C(\mathrm{Id} + EG)^{-1},$$

using $GA = \mathrm{Id} - \pi_N$ and $AG = \mathrm{Id} - \pi_C$. By Theorem A.3.3, $A + E$ is Fredholm. The first identity, Lemma A.3.4 and Proposition A.3.5 (invertible operators having index zero) give $\mathrm{ind}(G) + \mathrm{ind}(A + E) = 0$, and $\mathrm{ind}(G) = -\mathrm{ind}(A)$.

(b) A parametrix for A is a parametrix for $A + tK$ modulo compact operators, so $A + tK$ is Fredholm for $t \in [0, 1]$, and by (a) the index is locally constant, hence constant, in t. □

Theorem A.3.7 (Riesz theory)**.** *Let $K \in \mathcal{K}(H)$ and $\mu \in \mathbb{C} \setminus \{0\}$. Then $\mu - K$ is Fredholm of index zero, so $\mu \in \operatorname{Spec}(K)$ if and only if μ is an eigenvalue. For each $\epsilon > 0$ there are only finitely many eigenvalues with $|\mu| \geq \epsilon$; thus $\operatorname{Spec}(K) \setminus \{0\}$ is countable and discrete in $\mathbb{C} \setminus \{0\}$. For each eigenvalue $\mu \neq 0$ the generalized eigenspace $\bigcup_N \operatorname{Nul}((\mu - K)^N)$ is finite-dimensional.*

Proof. $\mu - K = \mu(\mathrm{Id} - K/\mu)$ is Fredholm of index zero by Theorem A.3.6(b); if it is injective it is therefore surjective, hence invertible. Suppose there were infinitely many distinct eigenvalues μ_k with $|\mu_k| \geq \epsilon$ and eigenvectors x_k; these are linearly independent. Let $M_k = \operatorname{span}\{x_1, \dots, x_k\}$ and choose unit vectors $y_k \in M_k \ominus M_{k-1}$. Since $(K - \mu_k)M_k \subset M_{k-1}$, for $j < k$

$$Ky_k - Ky_j = \mu_k y_k + \big((K - \mu_k)y_k - Ky_j\big), \quad (K - \mu_k)y_k - Ky_j \in M_{k-1},$$

so $\|Ky_k - Ky_j\| \geq |\mu_k| \geq \epsilon$, contradicting compactness. Finally the spaces $N_j = \operatorname{Nul}((\mu - K)^j)$ increase; if $N_{j-1} \neq N_j$ for all j, choose unit $u_j \in N_j \ominus N_{j-1}$. For $i < j$, $Ku_j - Ku_i = \mu u_j - \big((\mu - K)u_j + Ku_i\big)$ with the bracket in N_{j-1}, so $\|Ku_j - Ku_i\| \geq |\mu|$, again a contradiction. So $N_j = N_{j_0}$ for $j \geq j_0$, and $N_{j_0} = \operatorname{Nul}((\mu - K)^{j_0})$ is finite-dimensional because $(\mu - K)^{j_0} = \mu^{j_0}(\mathrm{Id} - K')$ with K' compact. □

A.4 Hilbert–Schmidt and trace-class operators

Definition A.4.1. An operator $A \in \mathcal{B}(H_1, H_2)$ is *Hilbert–Schmidt* if $\|A\|_2^2 = \sum_j \|Ae_j\|^2 < \infty$ for an orthonormal basis (e_j) of H_1.

For orthonormal bases (e_j) of H_1 and (f_k) of H_2, $\sum_j \|Ae_j\|^2 = \sum_{j,k} |(Ae_j, f_k)|^2 = \sum_k \|A^* f_k\|^2$, so $\|A\|_2$ does not depend on the basis and $\|A^*\|_2 = \|A\|_2$. The Hilbert–Schmidt operators form a Hilbert space with inner product $\langle A, B\rangle_2 = \sum_j (Ae_j, Be_j)$, also independent of the basis, and:

- $\|A\| \leq \|A\|_2$ (expand x in a basis and use Cauchy–Schwarz);
- $\|SAT\|_2 \leq \|S\|\,\|A\|_2\,\|T\|$ for bounded S, T;
- Hilbert–Schmidt operators are compact: if P_N is the projection onto $\operatorname{span}\{e_1, \dots, e_N\}$, $\|A - AP_N\|_2^2 = \sum_{j>N} \|Ae_j\|^2 \to 0$.

Proposition A.4.2. *Let (M_i, μ_i), $i = 1, 2$, be σ-finite measure spaces with $L^2(M_i)$ separable. An operator $A : L^2(M_1) \to L^2(M_2)$ is Hilbert–Schmidt if and only if it is given by a kernel $k \in L^2(M_2 \times M_1)$, $Au(x) = \int k(x, y)u(y)\, d\mu_1(y)$, and then $\|A\|_2 = \|k\|_{L^2}$. The same holds for vector-valued functions.*

Proof. If (e_j) and (f_k) are orthonormal bases of $L^2(M_1)$ and $L^2(M_2)$, the products $f_k(x)\overline{e_j(y)}$ form an orthonormal basis of $L^2(M_2 \times M_1)$. If A has kernel $k \in L^2$, then $(Ae_j, f_k) = \langle k, f_k \otimes \bar{e}_j\rangle$, so $\|A\|_2^2 = \sum_{j,k} |(Ae_j, f_k)|^2 = \|k\|_{L^2}^2$. Conversely, if A is Hilbert–Schmidt then $k = \sum_{j,k} (Ae_j, f_k) f_k \otimes \bar{e}_j \in L^2$ is a kernel for A, as one checks on the e_j. □

Definition A.4.3. An operator $A \in \mathcal{B}(H)$ is of *trace class* if it is a finite sum $A = \sum_k B_k C_k$ of products of Hilbert–Schmidt operators $C_k : H \to H_k'$, $B_k : H_k' \to H$. The trace norm $\|A\|_1$ is the infimum of $\sum_k \|B_k\|_2 \|C_k\|_2$ over such representations.

The trace-class operators form a two-sided ideal in $\mathcal{B}(H)$, contained in the compact operators, with $\|SAT\|_1 \le \|S\|\,\|A\|_1\|T\|$ and $\|A\| \le \|A\|_1$; $\|\cdot\|_1$ is a norm, and the ideal is complete with respect to it. Every finite-rank operator is of trace class. (Using the polar decomposition $A = U|A|$ one can show that $\|A\|_1 = \sum_j (|A|e_j, e_j)$, the sum of the singular values of A.)

Proposition A.4.4. *If $A \in \mathcal{B}(H)$ is of trace class then $\operatorname{Tr}(A) = \sum_j (Ae_j, e_j)$ converges absolutely, is independent of the orthonormal basis (e_j), and $|\operatorname{Tr}(A)| \le \|A\|_1$. Moreover:*

(a) $\operatorname{Tr}(XY) = \operatorname{Tr}(YX)$ *if $X \in \mathcal{B}(H', H)$ and $Y \in \mathcal{B}(H, H')$ are Hilbert–Schmidt;*

(b) $\operatorname{Tr}(ST) = \operatorname{Tr}(TS)$ *if S is of trace class and T is bounded;*

(c) *for finite-rank A, $\operatorname{Tr}(A)$ is the trace of the matrix of A restricted to any finite-dimensional space containing $\operatorname{Ran}(A)$ and $\operatorname{Ran}(A^*)$;*

(d) *if $A = BC$ on $L^2(M)$, with B, C Hilbert–Schmidt with kernels b, c, then $\operatorname{Tr}(A) = \iint b(x,y)c(y,x)\,d\mu(y)\,d\mu(x)$.*

Proof. For $A = BC$, $\sum_j (BCe_j, e_j) = \sum_j (Ce_j, B^*e_j) = \langle C, B^*\rangle_2$, which converges absolutely by Cauchy–Schwarz with modulus at most $\|B\|_2\|C\|_2$, and is independent of the basis. Summing over the terms of a representation gives the first statements.

(a) For orthonormal bases (e_j) of H and (f_k) of H',

$$\operatorname{Tr}(XY) = \sum_j (Ye_j, X^*e_j) = \sum_{j,k} (Ye_j, f_k)(f_k, X^*e_j) = \sum_{j,k} (Xf_k, e_j)(e_j, Y^*f_k) = \operatorname{Tr}(YX),$$

the double sum converging absolutely by Cauchy–Schwarz.

(b) It suffices to take $S = BC$; by (a), $\operatorname{Tr}(BCT) = \operatorname{Tr}(CTB) = \operatorname{Tr}(TBC)$.

(c) Take a basis containing an orthonormal basis of that space; A vanishes on its orthocomplement.

(d) $\operatorname{Tr}(BC) = \langle C, B^*\rangle_2$, and the kernel of B^* is $\overline{b(y,x)}$, so by Proposition A.4.2 this is $\iint c(y,x)b(x,y)$. □

Remark A.4.5 (Schatten classes). For a compact operator A let $s_1 \ge s_2 \ge \cdots \ge 0$ be its singular values, the eigenvalues of $(A^*A)^{1/2}$. For $1 \le p < \infty$ the Schatten class $\mathcal{S}_p$ consists of those A with $\|A\|_p = (\sum_j s_j^p)^{1/p} < \infty$. These are two-sided ideals in $\mathcal{B}(H)$, Banach spaces under $\|\cdot\|_p$, with $\mathcal{S}_1$ the trace class, $\mathcal{S}_2$ the Hilbert–Schmidt operators, $\mathcal{S}_p \subset \mathcal{S}_q$ for $p \le q$, and Hölder's inequality $\|AB\|_r \le \|A\|_p\|B\|_q$ for $\frac{1}{r} = \frac{1}{p} + \frac{1}{q}$. By Lidskii's theorem the trace of an operator of trace class is the sum of its eigenvalues, counted with algebraic multiplicity.

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